Séminaire de géométrie algébrique du Bois Marie

SGA 7

Monodromy Groups in Algebraic Geometry

Volumes I and II

Plain reading text

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The tables of contents in these readers are newly generated from the transcribed text. They do not purport to transcribe the printed tables. The SGA 7 II supplied layers begin with Exposé X; its preceding printed title, publication and contents pages are unavailable in these layers. SGA 7 I preserves its transcribed general introduction but has no separately transcribed printed contents table; no absent printed title or publication pages have been reconstructed. Exposés III–V are not missing transcriptions: the source-authority record explains their historical status.

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Part I

SGA 7 I

INTRODUCTION

Let S be a scheme and $f : X \longrightarrow S$ a morphism of schemes. If f is proper and smooth, and if the prime number ℓ is invertible on S, the ℓ-adic cohomology groups $H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$ of the geometric fibres of f form an ℓ-adic local system on S. When f is only assumed proper, these groups are the fibres of an ℓ-adic sheaf $R^i f_* \mathbb{Z}_\ell$ on S. The theory of vanishing cycles relates the ramification of this sheaf on S to the singularities of f.

We shall consider only the case where S is a henselian trait (= the spectrum of a henselian discrete valuation ring). In practice, this is the essential case. I refer to (I 2.1) for a heuristic description of the theory. Let $S = \mathrm{Spec}(V)$, let $k(\bar{\eta})$ be an algebraic closure of the fraction field $k(\eta)$ of V, and let $k(\bar{s})$ be the corresponding algebraic closure of the residue field $k(s)$ ($k(\bar{s})$ is the residue field of the normalization $\bar{V}$ of V in $k(\bar{\eta})$). In the special case where X is proper and flat over S, of relative dimension n, and where f has only one point of non-smoothness $x \in X_s$, one defines $\mathrm{Gal}(\bar{\eta}/\eta)$-modules ϕ^i, of a purely local nature in a neighbourhood of x, vanishing for $i \notin [0, n]$ (I 4.2), and one constructs a long exact sequence of $\mathrm{Gal}(\bar{\eta}/\eta)$-modules

$$\cdots \longrightarrow H^i(X_{\bar{s}}, \mathbb{Z}_\ell) \xrightarrow{\mathrm{sp}} H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell) \longrightarrow \phi^i \longrightarrow H^{i+1}(X_{\bar{s}}, \mathbb{Z}_\ell) \longrightarrow \cdots$$

(the $\mathrm{Gal}(\bar{s}/s)$-module $H^i(X_{\bar{s}}, \mathbb{Z}_\ell)$ is regarded as a $\mathrm{Gal}(\bar{\eta}/\eta)$-module on which inertia acts trivially; sp is the specialization map).

Criteria are also given, valid for the moment only in characteristic 0, for the ϕ^i with small i to vanish (for example, if X is moreover locally a complete intersection, $\phi^i = 0$ for $i \neq n$ (I. 4.6)).

The monodromy theorem asserts that the action of the inertia subgroup I of $\mathrm{Gal}(\bar{\eta}/\eta)$ is quasi- I VI
unipotent: for $T \in I$, there exist integers $N > 0$, $M > 0$ such that the endomorphism $(T^M - I)^N$ of $H^i(X_{\bar{\eta}}, \mathbb{Z}_\ell)$ is zero. Two proofs of this theorem are given. The first (I.1.2), which is arithmetic in nature, applies as soon as the cohomology groups under consideration have reasonable finiteness properties. The key point is that, when $k(s)$ is of finite type over its prime subfield, the action of I is quasi-unipotent for every ℓ-adic representation of $\mathrm{Gal}(\bar{\eta}/\eta)$. The second proof, which is more geometric, requires purity and resolution of singularities: for the moment it is valid only in characteristic 0 or for an H^1. It provides valuable information about the nilpotence exponent N ($N \leq i+1$ for an H^i). As will be seen later, over $\mathbb{C}$ it is well suited to comparison with transcendental theory.

Applied to the H^1 of abelian varieties, i.e. to their Tate module, these results make it possible to study their reduction modulo p. Thus two proofs are given of the stable reduction theorem for abelian varieties, according to which, after ramification, the connected special fibre of the Néron model is an extension of an abelian variety by a torus. The first (I. 6) follows the method of the arithmetic proof of the monodromy theorem. The second (IX 3.6) rests on a much finer analysis of the Néron model and of its polarization properties.

Grothendieck's Exposés I through V were not written up. They were summarized in one Exposé I. The results stated there are proved succinctly, but essentially completely.

Exposé II applies the method of Lefschetz pencils and (I 5.3) to the study of the fundamental group. If S is a surface over an algebraically closed field k of characteristic exponent p, it is shown that the profinite group $\pi_1^{(p)}(S, s)$, the completion away from p of the fundamental group, admits a finite pro-(p) presentation.

As explained above, Exposés III through V do not exist.

Exposé VI contains, with a few additions, Schlessinger's theory of deformations. Care is taken I VII
there to account for the infinitesimal automorphisms of the objects being classified, and not to pass too abruptly to the functor of isomorphism classes of objects. It also gives a new construction of $\underline{\mathrm{Ext}}^i(L_{X/S}, -)$ ($L_{X/S}$ being the relative cotangent complex of X/S). In the rest of the seminar this Exposé is used chiefly through its study of deformations of ordinary quadratic singularities.

Exposés VII and VIII are devoted to the theory of biextensions. This theory plays an essential role in Exposé IX, in expressing what becomes of a polarization when one passes from an abelian

variety to its Néron model. Exposé IX contains the proof of the stable reduction theorem for abelian varieties, as well as various applications.

The continuation of this seminar, SGA 7 II, by P. Deligne and N. Katz, will appear later.

Bures-sur-Yvette, May 1972,
P. DELIGNE

SGA 7 I1

Exposé I

Summary of the First Lectures by A. Grothendieck

written up by P. Deligne

Contents

0. Preliminaries
1. Arithmetic proof of the monodromy theorem
2. Vanishing cycles
3. Geometric proof of the monodromy theorem
4. Vanishing criteria for the sheaves of vanishing cycles
5. Action of monodromy on the π_1
6. Appendix by P. Deligne: arithmetic proof of the stable reduction theorem

0. Preliminaries

0.0. Terminology. Recall the following definitions:

(0.0.1) trait: the spectrum of a discrete valuation ring. The generic and closed points of a trait are often denoted by η and s.

(0.0.2) strictly local: for a ring, local henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.0.3) geometric point of S (in this exposé): a morphism $\bar{x} : \mathrm{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) \underset{\text{dfn}}{=} \bar{k}$ is a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, a generic geometric point of S (that is, one with image η) is often denoted by $\bar{\eta}$. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S centered at s.

(0.0.4) strict localization of X at the geometric point $\bar{x}$: SGA 4 VIII 4.4. I≈2

(0.0.5) essentially of finite type over S: for a local (respectively, henselian local; respectively, strictly local) S-scheme, the spectrum of a local ring (respectively, the henselization of a local ring; respectively, a strict localization) of a scheme of finite type over S.

(0.0.6) $F \longmapsto F(n)$: Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, that is, if there exist $N \geqslant 1$ and $M \geqslant 1$ such that $(T^N - 1)^M = 0$.

0.1. Review of ℓ-adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p, and let ℓ be a prime different from p.

The ℓ-adic cohomology groups of X were defined in SGA 4 and SGA 5.

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i-th cohomology group of the ètale site X_{et} of X with coefficients in $\mathbb{Z}/\ell^n$;

b) by definition,

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n);$$

c) by definition,

$$H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The ℓ-adic cohomology groups of X with proper supports are defined in the same way from the groups $H^i_c(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases, one knows how to prove that these groups $H^i(X)$ have reasonable finiteness properties:

(0.1.1) for cohomology with proper supports;

(0.1.2) when X is proper over k;

(0.1.3) when X is smooth over k (by (0.1.1) and Poincaré duality, SGA 4 XVIII);

(0.1.4) when Hironaka's resolution of singularities is available for schemes of finite type over k of dimension at most $\dim(X)$;

(0.1.5) when $i = 0, 1$.

I≈3

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are finitely generated, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and (except perhaps in case (0.1.5)) there are split short exact sequences

$$0 \longrightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \longrightarrow H^i(X, \mathbb{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \longrightarrow 0.$$

Moreover, in each case the proof of the finiteness theorem also proves a generic constructibility theorem: if k is the field of fractions of an integral noetherian scheme S, and X is the generic fibre of a morphism $f_1 : X_1 \longrightarrow S$ of finite type, then there exists a nonempty open subset $U \subset S$ such that over U the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and their formation commutes with every base change.

0.2. We no longer suppose that k is algebraically closed. Let $\bar{k}$ be an algebraic closure of k (or a separable closure, which amounts to the same thing here). We consider the "geometric" cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extension of scalars from k to $\bar{k}$. By transport of structure, the Galois group $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$-cohomology, in any of the cases (0.1.1)–(0.1.5), this gives continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Review of traits (0.0.1)

For the proofs, we refer to [4]. Let S be a henselian trait, and let η, s, V, $\bar{V}$, $\bar{\eta}$, and $\bar{s}$ be as in (0.0.1) and (0.0.3). Let p denote the characteristic exponent of $k(s)$, and let $\mathrm{Gal}(\bar{\eta}/\eta)$ (respectively, $\mathrm{Gal}(\bar{s}/s)$) denote the Galois group $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (respectively, the corresponding group for $\bar{s}/s$). By transport of structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ acts on $k(\bar{s})$; hence there is a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ onto $\mathrm{Gal}(\bar{s}/s)$ whose kernel is the inertia group I. The subfield of $k(\bar{\eta})$ fixed by I is a maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro-p subgroup P. The quotient I/P is the tame inertia group. Canonically,

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) \underset{\text{dfn}}{=} \widehat{\mathbb{Z}}(1)(k(\bar{s})).$$

If ϕ_n is the canonical map from I/P to $\mu_n(\bar{k})$ and u is an element of $\bar{V}$ of valuation $1/n$, then, I≈4 for $\sigma \in I$,

$$\sigma(u) \equiv \phi_n(\sigma)\, u \qquad \text{(modulo elements of valuation greater than } 1/n).$$

In summary,

$$\begin{aligned} \mathrm{Gal}(\bar{\eta}/\eta) &\supset I \supset P, \\ \mathrm{Gal}(\bar{\eta}/\eta)/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ a pro-}p \text{ group} \quad (\{e\} \text{ if } p = 1), \end{aligned} \tag{0.3.1}$$

and the action (by inner automorphisms in $\mathrm{Gal}(\bar{\eta}/\eta)$) of $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. More generally, let S be strictly local and regular, let D be a regular divisor in S, and let p' be the characteristic exponent of the generic point of D. Let $\bar{\eta}$ be a geometric generic point of S, and let $\bar{s}$ be the resulting geometric point centered at the closed point s. By Abhyankar's lemma, the fundamental group $\pi_1(S - D, \bar{\eta})$ has a filtration analogous to (0.3.1):

$$\begin{aligned} \pi_1(S - D, \bar{\eta}) &\supset I \supset P, \\ \pi_1(S - D, \bar{\eta})/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ with no quotient of order prime to } p'. \end{aligned} \tag{0.4.1}$$

The action by inner automorphisms of $\pi_1(S-D,\bar\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar s/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar s))$.

0.5. Néron desingularization

The method of Néron desingularization gives the following result.

Lemma 0.5.1. Let $S \longrightarrow S_o$ be a morphism of henselian traits, with $S = \mathrm{Spec}(V)$ and $S_o = \mathrm{Spec}(V_o)$, and let π be a uniformizer of V_o. Suppose that π is a uniformizer of V, and that the extensions $k(s)/k(s_o)$ and $k(\eta)/k(\eta_o)$ are separable. Then V is a filtered direct limit of smooth henselian V_o-algebras B_i essentially of finite type over V_o.

To construct the desired V_o-algebras, one takes $V_o \subset B \subset V$, with B of finite type, and desingularizes $\mathrm{Spec}(B)$ by Néron's method ([1], § 4). I≈5

Let $S = \mathrm{Spec}(V)$ be a henselian trait.

(0.5.2) If S is of equal characteristic, with uniformizer π, one may apply (0.5.1) with $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. The extension s/s_o is separable because $\mathbf{F}_p$ is perfect; the extension η/η_o is separable because π, being a uniformizer, is not a p-th power (if $p \neq 1$), and hence belongs to a p-basis. It follows that S is a limit of spectra of smooth henselian $\mathbf{F}_p[\pi]_{(\pi)}$-algebras essentially of finite type.

(0.5.3) If S is complete and of mixed characteristic, with perfect residue field k, then V is an Eisenstein extension of the ring of Witt vectors (that is, the Cohen ring) $W(k)$:

$$V \simeq W(k)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i\pi^i\right).$$

If the a_i are perturbed slightly, one obtains an isomorphic extension. We may therefore suppose that the a_i belong to $W(k')$, where k' is of finite type over $\mathbf{F}_p$. Let k_o be the perfect closure of k' in k, and set

I 4

$$V_o = W(k_o)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i\pi^i\right).$$

Lemma 0.5.1 applies to V/V_o, and the residue field of V_o is a radicial extension of a field of finite type over $\mathbf{F}_p$.

1. Arithmetic proof of the monodromy theorem I≈5

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime different from the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in $\mathrm{GL}(n,\mathbb{Q}_\ell)$. Suppose that the following condition holds:

$(*_\ell)$ No finite extension of $k(s)$ contains all roots of unity of ℓ-power order.

Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for every $\sigma \in I_1$. I≈6

Condition $(*_\ell)$ is automatically satisfied when $k(s)$ is of finite type over the prime field (or is radicial over such a field).

The following theorem follows immediately from 1.1.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η, and let H be a cohomology space of $X_{\bar\eta}$ with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If condition $(*_\ell)$ is satisfied, the action on H of every element of I is quasi-unipotent.

The limit arguments of 0.5 allow us to dispense with hypothesis $(*_\ell)$.

Variant 1.3. Condition $(*_\ell)$ in 1.2 is superfluous.

Let H'_o be a $\mathbb{Z}_\ell$-cohomology space of $X_{\bar\eta}$ giving rise to H, and put $H_o = H'_o/\text{torsion}$. By hypothesis, $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and the action of $\mathrm{Gal}(\bar\eta/\eta)$ is induced by

$$\rho : \mathrm{Gal}(\bar\eta/\eta) \longrightarrow \mathrm{GL}(H_o).$$

Let $I_{\ell'}$ be the largest prime-to-ℓ subgroup of I. Its image in $\mathrm{GL}(H_o)$ is an ℓ-adic Lie group of order prime to ℓ, and is therefore finite.

Make an extension of traits $S' \longrightarrow S$. This does not change H_o (SGA 4 XVI 6.6), and $\rho(I')$ has finite index in $\rho(I)$, because I'/P' has finite index in I/P, by what precedes. It therefore suffices to prove the theorem over S'; we may suppose that

a) $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on $H_o/\ell H_o$;

b) if S is of mixed characteristic, then S is complete with perfect residue field (EGA O_{III} 10.3.1).

Put $S = \mathrm{Spec}(V)$, define V_o as in (0.5.2) and (0.5.3), and apply (0.5.1). We have

I≈7

$$V = \varinjlim_i B_i.$$

Here $S_i = \mathrm{Spec}(B_i)$ is essentially of finite type and smooth over V_o. Let $D_i \subset S_i$ be defined by the equation $\pi = 0$. For sufficiently large i, H_o comes from a twisted constant $\mathbb{Z}_\ell$-sheaf $\mathcal{H}_i$ on $S_i - D_i$ (by "generic constructibility," (0.1)), and $\mathcal{H}_i/\ell\mathcal{H}_i$ is constant.

The filtrations (0.3.1) and (0.4.1) give rise to commutative diagrams:

(1.3.1)
$$\begin{array}{ccccc} \mathrm{Gal}(\bar\eta/\eta) & \supset & I & \supset & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar\eta) & \supset & I_i & \supset & P_i \end{array} \qquad \text{and} \qquad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \\ \downarrow & & \| \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \end{array}$$

The representation on H_o of $\mathrm{Gal}(\bar\eta/\eta)$ is induced by an action of $\pi_1(S_i - D_i, \bar\eta)$ on H_o that is trivial on $H_o/\ell H_o$. The group

$$\mathrm{Ker}\big(\mathrm{GL}(H_o) \longrightarrow \mathrm{GL}(H_o/\ell H_o)\big)$$

is a pro-ℓ group. Thus P_i acts trivially, and we apply the following variant of 1.1.

Variant 1.4. Retain the notation of 0.4, and let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in $\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies $(*_\ell)$ and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

2.1. Let S be a strictly local trait, and retain the notation of 0.3. Thus, by hypothesis, $s = \bar s$. Let $f : X \longrightarrow S$ be a scheme of finite type over S. The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar\eta}$, as well as the action of the inertia group on the latter, using information about f that is local on X.

A key result of this kind is the specialization theorem (SGA 4 XVI 2.2), according to which X_s and $X_{\bar\eta}$ have the same cohomology when f is proper and smooth.

In the transcendental theory, if $f : X \longrightarrow D$ is a proper morphism from an analytic space X to the disk D, vanishing cycles arise as follows:

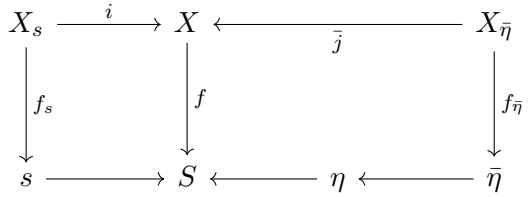
a) After shrinking D if necessary, one may suppose that X is a topological fibre bundle over the punctured disk D^* (Thom's isotopy theorem). In particular, the fibres

X_t $(t \in D^*)$ are all isomorphic. I≈8

b) It is not unreasonable to picture the special fibre X_s as obtained from "the" general fibre X_t by contracting polyhedra in X_t; one would then have a map $\pi : X_t \longrightarrow X_s$. The method of vanishing cycles consists in studying the difference between the cohomology of X_s and that of X_t by means of the Leray spectral sequence for π.

The geometric theory explained below is very close to this transcendental theory: the general point $t \in D$ must be replaced by $\bar\eta$, the geometric generic point of S. Conversely, the transcendental theory can be modeled on the geometric theory by replacing t or $\bar\eta$ with the universal cover $\widetilde{D}^*$ of D^*, and X_t or $X_{\bar\eta}$ with $X \times_D \widetilde{D}^*$. This eliminates many ε's and η's, but sometimes loses information.

2.2. Fix a torsion ring Λ in which p is invertible (for example, $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ prime to p). Consider the morphisms



The sheaves of vanishing cycles ψ^i on X_s are

$$\psi^i = i^* R^i \bar{j}_* \Lambda. \tag{2.2.1}$$

If f is proper, then (SGA 4 XII 5.5)

$$i^* : H^p(X, R^q \bar{j}_* \Lambda) \xrightarrow{\sim} H^p(X_s, i^* R^q \bar{j}_* \Lambda), \tag{2.2.2}$$

and the Leray spectral sequence for $\bar{j}$ becomes

$$H^p(X_s, \psi^q) \Longrightarrow H^{p+q}(X_{\bar{\eta}}, \Lambda). \tag{2.2.3}$$

When nothing happens at infinity, this spectral sequence can sometimes be obtained without a properness hypothesis. That some hypothesis at infinity is necessary is already clear from the trivial case $X_s = \varnothing$.

The definitions immediately give: I≈9

Proposition 2.3. Let $\bar{x}$ be a geometric point of X_s (for example, a closed point), let $X_{(\bar{x})}$ be the strict henselization of X at $\bar{x}$, and let $X_{(\bar{x})\bar{\eta}}$ be the geometric generic fibre of the projection $X_{(\bar{x})} \longrightarrow S$. Then

$$\psi^i_{\bar{x}} = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda).$$

Corollary 2.4. Wherever f is smooth, $\psi^i = 0$ for $i > 0$ and $\psi^0 = \Lambda$.

This is SGA 4 XV 2.1.

The group $\mathrm{Gal}(\bar{\eta}/\eta)$ (which, by hypothesis, equals the inertia group) acts on the ψ^i by transport of structure, and

Scholium 2.5. The spectral sequence (2.2.3) is $\mathrm{Gal}(\bar{\eta}/\eta)$-equivariant.

Variant 2.6. Put $\bar{S} = \mathrm{Spec}(\bar{V})$ and $\bar{X} = X \times_S \bar{S}$. The scheme $X_{\bar{\eta}}$ is the complement of $X_s = X_{\bar{s}}$ in $\bar{X}$. Put

$$\phi^i = \underline{H}^{i+1}_{X_s}(\bar{X}, \Lambda)$$

(the local-cohomology sheaves). There is an exact sequence of sheaves

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^0 \longrightarrow \phi^0 \longrightarrow 0,$$

and, for $i > 0$, isomorphisms $\psi^i \xrightarrow{\sim} \phi^i$. If f is smooth, the ϕ^i vanish.

Put $\phi^i_{\mathrm{gl}} = H^{i+1}_{X_s}(\bar{X}, \Lambda)$, the global analogue of ϕ^i. There is a spectral sequence

$$H^p(X_s, \phi^q) \Longrightarrow \phi^{p+q}_{\mathrm{gl}}, \tag{2.6.1}$$

and a long exact cohomology sequence which, when f is proper, becomes

$$\cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar{\eta}}, \Lambda) \longrightarrow \phi^i_{\mathrm{gl}} \longrightarrow \cdots . \tag{2.6.2}$$

Variant 2.7. Let K_t be the maximal tamely ramified extension of K in $\bar{K}$. Put $X_t = X \times_S \mathrm{Spec}(K_t)$, and let $\bar{j}_t$ be the projection from X_t to X. Define tame variants of the ψ^i by

I≈10

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda.$$

Since P is a pro-p group and p is invertible in Λ, one obtains

$$H^*(X_t, \Lambda) = H^*(X_{\bar{\eta}}, \Lambda)^P, \tag{2.7.1}$$

$$\psi_t^i = (\psi^i)^P. \tag{2.7.2}$$

When f is proper, the Leray spectral sequence for $\bar{j}_t$ is obtained from (2.2.3) by passing to P-invariants:

$$H^p(X_s, \psi_t^q) \Longrightarrow H^{p+q}(X_t, \Lambda) = H^{p+q}(X_{\bar{\eta}}, \Lambda)^P. \tag{2.7.3}$$

3. Geometric proof of the monodromy theorem

Let Λ be as in 2.2.

Conjecture 3.1 (purity). Let A be an excellent regular strictly local ring (0.0.2) and let D be a regular divisor in $\mathrm{Spec}(A)$ defined by a parameter x. Put $U = \mathrm{Spec}(A) - D = \mathrm{Spec}(A[1/x])$. Then

$$H^q(U, \Lambda) = \begin{cases} \Lambda & \text{if } q = 0, \\ \Lambda(-1) & \text{if } q = 1 \quad \text{(notation (0.0.6))}, \\ 0 & \text{if } q > 1. \end{cases} \tag{3.1.1}$$

Here are some cases in which this conjecture is known (SGA 4 XIX): for $q = 0$ or 1; when A has characteristic 0; when A has equal characteristic p and resolution of singularities is available in characteristic p in dimensions $\leqslant \dim(A)$; or when $(\mathrm{Spec}(A), D)$ is a smooth pair (the relative purity theorem, SGA 4 XVI 3.7).

When purity is available and $D = \sum_{i \in B} D_i$ is a normal-crossings divisor in $\mathrm{Spec}(A)$ defined by part of a regular system of parameters, the cohomology of $U = \mathrm{Spec}(A) - D$ is given by

$$\begin{aligned} &H^0(U, \Lambda) = \Lambda, \\ &H^1(U, \Lambda) = \Lambda(-1)^B, \\ &H^q(U, \Lambda) = \textstyle\bigwedge^q H^1(U, \Lambda) \qquad \text{(via the cup product)}. \end{aligned} \tag{3.1.2}$$

These formulas are identical to those which, over $\mathbb{C}$, give the cohomology of $D^{*B} \times D^k$.

3.2. Let S be a strictly local trait, and retain the notation of (0.0.1) and (0.0.3). Let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that X is regular, that X_η is smooth, and that X_s is a normal-crossings divisor in X. We shall suppose that this divisor is, globally and not merely locally for the étale topology, a sum of regular divisors: I 11

$$(X_s)_{\mathrm{red}} = \sum_{i \in B} D_i$$

(the general case can be recovered by localization). Denote their multiplicities by m_i, so that $X_s = \sum_{i \in B} m_i D_i$.

Let x be a point of X_s, let $\bar{x}$ be a geometric point of X lying over x (for example, x may be closed and equal to $\bar{x}$), put $C = \{i \mid x \in D_i\}$, and let K be the complex concentrated in degrees 0 and -1

$$K : \mathbf{Z}^C \xrightarrow{d} \mathbf{Z}, \qquad d((n_j)) = \sum_j m_j n_j.$$

Theorem 3.3 (uses purity). The groups of tame vanishing cycles $\psi^i_{t\bar{x}}$ are as follows:
(i) There is a canonical isomorphism of graded Λ-algebras

$$\psi^*_{t\bar{x}} \simeq \bigwedge\nolimits^*(H_1(K) \otimes \Lambda(-1)) \otimes_\Lambda \psi^0_{t\bar{x}}.$$

(ii) $\psi^0_{t\bar{x}}$ is the Λ-algebra Λ^E, where E is a set on which the tame inertia group $I_t = I/P = \widehat{\mathbf{Z}}(1)(k(s))$ acts transitively, and whose cardinality is the largest prime-to-p divisor of $\#H_0(K)$.

Here is the method of proof.

a) Let $X_{(\bar{x})}$ be the strict localization of X at $\bar{x}$, let $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (the complement of a normal-crossings divisor), let t_j be a local equation for D_j, put

$$X_n = X_{(\bar{x})}\big[(t_j^{1/n})\big] \qquad ((n,p) = 1),$$

which is a regular scheme, and let U_n be the inverse image of U in X_n. By (3.1.2),

$$H^q(U_n, \Lambda) = \bigwedge\nolimits^q(\Lambda(-1)^C).$$

For $m = nd$, the pullback map from $H^q(U_n, \Lambda)$ to $H^q(U_m, \Lambda)$ is multiplication by d^q. Putting $\widetilde{U} = \varprojlim_{(n,p)=1} U_n$, one therefore has I≈12

$$\text{(3.3.1)} \qquad \begin{cases} H^0(\widetilde{U}, \Lambda) = \Lambda, \\ H^q(\widetilde{U}, \Lambda) = \varinjlim_n H^q(U_n, \Lambda) = 0 \quad \text{if } q \neq 0. \end{cases}$$

b) $\widetilde{U}$ is a pro-covering of U with group $\widehat{\mathbf{Z}}(1)(k(s))^C$. By definition,

$$\psi^*_{t\bar{x}} = H^*(U_{\eta_t}, \Lambda).$$

$\widetilde{U}$ is a pro-covering, with group $H_1(K) \otimes \widehat{\mathbf{Z}}(1)(k(s))$, of each connected component of U_{η_t}, and these components are permuted transitively by the tame inertia group. Formula (3.3.1) and the Hochschild–Serre spectral sequence now give the formulas in 3.3.

Pictorially, one may say that $U_{\bar{\eta}}$ is the fibre of a morphism δ from the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s))^C, 1)$ represented by U to the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s)), 1)$ represented by η; the morphism δ induces

$$d : (n_j) \longmapsto \sum_j m_j n_j : \widehat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \widehat{\mathbf{Z}}(1)(k(s)).$$

Corollary 3.4 (uses purity). Suppose that f is proper. Let N be the least common multiple of the multiplicities m_j ($j \in B$). Let $q \geqslant 0$, and let q' be the minimum of q and the greatest number of divisors D_j passing through one point. For T in the tame inertia group $I_t = I/P$, one has

$$(T^N - 1)^{q'+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \Lambda)^P.$$

This follows at once from 3.3, Scholium 2.5, and Variant 2.7.

The required purity results are available for H^1, in characteristic 0, or when S is essentially of finite type over a field.

Theorem 3.5. Let C be a smooth projective curve over the fraction field $k(\eta)$ of a discrete valuation ring V. The inertia group I acts on $H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell)$ by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0 \quad \text{on} \quad H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell).$$

We reduce to the case in which V is strictly local. The assertion is that there exists an open subgroup I_1 of I such that $(T-1)^2 = 0$ on H^1 for $T \in I_1$.

We may extend the scalars of $k(\eta)$ to a finite extension, replacing V by its normalization in that I≈13
extension. Since the image of the pro-p group P in an ℓ-adic Lie group $\mathrm{GL}(n, \mathbf{Q}_\ell)$ is finite, we may in particular suppose that P acts trivially.

Let C_o be a minimal model of C over V (for the existence of C_o, which is based on Abhyankar's theorem, see the discussion in [3], § 2, p. 87; one could instead reduce to the case where V is excellent by passing to its completion and invoking SGA 4 XVI 1.6).

Let C_1 be a regular model of C, obtained from C_o by blowups, whose special fibre is a normal-crossings divisor. The arguments of 3.3 and 3.4 apply to C_1, ψ_t^0, ψ_t^1, and

$$H^1(C_{\bar\eta}, \mathbb{Z}/\ell^n)^P = H^1(C_{\bar\eta}, \mathbb{Z}/\ell^n),$$

with N independent of n; Theorem 3.5 follows.

Remark 3.6. Let A be an abelian variety over $k(\eta)$. Since A is a quotient of a Jacobian, it follows again that the action of I on $H^1(A_{\bar\eta}, \mathbb{Q}_\ell)$, or on $T_\ell(A)$, is quasi-unipotent of level 2.

Remark 3.7. The action of I on $H^1(X_{\bar\eta}, \mathbb{Q}_\ell)$ is in fact quasi-unipotent of level 2 for every scheme X of finite type over η.

Theorem 3.5 (or Remark 3.6) is the basis for the proof of the stable reduction theorem for abelian I 12
varieties that will be given in Exposé IX, § 3.

3.8. It is clear that whenever purity and resolution of singularities are available (for example, I 13
in characteristic 0), the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. Thus, in equal characteristic 0, if X is proper and smooth over η and T belongs to an open subgroup I_1 of I, one has

$$(T-1)^{q+1} = 0 \quad \text{on} \quad H^q(X_{\bar\eta}, \mathbb{Z}_\ell) \qquad (T \in I_1 \subset I).$$

4. Vanishing criteria for the sheaves of vanishing cycles

4.1. Let S be a strictly local trait and let $f : X \longrightarrow S$ be a scheme of finite type over S. Put $n = \dim(X_\eta)$.

If X' is the scheme-theoretic closure of X_η in X, then:

a) On $X_s - X'_s$, one has $\phi^{-1} = \Lambda$, and the other ϕ^i vanish;

b) on X'_s, the ϕ^i and ψ^i relative to X/S or to X'/S coincide, and $\phi^{-1} = 0$. I≈14

Since X' is flat over S, this often permits a reduction to the case in which X is flat over S.

Theorem 4.2.

(i) One has $\psi^i = 0$ for $i > n$.

(ii) More precisely, let x be a point of X_s whose closure has dimension k ($k = \mathrm{trdeg}_{k(s)}\, k(x)$), and let $\bar x$ be a geometric point lying over x. Then

$$\psi^i_{\bar x} = 0 \quad \text{for} \quad i > n-k, \qquad \text{i.e.} \qquad d(\psi^i) \leqslant n-i \quad (\text{notation of SGA 4 XIV 2.1}).$$

Indeed,

$$\psi^i_{\bar x} = H^i(X_{(\bar x)\bar\eta}, \Lambda),$$

and $X_{(\bar x)\bar\eta}$ is an inverse limit of affine schemes of dimension at most n over $k(\bar\eta)$. Assertion (i) therefore follows from the affine Lefschetz theorem (SGA 4 XIV 3.2) by passage to the limit.

To prove (ii), we may suppose that X is flat over S (4.1) and that there is a quasi-finite flat S-morphism

$$g : X \longrightarrow \mathbb{A}^k_S$$

such that $g(x)$ is the generic point of $\mathbb{A}^k_s$. Let S' be the strict localization of $\mathbb{A}^k_S$ at $\bar x$. Then:

a) $S'_{\bar\eta}$ is the spectrum of a field, and $\mathrm{Gal}(\bar\eta'/s'_{\bar\eta})$ is a p-group Q;

b) by (i), applied to g,

$$H^i(X_{(\bar x)\bar\eta'}, \Lambda) = 0 \qquad \text{for } i > n-k;$$

c) one has

$$H^i(X_{(\bar{x})\bar{\eta}}, \Lambda) = H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda)^Q,$$

which proves the theorem.

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension at most d, the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

4.4. To prove, in certain cases, that the ϕ^i vanish for small i, we shall use local duality, either directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities à la Hironaka is available in the dimensions under consideration.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \longrightarrow S$ be a morphism of finite type and let x be a closed point of X_s. Suppose that: 115

(a) x is an isolated non-smooth point in its fibre;

(b) X has relative geometric depth at least n at x: in a neighbourhood of x, the scheme X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x, with $n \leqslant N - k$.

Then $\phi^i_x = 0$ for $i < n$.

Let X_x be the strict localization of X at x, and put $V = X_x - \{x\}$. For every trait S' finite over S, put similarly

$$X'_x = X_x \times_S S', \qquad V' = V \times_S S'.$$

Let $\bar{S}$ be the normalization of S in $\bar{\eta}$, and put

$$\bar{X}_x = X_x \times_S \bar{S}, \qquad \bar{V} = V \times_S \bar{S}.$$

We have:

(1) If $\phi^i = 0$ on $(X_x)_s$ (i.e. at every generization of x in X_s) for $i \leqslant m$, then

$$H^{i-1}_{\{x\}}(\bar{X}_x) = \widetilde{H}^{\cdot,i}(\bar{V}) \xrightarrow{\sim} \widetilde{H}^{\cdot,i}(\bar{V}_{\bar{\eta}}) = \phi^i \qquad \text{for } i \leqslant m.$$

The ϕ^i are indeed the cohomology sheaves of $\bar{V}$ with support in V_s; under the hypotheses of 4.5, condition (a) makes (1) applicable with $m = \infty$ (SGA 4 XV 2.1).

(2) Hypothesis (b) of 4.5 is stable under base change. By SGA 2 XIV 5.6, it implies

$$\widetilde{H}^{\cdot,i}(V') = H^{i-1}_{\{x\}}(X'_x) = 0 \qquad \text{for } i < n.$$

We conclude by observing that $\widetilde{H}^{\cdot,i}(\bar{V})$ is the filtered colimit of the $\widetilde{H}^{\cdot,i}(V')$.

Combining 4.2 and 4.5 gives:

Corollary 4.6. Let S be a henselian trait of characteristic 0, let X/S be a relative complete intersection of relative dimension n, and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then $\phi^i_x = 0$ for $i \neq n$.

Corollary 4.7. Under the hypotheses of 4.6, ϕ^n_x is a flat Λ-module.

Indeed, if Λ' is a finite Λ-algebra, a general nonsense (SGA 4 XVII 5.2.11) gives, for the $\phi^i_{\Lambda'}$,

$$\phi^{n-i}_{\Lambda'} = \mathrm{Tor}^{\Lambda}_i(\Lambda', \phi^n_{\Lambda}),$$

and we apply 4.6 to Λ'.

Variant 4.8. In 4.5, omit condition (a), and let Z be the non-smooth locus of f. One may still conclude that

$$\phi^i_x = 0 \qquad \text{for } i < n - \dim Z_s.$$

We proceed by descending induction on $\dim \overline{\{x\}}$. The induction hypothesis permits us to apply 4.5 (1) with $m = n - \dim Z_s$, and the proof is completed as before.

4.9. It is possible to obtain a vanishing criterion using only information about the geometric fibres of f. For complete intersections, however, the result is one degree weaker than 4.5. I 16

Proposition 4.10. Under the general hypotheses of 4.5, suppose that:

(a) x is an isolated non-smooth point in its fibre;

(b) for every point y of $X_{\bar\eta}$ whose closure in $X_{\bar\eta}$ has dimension $d(y)$ and which is a generization of x, the étale depth satisfies

$$\operatorname{depth}_{\text{ét},y}(X_{\bar\eta}) \geqslant n - d(y), \quad \text{i.e.} \quad H^i_{\{y\}}(X_{\bar\eta}) = 0 \quad \text{for } i \leqslant n - d(y);$$

(c) $\operatorname{depth}_{\text{ét},x}(X_s) \geqslant n$.

Then $\phi^i_x = 0$ for $i < n - 1$.

Retain the notation of 4.5. We may apply 4.5 (1) with $m = \infty$. By the local Lefschetz theorem (SGA 2 XIV), hypothesis (b) gives

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \qquad (\ i < n\text{-}1\ )$$

$$H^i(V') \hookrightarrow H^i(V_s) \qquad (\ i = n\text{-}1\ )$$

Condition (c) then gives

$$H^i(\bar V) = \varinjlim H^i(V') = 0 \qquad \text{for } i \leqslant n - 1,$$

which proves 4.10.

4.11. As in 4.8, one can give a variant of 4.10 in which (a) is not assumed.

5. Action of monodromy on π_1 I ≈17

In this section (Grothendieck's exposé of 19 November 1968), we shall use the semistable reduction theorem for curves. Artin and Winters [2] gave a direct proof of this theorem; it can also be deduced [3] from the analogous theorem for abelian varieties (IX § 3, using 3.6). We shall also use Schlessinger's theory (Rim's Exposé VI).

5.1. Let k be an algebraically closed field, let C be a projective curve over k, and let $x_1, \ldots, x_n$ be a finite set of closed points of C. The pointed curve

$$X = (C; x_1, \ldots, x_n)$$

is called stable if:

(a) C is reduced, and its only singularities are double points with distinct tangents;

(b) the x_i are distinct smooth points of C;

(c) if C_1 is an irreducible component of C, and E is the set of points of the normalization C_1' of C_1 lying above a singular point of C or above one of the x_i, then, if C_1' has genus 0 (respectively 1), the set E has at least 3 (respectively 1) elements.

Condition (c) is equivalent to each of the following:

(c') X has no nontrivial infinitesimal automorphisms.

(c") If ω is the invertible dualizing sheaf, the invertible sheaf

$$\omega' = \omega\Big(\sum_i x_i\Big)$$

is ample.

An S-scheme C equipped with sections $x_1, \ldots, x_n$ is a stable curve over S if its geometric fibres are stable curves. In [3], §§ 1–2, only the case $n = 0$ is considered. The extension of the results of loc. cit. to the general case is immediate:

(1) If X/k is stable, then $H^1(C, \omega'^{\otimes n}) = 0$ for $n \geqslant 2$, and $\omega'^{\otimes n}$ is very ample for $n \geqslant 3$ (cf. [3] 1.2).

(2) The formal moduli scheme M of X is smooth, and C remains singular over a normal-crossings divisor of M.

(3) If X and Y are stable over S, then $\mathrm{Isom}_S(X,Y)$ is finite and unramified over S (isomorphisms I≈18
are required to respect the marked points).

(4) If X/k is stable, the map

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k \mathrm{Pic}^\circ(C)$$

is injective.

Finally, we have:

Proposition 5.1.1. Let $X_\eta = (C_\eta; x_1, \ldots, x_n)$ be stable over the generic point η of a trait S, with C_η smooth. There exist a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'.$$

This follows from the usual semistable reduction theorem, which permits one to choose η' so that the geometric special fibre of the minimal model C'' of $C_{\eta'}$ over S' satisfies (a). One then blows up, iteratively if necessary, smooth points of C'' so that (b) is satisfied. Next one contracts the chains of smooth rational components of self-intersection -2 which contain none of the $\bar{x}_i$ in the special fibre; the result is C' (cf. [3], proof of 2.3, p. 88).

As in loc. cit., one can check that C' exists as soon as the Jacobian of C_η has semistable reduction. The smoothness hypothesis on C_η is moreover unnecessary.

5.2. Let X_η be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S. For a geometric point ξ of $X_{\bar{\eta}}$, there is an exact sequence

$$0 \longrightarrow \pi_1(X_{\bar{\eta}}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow 0. \tag{5.2.1}$$

It induces a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the largest prime-to-p quotient $\pi_1^{(p)}(X_{\bar{\eta}})$ of $\pi_1(X_{\bar{\eta}})$, one obtains

$$\mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{Aut}_{\mathrm{ext}}(\pi_1^{(p)}(X_{\bar{\eta}})). \tag{5.2.2}$$

Suppose that X_η has a rational point x, and let $\bar{x}$ be the geometric point

$$\bar{\eta} \longrightarrow \eta \xrightarrow{x} X_\eta.$$

For $\xi = \bar{x}$, the exact sequence (5.2.1) splits canonically, giving

$$[x] : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{Aut}(\pi_1(X_{\bar{\eta}}, \bar{x})) \longrightarrow \mathrm{Aut}(\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})). \tag{5.2.3}$$

Theorem 5.3. The image of P under (5.2.2) (or under (5.2.3), which amounts to the same thing) I19
is finite.

5.3.1. Reduction to the case in which X_η is geometrically normal and integral.

Let X'_η be the normalization of X_η, let $(X'_{i\eta})_i$ be the family of connected components of X'_η, put

$$X''_\eta = X'_\eta \times_{X_\eta} X'_\eta,$$

and let $(X''_{j\eta})_j$ be the family of connected components of X''_η. After a finite extension of $k(\eta)$, we may suppose that the $X'_{i\eta}$ are geometrically normal and integral and have rational points x_i, and that the $X''_{j\eta}$ are geometrically connected and have rational points y_j.

For $k = 1$ or 2, and whenever $p_k(y_j) \in X'_{i\eta}$, choose a class of paths ℓ_{kij} from $\mathrm{pr}_k(\bar{y}_j)$ to $\bar{x}_i$. It is an element of a torsor under $\pi_1(X'_{i\bar{\eta}}, \bar{x}_i)$. Let $\bar{\ell}_{kij}$ be the corresponding prime-to-p path (the same path modulo $\mathrm{Ker}(\pi \longrightarrow \pi^{(p)})$). We may choose $\bar{\ell}_{kij}$ to be invariant under P: the set of prime-to-p paths is an inverse limit of finite sets of cardinality prime to p on which the pro-p group P acts.

Consequently, if a finite-index subgroup P' of P acts trivially on the groups $\pi_1^{(p)}(X'_{i\bar{\eta}}, \bar{x}_i)$, the Van Kampen formulas—equivalently, the fact that the normalization map $X'_{\bar{\eta}} \longrightarrow X_{\bar{\eta}}$ is effective descent for étale coverings—show that P' acts trivially on $\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})$.

5.3.2. If $X_{\bar{\eta}}$ is normal and U is an open subset, then $\pi_1(U)$ maps onto $\pi_1(X_{\bar{\eta}})$; we may therefore suppose that $X_{\bar{\eta}}$ is smooth and affine. By cutting with generic hyperplane sections (which replaces S by its strict localization at the generic point of the special fibre of an $\mathbb{A}^n_S$) and applying Bertini, we reduce to the case in which X_η is smooth, geometrically connected, and of dimension one.

After a finite extension of $k(\eta)$, we may suppose that X_η is the complement in a smooth projective curve $\bar{X}_\eta$ of a finite set of rational points x_i, whose number we are free to increase. By the semistable reduction theorem (5.1.1), we reduce to the case in which there is a stable curve $X = (C; x_1, \ldots, x_n)$ over S such that

$$X_\eta = C_\eta - \{x_1, \ldots, x_n\}.$$

We then apply the following result.

Theorem 5.4. Let S be a strictly local scheme, let $f : \bar{X} \longrightarrow S$ be a proper flat morphism whose special fibre is a curve with ordinary double points as its only non-smooth points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Let x be a rational point of $X = \bar{X} - Y$. Let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U. Then the image of $\pi_1(U, \xi)$ in $\mathrm{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ is abelian and prime to p. I 20

Observe that $\pi_1(U, \xi)$ can act on $\pi_1^{(p)}(X_\xi, x\xi)$ only because the groups $\pi_1^{(p)}$ satisfy the specialization theorem.

We reduce first to the case in which S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is the spectrum of a formal power-series ring over a Cohen ring:

$$S = \mathrm{Spec}(W[[t_1, \ldots, t_n]]),$$

and U is the complement of a normal-crossings divisor with equation $t_1 \cdots t_m = 0$. In this universal case, Abhyankar's lemma shows that $\pi_1(U, \xi)$ is abelian and prime to p, which proves the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by taking a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem

(by P. Deligne)

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \longrightarrow S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ obtained from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Néron model of $A_{\eta'}$ over S', let $\mathcal{A}_{S',s'}$ be its special fibre, and let $\mathcal{A}^\circ_{S',s'}$ be the identity component of $\mathcal{A}_{S',s'}$. The stable reduction theorem is as follows.

Theorem 6.1. For a suitable S', $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}^\circ_{S',s'}$ is an extension of an abelian variety by a torus.

It follows fairly readily from 6.1 that one may choose S' such that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field (the same argument would work if $k(s)$ were radicial over such a field). I ≈21

Let ℓ be a prime different from the residual characteristic exponent p, let $T_\ell(A)$ be the Tate module, let $V_\ell(A) = T_\ell(A) \otimes \mathbf{Q}_\ell$, and let

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(T_\ell(A))$$

be the natural representation. After a preliminary extension of scalars, we are reduced to the case in which the inertia group I acts unipotently (1.2), and hence through its largest pro-ℓ quotient $\mathbf{Z}_\ell(1)$ (0.3). Let T be the image in $\mathrm{GL}(T_\ell(A))$ of a generator of $\mathbf{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T-1)^r \neq 0$. We have:

Lemma 6.2. The map

$$(\sigma, v) \in I \times V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces the morphism and the isomorphism in the commutative diagram

$$\begin{array}{ccc} V_\ell(A)_I \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\quad M \quad} & V_\ell(A)^I \\ \downarrow & & \uparrow \\ (V_\ell(A)/\operatorname{Ker}(T-1)^r) \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\ \sim\ } & \operatorname{Im}(T-1)^r \end{array} \qquad (6.2.1)$$

$V_\ell(A)_I$ and $V_\ell(A)^I$ are $\mathrm{Gal}(\bar{s}/s)$-modules, and M is Galois-equivariant.

6.3. An ℓ-adic representation

$$\tau : \mathrm{Gal}(\bar{s}/s) \longrightarrow \mathrm{GL}(V)$$

is said to be of weight n ($n \in \mathbf{Z}$) if the following condition is satisfied:

(*) For a suitable model X of $k(s)$ (that is, an irreducible variety of finite type over the prime field such that $k(s) = k(X)$), τ factors through $\pi_1(X, \bar{s})$ and, for every closed point x of X, the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value $|k(x)|^{n/2}$ (here F_x denotes the geometric Frobenius ϕ_x^{-1}).

If V (respectively W) has weight n (respectively m) and $n \neq m$, then $\mathrm{Hom}_{\mathrm{Gal}}(V, W) = 0$. I≈**22**

6.4. It follows from the universal property of the Néron model that $V_\ell(A)^I = V_\ell(\mathcal{A}^\circ_{S,s})$. If a is the dimension of the largest abelian quotient of $\mathcal{A}^\circ_{S,s}$ and m is the dimension of its multiplicative part, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2.

6.5. Let A^* be the abelian variety dual to A. Recall that $V_\ell(A^*)$ and $V_\ell(A)$ (and hence $V_\ell(A^*)^I$ and $V_\ell(A)_I$) are in duality with values in $\mathbf{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m are for A, it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1.

6.6. The Galois module $\mathbf{Q}_\ell(r)$ has weight $-2r$. Since $\operatorname{Im}(T-1)^r \neq 0$, it follows from 6.4, 6.5, and the isomorphism in 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

so $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. It then follows from 6.2 that $\operatorname{Im}(T-1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0; these spaces have the same dimension $m_1 \leq m$.

We then have

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}_{S,s} \geq 2(a+m) \geq 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

so A_η has stable reduction ($\dim \mathcal{A}_{S,s} = a + m$). At the same time, we have obtained $m_1 = m$; that is,

$$\operatorname{Im}(T-1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}^\circ_{S,s})$$

is the V_ℓ of the multiplicative part of $\mathcal{A}^\circ_{S,s}$.

6.7. To treat the general case, we shall use (0.5) (cf. 1.3). We reduce to the case in which:

(a) the conditions of (0.5.2) or (0.5.3) are satisfied; I≈**23**

(b) $\mathrm{Gal}(\bar{\eta}/\eta)$ acts trivially on A_ℓ;

(c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses, we shall prove that A_η has stable reduction. Hypothesis (a) allows us to apply (0.5.1) and write $S = \varprojlim S_i$ as in 1.3, whose notation we retain. Let s_i be the closed point of S_i.

For sufficiently large i, the identity component $\mathcal{A}^\circ$ of the Néron model $\mathcal{A}_S$ descends to a smooth group scheme $\mathcal{A}_i$ with connected fibres over S_i, and $(\mathcal{A}_i)_\ell$ is constant over $S_i - D_i$. The group

$\pi_1(S_i - D_i, \bar{\eta})$ acts on $T_\ell(A)$ and acts trivially on $T_\ell(A)/\ell T_\ell(A) \simeq A_\ell$. As in 1.3, one sees that P_i acts trivially. We then have, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}_s^\circ) = T_\ell(A)^I = T_\ell(A)^{I_i},$$

and (6.2.1) is a diagram of $\mathrm{Gal}(\bar{s}_i/s_i)$-modules to which the arguments of 6.4–6.6 may be applied.

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Exposé II

Finiteness properties of the fundamental group

by Michèle Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k, and let $\Pi_1^{(p)}(X)$ be the largest "prime-to-p" quotient of $\Pi_1(X)$. In this exposé we prove that, when X is a surface, $\Pi_1^{(p)}(X)$ is topologically finitely presented, and that the same holds for arbitrary X if resolution of singularities is assumed.

For a surface, the proof method, due to J.-P. Murre, consists in reducing to a fibration over P^1 having properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k. Then there exists a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow P^1,$$

such that f has a section σ and the following conditions hold:

(i) f is smooth at the points of $\sigma(P^1)$.

(ii) The fibres of f are geometrically integral curves having only ordinary singularities.

(iii) There exists a nonempty open subset U of P^1 such that the fibres of f over the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \longrightarrow P^r$ for which a Lefschetz pencil exists (XVII 2.5). Indeed, let $D = \{H_t\}_{t \in P^1}$ be such a pencil of hyperplanes. Recall that it has the following properties (loc. cit. 2.2):

(a) The axis Δ of D meets S transversely; the closed subscheme $\Delta \cap S$ is therefore reduced and consists of finitely many closed points $x_1, \ldots, x_n$ of S. I ≈26

(b) There exists a nonempty open subset U of P^1 such that, for every $t \in U$, H_t meets S transversely.

(c) For every $t \in P^1 - U$, H_t meets S transversely except at one point, which is an ordinary quadratic singularity.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t. The map assigning $t \in P^1$ to x is a rational map $a : S \dashrightarrow P^1$ with domain of definition $S - (\Delta \cap S)$.

Consider the blow-up $g : S' \longrightarrow S$ of the closed subscheme $S \cap \Delta$. The scheme S' is regular, and the fibre S'_{x_i} over x_i is the projective bundle defined by the tangent space of S at x_i, hence is isomorphic to P^1 (EGA IV 19.4.2, 19.4.4). Moreover, the rational map $f = ag$ is everywhere defined (XVIII 3.1). Finally, to each x_i one can associate the section of f that sends $t \in P^1$ to the tangent vector at x_i of $H_t \cap S$.

Let us verify conditions (i), (ii), and (iii). For each $t \in P^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic. Indeed, the morphism $S'_t \longrightarrow S_t$ is an isomorphism except possibly at the points x_i, where S_t is regular; moreover, since S'_t is a complete intersection in S', it is integral, and is therefore isomorphic to S_t.

Conditions (ii) and (iii) are precisely conditions (b) and (c). Finally, f is smooth at the points of σ, since f is flat and the fibres S'_t are geometrically regular at $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S, and let $Y' = Y \times_S S'$. Then f, g, and U may be chosen so that $Y' \times_{P^1} U$ is a divisor with normal crossings relative to U. I 27

It is known that, if a Lefschetz pencil exists for an embedding $S \longrightarrow P^r$, the Lefschetz pencils form an open subset of the variety of lines in P^r (XVII 3.2.1). The same is true of the set of Lefschetz

pencils whose axis does not meet Y and for which there is a nonempty open subset U_1 of P^1 such that $Y \cap H_t$ is smooth for $t \in U_1$. It is then enough to construct f and g from a pencil satisfying these two conditions and replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is said to be topologically finitely generated if there is an integer n such that G is isomorphic to a quotient of the free pro-group $F(n)$ on n generators. Put $K = \mathrm{Ker}(F(n) \longrightarrow G)$. The group G is said to be topologically finitely presented if, in addition, there are finitely many elements $k_1, \ldots, k_r$ of K such that the smallest closed normal subgroup of K containing $k_1, \ldots, k_r$ is K.

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k. Suppose that schemes of finite type and dimension at most $\dim X$ over an algebraic closure of k admit strong resolution of singularities (SGA 5 I 3.1.5). This condition is satisfied when $\dim X = 2$, by [1], or when $\mathrm{char}\, k = 0$, by [2]. Then the fundamental group $\Pi_1^{(p)}(X)$ is topologically finitely presented.

By SGA 1 IX 4.10, we may suppose that k is algebraically closed.

1) Reduction to the case in which X is a regular surface.

If $f : X' \longrightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and let $(X'_a)_{a\in A}$ and $(X''_b)_{b\in B}$ be the families of connected components of X' and X'', respectively. When f is an effective descent morphism for the category of finite étale coverings, the fundamental group $\Pi_1(X)$ can be computed explicitly

in terms of the fundamental groups $\Pi_1(X'_a)$ and $\Pi_1(X''_b)$. It follows (SGA 1 IX 5.2, 5.3) that if I≈28
the $\Pi_1(X'_a)$ are topologically finitely generated, then so is $\Pi_1(X)$; and if the $\Pi_1(X'_a)$ are topologically finitely presented while the $\Pi_1(X''_b)$ are topologically finitely generated, then $\Pi_1(X)$ is topologically finitely presented.

Since X admits resolution of singularities, there is a regular scheme X' and a proper birational morphism $f : X' \longrightarrow X$. Since f is an effective descent morphism for finite étale coverings, it suffices to prove the theorem under the additional assumption that X is regular. One first deduces that $\Pi_1(X)$ is topologically finitely generated; knowing that $\Pi_1(X')$ is topologically finitely presented and that the $\Pi_1(X''_b)$ are topologically finitely generated then shows that $\Pi_1(X)$ is topologically finitely presented.

Now let $(U_i)_{i\in I}$ be a finite affine open covering of X, and let $f : \coprod U_i \longrightarrow X$ be the canonical morphism. Since f is an effective descent morphism for finite étale coverings, it is enough to prove the theorem for the U_i. We are thus reduced to the case in which X is affine and smooth.

Assume henceforth that X is affine and smooth over k, with $\dim X > 2$. The Lefschetz theorem for quasi-projective schemes ([3] V 7.4) applies: if Y is a sufficiently general hyperplane section of X, then

$$\Pi_1(Y) \simeq \Pi_1(X).$$

It therefore suffices to prove the theorem for Y, reducing us to the case $\dim X = 2$.

2) The case in which X is a smooth surface over k.

By the strong resolution theorem for surfaces, there exist a smooth, integral, projective surface S and an open immersion $i : X \longrightarrow S$ such that $Y = S - X$ is a normal-crossings divisor. We apply Corollary 2.2 and retain its notation.

We show that it is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically I≈29
finitely presented. Since X and X' are regular and

$$\mathrm{codim}\left(\bigcup_{1\leq i\leq n} \{x_i\}, X\right) \geq 2,$$

every étale covering of X' induces on $X - \bigcup_{1\leq i\leq n}\{x_i\}$ an étale covering which extends uniquely to an étale covering of X (SGA 2 1.11). This gives an isomorphism

$$\Pi_1(X') \simeq \Pi_1(X).$$

Consider the cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X'_U \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h_U} \\ P^1 & \longleftarrow & U \end{array}$$

where $h = f|_{X'}$. Let L be the set of primes different from p, and let $\bar{\eta}$ be a geometric point over the generic point η of P^1. We verify the hypotheses of SGA 1 XIII 4.8. The morphism h has a section σ, because f does. To prove condition (a) of SGA 1 XIII 4.7, note first that, since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic (SGA 4 XV 4.1, 1.16). It remains to show that, for every locally constant sheaf of sets F on X', (F, h) is cohomologically proper in dimensions ≤ 0. If ξ is a closed point of P^1 and Z denotes the integral closure of $\operatorname{Spec} \mathcal{O}_{P^1,\xi}$ in $\bar{\eta}$, then $X'_Z = X' \times_{P^1} Z$ satisfies condition (S_2) at its closed points and is regular at every other point, hence is normal. Consequently, an étale covering of X'_Z whose restriction to $X'_{\bar{\eta}}$ is connected is necessarily connected. This proves the assertion, and hence condition (a).

Since X'_U is the complement in S'_U of a divisor with normal crossings relative to U, h_U is locally 1-aspherical for L (SGA 4 XV 2.1); and, for every finite constant sheaf of L-groups F on X'_U, (F, h_U) is cohomologically proper in dimensions ≤ 1 (SGA 1 XIII 2.4), which gives condition (b). Since P^1 is normal and $T = P^1 - U$ has codimension 1, we have

I ≈30

$$\operatorname{prof} \text{ét}_T(P^1) \geq 2,$$

which gives condition (c). The condition

$$\operatorname{prof} \operatorname{hop}_x(X') \geq 3$$

holds at every closed point of X' by the purity theorem, which gives condition (e). Finally, for every $t \in P^1$, h is smooth at $\sigma(t)$, and hence is locally 1-aspherical for L at $\sigma(t)$, which gives condition (d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, there is an exact sequence

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}}, a) \longrightarrow \Pi'_1(X'_U, a) \longrightarrow \Pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\Pi_1(U, a)$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$. The group of coinvariants under K,

$$\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ by the closed normal subgroup generated by the elements $x^{-1}q(x)$, where $x \in \Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. (4.7.4), there is an isomorphism

$$\Pi_1^{(p)}(X', a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K.$$

Write $P^1 - U = \coprod_{1 \leq i \leq n} \{t_i\}$. For each i, let $\bar{T}_i$ be the strict localization of P^1 at a geometric point $\bar{t}_i$ over t_i, and let s_i be the generic point of $\bar{T}_i$. An étale covering of U whose restriction to every s_i is trivial extends to an étale covering of P^1, and is therefore trivial. For each i, let $\overline{k(s_i)}$ be an algebraic closure of $k(s_i)$. The preceding assertion says that $\Pi_1(U, a)$ is the smallest closed normal subgroup containing the images of the Galois groups $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\Pi_1(U, a)$. By I 5.3, the image K_i of $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed normal subgroup containing all the K_i, and since $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ is topologically finitely presented (SGA 1 XIII 3.2), this proves that $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K$, and hence also the isomorphic group $\Pi_1^{(p)}(X)$, is topologically finitely presented.

Remark 2.3.2. Resolution of singularities is not needed to prove Theorem 2.3.1 when X is the open complement of a normal-crossings divisor in a projective scheme smooth over k. I 31

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S G A 7 I32

Exposé VI

FORMAL DEFORMATION THEORY

by D. S. RIM

1. Formal existence theorem

We recall the notion of cofibered category (S.G.A.1.IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \to \underline{C}$ such that

Ax.1. For any map f in $\underline{C}$ and any object a in $\underline{A}$ with $P(a) =$ the source of f , there exists an f-morphism $\alpha : a \to b$ which is cocartesian (i.e. for any f-morphism $\alpha' : a \to b'$ there exists a unique T-morphism $\tau : b \to b'$ in $\underline{A}$ such that $\alpha' = \tau \cdot \alpha$ where $T =$ the target of f).

Ax.2. A composite of cocartesian morphisms in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \to \underline{C}$ follow immediately from the definitions:

(cancellation) Let $a \underset{\beta}{\overset{\alpha}{\rightrightarrows}} a'$ be any two cocartesian maps such that $P(\alpha) = P(\beta)$. If there exists a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$, $a \xrightarrow{\beta} a''$, there exists a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there exists $f : P(a') \to P(a'')$ in $\underline{C}$ such that $f \cdot P(\alpha) = P(\beta)$. Furthermore γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S in $\underline{C}$ we denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a in $\underline{A}$ such that $P(a) = S$, and $\mathrm{Hom}_{\underline{A}(S)}(a, a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a, a') \mid P(\alpha) = \mathrm{id}_S\}$. A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$ is a groupoid for every object S in $\underline{C}$. We recall that a groupoid is a category in I33 which every morphism is an isomorphism. Thus, a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \to \underline{C}$ such that every map in $\underline{A}$ is cocartesian. Furthermore, for every map $\alpha \in F\ell(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, we fix a complete noetherian local ring Λ with the residue field k , and we set $\underline{C}_\Lambda$ to be the category of artinian local Λ-algebras with the same residue field k , where the maps are local homomorphisms over Λ . From 1.2 on, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to have the property that $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a,\ b)$ consists of one and only one element for any two objects $a,\ b$ in $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called simply a $\underline{C}_\Lambda$ -map if $p(a') = p(a)$ and $p(\alpha) = 1_{p(a)}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called surjective, by abuse of language, if $p(\alpha) : p(a') \longrightarrow p(a)$ is surjective in $\underline{C}_\Lambda$.

Ex.1.1.(a) Let $F : \underline{C} \longrightarrow$(categories) be a functor. Define the category $\underline{F}$ as follows: an object in $\underline{F}$ is a pair (S, a) where $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$, and a map $(S, a) \to (S', a')$ is a pair (f, u') where $f : S \to S'$ is a map in $\underline{C}$ and $u' : F(f)a \to a'$ is a map in $F(S')$. Composition of two maps $(S, a) \xrightarrow{(f,u')} (S', a') \xrightarrow{(g,u'')} (S'', a'')$ is given by $(gf,\ u'' \cdot (F(g)u')) : (S, a) \to (S'', a'')$. With respect to the functor $P : \underline{F} \to \underline{C}$ given by $(S, a) \mapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \to$ (groups) yields a cofibered group, and a functor $F : \underline{C} \to$ (Ens) yields a cofibered discrete groupoid. We also note that a natural transformation $\varphi : F \to G$, where $F, G : \underline{C} \to$ (categories) are functors, induces a cocartesian functor $\varphi : \underline{F} \to \underline{G}$ over $\underline{C}$.

Ex.1.1.(b) Let X , Y be schemes over Λ together with a fixed isomorphism $X \underset{\Lambda}{\otimes} k \xleftrightarrow{\tau} Y \underset{\Lambda}{\otimes} k$ I34 . Then the functor $\mathrm{Isom}_{X,Y} : \underline{C}_\Lambda \to$ (groupoids) given by $\mathrm{Isom}_{X,Y}(S) =$ the set of isomorphisms $X \underset{\Lambda}{\otimes} S \xleftrightarrow{\sigma} Y \underset{\Lambda}{\otimes} S$ such that $\sigma \underset{\Lambda}{\otimes} k = \tau$ yields a cofibered groupoid $P : \underline{\mathrm{Isom}}_{X,Y} \to \underline{C}_\Lambda$.

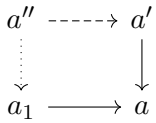
Ex.1.1.(c) Let X_0 be a fixed scheme over k , and let $\underline{M}_{X_0}$ be the category consisting of deformations of X_0 over $\underline{C}_\Lambda$ (see §4). Thus an object in $\underline{M}_{X_0}$ is a pair (R, X) where R is in $\underline{C}_\Lambda$ and X is a deformation of X_0 to R . Then the functor $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$ given by $(R, X) \rightsquigarrow R$ defines a cofibered groupoid over $\underline{C}_\Lambda$.

Ex.1.1.(d) For any groupoid X we denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects in X . If $P : \underline{F} \to \underline{C}_\Lambda$ is a cofibered groupoid, we obtain a functor $[\underline{F}] : \underline{C}_\Lambda \to$

(Ens) given by $[\underline{F}](S) = [\underline{F}(S)]$, and in turn a cofibered discrete groupoid, which is, by abuse of notation, denoted by $[\underline{F}]$.

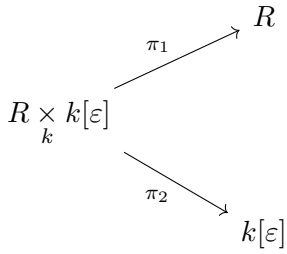
Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called semi-homogeneous if the following properties hold:

(1) every diagram

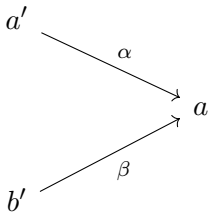


of solid arrows in $\underline{A}$ where $a' \to a$ is surjective can be completed into a commutative diagram including dotted arrows.

(2) Let



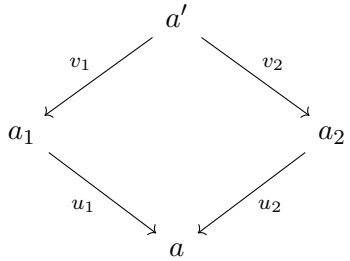
be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k , and let



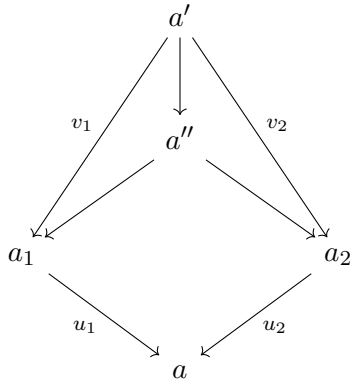
be π_1-maps in $\underline{A}$. Then there exists a $\underline{C}_\Lambda$ -map $\gamma : a' \longrightarrow b'$ with $\alpha = \beta \cdot \gamma$ if and only if $(\pi_2)_*(a') \approx (\pi_2)_*(b')$.

Remark 1.3.(a) A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). We note that if $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$, and in particular 1.2(2) entails the canonical map $[\underline{A}(R \times_k k[\epsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\epsilon])]$ to be bijective. I 35

Remark 1.3.(b) Consider a commutative diagram



in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Set $R = P(a)$, $R' = P(a')$ and $R_i = P(a_i)$ for $i = 1, 2$. If we set a'' to be the "direct image" under the map $(P(v_1), P(v_2)) : R' \to R_1 \times_R R_2$, we obtain a commutative diagram

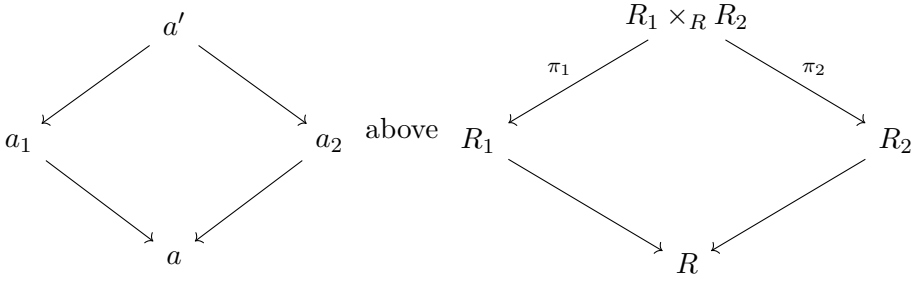


Thus 1.2(1) can be replaced by

(1)': every diagram

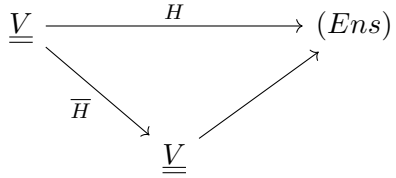
$a_1 \searrow \quad \swarrow a_2$ in $\underline{A}$ above $R_1 \searrow \quad \swarrow R_2$ in $\underline{C}_\Lambda$, where

$R_2 \to R$ is surjective can be completed to a commutative diagram



where π_i $(i = 1, 2)$ are projections.

Remark 1.3.(c) If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\epsilon])]$ is canonically provided with a vector space structure over k , where $k[\epsilon]$ is the ring of dual numbers over k . Indeed, let $\underline{\underline{V}}$ be the category of finite-dimensional vector spaces over k . Then any functor **136**
$H : \underline{\underline{V}} \to (Ens)$ preserving the product is canonically factored through

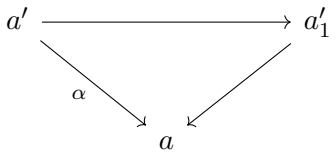


where $\underline{\underline{V}} \to (Ens)$ is the forgetful functor. On the other hand, the functor $G : \underline{\underline{V}} \longrightarrow \underline{C}_\wedge$ given by $W \longmapsto k[W] = k \oplus W$, with $W^2 = 0$, transforms products in $\underline{\underline{V}}$ into fibre products over k in $\underline{C}_\wedge$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$, then $[\underline{A}(k[V] \times_k k[W])] \longrightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$ is bijective by 1.3(a), and the functor $V \longmapsto [\underline{A}(k[V])]$ commutes with products, hence the result.

Definition 1.4. (1) Let $R \in ob(\underline{C}_\wedge)$. A small extension of R in $\underline{C}_\wedge$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{C}_\wedge$ such that $(\mathrm{Ker}\ f)^2 = 0$ and $\mathrm{Ker}\ f \simeq k$. In other words, $R' \xrightarrow{f} R$ in $\underline{C}_\wedge$ is a small extension of R if

$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$ is exact for some $t \in R'$ with $t^2 = 0$.

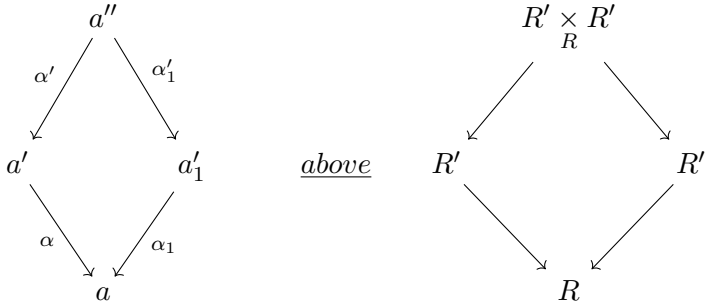
(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, and let a be an object in $\underline{A}$. A small prolongation of a is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{C}_\wedge$. An arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called "versal for small prolongations of a " if, for every small prolongation $a'_1 \longrightarrow a$ in $\underline{A}$, there exists a commutative diagram



Proposition 1.5. Let $\underset{\pi_1}{\nearrow} \underset{\pi_2}{\searrow}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\wedge$.

(1) Let $R \times_k k[\epsilon]$ $\begin{matrix} \nearrow R \\ \searrow k[\epsilon] \end{matrix}$ be projection maps in $\underline{C}_\wedge$, and let $a' \xrightarrow{\alpha} a$ be a π_1-map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\,\beta = 1_a$ if and only if there exists a map $a \longrightarrow (\pi_2)_* a'$. 137

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$, and consider a commutative diagram

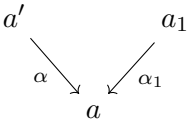


Define the canonical map $\tilde{f} : R' \underset{R}{\times} R' \longrightarrow k[\mathrm{Ker}\ f]$ by $\tilde{f}(r_1',\ r_2') = r_1'(0) + (r_1' - r_2')$. If there exists a map $a' \longrightarrow (\tilde{f})_*(a'')$, then there exists a map $\gamma : a' \longrightarrow a_1'$ such that $\alpha = \alpha_1 \cdot \gamma$.

Proof (1) The part "only if" is clear. Now assume that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. If we set $h : R \longrightarrow R \underset{k}{\times} k[\epsilon]$ by $h = 1 \times P(\varphi)$, then the composite map $R \xrightarrow{h} R \underset{k}{\times} k[\epsilon] \xrightarrow{\pi_1} R$ is the identity, and therefore we get maps $a \underset{\alpha_1}{\overset{\gamma}{\rightleftarrows}} h_* a$ such that $\alpha_1 \cdot \gamma = 1_a$ where $P(\gamma) = h$ and $P(\alpha_1) = \pi_1$. On the other hand, $\pi_2 \cdot h = P(\varphi)$ and hence there exists a π_2 -map $h_* a \longrightarrow (\pi_2)_* a'$. This means that $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and therefore by 1.2(2) we have a $\underline{C}_\Lambda$ -map $u : h_* a \longrightarrow a'$ such that $\alpha \cdot u = \alpha_1$. Define $\beta : a \longrightarrow a'$ by $\beta = u \cdot \gamma$. Then $\alpha \cdot \beta = \alpha \cdot u \cdot \gamma = \alpha_1 \cdot \gamma = 1_a$.

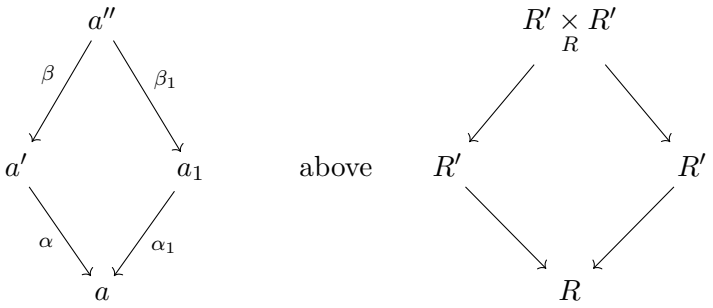
(2) The map $1 \times \tilde{f} : R' \underset{R}{\times} R' \longrightarrow R' \underset{k}{\times} k[\mathrm{Ker}\ f]$ is an isomorphism compatible with the projections on the first factor, and $\pi_2 \cdot (1 \times \tilde{f}) = \tilde{f}$ where $\pi_2 : R' \underset{k}{\times} k[\mathrm{Ker}\ f] \longrightarrow k[\mathrm{Ker}\ f]$ is the projection map. We note that $k[\mathrm{Ker}\ f] \simeq k[\epsilon]$ since $R' \xrightarrow{f} R$ is a small extension. Therefore, if there exists a map $a' \longrightarrow (\tilde{f})_* a'' = (\pi_2)_*[(1 \times \tilde{f})_* a'']$, then by (1) above there exists a map $\gamma' : a' \longrightarrow a''$ such that $\alpha' \cdot \gamma' = 1_{a'}$. Define $\gamma : a' \longrightarrow a_1'$ by $\gamma = \alpha_1' \cdot \gamma'$. We then have $\alpha_1 \cdot \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$ and consider a diagram 138



in $\underline{A}$ where α_1 is a small prolongation. Assume that a is versal for small prolongations of e . Then there exists a map $a' \xrightarrow{\rho} a_1$ such that $\alpha = \alpha_1 \rho$ in $\underline{A}$ if and only if there exists a map $f : P(a') \to P(a_1)$ such that $P(\alpha) = P(\alpha_1) \cdot f$ in $\underline{C}_\Lambda$.

Proof: Assume that $P(\alpha) = P(\alpha_1) \cdot f$ where $f : P(a') \to P(a_1)$. Replacing a' by its direct image under f , we may and shall assume that $P(\alpha) = P(\alpha_1)$. Set $R' = P(a_1)$, $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we get a commutative diagram



Since a is versal for small prolongations of e , so is a' and therefore our statement follows from 1.5(2).

Definition 1.7. For each object R in $\underline{C}_\Lambda$ we put $\overline{\Omega}_R = M/(m_\Lambda R + M^2)$ where m_Λ , M are the maximal ideals of Λ , R respectively. It is a finite-dimensional vector space over k , and $R \longmapsto \overline{\Omega}_R$ defines a functor $\overline{\Omega} : \underline{C}_\Lambda \to \underline{V}$ where $\underline{V}$ is the category of finite-dimensional vector spaces over k . If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, then the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \to \underline{V}$ is, by abuse of notation denoted by $\overline{\Omega}$ again.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Then, given an object a in $\underline{A}$ versal for small prolongations of e , there exists $a' \to a$ such that (i) a' is versal for small prolongations of a I 39

(ii) $p(a') \longrightarrow p(a)$ is surjective; and the kernel is annihilated by the maximal ideal of $p(a)$.

(iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

Proof: Set $R = P(a)$ and consider an exact sequence $0 \to J \to S \to R \to 0$ where S is a formal power-series ring over $\wedge$ in a finite number of variables, and $J \subset M^2$ where M is the maximal ideal of S . Let Σ be the set of ideals J_1 in S such that (i) $J \supset J_1 \supset MJ$ and (ii) there exists π_1-morphism $a_1 \to a$ in $\underline{A}$ where $\pi_1 : S/J_1 \to R$ is the canonical map. The property 1.2(1) of $\underline{A}$ together with the fact that J/MJ is a finite dimensional vector space over k entails that there exists a smallest ideal in the family Σ . Let it be J' and choose a π'-morphism $a' \xrightarrow{g} a$ where $\pi' : S/J' \to R$ is the canonical map. Note that $\overline{\Omega}_{a'} \to \overline{\Omega}_a$ is an isomorphism since $J' \subset J \subset M^2$. We claim that $a' \xrightarrow{g} a$ is versal for small prolongations of a . Indeed, let $a_1 \to a$ be a small prolongation. If we put $R_1 = P(a_1)$, we obtain an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & J/MJ & \longrightarrow & S/MJ & \longrightarrow & R & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle h_1} & & \| & & \\ 0 & \longrightarrow & k & \longrightarrow & R_1 & \longrightarrow & R & \longrightarrow & 0 \end{array}$$

If $h = 0$, then $\pi_1 : R_1 \to R$ admits a cross-section so that h_1 induces $f : S/J' \to R_1$. If $h \neq 0$, then $h_1 : S/MJ \to R_1$ is surjective and hence h_1 induces $f : S/J' \to R_1$ by the definition of the ideal J' . Therefore in either case we obtain a commutative diagram

$$\begin{array}{ccc} S/J' & \xrightarrow{\quad f \quad} & R_1 \\ & {\scriptstyle \pi'} \searrow \quad \swarrow {\scriptstyle \pi_1} & \\ & R & \end{array}$$

Since a is versal for small prolongations of e , it follows from 1.6 that we obtain a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\quad \rho \quad} & a_1 \\ & \searrow \quad \swarrow & \\ & a & \end{array}$$

I 40

Let $P : \underline{A} \to \underline{C}_\wedge$ be a cofibered category (in groupoids). It is then clear that we obtain a cofibered category $\mathrm{pro}(P) : \mathrm{pro}(\underline{A}) \to \mathrm{pro}(\underline{C}_\wedge)$ (in groupoids) (pour la définition de la catégorie de pro-objets $\mathrm{pro}(\underline{A})$, voir A. Grothendieck, Sém. Bourbaki 195).

$$\begin{array}{ccc} \underline{A} & \longrightarrow & \mathrm{pro}(\underline{A}) \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle \mathbf{pro}(P)} \\ \underline{C}_\wedge & \longrightarrow & \mathrm{pro}(\underline{C}_\wedge) \end{array}$$

Now let $\widehat{\underline{C}}_\wedge$ be the full subcategory of pro-$\underline{C}_\wedge$ in which $\mathrm{ob}(\widehat{\underline{C}}_\wedge) = \{(R_i)_{i\in\underline{N}}\ ,\ R_i \in \mathrm{ob}(\underline{C}_\wedge) \mid \varphi_i : R_i \to R_{i-1}$ is surjective for all i and induces isomorphisms $\underline{m}_i/\underline{m}_\wedge + \underline{m}_i^2 \to \underline{m}_{i-1}/\underline{m}_\wedge + \underline{m}_{i-1}^2$ for all sufficiently large $i\ \}$ where $\underline{N}$ stands for the ordered category of natural numbers, and $\underline{m}_i$ is the maximal ideal of R_i . We set $\widehat{\underline{A}}$ to be the full subcategory of pro-$\underline{A}$ in which

$$\mathrm{ob}(\widehat{\underline{A}}) = \{a \in \mathrm{ob}(\mathrm{pro}(\underline{A})) \mid \mathrm{pro}(P)\ (a) \in \mathrm{ob}(\widehat{\underline{C}}_\wedge)\}$$

We then obtain a cofibered category $\widehat{P} : \widehat{\underline{A}} \to \widehat{\underline{C}}_\wedge$ (in groupoids) with a commutative diagram

$$\begin{array}{ccc} \underline{A} & \hookrightarrow & \widehat{\underline{A}} \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle \widehat{P}} \\ \underline{C}_\wedge & \hookrightarrow & \widehat{\underline{C}}_\wedge \end{array}$$

We note that the category $\widehat{\underline{C}}_\wedge$ is canonically equivalent to the category of complete local $\wedge$-algebras of formally finite type over $\wedge$ with the fixed residue field k . Henceforth we shall make such identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$. An object ξ in $\widehat{\underline{A}}$ is called a quasi-initial object of $\widehat{\underline{A}}$ if for every object a in $\widehat{\underline{A}}$ there exists a map $\xi \to a$ in $\widehat{\underline{A}}$. A quasi-initial object ξ of $\widehat{\underline{A}}$ is called minimal if the morphism $\xi : h_R \to [\widehat{\underline{A}}]$ induces a bijection $\xi(k[\epsilon])$: $\mathrm{Hom}_{\wedge\text{-alg}}(R, k[\epsilon]) \to [\underline{A}(k[\epsilon])]$ 141
, where $R = P(\xi)$. An object ξ in $\hat{\underline{A}}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \to a$ in $\underline{A}$ induces a surjective map $\mathrm{Hom}_{\underline{A}}(\xi, a') \to \mathrm{Hom}_{\underline{A}}(\xi, a)$.

A versal formal object ξ of $\underline{A}$ is called minimal if the morphism $\xi : h_R \to [\hat{\underline{A}}]$ induces a bijection $\xi(k[\epsilon]) : \mathrm{Hom}_{\wedge\text{-alg}}(R, k[\epsilon]) \to [\underline{A}(k[\epsilon])]$ where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$.

(i) every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\hat{\underline{A}}$.

(ii) If ξ is an object in $\hat{\underline{A}}$ such that $\widetilde{\xi}(k[\epsilon])$: $\mathrm{Hom}_{\wedge\text{-alg}}(R, k[\epsilon]) \to [\underline{A}(k[\epsilon])]$ is injective, where $P(\xi) = R$, then every endomorphism of ξ in $\hat{\underline{A}}$ is an automorphism. In particular a quasi-initial minimal object of $\hat{\underline{A}}$, if it exists, is unique up to (non-canonical) isomorphism.

Proof: (i) let ξ be a versal formal object of $\underline{A}$. We must show that, for any object a in $\hat{\underline{A}}$, we have a map $\xi \to a$. By the definition of $\hat{\underline{A}}$, there exists a sequence of surjective maps in $\underline{A}$: $- - - \to a_n \xrightarrow{\alpha_n} a_{n-1} \to - - - \to a_1 \xrightarrow{\alpha_1} a_0$ such that $a = \varprojlim_n a_n$. Assume that there exist maps $\varphi_i : \xi \to a_i$ such that $\alpha_i\varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is a versal formal object of $\underline{A}$ and $\alpha_n : a_n \to a_{n-1}$ is surjective in $\underline{A}$, we obtain a map $\xi \xrightarrow{\varphi_n} a_n$ such that $\alpha_n\varphi_n = \varphi_{n-1}$. If we set $\varphi = \varprojlim_n \varphi_n$, we have $\varphi : \xi \to a$.

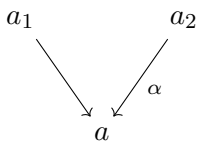
(ii) Let $\xi \xrightarrow{\alpha} \xi$ be a map in $\hat{A}$. Then for every $\lambda \in \mathrm{Hom}_{\wedge\text{-alg}}(R, k[\epsilon])$ we have $\lambda_*\xi = \lambda_*\alpha\xi \simeq (\lambda \cdot P(\alpha))_*\xi$ and hence by the hypothesis on ξ we have that $\lambda = \lambda \cdot P(\alpha)$ for all $\lambda \in \mathrm{Hom}_{\wedge\text{-alg}}(R, k[\epsilon])$, and therefore $P(\alpha) : R \to R$ is surjective. However it is trivial to see that every surjective endomorphism of a noetherian ring is always an automorphism and hence $P(\alpha)$ is an automorphism of R and therefore α is an automorphism of ξ .

Theorem 1.11. (Schlessinger) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$ admits a 142
minimally-versal formal object if and only if

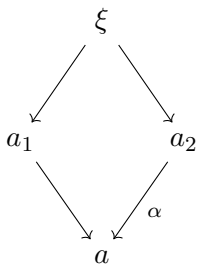
(i) $\underline{A}$ is semi-homogeneous

(ii) $[\underline{A}(k[\epsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

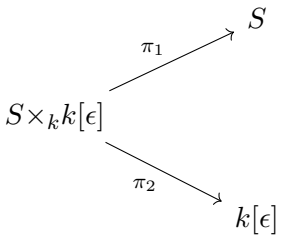
Proof; (Necessity) Let ξ in $\hat{\underline{A}}$ be a minimally-versal formal object of $\underline{A}$, and consider a diagram



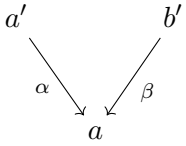
in $\underline{A}$ where α is surjective. Since ξ is a quasi-initial object of $\hat{\underline{A}}$, there exists a map $\xi \to a_1$. Since α is surjective, the composite map $\xi \to a_1 \to a$ can be lifted to a_2 i.e., we obtain a commutative diagram



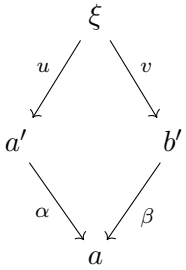
which proves the condition 1.2(1). We now prove the condition 1.2(2). Let



be the projections,
and let



be π_1-maps in $\underline{A}$. By the definition of ξ , we obtain a commutative diagram



and in particular we have $\pi_1 \cdot P(u) = \pi_1 \cdot P(v)$. Assume now that $(\pi_2)_* a' \approx (\pi_2)_* b'$. We then have $\pi_2 \cdot P(u) = \pi_2 \cdot P(v)$ because of the minimality of ξ , and hence $P(u) = \pi_1 P(u) \times \pi_2 P(u) =$ 143
$\pi_1 P(v) \times \pi_2 P(v) = P(v)$. Since $\underline{A}$ is a cofibered groupoid, we obtain a unique isomorphism $a' \xrightarrow{\sigma} b'$ in $\underline{A}(S \times_k k[\epsilon])$ such that $\sigma u = v$. Then $\beta\sigma u = \alpha u$ entails that $\beta\sigma = \alpha$ since $\underline{A}$ is a cofibered category.

(Sufficiency) Let $c_1, c_2, \ldots, c_n$ be objects in $\underline{A}(k[\epsilon])$ which form a k-basis of the vector space $[\underline{A}(k[\epsilon])]$, and choose a_1 in $\underline{A}(k[\epsilon] \times_k \cdots \times_k k[\epsilon])$ such that $(\pi_j)_* a_1 \approx c_j$. Then $\tilde{a}_1(k[\epsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\epsilon]) \to [\underline{A}(k[\epsilon])]$ is bijective, where $R_1 = P(a_1)$. In particular $a_1 \to e$ is

versal for small prolongations of e . Choose $a_2 \to a_1$ which is versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \to \overline{\Omega}_{a_1}$ is bijective. We then choose $a_3 \to a_2$, versal for small prolongations of a_2 such that $\overline{\Omega}_{a_3} \to \overline{\Omega}_{a_2}$ is bijective etc., and set $\xi = \varprojlim_n a_n$. We claim that ξ is a minimally-versal formal object of $\underline{A}$. Indeed, since $\overline{\Omega}_{a_n} \to \overline{\Omega}_{a_{n-1}}$ is bijective, for all n , $\overline{\Omega}_\xi \to \overline{\Omega}_{a_1}$ is bijective so that $\xi(k[\epsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\epsilon]) \to [\underline{A}(k[\epsilon])]$ is bijective. Now consider a diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow{\scriptstyle \alpha} \\ \xi & \xrightarrow{h} & a \end{array}$$

where α is surjective in $\underline{A}$. We must show that h can be lifted to a' . For this, one may assume, by induction, that a' is a small prolongation of a . Since $P(a)$ is an artinian ring, $h : \xi \to a$ is factored as a composite map $\xi \to a_q \longrightarrow a$ for some q . Since $\underline{A}$ is semi-homogeneous, we get a diagram

$$\begin{array}{ccccc} & & a'' & \longrightarrow & a' \\ & & \downarrow{\scriptstyle \alpha'} & & \downarrow{\scriptstyle \alpha} \\ \xi & \longrightarrow & a_q & \longrightarrow & a \end{array}$$

where α' is also a small prolongation. We then obtain a commutative diagram

$$\begin{array}{ccccc} & & & & a'' \\ & & & \nearrow & \downarrow{\scriptstyle \alpha'} \\ \xi & \longrightarrow & a_{q+1} & \longrightarrow & a_q \end{array}$$

and therefore we are done.

Remark 1.12. Let $\ldots\ldots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \ldots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_o = e$ **I 44**
be an infinite sequence of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1} \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective} \\ \overline{\Omega}_{a_i} \dashrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i >> 0 \end{cases}$$

Then the above proof shows that $\varprojlim a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$. If $\underline{A}$ has the property 1.11(i) and (ii), then so does $[\underline{A}]$, and $\xi \in \mathrm{ob}(\hat{\underline{A}})$ is a minimally-versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$ by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, admitting a minimally-versal formal object ξ . Then an invariant of $R = P(\xi)$, which is a complete local $\wedge$-algebra of formally finite type over $\wedge$, is automatically an invariant of $\underline{A}$ over $\underline{C}_\wedge$. Therefore, for example, we shall say that $\underline{A}$ is (formally) smooth over $\underline{C}_\wedge$ if R is (formally) smooth over $\wedge$ i.e., a formal power-series algebra over $\wedge$, or we say that $\underline{A}$ is irreducible over $\underline{C}_\wedge$ if $\mathrm{Spec}(R)$ is irreducible etc. In the next section we introduce the notion of "smoothness" of a functor $\underline{A} \to \underline{B}$ over $\underline{C}_\wedge$, where $\underline{A}, \underline{B}$ are cofibered categories in groupoids over $\underline{C}_\wedge$. It turns out (cf. 2.4) that "$\underline{A}$ is (formally) smooth over $\underline{C}_\wedge$" is equivalent to " $P : \underline{A} \to \underline{C}_\wedge$ is a smooth functor" , and therefore no confusion arises. We also define the dimension of $\underline{A}$ over $\underline{C}_\wedge$ by $\dim_{\underline{C}_\wedge} \underline{A} = \dim\, k \otimes_\wedge R$. For this we note the following fact: Let $\underline{C}_k$ be the category of artinian local k-algebras with the same residue field k . Then $\underline{C}_k \to \underline{C}_\wedge$ is a fully-faithful imbedding, whose (left) adjoint functor $\underline{C}_\wedge \to \underline{C}_k$ is given by $R \to k \otimes_\wedge R$. Let $\underline{A}|\underline{C}_k$ be the pull-back of $P : \underline{A} \to \underline{C}_\wedge$ under $\underline{C}_k \to \underline{C}_\wedge$. Then it is trivial to see that if ξ is a minimally-versal formal object for $\underline{A}$ over $\underline{C}_\wedge$, then $k \otimes_\wedge \xi$ is a minimally-versal formal object for $\underline{A}|\underline{C}_k$ over $\underline{C}_k$, where $\xi \to k \otimes_\wedge \xi$ is a cocartesian arrow above $R \to k \otimes_\wedge R$. It follows that $\dim_{\underline{C}_\wedge} \underline{A} = \dim_{\underline{C}_k} (\underline{A}|\underline{C}_k)$.

1.15. In the remaining of this paragraph, we briefly explain how to modify the preceding theory in order to allow for residue field extensions. I 45

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. We assume that

(a) the natural map from Λ to K is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ .

The most interesting case is when $K = K'$ and K is a finite inseparable extension of k .

We denote by $\hat{C}_{\Lambda,K}$ the category whose objects are the complete noetherian local rings R , provided with a structure of Λ-algebra and with a Λ-epimorphism $s : R \longrightarrow K$.

We denote by $C_{\Lambda,K}$ the subcategory of $\hat{C}_{\Lambda,K}$ consisting of the (R, s) such that $\mathrm{Ker}(s)$ is a R-module of finite length. The category $\hat{C}_{\Lambda,K}$ can be identified with a full subcategory of $\mathrm{Pro}(C_{\Lambda,K})$.

If K is artinian, so is any object in $C_{\Lambda,K}$.

For any finite dimensional vector space V over K' we denote by $D_V(K)$ the algebra of dual numbers over K defined by V ; its elements are of the form $k + v\,\varepsilon$ (with $k \in K$, $v \in V$ and $\varepsilon^2 = 0$: $(k + v\,\varepsilon)\,(k' + v'\,\varepsilon) = (k\,k') + (k\,v' + k'\,v)\varepsilon$). The map $s : D_V(K) \to K : s(k + v\,\varepsilon) = k$, turns $D_V(K)$ into an object of $C_{\Lambda,K}$. The role played before by $k[\varepsilon]$ will be played here by $D_{K'}(K)$.

As explained in 1.8, any cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends in a natural way to a cofibered category $\hat{\underline{A}}$ over $\hat{C}_{\Lambda,K}$. A map in $\underline{A}$ or $\hat{\underline{A}}$ will be called surjective if the corresponding homomorphism of algebras in $C_{\Lambda,K}$ or in $\hat{C}_{\Lambda,K}$ is.

We assume that $\underline{A}(K) = \{e\}$. I 46

Definition 1.16. A cofibered groupoïd $\underline{A}$ over $C_{\Lambda,K}$ is called semi-homogeneous if the following properties hold:

(I) every diagram

$$\begin{array}{ccc} a'' & \dashrightarrow & a' \\ \vdots & & \downarrow \\ a_1 & \longrightarrow & a \end{array}$$

of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed into a commutative diagram including dotted arrows;

(II) For any R , the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \to [\underline{A}(R) \times \underline{A}\,(D_{K'}(K))] \simeq [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, so is $[\underline{A}]$.

1.17. Let us first consider the restriction of $\underline{A}$ to the full subcategory of $C_{\Lambda,K}$ whose objects are the $D_V(K)$. Let Ω be the vector space $\Omega^1_{K/\Lambda} \otimes_K K'$, and $d : K \to \Omega$ the derivative. A map $f : D_V(K) \to D_W(K)$ can be written:

$$f(k + v\,\varepsilon) = k + (u(v) + D\,d\,k)\,\varepsilon$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$. If we identify f with (u, D) , the composition law is

$$(u, D) \circ (u', D') = (uu', D + u\,D') \ .$$

The condition (II) entails

(II') the functor from K'-vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume condition (II') is satisfied by $\underline{A}$. Then (1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector space structure, such that $[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))]$ as a functor in V . I 47

(2) There is a linear map $\gamma : \mathrm{Hom}(\Omega, K') \to [\underline{A}(D_{K'}(K))]$ such that, for any $D \in \mathrm{Hom}(\Omega, K')$,

$$[\underline{A}((1, D))] : [\underline{A}(D_{K'}(K))] \to [\underline{A}(D_{K'}(K))] \quad \text{is} \quad x \longmapsto x + \gamma(D) \quad .$$

(3) For any $(u, D) : D_V(K) \longrightarrow D_W(K)$, the map

$$(u, D) : [\underline{A}(D_V(K))] \to [\underline{A}(D_W(K))] \quad \text{i.e.}$$
$$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \to W \otimes [\underline{A}(D_{K'}(K))] \quad \underline{\text{is}}$$
$$x \longmapsto u(x) + \gamma(D) \quad .$$

Let us introduce the notations

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))] \\ A &= F(K') \\ F(u, D) : F(V) \to F(W) &= \text{the map induced by } (u, D) : D_V(K) \to D_W(K) \\ F(D) : F(V) \to F(V) &: F(D) = F(1, D) \quad . \end{aligned}$$

By (II') and the fact that K' is a K'-vector space object in the category of K'-vector spaces, $F(K')$ is a K'-vector space in (Ens), hence a K'-vector space, and (1) is clear. Let us now identify $F(V)$ and $V \otimes A$.

We deduce from the identity

$$(u, D) = (1, D) \circ (u, 0) \quad \text{that}$$
$$F(u, D) = F(D) \circ (u \otimes 1_A) \quad .$$

The functoriality of F amounts to

$$\begin{aligned} F(0) &= \mathrm{Id} \\ F(D) \circ F(D') &= F(D + D') \\ (u \otimes 1) \circ F(D) &= F(u\, D) \circ (u \otimes 1_A) \end{aligned}$$

For any $D : \Omega \to V$ and any $n \in N$, if i and p are the maps

$$i \ ; \ V \hookrightarrow V \oplus K'^n \qquad\qquad p : V \oplus K'^n \longrightarrow K'^n \quad ,$$

then $piD = 0$ and the following diagram is commutative I 48

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \\ \downarrow{\scriptstyle F(D)} & & \downarrow{\scriptstyle F(i\,D)} & & \| \\ V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \end{array}$$

Let $x : K'^n \to V$ be a map, $x_i = x(e_i)$, and let a_i be a family of elements of A . We have $(1_V + x) \circ i \circ D = D$. Hence, if

$$F(i\,D)\ (\Sigma\ e_i\ a_i) = \Sigma\ e_i\ a_i + a \qquad (a \in V \otimes A) \quad , \text{ then}$$
$$F(D)\ (\Sigma\ x_i\ a_i)\ = \Sigma\ x_i\ a_i + a$$

and a does not depend on the x_i . Hence,

$$\begin{aligned} F(D)\ (y) &= y + G(D) && \text{, with} \\ G(D+D') &= G(D) + G(D') && \\ u\ G(D) &= G(u\,D) && \text{(for any } u : V \to W) \quad . \end{aligned}$$

One easily checks that any such G is defined, for some $\gamma \in \Omega \otimes A$, by the formula

$$G(D) = D \otimes 1_A\ (\gamma) \quad ,$$

as promised in (2) (3).

Definition 1.19. (1) An object ξ in $\widehat{\underline{A}}$ is called versal formal object if every surjective map $\alpha : a' \longrightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{\underline{A}}}\ (\xi, a) \quad .$$

(2) A formal versal object ξ in $\widehat{A}$, with image (R, s) in $\widehat{C}_{\Lambda,K}$, is called minimal if for any two maps φ_1 , $\varphi_2 : R \rightrightarrows D_{K'}(K)$, if $\varphi_1(\xi)$ is iso- morphic to $\varphi_2(\xi)$, then there exists a derivation D of K into K' such that $(1, D)\ (\varphi_1) = \varphi_2$, i.e. φ_2-$\varphi_1 = D\,s$.

Theorem 1.20. If $\underline{A}$ is semi-homogeneous and if ,

$$\dim_{K'}\ ([A(D_{K'}(K))]) < \infty \quad ,$$

then $\underline{A}$ admits a minimally versal formal object, and any two of them are isomorphic. I 49

With the notations of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$, supplementary to the image of γ , and let H be the dual of H^* . We define $R_o = D_H(K)$. Let ξ_o be an object in $\underline{A}(R_o)$, whose image in $[\underline{A}(R_o)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}(H^*, [\underline{A}(D_{K'}(K))])$ is the inclusion of H^* in $[\underline{A}(D_{K'}(K))]$. Then

(1) The maps $\mathrm{Hom}_{C_{\wedge,K}}(D_H(K),\ D_V(K)) \longrightarrow [\underline{A}(D_V(K))]$ are surjective .

(2) The minimality condition 1.19(2) is valid for $(R_o,\ \xi_o)$.

Choose now some $S \in \hat{C}_{\wedge,K}$, formally smooth over $\wedge$, and an epimorphism $t : S \longrightarrow R_o$ such that

$$\Omega_{R_o/\wedge} \otimes K' \xrightarrow{\sim} \Omega_{S/\wedge} \otimes K' \quad .$$

If M is the maximal ideal of S , we inductively define, for $n \geq 1$, $R_n = S/I_n$ and $\xi_n \in \mathrm{Ob}(\underline{A}(R_n))$ by the properties that

(a) I_{n+1} is the intersection of the ideals $I : I_n M \subset I \subset I_n$, such that ξ_n extends to an object ξ in $\underline{A}(S/I)$.

(b) ξ_{n+1} is an extension of ξ_n .

If $R = \varprojlim R_n$ and $\xi = \varprojlim \xi_n$, then one checks that $(R,\ \xi)$ is a minimally-versal formal object, and the only one up to isomorphism.

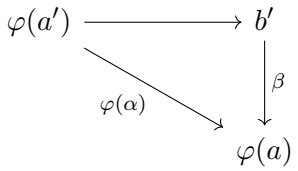
2. Prorepresentable cofibered groupoids.

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, admitting a minimally-versal formal object ξ . In general we can not recover $\underline{A}$ (up to $\underline{C}_\wedge$-equivalence) out of ξ alone. One has to know when two maps $R \overset{f}{\underset{g}{\rightrightarrows}} S$ yield a $\underline{C}_\wedge$-isomorphism $\xi(f) \longrightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12 we now consider the question of "recovering $\underline{A}$ " up to $\underline{C}_\wedge$-equivalence. Indeed, we shall have a quite I 50 satisfactory answer applicable to a wide class of cofibered groupoids.

A number of cofibered groupoids which usually occur in practice enjoy a slightly stronger property than those of semi-homogeneous ones, as it will be defined explicitly in 2.5. To investigate them it is convenient to rephrase what we have done so far in terms of "smooth functors" . All the notations and terminologies in the previous section remain in force throughout in this section.

Definition 2.1. Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\wedge$. We say that φ is smooth if every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

(*)



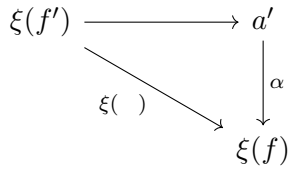
for some $\alpha : a' \longrightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and $\varphi(k[\epsilon]) : [\underline{A}(k[\epsilon])] \longrightarrow [\underline{B}(k[\epsilon])]$ is bijective.

Remark 2.2(a) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\wedge$. We note that " φ is smooth" $\Leftrightarrow$ "every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram (*) such that $\varphi(a') \longrightarrow b'$ is a $\underline{C}_\wedge$-isomorphism" $\Leftrightarrow$ "every small prolongation $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed into a commutative diagram (*) ".

Remark 2.2(b) For any cofibered groupoid $\underline{A}$ over $\underline{C}_\wedge$, the natural functor $\underline{A} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.2(c) For each object R in $\hat{C}_\wedge$, we set $h_R : \underline{C}_\wedge \longrightarrow$ (Ens) to be the functor defined by $h_R(S) = \mathrm{Hom}_{\wedge\text{-alg}}(R,\ S)$ for all S in $\underline{C}_\wedge$. It associates a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\wedge$, and a map $R \longrightarrow S$ in $\underline{C}_\wedge$ induces a functor $\underline{h}_S \longrightarrow \underline{h}_R$ of cofibered groupoids. The functor I 51
$\underline{h}_S \longrightarrow \underline{h}_R$ induced by a map $R \longrightarrow S$ in $\underline{C}_\wedge$ is smooth if and only if S is a formal power-series ring over R .

Remark 2.2(d) Given an object R in $\widehat{\underline{C}}_\wedge$, let us denote by $(R,\ \underline{C}_\wedge)$ the category of arrows in $\widehat{\underline{C}}_\wedge$ with the source R and a target in $\underline{C}_\wedge$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$. Given $\xi \in \mathrm{ob}\ \underline{A}$ with $P(\xi) = R$, let $\underline{A}' \longrightarrow (R,\ \underline{C}_\wedge)$ be the pull-back of $\underline{A} \longrightarrow \underline{C}_\wedge$ under the canonical functor $(R,\ \underline{C}_\wedge) \longrightarrow \underline{C}_\wedge$ and choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$. (This simply means that we choose, for each map $R \xrightarrow{f} S$, a f-map $\xi \longrightarrow \xi(f)$). We then obtain a functor $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\wedge$ by setting $(R \xrightarrow{f} S) \longmapsto \xi(f)$. Of course this functor does not depend on the choice of $\xi^\#$ up to natural equivalence of functors. Now " $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ is (minimally) smooth" means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed into a commutative diagram



(and $\underline{h}_R(k[\epsilon]) \longrightarrow \underline{A}(k[\epsilon])$ is bijective), which means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\wedge$. Thus we conclude that the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor $\underline{h}_R \longrightarrow \underline{A}$ over $\underline{C}_\wedge$ for some $R \in \mathrm{ob}(\widehat{\underline{C}}_\wedge)$.

The following general properties on smoothness are trivial to verify.

Proposition 2.3. Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\wedge$.

(a) If φ, ψ are both smooth, then so is $\psi.\varphi$.

(b) If $\psi.\varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth. I 52

Remark 2.4. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, admitting a versal formal object ξ . This means, by 2.2(d), that $\tilde{\xi} : \underline{h}_R \to \underline{A}$ is smooth, where $R = P(\xi)$. Therefore we have, " $\underline{A}$ is (formally) smooth over $\underline{C}_\wedge$ (c.f. 1.14) i.e., R is a formal power-series algebra over $\wedge$ " $\Leftrightarrow$ "the composite functor $\underline{h}_R \xrightarrow{\tilde{\xi}} \underline{A} \xrightarrow{P} \underline{C}_\wedge\ (= \underline{h}_\wedge)$ is smooth" $\Leftrightarrow$ " $\underline{A} \xrightarrow{P} \underline{C}_\wedge$ is a smooth functor" by 2.3(b).

We now consider a stronger property than being semi-homogeneous.

Definition 2.5. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$. It is called homogeneous if every diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \\ b' & \longrightarrow & a \end{array}$$

where $a' \to a$ is surjective admits a fibre-product $a' \underset{a}{\times} b'$ in $\underline{A}$ such that

$$P(a' \times b') = P(a') \underset{P(a)}{\times} P(b') \ .$$

Remark 2.6(a) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$, and consider a diagram

(*) $\begin{array}{ccccc} & & a' & & \\ & \swarrow & & \searrow & \\ a_1 & & & & a_2 \\ & \searrow & & \swarrow & \\ & & a & & \end{array}$ in $\underline{A}$ above the diagram $\begin{array}{ccccc} & & R_1 \underset{R}{\times} R_2 & & \\ & \swarrow^{\pi_1} & & \searrow^{\pi_2} & \\ R_1 & & & & R_2 \\ & \searrow & & \swarrow & \\ & & R & & \end{array}$

in $\underline{C}_\wedge$ where π_i are the projections. If a fibre-product $a_1 \underset{a}{\times} a_2$ exists in $\underline{A}$, we obtain a unique map $\varphi : a' \to a_1 \underset{a}{\times} a_2$ compatible with the projections. If $P(a_1 \underset{a}{\times} a_2) = R_1 \underset{R}{\times} R_2$, then $P(\varphi)$ is the identity map on $R_1 \underset{R}{\times} R_2$ and therefore φ is a $\underline{C}_\wedge$-isomorphism i.e., $a' = a_1 \underset{a}{\times} a_2$. In other words, if $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\wedge$, every diagram (*) defines a fibre-product $a_1 \underset{a}{\times} a_2$ provided $R_1 \to R$ is surjective in $\underline{C}_\wedge$. This fact entails immediately that a homogeneous cofibered groupoid is, a fortiori, semi-homogeneous. 153

Remark 2.6(b) Consider a diagram

$$\begin{array}{ccccc} X & & & & Y \\ & \searrow^{\varphi} & & \swarrow^{\psi} & \\ & & Z & & \end{array}$$

of categories and functors. The ordinary fibre-product $X \times_Z Y$ is a category in which an object is a pair (x, y) such that $\varphi(x) = \psi(y)$, and a map $(x, y) \to (x', y')$ consists of a pair $\alpha : x \to x'$, $\beta : y \to y'$ such that $\varphi(\alpha) = \psi(\beta)$. The fibre-product $X \times^2_Z Y$ in the sense of 2-categories is a category in which an object is a triple $(x, y; \gamma)$, where $\varphi(x) \xrightarrow{\gamma} \psi(y)$ is an isomorphism, and a map $(x, y; \gamma) \to (x', y'; \gamma')$ consists of a pair $x \xrightarrow{\alpha} x'$ $y \xrightarrow{\beta} y'$ with a commutative diagram

$$\begin{array}{ccc} \varphi(x) & \xrightarrow{\varphi(\alpha)} & \varphi(x') \\ \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \gamma'} \\ \psi(y) & \xrightarrow[\psi(\beta)]{} & \psi(y') \end{array}$$

One can rephrase the notions of (semi)homogeneous cofibered groupoid or smooth functor in terms of fibre-product of categories (in the sense of 2-category). Now let $P : \underline{A} \to \underline{C}_\wedge$ be a cofibered category, and consider an object $R_1 \times_R R_2$ in $\underline{C}_\wedge$. If we choose a direct image functor $(\pi_i)_* : \underline{A}(R_1 \times_R R_2) \to \underline{A}(R_i)$, where $\pi_i : R_1 \times_R R_2 \to R_i$ is the projection for $i = 1, 2$, we obtain a functor $\underline{A}(R_1 \times_R R_2) \to \underline{A}(R_1) \times^2_{\underline{A}(R)} \underline{A}(R_2)$. It is clear from the definition of cofibered category that this functor, up to canonical natural equivalence, does not depend on the choice of direct image functors. It is now clear from the definition that a cofibered groupoid $\underline{A}$ over $\underline{C}_\wedge$ is homogeneous over $\underline{C}_\wedge$ if and only if $\underline{A}(R_1 \times_R R_2) \to \underline{A}(R_1) \times^2_{\underline{A}(R)} \underline{A}(R_2)$ is an equivalence of categories whenever $R_1 \to R$ is surjective in $\underline{C}_\wedge$, and $\underline{A}$ is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \longrightarrow [\underline{A}(R_1) \times^2_{\underline{A}(R)} \underline{A}(R_2)]$$

is surjective whenever $R_1 \longrightarrow R$ is surjective in $\underline{C}_\wedge$, and is bijective whenever $R = k$ and 154
$R_1 = k[\epsilon]$. Now let $\varphi : \underline{A} \to \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\wedge$. For each map $R' \to R$ in $\underline{C}$, φ induces a functor $\underline{A}(R') \to \underline{A}(R) \times^2_{\underline{B}(R)} \underline{B}(R')$ uniquely determined up to a (canonical) natural equivalence. It is again trivial to see that φ is smooth if and only if the functor $A(R') \to \underline{A}(R) \times^2_{\underline{B}(R)} \underline{B}(R')$ is surjective on isomorphism classes of objects, whenever $R' \to R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is a homogeneous cofibered groupoid over $\underline{C}_\wedge$, the functor $A(k[V \times W]) \to \underline{A}(k[V]) \times \underline{A}(k[W])$ is an equivalence of categories for any finite-dimensional vector spaces V, W over k (We recall our convention $\underline{A}(k) = \{e\}$). In particular, we have

$$\mathrm{Aut}_{\underline{A}(k[V\times W])}(1_{V\times W}) \xrightarrow{\sim} \mathrm{Aut}_{\underline{A}(k[V])}(1_V) \times \mathrm{Aut}_{\underline{A}(k[W])}(1_W)$$

where 1_V is the direct image of e under the unique map $k \to k[V]$ in $\underline{C}_\wedge$. It follows that if $\underline{A}$ is homogeneous over $\underline{C}_\wedge$, $\mathrm{Aut}(1_\epsilon)$ as well as $[\underline{A}(k[\epsilon])]$ are canonically provided with vector space structures over k .

Remark 2.6(d). For any object R in $\hat{\underline{C}}_\wedge$, the cofibered (discrete) groupoid $\underline{h}_R$ over $\underline{C}_\wedge$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. However, the homogeneity of $\underline{A}$ does not imply in general the homogeneity of $[\underline{A}]$. Indeed we have

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\wedge$. The following statements are equivalent.

(i) $[\underline{A}]$ is homogeneous i.e. $[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$ is bijective for every small extension $R' \to R$.

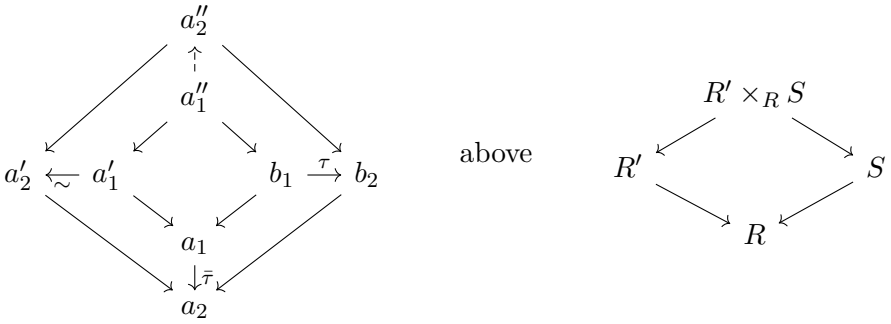
(ii) For every small prolongation $a' \to a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{\underline{A}(R')}(a') \longrightarrow \mathrm{Aut}_{\underline{A}(R)}(a)$$

is surjective, where $R' = P(a')$, $R = P(a)$. 155

Proof. (i) $\Rightarrow$ (ii) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\wedge$ and let $a' \to a$ be a f-map in $\underline{A}$. Given $\tau \in \mathrm{Aut}_{\underline{A}(R)}(a)$, we consider $a' \times_a a'$ and $a' \times_a a'_\tau$ where $a'_\tau \longrightarrow a$ denotes the composite map $a' \longrightarrow a \xrightarrow{\tau} a$. Since $[A(R' \times_R R')] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$ is bijective, we conclude the existence of an isomorphism $a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau$, and u induces, via the projection on the second factor, an isomorphism $a' \xrightarrow{\tau'} a'$ which is a lifting of τ .

(ii) $\Rightarrow$ (i) Let a''_i $(i = 1, 2)$ be objects in $\underline{A}(R' \times_R S)$ which have the same image under $[\underline{A}(R' \times_R S)] \to [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$. We then obtain a diagram of solid arrows



where $\bar{\tau}$; $a_1 \to a_2$ is the induced isomorphism from τ so that the square box on the lower right corner is commutative. If the condition (ii) is verified, one can always replace the given isomorphism $a'_1 \xrightarrow{\sim} a'_2$ so that the square box on the left corner is also commutative. Since $a''_i = a'_i \times_{a_i} b_i$ for

$i = 1, 2$ by 2.6(a), we obtain the isomorphism $a_1'' \xrightarrow{\sim} a_2''$. This proves that $[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$ is injective, and hence is bijective since $\underline{A}$ is homogeneous by hypothesis.

Now let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$. We have a canonical map $\tilde{f} : R' \times_R R' \longrightarrow k[\mathrm{Ker}\ f]$ given by $(r_1' , r_2') \longrightarrow r_1'(0) + (r_1' - r_2')$, **156**
and $1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\mathrm{Ker}\ f]$ is an isomorphism. Given any two f-maps $a_i \to a$ $(i = 1, 2)$ in $\underline{A}$, $a_1 \times_a a_2$ is an object over $R' \times_R R'$ and in turn we may consider $\tau(a_1,\ a_2) = (\tilde{f})_*(a_1 \times_a a_2)$ which is an object in $\underline{A}(k[\mathrm{Ker}\ f])$. Since $1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\mathrm{Ker}\ f]$ is an isomorphism, we have the isomorphism $a_1 \times_a a_2 \simeq a_1 \times \tau(a_1,\ a_2)$ compatible with the projections on the first factor. The statement 1.5(2) takes a more definite form for a homogeneous groupoid.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and $a_i \xrightarrow{\alpha_i} a$ $(i = 1, 2)$ f-maps in $\underline{A}$. Then

(1) There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ if and only if there exists an arrow $a_1 \to \tau(a_1,\ a_2)$ in $\underline{A}$.

(2) There exists $\rho : a_1 \to a_2$ with $\alpha_2 \cdot \rho = \alpha_1$ such that $P(\rho) = \mathrm{id}_{R'}$ if and only if $\tau(a_1,\ a_2) = 0$ in $[\underline{A}(k[\mathrm{Ker}\ f])]$.

Proof: We prove both at the same time. There exists $\rho : a_1 \to a_2$ in $\underline{A}$ with $\alpha_2 \cdot \rho = \alpha_1$ (such that $P(\rho) = \mathrm{id}_{R'}$) $\Leftrightarrow$ there exists $h : a_1 \to a_1 \times_a a_2$ in $\underline{A}$ with $\mathrm{pr}_1 \circ h = \mathrm{id}_{a_1}$ (such that $P(h) =$ diagonal map) $\Leftrightarrow$ there exists $g : a_1 \to a_1 \times \tau(a_1,\ a_2)$ in $\underline{A}$ with $\mathrm{pr}_1 \circ g = \mathrm{id}_{a_1}$ (such that $P(g) : R' \to R' \times_k k[\mathrm{Ker}\ f]$ is given by $r' \to (r',\ r'(0))$) $\Leftrightarrow$ there exists an arrow $\gamma : a_1 \to \tau(a_1,\ a_2)$ in $\underline{A}$ (such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is given by $r' \to r'(0)$) . However, there exists an arrow $\gamma : a_1 \to \tau(a_1,\ a_2)$ in $\underline{A}$ such that $P(\gamma) : R' \to k[\mathrm{Ker}\ f]$ is factored through $R' \to k \to k[\mathrm{Ker}\ f]$ if and only if $\tau(a_1,\ a_2) = 0$.

Proposition 2.9. (a) Let $\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$ be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

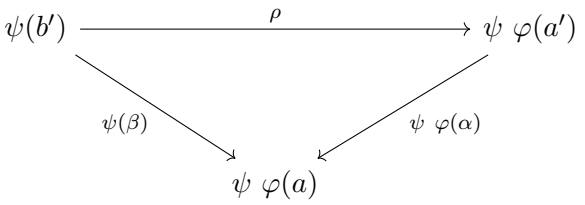
(i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous

(ii) $\psi(k[\epsilon]) : [\underline{B}(k[\epsilon])] \longrightarrow [\underline{C}(k[\epsilon])]$ is injective.

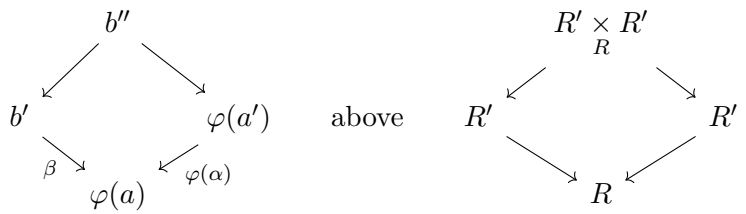
Then the composite functor $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth. **157**

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then $\widehat{\varphi} : \widehat{\underline{A}} \to \widehat{\underline{B}}$ is smooth (i.e., for any surjective map $b' \xrightarrow{\beta} \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$, there exists $\alpha : a' \to a$ in $\widehat{\underline{A}}$ and a $\widehat{\underline{C}}_\Lambda$-isomorphism $\rho : \widehat{\varphi}(a') \xrightarrow{\sim} b'$ such that $\beta \cdot \rho = \varphi(\alpha)$).

Proof: (a) Assume that the composite functor $\psi \circ \varphi$ is smooth. If φ is smooth, then so is ψ by 2.3(b), and therefore it suffices to show that φ is smooth. Let a small prolongation $b' \xrightarrow{\beta} \varphi(a)$ in $\underline{B}$ be given. Since the composite functor $\psi \circ \varphi$ is smooth, there exists an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism $\rho : \psi(b') \longrightarrow \psi\ \varphi(a')$ with a commutative diagram



Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

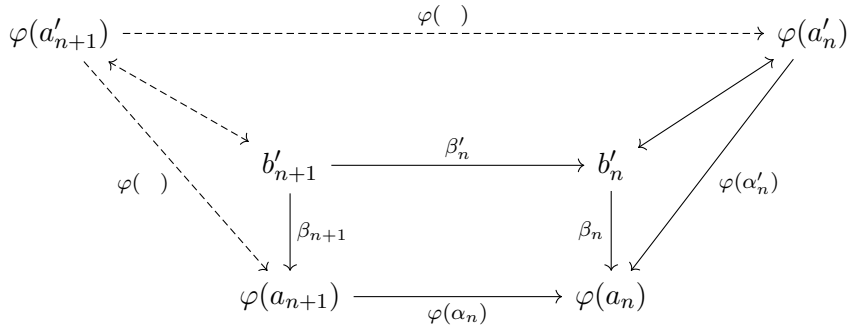


where $R = P(a)$, $R' = P(b') = P(a')$. Since $\underline{C}$ is homogeneous, $\psi(b'') = \psi(b') \underset{\psi\ \varphi(a)}{\times} \psi\varphi(a')$ by 2.6(a), and it follows from 2.8(2) that $0 = \tau(\psi\ \varphi(a'), \psi(b')) = (\widetilde{f})_*\psi(b'')$, and hence $(\widetilde{f})_*(b'') = 0$ by the hypothesis (ii). It follows from 1.5(2) that there exists a map $\gamma : b' \longrightarrow \varphi(a')$ such that $\beta = \varphi(\alpha) \cdot \gamma$.

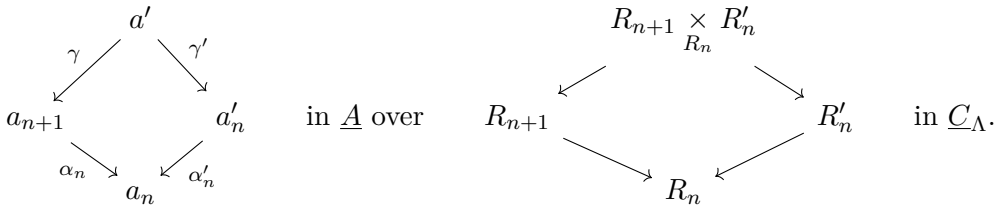
(b) Let a surjective map $\beta : b' \to \varphi(a)$ in $\widehat{\underline{B}}$ be given. This means that we are given a sequence of commutative diagrams

$$\begin{array}{ccccc} \cdots\cdots\rightarrow & b'_3 & \xrightarrow{\beta'_2} & b'_2 & \xrightarrow{\beta'_1} & b'_1 \\ & \downarrow{\scriptstyle \beta_3} & & \downarrow{\scriptstyle \beta_2} & & \downarrow{\scriptstyle \beta_1} \\ \cdots\cdots\rightarrow & \varphi(a_3) & \xrightarrow[\varphi(\alpha_2)]{} & \varphi(a_2) & \xrightarrow[\varphi(\alpha_1)]{} & \varphi(a_1) \end{array} \qquad \text{I 58}$$

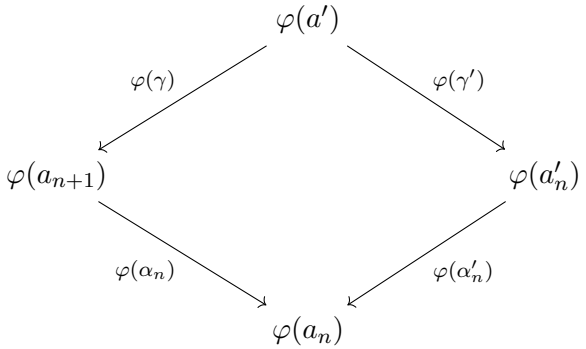
in $\underline{B}$, where α_v , β_v , β'_v are surjective for all v , such that $\beta = \varprojlim_v \beta_v$. To show that $\widehat{\varphi}$ is smooth, it suffices to prove that a commutative diagram below of solid arrows in which α_n , β_n , β_{n+1} , β'_n are surjective can be completed into a commutative diagram including dotted arrows.



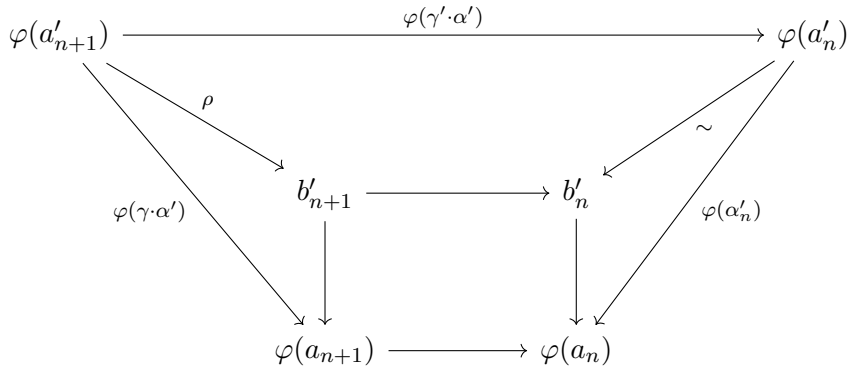
Now $\underline{A}$ is, by hypotheses, semi-homogeneous and $\alpha'_n : a'_n \to a_n$ is surjective, and therefore there exists a commutative diagram



The hypothesis that $\underline{B}$ is homogeneous entails that



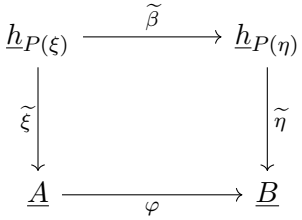
is a diagram of a fibre-product in $\underline{B}$, and therefore we obtain a unique map $\delta : b'_{n+1} \to \varphi(a')$ in $\underline{B}$ with an appropriate commutativity. Since β_{n+1} , β'_n are both surjective, so is δ and therefore, since φ is smooth, there exists $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism $\varphi(a'_{n+1}) \xrightarrow[\sim]{\rho} b'_{n+1}$ such that $\delta \cdot \rho = \varphi(\alpha')$. We then obtain the commutative diagram I 59



Corollary 2.10. (a) Let $\varphi : \underline{A} \to \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and η a minimally-versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for any map $\beta : \eta \to \varphi(\xi)$ in $\hat{\underline{B}}$, $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\beta) : P(\eta) \to P(\varphi(\xi)) = P(\xi)$.

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ , η be versal formal objects of $\underline{A}$. If η is minimal, then for any map $\alpha : \eta \to \xi$ in $\hat{A}$, $P(\xi)$ is a formal power-series ring over $P(\eta)$ through $P(\alpha) : P(\eta) \to P(\xi)$.

Proof: (a) A map $\beta : \eta \to \varphi(\xi)$ induces a commutative diagram



Since $\widetilde{\xi}$, φ are smooth, $\widetilde{\eta}\cdot\widetilde{\beta} = \varphi\cdot\widetilde{\xi}$ is smooth. On the other hand, $\widetilde{\eta}(k[\epsilon]) : \underline{h}_{P(\eta)}(k[\epsilon]) \to [\underline{B}(k[\epsilon])]$ is an isomorphism and therefore $\widetilde{\beta}$ is smooth by 2.9(a), which means that $P(\xi)$ is a formal power-series ring over $P(\eta)$. 160

(b) follows from (a) by setting $\underline{A} = \underline{B}$.

Finally we consider the question of "representability" of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Each object R in $\underline{C}$ defines a functor $h_R : \underline{C} \to$ (Ens) given by $h_R(S) = \mathrm{Hom}_{\underline{C}}(R, S)$, and in turn we obtain a functor $h : \underline{C}^0 \to \underline{\mathrm{Hom}}(\underline{C}, (\mathrm{Ens}))$ which is fully-faithful. A "cocategory in $\underline{C}$ " is a diagram

$$h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1} \underset{s_2}{\overset{s_1}{\underset{\longleftarrow \epsilon \longrightarrow}{\overset{\longrightarrow}{\longrightarrow}}}} h_{R_0} \qquad \text{with } s_i \cdot \epsilon = \text{identity for } i = 1, 2$$

$$(\text{where } \; h_{R_1} \times_{h_{R_0}} h_{R_1} \; = \; (h_{R_1})_{s_1} \times_{h_{R_0}} {}_{s_2}(h_{R_1})) \;\; \text{such that}$$

(i) c is associative and is compatible with s_i for $i = 1, 2$

(ii) $$h_{R_1} \xrightarrow{\sim} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\epsilon\times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

$$h_{R_1} \xrightarrow{\sim} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1\times\epsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

are both identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for an object S in $\underline{C}$, we can define a category $h_{(R_0,\ R_1,\ \ldots)}(S)$ by setting ob $h_{(R_0,\ R_1,\ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_0,\ S)$, and Fl $h_{(R_0,\ R_1,\ldots)}(S) = \mathrm{Hom}_{\underline{C}}(R_1, S)$ in which the composition rule of arrows is given by $c : h_{R_1} \times_{h_{R_0}} h_{R_1} \to h_{R_1}$. Then $h_{(R_0,\ R_1,\ldots)} : \underline{C} \to$ (categories) is a functor and in turn defines a cofibered category $\underline{h}_{(R_0,\ R_1,\ldots)}$ over $\underline{C}$. For instance,

any object R in $\underline{C}$ defines a discrete cocategory $\underline{h}_R = \underline{h}_{(R,\ R,\ s_1,\ s_2,\ \epsilon,\ c)}$ by setting s_1, s_2, ϵ, c to be all equal to the identity map. A homomorphism $(R_0,\ R_1, \ldots) \to (R_0',\ R_1', \ldots)$ of cocategories in $\underline{C}$ **I 61** is a pair of maps $h_{R_0} \to h_{R_0'}$, $h_{R_1} \to h_{R_1'}$ in $\underline{C}$ such that the induced map

$$h_{(R_0,\ R_1,\ldots)}(S) \to h_{(R_0',\ R_1',\ldots)}(S)$$

is a functor for every object S in $\underline{C}$. For instance, each cocategory $(R_0,\ R_1, \ldots)$ is provided with a homomorphism $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,\ R_1,\ldots)}$ of cocategories. A cogroupoid in $\underline{C}$ is by definition a cocategory $(R_0,\ R_1, \ldots)$ in $\underline{C}$ such that $h_{(R_0,\ R_1,\ldots)}(S)$ is a groupoid for every object S in $\underline{C}$. In other words, a cogroupoid in $\underline{C}$ is a cocategory $(R_0,\ R_1, \ldots)$ in $\underline{C}$ such that $\underline{h}_{(R_0,\ R_1,\ldots)}$ is a cofibered category in groupoids over $\underline{C}$.

In practice, a cocategory or cogroupoid occur in a following fashion: Let $P : \underline{F} \to \underline{C}$ be a cofibered category, and let ξ be a fixed object in $\underline{F}$. Set $R_0 = P(\xi)$ and let $(R_0,\ \underline{C})$ be the category of arrows in $\underline{C}$ with the source R_0 . Let $P : \underline{F}' \to (R_0,\ \underline{C})$ be the pull-back of $\underline{F}$ under the canonical functor $(R_0,\ \underline{C}) \to \underline{C}$, and choose a cocartesian section $\xi^{\#}$ of $\underline{F}'$ over $(R_0,\ \underline{C})$ such that $\xi^{\#}(\mathrm{id}_{R_0}) = \xi$. (This simply means that we choose, for each arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, a cocartesian f-map $\xi \to \xi^{\#}(f)$). Define the functor

$$\mathrm{Hom}_{\xi^{\#}} : (R_0 \amalg R_0,\ \underline{C}) \to (\mathrm{Ens}) \qquad \text{given by}$$

$$X = (f_1,\ f_2) \longmapsto \mathrm{Hom}_{\underline{F}(S)}(\xi^{\#}(f_1),\ \xi^{\#}(f_2)) \qquad \text{where}$$

$S =$ the target of f_1 ($=$ the target of f_2). We note that this functor does not depend on the choice of $\xi^{\#}$ up to (canonical) natural equivalence. Henceforth, by abuse of notation, we shall simply write ξ for $\xi^{\#}$. If this functor is representable by an object $X_0 = (R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_1)$ in $(R_0 \amalg R_0,\ \underline{C})$ i.e. if there exists R_0-map $\varphi : \xi(s_1) \to \xi(s_2)$ in $\underline{F}$ such that the induced map $\widetilde{\varphi} : \mathrm{Hom}_{(R_0 \amalg R_0,\ \underline{C})}(X_0,\ X) \to \mathrm{Hom}_{\xi}(X)$ is bijective for all X in $(R_0 \amalg R_0,\ \underline{C})$, then the object X_0 defines a cocategory $(R_0,\ R_1, \ldots)$ in $\underline{C}$ in a natural way. Indeed, a triple of arrows $R_0 \overset{u}{\underset{w}{\overset{\overset{v}{\longrightarrow}}{\longrightarrow}}} S$ **I 62**

in $\underline{C}$ defines a composition

$$\mathrm{Hom}_{\underline{F}(S)}(\xi(v),\ \xi(w)) \times \mathrm{Hom}_{\underline{F}(S)}(\xi(u),\ \xi(v)) \to \mathrm{Hom}_{\underline{F}(S)}(\xi(u),\ \xi(w))$$

and in turn yields the composition rule

$$h_{R_1}(S) \underset{h_{R_0}(S)}{\times} h_{R_1}(S) \to h_{R_1}(S)$$

depending functorially on S . Consequently the functor Hom_{ξ} defines the cocategory $(R_0,\ R_1, \ldots)$ in $\underline{C}$ and in turn defines a cofibered category $\underline{h}_{(R_0,\ R_1,\ldots)}$ in $\underline{C}$. By definition, $\mathrm{ob}\ \underline{h}_{(R_0,\ R_1,\ldots)} = \coprod_S \mathrm{Hom}_{\underline{C}}(R_0,\ S)$ (disjoint union for $S \in \mathrm{ob}(\underline{C})$), and an arrow $(R_0 \xrightarrow{f} S) \to (R_0 \xrightarrow{f'} S')$ is a pair $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that $u \cdot s_1 = h \cdot f$ and $u \cdot s_2 = f'$, and the structural functor $P : \underline{h}_{(R_0,\ R_1,\ldots)} \to \underline{C}$ is given by $(R_0 \xrightarrow{f} S) \longmapsto S$. Now the pair $(\xi,\ \varphi)$ defines the functor

$$(\xi,\ \varphi) : \underline{h}_{(R_0,\ R_1,\ldots)} \to \underline{F} \quad \text{over} \quad \underline{C} \quad \text{by}$$

$$(R_0 \xrightarrow{f} S) \ \longmapsto \ \xi(f)$$

$$[(R_0 \xrightarrow{f} S) \xrightarrow{(h,u)} (R_0 \to S')] \longmapsto \ \text{the composite map} \ \ \xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f') \ .$$

We note that, for each object S in $\underline{C}$, the functor $(\xi, \varphi)(S) : \underline{h}_{(R_0,R_1,\ldots)}(S) \to \underline{F}(S)$ is fully-faithful by virtue of the definition of $(R_0 \rightrightarrows R_1)$ coming from the representability of the functor Hom_{ξ} . Consequently the functor $(\xi, \varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$ is a $\underline{C}$-equivalence if and only if ξ is a quasi-initial object of $\underline{F}$ over $\underline{C}$ i.e., for every object a in $\underline{F}$ there exists a cocartesian arrow $\xi \to a$. Instead of this functor Hom_{ξ} one may also consider the functor

$$\mathrm{Isom}_\xi : (R_0 \amalg R_0,\ \underline{C}) \to (\mathrm{Ens})$$

given by $X = (f_1, f_2) \longmapsto \mathrm{Isom}_{\underline{F}(S)}(\xi(f_1), \xi(f_2))$ where S is the target of f_1 (= the target of f_2). 163
If this functor is representable by an object $(R_0 \rightrightarrows R_1)$ in $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0, R_1, \ldots)$ in $\underline{C}$ together with the functor $(\xi, \varphi) : \underline{h}_{(R_0,R_1,\ldots)} \to \underline{F}$, which is, when $\underline{F}$ is a cofibered groupoid, a $\underline{C}$-equivalence if and only if ξ is a quasi-initial object of $\underline{F}$ over $\underline{C}$.

Definition and proposition 2.11. A cogroupoid $(R_0, R_1, \ldots)$ in $\hat{\underline{C}}_\Lambda$ is called smooth if any one of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1 (and s_2)

(ii) $\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\Lambda$.

A minimally smooth cogroupoid or a normalized smooth cogroupoid in $\hat{\underline{C}}_\Lambda$ is a smooth cogroupoid $(R_0, R_1, \ldots)$ in $\hat{\underline{C}}_\Lambda$ such that $\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])$ is totally disconnected i.e., $s_1(k[\epsilon]) = s_2(k[\epsilon])$.

Proof: $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,R_1,\ldots)}$ is smooth over $\underline{C}_\Lambda \Leftrightarrow$ given any surjective map $(S', f') \xrightarrow{(h,u)} (S,\ f)$ in $\underline{h}_{(R_0,R_1,\ldots)}$, there exists a lifting f'' of f to S' together with an isomorphism $(S', f'') \xrightarrow{(1,u')} (S', f') \Leftrightarrow$ given $u \in h_{R_1}(S)$ and a lifting f' of $s_1(u)$ to S' , there exists a lifting u' of u to S' such that $s_1(u') = f'$, where $S' \to S$ is an arbitrary surjective map in $\underline{C}_\Lambda \Leftrightarrow s_1 : \underline{h}_{R_1} \to \underline{h}_{R_0}$ is smooth over $\underline{C}_\Lambda$.

Proposition 2.12. Let $(R_0, R_1, \ldots)$ be a cogroupoid in $\hat{\underline{C}}_\Lambda$.

(a) If $(R_0, R_1, \ldots)$ is smooth, then $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\epsilon]) = s_2(k[\epsilon])$, then the following statements are equivalent

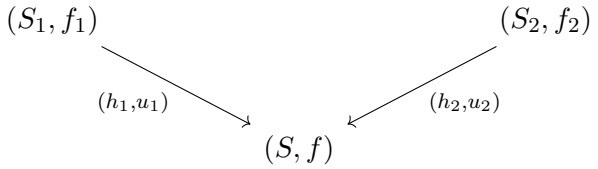
(i) $(R_0, R_1, \ldots)$ is smooth

(ii) $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous

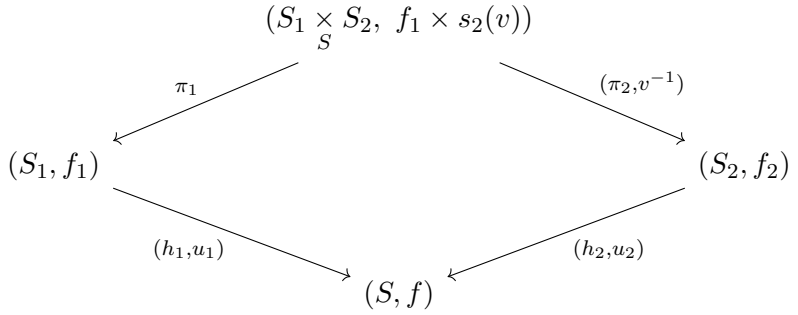
(iii) $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous

(c) If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$ for some minimally smooth 164
cogroupoid S_* .

Proof: (a) Consider a diagram



in $\underline{h}_{(R_0,R_1,\ldots)}$ where $h_2 : S_2 \longrightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$ is smooth by hypothesis, $u_1^{-1} u_2$ can be lifted to some $v \in h_{R_1}(S_2)$ such that $s_1(v) = f_2$. Since $h_2 \cdot s_2(v) = h_1 \cdot f_1$, $f_1 \times s_2(v) \in h_{R_0}(S_1 \underset{S}{\times} S_2)$ is defined and we obtain the commutative diagram



We claim that the above diagram defines a fibre-product $(S_1, f_1) \underset{(S,f)}{\times} (S_2, f_2)$ in $\underline{h}_{(R_0,R_1,\ldots)}$. Indeed, for any object $(T,\ g)$ in $\underline{h}_{(R_0,R_1,\ldots)}$ we have

$$\mathrm{Hom}((T,g),(S_1\underset{S}{\times}S_2, f_1\times s_2(v))) = \{(h_1'\times h_2', w_1\times w_2)\in h_T(S_1\underset{S}{\times}S_2)\times h_{R_1}(S_1\underset{S}{\times}S_2)\,|\,s_1(w_1\times w_2) = (h_1'\times h_2')\,g\,,$$
$$s_2(w_1\times w_2) = f_1\times s_2(v)\} = \{(h_1'\times h_2', w_1, w_2)\in h_T(S_1\underset{S}{\times}S_2)\ \times h_{R_1}(S_1)\times h_{R_1}(S_2)\,|\,h_1w_1 = h_2w_2,$$
$$s_1(w_1) = h_1'\ g,\ s_1(w_2) = h_2'\ g,\ s_2(w_1) = f_1,\ s_2(w_2) = s_2(v)\} =$$
$$\{(h_1'\times h_2', w_1, w_2)\in h_T(S_1\underset{S}{\times}S_2)\times h_{R_1}(S_1)\times h_{R_1}(S_2)\,|\,u_1^{-1}\ u_2\ h_2(v^{-1}w_2) = h_1\ w_1\ ,\ s_1(w_1) = h_1'\ \ g,$$

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$$s_2(w_1) = f_1,\quad s_1(v^{-1}w_2) = h_2'g,\quad s_2(v^{-1}w_2) = f_2\}\ \xrightarrow{\sim}\ \{(h_1'\times h_2', w_1, w_2')\in h_T(S_1\times_S S_2)\times h_{R_1}(S_1)\times h_{R_1}(S_2)\ \big|$$
$$u_2\,h_2\,w_2'\ =\ u_1\,h_1\,w_1,\quad s_1(w_1)\ =\ h_1'g,\quad s_2(w_1) = f_1,\quad s_1(w_2')\ =\ h_2'g,\quad s_2(w_2')\ =\ f_2\}\ =$$
$$\mathrm{Hom}((T,\ g),\ (S_1,\ f_1))\ \times_{\mathrm{Hom}((T,g),\ (S,f))}\ \mathrm{Hom}((T,\ g),\ (S_2,\ f_2))\ .$$

(b) It suffices to show (iii) ⇒ (i) . Since $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous and $\dim[\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])] = \dim h_{R_0}(k[\epsilon]) < \infty$, $\underline{h}_{(R_0,R_1,\ldots)}$ admits a minimally versal formal object. On the other hand, $(R_0,\ 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}}_{(R_0,R_1,\ldots)}$ since $\underline{h}_{R_0}(k[\epsilon])\to[\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])]$ is bijective, and therefore $(R_0,\ 1_{R_0})$ is a minimally versal formal object of $\underline{h}_{(R_0,R_1,\ldots)}$ by 1.10(ii), which means that $\epsilon:\underline{h}_{R_0}\longrightarrow\underline{h}_{(R_0,R_1,\ldots)}$ is minimally smooth.

(c) Let $(S_0,\ \xi_0)$ be a minimally versal formal object for $\underline{h}_{(R_0,R_1,\ldots)}$ (1.11). For any object $\xi: R_0\to T$ in $\underline{h}_{(R_0,\ R_1,\ldots)}(T)$, the functor $\mathrm{Isom}(\xi,\xi)$ from $\underline{C}_\Lambda/T\hat{\otimes}_\Lambda T$ to (sets) which to any R associates the set of isomorphisms between the two images of ξ on $R: R_0\xrightarrow{\xi}T\rightrightarrows T\hat{\otimes}_\Lambda T\longrightarrow R$, is represented by $R_1\otimes_{R_0\hat{\otimes}_\Lambda R_0}(T\otimes_\Lambda T)$. By a passage to limit, one gets that $\mathrm{Isom}(\xi_0,\ \xi_0)$ is pro-represented by some S_1 , and $\underline{h}_{(R_0,R_1,\ldots)}$ is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$. By (b), S_* is minimally smooth.

Definition 2.13. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (minimally) pro-representable if it is $\underline{C}_\Lambda$-equivalent to $\underline{h}_{(R_0,R_1,\ldots)}$ for some (normalized) cogroupoid $(R_0,R_1,\ldots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 2.14. If $(R_0,\ R_1,\ldots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism $(\alpha_0,\alpha_1)\ :\ (R_0,R_1,\ldots)\to(R_0,R_1,\ldots)$ which induces $\underline{C}_\Lambda$-equivalence $\underline{h}_{(\alpha_0,\alpha_1)}\ :\ \underline{h}_{(R_0,R_1,\ldots)}\longrightarrow\underline{h}_{(R_0,\ R_1,\ldots)}$ is an isomorphism. In particular, for any cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, a normalized prorepresentation, if it exists, is unique up to an isomorphism. I66

Proof: Consider the commutative diagram

$$\begin{array}{ccc}
\underline{h}_{(R_0,R_1,\ldots)} & \xrightarrow{\underline{h}_{(\alpha_0,\alpha_1)}} & \underline{h}_{(R_0,R_1,\ldots)} \\
\uparrow{\scriptstyle\epsilon} & & \uparrow{\scriptstyle\epsilon} \\
\underline{h}_{R_0} & \xrightarrow[\underline{h}_{\alpha_0}]{} & \underline{h}_{R_0}
\end{array}$$

where $\underline{h}_{(\alpha_0,\alpha_1)}$ is a $\underline{C}_\Lambda$-equivalence. Since $\underline{h}_{(R_0,R_1,\ldots)}(k[\epsilon])$ is totally disconnected, $[\underline{h}_{(\alpha_0,\alpha_1)}(k[\epsilon])]$ and $[\epsilon(k[\epsilon])]$ are both bijections, and hence so is $\underline{h}_{\alpha_0}(k[\epsilon])$.

Consequently, $\alpha_0: R_0\to R_0$ is a surjective endomorphism and hence is an automorphism. It then follows that α_1 is also an automorphism of R_1 .

Lemma 2.15. Let $A,\ B$ be discrete groupoids over $\underline{C}_\wedge$ and $\varphi: A\to B$ a minimally smooth functor. If $A,\ B$ are homogeneous, then φ is a $\underline{C}_\wedge$-equivalence.

Proof: We must show that $\varphi(S): A(S)\longrightarrow B(S)$ is bijective for all $S\in\mathrm{ob}(\underline{C}_\wedge)$. We note that $\varphi(S)$ is surjective for all $S\in\mathrm{ob}(\underline{C}_\wedge)$ and bijective for $S = k[\epsilon]$. We argue by induction on the length of S . Let $S'\xrightarrow{f}S$ be a small extension in $\underline{C}_\wedge$. Since $A,\ B$ are homogeneous discrete groupoids, the canonical isomorphism $1\times\tilde{f}: S'\times_S S'\longrightarrow S'\times_k k[\mathrm{Ker}\ f]$ induces the commutative diagram

$$\begin{array}{ccc} A(S') \underset{A(S)}{\times} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\mathrm{Ker}\ f]) \\ \downarrow & & \downarrow \\ B(S') \underset{B(S)}{\times} B(S') & \xrightarrow{\sim} & B(S') \times B(k[\mathrm{Ker}\ f]) \end{array}$$

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Therefore $A(S')$, $B(S')$ are principal homogeneous sets over $A(S)$, $B(S)$ with the structural group $A(k[\mathrm{Ker}\ f]) \xrightarrow{\sim} B(k[\mathrm{Ker}\ f])$. Consequently if $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence is bijective.

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\wedge$ such that $\mathrm{Aut}_{\underline{A}(k[\epsilon])}(1_\epsilon)$ is finite-dimensional. Then for any $\xi \in \mathrm{ob}(\hat{\underline{A}})$, the functor $\mathrm{Isom}_\xi : \underline{C}_{R\hat{\otimes}_\wedge R} \longrightarrow (\mathrm{Ens})$ is prorepresentable, where $R = P(\xi)$.

Proof: Assume that Isom_ξ is homogeneous. We then have, by 1.11, a minimally smooth functor $\Phi : \underline{h}_{R_1} \to \underline{\mathrm{Isom}}_\xi$, which is a $\underline{C}_{R\hat{\otimes}_\wedge R}$-equivalence by 2.15 i.e., Isom_ξ is prorepresentable. Therefore it suffices to show that Isom_ξ is homogeneous. Consider a diagram

$$\begin{array}{ccc} T_1 & & T_2 \\ & \searrow \quad \swarrow & \\ & T & \end{array}$$

in $\underline{C}_{R\hat{\otimes}_\wedge R}$ where $T_2 \longrightarrow T$ is surjective. Since $\underline{A}$ is homogeneous, $\xi_i(T_1) \underset{\xi_i(T)}{\times} \xi_i(T_2)$ exists in $\underline{A}$, and we have a canonical isomorphism

$\xi_i(T_1 \underset{T}{\times} T_2) \simeq \xi_i(T_1) \underset{\xi_i(T)}{\times} \xi_i(T_2)$ for $i = 1, 2$ (cf. 2.6(a)). It follows that

$\mathrm{Isom}_\xi(T_1 \underset{T}{\times} T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \underset{\mathrm{Isom}_\xi(T)}{\times} \mathrm{Isom}_\xi(T_2)$ is an isomorphism.

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\wedge$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if (1) $\underline{A}$ is homogeneous I 68

(2) $[A(k[\epsilon])]$ and $[\mathrm{Aut}(1_\epsilon)]$ are both finite-dimensional vector spaces over k .

Proof: Assume that $\underline{A}$ is prorepresentable by a smooth cogroupoid i.e., $\underline{A}$ is $\underline{C}_\wedge$-equivalent to $\underline{h} = \underline{h}_{(R_0, R_1, \ldots)}$ for some smooth cogroupoid $(R_0, R_1, \ldots)$ in $\hat{\underline{C}}_\wedge$. Since $\underline{h}$ is homogeneous by 2.12(a), and

$$[\underline{h}(k[\epsilon])] = \mathrm{Coker}\ (\mathrm{Hom}_{\wedge\text{-alg}}(R_1, k[\epsilon]) \underset{s_2}{\overset{s_1}{\rightrightarrows}} \mathrm{Hom}_{\wedge\text{-alg}}(R_0, k[\epsilon])) \qquad \text{as well as}$$

$$\mathrm{Aut}_{\underline{h}}(1_\epsilon) = \mathrm{Ker}(\mathrm{Hom}_{\wedge\text{-alg}}(R_1, k[\epsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\wedge\text{-alg}}(R_0, k[\epsilon]))$$

$$= \mathrm{Ker}(\mathrm{Hom}_{\wedge\text{-alg}}(R_1, k[\epsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\wedge\text{-alg}}(R_0, k[\epsilon])) \qquad \text{are both finite-dimensional}$$

vector spaces over k , we must have the same properties on $\underline{A}$. Conversely assume (1) and (2) on $\underline{A}$. We claim that $\underline{A}$ is prorepresentable by a normalized smooth cogroupoid. Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\epsilon])]$ is a finite-dimensional vector space over k , $\underline{A}$ admits a minimally versal formal object ξ . Set $P(\xi) = R_0$ and consider the functor $\mathrm{Isom}_\xi : \underline{C}_{R_0\hat{\otimes}_\wedge R_0} \to (\mathrm{Ens})$ given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T), \xi_2(T))$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \to T$ denotes the composite maps

$$R_0 \underset{s_2}{\overset{s_1}{\rightrightarrows}} R_0 \otimes_\wedge R_0 \longrightarrow T \ .$$

This functor is pro-representable by 2.16, i.e. there exists $R_1 \in \mathrm{ob}(\hat{\underline{C}}_{R_0\hat{\otimes}_\wedge R_0})$ and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ such that $\Phi : h_{R_1} \to \mathrm{Isom}_\xi$ is an isomorphism over $\underline{C}_{R_0\hat{\otimes}_\wedge R_0}$. Then $(R_0, R_1, \dots)$ defines a cogroupoid in $\hat{\underline{C}}_\wedge$ and the functor $(\xi, \varphi) : \underline{h}_{(R_0,R_1,\dots)} \to \underline{A}$ is a $\underline{C}_\wedge$-equivalence. Since ξ is a minimally versal formal object of $\underline{A}$, the composite functor $\underline{h}_{(R_0,R_0,\dots)} \xrightarrow{\epsilon} \underline{h}_{(R_0,R_1,\dots)} \xrightarrow{(\xi,\varphi)} \underline{A}$ is, by definition, minimally smooth and therefore so is ϵ since (ξ, φ) is a $\underline{C}_\wedge$-equivalence i.e., $(R_0, R_1, \dots)$ is a normalized smooth cogroupoid in $\hat{\underline{C}}_\wedge$.

Corollary 2.18. A functor $F : \underline{C}_\Lambda \longrightarrow$ (Ens) is pro-representable if and only if it is homogeneous and $\dim F(k[\epsilon]) < \infty$. 169

Corollary 2.19. Every smooth cogroupoid in $\hat{\underline{C}}_\Lambda$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0, R_1, \dots)$ in $\hat{\underline{C}}_\Lambda$, we set $\bar{R}_1 = R_1/J$ where J is the ideal in R_1 generated by $\{s_1(r) - s_2(r) \mid r \in R_0\}$. Then $(R_0, \bar{R}_1, \dots)$ defines a formal group scheme over R_0 with respect to the induced composition rule. Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) . Let $\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0\hat{\otimes}_\Lambda R_0}$ be the embeddings corresponding to the surjective maps $R_0\hat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k$, and let Aut_ξ , Aut^o be the restriction functors of Isom_ξ on $\underline{C}_{R_0}$, $\underline{C}_k$ respectively. In other words, $\mathrm{Aut}_\xi(S) =$ the automorphism group of $\xi(S)$ for every $S \in \mathrm{ob}(\underline{C}_{R_0})$, and $\mathrm{Aut}^o(S) =$ the automorphism group of 1_S for every $S \in \mathrm{ob}(\underline{C}_k)$, where $1_S =$ the direct image of $1 \in \underline{A}(k)$ under the map $k \to S$. Since R_1 prorepresents the functor Isom_ξ on $\underline{C}_{R_0\hat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents the functor Aut_ξ on $\underline{C}_{R_0}$, and $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^o on $\underline{C}_k$. Though $\underline{A}$ is prorepresented by a smooth cogroupoid $(R_0, R_1, \dots)$, neither $(R_0, \bar{R}_1, \dots)$ nor $(k,\ k \otimes_{R_0} \bar{R}_1, \dots)$ are in general smooth. Indeed we have

Corollary 2.20. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$ prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) .

(a) $(R_0, \bar{R}_1, \dots)$ and $(k,\ k \otimes_{R_0} \bar{R}_1, \dots)$ prorepresent the functors Aut_ξ and Aut^o respectively.

(b) The following statements are equivalent

(i) the formal group scheme $\underline{h}_{(R_0,\bar{R}_1,\dots)}$ is formally smooth over R_0 ;

(ii) for every surjective arrow $a' \to a$ in $\underline{A}$, $\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a)$ is surjective,

(iii) R_0 prorepresents the functor $[\underline{A}]$ over $\underline{C}_\Lambda$.

Proof: We have already noted (a). As for (b), (i) $\Leftrightarrow$ (ii) is clear whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since $\tilde{\xi} : h_{R_0} \longrightarrow [\underline{A}]$ is minimally smooth. 170

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pull-back of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ under $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. Then it is trivial to see that if $\underline{A}$ is prorepresented by $(R_0, R_1, \dots)$ via (ξ, φ) , then $\underline{A}|\underline{C}_k$ is prorepresented by $(k \otimes_\Lambda R_0\ ,\ k \otimes_\Lambda R_1, \dots)$ via $(k \otimes_\Lambda \xi\ ,\ k \otimes_\Lambda \varphi)$ where $k \otimes_\Lambda \xi =$ the direct image of ξ under the map $R_0 \longrightarrow k \otimes_\Lambda R_0$, $k \otimes_\Lambda \varphi =$ the image of φ under $\mathrm{Isom}_{\underline{A}(R_1)}(\xi(s_1),\ \xi(s_2)) \longrightarrow \mathrm{Isom}_{\underline{A}(k\otimes_\Lambda R_1)}(k \otimes_\Lambda \xi(s_1),\ k \otimes_\Lambda \xi(s_2))$. If $(R_0, R_1, \dots)$ is normalized, then so is $(k \otimes_\Lambda R_0,\ k \otimes_\Lambda R_1, \dots)$, and the relative dimension of R_1 over R_0 is equal to the relative dimension of $k \otimes_\Lambda R_1$ over $k \otimes_\Lambda R_0$ since R_1 is smooth over R_0 .

Remark 2.22. Let $\underline{A}$ be prorepresented by smooth cogroupoids $(R_0, R_1, \dots)$ and $(R'_0,\ R'_1, \dots)$ where we assume that $(R_0, R_1, \dots)$ is normalized. We then obtain a homomorphism $(i_0, i_1) : (R_0, R_1, \dots) \longrightarrow (R'_0,\ R'_1, \dots)$. Now R'_0 is a formal power-series ring over R_0 by 2.10(b), and the homomorphism (i_0, i_1) induces an isomorphism $R'_0 \otimes_{R_0} (R_0, R_1, \dots) \xrightarrow{\sim} (R'_0,\ R'_1, \dots)$.

3. The coherent sheaves $\underline{D}^i_X$

One of the nice functors which is usually not pro-representable but nevertheless admits a formal versal object is the deformation functor. To this end we recall a few basic facts on commutative ring extensions and its obstruction theory. This section contains nothing new (possibly except in its systematic approach via Grothendieck cohomology on site). A purpose of this section is to make this expose as much self-contained. (Indeed we prove here more than we need later). The basic facts are stated in theorem 3.12. Throughout rings are meant to be commutative rings with unit.

Given a ring R we denote by $\underline{C}_R$ the category of R-algebras. For each object A in $\underline{C}_R$ we set 171

$\mathrm{Cov}(A) =$ the set of all surjective arrows $\varphi : A' \to A$ in $\underline{C}_R$ with target A .

It clearly defines a pretopology and in turn we obtain a site $\underline{C}_R$. For each object A in $\underline{C}_R$, we shall denote by $\underline{C}_{A|R}$ the category of arrows in $\underline{C}_R$ with target A . We may and shall provide

the category $\underline{C}_{A|R}$ with the topology induced from the forgetful functor $j_A : \underline{C}_{A|R} \to \underline{C}_R$. We note that the category $\underline{C}_{A|R}$ has a final object $(1_A : A \to A)$ and an initial object $(i_A : R \to A)$, and admits a fibre-product as well as coproduct. Henceforth we shall denote the object $(B \to A)$ in $\underline{C}_{A|R}$, by abuse of notation, simply by B in case when there is no possible confusion. Given an abelian category $\underline{C}$, a $\underline{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor $F : \underline{C}^O_{A|R} \to \underline{C}$ such that, for any surjective arrow $B' \to B$ in $\underline{C}_{A|R}$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \underset{B}{\times} B')$$

is exact. The additive presheaves are those which preserve coproduct i.e., $F(B_1 \underset{R}{\otimes} B_2) = F(B_1) \oplus F(B_2)$. For example, an A-module M defines the functor

$$\mathrm{Der}_R(-, M) : \underline{C}^O_{A|R} \to (A\text{-mod})$$

given by $B \to \mathrm{Der}_R(B, M) = R$-linear derivations of B with values in M regarded as B-module via the structural map $B \to A$, and it is clearly an additive sheaf on $\underline{C}^O_{A|R}$, where $(A$-mod$)$ denotes the category of A-modules. A presheaf F on $\underline{C}_{A|R}$ with values in $(A$-mod$)$ is called "finite type" if $F(B)$ is an A-module of finite type whenever B is an R-algebra of finite type. For example, if A is noetherian and M is an A-module of finite type, then $\mathrm{Der}_R(-, M)$ is an additive sheaf of finite type on the site $\underline{C}_{A|R}$.

A few more words on notations and basic facts in homological algebra:

3.1(a) Throughout the rest of this section, we shall denote by $\widetilde{\underline{C}}_{A|R}$ (or $\overset{\vee}{\underline{C}}_{A|R}$) the category of sheaves (or presheaves) with values in $(A$-mod$)$. We have the (left exact) injection functor $i_{A|R} : \widetilde{\underline{C}}_{A|R} \to \overset{\vee}{\underline{C}}_{A|R}$, and the associated-sheaf (exact) functor $G_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \to \widetilde{\underline{C}}_{A|R}$ which is adjoint to the functor $i_{A|R}$. An object $B \in ob(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \to \underline{C}_{A|R}$ and in turn a commutative diagram 172

(*)
$$\begin{array}{ccc} \widetilde{\underline{C}}_{A|R} & \xrightarrow{\widetilde{j}} & \widetilde{\underline{C}}_{B|R} \\ {\scriptstyle i_{A|R}}\downarrow & & \downarrow \\ \overset{\vee}{\underline{C}}_{A|R} & \xrightarrow[j^\vee]{} & \overset{\vee}{\underline{C}}_{B|R} \end{array}$$

where the restriction functor $\widetilde{j}$ (as well as $j^\vee$) is an exact functor, since the site $\underline{C}_{B|R}$ is nothing but the category of arrows in $\underline{C}_{A|R}$ having target B , with the induced topology. A ring homomorphism $S \to R$ induces a continuous functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$, and in turn a commutative diagram

(**)
$$\begin{array}{ccc} \widetilde{\underline{C}}_{A|S} & \xrightarrow{\widetilde{\beta}} & \widetilde{\underline{C}}_{A|R} \\ {\scriptstyle i_{A|S}}\downarrow & & \downarrow \\ \overset{\vee}{\underline{C}}_{A|S} & \xrightarrow[\beta^\vee]{} & \overset{\vee}{\underline{C}}_{A|R} \end{array}$$

Note that $\widetilde{\beta}$ is not necessarily an exact functor.

3.1(b) The right derived functor of the injection functor $i_{A|R} : \widetilde{\underline{C}}_{A|R} \to \overset{\vee}{\underline{C}}_{A|R}$ will be denoted by $\underline{H}^\circ_{A|R}$. For each sheaf F on $\underline{C}_{A|R}$, we set

$$H^\circ(A|R, F) = [H^\circ_{A|R}(F)](A)$$

For any $B' \to B$ in $\mathrm{Cov}(B)$, let us define the functor $H^0((B' \to B), -) : \overset{\vee}{\underline{C}}_{A|R} \to (A\text{-mod})$ by $H^0((B' \to B), F) = Ker(F(B') \rightrightarrows F(B' \underset{B}{\times} B'))$. The right derived functors of $H^0((B' \to B), -)$ will be denoted by $H^\circ((B' \to B), -)$. One knows that the $H^\circ((B' \to B), F)$ are the cohomology groups of the (Čech) semi-simplicial module 173

$$F(B') \;\substack{\longrightarrow\\ \longrightarrow}\; F(B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; F(B' \times_B B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots$$

The right derived functor of $\underline{\check{H}}^0_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \to \overset{\vee}{\underline{C}}_{A|R}$ will be denoted by $\underline{\check{H}}^\bullet_{A|R}$, where

$$[\underline{\check{H}}^0_{A|R}(F)](B) \;=\; \lim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \to B\}, F) \ .$$

For each presheaf F on $\underline{C}_{A|R}$, we set

$$\check{H}^\bullet(A|R, F) \;=\; [\check{H}^\bullet_{A|R}(F)](A) \;=\; \lim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^\bullet(\{A' \to A\}, F)$$

for any sheaf F on $\underline{C}_{A|R}$, we have the usual spectral sequence

$$\check{H}^p(A|R, \underline{H}^q_{A|R}(F)) \;\Rightarrow\; H^{p+q}(A|R, F)$$

where we note that $\check{H}^0(A|R, \underline{H}^q(F)) = 0$ for all $q > 0$.

In particular

$$\begin{cases} \check{H}^1(A|R, F) \;=\; H^1(A|R, F) \\ 0 \to \check{H}^2(A|R, F) \to H^2(A|R, F) \to \check{H}^1(A|R, \underline{H}^1_{A|R}(F)) \to \check{H}^3(A|R, F) \quad \text{is exact.} \end{cases}$$

3.1(c) If B is an object in $\underline{C}_{A|R}$, the commutative diagram (*) in 3.1(a) together with the fact that $\tilde{j}$ as well as $j^\vee$ are exact functors yields the canonical isomorphism

$$H^\bullet(B|R, \tilde{j}F) \;=\; [\underline{H}^\bullet_{A|R}(F)](B)$$

On the other hand, a ring homomorphism $S \to R$ induces the commutative diagram (**) in 3.1(a). Now $\tilde{\beta}$ is not necessarily an exact functor. However, β preserves a fibre-product, and thus we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\tilde{\beta}F) \;\Rightarrow\; \check{\beta}\underline{H}^{p+q}_{A|S}(F) \qquad \text{i.e.,} \qquad H^p(A|R, R^q\tilde{\beta}F) \;\Rightarrow\; H^{p+q}(A|S, F)$$

Since we have the canonical isomorphism $\tilde{\beta} = G_{A|R} \circ \check{\beta} \circ i_{A|S}$ and $\beta^\# = G_{A|R} \circ \check{\beta}$ is an exact functor, 174 we have $R^\bullet\tilde{\beta}F = \beta^\#\underline{H}^\bullet_{A|S}(F)$. Thus the spectral sequence above can now be written as

$$H^p(A|R, \beta^\#\underline{H}^q_{A|S}(F)) \;\Rightarrow\; H^{p+q}(A|S, F) \ .$$

Lemma 3.2. Given $B \in ob(\underline{C}_{A|R})$, choose a surjective arrow $P \to B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R . Then for any $F \in ob(\check{\underline{C}}_{A|R})$, the natural map $H^\bullet(\{P \to B\}, F) \to \underline{\check{H}}^\bullet(F)\ (B)$ is an isomorphism.

Proof: It suffices to see that the natural map

$$H^0(\{P \to B\}, F) \to \underline{\check{H}}^0(F)\ (B) \;=\; \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \to B\}, F)$$

is an isomorphism, which is clear since, for any $\{B' \to B\} \in \mathrm{Cov}(B)$, there exists
an arrow $\{P \to B\} \to \{B' \to B\}$.

Corollary 3.3.

(a) Assume that A is noetherian. If $F \in ob(\widetilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for all n .

(b) For any polynomial algebra P over R , we have $H^n(P|R, F) = 0$ for all $n > 0$ and for all $F \in ob(\widetilde{\underline{C}}_{P|R})$.

(c) A diagram $0 \to F' \to F \to F'' \to 0$ in $\widetilde{\underline{C}}_{A|R}$ is exact if and only if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{C}_{A|R})$ which is a polynomial algebra over R .

Proof:

(a) In view of the spectral sequence

$$\check{H}^p(B|R, \underline{H}^q(F)) \;\Rightarrow\; H^{p+q}(B|R, F)$$

together with the fact that $\check{H}^0(\underline{H}^q(F)) = 0$ for all $q > 0$, it suffices to show that $\underline{H}^n(G)$ is of finite type for all n and for all presheaves G of finite type. However $\check{H}^\bullet(B|R, G) = H^\bullet(\{P \to B\}, G)$, where P is a polynomial algebra over R and $\{P \to B\} \in \mathrm{Cov}(B)$, and furthermore $H^\bullet(\{P \to B\}, G)$ is simply the cohomology gotten from the complex

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$$\ldots\ldots\ldots\ldots G(P \underset{B}{\times} P \underset{B}{\times} P) \quad \begin{matrix}\leftarrow\\ \leftarrow\\ \leftarrow\end{matrix} \quad G(P \underset{B}{\times} P) \begin{matrix}\leftarrow\\ \leftarrow\end{matrix} \quad G(P) \quad .$$

If B is an R-algebra of finite type, then we may take P to be of finite type over R so that $P \underset{B}{\times} \ldots \underset{B}{\times} P$ are all R-algebras of finite type. If G is of finite type, then $G(P \underset{B}{\times} \ldots \underset{B}{\times} P)$ are all A-modules of finite type, so that $H^\bullet(\{P \to B\}, G)$ is an A-module of finite type since A is noetherian.

(b) Let P be a polynomial algebra over R . If $B \to P$ is any surjective R-algebra map, then it admits a section and hence $H^i(\{B \to P\}, G) = 0$ for all $i > 0$, so that $\check{H}^i(P|R,\ G) = 0$ for all $i > 0$ and for all presheaves G on $\underline{C}_{P|R}$. It follows that the spectral sequence $\check{H}^p(P|R,\ \underline{H}^q_{P|R}(F)) \Rightarrow H^{p+q}(P|R,\ F)$ degenerates and hence $\check{H}^0(P|R,\ \underline{H}^n(F)) \simeq H^n(P|R,\ F)$ for all n . However $\check{H}^0(P|R,\ \underline{H}^n(F)) = 0$ for all $n > 0$ whenever F is a sheaf, and hence $H^n(P|R,\ F) = 0$ for all $n > 0$ and for all $F \in ob(\widetilde{\underline{C}}_{P|R})$.

(c) If $0 \to F' \to F \to F'' \to 0$ is an exact sequence in $\widetilde{\underline{C}}_{A|R}$, then

$$0 \to F'(B) \to F(B) \to F''(B) \to H^1(B|R,\ F') \to H^1(B|R,\ F) \to \ \ldots.$$

is exact. If $P \in ob(\underline{C}_{A|R})$ is a polynomial algebra over R , then $H^1(P|R,\ F') = 0$ by 3.3(b) and hence

$$0 \to F'(P) \to F(P) \to F''(P) \to 0$$

is exact. As for the other direction, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ in which

$$ob(\underline{P}_{A|R}) = \{P \in ob(\underline{C}_{A|R}) |\ \ P \text{ is a polynomial algebra over } R\} \ ,$$

provided with the induced topology, and let $p : \underline{P}_{A|R} \to \underline{C}_{A|R}$ be the inclusion functor. Since every object in $\underline{C}_{A|R}$ can be covered by an object in $\underline{P}_{A|R}$, it follows from a comparison lemma ( ) that $\widetilde{p} : \widetilde{\underline{C}}_{A|R} \to \widetilde{\underline{P}}_{A|R}$ is an equivalence of categories. Consequently, if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{P}_{A|R})$, then $0 \to \widetilde{p}F' \to \widetilde{p}F \to \widetilde{p}F'' \to 0$ is exact in $\widetilde{\underline{P}}_{A|R}$ and hence $0 \to F' \to F \to F'' \to 0$ is exact in $\widetilde{\underline{C}}_{A|R}$.

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Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows immediately that every presheaf on $\underline{P}_{A|R}$ is actually a sheaf i.e., $\widetilde{\underline{P}}_{A|R} = \check{\underline{P}}_{A|R}$. Consequently we find that the restriction functor $\widetilde{\underline{C}}_{A|R} \to \check{\underline{P}}_{A|R}$ is an equivalence of categories.

Lemma 3.5. Consider a diagram of R-algebras

$$\begin{array}{ccc} & & C \\ & & \downarrow \\ B' & \xrightarrow[\text{surjective}]{} & B \end{array}$$

and let S be an R-algebra.

(i) If $\mathrm{Tor}_1^R(B,S) = 0$, then the canonical map

$$(B' \underset{B}{\times} C) \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S)$$

is an isomorphism.

(ii) Assume that B' is R-flat. If $\mathrm{Tor}_i^R(C,S) = 0$ for all $0 < i < n$ and $\mathrm{Tor}_i^R(B,S) = 0$ for all $0 < i \leq n$, then $\mathrm{Tor}_i^R(B' \underset{B}{\times} C,\ S) = 0$ for all $0 < i < n$.

Proof: Let $0 \to J \to B' \to B \to 0$ be exact.

(i) If $\mathrm{Tor}_1^R(B,\ S) = 0$, then

$$0 \to J \underset{R}{\otimes} S \to B' \underset{R}{\otimes} S \to B \underset{R}{\otimes} S \to 0$$

is exact so that

$$0 \to J \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \to C \underset{R}{\otimes} S \to 0$$

is exact. We thus obtain an exact commutative diagram

$$\begin{array}{ccccccccc} & & J \underset{R}{\otimes} S & \longrightarrow & (B' \underset{B}{\times} C) \underset{R}{\otimes} S & \longrightarrow & C \underset{R}{\otimes} S & \longrightarrow & 0 \\ & & \| & & \downarrow & & \| & & \\ 0 & \longrightarrow & J \underset{R}{\otimes} S & \longrightarrow & (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) & \longrightarrow & C \underset{R}{\otimes} S & \longrightarrow & 0 \end{array}$$

and therefore $(B' \times_B C) \otimes_R S \to (B' \otimes_R S)\ \times_{(B \otimes_R S)}\ (C \otimes_R S)$ is an isomorphism. I 77

(ii) Since B' is R-flat and $\mathrm{Tor}_i^R(B,\ S) = 0$ for all $0 < i \leq n$ by hypothesis, we have $\mathrm{Tor}_i^R(J,\ S) = 0$ for all $0 < i < n$. Then the exact sequence $0 \to J \to B' \times_B C \to C \to 0$ together with the hypothesis that $\mathrm{Tor}_i^R(C,\ S) = 0$ for all $0 < i < n$ entails that $\mathrm{Tor}_i^R(B' \times_B C,\ S) = 0$ for all $0 < i < n$.

Corollary 3.6.

(a) Let A, S be R-algebras. If $\mathrm{Tor}_1^R(A,\ S) = 0$, then the canonical map $\check{H}^{\cdot}(A \otimes_R S|S,\ G) \to \check{H}^{\cdot}(A|R,\ \check{S}G)$ is an isomorphism for any $G \in \mathrm{ob}(\check{\underline{C}}_{A \otimes_R S|S})$ where the functor $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ is given by $X \longmapsto X \otimes_R S$. In particular if S is R-flat, then the canonical map $\check{S}\underline{\check{H}}^{\cdot}(G) \to \underline{\check{H}}^{\cdot}(\check{S}G)$ is an isomorphism for every $G \in \mathrm{ob}(\check{\underline{C}}_{A \otimes_R S|S})$.

(b) Let $P \to B$ be a surjective R-algebra map, where P is a polynomial algebra over R . If $\mathrm{Tor}_i^R(B,\ S) = 0$ for all $0 < i \leq n$, then $\mathrm{Tor}_i^R(P \times_B P \times_B \ldots \times_B P,\ S) = 0$ for all $0 < i < n$.

Proof:

(a) Given $B \in \mathrm{ob}(\underline{C}_{A|R})$, choose $\{P \to B\} \in \mathrm{Cov}(B)$ where P is a polynomial algebra over R . If $\mathrm{Tor}_1^R(B,\ S) = 0$, the natural map

$$(\overbrace{P \times_B \ \ldots \ \times_B P}^{n}) \otimes_R S \to (P \otimes_R S) \times_{(B \otimes_R S)} \ldots \times_{(B \otimes_R S)} (P \otimes_R S)$$

is an isomorphism for all n by 3.5(i), and hence $\check{H}^{\cdot}(B|R,\ \check{S}G) = H^{\cdot}(\{P \to B\},\ \check{S}G) = H^{\cdot}(\{P \otimes_R S \to B \otimes_R S\},\ G) = \check{H}^{\cdot}(B \otimes_R S|S,\ G)$. If S is R-flat, then $\mathrm{Tor}_1^R(X,\ S) = 0$ for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\check{H}^{\cdot}(X \otimes_R S|S,\ G) \to \check{H}^{\cdot}(X|R,\ \check{S}G)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ i.e. $\check{S}\underline{\check{H}}^{\cdot}(G) \to \underline{\check{H}}^{\cdot}(\check{S}G)$ is an isomorphism.

(b) Set $P^{(m)} = \overbrace{P \times_B \ \ldots \ \times_B P}^{m+1}$. If $m = 0$ there is nothing to prove since P is R-flat. Assume that $\mathrm{Tor}_i^R(P^{(m-1)},\ S) = 0$ for all $0 < i < n$. Since $P \to B$ is surjective and $P^{(m)} = P \times_B P^{(m-1)}$, it follows from 3.5(ii) that $\mathrm{Tor}_i^R(P^{(m)},\ S) = 0$ for all $0 < i < n$.

Proposition 3.7. 178

(a) Let A, S be R-algebras. If $\mathrm{Tor}_i^R(A,\ S) = 0$ for all $0 < i \leq n$, then for any sheaf F on $\underline{C}_{A \otimes_R S|S}$, the canonical map $H^i(A \otimes_R S|S,\ F) \to H^i(A|R,\ \widetilde{S}F)$ is an isomorphism for all $i \leq n$, where $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ is given by $X \longmapsto X \otimes_R S$.

(b) Let P be a polynomial algebra over R . Then for any additive sheaf F on $\underline{C}_{P \otimes_R A|R}$, the canonical map $H^n(P \otimes_R A|R,\ F) \to H^n(A|R,\ F)$ is an isomorphism for all $n > 0$.

(c) Let $S \to R \to A$ be ring homomorphisms. Then for any additive sheaf F on $\underline{C}_{A|S}$, we have an exact sequence

$$0 \to H^0(A|R,\ F_R) \to H^0(A|S,\ F) \to H^0(R|S,\ F) \to H^1(A|R,\ F_R) \to H^1(A|S,\ F) \to H^1(R|S,\ F) \to$$
$$\ldots\ldots\ldots \to H^n(A|R,\ F_R) \to H^n(A|S,\ F) \to H^n(R|S,\ F) \to H^{n+1}(A|R,\ F_R) \to \ldots.$$

where F_R is a sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \mathrm{Ker}(F(X) \to F(R))$$

for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ via the forgetful functor $\underline{C}_{A|R} \to \underline{C}_{A|S}$.

Proof:

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$$

where $S(X) = X \otimes_R S$. Since the functor S sends polynomial algebras over R to polynomial algebras over S , $\widetilde{S} : \underline{\widetilde{C}}_{A \otimes_R S|S} \to \underline{\widetilde{C}}_{A|R}$ is an exact functor by 3.3(c) and in turn we obtain the canonical map $\check{S}\underline{H}^{\cdot}_{A \otimes_R S|S}(F) \to \underline{H}^{\cdot}_{A|R}(\widetilde{S}F)$, and in turn we obtain the commutative diagram of spectral sequences:

$$\begin{array}{ccc}
E_S^{p,q}(S(X)) = \check{H}^p(X \otimes_R S|S,\ \underline{H}^q_{A \otimes_R S|S}(F)) & \Longrightarrow & H^{p+q}(X \otimes_R S|S,\ F) \\
\downarrow & & \downarrow \\
\check{H}^p(X|R,\ \check{S}\underline{H}^q_{A \otimes_R S|S}(F)) & & \\
\downarrow & & \\
E_R^{p,q}(X) = \ \check{H}^p\ (X|R,\ \underline{H}^q_{A|R}(\widetilde{S}F)) & \Longrightarrow & H^{p+q}(X|R,\ \widetilde{S}F)
\end{array}$$

We note that $E_S^{p,q}(S(X)) \to \check{H}^p(X|R,\ \check{S}\underline{H}^q_{A \underset{R}{\otimes} S|S}(F))$ is an isomorphism whenever $\mathrm{Tor}_1^R(X,\ S) = 0$ 179
by 3.6(a). Assume now that $[\check{S}\underline{H}^q_{A \underset{R}{\otimes} S|S}(F)](X) \to [\underline{H}^q_{A|R}(\widetilde{S}F)](X)$ is an isomorphism for all $q < n$ and for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ such that $\mathrm{Tor}_i^R(X,\ S) = 0$ for all $0 < i < n$. It follows immediately from 3.6(b) that $E_S^{p,q}(S(X)) \to E_R^{p,q}(X)$ is an isomorphism for all $q < n$ and for all p , provided $\mathrm{Tor}_i^R(X,\ S) = 0$ for all $0 < i \leq n$. Since $E_S^{0,n}(S(X)) \to E_R^{0,n}(X)$ is an isomorphism (since both sides are zero), we get that $H^n(X \underset{R}{\otimes} S|S,\ F) \to H^n(X|R,\ \widetilde{S}F)$ is an isomorphism whenever $\mathrm{Tor}_i^R(X,\ S) = 0$

for all $0 < i \leq n$ i.e., $[\check{S}\underline{H}^i_{A\underset{R}{\otimes}S|S}(F)](X) \to [\underline{H}^i_{A|R}(\widetilde{S}F)](X)$ is an isomorphism for all $i \leq n$ whenever $\mathrm{Tor}^R_i(X,\ S) = 0$ for all $0 < i \leq n$.

(b) Let P be a polynomial algebra over R and let $f : \underline{C}_{A|R} \to \underline{C}_{A\underset{R}{\otimes}P|R}$ be the continuous functor given by $f(X) = X \underset{R}{\otimes} P$. Let $P' \to X$ be a surjective arrow in $\underline{C}_{A|R}$ where P' is a polynomial algebra over R . Then $P' \underset{R}{\otimes} P \to X \underset{R}{\otimes} P$ is a surjective arrow in $\underline{C}_{A\underset{R}{\otimes}P|R}$, and $P' \underset{R}{\otimes} P$ is again a polynomial algebra over R . Since P is flat over R , it follows from 3.5(i) that $\check{H}^{\cdot}(X|R,\ \check{f}G) = H^{\cdot}(\{P' \to X\},\ \check{f}G) = H^{\cdot}\ (\{P' \underset{R}{\otimes} P \to X \underset{R}{\otimes} P\},\ G) = \check{H}^{\cdot}(X \underset{R}{\otimes} P|R,\ G)$ for every $G \in \mathrm{ob}(\underline{\check{C}}_{A\underset{R}{\otimes}P|R})$. It follows that $\widetilde{f} : \underline{\widetilde{C}}_{A\underset{R}{\otimes}P|R} \to \underline{\widetilde{C}}_{A|R}$ sends acyclic ones to acyclic ones. On the other hand, f sends polynomial algebras over R to polynomial algebras over R , and hence $\widetilde{f} : \underline{\widetilde{C}}_{A\underset{R}{\otimes}P|R} \to \underline{\widetilde{C}}_{A|R}$ is an exact functor by 3.3(c). It follows that $H^{\cdot}(A \underset{R}{\otimes} P|R,\ F) \to H^{\cdot}(A|R,\ \widetilde{f}F)$ is an isomorphism for every $F \in \mathrm{ob}(\underline{\widetilde{C}}_{A\underset{R}{\otimes}P|R})$. Now for each $X \in \mathrm{ob}(\underline{C}_{A|R})$, we have canonical maps

$$(\widetilde{f}F)\ (X) = F(X \underset{R}{\otimes} P) \to F(X) \oplus F(P) \quad ,$$

and hence a canonical map $\widetilde{f}F \to F \oplus F(P)$ where $F(P)$ denotes a constant sheaf (and F on the right hand side denotes, strictly speaking, the restriction of F on $\underline{C}_{A|R}$) . If F is an additive sheaf, then $(\widetilde{f}F)\ (X) = F(X \underset{R}{\otimes} P) \xrightarrow{\sim} F(X) \oplus F(P)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\widetilde{f}F \xrightarrow{\sim} F \oplus F(P)$ is an isomorphism. Therefore

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$$H^n(A|R,\ \widetilde{f}F) \xrightarrow{\sim} H^n(A|R,\ F \oplus F(P)) = H^n(A|R,\ F)$$

for all $n > 0$ since $F(P)$, being a constant sheaf, is flask. It follows that $H^n(A \underset{R}{\otimes} P|R,\ F) \to H^n(A|R,\ F)$ is an isomorphism for all $n > 0$, provided F is an additive sheaf on $\underline{C}_{A\underset{R}{\otimes}P|R}$.

(c) (communicated from Rinehart) Consider the forgetful functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$. We have a spectral sequence

$$E^{p,q} = H^p(A|R,\ \beta^{\#}\underline{H}^q_{A|S}(F)) \implies H^{p+q}(A|S,\ F)$$

where $\beta^{\#} = G_{A|R} \circ \check{\beta}$, $G_{A|R} : \underline{\check{C}}_{A|R} \to \underline{\widetilde{C}}_{A|R}$ is the "associated-sheaf" functor. Now let F be an additive sheaf. We claim that $\beta^{\#}\underline{H}^q_{A|S}(F)$ is a constant sheaf for all $q > 0$. Indeed, if P is a polynomial algebra over R , then $P = R \underset{S}{\otimes} P'$ for a polynomial algebra P' over S and hence

$$[\beta^{\#}\underline{H}^q_{A|S}(F)](P) = H^q(P|S,\ F) = H^q(R \underset{S}{\otimes} P'|S,\ F) \to H^q(R|S,\ F)$$

is an isomorphism for all $q > 0$ by 3.7(b), and surjective for $q = 0$. It follows from 3.3(c) that the canonical map $\beta^{\#}\underline{H}^q_{A|S}(F) \to H^q(R|S,\ F) =$ constant sheaf is an isomorphism for all $q > 0$, and is surjective for $q = 0$. It follows that $E^{p,q} = H^p(A|R,\ \beta^{\#}\underline{H}^q_{A|S}(F)) = 0$ for all $p,\ q > 0$, and hence the above spectral sequence yields the exact sequence

$$(*) \quad 0 \to E^{1,0} \to H^1 \to E^{0,1} \to\ E^{2,0} \to H^2 \to E^{0,2} \to E^{3,0} \to\ H^3 \to\ ....$$

where we note that $E^{0,q} = [\beta^{\#}\underline{H}^q_{A|S}(F)](A) \xrightarrow{\sim} H^q(R|S,\ F)$ for all $q > 0$ and $E^{p,0} = H^p(A|R,\ \widetilde{\beta}F)$. On the other hand, the exact sequence $0 \to F_R \to \widetilde{\beta}F \to F(R) \to 0$ gives us the exact sequence

$$(**) \qquad \begin{array}{ccccccccccc} 0 \to H^0(A|R,F_R) & \to & H^0(A|R,\widetilde{\beta}F) & \to & H^0(A|R,F(R)) & \to & H^1(A|R,F_R) & \to & H^1(A|R,\widetilde{\beta}F) & \to & 0 \\ & & \| & & \| & & & & & & \\ & & H^0(A|S,F) & & H^0(R|S,F) & & & & & & \end{array}$$

The exact sequences (*) and (**) combined give the desired exact sequence.

Corollary 3.8. I 81

(a) If A, B are R-algebras such that $\mathrm{Tor}_i^R(A,B) = 0$ for all $0 < i \leq n$, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A \underset{R}{\otimes} B|R})$, the canonical map

$$H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(A|R,\ F) \oplus H^i(B|R,\ F)$$

is an isomorphism for all $i \leq n$.

(b) Let A be an R-algebra such that $\mathrm{Tor}_i^R(A,A) = 0$ for all $i > 0$. If $A \underset{R}{\otimes} A \to A$ is an isomorphism, then for any additive sheaf $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A|R})$, we have $H^i(A|R,\ F) = 0$ for all i .

Proof:

(a) Let $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A \underset{R}{\otimes} B|R})$ be additive. By 3.7(c), the ring homomorphisms $R \to A \to A \underset{R}{\otimes} B$ give us the exact sequence

$$(*) \qquad \ldots \to H^i(A \underset{R}{\otimes} B|A,\ F_A) \to H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(A|R,\ F) \to \ldots.$$

consider now the commutative diagram

$$\begin{array}{ccc} H^i(A \underset{R}{\otimes} B|A,\ F_A) & \longrightarrow & H^i(A \underset{R}{\otimes} B|R,\ F) \\ \downarrow & & \downarrow \\ H^i(B|R,\ \widetilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R,\ F) \end{array}$$

where the functor $A : \underline{C}_{B|R} \to \underline{C}_{A \underset{R}{\otimes} B|A}$ is given by $X \longmapsto A \underset{R}{\otimes} X$, and the map $H^i(B|R,\ \widetilde{A}F_A) \to H^i(B|R,\ F)$ is induced from the map $\widetilde{A}F_A \to F$, which is an isomorphism since F is additive. Note that the commutativity of the above diagram, because of its functorial nature, is reduced to the case $i = 0$.

If $\mathrm{Tor}_i^R(A,B) = 0$ for all $0 < i \leq n$, then $H^i(A \underset{R}{\otimes} B|A,\ F_A) \to H^i(B|R,\ \widetilde{A}F_A)$ is an isomorphism for all $i \leq n$ by 3.7(a), and hence the composite map

$$H^i(A \underset{R}{\otimes} B|A,\ F_A) \to H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(B|R,\ F)$$

and (because of the symmetry)

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$$H^i(A \underset{R}{\otimes} B|B,\ F_B) \to H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(A|R,\ F)$$

are isomorphisms for all $i \leq n$. This implies that the exact sequence (*) breaks up to short split-exact sequences

$$0 \to H^i(A \underset{R}{\otimes} B|A,\ F_A) \to H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(A|R,\ F) \to 0$$

for all $i \leq n$, and that

$$H^i(A \underset{R}{\otimes} B|R,\ F) \to H^i(A|R,\ F) \oplus H^i(B|R,\ F)$$

is an isomorphism for all $i \leq n$.

(b) Since $\mathrm{Tor}_i^R(A,A) = 0$ for all $i > 0$, it follows from 3.8(a) that

$$H^i(A \underset{R}{\otimes} A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$$

is an isomorphism for all i . On the other hand, $A \underset{R}{\otimes} A \to A$ is an isomorphism by hypothesis and hence $H^i(A|R,\ F) \to H^i(A \underset{R}{\otimes} A|R,\ F)$ is an isomorphism for all i . This means that the diagonal map $H^i(A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$ is an isomorphism for all i i.e., $H^i(A|R,\ F) = 0$ for all i .

Corollary 3.9: Let S be a multiplicative subset of an R-algebra A . If $F \in \mathrm{ob}\left(\widetilde{\underline{C}}_{S^{-1}A|R}\right)$ is additive such that F_A is also additive, then the canonical map

$$H^i(S^{-1}A|R,\ F) \to H^i(A|R,\ F)$$

is an isomorphism for all i , where $F_A \in \mathrm{ob}\left(\widetilde{\underline{C}}_{S^{-1}A|A}\right)$ is defined by $F_A(X) = \mathrm{Ker}(F(X) \to F(A))$ for all $X \in \mathrm{ob}\left(\underline{C}_{S^{-1}A|A}\right)$.

Proof: Since F is additive, the ring homomorphisms $R \to A \to S^{-1}A$ give us the exact sequence

$$.... \to H^n(S^{-1}A|A,\ F_A) \to H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F) \to ....$$

On the other hand, since $S^{-1}A \underset{A}{\otimes} S^{-1}A \to S^{-1}A$ is an isomorphism and F_A is additive by hypothesis, it follows from 3.8(b) that $H^n(S^{-1}A|A,\ F_A) = 0$ for all n , and hence $H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F)$ is an isomorphism for all n .

Now let A be an R-algebra. An A-module M defines an additive sheaf $\mathrm{Der}_R(-,\ M) \in \mathrm{ob}(\widetilde{\underline{C}}_{A|R})$ **183** by $X \to \mathrm{Der}_R(X,\ M)$. We now make the following definition:

Definition 3.10: Let A be an R-algebra. For each A-module M we set

$$D^i(A|R,\ M) = H^i(A|R,\ \mathrm{Der}_R(-,\ M)) \quad .$$

We want to make an explicit computation of low-dimensional ones i.e., $D^i(A|R,\ M)$ for $i = 1,\ 2$: Choose a surjective arrow $P \to A$ in $\underline{C}_{A|R}$ where P is a polynomial algebra over R , and let $0 \to I \to P \to A \to 0$ be exact. Then

Lemma 3.10:

$$D^1(A|R,\ M) = \mathrm{Coker}(\mathrm{Der}_R(P,\ M) \to \mathrm{Hom}_P(I,\ M))$$
$$D^2(A|R,\ M) = H^1(A|R,\ \underline{H}^1_{A|R}(\mathrm{Der}_R(-,\ M)))$$

Proof: Consider the simplicial algebra

$$(*) \quad ..... \qquad P_A^{(2)} \quad \begin{matrix}\to\\ \to\\ \to\end{matrix} \quad P_A^{(1)} \quad \begin{matrix}\to\\ \to\end{matrix} \quad P_A^{(0)}$$

where $P_A^{(n)} = \overbrace{P \times_A \ldots \times_A P}^{n+1}$. For each n we have the exact sequence

$$0 \to I^{(n)} \to P_A^{(n)} \to A \to 0$$

where $I^{(n)} = \overbrace{I \times I \times \ldots \times I}^{n+1}$, and it is easy to see that

(a):

$$0 \to \mathrm{Hom}_A(\overline{I}^{(n)}/\Delta(\overline{I}),\ M) \to \mathrm{Der}_R(P_A^{(n)},\ M) \xrightarrow{\mathrm{Der}(\Delta,\ M)} \mathrm{Der}_R(P,\ M) \to 0$$

is exact for all n , where $\overline{I} = I/I^2$ and Δ stands for the diagonal map. We also note the exact sequence

(b):

$$0 \to \mathrm{Hom}_A(\overline{I}^{(n)}/\Delta(\overline{I}),\ M) \to \mathrm{Hom}_A(\overline{I}^{(n)},\ M) \xrightarrow{\mathrm{Hom}(\Delta,\ M)} \mathrm{Hom}_A(\overline{I},\ M) \to 0$$

The exact sequences (a) and (b) yield the respective exact sequences of simplicial A-modules. Since $\{I \to 0\}$, $\{I \to I\}$ admit sections, we have

$$H^i(\{I \to 0\},\ \mathrm{Hom}_A(-,\ M)) = H^i(\{I \to I\},\ \mathrm{Hom}_A(-,\ M)) = 0$$

for all $i > 0$, and it follows readily that

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$$\begin{cases} 0 \to \mathrm{Der}_R(A,\ M) \to \mathrm{Der}_R(P,\ M) \to \mathrm{Hom}_A(\overline{I},\ M) \to H^1(\{P \to A\},\ \mathrm{Der}_R(-,\ M)) \to 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{is exact} \\ H^i(\{P \to A\},\ \mathrm{Der}_R(-,\ M)) = 0 \ \text{ for all } \ i > 1 \end{cases}$$

It follows from 3.1(b) that

$$\begin{aligned} D^1(A|R,\ M) &= \check{H}^1(A|R,\ \mathrm{Der}_R(-,\ M)) = H^1(\{P \to A\},\ \mathrm{Der}_R(-,\ M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P,\ M) \to \mathrm{Hom}_A(\overline{I},\ M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P,\ M) \to \mathrm{Hom}_P(I,\ M)) \ , \\ D^2(A|R,\ M) &= H^1(A|R,\ \underline{H}^1_{A|R}(\mathrm{Der}_R(-,\ M))) \ . \end{aligned}$$

Now choose a free P-module F and a P-linear map $F \xrightarrow{\gamma} P$ such that $\gamma(F) = I$. Associated to P-linear map $\gamma : F \to P$, there is a Koszul complex, which is simply denoted, by abuse of notation, by K_I .

Lemma 3.11: $D^2(A|R,\ M) \simeq \mathrm{Coker}(H^1(K_I,\ M) \to \mathrm{Hom}_A(H_1(K_I),\ M))$.

Proof: We have

$$\begin{aligned} D^2(A|R,\ M) &= H^1(A|R,\ H^1_{A|R}(\mathrm{Der}_R(-,\ M))) \\ &= \mathrm{Ker}(H^1(P_A^{(1)}|R,\ M) \overset{\to}{\underset{\to}{\rightrightarrows}} H^1(P_A^{(2)}|R,\ M)) \end{aligned}$$

since $[\underline{H}^1_{A|R}(\mathrm{Der}_R(-,\ M))](P^0_A) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-,\ M))](P) = 0$.

Let $0 \to K \to F \xrightarrow{\gamma} I \to 0$ be exact. If we set $P[F]$ to be the symmetric algebra over P generated by F , then $P[F]$ and $P[F] \otimes P[F]$ are polynomial algebras over R and we obtain the surjective arrows

$$\begin{aligned} P[F] \to P_A^{(1)} = P \times_A P \qquad &\text{given by} \quad X \to (0,\ \gamma(X)) \\ P[F] \otimes_P P[F] \to P_A^{(2)} = P \times_A P \times_A P \qquad &\qquad\quad X \otimes 1 \to (0,\ \gamma(X),\ 0),\ 1 \otimes X \to (0,\ 0,\ \gamma(X)) \end{aligned}$$

where $X \in F$. If we set $J = \mathrm{Ker}(P[F] \to P_A^{(1)})$ and $J' = \mathrm{Ker}(P[F] \otimes_P P[F] \to P_A^{(2)})$, then

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$$\begin{aligned} J &= (F) \cap (F_\gamma) = (K) + (F \cdot F_\gamma) \\ J' &= (F \otimes 1 + 1 \otimes F) \cap (F_\gamma \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_\gamma) \\ &= (K \otimes 1 + 1 \otimes K) + (F \cdot F_\gamma \otimes 1 + 1 \otimes F \cdot F_\gamma) + (F \otimes F) \end{aligned}$$

where $F_\gamma =$ the ideal generated by $\{X - \gamma(X) | X \in F\}$. Now $\pi_i : P_A^{(2)} \to P_A^{(1)}$ $i = 0, 1, 2$ can be lifted to $\hat{\pi}_i : P[F] \underset{P}{\otimes} P[F] \to P[F]$ where, for each $X \in F$,

$$\pi_0 = \begin{cases} X \otimes 1 \to \gamma(X) - X \\ \\ 1 \otimes X \to X \end{cases} \qquad \pi_1 = \begin{cases} X \otimes 1 \to 0 \\ \\ 1 \otimes X \to X \end{cases} \qquad \pi_2 = \begin{cases} X \otimes 1 \to X \\ \\ 1 \otimes X \to 0 \end{cases} \quad .$$

A trivial computation shows that $\pi_0-\pi_1+\pi_2$ sends $K\otimes 1+1\otimes K+1\otimes F\cdot F_\gamma$ to zero, whereas it sends $(F\cdot F_\gamma\otimes 1+F\otimes F)$ onto $(F\cdot F_\gamma)$. Now for any $D\in \mathrm{Der}_R(P[F]\underset{P}{\otimes}P[F],M)$, $D(F\cdot F_\gamma\otimes 1+F\otimes F)=0$ since $I\cdot M=0$, and consequently

$$\begin{aligned} D^2(A|R,M) &= \mathrm{Ker}(D^1(P_A^{(1)}|R,M)\overset{\to}{\underset{\to}{\rightrightarrows}}D^1(P_A^{(2)}|R,M)) \\ &= \{[f]\in D^1(P_A^{(1)}|R,M)|f(F\cdot F_\gamma)=0\} \\ &= \mathrm{Coker}(\mathrm{Der}_R(P[F],M)\to \mathrm{Hom}_{P[F]}((F)\cap(F_\gamma)/(F\cdot F_\gamma),M)) \\ &= \mathrm{Coker}(\mathrm{Der}_P(P[F],M)\to \mathrm{Hom}_{P[F]}((F)\cap(F_\gamma)/(F)\cdot(F_\gamma),M)) \quad . \end{aligned}$$

Now

$$\begin{aligned} 0\to(F)\to P[F]\xrightarrow{P[0]}P\to 0 \\ 0\to(F_\gamma)\to P[F]\xrightarrow{P[\gamma]}P\to 0 \quad . \end{aligned}$$

are exact and hence $(F)\cap(F_\gamma)/(F)\cdot(F_\gamma)=\mathrm{Tor}_1^{P[F]}(P,P_\gamma)=H_1(K_I)$ where K_I is the Koszul complex associated to the P-linear map $F\overset{\gamma}{\to}I(\subset P)$. We thus have

$$\begin{aligned} D^2(A|R,M) &= \mathrm{Coker}(\mathrm{Hom}_P(F,M)\to \mathrm{Hom}_P(H_1(K_I),M)) \\ &= \mathrm{Coker}(H^1(K_I,M)\to \mathrm{Hom}_A(H_1(K_I),M)) \quad . \end{aligned}$$

We can now state all the basic facts needed for our purpose, summarizing the various results we have already established.

Theorem 3.12. I86

(1) If $0\to M'\to M\to M''\to 0$ is an exact sequence of A-modules, then

$$\begin{gathered} 0\to D^0(A|R,M')\to D^0(A|R,M)\to D^0(A|R,M'')\to D^1(A|R,M')\to \\ D^1(A|R,M)\to \ D^1(A|R,M'')\to \ \ldots\ldots\ \to D^n(A|R,M')\to \\ D^n(A|R,M)\to D^n(A|R,M'')\to \ \ldots \end{gathered}$$

is exact.

(2) Let $S\to R\to A$ be ring homomorphisms. Then for any A-module M we have the exact sequence

$$\begin{gathered} 0\to D^0(A|R,M)\to D^0(A|S,M)\to D^0(R|S,M)\to D^1(A|R,M)\to \\ D^1(A|S,M)\to D^1(R|S,M)\to \ \ldots\ldots\ \to D^n(A|R,M)\to \\ D^n(A|S,M)\to D^n(R|S,M)\to \ \ldots. \end{gathered}$$

(3) Let A,S be R-algebras such that $\mathrm{Tor}_i^R(A,S)=0$ for all $i>0$. Then for any $A\underset{R}{\otimes}S$-module M , the canonical map $D^i(A\underset{R}{\otimes}S|S,M)\to D^i(A|R,M)$ is an isomorphism for all i . In particular, if A is R-flat we have

$$D^i(A\otimes S|S,M)\overset{\sim}{\to}D^i(A|R,M)$$

for all i and for all R-algebras S .

(4) Let S be a multiplicative subset of A . Then for any $S^{-1}A$-module N , the canonical map

$$D^i(S^{-1}A|R,N)\to D^i(A|R,N)$$

is an isomorphism for all i . Consequently, for any A-module M we have the canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$$

for all i .

(5) If A, B are R-algebras such that $\mathrm{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then for any $A \underset{R}{\otimes} B$-module M , the canonical map

$$D^i(A \underset{R}{\otimes} B|R, M) \to D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for all $i \leq n$. In particular, if $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$ and 187

$$\Delta \ : \ \mathrm{Spec}(A) \ \to \ \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion (for instance A/R etale), then $D^i(A|R, M) = 0$ for all $i > 0$ and for all A-modules M .

(6) Let A be an R-algebra of essentially finite type over R , where R is noetherian, and let E be a flat A-module. Then we have a canonical isomorphism

$$E \underset{A}{\otimes} D^i(A|R, \ M) \ \xrightarrow{\sim} \ D^i(A|R, \ E \underset{A}{\otimes} M)$$

for all $i \geq 0$ and for all A-modules M .

(7) If R is noetherian and A is an R-algebra of essentially finite type, then $D^i(A|R, M)$ is an A-module of finite type for all i , provided M is an A-module of finite type.

(8) Let P be a polynomial algebra over R and let $0 \to I \to P \to A \to 0$ be exact. Then $D^0(A|R, M) = \mathrm{Der}_R(A, M)$,

$$\begin{aligned} D^1(A|R, \ M) \ &= \ \mathrm{Coker}(\mathrm{Der}_R(P, \ M) \ \to \ \mathrm{Hom}_A(I/I^2, \ M)) \\ &= \ \text{the set of isomorphism classes of} \\ &\qquad \text{R-algebra extensions of } A \text{ by } M \ , \\ D^2(A|R, \ M) \ &= \ \mathrm{Coker}(H^1(K_I, \ M) \ \to \ \mathrm{Hom}_A(H_1(K_I), \ M)) \ . \end{aligned}$$

Proof:

(1) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any polynomial algebra P over R , $0 \to \mathrm{Der}_R(P, M') \to \mathrm{Der}_R(P, M) \to \mathrm{Der}_R(P, M'') \to 0$ is exact and hence $0 \to \mathrm{Der}_R(-, M') \to \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, M'') \to 0$ is an exact sequence of sheaves by 3.3(c).

(2) $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$, and $\mathrm{Der}_R(B, M) = \mathrm{Ker}(\mathrm{Der}_S(B, M) \to \mathrm{Der}_S(R, M))$ for every $B \in \mathrm{ob}(\underline{C}_{A|R})$. Therefore our assertion follows from 3.7(c).

(3) Let $S : \underline{C}_{A|R} \to \underline{C}_{A \underset{R}{\otimes} S|S}$ be the functor given by $X \longmapsto X \underset{R}{\otimes} S$. Then $\widetilde{S}\mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$ and hence our assertion follows from 3.7(a).

(4) Follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and 188
$F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is also additive since $F_A(X) = \mathrm{Ker}(F(X) \to F(A)) = \mathrm{Ker}(\mathrm{Der}_R(X, N) \to \mathrm{Der}_R(A, N)) = \mathrm{Der}_A(X, N)$ i.e., $F_A = \mathrm{Der}_A(-, N)$.

(5) The first assertion follows from 3.8(a). Now assume that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$ and that

$$\Delta \ : \ \mathrm{Spec}(A) \ \to \ \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion i.e., $(A \underset{R}{\otimes} A)_{\varphi^{-1}(\underline{m})} \to A_{\underline{m}}$ is an isomorphism for all maximal ideals $\underline{m}$ of A , where $\varphi : A \underset{R}{\otimes} A \to A$ is the multiplication map. Then the first assertion combined with 3.12(4) entails that the diagonal map

$$\Delta_{\underline{m}} \ : \ D^i(A|R, \ M)_{\underline{m}} \ \to \ D^i(A|R, \ M)_{\underline{m}} \ \oplus \ D^i(A|R, \ M)_{\underline{m}}$$

is an isomorphism for every maximal ideal $\underline{m}$ of A and hence

$$\Delta \ : \ D^i(A|R,\ M) \ \to \ D^i(A|R,\ M) \ \oplus \ D^i(A|R,\ M)$$

is an isomorphism, which entails that $D^i(A|R, M) = 0$ for all i .

(6) In view of 3.12(4), we may assume that A is an R-algebra of finite type. Now consider the map

$$E \underset{A}{\otimes} \mathrm{Der}_R(-,\ M) \ \to \ \mathrm{Der}_R(-,\ E \underset{A}{\otimes} M)$$

in $\underline{C}^{\sim}_{A|R}$. Since E is A-flat, we have an isomorphism

$$E \underset{A}{\otimes} \mathrm{Der}_R(B,\ M) \ \xrightarrow{\sim} \ \mathrm{Der}_R(B,\ E \underset{A}{\otimes} M)$$

whenever $B \in \mathrm{ob}(\underline{C}_{A|R})$ is of finite type over R . It follows from 3.3(b) that

$$D^i(A|R,\ E \underset{A}{\otimes} \mathrm{Der}_R(-,\ M)) \ \to \ D^i(A|R,\ E \underset{A}{\otimes} M)$$

is an isomorphism. On the other hand, the functor $E \underset{A}{\otimes} - \ : \ (A\text{-mod}) \to (A\text{-mod})$ is exact, and therefore we have

$$E \underset{A}{\otimes} D^i(A|R,\ M) \ = \ D^i(A|R,\ E \underset{A}{\otimes} \mathrm{Der}_R(-,\ M)) \ ,$$

and hence our assertion.

(7) follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that $\mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) = \mathrm{Exalcomm}(A|R, M)$, where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of A by M . Let $0 \to M \to A' \to A \to 0$ be an R-algebra extension of A by M . Since P is a polynomial algebra over R , we obtain a commutative diagram 189

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

where $I \to M$ is uniquely determined by the extension A' up to $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi \ : \ \mathrm{Exalcomm}(A|R, M) \to \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \ .$$

Conversely a P-linear map $I \to M$ yields an extension by push-out

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

Two P-linear maps $I \rightrightarrows M$ which differ by $\mathrm{Der}_R(P, M)$ are trivially seen to yield isomorphic extensions and therefore we obtain a map

$$\Psi \ : \ \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \to \mathrm{Exalcomm}(A|R, M) \ .$$

It is straightforward to verify that Φ, Ψ are inverse processes to each other.

Corollary 3.13. (a) The R-algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A-module M .

(b) Let R be noetherian and A an R-algebra of finite type. Then A is a relative complete-intersection over R if and only if $D^2(A|R,M)=0$ for every A-module M .

(c) Let R be noetherian and A an R-algebra of finite type. Then for any multiplicative subset S in A , we have a canonical isomorphism

$$D^i(S^{-1}A|R,\ S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R,M) \text{ for all } i\geq 0 .$$

(d) Let R be noetherian and A a local R-algebra of essentially finite type. We denote by $\hat{A}$ the m-adic completion of A where m = the maximal ideal of A . Then for any A-module E of finite type we have a canonical isomorphism I 90

$$D^1(\hat{A}|R,\ \hat{A}\underset{A}{\otimes}E) \xrightarrow{\sim} \hat{A}\underset{A}{\otimes}D^1(A|R,E) .$$

Proof: (a) follows immediately from $D^1(A|R,M)=\mathrm{Exalcomm}(A|R,M)$.

(b) Let P be a polynomial algebra over R in a finite number of variables and let $0\to I\to P\to A\to 0$ be exact. Localizing if necessary, we may assume that I is generated by a P-regular sequence which would imply that $H_1(K_I)=0$ and therefore $D^2(A|R,M)=\mathrm{Coker}(H^1(K_I,M)\to \mathrm{Hom}_A(H_1(K_I),M))=0$. Conversely assume that $D^2(A|R,M)=0$ for every A-module M .

Let $0\to I\to P\to A\to 0$ be exact where P is a polynomial algebra over R in a finite number of variables. We have

$$0=D^2(A|R,M)=\mathrm{Coker}(H^1(K_I,M)\to \mathrm{Hom}_A(H_1(K_I),M))$$

i.e., $H^1(K_I,M)\to \mathrm{Hom}_A(H_1(K_I),M)$ is surjective for every A-module M . Choose an exact sequence $0\to K\to F\to I\to 0$ where F is a free P-module of finite type. We note that $H_1(K_I)=K/K_0$ where $K_0=\mathrm{Im}(\overset{2}{\wedge}F\to\overset{1}{\wedge}F)$. The fact that $D^2(A|R,H_1(K_I))=0$ i.e.,

$$H^1(K_I,\ H_1(K_I))\to \mathrm{Hom}_A(H_1(K_I),\ H_1(K_I)) \text{ is surjective}$$

entails that $0\to K/K_0\to F/IF\to I/I^2\to 0$ is exact and splits as A-modules and in particular I/I^2 is projective as an A-module. Therefore, localizing A if necessary, we may assume that I/I^2 is A-free, and that $F/IF\to I/I^2$ is an isomorphism. Then the fact that $0\to K/K_0\to F/IF\to I/I^2\to 0$ is exact and splits as A-modules entails that $H_1(K_I)=K/K_0=0$. This means that I is generated by a P-regular sequence i.e., A is a relative complete intersection over R .

(c) follows immediately from 3.12(4) and (6).

(d) Firstly it follows from 3.12(6) that

$$D^i(A|R,\ \hat{A}\underset{A}{\otimes}E) \xrightarrow{\sim} \hat{A}\underset{A}{\otimes}D^i(A|R,E)$$

is an isomorphism for all i , and therefore it suffices to show that $D^1(\hat{A}|R,\ \hat{A}\otimes_A E)\to D^1(A|R,\ \hat{A}\otimes_A E)$ is an isomorphism. Our proof is based on the description of D^1 in terms of algebra extensions. Let $0\to \hat{A}\otimes_A E\to A'\xrightarrow{\varphi}\hat{A}\to 0$ be an R-algebra extension whose restriction to A splits. The fact that $\hat{A}\otimes_A E$ is an $\hat{A}$-module of finite type entails readily that A' is a local R-algebra, complete with respect to m'-adic topology where m' = the maximal ideal of A' . Therefore if there is an R-algebra map $\psi: A\to A'$ lifting the canonical map $A\to\hat{A}$, then ψ extends uniquely to $\hat{\psi}:\hat{A}\to A'$ such that $\varphi\circ\hat{\psi}=\mathrm{id}_{\hat{A}}$. This proves that I 91

$$D^1(\hat{A}|R,\ \hat{A}\otimes_A E)\to D^1(A|R,\ \hat{A}\otimes_A E)$$

is injective. Now let $0\to \hat{A}\otimes_A E\to B\xrightarrow{\varphi} A\to 0$ be an R-algebra extension. Firstly we claim that the topology on $\hat{E}=\hat{A}\otimes_A E$ induced from m_B-adic topology on B coincides with the topology on $\hat{E}$ as $\hat{A}$-module. Indeed, since A is essentially of finite type over R , one can find a subalgebra B_0 of B such that B_0 is a local R-algebra of essentially finite type over R and that $\varphi(B_0)=A$. Set $E_0=\hat{E}\cap B_0$ and denote by m_B , m_0 , m the maximal ideals of B , B_0 , A respectively. Then

$\underline{m}_B = \underline{m}_0 + \hat{E}$ and hence $\underline{m}_B^{n+1} = \underline{m}_0^{n+1} + \underline{m}^n\hat{E}$ for all n . Since B_0 is noetherian, there exists an integer $r > 0$ by Artin-Rees Theorem such that

$$\begin{aligned}\hat{E} \cap \underline{m}_B^{n+1} &= \underline{m}^n\hat{E} + \hat{E} \cap \underline{m}_0^{n+1} = \underline{m}^n\hat{E} + E_0 \cap \underline{m}_0^{n+1} \\ &= \underline{m}^n\hat{E} + \underline{m}^{n+1-r}(E_0 \cap \underline{m}_0^r) \subset \underline{m}^{n+1-r}\hat{E}\end{aligned}$$

for all $n \geq r$, which proves our assertion about the two topologies on $\hat{E}$. In particular $\hat{E} = \hat{A} \otimes_A E$ is complete with respect to the topology induced from $\underline{m}_B$-adic topology on B , and hence we obtain the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \hat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \hat{E} & \longrightarrow & \hat{B} & \longrightarrow & \hat{A} & \longrightarrow & 0 \end{array}$$

where $\hat{B}$ stands for $\underline{m}_B$-adic completion of B . This proves that $D^1(\hat{A}|R,\ \hat{E}) \to D^1(A|R,\ \hat{E})$ is surjective and hence we are done.

Corollary 3.14. Let R be noetherian and A an R-algebra of finite type, smooth everywhere except at one closed point $x \in \mathrm{Spec}(A)$. We then have isomorphisms **192**

$$D^1(A|R,\ A) \simeq D^1(A_x|R,\ A_x) \simeq D^1(\hat{A}_x|R,\ \hat{A}_x) .$$

Proof: Since A is smooth over R except at $x \in \mathrm{Spec}(A)$, $D^1(A|R,\ A)$ is an A-module of finite type with its support on the set $\{x\}$, and therefore we have

$$D^1(A|R,\ A) = D^1(A|R,\ A)_x = D^1\widehat{(A|R,\ A)}_x .$$

Therefore our assertion follows from 3.13.

Remark 3.15. Let R be a topological ring and A a topological R-algebra. We then define, for any A-module E ,

$$D^i_{\mathrm{top}}(A|R,\ E) = \varinjlim_{\alpha} D^i(A_\alpha|R_\alpha,\ A_\alpha \otimes_A E)$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α , I_α are both open in A, R respectively. Then to say that A is formally smooth as a topological R-algebra means that $D^1_{\mathrm{top}}(A|R,\ E) = 0$ for every A-module E .

Now let X be a scheme over a ring R and $\underline{E}$ a quasi-coherent X-module. We then define the sheaves $\underline{D}^i_{X|R}(\underline{E}) = \underline{D}^i(X|R,\ \underline{E})$ on X via presheaves on X given by

$$[\underline{D}^i_{X|R}(\underline{E})](U) = D^i(\Gamma(U,\ \underline{O}_X)|R,\ \Gamma(U,\ \underline{E}))$$

for every affine open set $U \subset X$. If R is noetherian and X is of finite type over R , then the property 3.13(c) shows that $\underline{D}^i_{X|R}(\underline{E})$ can actually be constructed canonically, as a quasi-coherent X-module.

In case when there is no possible confusion we shall abbreviate by setting $\underline{D}^i_{X|R} = \underline{D}^i_{X|R}(\underline{O}_X)$. All the statements of 3.13 can now be translated in terms of these quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$, which we leave to the readers. We list below a few properties which are most relevant to our purpose.

Corollary 3.15. Let R be noetherian and X an R-scheme of finite type. Then (a) For any coherent X-module $\underline{E}$, $\underline{D}^i_{X|R}(\underline{E})$ is again a coherent X-module for all i , and furthermore $\mathrm{Supp}\,\underline{D}^i_{X|R}(\underline{E}) \subset X^{\mathrm{Sing}}$ for all $i > 0$ where $X^{\mathrm{Sing}} = \{x \in X | X$ is non-smooth over R at the point $x \in X\}$. **193**

(b) If X is flat over R , then for any R-algebra S , we have the canonical isomorphism

$$\underline{D}^i_{X \underset{R}{\otimes} S\,|\,S}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for all i .

(c) If X is a relative complete intersection over R , then for every surjective ring homomorphism $S \to R$ we have the exact sequence

$$0 \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2 \underset{R}{\otimes} \underline{O}_X, \underline{E}) \to 0$$

where $I = \mathrm{Ker}(S \to R)$. If, furthermore, R is local with the residue field k and X is R-flat, then we have the exact sequence

$$0 \to \underline{D}^1_{X_0|k}(\underline{O}_{X_0}) \to \underline{D}^1_{X|S}(\underline{O}_{X_0}) \to D^1(R|S,\ k) \underset{k}{\otimes} \underline{O}_{X_0} \to 0$$

where $X_0 = X \underset{R}{\otimes} k$.

Proof: (a) Follows from 3.12(7) and the vanishing for smooth algebras (cf. 3.13(a)), whereas (b) follows from 3.12(3).

(c) Since X is a relative complete-intersection over R , we have $\underline{D}^i_{X|R}(\underline{E}) = 0$ for all $i \geq 2$ by the complete-intersection vanishing (cf. 3.13(b)) and hence

$$0 \to \underline{D}^0_{X|R}(\underline{E}) \to \underline{D}^0_{X|S}(\underline{E}) \to \underline{D}^0_{R|S}(\underline{E}) \to \underline{D}^1_{X|R}(\underline{E}) \to \underline{D}^1_{X|S}(\underline{E}) \to \underline{D}^1_{R|S}(\underline{E}) \to 0$$

is exact by 3.12(2), where $\underline{D}^i_{R|S}(\underline{E})$ are quasi-coherent sheaves of X defined by

$$[\underline{D}^i_{R|S}(\underline{E})](U) = D^i(R|S,\ \underline{E}(U))$$

for every affine open subset $U \subset X$. In particular, $\underline{D}^0_{R|S}(\underline{E}) = 0$ since $S \to R$ is surjective, and

$$\underline{D}^1_{R|S}(\underline{E}) = \underline{\mathrm{Hom}}_{\underline{O}_X}(I/I^2 \underset{R}{\otimes} \underline{O}_X, \underline{E})$$

by 3.12(8), and hence the first exact sequence. If X is R-flat, we have $\underline{D}^i_{X_0|k}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$ for all **194**
i and for every quasi-coherent X_0-module $\underline{E}$ and in particular

$$\underline{D}^1_{X_0|k}(\underline{O}_{X_0}) \xrightarrow{\sim} \underline{D}^1_{X|R}(\underline{O}_{X_0})$$

from which the second exact sequence follows.

Remark 3.16: The quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$ we have defined should appear as $E_2^{0,i}$ of a spectral sequence relating the local ones and the global ones. L. Illusie has obtained such a theory (Lecture Notes in Mathematics 239). His method is via "homotopical algebra" generalizing the methods of M. André and D. Quillen rather than "cohomological algebra" . It seems that the approach we have taken here can be globalized via Grothendieck cohomology on the category of schemes with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

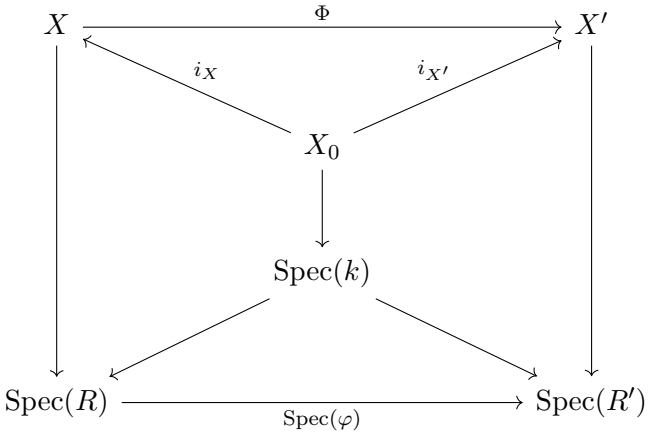
One amongst the nice cofibered categories in which the isomorphism classes of objects are usually not prorepresentable, but nevertheless admit a formal versal object is the cofibered category of deformations. We now come back to the situations and notations of paragraph 1. Thus, throughout the section, Λ is a fixed complete local noetherian ring with the residue field k , and $\underline{C}_\Lambda$ is the category of artinian local Λ-algebras with the same residue field k .

Let X_0 be a fixed scheme over the field k . By an (infinitesimal) deformation of X_0 , we mean a pair (R, X) where $R \in \mathrm{ob}(\underline{C}_\Lambda)$ and X is a flat R-scheme together with a commutative diagram

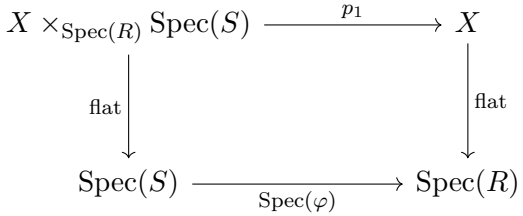
$$\begin{array}{ccc} X_0 & \xrightarrow{\quad i_X \quad} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that the induced morphism $X_0 \to X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism. We note that, since **195**
R is an artinian local ring, $i_X : X_0 \to X$ is necessarily a closed immersion which is a homeomorphism

on underlying topological spaces. If (R, X) , (R', X') are deformations of X_0 to R, R' respectively, we define an arrow $(R', X') \to (R, X)$ to consist of a map $R' \xrightarrow{\varphi} R$ in $\underline{C}_\Lambda$ and a morphism $\Phi : X \to X'$ of schemes with a commutative diagram



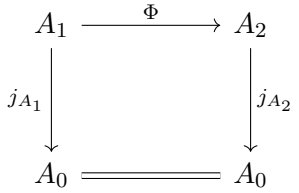
We thus obtain the category $\underline{M}_{X_0}$ of (infinitesimal) deformations of X_0 in which the objects and the arrows are defined as above. The category $\underline{M}_{X_0}$ is provided with a functor $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$ defined by $(R, X) \to R$. Now let $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$ and a map $R \xrightarrow{\varphi} S$ in $\underline{C}_\Lambda$ be given. We then obtain a commutative diagram



and φ-arrow $(\varphi, p_1) : (R, X) \to (S, X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S))$ is obviously cocartesian in $\underline{M}_{X_0}$. It follows that the category $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_\Lambda$ via $P : \underline{M}_{X_0} \to \underline{C}_\Lambda$. We observe the following facts:

4.1(a): Let $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ be an arrow in $\underline{M}_{X_0}$ i.e., $\Phi : X_2 \to X_1$ is a morphism **196** over R such that $X_2 \otimes_R k \to X_1 \otimes_R k$ is an isomorphism. Since R is artinian and X_i is R-flat, it follows immediately that Φ is an isomorphism so that $(1_R, \Phi) : (R, X_1) \to (R, X_2)$ is an isomorphism in $\underline{M}_{X_0}$. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b): Suppose that X_0 is affine, say $X_0 = \mathrm{Spec}(A_0)$. Then for any object $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, X is also affine. Consequently, for any $R \in \mathrm{ob}(\underline{C}_\Lambda)$, the category $\underline{M}_{X_0}(R)$ is canonically equivalent, via the sections, to the category in which an object is a flat algebra A over R together with an R-algebra map $j_A : A \to A_0$ inducing an isomorphism $A \otimes_R k \xrightarrow{\sim} A_0$, and a map $\Phi : A_1 \to A_2$ is an R-algebra map with a commutative diagram



4.1(c): Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, the category $\widehat{\underline{M}}_{X_0}(R)$ for any $R \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over the formal spectrum $\mathrm{Spf}(R)$, such that $\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\underline{m}^{n+1}$ is flat over $R/\underline{m}^{n+1}$ for every $n \geq 0$ (or what amounts to the same, if X_0 is noetherian, such that $\mathfrak{X}$ is flat over $\mathrm{Spf}(R)$) . Consequently,

if X_0 is noetherian, $\widehat{\underline{M}}_{X_0}$ is canonically $\widehat{\underline{C}}_\Lambda$-equivalent to the category of deformations of X_0 over the objects in $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d): Let $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$ i.e., we have a commutative diagram 197

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $X_0 \to X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism. Now let $U_0 \subset X_0$ be a fixed open subscheme of X_0 . Since $i_X : X_0 \to X$ is a closed immersion which is a homeomorphism on the underlying topological spaces, we obtain a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|U_0} & X|U_0 \\ \downarrow & & \downarrow \text{flat} \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $U_0 \to (X|U_0) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k) = X \otimes_R k|U_0$ is an isomorphism, so that $(R, X|U_0)$ is a deformation of U_0 to R . Furthermore, for any arrow $(\varphi, \Phi) : (R, X) \to (R', X')$ in $\underline{M}_{X_0}$, $(\varphi, \Phi|U_0) : (R, X|U_0) \to (R', X'|U_0)$ is an arrow in $\underline{M}_{U_0}$. Consequently we have the restriction functor $\underline{M}_{X_0} \to \underline{M}_{U_0}$ over $\underline{C}_\Lambda$. One may also consider a localization as well as a formalization at a point. Namely let x_0 be a fixed point in X_0 . If $(R, X) \in \mathrm{ob}(\underline{M}_{X_0})$, then $\mathrm{Spec}(\underline{O}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\underline{O}_{X_0,x_0})$ to R and $\mathrm{Spec}(\hat{\underline{O}}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})$ over R , where $\hat{}$ stands for the completion with respect to the linear topology given by the powers of the maximal ideal. We thus obtain functors $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,x_0})} \to \underline{M}_{\mathrm{Spec}(\hat{\underline{O}}_{X_0,x_0})}$.

Firstly we establish that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow[\text{surjective}]{} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E, E' be flat modules over R, R' respectively. Then for any R'-homomorphism $E' \to \bar{E} = E \otimes_R \bar{R}$, the fibre-product $E \times_{\bar{E}} E'$ is flat as $R \times_{\bar{R}} R'$-module

Proof: Let $0 \to I \to R \to \bar{R} \to 0$ be exact so that $0 \to I \to R \times_{\bar{R}} R' \to R' \to 0$ is exact. Since E 198 is R-flat, $0 \to I \otimes_R E \to E \to \bar{E} \to 0$ is exact and hence we obtain the exact sequence

$$(*) \qquad 0 \to I \otimes_R E \to E \times_{\bar{E}} E' \to E' \to 0 \ .$$

Set $S = R \times_{\bar{R}} R'$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S-module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R'-module and $\mathrm{Tor}_1^S(E \times_{\bar{E}} E', R') = 0$. However, we note that $I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E')$ and $E' = R' \otimes_S (E \times_{\bar{E}} E')$ and therefore the exact sequence (*) can be rewritten as

$$0 \ \to \ I \otimes_S (E \times_{\bar{E}} E') \ \to \ E \times_{\bar{E}} E' \ \to \ R' \otimes_S (E \times_{\bar{E}} E') \ \to \ 0 \ .$$

This means that $\mathrm{Tor}_1^S(R', E \times_{\bar{E}} E') = 0$ and $R' \otimes_S (E \times_{\bar{E}} E') = E'$ is flat as R'-module, and therefore $E \times_{\bar{E}} E'$ is S-flat.

Corollary 4.3. Given a diagram

$$\begin{array}{ccc} (R,\ X) & & (R',\ X') \\ & {\scriptstyle (\varphi,\ \Phi)} \searrow \quad \swarrow {\scriptstyle (\varphi',\ \Phi')} & \\ & (\bar{R},\ \bar{X}) & \end{array}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \underset{\bar{X}}{\amalg} X'$ (in the sense of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided $\varphi : R \to \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof: For each affine open $U_0 \subset X_0$, we have by definition that $X \underset{\bar{X}}{\amalg} X'|U_0 \simeq \mathrm{Spec}(\Gamma(U_0,\ X) \times_{\Gamma(U_0,\bar{X})} \Gamma(U_0,\ X'))$. Consequently, if $\varphi : R \to \bar{R}$ is surjective, $X \underset{\bar{X}}{\amalg} X'$ is a flat scheme over $R \times_{\bar{R}} R'$ by 4.2, and the morphism $i_X \underset{i_{\bar{X}}}{\amalg} i_{X'} : X_0 \to X \underset{\bar{X}}{\amalg} X'$ induces an isomorphism $X_0 \to (X \underset{\bar{X}}{\amalg} X') \otimes_S k$, where $S = R \times_{\bar{R}} R'$. Therefore $(R \times_{\bar{R}} R',\ X \underset{\bar{X}}{\amalg} X') \in \mathrm{ob}(\underline{M}_{X_0})$ and we obviously have that $(R \times_{\bar{R}} R',\ X \underset{\bar{X}}{\amalg} X') = (R,\ X) \times_{(\bar{R},\bar{X})} (R',\ X')$ in the category $\underline{M}_{X_0}$, which means that $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$. I 99

Lemma 4.4. For any scheme X_0 over k , we have a canonical exact sequence

$$0 \to H^1(X_0,\ \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\varepsilon])] \to H^0(X_0,\ \underline{D}^1_{X_0})$$

where $\underline{D}^i_{X_0}$ are as defined in paragraph 3.

Proof: (Special case) Assume that X_0 is affine and let $\Gamma(X_0,\ \underline{O}_{X_0}) = A_0$. Then as it was noted in 4.1(b), the category $\underline{M}_{X_0}(k[\varepsilon])$ is canonically equivalent to the category $\underline{E}$, where

$$\mathrm{ob}(\underline{E}) : \text{object in } \underline{E} \text{ is a flat algebra } A \text{ over } k[\varepsilon]$$

together with a $k[\varepsilon]$ -algebra map $j_A : A \to A_0$ inducing the isomorphism $k \otimes_{k[\varepsilon]} j_A : k \otimes_{k[\varepsilon]} A \xrightarrow{\sim} A_0$. Alternatively, an object in $\underline{E}$ is a k-algebra A together with an exact sequence $0 \to A_0 \xrightarrow{\varepsilon} A \xrightarrow{j_A} A_0 \to 0$ in which $\varepsilon(A_0)^2 = 0$ as an ideal in A .

F$\ell(\underline{E})$: An arrow $A_1 \to A_2$ in $\underline{E}$ with respect to first description of the object in $\underline{E}$ is a $k[\varepsilon]$-algebra map $\Phi : A_1 \to A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. Therefore, an arrow $A_1 \to A_2$ in $\underline{E}$ with respect to the alternative description of objects in $\underline{E}$ is a k-algebra map $\Phi : A_1 \to A_2$ yielding a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\varepsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 & \longrightarrow & 0 \\ & & \| & & \downarrow{\scriptstyle \Phi} & & \| & & \\ 0 & \longrightarrow & A_0 & \longrightarrow & A_2 & \xrightarrow{j_{A_2}} & A_0 & \longrightarrow & 0 \end{array} .$$

It follows that $[\underline{M}_{X_0}(k[\varepsilon])]$ is canonically isomorphic, via $\Gamma(X_0,\ -)$, to the set of isomorphism classes of k-algebra extensions of A_0 by A_0 . In other words, we have a canonical bijection $[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(X_0,\ \underline{D}^1_{X_0})$ via $\Gamma(X_0,\ -)$, which is trivially seen to respect the vector space structures on both sides.

(General case) For each affine open $U_0 \subset X_0$, we get a canonical map $[\underline{M}_{X_0}(k[\varepsilon])] \to H^0(U_0,\ \underline{D}^1_{X_0})$ I 100 defined as the composite map $[\underline{M}_{X_0}(k[\varepsilon])] \to [\underline{M}_{U_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(U_0,\ \underline{D}^1_{X_0})$ and hence we obtain a canonical map $[\underline{M}_{X_0}(k[\varepsilon])] \to H^0(X_0,\ \underline{D}^1_{X_0})$. Its kernel is the set of isomorphism classes of locally-trivial deformations of X_0 to $k[\varepsilon]$, i.e., the set of isomorphism classes of deformations of X_0 to $k[\varepsilon]$, locally (on X) isomorphic to $X_0 \otimes_k k[\varepsilon]$. Since $\underline{D}^0_{X_0} = \underline{\mathrm{Der}}_k(\underline{O}_{X_0},\ \underline{O}_{X_0}) = \underline{\mathrm{Aut}}(X_0 \otimes_k k[\varepsilon]|X_0) =$ the sheaf of germs of automorphisms of $X_0 \otimes_k k[\varepsilon]$ inducing the identity on the subscheme X_0 , it is canonically isomorphic to $H^1(X_0,\ \underline{D}^0_{X_0})$, so that we obtain the desired exact sequence.

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either (i) X_0 is proper over k or (ii) X_0 is affine but $X_0^{\mathrm{sing}} = \{x \in X_0 | X_0$ is non-smooth over k at the point $x\}$ is a finite set. Then

(a) $\underline{M}_{X_0}$ admits a formal versal object (called a formal versal deformation of X_0) over $\underline{C}_\Lambda$. If $(R,\ \mathfrak{X})$ is a formal versal deformation of X_0 , then for any $(S,\ \mathfrak{Y}) \in \mathrm{ob}(\widehat{\underline{M}}_{X_0})$, there is a map $(\varphi, \Phi) : (R, \mathfrak{X}) \to (S, \mathfrak{Y})$ in $\widehat{\underline{M}}_{X_0}$ such that the diagram below is cartesian

$$\begin{array}{ccc} \mathfrak{Y} & \xrightarrow{\Phi} & \mathfrak{X} \\ \downarrow & & \downarrow \\ \mathrm{Spf}(S) & \xrightarrow[\mathrm{Spf}(\varphi)]{} & \mathrm{Spf}(R) \end{array} \quad .$$

Furthermore a formal versal deformation $(R, \mathfrak{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \to R$ in $\underline{C}_\Lambda$ and a deformation $(R',\ X')$ of X_0 to R' , the homomorphism

$$\mathrm{Aut}_{R'}(X'|X_0) \to \mathrm{Aut}_R(X' \underset{R'}{\otimes} R|X_0)$$

is surjective.

(b) If X_0 is proper over k , then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$. I 101

Proof: (a) Since $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$ by 4.3, it suffices to see that $[\underline{M}_{X_0}(k[\epsilon])]$ is a finite-dimensional vector space over k by 1.11 and 2.9(b).

Now consider the exact sequence

$$(*) \qquad 0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to H^0(X_0, \underline{D}^1_{X_0}) \ .$$

If X_0 is proper over k , then $H^1(X_0, \underline{D}^0_{X_0})$ and $H^0(X_0, \underline{D}^1_{X_0})$ are both finite dimensional since $\underline{D}^i_{X_0}$ are coherent for all $i \geq 0$ by 3.15(a). Now assume that X_0 is affine such that X_0^{sing} is a finite set. Since $\underline{D}^1_{X_0}$ is a coherent sheaf on X_0 such that $\mathrm{Supp}\ \underline{D}^1_{X_0} \subset X_0^{\mathrm{sing}}$ by 3.15(a), we find that $H^0(X_0, \underline{D}^1_{X_0})$ is a finite-dimensional vector space over k , whereas $H^1(X_0, \underline{D}^0_{X_0}) = 0$ since X_0 is affine. Therefore, in either case, it follows from the exact sequence (*) that $[\underline{M}_{X_0}(k[\epsilon])]$ is a finite dimensional vector space over k . The second statement of (a) follows from 2.7.

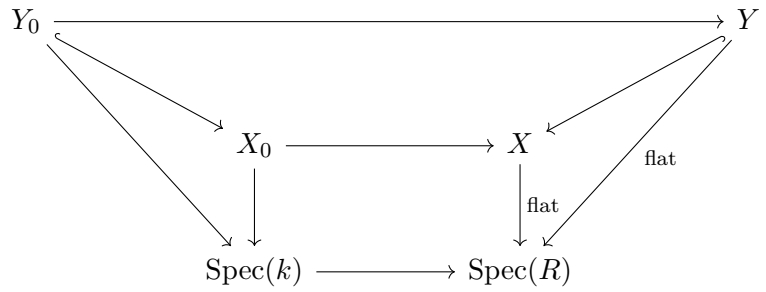
(b) If X_0 is proper, $\mathrm{Aut}\ (X_0 \otimes_k k[\epsilon]|X_0) = H^0(X_0, \underline{D}^0_{X_0})$ is a finite dimensional vector space and hence our statement follows from 2.13.

Remark 4.6(a): Let X_0 be affine. Then a formal versal deformation of X_0 exists if and only if $\underline{D}^1_{X_0}$ has its support on a finite number of points. However this does not imply that non-smooth points of X_0 are all isolated. There are affine schemes X_0 over k having a non-isolated non-smooth point but supp $\underline{D}^1_{X_0}$ is a finite set.

Remark 4.6(b): Henceforth a deformation of X_0 is meant by an object in $\underline{M}_{X_0}$. In case when there is a possible confusion, an object in $\underline{M}_{X_0}$ will be referred as an infinitesimal deformation of X_0 .

Remark 4.6(c): Let X_0 be a scheme over k . We have fixed a complete local noetherian ring Λ with the residue field k , and considered the deformations of X_0 over the category $\underline{C}_\Lambda$. (In practice I 102 one usually takes $\Lambda = k$ or $W(k)$ if k is a perfect field of positive characteristic). Now $\underline{C}_k \subset \underline{C}_\Lambda$ is a fully faithful imbedding, and that $\underline{M}_{X_0}|\underline{C}_k$ is simply the category of deformations of X_0 over S where $S \in \mathrm{ob}(\underline{C}_k)$. Therefore if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then $(k \otimes_\Lambda R, k \otimes_\Lambda \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that $\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim k \otimes_\Lambda R$ does not depend on the choice of Λ but depends only on X_0 .

One can generalize the above theorem in many ways. Let us for instance consider a pair $Y_0 \to X_0$ where X_0 is a scheme of finite type over k and Y_0 a fixed closed subscheme of X_0 . For each $R \in \mathrm{ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) to R is meant to be a commutative diagram



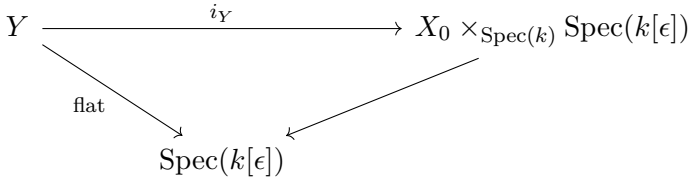
such that $(X_0, Y_0) \to (X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k), Y \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k))$ is an isomorphism. Defining a morphism $(R'; X', Y') \to (R; X, Y)$ of deformations of (X_0, Y_0) to be the obvious ones, we obtain the cofibered category $\underline{M}_{X_0,Y_0}$ in groupoids over $\underline{C}_\Lambda$. (This is a process often used in rigidification of automorphisms in moduli problems of deformations.) One sees immediately, by the same argument as in 4.3 based on 4.2, that $\underline{M}_{X_0,Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. $0 \to H^0(Y_0, \underline{\mathrm{Hom}}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) \to [\underline{M}_{X_0,Y_0}(k[\epsilon])] \to [\underline{M}_{X_0}(k[\epsilon])]$ is exact, where $\underline{I}$ is the ideal sheaf of $\underline{O}_{X_0}$ defining Y_0 i.e.

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$$0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0 \qquad \text{is exact,}$$

Proof: Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0,Y_0}(k[\epsilon])$ whose objects are of the form $(k[\epsilon]; X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\epsilon]), Y)$. We then have $[B] = \mathrm{Ker}\,([\underline{M}_{X_0,Y_0}(k[\epsilon])] \to [\underline{M}_{X_0}(k[\epsilon])])$. The category $\underline{B}$ is the groupoid of deformations of Y_0 to $k[\epsilon]$ "inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\epsilon])$ ", i.e., an object in $\underline{B}$ is a commutative diagram



such that $\mathrm{Spec}(k) \times_{\mathrm{Spec}(k[\epsilon])} i_Y = i_{Y_0}$, and an arrow $(Y_1, i_{Y_1}) \to (Y_2, i_{Y_2})$ is a $k[\epsilon]$-morphism $\Phi : Y_1 \to Y_2$ (necessarily an isomorphism) such that $\mathrm{Spec}(k) \times_{\mathrm{Spec}(k[\epsilon])} \Phi = \mathrm{id}_{Y_0}$ and $i_{Y_2} \circ \Phi = i_{Y_1}$. Alternatively, an object in $\underline{B}$ is a pair (Y, j_Y) , where Y is a deformation of Y_0 to $k[\epsilon]$ and $j_Y : Y \to X_0$ is a morphism over k such that the composite morphism $Y_0 \to Y \xrightarrow{j_Y} X_0$ is i_{Y_0} , and an arrow $(Y_1, j_{Y_1}) \to (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \to Y_2$ such that $k \otimes_{k[\epsilon]} \Phi = \mathrm{id}_{Y_0}$ and $j_{Y_2} \circ \Phi = j_{Y_1}$. Therefore the association

$$(Y, j_Y) \longmapsto \begin{array}{ccccccccc} 0 & \longrightarrow & \underline{I} & \longrightarrow & \underline{O}_{X_0} & \longrightarrow & \underline{O}_{Y_0} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow{\scriptstyle j_Y} & & \| & & \\ 0 & \longrightarrow & \underline{O}_{Y_0} & \xrightarrow{\epsilon} & \underline{O}_Y & \longrightarrow & \underline{O}_{Y_0} & \longrightarrow & 0 \end{array}$$

gives rise to an isomorphism

$$[B] \xrightarrow{j|\underline{I}} \mathrm{Hom}_{\underline{O}_{X_0}}(\underline{I}, \underline{O}_{Y_0}) = \mathrm{Hom}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0}) = H^0(Y_0, \underline{\mathrm{Hom}}_{\underline{O}_{Y_0}}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) .$$

Remark 4.8. For a more general formula implying the above lemma, see SGA3 III 4.4. One may also note the following fact. One can define for any S-scheme X and a quasi-coherent X-module $\underline{E}$ a quasi-coherent X-module $\underline{D}^i_{X|S}(\underline{E})$ for all $i \geq 0$, generalizing the case when S is affine. As usual we simply write $\underline{D}^i_{X|S}(\underline{O}_X) = \underline{D}^i_{X|S}$. We then have I 104

$$H^0(Y_0, \underline{\mathrm{Hom}}_{Y_0}(\underline{I}/\underline{I}^2, \underline{O}_{Y_0})) = H^0(Y_0, \underline{D}^1_{Y_0|X_0})$$

where $0 \to \underline{I} \to \underline{O}_{X_0} \to \underline{O}_{Y_0} \to 0$ is exact.

Theorem 4.9. Let X_0 be k-scheme of finite type and Y_0 a closed subscheme of X_0 . Assume that either (i) X_0 is proper over k or (ii) X_0 is affine and smooth over k outside a finite number of points and Y_0 is proper over k . We then have

(a) $\underline{M}_{X_0,\,Y_0}$ admits a formal versal deformation over $\underline{C}_\wedge$, and it pro-represents the functor $[\underline{M}_{X_0,\,Y_0}]$ if and only if, for every surjective map $R' \to R$ in $\underline{C}_\wedge$ and a deformation $(R'; X', Y')$,

$$\mathrm{Aut}_{R'}((X', Y')|X_0) \to \mathrm{Aut}_R((X' \otimes_{R'} R,\, Y' \otimes_{R'} R)|X_0)$$

is surjective.

(b) If furthermore $H^0(X_0,\, \underline{D}^0_{Y_0|k} \times_{\underline{D}^0_{X_0|k}(\underline{O}_{Y_0})} \underline{D}^0_{X_0|k})$ is finite dimensional, then $\underline{M}_{X_0,\,Y_0}$ is smoothly pseudo-representable.

Proof: It is obvious from 4.4 and 4.6 together with 1.11, 2.7, and 2.13.

We now consider the passage to localization and completion in the deformation theory of schemes. First, some notation: Let X_0 be a k-scheme of finite type. Given a fixed point $x_0 \in X_0$, we have natural functors

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,\,x_0})} \to \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,\,x_0})}$$

and in turn we obtain the functors I 105

$$\widehat{\underline{M}}_{X_0} \to \widehat{\underline{M}}_{\mathrm{Spec}(\underline{O}_{X_0,\ x_0})} \to \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,\ x_0})} \ .$$

As was remarked in 4.1(c), $\widehat{\underline{M}}_{X_0}$ is canonically equivalent to the category in which an object is a pair $(R,\ \mathfrak{X})$ where $R \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda)$ and $\mathfrak{X}$ is a flat R-adic formal scheme together with a fixed isomorphism $\mathfrak{X}_0 \longleftrightarrow X_0$. Henceforth we shall denote, when there is no possible confusion, the images of $(R,\ \mathfrak{X})$ under $\widehat{\underline{M}}_{X_0} \to \widehat{\underline{M}}_{\mathrm{Spec}(\underline{O}_{X_0,\ x_0})}$ and $\widehat{\underline{M}}_{X_0} \to \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,\ x_0})}$ by $(R,\ \mathfrak{X}_{(x_0)})$ and $(R,\ \mathfrak{X}_{(\hat{x}_0)})$ respectively. Therefore if $\mathfrak{X} = \varinjlim_\nu \mathfrak{X}_\nu$ where $\mathfrak{X}_\nu = \mathfrak{X} \underset{\mathrm{Spf}(R)}{\times} \mathrm{Spec}(R/\underline{m}^{\nu+1})$ then

$$\mathfrak{X}_{(x_0)} = \varinjlim_\nu \mathrm{Spec}(\underline{O}_{\mathfrak{X}_\nu,\ x_0})$$

$$\mathfrak{X}_{(\hat{x}_0)} = \varinjlim_\nu \mathrm{Spec}(\widehat{\underline{O}}_{\mathfrak{X}_\nu,\ x_0}) \ .$$

Theorem 4.10. Let X_0 be an affine scheme of finite type over k .

(a) If X_0 is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.

(b) Assume that, for a fixed closed point $x \in X_0$, $X_0 - \{x\}$ is smooth over k . Then

$$\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,\ x})} \to \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0,\ x})}$$

are both minimally smooth functors. In particular, if $(R,\ \mathfrak{X})$ is a formal versal deformation of X_0 , then $(R,\ \mathfrak{X}_{(x)})$ and $(R,\ \mathfrak{X}_{(\hat{x})})$ are formal versal deformations of $\mathrm{Spec}(\underline{O}_{X_0,\ x})$ and $\mathrm{Spec}(\widehat{\underline{O}}_{X_0,\ x})$ respectively.

Proof: In this proof we go back to the notations of the cohomology of commutative algebras defined in paragraph 3. We set $X_0 = \mathrm{Spec}(A_0)$

(a) It suffices to show that $P : [\underline{M}_{X_0}] \to \underline{C}_\Lambda$ is smooth. I 106

Let $(R,\ \mathrm{Spec}(A))$ be a deformation of $\mathrm{Spec}(A_0)$ to R , where $R \in \mathrm{ob}(\underline{C}_\Lambda)$. Given a small extension $R' \to R$ in $\underline{C}_\Lambda$ (i.e., $0 \to k \to R' \to R \to 0$ is exact), we must show that $(R,\ \mathrm{Spec}(A))$ can be lifted to R' . The exact sequence for D^i with coefficients in A_0 applied to the ring homomorphisms $R' \to R \to A$ gives us the exact sequence

$$\ldots \to D^1(A|R,\ A_0) \to D^1(A|R',\ A_0) \to D^1(R|R',\ k) \underset{k}{\otimes} A_0 \to D^2(A|R,\ A_0) \to \ldots$$

Since $R' \to R$ is a small extension, $D^1(R|R',\ k)$ is a 1-dimensional vector space over k generated by the element (denoted by $[R']$) represented by the extension $0 \to k \to R' \to R \to 0$. On the other hand, A is R-flat and hence $D^i(A|R,\ A_0) = D^i(A_0|k, A_0)$ for all $i > 0$ and in particular

$D^2(A|R,\ A_0) = 0$ by 3.12(8) since A_0 is locally a complete-intersection. Consequently we get the exact sequence

$$\to D^1(A_0|k,\ A_0) \to D^1(A|R',\ A_0) \to D^1(R|R',\ k) \underset{k}{\otimes} A_0 \to 0$$

This means that there exists an R'-algebra extension of A by A_0 , denoted by A' , such that $[A'] \to [R'] \otimes 1$ under the map $D^1(A|R',\ A_0) \to D^1(R|R',\ k) \underset{k}{\otimes} A_0$, which means that the deformation $(R,\ \mathrm{Spec}(A))$ can be lifted to a deformation $(R',\ \mathrm{Spec}(A'))$.

(b) By 3.14 we have the isomorphisms

$$D^1(A_0|k,\ A_0) \simeq D^1((A_0)_x|k,\ (A_0)_x) \simeq D^1(\widehat{(A_0)_x}|k,\ \widehat{(A_0)_x})$$

i.e., $[\underline{M}_{X_0}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,\ x})}(k[\epsilon])] \simeq [\underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,\ x})}(k[\epsilon])]$.

It suffices to show, by 2.7(a), that

$$[\underline{M}_{X_0}] \to [\underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,\ x})}] \to [\underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,\ x})}]$$

are both smooth functors. Let $(R,\ \mathrm{Spec}(A))$ be a deformation of $X_0 = \mathrm{Spec}(A_0)$, and let $0 \to k \to R' \to R \to 0$ be a small extension in $\underline{C}_\Lambda$. As above, the ring homomorphisms $R' \to R \to A$ give us the exact commutative diagram I 107

$$\begin{array}{ccccccccc}
\longrightarrow & D^1(A_0|k,A_0) & \longrightarrow & D^1(A|R',A_0) & \xrightarrow{p} & D^1(R|R',k) \underset{k}{\otimes} A_0 & \longrightarrow & D^2(A_0|k,A_0) & \longrightarrow \dots \\
 & \downarrow\approx & & \downarrow & & \downarrow & & \downarrow\approx & \\
\longrightarrow & D^1(A_0|k,(A_0)_x) & \longrightarrow & D^1(A|R',(A_0)_x) & \longrightarrow & D^1(R|R',k) \underset{k}{\otimes} (A_0)_x & \longrightarrow & D^2(A_0|k,(A_0)_x) & \longrightarrow \dots \\
 & \uparrow & & \uparrow & & \uparrow & & \uparrow & \\
\longrightarrow & D^1((A_0)_x|k,(A_0)_x) & \longrightarrow & D^1(A_x|R',(A_0)_x) & \xrightarrow{p_x} & D^1(R|R',k) \underset{k}{\otimes} (A_0)_x & \longrightarrow & D^2((A_0)_x|k,(A_0)_x) & \longrightarrow \dots
\end{array}$$

where the indicated isomorphisms come from the fact that A_0 is smooth over k except at one closed point x so that

$$D^i(A_0|k,\ (A_0)_x) = D^i(A_0|k,\ A_0)_x = D^i(A_0|k,\ A_0)$$

for all $i > 0$. Now suppose that we are given an arrow $(R',\ \mathrm{Spec}(B)) \to (R,\ \mathrm{Spec}(A_x))$ in $[\underline{M}_{\mathrm{Spec}((A_0)_x)}]$. It means that we are given a class $[B] \in D^1(A_x|R',\ (A_0)_x)$ such that $p_x([B]) = [R']$. Then a diagram chase shows that there exists $[A'] \in D^1(A|R',\ A_0)$ such that $p([A']) = [R']$ and $A'_x = B$ i.e., $(R',\ \mathrm{Spec}(A'))$ is a deformation of $\mathrm{Spec}(A_0)$ to R' such that $\mathrm{Spec}(A')_x = \mathrm{Spec}(B)$. This proves that $\underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{O}_{X_0,\ x})}$ is smooth. The smoothness of the functor $\underline{M}_{\mathrm{Spec}((A_0)_x)} \to \underline{M}_{\mathrm{Spec}(\widehat{(A_0)_x})}$ comes from the similar diagram together with the fact that

$$D^i((A_0)_x|k,\ \widehat{(A_0)_x}) = D^i(\widehat{(A_0)_x}|k,\ (A_0)_x) = D^i((A_0)_x|k,\ (A_0)_x)$$

for all $i > 0$ since x is the only possible non-smooth point.

Theorem 4.11. Let X_0 be a (separated) k-scheme of finite type, smooth outside a finite closed set $X_0^{\mathrm{Sing}} = \{x_1, \dots, x_r\}$ of X_0 . Let $U_0^1, \dots, U_0^r$ be affine open neighbourhoods of $x_1, \dots, x_r$ such that $U_0^i \cap X_0^{\mathrm{Sing}} = \{x_i\}$. If $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$, then the natural functor $\underline{M}_{X_0} \to \underline{M}_{U_0^1} \times \cdots \times \underline{M}_{U_0^r}$ is smooth.

Proof: We can complete $U_0^1, \dots, U_0^r$ into an affine open covering of X_0 by choosing $U_0^{r+1}, \dots, U_0^N$ such that $U_0^j \cap X_0^{\mathrm{Sing}} = \emptyset$ for all $j > r$. Now let $R' \to R$ be a small extension in $\underline{C}_\Lambda$, and let $(R',\ U'^1), \dots, (R',\ U'^r)$ be deformations of $U_0^1, \dots, U_0^r$ to R' such that $U'^i \underset{R'}{\otimes} R = X|U_0^i$, $1 \le i \le r$, for some deformation $(R,\ X)$ of X_0 to R . We must show that there exists a deformation $(R',\ X')$ I 108

of X_0 to R' such that $X' \underset{R'}{\otimes} R = X$ and $X'|U_0^i = U'^i$ for $1 \leq i \leq r$. Since U_0^j for $j > r$ is smooth over k , $X|U_0^j$ can be lifted, by 4.9(a), to a deformation $(R',\ U'^j)$ i.e., there exist deformations $(R',\ U'^j)$ of U_0^j to R' such that $U'^j \underset{R'}{\otimes} R = X|U_0^j$ for all $j > r$. Since, for any pair $i \neq j$, $U_0^i \cap U_0^j$ is affine and smooth over k , we obtain an isomorphism

$$\tau_{ij} : U'^i|U_0^i \cap U_0^j \xrightarrow{\sim} U'^j|U_0^i \cap U_0^j$$

such that $\tau_{ij} \underset{R'}{\otimes} R = \mathrm{id}_{X|U_0^i \cap U_0^j}$, and in turn defines a cohomology class in $H^2(X_0,\ \underline{D}^0_{X_0})$ given by $U_0^i \cap U_0^j \cap U_0^k \to \tau_{ij}\tau_{jk}\tau_{ik}^{-1}$. Since $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$ by hypothesis, the obstruction for glueing these to obtain a global deformation vanishes, and hence we get a deformation $(R',\ X')$ of X_0 to R' such that $X' \underset{R'}{\otimes} R = X$ and $X'|U_0^j \simeq U'^j$. This completes our proof.

Remark 4.12. The previous proof assumes X_0 to be separated, which is unnecessary. In fact, for a fixed deformation $(R,\ X)$ of X_0 and deformations $(R',\ U'^j)$ of U_0^j , the deformations $(R',\ V')$ of open set V of X to R' compatible with the deformation $(R',\ U'^j)$ already given, form a "gerbe" (in Giraud's terminology) whose structure sheaf is $\underline{D}^0_{X_0}$. Hence the obstruction to the existence of a section of this "gerbe" (i.e., a global deformation $(R',\ X')$ of X compatible with the given datum $(R',\ U'^j)$) is in $H^2(X_0,\ \underline{D}^0_{X_0})$ which is zero by assumption.

Corollary 4.13. Let X_0 be a k-scheme of finite type, smooth outside a finite subset $X_0^{\mathrm{Sing}} = \{x_1, \ldots, x_r\}$ of X_0 such that $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$. Then

$$\text{(a)} \quad \underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_1})} \times \cdots \times \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_r})} \quad \text{is smooth.}$$

$$\text{In particular,} \quad \dim \underline{M}_{X_0} = \dim H^1(X_0,\ \underline{D}^0_{X_0}) + \sum_{c=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_c})} \ .$$

(b) If furthermore X_0 is locally a complete-intersection, then $\underline{M}_{X_0}$ is smooth.

Proof: I 109

(a) The first statement follows from 4.10 and 4.9(b). Now let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), ..., (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and $\mathrm{Spec}(\underline{O}_{X_0,x_1}), ..., \mathrm{Spec}(\underline{O}_{X_0,x_r})$ respectively. We then obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1 \widehat{\otimes}_\Lambda \ldots \widehat{\otimes}_\Lambda R_r} \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_r})} \end{array}$$

Since the composite functor

$$\underline{h}_R \to \underline{M}_{X_0} \to \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_r})}$$

is smooth whereas

$$\underline{h}_{R_1 \widehat{\otimes}_\Lambda \ldots \widehat{\otimes}_\Lambda R_r} \to \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_1})} \times \ldots \times \underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_r})}$$

is minimally smooth, it follows from 2.7(a) that

$\underline{h}_R \to \underline{h}_{R_1 \widehat{\otimes}_\Lambda \ldots \widehat{\otimes}_\Lambda R_r}$ is smooth i.e., R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \ldots \widehat{\otimes}_\Lambda R_r$. Since

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to [\underline{M}_{X_0}(k[\epsilon])] \to [\underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_1})}(k[\epsilon])] \times \ldots \times [\underline{M}_{\mathrm{Spec}(\underline{\widehat{O}}_{X_0,x_r})}(k[\epsilon])]$$

is exact by 4.4 i.e.,

$$0 \to H^1(X_0, \underline{D}^0_{X_0}) \to \mathrm{Hom}_{\Lambda-\mathrm{alg}}(R, k[\epsilon]) \to \mathrm{Hom}_{\Lambda-\mathrm{alg}}(R_1 \widehat{\otimes}_\Lambda ... \widehat{\otimes}_\Lambda R_r, k[\epsilon])$$

is exact, it follows that R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda ... \widehat{\otimes}_\Lambda R_r$ in d-variables where $d = \dim_k H^1(X_0, \underline{D}^0_{X_0})$.

Therefore

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim k \otimes_\Lambda R = d + \dim k \otimes_\Lambda (R_1 \widehat{\otimes}_\Lambda ... \widehat{\otimes}_\Lambda R_r) \\ &= d + \dim (k \otimes_\Lambda R_1) \widehat{\otimes}_k ... \widehat{\otimes}_k (k \otimes_\Lambda R_r) \\ &= d + \sum_{i=1}^{r} \dim k \otimes_\Lambda R_i = d + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\underline{O}}_{X_0, x_i})} \end{aligned}$$

(b) follows from 4.10 and 4.9(a). I 110

4.14. We close this paragraph with an explicit construction of a formal versal deformation in a simple situation. In fact, if X_0 is affine and is a complete-intersection, it is trivial to write down explicitly. Thus let

$$A_0 = k[x_1, ..., x_n]/(f_1, ..., f_r) = S/J$$

where $S = k[x_1, ..., x_n]$, $J = (f_1, ..., f_r)$, and $f_1, ..., f_r$ is an S-regular sequence. We then obtain the exact sequence

$$... \to \mathrm{Der}_k(S, A_0) \to \mathrm{Hom}_{A_0}(J/J^2, A_0) \to D^1(A_0|k, A_0) \to 0$$

We note that J/J^2 is A_0-free module of rank r generated by $f_1 (\mathrm{mod}\ J^2), ..., f_r (\mathrm{mod}\ J^2)$, since $f_1, ..., f_r$ is an S-regular sequence. We now assume that $D^1(A_0|k, A_0)$ is a finite-dimensional vector space over k , say A_0 has isolated non-smooth points only. Let $d = \dim_k D^1(A_0|k, A_0)$ and choose $p_j : J/J^2 \to A_0$ $(j = 1, 2, ..., d)$ which form a k-basis of $D^1(A_0|k, A_0)$. Now choose $P_{jk} \in \Lambda[x_1, ..., x_n]$ such that $p_j(f_i) = P_{ji} \ (\mathrm{mod}\ \underline{m}_\Lambda + J)$ where $\underline{m}_\Lambda$ is the maximal ideal of Λ . We then consider

$$\begin{aligned} R &= \Lambda[[t_1, ..., t_d]] \\ B &= R[x_1, ..., x_n]/(G_1, ..., G_r) \end{aligned}$$

where $G_j = F_j - \sum_{i=1}^{d} t_i P_{ij}$ and F_j is a polynomial with coefficients in Λ gotten from f_j by lifting the coefficients from k to Λ . Since $R[x_1, ..., x_n]$ is R-flat and $G_1, ..., G_r$ modulo $\underline{m}_R$ (which becomes $f_1, ..., f_r$) is a regular sequence in $k \otimes_R R[x_1, ..., x_n]$, it follows that B is R-flat, and $k \otimes_R B = A_0$. Furthermore,

$$\mathrm{Hom}_{\Lambda-\mathrm{alg}}(R, k[\epsilon]) \to D^1(A_0|k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\epsilon])]$$

is bijective by our construction of B . Therefore, if we set

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$$\mathrm{Spf}(B) \ = \ \varinjlim_{v} \ \mathrm{Spec}(R/\underline{m}^{v+1} \underset{R}{\otimes} B)$$

then $(R,\ \mathrm{Spf}(B))$ is an object in $\widehat{\underline{M}}_{\mathrm{Spec}(A_0)}$ and the functor $\underline{h}_R \to \underline{M}_{\mathrm{Spec}(A_0)}$ induced by $(R,\ \mathrm{Spf}(B))$ is smooth by 1.6, since R is a formal power-series ring over Λ . In other words, $(R,\ \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

We note, by 4.10(b), that if we set

$$B = R[[X_1, \ldots, X_n]]/(G_1, \ldots, G_r) \ \ ,$$

then $(R,\ \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ where $A_0 = k[[X_1, \ldots, X_n]]/(f_1, \ldots, f_r)$.

Remark 4.15. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R,\ \mathfrak{X})$ be a formal versal deformation of X_0 . The above construction shows that if X_0 is affine and

locally a complete-intersection, then $\mathfrak{X}$ is a formalization of an R-scheme of finite type. This fact can easily be generalized to the case when X_0 is not necessarily affine but $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$.

Remark 4.16. The global theory of the relative cotangent complex (the latter theory being reduced to the truncated complex of length 1 , which is sufficient however for many purposes of deformation theory) allows to generalize and simplify certain results, such as 4.10. Namely let S be a scheme, $S_0 \subset S' \subset S$ two subschemes of S defined by ideals $J \supset I$ such that $JI = 0$, X' a scheme flat over S' , $L. = L^{X_0/S_0}$ the relative cotangent complex of X_0 over S_0 , which is an element of the derived category $D(X_0,\ \underline{O}_{X_0})$. We consider the category $\underline{F}$ of all flat schemes X over S which extend X'/S' , i.e. together with an S'-isomorphism $X \underset{S}{\times} S' \simeq X'$. Then

(a) For an object X of $\underline{F}$, the group of automorphisms is canonically isomorphic to $\mathrm{Ext}^0(X_0;\ L., I_{X_0})$ (where I_{X_0} is the inverse image of I , viewed as a module on S_0 , on X_0) .

(b) The set of isomorphism classes of objects of $\underline{F}$ is empty or a torsor (= a principal homogeneous space) under the group $\mathrm{Ext}^1(X_0;\ L., I_{X_0})$. I112

(c) One can define a canonical element $c \in \mathrm{Ext}^2(X_0;\ L., I_{X_0})$ whose vanishing is necessary and sufficient for $\underline{F}$ to be non empty.

In order to study the $\mathrm{Ext}^i(X_0;\ L., I_{X_0})$, it is often convenient to write

$$\mathrm{Ext}^i(X_0; L., I_{X_0}) = H^i(X_0,\ \underline{RHom}(L., I_{X_0}))\quad .$$

and to notice that the complex $\underline{RHom}(L., I_{X_0})$ has as cohomology sheaves the sheaves $\underline{D}^i_{X_0}(I_{X_0})$ introduced in §3. For $i = 0, 1, 2$, these sheaves may be viewed respectively as the sheaf of automorphisms of any object X of $\underline{F}$ (understood: automorphisms inducing the identity on X') , the sheaf of indeterminacy of isomorphism classes of local solutions to the flat deformation problem, and the sheaf of obstructions to the existence of local solutions. On the other hand, the standard spectral sequence gives us

$$\mathrm{Ext}^*(X_0;\ L., I_{X_0}) \leftarrow E_2^{p,q} = H^p(X_0,\ \underline{D}^q_{X_0}(I_{X_0}))\quad ,$$

which yields the well-known isomorphism

$$\mathrm{Aut}(X/X') \simeq H^0(X_0, \underline{D}^0_{X_0}(I_{X_0})) \simeq H^0(X_0, \underline{Hom}(\underline{\Omega}^1_{X_0/S_0}, I_{X_0})) = \mathrm{Hom}(\underline{\Omega}^1_{X_0/S_0}, I_{X_0})\quad ,$$

and the exact sequence in low degrees

$$0 - H^1(X_0, \underline{D}^0_{X_0}) - \mathrm{Ext}^1 - H^0(X_0, \underline{D}^1_{X_0}) - H^2(X_0, \underline{D}^0_{X_0}) - \mathrm{Ext}^2$$

This shows in particular that if $H^2(X_0, \underline{D}^0_{X_0}) = 0$, and if there exists any local flat deformation X of X'/S' , then there exists a flat global deformation which corresponds to given local deformations. More precisely and generally, for given local deformations (i.e. a section of the "sheaf of isomorphism classes of local flat deformations"), the argument 4.10 shows that the obstruction to finding a global flat deformation compatible with these is in $H^2(X_0,\ \underline{D}^0_{X_0})$, hence such a global flat deformation exists if $H^2(X_0,\ \underline{D}^0_{X_0}) = 0$: this is essentially 4.10.

On the other hand, the spectral sequence shows that the Ext^2 is zero whenever we have the relations I113

$$(*)\qquad H^0(X_0,\ \underline{D}^2_{X_0}\ ) = 0\quad ,\quad H^1(X_0,\ \underline{D}^1_{X_0}\ ) = 0\quad ,\quad H^2(X_0,\ \underline{D}^0_{X_0}\ ) = 0\quad .$$

The first relation is always satisfied if X_0 is a relative complete intersection over S_0 (as then $\underline{D}^2_{X_0} = 0$ for any module M on X_0), the second is satisfied provided the support of $\underline{D}^1_{X_0}$ is discrete, as is the case if X_0 is smooth over S_0 except on a discrete subset of X_0 over S_0 . When the previous three relations $(*)$ are all satisfied, then by (c) above, $\underline{F}$ is non empty, which generalizes 4.13(b).

More geometrically, and without using the global theory of the relative cotangent complex (but only the globalizing of the functors $\underline{D}^i_{X_0}$), one can get a simple geometric interpretation of the three

successive obstructions to deforming flatly X' to X , lying in the three left hand terms of $(*)$. Namely by the local theory, the first obstruction c_1 in $H^0(X_0,\ \underline{D}^2_{X_0})$ can be defined as the obstruction to finding local solutions, when this obstruction vanishes, the sheaf $\underline{F}$ of isomorphism classes of local solutions of the problem is (by the local theory) in a canonical way a torsor under $\underline{D}^1_{X_0}$, and the second obstruction c_2 , lying in $H^1(X_0,\ \underline{D}^1_{X_0})$, is the obstruction to finding a section of this torsor, i.e. of finding a "coherent system of isomorphism classes of local solutions" ; if this second obstruction also vanishes, then, choosing arbitrarily a section of $\underline{F}$, one gets, as noted above, an obstruction c_3 in $H^2(X_0,\ \underline{D}^0_{X_0})$ to finding a global solution corresponding to the given system of (isomorphism classes of) local solutions. Using the fact that the section of $\underline{F}$ is determined up to adding a section of $\underline{D}^1_{X_0}$, we get a map $d : H^0(X_0,\ \underline{D}^1_{X_0}) \to H^2(X_0,\ \underline{D}^0_{X_0})$, and a solution to the deformation problem exists if and only if $c_1 = 0$, $c_2 = 0$, and c_3 is zero as an element of Coker d (each c_i being defined when the previous ones are zero). Of course, the groups $H^0(X_0,\ \underline{D}^2_{X_0})$, $H^1(X_0,\ \underline{D}^1_{X_0})$ and $H^2(X_0,\ \underline{D}^0_{X_0})/\mathrm{Im}\ d$ are just the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the spectral sequence above, d being the differential (up to sign ?) I 1114

$$d_2 : E_2^{0,1} \to E_2^{2,0} \quad .$$

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k , and let $(R,\ \mathfrak{X})$ be a formal versal deformation of X_0 . Then $\mathfrak{X}$ is not an ordinary scheme over R but an R-adic formal scheme. The purpose of this section is to clarify "the non-smooth locus of $\mathfrak{X}$ over R ".

We first recall the definition of Fitting ideals associated to a module of finite type: Let A be a commutative ring and E an A-module of finite type. Choose a presentation

$$K \to F \to E \to 0$$

where F is locally-free of constant rank n . We define

$$I^p_A(E) = \mathrm{Im}(\overset{n-p}{\wedge} K \otimes \overset{n-p}{\wedge} F^* \to A) \qquad \text{for} \quad p = 0,\ 1,\ 2,\ \ldots$$

The ideals $I^p_A(E)$ thus defined do not depend on the choice of a presentation and therefore are invariants of the A-module E , called the Fitting ideals of the A-module E . The following properties readily follow from the definition.

5.1. (a) For any ring homomorphism $A \to B$, the canonical map

$$B \underset{A}{\otimes} A/I^p_A(E) \xrightarrow{\sim} B/I^p_B(B \underset{A}{\otimes} E)$$

is an isomorphism for all p . In particular we have

$$I^p_{S^{-1}A}(S^{-1}E) = S^{-1}I^p_A(E)$$

for any multiplicative subset S in A .

(b) We have $I^0_A(E) \subset I^1_A(E) \subset I^2_A(E) \subset\ \ldots$, and, for each point $x \in \mathrm{Spec}(A)$, $I^p_A(E)_x = A_x$ for all $p \geq \dim_{\varkappa(x)} E(x)$, where $E(x) = \varkappa(x) \underset{A}{\otimes} E$. I 1115

(c) $x \in \mathrm{Spec}(A) - \mathrm{Supp}\ A/I^p_A(E)$ if and only if, for any (or for some) presentation

$$K \to F \to E \to 0$$

where $F \approx A^n$, K_x contains A_x^{n-p} which is isomorphic to a direct summand of F_x . In particular, we have that

$$\mathrm{Supp}\ A/I^0_A(E) = \mathrm{Supp}\ E\ .$$

(d)

$$I^r_A(E_1 \oplus E_2) = \sum_{p+q=r} I^p_A(E_1) \cdot I^q_A(E_2)$$

(e) If A is a noetherian adic ring, then

$$A/I^p_A(E) = \varprojlim_\nu A_\nu / I^p_{A_\nu}(E_\nu)$$

where $A = \varprojlim_\nu A_\nu$ and $E_\nu = E \underset{A}{\otimes} A_\nu$.

In view of the property 5.1(a), our definition of Fitting ideals is readily globalized. Thus let X be any scheme, and $\underline{E}$ a quasi-coherent X-module of finite presentation. We then obtain a sequence of quasi-coherent ideal sheaves $\underline{I}^p_X(\underline{E})$ for $p \geq 0$ defined by

$$\Gamma(U,\ \underline{I}^p_X(\underline{E})) \xrightarrow{\sim} I^p_{\Gamma(U,\ \underline{O}_X)}(\Gamma(U,\ \underline{E}))$$

for every affine open subset $U \subset X$.

Now let $X \to Y$ be a morphism of schemes. The relative module of Kähler differentials $\underline{\Omega}_{X|Y} (= \underline{\Omega}^1_{X|Y})$ is a quasi-coherent X-module. If the morphism $X \to Y$ is of essentially finite presentation, then $\underline{\Omega}_{X|Y}$ is a quasi-coherent X-module of finite presentation, and in turn we may consider the ideal sheaves $\underline{I}^p_X(\underline{\Omega}_{X|Y})$.

Definition 5.2. Let $X \to Y$ be a morphism of essentially finite presentation. For each integer $p \geq 0$, we define $J^p_X(X|Y)$ to be the closed subscheme of X defined by the ideal sheaf $\underline{I}^p_X(X|Y) = \underline{I}^p_X(\underline{\Omega}_{X|Y})$ I 116
, called the Jacobian subschemes of X over Y . For each point $x \in X$, we set $X_x = \mathrm{Spec}(\underline{O}_{X,x})$.

Theorem 5.3. Let $X \to Y$ be a morphism of essentially finite presentation. Then

(1) For any base change $Y \leftarrow Y'$, the canonical map $J^p_{X'}(X'|Y') \xrightarrow{\sim} J^p_X(X|Y) \underset{Y}{\times} Y'$ is an isomorphism for all p where $X' = X \underset{Y}{\times} Y'$.

(2) For each point $x \in X$, we have the canonical isomorphism $J^p_{X_x}(X_x|Y) \xrightarrow{\sim} J^p_X(X|Y)_x$.

(3) $J^0_X(X|Y) \supset J^1_X(X|Y) \supset J^2_X(X|Y) \supset \ldots$, and $x \notin J^p_X(X|Y)$ for all $p \geq \dim_{\varkappa(x)} \Omega_{X|Y}(x)$.

(4) $x \notin J^p_X(X|Y)$ if and only if there exists an open neighbourhood U of x which admits a closed immersion into a Y-scheme U' such that U' is smooth over Y at x' and $\dim_{x'} U'_{f(x)} = p$ where x' is the image of x under $U \to U'$.

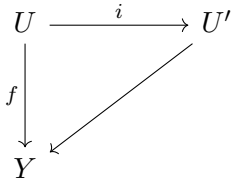
$$(5) \quad \underline{I}^r_{X_1 \underset{Y}{\times} X_2}(X_1 \underset{Y}{\times} X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}^p_{X_1}(X_1|Y) \cdot \pi_2^* \underline{I}^q_{X_2}(X_2|Y)$$

where $\pi_i : X_1 \underset{Y}{\times} X_2 \to X_i$ is the projection.

Proof: (3) follows from 5.1(b) whereas (1) follows from 5.1(a) together with the fact that $P^* \underline{\Omega}_{X|Y} \xrightarrow{\sim} \underline{\Omega}_{X'|Y'}$ is an isomorphism where $P : X \underset{Y}{\times} Y' \to X$ is the projection. (2) also follows from 5.1(a) since $\underline{O}_{X,x} \underset{\underline{O}_X}{\otimes} \underline{\Omega}_{X|Y} \xrightarrow{\sim} \underline{\Omega}_{X_x|Y}$ is an isomorphism. (5) follows from 5.1(d) and the isomorphism

$$\pi_1^* \underline{\Omega}_{X_1|Y} \oplus \pi_2^* \underline{\Omega}_{X_2|Y} \xrightarrow{\sim} \underline{\Omega}_{X_1 \underset{Y}{\times} X_2|Y}$$

(4) Assume the existence of a diagram



where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_{f(x)} = p$. Then $\dim_{\varkappa(x')} \underline{\Omega}_{U'|Y}(x') = p$ and I 117
$i^* \underline{\Omega}_{U'|Y} \to \underline{\Omega}_{U|Y} \to 0$ is exact, so that $\dim_{\varkappa(x)} \underline{\Omega}_{U|Y}(x) \leq p$ which implies that $x \notin J^p_U(U|Y)$ by

5.3(3). Since $J^p_U(U|Y) = J^p_X(X|Y)|U$, this means that $x \notin J^p_X(X|Y)$. Now assume conversely that $x \notin J^p_X(X|Y)$.

The question being local, we may assume that X, Y are both affine, say $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$. Choose a presentation over A

$$0 \to J \to P \to B \to 0$$

where $P = A[x_1, \ldots, x_n]$. Then $J/J^2 \xrightarrow{d} B \underset{P}{\otimes} \Omega_{P|A} \to \Omega_{B|A} \to 0$ is exact and $B \underset{P}{\otimes} \Omega_{P|A}$ is B-free of rank n , and therefore $I^p_B(\Omega_{B|A}) = \mathrm{Im}(\overset{n-p}{\wedge} J/J^2 \underset{B}{\otimes} (B \underset{P}{\otimes} \overset{n-p}{\wedge} \Omega_{P|A})^* \to B)$. Therefore, $x \notin J^p_X(X|Y)$ $\Leftrightarrow$ there exists $f_1, \ldots, f_{n-p}$ in J such that $\left[\det\left(\dfrac{\partial f_i}{\partial x_j}\right)\right](x) \neq 0$ for some choice of $n-p$ variables among $x_1, x_2, \ldots, x_n$ there exists $f_1, f_2, \ldots, f_{n-p}$ in J such that $\mathrm{Spec}(P/(f_1, \ldots, f_{n-p})) \to \mathrm{Spec}(A)$ is smooth at x' where x' is the image of x under $\mathrm{Spec}(B) \to \mathrm{Spec}(P/(f_1, \ldots, f_{n-p}))$. Thus it suffices to take $U' = \mathrm{Spec}(P/(f_1, \ldots, f_{n-p}))$.

The following is a generalization of the usual Jacobian criteria for relative situation.

Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. Assume that there exists a fixed integer $n \geq 0$ with either conditions:

(1) f is flat and $\dim_x X_{f(x)} = n$ for every point $x \in X$.

(2) f is a relative complete-intersection with virtual dimension n (SGA6 VIII 1.9).

Then $x \notin J^n_X(X|Y)$ if and only if f is smooth at x .

Proof: We have, by 5.3(4) that $x \notin J^n_X(X|Y)$ if and only if there exists a diagram $\begin{array}{ccc} U & \xrightarrow{i} & U' \\ {\scriptstyle f}\downarrow & \swarrow_{f'} & \\ Y & & \end{array}$ **1118**

such that f' is smooth at x' and $\dim_{x'} U'_{f(x)} = n$, where i is a closed immersion and $x' = i(x)$. Thus it suffices to show that either conditions (1) or (2) entails the isomorphism of i at the point x . Set $A = \underline{O}_{Y,f(x)}$, $B' = O_{U',x'}$, $B = O_{U,x}$ and let

$$0 \to I \to B' \to B \to 0$$

be exact. We must show that $I = 0$. Assume (1). Since B is A-flat,

$$0 \to I/\underline{m}I \to B'/\underline{m}B' \to B/\underline{m}B \to 0$$

is exact where $\underline{m}$ is the maximal ideal of A . By hypothesis, $\dim_x U_{f(x)} = \dim_{x'} U'_{f(x)}$ so that $\dim B'/\underline{m}B' = \dim B/\underline{m}B$. However $B'/\underline{m}B'$ is a regular local ring and therefore it entails that $I/\underline{m}I = 0$ so that $I = 0$. Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \underset{B'}{\otimes} \Omega_{B'|A} \to \Omega_{B|A} \to 0$$

Since X is a relative complete-intersection over Y and $U' \to Y$ is smooth at x , it follows that $d : I/I^2 \to B \underset{B'}{\otimes} \Omega_{B'|A}$ is injective at x and I/I^2 and $B \underset{B'}{\otimes} \Omega_{B'|A}$ are both free at x . Therefore by the definition of virtual dimension of relative complete-intersection we must have that

$$n = \dim_{\varkappa(x)} \Omega_{B'|A}(x') - \dim_{\varkappa(x)} I/I^2(x) = n - \dim_{\varkappa(x)} I/I^2(x)$$

i.e., $I/I^2 = 0$ i.e., $I = 0$.

Definition 5.5. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. In case when there exists a fixed integer n with the property (1) or (2) in 5.4, we set $\underline{I}_X(X|Y) = \underline{I}^n_X(X|Y)$ and $J_X(X|Y) = J^n_X(X|Y)$. **1119**

We can extend this notion to formal schemes in an obvious manner.

Thus let $\mathfrak{X}$ be a noetherian formal scheme and $\underline{E}$ a coherent $\mathfrak{X}$-module. Then we can define the ideal sheaf $\underline{I}^p_{\mathfrak{X}}(\underline{E})$ by setting

$$\underline{I}^p_{\mathfrak{X}}(\underline{E})(U) \;=\; I^p_{\mathfrak{X}(U)}(\underline{E}(U))$$

for every affine open $U \subset \mathfrak{X}$. If $\mathfrak{X} = \varinjlim_v X_v$, then $\underline{E} = \varprojlim_v \underline{E}_v$ where $\underline{E}_v = \underline{E} \otimes_{\underline{O}_{\mathfrak{X}}} \underline{O}_{X_v}$ and it follows from the fact that $\mathfrak{X}$ is noetherian that

$$\underline{I}^p_{\mathfrak{X}}(\underline{E}) \;=\; \varprojlim_v \underline{I}^p_{X_v}(\underline{E}_v)$$

Now let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme of essentially finite type. Let $\underline{I}_{\mathfrak{X}}$ be the ideal sheaf defining the closed immersion $\Delta : \mathfrak{X} \to \mathfrak{X} \underset{S}{\times} \mathfrak{X}$ (= the fibre-product in the category of formal schemes) i.e.,

$$0 \to \underline{I}_{\mathfrak{X}} \to \underline{O}_{\mathfrak{X} \underset{S}{\times} \mathfrak{X}} \to \underline{O}_{\mathfrak{X}} \to 0$$

is exact. We define $\underline{\Omega}_{\mathfrak{X}|S} = \underline{I}_{\mathfrak{X}}/\underline{I}^2_{\mathfrak{X}}$. Since $\mathfrak{X}$ is an S-adic formal scheme of essentially finite type, it follows that $\mathfrak{X} = \varinjlim_v X_v$ where $X_v = \mathfrak{X} \underset{S}{\times} S_v$, and hence $\underline{\Omega}_{\mathfrak{X}|S} \simeq \varprojlim_v \underline{\Omega}_{X_v|S_v}$ is a coherent $\mathfrak{X}$-module. We note that for any formal S-scheme S' of finite type, we have a canonical isomorphism

$$\underline{\Omega}_{\mathfrak{X} \underset{S}{\times} S'|S'} \;\simeq\; \underline{O}_{S'} \otimes_{\underline{O}_S} \underline{\Omega}_{\mathfrak{X}|S} \quad .$$

Definition 5.6. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme of essentially finite type. We define the ideal sheaf $\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S)$ by

$$\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S) \;=\; \underline{I}^p_{\mathfrak{X}}(\underline{\Omega}_{\mathfrak{X}|S})$$

and we set $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ the formal closed subscheme of $\mathfrak{X}$ defined by $\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S)$.

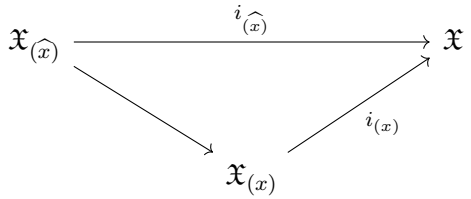
Since $\Omega_{\mathfrak{X}|S} = \varprojlim_n \Omega_{X_n|S_n}$ we have

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$$I^p_{\mathfrak{X}}(\mathfrak{X}|S) \;=\; \varprojlim_n I^p_{X_n}(X_n|S_n)$$

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) \;=\; \varinjlim_n J^p_{X_n}(X_n|S_n) \quad .$$

$\mathfrak{X}_{(x)}$ is an S-adic formal scheme of essentially finite type and we have a commutative diagram



We note that if x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$, $i_{\widehat{(x)}}$ are both isomorphisms to the connected component of $\mathfrak{X}$ containing the point x .

In particular, if $|\mathfrak{X}| = \{x_1, \ldots, x_m\}$ is a finite discrete space, then we have a canonical identification $\mathfrak{X} = \coprod_{1 \le i \le m} \mathfrak{X}_{(x_i)}$ (disjoint union).

Corollary 5.7. Let S be a noetherian formal scheme, and $\mathfrak{X}$ an S-adic formal scheme of essentially finite type. We then have

(1) $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ commutes with base change i.e., $J^p_{\mathfrak{X}'}(\mathfrak{X}'|S') = J^p_{\mathfrak{X}}(\mathfrak{X}|S) \underset{S}{\times} S'$ where $\mathfrak{X}' = \mathfrak{X} \underset{S}{\times} S'$.

(2) The connected components of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ are in bijective correspondence with those of $J^p_{X_0}(X_0|S_0)$. For each point $x \in |J^p_{X_0}(X_0|S_0)|$ we have a canonical isomorphism $J^p_{\mathfrak{X}}(\mathfrak{X}|S)_{(x)} \simeq J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$. In particular if

$$|J^p_{X_0}(X_0|S_0)| \;=\; \{x_1, \ldots, x_m\}$$

is discrete, then

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) \;=\; \coprod_{1\leq i\leq m} J^p_{\mathfrak{X}_{(x_i)}}(\mathfrak{X}_{(x_i)}|S)$$

(disjoint union).

(3) Let S be an ordinary noetherian scheme and X an S-scheme of essentially finite type. Given a closed subscheme $S_0 \subset S$, let $\widehat{S}$ be the formal completion of S along S_0 . I121

If we set

$$\widehat{X} \;=\; X \underset{S}{\times} \widehat{S} \;=\; \text{the formal completion of } X \text{ along } X_0 \;=\; X \underset{S}{\times} S_0 \;,\; \text{then}$$

$$J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) \;=\; J^p_X(X|S) \underset{S}{\times} \widehat{S} \;=\; \text{the formal completion of } J^p_X(X|S) \text{ along } J^p_{X_0}(X_0|S_0) \;.$$

(4) If there exists a fixed integer $n \geq 0$ with the property 5.4(1) or (2) for $X_\nu \to S_\nu$ for all ν , then $x \notin J^n_{\mathfrak{X}}(\mathfrak{X}|S)$ if and only if $\mathfrak{X}$ is formally smooth over S at x .

Proof: (1) follows from $\underline{\Omega}_{\mathfrak{X}'|S'} \simeq \underline{O}_{S'}\widehat{\otimes}_{\underline{O}_S}\underline{\Omega}_{\mathfrak{X}|S}$ whereas (2) follows from

$$\underline{O}_{\mathfrak{X}_{(x)}} \,\widehat{\otimes}_{\underline{O}_{\mathfrak{X}}}\, \underline{\Omega}_{\mathfrak{X}|S} \;\simeq\; \underline{\Omega}_{\mathfrak{X}_{(x)}|S} \;.$$

$$(3)\quad J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) \;=\; \varinjlim_{\nu} J^p_{X\underset{S}{\times}S_\nu}\left(X \underset{S}{\times} S_\nu \,\middle|\, S_\nu\right) \;=\; \varinjlim_{\nu} J^p_X(X|S) \underset{S}{\times} S_\nu \;=\; J^p_X(X|S) \underset{S}{\times} \widehat{S} \;.$$

(4) Since $\mathfrak{X}$ is an S-adic formal scheme, $\mathfrak{X} \to S$ is formally smooth at

$$x \Leftrightarrow X_\nu \to S_\nu \text{ is formally smooth at } x \text{ for all } \nu \;\Leftrightarrow\; x \notin J^n_{X_\nu}(X_\nu|S_\nu) \text{ for all } \nu$$
$$\Leftrightarrow\; x \notin \varinjlim_{\nu} J^n_{X_\nu}(X_\nu|S_\nu) \;=\; J^n_{\mathfrak{X}}(\mathfrak{X}|S) \;.$$

Remark 5.8. Let $\mathfrak{X}$ be an S-adic formal scheme of essentially finite type. Given a point $x \in \mathfrak{X}$ we also consider the S-adic formal scheme $\mathfrak{X}_{(\hat{x})} = \varinjlim_{\nu} \mathrm{Spec}(\widehat{\underline{O}_{X_\nu,x}})$. We thus have the natural morphisms $\mathfrak{X}_{(\hat{x})} \to \mathfrak{X}_{(x)} \to \mathfrak{X}$. Now $\widehat{\underline{O}_{X_0,x}}$ is in general not essentially finite type over $S_{(y)}$ where $y =$ the image of x under $\mathfrak{X} \to S$. For simplicity, let us set $A_\nu = \underline{O}_{X_\nu,x}$, $R_\nu = \underline{O}_{S_\nu,y}$. Now the natural map $\Omega_{A_\nu|R_\nu} \to \Omega_{\widehat{A}_\nu|R_\nu}$ induces an isomorphism $\widehat{\Omega_{A_\nu|R_\nu}} \to \widehat{\Omega_{\widehat{A}_\nu|R_\nu}}$, and therefore $\varprojlim_\nu \widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \varprojlim_\nu \widehat{\Omega_{\widehat{A}_\nu|R_\nu}}$ where $\widehat{\ }$ stands for the separated completion of a module with respect to the topology induced from the ring. Therefore if we define $\widehat{\Omega_{\mathfrak{X}_{(\hat{x})}|S}} = \varprojlim_\nu \widehat{\Omega_{\widehat{\underline{O}}_{X_\nu,x}|\underline{O}_{S_\nu,y}}}$, then $\widehat{\Omega_{\mathfrak{X}_{(\hat{x})}|S}}$ is a coherent $\mathfrak{X}_{(\hat{x})}$-module. Thus if we set $J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S)$ to be the formal closed subscheme of $\mathfrak{X}_{(\hat{x})}$ defined by the I122
ideal sheaf $\underline{I}^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) = \underline{I}^p_{\mathfrak{X}_{(\hat{x})}}(\widehat{\Omega}_{\mathfrak{X}_{(\hat{x})}|S})$, then we have

$$J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)_{(\hat{x})} \;\simeq\; J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) \;.$$

We note that if x is an isolated point of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, then $J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) \to J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$ is an isomorphism.

To consider the "image" of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ under the morphism $\mathfrak{X} \xrightarrow{f} S$, one may simply take the formal subscheme of S defined by the ideal sheaf $\mathrm{Ker}(\underline{O}_S \to \underline{O}_{\mathfrak{X}}/\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S))$. However, it is more convenient for our purpose to use the Fitting ideal again.

Definition 5.9. Let S be a noetherian formal scheme and $f : \mathfrak{X} \to S$ an S-adic formal scheme of essentially finite type. Assume that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S , so that $f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}$ is a coherent $\underline{O}_S$-module. We then define the ideal sheaf $\underline{I}^p_S(\mathfrak{X}|S)$ of $\underline{O}_S$ by

$$\underline{I}^p_S(\mathfrak{X}|S) \;=\; \underline{I}^0_S(f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)})$$

and set $J^p_S(\mathfrak{X}|S)$ to be the closed formal subscheme of S defined by the ideal sheaf $\underline{I}^p_S(\mathfrak{X}|S)$. In case when there exists a fixed integer n having the property (1) or (2) in 5.4 for $X_\nu \to S_\nu$ for all ν , we set

$$\underline{I}_{\mathfrak{X}}(\mathfrak{X}|S) = \underline{I}^n_{\mathfrak{X}}(\mathfrak{X}|S) \ , \qquad J_{\mathfrak{X}}(\mathfrak{X}|S) = J^n_{\mathfrak{X}}(\mathfrak{X}|S)$$
$$\underline{I}_S(\mathfrak{X}|S) = \underline{I}^n_S(\mathfrak{X}|S) \ , \qquad J_S(\mathfrak{X}|S) = J^n_S(\mathfrak{X}|S)$$

Corollary 5.10. Let S be a noetherian formal scheme and $\mathfrak{X} \xrightarrow{f} S$ an S-adic formal scheme, essentially of finite type such that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S . Then

$$(1) \quad |J^p_S(\mathfrak{X}|S)| = \text{the image of } |J^p_{\mathfrak{X}}(\mathfrak{X}|S)| \text{ under } f : \mathfrak{X} \to S \ .$$

(2) $J^p_S(\mathfrak{X}|S)$ commutes with base change i.e., $J^p_{S'}(\mathfrak{X}'|S') = J^p_S(\mathfrak{X}|S) \times_S S'$. I 123

(3) If $J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1\le i\le m} \mathfrak{Y}^{(i)}$ (disjoint union), then
$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\le i\le m} \underline{I}^0_S(f_*\underline{O}_{\mathfrak{Y}(i)})$. In particular if

$$|J^p_{X_0}(X_0|S_0)| = \{x_1, \dots, x_m\}$$

is discrete, then

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathfrak{X}_{(x_i)}|S) = \prod_{1\le i\le m} \underline{I}^p_S(\mathfrak{X}_{(\hat{x}_i)}|S)$$

(4) Let S be an ordinary noetherian scheme and X an S-scheme of essentially finite type. Given a closed subscheme $S_0 \to S$, let $\hat{S}$ be the formal completion of S along S_0 . If we set $\hat{X} = X \times_S \hat{S}$ = the formal completion of X along $X_0 = X \times_S S_0$, then $J^p_{\hat{S}}(\hat{X}|\hat{S}) = J^p_S(X|S) \times_S \hat{S}$ = the formal completion of $J^p_S(X|S)$ along $J^p_{S_0}(X_0|S_0)$.

Proof: Supp $f_*\underline{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)} = |J^p_S(\mathfrak{X}|S)|$ by 5.1(c), and hence (1) follows. (2) and (4) are clear from 5.1(a), and (3) follows from 5.1(d), and the isomorphism $J^p_{\mathfrak{X}_{(\hat{x})}}(\mathfrak{X}_{(\hat{x})}|S) \xrightarrow{\sim} J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$ for an isolated point x in $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$.

Remark 5.11. Our definition of Jacobian ideal, adopted from the classical one over base field, commutes with base change and gives a smoothness criteria under a mild condition of 5.4. However one can define, probably in many different ways, a canonical subscheme which gives the non-smooth locus in a general setting. We shall outline one here: Given an A-algebra B , define $I_{B|A} = \bigcap_E \operatorname{Ann} D^1(B|A, E)$ where E runs through all B-modules of finite type. This ideal $I_{B|A}$ commutes with localizations by 3.13(c), and therefore, given a morphism $X \xrightarrow{f} Y$, we obtain the ideal sheaf $\underline{I}_{X|Y}$. If f is a morphism of essentially finite type, then f is smooth at x if and only if $x \notin \operatorname{Supp} \underline{O}_X/\underline{I}_{X|Y}$ by 3.13(a). This notion is in an obvious way carried over to adic formal schemes of essentially finite type and thus it is equally suitable for deformation theory. I 124

6. Non-degenerate quadratic singularities

The objective in this section is to investigate the deformations of a k-scheme having non-degenerate quadratic singularities only. In view of 4.13, it becomes purely a local problem in an appropriate situation.

We go back to the notations of §4. Thus Λ is a fixed complete noetherian local ring with the residue field k , and $\underline{C}_\Lambda$ stands for the category of local artinian Λ-algebras with the same residue field k .

Let X be a k-scheme of finite type, and let x be a closed point of X . We recall that the point x is said to be a non-degenerate quadratic singularity rational over k if $\hat{O}_{X,x} \simeq k\,[[x_1, \dots, x_n]]/(f)$ with $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = (x_1, \dots, x_n)$. One can always assume f to be a quadratic form. Indeed we have

Lemma 6.1. Let $f \in k[[x_1, \dots, x_n]]$. If $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M$ $(= (x_1, \dots, x_n))$, then one can find $x'_1, \dots, x'_n$ with $x_i = x'_i \pmod{M^2}$ such that $f = q(x'_1, \dots, x'_n)$ where q is a quadratic form.

Proof: We do this by a successive approximation. Write $f = f_2 + h_3$ where f_2 is a quadratic form and $h_3 \in M^3$. Since $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M$, there exists $\epsilon_1^{(1)}, \dots, \epsilon_n^{(1)}$ in M^2 such that $\Sigma\ \epsilon_i^{(1)}\ \partial f/\partial x_i \equiv h_3 \pmod{M^4}$. If we set $x_i^{(1)} = x_i - \epsilon_i^{(1)}$, then $f(x_1^{(1)}, \dots, x_n^{(1)}) = f(x_1, \dots, x_n) - \Sigma\ \epsilon_i^{(1)}\ \partial f/\partial x_i + \dots \equiv f_2(x_1, \dots, x_n) \pmod{M^4}$ i.e., $f(x_1^{(1)}, \dots, x_n^{(1)}) \equiv f_2(x_1, \dots, x_n) + h_4$ with $h_4 \in M^4$. Now $(\partial f/\partial x_1^{(1)}, \dots, \partial f/\partial x_n^{(1)}) = (\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M$ and therefore we can choose $\varepsilon_1^{(2)}, \dots, \varepsilon_n^{(2)}$ in M^3 such that $\Sigma\ \varepsilon_i^{(2)}\ \partial f/\partial x_i^{(1)} \equiv h_4 \pmod{M^5}$. Then $f(x_1^{(1)} - \varepsilon_1^{(2)}, \dots, x_n^{(1)} - \varepsilon_n^{(2)}) =$ I 125
$f(x_1^{(1)}, \dots, x_n^{(1)}) - \Sigma\ \varepsilon_i^{(2)} \partial f/\partial x_i^{(1)} + \dots \equiv f_2(x_1, \dots, x_n) \pmod{M^5}$. Continuing this process we can find $\varepsilon_i^{(1)}$, $\varepsilon_i^{(2)}$, $\varepsilon_i^{(3)}, \dots$ $(i = 1, 2, \dots, n)$ with $\varepsilon_i^{(v)} \in M^{v+1}$ such that if we set $x_i^{(v)} = x_i - (\varepsilon_i^{(1)} + \dots + \varepsilon_i^{(v)})$ then $f(x_1^{(v)}, \dots, x_n^{(v)}) \equiv f_2(x_1, \dots, x_n) \pmod{M^{v+3}}$. Thus if we set $\varepsilon_i = \sum_v \varepsilon_i^{(v)}$ then $f(x_1 - \varepsilon_1, \dots, x_n - \varepsilon_n) = = f_2(x_1, \dots, x_n)$ i.e., $f(x_1, \dots, x_n) = f_2(x_1 + \varepsilon_1, \dots, x_n + \varepsilon_n)$.

Lemma 6.2. Let A_0 be a local k-algebra, essentially of finite type such that

$$\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$$

where f is a non-degenerate quadratic form, and let $(R,\ \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then

(1) $J_{\mathfrak{X}}(\mathfrak{X}|R) \to J_R(\mathfrak{X}|R) \to \mathrm{Spec}(\Lambda)$ are both isomorphisms.

(2) $I_R(\mathfrak{X}|R)$ is a principal ideal in R generated by a free formal generator of R over Λ i.e. $I_R(\mathfrak{X}|R) = (t)$ and $R = \Lambda[[t]]$.

(3) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathfrak{X} = \mathrm{Spf}(A)$. By 4.10(b), $(R,\ \mathrm{Spf}(\hat{A}))$ is a formal versal deformation of $\hat{X}_0 = \mathrm{Spec}(\hat{A}_0)$ where $\hat{A} = \varprojlim_v A \otimes_R R_v$. Now $\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$ and $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = (x_1, \dots, x_n)$ in $k[[x_1, \dots, x_n]]$. It follows readily that $[\underline{M}_{\hat{X}_0}(k[\varepsilon])] \simeq k$ and, by 4.14, we get

$$R \simeq \Lambda[[t]] \quad \text{and} \quad \hat{A} \simeq R[[x_1, \dots, x_n]]/(F - t)$$

where F is a quadratic polynomial in $\Lambda[x_1, \dots, x_n]$ $(\subset R[[x_1, \dots, x_n]])$ obtained from f by lifting the coefficients to Λ . Now

$$0 \to \hat{A} \xrightarrow{dF} \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n \longrightarrow \hat{\Omega}_{\hat{A}|R} \to 0$$

is exact where $dF : \hat{A} \to \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n$ is given by $1 \to (\partial F/\partial x_1)dx_1 + \dots + (\partial F/\partial x_n)dx_n$ I 126
, and therefore $I_{\hat{A}}(\hat{A}|R) = (\partial F/\partial x_1, \dots, \partial F/\partial x_n) = (x_1, \dots, x_n)$, and hence $\hat{A}/I_{\hat{A}}(\hat{A}|R) = \hat{A}/(x_1, \dots, x_n) = R[[x_1, \dots, x_n]]/(x_1, \dots, x_n, F - t) \simeq \Lambda$ i.e., the canonical map $\Lambda \to \hat{A}/I_{\hat{A}}(\hat{A}|R)$ is an isomorphism. It follows that $\hat{A}/I_{\hat{A}}(\hat{A}|R) \to A/I_A(A|R)$ is an isomorphism, and consequently $\Lambda \to R/I_R(A|R)$ is also an isomorphism. This proves (1) and (2). On the other hand, (3) follows from $\hat{A} \simeq R[[x_1, \dots, x_n]]/(F - t) \simeq \Lambda[[t, x_1, \dots, x_n]]/(F - t) \simeq \Lambda[[x_1, \dots, x_n]]$ as Λ-algebras.

Corollary 6.3. Let X_0 be a k-scheme of finite type, and let x in X_0 be a non-degenerate quadratic singularity rational over k .

(1) Let $(R,\ \mathfrak{X})$, $(R,\ \mathfrak{Y})$ be any two deformations of X_0 where $R \in \mathrm{ob}(\hat{C}_\Lambda)$. Assume that $k = k^2$. Then $\mathfrak{X}_{(\widehat{x})} \simeq \mathfrak{Y}_{(\widehat{x})}$ as R-adic formal schemes if and only if $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$.

(2) Let $R \in \mathrm{ob}(\hat{C}_\Lambda)$ be a regular local ring, and let $(R,\ \mathfrak{X})$ be a deformation of X_0 . Then $O_{\mathfrak{X},x}$ is a regular local ring if and only if $I_R(\mathfrak{X}_{(x)}|R) \not\subset \underline{m}_R^2$ i.e., if and only if $I_R(\mathfrak{X}_{(x)}|R)$ is a regular divisor in $\mathrm{Spf}(R)$.

Proof: Let $(R_0,\ Z)$ be a formal versal deformation of $\mathrm{Spec}(\hat{O}_{X_0,x})$. Since x in X_0 is a non-degenerate quadratic singularity rational over k , we may assume that $R_0 \simeq \Lambda[[t]]$ and $Z \simeq \mathrm{Spf}(R_0[[x_1, \dots, x_n]]/(F - t))$ as in 6.2.

(1) $(R,\ \mathfrak{X}_{(\widehat{x})})$, $(R,\ \mathfrak{Y}_{(\widehat{x})})$ are deformations of $\mathrm{Spec}(\hat{O}_{X_0,x})$, and therefore they are induced from $(R_0,\ Z)$ by continuous local Λ-algebra maps $\varphi,\ \psi : R_0 \to R$ respectively, i.e., if we set $a = \varphi(t)$, $b = \psi(t)$, then

$$\mathfrak{X}_{(\widehat{x})} \simeq \mathrm{Spf}(R[[X_1, \dots, X_n]]/(F - a)) \quad ,$$

$$\mathfrak{Y}_{\widehat{(x)}} \simeq \mathrm{Spf}(R[[X_1,\dots,X_n]]/(F-b))$$

as R-adic formal schemes. Assume that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$ i.e., $(a)=(b)$ in R . Then $a = ub$ for some unit u in R . We construct a matrix $T \in \mathrm{GL}_n(R)$ such that $F(TX) = uF(X)$.[1] Write $\mathfrak{m} = \mathfrak{m}_R$, and interpret polynomial congruences coefficientwise. Since $k = k^2$, choose $c \in R^\times$ whose residue satisfies $\bar{c}^2 = \bar{u}$, and put $T_1 = cI_n$. Then $F(T_1X) \equiv uF(X) \pmod{\mathfrak{m}}$. By the nondegeneracy condition in Lemma 6.2, the coefficient matrix of the linear forms $\partial F/\partial X_i$ is invertible modulo $\mathfrak{m}$, hence over R. Suppose $T_\nu \in \mathrm{GL}_n(R)$ satisfies $F(T_\nu X) \equiv uF(X) \pmod{\mathfrak{m}^\nu}$, with $\nu \geq 1$. The forms $(\partial F/\partial X_i)(T_\nu X)$ form an R-basis of the linear forms. Consequently, there are linear forms L_i with coefficients in $\mathfrak{m}^\nu$ such that

$$E_\nu(X) := uF(X) - F(T_\nu X) = \sum_i L_i(X)\frac{\partial F}{\partial X_i}(T_\nu X).$$

Indeed, every quadratic monomial is a product of two linear forms, and expressing one factor in this basis preserves its coefficient ideal. Define H_ν by $H_\nu X = (L_i(X))_i$, and set $T_{\nu+1} = T_\nu + H_\nu$. The quadratic identity gives

$$F(T_{\nu+1}X) = F(T_\nu X) + E_\nu(X) + F(H_\nu X) \equiv uF(X) \pmod{\mathfrak{m}^{\nu+1}},$$

since the coefficients of $F(H_\nu X)$ lie in $\mathfrak{m}^{2\nu} \subseteq \mathfrak{m}^{\nu+1}$. Moreover, $T_{\nu+1} \equiv T_\nu \pmod{\mathfrak{m}^\nu}$. Completeness and separatedness therefore give a limit $T \equiv cI_n \pmod{\mathfrak{m}}$ with $F(TX) = uF(X)$; in particular, T is invertible. The substitution $X \mapsto TX$ sends $F-a$ to $u(F-b)$ and induces the required isomorphism

$$R[[X_1,\dots,X_n]]/(F-a) \xrightarrow{\sim} R[[X_1,\dots,X_n]]/(F-b).$$

The converse follows from invariance of I_R under R-isomorphisms.

(2) There exists a continuous local Λ-algebra map $\varphi : R_0 = \Lambda[[t]] \to R$ such that $\mathfrak{X}_{\widehat{(x)}} \simeq \mathrm{Spf}(R[[X_1,\dots,X_n]]/(F-\varphi(t)))$. We note that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{X}_{\widehat{(x)}}|R) = (\varphi(t))$, and that $R[[X_1,\dots,X_n]]/(F-\varphi(t)) = \hat{\underline{O}}_{\mathfrak{X},x}$ is regular if and only if $\underline{O}_{\mathfrak{X},x}$ is regular. Now R is regular and hence $R[[X_1,\dots,X_n]]$ is also a regular local ring. Thus if we set M to be the maximal ideal of $R[[X_1,\dots,X_n]]$, then $R[[X_1,\dots,X_n]]/(F-\varphi(t))$ is regular if and only if $F-\varphi(t)$ is not in M^2 , i.e. if and only if $\varphi(t) \notin \underline{m}_R^2$, i.e., if and only if $I_R(\mathfrak{X}_{(x)}|R) \not\subset \underline{m}_R^2$.

Theorem 6.4. Let X_0 be a k-scheme of finite type with isolated non-smooth points only, and let $(R,\mathfrak{X})$ be a formal versal deformation of X_0 . Denote by $\{z_1,\dots,z_m\}$ the set of non-smooth points in X_0 . Assume that $H^2(X_0, \underline{D}^0_{X_0}) = 0$, and all z_i's are non-degenerate quadratic singularity rational over k . Then

(1) R is formally smooth over Λ i.e., R is a formal power-series ring over Λ ;

(2) $J_{\mathfrak{X}}(\mathfrak{X}|R) = \coprod_{1\leq i\leq m} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R)$ (disjoint union), and
$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \ -\simeq- \ J_R(\mathfrak{X}_{(z_i)}|R)$ is an isomorphism;

(3) $I_R(\mathfrak{X}_{(z_i)}|R)$ is a principal ideal and in turn so is
$I_R(\mathfrak{X}|R) = \prod_{1\leq\ i\leq\ m} I_R(\mathfrak{X}_{(z_i)}|R)$. Furthermore if we set $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i)$, then $t_1,\dots,t_m$ can be extended to a system of free formal generators of R over Λ (as formal power-series ring over Λ).

(4) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is smooth, i.e. $J_{\mathfrak{X}}(\mathfrak{X}|\Lambda) = \varphi$.

(5) If X_0 is a complete irreducible curve, then $\dim \underline{M}_{X_0} = 3g-3+a$ where $g = \dim_k H^1(X_0, \underline{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$.

Proof: (1) follows from (2) and (3): The first statement of (2) follows from 5.7(2). Now let $(R_i,\mathfrak{Y}_i)$ be a formal versal deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$. Since $(R,\mathfrak{X}_{\widehat{(z_i)}})$ is a deformation of $\mathrm{Spec}(\hat{\underline{O}}_{X_0,z_i})$, there exists a continuous local Λ-algebra maps $\varphi_i : R_i \to R$ with a cartesian diagram

[1] Editorial proof repair: the printed inference that every unit of R has a square root need not hold in characteristic 2. The following argument retains the original hypotheses.

$$\begin{array}{ccc} \mathfrak{X}_{(\widehat{z_i})} & \longrightarrow & \mathfrak{Y}_i \\ \downarrow & & \downarrow \\ \mathrm{Spf}(R) & \xrightarrow[\mathrm{Spf}(\varphi_i)]{} & \mathrm{Spf}(R_i) \end{array}$$

Therefore we have

$$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) = J_{\mathfrak{X}_{(\widehat{z_i})}}(\mathfrak{X}_{(\widehat{z_i})}|R) \simeq J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \underset{\mathrm{Spf}(R_i)}{\times} \mathrm{Spf}(R)$$
$$J_R(\mathfrak{X}_{(z_i)}|R) = J_R(\mathfrak{X}_{(\widehat{z_i})}|R) \simeq J_{R_i}(\mathfrak{Y}_i|R_i) \underset{\mathrm{Spf}(R_i)}{\times} \mathrm{Spf}(R) \quad .$$

Since $J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \to J_{R_i}(\mathfrak{Y}_i|R_i)$ is an isomorphism by 6.2(1), we have that the canonical morphism $J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \to J_R(\mathfrak{X}_{(z_i)}|R)$ is an isomorphism. Now $I_{R_i}(\mathfrak{Y}_i|R_i) = (u_i) = (u_i)$ with $R_i = \Lambda[[u_i]]$ by 6.2(2), and therefore

$$I_R(\mathfrak{X}_{(z_i)}|R) = I_R(\mathfrak{X}_{(\widehat{z_i})}|R) = (t_i)$$

where $t_i = \varphi_i(u_i)$, and therefore $I_R(\mathfrak{X}|R) = \prod_{1\leq\, i\leq\, m} I_R(\mathfrak{X}_{(z_i)}|R) = (\ \prod_{i=1}^m t_i\ )$.

Now $\underline{M}_{X_0} \to \underline{M}_{X_0,(\widehat{z_1})} \times \ldots \times \underline{M}_{X_0,(\widehat{z_m})}$ is smooth by and therefore $R_1 \hat{\otimes}_\Lambda \ldots \hat{\otimes}_\Lambda R_m \to R$ is formally smooth by 4.13, which implies that $t_1, \ldots, t_m$ is extended to a system of free formal generators of R over Λ .

(4) The projection morphism $\mathfrak{X}_{(\widehat{z_i})} \overset{\sim}{\to} \mathfrak{Y}_i \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R) \to \mathfrak{Y}_i$ is formally smooth since I 129
$\mathrm{Spf}(R) \xrightarrow{\mathrm{Spf}(\varphi_i)} \mathrm{Spf}(R_i)$ is formally smooth. On the other hand, $\mathfrak{Y}_i \to \mathrm{Spec}(\Lambda)$ is formally smooth by 6.2(3) and therefore $\mathfrak{X}_{(\widehat{z_i})} \to \mathrm{Spec}(\Lambda)$ is formally smooth. Consequently $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth since $\{z_1, \ldots, z_m\}$ is the set of non-smooth points of X_0 .

(5) Since R is formally smooth over Λ and

$$0 \to H^1(X_0,\ \underline{D}^0_{X_0}) \ \to [\underline{M}_{X_0}(k[\varepsilon])] \to H^0(X_0,\ \underline{D}^1_{X_0}) \to 0$$

is exact by 4.4 and 4.11, we have

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim k \otimes R \ = \ \dim_k H^1(X_0,\ \underline{D}^0_{X_0}) + \dim H^0(X_0,\ \underline{D}^1_{X_0}) \\ &= \dim_k H^1(X_0,\ \underline{D}^0_{X_0}) + m \ . \end{aligned}$$

Then it suffices to show that

$$\dim_k H^1(X_0,\ \underline{D}^0_{X_0}) + m = 3g - 3 + \dim_k H^0(X_0,\ \underline{D}^0_{X_0})$$

i.e., $\chi(\underline{D}^0_{X_0}) = 3 - 3g + m$

where we set $\chi(\underline{E}) = \dim_k H^0(X_0,\ \underline{E}) - \dim_k H^1(X_0,\ \underline{E})$ for any coherent X_0-module $\underline{E}$. Since $\chi(\underline{D}^0_{X_0})$ does not change when one replaces the base field k by its algebraic closure, we may assume that k is algebraically closed. Set $A = \underline{O}_{X_0,z}$, where z is a non-smooth point of X_0 , and let A' be its normalization. By hypothesis, we have $A \simeq k[[x,y]]/(f)$ where f is a non-degenerate quadratic form. Since k is algebraically closed, we may assume that $f = xy$, i.e., $A \simeq k[[x,y]]/(xy)$. From this we conclude immediately that $\ell(A'/A) = 1$ and $\mathrm{Der}_k(A,\ \underline{m}_A) = \mathrm{Der}_k(A,\ A)$. Let C be the conductor of A in A' . Since A is a complete-intersection we must have that $\ell(A/C) = \ell(A'/A) = 1$ so that $C = \underline{m}_A$. We thus obtain that $A' = \{a' \in K |\ a'\underline{m}_A \subset \underline{m}_A\}$ where K is the fraction field of A . We now claim that $\mathrm{Der}_k(A,\ A) \subset \mathrm{Der}_k(A',\ A')$ as A-modules contained in $\mathrm{Der}_k(K,\ K)$. Indeed, let $X \in \mathrm{Der}_k(A,\ A)$. Then for any $a' \in A'$ and $x \in \underline{m}_A$ we have $xa' \in \underline{m}_A$ so that $X(xa') \in \underline{m}_A$ and I 130
$a'X(x) \in \underline{m}_A$ since $\mathrm{Der}_k(A,\ \underline{m}_A) = \mathrm{Der}_k(A,\ A)$. Consequently $X(xa') = a'X(x) + xX(a')$ entails that $xX(a') \in \underline{m}_A$ for all $x \in \underline{m}_A$, and therefore $X(a') \in A'$. Furthermore, the exact sequence

$$0 \ \to \ \mathrm{Der}_k(A,\ A) \to \mathrm{Der}_k(A',\ A') \to A'/\underline{m}_A A' \to \ 0$$

entails the exact sequence

$$0 \ \to \ \mathrm{Der}_k(A,\ A) \to \mathrm{Der}_k(A',\ A') \to A'/C \to 0$$

Therefore if we set X_0' to be the normalization of X_0 , then

$$0 \to \underline{D}^0_{X_0} \to \underline{D}^0_{X_0'} \ \to \underline{O}_{X_0'}/C \to 0$$

is exact. Since X_0 is locally a complete-intersection, we have $\ell(\underline{O}_{X_0'}/C) = 2\ell(\underline{O}_{X_0'}/\underline{O}_{X_0}) = 2m$, whereas $\chi(\underline{D}^0_{X_0'}) = 3 - 3(g-m)$ by Riemann-Roch theorem. Consequently

$$\chi(\underline{D}^0_{X_0}) = \chi(\underline{D}^0_{X_0'}) - \ell(\underline{O}_{X_0'}/C) = 3 - 3(g-m) - 2m = 3 - 3g + m \ .$$

This completes our proof.

Remark 6.5. In the formula 6.4(5), the number $a = H^0(X_0,\ \underline{D}^0_{X_0})$ can be easily computed in terms of Riemann-Roch theorem. Thus

$$a = \begin{cases} 0 & \text{if} \quad g \geq 2 \\ 1 & \text{if} \quad g = 1 \\ 3 & \text{if} \quad g = 0 \end{cases}$$

Finally we add a remark on characteristic 2. If the characteristic of the base field k is 2 , there is no non-degenerate quadratic form on n variables if n is odd. In other words there does not exist a non-degenerate quadratic singularity of even dimension if $\mathrm{char}\, k = 2$. In this case one may consider a quadratic form of maximum non-degeneracy:

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity rational over k if the following holds: I131

(i) If either $chark \neq 2$ or $chark = 2$ and $dimO_{X,z}$ is odd, then z is a non-degenerate quadratic singularity rational over k .

(ii) If $chark = 2$ and $dimO_{X,z}$ is even, then $\bar{k} \underset{k}{\otimes} \hat{O}_{X,z} \simeq \bar{k}[[X_0, X_1, \ldots, X_{2n}]]/(f)$ with

$$f = X_0^2 + X_1X_2 + X_3X_4 + \ldots + X_{2n-1}X_{2n}$$

where $\bar{k}$ is the algebraic closure of k .

Lemma 6.7. Let k be a field of characteristic 2 and set $A_0 = k[[X_0, X_1, \ldots, X_{2n}]]/(f)$ where

$$f = X_0^2 + X_1X_2 + X_3X_4 + \ldots + X_{2n-1}X_{2n} \quad ,$$

and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then $R = \Lambda[[t, t_1]]$, where t, t_1 are free formal generators of R over Λ , such that $I_R(\mathfrak{X}|R) = (t^2)$, and $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathfrak{X} = \mathrm{Spf}(A)$. Since

$$(f, \frac{\partial f}{\partial X_0}, \ldots, \frac{\partial f}{\partial X_{2n}}) = (X_0^2, X_1, \ldots, X_{2n}) \quad ,$$

it follows that $D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2)$ and therefore $[\underline{M}_{X_0}(k[\epsilon])]$ is 2-dimensional vector space over k . Furthermore, it follows from 4.14 that

$$R \simeq \Lambda[[t, t_1]] \quad \text{and} \quad A \simeq R[[X_0, X_1, \ldots, X_{2n}]]/(F)$$

where $F = f + t_1 + tX_0$. Now the exact sequence

$$0 \longrightarrow A \xrightarrow{dF} AdX_0 \oplus \ldots \oplus AdX_{2n} \longrightarrow \hat{\Omega}_{A|R} \longrightarrow 0$$

where

$$dF = 1 \to \sum_{0 \le i \le 2n} \frac{\partial f}{\partial X_i} dX_i + t\, dX_0 \quad ,$$

entails that I 132

$$I_A(A|R) = (t, x_1, x_2, \ldots, x_{2n}) = (x_0^2 + t_1, t, x_1, \ldots, x_{2n})$$

so that

$$A/I_A(A|R) \simeq A/(X_0^2 + t_1, t, x_1, \ldots, x_{2n}) \simeq \Lambda[[t_1, X_0]]/(X_0^2 - t_1) \ .$$

Since $\Lambda[[t_1, X_0]]/(X_0^2 - t_1) \simeq R/(t) \oplus R/(t)$ as R-modules, it follows that $I_R(A|R) = (t^2)$, which proves our first assertion. On the other hand,

$$\begin{aligned} A &\simeq \Lambda[[t, t_1, x_o, x_1, \ldots, x_{2n}]]/(X_0^2 + X_1X_2 + \ldots + X_{2n-1}X_{2n} + tX_0 + t_1) \\ &\simeq \Lambda[[t, X_0, X_1, \ldots, X_{2n}]] \end{aligned}$$

and therefore A is formally smooth over Λ .

The above statement 6.7 yields a similar result as 6.4, proof of which we leave to the readers.

Corollary 6.8. Let X_0 be a k-scheme of finite type with isolated non-smooth points only such that $H^2(X_0, \underline{D}^0_{X_0}) = 0$, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Denote by $\{z_1, \ldots, z_m\}$ the set of non-smooth points in X_0 , and assume that $chark = 2$, $dimX_0$ is even, and all z_i's are ordinary quadratic singularities. Then

(1) R is formally smooth over Λ

(2) $I_R(\mathfrak{X}_{(z_i)}|R)$ is a principal ideal and in turn so is

$I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R)$. Furthermore, $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2)$ and $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over Λ .

(3) $\mathfrak{X} \to \mathrm{Spec}(\Lambda)$ is formally smooth

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then $dim\underline{M}_{X_0} = 3g - 3 + a$ where $g = dim_k H^1(X_0, \underline{O}_{X_0})$ and $a = dim_k H^0(X_0, \underline{D}^0_{X_0})$.

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SGA 7

Exposé VII

BIEXTENSIONS OF SHEAVES OF GROUPS

by A. Grothendieck

0. Introduction

The notion of a biextension was introduced by D. Mumford [6], in the context of formal groups, as a convenient way of expressing the relations between the formal groups associated with an abelian scheme and the dual abelian scheme. In the present exposé we undertake a systematic study of this notion in the general context of sheaves of groups on a given site. In particular, we show that it fits remarkably simply into the classical cohomological formalism of $\overset{\mathrm{L}}{\otimes}$ and R$\underline{\mathrm{Hom}}$ (3.6.4 and 3.6.5). In the following exposé we shall make explicit the most interesting special cases, notably that of biextensions of group schemes by the multiplicative group $\underline{G}_m$ (working with such sites as the Zariski site, the fpqc site, or an intermediate site). We shall see, in particular, that if A and B are two abelian schemes over a scheme S, biextensions of (A, B) by $\underline{G}_m$ admit a remarkable interpretation as, essentially, the classical divisorial correspondences on A, B. This gives a particularly convenient explicit cohomological meaning to the notion of divisorial correspondence (already implicit in Tate duality for abelian varieties over p-adic fields). This interpretation of divisorial correspondences is also convenient for following the fate of such a correspondence when A and B (assumed to be defined over the fraction field of a discrete valuation ring) are made to degenerate by means of Néron theory.

The results of Exposé VIII will show, for example, that one obtains a biextension of the pair of identity components of the Néron models, hence a biextension of their connected special fibres, which plays the role of a specialization of the original divisorial correspondence. These results will be used in Exposé IX, in conjunction with the monodromy theorem (I 3), to prove certain results about the Néron models of abelian varieties, the most important being the "semistable reduction theorem" for the Néron model. I ≈134

Throughout what follows, unless expressly stated otherwise, we work in a fixed $\underline{U}$-topos denoted by $\underline{T}$; that is (SGA 4, Exposé IV), $\underline{T}$ is a category equivalent to the category of sheaves on a suitable $\underline{U}$-site (which may, if desired, be taken to be $\underline{T}$ itself, endowed with its canonical topology). In our terminology we shall often identify the objects of $\underline{T}$ with the sheaves of sets on $\underline{T}$. All groups, extensions of groups, and so forth considered below are relative to this topos $\underline{T}$.

1. Complements on extensions of groups

1.1. Let P and G be two groups of $\underline{T}$, and let

$$1 \longrightarrow G \xrightarrow{\ i\ } E \xrightarrow{\ j\ } P \longrightarrow 1 \tag{1.1.1}$$

be an extension of P by G. Thus E is a group, $j : E \longrightarrow P$ is an epimorphism of groups, and i induces an isomorphism of G with $\ker j$. We shall reinterpret the datum of such an extension in different terms, which will be convenient for the definition and study of biextensions.

1.1.2. First, the homomorphism i makes G act on E in two ways, on the left and on the right, by setting I 135

$$gxg' = i(g)xi(g')$$

for $g, g' \in G(S)$ and $x \in E(S)$, where S is any object of $\underline{T}$. If, by means of j, we regard E as an object "over P", that is, as an object of the induced topos $\underline{T}_{/P}$, this amounts to saying that the group $G_P = G \times P$ induced by G in $\underline{T}_{/P}$ acts on E on the left and on the right. Moreover, it is well known that each of these actions makes E a torsor (a left torsor and a right torsor, respectively) over P, with group G_P; more precisely still, it makes E a bitorsor under (G_P, G_P). Recall [2] that this

means that E is a right torsor E_d under the right action of G_P, and that the left action induces an isomorphism of G_P with the group of automorphisms of E_d (the sheaf of groups of automorphisms of E that commute with the right action of G_P on E). This condition is equivalent to the symmetric condition: the left action of G_P on E makes E a left torsor E_s, and the right action induces an isomorphism of the opposite group G_P^o with the group of automorphisms of E_s.

1.1.2.1. When $\underline{T}$ is the "point topos", that is, the category of sets (equivalent to the category of sheaves on the one-point topological space), so that P and G are ordinary groups, the datum of a bitorsor E under (G_P, G_P) is plainly equivalent to the datum, for each $p \in P$, of a bitorsor E_p under (G, G), namely the fibre of E at p. An analogous interpretation is available in the general case. Indeed, giving a bitorsor E under (G_P, G_P) is equivalent to giving, for every "point" $p \in P(S)$, where S is any object of $\underline{T}$, a bitorsor E_p under (G_S, G_S) over S, in such a way that the formation of E_p is "compatible with base change along every morphism $S' \longrightarrow S$". Thus, if $p' : S' \longrightarrow P$ is the composite morphism, one is also given an isomorphism of bitorsors from $E_{p'}$ to the inverse image of E_p under $S' \longrightarrow S$, and these isomorphisms satisfy the usual transitivity condition for a composite $S'' \longrightarrow S' \longrightarrow S$.

When stated in canonical language (SGA 1, Exposé VI), this reduces to the following tautology: giving a bitorsor under (G_P, G_P) over P is equivalent to giving a cartesian section, over the category $\underline{T}_{/P}$, of the fibred category of bitorsors under (G_S, G_S) over a variable base $S \in \mathrm{ob}\,\underline{T}_{/P}$. More precisely, the "value at P" functor induces an equivalence between the category of the cartesian sections in question and the category of bitorsors under (G_P, G_P). This follows trivially from the fact that P is a final object of the base category $\underline{T}_{/P}$ of our fibred category: for <u>every</u> category fibred over $\underline{T}_{/P}$, the "value at P" functor is an equivalence between the category of cartesian sections of that fibred category and its fibre category over P. Despite its tautological nature, however, this interpretation will be useful for expressing certain compatibilities in language close to the set-theoretic case, using points with values in an object S in place of ordinary points.

1.1.3. In terms of the extension structure (1.1.1), the bitorsor structure of 1.1.2 on E under (G_P, G_P) used only part of the multiplicative structure of E: namely, multiplication of pairs of "points" of which at least one comes from a point of G (via i). We shall now interpret the multiplicative structure of E as an additional structure on the bitorsor under consideration. I 136

For this purpose, first consider two sections p, p' of P. With the notation of 1.1.2.1, these give two bitorsors $E_p, E_{p'}$ under (G, G), namely the inverse images in E of the sections $p, p' : e \longrightarrow P$ (where e denotes the final object of $\underline{T}$). It is then clear that multiplication in E induces a morphism of objects of $\underline{T}$ I ≈137

$$\varphi : E_p \times E_{p'} \longrightarrow E_{pp'},$$

which, in terms of the left and right structures of the three bitorsors involved (and, as usual, using points with values in an arbitrary object $S \in \mathrm{ob}\,\underline{T}$), satisfies

$$\begin{cases} \varphi(gx, x') = g\varphi(x, x'), & \varphi(x, x'g') = \varphi(x, x')g', \\ \varphi(xg, x') = \varphi(x, gx'). \end{cases} \tag{1.1.3.0}$$

Here $g, g' \in G(S)$, $x \in E_p(S)$, and $x' \in E_{p'}(S)$. In terms of the contracted product [2]

$$E_p \overset{G}{\times} E_{p'},$$

the third relation says that φ factors as

$$E_p \times E_{p'} \longrightarrow E_p \overset{G}{\times} E_{p'} \xrightarrow{\varphi_{p,p'}} E_{pp'},$$

where the first arrow is the canonical quotient morphism. The first two relations, on the other hand, say that the morphism thus obtained,

$$\varphi_{p,p'} : E_p \overset{G}{\times} E_{p'} \longrightarrow E_{pp'}, \tag{1.1.3.1}$$

is a homomorphism of bitorsors, where the bitorsor structure on the source is the one for which the left (respectively, right) operations of G come from the left (respectively, right) operations of G on the first factor E_p (respectively, on the second factor $E_{p'}$).

Notice that, like every homomorphism of bitorsors under a fixed pair of groups, $\varphi_{p,p'}$ is necessarily an isomorphism. I≈138

The definition of the preceding isomorphism (1.1.3.1) immediately extends to the case in which p, p' are points of P with values in an arbitrary object S of $\underline{T}$, rather than sections of P: one simply applies the preceding discussion to the induced topos $\underline{T}_{/S}$ and to the extension of groups over S obtained from (1.1.1) by base change. The formation of the isomorphisms $\varphi_{p,p'}$ is then compatible with every base change $S' \longrightarrow S$. In other words, it corresponds to a homomorphism of cartesian sections of the fibred category over the category $\underline{T}_{/P\times P}$ of pairs of arrows $p, p' : S \rightrightarrows P$ with target P.

Giving a system of $\varphi_{p,p'}$ satisfying this condition is, of course, equivalent to giving $\varphi_{p,p'}$ in the "universal" case $S = P \times P$, where p, p' are the two projections pr_i $(i = 1, 2)$ from $P \times P$ to P. It is then an isomorphism of bitorsors under $(G_{P\times P}, G_{P\times P})$:

$$\varphi : \mathrm{pr}_1^*(E)\ \mathrm{pr}_2^*(E) \xrightarrow{\sim} \pi^*(E), \tag{1.1.3.2}$$

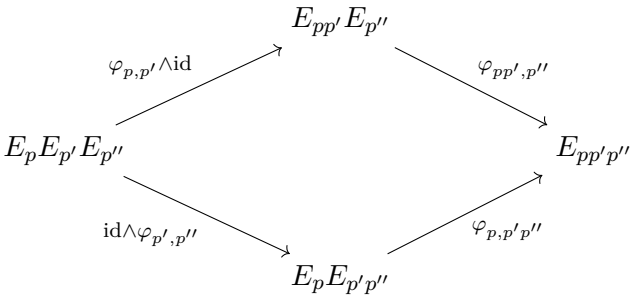
where

$$\pi : P \times P \longrightarrow P$$

is the morphism of objects of $\underline{T}$ defining the multiplicative structure of the group P. The compatibility properties below are most conveniently stated, however, in terms of the data (1.1.3.1), avoiding diagrams that are less close to intuition.

1.1.4. In writing the relations (1.1.3.0), which enabled us to express the given composition law on E in terms of isomorphisms of bitorsors (1.1.3.1), we used associativity of this composition law only for a product of three "points" of E of which at least one comes from a point of G. To express full associativity in terms of the $\varphi_{p,p'}$, one must require, for three points p, p', p'' of P with values in an object S of $\underline{T}$, that the following diagram be commutative: I139

(1.1.4.1)



In this diagram we have written the contracted product of bitorsors without the signs $\overset{G_S}{\times}$. It is understood that the contracted product $Q.R$ of two bitorsors is formed using, in passing to the quotient in $Q \times R$, the action of the structural group G on the right of Q and on the left of R. This contracted-product operation is associative up to a canonical isomorphism [2], and we make the identification

$$(E_pE_{p'})E_{p''} \simeq E_p(E_{p'}E_{p''}),$$

writing both sides simply as $E_pE_{p'}E_{p''}$. Obvious conventions of this kind will be used throughout the rest of this exposé without being expressly noted each time.

1.1.4.2. Just as the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) are determined by the "universal" case $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$ considered in (1.1.3.2), so the associativity condition (1.1.4.1) may likewise be checked only in the universal case $S = P \times P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, $p'' = \mathrm{pr}_3$. We leave it to the reader to write the diagram in whatever notation is preferred.

1.1.5. Conversely, suppose that the groups P and G of $\underline{T}$ are given, together with a bitorsor

$$j : E \longrightarrow P \tag{1.1.5.1}$$

under (G_P, G_P) (the two actions of G_P on E, or equivalently of G on E, being understood in the notation), and a system of isomorphisms of bitorsors over a variable base S,

$$\varphi_{p,p'} : E_p E_{p'} \xrightarrow{\sim} E_{pp'} \qquad (p, p' \in P(S)), \tag{1.1.5.2}$$

compatible with base change $S' \longrightarrow S$; in other words, an isomorphism of bitorsors (1.1.3.2). Suppose that the corresponding diagrams (1.1.4.1) commute. I claim that there is a unique extension structure on E of P by G such that the preceding data are obtained from it by the constructions of 1.1.2 and 1.1.3.

First observe that, on taking p, p' in (1.1.5.2) to be the identity section 1 of P, one obtains an isomorphism I≈140

$$\varphi_{1,1} : E_1 \xleftarrow{\sim} E_1 \overset{G}{\times} E_1. \tag{1.1.5.3}$$

Taking the contracted product, on the left say, of both sides with the inverse bitorsor E_1^{-1} of E_1 [2] therefore gives a canonical isomorphism

$$E_1 \simeq \text{the unit bitorsor } {}_sG_d \tag{1.1.5.4}$$

(that is, G acting on itself by left and right translations), in view of the canonical isomorphism I≈141

$$E_1^{-1}.E_1 \simeq \text{the unit bitorsor } {}_sG_d$$

of loc. cit., valid for every bitorsor. Equivalently, one has a section 1_E of E_1, hence of E,

$$1_E \in \Gamma(E_1) = \mathrm{Hom}(e, E_1) \subset \Gamma(E), \tag{1.1.5.5}$$

namely the image under (1.1.5.4) of the distinguished section of ${}_sG_d$, satisfying

$$g.1_E = 1_E.g \qquad (g \in G(S)).$$

By construction this section is characterized by

$$\varphi_{1,1}(1_E, 1_E) = 1_E. \tag{1.1.5.6}$$

On the other hand, giving the morphism of torsors φ of (1.1.3.2) is plainly equivalent to giving a morphism

$$\pi_E : E \times E \longrightarrow E, \qquad \text{also denoted } (x, y) \longmapsto xy, \tag{1.1.5.7}$$

satisfying

$$j(xy) = j(x)j(y) \qquad (x, y \in E(S)), \tag{1.1.5.8}$$

so that j is compatible with the multiplicative structures π_E on E and π on P, together with the three relations (1.1.3.0) (preferably beginning with the last of these). Finally define a morphism of objects of $\underline{T}$,

$$i : G \longrightarrow E,$$

by the formula

$$i(g) = g.1_E = 1_E.g. \tag{1.1.5.9}$$

The equality of the last two terms was already observed in constructing 1_E. I now claim that I≈142 the multiplication law π_E on E makes E a group with identity section 1_E, and that $j : E \longrightarrow P$ and $i : G \longrightarrow E$ are homomorphisms of groups making E an extension of P by G.

Indeed, associativity of the composition law on E is precisely the commutativity of the diagrams (1.1.4.1). The construction of π_E by means of the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) also shows that, for

every $x \in E(S)$, left and right translation by x in $E(S)$ are injective. Since $1_E 1_E = 1_E$ by (1.1.5.6), it follows at once that 1_E is a two-sided identity of E. We leave to the reader the verification that an $x \in E(S)$ lying over $p \in P(S)$ has an inverse; one uses the canonical isomorphism deduced from $\varphi_{p,p^{-1}}$,

$$E_{p^{-1}} \simeq (E_p)^{\mathrm{o}}. \tag{1.1.5.10}$$

Thus E is indeed a group. We have already observed that j is compatible with the composition laws (1.1.5.8), and the same is true of i, since, for $g, g' \in G(S)$,

$$\begin{aligned} i(g)i(g') &= (g.1_E)(g'.1_E) \\ &= (g.1_E.g')1_E && \text{(the last relation of (1.1.3.0))} \\ &= g.1_E.g' && \text{(since } 1_E \text{ is an identity)} \\ &= g.(1_E.g') = g.(g'.1_E) \\ &= (gg').1_E && \text{by (1.1.5.9)} \\ &= i(gg'). \end{aligned}$$

Since j is the structural projection of a torsor, it is an epimorphism of objects of $\underline{T}$. It remains only to see that i induces an isomorphism of G with $\ker j = E_1$, which is clear from the definition (1.1.5.9) and the construction of 1_E.

1.1.6. We shall accept that the preceding arguments prove the following. For a fixed topos $\underline{T}$, the preceding constructions give an equivalence between the category of extensions of groups of $\underline{T}$ as in (1.1.1), where P, G, E are all variable, and the category of systems (P, G, E, φ) in which P and G are groups of $\underline{T}$, E is a bitorsor under (G_P, G_P) over P, and φ is an isomorphism of bitorsors (1.1.3.2) (or, equivalently, a system of base-change-compatible isomorphisms of bitorsors (1.1.3.1)), satisfying the associativity condition expressed by commutativity of diagram (1.1.4.2) (or, equivalently, of the diagrams (1.1.4.1)). I 143

We leave it to the reader to spell out, using the preceding dictionary, the evident notion of inverse image (respectively, direct image) of an extension (1.1.1) under a homomorphism of groups

$$P' \longrightarrow P \qquad (\text{respectively, } G \longrightarrow G'),$$

or under both operations simultaneously. We likewise leave to the reader the compatibility of the preceding equivalence of categories with the inverse-image functor f^* associated with a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

For this one should use the datum φ in the form (1.1.3.2), in preference to the $\varphi_{p,p'}$ of (1.1.3.1); it then defines the corresponding datum $f^*(\varphi)$ on the topos $\underline{T}'$.

1.2. The commutative case. From 1.4 onward we shall be concerned only with commutative extensions of groups. We therefore assume that P and G are commutative, and spell out here, in terms of the bitorsor description (P, G, E, φ) of an extension, the condition that E be a commutative group.

An obviously necessary first condition—which in fact expresses that $i(G)$ be a central subgroup of E—is that the bitorsors E_p, for $p \in P(S)$, be ordinary torsors; that is, the left and right actions of G on E_p must coincide. Equivalently, E over P is an ordinary torsor under G_P. (Thus there will henceforth be no need to speak of bitorsors, and we shall consider only ordinary torsors.) Assuming this preliminary condition, commutativity of E is expressed by commutativity of the following diagrams, for $p, p' \in P(S)$: I ≈144

$$\begin{array}{ccc} E_p E_{p'} & \xrightarrow{\varphi_{p,p'}} & E_{pp'} \\ {\scriptstyle \mathrm{sym}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ E_{p'} E_p & \xrightarrow{\varphi_{p',p}} & E_{p'p}. \end{array} \tag{1.2.1}$$

Here the first vertical arrow is the canonical commutativity isomorphism (defined because G is now commutative); moreover $pp' = p'p$ because P is commutative. This condition is plainly equivalent to the corresponding condition in the universal situation $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, which we need not write out again.

1.3. Relations with rigidified or birigidified torsors

1.3.1.0. Let P be an object of $\underline{T}$ equipped with a section e_P, let G be a group of $\underline{T}$, and let E be a torsor under G_P. A rigidification of E, relative to the given section e_P of P, is a trivialization of the torsor $E_1 = e_P^*(E)$ under G;

that is, a section of this torsor. This terminology is inspired by the case where G is commutative I≈145
and

$$\mathrm{Hom}_{\mathrm{pt}}(P, G) \simeq (e), \tag{1.3.1.1}$$

where the left-hand side denotes the set of morphisms carrying e_P to the identity section e_G of G. This relation is equivalent to saying that the natural homomorphism

$$\Gamma(G) \xrightarrow{\sim} \mathrm{Hom}(P, G), \tag{1.3.1.2}$$

which associates with a section g of G the constant morphism $P \longrightarrow G$ with value g, is an isomorphism. It is immediate that these equivalent conditions are also equivalent to saying that every automorphism of the torsor E preserving the given rigidification is the identity automorphism (such automorphisms correspond precisely to the elements of the left-hand side of (1.3.1.1)). In particular, under condition (1.3.1.1), the category of torsors under G_P rigidified relative to e_P is "rigid".

1.3.2. Let, in addition, Q be an object of $\underline{T}$ equipped with a section e_Q, and now let E denote a torsor under $G_{P\times Q}$. Since e_Q defines a section $e_{P\times Q/P}$ of $P \times Q$ over P, one may consider rigidifications α of E relative to this section, working in the induced topos $\underline{T}_{/P}$; explicitly, these are sections of the torsor induced by E on $P \times e_Q$. Interchanging the roles of P and Q, one may likewise consider rigidifications β of E relative to the section $e_{P\times Q/Q}$ of $P \times Q$ over Q defined by e_P. Two such partial rigidifications are said to be compatible if the sections α and β agree on $e_P \times e_Q$, that is, if they define the same section of the torsor $E_{1,1}$ under G, the inverse image

of E under the section (e_P, e_Q) of $P \times Q$. A birigidification of the torsor E relative to the pair of I≈146
sections (e_P, e_Q) is a pair of compatible partial rigidifications.

If G is commutative, then E admits a birigidification (α, β) if and only if it admits partial rigidifications α and β; moreover one may find a birigidification in which α (or β) is prescribed in advance. Indeed, it suffices to correct β by the morphism $Q \longrightarrow G$ obtained as the composite

$$Q \longrightarrow e \xrightarrow{\lambda} G,$$

where λ is defined by $\beta e_Q = (\alpha e_P)\lambda$.

Still in the commutative case, if E, E' are two birigidified torsors under $G_{P\times Q}$, saying that they are isomorphic merely as $G_{P\times Q}$-torsors is equivalent to saying that they are isomorphic as birigidified $G_{P\times Q}$-torsors. In other words, if there is an isomorphism $u : E \longrightarrow E'$ of $G_{P\times Q}$-torsors, then there is one $v : E \longrightarrow E'$ that preserves the birigidifications. Indeed, measuring the discrepancy of u from the given birigidifications gives morphisms $f : P \longrightarrow G$ and $g : Q \longrightarrow G$ such that $f(e_P) = g(e_Q) = a \in G(e)$. There is then a morphism $P \times Q \longrightarrow G$ whose restrictions along $e_{P\times Q/P}$ and $e_{P\times Q/Q}$ are respectively f and g, namely

$$h(p, q) = f(p)g(q)a^{-1},$$

and one may take $v = uh$.

1.3.3. The terminology of 1.3.1 extends immediately to the case of a bitorsor E under (G_P, G_P). A rigidification of the bitorsor E is then required to be an isomorphism from $e_P^*(E)$ to the trivial bitorsor ${}_sG_d$; equivalently, it is defined by a central section e_E of $e_P^*(E)$, one satisfying

$$g.e_E = e_E.g \qquad (g \in G(S)).$$

The terminology of birigidifications extends in the same way to bitorsors.

1.3.4. Return to the case in which two groups P and G are given, and choose the identity section **I 147** e_P in order to define rigidifications of bitorsors over P and birigidifications of bitorsors over $P \times P$. There are natural functors

$$\textbf{EXT}(P;G) \xrightarrow{i} \mathrm{BITORSRIG}(P,G) \xrightarrow{\delta} \textbf{BITORSBIRIG}(P,P;G), \tag{1.3.4.1}$$

where the three expressions denote, respectively, the category of extensions of the group P by G, the category of bitorsors under (G_P, G_P) rigidified at e_P, and the category of bitorsors under $(G_{P\times P}, G_{P\times P})$ birigidified at (e_P, e_P). The functor i associates with an extension E of P by G the corresponding bitorsor over P (1.1.6), rigidified by the identity section of E. The functor δ is defined by

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}, \tag{1.3.4.2}$$

where $\pi : P \times P \longrightarrow P$ is the composition law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections. There is a canonical birigidification of $\delta(E)$. Indeed, the restriction of $\delta(E)$ to $P \times e_P$ is canonically isomorphic to

$$\mathrm{id}_P^*(E)\,\mathrm{id}_P^*(E)^{-1}\,e^*(E)^{-1},$$

where $e = \mathrm{pr}_2 \mathrm{inj}_1$ is the constant morphism $P \longrightarrow P$ with value e_P. The given rigidification of E provides an isomorphism from $e^*(E)$ to the trivial bitorsor, and hence an isomorphism of the restriction of $\delta(E)$ to $P \times e_P$ with the trivial bitorsor. Symmetrically one defines a partial rigidification along $e_P \times P$, and the two partial rigidifications are immediately seen to be compatible.

1.3.4.3. Observe that the composite δi of the functors in (1.3.4.1) is canonically isomorphic to **I 148** the trivial functor, that is, to the constant functor whose value is the trivial birigidified bitorsor over $P \times P$. Indeed, for every extension E of P by G, the isomorphism φ of (1.1.3.2) plainly defines a trivialization of $\delta i(E)$ compatible with its canonical birigidification.

Proposition 1.3.5. Let P and G be two groups of $\underline{T}$.

a) Suppose that every morphism of sheaves of sets $P \times P \longrightarrow G$ which is trivial on $P \times e_P$ and on $e_P \times P$ is trivial. Then the functor i of (1.3.4.1) is fully faithful. In particular, on a bitorsor E under (G_P, G_P) rigidified relative to e_P, that is, equipped with a central section e_E of $E_1 = e_P^*(E)$, there is at most one extension structure of P by G compatible with its bitorsor structure and having e_E as identity section.

b) Suppose in addition that every morphism of sheaves of sets $P \times P \times P \longrightarrow G$ which is trivial on the partial products $P \times P \times e_P$, $P \times e_P \times P$, and $e_P \times P \times P$ is trivial. Let E be a bitorsor under (G_P, G_P). In order that E admit an extension structure of P by G compatible with its bitorsor structure, it is necessary and sufficient that the bitorsor under $(G_{P\times P}, G_{P\times P})$ underlying

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}$$

be trivial. When this is so, for every central section e_E of $E_1 = e_P^*(E)$, that is, every rigidification of E relative to e_P, there is a unique extension structure on E compatible with its bitorsor structure and having e_E as identity section. In other words, E, regarded as a bitorsor rigidified relative to e_P, lies in the essential image of i. Moreover, the birigidified bitorsor $\delta(E)$ has one and only one trivialization.

Thus, in terms of diagram (1.3.4.1), the essential image of the fully faithful functor i is "the **I ≈149** kernel" of δ. More picturesquely, diagram (1.3.4.1) is an "exact sequence of pointed categories," and even of categories endowed with pseudo-group structures (in the terminology introduced in 2.5.1 below).

Let us prove 1.3.5. For a), let E, E' be two extensions of P by G, and let $f : E \longrightarrow E'$ be a morphism of the underlying rigidified bitorsors. We show that f is a morphism of extensions, or, what plainly amounts to the same thing, a homomorphism of groups. We must prove $f(xy) = f(x)f(y)$. If $j' : E' \longrightarrow P$ denotes the structural morphism of the torsor E' over P, then $j'f(xy) = j'(f(x)f(y))$: the two sides are respectively $j(xy)$ and $j(x)j(y)$, which are equal because j is multiplicative. Hence

$$f(xy) = u(j(x), j(y))f(x)f(y),$$

where

$$u : P \times P \longrightarrow G$$

is a uniquely determined morphism. Setting first $x = 1$ and then $y = 1$ in the preceding relation, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

Thus u is trivial on $P \times e_P$ and on $e_P \times P$; by hypothesis it is trivial, and therefore $f(xy) = f(x)f(y)$, as required.

For b), necessity of the stated condition for E to admit an extension structure compatible with its bitorsor structure is immediate from 1.3.4.3. It remains to prove sufficiency for the first assertion. Pulling a trivialization φ of $\delta(E)$ back along the identity section of $P \times P$ gives a trivialization of $E_1 = e_P^*(E)$ (compare 1.5.1), that is, a rigidification of E relative to e_P. Fix this rigidification. A given trivialization φ of $\delta(E)$ need not be compatible with the corresponding birigidification. Relative to the natural partial rigidifications of $\delta(E)$, its restrictions to $P \times e_P$ and $e_P \times P$ are described by central morphisms

$$f : P \longrightarrow G, qquad g : P \longrightarrow G,$$

whose values at the identity section of P coincide: $f(1) = g(1) = a \in \Gamma G$. The central morphism

$$h : P \times P \longrightarrow G, qquad h(p, q) = f(p)g(q)a^{-1},$$

has restrictions f and g to $P \times e_P$ and $e_P \times P$. Correcting φ by h, that is, replacing it by $\varphi(p, q)h(p, q)^{-1}$, therefore gives a trivialization of $\delta(E)$ compatible with its birigidification. Such a trivialization is unique: two of them differ by a central morphism $P \times P \longrightarrow G$ trivial on $P \times e_P$ and $e_P \times P$, hence trivial by hypothesis. This proves the last assertion of 1.3.5 b).

It remains to prove that φ, viewed as an isomorphism of the form (1.1.3.2), satisfies the associativity condition (1.1.4.1); by our choice of φ, e_E is already a two-sided identity for the composition law defined by φ. The two composites in (1.1.4.1) are isomorphisms of bitorsors under $(G_{P\times P\times P}, G_{P\times P\times P})$, so they differ by a central morphism I≈150

$$u : P \times P \times P \longrightarrow G$$

through the formula

$$(xy)z = u(jx, jy, jz)\, x(yz).$$

Setting successively $z = 1$, $x = 1$, and $y = 1$, and using the fact that 1 is a two-sided identity for the composition law, shows that u is trivial on the three partial products $P \times P \times e_P$, $e_P \times P \times P$, and $P \times e_P \times P$. The hypothesis then implies that u is trivial, so $(xy)z = x(yz)$. This completes the proof. I≈151

Corollary 1.3.6. Suppose that every morphism of sheaves of sets $P \times G \longrightarrow G$ which is trivial on $P \times e_G$ and on $e_P \times G$ is trivial. Then, if G is commutative, G is central in every extension of P by G. If P is also commutative and the hypothesis of 1.3.5 a) holds, then every extension E of P by G is commutative.

For the first assertion, we must prove that the action of P on G defined by the extension E is trivial. This action is expressed by a morphism $u : P \times G \longrightarrow G$. The restrictions to $P \times e_G$ and $e_P \times G$ of the corresponding morphism

$$(p, g) \longmapsto u(p, g)g^{-1}$$

are plainly trivial. Hence this last morphism is trivial, so $u(p, g) = g$, as required. Now, if E is an extension of P by G, its opposite group E^o is an extension of P^o by G^o, and the bitorsor structure underlying $i(E^o)$ is obtained from $i(E)$ by interchanging the left and right actions. When E is a central extension, one therefore has $i(E) = i(E^o)$; if, in addition, P is commutative, so that $P = P^o$, the uniqueness assertion 1.3.5 a) gives $E = E^o$. Thus E is commutative, proving the corollary.

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1.3.7. One interesting special case in which all the hypotheses of 1.3.5 and 1.3.6 hold is

$$\underline{\mathrm{Hom}}_{\mathrm{pt}}(P,G) = e \quad \text{(the final sheaf)}, \tag{1.3.7.1}$$

where the left-hand side denotes the sheaf of morphisms of pointed sheaves of sets from P to G. Independently of the marked sections, this condition may also be written in the form, analogous to (1.3.1.2),

$$G \simeq \underline{\mathrm{Hom}}_{\mathrm{sets}}(P,G), \tag{1.3.7.2}$$

where the second member denotes the sheaf of morphisms between the underlying sheaves of sets of P and G. These conditions also mean that, for every object Q of $\underline{T}$, the map $u \longmapsto u \circ \mathrm{pr}_2$

$$\mathrm{Hom}_{\mathrm{sets}}(Q,G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{sets}}(P \times Q, G) \tag{1.3.7.3}$$

is bijective. This property of P (with G fixed) is stable under Cartesian products; in particular it implies the same property for $P \times P$, $P \times P \times P$, and so forth. It follows immediately that it implies the hypotheses made in 1.3.5 and 1.3.6. It also implies that all the categories occurring in (1.3.4.1) are rigid, that is, every automorphism of an object of any one of them is the identity.

Moreover, since condition (1.3.7.1) on P, G is stable under every base change $S \longrightarrow e$, descent together with the uniqueness result 1.3.5 a) shows that a rigidified bitorsor E over P comes from an extension of P by G as soon as it does so locally over e. By 1.3.5 b), this may also be expressed by the vanishing of the section

$$\underline{\delta}(E) \in \Gamma R^1 h_*(G_{P\times P})$$

defined by $\delta(E)$, where

$$h : P \times P \longrightarrow e$$

is the structural morphism.

1.3.8. Let $p : P \longrightarrow S$ be a morphism of schemes which is coherent (that is, quasi-compact and quasi-separated) and flat, and suppose that I 153

$$p_*(\mathcal{O}_P) \xleftarrow{\sim} \mathcal{O}_S. \tag{1.3.8.1}$$

Since the formation of $p_*(\mathcal{O}_P)$ commutes with every flat base change (by coherence of p), the stated conditions on the relative scheme P/S are stable under Cartesian products and hence hold, in particular, for the Cartesian powers $P \times P$, $P \times P \times P$, and so forth. They are also stable under every flat base change.

Now let G be an affine scheme over S. Relation (1.3.7.2) follows at once from (1.3.8.1), since

$$\mathrm{Hom}_S(P,G) = \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, p_*(\mathcal{O}_P)) \simeq \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, \mathcal{O}_S),$$

where $\underline{A}$ is the quasi-coherent algebra over S such that $G = \mathrm{Spec}(\underline{A})$. By the preceding observation, the same conclusion applies to $P \times P$, $P \times P \times P$, and so on. Likewise, if G is flat over S, the same conclusion applies to the relative scheme P_G over G with respect to the relative scheme G_G over G.

When P and G are group schemes over S, and one works with a topology on $(\mathrm{Sch})_{/S}$ coarser than the canonical topology, so that P and G may be identified with sheaves on this site, every morphism from P, $P \times P$, $P \times P \times P$, and so on to G which is compatible with the marked sections is trivial. Similarly, if G is flat, every S-morphism $P \times G \longrightarrow G$ compatible with the marked sections is trivial, as one sees by interpreting it as a G-morphism from P_G to G_G. Thus the hypotheses of 1.3.5 hold, as do those of 1.3.6 when G is flat. Moreover, the three categories occurring in (1.3.4.1) are rigid.

1.3.8.2. The most interesting case in which (1.3.8.1) holds, and continues to hold after every extension of the base, is that in which P is an abelian scheme over S; then the conditions of 1.3.7 themselves hold. In this case the conclusions of 1.3.5 and 1.3.6 are therefore valid for every affine group G over S. This is the case originally considered by J. P. Serre, and the preceding developments were plainly inspired by it. An interesting case in which (1.3.8.1) holds without holding universally is I 154

that in which P is the Néron model of an abelian scheme over the fraction field of a discrete valuation ring with spectrum S.

1.4. Extensions of a free commutative group

Let I be an object of $\underline{T}$, and put

$$P = \mathbb{Z}[I], \tag{1.4.1}$$

the free $\mathbb{Z}$-module generated by I (SGA 4, IV, 11). With every extension E of P by a given $\mathbb{Z}$-module G (from now on only commutative group extensions will occur), associate the corresponding torsor under G_P, and then restrict it to I along the canonical morphism

$$I \longrightarrow P = \mathbb{Z}[I].$$

This gives a canonical functor I≈155

$$\mathrm{EXT}(P, G) \longrightarrow \mathrm{TORS}(I, G_I). \tag{1.4.2}$$

I claim that this functor is an equivalence of categories. Equivalently, for every G, the maps

$$\mathrm{Hom}_{\mathbb{Z}}(P, G) \longrightarrow H^0(I, G_I) = G(I) \tag{1.4.3}$$

and

$$\mathrm{Ext}^1_{\mathbb{Z}}(P, G) \longrightarrow H^1(I, G_I) \tag{1.4.4}$$

are isomorphisms. More generally, if A is a ring (not necessarily commutative) of $\underline{T}$ and $P = A[I]$ is the free A-module generated by I, there are canonical isomorphisms, functorial in the variable A-module G,

$$\mathrm{Ext}^i_A(A[I], G) \xrightarrow{\sim} H^i(I, G_I). \tag{1.4.5}$$

For $A = \mathbb{Z}$ and $i = 0, 1$, these coincide with (1.4.3) and (1.4.4). To define (1.4.5), recall that its left-hand side is the derived functor of

$$G \longmapsto \mathrm{Hom}_A(A[I], G),$$

which, by the definition of $A[I]$, is isomorphic to $G \longmapsto G(I) = H^0(I, G_I)$. Since $G \longmapsto G_I$ carries injective modules to injective modules of $\underline{T}_{/I}$ (SGA 4, V), the assertion follows. We leave to the reader the verification of the stated compatibility with (1.4.3) and (1.4.4).

Finally, the functor (1.4.2) is compatible with inverse images under a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

2. The notion of a biextension of abelian sheaves I156

2.0. Throughout this section, P, Q, G denote commutative groups of the topos $\underline{T}$. We shall frequently use the Cartesian diagram

$$\begin{array}{ccc} P & \longleftarrow & P \times Q \\ \downarrow & & \downarrow \\ e & \longleftarrow & Q \end{array} \tag{2.0.1}$$

and the groups G_P (respectively G_Q, $G_{P\times Q}$) over P (respectively Q, $P \times Q$), obtained by inverse image from the group G over e. We identify $G_{P\times Q}$ either with the inverse image $(G_P)_{P\times Q}$ under pr_1, or with the inverse image $(G_Q)_{P\times Q}$ under pr_2. In this notation the group structures of $P, Q, P \times Q$

play no role; indeed, we shall almost never use the product-group structure on $P \times Q$. We shall instead use on $P \times Q$ the structure of a group over P obtained by inverse image from Q over e,

$$P \times Q = Q_P, \tag{2.0.2}$$

and, similarly, the structure of a group over Q obtained by inverse image from P over e,

$$P \times Q = P_Q. \tag{2.0.3}$$

Thus over P we have the two groups G_P, Q_P, and over Q the two groups G_Q, P_Q.

If E is a torsor over $P \times Q$ under $G_{P\times Q}$, it may be viewed either as a torsor over Q_P under $G_{P\times Q} = (G_P)_{Q_P}$, or as a torsor over P_Q under $G_{P\times Q} = (G_Q)_{P_Q}$. It therefore makes sense to equip E with a commutative extension structure in either variable. Applied in the induced topoi $\underline{T}_{/P}$ and $\underline{T}_{/Q}$, 1.1.6 gives a convenient explicit description, using the canonical isomorphisms

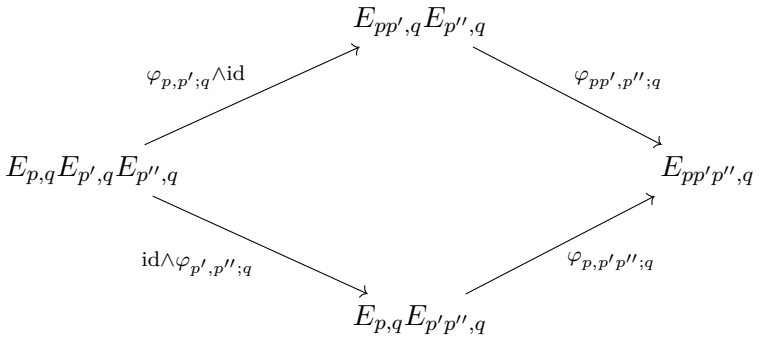
$$\underline{T}_{/P\times Q} \simeq (\underline{T}_{/P})_{/P\times Q}, \qquad \underline{T}_{/P\times Q} \simeq (\underline{T}_{/Q})_{/P\times Q}.$$

An extension structure in the P-direction—that is, of P_Q by G_Q—is equivalent to giving, for every object S of $\underline{T}$ and elements $p, p' \in P(S)$, $q \in Q(S)$, an isomorphism of torsors under G_S I≈157

$$\varphi_{p,p';q} : E_{p,q}E_{p',q} \xrightarrow{\sim} E_{pp',q}. \tag{2.0.4}$$

As usual (cf. 1.1.2.1), for $(p, q) \in (P \times Q)(S)$, $E_{p,q}$ denotes the torsor under G_S obtained from E by inverse image along $(p, q) : S \longrightarrow P \times Q$. The data (2.0.4) must be functorial in S, associative, and commutative. The associativity condition is the commutativity of

(2.0.5)



for $p, p', p'' \in P(S)$ and $q \in Q(S)$. Commutativity is expressed by

(2.0.6)

$$\begin{array}{ccc} E_{p,q}E_{p',q} & \xrightarrow{\varphi_{p,p';q}} & E_{pp',q} \\ \Big\downarrow{\scriptstyle \mathrm{sym}} & & \Big\downarrow{\scriptstyle \mathrm{id}} \\ E_{p',q}E_{p,q} & \xrightarrow[\varphi_{p',p;q}]{} & E_{p'p,q}. \end{array}$$

Here $p, p' \in P(S)$ and $q \in Q(S)$.

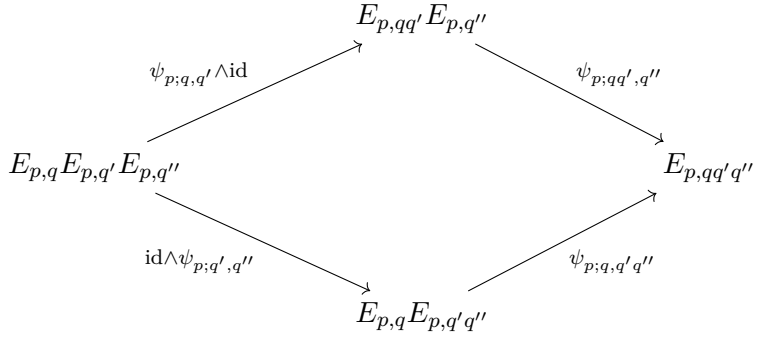
Symmetrically, an extension structure in the Q-direction—that is, of Q_P by G_P—is described by isomorphisms of torsors under G_S

$$\psi_{p;q,q'} : E_{p,q}E_{p,q'} \xrightarrow{\sim} E_{p,qq'}, \tag{2.0.7}$$

for $p \in P(S)$ and $q, q' \in Q(S)$. These are functorial in S and satisfy the symmetric associativity and

commutativity conditions

(2.0.8)



for $p \in P(S)$ and $q, q', q'' \in Q(S)$, and

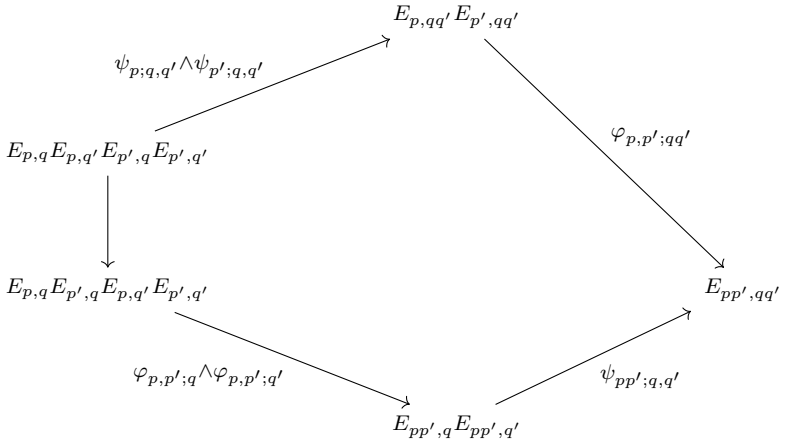
(2.0.9)

$$\begin{array}{ccc} E_{p,q}E_{p,q'} & \xrightarrow{\psi_{p;q,q'}} & E_{p,qq'} \\ \downarrow{\scriptstyle \mathrm{sym}} & & \downarrow{\scriptstyle \mathrm{id}} \\ E_{p,q'}E_{p,q} & \xrightarrow[\psi_{p;q',q}]{} & E_{p,q'q}. \end{array}$$

Here $p \in P(S)$ and $q, q' \in Q(S)$.

Definition 2.1. Let P, Q, G be abelian groups of $\underline{T}$, and let E be a torsor under $G_{P\times Q}$ equipped with an extension structure in the P-direction, defined by the isomorphisms (2.0.4), and an extension structure in the Q-direction, defined by the isomorphisms (2.0.7), each functorial in S and subject to the associativity and commutativity conditions above. The two structures are called compatible if, for every object S of $\underline{T}$ and $p, p' \in P(S)$, $q, q' \in Q(S)$, the following diagram commutes:

(2.1.1)



where the vertical arrow is induced by the symmetry isomorphism

$$E_{p,q'}E_{p',q} \xrightarrow{\sim} E_{p',q}E_{p,q'}.$$

A *biextension* of (P, Q) by G is a torsor E under $G_{P\times Q}$ equipped with compatible extension structures in the P- and Q-directions in the preceding sense.

2.1.1. Equivalently, a biextension structure on the $G_{P\times Q}$-torsor E is given by two partially defined composition laws $\varphi(x, y)$ and $\psi(x, y)$ on E. The first is defined when $\mathrm{pr}_2\, j(x) = \mathrm{pr}_2\, j(y)$, and the second when $\mathrm{pr}_1\, j(x) = \mathrm{pr}_1\, j(y)$, where $j : E \longrightarrow P \times Q$ is the structural morphism of the torsor. Besides associativity relations of the type I 160

(2.1.1.1) $$\varphi(gx, y) = \varphi(x, gy) = g\varphi(x, y) \qquad (x, y \in E(S),\ g \in G(S))$$

and the analogous relations for ψ, the axioms say that these partial composition laws are associative and commutative and satisfy

$$\varphi(\psi(x,y),\psi(z,t)) = \psi(\varphi(x,z),\varphi(y,t)), \tag{2.1.1.2}$$

where x, y, z, t lie respectively in $E_{p,q}(S), E_{p,q'}(S), E_{p',q}(S), E_{p',q'}(S)$.

2.2. Identity sections. Consider first the identity section of E regarded as an extension of Q_P by G_P,

$$e_{E/P} \in \Gamma(E/P) = \mathrm{Hom}_P(P,E), \tag{2.2.1}$$

and, symmetrically, the identity section of E regarded as an extension of P_Q by G_Q,

$$e_{E/Q} \in \Gamma(E/Q) = \mathrm{Hom}_Q(Q,E). \tag{2.2.2}$$

In particular, the torsor E over $P \times Q$ under $G_{P\times Q}$ is canonically trivialized over $P \times 1_Q$ by $e_{E/P}$, and over $1_P \times Q$ by $e_{E/Q}$. These trivializations coincide over the identity section $1_P \times 1_Q$ of $P \times Q$, and hence define the same section

$$e_E \in \Gamma(E), \tag{2.2.3}$$

which, by abuse of language, will be called the identity section (or neutral section) of E.

To prove the asserted equality, let e, e' be the two sections of E deduced respectively from $e_{E/P}$ and $e_{E/Q}$. They are characterized as the identity sections in the inverse image $E_{1_P,1_Q}$ of E along the identity section of $P \times Q$, for the two composition laws φ and ψ, respectively. Applying (2.1.1.2) to the four sections e, e', e', e of $E_{1,1}$ gives e on the left and e' on the right; hence $e = e'$. It follows moreover that the two composition laws φ and ψ on $E_{1,1}$ are identical. Indeed, under transport of structure by the morphisms $g \mapsto g.e$ and $g \mapsto g.e'$, respectively, both correspond to the group law of G, and $e = e'$. I≈161

The extension of Q_P by G_P over the base P, restricted over the identity section 1_P of P, gives an extension of Q by G on the inverse image of $1_P \times Q$ in E. The homomorphism $e_{E/Q}$ is immediately seen to be multiplicative for this extension structure, so the extension is canonically split. Symmetrically, the extension of P by G obtained by restricting the extension of P_Q by G_Q over Q to the identity section 1_Q is canonically split by $e_{E/P}$.

2.3. Morphisms of biextensions. Let E be a biextension of (P,Q) by G, and let E' likewise be a biextension of (P',Q') by G', where P', Q', G' are again commutative groups of $\underline{T}$. A morphism of biextensions from E to E' is a system (u,v,w,f), where I162

$$u : P \longrightarrow P', \qquad v : Q \longrightarrow Q', \qquad w : G \longrightarrow G' \tag{2.3.1}$$

are morphisms of groups, and

$$f : E \longrightarrow E' \tag{2.3.2}$$

is a morphism of the sheaves of sets underlying E and E', such that f is simultaneously a homomorphism of group extensions over P and P', respectively, associated with the relative group homomorphisms

$$u \times v : Q_P \longrightarrow Q'_{P'}, \qquad u \times w : G_P \longrightarrow G'_{P'},$$

and a homomorphism of group extensions over Q and Q', respectively, associated with the relative group homomorphisms

$$u \times v : P_Q \longrightarrow P'_{Q'}, \qquad v \times w : G_Q \longrightarrow G'_{Q'}.$$

Notice moreover that u, v, w are determined by f alone. Indeed, passing to the quotient recovers $u \times v$ from f, and then recovers u and v by restriction to $P \times 1_Q$ and $1_P \times Q$, respectively; restricting f to $G.e_E$ recovers w. Thus no ambiguity arises from identifying the morphism (u,v,w,f) with f.

After taking the usual precaution of adjoining their sources and targets to morphisms (the canonical genuflection), one sees that the biextensions of the topos $\underline{T}$ are the objects of a category, whose objects and morphisms are those defined in 2.1 and 2.3 and whose composition of morphisms is defined in the evident way. Moreover, $(P, Q, G, E) \mapsto (P, Q, G)$ gives a canonical functor

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.3}$$

where $\text{BIEXT}(\underline{T})$ denotes the category of biextensions over $\underline{T}$, and $\text{GRAB}(\underline{T})$ the category of abelian groups over $\underline{T}$. It is useful to decompose this canonical functor into the two functors $E \mapsto (P, Q)$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.4}$$

and $E \mapsto G$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}). \tag{2.3.5}$$

2.4. The category $\text{BIEXT}(P, Q; G)$ for variable P, Q, G. I **163**

I claim that the canonical functor (2.3.4) is fibred, and that the canonical functor (2.3.5) is cofibred. The first assertion means, in particular, that if E is a biextension of (P, Q) by G, and if

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q$$

are morphisms of abelian groups, then there is a biextension E' of (P', Q') by G and a morphism of biextensions

$$f : E' \longrightarrow E$$

compatible with u, v, id_G, having the usual universal property for morphisms of biextensions into E compatible with u, v, id_G.

The second assertion is expressed similarly. In particular, if E is a biextension of (P, Q) by G, and if

$$w : G \longrightarrow G'$$

is a morphism of abelian groups, then there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$g : E \longrightarrow E'$$

compatible with $\text{id}_P, \text{id}_Q, w$, having the usual universal property for morphisms of biextensions from E compatible with $\text{id}_P, \text{id}_Q, w$. We leave it to the reader to spell out the two constructions whose existence has just been asserted; they are obtained easily by paraphrasing the familiar analogous constructions for extensions of ordinary groups.

Thus, for three abelian groups P, Q, G of $\underline{T}$, write

$$\text{BIEXT}(P, Q; G) \tag{2.4.1}$$

for the category of biextensions of (P, Q) by G, defined as the fibre category of (2.3.3) over (P, Q, G) (so its morphisms are the morphisms of biextensions compatible with the identity morphisms of P, Q, G). Then this category depends covariantly on G and contravariantly on P, Q. Moreover, with these variances, the category $\text{BIEXT}(P, Q; G)$ depends additively on each of the three arguments P, Q, G. For example, for the argument G, this means precisely:

a) $\text{BIEXT}(P, Q; 0)$ is equivalent to the one-point category;

b) for two abelian groups G, G' of $\underline{T}$, the functor

$$\text{BIEXT}(P, Q; G \times G') \longrightarrow \text{BIEXT}(P, Q; G) \times \text{BIEXT}(P, Q; G') \tag{2.4.2}$$

deduced from the two projections $G \times G' \longrightarrow G$ and $G \times G' \longrightarrow G'$ is an equivalence of categories.

In the language of [4], one also says that, for fixed P, Q, the category of biextensions of (P, Q) by variable groups G is, via the functor $E \mapsto G$ induced by (2.3.5), an additive cofibred category over GRAB($\underline{T}$).

Remark 2.4.3. In the terminology of *loc. cit.*, the preceding cofibred category is not merely additive but even left exact [4, 2.2]. We shall not verify this property here, since it will reappear below in 3.6 as a consequence of more precise facts. We only point out that the general theory of *loc. cit.* would then describe the structure of this cofibred category, up to equivalence of fibred categories, in terms of a chain complex $L_\bullet$ in GRAB($\underline{T}$) depending only on P and Q. We shall also give below a direct construction of this complex as I 165

$$P \overset{\mathbb{L}}{\otimes} Q,$$

the tensor product taken in a derived category.

2.5. Structure of the category BIEXT$(P, Q; G)$.

The general formal considerations of [4, no. 1] give several useful consequences of the additive dependence of the categories BIEXT$(P, Q; G)$ on G.

2.5.1. First, for fixed G, the category BIEXT$(P, Q; G)$ has a composition law which assigns functorially to two biextensions E, E' of (P, Q) by G a product biextension, denoted $E.E'$. This construction is simply the analogue of the familiar product of two extensions of one commutative group by another. It is compatible with that construction whether E, E' are viewed as group extensions of Q_P by G_P, or as group extensions of P_Q by G_Q. In particular, as a torsor under $G_{P\times Q}$, EE' is the product of the torsors under $G_{P\times Q}$ defined by E and E'.

The formation of $E.E'$ is moreover associative and commutative up to canonical isomorphisms, I ≈166 and this composition law on BIEXT$(P, Q; G)$ has a two-sided neutral object, called the trivial biextension of (P, Q) by G. It may be constructed, as in *loc. cit.*, from the unique biextension E_0 of (P, Q) by the zero group 0, up to isomorphism, by means of the homomorphism $0 \longrightarrow G$. From this description one immediately obtains an alternative description of the trivial biextension as the trivial torsor

$$E = P \times Q \times G.$$

Its two group structures over P and over Q are obtained, respectively, from the product group structures on $Q \times G$ and $P \times G$ by the base changes $P \longrightarrow e$ and $Q \longrightarrow e$.

Finally, for every biextension E of (P, Q) by G, one can define the inverse biextension, denoted E^{-1}, together with an isomorphism from EE^{-1} to the trivial biextension. This determines E^{-1} up to unique isomorphism and makes its dependence on E functorial.

In short, the category BIEXT$(P, Q; G)$ behaves like a group object in the category **Cat** of categories, except that associativity, commutativity, unit, and inverse relations hold only up to specified isomorphisms. One may call it a pseudo-group in the 2-category **Cat**.

2.5.2. For fixed P, Q, G, the preceding properties imply the following facts. For every biextension I 167 E of (P, Q) by G, the monoid of endomorphisms of E is canonically isomorphic to the monoid of endomorphisms of the trivial biextension $\underline{1}$: an endomorphism u of $\underline{1}$ is sent to $u.\mathrm{id}_E$ on $\underline{1}.E \simeq E$. The endomorphism monoid of $\underline{1}$ is in fact a commutative group, denoted

$$\mathrm{Biext}^0(P, Q; G). \tag{2.5.2.1}$$

Its composition law is also given by $(u, v) \mapsto u.v \in \mathrm{End}(\underline{1}.\underline{1}) \simeq \mathrm{End}(\underline{1})$.

Likewise, the composition law $(E, E') \mapsto EE'$ equips the set

$$\mathrm{Biext}^1(P, Q; G) \tag{2.5.2.2}$$

of isomorphism classes of biextensions of (P, Q) by G with the structure of a commutative group; the inverse of the class of E is the class of E^{-1}.

2.5.3. Given a group homomorphism $G \longrightarrow G'$, the corresponding functor

$$\mathrm{BIEXT}(P, Q; G) \longrightarrow \mathrm{BIEXT}(P, Q; G') \tag{2.5.3.1}$$

is compatible, up to canonical isomorphisms, with the operations $E.E'$ and E^{-1} of 2.5.1. Consequently, the two evident maps

$$\text{Biext}^i(P,Q;G) \longrightarrow \text{Biext}^i(P,Q;G') \qquad (i=0,1) \tag{2.5.3.2}$$

are group homomorphisms. Moreover, if E is a biextension of (P,Q) by G and E' the corresponding biextension by G', the natural homomorphism

$$\text{Hom}(E,E) = \text{Aut}(E) \longrightarrow \text{Hom}(E',E') = \text{Aut}(E') \tag{2.5.3.3}$$

is precisely (2.5.3.2) for $i=0$.

2.5.4. The group $\text{Biext}^0(P,Q;G)$ admits a simple explicit description. If E is a biextension of I 168
(P,Q) by G, an automorphism u of E is first of all an automorphism of the underlying torsor under $G_{P\times Q}$, and hence is given by a morphism

$$u_0 : P\times Q \longrightarrow G$$

through the formula

$$u(x) = u_0(j(x))x.$$

Compatibility of u with the first, respectively the second, extension structure on E is equivalent to additivity of u_0 in the second, respectively the first, argument. Thus u is an automorphism of the biextension structure of E if and only if u_0 is bilinear. This gives a canonical bijection

$$\text{Biext}^0(P,Q;G) \simeq \text{Hom}(P\otimes Q, G). \tag{2.5.4.1}$$

It is readily checked that this bijection is independent of E, respects the group structures, and is functorial in G for the functoriality on the left described in (2.5.3.2).

Below we shall determine $\text{Biext}^1(P,Q;G)$ and find that, canonically and functorially in G, it is isomorphic to

$$\text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G).$$

2.6. In 2.5 only G was varied, while P and Q were held fixed. We now say a few words about variation in P,Q. The essential fact, from which the others follow formally, is that the contravariant dependence of $\text{BIEXT}(P,Q;G)$ on P,Q commutes with its covariant dependence on G. Thus the categories in question may be viewed as the fibre categories of a cofibred category

$$\text{BIEXT}'(\underline{T}) \longrightarrow \text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T}), \tag{2.6.1}$$

where the superscript op denotes the opposite category and the structural functor is $E \mapsto (P,Q,G)$. This cofibred category is additive in each of its three arguments, or, as we shall say, tri-additive.

By paraphrasing standard arguments, which we shall not reproduce, it follows that the composi- I ≈169
tion $E.E'$ in the fibre category $\text{BIEXT}(P,Q;G)$ may be described using the additive structure of any one of the three arguments: the three resulting pseudo-group structures are canonically isomorphic. Consequently, the abelian groups

$$\text{Biext}^i(P,Q;G), \qquad i=0,1, \tag{2.6.2}$$

may be regarded as two trilinear functors in P,Q,G, respectively in $\text{GRAB}(\underline{T})^{\text{op}}$, $\text{GRAB}(\underline{T})^{\text{op}}$, and $\text{GRAB}(\underline{T})$. Equivalently, they are trilinear functors of variable P,Q,G in $\text{GRAB}(\underline{T})$, contravariant in P,Q and covariant in G, with values in the category of abelian groups. For this variance, the isomorphism (2.5.4.1) is functorial in all of P,Q,G, and not merely in G. The analogous assertion will hold below for the description of Biext^1 by an Ext^1.

2.7. The symmetric biextension of a given biextension.

If E is a biextension of (P,Q) by G, define its symmetric biextension ${}^sE$, which is a biextension of (Q,P) by G. The underlying sheaf of sets of ${}^sE$ is that of E, and the actions of G on E and ${}^sE$ are the same. The structural morphism

$${}^sE \longrightarrow Q\times P$$

is the composite of $E \longrightarrow P \times Q$ with the symmetry $P \times Q \longrightarrow Q \times P$; finally, the left and right extension structures of ${}^sE$ are, respectively, the right and left extension structures of E. Evidently $E \mapsto {}^sE$ is an automorphism of the category $\mathrm{BIEXT}(\underline{T})$ of biextensions, and indeed an involutive automorphism:

$$ {}^s({}^sE) = E. $$

Every statement about bitorsors $E, \ldots$ therefore gives a symmetric statement upon being applied to ${}^sE, \ldots$. We may call it the statement obtained by *exchanging left and right*. In general we shall spell out only one of the two symmetric statements, choosing between left and right, for example, according to our political preferences.

2.8. Change of topos. Localization. I 170

Let

$$ f : \underline{T}' \longrightarrow \underline{T} $$

be a morphism of topoi. The inverse-image functor f^* transforms a biextension of (P, Q) by G into a biextension $f^*(E)$ of (f^*P, f^*Q) by f^*G, as always for a structure expressible in terms of finite projective limits and arbitrary inductive limits. To prove compatibility of the two extension structures on $f^*(E)$ induced by those on E, one expresses compatibility by commutativity of (2.1.1) in the universal case

$$ S = P \times P \times Q \times Q, $$

where p, p', q, q' are the four projections. As expected, $E \mapsto f^*(E)$ gives a canonical functor

$$ f^* : \mathrm{BIEXT}(\underline{T}) \longrightarrow \mathrm{BIEXT}(\underline{T}'). $$

In what follows we shall use without further mention the fact that every construction made with biextensions is compatible with inverse images under morphisms of topoi. This applies in particular to the product EE' and inverse E^{-1} considered in 2.5. I ≈171

One may take for $\underline{T}'$ an induced topos $\underline{T}_{/S'}$, and for f the canonical morphism, called the localization morphism (SGA 4 IV 5),

$$ f = i_{S'} : \underline{T}_{/S'} \longrightarrow \underline{T}. $$

Then $f^*(E)$ is called the biextension induced by E on S', and is also denoted, as usual, by $E|_{S'}$ or $E_{S'}$.

Since restriction has the usual transitivity properties, the categories $\mathrm{BIEXT}(\underline{T}_{/S})$, for variable $S \in \underline{T}$, form the fibres of a fibred category over $\underline{T}$ (SGA 1 VI 8). In fact, this category is even a stack over $\underline{T}$ in the sense of Giraud [2]: morphisms and objects in its fibres can be glued. For example, to give a biextension over S up to canonical isomorphism, it suffices to give a covering family (that is, an epimorphic family) $S_i \longrightarrow S$, a biextension E_i over each S_i (an object of $\underline{T}_{/S_i}$), and, for every pair (i, j), an isomorphism between $E_i|_{S_j}$ and $E_j|_{S_i}$, subject to the usual cocycle condition.

We call this the stack of biextensions over $\underline{T}$. Replacing $\underline{T}$ in (2.3.3), and hence also in (2.3.4) and (2.3.5), by $\underline{T}_{/S}$ for variable S gives morphisms of stacks over $\underline{T}$. The same applies to the operations $E.E'$ and E^{-1} of (2.5.1) in the categories $\mathrm{BIEXT}(P_S, Q_S; G_S)$ for variable S, where P, Q, G are fixed commutative groups over $\underline{T}$.

2.9. Relations with birigidified torsors. I 172

In the terminology of 1.3.2, we saw in 2.2 that, for every biextension E of (P, Q) by G, the underlying torsor structure under $G_{P\times Q}$ is canonically birigidified relative to the pair of identity sections of P and Q. This gives a canonical functor

$$ \mathrm{BIEXT}(P, Q; G) \xrightarrow{i} \mathrm{TORSBIRIG}(P, Q; G), \tag{2.9.1} $$

where the category on the right consists of birigidified torsors over $P \times Q$ under $G_{P\times Q}$.

Using the identity sections of P and Q, for every product of n objects each isomorphic to P or Q one obtains a canonical morphism inj_i from the i-th factor into that product. Thus, for a morphism

from such a product to G, one may speak of its restriction to the i-th factor. Consider the products

$$\text{(2.9.2)} \qquad \begin{cases} P \times Q, \\ P \times Q \times Q, \quad P \times Q \times Q \times Q, \\ P \times P \times Q, \quad P \times P \times P \times Q, \\ P \times P \times Q \times Q. \end{cases}$$

Proposition 2.9.3. Suppose that every morphism from one of the products (2.9.2) to G which is trivial on each factor of that product (embedded by the identity sections as just explained) is trivial. Then:

a) The canonical functor (2.9.1) is fully faithful, and both sides of (2.9.1) are rigid categories. In particular, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), there is at most one biextension structure on E compatible with its birigidified torsor structure.

b) Such a biextension structure exists on E if and only if E admits an extension structure of Q_P by G_P compatible with its torsor structure under G_P, and an extension structure of P_Q by G_Q compatible with its torsor structure under G_Q (cf. 1.3.5(b)). In that case there is a unique extension structure of Q_P by G_P compatible with the torsor structure under G_P and the given rigidification over $P \times e_Q$, and a unique extension structure of P_Q by G_Q compatible with the torsor structure under G_Q and the given rigidification over $e_P \times Q$; these two extension structures are compatible in the sense of 2.1. I≈173

Proof. a) If E is a birigidified torsor over $P \times Q$, its automorphism group identifies with the group of morphisms $P \times Q \longrightarrow G$ trivial on the two partial factors. By the hypothesis applied to $P \times Q$, this group contains only the identity automorphism, proving rigidity. Now let E, E' be biextensions of (P, Q) by G. To see that every morphism $f : E \longrightarrow E'$ of birigidified torsors is a morphism of biextensions, apply 1.3.5(a) to the two extension structures defined by E, E' over P and over Q, respectively. This uses the hypothesis for $P \times Q \times Q$ and $P \times P \times Q$ in (2.9.2).
b) Necessity is clear. For sufficiency, 1.3.5(b) first shows that the stated condition gives unique extension structures of Q_P by G_P and of P_Q by G_Q, compatible with the torsor structure and the rigidifications. Here the hypothesis is used for $P \times Q \times Q \times Q$ and $P \times P \times P \times Q$. It remains to prove that the two extension structures are compatible, that is, that diagram (2.1.1) commutes. Its two composites are isomorphisms of torsors over $P \times P \times Q \times Q$, and hence differ by a morphism

$$u : P \times P \times Q \times Q \longrightarrow G.$$

Because the partial trivializations of E are identity sections for the laws φ and ψ, respectively, u is trivial on each of the four factors. The hypothesis for $P \times P \times Q \times Q$ therefore makes u trivial. This proves the proposition.

Corollary 2.9.4. Let P, Q, G be commutative groups satisfying

$$\text{(2.9.4.1)} \qquad \underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e, \qquad \underline{\mathrm{Hom}}_{\mathrm{pt}}(Q, G) = e$$

(cf. 1.3.7). Then the hypothesis of Proposition 2.9.3 is satisfied, and conclusions (a) and (b) of that proposition hold. Moreover, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), it admits a biextension structure of (P, Q) by G compatible with its given structures if and only if the two morphisms of pointed sheaves which it defines,

$$\text{(2.9.4.2)} \qquad Q \longrightarrow R^1 p_*(G_P), \qquad P \longrightarrow R^1 q_*(G_Q),$$

where $p : P \longrightarrow e$ and $q : Q \longrightarrow e$ are the structural morphisms, are morphisms of groups.

The first assertion was proved in 1.3.7. For the second, necessity is clear. To prove sufficiency, apply Proposition 2.9.3(b) and 1.3.5(b). Multiplicativity of the first morphism in (2.9.4.2) says that the rigidified torsor E under G_P satisfies the condition of 1.3.5(b) locally over P. As observed in 1.3.7, that condition is then true globally over P, so E admits an extension structure of Q_P by G_P

compatible with its rigidified torsor structure. Symmetrically, E admits an extension structure of P_Q by G_Q compatible with its rigidified torsor structure under G_Q. The conclusion follows from Proposition 2.9.3(b).

Example 2.9.5. Let S be a scheme, let P and Q be abelian schemes over S, and let $G = \underline{G}_{m,S}$ be the multiplicative group scheme over S. We noted in 1.3.8 that (2.9.4.1) then holds. A birigidified torsor E over $P \times Q$ is equivalently a birigidified invertible sheaf over $P \times Q$. Recall that a divisorial correspondence on P, Q is the isomorphism class of such a birigidified invertible sheaf (see, for example, [1]). In the fppf topology, the morphisms (2.9.4.2) are

$$Q \longrightarrow \underline{\mathrm{Pic}}_{P/S}, \qquad P \longrightarrow \underline{\mathrm{Pic}}_{Q/S}, \tag{2.9.5.1}$$

defined by the divisorial correspondence. The theorem of the square (*loc. cit.*) says that these morphisms necessarily respect the group structures. Thus 2.9.4 shows that the category of biextensions of (P, Q) by $G = \underline{G}_{m,S}$ is equivalent to the category of birigidified invertible sheaves on $P \times Q$; it is therefore a rigid category, and its set of isomorphism classes identifies with the set of divisorial correspondences on (P, Q). This identification plainly respects the natural group laws on the two sets, the first being that of (2.5.2.2).

Remark 2.9.6. Additivity of the first morphism in (2.9.4.2) is equivalent to factorization of the second through the subsheaf $\underline{\mathrm{Ext}}^1(Q, G)$; note that, by 1.3.5, the canonical morphism

$$\underline{\mathrm{Ext}}^1(Q, G) \longrightarrow R^1 q_*(G_Q)$$

is injective. Likewise, additivity of the second morphism in (2.9.4.2) means that the first factors through $\underline{\mathrm{Ext}}^1(P, G)$. This gives group isomorphisms

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(Q, \underline{\mathrm{Ext}}^1(P, G)) \simeq \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q, G)). \tag{2.9.6.1}$$

In the situation of 2.9.5 these reduce to the familiar isomorphisms

$$\mathrm{Corr}(P, Q) \simeq \mathrm{Hom}(P, Q') \simeq \mathrm{Hom}(Q, P'), \tag{2.9.6.2}$$

where

$$P' = \underline{\mathrm{Ext}}^1(P, \underline{G}_m), \qquad Q' = \underline{\mathrm{Ext}}^1(Q, \underline{G}_m)$$

are the dual abelian schemes of P and Q, identified respectively with $\underline{\mathrm{Pic}}^0_{P/S}$ and $\underline{\mathrm{Pic}}^0_{Q/S}$. In both formulas, Hom denotes the group of group homomorphisms.

2.10. Various generalizations of the notion of biextension. We mention them for completeness, I 176
although we shall not use them.

2.10.1. One can define a biextension of (P, Q) by G without assuming P or Q commutative, only G being commutative. It is a torsor under $G_{P\times Q}$ equipped with central extension structures of Q_P by G_P and of P_Q by G_Q, each compatible with the torsor structure and compatible with one another, meaning that they make (2.1.1) commute. Most of the preceding developments remain valid, *mutatis mutandis*, for this more general notion. Even when P, Q are commutative, it is more general than the notion of 2.1, which is the case in which the two underlying relative group structures on E are commutative. In the noncommutative case, the category of biextensions of (P, Q) by G still depends additively on G, but not on P and Q.

2.10.2. Paraphrasing 2.1, for a fixed integer $n \geq 1$ one defines an n-extension of a sequence of I 177
n commutative groups $(P_1, P_2, \ldots, P_n)$ by a commutative group G. It consists of a torsor E under $G_{P_1\times\cdots\times P_n}$, equipped with n partial extension structures over the partial products of $n-1$ factors, with a compatibility relation of type (2.1.1) for every pair of distinct indices between 1 and n.

If $n \geq 3$, the P_i are abelian schemes over a scheme S, and $G = \underline{G}_{m,S}$ is the multiplicative group over S (compare 2.9.3), every n-extension of $(P_1, \ldots, P_n)$ by G is trivial.

2.10.3. The notion of biextension in 2.1 may be viewed as a variant of the notion of a commutative extension of commutative groups, that is, of an extension of $\mathbb{Z}$-modules. Let A be a ring of $\underline{T}$, assumed commutative for simplicity, and let P, Q, G be three A-modules. More generally one can define an A-module biextension of (P, Q) by G, a variant of an extension of A-modules.

Such a biextension is a biextension in the sense of 2.1 whose underlying partial extension structures are enriched to extension structures of A_P-modules and A_Q-modules, respectively, compatible with the relative module structures already present on Q_P, G_P and P_Q, G_Q. The two relative module structures on E may be interpreted as two actions of the underlying multiplicative monoid of A on the sheaf of sets underlying E, and these two actions are required to commute. This condition is added to the compatibility expressed by (2.1.1), and is itself expressed by an analogous diagram, which we leave to the reader. Finally, the first action of A on E is required to be distributive with respect not only to the first composition law on E but also to the second; the same is required of the second action. I≈178

The dictionary of 1.1.6 also has an A-linear variant. An extension of the A-module P by the A-module G amounts to a family of torsors E_p under G_p, parametrized by points $p \in P(S)$ for variable $S \in \mathrm{ob}\,\underline{T}$, with E_p varying $\underline{A\text{-linearly}}$, rather than merely additively, in p. Besides the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1), this is expressed by isomorphisms

$$\mu_{a,p} : {}^aE_p \xrightarrow{\sim} E_{ap} \qquad (p \in P(S),\ a \in A(S)),$$

where ${}^aE_p$ denotes the image of the G_S-torsor E_p under extension of the structure group along

$$g \longmapsto ag : G_S \longrightarrow G_S.$$

We leave to the reader the conditions on φ and μ which make these data correspond to an extension of A-modules E of P by G (cf. 3.5.4(b)).

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3. Supplements on homological algebra

The principal purpose of this section is to define the isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

announced above. For this we shall need certain developments in homological algebra which are of independent interest and will occupy sections 3.1 through 3.4. The developments in 3.1 and 3.3 may be regarded as supplements to [4], especially to section 6.9 therein.

3.1. Construction of an additive cofibred category $\underline{\Psi}_{L_\bullet}$ in terms of a chain complex $L_\bullet$ in an abelian category $\underline{C}$

Let $\underline{C}$ be an abelian category and let $L_\bullet$ be a chain complex in $\underline{C}$,

$$L_2 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \tag{3.1.1}$$

written with lower-degree conventions, so that the differential has degree -1. Thus $L_i = 0$ for $i < 0$. We shall define a cofibred category

$$\underline{\Psi} = \underline{\Psi}_{L_\bullet}$$

over $\underline{C}$; up to canonical equivalence of cofibred categories it is the category of the same name in [4, 6.9]. The objects of $\underline{\Psi}_{L_\bullet}$ are pairs

$$(X, \alpha),$$

where X is an extension of L_0 by an object A of $\underline{C}$, depending on X,

$$0 \longrightarrow A \xrightarrow{i} X \xrightarrow{j} L_0 \longrightarrow 0, \tag{3.1.2}$$

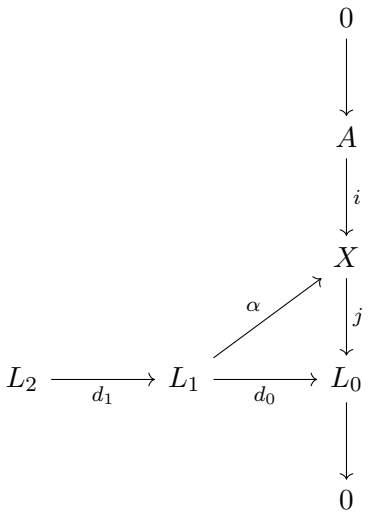
and α is a trivialization of the inverse image $d_0^*(X)$ of this extension by $d_0 : L_1 \longrightarrow L_0$. It is required that the corresponding trivialization of the inverse image $d_1^* d_0^*(X)$ by $d_1 : L_2 \longrightarrow L_1$ be the evident one, obtained from the transitivity isomorphism

$$d_1^* d_0^*(X) \simeq (d_0 d_1)^*(X)$$

and the relation $d_0 d_1 = 0$.

Equivalently, α is a lifting of $d_0 : L_1 \longrightarrow L_0$ to a morphism $\alpha : L_1 \longrightarrow X$, subject to $\alpha d_1 = 0$: I≈180

(3.1.3)



, $\alpha d_1 = 0$. (∗)

Here is a simpler description, pointed out by Deligne: $\underline{\Psi}_{L_\bullet}(A)$ is the category of extensions of the complex $L_\bullet$ by A, regarded as a complex concentrated in degree zero.

In this form one sees that the description of the objects (X, α) depends only on the chain complex $[L_\bullet]$ truncated in degree 1, obtained from $L_\bullet$ by "killing $H_i(L_\bullet)$ for $i \geq 2$": its terms are zero in degrees at least 2, agree with those of $L_\bullet$ in degrees at most 0, and its degree-1 term is the cokernel of $d_1 : L_2 \longrightarrow L_1$.

A homomorphism from (X, α) to (X', α') is a homomorphism of extensions $u : X \longrightarrow X'$, compatible with the identity on L_0, such that $u\alpha = \alpha'$. Homomorphisms are composed by composing the corresponding homomorphisms of extensions. This gives the announced category $\underline{\Psi}_{L_\bullet}$. Up to canonical isomorphism, it depends only on the degree-1 truncation $[L_\bullet]$ of $L_\bullet$.

There is an evident functor $(X, \alpha) \longmapsto A$, I≈181

(3.1.4) $$\pi : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{C}.$$

This functor is cofibred. Indeed, let $u : A \longrightarrow B$ be a morphism of $\underline{C}$, and let (X, α) be an object of the fibre $\underline{\Psi}_{L_\bullet}(A)$. Thus X is an extension of L_0 by A. Form the extension $u_*(X)$ of L_0 by B, the direct image of X under u, and equip it with the partial trivialization $\beta = f\alpha$, where $f : X \longrightarrow u_*(X)$ is the canonical morphism. It follows immediately that

$$f : (X, \alpha) \longrightarrow (u_*(X), \beta)$$

is cocartesian for (3.1.4): it has the usual universal property for u-morphisms in $\underline{\Psi}_{L_\bullet}$ with source (X, α). This is just the familiar and elementary corresponding universal property of $f : X \longrightarrow u_*(X)$.

Finally, the cofibred category $\underline{\Psi}_{L_\bullet}$ is additive; that is, it satisfies

(3.1.5) $\underline{\Psi}_{L_\bullet}(0)$ is equivalent to the one-point category,

and, for arbitrary A, B in $\underline{C}$,

(3.1.6) $\underline{\Psi}_{L_\bullet}(A \times B) \longrightarrow \underline{\Psi}_{L_\bullet}(A) \times \underline{\Psi}_{L_\bullet}(B)$ is an equivalence.

This follows immediately from the corresponding properties of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$. We shall freely use the consequences of the additivity of $\underline{\Psi}_{L_\bullet}$ established in [4, 1] and recalled above in 2.5.

3.1.7. Among these consequences, for every object A of $\underline{C}$ the category $\underline{\Psi}_{L_\bullet}(A)$ has a zero, or I182
trivial, object (2.5.1). It is the trivial extension $X = L_0 + A$ of L_0 by A, equipped with the partial trivialization $\alpha : L_1 \longrightarrow X$ whose components are d_0 and 0.

Moreover, by 2.5.2, for every (X, α) in $\underline{\Psi}_{L_\bullet}(A)$ the automorphism group of this object is canonically isomorphic to the automorphism group of the trivial object. Denote the latter commutative group by $\Psi^0_{L_\bullet}(A)$. The explicit description above gives the canonical isomorphisms

$$\Psi^0_{L_\bullet}(A) = \mathrm{Ker}\big(\mathrm{Hom}(L_0, A) \longrightarrow \mathrm{Hom}(L_1, A)\big) \simeq \mathrm{Hom}(H_0(L_\bullet), A) \simeq \mathrm{Ext}^0(L_\bullet, A), \tag{3.1.8}$$

induced by the usual identification of the automorphism group of the extension X with $\mathrm{Hom}(L_0, A)$. The isomorphism (3.1.8) is plainly functorial in A.

3.1.9. Finally, recall from 2.5.2 that the natural multiplicative structure on the fibre category $\underline{\Psi}_{L_\bullet}(A)$ endows its set of isomorphism classes with a natural group structure. We denote this group by $\Psi^1_{L_\bullet}(A)$; it is an additive functor of A.

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3.2. Determination of $\Psi^1_{L_\bullet}(A)$

Consider an object (X, α) (3.1.3) of $\underline{\Psi}_{L_\bullet}(A)$. Consider the cochain complex of length one

$$[X \longrightarrow L_0]: \quad 0 \longrightarrow X \longrightarrow L_0 \longrightarrow 0 \longrightarrow \cdots \tag{3.2.1}$$

defined by the morphism $j : X \longrightarrow L_0$, with X as its degree-zero component. It is a resolution of A, and hence may be identified with A as an object of the derived category $D(\underline{C})$ [5]. The diagram (3.1.3) then gives a degree-one morphism of complexes

$$L_\bullet \longrightarrow [X \longrightarrow L_0], \tag{3.2.2}$$

made explicit by the following diagram with anticommutative squares:

$$\begin{array}{ccccccccc} \cdots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 & \longrightarrow 0 & \cdots \\ & & & & \downarrow{\scriptstyle u^{-1}=-\alpha} & & \downarrow{\scriptstyle u^0=\mathrm{id}} & & \\ \cdots & & 0 & \longrightarrow & X & \xrightarrow{j} & L_0 & \longrightarrow 0 & \cdots \end{array} \tag{3.2.3}$$

(We use the usual convention which associates with the chain complex $L_\bullet$ a complex whose differential has degree 1 and whose degree-i component is L_{-i}.) We thus obtain a degree-one morphism in the derived category

$$c(X, \alpha) : L_\bullet \longrightarrow A \qquad (\text{a degree-one morphism in } D(\underline{C})), \tag{3.2.3}$$

which plainly depends only on the isomorphism class of (X, α). We have thereby defined a canonical map

$$\Psi^1_{L_\bullet}(A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A) \overset{\mathrm{def}}{=} \mathrm{Hom}_{D(\underline{C})}(L_\bullet, A[1]). \tag{3.2.4}$$

Theorem 3.2.5. The map (3.2.4) is an isomorphism of groups, functorial in the object A of $\underline{C}$. I 184

We leave to the reader the verification that (3.2.4) is additive and functorial in A, and shall confine ourselves to proving that it is bijective. For injectivity, let us first show that if the morphism of complexes u in (3.2.3) is zero in $D(\underline{C})$, then it is already zero in the homotopy category $K(\underline{C})$; that is, there exists a morphism

$$s : L_0 \longrightarrow X$$

such that

$$js = u^0 = \mathrm{id}_{L_0}, \qquad sd_0 = -u^{-1} = \alpha. \tag{3.2.5.1}$$

Indeed, the hypothesis says that the conclusion becomes true after the morphism of complexes under consideration is composed with a suitable quasi-isomorphism v

$$(*)\qquad \begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{j} & L_0 & \longrightarrow & 0 & & \\ & & \downarrow & & \downarrow & & & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{j'} & C^1 & \longrightarrow & C^2 & \longrightarrow & \cdots \end{array}$$

(and, after replacing C^i by 0 for $i \geq 2$ and C^1 by $\mathrm{Im}(C^0 \longrightarrow C^1) = \mathrm{Ker}(C^1 \longrightarrow C^2)$ if necessary, we may suppose that $C^i = 0$ for $i \geq 2$). But the morphism (*) induces the identity on A, and therefore identifies X with the fibre product of L_0 and C^0 over C^1. A morphism $s' : L_0 \longrightarrow C^0$ defining a homotopy $vu \sim 0$, and hence satisfying $j's' = v^1u^0$, consequently factors through a morphism $s : L_0 \longrightarrow X$ satisfying $js = u^0 = \mathrm{id}_{L_0}$. Thus s splits the extension X of L_0 by A, so the complex $X \longrightarrow L_0$ is isomorphic in $K(\underline{C})$ to the complex $K^\bullet$ concentrated in A. For the latter complex it is clear that

$$\mathrm{Hom}_{K(\underline{C})}(L_\bullet, K^\bullet) \longrightarrow \mathrm{Hom}_{D(\underline{C})}(L_\bullet, K^\bullet)$$

is an isomorphism, proving the assertion.

Now the choice of a morphism $s : L_0 \longrightarrow X$ satisfying (3.2.5.1) is precisely equivalent to a trivialization of the extension X of L_0 by A compatible with the partial trivialization α. This proves the injectivity of (3.2.4). I≈185

For surjectivity, start with an element of the right-hand side. By definition it may be described by a degree-one morphism of complexes

$$(3.2.5.2)\qquad \begin{array}{ccccccccc} \ldots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 & \longrightarrow & 0 \\ & & & & \downarrow{\scriptstyle u^{-1}} & & \downarrow{\scriptstyle u^0} & & \\ & & 0 & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & \ldots \end{array}$$

with anticommutative squares, where $C^\bullet$ is a resolution of A and, as above, we may suppose that $C^i = 0$ for $i \geq 2$. Thus C^0 is an extension of C^1 by A, and we may form the extension X obtained by pulling it back along $L_0 \longrightarrow C^1$. Anticommutativity in (3.2.5.2) shows that $-u^{-1}$ factors through a morphism $\alpha : L_1 \longrightarrow X$ which is a partial splitting, that is, $j\alpha = d_0$ (with j as in (3.1.2)). Moreover, the relation $u^{-1}d_1 = 0$ in (3.2.5.2) implies $\alpha d_1 = 0$. Hence (X, α) is indeed an object of $\underline{\Psi}_{L_\bullet}(A)$. By construction, the morphism (3.2.5.2) is the composite of the morphism (3.2.3) associated with (X, α) and the morphism

$$[X \longrightarrow L_0] \longrightarrow C^\bullet = [C^0 \longrightarrow C^1]$$

whose components are (f, u^0), where $f : X \longrightarrow C^0$ is the canonical morphism obtained from the definition of X as the pullback of the extension C^0.

The preceding morphism is a morphism of resolutions of A and induces the identity on A. It follows, by the definition $\mathrm{Ext}^1 = \mathrm{Hom}^1_{D(\underline{C})}$, that the given element of $\mathrm{Ext}^1(L_\bullet, A)$ described by (3.2.5.2) is precisely $c(X, \alpha)$. This proves the surjectivity of (3.2.4) and completes the proof of 3.2.5. I≈186

3.3. Properties of the cofibred category $\underline{\Psi}_{L_\bullet}$

In this section, which is not essential for understanding what follows, we freely use the notions of [4], our purpose being to compare the preceding results with those of that work.

Theorem 3.3.1. The additive cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ is left exact ([4, 2.2]), admits a quasi-maximal object ([4, 4.4]), and, for every object $E = (X, \alpha)$ of this category, there exists a typical complex $L^E_\bullet$ for E ([4, 3.1]). Moreover, the typical pro-complex of $\underline{\Psi}_{L_\bullet}$ ([4, 3.7 and 4.4]) is canonically isomorphic in $D(\underline{C})$ to the truncated complex $[L_\bullet]$ considered in 3.1.

We now spell out the meaning of the several assertions in 3.3.1.

3.3.2. To say that the cofibred category $\underline{\Psi}_{L_\bullet}$ is left exact means that it is additive and also satisfies the following condition:

3.3.2.1. For every exact sequence

$$0 \longrightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \longrightarrow 0$$

in $\underline{C}$, the natural functor (in the notation of [4, 1.8])

$$\underline{\Psi}_{L_\bullet}(A') \longrightarrow \underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$$

is an equivalence of categories.

Recall that if $\underline{\Psi}$ is a cofibred category over $\underline{C}$ and $v : A \longrightarrow A''$ is a morphism in $\underline{C}$, then $\underline{\Psi}([A \longrightarrow A''])$ is the category of pairs (E, ρ) in which E is an object of $\underline{\Psi}(A)$ and ρ is a "trivialization" of the corresponding object $v_*(E)$ of $\underline{\Psi}(A'')$; that is, an isomorphism from it to the zero object (2.5.1) of $\underline{\Psi}(A'')$. The definition of the functor

$$\underline{\Psi}(A') \longrightarrow \underline{\Psi}([A \longrightarrow A''])$$

in 3.3.2.1 is evident. When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the objects of $\underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$ are explicitly the triples (X, α, ρ), where (X, α) is as in (3.1.3) and $\rho : X \longrightarrow A''$ is a morphism in $\underline{C}$ satisfying

$$(*) \qquad \rho i = v, \qquad \rho\alpha = 0.$$

If $v : A \longrightarrow A''$ belongs to a short exact sequence as in 3.3.2.1, then, as is well known, giving a morphism satisfying the first relation (*) is equivalent, by the left exactness of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$, to giving an extension X' of L_0 by $A' \simeq \operatorname{Ker} v$ which is a subextension of X: one takes $X' = \operatorname{Ker} \rho$. The second relation (*) says that α factors through a morphism $\alpha' : L_1 \longrightarrow X'$. Up to the compatibilities whose verification is immediate, this proves the left exactness stated in 3.3.2.1.

3.3.3. Recall that an object E_0 of a cofibred category $\underline{\Psi}$ over $\underline{C}$ is called quasi-maximal if, for every object E of $\underline{\Psi}$ lying over an object A of $\underline{C}$, there is a morphism $E \longrightarrow E'$ in $\underline{\Psi}$ inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that E' is dominated by E_0, meaning that there is a morphism $E_0 \longrightarrow E'$. If $\underline{C}$ has enough injective objects, this is equivalent to saying that every object of $\underline{\Psi}$ over an injective object of $\underline{C}$ is dominated by E_0.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ there is the following canonical quasi-maximal object. Put

$$L_1' = \operatorname{Coker}(d_1 : L_2 \longrightarrow L_1)$$

and let $\bar{d}_0 : L_1' \longrightarrow L_0$ be the morphism induced by $d_0 : L_1 \longrightarrow L_0$. Let $E_0 = (X_0, \alpha_0)$ be the object of $\underline{\Psi}_{L_\bullet}(L_1')$ for which $X_0 = L_0 \oplus L_1'$ is the trivial extension of L_0 by L_1' and α_0 is the composite

$$L_1 \twoheadrightarrow L_1' \xrightarrow{(\bar{d}_0, \mathrm{id}_{L_1'})} L_0 \oplus L_1'.$$

To see that E_0 is quasi-maximal, observe that for every object (X, α) of $\underline{\Psi}_{L_\bullet}$ one can find a morphism from it to an object (X', α'), inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that the extension X' of L_0 by A' is split. Indeed, it suffices to take $A' = X$ and $(X', \alpha') = i_*(X, \alpha)$, with the notation of (3.1.2). On the other hand, it is immediate that any object (X', α') of $\underline{\Psi}$ for which the extension X' is split is dominated by E_0. I≈188

3.3.4. Let E be an object of an additive cofibred category $\underline{\Psi}$ over $\underline{C}$, lying above an object A of $\underline{C}$, and consider the following functor of the variable object B of $\underline{C}$:

$$(3.3.4.1) \qquad B \longmapsto \operatorname{Hom}(E, \theta_B),$$

where θ_B denotes the trivial object of $\underline{\Psi}(B)$. We say that E admits a typical complex if this functor is representable by an object Ω_E of $\underline{C}$. Since the functor in question maps canonically to the functor $B \longmapsto \mathrm{Hom}(A, B)$ represented by A, one obtains a canonical morphism

$$A \longrightarrow \Omega_E, \tag{3.3.4.2}$$

and the corresponding length-one chain complex $A \longrightarrow \Omega_E$ in $\underline{C}$ is called the typical complex of E.

When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ and $E = (X, \alpha)$, one checks immediately that the functor (3.3.4.1) is represented by $\mathrm{Coker}\,\alpha$, and that the canonical morphism (3.3.4.2) is simply the composite I≈189

$$A \xrightarrow{i} X \xrightarrow{\mathrm{can}} \mathrm{Coker}\,\alpha.$$

3.3.5. If an additive cofibred category $\underline{\Psi}$ over $\underline{C}$ admits a quasi-maximal object (3.3.3), and the typical complexes of its objects exist, then the typical complex $T_\bullet$ of a quasi-maximal object E_0, regarded as an object of the derived category $D(\underline{C})$, is independent, up to canonical isomorphism, of the choice of E_0 ([4, 4.4]). It is called the typical complex of the additive cofibred category $\underline{\Psi}$. If, in addition, $\underline{\Psi}$ is left exact (3.3.2), then $\underline{\Psi}$ can be reconstructed from the typical complex $T_\bullet$, up to an equivalence of fibred categories over $\underline{C}$ unique up to unique isomorphism ([4, 6.10]). There are also canonical isomorphisms, functorial in $A \in \underline{C}$ ([4, 5.4]),

$$\Psi^0(A) \xrightarrow{\sim} \mathrm{Ext}^0(T_\bullet, A), \qquad \Psi^1(A) \xrightarrow{\sim} \mathrm{Ext}^1(T_\bullet, A). \tag{3.3.5.1}$$

To complete the comparison with [4], we shall prove that when $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ its typical complex is the truncation $[L_\bullet]$. This gives another definition of the isomorphisms (3.1.8) and (3.2.4) in terms of (3.3.5.1), and one checks that it is compatible with the definitions in 3.1 and 3.2.

For this assertion we may plainly restrict to the case where $L_\bullet$ is already truncated in degree 1, since 3.1 noted that truncation does not change $\underline{\Psi}_{L_\bullet}$ up to an isomorphism of cofibred categories. The explicit construction in 3.3.4 then shows immediately that the typical complex of the quasi-maximal object described in 3.3.3 is canonically isomorphic to $L_\bullet$. I≈190

Remark 3.3.6. Let

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $\underline{C}$. It gives an exact sequence of Ext^i groups

$$0 \longrightarrow \mathrm{Ext}^0(L_\bullet, A') \longrightarrow \mathrm{Ext}^0(L_\bullet, A) \longrightarrow \cdots \longrightarrow \mathrm{Ext}^1(L_\bullet, A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A''),$$

which could of course be continued to the right by introducing Ext^i for $i \geq 2$. Using the isomorphisms (3.1.8) and (3.2.4), we obtain the exact sequence

$$\begin{aligned} 0 \longrightarrow \Psi^0_{L_\bullet}(A') \longrightarrow \Psi^0_{L_\bullet}(A) \longrightarrow \Psi^0_{L_\bullet}(A'') \xrightarrow{\partial} \Psi^1_{L_\bullet}(A') \\ \longrightarrow \Psi^1_{L_\bullet}(A) \longrightarrow \Psi^1_{L_\bullet}(A''). \end{aligned} \tag{3.3.6.1}$$

All arrows other than ∂ are defined simply by functoriality of $\Psi^0_{L_\bullet}$ and $\Psi^1_{L_\bullet}$. On the other hand, [4, 2.5.2] defines an exact sequence (3.3.6.1), for every left-exact cofibred category $\underline{\Psi}$ over $\underline{C}$, through a morphism

$$\partial : \Psi^0(A'') \longrightarrow \Psi^1(A'). \tag{3.3.6.2}$$

It associates with an element u of the left-hand side the class, in the right-hand side, of the object of $\underline{\Psi}(A')$, determined up to canonical isomorphism, whose image in $\underline{\Psi}([A \longrightarrow A''])$ (notation of 3.3.2) is the pair (X_0, u), where X_0 is the trivial object of $\underline{\Psi}(A)$.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the morphism (3.3.6.2) defined by this "geometric" construction is the negative of the boundary morphism ∂ in (3.3.6.1). Thus, after the identifications above and a change of sign in ∂, the latter exact sequence is precisely that of [4, 2.5.2]. This follows from the final assertion of [4, 5.4], in view of the compatibilities noted in 3.3.5. I≈191

Remark 3.3.6.3. We take this opportunity to clarify a point in [4, no. 3.11] concerning the fundamental isomorphisms (3.11.3) and (3.11.4), which play an essential role in the proof of [4, 5.4]. The diagram on page 35 of [4] is commutative, not anticommutative, and therefore does not as it stands define a morphism

$$u : [L_1^X \longrightarrow L_0^X] \longrightarrow K^\bullet[1].$$

It must be made anticommutative by replacing u^1 by $-u^1$. One could instead adopt the opposite convention and replace u^0 by $-u^0$; however, the stated convention, in which u^1 becomes $-u^1$, is the one for which (3.11.3) generalizes the isomorphism (3.5.3) of [4], as claimed there. With this clarification, the claim made in [4, 5.4] must also be corrected: the two boundary morphisms considered there differ by a sign.

3.3.7. In [4, 6.13] the variance of the cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ as $L_\bullet$ varies was also made explicit, in particular by reconstructing the category of cocartesian functors from $\underline{\Psi}_{L_\bullet}$ to $\underline{\Psi}_{L'_\bullet}$ in terms of the chain complexes $L_\bullet$ and $L'_\bullet$. The set of isomorphism classes of such cocartesian functors is canonically in bijection with I 192

$$\mathrm{Hom}_{D(\underline{C})}(L'_\bullet, L_\bullet).$$

We shall use only a small part of these results, which we develop in the next section using the simple description of $\underline{\Psi}_{L_\bullet}$ given in 3.1.

3.4. Variance of $\underline{\Psi}_{L_\bullet}$ with $L_\bullet$

Consider a morphism of complexes

$$f : L'_\bullet \longrightarrow L_\bullet \tag{3.4.1}$$

given by a commutative diagram

$$\begin{array}{ccccccccc} \longrightarrow & L'_2 & \longrightarrow & L'_1 & \longrightarrow & L'_0 & \longrightarrow & 0 \\ & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \\ \longrightarrow & L_2 & \longrightarrow & L_1 & \longrightarrow & L_0 & \longrightarrow & 0. \end{array} \tag{3.4.2}$$

We associate with it a canonical functor

$$f^* : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{\Psi}_{L'_\bullet}. \tag{3.4.3}$$

It is in fact a cocartesian functor of cofibred categories over $\underline{C}$: it is compatible with the structural functors to $\underline{C}$ and carries cocartesian morphisms to cocartesian morphisms; see SGA 1, Exposé VI.

If (X, α) is an object of $\underline{\Psi}_{L_\bullet}$, define $f^*(X, \alpha) = (X', \alpha')$ as follows. The extension X' of L'_0 by A is the pullback, along f_0, of the given extension X of L_0 by A, and α' is the partial trivialization deduced from α by transitivity in the second square of (3.4.2). The compatibility of α' with the relation $d'_0 d'_1 = 0$ follows immediately from the analogous compatibility for α. By construction, (3.4.3) commutes with the structural functors to $\underline{C}$. Its cocartesian character follows at once from the fact that, in the category of extensions of objects of $\underline{C}$, direct images of extensions commute with inverse images of extensions.

The functor (3.4.3) induces morphisms of functors

$$f^i : \Psi^i_{L_\bullet} \longrightarrow \Psi^i_{L'_\bullet}, \qquad i = 0, 1 \tag{3.4.4}$$

(compare 2.6). The morphism (3.4.1), on the other hand, induces morphisms

$$f^i : \mathrm{Ext}^i(L_\bullet, -) \longrightarrow \mathrm{Ext}^i(L'_\bullet, -), \qquad i \in \mathbb{Z}. \tag{3.4.5}$$

For $i = 0, 1$, the morphisms (3.4.4) and (3.4.5) are compatible with the isomorphisms (3.1.8) and (3.2.4). In other words, for $A \in \mathrm{Ob}\,\underline{C}$ and $i = 0, 1$, the following diagram is commutative:

$$\begin{array}{ccc} \Psi^i_{L_\bullet}(A) & \xrightarrow{\sim} & \mathrm{Ext}^i(L_\bullet, A) \\ \downarrow & & \downarrow \\ \Psi^i_{L'_\bullet}(A) & \xrightarrow{\sim} & \mathrm{Ext}^i(L'_\bullet, A). \end{array} \tag{3.4.6}$$

The verification of this compatibility is essentially immediate and is left to the reader.

Proposition 3.4.7. The functor (3.4.3) is faithful (respectively fully faithful, respectively an equivalence of categories) if and only if 1194

$$H_0(f) : H_0(L'_\bullet) \longrightarrow H_0(L_\bullet)$$

is an epimorphism (respectively, if $H_0(f)$ is an isomorphism and $H_1(f)$ is an epimorphism; respectively, if $H_0(f)$ and $H_1(f)$ are isomorphisms).

Indeed, with f^i denoting the functorial morphism (3.4.4), the required condition is that f^0 be a monomorphism (respectively, that f^0 be an isomorphism and f^1 a monomorphism; respectively, that f^0 and f^1 be isomorphisms). By the commutativity of (3.4.6), these conditions may be expressed by the analogous conditions on the functors Ext^i, $i = 0, 1$. The proposition then reduces to a straightforward statement of homological algebra, which we leave to the reader as an exercise.

3.4.8. Consider now an exact sequence of chain complexes

$$0 \longrightarrow L_\bullet \xrightarrow{f} L'_\bullet \xrightarrow{g} L''_\bullet \longrightarrow 0. \tag{3.4.8.1}$$

It gives functors of the form (3.4.3),

$$\underline{\Psi}_{L''_\bullet} \xrightarrow{g^*} \underline{\Psi}_{L'_\bullet} \xrightarrow{f^*} \underline{\Psi}_{L_\bullet}.$$

The transitivity property implicit in functors of the form (3.4.3) gives an isomorphism of functors

$$f^* g^* \simeq (gf)^* = (0_{L_\bullet \to L''_\bullet})^*.$$

The last member is evidently canonically isomorphic to the cocartesian zero functor from $\underline{\Psi}_{L''_\bullet}$ to $\underline{\Psi}_{L_\bullet}$. The preceding isomorphism therefore defines a canonical functor

$$\underline{\Psi}_{L''_\bullet} \longrightarrow \underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}. \tag{3.4.8.2}$$

Here $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$ denotes the category of pairs (X', t) in which X' is an object of $\underline{\Psi}_{L'_\bullet}$ and t is a trivialization of its inverse image $f^*(X')$, that is, an isomorphism

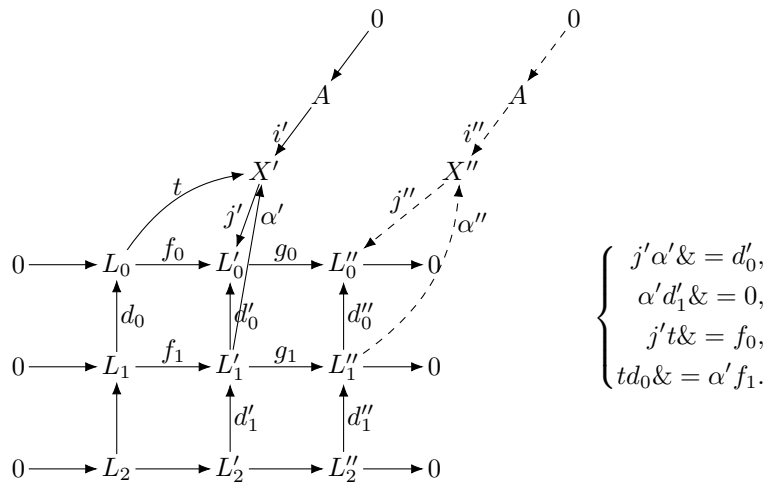
$$f^*(X') \simeq 0_{L_\bullet, A},$$

where $0_{L_\bullet, A}$ denotes the zero object of $\underline{\Psi}_{L_\bullet}(A)$ and A is the object of $\underline{C}$ underlying X'. We claim that the functor (3.4.8.2) is an equivalence of categories. Together with the left exactness in A established in 3.3.1, this completes the exactness property in $L_\bullet$ of the expression $\underline{\Psi}_{L_\bullet}(A)$.

Let us sketch the proof. To give an object of the target of (3.4.8.2) is to give an extension

$$0 \longrightarrow A \xrightarrow{i'} X' \xrightarrow{j'} L'_0 \longrightarrow 0$$

of L'_0 by an object A of $\underline{C}$, together with a morphism $\alpha' : L'_1 \longrightarrow X'$ satisfying $j'\alpha' = d'_0$ and $\alpha' d'_1 = 0$, and a morphism $t : L_0 \longrightarrow X'$ satisfying $j't = f_0$ and $td_0 = \alpha' f_1$. These data are displayed in the following diagram:



The familiar left exactness, in the first argument, of extensions of an object L by an object A, applied to the first row of the diagram, shows that a morphism t satisfying $j't = f_0$ allows one to interpret X' as the inverse image of an extension X'' of L''_0 by A. Under this interpretation, t corresponds to the natural trivialization of the inverse image of that extension along $g_0 f_0 = 0$.

The morphism α' satisfying $j'\alpha' = d'_0$ and $\alpha' f_1 = t d_0$ is interpreted by essentially the same dictionary, now applied to the second row of the diagram, as arising from a morphism $\alpha'' : L''_1 \longrightarrow X''$ satisfying $j''\alpha'' = d''_0$. Here α' and α'' are regarded as trivializations of the inverse images of X' and X'' along d'_0 and d''_0, respectively. Finally, the relation $\alpha' d'_1 = 0$ is then equivalent to $\alpha'' d''_1 = 0$. Thus the data (X', α', t), an object of $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$, are equivalent to the data (X'', α''), an object of $\underline{\Psi}_{L''_\bullet}$. This proves, in substance, that (3.4.8.2) is indeed an equivalence of categories.

3.4.9. Continue to consider the exact sequence (3.4.8.1), and the exact sequence of Ext^i groups that it defines:

$$(3.4.9.1)\quad 0 \longrightarrow \mathrm{Ext}^0(L''_\bullet, A) \xrightarrow{g^0} \mathrm{Ext}^0(L'_\bullet, A) \xrightarrow{f^0} \mathrm{Ext}^0(L_\bullet, A) \xrightarrow{\partial^0} \mathrm{Ext}^1(L''_\bullet, A) \xrightarrow{g^1} \mathrm{Ext}^1(L'_\bullet, A) \xrightarrow{f^1} \mathrm{Ext}^1(L_\bullet, A) \longrightarrow \cdots .$$

The canonical isomorphisms of the forms (3.1.8) and (3.2.4) give a geometric interpretation of the first six terms of this exact sequence in terms of the categories $\underline{\Psi}_{L_\bullet}$, $\underline{\Psi}_{L'_\bullet}$, and $\underline{\Psi}_{L''_\bullet}$. Compatibility (3.4.6) likewise gives a geometric interpretation of the arrows f^0, g^0, f^1, g^1. We shall now give such an interpretation of ∂^0, using the result just proved that (3.4.8.2) is an equivalence of categories.

Let

$$u \in \mathrm{Ext}^0(L_\bullet, A), \qquad \text{i.e.} \qquad u : L_0 \longrightarrow A, \quad u d_0 = 0.$$

Consider the object $(0, \overline{u})$ of $\underline{\Psi}_{[L_\bullet \to L'_\bullet]}$. Here 0 is the zero object of $\underline{\Psi}_{L'_\bullet}$, whose inverse image along f_0 is therefore identified with the zero object of $\underline{\Psi}_{L_\bullet}$, and $\overline{u}$ is the automorphism of the latter defined by u through the canonical isomorphism (3.1.8). Since (3.4.8.2) is an equivalence, $(0, \overline{u})$ comes from an object of $\underline{\Psi}_{L''_\bullet}(A)$, determined up to unique isomorphism. Its isomorphism class is an element of

$$\Psi^1_{L''_\bullet}(A) \simeq \mathrm{Ext}^1(L''_\bullet, A).$$

We claim that this element is precisely $\partial^0(u)$.

We prove this compatibility by reducing it to the analogous one in [4, Cor. 5.5], where the sign must be corrected as indicated in 3.3.6.3. The argument also gives another proof of the equivalence (3.4.8.2), by reducing it to [4, Prop. 2.7]. Consider the opposite category $\underline{C}^{\mathrm{o}}$ and interpret the construction of $\underline{\Psi}_{L_\bullet}(A)$, for fixed A, together with the constructions of 3.4.8 and the present section, in terms of $\underline{C}^{\mathrm{o}}$ and the category $\underline{E}$ cofibred over $\underline{C}^{\mathrm{o}}$ whose objects are extensions of the fixed object A of $\underline{C}^{\mathrm{o}}$ by a variable object L of $\underline{C}^{\mathrm{o}}$. This is an additive cofibred category over $\underline{C}^{\mathrm{o}}$, and is even left exact there (in the sense of [4, 2.2]). I≈197

A chain complex $L_\bullet$ of $\underline{C}$ may be interpreted as a cochain complex of $\underline{C}^{\mathrm{o}}$. Under this interpretation, the construction of $\underline{\Psi}_{L_\bullet}(A)$ coincides with the opposite of the category $\underline{\widehat{E}}(L_\bullet)$ in the sense

of [4, 1.8]. The functor (3.4.8.2) is then identified with the opposite of the functor of the form [4, (2.1.2)], with E replaced by its extension $\underline{\widehat{E}}$ to the category of cochain complexes. Consequently, the assertion that this functor is an equivalence is the special case [4, 2.7, a)⇒d)].

On the other hand, the exact sequence (3.4.9.1) is identified with the exact sequence of the groups $\mathrm{Ext}^i(A, L''_\bullet)$, $\mathrm{Ext}^i(A, L'_\bullet)$, and $\mathrm{Ext}^i(A, L_\bullet)$, apart from a factor $(-1)^i$ on ∂^i (Verdier dixit). Similarly, the isomorphisms (3.1.8) and (3.2.4) are identified with the analogous isomorphisms (3.11.5) and (3.11.4). The asserted compatibility is therefore the special case of [4, Cor. 5.5] corrected as above. I≈198

Remark 3.4.9.2. The editor, being very afraid of signs, confesses that he does not have complete confidence in the preceding proof: he fears that signs may have been omitted in passing from C to C^{o}. He recommends that a courageous reader verify the asserted compatibility directly (or, respectively, its opposite). Fortunately, these sign questions will play no role in the following exposés.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$-module

We apply the preceding results when $\underline{C}$ is the category of commutative groups of the topos $\underline{T}$, that is, the category of $\mathbb{Z}$-modules of $\underline{T}$. Let P be an object of $\underline{C}$. We shall define, functorially in P, a chain complex $L_\bullet(P)$ augmented to P:

$$0 \longrightarrow \overbrace{\mathbb{Z}[P\times P]\times\mathbb{Z}[P\times P\times P]}^{L_2} \xrightarrow{d_1} \overbrace{\mathbb{Z}[P\times P]}^{L_1} \xrightarrow{d_0} \overbrace{\mathbb{Z}[P]}^{L_0} \longrightarrow 0 \qquad (3.5.1)$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \downarrow{\scriptstyle\varepsilon}$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad P$$

with $L_i(P) = 0$ for $i \geq 3$. The components are displayed in (3.5.1), using the notation of 1.4. To describe the differentials d_0, d_1 and the augmentation ε, write $[p]$ for the section of $\mathbb{Z}[P](S)$ defined by $p \in P(S)$, and similarly write $[p,q]$ and $[p,q,r]$ for the corresponding sections of $\mathbb{Z}[P\times P](S)$ and $\mathbb{Z}[P\times P\times P](S)$. Then set

$$\begin{cases} \varepsilon[p] = p, \\ d_0[p,q] = [p+q] - [p] - [q], \\ d_1[p,q] = [p,q] - [q,p], \\ d_1[p,q,r] = [p+q,r] - [p,q+r] + [p,q] - [q,r]. \end{cases} \qquad (3.5.2)$$

These homomorphisms of commutative groups do define a complex $L_\bullet(P)$ augmented to P:

$$\varepsilon d_0 = 0, \qquad d_0 d_1 = 0.$$

The first relation is immediate. The second splits into two relations according to the two factors of $L_2(P)$; they express, respectively, commutativity and associativity of the composition law on P.

The augmented complex depends functorially on P. A morphism $u : P \longrightarrow P'$ gives a commutative diagram I≈200

$$\begin{array}{ccc} L_\bullet(P) & \xrightarrow{L_\bullet(u)} & L_\bullet(P') \\ \downarrow & & \downarrow \\ P & \xrightarrow{u} & P'. \end{array}$$

Its components are flat, as is every free $\mathbb{Z}$-module.

Proposition 3.5.3. One has $H_1(L_\bullet(P)) = 0$, and the augmentation ε induces an isomorphism

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

For the proof one can make the standard reduction to the case in which $\underline{T}$ is the category of sets and P is an ordinary $\mathbb{Z}$-module, where (3.5.1) is likely familiar. We instead give a direct proof that does not use this reduction, applying 3.4.7 to the augmentation

$$\varepsilon : L_\bullet(P) \longrightarrow P,$$

where P is regarded, as usual, as a chain complex concentrated in degree zero. It remains to prove that, for every object G of $\underline{C}$, the functor

$$(*) \qquad \varepsilon^* : \underline{\Psi}_P(G) \longrightarrow \underline{\Psi}_{L_\bullet(P)}(G)$$

is an equivalence of categories.

The source of (*) is plainly isomorphic to the category of extensions of the $\mathbb{Z}$-module P by G. To describe its target, use 1.4 to identify the categories of extensions of L_0, L_1, L_2 by G, respectively, with the category of torsors under G_P, the category of torsors under $G_{P\times P}$, and the category of pairs consisting of a torsor under $G_{P\times P}$ and a torsor under $G_{P\times P\times P}$.

Thus an object of the target of (*) is a pair (E, φ), where E is a torsor under G_P and φ is a trivialization of the torsor E' on $P \times P$ corresponding to the inverse image along d_0 of the extension of $\mathbb{Z}[P]$ by G defined by E. The trivialization is subject to the compatibility with $d_0 d_1 = 0$ described below. Formula (3.5.2) immediately gives I≈**201**

$$E' = \pi^* E \ \mathrm{pr}_1^*(E)^{-1} \ \mathrm{pr}_2^*(E)^{-1},$$

and a trivialization of this torsor is the same as an isomorphism, also denoted by φ,

$$\varphi : \mathrm{pr}_1^* E \ \mathrm{pr}_2^* E \xrightarrow{\sim} \pi^* E.$$

Expanding the compatibility of φ with $d_0 d_1 = 0$ gives the commutativity condition (1.2.1) and the associativity condition (1.1.4.1), corresponding to the two factors of L_2. The assertion that (*) is an equivalence is therefore precisely the dictionary of 1.1.6. This proves 3.5.3.

Remark 3.5.4. a) It would be interesting to strengthen 3.5.3 by constructing a resolution $L_\bullet(P)$ of P, functorial in P, whose components are functorially isomorphic to sums of free $\mathbb{Z}$-modules generated by cartesian powers of P. In view of isomorphisms of the form (1.4.5), this would give a method for partially computing $\mathrm{Ext}^i(P, A)$ as the abutment of a spectral sequence whose initial term is expressed through groups

$$H^j(P \times P \times \cdots \times P, A).$$

Here we obtain only very partial information about Ext^1, and no information about the higher Ext^i. There is, in any case, a natural way to continue the partial resolution $L_\bullet(P)$ by one degree.[2]

b) There is a variant of (3.5.1) and 3.5.3 in the category of A-modules, where A is a fixed ring of $\underline{T}$. The form of $L_\bullet(P)$ is more complicated because the description 1.1.6 must be enlarged (compare 2.10.3). One obtains I≈**202**

$$(3.5.5) \qquad \begin{cases} L_0 = A[P], \\ L_1 = A[P \times P] + A[A \times P], \\ L_2 = A[P \times P \times P] + A[P \times P] + A[A \times A \times P] + A[A \times P \times P]. \end{cases}$$

The augmented-complex structure is given by (3.5.2), supplemented by

$$(3.5.6) \qquad \begin{cases} d_0[\lambda, p] = [\lambda p] - \lambda[p], \\ d_1[\lambda, \mu, p] = [\lambda, \mu p] - [\lambda\mu, p] + \lambda[\mu, p], \\ d_1[\lambda, p, q] = [\lambda, p+q] - [\lambda, p] - [\lambda, q] + \lambda[p, q] - [\lambda p, \lambda q]. \end{cases}$$

[2] L. Breen constructed such a resolution (editor's note).

The relation $d_0 d_1 = 0$ follows from the analogous relation on the components appearing in (3.5.2), which, as we saw, is equivalent to associativity and commutativity of addition on P, together with the relations

$$d_0 d_1[\lambda, \mu, p] = 0, \qquad d_0 d_1[\lambda, p, q] = 0.$$

The latter express, respectively, associativity of the given action of A on P,

$$\lambda(\mu p) = (\lambda\mu)p,$$

and its distributivity over addition,

$$\lambda(p+q) = \lambda p + \lambda q.$$

For a torsor E under G_P, a trivialization of $d_0^* E$ amounts to an internal composition law on E (satisfying the compatibility conditions already described for the internal law on E and its torsor structure), together with an external composition law of A on E, compatible with the action of A on P and on G. The compatibility of this trivialization with the evident trivialization of $d_1^* d_0^* E$ is equivalent to the conditions already described above, which say that the internal law on E is a commutative-group law extending P by G, together with associativity of the external law and distributivity of the latter with respect to the internal law. These conditions say exactly that E, with its two composition laws and its torsor structure, is an A-module extension of the A-modules P and G. I ≈**203**

It follows, by essentially the same method as in 3.5.3, that $L_\bullet$ is a partial resolution of the module P, that is, $H_1(L_\bullet) = 0$. It would again be interesting to extend this partial resolution to an infinite resolution of the same kind, with components of the form

$$A[A \times A \times \cdots \times P \times P \times \cdots].$$

Compare the analogous question in a) for commutative groups.

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3.6. Application to the description of the cofibred category of biextensions of a given pair (P, Q) of commutative groups

Let P and Q be two $\mathbb{Z}$-modules of the topos $\underline{T}$. Choose a flat resolution $L'_\bullet(P)$ of P that agrees in degrees at most 2 with the augmented complex $L_\bullet(P)$ of 3.5. This is possible by 3.5.3 and because every $\mathbb{Z}$-module on $\underline{T}$ is a quotient of a free, hence flat, $\mathbb{Z}$-module. Similarly choose a flat resolution $L'_\bullet(Q)$ of Q agreeing with $L_\bullet(Q)$ in degrees at most 2. We have canonical homomorphisms

$$L_\bullet(P) \longrightarrow L'_\bullet(P), \qquad L_\bullet(Q) \longrightarrow L'_\bullet(Q), \tag{3.6.1}$$

and hence a canonical homomorphism

$$\begin{aligned} L_\bullet(P,Q) &\overset{\text{def}}{=} L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow L'_\bullet(P,Q) \\ &\overset{\text{def}}{=} L'_\bullet(P) \otimes L'_\bullet(Q) \simeq P \overset{\mathbb{L}}{\otimes} Q. \end{aligned} \tag{3.6.2}$$

It is an isomorphism on components in degrees at most 2. Consequently, by definition,

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}. \tag{3.6.3}$$

Theorem 3.6.4. The additive cofibred category over the category

$$\underline{C} = \mathrm{GRAB}(\underline{T})$$

of commutative groups of $\underline{T}$, consisting of biextensions of (P, Q) by a variable abelian group G (2.4), is canonically equivalent to the additive cofibred category

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}$$

defined from $L_\bullet(P,Q) = L_\bullet(P) \otimes L_\bullet(Q)$ as in 3.1.

This and 3.2.5 give the principal result of the present section, announced above:

Corollary 3.6.5. For fixed P and Q there is a canonical isomorphism, functorial in G, I 205

$$\mathrm{Biext}^1(P,Q;G) \xrightarrow{\sim} \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G). \tag{3.6.5.1}$$

To prove 3.6.5, first make explicit the components of degrees $0,1,2$ of $L_\bullet(P,Q)$:

$$\begin{cases} L_0(P,Q) = L_0(P) \otimes L_0(Q) = \mathbb{Z}[P \times Q], \\ L_1(P,Q) = L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \\ \qquad\quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times P \times Q], \\ L_2(P,Q) = L_0(P) \otimes L_2(Q) + L_1(P) \otimes L_1(Q) + L_2(P) \otimes L_0(Q) \\ \qquad\quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times Q \times Q \times Q] \\ \qquad\qquad + \mathbb{Z}[P \times P \times Q \times Q] \\ \qquad\qquad + \mathbb{Z}[P \times P \times Q] + \mathbb{Z}[P \times P \times P \times Q]. \end{cases} \tag{3.6.6}$$

To describe the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ through its fibres $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, use 1.4 to express extensions of $L_i(P,Q)$, for $i = 0,1,2$, by G in terms of torsors. An extension of $L_0(P,Q)$ by G is a torsor under $G_{P\times Q}$. An extension of $L_1(P,Q)$ by G is a pair consisting of a torsor under $G_{P\times Q\times Q}$ and one under $G_{P\times P\times Q}$. An extension of $L_2(P,Q)$ by G is similarly a system of five torsors, under the groups obtained from G by base change to

$$P\times Q\times Q, \quad P\times Q\times Q\times Q, \quad P\times P\times Q\times Q,$$
$$P\times P\times Q, \quad P\times P\times P\times Q.$$

Under this dictionary, a pair (X,α) consisting of an extension of $L_0(P,Q)$ by G and a trivialization of d_0^*X on $L_1(P,Q)$ becomes a torsor E on $P\times Q$ equipped with two trivializations (φ,ψ) of the torsors (E_1,E_2) on $P\times Q\times Q$ and $P\times P\times Q$, respectively, obtained from E by inverse image along

$$d_0 : L_1(P,Q) \longrightarrow L_0(P,Q).$$

Expanding d_0, as in the proof of 3.5.3, identifies φ and ψ with isomorphisms

$$\varphi : q_1^*E\, q_2^*E \xrightarrow{\sim} \pi_Q^*E, \qquad \psi : p_1^*E\, p_2^*E \xrightarrow{\sim} \pi_P^*E.$$

Here $\pi_Q, q_1, q_2 : P\times Q\times Q \longrightarrow P\times Q$ are induced by the composition law and the first and second projections $Q\times Q \longrightarrow Q$; the maps

$$\pi_P, p_1, p_2 : P\times P\times Q \longrightarrow P\times Q$$

are defined symmetrically. Thus φ and ψ are data of the forms (2.0.4) and (2.0.7) appearing in 2.1.

Finally, the condition that (X,α) be an object of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, namely compatibility of the partial trivialization α with $d_0d_1 = 0$, becomes five compatibility relations for the structure (E,φ,ψ) on torsors over, respectively,

$$P\times Q\times Q, \quad P\times Q\times Q\times Q, \quad P\times P\times Q\times Q,$$
$$P\times P\times Q, \quad P\times P\times P\times Q.$$

They correspond to the five terms in the expression (3.6.6) for $L_2(P,Q)$. As in the proof of 3.5.3, the first two are precisely the commutativity relation (2.0.6) and the associativity relation (2.0.5); together they say that φ gives E the structure of a commutative extension of Q_P by G_P. Symmetrically, the last two say that ψ gives E the structure of a commutative extension of P_Q by G_Q. The middle relation, on torsors over $P\times P\times Q\times Q$, is exactly the commutativity of diagram (2.1.1). It

says that φ and ψ are compatible, hence, in view of the other conditions, define on E a biextension structure of (P, Q) by G.

The verification of these interpretations is immediate and is left to the reader. The preceding discussion indeed defines an equivalence of categories cofibred over $\underline{C} = \mathrm{GRAB}(\underline{T})$, as asserted in 3.6.4. I≈206

Remark 3.6.7. Each of the three generalizations of biextensions mentioned in 2.10 has a corresponding variant of 3.6.4 and hence of 3.6.5. The generalization to any number of factors (2.10.2) is immediate, as is the generalization to biextensions of A-modules (2.10.3), once the variant of $L_\bullet(P)$ described in 3.5.4 b) has been constructed. From the viewpoint of homological algebra, the case in which P and Q are not assumed commutative (2.10.1) is more interesting. I 207

Associate with a not necessarily commutative group P of $\underline{T}$ a complex $K_\bullet(P)$ truncated in degree 2. It is described as $L_\bullet(P)$ was in 3.5, except that

$$K_2(P) = \mathbb{Z}[P \times P \times P],$$

the first factor occurring in $L_2(P)$ being omitted; d_0 and d_1 are defined as in (3.5.2). Then

$$H_0(K_\bullet(P)) = P/[P, P] \simeq \underline{H}_1(P, \mathbb{Z}),$$

and similarly

$$H_1(K_\bullet(P)) \simeq \underline{H}_2(P, \mathbb{Z}).$$

Indeed, $K_\bullet(P)$ is precisely the truncation in degree 2 of the chain complex $C_\bullet(P, \mathbb{Z})$ of P with coefficients in $\mathbb{Z}$.

The argument proving 3.5.3 also shows that the cofibred category $\underline{\Psi}_{K_\bullet(P)}$ over $\underline{C} = \mathrm{GRAB}(\underline{T})$ is canonically equivalent to the category of central extensions of P by variable commutative groups G. Likewise, for two groups P, Q, the cofibred category of biextensions of (P, Q) in the sense of 2.10.1 is equivalent to $\underline{\Psi}_{K_\bullet(P,Q)}$, where

$$K_\bullet(P, Q) = K_\bullet(P) \overset{\mathbb{L}}{\otimes} K_\bullet(Q).$$

These considerations suggest that, when P is commutative, the canonical resolution sought in 3.5.4 a) might be obtained by a suitable modification of $C_\bullet(P, \mathbb{Z})$. I≈208

3.7. Various compatibilities

In this section we examine the behaviour, for variable P and Q, of the equivalence of cofibred categories in 3.6.4. The variance of the cofibred category of biextensions of (P, Q) by a variable G with respect to variable P, Q is defined in principle in 2.4; it may also be expressed by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ with respect to variable P, Q is also clear, since $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ depends functorially on the pair (P, Q), while $\underline{\Psi}_{L_\bullet}$ depends contravariantly on $L_\bullet$ (3.4). With this understood, a first compatibility is expressed, for every triple of morphisms

$$P' \longrightarrow P, \qquad Q' \longrightarrow Q, \qquad G \longrightarrow G', \tag{3.7.1}$$

by the fact that the diagram of functors

$$\begin{array}{ccc} \underline{\Psi}_{L_\bullet(P,Q)}(G) & \longrightarrow & \underline{\Psi}_{L_\bullet(P',Q')}(G') \\ \downarrow & & \downarrow \\ \mathrm{BIEXT}(P, Q; G) & \longrightarrow & \mathrm{BIEXT}(P', Q'; G') \end{array} \tag{3.7.2}$$

is commutative up to a canonical isomorphism. We shall admit this compatibility: making the commutativity isomorphism explicit could only be as obvious as it is tedious (as, indeed, could the complete explication it presupposes of the equivalence of cofibred categories in 3.6.4).

3.7.3. It follows in particular from the preceding compatibility that the isomorphism (3.6.5.1) I 209
is functorial not only in G, but also in P and Q. One may state, similarly and more trivially, the analogous functoriality of the isomorphism defined through (3.1.8), just as (3.6.5.1) was defined through (3.2.4):

$$\mathrm{Biext}^0(P,Q;G) \xrightarrow{\sim} \mathrm{Ext}^0(P \overset{\mathbb{L}}{\otimes} Q, G) \simeq \mathrm{Hom}(P \otimes Q, G). \tag{3.7.4}$$

But such an isomorphism was already obtained in (2.5.4.1), and was observed in 2.6 to be functorial in its three arguments. It is immediate to verify that the two isomorphisms (3.7.4) and (2.5.4.1) are equal.

3.7.5. The value of the isomorphism in 3.6.5 is that it expresses the remarkable group of geometric origin $\mathrm{Biext}^1(P,Q;G)$ in terms of the very simple homological-algebra invariant $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$. The latter belongs to the sequence of invariants $\mathrm{Ext}^i(P \overset{\mathbb{L}}{\otimes} Q, G)$, to which the usual formalism of homological algebra applies. In particular, the sequence of these Ext^i is a cohomological functor in the sense of [3] (or a connected sequence of functors in the terminology of [5]) in each of the three arguments P, Q, G. Thus an exact sequence

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{3.7.5.1}$$

(respectively,

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \qquad \text{or} \qquad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0) \tag{3.7.5.2}$$

gives rise to an infinite exact sequence of the Ext^i. Its first six terms, which involve only Ext^0 and Ext^1, may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated in 3.3.6 a geometric interpretation of the exact sequence deduced from (3.7.5.1), and especially of the boundary map

$$\partial : \mathrm{Biext}^0(P,Q;G'') \longrightarrow \mathrm{Biext}^1(P,Q;G'), \tag{3.7.5.3}$$

within the framework of a left exact cofibred category. One may seek an analogous interpretation of the six-term exact sequence associated with either short exact sequence in (3.7.5.2), say the first; equivalently, in view of 3.7.3, one may seek a geometric interpretation of the boundary homomorphism

$$\partial : \mathrm{Biext}^0(P',Q;G) \longrightarrow \mathrm{Biext}^1(P'',Q;G). \tag{3.7.5.4}$$

Indeed, we have the following result.

Proposition 3.7.6. The cofibred category over I 210

$$\mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})$$

of biextensions of (P,Q) by G, with P, Q, G variable (cf. (2.6.1)), is left exact not only in the argument G, but also in each of the arguments P and Q. Moreover, the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i associated with (3.7.5.2), by means of the isomorphisms 3.6.5 and 3.7.4, is the boundary homomorphism in the theory of left exact cofibred categories over $\underline{C} = \mathrm{GRAB}(\underline{T})^{\mathrm{op}}$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P,Q) by G, where Q and G are fixed and P varies (subject to a possible sign error in 3.4.9; cf. 3.4.9.2).

We sketch the proof of 3.7.6. By 3.6.4 and compatibility (3.7.2), we interpret biextensions of I ≈211
(P,Q) by G as objects of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$. Consider the homomorphism

$$L_\bullet(P,Q) = L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q)$$

defined by the augmentation $L_\bullet(P) \longrightarrow P$. It follows from 3.5.3 and the flatness of $L_\bullet(Q)$ that this homomorphism induces isomorphisms on H_0 and H_1, and hence by 3.4.7 an equivalence

$$\underline{\Psi}_{P \otimes L_\bullet(Q)} \longrightarrow \underline{\Psi}_{L_\bullet(P,Q)}.$$

We may therefore interpret biextensions of (P,Q) by G as objects of $\underline{\Psi}_{P\otimes L_\bullet(Q)}(G)$. The exact sequence (3.7.5.2) now gives an exact sequence of chain complexes

$$0 \longrightarrow P'\otimes L_\bullet(Q) \longrightarrow P\otimes L_\bullet(Q) \longrightarrow P''\otimes L_\bullet(Q) \longrightarrow 0. \tag{3.7.6.1}$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence of complexes. (A direct verification would in any event be immediate.) Moreover, the homomorphism (3.7.5.4) is precisely the boundary homomorphism $\mathrm{Ext}^0 \longrightarrow \mathrm{Ext}^1$ deduced from this exact sequence by taking the Ext^i with values in G. The last assertion of 3.7.6 is consequently a special case of 3.4.9.

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3.8. Biextensions of (P,Q) trivialized over extensions of P and Q

First consider two exact sequences of abelian groups

$$0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \qquad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0. \tag{3.8.1}$$

We shall describe the category of biextensions E of (P,Q) by an abelian group G for which the inverse image $(u,v)^*(E)$ is trivial; that is, those for which there exists a homomorphism s making the following triangle commutative:

(3.8.2)

$$\begin{array}{ccc} E & & \\ \downarrow & \nwarrow^{s} & \\ P\times Q & \xleftarrow{u\times v} & L_0\times M_0 \end{array}$$

and satisfying the additivity condition required in each of its two arguments. The choice of such an s therefore identifies $(u,v)^*(E)$ with the trivial biextension $E_0 = L_0\times M_0\times G$ of (L_0,M_0) by G. The left exactness in P,Q of the cofibred category of biextensions of (P,Q) by fixed G (3.7.6) then shows that to give the biextension E of (P,Q), together with the trivialization s of its inverse image, is essentially equivalent to giving two partial trivializations of the trivial biextension E_0 of (L_0,M_0) by G, over (L_1,M_0) and (L_0,M_1) respectively, which agree over (L_1,M_1). Since the biextensions induced by E_0 over (L_1,M_0), (L_0,M_1), and (L_1,M_1) identify with the trivial biextensions, a partial trivialization over (L_1,M_0) (respectively over (L_0,M_1)) is equivalent to giving a homomorphism (or pairing)

$$f : L_1\otimes M_0 \longrightarrow G \qquad (\text{respectively } g : L_0\otimes M_1 \longrightarrow G). \tag{3.8.3}$$

The compatibility in question simply means that the two homomorphisms agree over $L_1\otimes M_1$, namely

$$f(x_1\otimes y_1) = g(x_1\otimes y_1), \tag{3.8.4}$$

for sections x_1, y_1 of L_1, M_1 over an object of $\underline{T}$. In terms of the homomorphism s of (3.8.2), the forms f and g are characterized by

$$s(x_1,y_0) = f(x_1\otimes y_0)e_{E/Q}(y_0), \qquad s(x_0,y_1) = g(x_0\otimes y_1)e_{E/P}(x_0), \tag{3.8.5}$$

which again makes clear the necessity of condition (3.8.4).

Once the trivialization s has been chosen, the other trivializations are the morphisms $s' : L_0\times M_0 \longrightarrow E$ of the form I ≈213

$$s'(x_0,y_0) = s(x_0,y_0) + h(x_0\otimes y_0), \tag{3.8.6}$$

where

$$h : L_0\otimes M_0 \longrightarrow G$$

is an arbitrary form. It is then clear from (3.8.5) that the forms f, g defined by s are changed into

$$\begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned} \tag{3.8.7}$$

Thus we obtain:

Proposition 3.8.8. The subgroup of $\mathrm{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image under (u, v) is a trivial biextension of (L_0, M_0) (that is, the kernel of $\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)$) is canonically isomorphic, by the preceding description, to the quotient of the group of pairs of forms (f, g) in (3.8.3) satisfying compatibility condition (3.8.4), by the pairs of forms which arise by restriction from a form $h : L_0 \otimes M_0 \longrightarrow G$.

We now interpret this isomorphism in terms of homological algebra, taking into account the I≈214
canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

of 3.6.5. To this end, interpret the data (3.8.1) as two complexes resolving P and Q, respectively:

$$L_\bullet : \ 0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad M_\bullet : \ 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0, \tag{3.8.9}$$

and consider the tensor-product complex $L_\bullet \otimes M_\bullet$

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0.$$

There is, moreover, a canonical homomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes} Q \longrightarrow L_\bullet \otimes M_\bullet,$$

and hence an equally canonical homomorphism

$$H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)) \longrightarrow \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G). \tag{3.8.10}$$

The group of 1-cocycles of $\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f, g) as in (3.8.3) which satisfy compatibility condition (3.8.4): to such a pair one associates the homomorphism $F : L_1 \otimes M_0 + L_0 \otimes M_1 \longrightarrow G$ with components (f, g),

$$F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1). \tag{3.8.11}$$

Moreover, the coboundary of the 0-cochain $h : L_0 \otimes M_0 \longrightarrow G$ is precisely the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Consequently, the group of pairs (f, g) modulo the pairs (h, h) described in 3.8.8 is canonically isomorphic to the first term of (3.8.10). With this understood, one has the following compatibility.

Proposition 3.8.12. If I 215

$$\xi \in \mathrm{Ker}\Big(\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)\Big)$$

is described by the pair (f, g) as in 3.8.8, then the image of ξ in $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$ under the isomorphism (3.6.5.1) is the image under (3.8.10) of the element of its first term defined by the cocycle (3.8.11).

The verification of this compatibility is left to the reader.

3.8.13. We shall apply this especially in the case

$$L_0 = \mathbb{Z}[P], \qquad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are respectively the kernels of the canonical homomorphisms $\mathbb{Z}[P] \longrightarrow P$ and $\mathbb{Z}[Q] \longrightarrow Q$. Notice the canonical isomorphisms

$$L_0 \overset{\mathbb{L}}{\otimes} M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q].$$

Since L_0 and M_0 are flat, (1.4.5) gives canonical isomorphisms

$$\mathrm{Ext}^i(L_0 \overset{\mathbb{L}}{\otimes} M_0, G) \simeq H^i(P \times Q, G_{P\times Q}).$$

It follows from this and 3.6.5, or directly from the definitions and 1.4, that the functor which assigns to each biextension of (L_0, M_0) by G the $G_{P\times Q}$-torsor over $P \times Q$ that it induces is an equivalence of categories. (This remains valid without using an additive structure on P, Q.) Consequently, if E is a biextension of (P, Q) by G, giving a trivialization of its inverse image over (L_0, M_0) is equivalent to giving a section of E considered only as an object over $P \times Q$, that is, a trivialization of E considered as a torsor under $G_{P\times Q}$.

Biextensions equipped with this additional structure are therefore described by the pairs (f, g) I≈**216** of (3.8.3) satisfying compatibility (3.8.4), and their classification up to isomorphism is the one given in 3.8.8.

With the notation of 3.5, one may also regard L_1 as the cokernel of $L_2(P) \longrightarrow L_1(P)$, and similarly for M_1. This makes it possible to express the data (f, g) in terms of a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \longrightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \longrightarrow G, \end{cases}$$

that is, homomorphisms of sheaves of sets

$$P \times P \times Q \longrightarrow G, \qquad P \times Q \times Q \longrightarrow G,$$

satisfying five compatibility conditions. These express that f' and g' factor through f and g—which gives 2×2 equations, in view of the expression of $L_2(P)$ and $L_2(Q)$ as sums of two terms—together with condition (3.8.4). One then sees that f' and g' are precisely the partial multiplication data defining a biextension, and that the five conditions in question are exactly the five axioms for biextensions considered in 2.1 (cf. the proof of 3.6.4).

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SGA 7

Exposé VIII

COMPLEMENTS ON BIEXTENSIONS. GENERAL PROPERTIES OF BIEXTENSIONS OF GROUP SCHEMES

by A. Grothendieck

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0. Introduction

0.1. We continue here the general study of biextensions begun in the preceding exposé. First we give some complements in the case of an arbitrary base topos, developing in particular, with the detail it deserves, the formalism of the pairings associated with biextensions (Section 2), whose importance in the theory of abelian schemes is well known. We then begin the study of biextensions of general group schemes, the case of greatest interest for us being that of biextensions by the multiplicative group $\underline{G}_m$. We shall nevertheless be obliged, as a technical intermediate step, also to study biextensions by other types of group schemes (Sections 5 and 6).

The most interesting result for what follows is given in 7.1, which gives conditions under which, for I ≈219
P and Q smooth over a trait with generic point η, a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$ can be prolonged to a biextension of (P, Q) by $\underline{G}_m$, determined up to a unique isomorphism. This result will be very convenient in the following exposé, devoted to the theory of the Néron model of abelian varieties. It is likely that one could manage there at less expense, without the imposing paperweight of Exposés VII and VIII (which will no longer be used in the remainder of the Seminar); it nevertheless seems likely to us that the developments given in these exposés are destined to be useful elsewhere as well.

0.2. When we work with group schemes over a scheme S, it will be understood that we work with the fppf topos of S. This is the topos associated with the $\underline{U}$-site of schemes locally of finite presentation over S (and belonging to $\underline{U}$), endowed with the fppf topology (SGA 3 IV 6.3). The reader will readily verify that most of the statements we shall give would also be valid (and essentially equivalent to the fppf statements) for other topologies on S, whose underlying site contains at least the schemes locally of finite presentation over S, such as the Zariski, étale, and fpqc topologies. One reason is that, by descent theory, the category of extensions of a group scheme P locally of finite presentation over S by the group scheme $\underline{G}_m$ does not depend, up to equivalence, on the chosen topology; the same statement holds for the notion of biextension.

The technical reason for preferring the fppf topology (or a finer topology, such as fpqc) to coarser topologies such as the Zariski and étale topologies is that, for every integer $n > 0$, the homomorphism $n \cdot \mathrm{id}$ from $\underline{G}_m$ to itself is an fppf epimorphism (because it is faithfully flat and of finite presentation),

but in general is not a Zariski or an étale epimorphism. Notice that it is an étale epimorphism if and only if n is prime to the residue characteristics of S, and a Zariski epimorphism if and only if S is empty.

0.3. Finally, most of the statements concerning extensions or biextensions by $\underline{G_m}$ would remain valid if that group were replaced by a torus G. Since such a G is locally isomorphic (for the fppf, and even for the étale topology) to a group $\underline{G_m^r}$, this is an essentially trivial generalization. I 220

1. Various special cases over an arbitrary topos

1.1. In this section and the next, as in Exposé VII, T denotes a topos, and P, Q, G denote three abelian groups of T. Thus there is a canonical isomorphism (VII 3.6.5)

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big). \tag{1.1.1}$$

Recall the canonical isomorphisms (called the Cartan isomorphisms)

$$\begin{aligned}\mathrm{Ext}^1\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big) &\simeq \mathrm{Ext}^1(P, R\underline{\mathrm{Hom}}(Q,G)) \\ &\simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G))\,.\end{aligned} \tag{1.1.2}$$

All these isomorphisms are functorial in the three arguments P, Q, G.

Taking the standard truncation of the derived tensor product, whose homology objects are the $\mathrm{Tor}_i(P,Q)$, one obtains from (1.1.1) the following infinite exact sequence:

$$\begin{aligned}0 \longrightarrow\ & \mathrm{Ext}^1(P\otimes Q, G) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(\mathrm{Tor}_1(P,Q),G) \\ \longrightarrow\ & \mathrm{Ext}^2(P\otimes Q, G) \longrightarrow \mathrm{Ext}^2\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big) \longrightarrow \cdots .\end{aligned} \tag{1.1.3}$$

On the other hand, since the cohomology objects of $R\underline{\mathrm{Hom}}(Q,G)$ are the $\underline{\mathrm{Ext}}^i(Q,G)$, the first equality in (1.1.2) gives rise to the five-term exact sequence

$$\begin{aligned}0 \longrightarrow\ & \mathrm{Ext}^1(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)) \\ \longrightarrow\ & \mathrm{Ext}^2(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Ext}^2(P, R\underline{\mathrm{Hom}}(Q,G)),\end{aligned} \tag{1.1.4}$$

associated with the usual spectral sequence

$$\mathrm{Ext}^*(P, R\underline{\mathrm{Hom}}(Q,G)) \Longleftarrow \mathrm{Ext}^p(P, \underline{\mathrm{Ext}}^q(Q,G)).$$

We leave it to the reader, should the need arise, to make the first arrow of (1.1.3), respectively of (1.1.4), explicit by means of the natural fully faithful functor

$$\mathrm{EXT}(P\otimes Q, G) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.5}$$

respectively

$$\mathrm{EXT}(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.6}$$

where EXT and BIEXT denote the evident categories of extensions and biextensions.

Of course, the isomorphism I ≈221

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G))$$

gives rise to an exact sequence analogous to (1.1.3), with the roles of P and Q interchanged. Its first arrow can be made explicit by means of the fully faithful functor

$$\mathrm{EXT}(Q, \underline{\mathrm{Hom}}(P,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G) \tag{1.1.7}$$

analogous to (1.1.6).

1.2. Suppose that

$$\mathrm{Tor}_1(P,Q)=0. \tag{1.2.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G)\simeq \mathrm{Ext}^1(P\otimes Q,G). \tag{1.2.2}$$

This holds, for example, if P or Q is a flat $\mathbb{Z}$-module. More precisely, under condition (1.2.1), the functor (1.1.5) is an equivalence of categories.

1.3. Suppose that I 222

$$P\otimes Q=0. \tag{1.3.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G)\simeq \mathrm{Hom}(\mathrm{Tor}_1(P,Q),G). \tag{1.3.2}$$

The case of greatest interest to us in which (1.3.1) holds is that in which there is an integer $n>0$ such that

$$nP=P,\qquad Q=\varinjlim_i {}_{n^i}Q; \tag{1.3.3}$$

that is, P is divisible by n, and Q is the limit of subgroups annihilated by powers of n. Put, as usual,

$$T_n(P)=({}_{n^i}P)_{i\geq 0}\quad\text{(a projective system)},\qquad {}_{n^\infty}P=\varinjlim_i {}_{n^i}P.$$

From (1.3.3) one obtains the following formulas, more precise than (1.3.1):

$$\begin{aligned} P\otimes Q&=0,\\ \mathrm{Tor}_1(P,Q)&\simeq \varinjlim_i\big({}_{n^i}P\otimes {}_{n^i}Q\big)\simeq T_n(P)\otimes Q\\ &\simeq {}_{n^\infty}P\otimes T_n(Q). \end{aligned} \tag{1.3.4}$$

Thus the isomorphism (1.3.2) here assumes the more concrete form

$$\begin{aligned} \mathrm{Biext}^1(P,Q;G)&\simeq \mathrm{Hom}(T_n(P),D(Q))\\ &\simeq \mathrm{Hom}(Q,D(T_n(P)))\\ &\simeq \mathrm{Hom}(T_n(P)\otimes T_n(Q),T_n(G)). \end{aligned} \tag{1.3.5}$$

Let us specify the meaning of these formulas.

a) In the inductive system occurring in (1.3.4), the transition homomorphism

$${}_{n^i}P\otimes {}_{n^i}Q\longrightarrow {}_{n^j}P\otimes {}_{n^j}Q\qquad (i\leq j)$$

is described by identifying the first member with ${}_{n^j}P\otimes {}_{n^i}Q$ by means of $n^{j-i}\otimes \mathrm{id}$, and then using the homomorphism

$$\mathrm{id}\otimes \mathrm{inj}: {}_{n^j}P\otimes {}_{n^i}Q\longrightarrow {}_{n^j}P\otimes {}_{n^j}Q,$$

where $\mathrm{inj}: {}_{n^i}Q\longrightarrow {}_{n^j}Q$ denotes the inclusion.

b) By definition, put

$$T_n(P) \otimes Q \overset{\text{def}}{=} \varinjlim_i (T_n(P) \otimes {}_{n^i}Q),$$

where

$$T_n(P) \otimes M \overset{\text{def}}{=} {}_{n^i}P \otimes M \qquad \text{if } n^i M = 0.$$

The meaning of the third term written in (1.3.4) is specified in the same way when $nQ = Q$ is also assumed, so that $T_n(Q)$, like $T_n(P)$, is a strict projective system. The last term of (1.3.5) is a Hom between projective systems of groups.

c) The letter D denotes $\underline{\mathrm{Hom}}(-, G)$. The object $D(Q)$ is defined as the projective system of groups

$$D(Q) = (D({}_{n^i}Q))_{i\geq 0} = (\underline{\mathrm{Hom}}({}_{n^i}Q, G))_{i\geq 0},$$

the second member of (1.3.5) being the group of homomorphisms of projective systems of groups. The following definition gives a simpler interpretation of the next member:

$$D(T_n(P)) = \varinjlim_i D({}_{n^i}P),$$

where the transition morphisms of this inductive system are obtained by transposition from the transition morphisms

$$n^{j-i} : {}_{n^j}P \longrightarrow {}_{n^i}P$$

of $T_n(P)$.

We leave it to the reader to make explicit the calculation of $\mathrm{Tor}_1(P, Q)$ indicated in (1.3.4), and to derive (1.3.5) from it. We note only that one uses the relations, deduced from the second hypothesis in (1.3.3), I≈**224**

$$\mathrm{Tor}_p(P, Q) \simeq \varinjlim_i \mathrm{Tor}_p(P, {}_{n^i}Q).$$

For $p = 0$ this immediately gives $P \otimes Q = 0$, and it reduces the calculation of $\mathrm{Tor}_1(P, Q)$ to that of the $\mathrm{Tor}_1(P, {}_{n^i}Q)$ and their transition morphisms. Thus, when Q is annihilated by an integer $m = n^i$ such that $mP = P$, it remains to make explicit an isomorphism

$$\mathrm{Tor}_1(P, Q) \xleftarrow{\sim} {}_mP \otimes {}_mQ \qquad (mP = P,\ mQ = 0). \tag{1.3.6}$$

There are several ways to do this. One consists in using the exact Tor sequence associated with

$$0 \longrightarrow {}_mP \longrightarrow P \xrightarrow{m} P \longrightarrow 0.$$

One may also use the homomorphism (2.1.1) below; the fact that the transition morphisms are those specified in part (a) is then precisely 2.1.11. In any case, the choice of (1.3.6) amounts to the choice of a sign; see 2.1.10.

1.4. Suppose that

$$\underline{\mathrm{Hom}}(Q, G) = 0. \tag{1.4.1}$$

Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \xrightarrow{\sim} \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q, G)). \tag{1.4.2}$$

This case occurs in particular when Q is an abelian scheme and $G = \underline{G}_m$; see VII 1.3.8. Hence the functor in P,

$$P \longmapsto \mathrm{Biext}^1(P, Q; G),$$

is <u>representable</u> by $Q^* = \underline{\mathrm{Ext}}^1(Q, G)$. In particular, there is a canonical biextension E^Q of (Q^*, Q) by G, determined up to a unique isomorphism, since its automorphism group is

$$\mathrm{Hom}(Q^*, \underline{\mathrm{Hom}}(Q, G)) = 0.$$

This biextension, or its symmetric biextension ${}^sE^Q$ of (Q, Q^*) by G, may be called the <u>Weil biextension</u> of (Q^*, Q), respectively (Q, Q^*), by G.

1.5. Suppose that I **225**

$$\mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q, G)) = 0. \tag{1.5.1}$$

This holds in particular if $\underline{\mathrm{Ext}}^1(Q, G) = 0$; for example, if Q is a finite locally free group scheme or a group of multiplicative type (SGA 3 IX), and $G = \underline{G}_m$; see 3.4 below. Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \xleftarrow{\sim} \mathrm{Ext}^1(P, \underline{\mathrm{Hom}}(Q, G)), \tag{1.5.2}$$

where $\underline{\mathrm{Hom}}(Q, G)$ plays the role of a "dual group" of Q (relative to G). More precisely, it follows from (1.5.2) that the fully faithful functor (1.1.6) is an <u>equivalence of categories</u>.

1.6. Suppose that the two relations

$$\underline{\mathrm{Hom}}(P, G) = 0, \qquad \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q, G)) = 0 \tag{1.6.1}$$

hold simultaneously. By 1.4 and 1.5, respectively, these two relations give isomorphisms

$$\mathrm{Biext}^1(P, Q; G) \xrightarrow{\sim} \mathrm{Hom}(Q, \underline{\mathrm{Ext}}^1(P, G)),$$

$$\mathrm{Biext}^1(P, Q; G) \xleftarrow{\sim} \mathrm{Ext}^1(P, \underline{\mathrm{Hom}}(Q, G)).$$

Their composite is the canonical isomorphism

$$\mathrm{Ext}^1(P, \underline{\mathrm{Hom}}(Q, G)) \xrightarrow{\sim} \mathrm{Hom}(Q, \underline{\mathrm{Ext}}^1(P, G)) \quad (\simeq \mathrm{Biext}^1(P, Q; G)), \tag{1.6.2}$$

which plays the role of a duality formula.

Notice also that the first relation in (1.6.1) plainly implies $\mathrm{Hom}(P \otimes Q, G) = 0$. Hence the category $\mathrm{EXT}(P, \underline{\mathrm{Hom}}(Q, G))$ of commutative extensions of P by $\underline{\mathrm{Hom}}(Q, G)$ is rigid (every automorphism of an object is the identity), as is $\mathrm{BIEXT}(P, Q; G)$. The structure of these categories is therefore known once the group of isomorphism classes of objects is known; by (1.6.2), this group is canonically isomorphic to

$$\mathrm{Hom}(Q, \underline{\mathrm{Ext}}^1(P, G)).$$

The map (1.6.2) is in fact defined in all cases, without assuming (1.6.1), and can also be described directly, without passing through $\mathrm{Biext}^1(P, Q; G)$, in terms of the natural pairing I ≈**226**

$$Q \times Q' \longrightarrow G, \qquad Q' = \underline{\mathrm{Hom}}(Q, G).$$

Interpreting it as a homomorphism $Q \longrightarrow \underline{\mathrm{Hom}}(Q', G)$ gives a composite

$$\begin{aligned} \mathrm{Ext}^1(P, Q') \times Q &\longrightarrow \mathrm{Ext}^1(P, Q') \times \mathrm{Hom}(Q', G) \\ &\longrightarrow \mathrm{Ext}^1(P, G), \end{aligned} \tag{1.6.3}$$

which recovers (1.6.2).

For a useful case in which the considerations of 1.6 apply, see 3.7 below.

2. Pairings defined by a biextension

2.1. Let P and Q be two abelian groups of T, and let $n > 0$ be an integer. Recall that ${}_nP$ denotes the kernel of multiplication by $n \cdot \mathrm{id}_P$. We shall define a canonical homomorphism

$$\alpha = \alpha^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P, Q), \tag{2.1.1}$$

which will be the composite

$${}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1({}_nP, {}_nQ) \longrightarrow \underline{\mathrm{Tor}}_1(P, Q), \tag{2.1.2}$$

where the first arrow is $\alpha^n_{{}_nP,{}_nQ}$ and the second is induced by the inclusions ${}_nP \longrightarrow P$ and ${}_nQ \longrightarrow Q$. We are thus reduced to defining (2.1.1) when $nP = 0$ and $nQ = 0$, that is, when P and Q are $\mathbb{Z}/n\mathbb{Z}$-modules.

Suppose only that $nQ = 0$. Taking a flat resolution $L_\bullet$ of P (as a $\mathbb{Z}$-module), one obtains canonical isomorphisms I ≈227

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq L_\bullet \otimes_{\mathbb{Z}} Q \simeq (L_\bullet \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} Q,$$

which may be written as an isomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq \left(P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}\right) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}/n\mathbb{Z}} Q. \tag{2.1.3}$$

This gives rise to the homological Künneth-type spectral sequence (cf. SGA 1 IV 5.2)

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_*(P, Q) \Longleftarrow E^2_{p,q} = \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_p(\underline{\mathrm{Tor}}^{\mathbb{Z}}_q(P, \mathbb{Z}/n\mathbb{Z}), Q). \tag{2.1.4}$$

Recall also the canonical isomorphisms

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_0(P, \mathbb{Z}/n\mathbb{Z}) = P_n := P/nP, \qquad \underline{\mathrm{Tor}}^{\mathbb{Z}}_1(P, \mathbb{Z}/n\mathbb{Z}) \simeq {}_nP,$$

and

$$\underline{\mathrm{Tor}}^{\mathbb{Z}}_i(P, \mathbb{Z}/n\mathbb{Z}) = 0 \qquad \text{for } i \geq 2.$$

They follow by computing these objects from the canonical flat resolution of $(\mathbb{Z}/n\mathbb{Z})_T$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z}_T & \xrightarrow{\ n\ } & \mathbb{Z}_T & \longrightarrow & 0 \\ & & \| & & \| & & \\ & & L_1 & & L_0 & & \end{array}$$

whose augmentation $L_0 \longrightarrow (\mathbb{Z}/n\mathbb{Z})_T$ sends 1 to 1. Consequently, the spectral sequence (2.1.4) reduces essentially to the exact sequence

$$\begin{aligned} 0 \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_2(P_n, Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}^{\mathbb{Z}}_1(P, Q) \\ \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}/n\mathbb{Z}}_1(P_n, Q) \longrightarrow 0, \end{aligned} \tag{2.1.5}$$

where $P_n = P \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = P/nP$. The desired homomorphism α is the edge homomorphism $E^2_{0,1} \longrightarrow \underline{\mathrm{Tor}}^{\mathbb{Z}}_1$ in this exact sequence. Since the sequence is plainly functorial in P and Q, functoriality in P shows that the homomorphism in question is indeed the composite

$${}_nP \otimes Q \longrightarrow \underline{\mathrm{Tor}}_1({}_nP, Q) \longrightarrow \underline{\mathrm{Tor}}_1(P, Q),$$

where the first homomorphism is the α just defined, with (P, Q) replaced by $({}_nP, Q)$, and the second is induced by the inclusion ${}_nP \longrightarrow P$.

2.1.6. This therefore defines, for arbitrary P and Q, a homomorphism (2.1.1) satisfying the factorization condition (2.1.2). Moreover, when $nQ = 0$, the exact sequence (2.1.5) shows that α is I 228

an isomorphism whenever P_n is flat over $\mathbb{Z}/n\mathbb{Z}$. This holds in particular if $P_n = 0$, that is, if $nP = P$ and P is n-divisible, or if P itself is a flat $\mathbb{Z}/n\mathbb{Z}$-module. It also holds if Q is a flat $\mathbb{Z}/n\mathbb{Z}$-module.

2.1.7. Let us make the calculation of the homomorphism α in (2.1.5) explicit. To determine it, it suffices, for every section q of Q (over an object S of T; take the final object for simplicity), to determine the corresponding homomorphism

$$ {}_nP \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q). $$

The section q may be regarded as the image of the section 1 of $(\mathbb{Z}/n\mathbb{Z})_T$ under the uniquely determined homomorphism

$$ u_q : (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow Q. $$

Functoriality of α in Q therefore reduces us to the case $Q = (\mathbb{Z}/n\mathbb{Z})_T$, with q its unit section. In this case, identify

$$ \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q) = \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) $$

with ${}_nP$ as explained after (2.1.4). The homomorphism in question is then the identity. Equivalently, the edge homomorphism

$$ E^2_{0,q} = \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \longrightarrow \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) $$

is the identity. This follows immediately from the definitions. The analogous assertion remains valid when $\mathbb{Z}$ and its quotient $\mathbb{Z}/n\mathbb{Z}$ are replaced by any ring A on T and any quotient ring A/I.

2.1.8. The homomorphism α of (2.1.1) may also be determined by describing, for every pair of **I 229** sections p of ${}_nP$ and q of ${}_nQ$ (over an object S of T, again taken to be final to simplify the notation), the section of $\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q)$ determined by (p, q). By functoriality in P and Q, as above, we are reduced to the case

$$ P = Q = (\mathbb{Z}/n\mathbb{Z})_T, $$

with p and q the unit section. Thus we need only determine

$$ \alpha : (\mathbb{Z}/n\mathbb{Z})_T \otimes (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big). $$

The description in 2.1.7 shows that this is the isomorphism

$$ (\mathbb{Z}/n\mathbb{Z})_T \xrightarrow{\sim} \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big) \tag{2.1.8.1} $$

deduced from the canonical flat resolution

$$ 0 \longrightarrow \mathbb{Z}_T \xrightarrow{n} \mathbb{Z}_T \longrightarrow 0 \tag{2.1.8.2} $$

of the first factor $P = (\mathbb{Z}/n\mathbb{Z})_T$ occurring in $\underline{\mathrm{Tor}}_1$.

2.1.9. The definition given in (2.1.1) is not symmetric. By interchanging the roles of P and Q in **I 230** the preceding construction, one may define an analogous homomorphism

$$ \beta^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P, Q), \tag{2.1.9.1} $$

first reducing to the case $nP = 0$ and then using a flat resolution of Q. Equivalently, β may be defined in terms of α by requiring the following diagram to commute:

$$ \begin{array}{ccc} {}_nP \otimes {}_nQ & \xrightarrow{\ \sim\ } & {}_nQ \otimes {}_nP \\ \Big\downarrow{\scriptstyle \beta^n_{P,Q}} & & \Big\downarrow{\scriptstyle \alpha^n_{Q,P}} \\ \underline{\mathrm{Tor}}_1(P, Q) & \xrightarrow{\ \sim\ } & \underline{\mathrm{Tor}}_1(Q, P). \end{array} \tag{2.1.9.2} $$

Here the horizontal arrows are the symmetry isomorphisms. With this convention, the relation between α and β is

$$\beta^n_{P,Q} = -\alpha^n_{P,Q}. \tag{2.1.9.3}$$

In other words:

Lemma 2.1.10. Let P and Q be two commutative groups and let $n > 0$ be an integer. Then the following diagram is anticommutative, where α is defined in (2.1.1):

$$\begin{array}{ccc} {}_nP \otimes {}_nQ & \xrightarrow{\sim} & {}_nQ \otimes {}_nP \\ \downarrow{\scriptstyle \alpha^n_{P,Q}} & - & \downarrow{\scriptstyle \alpha^n_{Q,P}} \\ \underline{\mathrm{Tor}}_1(P,Q) & \xrightarrow{\sim} & \underline{\mathrm{Tor}}_1(Q,P). \end{array} \tag{2.1.10.1}$$

The horizontal arrows are the symmetry isomorphisms.

Proof. The procedure of 2.1.8 immediately reduces us to the case $P = Q = (\mathbb{Z}/n\mathbb{Z})_T$. We must then prove that the two isomorphisms (2.1.8.1), obtained by resolving respectively the first and second arguments of $\underline{\mathrm{Tor}}_1$ by (2.1.8.2), differ by a sign (equivalently, that the symmetry isomorphism on $\underline{\mathrm{Tor}}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is $-\mathrm{id}$). We may plainly suppose that T is the punctual topos, so that we work with ordinary commutative groups. Let $L_\bullet$ be the complex

$$L_\bullet = (0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0),$$

regarded as a flat resolution of $P = Q = \mathbb{Z}/n\mathbb{Z}$. Use $L_\bullet \otimes L_\bullet$ to compute

$$\mathrm{Tor}_1(P,Q) = \mathrm{Tor}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}).$$

The isomorphisms with $H_1(L_\bullet \otimes \mathbb{Z}/n\mathbb{Z})$, which defines α, and with $H_1(\mathbb{Z}/n\mathbb{Z} \otimes L_\bullet)$, which defines β, are induced respectively by the augmentations $L_\bullet \longrightarrow \mathbb{Z}/n\mathbb{Z}$ of the second and first factors of $L_\bullet \otimes L_\bullet$. It follows at once that $\alpha(1,1)$ is represented by the cycle

$$-1_0 \otimes 1_1 + 1_1 \otimes 1_0 \in (L_\bullet \otimes L_\bullet)_1 = L_0 \otimes L_1 + L_1 \otimes L_0,$$

where 1_0 and 1_1 denote the element 1 of L_0 and L_1 respectively, whereas $\beta(1,1)$ is represented by $1_0 \otimes 1_1 - 1_1 \otimes 1_0$. This proves 2.1.10.

To finish these generalities on the pairings α, we must still specify the relations among the homomorphisms $\alpha^n_{P,Q}$ as n varies. The main result is the following. I≈231

Lemma 2.1.11. Let P, Q be two commutative groups of T, and let n, n' be positive integers such that n' is a multiple of n, say $n' = mn$. Then the following diagram is commutative:

$$\begin{array}{ccccc} & & {}_{n'}P \otimes {}_nQ & & \\ & \swarrow{\scriptstyle P_m \otimes \mathrm{id}_{{}_nQ}} & & \searrow{\scriptstyle \mathrm{id}_{{}_{n'}P} \otimes i_m} & \\ {}_nP \otimes {}_nQ & & & & {}_{n'}P \otimes {}_{n'}Q \\ & \searrow{\scriptstyle \alpha^n_{P,Q}} & & \swarrow{\scriptstyle \alpha^{n'}_{P,Q}} & \\ & & \underline{\mathrm{Tor}}_1(P,Q) & & \end{array} \tag{2.1.11.1}$$

Here $P_m : {}_{n'}P \longrightarrow {}_nP$ is the homomorphism induced by $m \cdot \mathrm{id}_P$, and $i_m : {}_nQ \longrightarrow {}_{n'}Q$ is the inclusion. Likewise, the diagram (2.1.11.2) symmetric to (2.1.11.1) is commutative; it has the same vertices as (2.1.11.1), except that the upper vertex ${}_{n'}P \otimes {}_nQ$ is replaced by ${}_nP \otimes {}_{n'}Q$.

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It suffices to prove the first assertion; the second follows by symmetry in view of 2.1.10. By the procedure of 2.1.8, we are reduced to the punctual topos, where P and Q are ordinary commutative groups and

$$P = \mathbb{Z}/n'\mathbb{Z}, \qquad Q = \mathbb{Z}/n'\mathbb{Z}.$$

In fact, for P we use only the hypothesis $n'P = 0$. Using the resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n'} \mathbb{Z} \longrightarrow 0$$

of Q, identify $\mathrm{Tor}_1(P,Q)$ with ${}_{n'}P = P$. Likewise, using

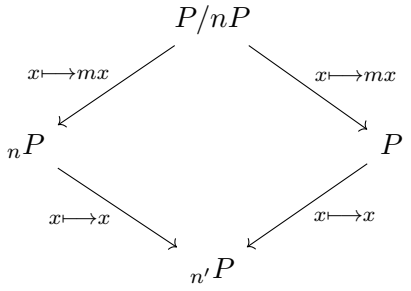
$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0$$

to resolve ${}_nQ \simeq \mathbb{Z}/n\mathbb{Z}$, identify $\mathrm{Tor}_1({}_nP, {}_nQ)$ with ${}_nP$. The augmentation in the latter resolution sends 1_0 to $m \cdot 1_Q$, where 1_Q denotes the element 1 of $Q = \mathbb{Z}/n'\mathbb{Z}$. Under these identifications the homomorphism

$$\begin{array}{ccc} \mathrm{Tor}_1({}_nP, {}_nQ) & \longrightarrow & \mathrm{Tor}_1(P,Q) \\ \wr\downarrow & & \wr\downarrow \\ {}_nP & & {}_{n'}P = P \end{array}$$

induced by the inclusions ${}_nP \hookrightarrow P$ and ${}_nQ \hookrightarrow Q$ is simply the natural inclusion ${}_nP \hookrightarrow {}_{n'}P$.

Consequently, identify the upper vertex of (2.1.11.1) with P/nP using the generator $m \cdot 1_Q$ of ${}_nQ$, which identifies ${}_nQ$ with $\mathbb{Z}/n\mathbb{Z}$. The diagram becomes



which is plainly commutative.

Corollary 2.1.12. With the notation of 2.1.11, the following diagram is commutative: I 233

(2.1.12.1)
$$\begin{array}{ccc} {}_{n'}P \otimes {}_{n'}Q & \xrightarrow{m\otimes m} & {}_nP \otimes {}_nQ \\ \downarrow{\scriptstyle \alpha^{n'}_{P,Q}} & & \downarrow{\scriptstyle \alpha^{n}_{P,Q}} \\ {}_{n'}\underline{\mathrm{Tor}}_1(P,Q) & \xrightarrow{m} & {}_n\underline{\mathrm{Tor}}_1(P,Q). \end{array}$$

In the description of the horizontal arrows, m denotes the homomorphisms ${}_{n'}E \longrightarrow {}_nE$ induced by $m \cdot \mathrm{id}_E$, for $E = P, Q$, and $\underline{\mathrm{Tor}}_1(P,Q)$ respectively.

Indeed, by 2.1.11, for $x \in {}_{n'}P(S)$ and $y \in {}_{n'}Q(S)$ one has

$$\alpha^n(mx \otimes my) = \alpha^{n'}(x \otimes my).$$

The right-hand side equals $m\alpha^{n'}(x \otimes y)$, since $x \otimes my = m(x \otimes y)$. This proves the commutativity of (2.1.12.1).

2.1.13. Let ℓ be a prime number. For every commutative group E, denote by $T_\ell(E)$ the inverse system I 234

(2.1.13.1)
$$T_\ell(E) = ({}_{\ell^\nu}E)_{\nu \geq 0},$$

which plainly depends functorially on E. The compatibilities of 2.1.12, for n, n' of the form $\ell^\nu, \ell^{\nu'}$ with $\nu' \geq \nu$, show that the homomorphisms $\alpha_{P,Q}^{\ell^\nu}$, as ν varies, define a homomorphism of inverse systems, functorial in P, Q:

$$\alpha_{P,Q}^{(\ell)} : T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell(\underline{\mathrm{Tor}}_1(P,Q)). \tag{2.1.13.2}$$

Knowledge of these homomorphisms for every prime ℓ is plainly equivalent to knowledge of the $\alpha_{P,Q}^n$ for every integer n. The antisymmetry relations of 2.1.10 then become the analogous ℓ-adic antisymmetry relations, for every ℓ, expressed by the anticommutativity of the diagrams

$$\begin{array}{ccc} T_\ell(P) \otimes T_\ell(Q) & \longrightarrow & T_\ell(Q) \otimes T_\ell(P) \\ \Big\downarrow{\scriptstyle \alpha_{P,Q}^{(\ell)}} & - & \Big\downarrow{\scriptstyle \alpha_{Q,P}^{(\ell)}} \\ T_\ell(\underline{\mathrm{Tor}}_1(P,Q)) & \longrightarrow & T_\ell(\underline{\mathrm{Tor}}_1(Q,P)). \end{array} \tag{2.1.13.3}$$

Here the horizontal arrows are the symmetry arrows.

Remark 2.1.14. The construction of the homomorphism (2.1.1), and the results 2.1.10 and 2.1.11 concerning it, may be generalized to the situation in which one is given a commutative ring $\underline{A}$ on T and two ideals $\underline{a}$ and $\underline{b}$ of $\underline{A}$. In the case treated above, both ideals are the ideal $n\mathbb{Z}_T$ of the constant ring $\mathbb{Z}_T$ on T. One then defines a homomorphism, functorial in the $\underline{A}$-modules P, Q,

$$ {}_{\underline{a}}P \otimes {}_{\underline{b}}Q \xrightarrow{\alpha} \underline{\mathrm{Hom}}_{\underline{A}}\left(\frac{\underline{a} \cap \underline{b}}{\underline{a} \cdot \underline{b}}, \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q)\right), \tag{2.1.14.1}$$

as the composite of the homomorphisms

$$\begin{aligned} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q &\longrightarrow \underline{\mathrm{Hom}}(\underline{A}/\underline{a}, P) \otimes \underline{\mathrm{Hom}}(\underline{A}/\underline{b}, Q) \\ &\longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}), \underline{\mathrm{Tor}}_1(P,Q)), \end{aligned}$$

using the well-known canonical isomorphism

$$\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}) \simeq \frac{\underline{a} \cap \underline{b}}{\underline{a} \cdot \underline{b}}. \tag{2.1.14.2}$$

In fact, the sign in 2.1.10 is "hidden" in the choice of this isomorphism, according as one works with the resolution $0 \longrightarrow \underline{a} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{a}$, or with the resolution $0 \longrightarrow \underline{b} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{b}$. Here it becomes the anticommutativity of the diagram

$$\begin{array}{ccc} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q & \longrightarrow & {}_{\underline{b}}Q \otimes {}_{\underline{a}}P \\ \Big\downarrow & -1 & \Big\downarrow \\ \underline{\mathrm{Hom}}\left(\dfrac{\underline{a} \cap \underline{b}}{\underline{a} \cdot \underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\right) & \longrightarrow & \underline{\mathrm{Hom}}\left(\dfrac{\underline{a} \cap \underline{b}}{\underline{a} \cdot \underline{b}}, \underline{\mathrm{Tor}}_1(Q,P)\right). \end{array} \tag{2.1.14.3}$$

The horizontal arrows are the symmetry arrows. When $\underline{a} = \underline{b}$, this anticommutativity may also be expressed by saying that the symmetry automorphism on $\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{a})$ is $-\mathrm{id}$.

In the special case where $\underline{a} = \underline{b} = n\underline{A}$, with n a section of $\underline{A}$, multiplication by n induces an epimorphism I≈235

$$\underline{A}/\underline{a} \longrightarrow \frac{\underline{a} \cap \underline{a}}{\underline{a} \cdot \underline{a}} = \underline{a}/\underline{a}^2,$$

and hence a functorial injective homomorphism, in the $\underline{A}$-module T,

$$\underline{\mathrm{Hom}}(\underline{a}/\underline{a}^2, T) \longrightarrow {}_{\underline{a}}T.$$

Thus in this case the homomorphism (2.1.14.1) may be interpreted as a homomorphism

$$ {}_nP \otimes {}_nQ \longrightarrow {}_n\underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.14.4}$$

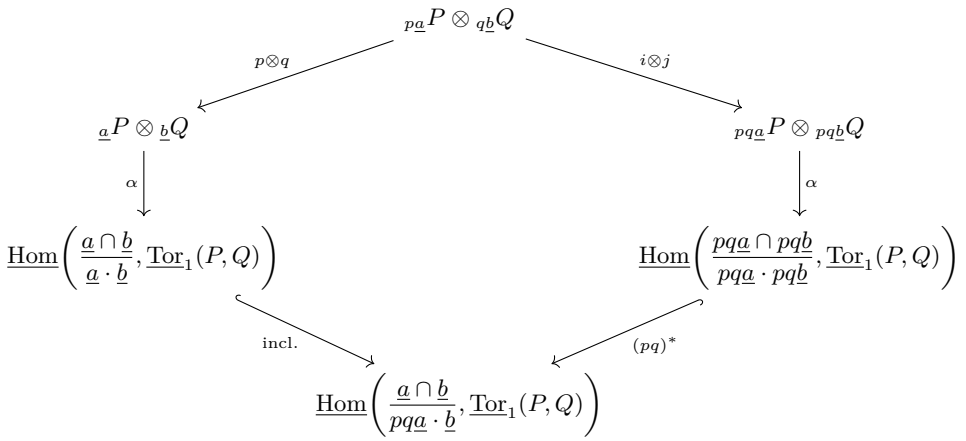
which is closer in form to (2.1.1).

Continue to suppose that $\underline{a} = \underline{b} = n\underline{A}$, and that n is a regular section of $\underline{A}$. The spectral sequence (2.1.4) still makes sense in the present setting and yields the following exact sequence of low-degree terms, generalizing (2.1.5), where Q is now an $\underline{A}$-module annihilated by $\underline{a}$: I≈236

$$\begin{aligned} \underline{\mathrm{Tor}}_2^{\underline{A}}(P,Q) \longrightarrow \underline{\mathrm{Tor}}_2^{\underline{A}/n\underline{A}}(P_n,Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q) \\ \longrightarrow \underline{\mathrm{Tor}}_1^{\underline{A}/n\underline{A}}(P_n,Q) \longrightarrow 0. \end{aligned} \tag{2.1.14.5}$$

Return now to arbitrary $\underline{a}$ and $\underline{b}$, and let p and q be two sections of $\underline{A}$. We then obtain the commutative diagram below; (2.1.11.1) is essentially the special case obtained by taking $q = 1$:

(2.1.14.6)



Here

$$ i : {}_{p\underline{a}}P \longrightarrow {}_{pq\underline{a}}P, \qquad j : {}_{q\underline{b}}Q \longrightarrow {}_{pq\underline{b}}Q $$

are the canonical inclusions, and $(pq)^*$ denotes the injective homomorphism obtained, by passing to $\underline{\mathrm{Hom}}$, from the surjective homomorphism

$$ \frac{\underline{a}\cap\underline{b}}{pq\underline{a}\cdot\underline{b}} \longrightarrow \frac{pq\underline{a}\cap pq\underline{b}}{pq\underline{a}\cdot pq\underline{b}} $$

induced by multiplication by pq. When $\underline{a} = \underline{b} = n\underline{A}$, this diagram may be rewritten in a simpler form, given in (2.1.11.1) when $q = 1$.

2.2. Consider now, in addition to P and Q, a commutative group G. In (1.1.3) we defined a canonical homomorphism, functorial in P, Q, and G, I237

$$ \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big). \tag{2.2.1}$$

On the other hand, if n is a positive integer (respectively, if ℓ is a prime number), the homomorphism $\alpha^n_{P,Q}$ of (2.1.1) (respectively, the homomorphism $\alpha^{(\ell)}_{P,Q}$ of (2.1.13.2)) gives a homomorphism

$$ \mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big) \longrightarrow \mathrm{Hom}({}_nP \otimes {}_nQ, G) $$

(respectively, a homomorphism

$$ \mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big) \longrightarrow \mathrm{Hom}\big(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)\big)\big). $$

These last homomorphisms are also functorial in P, Q, G. Composing them with (2.2.1), we obtain canonical homomorphisms, functorial in P, Q, G,

$$ \varphi^n = \varphi^n_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}({}_nP \otimes {}_nQ, G), \tag{2.2.2}$$

$$\varphi^{(\ell)} = \varphi^{(\ell)}_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(T_\ell(P)\otimes T_\ell(Q), T_\ell(G)\big). \tag{2.2.3}$$

For

$$\xi \in \mathrm{Biext}^1(P,Q;G), \tag{2.2.4}$$

denote its image under (2.2.2) (respectively, under (2.2.3)) by φ^n_ξ (respectively, by $\varphi^{(\ell)}_\xi$):

$$\varphi^n_\xi : {}_nP \otimes {}_nQ \longrightarrow G \qquad \text{(or, equivalently, } {}_nG), \tag{2.2.5}$$

$$\varphi^{(\ell)}_\xi : T_\ell(P)\otimes T_\ell(Q) \longrightarrow T_\ell(G). \tag{2.2.6}$$

2.2.7. Plainly, knowledge of $\varphi^{(\ell)}_\xi$ is equivalent to knowledge of the φ^n_ξ for all $n = \ell^\nu$, with $\nu \geq 0$. I 238
For fixed n, knowledge of φ^n_ξ is equivalent to knowledge, on the image of

$$\alpha^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q),$$

of the homomorphism

$$\underline{\mathrm{Tor}}_1(P,Q) \longrightarrow G \tag{2.2.8}$$

which is the image of ξ under (2.2.1).

Observe that if $P = nP$ and $Q = nQ$, then the preceding image is ${}_n\underline{\mathrm{Tor}}_1(P,Q)$. This follows readily by considering the exact sequence (1.3.5) and the analogous exact sequence for Q. Furthermore, if P and Q are divisible, that is, if $nP = P$ and $nQ = Q$ for every positive integer n, one readily checks that $\underline{\mathrm{Tor}}_1(P,Q)$ is torsion, that is,

$$\underline{\mathrm{Tor}}_1(P,Q) = \varinjlim_n {}_n\underline{\mathrm{Tor}}_1(P,Q).$$

Thus, in this case, knowledge of the φ^n_ξ for all n is equivalent to knowledge of (2.2.8), whereas knowledge of $\varphi^{(\ell)}_\xi$ for a given prime ℓ is equivalent to knowledge of the restriction of (2.2.8) to the ℓ-primary component of $\underline{\mathrm{Tor}}_1(P,Q)$.

We now make explicit the symmetry properties of the pairings (2.2.5) and (2.2.6) associated with biextensions. First observe that there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow & & \downarrow \\ \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) & \xrightarrow{\sim} & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big), \end{array} \tag{2.2.9}$$

where the horizontal arrows are the symmetry arrows (the first of which was defined in VII 2.7), and the vertical arrows are (2.2.1). The commutativity follows from that of the two constituent squares in the fuller diagram

$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow & & \downarrow \\ \mathrm{Ext}^1\Big(P \overset{\mathbb{L}}{\otimes} Q, G\Big) & \xrightarrow{\sim} & \mathrm{Ext}^1\Big(Q \overset{\mathbb{L}}{\otimes} P, G\Big) \\ \downarrow & & \downarrow \\ \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) & \xrightarrow{\sim} & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big). \end{array} \tag{2.2.10}$$

The first vertical arrow in each column is the canonical isomorphism of VII 3.6.5, and the second is the evident homomorphism. The second square in (2.2.10) is trivially commutative: by definition, the symmetry isomorphism

$$P \overset{\mathbb{L}}{\otimes} Q \xrightarrow{\sim} Q \overset{\mathbb{L}}{\otimes} P$$

induces the symmetry isomorphism $\underline{\mathrm{Tor}}_1(P,Q) \simeq \underline{\mathrm{Tor}}_1(Q,P)$, and, for a variable complex $L_\bullet$, the canonical isomorphism

$$\mathrm{Ext}^1(L_\bullet, G) \simeq \mathrm{Hom}(H_1(L_\bullet), G)$$

is functorial in $L_\bullet$.

As for the commutativity of the first square of (2.2.10), it follows by making explicit simultaneously the symmetry isomorphisms and the vertical isomorphisms defined in VII 3.6.5. We leave the details of the verification to the reader. One uses the fact that the symmetry isomorphism I ≈239

$$L_\bullet(P) \otimes L_\bullet(Q) \simeq L_\bullet(Q) \otimes L_\bullet(P)$$

induces in degree 1 the isomorphism

$$L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \simeq L_0(Q) \otimes L_1(P) + L_1(Q) \otimes L_0(P),$$

which, on corresponding summands, is given by the ordinary symmetry isomorphism of the underlying ungraded modules (and hence introduces no sign).

In view of the commutativity of (2.2.9), the anticommutativity of (2.1.10.1), in terms of the φ^n or $\varphi^{(\ell)}$, becomes the following relation.

Proposition 2.2.11. The following diagrams are anticommutative: I 240

(2.2.11.1)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow{\scriptstyle \varphi^n_{P,Q}} & - & \downarrow{\scriptstyle \varphi^n_{Q,P}} \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) & \xrightarrow{\sim} & \mathrm{Hom}({}_nQ \otimes {}_nP, G), \end{array}$$

(2.2.11.2)
$$\begin{array}{ccc} \mathrm{Biext}^1(P,Q;G) & \xrightarrow{\sim} & \mathrm{Biext}^1(Q,P;G) \\ \downarrow{\scriptstyle \varphi^{(\ell)}_{P,Q}} & - & \downarrow{\scriptstyle \varphi^{(\ell)}_{Q,P}} \\ \mathrm{Hom}(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)) & \xrightarrow{\sim} & \mathrm{Hom}(T_\ell(Q) \otimes T_\ell(P), T_\ell(G)), \end{array}$$

where the horizontal arrows are the symmetry arrows.

We record the important corollary.

Corollary 2.2.12. Suppose that $P = Q$, and suppose that ξ is the class of a symmetric (respectively, antisymmetric) biextension of (P,P) by G; that is, suppose ${}^s\xi = \xi$ (respectively, ${}^s\xi = -\xi$), where ${}^s\xi$ denotes the image of ξ under symmetry. Then the bilinear maps φ^n_ξ, on ${}_nP \times {}_nP$, and $\varphi^{(\ell)}_\xi$, on $T_\ell(P) \times T_\ell(P)$, are antisymmetric (respectively, symmetric).

Remark 2.2.13. Suppose that ξ is symmetric. If n (respectively, ℓ) is odd, then 2.2.12 shows that φ^n_ξ (respectively, $\varphi^{(\ell)}_\xi$) is in fact an alternating form. This conclusion is not valid in general without the I 241

restriction that n (respectively, ℓ) be odd, as is already seen when P is the ordinary group $\mathbb{Z}/2\mathbb{Z}$ and $G = \mathbb{Z}/2\mathbb{Z}$.

Suppose, however, that $2P = P$. This is the case in particular when P is an abelian scheme, working as usual in the fppf or fpqc topology. The preceding forms are then still alternating. It is enough to prove that $\varphi_\xi^{2^\nu}(x,x) = 0$. Since $2P = P$, a given element $x \in {}_{2^\nu}P(S)$ may be written as $x = 2y$, with $y \in {}_{2^{\nu+1}}P(S)$. Hence

$$\varphi_\xi^{2^\nu}(x,x) = 2\varphi_\xi^{2^{\nu+1}}(y,y) = 0,$$

because $\varphi_\xi^{2^{\nu+1}}$ is antisymmetric.

2.3. Relation with "Cartier duality". The description given in 2.2 of the pairings defined by a biextension, through the isomorphism theorem VII 3.6.5 and the homomorphism (2.2.1), is plainly not the definition formerly used for abelian schemes, at a time when the very notion of biextension did not yet exist. That definition proceeds through Cartier duality, and we now give its relation to the present one.

Consider again the homomorphism

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q,G)\big) \tag{2.3.1}$$

arising from the exact sequence (1.4). It is in fact an isomorphism when

$$\underline{\mathrm{Hom}}(Q,G) = 0. \tag{2.3.2}$$

This condition holds in particular when Q is an abelian scheme over a scheme S and G is the multiplicative group $\underline{G}_m$ (VII 1.3.8).

There is also an evident "restriction to nQ" homomorphism I≈242

$$\underline{\mathrm{Ext}}^1(Q,G) \longrightarrow \underline{\mathrm{Ext}}^1(nQ,G),$$

and hence a composite homomorphism

$$\begin{aligned}\mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q,G)\big) &\longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(nQ,G)\big)\\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big),\end{aligned} \tag{2.3.4}$$

where the second arrow is restriction to ${}_nP$. Composing with (2.3.1), we obtain a homomorphism

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big). \tag{2.3.5}$$

On the other hand, the exact sequence

$$0 \longrightarrow {}_nQ \longrightarrow Q \xrightarrow{n} nQ \longrightarrow 0 \tag{2.3.6}$$

gives rise to a long exact sequence of the $\underline{\mathrm{Ext}}^i(-,G)$, and hence to the short exact sequence

$$0 \longrightarrow \underline{\mathrm{Hom}}(Q,G)_n \longrightarrow \underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{\partial} {}_n\underline{\mathrm{Ext}}^1(nQ,G) \longrightarrow 0, \tag{2.3.7}$$

where, as usual, E_n denotes E/nE. We retain from this sequence the canonical surjective homomorphism

$$\underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{\quad\partial\quad} {}_n\underline{\mathrm{Ext}}^1(nQ,G), \tag{2.3.8}$$

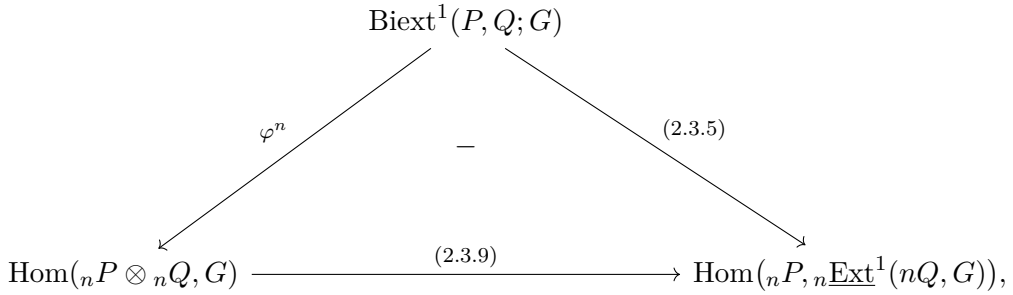
which is an isomorphism if $\underline{\mathrm{Hom}}(Q,G)_n = 0$, and a fortiori if (2.3.2) holds. From (2.3.8) we deduce a homomorphism

$$\begin{aligned}\mathrm{Hom}\big({}_nP, \underline{\mathrm{Hom}}({}_nQ,G)\big) &\simeq \mathrm{Hom}({}_nP \otimes {}_nQ, G)\\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ,G)\big),\end{aligned} \tag{2.3.9}$$

which is an isomorphism when (2.3.2) holds.

All the homomorphisms under consideration are functorial in P, Q, G. The homomorphisms (2.3.5), (2.3.9), and (2.2.2) form the diagram

(2.3.10)



where the horizontal arrow (2.3.9) is an isomorphism if (2.3.2) holds.

Proposition 2.3.11. The diagram (2.3.10) is anticommutative. I 243

When $\underline{\mathrm{Hom}}(Q,G) = 0$, this statement therefore recovers a characterization of the homomorphism φ^n of (2.2.2) in terms of the homomorphisms (2.3.5) and (2.3.9). This is the characterization formerly used in the theory of abelian schemes, where moreover $nQ = Q$.

Proof of 2.3.11.[3] Let ξ be an element of Biext$^1(P,Q;G)$, let

$$\varphi_\xi : {}_nP \otimes {}_nQ \longrightarrow G$$

be the associated pairing, and let

$$u_\xi : {}_nP \longrightarrow {}_n\underline{\mathrm{Ext}}^1(nQ, G)$$

be the image of ξ under (2.3.5). We must prove that $-u_\xi$ is obtained from φ_ξ by the arrow (2.3.9), that is, that it is the composite

$$(*) \qquad {}_nP \xrightarrow{\varphi_\xi} \underline{\mathrm{Hom}}({}_nQ, G) \xrightarrow{(2.3.8)} {}_n\underline{\mathrm{Ext}}^1(nQ,G).$$

To prove this, we may take a section x of ${}_nP$ and prove that its image under $-u_\xi$ equals its image under the composite (*). If desired, this would allow us to reduce to the case $P = (\mathbb{Z}/n\mathbb{Z})_T$, but we shall not need that reduction.

a) Choose flat resolutions $L_\bullet$ and $M_\bullet$,

$$0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0,$$

of P and Q, respectively. A flat resolution of P, for example, is obtained by writing P as a quotient of a flat $\mathbb{Z}_T$-module L_0 and taking L_1 to be the kernel of $L_0 \longrightarrow P$; this kernel is flat because it is torsion-free. We obtain the complex $L_\bullet \otimes M_\bullet$

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0,$$

whose sheaf $\underline{H}_1$ will serve as $\underline{\mathrm{Tor}}_1(P,Q)$.

Let L_0' (respectively, M_0') be the subsheaf of L_0 (respectively, M_0) which is the inverse image of ${}_nP$ (respectively, ${}_nQ$). Thus I ≈244

$$0 \longrightarrow L_1 \longrightarrow L_0' \longrightarrow 0 \qquad (\text{respectively, } 0 \longrightarrow M_1 \longrightarrow M_0' \longrightarrow 0)$$

is a flat resolution of ${}_nP$ (respectively, of ${}_nQ$). The homomorphism

$${}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q) = \underline{H}_1(L_\bullet \otimes M_\bullet)$$

[3] For a more elegant proof, see 2.3.19 below.

may then be described, by passage to the quotient, from the homomorphism

$$L'_0 \otimes M'_0 \longrightarrow \underline{H}_1(L_\bullet \otimes M_\bullet)$$

given by

$$x_0 \otimes y_0 \longmapsto \text{the class of the 1-cycle } [nx_0] \otimes y_0 - x_0 \otimes [ny_0].$$

Here the brackets indicate that the element in question is regarded as a section of the subsheaf L_1 of L_0 (respectively, M_1 of M_0). This description follows from the calculation made in the proof of 2.1.10.

b) Take an injective resolution

$$C^\bullet = (0 \longrightarrow C^0 \xrightarrow{d^0} C^1 \longrightarrow \cdots)$$

of G, and identify

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) = H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, C^{\cdot}))\,.$$

The element ξ may therefore be represented by a pair of homomorphisms f_1, f_0, giving the anticommutative diagram

$$\begin{array}{ccc} L_1 \otimes M_0 + L_0 \otimes M_1 & \xrightarrow{d_1} & L_0 \otimes M_0 \\ \downarrow{\scriptstyle f_1} & - & \downarrow{\scriptstyle f_0} \\ C^0 & \xrightarrow{d^0} & C^1. \end{array} \tag{2.3.12}$$

By definition, the pairing

$$\varphi_\xi : {}_nP \otimes {}_nQ \longrightarrow G$$

is obtained, by passage to the quotient, from the homomorphism

$$\varphi : L'_0 \otimes M'_0 \longrightarrow G$$

given by

$$\varphi(x_0 \otimes y_0) = f_1([nx_0] \otimes y_0 - x_0 \otimes [ny_0]) \in G = \ker(C^0 \longrightarrow C^1). \tag{2.3.13}$$

Since assertion 2.3.11 is local on T, we may suppose that the given section x of ${}_nP$ lifts to a section x_0 of L'_0. Regarding y_0 as variable, the preceding formula then defines a homomorphism

$$\varphi' : M'_0 \longrightarrow G, \qquad y_0 \longmapsto f_1([nx_0] \otimes y_0 - x_0 \otimes [ny_0]), \tag{2.3.14}$$

which passes to the quotient to give

$${}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x, y),$$

the homomorphism associated with φ_ξ and x.

c) We must take the image of this homomorphism under the connecting operator ∂ of (2.3.7), associated with the exact sequence (2.3.6). Now $nQ \simeq Q/{}_nQ$ is plainly also isomorphic to M_0/M'_0, so that we have a homomorphism of exact sequences

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$$\begin{array}{ccccccccc} 0 & \longrightarrow & M'_0 & \longrightarrow & M_0 & \longrightarrow & M_0/M'_0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow\wr & & \\ 0 & \longrightarrow & {}_nQ & \longrightarrow & Q & \longrightarrow & nQ & \longrightarrow & 0 \end{array} \quad ,$$

where the last vertical arrow is an isomorphism. This shows that the desired section of $\underline{\mathrm{Ext}}^1(nQ, G)$ may also be computed as the image of φ' in (2.3.14) under the connecting operator

$$\underline{\mathrm{Hom}}(M_0', G) \xrightarrow{\partial} \underline{\mathrm{Ext}}^1(M_0/M_0', G),$$

or, equivalently, as the section of the second term defined by the element of global Ext^1 which is the image of φ' under the connecting operator

$$\mathrm{Hom}(M_0', G) \longrightarrow \mathrm{Ext}^1(M_0/M_0', G).$$

The usual construction of the connecting operator gives the following recipe. Extend the composite (2.3.13)

$$M_0' \xrightarrow{\varphi'} G \longrightarrow C^0$$

to a homomorphism

$$\psi : M_0 \longrightarrow C^0,$$

which is possible since C^0 is injective, and consider the unique homomorphism

$$\lambda : M_0/M_0' \longrightarrow C^1 \tag{2.3.15}$$

which makes the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_0' & \longrightarrow & M_0 & \longrightarrow & M_0/M_0' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varphi'} & & \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \lambda} & & \\ 0 & \longrightarrow & G & \longrightarrow & C^0 & \longrightarrow & C^1 & & \end{array} \quad .$$

commutative. It is clear that (2.3.15) takes values in the 1-cocycles—that is, it is a 1-cocycle—and that this cocycle represents the desired element of

$$\mathrm{Ext}^1(nQ, G) = \mathrm{Ext}^1(M_0/M_0', G).$$

d) It remains to describe explicitly the section of $\underline{\mathrm{Ext}}^1(nQ, G)$ obtained from ξ and x by means of arrow (2.3.5) in diagram (2.3.10). For this purpose we may disregard the fact that the section x of P is annihilated by n. In terms of the chosen lift x_0 of x to a section of L_0, the description is as follows. This lift defines an augmentation homomorphism I ≈**247**

$$u : M_\bullet \longrightarrow L_\bullet \otimes M_\bullet, \qquad y_0 \longmapsto x_0 \otimes y_0, \quad y_1 \longmapsto x_0 \otimes y_1.$$

The section of $\underline{\mathrm{Ext}}^1(Q, G)$ obtained from (2.3.1) is therefore the one obtained from the element ξ of $\mathrm{Ext}^1(L_\bullet \otimes M_\bullet, G)$ by composition with the preceding homomorphism. Thus it is represented by the degree-one homomorphism

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_1 & \xrightarrow{d_1} & M_0 & \longrightarrow & 0 & & \\ & & \downarrow{\scriptstyle y_1 \mapsto f_1(x_0 \otimes y_1)} & - & \downarrow{\scriptstyle y_0 \mapsto f_0(x_0 \otimes y_0)} & & & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{d^0} & C^1 & & \ldots & & \end{array} \quad .$$

To restrict this class to $nQ \hookrightarrow Q$, use the isomorphism $nQ \simeq M_0/M_0'$, which gives a flat resolution

$$0 \longrightarrow M_0' \longrightarrow M_0 \longrightarrow 0$$

of nQ. The inclusion $nQ \hookrightarrow Q$ lifts to the following homomorphism of resolutions:

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_0' & \longrightarrow & M_0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle y_0' \mapsto [ny_0']} & + & \downarrow{\scriptstyle y_0 \mapsto ny_0} & & \\ 0 & \longrightarrow & M_1 & \longrightarrow & M_0 & \longrightarrow & 0 \end{array} \quad .$$

By composition, the desired section of $\underline{\mathrm{Ext}}^1(nQ, G)$ is therefore associated with the element of $\mathrm{Ext}^1(nQ, G)$ represented by the degree-one homomorphism of complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_0' & \xrightarrow{d_1} & M_0 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle g_1} & & \downarrow{\scriptstyle g_0} & & \\ 0 & \longrightarrow & C^0 & \longrightarrow & C^1 & \longrightarrow & \cdots \end{array} \quad ,$$

where

$$g_1(y_0') = f_1\big(x_0 \otimes [ny_0']\big), \qquad g_0(y_0) = nf_0(x_0 \otimes y_0).$$

e) Using now the fact that x is a section of ${}_nP$, we show that the preceding element of $\mathrm{Ext}^1(nQ, G)$ is the negative of the element defined by (2.3.15). For any homomorphism I≈**248**

$$h : M_0 \longrightarrow C^0,$$

the element of Ext^1 considered in d) is unchanged if g_1, g_0 are replaced, respectively, by

$$g_1' = g_1 + hd_1, \qquad g_0' = g_0 - d^0 h.$$

We have at our disposal the homomorphism $\psi : M_0 \longrightarrow C^0$ constructed in c), which by definition satisfies

$$\psi(y_0') = f_1\big([nx_0] \otimes y_0' - x_0 \otimes [ny_0']\big)$$

for every section y_0' of M_0'. Equivalently,

$$f_1\big(x_0 \otimes [ny_0']\big) (= g_1(y_0')) = f_1\big([nx_0] \otimes y_0'\big) - \psi(y_0').$$

Define $h : M_0 \longrightarrow C^0$ by

$$h(y_0) = -f_1\big([nx_0] \otimes y_0\big) + \psi(y_0).$$

By construction, then,

$$g_1' = 0, \qquad g_0'(y_0) = nf_0(x_0 \otimes y_0) + d^0 f_1\big([nx_0] \otimes y_0\big) - d^0\psi(y_0).$$

Since $d^0 f_1 = -f_0 d_1$, we have

$$d^0 f_1\big([nx_0] \otimes y_0\big) = -f_0(nx_0 \otimes y_0),$$

and hence

$$g_1' = 0, \qquad g_0' = -d^0\psi.$$

Thus the element in question is the one defined by the homomorphism

$$M_0/M_0' \simeq nQ \longrightarrow C^1$$

induced by $-d^0\psi$ upon passage to the quotient. We therefore obtain the negative of the element defined by (2.3.15).

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Remarks 2.3.16. The preceding proof actually gives a slightly more precise compatibility than the one stated in 2.3.11, obtained by choosing a section x of ${}_nP$. This section then defines a homomorphism

$$ {}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x,y), $$

and hence, by the connecting homomorphism associated with (2.3.6), an element of $\mathrm{Ext}^1(nQ,G)$. On the other hand, if we forget that x is annihilated by n, we consider the homomorphism $\mathbb{Z}_T \longrightarrow P$ defined by x. Its inverse image is a biextension of $(\mathbb{Z}_T, Q)$ by G, whose class may be interpreted as an element of $\mathrm{Ext}^1(Q,G)$ and therefore induces an element of $\mathrm{Ext}^1(nQ,G)$. With these conventions, the two elements obtained in $\mathrm{Ext}^1(nQ,G)$ are opposite.

2.3.17. Let us use the explicit cocycle description of the pairing

$$ \varphi_\xi^n : {}_nP \otimes {}_nQ \longrightarrow G $$

given in (2.3.12) to obtain another expression for it. First suppose that the biextension E of class ξ admits a section s over $P \times Q$. As in VII 3.8.13, take for the flat resolution of P the one with $L_0 = \mathbb{Z}[P]$ and L_1 the kernel of the canonical homomorphism $\mathbb{Z}[P] \longrightarrow P$, and define $M_\bullet$ similarly. There is then a unique biextension trivialization of the inverse image of E on (L_0, M_0) whose restriction to $P \times Q$ is given by s; denote this trivialization again by s (cf. VII 3.8.2). By VII 3.8.12, the class of ξ in

$$ \mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) $$

is represented by the cocycle with components (f,g) characterized by

$$ s(x_1,y_0) = f(x_1 \otimes y_0) e_{E/Q}(y_0), \qquad s(x_0,y_1) = g(x_0 \otimes y_1) e_{E/P}(x_0). $$

If x and y are sections of ${}_nP$ and ${}_nQ$, lift them to the corresponding sections $x_0 = [x]$ and $y_0 = [y]$ of L_0 and M_0. Applying formula (2.3.13), and putting $z = s(x,y)$, gives

$$ \varphi_\xi^n(x,y) = {}^n z / e_{E/Q}(y) - z^n / e_{E/P}(x). \tag{2.3.18} $$

Here the exponent n written on the left, respectively on the right, of z denotes the nth power for the left, respectively right, multiplicative structure on E. It therefore gives a section of E over $(0,y)$, respectively $(x,0)$; the meaning of the "quotient" of such an element by $e_{E/Q}(y)$, respectively $e_{E/P}(x)$, is clear.

The preceding formula (2.3.18) still makes sense whenever z is a global section of E lifting the sections x, y of ${}_nP$, ${}_nQ$. I claim that (2.3.18) is valid for every such section, without assuming that a section of E over $P \times Q$ exists. The section (x,y) of $P \times Q$ always lifts locally to a section of E, since $E \longrightarrow P \times Q$ is an epimorphism; thus the formula computes $\varphi_\xi^n(x,y)$ in every case. Moreover, if z is replaced by another section z' of E lifting (x,y), then $z' = z + g$ for a section g of G. The two terms on the right-hand side of (2.3.18) corresponding to z' are the terms corresponding to z, each increased by ng. Their difference is therefore independent of z, showing that (2.3.18) defines a well-determined homomorphism

$$ {}_nP \otimes {}_nQ \longrightarrow G. $$

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It remains to show that this homomorphism is indeed φ_ξ^n. For this we reduce at once to the case

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$$ P = Q = (\mathbb{Z}/n\mathbb{Z})_T. $$

Then $P \times Q$ is of the form I_T, where $I = \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ is a finite set. In this case the torsor E under $G_{P\times Q}$—which identifies with a family of torsors E_i under G, for $i \in I$—plainly becomes trivial locally on T. We are thus reduced to the already treated case in which E has a global section over $P \times Q$.

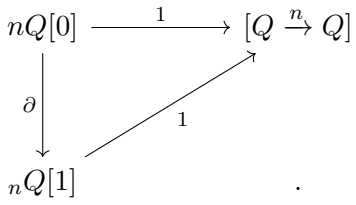
2.3.19. Another proof of 2.3.11 (by Deligne).

a) Reduction to $P = \mathbb{Z}/n\mathbb{Z}$ and to $T = (\mathrm{Sets})$.

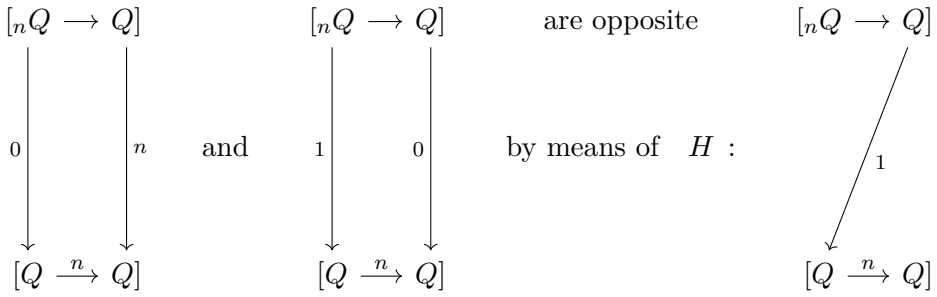
b) Let ∂ be the following arrow from nQ to ${}_nQ[1]$:

$$nQ \xleftarrow[\text{“}n\text{”}]{\sim} [\,{}_nQ \longrightarrow Q\,] \longrightarrow {}_nQ[1].$$

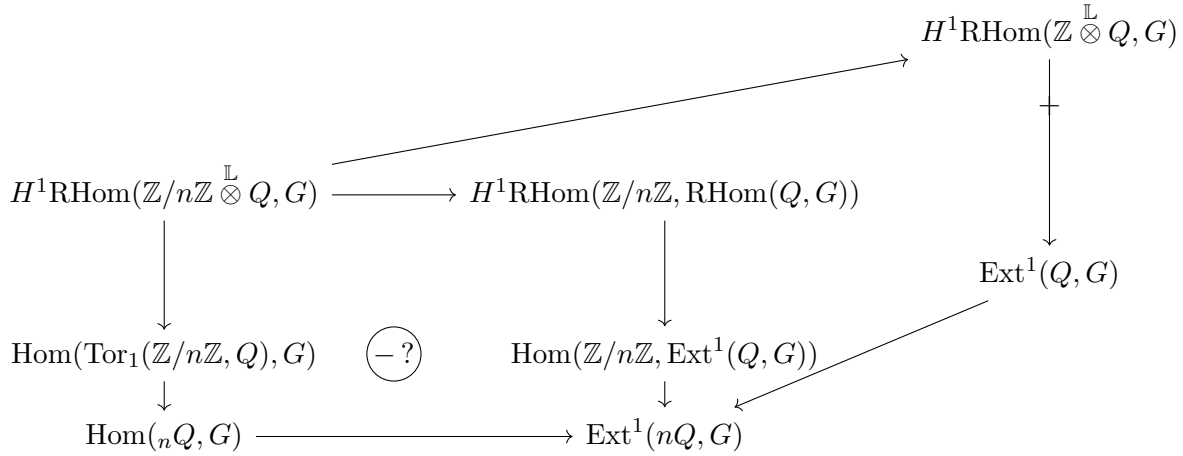
The following diagram is anticommutative:



Indeed, the arrows



c) Diagram 2.3.10 is I ≈**252**



and:

c1) We have an isomorphism

$$\mathbb{Z}/n\mathbb{Z} \overset{\mathbb{L}}{\otimes} Q \simeq [Q \xrightarrow{n} Q],$$

and, under this isomorphism, the first composite vertical arrow is induced by

$${}_nQ[1] \longrightarrow [Q \xrightarrow{n} Q].$$

c2) The second horizontal arrow is the transpose of the arrow ∂ displayed above in b).

c3) If the diagram is traversed clockwise, it is clear that one obtains the arrow induced by

$$nQ[0] \longrightarrow [Q \xrightarrow{n} Q].$$

This proves 2.3.11.

2.4. Let I 253

$$f : \underline{T}' \longrightarrow \underline{T}$$

be a morphism of topoi. There is then a canonical homomorphism (VII 2.8)

$$\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(P', Q'; G'),$$

where primes denote inverse-image groups. Since, for every commutative group E of $\underline{T}$, there is plainly a canonical isomorphism

$$({}_nE)' \simeq {}_n(E'),$$

the homomorphisms (2.2.2) over $\underline{T}$ and $\underline{T}'$ give a canonical diagram

$$\begin{array}{ccc} \mathrm{Biext}^1(P, Q; G) & \longrightarrow & \mathrm{Biext}^1(P', Q'; G') \\ \downarrow{\scriptstyle \varphi^n_{P,Q}} & & \downarrow{\scriptstyle \varphi^n_{P',Q'}} \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) & \longrightarrow & \mathrm{Hom}({}_nP' \otimes {}_nQ', G'). \end{array} \tag{2.4.1}$$

Similarly, the homomorphisms (2.2.3) give an analogous diagram in terms of the T_ℓ. I claim that these diagrams are commutative. The verification is essentially immediate from the definitions and is left to the reader. More generally, notice that all the canonical homomorphisms considered in this section and the preceding one are compatible with change of topos, in the evident sense. We shall henceforth use compatibilities of this kind without further mention.

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3. Biextensions of group schemes (P, Q) by $\underline{G}_m$: generalities

Throughout this section we fix a base scheme S and, as announced in Section 0, work in the fppf topos of S.

3.1. Let us first recall the most important example of a group $\mathrm{Biext}^1(P, Q; \underline{G}_m)$: the case where P and Q are two abelian schemes over S, and where Biext^1 is interpreted as the group of divisorial correspondences on (P, Q) (VII 2.9.4). The ℓ-adic pairings of the preceding section are then the familiar pairings [1] in the theory of abelian schemes.

3.2. More generally, suppose only that Q is an abelian scheme. It is well known (VII 1.3.8) that

$$\underline{\mathrm{Hom}}(Q, \underline{G}_m) = 0, \tag{3.2.1}$$

and hence (1.4) there is a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \underline{G}_m) \simeq \mathrm{Hom}(P, Q^*), \tag{3.2.2}$$

where

$$Q^* = \underline{\mathrm{Ext}}^1(Q, \underline{G}_m) \tag{3.2.3}$$

is the dual abelian scheme of Q. (For the representability of Q^*, due to M. Raynaud in the general case, see [1].)

When P is also an abelian scheme, symmetrically to (3.2.2), there is a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \underline{G}_m) \simeq \mathrm{Hom}(Q, P^*). \tag{3.2.4}$$

Thus the datum of a biextension ξ of (P, Q) by $\underline{G}_m$ may be interpreted equivalently as the datum of a homomorphism

$$u : P \longrightarrow Q^*$$

or of a homomorphism

$$v : Q \longrightarrow P^*.$$

Recall that a homomorphism of abelian schemes $w : A \longrightarrow B$ is an isomorphism if and only if, for every integer $n > 0$, the induced homomorphism ${}_nw : {}_nA \longrightarrow {}_nB$ is an isomorphism. In the case of u and v above, ${}_nu : {}_nP \longrightarrow {}_nQ^*$ is identified with the homomorphism ${}_nP \longrightarrow D({}_nQ)$ defined by the pairing I≈255

$$\varphi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m$$

(see 2.3.11), whereas ${}_nv : {}_nQ \longrightarrow {}_nP^*$ is identified with the homomorphism ${}_nQ \longrightarrow D({}_nP)$ defined by the pairing

$$\psi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m.$$

Now $\varphi_\xi^n = -\psi_\xi^n$ (2.1.10), so, up to sign, ${}_nu$ and ${}_nv$ are transposes of one another. Since ${}_nP$ and ${}_nQ$ are finite and locally free over S, and therefore satisfy Cartier duality, it follows that ${}_nu$ is an isomorphism if and only if ${}_nv$ is an isomorphism. Consequently u is an isomorphism if and only if v is an isomorphism.

An equivalent, though slightly less symmetric, formulation is the following. Consider the canonical biextension (the Weil biextension) of (P, P^*) by $\underline{G}_m$, corresponding to the identity homomorphism $v : P^* \longrightarrow P^*$. Then the corresponding canonical homomorphism

$$u : P \longrightarrow P^{**}$$

is an isomorphism. This is the biduality theorem of Barsotti and Cartier.

3.3. We now turn to the case where Q is finite locally free over S, or is a group scheme of multiplicative type over S. The key lemma in this case is the following well-known result. I 256

Proposition 3.3.1. Let Q be a commutative group scheme over S, and suppose that Q is finite locally free over S, or that Q is of finite type and of multiplicative type over S (SGA 3 IX). Then

$$\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0.$$

Proof. We distinguish the two cases.

a) Suppose that Q is finite locally free over S. We must prove that every extension E of Q by $\underline{G}_m$ splits locally for the fppf topology. We may plainly suppose S affine; there is then an integer $n > 0$ such that $nQ = 0$ (SGA 3 IX 6.1). Moreover, $n \cdot \mathrm{id}$ is an epimorphism of $\underline{G}_m$ for the fppf topology, and hence the Kummer exact sequence

$$0 \longrightarrow \mu_n \longrightarrow \underline{G}_m \xrightarrow{n} \underline{G}_m \longrightarrow 0$$

gives an exact sequence of the $\mathrm{Ext}^i(Q, -)$ and, in particular, an epimorphism

$$\mathrm{Ext}^1(Q, \mu_n) \longrightarrow {}_n\,\mathrm{Ext}^1(Q, \underline{G}_m) = \mathrm{Ext}^1(Q, \underline{G}_m).$$

The last equality follows because multiplication by n, being zero on Q, is also zero on $\mathrm{Ext}^1(Q, \underline{G}_m)$. Thus the extension E comes from an extension F of Q by μ_n.

The group F is represented by a finite locally free scheme over S (fpqc descent theory, SGA 1 VII). By the known properties of Cartier duality for finite locally free groups (SGA 3 VII 3),

$$D(F) = \underline{\mathrm{Hom}}(F, \underline{G}_m)$$

is an extension of $D(\mu_n)$ by $D(Q)$. Here one uses the following fact: if $A \longrightarrow B$ is a monomorphism of commutative finite locally free groups over S, then $D(B) \longrightarrow D(A)$ is faithfully flat. Since $D(B)$ and $D(A)$ are themselves finite locally free over S, the fibrewise criterion for faithful flatness reduces us to the case where S is the spectrum of a field k. In that case the assertion says that the homomorphism of affine coordinate rings

$$k(D(A)) \longrightarrow k(D(B))$$

is injective (SGA 3 VI$_B$ 11.14). This homomorphism is the transpose of $k(B) \longrightarrow k(A)$, which is surjective because $A \longrightarrow B$ is a closed immersion (SGA 3 VI$_B$ 1.4.2). Hence $k(D(A)) \longrightarrow k(D(B))$ is indeed injective.

The inclusion $\mu_n \longrightarrow \underline{G}_m$ identifies with a section of I≈257

$$D(\mu_n) = \underline{\mathrm{Hom}}(\mu_n, \underline{G}_m),$$

and it remains only to prove that, locally for the fppf topology on S, this inclusion extends to a homomorphism $F \longrightarrow \underline{G}_m$; equivalently, that the section of $D(\mu_n)$ just considered lifts fppf-locally to a section of $D(F)$. This is clear, since $D(F) \longrightarrow D(\mu_n)$ is faithfully flat and of finite presentation.

b) By SGA 3 X 4.5, after replacing S by an ètale cover, we may suppose that Q is diagonalizable, say $Q = D(M)$, where M is a finitely generated $\mathbb{Z}$-module. Let T be the torsion submodule of M, and put $L = M/T$, so that $L \simeq \mathbb{Z}^r$. The exact sequence

$$0 \longrightarrow T \longrightarrow M \longrightarrow L \longrightarrow 0$$

then gives an exact sequence (SGA 3 IX 3.1)

$$0 \longrightarrow D(L) \longrightarrow D(M) \longrightarrow D(T) \longrightarrow 0.$$

It is therefore enough to prove separately that the $\underline{\mathrm{Ext}}^1$ of $D(L)$ and of $D(T)$ by $\underline{G}_m$ vanish. The case of $D(T)$ follows from a), and the case of $D(L)$ reduces to $L = \mathbb{Z}$. Thus it remains to check that

$$\underline{\mathrm{Ext}}^1(\underline{G}_m, \underline{G}_m) = 0,$$

that is, that every extension E of $\underline{G}_m$ by $\underline{G}_m$ is fppf-locally trivial.

Such an extension is represented by an affine flat group scheme over S (fpqc descent theory, SGA 1 VII) whose fibres are tori; it is therefore a torus (SGA 3 X 4.9). By SGA 3 X 4.5 we may suppose that E is a split torus I≈258

$$E = D(R) \simeq \underline{G}_m^2.$$

After a Zariski localization, we may also suppose that the monomorphism $\underline{G}_m \hookrightarrow E = D(R)$ comes from a constant morphism $R \longrightarrow \mathbb{Z}_S$ (SGA 3 VIII 1.4), which is necessarily an epimorphism (SGA 3 VIII 3.2 b)). Choosing $r \in R$ above $1 \in \mathbb{Z}$ gives a homomorphism $E \longrightarrow \underline{G}_m$ splitting the extension, and proves the assertion.

Corollary 3.3.2. Let Q be as in 3.3.1. Then, for every P, there is a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \underline{G}_m) \xleftarrow{\sim} \mathrm{Ext}^1(P, D(Q)), \tag{3.3.2.1}$$

where $D(Q)$ denotes the Cartier dual $\underline{\mathrm{Hom}}(Q, \underline{G}_m)$ of Q. More precisely, the canonical functor (1.1.6) is an equivalence from the category of extensions of P by $D(Q)$ to the category of biextensions of (P, Q) by $\underline{G}_m$.

This is the special case of 1.5 obtained by applying 3.3.1.

Remarks 3.3.3. Under the hypotheses of 3.3.1, $D(Q)$ is represented by a separated group scheme, locally of finite presentation and locally quasi-finite over S. More precisely, if Q is finite locally free over S, then so is $D(Q)$, as is well known (SGA 3 VII 3). If Q is of multiplicative type, then $D(Q)$ is a finitely generated twisted constant group (SGA 3 X 5.1), as was seen in SGA 3 IX 4.5 and X 5.7. If P is a group scheme, every extension of P by $D(Q)$ is therefore representable, by a familiar lemma from fpqc descent theory (SGA 3 X 5.4). I≈259

3.3.4. It is false that $\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0$ for every affine flat group scheme Q of finite presentation over S. This already fails for

$$Q = \underline{G}_{a_S}, \qquad S = \mathrm{Spec}\, k[t]/(t^2),$$

where k is a field of characteristic $p > 0$.

Proposition 3.4. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let Q be a torus over S. Then the category of biextensions of (P, Q) by $\underline{G}_m$ is equivalent to the one-point category; that is,

$$\mathrm{Biext}^0(P, Q; \underline{G}_m) = \mathrm{Biext}^1(P, Q; \underline{G}_m) = 0. \tag{3.4.1}$$

Proof. Put $M = D(Q)$. This is a twisted constant group over S whose fibres are isomorphic to groups $\mathbb{Z}^r$ (3.3.3). By 3.3.2, the category in question is equivalent to the category of extensions of P by M. This category is rigid: the automorphism group of any object is the trivial group. This is the first equality in (3.4.1), which may also be written

$$\mathrm{Hom}(P, M) = 0.$$

It follows at once from the fact that M is etale over S, whereas P has connected fibres. It remains to prove

$$\mathrm{Ext}^1(P, M) = 0,$$

that is, that every extension E of P by M is trivial.

By the uniqueness of a splitting when one exists, it is enough to prove that E is locally trivial. I≈**260**
We may therefore reduce to the case $M = \mathbb{Z}_S^r$, and then to $M = \mathbb{Z}_S$. When S is the spectrum of a field, the conclusion follows from 5.5 (i) below. (Notice that we shall not use 3.4 before IX 6, so in particular there is no vicious circle.) We now reduce the general case to this special case.

It is enough to find a section f of E over P such that $f(0) = 0$, for such a section is necessarily multiplicative. Indeed,

$$f(x + y) - f(x) - f(y) = 0 :$$

the left-hand side defines a morphism $P \times_S P \longrightarrow \mathbb{Z}_S$ which vanishes on the zero section of $P \times_S P$. It therefore vanishes identically, since $P \times_S P$ has connected fibres (the geometric fibres of P being connected). The required conclusion now follows from the following lemma.

Lemma 3.4.2. Let S be a scheme, P a smooth S-scheme of finite presentation with geometrically connected fibres, Z a group, and E a Z_P-torsor (for the fppf, etale, or fpqc topology; these notions are equivalent here, cf. SGA 3 X 5.4). Let e be a section of P over S, and let e' be a section of E over S above e (that is, a section of $e^{-1}(E)$ over S).

The torsor E is trivial if and only if, for every $s \in S$, the induced torsor E_s over P_s is trivial. In I≈**261**
that case there is a unique section f of E over P such that $f \circ e = e'$.

Proof. Uniqueness is clear. Two such sections f and g "differ" by a morphism

$$h : P \longrightarrow Z_S, \qquad h \circ e = 1,$$

and such a morphism is constantly equal to 1, since P has connected fibres. Moreover, if a section f of E over P exists, it can always be corrected by a morphism $P \longrightarrow Z_S$ coming from a morphism $z : S \longrightarrow Z_S$, so as to obtain a section f' satisfying $f' \circ e = e'$. One need only choose z so that

$$f(e(t)) = e'(t)z(t).$$

Applying this observation fibrewise, for every $s \in S$ we obtain a well-determined section f_s of E_s over P_s taking e_s to e'_s. It remains to prove that the f_s arise from a section f of E over P. Equivalently (EGA IV 17.9.3), we must prove that

$$U = \bigcup_{s \in S} f_s(P_s) \subset E$$

is open in E.

Suppose first that S is noetherian. Then U is constructible. To see this, for each $s \in S$ let T be the closure of s, endowed with its reduced induced structure. There is an open neighbourhood T' of s in T and a section of $E_{T'}$ over $P_{T'}$ extending f_s and compatible with the marked sections e and e'

(as follows directly from EGA IV 8.8.2). By the criterion of EGA IV 1.10.1, it remains to show that U is stable under generization. The usual argument using EGA II 7.1.9 reduces this to the case in which S is the spectrum of a discrete valuation ring.

Since P is smooth over S, it is regular and hence normal; and since it has connected fibres, it is connected. The given section of E over the generic fibre of P over S therefore extends uniquely to a section f of E over P (EGA IV 18.10.19 and 18.10.20), necessarily compatible with e and e'. Thus in this case $U = f(P)$ is open. This proves 3.4 when S is noetherian. The general case reduces readily to this one by the usual limit arguments of EGA IV, 8 and 9; we leave the details to the reader.

Remark 3.4.3. Lemma 3.4.2 can be "crushed" under learned references, namely SGA 4 XV 1.15 (the case $n = 1$) and 4.1. I 262

Corollary 3.5. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0$$

be an exact sequence of commutative groups over S, with Q' a torus. Then the inverse-image functor on biextensions

$$\mathrm{BIEXT}(P, Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G}_m) \tag{3.5.1}$$

is an equivalence of categories. In particular, the following natural homomorphisms are isomorphisms:

$$\mathrm{Biext}^i(P, Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G}_m), \qquad i = 0, 1. \tag{3.5.2}$$

Indeed, up to equivalence, $\mathrm{BIEXT}(P, Q''; \underline{G}_m)$ identifies with the category of biextensions E of (P, Q) by $\underline{G}_m$ equipped with a trivialization of the induced biextension E' of (P, Q') by $\underline{G}_m$ (VII 3.7.6). By 3.4 there is exactly one such trivialization of E' for each given biextension E. One can of course also prove directly that (3.5.2) is an isomorphism for $i = 0, 1$ - which is equivalent to the conclusion of 3.5 - by using the exact sequence of the Biext^i in VII 3.7.5 associated with

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0.$$

Remark 3.6. The symmetric statement obtained from 3.5 by interchanging left and right is of course also valid; it also follows formally from 3.5 by passing to symmetric biextensions (VII 2.7). Suppose, in addition to the data and hypotheses of 3.5, that the symmetric data and hypotheses are given: namely, that Q has connected fibres (equivalently, that Q'' has connected fibres, since Q' has connected fibres), and that there is an exact sequence I 263

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

with P' a torus. Successive application of 3.5 and of its symmetric form, applied to this last sequence and to Q'', shows that the functor

$$\mathrm{BIEXT}(P'', Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G}_m) \tag{3.6.1}$$

is an equivalence of categories. In particular, the following homomorphism is an isomorphism:

$$\mathrm{Biext}^i(P'', Q''; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G}_m), \qquad i = 0, 1. \tag{3.6.2}$$

Proposition 3.7. Let A be an abelian scheme over S, let T be a group scheme of multiplicative type (SGA 3 IX) and of finite type over S, and let

$$\underline{M} = \underline{\mathrm{Hom}}(T, \underline{G}_m)$$

be the Cartier dual of T (a finitely generated twisted constant group over S, by 3.3.3). Then the categories $\mathrm{EXT}(A, T)$ and $\mathrm{BIEXT}(A, \underline{M}; \underline{G}_m)$, formed respectively by commutative extensions of A

by T and by biextensions of $(A, \underline{M})$ by $\underline{G}_m$, are rigid and canonically equivalent. Their groups of isomorphism classes are canonically isomorphic to $\mathrm{Hom}(\underline{M}, A^*)$, where

$$A^* = \underline{\mathrm{Ext}}^1(A, \underline{G}_m)$$

is the dual abelian scheme of A (3.2). In particular, there is a canonical isomorphism

$$\mathrm{Ext}^1(A, T) \xrightarrow{\sim} \mathrm{Hom}(\underline{M}, A^*). \tag{3.7.1}$$

This is the special case of 1.6 obtained by taking $P = A$ and $Q = \underline{M}$. By the biduality theorem (SGA 3 X 5.7), we do indeed have I≈**264**

$$T \simeq \underline{\mathrm{Hom}}(\underline{M}, \underline{G}_m),$$

and it remains only to verify the conditions (1.6.1), which here read

$$\underline{\mathrm{Hom}}(A, \underline{G}_m) = 0, \qquad \mathrm{Hom}\big(A, \underline{\mathrm{Ext}}^1(\underline{M}, \underline{G}_m)\big) = 0.$$

The first equality was recalled in (3.2.1); the second may be strengthened to

$$\underline{\mathrm{Ext}}^1(\underline{M}, \underline{G}_m) = 0.$$

This assertion is local on S, so we may suppose that $\underline{M}$ is constant, $\underline{M} = M_S$. Decomposing M as a direct sum of cyclic $\mathbb{Z}$-modules reduces us to $M = \mathbb{Z}$, which is trivial, and to $M = \mathbb{Z}/n\mathbb{Z}$, which follows from the n-divisibility of $\underline{G}_m$ (or, more learnedly, from 3.3.1).

When $T = \underline{G}_m$, and hence $\underline{M} = \mathbb{Z}_S$, (3.7.1) reduces to the identity isomorphism (3.2.3). Of course, a direct proof of (3.7.1) would be essentially trivial, without introducing the group

$$\mathrm{Ext}^1(A \overset{\mathbb{L}}{\otimes} \underline{M}, \underline{G}_m) = \mathrm{Biext}^1(A, \underline{M}; \underline{G}_m).$$

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4. Extensions and biextensions of smooth connected group schemes over a field

We recall the following lemma due to Rosenlicht [3 bis, Proposition 3].

Lemma 4.1. Let k be a field, and let X and Y be two separable, geometrically connected k-schemes of finite type, equipped respectively with k-rational points e_X and e_Y. Then every morphism

$$X \times Y \longrightarrow \underline{G}_m$$

which is trivial on the two subschemes $X \times e_Y$ and $e_X \times Y$ is trivial.

We shall use this result only when X and Y are group schemes over k; in that case the hypotheses imply that X and Y are in fact smooth over k. By induction, 4.1 also implies the analogous result for any number of factors. We may therefore apply the results of VII 1.3 and VII 2.9, and obtain the following two propositions.

Proposition 4.2. Let P be a smooth connected group over the field k. The natural functor from the category of group extensions of P by $\underline{G}_m$ to the category of line bundles on P rigidified at the identity element e_P is fully faithful. Its essential image consists of the rigidified line bundles $\mathbb{L}$ on P for which

$$\pi^*\mathbb{L} \simeq \mathrm{pr}_1^*\,\mathbb{L} \otimes \mathrm{pr}_2^*\,\mathbb{L},$$

where $\pi : P \times P \longrightarrow P$ is the group law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections.

Proposition 4.3. Let P and Q be two smooth connected algebraic groups over the field k. The natural functor from the category of biextensions of (P, Q) by $\underline{G}_m$ to the category of line bundles on $P \times Q$ birigidified with respect to the identity sections e_P and e_Q of P and Q is fully faithful; moreover, both categories are rigid. A birigidified line bundle $\mathbb{L}$ on $P \times Q$ comes from a biextension if and only

if its pullbacks to $P \times P \times Q$ and $P \times Q \times Q$ by $\pi_P \times \mathrm{id}_Q$ and $\mathrm{id}_P \times \pi_Q$ satisfy the "theorem of the cube" ([1] or [2]).

This last condition is simply the translation of the conditions in VII 2.9.3 b), in the explicit form given in VII 3.5 b). I≈266

4.4. I do not know whether the cube condition in 4.3 is automatically satisfied. At least it is known [2, IV 2.6] that, for a given $\mathbb{L}$, there is always an integer $n \geq 1$, a power of the characteristic exponent of k, such that $\mathbb{L}^{\otimes n}$ satisfies that condition and hence comes from a biextension of (P, Q) by $\underline{G}_m$. It is also always possible to find a purely inseparable extension k' of k such that the required condition becomes true after the base extension $k \longrightarrow k'$; after this extension, therefore, $\mathbb{L}$ is a birigidified line bundle arising from a biextension of (P, Q) by $\underline{G}_m$.

In particular, when k is perfect, the condition in 4.3 is always satisfied. This last fact, which formally implies the preceding statements by a trace argument, can also be seen by applying 4.7 below; cf. 4.9.

We recall, in the form of a lemma, the following result from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.13.

Lemma 4.5. Let $f : X \longrightarrow Y$ be a smooth surjective morphism with integral fibres, where Y is normal and integral with generic point y. Let $\mathbb{L}$ be a line bundle on X whose restriction to X_y is trivial. Then there is a line bundle $\mathbb{M}$ on Y such that I 267

$$\mathbb{L} \simeq f^*\mathbb{M}.$$

Proposition 4.6. Let P, Q be two smooth connected algebraic groups over the field k, with P affine. Suppose in addition either that P admits a composition series each of whose factors is a torus or is isomorphic to the additive group $\underline{G}_a$ (a condition satisfied in particular when k is perfect [SGA 3, XVII 7.2.1, 4.1.1 (i)$\Longleftrightarrow$ (ii), and 4.1.5]), or that Q is an extension of an abelian variety B by a smooth connected affine algebraic group satisfying the preceding condition. Under either hypothesis, every birigidified line bundle $\mathbb{L}$ on $P \times Q$ is trivial; a fortiori, by 4.3, every biextension of (P, Q) by $\underline{G}_m$ is trivial.

Proof. First observe that every automorphism of a birigidified line bundle is the identity (4.3). Descent theory therefore shows that it is enough to prove that $\mathbb{L}$ becomes trivial after a finite separable extension of k. Passing to the limit immediately reduces us to the case where k is separably closed.

Every torus over such a field is split. Hence, if P admits a composition series as in 4.6, it also admits one whose quotients are isomorphic to $\underline{G}_a$ or $\underline{G}_m$. Repeated application of 4.5, together with

$$\mathrm{Pic}(\underline{G}_{a_K}) = \mathrm{Pic}(\underline{G}_{m_K}) = 0$$

for every field K, gives

$$\mathrm{Pic}(P) = 0.$$

This equality remains true after every extension of the base field k. Applying 4.5 to

$$\mathrm{pr}_2 : P \times Q \longrightarrow Q$$

and to $\mathbb{L}$, we find a line bundle $\mathbb{M}$ on Q such that $\mathbb{L} \simeq \mathrm{pr}_2^*\,\mathbb{M}$. Since the restriction of $\mathbb{L}$ to $e_P \times Q$ is trivial, $\mathbb{M}$ is trivial, and so is $\mathbb{L}$. This proves the first case.

In the second case, let Q be an extension of an abelian scheme B by a group M admitting a composition series with quotients isomorphic to $\underline{G}_m$ or $\underline{G}_a$. Apply 4.5 to I≈268

$$f : X = P \times Q \longrightarrow Y = P \times B.$$

The fibre over the generic point y of $P \times B$ is an M_y-torsor. It is necessarily trivial by the particular structure of M and the classical equalities

$$H^1(K, \underline{G}_a) = 0, \qquad H^1(K, \underline{G}_m) = 0,$$

valid for every field K, here taken to be $k(y)$. Since $\mathrm{Pic}(M) = 0$, we have $\mathrm{Pic}(X_y) = 0$; thus 4.5 applies and shows that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{N}$ on $Y = P \times B$.

Let $\mathbb{N}_1$ and $\mathbb{N}_2$ be the restrictions of $\mathbb{N}$ to $P \times e_B$ and $e_P \times B$, respectively. Replacing $\mathbb{N}$ by

$$\mathbb{N} \otimes \mathrm{pr}_1^*(\mathbb{N}_1)^{-1} \otimes \mathrm{pr}_2^*(\mathbb{N}_2)^{-1},$$

we may suppose that $\mathbb{N}$ is birigidified and that $\mathbb{L}$ is its birigidified pullback. We are thereby reduced to the case in which Q itself is an abelian scheme.

Then $\mathbb{L}$ defines a canonical morphism

$$(*) \qquad P \longrightarrow \underline{\mathrm{Pic}}_{Q/k}$$

taking the identity to the identity. Since P is connected, this morphism factors through the dual abelian scheme

$$Q' = \underline{\mathrm{Pic}}^0_{Q/k}.$$

It is well known that every morphism of schemes from a smooth connected affine algebraic group to an abelian scheme is constant. This is a geometric assertion, so we may suppose k algebraically closed; a devissage then reduces it immediately to $P = \underline{G}_a$ or $P = \underline{G}_m$, and every rational map from $\mathbb{P}^1_k$ to an abelian scheme over k is constant.

The constancy, hence vanishing, of (*) means that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{M}$ on P. I≈**269**
Since $\mathbb{L}$ is trivial on $P \times e_Q$, $\mathbb{M}$ is trivial, and therefore so is $\mathbb{L}$. This completes the proof.

Proposition 4.7. Let P and Q be two smooth connected algebraic groups over the field k. Suppose that P is an extension of an abelian scheme A by a smooth connected affine algebraic group L admitting a composition series each of whose factors is isomorphic to a torus or to $\underline{G}_a$ (a condition always satisfied when k is perfect). Then the functor considered in 4.3 is an equivalence of categories; that is, every birigidified line bundle $\mathbb{L}$ on $P \times Q$ comes from a biextension of (P, Q). Moreover, the functor between the rigid categories

$$\mathrm{BIEXT}(A, Q; \underline{G}_m) \longrightarrow \mathrm{BIEXT}(P, Q; \underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

$$(4.7.1) \qquad \mathrm{Biext}^1(A, Q; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, Q; \underline{G}_m).$$

For the first assertion, descent and 4.3 again reduce us to the case where k is separably closed. Proceeding as in the proof of 4.6, we deduce from 4.5 and the hypothesis on L that $\mathbb{L}$ is isomorphic to the pullback of a birigidified line bundle $\mathbb{M}$ on $A \times Q$. For the first assertion of 4.7 we are thus reduced to the case $A = P$. The line bundle $\mathbb{L}$ then defines a morphism of pointed schemes

$$f : Q \longrightarrow \underline{\mathrm{Pic}}_{P/k},$$

which necessarily factors through the dual abelian scheme P' of P. Let $\mathbb{L}'$ be the pullback of the Weil sheaf on $P \times P'$ along

$$\mathrm{id}_P \times f : P \times Q \longrightarrow P \times P'.$$

This is a birigidified line bundle which, by the definition of f, differs from $\mathbb{L}$ only by the pullback of a line bundle $\mathbb{M}$ on P. Since both $\mathbb{L}'$ and $\mathbb{L}$ have trivial restriction to $P \times e_Q$, it follows that $\mathbb{M}$ is trivial and hence that $\mathbb{L} \simeq \mathbb{L}'$. We noted in VII 2.9.4 that divisorial correspondences on a product of two abelian schemes arise from biextensions. Consequently $\mathbb{L}'$, and therefore $\mathbb{L}$, comes from a biextension.

Returning to the general case, where P is an extension of A by L, it remains to prove that (4.7.1) I≈**270**
is an isomorphism. This follows from the exact sequence of the Biext^i, for $i = 0, 1$, associated with

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0,$$

and from the relations

$$\mathrm{Biext}^0(L,Q;\underline{G}_m)=0, \qquad \mathrm{Biext}^1(L,Q;\underline{G}_m)=0.$$

The first is a special case of the rigidity assertion in 4.3, and the second is precisely the first case of 4.6.

I 271 Corollary 4.8. Under the hypotheses of 4.7, suppose that Q is an extension of an algebraic group B (necessarily smooth and connected) by a smooth connected affine algebraic group M. Then the functor between the rigid categories

$$\mathrm{BIEXT}(A,B;\underline{G}_m)\longrightarrow \mathrm{BIEXT}(P,Q;\underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

$$\mathrm{Biext}^1(A,B;\underline{G}_m)\xrightarrow{\sim}\mathrm{Biext}^1(P,Q;\underline{G}_m). \tag{4.8.1}$$

In view of (4.7.1), it remains to prove that

$$\mathrm{Biext}^1(A,B;\underline{G}_m)\xrightarrow{\sim}\mathrm{Biext}^1(A,Q;\underline{G}_m)$$

is an isomorphism. As in the proof of (4.7.1), this follows from the exact sequence of the Biext^i associated with

$$0\longrightarrow M\longrightarrow Q\longrightarrow B\longrightarrow 0,$$

together with the relations

$$\mathrm{Biext}^0(P,M;\underline{G}_m)=0, \qquad \mathrm{Biext}^1(P,M;\underline{G}_m)=0.$$

Indeed, (4.7.1), applied with M in place of Q, identifies these groups with the corresponding groups having A in the first variable, as required by the preceding exact sequence. The second vanishing displayed above is a special case of the second case of 4.6, after interchanging the two factors.

I 272 Remark 4.9. Let P and Q be two smooth connected algebraic groups over a perfect field k. It is then known [3] that each is an extension of an abelian scheme by a smooth connected affine group, the latter being the product of a torus with an iterated extension of copies of $\underline{G}_a$ (cf. the reference given in 4.6):

$$0\longrightarrow L\longrightarrow P\longrightarrow A\longrightarrow 0, \qquad 0\longrightarrow M\longrightarrow Q\longrightarrow B\longrightarrow 0.$$

Thus 4.8 applies and shows that the theory of biextensions of (P,Q) by $\underline{G}_m$ reduces to the analogous (and much more classical!) theory for the pair (A,B) of abelian varieties.

Corollary 4.10. Let P and Q be two smooth connected algebraic groups over k, let L be a smooth connected affine algebraic subgroup of P, and let E be a biextension of (P,Q) by $\underline{G}_m$. By 2.2, E induces a pairing

$$T_\ell(P)\times T_\ell(Q)\longrightarrow T_\ell(\underline{G}_m).$$

Then the induced pairing

$$T_\ell(L)\times T_\ell(Q)\longrightarrow T_\ell(\underline{G}_m)$$

is zero.

Indeed, we may plainly suppose that k is algebraically closed. We are then under the hypotheses of 4.6, which imply that the restriction of E to (L,Q) is the trivial biextension. The asserted conclusion follows, since the induced pairing in question is the pairing defined by this restriction.

I 273 4.11. Let P and Q be algebraic groups smooth over k, let E be a biextension of (P,Q) by $\underline{G}_m$, and let T be the largest subtorus of P. We shall determine the left annihilator of the pairing

$$T_\ell(P)\times T_\ell(Q)\longrightarrow T_\ell(\underline{G}_{m,k}), \tag{4.11.1}$$

where ℓ is again a fixed prime number. By 4.10 this annihilator always contains the subgroup $T_\ell(T)$ of $T_\ell(P)$.

Choose an extension k' of k over which P and Q can each be written as an extension of an abelian scheme by an affine group scheme that is an iterated extension of tori and copies of $\underline{G}_a$. If P and Q have unipotent rank zero, we may take $k' = k$; in general, we may take for k' the perfect closure of k (or a finite radicial extension). Let A' and B' be the abelian parts of $P_{k'}$ and $Q_{k'}$, respectively. By 4.7, the biextension $E_{k'}$ of $(P_{k'}, Q_{k'})$ comes from a biextension F' of (A', B'), uniquely determined up to unique isomorphism. Consequently the pairing

$$T_\ell(P_{k'}) \times T_\ell(Q_{k'}) \longrightarrow T_\ell(\underline{G}_{m,k'}), \tag{4.11.2}$$

obtained from (4.11.1) by the base-field extension $k \to k'$, and also defined by the biextension $E_{k'}$, is obtained from the pairing

$$T_\ell(A') \times T_\ell(B') \longrightarrow T_\ell(\underline{G}_{m,k'}) \tag{4.11.3}$$

defined by F', by composing with the canonical homomorphisms (which are in fact epimorphisms if P and Q are ℓ-divisible)

$$T_\ell(P_{k'}) \longrightarrow T_\ell(A'), \qquad T_\ell(Q_{k'}) \longrightarrow T_\ell(B'). \tag{4.11.4}$$

It follows that the left annihilator of (4.11.2), which is obtained from the left annihilator of (4.11.1) by base change, is the inverse image under the first homomorphism in (4.11.4) of the left annihilator of (4.11.3). Let C' be the largest abelian subscheme of A' such that the restriction of F' to (C', B') is the trivial biextension. Equivalently, C' is the reduced identity component of the kernel of the homomorphism

$$A' \longrightarrow (B')^* \tag{4.11.5}$$

defined by F', where $(B')^*$ is the dual abelian scheme of B'. The left annihilator of (4.11.3) identifies with $T_\ell(C')$; therefore the left annihilator of (4.11.1) is precisely the inverse image of

$$T_\ell(C') \hookrightarrow T_\ell(A')$$

under the first morphism in (4.11.4).

In particular, to say that the left annihilator of (4.11.1) is exactly $T_\ell(T)$ – equivalently, that the left annihilator of (4.11.2) is the kernel $T_\ell(T')$ of the first morphism in (4.11.4) – is to say that (4.11.3) is nondegenerate on the left, that is, that $C' = 0$, or equivalently that (4.11.5) has finite kernel. In this last form the condition is visibly independent of ℓ, and in any of its forms it is plainly independent of the choice of k'. When it holds, we shall simply say that the given biextension E of (P, Q) by $\underline{G}_m$ is nondegenerate on the left. By duality, this also means that the homomorphism I≈274

$$B' \longrightarrow (A')^* \tag{4.11.6}$$

defined by F' is an epimorphism, that is, is surjective.

The notion of a biextension nondegenerate on the right is defined symmetrically: (4.11.6) must have finite kernel, or, equivalently, (4.11.5) must be an epimorphism. The biextension E of (P, Q) by $\underline{G}_{m,k}$ is called nondegenerate if it is nondegenerate on both the left and the right. This is equivalent to saying that (4.11.5), or equivalently (4.11.6), is an isogeny.

It is immediate from the definition that these notions are invariant under every extension of the base field k. I≈275

4.12. Let P be a smooth connected group scheme, let $\mathbb{L}$ be a line bundle on P, and put

$$\delta(\mathbb{L}) = \pi^*\mathbb{L} \otimes \mathrm{pr}_1^*(\mathbb{L})^{-1} \otimes \mathrm{pr}_2^*(\mathbb{L})^{-1}, \tag{4.12.1}$$

with the notation of 4.2. As already observed in VII 1.3.4, this is a birigidified line bundle on $P \times P$. By 4.3, if $\delta(\mathbb{L})$ comes from a biextension, that biextension is determined up to unique isomorphism.

In particular, by 4.7, it is defined when P is an extension of an abelian scheme A by an iterated extension of tori and copies of $\underline{G}_a$. Whenever it is defined, this biextension is plainly symmetric.

In the preceding favourable case, and assuming that the maximal torus of P is split, the construction of $\delta(\mathbb{L})$ can be reduced to the case where P is an abelian scheme. Indeed, the construction is functorial in P, while the homomorphism

$$\mathrm{Pic}(A) \longrightarrow \mathrm{Pic}(P) \tag{4.12.2}$$

induced by the canonical quotient homomorphism $f : P \longrightarrow A$ is surjective by 4.5 (cf. the beginning of the proof of 4.6). We can therefore find an isomorphism

$$\mathbb{L} \simeq f^*\mathbb{M},$$

where $\mathbb{M}$ is a line bundle on A. Then $\delta(\mathbb{L})$ is the pullback of the biextension $\delta(\mathbb{M})$ of (A, A) by $\underline{G}_{m,k}$.

Proposition 4.13. Under the hypotheses of 4.12, suppose that $\mathbb{L}$ is an ample line bundle on P. If the biextension $\delta(\mathbb{L})$ of (P, P) by $\underline{G}_{m,k}$ is defined, then it is nondegenerate in the sense of 4.11. If P is an extension of an abelian scheme A by a smooth connected affine group L satisfying the condition of 4.7, then the biextension of (A, A) from which $\delta(\mathbb{L})$ comes by 4.8 is of the form $\delta(\mathbb{M})$, where $\mathbb{M}$ is an ample line bundle on A. I 276

We reduce to the case where k is algebraically closed. By the definition, it is then enough, with the notation of 4.12, to prove that the biextension $\delta(\mathbb{M})$ of (A, A) is nondegenerate. M. Raynaud proves in his thesis [2, XI 1.11] that if $\mathbb{L} = f^*\mathbb{M}$ is ample, then $\mathbb{M}$ is ample as well. The theory of A. Weil then gives the familiar conclusion that $\delta(\mathbb{M})$ is nondegenerate [1, 2], that is, that the homomorphism from A to its dual abelian scheme is an isogeny.

Remark 4.14. Suppose we are in the favourable case where P is an extension of an abelian scheme A by a group L which is an iterated extension of tori and copies of $\underline{G}_a$, and consider the canonical composite

$$\mathrm{NS}(A) \xrightarrow{\delta} \mathrm{Biext}^1(A, A; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, P; \underline{G}_m). \tag{4.14.1}$$

By 4.8 the second arrow is bijective. On the other hand, the theory of abelian varieties shows that the first arrow identifies the Neron–Severi group $\mathrm{NS}(A)$ of A with the subgroup of

$$\mathrm{Biext}^1(A, A; \underline{G}_m) = \mathrm{Corr}(A, A)$$

consisting of the classes of symmetric biextensions, or equivalently symmetric divisorial correspondences. Thus (4.14.1) identifies $\mathrm{NS}(A)$ with the group of symmetric biextensions of (P, P) by $\underline{G}_m$. By 4.7 this group can equally be described as the group of classes of symmetric divisorial correspondences on (P, P).

It is natural to call a divisorial correspondence on (P, P) ample, and likewise to call the corresponding biextension of (P, P) by $\underline{G}_m$ ample, when it comes from an ample element of $\mathrm{NS}(A)$. This condition is plainly invariant under extension of the base field; the notion therefore still makes sense without the preceding restrictive hypothesis on P, assuming only that P is smooth and connected. I ≈277

Suppose further that L is "split solvable over k", in the sense that it admits a devissage by copies of $\underline{G}_a$ and $\underline{G}_m$. The preceding observation and the surjectivity of (4.12.2) show that the image of

$$\mathrm{Pic}(P) \xrightarrow{\delta} \mathrm{Biext}^1(P, P; \underline{G}_m) \tag{4.14.2}$$

consists of the symmetric biextensions, and that its kernel is the subgroup $\mathrm{Pic}^0(P)$ of $\mathrm{Pic}(P)$ which is the image of $\mathrm{Pic}^0(A)$. Consequently, if we put

$$\mathrm{NS}(P) = \mathrm{Pic}(P)/\mathrm{Pic}^0(P) \simeq \mathrm{NS}(A),$$

we may also interpret (4.14.1) as a canonical isomorphism

$$\mathrm{NS}(P) \xrightarrow{\sim} \mathrm{Biext}^1(P, P; \underline{G}_m)^{\mathrm{sym}}. \tag{4.14.3}$$

Finally, the theorem of Raynaud cited in 4.13 says that an element ξ of $\mathrm{Pic}(P)$ is ample if and only if its image $\delta(\xi)$ under (4.14.2) is the class of an ample biextension. I≈278

5. Extensions and biextensions by torsion-free truncated constant group schemes.

In this section and the next we give some auxiliary results that will be used in section 7 in the case $Z = \mathbb{Z}$.

Proposition 5.1. Let S be an irreducible geometrically unibranch scheme with generic point η, and let Z be a torsion-free group, hence giving a constant group scheme Z_S over S. Then every torsor E under Z_S is trivial, and the functor

$$E \longmapsto E(\eta)$$

from the category of torsors under Z_S to the category of torsors under $Z = Z_S(\eta)$ is an equivalence of categories.

A familiar lemma of descent theory (SGA 3 X 5.4) shows that every torsor E under Z_S is representable. In fact, that result applies to every group scheme over S which is separated, locally of finite presentation, and locally quasi-finite over S. Since Z_S is etale over S, fpqc descent shows that the same is true of every torsor E under Z_S; such a torsor is therefore locally trivial for the etale topology (and not merely for the fppf topology). To prove that E is trivial, it is consequently enough to prove

$$H^1(S_{\text{ét}}, Z_S) = e.$$

Likewise, the full-faithfulness assertion of 5.1 reduces to the relation

$$Z \xrightarrow{\sim} H^0(S_{\text{ét}}, Z_S).$$

The latter is immediate. To prove the former, consider the inclusion

$$i : \eta \longrightarrow S.$$

Because S is geometrically unibranch, we have

$$Z_S \xrightarrow{\sim} i_*(Z_\eta)$$

(SGA 4 IX 2.14.1). The noncommutative low-degree exact sequence that replaces the Leray spectral sequence for $i : \eta \to S$ (SGA 4 XII 3.2) therefore begins

$$e \longrightarrow H^1(S, i_*(Z_\eta)) \longrightarrow H^1(\eta, Z_\eta) \longrightarrow \cdots,$$

and shows that

$$H^1(S, Z_S) \longrightarrow H^1(\eta, Z_\eta)$$

is injective. We are thus reduced to the case in which S is the spectrum of a field k. If π is the Galois group of a separable closure of k, then

$$H^1(k, Z) = \mathrm{Hom}_{\mathrm{cont}}(\pi, Z)/Z = e,$$

because Z is torsion-free. This completes the proof.

Corollary 5.2. With the notation of 5.1, let P be a smooth scheme over S. Then every torsor E under Z_P is trivial, and the restriction functor

$$E \longmapsto E_\eta$$

from the category of torsors on P under Z_P to the category of torsors on P_η under Z_{P_η} is an equivalence of categories.

Indeed, P is geometrically unibranch (EGA IV 17.5.7). Moreover, every irreducible component of P dominates S, because P is flat over S, and the set of irreducible components of P is locally finite. Thus P is the disjoint sum of irreducible geometrically unibranch schemes P_i. The assertion

now follows immediately from 5.1 applied to the various P_i and their generic fibres $P_{i\eta}$, noting that the generic point p_i of P_i is also the generic point of $P_{i\eta}$.

I 280 5.2.1. The preceding corollary immediately gives the analogous statement for bitorsors on P under (Z_P, Z_P): restriction to the generic fibre induces an equivalence from their category to the category of bitorsors on P_η under (Z_{P_η}, Z_{P_η}). Indeed, it is enough to express the bitorsor structure on E in terms of its structure as a right torsor under Z_P, together with an isomorphism from Z_P to the commutant of Z_P for its action on E. Since E is necessarily trivial, that commutant is itself isomorphic to Z_P. We are thus reduced, on each irreducible component X of P, to the following immediate observation: if X is a nonempty connected scheme, the functor $Z \mapsto Z_X$ from sets to X-schemes is fully faithful.

Proposition 5.3. Let S and Z be as in 5.1, and let P be a smooth group scheme over S. Consider extensions E of group schemes of P by Z_S. Then restriction to the generic fibre

$$E \longmapsto E_\eta$$

is an equivalence from the category of these extensions to the category of extensions of P_η by Z_η. If P and Z are commutative, it also induces an equivalence between the corresponding categories of commutative extensions.

First observe that every extension of sheaves of groups of P by Z_S (and similarly of P_η by Z_η) I ≈281 gives, by 5.2, a trivial torsor, hence in particular a representable one. The categories in 5.3 may therefore equally well be interpreted as categories of extensions of sheaves of groups. The description in VII 1.1.6 and 1.2 expresses these in terms of bitorsors on P, $P \times P$, and $P \times P \times P$ (respectively on P_η, $P_\eta \times P_\eta$, and $P_\eta \times P_\eta \times P_\eta$), under the pairs of groups obtained from (Z, Z) by base change. Applying 5.2 in the commutative case and 5.2.1 in the general case to the smooth S-schemes P, $P \times P$, and $P \times P \times P$ gives the asserted conclusion. The same argument gives the following result.

Corollary 5.4. Let S, Z be as in 5.1, and let P, Q be two smooth group schemes over S. Suppose that P, Q, and Z are commutative, and consider biextensions E of (P, Q) by Z_S. Then restriction to the generic fibre

$$E \longmapsto E_\eta$$

is an equivalence from the category of these biextensions to the category of biextensions of (P_η, Q_η) by Z_η.

The preceding results reduce the determination of categories of extensions or biextensions by a torsion-free constant group to the case in which the base scheme is the spectrum of a field. In that case we obtain the following.

I 282 Proposition 5.5. Let k be a field, Z a torsion-free group, P a k-group scheme locally of finite type, P° its identity component, and $\Phi = P/P^\circ$ its group of connected components (SGA 3 VI$_A$ 4). Then:

(i) The natural functor from the category of group schemes which are extensions (respectively commutative extensions) of Φ by Z_k to the category of group schemes which are extensions (respectively commutative extensions) of P by Z_k is an equivalence. In particular, if P is connected, every extension of P by Z_k is trivial, and the category of such extensions is rigid.

(ii) Suppose that Φ is a torsion group and that P and Z are commutative. Then the category of extensions of P by Z_k is rigid: every automorphism of an object of this category is trivial. If, in addition, there is an integer $n > 0$ such that $n\mathrm{id}_\Phi = 0$—for example, if Φ is finite, as happens when P is of finite type—then the group of classes of commutative extensions of P by Z_k is given by the canonical isomorphism

$$\mathrm{Ext}^1(P, Z_k) \xleftarrow{\sim} \mathrm{Hom}\big(\Phi, (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z\big), \tag{5.5.1}$$

defined by the boundary operator in the exact sequence of Ext groups associated with the exact sequence

$$0 \longrightarrow Z \longrightarrow Z \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z \longrightarrow 0. \tag{5.5.2}$$

The canonical morphism

$$P \longrightarrow \Phi$$

plainly induces an isomorphism on the sets of irreducible components, since it is flat with irreducible fibres. By 5.1, the induced functor from torsors under Z_Φ to torsors under Z_P is therefore an equivalence. (Recall that a group scheme locally of finite type over k is geometrically unibranch.) The same conclusion applies to bitorsors and to the morphisms

$$P \times P \longrightarrow \Phi \times \Phi, \qquad P \times P \times P \longrightarrow \Phi \times \Phi \times \Phi.$$

The description in VII 1.1.6 of extensions of P by Z_k and of Φ by Z_k (and VII 1.2 in the commutative case) now gives the assertions in (i).

We restrict attention to commutative extensions. The automorphism group of an extension of P by Z_k is

$$\mathrm{Hom}(P, Z_k) \simeq \mathrm{Hom}(\Phi, Z_k),$$

which is plainly zero when Φ is a torsion group. Likewise, if $V = Z \otimes_{\mathbb{Z}} \mathbb{Q}$ and Φ is annihilated by the integer $n > 0$, then

$$\mathrm{Ext}^i(\Phi, V_k) = 0 \qquad \text{for every } i \in \mathbb{Z},$$

because the groups in question are $\mathbb{Q}$-vector spaces annihilated by n. The isomorphism (5.5.1) follows at once.

The same method proves, essentially, the following. I ≈**283**

Corollary 5.6. Let k be a field, Z a torsion-free commutative group, P and Q two commutative group schemes locally of finite type over k, and let $\Phi = P/P^\circ$ and $\Psi = Q/Q^\circ$ be their groups of connected components. Then:

(i) The natural functor from the category of biextensions of (Φ, Ψ) by Z_k to the category of biextensions of (P, Q) by Z_k is an equivalence. In particular, if P or Q is connected, the category of biextensions of (P, Q) by Z_k is equivalent to the terminal category.

(ii) Suppose that Φ or Ψ is a torsion group. Then the category of biextensions of (P, Q) by Z_k is rigid. If, in addition, there is an integer $n > 0$ such that $n\mathrm{id}_\Phi = 0$ or $n\mathrm{id}_\Psi = 0$, then the group of classes of biextensions of (P, Q) by Z_k is given by the canonical isomorphism I ≈**284**

$$\mathrm{Biext}^1(P, Q; Z_k) \xleftarrow{\sim} \mathrm{Hom}\big(\Phi \otimes \Psi, (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z\big), \tag{5.6.1}$$

defined by the boundary operator in the exact sequence of Biext^i groups $(i = 0, 1)$ associated with (5.5.2).

5.7. Let S be a scheme, T a closed subscheme, and Z a set. The constant scheme over S, truncated along T, with value Z, denoted $Z_{S,T}$, is the scheme obtained by taking the sum of copies S_z of S, one for each $z \in Z$, and gluing all these copies along the open subscheme $U = S \setminus T$. Thus

$$Z_{S,T} \times_S U \simeq U, \qquad Z_{S,T} \times_S T \simeq Z_T, \tag{5.7.1}$$

and $Z_{S,T}$ is etale over S. For a variable S-scheme S' there is an isomorphism, functorial in S',

$$\mathrm{Hom}_S(S', Z_{S,T}) \simeq \mathrm{Hom}_T(S'_T, Z_T). \tag{5.7.2}$$

The map is restriction to S'_T, using the second isomorphism in (5.7.1); we leave it to the reader to check that the map is bijective. If

$$i : T \longrightarrow S$$

denotes the inclusion, the functorial isomorphism (5.7.2) may also be written as an isomorphism of sheaves

$$Z_{S,T} \simeq i_*(Z_T). \tag{5.7.3}$$

This formula, or the definition directly, shows that as Z varies, $Z_{S,T}$ depends functorially on Z, and that this functor commutes with finite inverse limits. In particular, it takes groups to group schemes and commutative groups to commutative group schemes, as is also clear from the form (5.7.2) of the functor represented by $Z_{S,T}$.

It is also immediate from the definition, or from (5.7.3), that the formation of $Z_{S,T}$ commutes with every base change $f : S' \to S$. This gives an isomorphism, functorial in Z, I≈285

$$Z_{S,T} \times_S S' \simeq Z_{S',T'}, \qquad T' = T \times_S S' = T_{S'}. \tag{5.7.4}$$

5.8. We shall study torsors E under a group scheme of the form $Z_{S,T}$, where Z is a group, assumed commutative here for simplicity. There is a natural functor

$$E \longmapsto E_T \tag{5.8.0}$$

from the category of torsors under $Z_{S,T}$ to the category of torsors under Z_T. We claim that this functor is an equivalence of categories. Its full faithfulness means that for two torsors E, F under $Z_{S,T}$, the map

$$\mathrm{Hom}(E, F) \longrightarrow \mathrm{Hom}(E_T, F_T)$$

is bijective. By fpqc descent this reduces immediately to the case in which E and F are trivial, hence to $E = F = Z_{S,T}$, where the assertion is precisely the bijectivity of (5.7.2) for $S' = S$.

It remains to prove that (5.8.0) is essentially surjective. Let E_T be a torsor under Z_T. We already observed in the proof of 5.1 that E_T is representable and locally trivial for the etale topology. It therefore suffices to prove that I≈286

$$H^1(S_{\text{ét}}, Z_{S,T}) \longrightarrow H^1(T_{\text{ét}}, Z_T)$$

is bijective. This follows from (5.7.3) and the relation

$$R^1 i_{\text{ét},*}(Z_T) = 0,$$

which is the special case SGA 4 VIII 5.6.

5.8.1. Let now P be a commutative group scheme over S. We wish to determine the category of extensions, as sheaves of commutative groups, of P by $Z_{S,T}$. For this we use the description in VII 1.1.6 and 1.3 in terms of torsors on P, $P \times P$, and $P \times P \times P$, under the inverse images of $Z_{S,T}$. By (5.7.4), these inverse images are again of the same form as $Z_{S,T}$. Applying the preceding result to those torsors shows that they correspond to torsors on P_T, $P_T \times_T P_T$, and $P_T \times_T P_T \times_T P_T$ under the inverse images of Z_T. We therefore obtain the following.

Proposition 5.9. Let S be a scheme, T a closed subscheme, Z a commutative group, $Z_{S,T}$ the corresponding truncated constant group scheme over S defined in 5.7, and P a commutative group scheme over S. Then restriction to T induces an equivalence between the category of sheaves of commutative groups which are extensions of P by $Z_{S,T}$ and the category of sheaves of commutative groups which are extensions of P_T by Z_T.

An essentially identical argument gives the following. I≈287

Corollary 5.10. With the notation of 5.9, let Q be another commutative group scheme over S. Then restriction to T induces an equivalence between the category of biextensions of (P, Q) by $Z_{S,T}$ and the category of biextensions of (P_T, Q_T) by Z_T.

6. Extensions and biextensions by the Neron model of G_m.

6.1. Let S be a scheme, T a closed subscheme, and Z a set, giving a truncated constant scheme $Z_{S,T}$ over S (5.7). There is a canonical homomorphism, functorial in Z, of etale schemes over S,

$$Z_S \longrightarrow Z_{S,T}, \tag{6.1.1}$$

which is a surjective morphism, as follows at once from (5.7.1). Its functoriality in Z shows that when Z is a group, (6.1.1) is a homomorphism of etale group schemes over S. Its kernel is therefore an etale group scheme over S, denoted $Z_{S,U}$, and there is a canonical exact sequence, functorial in Z,

$$0 \longrightarrow Z_{S,U} \longrightarrow Z_S \longrightarrow Z_{S,T} \longrightarrow 0. \tag{6.1.2}$$

It follows immediately from (5.7.1) that $Z_{S,U}$ is the sum of a family of S-schemes S_n, $n \in Z$, where $S_n = S$ for $n = 0$ and $S_n = U$ for $n \neq 0$. This description of $Z_{S,U}$ readily shows that for every group F over S, restriction to U induces a bijection

$$\mathrm{Hom}_{\mathrm{gr}}(Z_{S,U}, F) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(Z_U, F|U). \tag{6.1.3}$$

I 288 6.1.4. Henceforth we suppose that Z is commutative, and we wish to examine extensions of commutative groups of $Z_{S,T}$ by a given commutative group G over S. By the exact sequence (6.1.2), the category of these extensions is equivalent to the category of extensions F of Z_S by G, furnished with a trivialization of the inverse image of the extension along $Z_{S,U} \to Z_S$. By (6.1.3), such a trivialization amounts to a trivialization of the extension $F|U$ of Z_U by $G|U$.

For what follows we take Z to be the group $\mathbb{Z}$ of integers. Thus $\mathbb{Z}_S$ is the free group on one generator, and VII 1.4 gives an equivalence between the category of extensions of $\mathbb{Z}_S$ by G and the category of torsors under G. Applying the same result to $\mathbb{Z}_U$ and $G|U$ gives the following.

Lemma 6.2. Let S be a scheme, T a closed subscheme, $U = S \setminus T$, and G a commutative group over S. The category of commutative groups over S which are extensions of $\mathbb{Z}_{S,T}$ (defined in (5.7)) by G is equivalent to the category of pairs (F, s) consisting of a torsor F under G and a section s of $F|U$. If $\alpha : S \to \mathbb{Z}_{S,T}$ is the section of $\mathbb{Z}_{S,T}$ which is the image of the section 1 of $\mathbb{Z}_S$ over S, then the torsor corresponding to an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is $F = \alpha^*(\overline{G})$ under G, furnished with the section of $F|U$ given by restricting the unit section of $\overline{G}$ to U.

I 289 6.2.1. If $\overline{G}$ corresponds to the pair (F, s), then the inverse image $f_n^*(\overline{G})$ in $\overline{G}$ of the section f_n of $\mathbb{Z}_{S,T}$ defined by the constant section of $\mathbb{Z}_S$ with value $n \in \mathbb{Z}$ is canonically identified, as a torsor under G, with the n-th power $F^{(n)}$ of the G-torsor F. The canonical isomorphism of G_U-torsors

$$f_n^*(\overline{G})|U \simeq G_U,$$

defined by the first relation in (5.7.1), is determined by the section $s^{(n)}$ of $F_U^{(n)}$, the n-th power of the given section s of F_U. This makes the following description of the sections of $\overline{G}$ over S explicit. A section $\overline{g}$ of $\overline{G}$ over S is determined by:

a) a section $\underline{n}$ of $\mathbb{Z}_{S,T}$ over S, which may be interpreted as a section of $\mathbb{Z}_T$, that is, as a locally constant function on T with values in $\mathbb{Z}$, or equivalently as a partition

$$T = \coprod_{i \in \mathbb{Z}} T_i$$

of T into a disjoint sum of open subsets indexed by $i \in \mathbb{Z}$;

b) a family of sections

$$\overline{g}_i \in \Gamma(F^{(i)}/S'_i), \qquad i \in \mathbb{Z},$$
$$S'_i = S \setminus T'_i, \qquad T'_i = \bigcup_{\substack{j \in \mathbb{Z} \\ j \neq i}} T_j,$$

subject to the following compatibility condition. For $i, j \in \mathbb{Z}$, $\overline{g}_i$ and $\overline{g}_j$ agree on $S'_i \cap S'_j = U$ after $F^{(i)}|U$ and $F^{(j)}|U$ have been identified with G_U by means of the sections $s^{(i)}$ and $s^{(j)}$, respectively. In other words, there must exist a necessarily unique section $h_{ij} \in \Gamma(G/U)$ such that

$$\overline{g}_i = h_{ij} s^{(i)}, \qquad \overline{g}_j = h_{ij} s^{(j)}.$$

Since the description in 6.2 of an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is plainly compatible with every base change $S' \to S$, the preceding gives a complete description of the sheaf $\overline{G}$ as a functor of the argument $S' \in \mathrm{Ob}(\mathrm{Sch}_{/S})$.

6.2.2. If T is a scheme which is a sum of nonempty connected schemes T_i, $i \in I$, the description in 6.2.1 of the sections of $\overline{G}$ over S may be simplified. Such a section is determined by a family of integers $(n_i)_{i\in I}$ indexed by I, and a family of sections I ≈290

$$\overline{g}_i \in \Gamma(F^{(n_i)}/S'_i), \qquad i \in I,$$

where

$$S'_i = S \setminus T'_i, \qquad T'_i = \coprod_{\substack{j\in I \\ j\neq i}} T_j.$$

The $\overline{g}_i$ satisfy the following compatibility condition: for every pair $i, j \in I$, the restrictions of $\overline{g}_i$ and $\overline{g}_j$ to $S'_i \cap S'_j = U$ agree under the identifications of $F^{(n_i)}|U$ and $F^{(n_j)}|U$ deduced from the respective sections $s^{(n_i)}$ and $s^{(n_j)}$ of these torsors.

6.2.3. Finally, if T is assumed nonempty and connected, giving a section $\overline{g}$ of $\overline{G}$ over S amounts to giving an integer n and a section $\overline{g}$ of $F^{(n)}$ over S. There is no longer any compatibility condition to impose, so the section s of $F|U$ does not occur in the description of the group of sections of $\overline{G}$ over S. (Of course, s occurs again as soon as one seeks to describe the points of $\overline{G}$ with values in a general S-scheme S'.)

6.2.4. Since $\overline{G}$ is the union of the "open subschemes" $f_n^*(\overline{G})$, $n \in \mathbb{Z}$, described in 6.2.1, it follows that $\overline{G}$ is representable if and only if the torsors $F^{(n)}$ are representable. For example, it suffices that G be representable and affine over S, by the descent theory of SGA 1 VIII 2.1.

6.3. In what follows we suppose that $G = \underline{G}_{mS}$. Every extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $G = \underline{G}_{mS}$ is then representable (6.2.4), and is of course smooth over S, being an extension of the etale group scheme $\mathbb{Z}_{S,T}$ by the smooth group scheme $\underline{G}_{mS}$. Moreover, giving a torsor F under $\underline{G}_m$ is equivalent to giving an invertible Module $\mathbb{L}$ over S (by associating to the latter the torsor of invertible sections of $\mathbb{L}$). Thus $\overline{G}$ is described by a pair $(\mathbb{L}, s)$, where $\mathbb{L}$ is an invertible Module over S and s is a trivialization of $\mathbb{L}$ over $U = S \setminus T$, that is, an invertible section of $\mathbb{L}|U$. I 291

When U is schematically dense in S, and S is locally noetherian, or is reduced with a locally finite set of irreducible components, giving such a pair is equivalent to giving a divisor D on S such that

$$\mathrm{supp}\, D \subset T, \tag{6.3.1}$$

by associating to such a divisor the pair

$$(\underline{O}_S(-D), s_{-D}|U) \tag{6.3.2}$$

formed by the invertible Module $\underline{O}_S(-D)$ associated with $-D$ (EGA IV 21.2.8.1) and the restriction to U of its canonical rational section s_{-D} (EGA IV 21.2.11 and 20.2.11(ii)). In any event, without any hypothesis on S, a divisor D on S satisfying (6.3.1) defines by (6.3.2) a pair $(\mathbb{L}, s)$ and hence an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $\underline{G}_m$. We call it the Neron–Raynaud model of $\underline{G}_m$ associated with the divisor D and the closed subscheme T of S satisfying (6.3.1). If $T = \mathrm{supp}\, D$, we omit T from the terminology.

It is immediate that the formation of this extension $\overline{G}$ commutes with every base change $f : S' \to S$ for which the inverse-image divisor $f^*(D)$ is defined (EGA IV 21.4.2), and in particular with every flat base change. I ≈292

6.3.3. The general description in 6.2.1 of the sections of $\overline{G}$ immediately shows that they may be identified with the sections f of the sheaf $\mathcal{M}_S^*$ of regular meromorphic sections on S (EGA IV 20.1.8) whose divisor $\mathrm{div}(f)$ is locally, for the Zariski topology, an integral multiple of D. If V is an open

subset of S on which the restriction of $\mathrm{div}(f)$ equals that of nD, where $n \in \mathbb{Z}$, then the section of $\mathbb{Z}_{S,T}$ defined by f is simply the image in (6.1.2) of the constant section of $\mathbb{Z}_S$ with value n. The preceding description of the group of sections of $\overline{G}$ over S remains valid over every open subset of S, and more generally for the group of points of $\overline{G}$ with values in any flat S-scheme S'. Since $\overline{G}$ itself is flat (indeed smooth) over S, this description again gives a complete characterization of the Neron–Raynaud model of $\underline{G}_m$ defined by D and $T \supset \mathrm{supp}\, D$.

6.4. Suppose now that S is normal, that its set of irreducible components is locally finite (so that S is a sum of normal integral schemes), and that D is a positive divisor with all multiplicities equal to 1 (EGA IV 21.6), that $T = \mathrm{supp}\, D$, and that T is locally irreducible, that is, its irreducible components are disjoint. Then for every rational function f on S which is regular on $U = S \setminus T$, the divisor $\mathrm{div}(f)$ is locally a multiple of D. Thus in this case there is a canonical isomorphism of groups

$$\Gamma(S, \overline{G}) \xrightarrow{\sim} \Gamma(U, \underline{G}_{mU}). \tag{6.4.1}$$

Suppose further that T is geometrically unibranch and locally integral, a hypothesis stable under smooth base change (EGA IV 17.5.7). Since all the hypotheses are then stable under base change, (6.4.1) implies that for every S' smooth over S there is a canonical group isomorphism, functorial in S',

$$\mathrm{Hom}_S(S', \overline{G}) \xrightarrow{\sim} \mathrm{Hom}_U(S'_U, \underline{G}_{mU}) \simeq \Gamma(S'_U, \underline{O}^*_{S'_U}). \tag{6.4.2}$$

Since $\overline{G}$ is itself smooth over S, this isomorphism characterizes the group scheme $\overline{G}$ over S up to unique isomorphism. It also shows that $\overline{G}$ plays the role of a "Neron model" for the group scheme $\underline{G}_{mU}$ over the open subset U of S. Together with the fact that it was first introduced and used by M. Raynaud, this justifies the terminology adopted in 6.3 for $\overline{G}$.

Theorem 6.5. Let S be a normal locally integral scheme and let D be a positive divisor on S, without multiple components, whose support is geometrically unibranch and locally integral. Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ over S relative to D (6.3), and consider the category of torsors E over S under the group scheme $\overline{G}$. The restriction functor I 293

$$E \longmapsto E_U = E|U$$

from this category to the category of torsors over U under the group scheme $\underline{G}_{mU}$ is fully faithful. If S is regular (more generally, if its local rings are factorial), this functor is an equivalence of categories.

Let us prove that the functor in question is fully faithful, that is, that for two torsors E, F under $\overline{G}$, the restriction map

$$\mathrm{Hom}(E, F) \longrightarrow \mathrm{Hom}(E_U, F_U) \tag{*}$$

is bijective, where Hom denotes the set of homomorphisms of torsors. First observe that this question is plainly local on S for the Zariski topology. Indeed, suppose that S is the union of open subsets S_i such that for every open subset S' of S contained in some S_i (and hence in particular for $S' = S_i$ or $S_i \cap S_j$), the map

$$\mathrm{Hom}(E_{S'}, F_{S'}) \longrightarrow \mathrm{Hom}(E_{S' \cap U}, F_{S' \cap U})$$

is bijective. Then (*) is bijective. On the other hand, it follows at once from 5.8 applied to $S, T, \mathbb{Z}$, and from 5.1 applied to $T, \mathbb{Z}$, that every torsor under $\overline{G}$ is locally trivial for the Zariski topology. These two observations reduce us to proving the bijectivity of (*) when $E = F = \overline{G}$. In that case the map is precisely (6.4.1), whose bijectivity has already been established.

We now prove that the functor $E \mapsto E_U$ is essentially surjective when the local rings of S are factorial. Since $\underline{G}_{mS}$ is identified with an open subgroup scheme of $\overline{G}$, it is enough to show that every torsor under $\underline{G}_{mU}$ is isomorphic to the restriction to U of a torsor under $\underline{G}_{mS}$. Equivalently, every invertible Module over U must be isomorphic to the restriction of an invertible Module over S. This follows from the hypothesis on S (EGA IV 21.6.11, where only the locally noetherian case is treated; that is the only case needed below). This proves 6.5. I ≈294

6.5.1. Suppose now, in addition, that S is locally noetherian and regular. Since S is then locally factorial, giving D is equivalent to giving a reduced, purely 1-codimensional closed subscheme, again denoted D, which is geometrically unibranch. Notice also that the hypotheses on S and D are stable under every smooth base change $S' \to S$, so that the conclusion of 6.5 applies to S' with the subscheme $D' = D \times_S S'$. The description in VII 1.1.6 of extension groups now immediately gives the following consequence. I 295

Corollary 6.6. Let S be a locally noetherian regular scheme, let D be a reduced, geometrically unibranch, purely 1-codimensional closed subscheme (thus defining a divisor on S), let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ relative to D, and let P be a smooth commutative group scheme over S. Consider the category of commutative extensions, as sheaves of groups, of P by $\overline{G}$. Restriction to U, $E \mapsto E|U$, induces an equivalence between this category and the category of extensions of P_U by $\underline{G}_{mU}$.

6.6.1. The same method gives the analogous statement for extensions which are not necessarily commutative (with P no longer assumed commutative). We leave the details of the proof to the reader; they are slightly more involved because bitorsors must be considered in place of torsors.

By essentially the same proof as for 6.6, one likewise obtains:

Corollary 6.7. Let $S, D, \overline{G}$ be as in 6.6, and let P, Q be two smooth commutative group schemes over S. Consider the category of biextensions of (P, Q) by $\overline{G}$. Then the restriction functor to U, $E \mapsto E|U$, from this category to the category of biextensions of (P_U, Q_U) by $\underline{G}_{mU}$ is an equivalence of categories. I 296

Remarks 6.8. a) We shall apply 6.6 and 6.7 when S is a trait and D is the divisor defined by the inclusion of the closed point s of S into S. In this case U is the spectrum of the fraction field K of S, and $\overline{G}$ is the usual Neron model of the multiplicative group over K, introduced by Raynaud. Notice that even in this special case, the proof of 6.6 and 6.7 uses 6.5 over bases more general than traits (such as P and its Cartesian powers). Nevertheless, in this case it is enough to know the existence of the Neron model $\overline{G}$ in the special situation under consideration: 6.5 for base schemes smooth over S, and hence 6.6 and 6.7, then follow formally. For us, the introduction of $\overline{G}$ will chiefly serve as a convenient intermediate step in studying, in the next section, the problem of extending extensions and biextensions by the multiplicative group.

b) A Neron–Raynaud model $\overline{G}$ of $\underline{G}_{mU}$, satisfying the fundamental condition (6.4.2) for every S' smooth over S, can be defined under much more general conditions than those considered above, with U merely a schematically dense open subset of a normal locally noetherian scheme S. In this case $\overline{G}$ is an fppf sheaf, not necessarily representable. Let $\underline{D}$ denote the sheaf, on the small etale site of S, of divisors on S whose support is contained in $T = S \setminus U$, and, by abuse of language, also denote by the same letter its canonical extension to a sheaf on $(\mathrm{Sch})_{/S}$. Its sections over a relative scheme $f : S' \to S$ are the sections of $f^*_{\text{ét}}(\underline{D})$ over S'. The desired model is an extension of groups

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \underline{D} \longrightarrow 0. \tag{6.8.1}$$

The construction of a canonical extension (6.8.1), without any normality hypothesis on S, presents no difficulty when one uses the general description VII 1.1.6; at the same time one obtains (6.4.2) for S' etale over S. For the same relation to hold for every S' smooth over S, it is equivalent to verify that the formation of the sheaf $\underline{D}$ commutes with every smooth base change $f : S' \to S$. For normal S, this follows from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9.

6.9. "Self-criticism" (observations of P. Deligne).[4] I ≈297

Sections 5 and 6 are really rather uninspiring. One periodically has the impression that the formalism of i_*, i^* is being redone "by hand", without seeing what is geometric and what is "general nonsense".

[4]The author of the present Expose having admitted defeat, the reader is invited, if necessary, to rewrite sections 5 and 6 according to the accompanying directions.

I propose extracting the "geometric" statements a), b), c):

a) (cf. 6.8 b)) Let $j : U \hookrightarrow S$ be a schematically dense open subset of a normal noetherian scheme S. The etale sheaf $\underline{G}_{mS}$ is then identified with a subsheaf of the etale sheaf $j_*\underline{G}_{mU}$. Let $\underline{D}$ be the quotient etale sheaf in the exact sequence

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow j_*\underline{G}_{mU} \longrightarrow \underline{D} \longrightarrow 0.$$

Then the formation of $\underline{D}$ commutes with every smooth base change (EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9); moreover, $R^1j_*\underline{G}_m = 0$ (Theorem 90) if the strictly local rings of S are factorial.

b) If $U = S \setminus Y$, where Y is a geometrically unibranch divisor, and if i is the inclusion of Y into S, then I≈298

$$\underline{D} = i_*\mathbb{Z}_Y.$$

c) If Y is a closed subscheme of a scheme X, with closed immersion $i : Y \hookrightarrow X$, and if E is a set, then the etale sheaf $i_*\underline{E}_Y$ is represented by an etale scheme over X, whose formation commutes with every base change.

Corollary. Let S be a normal noetherian scheme, let Y be a geometrically unibranch divisor, and let $U = S \setminus Y$:

$$U \xhookrightarrow[j]{} S \xleftarrow{\ i\ } Y \qquad .$$

There exists a unique smooth group scheme $\mathcal{G}_m$ over S such that after every smooth base change one has $\mathcal{G}_m|S_{\text{ét}} = j_*(\underline{G}_m)_U$.

Indeed, the preceding requirements determine $\mathcal{G}_m$ as a sheaf, for the etale topology, on the category of schemes smooth over S. By a) and b), the sequence

$$0 \longrightarrow \underline{G}_m \longrightarrow \mathcal{G}_m \xrightarrow{q} i_*\mathbb{Z}_Y \longrightarrow 0$$

is exact. By c), $i_*\mathbb{Z}_Y$ is representable. Finally, the morphism q is representable.

Use of the preceding "small smooth site" should make it possible to deduce the statements of section 6 formally from a).

7. Canonical extensions of extensions and biextensions by $\underline{G}_m$. I≈299

7.0. We now reap the fruit of our preparations in sections 5 and 6. First let the base scheme S be a trait, that is, the spectrum of a discrete valuation ring; let η be its generic point, s its closed point, and $k = k(s)$. Let P be a smooth commutative group scheme over S, and write $P_0 = P_s$ for its special fibre. In the present section we consider only commutative groups; in particular, all group extensions under consideration are commutative.

Theorem 7.1. The notation is that recalled in 7.0.

a) Suppose that the group of connected components of the special fibre of P,

$$\Phi = P_0/P_0^\circ,$$

is a torsion group—as is the case, in particular, if P is of finite type over S. Then the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$. Then every extension E_η of P_η by $\underline{G}_{m\eta}$ has an associated canonical homomorphism

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.1}$$

whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of an extension of P by $\underline{G}_{mS}$. In particular, if P_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.

b) Let P and Q be two smooth group schemes over S, put

$$\Phi = P_0/P_0^\circ, \qquad \Psi = Q_0/Q_0^\circ,$$

and suppose that either Φ or Ψ is a torsion group (for example, that either P or Q is of finite type over S). Then the functor $E \mapsto E_\eta$ from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P_η, Q_η) by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$ or $n\Psi = 0$. Then every biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ has an associated canonical pairing

$$d(E_\eta) : \Phi \otimes_{\mathbb{Z}} \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.2}$$

whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of a biextension E of (P, Q) by $\underline{G}_{mS}$. In particular, if either P_0 or Q_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.

Proof. a) Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$. The functor under consideration is the composite of the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" with the functor "restriction to η" defined on the category of extensions by $\overline{G}$. The latter functor is an equivalence of categories by 6.6, so it remains to study the first. For this purpose we use the exact sequence of group schemes

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \mathbb{Z}_{S,s} \longrightarrow 0, \tag{7.1.3}$$

where the notation $\mathbb{Z}_{S,s}$ is that of 5.7. The exact sequence of Ext^i, $i = 0, 1$, associated with (7.1.3) shows that the functor in question is fully faithful if and only if

$$\mathrm{Hom}(P, \mathbb{Z}_{S,s}) = 0, \tag{7.1.4}$$

and that the obstruction for an extension $\overline{E}$ of P by $\overline{G}$ to come from an extension E of P by $\underline{G}_{mS}$ lies in

$$\mathrm{Ext}^1(P, \mathbb{Z}_{S,s}). \tag{7.1.5}$$

By (5.7.2), the first group in (7.1.4) is isomorphic to $\mathrm{Hom}(P_0, \mathbb{Z}_k)$, which in turn is isomorphic to $\mathrm{Hom}(\Phi, \mathbb{Z}_k)$. Thus (7.1.4) is equivalent to

$$\mathrm{Hom}(\Phi, \mathbb{Z}_k) = 0. \tag{7.1.4 bis}$$

This relation certainly holds if Φ is a torsion group. Moreover, if $n\Phi = 0$, then 5.9 and 5.5 give canonical isomorphisms

$$\mathrm{Ext}^1(P, \mathbb{Z}_{S,s}) \simeq \mathrm{Ext}^1(P_0, \mathbb{Z}_k) \simeq \mathrm{Hom}\big(\Phi, (\mathbb{Q}/\mathbb{Z})_k\big), \tag{7.1.6}$$

which completes the proof of 7.1(a).

b) The proof is essentially the same. By 6.7, we reduce to studying the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P, Q) by $\overline{G}$. We use the exact sequence (7.1.3) and the associated exact sequence of Biext^i, $i = 0, 1$ (VII 3.7.5). Full faithfulness of the functor is equivalent to

$$\mathrm{Hom}(P \otimes Q, \mathbb{Z}_{S,s}) = 0. \tag{7.1.7}$$

By (5.7.2), the left side is isomorphic to $\mathrm{Hom}(P_0 \otimes Q_0, \mathbb{Z}_k)$ and hence to $\mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Z}_k)$; this group is indeed zero if either Φ or Ψ is a torsion group. On the other hand, for a given biextension $\overline{E}$ of (P, Q) by $\overline{G}$, the obstruction to its coming from a biextension E of (P, Q) by $\underline{G}_{mS}$ lies in

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_{S,s}).$$

By 5.10 this is isomorphic to $\mathrm{Biext}^1(P_0, Q_0; \mathbb{Z}_k)$, which, when $n\Phi = 0$ or $n\Psi = 0$, is itself isomorphic by 5.6 to

$$\mathrm{Hom}\big(\Phi \otimes \Psi, (\mathbb{Q}/\mathbb{Z})_k\big).$$

This completes the proof of 7.1.

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Remark 7.2. Given a smooth group scheme P over a connected regular locally noetherian scheme S with generic point η, one may ask to study the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$. This problem is readily reduced to the one treated in 7.1(a). For simplicity, suppose that P is of finite type over S. It follows easily from 7.1 that this functor is fully faithful.[5] One readily deduces that if E_η is an extension of P_η by $\underline{G}_{m\eta}$, then it is isomorphic to the generic fibre of an extension E of P by $\underline{G}_{mS}$ if and only if, for every $s \in S$ such that

$$\dim \mathrm{Spec}(\underline{O}_{S,s}) = 1,$$

the analogous extension problem obtained by base change $\mathrm{Spec}(\underline{O}_{S,s}) \to S$ has a solution. By 7.1(a), this is expressed by the vanishing of a certain homomorphism

$$d(E_\eta, s) : P_s/P_s^\circ \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k(s)}. \tag{7.2.1}$$

The preceding assertion plainly reduces to the following one. Let U be an open subset of S whose complement T satisfies

$$\mathrm{codim}(T, S) \geq 2.$$

Then the functor $E \mapsto E|U$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_U by $\underline{G}_{mU}$ is an equivalence of categories. In terms of the description VII 1.1.6 this assertion is essentially trivial: for every smooth S-scheme X (such as P, $P \times P$, $P \times P \times P$, and so on), X is locally factorial and $X_T = X \setminus X_U$ has codimension at least 2 in X. Consequently, the functor $F \mapsto F|X_U$ from the category of torsors under $\underline{G}_{mX}$ to the category of torsors under $\underline{G}_{mX_U}$ is an equivalence of categories.

There are entirely analogous results for the problem of extending biextensions; we leave their formulation to the reader. I ≈303

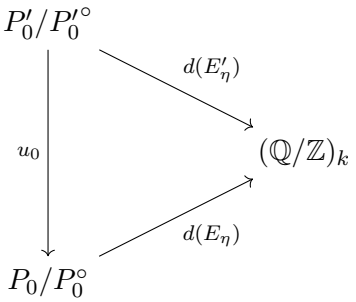
7.3. We shall make explicit some functorial properties of the obstructions defined in (7.1.1) and (7.1.2), with P, Q, S variable, and some relations among them. In what follows, we shall suppose that the conditions allowing these obstructions to be defined are satisfied. Since the verifications present no difficulty, we leave them to the reader.

7.3.1. Let

$$u : P' \longrightarrow P$$

be a morphism of smooth group schemes over S, and let E_η be an extension of P_η by $\underline{G}_{m\eta}$. Pullback by $u_\eta : P'_\eta \to P_\eta$ gives an extension E'_η of P'_η by $\underline{G}_{m\eta}$. With this notation, the following diagram is commutative:

(7.3.1.1)



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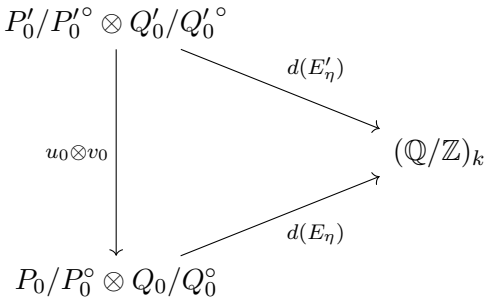
[5] For this it is enough that S be normal rather than regular.

Similarly, if we have morphisms of smooth group schemes over S

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q,$$

and a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, whose pullback by (u_η, v_η) is a biextension E'_η of (P'_η, Q'_η) by $\underline{G}_{m\eta}$, then the following diagram is commutative:

(7.3.1.2)



The essentially mechanical verification of these compatibilities is left to the reader.

7.3.2. It follows from the compatibility (7.3.1.1) that, if E_η is an extension of P_η by $\underline{G}_{m\eta}$, there is a largest group subscheme V of P such that the restriction of E_η to V_η extends to an extension of V by $\underline{G}_{mS}$. It is the open subgroup V of P containing P_η and defined by the condition

$$V_0/V_0^\circ = \ker d(E_\eta). \tag{7.3.2.1}$$

If $u : P' \to P$ is then a morphism of smooth group schemes over S, the extension E'_η, obtained by pulling E_η back along u_η, extends to an extension of P' by $\underline{G}_{mS}$ if and only if

$$u(P') \subset V. \tag{7.3.2.2}$$

In the case of a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, there is likewise a largest group subscheme V of P such that the restriction of E_η to (V_η, Q_η) extends to a biextension of (V, Q) by $\underline{G}_{mS}$. It is the open subgroup of P containing the generic fibre P_η and characterized by the fact that V_0/V_0° is the left annihilator in Φ of the pairing (7.1.2), that is, the kernel of the corresponding homomorphism from Φ to $\underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)$: I ≈**305**

$$V_0/V_0^\circ = \ker\left(\Phi \xrightarrow{d(E_\eta)} \underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)\right). \tag{7.3.2.3}$$

If $u : P' \to P$ is a morphism of smooth group schemes over S, the biextension of (P'_η, Q_η) by $\underline{G}_{m\eta}$ obtained by pulling E_η back along u_η extends to a biextension of (P', Q) by $\underline{G}_{mS}$ if and only if (7.3.2.2) holds. Symmetric results hold upon interchanging the roles of P and Q.

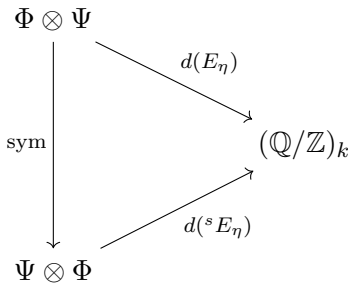
Of course, in general there is no largest pair of subgroups $P' \to P$, $Q' \to Q$ for which the restriction E'_η of E_η to a biextension of (P'_η, Q'_η) by $\underline{G}_{m\eta}$ extends to a biextension of (P', Q') by $\underline{G}_{mS}$. One can only say that, for a given pair (P', Q'), or more generally for a given pair of morphisms $P' \to P$, $Q' \to Q$ of smooth group schemes, the desired extension exists if and only if the restriction of the form $d(E_\eta)$ to

$$P'_0/P'^\circ_0 \otimes Q'_0/Q'^\circ_0$$

is zero.

7.3.3. Symmetry. Let E_η be a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, and consider the transposed biextension ${}^sE_\eta$ (VII 2.7), which is a biextension of (Q_η, P_η) by $\underline{G}_{m\eta}$. The following diagram is commutative: I **306**

(7.3.3.1)



,

where the vertical arrow is the canonical symmetry. (Compare this statement with the antisymmetry statement of 2.2.10.) It follows in particular that, if $P = Q$ and E_η is a symmetric biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, that is, if E_η is isomorphic to ${}^sE_\eta$, then the bilinear form $d(E_\eta)$ on $\Phi \times \Phi$ with values in $(\mathbb{Q}/\mathbb{Z})_k$ is symmetric (and not alternating, as in 2.2.12!). If it is zero, that is, if E_η extends to a biextension E of (P, P) by $\underline{G}_{mS}$, the latter is necessarily symmetric as well: by 7.1(a), every isomorphism ${}^sE_\eta \to E_\eta$ extends uniquely to an isomorphism ${}^sE \to E$.

7.3.4. Reduction of obstructions of type (7.1.2) to those of type (7.1.1). I 307

Suppose, for definiteness, that there is an integer $n > 0$ such that $n\Phi = 0$, and let

$$q \in Q(S)$$

be a section of Q. By an abuse of notation, write

$$q_0 \in \Psi(k)$$

for the image in $\Psi(k)$ of the value of q at s. Regarding E_η as an extension of $P_\eta \times_\eta Q_\eta$ by $\underline{G}_{mQ_\eta}$ over Q_η, pullback gives an extension

$$q_\eta^*(E_\eta), \qquad \text{an extension of } P_\eta \text{ by } \underline{G}_{m\eta},$$

and hence an obstruction

$$d_q \overset{\text{dfn}}{=} d(q_\eta^*(E_\eta)) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k. \tag{7.3.4.1}$$

With $d = d(E_\eta)$, one has the relation

$$d_q(x) = d(x, q_0), \tag{7.3.4.2}$$

valid for every point x of Φ with values in a k-scheme T.

7.3.4.3. For every $q \in Q(S)$, this formula determines d_q in terms of the pairing d; conversely, it plainly recovers d from the invariants d_q, at least when every point of Ψ comes from a section of Q. By Hensel's lemma, this condition holds automatically when S is strictly local. The pairing d is known once it is known over the separable closure of k, and its formation commutes with etale localization on S by 7.3.5.3 below. Thus, after replacing S by its strict localization if necessary, the determination of d reduces to that of the invariants d_q.

In particular, when every point of Ψ comes from a section q of Q, the pairing d is zero—that is, E_η extends to a biextension E of (P, Q) by $\underline{G}_m$—if and only if d_q is zero for every section q of Q, that is, if and only if $q_\eta^*(E_\eta)$ extends to an extension of P by $\underline{G}_m$. With no hypothesis on Ψ, the analogous criterion remains valid provided that the q are allowed to be points with values in local etale S-schemes S'. I ≈308

7.3.4.4. Conversely, starting from the situation of 7.1(a), the obstruction (7.1.1) may be interpreted as an obstruction of type (7.1.2): take $Q = \mathbb{Z}_S$ and the biextension of $(P_\eta, \mathbb{Z}_\eta)$ deduced from E_η. Indeed, it suffices to apply the preceding construction with q equal to the section 1 of $\mathbb{Z}_S$.

7.3.5. Effect of a base change $S' \to S$. Suppose given a dominant local morphism of traits

$$f : S' \longrightarrow S,$$

and denote by η' the generic point of S', by s' its closed point, and put $k' = k(s')$. If $S = \mathrm{Spec}(V)$ and $S' = \mathrm{Spec}(V')$, then V' is a discrete valuation ring dominating the discrete valuation ring V. Put

$$e(S'/S) = e(V'/V) = \mathrm{length}_{V'}(V'/\mathfrak{m}V'), \tag{7.3.5.1}$$

where $\mathfrak{m}$ is the maximal ideal of V.

Let now P be as in 7.1(a), let E_η be an extension of P_η by $\underline{G}_{m\eta}$, put $P' = P \times_S S'$, and let $E'_{\eta'}$ be the extension of $P'_{\eta'}$ by $\underline{G}_{m\eta'}$ obtained by pulling back E_η. With the notation of 4.1(a), there is a canonical isomorphism

$$\Phi' \overset{\text{dfn}}{=} P'_0/P'^\circ_0 \simeq \Phi \otimes_k k' \simeq \Phi_{k'},$$

and hence one may consider

$$d(E_\eta)_{k'} : \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}.$$

Then

$$d(E'_{\eta'}) = e(S'/S)\, d(E_\eta)_{k'}. \tag{7.3.5.2}$$

Likewise, under the hypotheses of 7.1(b) and with the evident notation, (7.3.5.2) holds with $d(E_\eta)$ now denoting the pairing (7.1.2), $d(E_\eta)_{k'}$ the pairing I≈309

$$\Phi' \otimes \Psi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}$$

obtained from it by base change $k \to k'$, and $E'_{\eta'}$ the biextension of $(P'_{\eta'}, Q'_{\eta'})$ by $\underline{G}_{m\eta'}$ obtained from the biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ through the base change $\eta' \to \eta$.

7.3.5.3. An interesting case is $e(S'/S) = 1$, that is,

$$\mathfrak{m}' = \mathfrak{m}V',$$

where $\mathfrak{m}'$ denotes the maximal ideal of V'. This holds in particular if V' is etale over V, or if V' is the henselization, strict henselization, or completion of V. In this case (7.3.5.2) simply says that the formation of obstruction (7.1.1), respectively (7.1.2), commutes with the base change under consideration $S' \to S$.

7.3.5.4. On the other hand, (7.3.5.2) in the general case shows that, for every extension E_η as in 7.1(a) (respectively every biextension E_η as in 7.1(b)), there is a finite morphism of traits $S' \to S$ such that $k(\eta')$ is a separable extension of $k(\eta)$ and such that the extension (respectively biextension) $E'_{\eta'}$ obtained from E_η by base change $\eta' \to \eta$ extends to an extension (respectively biextension) E' over all of S'. Indeed, it is enough to choose S'/S so that $e(S'/S)$ is a multiple of the integer $n > 0$ occurring in 7.1, and such an extension can always be found with $k(\eta')$ separable over $k(\eta)$.

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Proposition 7.4. Let S be a reduced scheme.

a) Let P be a smooth group scheme over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\mathrm{EXT}(P, \underline{G}_{mS}) \longrightarrow \mathrm{TORSRIG}(P, e_P; \underline{G}_{mS}) \tag{7.4.1}$$

from the category of extensions of P by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{mP}$ rigidified along the identity section of P is fully faithful. If S is normal and P has connected fibres, an object E of the target of (7.4.1) belongs to the essential image of this functor if and only if, for every maximal point η of S, the rigidified torsor E_η on P_η comes from an extension of P_η by $\underline{G}_{m\eta}$.

b) Let P, Q be two smooth group schemes over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\mathrm{BIEXT}(P, Q; \underline{G}_{mS}) \longrightarrow \mathrm{TORSBIRIG}(P, Q; \underline{G}_{mS}) \tag{7.4.2}$$

from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{m,P\times_S Q}$ birigidified along the pair of identity sections of P and Q is fully faithful. If S is normal and either P or Q has connected fibres, an object E of the target of (7.4.2) belongs to the essential image of the functor if and only if, for every maximal point η of S, E_η comes from a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$.

We prove (a); the essentially identical proof of (b) is left to the reader. For the full-faithfulness assertion, note that 4.1 remains valid if X and Y are flat schemes over a reduced scheme S whose I≈311

fibres at the maximal points satisfy the conditions stated in 4.1. Indeed, the family of maximal fibres of $X \times_S Y$ (which is flat over the reduced scheme S) is schematically dense in $X \times_S Y$. Hence a morphism from $X \times_S Y$ to the separated S-scheme $\underline{G}_{mS}$ is trivial as soon as it is trivial on the maximal fibres. Thus the full faithfulness of (7.4.1) is proved as in 4.2, and the characterization of its essential image given there remains valid.

It remains to prove that, if the rigidified torsor E on P is such that the E_η come from extensions, then so does E, assuming that S is normal and P has connected fibres. The latter condition implies that P is of finite presentation over S (SGA 3 VI$_B$ 5.5). Full faithfulness also shows that the question is local on S, so that we may suppose S affine and local. The usual limit argument, using Nagata's theorem on the finiteness of the integral closure of a reduced finitely generated $\mathbb{Z}$-algebra (EGA IV 7.8), reduces us further to the case in which S is noetherian.

Then 7.1(a) shows that there is an open subset $U \subset S$ whose complement has codimension at least 2 and such that E_U comes from an extension of P_U by $\underline{G}_{mU}$. In other words, the desired isomorphism

$$\pi^*(E) \simeq \mathrm{pr}_1^*(E)\,\mathrm{pr}_2^*(E)$$

can be constructed over $(P \times_S P)_U$. This is an open subset of the normal scheme $P \times_S P$ whose complement has codimension at least 2. The isomorphism on $(P \times_S P)_U$ therefore extends over all of $P \times_S P$, completing the proof.

Corollary 7.5. Let S be a normal scheme and let P and Q be two smooth group schemes over S, with connected fibres at the maximal points, either P or Q having connected fibres. Suppose in addition that, for every maximal point η of S, either P_η or Q_η is an extension of an abelian scheme by a torus (or, more generally, admits a composition series whose factors are abelian schemes, tori, or groups $\underline{G}_a$). Then the functor (7.4.2) is an equivalence of categories. I 312

This is an immediate consequence of 7.4(b) and 4.7.

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SGA 7

Exposé IX

NÉRON MODELS AND MONODROMY

by A. Grothendieck
(with an appendix by M. Raynaud)

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0. Introduction I≈314

0.1. In the present exposé we apply the results of the preceding exposés to the study of the Néron model (1.1) A of an abelian scheme A_K defined over the function field K of the Henselian trait S. We shall see that the most important properties of the Néron model can be interpreted in terms of the action of the local monodromy group I on the Tate module $T_\ell(A_K(\overline{K}))$ (where ℓ is a prime number distinct from p). Since the dual of the latter can be interpreted as the ℓ-adic cohomology group $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, and since, for every smooth projective scheme X over K, $H^1(X_{\overline{K}}, \mathbb{Z}_\ell)$ is identified with $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, where $A_{\overline{K}}$ is the Albanese variety of $X_{\overline{K}}$, the present exposé may also be viewed as a detailed study of local monodromy phenomena for ℓ-adic H^1 (or, better still, for "motivic" H^1) of smooth projective varieties over K. From this point of view, it seems clear that the principal results of the present exposé are destined to be subsumed within a "Néron theory" for motives of arbitrary weight, that is, for H^i (ℓ-adic, de Rham, Hodge, and so forth) with arbitrary i, of which at present we are only beginning to catch a glimpse. (See [9] in this connection, and more particularly Deligne's Conjectures 9.8 through 9.13 in the cited report.)

0.2. The most important result of the present exposé is the "semistable reduction theorem" 3.6, I 315 which asserts that, after a finite separable extension K' of K, the Néron model of $A_{K'}$ has a special fiber whose identity component is an extension of an abelian scheme by a torus. This result was conjectured by J.-P. Serre in 1964, and the principle of the proof given here [6] goes back to the same year. It appears in 3.5 that this property of "semistable reduction" is equivalent to the action of the monodromy group I on $T_\ell(A_K(\overline{K}))$ being "essentially unipotent of step 2" (where ℓ is a fixed prime distinct from the residual characteristic p of S). In this form it was proved, in terms of ℓ-adic H^1, in Exposé III as a special case of the "monodromy theorem", using resolution of singularities for excellent schemes of dimension 2 (due to S. Abhyankar). To prove the equivalence of the two properties, we shall also need an "orthogonality theorem" 2.4, due to Igusa [11] in the special case in which A_K is the Jacobian of a smooth curve X_K over K, the generic fiber of a regular connected projective scheme X over S whose geometric special fiber has no singular points other than ordinary quadratic singularities. This is indeed a semistable-reduction case, but it would not suffice for our purposes; it is chiefly in order to establish the orthogonality theorem in its general form that we use the convenient tool supplied by the theory of biextensions developed in the two preceding exposés. Let us also point out that D. Mumford has found a very different proof of the semistable reduction theorem, one which does not use resolution of singularities and instead uses his theory of theta functions (unpublished); unfortunately, he must restrict himself to residual characteristic different from 2.

0.3. The semistable reduction theorem is the principal tool used in the subsequent sections to I 316 study Néron models, which we undertake here from the "algebraic" point of view. It also seems indispensable for studying abelian varieties over K from the "rigid-analytic" point of view, or from the point of view of "theta functions" (these terms apparently being synonymous when abelian varieties over complete valued fields are in question), as developed recently by D. Mumford and M. Raynaud following work of Tate and Morikawa. (In Sections 13 and 14 we shall say a few words about the relations between the algebraic and transcendental points of view.) Finally, the same theorem is a key ingredient in the proof of the "semistable reduction theorem for algebraic curves" of Artin–Deligne–Mumford [3],[7] which we have already had occasion to use in the study of the action of monodromy on the fundamental group (the "essentially tame action theorem" of Exposé V), and which is used again in the proof of the integrality and positivity theorem 10.4 in the present exposé.[8]

0.4. The present exposé differs from the other exposés of the Seminar in that, in studying the Tate module $T_\ell(A_K)$, we have systematically taken into account also the case in which ℓ equals the residual characteristic p. It turns out that the language of Barsotti–Tate groups [33][9] makes it

[6] It is developed in a letter from the lecturer to J.-P. Serre dated 30 October 1964.

[7] P. Deligne gave the proof of this theorem in the oral Seminar, but it was not considered useful to reproduce it in the Seminar texts.

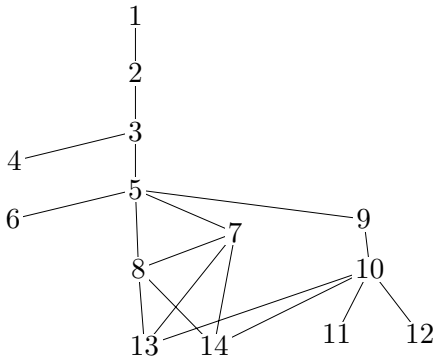
[8] There is another proof of the latter theorem, by way of rigid-analytic theory, which does not use the semistable reduction theorem.

[9] Or "p-divisible groups" in Tate's terminology.

possible to extend to this case all the most important phenomena established for $\ell \neq p$, subject on occasion to having available Tate's fundamental theorem [33, Theorem 4], which at present has been established only when K has characteristic zero. Of course, in the latter case the pro-p Barsotti–Tate group $T_p(A_K)$ may still be interpreted in terms of the action of the monodromy group I on the p-adic module $T_p(A_K(\overline{K}))$, but one should bear in mind that most of the statements suggested by this superficial analogy with the case $\ell \neq p$ are grossly false. (Thus it is no longer true that the monodromy action is essentially unipotent when $\ell = p$.) By using more judicious translations, however (such as 5.13 below), it is already clear that the study of pro-p Barsotti–Tate groups defined geometrically over the function field of a trait of residual characteristic $p > 0$, whether or not the trait has mixed characteristic, must be regarded as part of the theory of "local monodromy", in which these groups play the role of p-adic local systems on $\mathrm{Spec}(K)$. In fact, these are essentially the "local systems" that enter into the study, from the "p-adic" point of view, of H^1 of smooth projective varieties over K. For the still-to-be-developed study of higher H^i from the p-adic point of view, it appears that the category of pro-p Barsotti–Tate groups over K, respectively over S, will have to be suitably enlarged, taking inspiration from the viewpoints provided by "crystalline cohomology".

0.5. The exposé given here follows the oral lectures very closely up to Section 4, but is markedly more detailed and complete from the following sections onward, which had been only partially sketched. It will not be used in the subsequent exposés. Some results of the present exposé (10.4 and 12.5), on the other hand, use the Picard–Lefschetz formula, which will be established only later. I 318

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1. Neron models of abelian schemes: notation; the canonical pairing

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

and the canonical biextension W^o of (A^o, A'^o).

1.0. Throughout this exposé we fix a trait S, for which we use the usual notation s, η, k, K, and so forth, of I 1. Let A_K be an abelian scheme over K, and denote its dual abelian scheme [5] by A'_K. The duality between A_K and A'_K is given by a divisorial correspondence W_K on (A_K, A'_K), that is, an invertible sheaf on $A_K \times A'_K$, birigidified with respect to the identity sections of A_K and A'_K. It is determined up to unique isomorphism by its class (cf. VIII 3.1) I 319

$$w_K \in \mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK}). \tag{1.0.1}$$

An important role will be played by the corresponding pairing (VIII 2.2)

$$\varphi : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow \mathbb{Z}_\ell(1)_K \overset{\mathrm{def}}{=} T_\ell(\mathbb{G}_{mK}), \tag{1.0.2}$$

where ℓ is an arbitrary prime number. When $\ell \neq p = \mathrm{char}\, k$, giving φ is equivalent to giving a pairing of finite free $\mathbb{Z}_\ell$-modules

$$\varphi : T_\ell(A_K(\overline{K})) \times T_\ell(A'_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{1.0.3}$$

compatible with the action of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

on the three modules in question, where $\overline{K}$ is a separable closure of K. It is known (VIII 3.2) that this pairing is a perfect duality: it identifies each of the modules $T_\ell(A_K(\overline{K}))$ and $T_\ell(A'_K(\overline{K}))$ with the dual of the other, with values in the invertible module $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$.

At times we shall suppose that a polarization of A_K has been chosen; thus we are given an element I≈320

$$\xi \in \mathrm{NS}(A_K/K) = \underline{\mathrm{NS}}_{A_K/K}(K), \tag{1.0.4}$$

where

$$\underline{\mathrm{NS}}_{A_K/K} \overset{\mathrm{def}}{=} \underline{\mathrm{Pic}}_{A_K/K}/(\underline{\mathrm{Pic}}^o_{A_K/K} = A'_K).$$

In the usual way it defines a homomorphism

$$\psi_\xi : A_K \longrightarrow A'_K, \tag{1.0.5}$$

hence a homomorphism

$$T_\ell(\psi_\xi), \quad \text{or simply} \quad \psi_\xi : T_\ell(A_K) \longrightarrow T_\ell(A'_K), \tag{1.0.6}$$

and a bilinear form φ_ξ

$$\begin{aligned} \varphi_\xi(x,y) &= \varphi(x, \psi_\xi(y)), \\ \varphi_\xi : T_\ell(A_K) \times T_\ell(A_K) &\longrightarrow \mathbb{Z}_\ell(1)_K, \end{aligned} \tag{1.0.7}$$

which, for $\ell \neq p$, may also be interpreted as a pairing of finite free $\mathbb{Z}_\ell$-modules with operators:

$$\varphi_\xi : T_\ell(A_K(\overline{K})) \times T_\ell(A_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*). \tag{1.0.8}$$

These constructions make sense for every element of the Neron–Severi group of A_K, and yield an alternating form φ_ξ, in either interpretation (1.0.7) or (1.0.8) (VIII 2.2.11). The fact that ξ is a polarization implies that ψ_ξ in (1.0.5) is an isogeny; equivalently, since $\dim A_K = \dim A'_K$, it is an epimorphism of groups. This is also equivalent to the injectivity of $T_\ell(\psi_\xi)$ in (1.0.6) (or to its having finite cokernel), and finally to the form φ_ξ in (1.0.7) being separating.

1.1. Recall that there exists a smooth scheme A over S, called the Neron model of A_K, which represents the functor I321

$$S' \longmapsto \mathrm{Hom}(S'_\eta, A_K) \tag{1.1.1}$$

on the category of smooth S-schemes. Thus, by definition, there is an isomorphism, functorial in the smooth relative S-scheme S',

$$\mathrm{Hom}_S(S', A) \simeq \mathrm{Hom}_K(S'_\eta, A_K). \tag{1.1.2}$$

Neron [19] constructed A when the residue field k of S is perfect. Raynaud extended the construction to the general case in an unwritten seminar held at IHES in 1966–67 (see [21] for the statement of Raynaud's principal results, which complete those of Neron in loc. cit.). We shall admit this existence result here, together with the fact that A is a quasi-projective scheme over S. Applying (1.1.2) when S' is smooth over η, one also finds that the following canonical morphism is an isomorphism:

$$A \times_S \mathrm{Spec}(K) \xrightarrow{\sim} A_K. \tag{1.1.3}$$

This justifies the paired notation A, A_K; when convenient we shall also write A_η in place of A_K. Using (1.1.3) as an identifying isomorphism, the canonical bijection (1.1.2) is simply the map given by restriction to the generic fibers.

Since the functor (1.1.1) plainly has the structure of a commutative group, the Neron model A representing it itself has the structure of a smooth commutative group scheme over S. The remainder of this exposé is devoted to the study of this group scheme.

An important invariant arising from the Neron model is, of course, the smooth commutative group scheme of finite type over k I≈322

$$A_o = A \times_S s = A_k. \tag{1.1.4}$$

Following the general notation, we denote its identity component by A_o^o and also consider the group scheme

$$\Phi_o = A_o/A_o^o \tag{1.1.5}$$

of connected components of A_o, which is finite and etale over k. In terms of a separable closure $\overline{k}$ of k, it is therefore determined by the ordinary finite group

$$\Phi_o(\overline{k}), \tag{1.1.6}$$

together with the natural action of

$$\pi_o = \mathrm{Gal}(\overline{k}/k)$$

on this group.

Along with the Neron model A of A_K, we shall study

$$A' = \text{the Neron model of } A'_K, \tag{1.1.7}$$

which is likewise a smooth commutative group scheme of finite type over S. We shall denote by

$$\Phi'_o = A'_o/A'^o_o \tag{1.1.8}$$

the group of connected components of its special fiber A'_o.

1.2. Apply VIII 7.1(b) to the pair (A, A') and to the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ in (1.0.1). I323 We thereby obtain a canonical pairing

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{1.2.1}$$

that is, a pairing of π_o-modules

$$\Phi_o(\overline{k}) \times \Phi'_o(\overline{k}) \longrightarrow \mathbb{Q}/\mathbb{Z}. \tag{1.2.2}$$

Here the pairing appears as the obstruction to extending W_K to a biextension W of (A, A') by $\mathbb{G}_{mS}$; the meaning of this point of view is explained in VIII 7.3.2. We record at once the following conjecture concerning this pairing.

Conjecture 1.3. The pairing (1.2.1) is a perfect duality: it identifies each of the component groups Φ_o and Φ'_o with the dual of the other, with values in $(\mathbb{Q}/\mathbb{Z})_k$.

1.3.1. The formation of (1.2.1) commutes with formally etale base change (VIII 7.3.5.3), and the formation of Neron models has the analogous property. Consequently, to prove Conjecture 1.3 it would suffice to treat the case in which S is complete and has separably closed residue field. Artin and Mazur proved the conjecture when A_K is the Jacobian of a proper smooth curve over K, using their general, unpublished theory of the self-duality of the relative Jacobian (in the sense of derived categories) of a proper curve scheme over S. They also proved Conjecture 1.3 when the residue field is finite, using Tate duality [32]. We shall likewise see in 11.4 and 11.5 (subject to compatibility checks) that 1.3 holds on the ℓ-primary components for $\ell \neq p$, and that it holds in the case of "semistable reduction". These results make 1.3 extremely plausible. Moreover, 1.3 may also be regarded as a by-product of a general duality theory for group schemes over fraction fields of complete discrete

valuation rings, formally analogous to SGA 5 I. Such a theory remains heuristic at present; it would also contain local class field theory in the form given to it by Serre [27].

1.3.2. For a given abelian scheme A_K, the validity of 1.3 means that the pairing (1.2.1) is separating on both the left and the right. By VIII 7.3.2, separation on the left says precisely that the only open subgroup scheme V of A for which W_K can be extended to a biextension of (V, A') by $\mathbb{G}_{mS}$ is the identity component A^o of A. There is a symmetric interpretation of separation on the right. I 324

1.4. More generally, suppose that

$$U \subset A \qquad \text{and} \qquad U' \subset A' \tag{1.4.1}$$

are open subgroup schemes such that W_K extends to a biextension of (U, U') by $\mathbb{G}_{mS}$. By VIII 7.3.2, this says explicitly that the subgroups

$$F_o = U_o/U_o^o \subset \Phi_o, \qquad F'_o = U'_o/U'^o_o \subset \Phi'_o \tag{1.4.2}$$

defined by U and U' are orthogonal under (1.2.1). We denote this extension (unique up to unique isomorphism) of W_K by W. It is determined up to unique isomorphism by its class

$$w \in \mathrm{Biext}^1(U, U'; \mathbb{G}_{mS}), \tag{1.4.3}$$

which is itself characterized by the condition that its image in $\mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK})$ be the class w_K of (1.0.1).

For example, one may take for U and U' the identity components A^o and A'^o of A and A'. One thereby obtains a canonical biextension W^o, or W, of (A^o, A'^o) by $\mathbb{G}_{mS}$, giving rise to a class I ≈325

$$w = w^o \in \mathrm{Biext}^1(A^o, A'^o; \mathbb{G}_{mS}), \tag{1.4.4}$$

with which we shall work chiefly. Evidently, the restrictions to (A^o, A'^o) of the classes w in (1.4.3), for the various pairs (U, U'), are all equal to w^o in (1.4.4).

Proposition 1.5. Consider the special fiber W_s of the canonical biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$ in (1.4.4), so that W_s is a biextension of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$. This biextension is separating (VIII 4.11).

Indeed, choose an ample invertible sheaf $\mathbb{L}$ on A^o. Then $\mathbb{L}_\eta$ defines a homomorphism (cf. 1.0.5)

$$A_K \longrightarrow A'_K,$$

which extends to a homomorphism $A \longrightarrow A'$, and hence gives

$$\psi : A^o \longrightarrow A'^o.$$

We may consider the biextension $W_{\mathbb{L}}$ of (A^o, A^o) by $\mathbb{G}_{mS}$ obtained by pulling back W along $(\mathrm{id}_{A^o}, \psi)$. Its restriction to the generic fibers is precisely the biextension $\delta(\mathbb{L}_\eta)$ of VIII 4.12. Hence, by VIII 7.4, $W_{\mathbb{L}}$ is associated with the birigidified invertible sheaf $\delta(\mathbb{L})$, and consequently $(W_{\mathbb{L}})_s$ is defined by $\delta(\mathbb{L}_s)$. On the other hand, it may also be described as the pullback of W_s by the morphisms

$$\mathrm{id}_{A^o_o} : A^o_o \longrightarrow A^o_o, \qquad \psi_o : A^o_o \longrightarrow A'^o_o.$$

Since $\mathbb{L}$, and hence $\mathbb{L}_s$, is ample, VIII 4.13 shows that $W_{\mathbb{L}_s}$ is separating, in particular separating on the left. A fortiori W_s is therefore separating on the left. By symmetry it is also separating on the right, which proves 1.5.

We shall substantially sharpen 1.5 when A^o_o has unipotent rank zero (5.4). I ≈326

2. Fixed and toric parts of $T_\ell(A_K)$. Criteria for good reduction. Orthogonality theorem for $\ell \neq p$.

2.1. Like every commutative algebraic group over k, A_o has a largest subtorus, whose formation commutes with every extension of the base field [SGA 3 XVII 7.2.1, XII 1.12]:

$$T_o \subset A_o. \tag{2.1.1}$$

When k is perfect, T_o is characterized by the condition that A_o^o/T_o be an extension of an abelian scheme B_o over k by a smooth connected unipotent group [25]. The latter necessarily has a composition series all of whose quotients are isomorphic to $\mathbb{G}_{ak}$ (SGA 3 XVII 4.1.3), and one has an exact sequence

$$0 \longrightarrow U_o \longrightarrow A_o^o/T_o \longrightarrow B_o \longrightarrow 0. \tag{2.1.2}$$

In any case, whether or not k is perfect, if A_o^o is an extension of an abelian scheme B_o by a smooth connected affine scheme L_o,

$$0 \longrightarrow L_o \longrightarrow A_o^o \longrightarrow B_o \longrightarrow 0, \tag{2.1.3}$$

this extension structure is known to be unique [25]. Moreover, $T_o \subset L_o$, and T_o is the largest subtorus of L_o. Thus one has the exact sequence (2.1.2), with

$$U_o = L_o/T_o. \tag{2.1.4}$$

2.1.5. Let us note that A_o^o has unipotent rank zero (that is, after passage to an algebraic closure of k, it has no subgroup isomorphic to $\mathbb{G}_a$) if and only if A_o^o/T_o is an abelian scheme B_o. In that case the preceding hypotheses apply with $U_o = 0$. I 327

Let ℓ be a prime different from the characteristic of k, so that A_o^o is ℓ-divisible. The inclusion (2.1.1) then gives an exact sequence of pro-group schemes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_\ell(T_o) & \longrightarrow & T_\ell(A_o^o) & \longrightarrow & T_\ell(A_o^o/T_o) & \longrightarrow & 0 \\ & & & & \downarrow\wr & & \downarrow\wr & & \\ & & & & T_\ell(A_o) & & T_\ell(A_o/T_o) & & \end{array} \quad , \tag{2.1.6}$$

whose terms may also be interpreted as twisted constant free ℓ-adic sheaves. When an exact sequence (2.1.2) is available—for example, when k is perfect or when A_o^o has unipotent rank zero—one has

$$T_\ell(A_o^o/T_o) \xrightarrow{\sim} T_\ell(B_o), \tag{2.1.7}$$

so that the exact sequence (2.1.6) becomes

$$0 \longrightarrow T_\ell(T_o) \longrightarrow T_\ell(A_o^o) \longrightarrow T_\ell(B_o) \longrightarrow 0. \tag{2.1.8}$$

Put

$$n = \dim A_K = \dim A_o, \qquad \mu = \dim T_o, \qquad \lambda = \dim U_o, \qquad \alpha = \dim B_o. \tag{2.1.9}$$

Then

$$n = \mu + \lambda + \alpha, \tag{2.1.10}$$

and

$$\begin{cases} \operatorname{rank} T_\ell(T_o) = \mu, & \operatorname{rank} T_\ell(A_o^o/T_o) = \operatorname{rank} T_\ell(B_o) = 2\alpha, \\ \operatorname{rank} T_\ell(A_o^o) = \mu + 2\alpha. \end{cases} \tag{2.1.11}$$

The second relation is a well-known theorem of Weil [5, 15].

2.1.12. The integers n, μ, λ, α are plainly defined, after passing to an algebraic closure of k, even without assuming the existence of an exact sequence (2.1.2). They are respectively called the dimension of A_o, the reductive rank of A_o (equal to the dimension of T_o), the unipotent rank of A_o, and the abelian rank of A_o. They are still related by (2.1.10) and (2.1.11), with the term $\operatorname{rank} T_\ell(B_o)$ omitted from the latter. I 328

2.1.13. Suppose that k has characteristic $p > 0$ and that A_o^o is p-divisible, or equivalently that it has unipotent rank zero (cf. 2.1.5). One should then also consider the pro-group $T_p(A_o^o)$, which is a Barsotti–Tate group (or a p-divisible group, in the terminology of [33]). The exact sequence (2.1.8) remains valid for $\ell = p$ and is then an exact sequence of Barsotti–Tate groups. Here $\lambda = 0$, and hence

$$n = \mu + \alpha. \tag{2.1.13.1}$$

The formulas (2.1.11) also remain valid, provided that rank is interpreted as the rank (or "height") of the Barsotti–Tate groups in question.

2.1.14. For A'^o_o we likewise introduce the invariants I 329

$$T'_o \subset A'^o_o, \qquad U'_o, \qquad B'_o,$$

where T'_o is the toric part of A'^o_o, B'_o its abelian part (defined in particular when k is perfect or when A'^o_o has unipotent rank zero), and U'_o its unipotent part (defined, for example, when k is perfect, or trivially when A'^o_o has unipotent rank zero). There is a corresponding exact sequence of the form (2.1.6), or (2.1.8), which we shall not write again. One could introduce integers μ', λ', α' for A'^o_o as in (2.1.9), but we shall see below (2.2.7) that they are the same as those for A_o^o; this spares us from introducing new notation for them.

2.2. We propose to make more precise the relations between $T_\ell(A_K)$ and $T_\ell(A_o) = T_\ell(A_o^o)$ by bringing $T_\ell(A^o)$ into play. First recall the following.

Lemma 2.2.1. Let $m > 0$ be an integer. Then the following conditions are equivalent:
(i) $m\,\mathrm{id}_{A^o}$ is flat.
(ii) $m\,\mathrm{id}_{A^o}$ is surjective.
(iii) $m\,\mathrm{id}_{A^o}$ is quasi-finite.
(iv) m is prime to the characteristic of k, or A_o^o has unipotent rank zero (cf. 2.1.5).
Moreover, if these conditions hold for m, they hold for m^ν for every $\nu > 0$; the groups

$$_{m^\nu}A^o \underset{\text{dfn}}{=} \ker(m^\nu\,\mathrm{id}_{A^o})$$

are quasi-finite, separated, and flat over S, and, as ν varies, they form an inverse system of group schemes whose transition morphisms

$$x \longmapsto m^{\nu'-\nu}x : {}_{m^{\nu'}}A^o \longrightarrow {}_{m^\nu}A^o \qquad (\nu' \geq \nu)$$

are faithfully flat.

Indeed, the fiberwise criterion for flatness (EGA IV 11.3.5) shows that (i), like (ii) and (iii), is a condition on the fibers; these three conditions are equivalent by SGA 3 VI$_A$ 5.6, since the fibers in question are smooth and connected. In the form (ii), a classical theorem of Weil [5, 15] shows that the conditions hold on the generic fiber, which is an abelian scheme; it is therefore enough to state them on the special fiber. Their equivalence with (iv) is then also well known from the structure theory. In any of these forms, stability of the condition under $m \mapsto m^\nu$ is immediate. It follows that the ${}_{m^\nu}A^o$ are quasi-finite flat group schemes over S, and they are plainly separated over S because A^o is. Finally, faithful flatness of the transition morphisms—equivalently, the fact that they are epimorphisms for the fppf topology—follows formally from the fact that $m\,\mathrm{id}_{A^o}$ is faithfully flat, hence an epimorphism for that topology. I ≈330

2.2.2. Consequently, we shall take m to be a prime ℓ satisfying the conditions of 2.2.1. We then denote by

$$T_\ell(A^o) = ({}_{\ell^\nu}A^o)_{\nu>0} \tag{2.2.2.1}$$

the inverse system described in 2.2.1. Under the two base changes $\eta \longrightarrow S$ and $s \longrightarrow S$, it recovers the inverse systems $T_\ell(A_K)$ and $T_\ell(A_o^o) = T_\ell(A_o)$ considered in 1.0 and 2.1 respectively.

2.2.3. Suppose now that S is henselian. Every quasi-finite separated scheme X over S then has a canonical decomposition as a disjoint union I 331

$$X = X^f \coprod X', \tag{2.2.3.1}$$

where X^f is finite over S and $X'_o = \varnothing$, that is, X' is confined to its generic fiber X'_η; thus X^f has the same special fiber as X. This decomposition is plainly functorial in X. In particular, if X is a group scheme over S, then X^f is a subgroup scheme. If X is flat, respectively étale, over S, then so is X^f.

We apply this to the components ${}_{\ell^\nu}A^o$ of $T_\ell(A^o)$. This gives a projective subsystem of groups

$$\left(({}_{\ell^\nu}A^o)^f\right)_{\nu>0} \underset{\text{dfn}}{=} T_\ell(A^o)^f \subset T_\ell(A^o), \tag{2.2.3.2}$$

called the fixed part of the Tate module $T_\ell(A^o)$. By construction, there are canonical isomorphisms, functorial in A,

$$T_\ell(A^o)^f \times_S s \xrightarrow{\sim} T_\ell(A^o) \times_S s \xrightarrow{\sim} T_\ell(A_o^o). \tag{2.2.3.3}$$

On the other hand, the generic fiber of $T_\ell(A^o)^f$ defines a canonical sub-pro-group of $T_\ell(A_K)$, again called the "fixed part" of this Tate module:

$$T_\ell(A_K)^f \subset T_\ell(A_K). \tag{2.2.3.4}$$

2.2.4. When $\ell \neq p = \operatorname{char} k$, the fixed part $T_\ell(A^o)^f$ may be interpreted as a twisted constant ℓ-adic sheaf on S (the $({}_{\ell^\nu}A^o)^f$ being finite étale over S). Since the functor I 332

$$X \longmapsto X_o = X \times_S s$$

from finite étale schemes over S to finite étale schemes over k is an equivalence of categories when S is henselian, there is an analogous equivalence for the categories of twisted constant ℓ-adic sheaves over S and over k. Hence (2.2.3.3), which gives the special fiber of $T_\ell(A^o)^f$, characterizes this fixed part up to unique isomorphism. On the other hand, as an ℓ-adic subsheaf of $T_\ell(A^o)$ it is plainly characterized by its generic fiber (2.2.3.4), or equivalently by the corresponding $\mathbb{Z}_\ell$-submodule, stable under $\pi = \operatorname{Gal}(\overline{K}/K)$:

$$T_\ell(A_K)^f(\overline{K}) \underset{\text{dfn}}{=} T_\ell(A(\overline{K}))^f \subset T_\ell(A(\overline{K})). \tag{2.2.4.1}$$

In fact, one has the following result.

Proposition 2.2.5. Suppose, as above, that $\ell \neq p$ and that S is henselian. Then the submodule (2.2.4.1) of $T_\ell(A(\overline{K})) = T_\ell(A_K)(\overline{K})$ corresponding to the fixed part $T_\ell(A_K)^f$ in (2.2.3.4) is the submodule $T_\ell(A(\overline{K}))^I$ of invariants under the inertia group $I \subset \pi$.

This follows, by passage to the inverse limit, from the analogous relation valid for every integer $m > 0$ prime to $p = \operatorname{char} k$:

$$({}_mA)^f(\overline{K}) = ({}_mA(\overline{K}))^I. \tag{2.2.5.1}$$

To prove this relation, introduce the strict localization $\widetilde{S}$ of S, whose fraction field $\widetilde{K}$ is the maximal unramified subextension of $\overline{K}/K$. Since $({}_mA)^f$ is finite étale over S, its points with values in $\overline{K}$ already have values in $\widetilde{K}$. Thus the left-hand side of (2.2.5.1) is $\operatorname{Hom}_S(\widetilde{S}, ({}_mA)^f)$, whereas the right-hand side is, by definition of $\widetilde{K}$, equal to ${}_mA(\widetilde{K})$. By the definition of $({}_mA)^f$, and since $\widetilde{S}$ is an inverse limit of finite étale S-schemes S_i, the former is also equal to

$$\operatorname{Hom}_S(\widetilde{S}, {}_mA) = {}_m \operatorname{Hom}_S(\widetilde{S}, A).$$

By the defining property of the Néron model, $\mathrm{Hom}_S(\widetilde{S}, A) = A(\widetilde{K})$, which proves (2.2.5.1).

To deduce 2.2.5, observe that (2.2.5.1) gives I≈333

$$(*) \qquad T_\ell(A(\overline{K}))^I = \varprojlim_\nu (\ell^\nu A)^f(\overline{K}) \supset \varprojlim_\nu (\ell^\nu A^o)^f(\overline{K}).$$

If (x_ν) belongs to the rightmost term, then for every $\nu \geq 0$, x_ν is a section of A that is infinitely divisible by ℓ. Its value at s is therefore an element of $A_o(k)$ infinitely divisible by ℓ, and consequently lies in $A_o^o(k)$. This proves equality of the last two terms of (*) and completes the proof of 2.2.5.

Proposition 2.2.6. Let C_K be a second abelian scheme over K, with Néron model C, and consider a homomorphism $u_K : C_K \longrightarrow A_K$, which therefore extends to a homomorphism $u : C \longrightarrow A$. Suppose that u_K is an isogeny. Then, for every prime ℓ, $T_\ell(u_K) : T_\ell(C_K) \longrightarrow T_\ell(A_K)$ induces an isogeny of fixed parts

$$(2.2.6.1) \qquad T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f,$$

and likewise the map induced by $u_o : C_o \longrightarrow A_o$

$$(2.2.6.2) \qquad T_\ell(u_o^o) : T_\ell(C_o^o) \longrightarrow T_\ell(A_o^o)$$

is an isogeny. Finally, $u_o : C_o \longrightarrow A_o$ induces an isogeny from the largest subtorus of C_o to the largest subtorus T_o of A_o; and over an extension k'/k on which the "abelian parts" of C_o and A_o are defined (for example, over the perfect closure of k), u_o induces an isogeny between the abelian parts of $C_{ok'}$ and $A_{ok'}$.

This statement follows formally from the fact that $T_\ell(A_K)^f$, $T_\ell(A_o^o)$, the largest subtorus of A_o, I≈334 and the "abelian part" of $A_{ok'}$ are additive functors of the varying A_K, and from the fact that u_K is an isogeny precisely in the sense that there is a morphism $v_K : A_K \longrightarrow C_K$ such that

$$u_K v_K = n\,\mathrm{id}, \qquad v_K u_K = n\,\mathrm{id}$$

for some integer $n > 0$.

Corollary 2.2.7. If A_K and C_K are isogenous abelian schemes over K, then the corresponding invariants μ, λ, α in (2.1.11) are the same for A_K and C_K. In particular, they are the same for A_K and its dual A'_K.

Remarks 2.2.8. In the two preceding statements, 2.2.6 and 2.2.7, we assumed that S is henselian. It is immediate, however, that 2.2.7, as well as the assertions of 2.2.6 concerning the properties of the induced homomorphism $u_o : C_o \longrightarrow A_o$, holds without this hypothesis. Indeed, one reduces to the case already treated by the base change $S^h \longrightarrow S$, where S^h is the henselization of S, using the fact—immediate from the definitions—that formation of Néron models commutes with this base change.

Corollary 2.2.9. (Good reduction.) Let S be a trait, let A_K be an abelian scheme over its fraction I 335 field K, let A be the Néron model of A_K, and let ℓ be a prime different from $\mathrm{char}\, k$. The following conditions are equivalent.

(i) There exists an abelian scheme over S whose generic fiber is isomorphic to A_K.
(ii) A is an abelian scheme.
(ii^o) A^o is an abelian scheme.
(iii) A_o is an abelian scheme.
(iii^o) A_o^o is an abelian scheme, that is, with the notation of (2.1.11), $\lambda = \mu = 0$.
(iv) A is proper over S.
(iv^o) A^o is proper over S.
(v) $T_\ell(A_K)$ is "unramified over S", that is, it extends to a twisted constant ℓ-adic sheaf over S, or equivalently the inertia group I acts trivially on $T_\ell(A_K(\overline{K}))$.
(v bis) For every integer $\nu \geq 0$, ${}_{\ell^\nu}A_K$ is "unramified over S", that is, it extends to an étale covering of S, or equivalently the inertia group I acts trivially on ${}_{\ell^\nu}A_K(\overline{K})$.

Finally, an abelian scheme over S as in (i) is a Néron model of A_K, and in particular is determined up to unique isomorphism.

2.2.9.1. We say that A_K has good reduction over S if it satisfies condition (i) above. Criterion (v), or (v bis), is known as the Néron–Ogg–Shafarevich criterion for good reduction.

Let us prove 2.2.9. It is well known (cf., for example, [5]) that an abelian scheme over S satisfies the universal property of a Néron model; this reduces to the fact that a rational section of an abelian scheme over a regular base is defined everywhere. This proves the equivalence $(i) \iff (ii) \iff (ii^o)$, as well as the last assertion of 2.2.9. These conditions plainly imply (iii), (iv), and (iv^o), each of which implies (iii^o). Let us prove that (iii^o)$\Longrightarrow$(ii^o), which will establish the equivalence of (i) through (iv^o). This follows from the fact that a smooth group scheme over S whose fibers are abelian schemes is proper over S (and hence is an abelian scheme), by EGA IV 15.7.10.

On the other hand, in view of 2.2.5 and the fact that I ≈336

$$T_\ell(A_K(\overline{K}))/T_\ell(A_K(\overline{K}))^I$$

is plainly a torsion-free $\mathbb{Z}_\ell$-module, condition (v) says that $T_\ell(A_K(\overline{K}))$ and its fixed part have the same rank. By (2.1.11) and (2.2.3.3), this is the condition

$$\mu + 2\alpha = 2(\lambda + \mu + \alpha),$$

that is, $2\lambda + \mu = 0$, or $\lambda = \mu = 0$, which is precisely (iii^o). Finally, the equivalence of (v) and (v bis) is clear from the definitions. This completes the proof of 2.2.9.

2.3. Suppose once again that S is henselian and that $\ell \neq p = \operatorname{char} k$. We now use the inclusion

$$T_\ell(T_o) \hookrightarrow T_\ell(A_o^o) \tag{2.3.1}$$

of (2.1.6), together with the equivalence of categories recalled in 2.2.4. We obtain a canonical twisted constant ℓ-adic subsheaf of $T_\ell(A^o)^f$, called the toric part of $T_\ell(A^o)$,

$$T_\ell(A^o)^t \hookrightarrow T_\ell(A^o)^f \hookrightarrow T_\ell(A^o), \tag{2.3.2}$$

characterized by the condition that its special fiber is the inclusion (2.3.1):

$$T_\ell(A^o)^t \times_S s = T_\ell(T_o) \hookrightarrow T_\ell(A_o^o). \tag{2.3.3}$$

Later we shall give an interpretation of the toric part of $T_\ell(A^o)$ as the Tate module $T_\ell(T)$ of a torus T over S lifting the torus T_o over k. That interpretation will allow the definition of the toric part to be extended to the case $\ell = p$. For the moment, and to fix ideas, we confine ourselves to the case $\ell \neq p$. In this case the toric part of $T_\ell(A^o)$ may be described by the corresponding $\mathbb{Z}_\ell$-submodule of $T_\ell(A(\overline{K}))$:

$$T_\ell(A(\overline{K}))^t \hookrightarrow T_\ell(A(\overline{K}))^f \hookrightarrow T_\ell(A(\overline{K})). \tag{2.3.4}$$

It is stable under the action of π; its first term is defined as $T_\ell(A^o)^t(\overline{K})$ and is, as one would expect, again called the toric submodule of the Tate module $T_\ell(A(\overline{K}))$. Thus this Tate module carries the canonical three-step filtration (2.3.4). Applying the same considerations to the dual abelian scheme A'_K, we obtain the analogous three-step filtration

$$T_\ell(A'(\overline{K}))^t \hookrightarrow T_\ell(A'(\overline{K}))^f \hookrightarrow T_\ell(A'(\overline{K})). \tag{2.3.5}$$

The middle terms in (2.3.4) and (2.3.5), the "fixed parts," are described in terms of the π-module structures of the Tate modules of $A(\overline{K})$ and $A'(\overline{K})$ as their submodules of invariants under the inertia group I (2.2.5). The toric parts are then completely described in terms of the fixed parts and the pairing φ of (1.0.2), by the following theorem (a special case of which already appears in [11]).

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Theorem 2.4. (Orthogonality theorem.) Let S be a henselian trait, let A_K be an abelian scheme over its fraction field K, let A'_K be the dual abelian scheme, and let ℓ be a prime different from the residual characteristic of S. The Tate modules $T_\ell(A(\overline{K}))$ and $T_\ell(A'(\overline{K}))$ then carry the canonical filtrations (2.3.4) and (2.3.5), respectively, by the "fixed part" and the "toric part." The toric part $T_\ell(A(\overline{K}))^t$ is the intersection of the fixed part $T_\ell(A(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A'(\overline{K}))^f = T_\ell(A'(\overline{K}))^I$ with respect to the canonical pairing φ of (1.0.3). Symmetrically, the toric part $T_\ell(A'(\overline{K}))^t$ is the intersection of $T_\ell(A'(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A(\overline{K}))^f = T_\ell(A(\overline{K}))^I$. (Here the superscript I denotes the submodule of invariants under the action of the inertia group I.)

Proof. a) Let us prove that $T_\ell(A(\overline{K}))^t$ is orthogonal to $T_\ell(A'(\overline{K}))^f$. To do so, we interpret the restriction of φ to the fixed parts of the Tate modules using the canonical biextension $W^o = W$ of (1.4.4). This is the canonical extension of the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ to a biextension of (A^o, A'^o) by $\mathbb{G}_{mS}$. By VIII 2.2, W gives rise to a pairing

$$\varphi_W : T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.1}$$

from which the pairing φ of (1.0.3) is obtained by passing to $\overline{K}$-valued points. It follows that the restriction of φ to the fixed parts considered in 2.4 is obtained, by passing to $\overline{K}$-valued points, from the pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.2}$$

the restriction of (2.4.1). Since the two factors on the left of (2.4.2) are now twisted constant ℓ-adic sheaves, this pairing may be studied on the special fibers. The fibers of the two factors are precisely the Tate modules $T_\ell(A^o_o)$ and $T_\ell(A'^o_o)$ (2.2.3.3), and the pairing induced on special fibers by (2.4.2) is

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.3}$$

obtained from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$, the restriction of W to the special fibers.

Recall from 2.3 that the toric part of $T_\ell(A(\overline{K}))$ is obtained by passing to $\overline{K}$-valued points in the twisted constant ℓ-adic subsheaf $T_\ell(A^o)^t$ of $T_\ell(A^o)^f$. Thus the orthogonality relation to be proved says that the pairing induced by (2.4.2),

$$T_\ell(A^o)^t \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.4}$$

is identically zero. It is enough to verify this for the corresponding pairing on the special fibers, which by the definition of the toric part is

$$T_\ell(T_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.5}$$

induced by (2.4.3). By VIII 4.10 this pairing is necessarily zero, which proves the asserted orthogonality.

b) Let us prove that the inclusion

$$T_\ell(A(\overline{K}))^t \subset T_\ell(A(\overline{K}))^f \cap \left(T_\ell(A'(\overline{K}))^f\right)^\perp \tag{2.4.6}$$

is an equality, where $\perp$ denotes the orthogonal. Equivalently, in the pairing (2.4.2), the annihilator of the second factor—which, as we have just seen, contains the toric part $T_\ell(A^o)^t$—is equal to that toric part. Since these are again relations between twisted constant ℓ-adic sheaves, it is enough to check the assertion on the special fibers: namely, that the inclusion already obtained (2.4.7)

$$T_\ell(T_o) \subset T_\ell(A'^o_o)^\perp \tag{2.4.7}$$

is an equality. In other words, we must verify that the biextension W_o of (A^o_o, A'^o_o) is nondegenerate on the left (VIII 4.11). But this is precisely what we already observed in 1.5, as a consequence of a result from Raynaud's thesis.

2.5. The orthogonality theorem has an equivalent formulation using a polarization ξ of A_K and the alternating bilinear form (1.0.8), compatible with the operations of π,

$$\varphi_\xi : T_\ell(A(\overline{K})) \times T_\ell(A(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{2.5.1}$$

obtained from the form φ of (1.0.3) by transporting structure in the second argument along the isogeny of $\mathbb{Z}_\ell$-modules

$$T_\ell(A(\overline{K})) \longrightarrow T_\ell(A'(\overline{K}))$$

induced by the isogeny $\psi_\xi : A_K \longrightarrow A'_K$ defined by ξ. The equality in (2.4.6) then takes the simple form

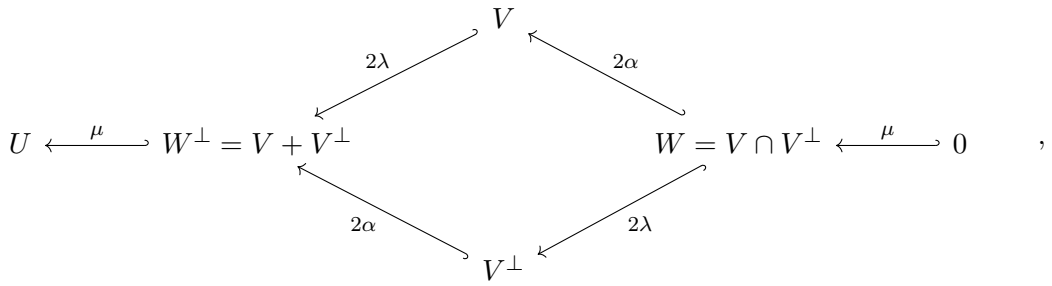
$$W = V \cap V^\perp, \tag{2.5.2}$$

where we have set

$$\begin{cases} U = T_\ell(A(\overline{K})), \qquad V = U^I = \text{the "fixed part" of } T_\ell(A(\overline{K})), \\ \qquad\qquad W = \text{the toric part of } T_\ell(A(\overline{K})), \end{cases} \tag{2.5.3}$$

and $\perp$ denotes the orthogonal with respect to the polarization form (2.5.1). These modules give the following diagram of inclusions:

(2.5.4)



where the integers 2α, 2λ, and μ are the ranks of the quotients of consecutive modules in the diagram, with the notation of (2.1.9). (That W has rank μ is the definition of μ; that V/W has rank 2α follows from the second relation in (2.1.11) together with the exact sequence (2.1.6). Hence, by orthogonality, $\operatorname{rank}(U/W^\perp) = \mu$ and $\operatorname{rank}(W^\perp/V^\perp) = 2\alpha$. Finally, the formulas

$$\operatorname{rank}(V^\perp/(V \cap V^\perp)) = \operatorname{rank}(V^\perp/W) = 2\lambda \tag{2.5.5}$$

and the dual formula follow from the preceding ones, in view of (2.1.10) and $\operatorname{rank} U = 2n$.)

Remarks 2.6. a) Here we obtained the orthogonality theorem 2.4 as an easy consequence of the theory of biextensions developed in the preceding sections. The inclusion (2.4.6) can be proved without using the theory of biextensions, by an arithmetic argument in the style of the proof of the "monodromy theorem" given in Exposé I. One reduces to a situation in which S has finite residue field, where the orthogonality result follows from a simple consideration of weights for the Frobenius operation (using A. Weil's result on the eigenvalues of that Frobenius). There is no need to invoke resolution of singularities here: one may state the desired result for group schemes A over schemes S under sufficiently general conditions. Unfortunately, it does not appear possible to obtain by this method equality in the preceding inclusion. That equality will apparently be used in an essential way in the proof of the "semistable reduction theorem" in the next section. I≈341

b) The orthogonality theorem 2.4 allows the most important properties of the Néron model of an abelian variety A_K to be expressed in terms of the representation of the inertia group I on the Tate module $T_\ell(A(\overline{K}))$, where ℓ is a fixed prime different from the residual characteristic. (The case $\ell = p$ will be considered separately later, in section 5.) As a first statement of this kind, though one that does not use the orthogonality theorem, recall the good-reduction criterion 2.2.9. We shall give many further examples below, beginning in the next section (3.5). I≈342

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3. The case of semistable reduction. The semistable reduction theorem.

We first give an auxiliary result in the style of SGA 3.

Proposition 3.1. Let S be a scheme, let G be a commutative S-group scheme that is flat, separated, and of finite presentation over S, let H be a quasi-separated group scheme locally of finite type over S, let $u : G \longrightarrow H$ be a homomorphism of S-group schemes, and put $K = \ker u$. For every $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $k(s)$, and denote by $\mu(s)$ (respectively, $\alpha(s)$) the reductive rank (respectively, the abelian rank) of K_s, that is, the dimension of the largest subtorus (respectively, of the largest abelian-variety quotient) of $(K_{\bar{s}})^o_{\mathrm{red}}$. Let $s \in S$, let $t \in S$ be a generalization of s, and let p be the characteristic exponent of $k(s)$.

a) For every integer $n > 0$ prime to p, ${}_nK$ is etale, of finite type, and separated over S in a neighborhood of s, and one has the divisibility relation

$$\mathrm{rank}({}_nK_s) \mid \mathrm{rank}({}_nK_t). \tag{3.1.1}$$

b) One has

$$\mu(s) + 2\alpha(s) \leq \mu(t) + 2\alpha(t). \tag{3.1.2}$$

c) If $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent, then so is $(K_{\bar{s}})^o_{\mathrm{red}}$.

d) Suppose that G_s has unipotent rank zero, that H is smooth over S, and that u_t is an isogeny. Then u is flat and quasi-finite over an open neighborhood U of s; a fortiori, $K|_U$ is flat and quasi-finite.

e) Under the hypotheses of d), suppose in addition that u_t is a monomorphism (and hence an isomorphism from G_t onto an open subgroup of H_t). Then u_s is an isomorphism from G_s onto an open subgroup of H_s. If, moreover, S is irreducible with generic point t, then there is an open neighborhood U of s such that u induces an isomorphism from $G|_U$ onto an open subgroup of $H|_U$.

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Proof.

a) We may suppose that n is invertible on S. Then $n\,\mathrm{id}_G$ is etale on every fibre (SGA 3, XV, 1.3); since G is flat and of finite presentation over S, the fibrewise criterion for etaleness (EGA IV, 17.8.2) shows that $n\,\mathrm{id}_G$ is etale. Its kernel ${}_nG$ is therefore etale; it is also separated and of finite presentation over S, since G is. Similarly, ${}_nH$ is locally of finite type, quasi-separated, and formally unramified over S, [10] so its unit section is an open immersion (EGA IV, 17.4.2 b')). Consequently, the kernel of

$${}_nu : {}_nG \longrightarrow {}_nH$$

is an open subscheme of ${}_nG$, hence is etale over S; it is quasi-compact in ${}_nG$ and therefore of finite presentation over S, and it is separated over S because G is. This kernel is plainly ${}_nK$.

Formula (3.1.1) follows from these properties of ${}_nK$. Indeed, after a base change we may reduce to the case where S is henselian local with closed point s, and then use the decomposition of ${}_nK$ into ${}_nK^f$ and a scheme with empty special fibre (cf. 2.2.3). More precisely, the same argument gives a monomorphism on geometric fibres

$${}_nK(\bar{s}) \longhookrightarrow {}_nK(\bar{t}).$$

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b) Choose an integer $n > 0$ prime to the characteristic exponent of $k(s)$ and to the ranks of the quotients of the geometric fibres of K at s and t by their reduced identity components. The two sides of (3.1.1) are then obtained by raising n, respectively, to the powers whose exponents are the two sides of (3.1.2), as follows from the known structure of commutative algebraic groups. Thus b) follows from a).

c) The hypothesis that $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent is equivalent to the equalities $\mu(t) = \alpha(t) = 0$, and likewise for $K_{\bar{s}}$. Assertion c) therefore follows formally from b).

[10] At least when H is commutative. In the general case, ${}_nH$ is still unramified over S along its unit section, which is all that is needed here.

d) Since u_t is an isogeny, K_t is finite. We are therefore under the hypotheses of c), and $(K_{\bar{s}})^o_{\mathrm{red}}$ is unipotent. As G_s has unipotent rank zero, it follows that K_s is finite. Moreover, the fact that u_t is an isogeny gives $\dim G_t = \dim H_t$. Since G and H are flat and locally of finite presentation, one has

$$\dim G_s = \dim G_t, \qquad \dim H_s = \dim H_t \quad \text{(SGA 3, VI}_B\text{, 4.3)},$$

and hence $\dim G_s = \dim H_s$. Since $\dim \ker u_s = 0$, it follows that $\dim H_s = \dim u_s(G_s)$ (SGA 3, VI$_B$, 1.2), which, because H_s is smooth, is equivalent to u_s being flat (SGA 3, VI$_A$, 5.6). There is therefore an open neighborhood U of s such that u is flat and quasi-finite at the points of the unit section over U (SGA 3, VI$_B$, 2.2(i) and 2.5(ii)). As for u_s, it follows that $u_{s'}$ is flat for every $s' \in U$; hence $u|_{G_U}$ is flat and quasi-finite, since it has these properties on every fibre.

e) For the first assertion, a base change immediately reduces us to the case where S is integral with generic point t, and hence to proving the second assertion of e). That assertion follows at once from d) and the following lemma.

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Lemma 3.1.3. Let S be a scheme, let K be a quasi-finite S-scheme of finite presentation that is separated and flat over S, let $s \in S$, and let $t \in S$ be a generalization of s. Then

$$\operatorname{rank} K_s \leq \operatorname{rank} K_t.$$

If $\operatorname{rank} K_\eta \leq 1$ for every maximal point η of S, then $K \longrightarrow S$ is an open immersion (and hence an isomorphism if it admits a section).

For the first assertion, strict localization reduces us to the case where S is strictly local. Decomposing K as the disjoint union of a finite scheme K^f and a scheme with empty special fibre, we obtain

$$\operatorname{rank} K_s = \operatorname{rank} K^f_s = \operatorname{rank} K^f_t \leq \operatorname{rank} K_t.$$

The second equality follows because K^f is flat, finite, and of finite presentation over S, and therefore locally free over S.

For the second assertion, the preceding argument already shows that $\operatorname{rank} K_s \leq 1$ for every $s \in S$. Thus K_s is empty or isomorphic to $\operatorname{Spec} k(s)$. Consequently, $K \longrightarrow S$ is a monomorphism (SGA 3, VI$_B$, 2.11), and, being flat and of finite presentation, it is an open immersion (EGA IV, 17.9.1). This proves the lemma.

Remark 3.1.4. Part 3.1 a), and the remainder of Proposition 3.1, give no information about ${}_nK_s$ when n is a power of p (for $p > 1$). At least when G is smooth (and S is locally noetherian or H is of finite presentation over S), essentially the same method as in the proof of a) gives such information. Indeed, reducing to the case where S is complete local noetherian and using SGA 3, XV, 1.6 c), X, 3.2, IX, 3.6, and IX, 5.2, one obtains

$$\operatorname{rank}_{\mathrm{mult}}({}_nK_s) \mid \operatorname{rank}_{\mathrm{mult}}({}_nK_t), \tag{3.1.4.1}$$

where $\operatorname{rank}_{\mathrm{mult}}$ denotes the rank of the largest subgroup of multiplicative type of the commutative algebraic group under consideration (SGA 3, XVII, 7.2.1). Notice that a torsion algebraic group of multiplicative type is finite. Formula (3.1.4.1) is valid for every integer $n \geq 1$. It follows from SGA 3, XVII, 4.6.1(vi) that, in c), if K_t is unipotent (SGA 3, XVII, 1.1), then K_s is unipotent as well.

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Proposition 3.2. Let S be a regular integral locally noetherian scheme of dimension 1, let K be its field of rational functions, let A_K be an abelian scheme over K, and let A be its Neron model over S, which is a smooth separated group scheme over S (and is quasi-projective over S if S is noetherian [23, VIII, 2, 3°]). The following conditions are equivalent:

(i) For every $s \in S$, A^o_s has unipotent rank zero; that is, it is an extension of an abelian scheme by a torus (2.1.5).

(ii) There exists a smooth separated group scheme G of finite type over S, extending A_K, whose fibres have unipotent rank zero.

Moreover, if G is as in (ii), then the unique morphism $G \longrightarrow A$ inducing the given isomorphism on the generic fibres is an isomorphism from G onto an open subgroup of A; it therefore induces an isomorphism of identity components

$$G^o \xrightarrow{\sim} A^o.$$

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3.2.1. Let us note at once that the definition of the Neron model A of A_K is the same as in the local case recalled in 1.1. The existence of this Neron model, and the fact that it is separated and of finite type over S, are well known and in fact reduce immediately to the local case. Of course, for every closed point $s \in S$,

$$S' = \mathrm{Spec}(\mathcal{O}_{S,s})$$

is a trait, and $A \times_S S'$ is precisely the Neron model of A_K over S' in the sense considered in 1.1.

Let us prove 3.2. It is trivial that (i) implies (ii), since it suffices to take $G = A$. It remains to prove that if G is as in (ii), then $G \longrightarrow A$ is an open immersion. This is indeed a special case of 3.1 e).

Corollary 3.3. With the notation of 3.2, suppose that the identity components of the fibres of the Neron model have unipotent rank zero. Let S' be a scheme satisfying the same hypotheses as S, let K' be its field of rational functions, let $f : S' \longrightarrow S$ be a dominant morphism, let

$$A_{K'} \simeq A_K \otimes_K K' \simeq \text{the generic fibre of } A \times_S S',$$

let A' be the Neron model of $A_{K'}$, and consider the unique morphism

$$u : A \times_S S' \longrightarrow A'$$

inducing the identity on the generic fibres. Then u is an open immersion and therefore induces an isomorphism on identity components

$$A^o \times_S S' \xrightarrow{\sim} A'^o. \tag{3.3.1}$$

Indeed, it suffices to apply 3.2 over S', taking $G \simeq A \times_S S'$.

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3.3.2. Thus the equivalent properties of 3.2 imply a partial "stability" property for Neron models. One should bear in mind that when A_K does not have good reduction, that is, when A is not an abelian scheme, it is not true that, for every S' as in 3.3 (even for S' finite over S and K'/K separable), the morphism

$$A \times_S S' \longrightarrow A'$$

itself is an isomorphism, or that it induces isomorphisms on the groups of connected components of the fibres (see 11.10 below for the behavior of these groups of connected components under extension of the base). These observations, reinforced by 3.9 below, justify the following terminology, introduced by Mumford (and which also appears to agree with the notions of the same name studied in his book [17]).

Definition 3.4. With the notation of 3.2, the abelian scheme A_K over K is said to have semistable reduction over S if the equivalent conditions (i) and (ii) of 3.2 are satisfied.

3.4.0. Notice that in the form (ii), this condition does not invoke the notion of a Neron model. Moreover, by localization it reduces immediately to the case where S is a trait, as follows for example from 3.2.1. Furthermore, since formation of the Neron model commutes with base change from S to its henselization or its strict henselization, the notion even reduces to the case where S is a strictly local trait.

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Proposition 3.5. (Galois criterion for semistable reduction). Suppose that S is a trait. The following conditions on the abelian scheme A_K are equivalent (the notation being that of (2.5.3), applied to the situation deduced from (A_K, S) by passing to the henselization of S):

(i) A_K has semistable reduction over S (3.4).

(ii) With the notation of (2.5.3) and (2.5.4), one has

$$V^{\perp} \subset V.$$

(iii) For the toric part W of the Tate module one has

$$W = V^{\perp}.$$

(iv) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2 (that is, there is an I-stable submodule U' of U such that I acts trivially on U' and on U/U').

We may suppose that S is henselian (3.4.0). With the notation of 2.1.9, (i) means that

$$\lambda = 0.$$

By the rank formulas following (2.5.4), this is equivalent to $V^{\perp} = V \cap V^{\perp}$, that is, to (ii). Since $W = V \cap V^{\perp}$ by the orthogonality theorem 2.4, conditions (ii) and (iii) are equivalent. Moreover, (iv) also says that I acts trivially on $U/U^I = U/V$. By duality this means that it acts trivially on $V^{\perp}$, that is, that $V^{\perp} \subset U^I = V$, which is (ii). This completes the proof.

Remarks 3.5.1. Conditions (ii) and (iv) plainly depend only on the action of I on the $\mathbb{Q}_\ell$-vector space $U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and could have been formulated in terms of that action. Let us also note that these conditions are manifestly stable under every base change by a morphism of traits $S' \longrightarrow S$. Finally, if (iv) holds for A_K, it plainly holds for every abelian subscheme and every quotient abelian scheme of A_K, and in particular for every abelian scheme isogenous to A_K (for example, for A'_K).

Corollary 3.5.2. Let g be a topological generator of $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$. The equivalent conditions of 3.5 are also equivalent to the following condition: I 351

(v) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ factors through the maximal pro-ℓ quotient $I(\ell)$ of I, which is canonically isomorphic to $\mathbb{Z}_\ell(1)$, and the action g_U of the generator g on U satisfies

$$(1 - g_U)^2 = 0. \tag{3.5.3}$$

Indeed, for every continuous unipotent action of a profinite group I on an ℓ-adic vector space, one sees at once that the image of I is a pro-ℓ group. It follows that (v) is equivalent to (iv).

We can now prove the central result of the present expose:

Theorem 3.6. (Semistable reduction theorem). Let S be a connected regular noetherian scheme of dimension 1, let K be the field of rational functions of S, and let A_K be an abelian scheme over K. Then there is a finite Galois extension K' of K such that the abelian scheme $A_{K'} = A_K \otimes_K K'$ has semistable reduction (3.4) over the normalization S' of S in K'.

Notice that S' is connected, regular, noetherian, and of dimension 1, so that the stated conclusion makes sense. To prove 3.6, choose a separable closure $\overline{K}$ of K and write $\overline{K}$ as the filtered union of its finite Galois subextensions K_i. Let S_i be the normalization of S in K_i, and let U be the largest open subset of S over which A_K has good reduction. Since S is connected, noetherian, and one-dimensional, and $U \neq \varnothing$, the complement $T = S - U$ is a finite set of closed points of S, and $\mathcal{O}_{S,t}$ is a discrete valuation ring for every $t \in T$. It is enough to find, for every $t \in T$, an index $i = i(t)$ such that A_{K_i} has semistable reduction at the points of S_i above t: one then takes for K' the compositum of the extensions $K_{i(t)}$, for $t \in T$. Thus we are reduced to the case where S is a trait. I ≈ 352

Moreover, since the Galois group permutes the points of the normalization lying above the closed point s of S, it is enough to find K' such that $A_{K'}$ has semistable reduction at *one* point of the normalization S'. Here it is unnecessary at first to require that K'/K be Galois: it is enough to find a finite etale extension K'/K satisfying the preceding condition, since its Galois closure satisfies the same condition. Using criterion 3.5(iv), we see that Theorem 3.6 reduces to the following assertion.

Corollary 3.7. Suppose again that S is a trait. Then there is an open subgroup π' of π such that the action of $I' = I \cap \pi'$ on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2; equivalently, there is an open subgroup I' of I whose action on U is unipotent of level 2.

There is a canonical isomorphism, compatible with the actions of π, I≈353

$$(*) \qquad H^1(A_{\overline{K}}, \mathbb{Z}_\ell) \simeq \mathrm{Hom}(T_\ell(A(\overline{K})), \mathbb{Z}_\ell),$$

which shows that assertion 3.7 is equivalent to the same assertion for the actions of I on the left-hand side of (*). But that assertion was proved in Expose III (using resolution of singularities for excellent schemes of dimension 2).

Corollary 3.8. The equivalent conditions (i)–(iv) of 3.5 are also equivalent to the following condition:

(vi) The action of I on $U = T_\ell(A(\overline{K}))$ is unipotent.

Indeed, to prove that (vi) implies (iv), observe that if a group I acting unipotently on a finite-dimensional vector space E over a field of characteristic zero (here $\mathbb{Q}_\ell$) has a subgroup I' of finite index acting with level m (here $m = 2$), then I itself acts with level m. In fact, since I' has finite index in I, the algebraic group $G' \subset \mathrm{GL}(E)$ that it defines has finite index in the group G defined by I; but G is unipotent and hence connected (in characteristic zero), so $G = G'$. Thus I and I' have the same "level of unipotence."

We also obtain a converse to 3.3:

Corollary 3.9. The notation is that of 3.2. The conditions (i) and (ii) of 3.2 are also equivalent to the following condition:

(iii) For every finite etale extension K' of K, let S' denote the normalization of S in K', and let A' denote the Neron model of $A_{K'} = A \otimes_S K' = A_K \otimes_K K'$. Then the canonical morphism

$$A \times_S S' \longrightarrow A'$$

induces an isomorphism on identity components.

The necessity of the condition was proved, in a stronger form, in 3.3. For sufficiency, choose I≈354
K' as in 3.6. Since the fibres of A'^o have unipotent rank zero, the same is true of the fibres of $(A \times_S S')^o = A^o \times_S S'$, and hence also of the fibres of A^o, from which the former are obtained by extension of the base field.

4. Application to a conjecture of Serre–Tate and to the conductor

4.1. We suppose that S is henselian, and retain the notation of (2.5.3), with $\ell \neq p$. [11] To study the action of $\pi = \mathrm{Gal}(\overline{K}/K)$ and of its inertia subgroup I on $U = T_\ell(A(\overline{K}))$, we shall use the fact that there exists an open subgroup I' of I, which we may moreover take to be normal in π, such that the action of I' on U is unipotent of level two; that is,

$$U^{I'} \supset (U^{I'})^\perp.$$

It is then clear that $U^{I'}$ does not depend on the particular choice of I'. We might call it the essentially fixed submodule of U, and shall denote it by U^{ef}. Its orthogonal will be called the essentially toric submodule of U, and will be denoted by U^{et}. We thus have a three-step filtration

$$(4.1.1) \qquad U \supset U^{\mathrm{ef}} \supset U^{\mathrm{et}} \supset 0,$$

which is plainly invariant under the action of π.

Let us also note that there is a largest subgroup of I acting unipotently (or, equivalently here, unipotently of level two) on U: it is the subgroup I' consisting of the elements acting unipotently. It is also the kernel of the representation of I on the graded module associated with the filtration (4.1.1), or, equivalently in the present case, the kernel of the induced representation of I on U^{ef}. By the Galois criterion for semistable reduction (3.5), this subgroup I corresponds to the smallest Galois extension $\widetilde{K'}$ of $\widetilde{K}$ (the maximal unramified extension of K) such that $A_{\widetilde{K'}}$ has semistable reduction. This description also shows that the subgroup I' of I is independent of the choice of the prime $\ell \neq p$.

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[11] The case $\ell = p$ will be studied in §§5 and 6 below.

Notice that, once the representation of π on U^{ef}, and hence on U^{et}, is known, the representation on U/U^{ef} is known up to isogeny, by means of the isogeny induced by the polarization pairing (2.5.1):

$$U/U^{\mathrm{ef}} \longrightarrow (U^{\mathrm{et}})^{\vee}(1) \qquad \text{an isogeny,} \tag{4.1.2}$$

where (1) denotes the Tate twist by $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$; this homomorphism is plainly compatible with the operations of π. More precisely, there is a canonical isomorphism

$$U/U^{\mathrm{ef}} \simeq ((U')^{\mathrm{et}})^{\vee}(1), \qquad U' = T_\ell(A'(\overline{K})).$$

It therefore remains to study more closely the action of π on U^{ef}.

By construction this action is trivial on the normal subgroup I' of I, and hence factors through a representation of the quotient group

$$\pi' = \pi/I', \tag{4.1.3}$$

an action of which we shall now give a description "independent of ℓ."

4.2. Let K' be a Galois extension of K corresponding to an open normal subgroup V of π such that $V \cap I = I'$. The Galois group G of K'/K then occurs in an exact sequence I 356

$$1 \longrightarrow \Gamma \longrightarrow G \longrightarrow G_o \longrightarrow 1, \tag{4.2.1}$$

where

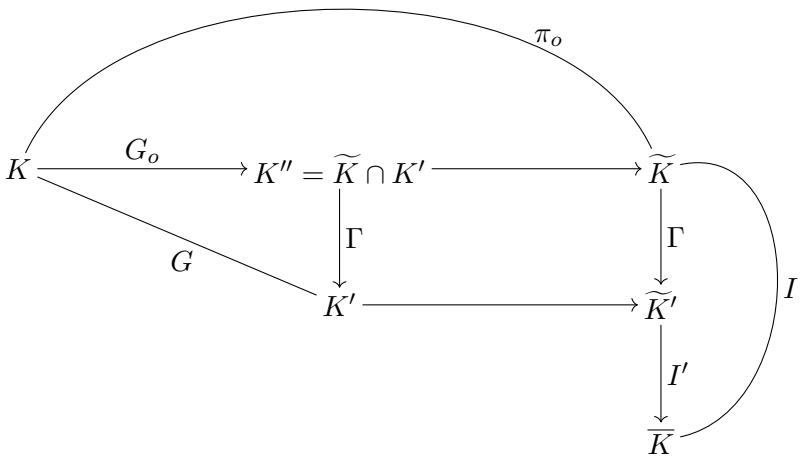
$$\Gamma = I/I' \tag{4.2.2}$$

is the inertia subgroup, and G_o is the Galois group of the largest unramified subextension K'' of K'/K. Thus (4.2.1) is the image of the analogous exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \pi' \longrightarrow \pi_o \longrightarrow 1, \tag{4.2.3}$$

where π' is the group (4.1.3) acting on U^{ef}. We therefore have the following diagram of field inclusions:

(4.2.4)



Let S' be the normalization of S in K', and let A^* be the Neron model of $A_{K'}$. Then, by the Galois criterion for semistable reduction (3.5), $A_{K'}$ has semistable reduction over S'; that is, A_o^{*o} is an extension of an abelian scheme by a torus. On the other hand, by transport of structure, G acts on A^*, and hence on A^{*o}, by automorphisms in a manner compatible with the given action of G on the base scheme S' of A^*. It is then clear that the action of π' on

$$U^{\mathrm{ef}} = T_\ell(A^{*o})^f(\overline{K})$$

is obtained by transport of structure from the action of π' on $(A^{*o}/K', \widetilde{K'})$, where π' acts on A^* through G. Taking into account the canonical isomorphism

$$U^{\mathrm{ef}} \simeq T_\ell(A^{*o})^f(\overline{K}) \simeq T_\ell(A_o^{*o})(\overline{k}) \tag{4.2.6}$$

(valid for every twisted constant ℓ-adic sheaf over S'; cf. Exposé I [reference omitted in the source]), we obtain the following description of the action of π' on U^{ef}.

Consider the smooth connected group scheme A_o^{*o} over the residue field $k' \hookrightarrow \overline{k}$ of S', which is a quasi-Galois extension of k with Galois group G_o. The quotient group G of π' acts on A_o^{*o} compatibly with its action on k' through G_o. Moreover, π' acts on $\overline{k}$ through its quotient π_o, and hence acts on the situation $(A_o^{*o}/k', \overline{k})$. Thus, by transport of structure, π' acts on I≈357

$$U^{\mathrm{ef}} = T_\ell(A_o^{*o})(\overline{k}),$$

and this is the desired action of π'. In particular, the subgroup Γ of G, which acts trivially on k', acts on A_o^{*o} by k'-automorphisms, and the corresponding representation of $\Gamma = I/I'$ on $T_\ell(A_o^{*o})(\overline{k})$ is the representation deduced from the natural representation of the inertia group I on U^{ef}.

On the other hand, the essentially toric part U^{et} of U corresponds simply to the submodule

$$U^{\mathrm{et}} \simeq T_\ell(T_o^*)(\overline{k}) \subset U^{\mathrm{ef}} \simeq T_\ell(A_o^{*o})(\overline{k}), \tag{4.2.7}$$

where T_o^* is the largest subtorus of A_o^{*o}. Likewise, put

$$B_o^* = A_o^{*o}/T_o^* \tag{4.2.8}$$

for the abelian part of A_o^{*o}. The action of G on A_o^{*o} induces separately actions of G on T_o^* and on B_o^*, and the action of π' on U^{et} and on $U^{\mathrm{ef}}/U^{\mathrm{et}}$ is obtained from these actions by means of the isomorphism (4.2.7) and the corresponding isomorphism

$$U^{\mathrm{ef}}/U^{\mathrm{et}} \simeq T_\ell(B_o^*)(\overline{k}). \tag{4.2.9}$$

It follows from this and the isogeny (4.1.2) that the character $g \longmapsto \operatorname{Tr} g_U$ of the action of π' (or of π) on U is given by

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \chi_\ell(g) \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.2.10}$$

Here U_ℓ has been written for U on the left to recall the chosen prime $\ell \neq p$; $g_{B_o^*,\ell}$ and $g_{T_o^*,\ell}$ denote, respectively, the actions of g on the Tate modules $T_\ell(B_o^*)(\overline{k})$ and $T_\ell(T_o^*)(\overline{k})$ induced by its actions on B_o^* and T_o^* just described; and

$$\chi_\ell : \pi' \longrightarrow \pi_o \longrightarrow \mathbb{Z}_\ell^* \tag{4.2.11}$$

is the character of π' (or of π_o) induced by the action on $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$—a character that is trivial on Γ.

We are now in a position to prove a special case of a conjecture of Serre–Tate [29, Appendix]. [12] I≈358

Theorem 4.3. Let S be a henselian trait with residue field k and function field K, and let A_K be an abelian scheme over K. Consider the representation of $\pi = \mathrm{Gal}(\overline{K}/K)$ on the Tate module $U_\ell = T_\ell(A(\overline{K}))$, where ℓ is a prime distinct from the residual characteristic. I 359

a) There exists an open subgroup I' of the inertia group that acts unipotently on U_ℓ, so that the character of the representation of π on U_ℓ is in fact a function on π/I'. Moreover, on $\Gamma = I/I'$ this function takes integral values and is independent of ℓ. More generally, for every $g \in I$, the coefficients of the characteristic polynomial of g_{U_ℓ} are rational integers independent of ℓ.

b) Suppose that k is a finite field with q elements, so that

$$\pi_o = \pi/I \simeq \mathrm{Gal}(\overline{k}/k)$$

[12] This is essentially the case $i = 1$ of those conjectures for the H^i.

is identified with the profinite completion $\widehat{\mathbb{Z}}$ of $\mathbb{Z}$, the generator 1 of $\mathbb{Z}$ corresponding to the Frobenius automorphism

$$f_q : x \longmapsto x^q$$

of $\overline{k}/k$. Then, for every $g \in \pi$ whose image in π_o lies in $\mathbb{Z}$, the trace (and, more generally, every coefficient of the characteristic polynomial) of g_{U_ℓ} is a rational number independent of ℓ. It is even an integer if the image of g in π_o is of the form f_q^n, with $n \geq 0$.

Let us prove, for example, the assertions concerning traces. The assertions concerning the coefficients of the characteristic polynomial could be proved in the same way. One may also note that they follow formally from the assertions about traces, by expressing these coefficients in terms of the traces of the iterates (although those formulas introduce denominators and therefore do not give the asserted integrality), together with a lemma of Dwork–Fatou, for which a pleasant proof is given by Kleiman in [14, 2.8].

a) We use formula (4.2.10), which here becomes I≈**360**

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.3.0}$$

Here $g_{B_o^*}$ and $g_{T_o^*}$ are now k'-automorphisms of B_o^* and T_o^*, respectively. It therefore suffices to show that the two corresponding traces are rational integers independent of ℓ, which is a general "geometric" assertion about algebraic groups. For $g_{B_o^*}$ this is a well-known result of Weil in the theory of abelian varieties [5] [15]. For $g_{T_o^*}$ it is practically immediate: one describes the representation of Γ on $T = T_o^*$ by means of the corresponding representation on the dual group $M' \simeq \mathbb{Z}^\mu$. The required trace is that of the contragredient of the latter representation, by the canonical isomorphism

$$T_\ell(T) \simeq \check{M}_\ell(1) \overset{\text{def}}{=} \check{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1).$$

b) If the image of g in π_o is f_q^n, formula (4.2.10) now becomes

$$\operatorname{Tr} g_{U,\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + q^n \operatorname{Tr} g_{T_o^*,\ell}^{-1}. \tag{4.3.1}$$

We are thus reduced to proving the following lemma, in which k is to be replaced by an algebraic closure of k', and A by $A_o^{*o} \otimes_{k'} \overline{k}$.

Lemma 4.3.2. Let k be the algebraic closure of a finite field, let A be a commutative group scheme of finite type over k, and let g be an automorphism of this group scheme compatible with the automorphism f of $\operatorname{Spec}(k)$ obtained by transport of structure from an iterated Frobenius automorphism

$$x \longmapsto x^q, \qquad q = p^r, \quad r \geq 0,$$

where q is a power of the characteristic. Thus the action of g on functions φ satisfies

$$g^*(\lambda\varphi) = f^*(\lambda)g^*(\varphi) = \lambda^{q^{-1}} g^*(\varphi) = \lambda^{p^{-r}} g^*(\varphi) \qquad (\lambda \in k).$$

Let g_ℓ be the automorphism induced by g on $T_\ell(A(k))$ by "transport of structure," where ℓ is a prime different from p. Then the trace of g_ℓ (and, more generally, every coefficient of its characteristic polynomial) is a rational integer independent of ℓ, and its eigenvalues are algebraic integers of absolute value $q^{1/2}$ or q. If A is an abelian scheme, their absolute values are $q^{1/2}$; if A is affine, the eigenvalues are of the form $q\varepsilon_i$, where the ε_i are roots of unity.

Indeed, consider the "q-th power" endomorphism f_q^A of the underlying absolute scheme of A, I≈**361** defined as the identity on points and as the q-th power map on the structure sheaf. It is an endomorphism of the group scheme A compatible with the endomorphism $f_q^{\operatorname{Spec}(k)} = f^{-1}$ of the base $\operatorname{Spec}(k)$. Consequently the composite

$$F = f_q^A g \tag{4.3.3}$$

is an endomorphism of the group scheme A compatible with the identity on $\mathrm{Spec}(k)$; in other words, it is a k-endomorphism of the group scheme A. I claim that the automorphism Γ_g of $A(k)$ defined by g through transport of structure, namely

$$\Gamma_g(u) = g \circ u \circ f^{-1} \qquad \text{for } u \in A(k), \tag{4.3.4}$$

may also be described by

$$\Gamma_g(u) = F(u) \qquad \text{for } u \in A(k). \tag{4.3.5}$$

To verify this, it is enough to observe that the underlying set maps of the morphisms $\mathrm{Spec}(k) \longrightarrow A$ represented by the two sides are the same. This reduces us to the equality obtained from (4.3.5) by composing its left-hand member on the left with f_q^A and its right-hand member on the right with f^{-1}, since f_q^A and f^{-1} induce the identity on the underlying sets of the source schemes. The resulting formula follows immediately from (4.3.3) and (4.3.4).

Applying (4.3.5) to $u \in {}_{\ell^\nu}A(k)$ and passing to the inverse limit over ν, we find that g_ℓ may also I≈362
be interpreted as the endomorphism $T_\ell(F)$ of $T_\ell(A(k))$ associated with the endomorphism F of A. It is known, on the other hand, that $T_\ell(F)$ has a characteristic polynomial with integral coefficients independent of ℓ. By devissage, this reduces to the case in which A is either an abelian scheme or a torus, where the required result was already used in part a) of the proof. The same argument also proves the assertions about the eigenvalues, in view, in the abelian-scheme case, of the classical results of Weil [15].

This proves 4.3.2 and completes the proof of 4.3. At the same time it proves the following result.

Corollary 4.4. Suppose that the hypotheses of 4.3 b) hold. Then, for every $g \in \pi'$ whose image in π_o is of the form f_q^n with $n \geq 0$, the eigenvalues of the induced automorphisms of

$$U^{\mathrm{et}}, \qquad U^{\mathrm{ef}}/U^{\mathrm{et}}, \qquad U/U^{\mathrm{ef}}$$

are algebraic integers whose absolute values are, respectively, q^n, $q^{n/2}$, and 1. For the two extreme pieces one may say more precisely that the corresponding eigenvalues are, respectively,

$$(q^n\varepsilon_i)_{1\leq i\leq \mu} \qquad \text{and} \qquad (\varepsilon_i)_{1\leq i\leq \mu},$$

where the ε_i $(1 \leq i \leq \mu)$ are roots of unity.

Indeed, formula (4.1.2) gives for the last piece the eigenvalues I≈363

$$\frac{q^n}{q^n\varepsilon_i} = \varepsilon_i^{-1}.$$

But since ε_i is a root of unity, one has $\varepsilon_i^{-1} = \overline{\varepsilon_i}$ (complex conjugate), and since the family of the ε_i is stable under complex conjugation (being the family of roots of an equation with integral coefficients), the family of the ε_i^{-1} is also the family of the ε_i. This proves the last assertion of 4.4.

4.5. We shall give an application of 4.3 a) to the theory of the local conductor of abelian schemes. Suppose that k is algebraically closed.[13] Recall (SGA 5 X 6.2, or [20]) that, for every commutative etale ℓ-group scheme F_K over K $(\ell \neq p)$, one defines an integer

$$\alpha_s(F_K) = \alpha_s(i_*F_K), \tag{4.5.1}$$

where $i : \eta \longrightarrow S$ denotes the inclusion, as follows. One chooses a finite Galois extension K' of K, with group Γ, that splits F_K, so that F_K is represented by a finite ℓ-group M on which Γ acts. On the other hand, one associates with the extension K'/K a certain projective $\mathbb{Z}_\ell[\Gamma]$-module, defined up to isomorphism,

$$\mathrm{Sw} = \mathrm{Sw}(K'/K, \mathbb{Z}_\ell), \tag{4.5.2}$$

[13] It would in fact suffice for k to be perfect, but, after passing to the strict henselization of S, one is immediately reduced to the case considered here.

and sets

$$\alpha_s(F_K) = \text{length}\,\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, M). \tag{4.5.3}$$

One checks (loc. cit.) that this expression is independent of the choice of the extension K'/K. Moreover, the projectivity of Sw as a $\mathbb{Z}_\ell[\Gamma]$-module implies that $\alpha_s(F_K)$ is additive in exact sequences.

We apply this definition to the sheaf ${}_\ell A_K$, where A_K is our abelian scheme over K. In this case one therefore obtains an integer I≈364

$$\alpha_s(A_K;\ell) = \text{length}\,\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell), \qquad U_\ell = T_\ell(A(\overline{K})), \tag{4.5.4}$$

where Γ is chosen as above (a choice that a priori depends on ℓ). The integer (4.5.4) is called the *wild conductor exponent* of the abelian scheme A_K relative to the discrete valuation ring $V \subset K$.[14] After passage to the henselization it is of course defined without assuming that V is henselian, and one is immediately reduced to the case in which S is strictly local. We intend to prove the following result.

Corollary 4.6. The wild conductor exponent (4.5.4) is independent of the choice of the prime $\ell \neq p$.

We may suppose that S is strictly local. We compute (4.5.4), replacing Sw by any other projective $\mathbb{Z}_\ell(\Gamma)$-module L. Observe that the filtration (4.1.1) of U_ℓ consists of $\mathbb{Z}_\ell$-submodules that are manifestly direct summands. It therefore induces a corresponding filtration of I≈365

$$U(\ell) = U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell,$$

whose successive factors are obtained, by tensoring with $\mathbb{F}_\ell$, from the factors V_i $(i = 1, 2, 3)$

$$U^{\text{et}}, \qquad U^{\text{ef}}/U^{\text{et}}, \qquad U/U^{\text{ef}}$$

appearing in (4.1.1). This filtration is, of course, stable under Γ. Additivity thus decomposes

$$\text{length}\,\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell))$$

as a sum of three terms, the lengths of the groups

$$\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) \simeq \text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i) \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell. \tag{*}$$

The isomorphism in (*) again follows from the assumed projectivity of L; it is only necessary to choose K' large enough for Γ to act on the V_i. Now $\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)$ is plainly a torsion-free $\mathbb{Z}_\ell$-module, so its rank is the length of the right-hand side of (*). If φ_i is the character of the representation of Γ on V_i, and ψ the character of the representation of I on $\check{L}$, we obtain

$$\text{length}\,\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell)) = \sum_{1\le i\le 3} N^{-1} \sum_{g\in\Gamma} \psi(g)\varphi_i(g) \tag{4.6.1}$$

$$= N^{-1} \sum_{g\in\Gamma} \psi(g)\varphi(g), \qquad N = \text{card}\,\Gamma,$$

where $\varphi(g) = \sum_i \varphi_i(g)$ is also the function on Γ obtained, by passage to the quotient, from the character of the monodromy representation of the inertia group I on U. By 4.3 a), this character is an integral-valued rational function independent of ℓ. Taking $L = \text{Sw}$, the same is true of the character $\psi(g)$ of L, which is

$$a(g) - N\delta(g) + 1,$$

[14] One should also define the *tame conductor exponent* of A_K relative to V as the difference between the lengths over $\mathbb{Z}/\ell\mathbb{Z}$ of the geometric generic and special fibres of ${}_\ell A^o$. It is independent of ℓ and equals $2\lambda + \mu$ in the notation of (1.1.11). The (total) conductor exponent of A_K is the sum of the tame and wild exponents; it is the local term occurring in the Euler–Poincare formula for ${}_\ell A^o$ in SGA 5 X 6.2 and [20].

where δ is the Dirac function on Γ and a is the Artin conductor of K'/K [26]. This proves 4.6 and at the same time gives the following expression, independent of ℓ, for the wild conductor exponent:

$$\alpha_s(A_K) = N^{-1} \sum_{g \in \Gamma} (a(g) - N\delta(g) + 1)\varphi(g). \tag{4.6.2}$$

Here Γ is the Galois group of a finite Galois extension K' of K that splits U^{ef} (for example the smallest such extension; cf. 4.1), $a(g)$ is the Artin conductor on Γ, and $\varphi(g)$ is the trace function on Γ described in (4.3.0). In the semistable-reduction case one may take $K' = K$, hence $\Gamma = \{e\}$, and obtains

$$\alpha_s(A_K) = 0 \qquad \text{(semistable-reduction case)}. \tag{4.6.3}$$

To conclude, let us state explicitly a criterion for semistable reduction that follows from the general considerations of 4.2. I≈366

Proposition 4.7. (M. Raynaud's criterion for semistable reduction.) Let S be a trait, let A_K be an abelian scheme over its function field K, and let $n > 1$ be an integer prime to the residual characteristic p. Suppose that ${}_nA_K$ is unramified relative to S, that is, that the inertia group acts trivially on ${}_nA(\overline{K})$. Then either A_K has semistable reduction over S, or $n = 2$; in the latter case, if S is strictly local, A_K acquires semistable reduction over a finite Galois extension K' of K whose Galois group is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$.[15]

We may plainly suppose that S is strictly local. With the notation of 4.1 and 4.2, take K' as small as possible; in other words, suppose that Γ acts faithfully on A^{*o}_o. By hypothesis Γ acts trivially on ${}_nA(\overline{K})$, and hence a fortiori on

$${}_nA(\overline{K})^{\mathrm{ef}} \simeq {}_nA^*_o(k) \supset {}_nA^{*o}_o(k).$$

The conclusion follows from the next two lemmas.

Lemma 4.7.1. (J.-P. Serre.) Let G be a commutative group scheme over a field k, which is an extension of an abelian scheme by a torus, and let $n \geq 2$ be an integer. Every finite-order automorphism of G that induces the identity on ${}_nG$ is the identity if $n \neq 2$, and is either the identity or has order two if $n = 2$. I367

Lemma 4.7.2. A finite group in which every element distinct from 1 has order 2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$ for a suitable $r \geq 0$.

4.7.3. Let us recall the well-known proof of 4.7.1. Since $n\,\mathrm{id}_G$ is an epimorphism, the hypothesis that u induces the identity on ${}_nG$, or equivalently that $1 - u$ induces zero on ${}_nG$, is expressed by the condition

$$1 - u = nv$$

in the endomorphism ring of G. To prove the asserted conclusion, we reduce, by decomposing the finite cyclic group generated by u into its p-primary components for the various primes p dividing the order of u, to the case in which u has order p^N.

Let E be the subring of $\mathrm{End}(G)$ generated by v. Since $u = 1 - nv$ and $u^{p^N} = 1$, it is a quotient of

$$\mathbb{Z}[T]/((1 - nT)^{p^N} - 1);$$

it is therefore of finite type over $\mathbb{Z}$. It is moreover torsion-free (a formal consequence of the divisibility of G), and thus embeds in

$$E_{\mathbb{Q}} = E \otimes_{\mathbb{Z}} \mathbb{Q}.$$

The latter is a quotient, after the change of variable $u = 1 - nT$, of $\mathbb{Q}[u]/(u^{p^N} - 1)$, which is a direct product of finite field extensions of $\mathbb{Q}$; the same is therefore true of $E_{\mathbb{Q}}$. Considering the images of E,

[15] As follows from an example communicated to me by J. Tate, 4.7 is no longer valid when n equals the residual characteristic p, even if K has characteristic zero and A_K is an elliptic curve with complex multiplication.

u, and v in the component fields K_i of $E_{\mathbb{Q}}$, it remains to prove the following assertion. Let $K = \mathbb{Q}(u)$ be generated by a primitive p^m-th root of unity u, and let $n \geq 2$ be an integer such that

$$u - 1 \equiv 0 \pmod{n}$$

in the ring of integers of K. Then $u = 1$, or $u = -1$ and $n = 2$.

This follows at once from the familiar fact that, if $u \neq 1$, that is, $m \geq 1$, the ideal $(u-1)$ is prime and (p) is a power of it, with exponent greater than 1 except when $p = 2$ and $m = 1$, that is, when $u = -1$. Since $(u-1)$ is prime and divides p, an integer $n \geq 2$ can divide $u - 1$ only if $n = p$; and p can divide $u - 1$ only if $(u-1) = (p)$, that is, if $u = -1$. Hence $n = 2$. I≈368

4.7.4. Let us prove 4.7.2. It is enough to prove that Γ is commutative. If $x, y \in \Gamma$, then $(xy)(xy) = 1$; since $x^2 = 1$ and $y^2 = 1$, this may also be written

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx,$$

as required.

Remark 4.7.5. The proof given in 4.7.3, together with 4.7.2, shows that if G is a divisible commutative group in a topos, $n \geq 2$ is an integer, and Γ is a finite group acting faithfully on G while inducing the identity on ${}_nG$, then Γ is the trivial group if $n \geq 3$, and is isomorphic to a group $(\mathbb{Z}/2\mathbb{Z})^r$ if $n = 2$.

5. The orthogonality theorem in the case $\ell = p$ (semistable case). Duality of the abelian schemes B_o and B'_o. Characterization of the fixed part. Criteria for good reduction, essentially good reduction, and semistable reduction. I≈369

5.1. Let us place ourselves in the situation of 2.2.3, where S is henselian and A^o is ℓ-divisible, so that the fixed part

$$T_\ell(A^o)^f \hookrightarrow T_\ell(A)$$

is defined as the pro-ℓ Barsotti–Tate group formed by the $({}_{\ell^\nu}A^o)^f$. We shall now give a definition of the toric part (2.3.2) of $T_\ell(A^o)$ which is independent of the hypothesis $\ell \neq p$ made in 2.3, but which instead assumes that S is complete.[16] For this purpose, for every integer $n \geq 0$ introduce the local Artinian scheme

$$S_n = \mathrm{Spec}(V/\mathfrak{m}^{n+1}), \tag{5.1.1}$$

the n-th infinitesimal neighbourhood of s in S, and, for every S-scheme X, put $X_n = X \times_S S_n$. Since $(A^o)_n = (A_n)^o$ is smooth over S_n, it is known (SGA 3 IX 3.6 bis) that T_o is the special fibre of a unique subtorus

$$T_n \subset A^o_n. \tag{5.1.2}$$

By the uniqueness of the T_n, they are obtained from one another by the base changes $S_{n'} \longrightarrow S_n$. In other words, they define a formal subtorus $\widehat{T}$ of the group $\widehat{A}^o$ (in the category of formal schemes over S), the formal completion of A^o along the special fibre:

$$\widehat{T} \subset \widehat{A}^o. \tag{5.1.3}$$

Since the functor $X \longmapsto \widehat{X}$ (formal completion along the special fibre) induces an equivalence between the category of finite schemes over S and that of finite formal schemes over S (EGA III 4.8), we shall simply identify a finite scheme over S with the corresponding formal scheme. With this identification, the definition of the *fixed parts* immediately gives, for every $\nu \geq 0$,

$$({}_{\ell^\nu}A^o)^f = {}_{\ell^\nu}(\widehat{A}^o), \tag{5.1.4}$$

which may also be written as an identity of projective systems:

$$T_\ell(A^o)^f = T_\ell(\widehat{A}^o). \tag{5.1.5}$$

[16] This hypothesis is in fact not essential, as is shown in 6.1 below.

We therefore obtain a Barsotti–Tate pro-subgroup of the fixed part of $T_\ell(A^o)$ by taking $T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o)$. We accordingly define the toric part of $T_\ell(A^o)$ by

$$T_\ell(A^o)^t \overset{\text{def}}{=} T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o) = T_\ell(A^o)^f. \tag{5.1.6}$$

Thus, by definition,

$$T_\ell(A^o)^t = ({}_{\ell^\nu}\widehat{T})_{\nu\geq 0}, \tag{5.1.7}$$

the finite flat S-group ${}_{\ell^\nu}\widehat{T}$ itself being described as the inductive limit of the finite flat S_n-groups

$${}_{\ell^\nu}(T_n), \qquad n \text{ variable;} \tag{5.1.8}$$

that is, its affine algebra is described as the projective limit of the affine algebras of the groups (5.1.8). Of course,

$$\big(T_\ell(A^o)^t\big)_n = T_\ell(T_n). \tag{5.1.9}$$

For $n = 0$ this is precisely formula (2.3.3), by which we characterized the toric part (as a profinite etale group) in the case $\ell \neq p$. This shows that definition (5.1.6) is compatible with that given in 2.3 when $\ell \neq p$.

Passing to generic fibres, the fixed and toric parts of $T_\ell(A^o)$ define fixed and toric parts in $T_\ell(A_K)$: I≈371

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \tag{5.1.10}$$

These are inclusions of pro-ℓ Barsotti–Tate groups. Consider the analogous filtration corresponding to the dual abelian scheme A'_K:

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K). \tag{5.1.11}$$

We then have the following result, which generalizes 2.4 (at least in the case where S is complete, to which one could easily have reduced in 2.4).

Theorem 5.2. Suppose that S is a complete trait, and let A_K be an abelian scheme over K, A'_K the dual abelian scheme, and ℓ a prime such that the connected Neron model A^o is ℓ-divisible (cf. 2.2.1), so that one then has the filtrations (5.1.10) and (5.1.11) of the pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$. Then the toric parts are recovered from the fixed parts by the formula

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap \big(T_\ell(A'_K)^f\big)^\perp, \tag{5.2.1}$$

and by the analogous formula obtained by interchanging the roles of A_K and A'_K, where $\perp$ denotes the orthogonal complement with respect to the canonical pairing (1.0.2).

Finally, when A_K has semistable reduction over S (which follows from the other hypotheses in the case $\ell = p$), one even has the formula I≈372

$$T_\ell(A_K)^t = \big(T_\ell(A'_K)^f\big)^\perp. \tag{5.2.2}$$

5.2.3. The meaning of formula (5.2.1) is quite clear when ℓ is prime to the characteristic of K: as in 2.4, it can then be interpreted by passing to the ordinary $\mathbb{Z}_\ell$-modules, with an action of $\pi = \mathrm{Gal}(\overline{K}/K)$, formed by the $\overline{K}$-points of the pro-groups involved. Let us spell it out in the general case, using the notion of the dual

$$D(G) = \mathrm{Hom}(G, T_\ell(\mathbb{G}_m)) \tag{5.2.3.1}$$

of a pro-ℓ Barsotti–Tate group $G = (G(\nu))$. By definition, this is the projective system of Cartier duals $D(G(\nu))$ (SGA 3 VII 3.3.1 and VIII 1), whose transition morphisms are defined in the evident

way (for example, by transposition from the inclusion morphisms among the $G(\nu)$). The duality theory of abelian varieties tells us that the pairing mentioned in 5.2 actually defines an isomorphism

$$T_\ell(A'_K) \simeq D(T_\ell(A_K)), \tag{5.2.3.2}$$

and hence, by composition, a morphism

$$T_\ell(A_K)^f \longrightarrow D(T_\ell(A'_K)^f). \tag{5.2.3.3}$$

Formula (5.2.1) says that the kernel of the preceding homomorphism (which a priori is a pro-subgroup of its first term) is precisely the toric part $T_\ell(A_K)^t$. Notice that this is an isomorphism of pro-objects, and not of projective systems: we do not assert (and it would in general be false) that, for every integer ν, $T_\ell(A_K)^t(\nu)$ equals the kernel of

$$T_\ell(A_K)^f(\nu) \longrightarrow D(T_\ell(A'_K)^f)(\nu);$$

the latter can a priori be larger. We shall see below, however, that it is not larger if A_K has semistable reduction; by the hypotheses made in 5.2, this is in particular the case when $\ell = p$.

5.2.4. The proof of (5.2.1) merely repeats, with practically no change, the proof of 2.4. Observe that **I 373** the induced pairing

$$T_\ell(A_K)^f \times T_\ell(A'_K)^f \longrightarrow T_\ell(\mathbb{G}_{mK}) \tag{5.2.4.1}$$

which gives rise to (5.2.3.3) is induced by a pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.2}$$

of pro-Barsotti–Tate groups over S, itself induced by the pairing of pro-groups

$$T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.3}$$

deduced from the biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$. Consequently, the pairing

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{5.2.4.4}$$

obtained from (5.2.4.2) by passing to the special fibres, is precisely the pairing deduced from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$. By VIII 4.11 and 1.5, the left annihilator of the pairing (5.2.4.4) is therefore $T_\ell(T_o)$, which is precisely the special fibre of the toric part $T_\ell(A^o)^t$. Returning to the definition of left annihilators given in 5.2.3, we conclude by means of the following lemma.

Lemma 5.2.5. Let $u : G \longrightarrow H$ be a homomorphism of pro-ℓ Barsotti–Tate groups over S. If **I 374** $u_o : G_o \longrightarrow H_o$ is zero, then u is zero. If $u_o : G_o \longrightarrow H_o$ is an isogeny (respectively an isomorphism), then u is an isogeny (respectively an isomorphism).

Indeed, apply the first assertion of 5.2.5 to the composite morphism

$$T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow D(T_\ell(A'^o)^f),$$

where the second morphism is deduced from the pairing (5.2.4.2). The composite is zero, so the pairing under consideration vanishes on the toric part on the left. By symmetry it also vanishes on the toric part on the right, and hence descends to a pairing on the quotients, or *abelian pieces*,

$$T_\ell(A^o)^{ab} \times T_\ell(A'^o)^{ab} \longrightarrow T_\ell(\mathbb{G}_m).$$

These quotients are again pro-Barsotti–Tate groups, as one checks readily. It remains to prove that the morphism corresponding to this pairing,

$$T_\ell(A^o)^{ab} \longrightarrow D(T_\ell(A'^o)^{ab}),$$

is an isogeny, which again follows from 5.2.5.

We leave to the reader the easy proof of 5.2.5, a result valid over any connected Noetherian base. Moreover, below in 7.4.7, by a somewhat more precise analysis of the biextensions W_n of $(A_n^o, A_n'^o)$, we shall find a quite evident direct proof of the two special cases in which we had to use 5.2.5.

5.2.6. Suppose now that A_K has semistable reduction over S, so that ℓ may be any prime whatsoever (equal or unequal to p). Moreover, by the Galois criterion for semistable reduction 3.5, for $\ell \neq p$ one has I 375

$$(T_\ell(A'_K)^f)^\perp \ (= T_\ell(A_K)^t) \subset T_\ell(A_K)^f. \tag{5.2.7}$$

To finish the proof of 5.2, it remains to show that this inclusion is valid without restriction on ℓ.

Indeed, the left-hand term is canonically isomorphic to

$$D\big(T_\ell(A'_K)/T_\ell(A'_K)^f\big)$$

and therefore has *rank* (or *height*) $2n - 2\alpha - \mu = \mu$, in the notation of (2.1.9), since by hypothesis $\lambda = 0$. This is also the rank of $T_\ell(A_K)^t$. Since the inclusion

$$T_\ell(A_K)^t \subset T_\ell(A_K) \tag{*}$$

is strict, in the sense that it already induces a monomorphism on every term of the projective systems under consideration, it follows readily that the left-hand term of (*) is maximal among the pro-Barsotti–Tate subgroups of $T_\ell(A_K)$ having the same rank. Thus equality holds in (5.2.7); by the orthogonality theorem this is equivalent to the inclusion between the two extreme terms of (5.2.7).

5.3. Using the orthogonality correspondence between strict pro-Barsotti–Tate subgroups (in the sense specified above) of the dual pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$, one formally deduces from (5.2.2) that the pairing (1.0.2) between these pro-groups induces a pairing

$$T_\ell(A_K)^{ab} \times T_\ell(A'_K)^{ab} \longrightarrow T_\ell(\mathbb{G}_{mK}) \tag{5.3.2}$$

which is a perfect duality, that is, it identifies each of the two factors on the left with the dual of the other (in the sense of 5.2.3).[17] It is also clear that (5.3.2) is induced by the pairing

$$T_\ell(A^o)^{ab} \times T_\ell(A'^o)^{ab} \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.3.3}$$

obtained from the pairing (5.2.4.2) by passing to the quotients. Since this is a perfect duality on the generic fibres, that is, since the corresponding homomorphism

$$T_\ell(A^o)^{ab} \longrightarrow D\big(T_\ell(A'^o)^{ab}\big) \tag{5.3.4}$$

is an isomorphism on the generic fibres, one naturally wishes to deduce that it is so over the whole of S, or equivalently on the special fibres. This deduction is trivially valid if $\ell \neq p$. It is also valid, at least in unequal characteristics when $\ell = p$, by a deep theorem of Tate [33, Th. 4]. We shall nevertheless establish it directly below in 7.4.6, without using Tate's theorem.

To make precise the geometric meaning of the result we wish to prove, observe that the pairing obtained from (5.3.3) by passage to the special fibres is precisely the pairing I ≈376

$$T_\ell(B_o) \times T_\ell(B'_o) \longrightarrow T_\ell(\mathbb{G}_{mk}) \tag{5.3.5}$$

deduced from the biextension W_o of $(A_o^o, A_o'^o)$ by passage to the quotient, where B_o and B'_o denote respectively the abelian parts of A_o^o and $A_o'^o$. By VIII 4.8, the biextension W_o comes from a biextension W_o^{ab} of (B_o, B'_o) by $\mathbb{G}_{mk}$, unique up to a unique isomorphism. Thus the pairing (5.3.5) may also be interpreted as the one deduced from the divisorial correspondence W_o^{ab}, and the corresponding homomorphism

$$T_\ell(B_o) \longrightarrow D\big(T_\ell(B'_o)\big) \simeq T_\ell\big(D(B'_o)\big) \tag{5.3.6}$$

[17] The superscripts *ab* denote the *abelian parts*, the quotient of the fixed part by the toric part.

(where, in the last term, D denotes the dual abelian scheme) is precisely the morphism obtained by applying T_ℓ functorially to the homomorphism

$$B_o \longrightarrow D(B'_o) \tag{5.3.7}$$

defined by W_o^{ab}. Consequently, by the standard dictionary (cf. [5]), saying that (5.3.5) is a perfect duality is also saying that the kernel of (5.3.7), which is a finite group over k by 1.5, has zero ℓ-primary component. Saying that this holds for every ℓ therefore means that the isogeny (5.3.7) is an isomorphism, that is, that the divisorial correspondence W_o^{ab} is a perfect duality between B_o and B'_o. The preceding considerations show only that the kernel of (5.3.7) is a p-group. We therefore state the more complete result.

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Theorem 5.4. Let S be a trait, and let A_K be an abelian scheme over K having semistable reduction over S. Let B_o and B'_o denote respectively the abelian parts of the Neron models of A_K and of the dual abelian scheme A'_K. Then the canonical divisorial correspondence W_o^{ab} on (B_o, B'_o), deduced by VIII 4.8 from the canonical biextension W_o of $(A_o^o, A_o'^o)$ (the special fibre of the biextension W of 1.4), is a perfect duality between B_o and B'_o; that is, it defines an isomorphism of each of these abelian schemes with the dual of the other.

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The proof of 5.4 in the general case will be given in section 7, as a consequence of a very natural and important construction due to Raynaud. Let us merely observe here that, in proving 5.4, one may assume that S is complete, by 3.3.

5.5. To express conveniently the extreme factors in the filtrations (5.1.10) and (5.1.11), in the case of semistable reduction, let us introduce the twisted constant group schemes over S

$$\underline{M'}_o = D(T_o), \qquad \underline{M}_o = D(T'_o), \tag{5.5.1}$$

dual to the maximal tori T_o, T'_o of the special fibres of A, A' (SGA 3, X 5.7). Since S is complete local, these group schemes may be regarded as the special fibres of twisted constant schemes

$$\underline{M'}, \qquad \underline{M} \quad \text{over } S. \tag{5.5.2}$$

Thus, if the torus T_o is split (for example, if k is separably closed), that is, if $\underline{M'}_o$ is the constant group defined by the ordinary group

$$M' = \mathrm{Hom}_{k\text{-gr}}(T_o, \mathbb{G}_{mk}), \tag{5.5.3}$$

then $\underline{M'}$ is the constant group defined by M':

$$\underline{M'} = M'_S.$$

One may moreover observe that, since T_o and T'_o are isogenous (2.2.6), T'_o is split whenever T_o is. In this case one should therefore also introduce

$$M = \mathrm{Hom}_{k\text{-gr}}(T'_o, \mathbb{G}_{mk}), \tag{5.5.4}$$

and then

$$\underline{M} = M_S.$$

With this notation, for every integer n there are evidently isomorphisms

$$\underline{M'}_n \simeq D(T_n), \qquad \underline{M}_n \simeq D(T'_n), \tag{5.5.5}$$

where T'_n is defined in terms of A' as T_n is in terms of A (5.1.2). Hence there follow canonical isomorphisms (where $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$)

$$T_\ell(T_n) \simeq \check{\underline{M}}'_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \qquad T_\ell(T'_n) \simeq \check{\underline{M}}_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \tag{5.5.6}$$

and, on passing to the limit over n, canonical isomorphisms

$$\text{(5.5.7)} \qquad \begin{cases} T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq \check{\underline{M}}' \otimes \mathbb{Z}_\ell(1), \\ T_\ell(A'^o)^t = T_\ell(\widehat{T}') \simeq \check{\underline{M}} \otimes \mathbb{Z}_\ell(1). \end{cases}$$

In the case of semistable reduction, duality, together with the orthogonality theorem 5.2 and relation (5.2.2), then gives canonical isomorphisms

$$\text{(5.5.8)} \qquad \begin{cases} T_\ell(A_K)/T_\ell(A_K)^f \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell, \\ T_\ell(A'_K)/T_\ell(A'_K)^f \simeq \underline{M}'_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell. \end{cases}$$

Likewise, the same formula (5.2.2) gives canonical isomorphisms

$$\text{(5.5.9)} \qquad \begin{cases} T_\ell(A_K)/T_\ell(A_K)^t \simeq D\big(T_\ell(A'_K)^f\big), \\ T_\ell(A'_K)/T_\ell(A'_K)^t \simeq D\big(T_\ell(A_K)^f\big), \end{cases}$$

where one should recall that the groups $T_\ell(A_K)^f$ and $T_\ell(A'_K)^f$ occurring on the right are the restrictions to η of the pro-ℓ Barsotti–Tate groups $T_\ell(A^o)^f$ and $T_\ell(A'^o)^f$, respectively, defined over S. Here D denotes duality in the sense of pro-Barsotti–Tate groups, namely $\underline{\mathrm{Hom}}(-, \mathbb{Z}_\ell(1))$. Thus we have:

I 380 Proposition 5.6. Suppose that S is complete and that A_K has semistable reduction over S. Then, for every prime number ℓ (which may equal the residue characteristic of S), the pro-ℓ Barsotti–Tate group $T_\ell(A_K)^t$ is the generic fibre of a pro-Barsotti–Tate group over S "of toric type", given by (5.5.7), whereas $T_\ell(A_K)/T_\ell(A_K)^f$ is the generic fibre of a pro-Barsotti–Tate group over S of pro-étale type, given by (5.5.8). Finally, the groups $T_\ell(A_K)^f$ and $T_\ell(A_K)/T_\ell(A_K)^t$ are the generic fibres of pro-Barsotti–Tate groups defined over S, equal respectively to $T_\ell(A^o)^f$ and to $D(T_\ell(A'^o)^f) = \underline{\mathrm{Hom}}(T_\ell(A'^o)^f, \mathbb{Z}_\ell(1))$.

5.6.1. Observe that, for an étale covering S' of S, splitting $T_\ell(A_K)^t$ is equivalent to splitting $T_\ell(A_K)/T_\ell(A_K)^f$: these conditions say respectively that S' splits the twisted constant groups $\underline{M}'$ and $\underline{M}$, and these are isogenous because T_o and T'_o are. At the same time one sees that there is a smallest étale extension S' of S which does the job.

We also record the following easy consequence of the definitions in 2.2.3.

I 381 Proposition 5.7. Suppose that A^o is ℓ-divisible and that S is henselian. Let F_K be a constructible ℓ-adic sheaf over K which is "unramified", that is, which extends to a twisted constant ℓ-adic sheaf F over S, and let $u : F_K \longrightarrow T_\ell(A_K)$ be a homomorphism of pro-groups. Then u factors through a homomorphism from F to $T_\ell(A_K)^f$.

Indeed, write

$$\text{(5.7.1)} \qquad u = (u(\nu))_{\nu \geqslant 0}, \qquad u(\nu) : F_K(\nu) \longrightarrow T_\ell(A_K)(\nu) = {}_{\ell^\nu}A_K.$$

By the universal property of A, since $F(\nu)$ is étale and hence smooth over S, the morphism $u(\nu)$ comes from a homomorphism

$$v(\nu) : F(\nu) \longrightarrow {}_{\ell^\nu}A.$$

By definition of the fixed part, this homomorphism factors as

$$v(\nu) : F(\nu) \longrightarrow ({}_{\ell^\nu}A)^f.$$

Since the image of the induced morphism on special fibres lies in the infinitely ℓ-divisible part of A_o, it follows that $v(\nu)$ in fact factors through $({}_{\ell^\nu}A^o)^f$. Passing to the limit over ν gives a homomorphism

$$v : F \longrightarrow T_\ell(A^o)^f,$$

and the induced morphism on generic fibres plainly gives the required factorization.

Remarks 5.7.2. When $\ell \neq p$, property 5.7 uniquely characterizes the sub-pro-group $T_\ell(A_K)^f$ of $T_\ell(A_K)$ as its largest unramified ℓ-adic subsheaf. When $\ell = p$, the statement of 5.7 ceases to be very satisfactory. We shall give a considerably stronger property, which can serve as an intrinsic characterization of the fixed part of $T_p(A_K)$ (in terms of the pro-Barsotti–Tate group structure of $T_p(A_K)$ alone), at least in unequal characteristics.

I 382 Theorem 5.8 (M. Raynaud). Let A_K be an abelian scheme with semistable reduction over the function field K of a henselian trait S. Let ℓ be a prime number (the interesting case being $\ell = p$), and consider the pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ and its "fixed part" X_η^f, which is the generic fibre of the pro-ℓ Barsotti–Tate group $X^f = T_\ell(A^o)^f$. If $\ell = p$, suppose that $\operatorname{char} K = 0$ (so that Tate's theorem [33, Th. 4] is available over S). Let F be a pro-ℓ Barsotti–Tate group over S. Then every homomorphism $F_\eta \longrightarrow X_\eta$ factors through X_η^f.

Passing to the ind-ℓ groups associated with the pro-ℓ Barsotti–Tate groups under consideration, the preceding statement becomes equivalent, by the cited theorem of Tate, to the following.

Corollary 5.9. If F is an ind-ℓ Barsotti–Tate group, then every homomorphism $u_\eta : F_\eta \longrightarrow A_\eta$ extends uniquely to a homomorphism $u : F \longrightarrow A$.

In other words, the characteristic Neron property (1.1.2) of the Neron model remains valid for homomorphisms from ind-group schemes of Barsotti–Tate type, with no smoothness hypothesis on them. (Only the case of ind-étale groups would follow directly from the definition.)

I 383 Remark 5.9.1. In fact, M. Raynaud proves 5.9 without assuming that A_K has semistable reduction, and even for every Neron group scheme A over S (whose generic fibre need not be an abelian scheme). Of course, the assumption that S is henselian is also unnecessary in the formulation of 5.9. Raynaud's proof of this more general case reduces it to the special case considered in 5.8, using an unpublished property in the construction of Neron models. Here we shall give only Raynaud's proof of 5.9.

Proof of 5.9. It is known [33] that F is an extension of an ind-étale Barsotti–Tate group F' by a Barsotti–Tate group F'' with connected special fibre, so that for every integer n there is an exact sequence

$$0 \longrightarrow F''(n) \longrightarrow F(n) \longrightarrow F'(n) \longrightarrow 0,$$

with $F'(n)$ étale and $F''(n)_o$ purely infinitesimal. The given homomorphism u_η is defined by a sequence of homomorphisms

$$u_\eta(n) : F(n)_\eta \longrightarrow A_\eta,$$

and it is enough to extend these separately to homomorphisms $u(n) : F(n) \longrightarrow A$. Lemma 5.9.2 below then reduces us to proving that the homomorphism $u''_\eta : F''_\eta \longrightarrow A_\eta$ induced by u_η extends to $u'' : F'' \longrightarrow A$. Consequently, we may suppose that F has connected special fibre.

We may regard u_η as a homomorphism from F_η to the ind-ℓ Barsotti–Tate group Y_η associated with $X_\eta = T_\ell(A_K)$, and then consider the composite

$$(*) \qquad v_\eta : F_\eta \longrightarrow Y_\eta \longrightarrow Y_\eta / Y_\eta^f.$$

By 5.6, the latter group extends to an ind-étale Barsotti–Tate group Z over S, namely

$$\underline{M} \otimes_{\mathbb{Z}} (\mathbb{Q}_\ell / \mathbb{Z}_\ell).$$

By Tate's theorem cited in 5.8, the homomorphism (*) then extends to a homomorphism

$$v : F \longrightarrow Z.$$

Since F has connected special fibre and Z is ind-étale, this morphism is zero. Hence v_η is zero as well, and u_η therefore factors through Y_η^f, as required.

It remains to prove the following lemma. I ≈384

Lemma 5.9.2. Let S be a trait, let A be a Neron group scheme over S (1.1.2), let

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

be an extension of group schemes over S, with G'' smooth, let $u : G' \longrightarrow A$ be a group homomorphism, and let $v_\eta : G_\eta \longrightarrow A_\eta$ be a group homomorphism extending u_η. Suppose that the extension of G'' by A obtained from the extension G by pushout along u is representable [18]—as is the case, for example, if $G \longrightarrow G''$ admits a section locally for the "finite locally free" topology (cf. SGA 3, IV 6.3) and if A is quasi-projective over S. Then there exists a unique group homomorphism $v : G \longrightarrow A$ extending both v_η and u.

Indeed, the last hypothesis reduces us to the case in which $G' = A$, u is the identity, and $v_\eta : G_\eta \longrightarrow A_\eta$ is a projection which we seek to extend to a projection $v : G \longrightarrow A$. But G is smooth, being an extension of two smooth group schemes, and the conclusion follows from the definition of the Neron property of A.

Corollary 5.10. Let A_K be an abelian scheme over the function field K of a trait S, and let ℓ be a prime number. When $\ell = p$, suppose that $\operatorname{char} K = 0$, so that Tate's theorem already invoked in 5.8 is available. Then A_K has good reduction over S if and only if the pro-ℓ Barsotti–Tate group $T_\ell(A_K)$ "has good reduction over S", that is, extends to a Barsotti–Tate group over S. I 385

(This is the result conjectured in [8, 4.9].)

Necessity being clear, it remains to prove sufficiency; observe that the case $\ell \neq p$ has already been treated in 2.2.9. We may suppose that S is henselian (2.2.9). First suppose that A_K has semistable reduction over S. Applying 5.8 with $F = X$ gives

$$T_\ell(A_K) = T_\ell(A_K)^f.$$

This also says that the maximal torus of A_o is zero, that is, that A_o^o is an abelian scheme; the conclusion then follows from 2.2.9 (iii^o).

We now treat the general case. By 2.2.9 we may plainly assume that S is strictly local. Let K'/K be a finite Galois extension such that $A_{K'}$ has semistable reduction over the normalization S' of S in K' (3.6). If A^* denotes the corresponding Neron model, $G = \operatorname{Gal}(K'/K)$ acts on the situation (A^*, S'), and hence on A_o^{*o}. By 4.1 it is enough to prove that G acts trivially on the latter; this indeed implies that A_K has semistable reduction, reducing us to the case already treated.

Let X be a Barsotti–Tate group over S extending $X_\eta = T_\ell(A_K)$. By Tate's uniqueness theorem already invoked, there is a unique isomorphism I ≈ 386

$$X \times_S S' \simeq T_\ell(A^{*o})$$

which induces the identity on generic fibres. By transport of structure, this isomorphism is compatible with the actions of G. Now G plainly acts trivially on the special fibre of $X \times_S S'$, and hence on $T_\ell(A_o^{*o})$. It follows at once that G acts trivially on A_o^{*o}, completing the proof.

Remark 5.11. Once Tate's theorem has been proved without any restrictive hypothesis on $\operatorname{char} K$, the preceding results 5.8, 5.9, and 5.10, as well as 5.13 below, will plainly hold without such a restriction.

We record a definition which will also be useful elsewhere.

Definition 5.12. Let S be a henselian trait, let ℓ be a prime number, and let X_η be a pro-ℓ Barsotti–Tate group over the generic point η of S. We say that X_η has *good reduction over S* if it extends to a Barsotti–Tate group over S (cf. 5.10). We say that it has *virtual good reduction over S* if there exists a filtration

$$\left(\operatorname{Fil}^i(X_\eta)\right)_{i \geq 0}$$

by strict Barsotti–Tate subgroups[19] such that the Barsotti–Tate groups

$$\operatorname{Gr}^i(X_\eta) = \operatorname{Fil}^i(X_\eta)/\operatorname{Fil}^{i+1}(X_\eta)$$

have good reduction over S. If the filtration can be chosen to have length at most n (that is, if at most n of the Gr^i are nonzero), we say that X_η has *virtual good reduction of level n over S*.

[18] Unpublished results of M. Raynaud imply, moreover, that this condition is always satisfied.

[19] That is, subobjects as projective systems, or equivalently subobjects term by term.

Finally, we say that X_η has *essential good reduction over* S (respectively, *essential virtual good* I≈387
reduction, respectively, *essential virtual good reduction of level* n) if there exists a finite extension K'/K such that $X_{K'}$ has good reduction (respectively, virtual good reduction, respectively, virtual good reduction of level n) over the normalization S' of S in K'.

5.12.1. When $\ell \neq p$, the categories of pro-ℓ Barsotti–Tate groups over η and over S may be interpreted in terms of ℓ-adic representations of

$$\pi = \pi_1(\eta), \qquad \pi_o = \pi_1(S),$$

respectively. The six notions just introduced can then be expressed in terms of the natural representation of the inertia group I on $X_\eta(\overline{K})$: the representation is, respectively, trivial, unipotent, unipotent of level n, essentially trivial, essentially unipotent, or essentially unipotent of level n.

5.12.2. Take $X_\eta = T_\ell(A_K)$, where A_K is an abelian scheme over K. We saw in 5.6 that, if A_K has semistable reduction over S, then X_η has virtual good reduction of level 2 (and we shall see the converse in a moment). Consequently, by the semistable reduction theorem 3.6, X_η has essential virtual good reduction of level 2. This statement should be regarded as the "right" formulation of the geometric monodromy theorem in Exposé III for H^1 of a proper smooth scheme over K, without excluding the case $\ell = p$ (which was not considered there). The geometric monodromy theorem—or, for the present, conjecture—for the higher H^i given in Exposé III should likewise be reformulated so as to include "p-adic" coefficients, as was observed in 0.4.

We now summarize the relations between the reduction properties of abelian schemes over K and I≈388
those of the associated Barsotti–Tate groups.

Proposition 5.13. Let S be a henselian trait, let A_K be an abelian scheme over its fraction field, and let ℓ be a prime number. When $\ell = p$, assume that K has characteristic zero (cf. 5.11).

a) The abelian scheme A_K has good reduction over S (2.2.9) if and only if $T_\ell(A_K)$ has good reduction over S (5.12).

b) The abelian scheme A_K has essential good reduction over S—that is, there exists a finite extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K'—if and only if $T_\ell(A_K)$ has essential good reduction over S (5.12). In that case K' may be chosen to be a finite separable extension of K.

c) The following conditions are equivalent:

1) A_K has semistable reduction over S (3.4);

2) $T_\ell(A_K)$ has virtual good reduction of level 2 over S (5.12);

3) $T_\ell(A_K)$ has virtual good reduction over S.

Proof. Part a) is just 5.10 and has been included for reference.

For b), the first assertion follows immediately from a) and the definitions. To prove the final assertion, we may suppose that $\ell \neq p$. In view of a), it then suffices to take for K' the finite separable extension of K corresponding to an open subgroup π' of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

whose intersection with the inertia group I is contained in the kernel of the representation of I on $T_\ell(A_K(\overline{K}))$; this kernel is open by hypothesis (cf. 5.12.1).

For c), we have already noted that 1) implies 2) (5.12.2), and it is immediate that 2) implies 3). I≈389
It remains to prove that 3) implies 1). For $\ell \neq p$, this is precisely the Galois criterion for semistable reduction 3.5. For $\ell = p$, argue as follows. Choose a prime $\ell' \neq p$ and apply the considerations of 4.1 to it. We may suppose that S is strictly local (3.4.0), and choose a finite Galois extension K'/K, with group G, such that $A_{K'}$ has semistable reduction. We must prove that the representation of G on each of the three quotients occurring in (4.1.1) is trivial. Equivalently, their direct-sum representation

is trivial, or again its trace is the "trivial" trace, that is, is constant. We saw in 4.3(a) that this trace function is independent of the choice of $\ell' \neq p$, and in 6.4 below we indicate a method of calculating it in terms of $T_p(A_K)$ as well. That calculation plainly gives a constant function when $T_p(A_K)$ has virtual good reduction, and this completes the proof.

Remark 5.13.1. It seems plausible that, if X_η is a pro-ℓ Barsotti–Tate group over η having virtual good reduction over S and essential virtual good reduction of level n, then it has virtual good reduction of level n. (This is true when $\ell \neq p$.)

The following result, an easy consequence of 5.7 and of the semistable reduction theorem 3.6, was pointed out to me by J.-P. Serre.

Proposition 5.14 (Criterion for essential good reduction). Let S be a trait with function field K, let A_K be an abelian scheme over K, and let ℓ be a prime number distinct from $\operatorname{char} K$. I 390

a) Suppose that $\ell \neq p = \operatorname{char} k$. Then the following conditions are equivalent:

(i) A_K has essential good reduction over S; that is, there exists a finite separable extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K';

(ii) the action of the inertia group I on $U_\ell = T_\ell(A(\overline{K}))$ is essentially trivial; that is, it is trivial on an open subgroup of I, or equivalently factors through a finite quotient Γ of I;

(iii) the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is semisimple.

b) Suppose that $\ell = p$. If $n = \dim A_K \neq 0$, then the image of I in the automorphism group of U_ℓ is not finite; more precisely, its image in the group $\mathbb{Z}_\ell^\times$ of automorphisms of $\bigwedge^{2n} U_\ell$ has finite index. A sufficient (but by no means necessary) condition for A_K to have essential good reduction is that the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ be semisimple.

Proof. When $\ell \neq p$, the equivalence of (i) and (ii) follows immediately from the good-reduction criterion 2.2.9, and (ii) implies (iii) because $\mathbb{Q}_\ell$ has characteristic zero. The homomorphism

$$\pi = \operatorname{Gal}(\overline{K}/K) \longrightarrow \mathbb{Z}_\ell^\times \simeq \operatorname{Aut}\left(\bigwedge^{2n} U_\ell\right)$$

is χ^n, where χ is induced by the character

$$\mathbb{Z}_\ell^\times = \operatorname{Aut}(\mathbb{Z}_\ell(1))$$

arising from the action on the ℓ^ν-th roots of unity. It is known that, when $\ell = p$, the image of I under this character has finite index [33]. The fact that the action of I on $U_p \otimes \mathbb{Q}_p$ need not be semisimple even when A_K is an elliptic curve with good reduction is exhibited in [28, IV A 2.4].

It remains to prove that (iii) implies (i), for every ℓ. We use the semistable reduction theorem 3.6, which immediately reduces us to the case where A_K has semistable reduction over S: hypothesis (iii) is preserved upon replacing K by a finite separable extension K', or equivalently I by an open subgroup I'. When $\ell \neq p$, the action of I on $U_\ell \otimes \mathbb{Q}_\ell$ is unipotent (3.5(v)) and semisimple, hence trivial, and the good-reduction criterion 2.2.9 gives the result. We now give an argument valid with no restriction on ℓ, which yields a slightly sharper statement. I ≈391

Corollary 5.15. With the notation of 5.14, suppose that A_K has semistable reduction over S, and that the fixed part

$$(U_{\ell\mathbb{Q}_\ell})^f \overset{\text{def}}{=} (U_\ell^f)_{\mathbb{Q}_\ell}$$

admits an I-stable complement in $U_{\ell\mathbb{Q}_\ell}$. Then A_K has good reduction.

It is enough to prove under these hypotheses that $U_\ell = U_\ell^f$. By 5.6 this says precisely that the toric part T_o of A_o^o is zero, so that A_o^o is an abelian scheme; hence A is an abelian scheme by 2.2.9. We may plainly suppose that k is separably closed. The intersection of the asserted complement in 5.15 with U_ℓ then defines a pro-ℓ subgroup F_K of $T_\ell(A_K)$ which is isogenous to the cotoric part

$$T_\ell(A_K)/T_\ell(A_K)^f.$$

The latter is pro-étale and unramified by 5.6, and the same is therefore true of F_K. It follows from 5.7 that F_K is contained in $T_\ell(A_K)^f$; thus $F_K = 0$, proving that U_ℓ is equal to its fixed part.

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6. Remarks on the construction of the toric part and the fixed part of $T_p(A)$ in the case of non-semistable reduction.

6.1. The construction of the toric part of $T_\ell(A_K)$ given in 5.1 still makes sense as soon as S is complete, without assuming that A^o is ℓ-divisible; that is (when $p = \ell$), without assuming semistable reduction. That hypothesis had been used only in order to formulate the orthogonality theorem 5.2. Moreover, one can construct $T_p(A_K)^t$ while assuming only that S is henselian: one shows that the toric part $T_p(A_{\widehat{K}})^t$ over the completion, constructed by the method of 5.1, comes from a Barsotti–Tate pro-subgroup of $T_p(A_K)$, necessarily determined uniquely.

More precisely, one can construct a pro-p Barsotti–Tate group $T_p(A)^t$ over S and a homomorphism

$$T_p(A)^t \longrightarrow T_p(A)$$

of projective systems which is a monomorphism term by term and whose pullback to $\widehat{S}$ gives the analogous objects constructed by the method of 5.1. This follows by paraphrasing the construction of 5.1, except that, instead of lifting the whole maximal torus T_o of A_o, for each integer $\nu \geq 0$ one merely lifts the subgroup ${}_{p^\nu}T_o$ to a multiplicative-type subgroup $({}_{p^\nu}A^o)^t$ of ${}_{p^\nu}A^o$.

For the existence and uniqueness of such a lifting, note that the functor

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$$(\mathrm{Sch})^o_{/S} \longrightarrow \mathrm{Sets}, \qquad S' \longmapsto \{\text{finite multiplicative-type subgroups of } A^o_{S'}\}$$

is represented by an S-scheme X of finite type (cf. the proof of SGA 3 XI 3.12(a)), which is étale over S by SGA 3 IX 3.6 bis. Consequently, there is exactly one section of X over S extending a given section of X_s over s, which proves the assertion.

The same argument shows more generally that, for every finite multiplicative-type subgroup H_o of A_o, there is a unique finite multiplicative-type subgroup H of A extending it. Consequently, for every toric Barsotti–Tate p-subgroup H_o of A_o (that is, one whose components are of multiplicative type), there is a unique Barsotti–Tate p-subgroup H of A extending it. This in turn gives a Barsotti–Tate p-subgroup of ${}_{p^\infty}A_K$, or, equivalently, a pro-p Barsotti–Tate subgroup of $T_p(A_K)$.

In particular, this procedure applies when H_o is the largest toric Barsotti–Tate p-subgroup of A_o. It is obtained by taking the largest p-divisible subgroup $\overline{A}_o$ of A^o_o ($\overline{A}_o = p^N A^o_o$ for N large) and the "toric part" of ${}_{p^\infty}\overline{A}_o$ (defined, for example, by SGA 3 XVII 7.2.1). We thus obtain a Barsotti–Tate subgroup of ${}_{p^\infty}A_K$ which might be called the toric part of the latter.

6.2. Again let $\overline{A}_o$ be the largest p-divisible subgroup scheme of A^o_o, and consider the Barsotti–Tate p-group

$$F_o = {}_{p^\infty}(\overline{A}_o).$$

It is immediate that F_o may be characterized as the largest Barsotti–Tate p-subgroup of A_o (or of A^o_o). It is natural to ask whether this Barsotti–Tate group can be extended to a Barsotti–Tate subgroup F of A, which would then play the role of the fixed part $T_\ell(A)^f$ of (2.2.3.2) in the case in which A^o is ℓ-divisible.

In the semistable case, where $\overline{A}_o = A^o_o$, the method of 2.2 answers the question affirmatively. I≈394
In the general case, whenever the problem has a solution, it is readily seen to be unique. The corresponding pro-subgroups of $T_p(A_K)$ and $T_p(A'_K)$ (which again deserve to be called the fixed parts of the pro-groups in question) still satisfy the orthogonality theorem 5.2, and the fixed part $T_p(A_K)^f$ may still be characterized as in 5.8.

6.2.1. We shall nevertheless show that in general the question has no affirmative answer; that is, the fixed part $T_p(A)^f$ cannot in general be defined in the case of non-semistable reduction. Indeed, by a lifting theorem of Serre–Tate [30], when S is complete, the existence of the fixed part is equivalent to the existence of infinitesimal liftings $\overline{A}_n$ of $\overline{A}_o$ in the A_n, or again to the existence of a smooth group

scheme $\overline{A}$ over S, an extension of an abelian scheme by a torus T, having special fibre $\overline{A}_o$, together with a formal homomorphism

$$\widehat{\overline{A}} \longrightarrow \widehat{A}$$

extending the inclusion on the special fibres.

Thus, when $\overline{A}_o$ is an abelian scheme, that is, when $T_o = 0$, this is equivalent to the existence of an abelian scheme $\overline{A}$ extending $\overline{A}_o$ and a homomorphism $\overline{A} \longrightarrow A$ extending the inclusion of the special fibres. It will therefore suffice to give an example in which $T_o = 0$, A_o^o is neither unipotent nor proper, and A_K is simple over K.

6.2.2. Here is an example, due to M. Raynaud. It is based on the observation that, if S is a trait, S' **I 395** is the normalization of S in a finite separable extension K' of K, and A' is an abelian scheme over S', then the group scheme

$$A = \prod_{S'/S} A'/S'$$

over S is a Néron model: it identifies with the Néron model of its generic fibre, which is indeed an abelian scheme. This follows immediately from the definitions and from the fact that A' itself is a Néron model over S'. Then

$$A_o = \prod_{S'_s/s} (A' \times_{S'} S'_s)/S'_s$$

is an extension of an abelian scheme, nonzero if $A' \neq 0$, by a connected unipotent scheme, nonzero if $A' \neq 0$ and S' is ramified over S.

It therefore suffices to give an example in which $A' \neq 0$, S' is ramified over S (a complete local trait), and

$$A_K = \prod_{K'/K} A'_{K'}/K'$$

is simple over K. Take K'/K to be a ramified quadratic extension (such an extension always exists), and take A' to be an elliptic curve over S'. Then A has relative dimension two, and saying that A_K is simple over K means that there is no nonzero homomorphism

$$B_K \longrightarrow A_K,$$

where B_K is an abelian scheme of dimension one over K. Giving such a homomorphism is equivalent to giving a nonzero homomorphism

$$B_{K'} \longrightarrow A'_{K'},$$

which would necessarily be an isogeny. Thus A_K is simple over K precisely when $A'_{K'}$ is not isogenous to an elliptic curve coming from K.

If the contrary held, the j-invariant of A' would be algebraically dependent (over the prime field) on its conjugate $\overline{j}$ under the nontrivial K-automorphism of K'. If j is chosen so that

$$a = j + \overline{j} = \mathrm{Tr}_{K'/K}\, j$$

is transcendental over the prime field, this is equivalent to saying that j is algebraically dependent on a. By cardinality considerations, however, there exists an element $a \in V$ transcendental over the prime subfield P of K, and then an element $b \in V'$ (the normalization of V in K') which is transcendental over $P(a)$ and satisfies

$$\mathrm{Tr}_{V'/V}\, b = 0.$$

It is enough to take $j = a + b \in V'$. We may moreover suppose that $b \in \mathfrak{r}(V')$, and that the image of a in $k = V/\mathfrak{m}$ is a prescribed element of k, chosen distinct from the exceptional values $j_o = 0$ and $j_o = 12^3$. This is possible even when k has only two elements, because then these two contemplated values of j_o coincide. Thus the image j_o of j in $k(s')$ is different from 0 and 12^3, and it is known that there then exists an elliptic curve over S' with j-invariant j.

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6.3. Return to the case in which S is merely assumed henselian. In analogy with the definitions of 4.1 for $\ell \neq p$, one can then define a filtration

$$T_p(A_K) \supset T_p(A_K)^{\mathrm{ef}} \supset T_p(A_K)^{\mathrm{et}} \supset 0 \tag{6.3.1}$$

by strict Barsotti–Tate pro-subgroups, called respectively the essentially fixed part and the essentially toric part of $T_p(A_K)$.

To do so, choose a finite Galois extension K' of K such that $A_{K'}$ has semistable reduction over the normalization S' of S in K'. The fixed part $T_p(A_{K'})^f$ of $T_p(A_{K'}) = (T_p(A_K))_{K'}$ is then defined (2.2.3), as is the toric part (6.1). The latter is moreover the orthogonal complement of the fixed part $T_p(A'_{K'})^f$ in $T_p(A'_{K'})$ (5.2). Plainly

$$T_p(A_{K'})^t \subset T_p(A_{K'})^f$$

(this follows from 5.3, or directly from the construction of the toric part in 6.1), and the Galois group of K'/K preserves these two strict pro-subgroups of $(T_p(A_K))_{K'}$. Descent therefore gives the desired filtration (6.3.1). It is independent of the choice of K', by the semistable-reduction property 3.3. Descent also plainly gives the orthogonality property

$$\begin{aligned} T_p(A_K)^{\mathrm{et}} &= (T_p(A'_K)^{\mathrm{ef}})^{\perp}, \\ T_p(A_K)^{\mathrm{ef}} &= (T_p(A'_K)^{\mathrm{et}})^{\perp}. \end{aligned} \tag{6.3.2}$$

One could give an intrinsic characterization of the filtration (6.3.1) in terms of the Barsotti–Tate pro-group $X = T_p(A_K)$ over the fraction field K of S, in the spirit of 5.8—at least, for the moment, in unequal characteristics. The essentially fixed part is the largest Barsotti–Tate pro-subgroup having "essentially good reduction over S": after a finite separable extension K' of K, it gives a Barsotti–Tate pro-group $X_{K'}$ having good reduction over the normalization S' of S in K', that is, extending to a Barsotti–Tate pro-group over S'. Such an extension is necessarily unique up to a unique isomorphism, by the already-cited theorem of Tate [33, Th. 4]. The essentially toric part is then defined in terms of the essentially fixed part by orthogonality (6.3.2).

Let us also record the following immediate consequence of 5.8 and the definitions; for $\ell = p$ it reduces to 5.13(b).

Proposition 6.3.3. Let ℓ be a prime number. If $\ell = p$, assume that $\operatorname{char} K = 0$. [20] Then A_K has essential good reduction over S (5.13(b)) if and only if its essentially fixed part $T_\ell(A_K)^{\mathrm{ef}}$ (4.1.1 and 6.3.1) is all of $T_\ell(A_K)$.

6.4. Recall that, when $\ell \neq p$, in 4.3(a) we exhibited a remarkable integer-valued function on the inertia group I of S, independent of the choice of ℓ. When the residue field is perfect, knowledge of this function suffices to reconstruct the conductor exponent of A (4.5): one takes, on a sufficiently large quotient $\Gamma = I/I'$ of I, the inner product of this function with the Artin conductor on Γ (cf. (4.6.2)). It is useful to observe that this integer-valued function on I can also be recovered from the knowledge of $T_\ell(A_K)$ for $\ell = p$—at least, for the time being, in unequal characteristics. I 398

When the residue field k is finite, an analogous observation applies to the $\mathbb{Q}$-valued function considered in 4.3(b), defined on the subgroup π' of π consisting of the elements whose image in π_o has the form f_q^n, $n \in \mathbb{Z}$. We indicate the principle of the verification. Using the filtration (6.3.1), one is reduced to attaching a suitable function (on I, respectively on π') to a pro-p Barsotti–Tate group Y over K having "essentially good reduction over S" in the sense of 5.12. Here Y will be one of the three quotients arising from the filtration (6.3.1).

In the case of 4.3(a), a suitable residue-field extension within S (EGA 0_{III} 10.3.1) reduces us to the case of a perfect residue field, or even an algebraically closed one if desired. Choose a finite Galois extension K' of K such that $Y_{K'}$ has good reduction over S'. We then have a Barsotti–Tate group $Y_{S'}$ over S' and an action of G on $Y_{S'}$ compatible with its action on S'. If k' is the residue field of

[20] This hypothesis is probably unnecessary; cf. 5.11.

S', then G acts, not necessarily faithfully, on k' (whose invariant field is k), and on $Y_{k'}$ compatibly with its action on k'.

By Dieudonné theory [16 bis], the Barsotti–Tate pro-p group $Y_{k'}$ has a canonically associated finite free module M (of rank equal to the rank of $Y_{k'}$) over the ring $W(k')$ of Witt vectors of k'. The group G plainly acts on M compatibly with its actions on $W(k')$, through its actions on k'. Hence the inertia subgroup Γ of G acts $W(k')$-linearly on M; since Γ is a quotient of the inertia group I, this gives a trace function on Γ, and hence on I, which is the desired function. I≈399

A priori this function takes values in $W(k')$, [21] but it agrees with the analogous function defined in terms of the T_ℓ for $\ell \neq p$ whenever the action of G on $Y_{S'}$ is obtained from an action of G on a smooth p-divisible group scheme C' over S' by taking $Y = T_p(C')$ (cf. [25 bis, Th. 7, p. 737]). It is therefore integer-valued in this case. The case of 4.3(b) is handled similarly: to define a trace function on π', one uses the method of the proof of 4.3.2 to "linearize" on M the action defined by an element $g \in \pi$ lying above an f_q^n, with $n \geq 0$.

7. The Raynaud extension $G^\natural$ over S associated with the abelian scheme A_K having semistable reduction. Duality of the abelian schemes B, B' over S.

7.1. Let S be a Noetherian scheme which is the spectrum of a ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let A be a smooth group scheme over S, let A^o be its identity component, and suppose that A_o/A_o^o is finite (hence finite etale) over S_o, and that A_o^o is an extension of an abelian scheme by a torus. As usual, we have put $A_n = A \times_S S_n$, where $S_n = \mathrm{Spec}(V/\underline{m}^{n+1})$. We shall be interested chiefly in the case where S is a trait, the generic fiber A_K is an abelian scheme having semistable reduction over S, and A is the Neron model of A_K.

Let T_o be the maximal torus of A_o, so that $B_o = A_o^o/T_o$ is an abelian scheme. The construction of 5.1 then applies and gives a formal subtorus $\widehat{T}$ of the formal completion $\widehat{A}$. For every n, the quotient I≈400

$$B_n = A_n^o/T_n \tag{7.1.1}$$

is representable (SGA 3 VI$_A$ 3.2); more precisely, it is a flat scheme of finite type over S_n whose pullback to S_o is the abelian scheme B_o. Consequently B_n is an abelian scheme over S_n. As n varies, the B_n are obtained from one another by base change and therefore define a formal abelian scheme

$$\widehat{B} = (B_n)_{n\geqslant 0}\,. \tag{7.1.2}$$

We shall give conditions implying that this formal abelian scheme is algebraizable (EGA III 5.4.1), more precisely, that it is the formal completion of a projective abelian scheme over S, which we shall then denote by B. By loc. cit., B is determined up to a unique isomorphism (as an S-scheme; the group structure of $\widehat{B}$ then comes uniquely from a group structure on B, again by loc. cit.). This abelian scheme B will then be called the abelian part of the scheme A, and will sometimes be denoted by A^{ab}.

7.1.3. By EGA III 5.4.5, projectivity of $\widehat{B}$ means that one can find an invertible sheaf $\widehat{\mathbb{M}}$ on $\widehat{B}$, that is, a sequence of invertible sheaves $\mathbb{M}_n$ on the B_n which glue as n varies, and which is relatively ample over S, that is, such that $\mathbb{M}_o$ is ample. The natural idea would be to start with an ample invertible sheaf $\mathbb{L}$ on A and show that, for every n, its restriction $\mathbb{L}_n$ to A_n is the pullback of an invertible sheaf $\mathbb{M}_n$ on B_n. I do not know whether such $\mathbb{M}_n$ necessarily exist (and strongly doubt it). We shall follow a slightly different method, constructing a polarization of $\widehat{B}$, that is, a projective system of polarizations ρ_n of the B_n which glue. It is known [5] that the doubles of these polarizations canonically define (symmetric) invertible sheaves $\mathbb{M}_n$ on the B_n. By their canonical nature these sheaves necessarily glue and furnish the desired $\widehat{\mathbb{M}}$. I 401

We shall interpret an element of $\mathrm{NS}(B_n/S_n)$ as a symmetric divisorial correspondence on (B_n, B_n) [5], that is, as a biextension of (B_n, B_n) by $\mathbb{G}_{mS_n}$. By VIII 3.5, these biextensions correspond bijectively to biextensions of (A_n, A_n) by $\mathbb{G}_{mS_n}$. To obtain a coherent system of such biextensions W_n, it suffices to start with

$$W, \quad \text{a symmetric biextension of } (A, A) \text{ by } \mathbb{G}_{mS}, \tag{7.1.4}$$

[21] In fact it is immediate that this function always takes values in $W(k)$, since it is invariant under $\mathrm{Gal}(k'/k)$.

and take the W_n obtained from W by the base changes $S_n \longrightarrow S$. We must in addition suppose that the element of $\mathrm{NS}(B_o)$ defined by W_o is a polarization of B_o; we thereby obtain a polarization of $\widehat{B}$.

Proposition 7.1.5. With the preceding notation for $S, A, \widehat{B}$, suppose that S is normal. Then $\widehat{B}$ is algebraizable, and the abelian scheme B over S from which it comes is projective over S. I 402

We may plainly suppose that $A = A^o$ and that S is integral. By Raynaud [23, XI 1.13], the hypothesis that S is normal implies that A is quasi-projective over S. Choose a relatively ample invertible sheaf $\mathbb{L}$ on A, and form the birigidified invertible sheaf

$$\delta(\mathbb{L}) = \pi^*(\mathbb{L})\,\mathrm{pr}_1^*(\mathbb{L})^{-1}\,\mathrm{pr}_2^*(\mathbb{L})^{-1}$$

on $A \times_S A$, as in VIII 4.12. Since the fibers of A_o have unipotent rank zero, the same is true of the fibers of A (this follows easily from 3.1). Therefore, by VIII 7.5, $\delta(\mathbb{L})$ defines a biextension W of (A, A), plainly symmetric, which by VIII 4.13 (that is, essentially by Raynaud [23, XI 1.11]) satisfies the required ampleness condition at the points of S_o.

Remark 7.1.6. Without assuming that S is normal, one can prove that $\widehat{B}$ is always algebraizable by reducing to the normal case as in 13.5.3(a).

7.2. We now record the following lemma.

Lemma 7.2.1. Let S be the spectrum of a Noetherian ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let C be the category of group schemes Λ over S, smooth and of finite type over S, such that Λ/Λ^o is a finite etale group scheme over S and Λ^o is an extension of an abelian scheme B by an isotrivial torus T (SGA 3 IX 1.1). Let C' be the category of group objects in the category of formal schemes of finite type (EGA I 10.13.3) over $\widehat{S} = \mathrm{Spf}(V)$, and consider the functor

$$A \longmapsto \widehat{A}$$

from C to C' which takes the formal completion along $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$. Then:

(a) The preceding functor is fully faithful. I ≈403

(b) Let $\mathcal{A}$ be an object of C', smooth over $\widehat{S}$; that is, suppose that the ordinary schemes

$$A_n = \mathcal{A} \times_{\widehat{S}} S_n, \qquad S_n = \mathrm{Spec}(V/\underline{m}^{n+1}),$$

are smooth over S_n. Suppose that A_o/A_o^o is a finite etale group scheme over S_o, and that A_o^o is an extension of an abelian scheme B_o by an isotrivial torus T_o. Put

$$\mathcal{A}^o = (A_n^o)_{n\geq 0}, \qquad \mathcal{A}/\mathcal{A}^o = (A_n/A_n^o)_{n\geq 0}.$$

Then $\mathcal{A}/\mathcal{A}^o$ is finite etale over $\widehat{S}$, and $\mathcal{A}^o$ is, in an essentially unique manner and functorially in $\mathcal{A}$, an extension of a formal abelian scheme $\mathcal{B}$ over $\widehat{S}$ by a formal torus $\mathcal{T}$ over $\widehat{S}$, reducing respectively to B_o and T_o. The formal group scheme $\mathcal{A}$ belongs to the essential image of $C \longrightarrow C'$—or, as we shall also say, is "algebraizable"—if and only if $\mathcal{B}$ is algebraizable.

7.2.1.1. First observe, over an arbitrary scheme S, that if A is a smooth group scheme such that A^o is an extension of an abelian scheme by a torus, this extension structure is essentially unique and is functorial in A in the evident sense. Indeed, the assertion reduces at once to the fact that every morphism from a torus T to an abelian scheme B is zero. This is well known on geometric fibers [15], and SGA 3 IX 5.3 then gives the conclusion. Applying the result to the compatible systems of smooth group schemes A_n over the S_n occurring in 7.2.1 gives the analogous result for formal group schemes over S. Moreover, we already observed in 7.1 that, if A_o^o is an extension of an abelian scheme by a torus, the same is true of A_n^o. This proves the first assertion of (b).

7.2.1.2. Suppose next that the formal group scheme $\mathcal{A}$ over S is of the form $\widehat{A}$, with $A \in \mathrm{ob}\,C$. If A^o is an extension of an abelian scheme B by a torus T, then plainly $\mathcal{B} \simeq \widehat{B}$. Thus algebraizability of $\mathcal{A}$ implies algebraizability of $\mathcal{B}$. I 404

Conversely, suppose that $\mathcal{B}$ is algebraizable, that is, that it comes from an abelian scheme B, and let us prove that $\mathcal{A}$ is algebraizable. We first prove that $\mathcal{A}^o = (A_n^o)$ is algebraizable. By VIII 3.7, the extensions A_n^o of B_n by T_n may be interpreted as homomorphisms

$$(*) \qquad \underline{M}_n \longrightarrow B'_n,$$

where $\underline{M}_n$ is the twisted constant group over S_n dual to T_n (SGA 3 X 5.7), and B'_n is the abelian scheme dual to B_n. Restriction to S_o, or to any S_n, induces an equivalence from the category of isotrivial tori over S to the corresponding category over S_o (SGA 3 X 3.2). Hence the T_n come from a torus T over S, determined up to a unique isomorphism. Denote its dual group by $\underline{M}$; its restriction to S_n is canonically isomorphic to $\underline{M}_n$.

The isotriviality of T means that $\underline{M}$ is a disjoint union of S-schemes $\underline{M}^i$ which are finite over S (SGA 3 X 5.11). It follows at once that the homomorphisms $(*)$ come from a unique homomorphism

$$(**) \qquad \underline{M} \longrightarrow B',$$

where B' is the abelian scheme dual to B. In turn, this homomorphism is interpreted as an extension of B by T, say A^o, which plainly supplies an algebraization of $\mathcal{A}^o$.

7.2.1.3. On the other hand, by hypothesis A_o/A_o^o is a finite group scheme Φ_o over S_o, and the A_n/A_n^o are likewise finite group schemes Φ_n over S_n; they are necessarily etale over S_n. There is therefore an etale group scheme Φ over S, determined up to a unique isomorphism, whose formal completion is the formal group scheme $\mathcal{A}/\mathcal{A}^o$ defined by the Φ_n. I≈405

Since $G = A^o$ is divisible, the category of extensions, as fppf sheaves, of a finite locally constant group over S such as Φ by the group scheme A^o is canonically equivalent to the category of torsors under the sheaf

$$H = \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi, G),$$

and this equivalence is compatible with every base change. Applying the same description to extensions of Φ_n by G_n in terms of torsors under

$$H_n \simeq \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi_n, G_n),$$

the A_n define a coherent system of torsors P_n under the H_n. Because G is an extension of an abelian scheme by a torus, it follows readily that H is a group scheme finite and locally free over S. Consequently, the functor

$$P \longmapsto \widehat{P} = (P_n)$$

is an equivalence between the category of torsors over S under H and the category of "formal" torsors $\widehat{P}$ under $\widehat{H}$, that is, coherent systems of torsors P_n under the H_n.

Thus the P_n defined by the extensions A_n come from a single torsor P over S under H. We obtain an extension A of Φ by $G = A^o$ which algebraizes $\mathcal{A}$, except that its representability remains to be checked. Regard A as a torsor over Φ under A_Φ^o. This torsor becomes trivial after base change by the finite, surjective, locally free morphism I≈406

$$P \times_S \Phi \longrightarrow \Phi,$$

since the base change $P \longrightarrow S$ splits the extension A of Φ by A^o. One may therefore conclude by one of Raynaud's many criteria for effectiveness of descent [23, XI 3.1(6)].

7.2.1.4. This completes the proof of (b), and only the full faithfulness assertion (a) remains. We leave it to the reader to verify that this assertion is essentially contained in the method of proof of (b), or, if preferred, to find an independent direct proof.

7.2.1.5. Notice that the appeal to Raynaud's delicate descent result becomes unnecessary if A^o is assumed quasi-projective over S, or, equivalently, if B is projective over S. This condition holds in particular when one starts from the formal group scheme $\widehat{A}$ over S, with A as in 7.1.5. This is the only case in which we shall use 7.2.1.

7.2.2. We apply 7.2.1 to the formal group scheme $\widehat{A}$ considered in 7.1, in the favorable case treated there in which the formal abelian scheme $\widehat{B}$ is algebraizable (cf. 7.1.5), and we suppose in addition that the torus T_o over S_o is isotrivial. We thereby obtain a group scheme $A^\natural$ over S, called the Raynaud group of A over S, characterized up to isomorphism by a canonical isomorphism of formal group schemes over $\mathrm{Spf}(V) = \widehat{S}$:

$$(A^\natural)^\wedge \simeq \widehat{A}. \tag{7.2.3}$$

It follows from 7.2.1 that

$$\Phi = A^\natural / A^{\natural o} \tag{7.2.4}$$

is a finite etale group over S, and that the identity component

$$A^{\natural o} = A^{o\natural}$$

is endowed with the structure of an extension

$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0, \tag{7.2.5}$$

where B is the abelian scheme over S whose formal completion is the formal abelian scheme $\widehat{B}$ constructed in 7.1, and T is characterized, up to a unique isomorphism, as the isotrivial torus over S lifting T_o. On restriction to the S_n, the extension structure (7.2.5) gives the extension structures on $A_n^{\natural o} = A_n^o$ defined by (7.1.1).

7.2.6. It also follows immediately from 7.2.1 that the Raynaud group $A^\natural$, as well as the exact sequence (7.2.5), depends functorially on A (satisfying the conditions under consideration), and that its formation commutes with every base change $S' \longrightarrow S$ associated with a homomorphism of adic topological rings $V \longrightarrow V'$. I 407

7.3. Let us now consider, for a fixed prime number ℓ, the pro-ℓ group $T_\ell(A^o)$. Identifying a finite scheme over S with the corresponding formal scheme, and defining the "fixed part" $T_\ell(A^o)^f$ as usual, we obtain an isomorphism as in (5.1.5):

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$$T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad , \tag{7.3.1}$$

and, defining the "toric part" as in (5.1.6),

$$T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq T_\ell(T) \subset T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad , \tag{7.3.2}$$

we immediately obtain, for the "abelian part," an isomorphism

$$T_\ell(A^o)^{ab} \overset{\mathrm{def}}{=} T_\ell(A^o)^f / T_\ell(A^o)^t \simeq T_\ell(\widehat{B}) \quad . \tag{7.3.3}$$

We have not yet used here the hypothesis that $\widehat{B}$ is algebraizable, the hypothesis that allowed us to construct the abelian scheme B and the Raynaud extension of B by T. Since there is, of course, a canonical isomorphism (the special case of (7.3.1) applied to the abelian scheme B!) $T_\ell(\widehat{B}) \simeq T_\ell(B)$, the isomorphism (7.3.3) also gives us

$$T_\ell(A^o)^{ab} \simeq T_\ell(B) \quad . \tag{7.3.4}$$

Let us also note in passing that (7.2.3) and (7.3.1) imply a canonical isomorphism

$$T_\ell(A^o)^f \simeq T_\ell(A^{\natural o}) \quad , \tag{7.3.5}$$

since the right-hand side is already a profinite group, that is, identical to its fixed part. The canonical exact sequence

$$0 \longrightarrow T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow T_\ell(B) \longrightarrow 0 \quad , \tag{7.3.6}$$

which can in any case (without the algebraizability hypothesis on $\widehat{B}$) be interpreted as the exact sequence obtained by applying the functor T_ℓ to the exact sequence

$$0 \longrightarrow \widehat{T} \longrightarrow \widehat{A}^o \longrightarrow \widehat{B} \longrightarrow 0 \quad ,$$

can also be regarded as obtained by applying the functor T_ℓ to the canonical exact sequence of ordinary group schemes

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$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0 \qquad .$$

7.4. Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, and suppose that A and A' have connected fibres. Let W be a biextension of (A, A') by $\mathbb{G}_{mS}$. We thereby obtain a pairing (VIII 2.2)

$$T_\ell(A) \times T_\ell(A') \longrightarrow T_\ell(\mathbb{G}_{mS}) = \mathbb{Z}_\ell(1) \tag{7.4.1}$$

and, by restriction to the fixed parts, a pairing

$$T_\ell(A)^f \times T_\ell(A')^f = T_\ell(\widehat{A}) \times T_\ell(\widehat{A'}) \longrightarrow \mathbb{Z}_\ell(1) \qquad . \tag{7.4.2}$$

This can also be interpreted as the pairing defined by the biextension $\widehat{W}$ of $(\widehat{A}, \widehat{A'})$ by $\widehat{\mathbb{G}}_{mS}$ obtained from W by completion (here, for simplicity, defining a biextension of formal group schemes as a coherent family of biextensions W_n of (A_n, A'_n) by $\mathbb{G}_{mS_n}$). Now we know (VIII 3.5) that W_n comes from a biextension W_n^{ab} of (B_n, B'_n), where B'_n is the abelian part of A'_n, so that (7.4.2) factors through

$$T_\ell(A)^{ab} \times T_\ell(A')^{ab} = T_\ell(\widehat{B}) \times T_\ell(\widehat{B'}) \longrightarrow \mathbb{Z}_\ell(1) \qquad . \tag{7.4.3}$$

This is also the pairing defined by the biextension

$$\widehat{W}^{ab} = (W_n^{ab})$$

of $(\widehat{B}, \widehat{B'})$ by $\widehat{\mathbb{G}}_{mS}$. When $\widehat{B}$ and $\widehat{B'}$ are algebraizable, so that abelian schemes B, B' over S are available, EGA III 5.1.4 shows that $\widehat{W}^{ab}$ comes from a biextension, defined up to a unique isomorphism,

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$$W^{ab} \quad \text{of} \quad (B, B') \quad \text{by} \quad \mathbb{G}_{mS},$$

and the corresponding pairing

$$T_\ell(B) \times T_\ell(B') \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{7.4.4}$$

is canonically isomorphic to (7.4.3).

Writing $D(B')$ for the abelian scheme dual to B', W^{ab} defines a homomorphism

$$u : B \longrightarrow D(B') \qquad , \tag{7.4.5}$$

whose properties reflect those of the pairings (7.4.4), that is, (7.4.3). For example, the pairing (7.4.3) is a perfect duality at a point $t \in S$ if and only if $u_t : B_t \longrightarrow D(B')_t$ is an isogeny whose kernel has rank prime to ℓ. This implies that u itself is an isogeny (since S is local, hence connected), and $\ker u$ is flat and finite over S, so that $\ker u$ has the same rank at every point. Thus the condition just considered is independent of the chosen point t.

7.4.6. This completes in particular the proof of 5.4. Indeed, under the conditions of that theorem (where S may be assumed complete), applying the preceding discussion to A^o and A'^o shows that, for every ℓ, the pairing (7.4.3) is a perfect duality at the generic point η, and hence also at the point s.

7.4.7. The preceding considerations likewise make the orthogonality theorem 5.2 clear, without recourse to Lemma 5.2.5. Indeed, along the way in 7.4 we established that the pairing (5.2.4.2) = (7.4.2) vanishes on the toric parts on both sides, and therefore factors through the pairing (7.4.3). It remains only to establish that this pairing is nondegenerate on the generic fibres of the pro-groups under consideration, knowing that it is so on the special fibres. This follows from the fact that if (7.4.5) induces an isogeny on the special fibres, then it is an isogeny. I 411

7.4.8. Return to the general conditions of 7.4, and suppose that an open subset $U \subset S - S_o$ is given such that A_U and A'_U are abelian schemes, dual to one another through the biextension W_U. Restricting (7.3.6), and the analogous exact sequence for A', to U, we obtain canonical filtrations on the Barsotti–Tate pro-ℓ groups $T_\ell(A_U)$ and $T_\ell(A'_U)$:

$$\text{(7.4.9)} \qquad \begin{cases} T_\ell(A_U) \supset T_\ell(A_U)^f \supset T_\ell(A_U)^t \supset 0, \\ \\ T_\ell(A'_U) \supset T_\ell(A'_U)^f \supset T_\ell(A'_U)^t \supset 0 \qquad . \end{cases}$$

I claim that these two filtrations are dual to one another with respect to the pairing (7.4.1), which here is a perfect duality (VIII 3.2); that is, relation (5.2.2) and its symmetric counterpart still hold, with A_K, A'_K replaced by A_U, A'_U. We saw above that this orthogonality assertion is also equivalent to saying that the biextension W^{ab} of (B, B') by $\mathbb{G}_{mS}$ is a perfect duality over U (or, equivalently when U is dense in S, over S). It is enough to verify this property at every point η of U; the usual argument then reduces us to the already known case in which S is a trait with generic point η.

The orthogonality relation also allows us to extend formulas (5.5.7), (5.5.8), and (5.5.9) to the present case, replacing restriction to the generic point η by restriction to the open subset U. Consequently, 5.6 likewise extends to the present case. We refrain from rewriting the preceding formulas and statement here in this context. I ≈412

Let us summarize, from the point of view of Néron models, the results obtained concerning the abelian parts and their pairings.

Proposition 7.5. Let S be a complete trait with function field K. To every abelian scheme A_K over K having semistable reduction over S, the procedure of 7.1, applied to the identity component A^o of the Néron model A of A_K, associates its abelian part $B = A^{ab}$, which is an abelian scheme over S depending functorially on A_K. For every other abelian scheme A'_K with semistable reduction, one can associate to every divisorial correspondence W_K on (A_K, A'_K) a unique divisorial correspondence W^{ab} on (A^{ab}, A'^{ab}) such that, for every integer $n \geq 0$, the pullback of W_n^{ab} along $A_n^o \longrightarrow B_n = A_n^{oab}$ and $A'^o_n \longrightarrow B'_n = A'^{oab}_n$ is the biextension W_n of (A_n^o, A'^o_n) obtained from W by reduction, where W is the canonical extension (1.4) of W_K to (A^o, A'^o). The formation of W^{ab} is compatible with the formation of pullbacks of correspondences by morphisms of abelian schemes $\overline{A}_K \longrightarrow A_K$ and $\overline{A'}_K \longrightarrow A'_K$. If W_K is a perfect duality, then so is W^{ab}. If W_K is the divisorial correspondence on (A_K, A_K) associated with a polarization of A_K, then W^{ab} is associated with a polarization of A^{ab}.

The compatibility of the formation of the W^{ab} with pullbacks is immediate from the indicated characterization of the W^{ab}. The additional assertion concerning polarizations was established in 7.1.5 at the same time as the algebraizability of $\widehat{B}$. I ≈413

7.6. Continue to suppose that S is a complete trait, with A the Néron model of an abelian scheme A_K over K. Set

$$A_K^{\natural} = (A^{\natural})_K \quad , \tag{7.6.1}$$

where $A^{\natural}$ is the Raynaud group scheme (7.2.2): it is an extension of the finite etale group Φ by $A^{\natural o}$, while $A^{\natural o}$ is an extension of the abelian scheme B by the torus T. Thus $A_K^{\natural}$ is itself an extension of Φ_K by $(A_K^{\natural})^o = (A^{\natural o})_K = (A^{o\natural})_K$, which is an extension of the abelian scheme B_K by the torus T_K. For every prime number ℓ, (7.3.5) and (7.3.2) give

$$T_\ell(A_K^{\natural}) \simeq T_\ell(A_K^{\natural o}) \simeq T_\ell(A_K)^f \ , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^t \quad . \tag{7.6.2}$$

7.6.3. By 3.3, the formation of $A_K^{\natural o}$ commutes with every base change associated with a morphism of complete traits $S' \longrightarrow S$.

8. The Raynaud extension $A_K^{\natural o}$ in the non-semistable case, and the ind-group $A_K^{\flat}$ I≈414

8.1. Continue to suppose that the trait S is complete, but no longer suppose necessarily that A_K has semistable reduction. As usual, let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction relative to the normalization S' of S in K'. Consider the Raynaud extension $A_{K'}^{\natural}$ associated with $A_{K'}$ (7.6.1). It is clear that the Galois group of K'/K acts by transport of structure on this group scheme; hence, by Galois descent, it gives a group scheme over K, denoted $A_{K,K'}^{\natural}$, which depends on the choice of the extension K'. Its identity component, however, is independent of this choice up to canonical isomorphism, by 7.6.3, and will simply be denoted $A_K^{\natural o}$. Descent shows that $A_K^{\natural o}$ is an extension

$$0 \longrightarrow T_K \longrightarrow A_K^{\natural o} \longrightarrow B_K \longrightarrow 0 \quad , \tag{8.1.1}$$

where T_K is a torus and B_K an abelian scheme over K. Likewise, by the definition of the essentially fixed and essentially toric parts of $T_\ell(A_K)$ (4.1 and 6.3), the isomorphisms (7.6.2) give canonical isomorphisms

$$T_\ell(A_K^{\natural}) \simeq T_\ell(A_K)^{ef} \quad , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^{et} \quad , \tag{8.1.2}$$

and hence, on passing to the quotient,

$$T_\ell(B_K) \simeq T_\ell(A_K)^{ef}/T_\ell(A_K)^{et} \quad . \tag{8.1.3}$$

This therefore determines two of the quotients occurring in the filtration I≈415

$$T_\ell(A_K) \supset T_\ell(A_K)^{ef} \supset T_\ell(A_K)^{et} \supset 0 \tag{8.1.4}$$

Thus the extension of group schemes (8.1.1) describes two quotients in this filtration. For the third quotient $T_\ell(A_K)/T_\ell(A_K)^{ef}$, (6.3.2) gives a canonical isomorphism

$$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(A'_K)^{et}) \quad , \tag{8.1.5}$$

where $D = \underline{\mathrm{Hom}}(-, T_\ell(\mathbb{G}_m))$ denotes Cartier duality (in the sense of Barsotti–Tate pro-groups), and A'_K denotes the abelian scheme dual to A_K. Consider then the exact sequence of type (8.1.1)

(8.1.6) $$0 \longrightarrow T'_K \longrightarrow {A'_K}^{\natural o} \longrightarrow B'_K \longrightarrow 0$$

associated with the abelian scheme A'_K over K, together with the corresponding isomorphisms of type (8.1.2). Formula (8.1.5) then gives

(8.1.7) $$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(T'_K)) \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad ,$$

where, generalizing the notation introduced in (5.5.2), we have set

(8.1.8) $$\underline{M}_K = D(T'_K) \quad , \qquad \underline{M}'_K = D(T_K) \quad ,$$

and where D now denotes Cartier duality $\underline{\mathrm{Hom}}(-, \mathbb{G}_{mK})$ in the sense of ordinary group schemes. Thus $\underline{M}_K$ and $\underline{M}'_K$ are twisted constant groups over K which become constant over K' and isomorphic to groups of the form $\mathbb{Z}^r$, where $r = \dim T = \dim T'$.

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8.2. The hypotheses are those of 8.1. It is clear that the extension (8.1.1) depends functorially on the abelian scheme A_K, and that its formation commutes with every morphism of traits $\widetilde{S} \longrightarrow S$. Choose a polarization of A_K, hence an isogeny $A_K \longrightarrow A'_K$. It induces a homomorphism from (8.1.1) to (8.1.6), which is immediately seen, by looking at the T_ℓ, to be an isogeny term by term. Likewise, using 7.5 and descent, one sees that every biextension by $\mathbb{G}_{mK}$ of abelian schemes A_K and $\overline{A}_K$ over K defines a biextension of $(B_K, \overline{B}_K)$ by $\mathbb{G}_{mK}$, where B_K and $\overline{B}_K$ denote the abelian quotients of the Raynaud extensions corresponding to A_K and $\overline{A}_K$, respectively. When W is a perfect duality, so is the corresponding biextension of $(B_K, \overline{B}_K)$. In particular, with the notation of 7.6, the abelian schemes B_K and B'_K are canonically dual to one another.

The correspondence between biextensions on $(A_K, \overline{A}_K)$ and biextensions on $(B_K, \overline{B}_K)$ has evident functoriality properties as A_K and $\overline{A}_K$ vary. These follow by descent from the analogous properties stated in 7.5, and we leave their formulation to the reader. We also leave to the reader the compatibility of the preceding construction with every change of traits $\widetilde{S} \longrightarrow S$.

8.3. Return to the group $A^\natural_{K,K'}$ and study its variation with K'. It is clear that two isomorphic extensions K' and K'_1 of K give canonically isomorphic groups, so it is enough to consider the subextensions K' of the separable closure $\overline{K}$ of K. If $K' \hookrightarrow K''$ are two such finite Galois subextensions of $\overline{K}$, it follows immediately from 3.3 that there is a canonical <u>open immersion</u>

(8.3.1) $$A^\natural_{K,K'} \hookrightarrow A^\natural_{K,K''} \quad ,$$

which identifies the first group with an open subgroup of the second. As K' varies, the $A^\natural_{K,K'}$ thus form an inductive system of smooth groups of finite type over K, whose transition morphisms are open immersions. If we set

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(8.3.2) $$\Phi_{K,K'} = A^\natural_{K,K'}/{A^\natural_{K,K'}}^{o} \quad ,$$

the $\Phi_{K,K'}$ themselves form an inductive system of finite etale groups over K, which will be studied in greater detail below.

We shall set

(8.3.3) $$A^\flat_K = \varinjlim_{K'} A^\natural_{K,K'} \quad ,$$

where the limit is taken in the category of fppf sheaves over K. Since the transition morphisms in this inductive system are open immersions, it is clear moreover that this limit is represented by a

group scheme over K (in general non-quasi-compact), which we shall naturally denote by the same symbol $A_K^\flat$. For this group scheme one has

$$A_K^{\flat\,o} = A_K^{\natural o} \quad , \qquad A_K^\flat / A_K^{\flat\,o} \simeq \varinjlim_{K'} \Phi_{K,K'} \qquad . \tag{8.3.4}$$

The latter group will be determined in 11.10: it is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, with the notation of (8.1.8).

9. Definition of the monodromy pairing $\underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_{\ell S}$. I≈418

9.1. In this section, suppose that S is henselian and that A_K has semistable reduction over S, or, equivalently, that A'_K has semistable reduction over S (3.5.1). We then have over S two twisted constant $\mathbb{Z}$-modules $\underline{M}$ and $\underline{M}'$ (5.5.2). Over a suitable finite etale covering of S (and hence over S itself if S is strictly local, that is, if k is separably closed), they split into constant groups with values M and M', respectively, noncanonically isomorphic to $\mathbb{Z}^r$, where r is the common dimension of the maximal tori of A_o and A'_o. For every prime number ℓ, we shall consider the corresponding ℓ-adic sheaves

$$\underline{M}_\ell = \underline{M} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad , \qquad \underline{M}'_\ell = \underline{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad , \tag{9.1.1}$$

whose geometric meaning is made explicit by formulas (5.5.7) and (5.5.8). In this section we shall define a canonical pairing, called the monodromy pairing,

$$u_\ell : \underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_\ell \qquad , \tag{9.1.2}$$

which will be defined in terms of the Barsotti–Tate pro-ℓ-groups $T_\ell(A^o)$ and $T_\ell(A'^o)$, and even in terms of the generic fiber $T_\ell(A_K)$ of the first of these groups (subject, when $\ell = p$, to the availability of Tate's theorem [33, Th. 4], that is, for the time being, provided we are then in unequal characteristic). When $\ell \neq p$, this is equivalent to saying that the pairing (9.1.2) is described in terms of the Galois π-module

$$U = T_\ell(A_K)(\overline{K}) = T_\ell(A_{\overline{K}}) \tag{9.1.3}$$

considered in (2.5.3). More precisely, we shall see that knowing the pairing is essentially equivalent to knowing the action of the monodromy group I on U.

9.2. The case $\ell \neq p$. Begin with the simpler case $\ell \neq p$, using the notation introduced in (2.5.3). I 419
Note that, up to torsion, the relation $V = U^I$ also means that $V^\perp$ is the module generated by the elements $gx - x$, where $x \in U$ and $g \in I$, and in any case one has

$$gx - x \in V^\perp \quad \text{for } x \in U,\ g \in I \qquad . \tag{9.2.1}$$

Thus, for $g \in I$,

$$gx = x + u(g)\,.\,x \qquad , \tag{9.2.2}$$

where $u(g)$ is a homomorphism from U to $V^\perp$ which vanishes on $V = U^I$, or equivalently

$$u \in \mathrm{Hom}(U/V, V^\perp) \simeq (U/V)^\vee \otimes V^\perp \qquad . \tag{9.2.3}$$

In the case of semistable reduction under consideration, that is, when $V^\perp \subset V$, formula (9.2.2) says that the action of g on U is a transvection parallel to $V^\perp = W$, and composition of these transvections is simply addition of the corresponding homomorphisms u in (9.2.3). Consequently, the action of the inertia group I on U is completely described by a group homomorphism

$$I \longrightarrow (U/V)^\vee \otimes V^\perp \quad . \tag{9.2.4}$$

The orthogonality theorem, applied both to A_K and to A'_K, gives I≈420

$$V^\perp = W \quad , \qquad (U/V)^\vee \simeq W'(-1) \quad ,$$

where W' is the toric part of $U' = T_\ell(A'(\overline{K}))$, and (-1) denotes the Tate twist. Moreover, the homomorphism (9.2.4) factors through $I(\ell) \simeq \mathbb{Z}_\ell(1)$, so that (9.2.4) may be interpreted as a canonical homomorphism

$$u : \mathbb{Z}_\ell(1) \longrightarrow W' \otimes W(-1) \quad , \tag{9.2.5}$$

or, equivalently, as an element

$$u \in W' \otimes W(-2) \quad . \tag{9.2.6}$$

Recall that W and W' are the groups of $\overline{K}$-valued points of the two twisted constant ℓ-adic sheaves $T_\ell(A^o)^t$ and $T_\ell(A'^o)^t$, whose special fibers are $T_\ell(T_o)$ and $T_\ell(T'_o)$, respectively. We obtain the canonical isomorphisms

$$\begin{cases} W \simeq T_\ell(T_o)(\overline{k}) \simeq \check{M} \otimes \mathbb{Z}_\ell(1), \\ W' \simeq T_\ell(T'_o(\overline{k})) \simeq \check{M}' \otimes \mathbb{Z}_\ell(1), \end{cases} \tag{9.2.7}$$

where M and M' denote the character groups over $\overline{k}$ of T_o and T'_o, respectively:

$$T_o = D(M) \quad , \qquad T'_o = D(M') \quad . \tag{9.2.8}$$

It follows that the element u of (9.2.6), which describes the action of the inertia group I on the Tate module U of A_K, is interpreted as a canonical element of $(\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. It is moreover invariant under $\pi_o = \pi/I = \mathrm{Gal}(\overline{k}/k)$, since by definition it is invariant under transport of structure:

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$$u \in \left((\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell\right)^{\pi_o} \simeq \mathrm{Hom}_{\mathbb{Z}}(\underline{M} \otimes \underline{M}', \mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(\underline{M}_\ell \otimes \underline{M}'_\ell, \mathbb{Z}_\ell) \quad . \tag{9.2.9}$$

Up to writing u in place of u_ℓ, this is the element (9.1.2) that we wished to define.

To define (9.1.2) in the general case, we shall need a few preliminaries, which will be developed in the next two sections.

9.3. Mixed extensions.

Let $\underline{C}$ be an abelian category, and let

$$P,\ R,\ Q \tag{9.3.1}$$

be three objects of $\underline{C}$. We are interested in classifying the objects X of $\underline{C}$ endowed with a three-step filtration

(*) $$X^0 = X \supset X^1 \supset X^2 \supset X^3 = 0$$

and with given isomorphisms

$$X^0/X^1 \xrightarrow{\sim} Q, \qquad X^1/X^2 \xrightarrow{\sim} R, \qquad X^2/X^3 = X^2 \xrightarrow{\sim} P.$$

Putting

$$E = X^0/X^2, \qquad F = X^1/X^3 = X^1,$$

we see that X gives rise to two ordinary extensions

(9.3.2) $$0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0,$$

of R by P and of Q by R, respectively. Our point of view here will be to start, in addition to the objects (9.3.1), with the extension structures (9.3.2), which are likewise assumed to be given, and to consider the corresponding three-step filtered structures X.

More precisely, a mixed extension of the extension E by the extension F will mean an object X of $\underline{C}$, endowed with a three-step filtration (*) and with given isomorphisms I≈422

(9.3.3) $$E \simeq X^0/X^2, \qquad F \simeq X^1/X^3 = X^1,$$

compatible with the filtrations: they carry X^1/X^2 to $\mathrm{Im}(R \longrightarrow E)$ and X^2 to $\mathrm{Im}(P \longrightarrow F)$, respectively, and the isomorphism $R \simeq F/P$ (and hence $R \simeq X^1/X^2$) deduced from (9.3.3) is required to equal the one deduced from the first exact sequence in (9.3.2).

For fixed P, Q, R and fixed extensions E and F, the mixed extensions of E by F form a category in the evident way; it is in fact a groupoid (every arrow is invertible). The automorphism group of a mixed extension X of E by F is canonically isomorphic to $\mathrm{Hom}(Q, P)$ through

(9.3.4) $$\mathrm{Hom}(Q, P) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{extpan}}(X), \qquad u \longmapsto \mathrm{id}_X + \bar{u},$$

where $\bar{u}$ is the endomorphism of the object of $\underline{C}$ underlying X obtained by composing $X \longrightarrow Q \xrightarrow{u} P \longrightarrow X$; the outer arrows are respectively the canonical epimorphism and the canonical monomorphism coming from the given mixed-extension structure on X.

One could make the structure of the category $\mathrm{EXTPAN}(E, F)$ of mixed extensions of E by F precise by regarding it as a "torsor" under the "group-category" (VII 2.5) $\mathrm{EXT}(Q, P)$ of extensions of Q by P, using a natural functor analogous to the Brauer composition of ordinary extensions:

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(9.3.5) $$\omega : \mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F) \longrightarrow \mathrm{EXTPAN}(E, F),$$
$$\omega(G, X) = G \wedge X.$$

For every object X of $\mathrm{EXTPAN}(E, F)$, the partial functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$.

We shall give only a brief description of ω. Regard a mixed extension X of E by F as defining an ordinary extension, also denoted X, of Q by F (viewed as an object of $\underline{C}$). For every extension G of Q by P, let $\bar{G}$ denote the extension of Q by F induced by the inclusion $P \longrightarrow F$. Set

(9.3.6) $$G \wedge X = \omega(G, X) \overset{\mathrm{dfn}}{=} \bar{G} \wedge X,$$

where $\wedge$ denotes Brauer composition in the category of extensions of Q by F. The three-step filtration of the right-hand side X' of (9.3.6) is clear, as is the second isomorphism of type (9.3.3) attached to X'.

To define the first isomorphism of type (9.3.3) for X', observe that the functor

$$Y \longmapsto \widetilde{Y}, \qquad \mathrm{EXT}(Q,F) \longrightarrow \mathrm{EXT}(Q,R),$$

induced by the given morphism $F \longrightarrow R$ in (9.3.2) is compatible with Brauer composition. It therefore sends X' to an extension $\widetilde{X}'$ canonically isomorphic to $\widetilde{\widetilde{G}} \wedge \widetilde{X}$. The first isomorphism (9.3.3) gives an isomorphism $\widetilde{X} \simeq E$, while $\widetilde{\widetilde{G}}$—the extension induced from G by the composite $P \longrightarrow F \longrightarrow R$—is canonically isomorphic to the trivial extension, since that composite $P \longrightarrow R$ is zero. Consequently there is a canonical isomorphism $\widetilde{X}' \simeq E$, which specifies on the right-hand side of (9.3.6) its mixed-extension structure of E by F.

We leave to the reader the definition of ω on the arrows of $\mathrm{EXT}(Q,P) \times \mathrm{EXTPAN}(E,F)$, the verification that (9.3.5) has the asserted property—namely, for every fixed mixed extension X, the functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q,P)$ to $\mathrm{EXTPAN}(E,F)$—and the proof of the associativity formula I≈**424**

$$G_1 \wedge (G_2 \wedge X) \simeq (G_1 \wedge G_2) \wedge X, \tag{9.3.7}$$

where $G_1 \wedge G_2$ denotes Brauer composition in $\mathrm{EXT}(Q,P)$.

The preceding discussion immediately gives parts (a) and (b) of the following result.

Proposition 9.3.8. Given two extensions E and F as in (9.3.2) in the abelian category $\mathbf{C}$, consider the category $\mathrm{EXTPAN}(E,F)$ of mixed extensions of E by F. Then:

a) $\mathrm{EXTPAN}(E,F)$ is a groupoid. For every one of its objects the automorphism group is canonically isomorphic to $\mathrm{Ext}^0(Q,P) = \mathrm{Hom}(Q,P)$.

b) The set $\mathrm{Extpan}(E,F)$ of isomorphism classes of objects of $\mathrm{EXTPAN}(E,F)$ is empty, or is naturally a torsor under the group $\mathrm{Ext}^1(Q,P)$.

c) One can canonically define a class

$$c(E,F) \in \mathrm{Ext}^2(Q,P), \tag{9.3.9}$$

whose vanishing is necessary and sufficient for $\mathrm{Extpan}(E,F) \neq \varnothing$, that is, for the existence of a mixed extension of E by F.

It remains to prove (c). A mixed extension of E by F may also be described as an extension X of E by P together with an isomorphism of extensions of R by P

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$$X^1 \simeq F,$$

where X^1 denotes the image of the extension X under the morphism $R \longrightarrow E$ in (9.3.2). Such a mixed extension therefore exists if and only if there is an element of $\mathrm{Ext}^1(E,P)$ whose image in $\mathrm{Ext}^1(R,P)$ is the class ξ of F. Consider the exact sequence of the $\mathrm{Ext}^i(-,P)$ associated with the second exact sequence in (9.3.2):

$$\cdots \longrightarrow \mathrm{Ext}^1(E,P) \longrightarrow \mathrm{Ext}^1(R,P) \xrightarrow{\partial} \mathrm{Ext}^2(Q,P) \longrightarrow \cdots .$$

It is enough to define $c(E,F)$ by

$$c(E,F) = \partial\xi, \qquad \xi = \mathrm{cl}(F) \in \mathrm{Ext}^1(R,P). \tag{9.3.10}$$

Now suppose that we are given an exact functor

(9.3.11) $$r : \underline{C} \longrightarrow \underline{C}',$$

where $\underline{C}'$ is another abelian category. Write P', Q', R', E', F' for the objects of $\underline{C}'$ obtained from P, Q, R, E, F by applying r, so that E' is an extension of Q' by R' and F' is an extension of R' by P'. Up to canonical isomorphism, the following diagram of functors is commutative:

(9.3.12)
$$\begin{array}{ccc} \mathrm{EXT}(Q,P)\times\mathrm{EXTPAN}(E,F) & \xrightarrow{\ \omega\ } & \mathrm{EXTPAN}(E,F) \\ \downarrow & & \downarrow \\ \mathrm{EXT}(Q',P')\times\mathrm{EXTPAN}(E',F') & \xrightarrow[\ \omega'\ ]{} & \mathrm{EXTPAN}(E',F') \end{array} ,$$

The vertical arrows are the evident functors induced by the exact functor r, and ω' is the arrow analogous to ω in (9.3.5), with $\underline{C}$ replaced by $\underline{C}'$. The lower arrow in the printed diagram is labelled ω without a prime; the explanatory sentence identifies it as ω', which is the notation used here. We obtain analogous compatibilities, with the functor r, of the structures considered in 9.3.8; their explicit formulation is left to the reader.

We shall use the following consequence of this commutativity. Suppose that $\mathrm{EXTPAN}(E,F)$ is nonempty and choose an object X in it. Then, by means of $G' \longmapsto G' \wedge X'$, where $X' = r(X)$, the category $\mathrm{EXTPAN}(E',F')$ is equivalent to $\mathrm{EXT}(Q',P')$. Thus an object Y' of $\mathrm{EXTPAN}(E',F')$ is known up to isomorphism when the isomorphism class of the corresponding object of $\mathrm{EXT}(Q',P')$ is known, that is, when the corresponding element I ≈426

$$c_X(Y') \in \mathrm{Ext}^1(Q',P')$$

is known.

If X is replaced by another object $X_1 = H \wedge X$, where H is an extension of Q by P, then one immediately obtains

$$c_{X_1}(Y') = c_X(Y') - \mathrm{cl}(H')$$

from the associativity formula (9.3.7). Hence, for fixed Y' and varying X, the elements $c_X(Y')$ run through exactly one coset

(9.3.13) $$c(Y') \in \mathrm{Ext}^1(Q',P')/\operatorname{Im}\mathrm{Ext}^1(Q,P),$$

which is therefore canonically associated with the mixed extension Y' of E' by F'. By construction, its vanishing is necessary and sufficient for Y' to be isomorphic to the transform under r of a mixed extension X of E by F.

9.4. Let S be a henselian trait with generic point η, let n be an integer > 0, put $\Lambda = \mathbb{Z}/n\mathbb{Z}$, and let $\underline{C}$ be the category of Λ-modules on S (for the fppf site on S; cf. VIII 0.2) and $\underline{C}'$ the category of Λ-modules on η. Let P, R, Q be objects of $\underline{C}$, and let E, F be extensions of Q by R and of R by P, respectively (9.3.2). Suppose that Q is a locally free Λ_S-module of finite type and that P is represented by a finite group of multiplicative type over S (SGA 3 IX), so that it is of the form I 427

$$P = D(Q^*),$$

where $D = \underline{\mathrm{Hom}}(-, \mathbb{G}_{mS})$ denotes Cartier duality. Let $P_\eta, R_\eta, Q_\eta, E_\eta, F_\eta$ denote the restrictions of P, R, Q, E, F to η, and suppose that a mixed extension X_η of E_η by F_η is given. Under these conditions we shall canonically define a pairing

(9.4.1) $$c(X_\eta) : Q \otimes_\Lambda Q^* \longrightarrow \Lambda_S \qquad (\Lambda = \mathbb{Z}/n\mathbb{Z}).$$

For this purpose, first suppose that S is strictly local. Then Q and Q^* are constant, say $Q = Q_{oS}$ and $Q^* = Q^*_{oS}$, and giving (9.4.1) amounts to giving an element

(9.4.2) $$c(X_\eta) \in \check{Q}_o \otimes \check{Q}^*_o,$$

where the check denotes the ordinary dual $\mathrm{Hom}(-, \Lambda)$. We apply the considerations of 9.3 with the restriction functor $r : \underline{C} \longrightarrow \underline{C}'$. First observe that a mixed extension of E by F exists. Indeed, the obstruction to its existence lies in $\mathrm{Ext}^2(Q, P)$ (9.3.8(c)), and it is enough to prove

$$\mathrm{Ext}^2(Q, P) = 0.$$

This is a special case of the following lemma.

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Lemma 9.4.3. Under the preceding hypotheses, with S strictly local, one has:

a) canonical isomorphisms

$$\mathrm{Ext}^i_\Lambda(Q, P) \simeq H^i(S, \check{Q} \otimes P), \qquad \mathrm{Ext}^i_\Lambda(Q_\eta, P_\eta) \simeq H^i(\eta, \check{Q} \otimes P);$$

b) a canonical isomorphism

(b)) $$\check{Q} \otimes P \simeq D(Q \otimes Q^*);$$

c) for every finite constant group $H = H_{oS}$ over S,

(c)) $$H^i(S, D(H)) = H^i(\eta, D(H)_\eta) = 0 \qquad \text{for } i \geq 2;$$

d) canonical isomorphisms

(d)) $$H^1(S, D(H)) \simeq \mathbb{G}_m(S) \otimes \check{H}_o = V^* \otimes \check{H}_o,$$
$$H^1(\eta, D(H)_\eta) \simeq \mathbb{G}_m(\eta) \otimes \check{H}_o = K^* \otimes \check{H}_o,$$

where the check in (d) denotes $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$.

The formulas in (a) and (b) follow immediately from the fact that Q is a locally free Λ_S-module of finite type. To prove (c), we plainly reduce to the case $H_o = \mathbb{Z}/m\mathbb{Z}$, that is, $D(H) = \mu_m$. The cohomology sequence associated with the Kummer exact sequence

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m\cdot\mathrm{id}} \mathbb{G}_m \longrightarrow 0$$

then reduces us to proving

$$H^i(S, \mathbb{G}_m) = H^i(\eta, \mathbb{G}_m) = 0 \qquad \text{for } i \geq 1.$$

These groups are the groups $H^i_{\text{ét}}$ computed on the étale site [6, Theorem 11.7], and therefore vanish for S, since S is strictly local. Their vanishing for η is likewise well known [6, Theorem 1.1].

It remains to describe the isomorphisms in (d). Over an arbitrary base U there is a canonical homomorphism I ≈429

(9.4.3.1) $$\mathbb{G}_m(U) \otimes \check{H}_o \longrightarrow H^1(U, D(H_{oU})).$$

If H_o is annihilated by an integer $m > 0$, this homomorphism is described using the boundary map $\mathbb{G}_m(U) \xrightarrow{\partial} H^1(U, \mu_m)$ associated with the Kummer sequence: it is the composite

$$\begin{aligned}\mathbb{G}_m(U) \otimes \check{H}_o &\longrightarrow H^1(U, \mu_m) \otimes \check{H}_o \\ &\longrightarrow H^1(U, \mu_m \otimes \check{H}_o) \longrightarrow H^1(U, D(H_{oU})).\end{aligned}$$

Here the second arrow is deduced from the evident homomorphism $\check{H}_o \longrightarrow \mathrm{Hom}(\mu_m, \mu_m \otimes \check{H}_o)$, and the third is a special case of (b), with m in place of n. We leave it to the reader to check that (9.4.3.1) is independent of the choice of the integer m such that $mH_o = 0$. We also claim that (9.4.3.1) is always injective and that its cokernel vanishes if $\mathrm{Pic}(U) = 0$. (In fact, the cokernel is always isomorphic to $\mathrm{Hom}(H_o, \mathrm{Pic}(U))$, but this is immaterial here.) Indeed, we reduce to the case $H_o = \mathbb{Z}/n\mathbb{Z}$, and the assertion then follows immediately from the cohomology sequence of the Kummer sequence above. This proves (d), and immediately yields:

Corollary 9.4.4. With the notation of 9.4.3, there is a canonical isomorphism
e)

$$H^1(\eta, D(H)_\eta)/\operatorname{Im} H^1(S, D(H)) \simeq \check{H}_o,$$

and hence, on combining (a) and (b), a canonical isomorphism

(9.4.5) $$\mathrm{Ext}^1(Q_\eta, P_\eta)/\operatorname{Im} \mathrm{Ext}^1(Q, P) \simeq \check{Q}_o \otimes \check{Q}_o^*.$$

Returning to the definition of (9.4.2) when S is strictly local, the relation $\mathrm{Ext}^2(Q, P) = 0$ makes it possible to define the canonical element (9.3.13), taking for Y' the given mixed extension X_η of E_η by F_η. Formula (9.4.5) identifies this element $c(X_\eta)$ with an element of $\check{Q}_o \otimes \check{Q}_o^*$, thereby completing the definition of (9.4.2). The general considerations of 9.3 also show that this element is the obstruction to extending X_η to a mixed extension X of E by F. I ≈430

9.4.6. The definition of (9.4.1) when S is henselian now reduces immediately to the strictly local case by Galois descent from the strict henselization of S.

Proposition 9.4.7. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

(9.4.7.1) $$X \longmapsto X_\eta$$

from the category of mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every mixed extension X_η of E_η by F_η, consider the pairing

(9.4.7.2) $$c_o(X_\eta) : Q_o \otimes Q_o^* \longrightarrow \Lambda_s$$

induced by the pairing of type (9.4.1) defined over the henselization S^h of S. Then the functor (9.4.7.1) is fully faithful, and a mixed extension X_η of E_η by F_η belongs to its essential image if and only if the pairing (9.4.7.2) is zero.

9.4.7.3. This statement still makes sense and remains valid if one assumes only that S is a connected regular noetherian scheme of dimension 1. Instead of the single pairing (9.4.7.2), one must then consider a pairing $Q_s \otimes Q_s^* \longrightarrow \Lambda_s$ for each closed point s of S. It will be most convenient to prove 9.4.7 in this more general form. I 431

a) For every mixed extension X, the map

$$\mathrm{End}(X) \longrightarrow \mathrm{End}(X_\eta)$$

is bijective. The two sides are respectively the sections over S and η of $\underline{\mathrm{Hom}}(Q,P) = D(Q\otimes Q^*)$; hence the assertion follows from the normality of S and the finiteness of $D(Q\otimes Q^*)$ over S. It follows at once that if X and Y are two mixed extensions already known to be isomorphic, then

$$\mathrm{Hom}(X,Y) \longrightarrow \mathrm{Hom}(X_\eta, Y_\eta)$$

is bijective.

b) If X and Y are two mixed extensions, every isomorphism $X_\eta \xrightarrow{\sim} Y_\eta$ comes from an isomorphism $X \longrightarrow Y$. By (a), the question is local for the étale topology, so we may reduce to the case in which S is strictly local. In that case 9.4.3 shows that the map

$$\mathrm{Ext}^1_\Lambda(Q,P) \longrightarrow \mathrm{Ext}^1_\Lambda(Q_\eta, P_\eta)$$

identifies with

$$V^*\otimes \check{H}_o \longrightarrow K^*\otimes \check{H}_o, \qquad H_o = Q_o\otimes Q_o^*.$$

The latter is injective, since the exact sequence $0 \longrightarrow V^* \longrightarrow K^* \longrightarrow \mathbb{Z} \longrightarrow 0$ is split. By the discussion of 9.3 this implies that if X_η and Y_η are isomorphic, then so are X and Y; the conclusion now follows from (a).

Thus (a) and (b) show that $X \longmapsto X_\eta$ is <u>fully</u> faithful. I≈**432**

c) Let X_η be a mixed extension over η. We prove that it extends to S if and only if the pairings (9.4.7.2) vanish. Here again, the uniqueness already proved shows that the question is local for the étale topology, reducing us once more to the strictly local case. The conclusion then holds by construction, as observed immediately before 9.4.6.

This completes the proof of 9.4.7.

9.5. Fix a prime number ℓ, pro-ℓ Barsotti–Tate groups over S

$$P = (P(n))_{n\geq 0}, \qquad R = (R(n))_{n\geq 0}, \qquad Q = (Q(n))_{n\geq 0}, \tag{9.5.1}$$

and extensions of pro-ℓ Barsotti–Tate groups

$$0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0. \tag{9.5.2}$$

Thus, for every integer $n \geq 0$, these define extensions of Λ_n-modules, where $\Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}$:

$$\begin{aligned} &0 \longrightarrow P(n) \longrightarrow F(n) \longrightarrow R(n) \longrightarrow 0,\\ &0 \longrightarrow R(n) \longrightarrow E(n) \longrightarrow Q(n) \longrightarrow 0. \end{aligned} \tag{9.5.3}$$

We assume that Q is of étale type and P of multiplicative type. Finally, suppose that a pro-ℓ Barsotti–Tate group X_η is given over η which is a <u>mixed extension</u> of E_η by F_η. By this we mean that it is endowed, as a pro-ℓ Barsotti–Tate group, with a three-step filtration whose partial extensions are supplied with isomorphisms to E_η and F_η, respectively. It follows that, for every n, $X_\eta(n)$ is endowed with the structure of a mixed extension of $E(n)_\eta$ by $F(n)_\eta$. We may therefore associate with it a pairing of type (9.4.1):

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$$c(X(n)_\eta) : Q(n)\otimes Q^*(n) \longrightarrow \Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}, \tag{9.5.4}$$

where the $Q^*(n)$ are the finite étale groups over S defined by

$$P(n) \simeq D(Q^*(n)), \qquad \text{i.e.} \qquad Q^*(n) = D(P(n)),$$

and D again denotes Cartier duality. The $Q^*(n)$ form a pro-ℓ group in the evident way, denoted by Q^*. We leave it to the reader to verify that, as n varies, the homomorphisms (9.5.4) define a homomorphism of projective systems, that is, a homomorphism of twisted constant ℓ-adic sheaves over S:

$$c(X_\eta) : Q \otimes Q^* \longrightarrow \mathbb{Z}_{\ell S}. \tag{9.5.5}$$

We mention in passing the following result, which will not be used below.

Proposition 9.5.6. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

$$X \longmapsto X_\eta \tag{9.5.6.1}$$

from the category of Barsotti–Tate mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every Barsotti–Tate group X_η which is a mixed extension of E_η by F_η, consider the corresponding pairing

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$$Q_o \otimes Q_o^* \longrightarrow \mathbb{Z}_{\ell s} \tag{9.5.6.2}$$

induced by the pairing of type (9.5.5) defined over the henselization of S. Then the functor (9.5.6.1) is fully faithful, and its essential image consists of those Barsotti–Tate groups X_η which are mixed extensions of E_η by F_η and for which the corresponding pairing (9.5.6.2) is zero. Moreover, when $\operatorname{char} K = 0$ (so that Tate's theorem [33, Theorem 4] applies), X_η belongs to the essential image if and only if it extends to a Barsotti–Tate group over S.

9.5.6.3. We leave it to the reader to extend this statement to the case in which S is assumed only to be a connected regular noetherian scheme of dimension 1.

The first assertion of 9.5.6 follows immediately from 9.4.7, applied to the components of the Barsotti–Tate groups under consideration.

It remains to prove that if a Barsotti–Tate group X_η, which is a mixed extension of E_η by F_η, extends to a Barsotti–Tate group X over S, then it also extends as a mixed extension.

By Tate's theorem [33], the homomorphism $X_\eta \longrightarrow Q_\eta$ extends uniquely to a homomorphism

$$X \longrightarrow Q.$$

For every integer n, the corresponding homomorphism $X(n) \longrightarrow Q(n)$ is surjective on the generic fibres and hence is surjective. Since $Q(n)$ is étale over S, the map $X(n) \longrightarrow Q(n)$ is flat, and consequently it is an epimorphism. Thus $X \longrightarrow Q$ is an epimorphism component by component.

Let $\overline{F}$ be the kernel of $X \longrightarrow Q$. Then $\overline{F}$ is a Barsotti–Tate group and X is an extension of Q by $\overline{F}$. The isomorphism $F_\eta \xrightarrow{\sim} \overline{F}_\eta$ extends, by Tate's theorem, to an isomorphism $F \simeq \overline{F}$. Hence X is an extension of Q by F. Consider now the quotient $\overline{E} = X/P$. A further application of Tate's theorem gives an isomorphism $E \simeq \overline{E}$. It follows that X is a mixed extension of E by F extending the mixed extension X_η, as required.

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Remarks 9.5.6.4.

a) Of course, the last assertion of 9.5.6 should hold without restriction on K, since the same should be true of the invoked theorem of Tate.

b) One can give another proof of the good-reduction criterion 5.10. As in 5.10, reduce to the case where A_K has semistable reduction, and then apply 9.5.6 to $T_\ell(A_K)$ regarded as a mixed extension (cf. 9.6). This implies $u_\ell = 0$; using 11.5.2(a) below then gives $T_\ell(A_K)^t = 0$.

9.5.8. When $\ell \neq p$, the Barsotti–Tate groups under consideration over S and over K, respectively, may simply be interpreted in terms of twisted constant ℓ-adic sheaves over S, or equivalently in terms

of representations of $\pi_o = \mathrm{Gal}(\overline{k}/k)$ and $\pi = \mathrm{Gal}(\overline{K}/K)$, respectively, on free $\mathbb{Z}_\ell$-modules of finite type. One then sees that the homomorphism (9.5.5) is the canonical homomorphism expressing the action of the inertia subgroup $I \subset \pi$ on $X_\eta(\overline{K})$, as defined by the method of 9.2.

To verify this, one reduces immediately to the situation of 9.4, taking n there to be a power of ℓ and R to be a locally constant Λ-module (as are Q and P); the same is then true of E, F, and X_η. Verification that the two definitions agree is left to the reader.

9.6. We are now in a position to define the announced canonical pairing (9.1.2), without any restriction on ℓ, in such a way that, for $\ell \neq p$, this definition agrees with the one given in 9.2. For this purpose, we apply the considerations of 9.5 to the following situation: I 436

$$(9.6.1)\qquad \begin{cases} P = T_\ell(A^o)^t = \check{\underline{M}}'_\ell(1), \quad Q = \underline{M}_\ell, & R = T_\ell(A^o)^f/T_\ell(A^o)^t, \\ F = T_\ell(A^o)^f, & E = D(F') = D\big(T_\ell(A'^o)^f\big). \end{cases}$$

In the last formula, D denotes Cartier duality in the sense of pro-ℓ Barsotti–Tate groups; that is, $D(Y)$ is the projective system of the $D(Y(n))$, with the usual transition morphisms. The structure of F as an extension of R by P is clear. Likewise,

$$F' = T_\ell(A'^o)^f$$

has the structure of an extension of R' by $P' = T_\ell(A'^o)^t = \check{\underline{M}}_\ell(1)$. Applying Cartier duality, we conclude that $E = D(F')$ is endowed with the structure of an extension of $D(P') = \underline{M}_\ell$ by $D(R')$. Now $D(R')$ is canonically isomorphic to R by 5.5 (proved in 7.4.6); this specifies the structure of E as an extension of Q by R.

As the mixed extension X_η of E_η by F_η, we take I ≈437

$$(9.6.2)\qquad X_\eta = T_\ell(A_K) = X_\eta^0 \supset X_\eta^1 = T_\ell(A_K)^f \supset X_\eta^2 = T_\ell(A_K)^t \supset 0.$$

The isomorphism of extensions of R_η by P_η

$$X_\eta^1 \simeq F_\eta$$

is clear. The isomorphism of extensions of Q_η by R_η

$$X_\eta/X_\eta^2 \simeq E_\eta = D(F'_\eta)$$

is clear from the orthogonality theorem 5.2.

We therefore have the data required to define a pairing (9.5.5), which here, since $Q^* = \underline{M}'_\ell$, takes the form (9.1.2).

Its compatibility with the definition of 9.2 follows from 9.5.8. The considerations of 5.8 also show, subject to the validity of Tate's theorem [33] (hence at least when $\mathrm{char}\, K = 0$), that the pairing in question can be described in terms of the single pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ over η: its three-step filtration, and the extensions F and E extending X_η^1 and X_η/X_η^2, can be defined canonically in terms of X_η alone.

Remark 9.7. We no longer necessarily assume that A_K has semistable reduction over the henselian trait S. Let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction over the normalization S' of S in K' (3.6). With the notation of (8.1.8), Galois descent from the monodromy pairing for $A_{K'}$ then defines a pairing

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$$(*)\qquad v_\ell^{K'} : \underline{M}_K \otimes \underline{M}'_K \longrightarrow \mathbb{Z}_{\ell K}.$$

A priori this pairing depends on the choice of K', but it follows from (10.4.3) below that the pairing

(9.7.1) $$u_\ell : \underline{M}_K \otimes \underline{M'}_K \longrightarrow \mathbb{Q}_{\ell K}$$

given by

$$u_\ell = \frac{1}{e(S'/S)}\, v_\ell^{K'}$$

is independent of it, where

$$e(S'/S) = \text{length}(V'/\mathfrak{m}V').$$

We may again call (9.7.1) the monodromy pairing for the abelian scheme A_K over K and the prime number ℓ under consideration: when A_K itself has semistable reduction, it plainly coincides with the restriction to η of the monodromy pairing (9.1.2). It follows from 10.4 below that this pairing in fact always has values in $\mathbb{Q}_K$ and is independent of ℓ.

One may ask whether it has values in $\mathbb{Z}_K$, which is equivalent to saying that (9.7.1) has values in $\mathbb{Z}_{\ell K}$ for every prime number ℓ. This is so at least when A_K is an elliptic curve. The only case requiring verification is then the one in which $\underline{M}_K \simeq \underline{M'}_K$ has rank 1, so that $\underline{M}_K \otimes \underline{M'}_K$ is canonically isomorphic to $\mathbb{Z}_K$ and the monodromy pairing (9.7.1) can simply be interpreted as an ordinary rational number. It follows from the results of Néron [19] that this number is $v(j)$, where j is the absolute invariant of A_K and v is the normalized valuation of K.

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10. Properties of the monodromy pairing. Integrality and positivity theorem

In this section we shall first give certain essentially trivial formal properties of the monodromy pairings, leaving the (somewhat tedious) details of the verifications to the reader. We shall also state a deeper theorem (10.4) concerning these pairings and indicate how its proof may be reduced to a special case of Jacobians, to be treated (by transcendental methods) in section 12, which is in fact logically independent of section 11.

10.1. Transport of structure. It is clear from the definition of the monodromy pairing (9.1.2) that this pairing is compatible with transport of structure in the situation (S, A_K, A'_K).

10.2. Symmetry. Starting with the dual scheme A'_K of A_K and applying the construction of section 9 to it, we must introduce the dual scheme $(A'_K)'$ of A'_K, which is canonically isomorphic to A_K by the biduality theorem ([5] or VIII 3.2). Denote by $u_\ell^{A_K}$ the monodromy pairing (9.1.2) relative to A_K. In the same way we obtain a pairing

(10.2.1) $$u_\ell^{A'_K} : \underline{M'}_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

I claim that the latter is obtained from $u_\ell^{A_K}$ by the symmetry

$$s : \underline{M'}_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \underline{M'}_\ell;$$

that is,

(10.2.2) $$u_\ell^{A'_K} = u_\ell^{A_K} \circ s.$$

Taking account of the known anticommutativity property of the pairings φ_ξ ([5] or VIII 2.2.11), the symmetry formula (10.2.2) may be reformulated as follows in terms of a fixed polarization ξ of A_K. It defines a homomorphism I ≈440

$$\widetilde{\xi} : A_K \longrightarrow A'_K$$

and hence a corresponding homomorphism (t standing for "toric")

(10.2.3) $$\xi_t : \underline{M} \longrightarrow \underline{M}'.$$

We may therefore form the pairing

(10.2.4) $$\begin{aligned} \varphi^t_{\xi,\ell} : \underline{M}_\ell \otimes \underline{M}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi,\ell}(x,y) &= u_\ell^{A_K}(x, \xi_t(y)). \end{aligned}$$

The preceding form $\varphi^t_{\xi,\ell}$ is symmetric:

(10.2.5) $$\varphi^t_{\xi,\ell}(x,y) = \varphi^t_{\xi,\ell}(y,x).$$

In fact, this assertion is valid for every element ξ of the Néron–Severi group of A_K, not necessarily a polarization. But only when ξ is nondegenerate—that is, defines an isogeny $A_K \longrightarrow A'_K$, and hence also an isogeny (10.2.3)—is the symmetry relation (10.2.5) equivalent to (10.2.2). When ξ is a polarization, we shall moreover sharpen (10.2.5) considerably in 10.4(b) below.

Let us finally mention a third interpretation of the symmetry relation. Consider a biextension (that is, a divisorial correspondence) ξ on a pair $(A_K, \overline{A}_K)$ of abelian schemes with semistable reduction over K. It defines a homomorphism

$$A_K \longrightarrow \overline{A}'_K.$$

Passing to the Néron models, this defines a homomorphism on the maximal tori of the special fibres, and therefore, in the reverse direction, a homomorphism on their character groups:

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(10.2.6) $$\xi_t : \overline{\underline{M}} \longrightarrow \underline{M}',$$

where $\overline{\underline{M}}$ is defined from $\overline{A}_K$ as $\underline{M}$ is defined from A_K. We can therefore form the pairing

(10.2.7) $$\begin{aligned} \varphi^t_{\xi} : \underline{M}_\ell \otimes \overline{\underline{M}}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi}(x,y) &= u_\ell^{A_K}(x, \xi_t(y)). \end{aligned}$$

Notice in passing that, for a variable abelian scheme A_K with semistable reduction, the twisted constant group $\underline{M}$ is a covariant functor of A_K. Formation of the pairings (10.2.7) associated with biextensions of abelian schemes with semistable reduction over (K, S) is then compatible, in the evident sense, with taking inverse images of biextensions.

When ξ is the canonical ("Weil") biextension of (A_K, A'_K), the pairing (10.2.7) is of course precisely the monodromy pairing (9.1.2).

Now consider the transpose ${}^s\xi$ of ξ, a biextension of $(\overline{A}_K, A_K)$, and hence the pairing

$$\varphi^t_{{}^s\xi} : \overline{\underline{M}}_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

The symmetry relation is

(10.2.8) $$\varphi^t_{{}^s\xi} = \varphi^t_\xi \circ s, \qquad s : \overline{\underline{M}}_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \overline{\underline{M}}_\ell,$$

where s is again the symmetry isomorphism. Compare this relation with the antisymmetry relation of VIII 2.2.11.

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10.3. Change of trait. Let

(10.3.1) $$g : S' \longrightarrow S$$

be a dominant morphism of henselian traits, associated with an injective local homomorphism of discrete valuation rings

$$V \longrightarrow V'.$$

As usual, put

(10.3.2) $$e(S'/S) = \mathrm{length}_V(V'/\underline{m}V'),$$

where $\underline{m}$ is the maximal ideal of V. It is then known that the homomorphism induced by (10.3.1) on the "tame" parts of the inertia groups,

(10.3.3) $$g_* : I'_t \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k') \longrightarrow I_t = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k),$$

is given by

(10.3.4) $$g_*(x) = e(S'/S)x,$$

after the two sides of (10.3.3) have been identified by means of the evident isomorphisms $\mathbb{Z}_\ell(1)(k) \xrightarrow{\sim} \mathbb{Z}_\ell(1)(k')$.

On the other hand, if A_K is an abelian scheme over K with semistable reduction over S, formation of the twisted lattices $\underline{M}$ and $\underline{M}'$ commutes with every base change $S' \longrightarrow S$ as above, as follows immediately from the definitions (especially 3.4). We may therefore regard the pairing $u_\ell^{A_{K'}}$ defined by $A_{K'}$, where K' is the fraction field of V', as a pairing

$$u_\ell^{A_{K'}} : \underline{M}_{\ell S'} \otimes \underline{M}'_{\ell S'} \longrightarrow \mathbb{Z}_{\ell S'},$$

and compare it with the pairing $u_{\ell S'}^{A_K}$ obtained from $u_\ell^{A_K}$ by base change. When $\ell \neq p$, the construction of the pairings u_ℓ given in 9.2, together with formula (10.3.4), shows at once that

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(10.3.5) $$u_\ell^{A_{K'}} = e(S'/S)u_\ell^{A_K}.$$

In fact this formula remains valid for $\ell = p$. To see this, return to the general construction of 9.3–9.6 and reduce to the analogous compatibility assertion for the isomorphism 9.4.4(e), under a change of strictly local traits $S' \longrightarrow S$. Notice that the isomorphism in question is induced by the normalized valuation $K^*/V^* \simeq \mathbb{Z}$, and therefore gives the commutative diagram

$$\begin{array}{ccc} K^*/V^* & \xrightarrow{\sim} & \mathbb{Z} \\ \downarrow & & \downarrow{\scriptstyle e(S'/S)} \\ K'^*/V'^* & \xrightarrow{\sim} & \mathbb{Z}. \end{array}$$

Here is a deeper result concerning the monodromy pairings. We shall prove it by transcendental methods in section 12, relying on results that will be proved later in the Seminar. (We shall not, in fact, use 10.4 in the remainder of the Seminar, any more than we shall use the other results of the present exposé.)

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Theorem 10.4. Let S be a henselian trait, and let A_K be an abelian scheme over K with semistable reduction over S. Consider the lattices $\underline{M}$ and $\underline{M}'$ associated as in 9.1 (cf. 5.5.2). Then the following assertions hold:

a) There exists a unique pairing

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S \tag{10.4.1}$$

such that, for every prime number ℓ, $u \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is the monodromy pairing u_ℓ of (9.1.2). This pairing is separating.

b) Let ξ be a polarization of A_K, thus defining a homomorphism (10.2.3)

$$\xi_t : \underline{M} \longrightarrow \underline{M}',$$

and consider the form

$$\begin{aligned} \varphi_\xi^t &: \underline{M} \otimes \underline{M} \longrightarrow \mathbb{Z}_S, \\ &\varphi_\xi^t(x,y) = u(x, \xi_t(y)). \end{aligned} \tag{10.4.2}$$

This form is symmetric, positive, and nondegenerate.

Of course, in terms of the geometric fibre M_o of $\underline{M}$ at a given geometric point of S, the preceding assertion says that the corresponding form

$$M_o \otimes M_o \longrightarrow \mathbb{Z}$$

is symmetric, positive, and nondegenerate. The pairing (10.4.1) will again be called the monodromy pairing.

10.5. Reduction of 10.4 to a particular case **I 445**

10.5.1. The uniqueness of a u satisfying the stated condition (even for a single ℓ) is clear. On the other hand, existence simply means that every u_ℓ in fact maps $\underline{M} \otimes \underline{M}'$ into $\mathbb{Z}_S$, and that the homomorphism thereby obtained

$$\underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S$$

is independent of ℓ. In fact it even suffices to prove the analogous assertion for the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and for a homomorphism

$$u_{\mathbb{Q}} : \underline{M}_{\mathbb{Q}} \otimes \underline{M}'_{\mathbb{Q}} \longrightarrow \mathbb{Q}_S.$$

Suppose, then, that existence has been established after a base change $S' \longrightarrow S$ as in 9.3. It follows immediately from (10.3.5) that there exists a homomorphism

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \frac{1}{e(S'/S)} \mathbb{Z}_S \subset \mathbb{Q}_S$$

compatible with the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$; this proves a). As for b), it is then plain that, if it holds for $A_{K'}$, it also holds for A_K.

Consequently, in proving 10.4 we may make any dominant change of traits $S' \longrightarrow S$. In particular, this allows us to suppose that S is complete with algebraically closed residue field. In that case $\underline{M}$ and $\underline{M}'$ are constant group schemes, with values M and M', finitely generated free $\mathbb{Z}$-modules, and a) and b) may be reformulated more conveniently in terms of the latter.

10.5.2. Suppose now that A_K is isomorphic to a direct factor of an abelian scheme $\overline{A}_K$ over K with semistable reduction, and that 10.4 has been proved for the latter. It follows at once that 10.4 also holds for A_K, in view of the evident functorial behaviour of the lattices $\underline{M}, \underline{M}'$ and of the monodromy pairings with respect to direct sums of abelian schemes with semistable reduction.

It is equally immediate from 10.5.1 that the validity of 10.4 depends only on the isogeny class of A_K. Together with the preceding remark and the "Poincaré reducibility theorem" [15], this shows that, if A_K is a quotient of an abelian scheme $\overline{A}_K$ for which 10.4 holds, then 10.4 also holds for A_K. I≈446

10.5.3. We now use the well-known fact [31, Th. 11 (p. 20)] that, after making a finite extension of K if necessary (which is permissible by 10.5.1), every abelian scheme A_K is a quotient of the Jacobian

$$\underline{\mathrm{Pic}}^0_{X_K/K}$$

of a proper, smooth, geometrically connected curve X_K over K. Thus, by 10.5.2, we are reduced to proving 10.4 for such a Jacobian.

10.5.4. By a theorem of Artin–Mumford [3] (already cited and used in Exposé V), after making a further finite extension K' of K if necessary, we may suppose that X_K extends to a projective and flat scheme over S which is regular and whose geometric special fibre $X_{\bar{s}}$ has no singularities other than ordinary quadratic singularities (that is, here, normal crossings of two branches).

In this case we shall prove 10.4 a) in 12.5 below, by describing the monodromy pairing (10.4.1) completely in terms of the geometric data of the situation; at the same time we shall verify 10.4 b) for the canonical polarization of the Jacobian.

10.5.5. It remains to see that, if for a given A_K with semistable reduction assertion 10.4 a) holds and assertion 10.4 b) holds for one given polarization ξ_o of A_K, then the same conclusion holds for every polarization ξ of A_K. Indeed, let I 447

$$\widetilde{\xi}_o, \widetilde{\xi} : A_K \longrightarrow A'_K$$

be the homomorphisms defined by ξ_o and ξ; these are isogenies. Then

$$\widetilde{\xi} = \widetilde{\xi}_o \alpha = {}^t(\alpha^*)\widetilde{\xi}_o, \qquad \alpha \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

where $\alpha \longmapsto \alpha^*$ denotes the canonical involution of the algebra $\mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q}$ defined by the polarization ξ_o, and the superscript t in ${}^t(\alpha^*)$ denotes transpose (associating with an endomorphism of A_K an endomorphism of A'_K). On the other hand, it is known [15, Chap. V, Th. 3] that

$$\alpha = \sum_i \beta_i^* \beta_i, \qquad \beta_i \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

and hence

$$\widetilde{\xi} = \sum_i {}^t\beta_i \, \widetilde{\xi}_o \, \beta_i.$$

Equivalently, in terms of the biextensions (that is, divisorial correspondences) associated with the polarizations under consideration (VIII 4.14),

$$\delta(\xi) = \sum_i \delta(\xi_o) \circ (\beta_i, \beta_i). \tag{10.5.5.1}$$

By what was said after (10.2.7), this formula implies (with the notation of (10.4.2))

$$\varphi^t_\xi = \sum_i \varphi^t_{\xi_o}(\beta_i, \beta_i) \in \mathrm{Hom}(M_{\mathbb{Q}} \otimes_{\mathbb{Q}} M_{\mathbb{Q}}, \mathbb{Q}). \tag{10.5.5.2}$$

It indeed follows that, if $\varphi^t_{\xi_o}$ is symmetric and positive, then so is φ^t_ξ.

Finally, the nondegeneracy of $\varphi^t_{\xi_o}$ is plainly equivalent to the assertion that the pairing (10.4.1) is separating (since $\xi_{ot} : M \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow M' \otimes_{\mathbb{Z}} \mathbb{Q}$ is an isomorphism, $\widetilde{\xi}_o$ being an isogeny), and is therefore likewise independent of the chosen polarization. One may also observe that, in order to verify that (10.4.1) is separating, it suffices to verify the same assertion for the monodromy pairing u_ℓ of (9.1.2) I≈448

for one prime number ℓ. Taking $\ell \neq p$, one may use the description of u_ℓ in 9.2; the fact that u_ℓ is separating on the left (and hence on both sides, for reasons of rank) is then immediate.

11. Connected components of the Néron model: duality relation and asymptotic behaviour

11.0. The purpose of the present section is to study the finite étale k-group

$$\Phi_o = A_o/A_o^o \tag{11.0.1}$$

already considered in section 1. For every prime number ℓ, denote by

$$\Phi_o(\ell)$$

its ℓ-primary component. Thus knowledge of Φ_o is equivalent to knowledge of the $\Phi_o(\ell)$, by the formula

$$\Phi_o = \coprod_\ell \Phi_o(\ell). \tag{11.0.2}$$

Notice that the groups Φ_o and $\Phi_o(\ell)$ are determined, respectively, by the ordinary finite groups I≈449
$\Phi_o(\bar{k})$ and $\Phi_o(\ell)(\bar{k})$ together with the natural action of

$$\pi_o = \mathrm{Gal}(\bar{k}/k)$$

on them. On the other hand, it is clear that $\Phi_o(\bar{k})$ is also the group of connected components of the Néron model of $A_{\widetilde{K}}$, where $\widetilde{K}$ is the fraction field of the strict henselization of S relative to the chosen separable closure $\bar{k}$ of k. Thus, in studying Φ_o, we may reduce to the case where S is strictly local; the actions of

$$\pi_o \simeq \mathrm{Gal}(\widetilde{K}/K)$$

on $\Phi_o(\bar{k})$ are then recovered automatically by transport of structure. We could likewise reduce further to the case where S is complete, using the base change $\widehat{S} \longrightarrow S$ and the fact that formation of the Néron model commutes with it [21, Th. 3.4 b)].

11.1. Suppose, then, that S is strictly local, so that Φ_o may be identified with the ordinary finite group $\Phi_o(k)$. We obtain

$$\Phi_o(k) \simeq A_o(k)/A_o^o(k) \simeq A(S)/A^o(S) \simeq A(K)/A(K)^o, \tag{11.1.1}$$

where the first equality follows from the fact that A_o is smooth over k, the second from Hensel's lemma since A is smooth over S and S is henselian, and the third from the relation $A(S) \xrightarrow{\sim} A(K)$ and from the convention

$$A(K)^o := A^o(S). \tag{11.1.2}$$

Thus $A(K)^o$ appears as a natural subgroup of finite index in $A(K)$ (and is not to be confused with $A^o(K) = A(K)$). The isomorphisms (11.1.1) are plainly functorial in A and compatible with transport of structure in (S, A_K), and in particular with the action of any Galois groups.

Now let ℓ be a prime number different from p. Then A^o is ℓ-divisible; more precisely, $\ell\,\mathrm{id}_{A^o}$ is an I≈450
étale surjective morphism. It consequently induces a surjective endomorphism of $A^o(S) = A(K)^o$, which is therefore an ℓ-divisible group. It follows at once that, for every integer $n \geq 0$ prime to $p = \mathrm{char}(k)$,

(11.1.3) $$ {}_n\Phi_o(k) \simeq {}_nA(K)/{}_nA(K)^o, \qquad (n,p)=1. $$

By the definition (2.2.3) of the fixed part of ${}_nA_K$, we have

(11.1.4) $$ {}_nA(K)^o = ({}_nA_K)^f(\bar{K}) \simeq T_n(A_K)^f(\bar{K}) \otimes_{\mathbb{Z}_n} \mathbb{Z}/n\mathbb{Z}. $$

Taking n to be a power of ℓ, and putting as in (2.5.3)

(11.1.5) $$ U = T_\ell(A(\bar{K})), \qquad T_\ell(A(\bar{K}))^f = U^I, $$

where U^I denotes the submodule of U consisting of invariants under the inertia group I, we may write (11.1.3) in the form

(11.1.6) $$ {}_n\Phi_o(k) \simeq (U_n)^I/(U^I)_n, \qquad n = \ell^{r+1}, $$

where the subscript n denotes reduction modulo n, that is, tensor product over $\mathbb{Z}_\ell$ with $\mathbb{Z}_\ell/n\mathbb{Z}_\ell$. The transition morphisms between the groups occurring on the two sides of (11.1.6) being evident, passage to the inductive limit over the powers of ℓ gives the following result.

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Proposition 11.2. Suppose that S is strictly local, and let ℓ be a prime number different from p. Then there is a canonical isomorphism

(11.2.1) $$ \Phi_o(\ell) \simeq \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell}, $$

where U is defined in (11.1.5) and $D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell$.

Thus the group $\Phi_o(\ell)$ can be described solely in terms of the Tate module $U = T_\ell(A(\bar{K}))$, regarded as a Galois module under the action of the inertia group I.

11.3. We shall use the explicit expression (11.2.1) to sketch the definition of a canonical pairing

(11.3.1) $$ \Phi_o(\ell) \otimes \Phi'_o(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z} $$

which is a perfect duality; here Φ'_o denotes, as in section 1, the group of connected components associated with the Néron model of the dual A'_K of A_K. There is little doubt (although the author has dispensed with the verification) that this pairing is none other than, possibly up to sign, the one defined in (1.2.1), and the reader is invited, as an exercise, to establish the required compatibility (with the correct sign!).

Put

$$ U' = T_\ell(A'(\bar{K})), \qquad T = T_\ell(1)(\bar{K}) = T_\ell(\bar{K}^*). $$

There is a perfect pairing of finitely generated free $\mathbb{Z}_\ell$-modules, compatible with the actions of I,

(11.3.2) $$ \varphi : U \otimes U' \longrightarrow T, \qquad (T \simeq \mathbb{Z}_\ell \text{ noncanonically}), $$

and, by the known structure of I, an exact sequence

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(11.3.3) $$ 1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1, $$

where R is a profinite group prime to ℓ. Starting now from the data (11.3.2) and (11.3.3), introduce the groups

$$\Phi = \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell}, \qquad \Phi' = \frac{(U' \otimes D_\ell)^I}{U'^I \otimes D_\ell}, \qquad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.3.4}$$

We shall define a perfect duality

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}/\mathbb{Z}, \tag{11.3.5}$$

which will in particular give the announced pairing (11.3.1). Consider the exact sequence of I-modules

$$0 \longrightarrow U \xrightarrow{i} U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \xrightarrow{j} U \otimes D_\ell \longrightarrow 0 \tag{11.3.6}$$

and the associated exact cohomology sequence, where the cohomology of I is defined by continuous cochains with coefficients in the topological groups under consideration:

(11.3.7)

$$\begin{array}{ccccccccc} 0 \longrightarrow & H^o(I,U) & \xrightarrow{i^o} & H^o(I, U\otimes\mathbb{Q}_\ell) & \xrightarrow{j^o} & H^o(I, U\otimes D_\ell) & \xrightarrow{\partial} & H^1(I,U) & \xrightarrow{i^1} & H^1(I, U\otimes\mathbb{Q}_\ell) \\ & \big\downarrow\wr & & \big\downarrow\wr & & \big\downarrow\wr & & & & \big\downarrow\wr \\ & U^I & & (U\otimes\mathbb{Q}_\ell)^I \simeq U^I\otimes_{\mathbb{Z}}\mathbb{Q} & & (U\otimes D_\ell)^I & & & & H^1(I,U)\otimes_{\mathbb{Z}}\mathbb{Q}. \end{array}$$

It is immediate that the group Φ defined by (11.3.4) is precisely $\operatorname{coker} j^o$, and is therefore canonically isomorphic to $\ker i^1$. Thus we obtain a canonical exact sequence

$$0 \longrightarrow \Phi \longrightarrow H^1(I,U) \longrightarrow H^1(I,U)\otimes_{\mathbb{Z}}\mathbb{Q}, \tag{11.3.8}$$

which shows that Φ is canonically isomorphic to the torsion subgroup of $H^1(I,U)$.

On the other hand, from the exact sequence (11.3.3) we deduce, for every $n = \ell^r$, an isomorphism I≈453

$$H^1(I,T_n) \simeq \operatorname{Hom}(I,T_n) \xleftarrow{\sim} \operatorname{Hom}(T,T_n) \xleftarrow{\sim} \mathbb{Z}/n\mathbb{Z}. \tag{11.3.9}$$

(Notice that I acts trivially on T, and hence on $T_n = T/nT$.) It follows that, for every finite I-$\mathbb{Z}_\ell$-module M, if

$$M' = \operatorname{Hom}(M, T\otimes D_\ell) \tag{11.3.10}$$

denotes its "Cartier dual," there is a canonical pairing

$$H^i(I,M)\otimes H^{1-i}(I,M') \longrightarrow H^1(I, T\otimes D_\ell) \simeq D_\ell, \tag{11.3.11}$$

where the last isomorphism is deduced from the isomorphisms (11.3.9) by passage to the inductive limit over the powers n of ℓ. It is well known that these pairings are perfect dualities (see, for example, SGA 5 I 5). Taking M in these pairings to be of the form U_n, so that $M' = U'_n$, and passing to the limit, we obtain a perfect pairing

$$H^o(I, U\otimes D_\ell) \times H^1(I,U') \longrightarrow D_\ell, \tag{11.3.12}$$

where the second factor on the left is a finitely generated $\mathbb{Z}_\ell$-module. A further passage to the limit in the pairings (11.3.12), for transition morphisms of the form ℓ^r id, gives a second perfect pairing

$$H^o(I, U \otimes \mathbb{Q}_\ell) \times H^1(I, U' \otimes \mathbb{Q}_\ell) \longrightarrow D_\ell \tag{11.3.13}$$

of finite-dimensional vector spaces over $\mathbb{Q}_\ell$. Let i' and j' denote the morphisms defined as in (11.3.6), with U replaced by U'. It is then easy to see that the following homomorphisms

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$$H^o(I, U \otimes \mathbb{Q}_\ell) \xrightarrow{\quad j^o \quad} H^o(I, U \otimes D_\ell)$$

$$H^1(I, U' \otimes \mathbb{Q}_\ell) \xleftarrow{\quad {i'}^1 \quad} H^1(I, U')$$

are transposes of one another under the pairings (11.3.12) and (11.3.13). It follows immediately that $\operatorname{coker} j^o$ and $\ker {i'}^1$ are dual to one another by a canonical pairing with values in D_ℓ; this is the announced pairing (11.3.5).

11.4. The preceding description of a pairing (11.3.1), by means of the expression (11.2.1) for the ℓ-primary component of the group of connected components, is valid only when $\ell \neq p$. Suppose now that A_K has semistable reduction over S. We shall again define pairings that are perfect dualities

$$\Phi_o(\ell) \otimes \Phi'_o(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z} \tag{11.4.1}$$

without any restriction on ℓ. As in 11.3, one should establish that this recovers (up to a sign, perhaps, which ought to be specified) the pairing defined in (1.2.1). This would therefore establish Conjecture 1.3 in the case of semistable reduction, just as it does when one restricts to the ℓ-primary components for $\ell \neq p$.

To define (11.4.1), we shall give, essentially, an expression for $\Phi_o(\ell)$ in terms of the Barsotti–Tate group $T_\ell(A_K)$ over K; in the present case this will play the same role as 11.2 did in the case treated above.

Theorem 11.5. Suppose that S is henselian and that A_K has semistable reduction over S. Let ℓ be a prime and consider the monodromy pairing (9.1.2), and hence the canonical homomorphism I 455

$$\widetilde{u}_\ell : \underline{M}_\ell \longrightarrow \check{\underline{M}}'_\ell.$$

Put $\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell$. Then there is a canonical isomorphism

$$\Phi_o(\ell) \simeq \Phi(\ell) \times_S s. \tag{11.5.1}$$

This statement immediately gives the definition of a perfect pairing (11.4.1). Indeed, applying 11.5 to A'_K in place of A_K and using (10.2.2), we also obtain a canonical isomorphism

$$\Phi'_o(\ell) \simeq \Phi'(\ell) \times_S s, \qquad \Phi'(\ell) = \operatorname{coker}({}^t\widetilde{u}_\ell : \underline{M}'_\ell \longrightarrow \check{\underline{M}}_\ell).$$

Remarks 11.5.2. a) Theorem 11.5 implies in particular that $\widetilde{u}_\ell$ is an isogeny, that is, that u_ℓ is separating, including in the case $\ell = p$, where this assertion is not immediate from the definition in 9.6. Notice that we do not use here the (considerably deeper) Theorem 10.4, which will be proved in the next section.

b) Using 10.4, which supplies us with a monodromy pairing

$$u : \underline{M} \otimes \underline{M'} \longrightarrow \mathbb{Z}_S,$$

and denoting by Φ the cokernel of the corresponding homomorphism

$$\widetilde{u} : \underline{M} \longrightarrow \underline{\check{M}}',$$

we obtain an equivalent formulation of 11.5, independent of the choice of ℓ, in the canonical isomorphism

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$$\Phi_o \simeq \Phi \times_S s. \tag{11.5.3}$$

11.6. To define the isomorphism (11.5.1), we may restrict ourselves to the case in which S is strictly local (11.0), so that $\underline{M}$ and $\underline{M'}$ are constant, with values

$$M = D(T'_o), \qquad M' = D(T_o), \tag{11.6.1}$$

where T_o and T'_o are respectively the maximal tori of A_o and A'_o. Likewise identifying Φ_o with an ordinary finite group, we must define a canonical isomorphism between its component $\Phi_o(\ell)$ and

$$\Phi(\ell) = \operatorname{coker}(\widetilde{u}_\ell : M_\ell \longrightarrow \check{M}'_\ell).$$

In fact, we shall directly define a canonical isomorphism

$$\Phi_o(\ell) \simeq \ker(\widetilde{u}_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell). \tag{11.6.2}$$

Since $\Phi_o(\ell)$ is finite and M_ℓ and $\check{M}'_\ell$ have the same rank, it will first follow that $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ is an isogeny. It then follows that the right-hand side of (11.6.2) is canonically isomorphic to $\Phi(\ell)$:

$$\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell \simeq \ker(\widetilde{u}_\ell \otimes D_\ell). \tag{11.6.3}$$

Indeed, identifying M_ℓ by means of $\widetilde{u}_\ell$ with a finite-index submodule of $\check{M}'_\ell$, we may write the source and target of $\widetilde{u}_\ell \otimes D_\ell$ as

$$M_\ell \otimes D_\ell = V/M_\ell, \qquad \check{M}'_\ell \otimes D_\ell = V/\check{M}'_\ell,$$

where

$$V = M_\ell \otimes \mathbb{Q}_\ell \simeq \check{M}'_\ell \otimes \mathbb{Q}_\ell.$$

Composing (11.6.2) with (11.6.3) now immediately gives the isomorphism (11.5.1) to be defined, so everything reduces to defining the canonical isomorphism (11.6.2).

Writing $M_\ell \otimes D_\ell$ (respectively, $\check{M}'_\ell \otimes D_\ell$) as the inductive limit of the M_n (respectively, the $\check{M}'_n$), where n ranges over the powers of ℓ, it is equivalent to define canonical isomorphisms I≈**457**

$${}_n\Phi_o \xrightarrow{\sim} \ker(\widetilde{u}_n : M_n \longrightarrow \check{M}'_n), \qquad n = \ell^{r+1}, \tag{11.6.4}$$

compatible with the evident transition morphisms as n varies. Here, by an abuse of notation, $\widetilde{u}_n$ now denotes the homomorphism obtained by reducing $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ modulo n.

For this purpose, consider the group scheme ${}_nA$ over S. By the semistable-reduction hypothesis it is flat and quasi-finite over S (2.2.1). There is a canonical filtration

(11.6.5) $$ {}_nA \supset ({}_nA^o)^f \supset ({}_nA^o)^t \supset 0, $$

where the superscript f (respectively, t) denotes the "fixed part" (2.2.3) (respectively, the "toric part" (5.1)) of the group scheme to which it is applied. In the definition of the toric part in 5.1, we had assumed S complete; this is permissible here (11.0). (We saw moreover in 6.1 that the construction generalizes to the henselian case, if one insists.) Put

(11.6.6) $$ {}_n\Psi = {}_nA/({}_nA^o)^f. $$

This is a flat, separated, quasi-finite group scheme over S (SGA 3 VI$_A$ 3.2 and VI$_B$ 9.2), with generic fibre

$$ {}_n\Psi_K = ({}_nA_K)/({}_nA_K)^f, $$

where, by definition, $({}_nA_K)^f = ({}_nA^o)^f_K$ denotes the r-th constituent ($n = \ell^{r+1}$) of the Barsotti–Tate pro-group $T_\ell(A_K)^f$. The first of relations (5.5.8) therefore gives a canonical isomorphism

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(11.6.7) $$ {}_n\Psi_K \xrightarrow{\sim} \underline{M}_{nK}, \qquad \underline{M}_n := \underline{M} \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}). $$

In fact, we have:

Lemma 11.6.8. The group scheme ${}_n\Psi$ over S defined in (11.6.6) is *étale* over S, and the isomorphism (11.6.7) extends uniquely to a morphism

(11.6.8.1) $$ {}_n\Psi \hookrightarrow \underline{M}_n $$

which is an open immersion.

To see that ${}_n\Psi$ is étale, in view of (11.6.7) it remains only to verify this at the points of its special fibre, or equivalently to verify that this fibre is étale. This is immediate, since the fibre in question is

$$ {}_nA_o/{}_nA_o^o \subset A_o/A_o^o = \Phi_o. $$

It follows that ${}_n\Psi$ is normal; and since $\underline{M}_n$ is finite over S, the existence of the extension (11.6.8.1) follows at once. This is a morphism of étale group schemes over S, so its kernel is étale over S. Since ${}_n\Psi$, and hence this kernel, is separated, while its generic fibre is the unit group by (11.6.7), the kernel is the unit group (3.1.3). Thus (11.6.8.1) is an étale monomorphism, hence an open immersion (EGA IV 17.9.1), proving the lemma.

Now put

(11.6.9) $$ {}_n\Phi = ({}_nA)^f/({}_nA^o)^f \subset {}_n\Psi. $$

Thus ${}_n\Phi$ is a finite open subscheme of ${}_n\Psi$, since $({}_nA)^f$ is a finite open subgroup scheme of ${}_nA$. More precisely, it is immediate that

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(11.6.10) $$ {}_n\Phi \xrightarrow{\sim} ({}_n\Psi)^f. $$

Composing (11.6.9) with (11.6.8.1), we obtain an open immersion

(11.6.11) $$ {}_n\Phi \hookrightarrow \underline{M}_n. $$

On the other hand,

$$ {}_n\Phi \times_S s = {}_n\Psi \times_S s = {}_nA_o/{}_nA_o^o, $$

and, since A_o^o is n-divisible (2.2.1), the last member is immediately identified with ${}_n\Phi_o$. We therefore obtain an isomorphism

$$ {}_n\Phi \times_S s \xrightarrow{\sim} {}_n\Phi_o, \tag{11.6.12} $$

which shows that ${}_n\Phi$ is characterized, up to unique isomorphism, as the finite étale scheme over S whose special fibre is ${}_n\Phi_o$. Thus, to define the desired isomorphism (11.6.4) by means of the open immersion (11.6.11), it is enough to prove the following.

Lemma 11.6.13. The image of the canonical homomorphism (11.6.11) is equal to the kernel of

$$ \widetilde{u}_n : \underline{M}_n \longrightarrow \underline{\check{M}}'_n. $$

The required compatibility of the homomorphisms (11.6.11) with the transition homomorphisms as n varies is immediate from the definitions. Consequently Lemma 11.6.13 will complete the proof of Theorem 11.5. I 455

To prove the lemma, note that I ≈460

$$ K_n = \ker \widetilde{u}_n $$

is also the left annihilator of the canonical pairing

$$ u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S, $$

whose explicit definition is given in 9.4 and 9.6 in terms of the extensions

(11.6.14)

$$ 0 \longrightarrow \underline{\check{M}}'_n(1) \longrightarrow F_n \longrightarrow R_n \longrightarrow 0 \qquad , \qquad 0 \longrightarrow \underline{\check{M}}_n(1) \longrightarrow F'_n \longrightarrow R'_n \longrightarrow 0 $$

$$ \Big\| \qquad\qquad\qquad\qquad\qquad\qquad \Big\| $$

$$ ({}_nA^o)^f \qquad\qquad\qquad\qquad ({}_nA'^o)^f $$

(where R_n and R'_n are Cartier dual to one another), and in terms of the mixed extension $X_{n\eta}$ of $E_{n\eta}$ by $F_{n\eta}$, which is the generic fibre of (11.6.5), where we put

(11.6.15)

$$ E_n = D(F'_n), \qquad 0 \longrightarrow R_n \longrightarrow E_n \longrightarrow \underline{M}_n \longrightarrow 0 \qquad . $$

$$ \wr \Big| $$

$$ D(R'_n) $$

It follows immediately from Proposition 9.4.7 that K_n is the largest among the finite étale subgroups K'_n of $\underline{M}_n$ for which the inverse image $\overline{X}_{n\eta}$ of K'_n in the mixed extension $X_{n\eta}$, regarded as a mixed extension of $\overline{E}_{n\eta}$ by $F_{n\eta}$, extends to a mixed extension of $\overline{E}_n$ by F_n. Here $\overline{E}_n$ is the evident subgroup extension of K'_n by R_n.

This condition is plainly satisfied for the subgroup ${}_n\Phi$ of $\underline{M}_n$ in (11.6.11), since the corresponding mixed extension $({}_nA)^f$ over S is available. Hence ${}_n\Phi \subset K_n$. To prove the opposite inclusion, consider the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n extending $\overline{X}_{n\eta}$, defined as above by taking $K'_n = K_n$. Thus it is an extension

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$$0 \longrightarrow F_n \longrightarrow \overline{X}_n \longrightarrow K_n \longrightarrow 0,$$

and there is a canonical homomorphism

(*) $$F_n = ({}_nA^o)^f \hookrightarrow A,$$

which on the generic fibre is induced by the canonical homomorphism

(**) $$\overline{X}_{n\eta} \hookrightarrow X_{n\eta} = {}_nA_K \hookrightarrow A_K.$$

It follows immediately from (5.9.2) that the homomorphism (∗∗) extends to a unique homomorphism

(***) $$\overline{X}_n \longrightarrow A$$

inducing (∗). Since this last homomorphism maps the generic fibre into ${}_nA$, which is a closed subscheme of A, it factors through $\overline{X}_n \to {}_nA$; and since $\overline{X}_n$ is finite, it factors through $({}_nA)^f$. Looking at the induced homomorphism on generic fibres, we conclude that

$$\overline{X}_{n\eta} \subset ({}_nA)^f_\eta, \quad \text{i.e.} \quad K_{n\eta} \subset {}_n\Phi_\eta, \quad \text{i.e.} \quad K_n \subset {}_n\Phi.$$

This proves the lemma.

Remarks 11.7. The construction made in 11.6 shows that, from the extensions (11.6.14) over S and the mixed extension $X_{n\eta} = {}_nA_K$ of E_n by F_n, one can reconstruct, up to unique isomorphism, the mixed extension (11.6.5), and hence also the subgroup

$$({}_nA)^f \supset ({}_nA^o)^f$$

of ${}_nA$. Indeed, by 9.4 these data determine the pairing

$$u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S,$$

and the preceding construction gives $({}_nA)^f$ as the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n, extending the mixed extension $\overline{X}_{n\eta}$ induced by $X_{n\eta}$. Here $\overline{E}_n$ is the inverse image of $\ker \widetilde{u}_n$ under $E_n \to \underline{M}_n$. By (9.4.7), this mixed extension $\overline{X}_n$ is thus determined up to unique isomorphism. We then recover ${}_nA$ by gluing $\overline{X}_n$ and $X_{n\eta}$ along their common open subscheme $\overline{X}_{n\eta}$.

Thus, if the structural data (9.6.1) and the mixed extension I≈462

$$X_\eta = T_\ell(A_K)$$

of E_η by F_η are known—equivalently, if the extensions (11.6.14) and $X_{n\eta}$ are known for every $n = \ell^{r+1}$, $r \geq 0$—then one can reconstruct the inductive (or, if preferred, projective) system of mixed extensions (11.6.5).

11.8. As in 8.3, regard the separable closure $\overline{K}$ of K as the inductive limit of its finite subextensions K'. For every such K', let $A[K']$ denote the Néron model of

$$A_{K'} = A_K \otimes_K K'.$$

It gives a finite étale group scheme

(11.8.1) $$\Phi_o[K'] = A[K']_o/A[K']^o_o$$

of connected components, defined over the residue field of the normalization $S[K']$ of S in K'. For simplicity, suppose that the residue field of S is separably closed, so that the same is true of the

residue fields of the $S[K']$. The groups (11.8.1) may then be identified with ordinary finite groups $\Phi[K']$, and it is immediate that, as K' varies, these form an inductive system, which we now propose to study. We shall be interested in particular in the group

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$$\Psi = \varinjlim_{K'} \Phi[K']. \tag{11.8.2}$$

We know from 3.6 that, for K' sufficiently large, say $K' \supset K'_o$, $A[K']$ has semistable reduction. Hence for $K' \supset K'_o$ the transition morphisms are injective (3.3). Moreover, with the notation of 5.5, retained in the present section, there is for $K' \supset K'_o$ a canonical isomorphism

$$\Phi[K'](\ell) \simeq \ker(\widetilde{u}[K']_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell), \tag{11.8.3}$$

coming from (11.6.2) applied to $A[K']$. One also verifies immediately that the corresponding immersions

$$\Phi[K'](\ell) \hookrightarrow M_\ell \otimes D_\ell$$

are compatible with the transition morphisms as K' varies. Passing to the limit therefore gives a canonical monomorphism

$$\Psi(\ell) \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'](\ell) \hookrightarrow M_\ell \otimes D_\ell \tag{11.8.4}$$

from the ℓ-primary component of Ψ; summing over all ℓ gives

$$\Psi \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'] \longrightarrow M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.8.5}$$

Theorem 11.9. The canonical homomorphism (11.8.5) is an isomorphism.

We already know that it is a monomorphism. To prove that it is an isomorphism, we must use the characterization (11.8.3) of the image of $\Phi[K'](\ell)$ and the behaviour of $\widetilde{u}[K']$ as K' varies. That behaviour is given by (10.3.4).

We may reduce to the case in which S is strictly local. To prove 11.9 we are then reduced to the following familiar assertion, applied to $S[K'_o]$: if S is a strictly local trait with fraction field K, then for every integer $n \geq 1$ there is a separable extension K' of K whose degree is divisible by n. I≈464

Corollary 11.10. Under the hypotheses (S complete) and with the notation of 8.3, the ind-finite group scheme over K of connected components of the group scheme $A^\flat_K$, namely $A^\flat_K/A^{\flat\,o}_K$, is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, where $\underline{M}_K$ is the twisted constant group over K defined in (8.1.8).

This follows immediately from (8.3.4) and 11.9, in view of the definition of $\underline{M}_K$.

Remarks 11.11. a) If one restricts to the ℓ-primary components for $\ell \neq p$, Theorem 11.9 is immediate in terms of 11.2 and does not require the more delicate considerations of 11.6 (via 9.4). Likewise, if K is assumed to be of characteristic zero, 11.9 can be proved without using (11.8.3): it is immediate that the image of ${}_n\Phi[K'](\ell)$ in

$$M_n = {}_n(M_\ell \otimes D_\ell)$$

contains the image of the group ${}_nA(K')$ of K'-valued points of ${}_nA$. For K' sufficiently large this group is equal to ${}_nA(\overline{K})$, and therefore maps onto M_n. If $\operatorname{char} K = p$, on the other hand, it appears that there is no "trivial" proof—that is, no proof avoiding 9.4—of 11.10 for the p-primary component.

b) The existence of an isomorphism

$$\Psi \simeq (\mathbb{Q}/\mathbb{Z})^r,$$

together with the correct value of r, was conjectured in 1964 by J.-P. Serre.

12. Néron model of a Jacobian and the Picard–Lefschetz formula I≈465

Let us first recall the following general results of M. Raynaud [22, Ths. 5 and 7; 22 bis].

Theorem 12.1 (M. Raynaud). Let S be a strictly local trait, and let $f : X \to S$ be a scheme proper and flat over S, with one-dimensional fibres, such that

$$f_*\underline{\mathcal{O}}_X \simeq \underline{\mathcal{O}}_S, \tag{12.1.1}$$

and such that the local rings of X are factorial. For every maximal point x_i of X_o, put

$$d_i = \text{length}\,\underline{\mathcal{O}}_{X_o,x_i}, \qquad d_i^t = \mu_i d_i, \tag{12.1.2}$$

where μ_i is the radicial multiplicity of $k(x_i)$ over k (EGA IV, 4.7.4) (thus a power of the characteristic exponent of k). Let d (respectively d^t) be the gcd of the integers d_i (respectively d_i^t); hence $d = d^t$ if k is perfect. Put

$$P = \underline{\text{Pic}}_{X/S} : (\text{Sch})^o_{/S} \longrightarrow (\text{Ab}). \tag{12.1.3}$$

a) Consider the following conditions:

1) $d^t = 1$.

2) X has a zero-cycle of degree 1.

3) There exists an invertible sheaf L on $X \times P^o$ such that the corresponding homomorphism $P^o_\eta \to \underline{\text{Pic}}_{X_\eta/\eta} = P_\eta$ is the canonical inclusion. (It is known, by J.-P. Murre, that P_η is representable by a group scheme locally of finite type over η; see, for example, SGA 6, XII, 1.5.)

4) X is cohomologically flat over S (EGA III, 7.8.1) (equivalently, cohomologically flat in dimension zero, or equivalently $k \xrightarrow{\sim} H^o(X_o, \underline{\mathcal{O}}_{X_o})$), and $d = 1$. I≈466

5) P is representable (equivalently, representable by a group scheme smooth over S).

6) P^o is representable (equivalently, representable by a group scheme smooth over S). (For the definition of P^o, see SGA 3, VI_B, 3.1.)

7) P^o is separated over S, that is, it satisfies the valuative criterion of separatedness of EGA II, 7.2.3 (equivalently, every section s of P^o over S for which $s(\eta) = 0$ is zero).

8) $d = 1$.

Then one has the implications

$$1) \Longleftrightarrow 2) \Longrightarrow 3) \Longrightarrow 4) \Longleftrightarrow 5) \Longleftrightarrow 6) \Longleftrightarrow 7) \Longrightarrow 8). \tag{12.1.4}$$

In particular, if $d = d^t$, for example if k is perfect, the conditions 1) through 8) are equivalent.

b) Suppose that $d^t = 1$, so that P is representable by a group scheme smooth over S. Then the schematic closure of the identity section of P is a group subscheme E which is ètale over S. If I is the set of reduced irreducible components X_o^i of X_o, one obtains an isomorphism

$$\mathbb{Z}^I \simeq \Gamma(S, E) \tag{12.1.5}$$

by associating to every $i \in I$ the section S_i of P over S (zero over η) defined by the invertible sheaf $\underline{\mathcal{O}}_X(X_o^i)$ on X. (Because the local rings of X are factorial by hypothesis, the cycles X_o^i are indeed I≈467

divisors; thus $\underline{\mathcal{O}}_X(X_o^i)$ has its stated meaning as the invertible sheaf associated with that divisor; see EGA IV, 21.2.8.1.) One has

(12.1.6) $$P^o \cap E = \text{the zero section of } P \quad .$$

c) Under the conditions of b), the quotient

(12.1.7) $$Q = P/E$$

is representable by a smooth separated group scheme over S, and this group scheme is Néron, that is, it satisfies the universal property (1.1.2) in terms of its generic fibre.

d) Under the conditions of c), suppose in addition that the geometric generic fibre of X is integral. Then there is a canonical exact sequence

(12.1.8) $$0 \longrightarrow A \longrightarrow Q \xrightarrow{\deg} \mathbb{Z}_S \longrightarrow 0,$$

where deg is induced by the "degree" homomorphism $P \to \mathbb{Z}_S$, and where A is a smooth separated group scheme of finite type over S, which is likewise Néron, that is, a Néron model of

(12.1.9) $$A_\eta = P_\eta^o = \underline{\mathrm{Pic}}^o_{X_\eta/\eta} \quad .$$

Moreover, the composite homomorphism $P^o \to P \to Q$ induces an isomorphism

(12.1.10) $$P^o \xrightarrow{\sim} A^o.$$

I 468 12.1.11. The case of interest to us is that in which X_η is smooth over η, so that, in view of (12.1.1), the geometric generic fibre of X is smooth and connected (hence integral), and in which X_η has a zero-cycle of degree one. All the conditions considered in 12.1 (and in particular conditions 1) through 8) of a)) then hold. Moreover, $A_\eta = P_\eta^o$ is then an abelian scheme, and d) gives us a very explicit construction of its Néron model A. In particular, (12.1.10) gives an isomorphism

(12.1.12) $$A_o^o \simeq \underline{\mathrm{Pic}}^o_{X_o/k},$$

which shows in particular that A has semistable reduction over S if and only if $\underline{\mathrm{Pic}}^o_{X_o/k}$ has unipotent rank zero. When X is regular and X_o is separable over k, this is known to hold if and only if the geometric special fibre $\overline{X}_o$ has no singular points other than ordinary double points (that is, points whose completed local ring is the same as that of the union of the two coordinate axes of affine 2-space at the origin). Verification of this fact, together with the "well-known" description of the Picard scheme of a proper separable curve over a field k, using EGA IV, 21.8.5, is left to the reader as an exercise. Notice also that the implication "$\overline{X}_o$ has ordinary double points" $\Longrightarrow$ "$A_\eta = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ has semistable reduction" follows immediately from the "geometric monodromy theorem" III, in view of the Galois criterion for semistable reduction in 3.5. This argument, however, does not give the important isomorphism (12.1.10) and its consequence (12.1.12).

I 469 12.2. For the proof of 12.1, we refer to [22 bis]; in fact we shall need only the case where X_η is smooth over η. Let us also note that [22 bis] proves relations between Picard schemes and Néron models under conditions substantially more general than those of 12.1 c) and d). As their statements are more technical, we have chosen not to include them in this review. We mention only that b), c), and d) are easy consequences of a), and that in a) the only delicate points are the implications

3) $\Longrightarrow$ 4) $\Longrightarrow$ 5) and 7) $\Longrightarrow$ 4), the first being the most difficult. This point is trivial, however, if one assumes that X_o is separable over k (which implies that X has a section over S, since S is strictly local, and hence that condition 1), $d^t = 1$, holds). We advise the reader to give the proof in this particularly simple case, assuming in addition, if necessary, that X_η is smooth, so that P_η^o is an abelian scheme. One may then use its Néron model A, and A^o represents $\underline{\mathrm{Pic}}^o_{X/S}$. We shall use 12.1 only in this particular case, and under the further assumption that $\overline{X}_o$ has no singularities other than ordinary double points, that is, by (12.1.11), that A_η has semistable reduction. This is the case considered in 10.5.4, to which we restrict ourselves from now on.

12.3. Let C be a proper separable curve over a field k, which for simplicity we suppose separably closed (or even algebraically closed, if the reader prefers). If $\widetilde{C}$ is the normalization of C, it is known that $\underline{\mathrm{Pic}}^o_{\widetilde{C}/k}$ has "reductive rank" zero (that is, its maximal torus is zero). Consequently the maximal torus of $\underline{\mathrm{Pic}}^o_{C/k}$ is contained in

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$$N = \ker\big(\underline{\mathrm{Pic}}_{C/k} \longrightarrow \underline{\mathrm{Pic}}_{\widetilde{C}/k}\big),$$

and is therefore canonically isomorphic to the maximal torus of this last group. The structure of N is easily made explicit by EGA IV, 21.8.5: N is canonically an extension of a torus T, which we shall describe, by a connected unipotent group U (which itself has, over k, a composition series whose quotients are isomorphic to the additive group $\mathbb{G}_a$). Moreover, if T' is the maximal torus of N, then $T' \cap U$ is both unipotent and of multiplicative type, and is therefore zero (SGA 3, XVII, 2.4), while the image of T' in T is all of T (SGA 3, XII, 7.1 e)). Thus $T' \to T$ is an isomorphism. Hence T is canonically isomorphic to the maximal torus of N, and therefore to that of $\underline{\mathrm{Pic}}^o_{C/k}$.

To make the structure of N explicit, it is convenient to suppose that k is separably closed. The irreducible components C_i of C are then geometrically irreducible, and the nonnormal points of C are radicial over k. Let J be the set of irreducible components of $\widetilde{C}$, and for every $x \in C$ let $J(x)$ be the set of "branches" of C through x (that is, the points of $\widetilde{C}$ above x). Finally put

$$R(x) = \mathbb{Z}^{J(x)} / \operatorname{Im} \mathbb{Z}, \tag{12.3.1}$$

where $\mathbb{Z} \to \mathbb{Z}^{J(x)}$ is the diagonal map, and

$$R = \left(\bigoplus_x R(x) \right) / \operatorname{Im} \mathbb{Z}^J. \tag{12.3.2}$$

Clearly it is enough to take the sum over the finite set of points of C which belong to at least two branches of C. Thus R is a finitely generated $\mathbb{Z}$-module. There is then a canonical isomorphism

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$$T \ (\simeq \text{ the maximal torus of } \underline{\mathrm{Pic}}^o_{C/k}) \simeq R \otimes \mathbb{G}_{m,k}. \tag{12.3.3}$$

In other words, the character group M of T is

$$M \overset{\text{def}}{=} D(T) = \check{R} \simeq \ker\left(\bigoplus_x \check{R}(x) \longrightarrow \mathbb{Z}^J \right), \tag{12.3.4}$$

where $\check{R}(x)$ may also be interpreted as the set of integer-valued functions on $J(x)$ whose values have sum zero.

12.3.5. We shall need only the case in which at most two branches of C pass through any point x (as happens, in particular, for ordinary double points). If $\operatorname{card} J(x) = 2$, then $R(x)$ is isomorphic to $\mathbb{Z}$, but the isomorphism depends on an ordering of the two branches (C', C'') through x, or equivalently

on the choice of one branch C': to C' one associates the element of $R(x)$ which is the canonical image of $0_{C'} + 1_{C''}$. One then obtains an isomorphism

$$R = \mathbb{Z}^I / \operatorname{Im} \mathbb{Z}^J, \qquad T \simeq \mathbb{G}_m^I / \operatorname{Im} \mathbb{G}_m^J, \qquad M = \ker(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J), \tag{12.3.6}$$

where I is the set of points $x \in C$ which belong to two branches of C, after a choice of signs as just explained.

Remarks 12.3.7. A sometimes convenient way to describe the character group M of the maximal torus T of $\underline{\mathrm{Pic}}^o_{C/k}$ is the following. Consider the "unoriented one-dimensional graph" $\Gamma(C)$ (N. Bourbaki, *Groupes et algèbres de Lie*, Chapter IV, Appendix), whose vertices are the irreducible components of $\widetilde{C}$. Two vertices $\widetilde{C}_i$ and $\widetilde{C}_j$ are joined by a set of edges in bijective correspondence with the pairs of distinct branches of C, contained respectively in $\widetilde{C}_i$ and $\widetilde{C}_j$, which pass through the same point of C. This set is $\widetilde{C}_i \times_C \widetilde{C}_j$ when $i \neq j$; when $i = j$, it is

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$$(\widetilde{C}_i \times_C \widetilde{C}_i - \Delta(\widetilde{C}_i))/(\text{symmetry}),$$

where $\Delta : \widetilde{C} \to \widetilde{C} \times_C \widetilde{C}$ is the diagonal. (It is therefore empty if C_i is geometrically unibranch.) One then has canonical isomorphisms

$$M = D(T) \xleftarrow{\sim} H_1(\Gamma(C), \mathbb{Z}), \qquad \check{M} = R \simeq H^1(\Gamma(C), \mathbb{Z}). \tag{12.3.7}$$

We shall not need this combinatorial interpretation of M, and shall confine ourselves to defining the canonical homomorphism

$$H_1(\Gamma(C), \mathbb{Z}) \longrightarrow M, \tag{12.3.8}$$

leaving it to the reader to prove that this is an isomorphism. Let I be the set of points of C which belong to at least two branches of C, let J, as above, be the set of irreducible components of $\widetilde{C}$ (or of C, which amounts to the same thing), and let K be the set of "branches" of C at the points of I, that is, the inverse image of I in $\widetilde{C}$. There is a canonical map

$$h : K \longrightarrow I \times J, \qquad h = (f, g), \tag{12.3.9}$$

and the description (12.3.4) of M is expressed in terms of h as

$$M \simeq \ker\left(\mathbb{Z}^{(h)} : \mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I \times J)}\right). \tag{12.3.10}$$

Indeed, the expression (12.3.2) for $R = \check{M}$ may plainly be interpreted as the cokernel of the homomorphism $\mathbb{Z}^{I \times J} \to \mathbb{Z}^K$ induced by h, which is the transpose of $\mathbb{Z}^{(h)}$. The graph $\Gamma(C)$ is likewise described in terms of h by

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$$\Gamma_0 = J, \qquad \Gamma_1 = (K \times_I K - \Delta(K))/(\text{symmetry}).$$

Choose an orientation on each edge of $\Gamma(C)$, and denote by Γ_1 the set of these oriented edges (x, C', C''). Then

$$H_1(\Gamma(C), \mathbb{Z}) = \ker\left(d : \mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right), \tag{12.3.11}$$

where the differential is

(12.3.12) $$d(x, C', C'') = g(C'') - g(C').$$

The homomorphism (12.3.8) is induced by the following homomorphism v of chain complexes:

(12.3.13) $$v = (v_0, v_1) : C_\bullet(\Gamma(C)) = \left[\mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right] \longrightarrow \left[\mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I\times J)}\right],$$

given by

(12.3.14) $$v_1(x, C', C'') = (x, C'') - (x, C'), \qquad v_0(\widetilde{C}_j) = (0, \widetilde{C}_j).$$

12.4. We now place ourselves in the case considered at the end of 12.2. By (12.1.12) and (12.3.6), applied with $C = X_o$, the character group M of the maximal torus T_o of the special fibre A_o of the Néron model A of $A_\eta = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ is given by

(12.4.1) $$M = \ker\left(\mathbb{Z}^I \xrightarrow{d} \mathbb{Z}^J\right),$$

where I is the set of singular points of X_o, and J is the set of its irreducible components, the homomorphism $\mathbb{Z}^I \to \mathbb{Z}^J$ being given by

(12.4.2) $$d(x_i) = C_i' - C_i''.$$

Here C_i' and C_i'' are the two irreducible components of $\widetilde{C}$ corresponding to the branches through the double point x_i. The operator d depends on the choice of an order (C_i', C_i'') on these two branches, and the isomorphism (12.4.1) plainly depends on this system of choices. Nevertheless, if, for $i \in I$, we denote by I≈474

(12.4.3) $$\varphi_{x_i} : M \longrightarrow \mathbb{Z}$$

the composite

$$M \longrightarrow \mathbb{Z}^I \xrightarrow{\mathrm{pr}_i} \mathbb{Z},$$

where the first homomorphism comes from (12.4.1), then, on returning to the expression (12.3.4), one sees at once that φ_{x_i} is well-defined up to sign. More precisely, it depends only on the order chosen between the two branches C_i', C_i'' through x_i, and changes sign when that order is reversed. It follows in particular that the bilinear form

(12.4.4) $$\varphi_{x_i} \otimes \varphi_{x_i} \in \check{M} \otimes_{\mathbb{Z}} \check{M} \simeq \mathrm{Hom}(M \otimes M, \mathbb{Z})$$

is determined canonically by the singular point x_i, without any sign ambiguity. We may therefore put

(12.4.5) $$u = \sum_i \varphi_{x_i} \otimes \varphi_{x_i} : M \otimes_{\mathbb{Z}} M \longrightarrow \mathbb{Z}.$$

This gives a well-defined pairing, independent of every choice of signs. It is plainly positive definite, since it is the restriction to $M \hookrightarrow \mathbb{Z}^I$ of the standard quadratic form $\sum_i X_i^2$ on $\mathbb{Z}^I$.

The notation M for the character group of T_o is not, a priori, consistent with that of the preceding sections, where this group was written M', while M denoted the analogous group associated with the dual abelian scheme A'_K of A_K. But since A_K is a Jacobian, it has a canonical polarization ξ [5], which induces a canonical isomorphism I≈475

$$\psi_\xi : A_K \xrightarrow{\sim} A'_K. \tag{12.4.6}$$

By means of this isomorphism we shall identify A_K and A'_K, so that there is no longer any need to distinguish M from M'. In particular, for every prime number ℓ, the monodromy pairing (9.1.2) may be regarded as a pairing

$$u_\ell : M_\ell \otimes_{\mathbb{Z}_\ell} M_\ell \longrightarrow \mathbb{Z}_\ell. \tag{12.4.7}$$

With this understood, we can state the principal result of the present section:

Theorem 12.5 (Picard–Lefschetz formula). Let S be a strictly local trait, and let X be a regular scheme, projective and flat over S, with one-dimensional fibres, whose geometric generic fibre is smooth and connected and whose geometric special fibre has no singularities other than ordinary quadratic singularities. With the notation introduced in 12.4, for every prime number ℓ, the monodromy pairing (12.4.7) is equal to the pairing obtained from (12.4.5) by extension of scalars from $\mathbb{Z}$ to $\mathbb{Z}_\ell$.

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12.6. Notice that this statement proves 10.4 in the particular case under consideration when, in 10.4 b), one takes for ξ the canonical polarization considered there. As we showed in 10.5, this suffices to imply 10.4 in the general case.

12.7. Proof of 12.5. The case $\ell \neq p$. It relies essentially on the cohomological Picard–Lefschetz formula in the situation considered in 12.5, a formula giving the action of the monodromy group I on the following group:

$$H^1(X_{\bar\eta}, \mathbb{Z}_\ell) \simeq H^1(A_{\bar\eta}, \mathbb{Z}_\ell) \simeq \mathrm{Hom}(T_\ell(A_{\bar\eta}), \mathbb{Z}_\ell) \tag{12.7.1}$$

when $\ell \neq p$. This formula, which will be proved in a later exposé (Exposé XV), and indeed in a substantially more general case, is essentially identical to assertion 12.5 for the prime ℓ in question, in view of the definition in 9.2 of u_ℓ for $\ell \neq p$. It therefore remains to treat the case $\ell = p$. We shall do this by a method which reduces it to characteristic zero (and, moreover, to the case where X_o has only one singular point), which we now explain.

Remarks 12.7.2. Notice also that, once the announced reduction has been made, the usual arguments would allow us to reduce to the case in which the base field is the field $\mathbb{C}$ of complex numbers, and then to invoke the Picard–Lefschetz formula in its classical form—except that there does not seem to be a satisfactory reference for that formula. The proof of the Picard–Lefschetz formula given in our Seminar will proceed similarly, by reduction to the transcendental case, which is amenable to properly transcendental methods. One may therefore say that, in any event, the proof of 12.5 (and hence of 10.4) proposed here is transcendental in nature. Let us also point out that the only other known proof of 10.4 comes from rigid-analytic theory [24], and must for that reason likewise be regarded as essentially "transcendental." It would be interesting to find purely algebraic proofs of these results (although it is by no means evident that this is possible!).

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12.8. Proof of 12.5: reduction to the case in which S has characteristic zero. We shall use an argument based on the same principle as the proof of the "theorem on the essentially tame action of monodromy on π_1" in Exposé V. Notice that we may assume S complete, by 10.3. Let W be a Cohen p-ring with residue field k, and choose a homomorphism $W \longrightarrow V$ inducing the identity on

residue fields (EGA 0_{IV}, 19.8.6). Let $(x_\lambda)_{\lambda\in L}$ be the set of singular points of X_o, and consider the formal moduli scheme $\underline{S}$ of X_o over W, together with the "universal curve"

$$\underline{f} : \underline{X} \longrightarrow \underline{S}$$

whose special fibre is X_o (Exposé VI). We saw in the cited place that $\underline{S}$ is isomorphic to the formal spectrum of a power-series ring $W[[T_1, \ldots, T_N]]$, and that in $\underline{S}$ one can find a normal-crossings divisor

$$D = \sum_{\lambda\in L} D_\lambda, \tag{12.8.1}$$

whose irreducible components D_λ are indexed by L, each D_λ being described by an equation of the form $T_{i_\lambda} = 0$. (If desired, one may suppose $i_\lambda = \lambda$ after indexing L by a set of integers $[1, r]$.) The component D_λ is the image under $\underline{f} : \underline{X} \longrightarrow \underline{S}$ of the component D'_λ through the singular point x_λ of X_o in the non-smoothness locus

$$D' = \bigcup_{\lambda\in L} D'_\lambda$$

of $\underline{f}$. By the construction itself, $f : X \longrightarrow S$ is isomorphic to the pullback of $\underline{f} : \underline{X} \longrightarrow \underline{S}$ by a suitable W-morphism $g : S \longrightarrow \underline{S}$:

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$$\begin{array}{ccc} \underline{X} & \longleftarrow & X \\ {\scriptstyle \underline{f}}\downarrow & & \downarrow{\scriptstyle f} \\ \underline{S} & \xleftarrow{\ g\ } & S \end{array} \quad . \tag{12.8.2}$$

Set

$$P = \underline{\mathrm{Pic}}^o_{\underline{X}/\underline{S}}. \tag{12.8.3}$$

Since the special fibre of $\underline{f}$ is geometrically reduced of dimension 1, it follows that $\underline{f}$ is cohomologically flat in dimension zero, by EGA III, 7.8.1, and that P is formally smooth over $\underline{S}$ (the latter fact follows from $H^2(X_o, \underline{\mathcal{O}}_{X_o}) = 0$, since X_o has dimension 1). For simplicity, we shall assume that P is representable by a group scheme over $\underline{S}$; it will necessarily be a smooth group scheme with connected fibres. This is a special case of an unpublished result of Mumford cited in [22] (and considerably generalized by Raynaud in the note cited there, when the base is a trait). Since no proof of either Mumford's or Raynaud's result has been published, the reader may prefer to use the general results of M. Artin [1]. They give the representability of P—under the sole assumption that $\underline{f}$ is proper, flat, and cohomologically flat in dimension zero—by an "algebraic space" in Artin's sense, which would suffice for our purpose. Put

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$$U = \underline{S} - \operatorname{supp} D. \tag{12.8.4}$$

Thus U is the largest open subset of $\underline{S}$ over which $\underline{X}$ is smooth, and hence

$$P|_U = \underline{\mathrm{Pic}}^o_{\underline{X}_U/U}$$

is an abelian scheme over U. Its pullback to η by $g : \eta \longrightarrow U$ is the abelian scheme A_η over η that we wish to study. More precisely, there is a canonical isomorphism over S (12.1.10):

$$P \times_{\underline{S}} S \simeq A^o. \tag{12.8.5}$$

Since the special fibre $P_o = A_o^o$ is ℓ-divisible for every ℓ, the same holds for P itself; all its fibres are therefore extensions of an abelian variety by a torus. Consider the pro-ℓ group $T_\ell(P)$ over $\underline{S}$, whose pullback by g is precisely $T_\ell(A^o)$. Applying the constructions of 7.3, one defines a filtration of this pro-group by a "fixed part" and a "toric part":

$$T_\ell(P) \supset T_\ell(P)^f = T_\ell(\widehat{P}) \supset T_\ell(P)^t = T_\ell(T) \supset 0. \tag{12.8.6}$$

Its pullback by g is the analogous filtration of $T_\ell(A^o)$ studied in sections 2 and 5.

Consider the restriction of this filtration over the open subset U of $\underline{S}$. There we have the canonical self-duality of P_U, as the Jacobian of a smooth curve scheme over U, and hence a corresponding pairing

$$\varphi : T_\ell(P_U) \otimes T_\ell(P_U) \longrightarrow T_\ell(\mathbb{G}_{mU}). \tag{12.8.7}$$

This is a self-duality, in the sense of Barsotti–Tate groups, of $T_\ell(P_U)$. We saw in 7.4.8 that, for this pairing, $T_\ell(P_U)^f$ and $T_\ell(P_U)^t$ are each other's orthogonal complements. It follows that I≈480

$$E_U = T_\ell(P_U)/T_\ell(P_U)^t$$

is canonically isomorphic to the Barsotti–Tate pro-group dual to $T_\ell(P_U)^f = F_U$. It therefore extends canonically to a Barsotti–Tate group E over $\underline{S}$, namely

$$E = D(F), \qquad \text{where} \qquad F = T_\ell(P)^f. \tag{12.8.8}$$

We are now in a typical situation of mixed extensions (9.3), and can repeat the constructions of 9.4–9.6 (for whose having been written in the stale and piss-poor setting of a base scheme reduced to a trait we apologize!): $T_\ell(P_U)$ is a mixed extension of E_U by F_U, where E and F are Barsotti–Tate groups over $\underline{S}$ which are extensions, dual to one another:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & P & \longrightarrow & F & \longrightarrow & R & \longrightarrow & 0 \\ & & \wr\downarrow & & \wr\downarrow & & & & \\ & & T_\ell(P)^t \simeq \underline{\mathrm{Hom}}(M, T_\ell(\mathbb{G}_{m\underline{S}})) & & T_\ell(P)^f & & & & \end{array} \tag{12.8.9}$$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & E & \longrightarrow & Q & \longrightarrow & 0 \\ & & \wr\downarrow & & & & \wr\downarrow & & \\ & & D(R) & & & & M \otimes \mathbb{Z}_{\ell\,\underline{S}} & & \end{array} \quad .$$

Notice that R is canonically self-dual, as follows from the self-duality (12.8.7) and the orthogonality statement noted above. The only difference from the cited construction is that the pairing obtained on $M_\ell \times M_\ell$ takes values in the group

$$\big(\mathbb{G}_m(U)/\mathbb{G}_m(\underline{S})\big) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is no longer canonically isomorphic to $\mathbb{Z}_\ell$, but rather to $\mathbb{Z}_\ell^L$, by means of the isomorphism I≈481

$$\mathbb{Z}^L \simeq \mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}). \tag{12.8.10}$$

It maps the element indexed by λ in the canonical basis of $\mathbb{Z}^L$ to the class of T_{i_λ} in the right-hand side, where T_{i_λ} is an equation for the divisor D_λ. We thus obtain a canonical pairing

(12.8.11) $$\underline{u}_\ell : M_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell^L \qquad .$$

It is essentially immediate that the monodromy pairing u_ℓ (12.4.7) over S which we propose to study is the composite of $\underline{u}_\ell$ with the homomorphism $\mathbb{Z}_\ell^L \longrightarrow \mathbb{Z}_\ell$ obtained, by tensoring with $\mathbb{Z}_\ell$ over $\mathbb{Z}$, from the homomorphism

$$\begin{array}{ccc} \mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}) & \longrightarrow & \mathbb{G}_m(\eta)/\mathbb{G}_m(S) \\ \wr\big| & & \wr\big| \\ \mathbb{Z}^L & & \mathbb{Z} \end{array}$$

induced by $g : S \longrightarrow \underline{S}$ (which sends η into U). This homomorphism is given by an element

(12.8.12) $$(d_\lambda)_{\lambda\in L} \in \mathbb{Z}^L, \qquad d_\lambda \geqslant 1 \quad \text{for every } \lambda \in L,$$

where, for every $\lambda \in L$, d_λ is the multiplicity of the inverse image in S of the divisor D_λ on $\underline{S}$. It I≈**482** in fact depends only on the completion $\widehat{\underline{\mathcal{O}}}_{X,x_\lambda}$ as a V-algebra (Exposé VI). Moreover, it is known (Exposé VI) that, for a given λ, one has $d_\lambda = 1$ if and only if X is regular at x_λ. Since in 12.5 we assumed X regular, that is, all the d_λ equal to 1, assertion 12.5 will therefore follow from the following lemma, in which X, S, g no longer occur.

Lemma 12.8.13. If $\lambda \in L$, the composite of (12.8.11) with pr_λ is equal to the form $\varphi_{x_\lambda} \otimes \varphi_{x_\lambda}$ of (12.4.4).

To prove this last assertion, consider the completed strict localization $\underline{S}^\lambda$ of $\underline{S}$ at the generic point of D_λ, and put

$$\underline{X}^\lambda = \underline{X} \times_{\underline{S}} \underline{S}^\lambda.$$

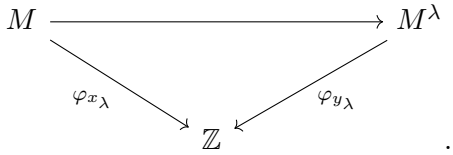
The special fibre $\underline{X}_o^\lambda$ then has only one singular point y_λ, which is again an ordinary quadratic singularity. In the corresponding monodromy pairing

(12.8.14) $$u_\ell^\lambda : M_\ell^\lambda \otimes M_\ell^\lambda \longrightarrow \mathbb{Z}_\ell \qquad ,$$

M^λ consequently has rank 1 or 0. We leave to the reader the verification of the following easy facts, which make explicit, in the special case in which we need them, the behaviour under localization of the preceding constructions.

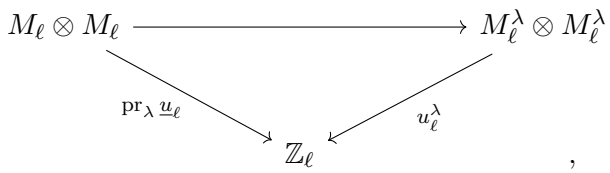
a) There is a unique subtorus T_λ of $P \times_{\underline{S}} D_\lambda$ whose restriction to the generic point s_λ of D_λ is the maximal torus of P_{s_λ} (which has dimension 0 or 1).

b) Consider the transpose homomorphism $M \longrightarrow M^\lambda$ on character groups of the inclusion $T_{\lambda o} \hookrightarrow T_o$, I≈**483** where T_o is the maximal torus of P_o, and the form $\varphi_{y_\lambda} : M^\lambda \longrightarrow \mathbb{Z}$, defined up to sign as in 12.4.3 (with X replaced by $\underline{X}^\lambda$). Then, up to sign, the following diagram is commutative:

(12.8.15)


(In fact, choosing an ordering of the two branches of X_o through x_λ determines an analogous choice at the point y_λ of $\underline{X}_o^\lambda$, and for choices corresponding in this way the preceding diagram is commutative.)

c) The following diagram is commutative:



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where the notation is that of 12.8.13 and (12.8.14). These relations show at once that, to prove 12.8.13, it is enough to prove the analogous relation for $\underline{X}^\lambda$:

$$u_\ell^\lambda = (\varphi_{y_\lambda} \otimes \varphi_{y_\lambda}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is nothing other than 12.5 for $\underline{X}^\lambda$ over $\underline{S}^\lambda$. Now $\underline{S}^\lambda$ is a trait of characteristic zero. This completes the announced reduction.

Remarks 12.9. Notice that this reduction simultaneously reduces us to the case in which X_o has only one singular point, the case classically considered in the Picard–Lefschetz formula. Notice also that, in this case, X_o has one or two irreducible components, corresponding respectively to rank $M = 1$ and rank $M = 0$ (the latter being the case in which the Jacobian of the generic fibre has good reduction). I≈484

12.10. The proof given in 12.8 yields a result more general than 12.5, valid without assuming that X is regular, by introducing the integers $d_\lambda \geqslant 1$ of (12.8.12): all the monodromy pairings are then deduced from the pairing

$$u = \sum_{\lambda \in L} d_\lambda \, \varphi_{x_\lambda} \otimes \varphi_{x_\lambda} : M \otimes M \longrightarrow \mathbb{Z} \qquad . \tag{12.10.1}$$

13. Links with transcendental theory: the complex analytic case.
For the general notions concerning analytic spaces, we refer to [2].

13.1. Review of analytic groups over S. Let S be an analytic space, and let A be an analytic space over S, endowed with the structure of a commutative group over S (that is, A is a commutative group in the category of analytic spaces over S). We assume that A is smooth over S, and denote by I≈485

$$\underline{E} = \underline{\mathrm{Lie}}(A/S) \tag{13.1.1}$$

the locally free module on S obtained by pulling the relative tangent sheaf of A over S back along the zero section. We shall admit here the definition of the exponential homomorphism

$$\exp : E \longrightarrow A \qquad , \tag{13.1.2}$$

where E is the analytic vector bundle on S whose module of holomorphic sections is $\underline{E}$. In a neighbourhood of a point $s \in S$, this homomorphism is first defined near the zero section of E by the familiar methods of the theory of formal groups in characteristic zero, and is then extended in the evident way, by multiplicativity, to all of E. It is a homomorphism of relative analytic groups over S; it is étale, and its image is the open subspace A^o of A which is the union of the identity components of the fibres of A over S. If R is the analytic subspace which is the kernel of exp, we therefore have an exact sequence of relative analytic groups

$$0 \longrightarrow R \longrightarrow E \longrightarrow A^o \longrightarrow 0 \qquad . \tag{13.1.3}$$

Here the morphism $E \longrightarrow A^o$ is surjective and étale, so R is a relative analytic group étale over S. It is an analytic subspace of E, and is closed in E if and only if A^o is separated over S, that is, if and only if the zero section of A^o is closed in A^o.

13.1.4. Of course, the exact sequence (13.1.3) is functorial in A, and commutes with every base change $S' \longrightarrow S$. In this way one obtains an equivalence of categories, compatible with base changes, between the category of relative analytic groups over S which are smooth and have connected fibres, and the category of pairs (E, R) consisting of a locally free vector bundle E over S and an analytic subgroup R of E which is étale over S. I 486

13.1.5. When A is an "abeloid" group over S, by which we mean that it is smooth, proper, and separated over S, with connected fibres—so that in particular $A = A^o$—then R is a locally constant group over S, whose rank at each point s (as a $\mathbb{Z}$-module) is equal to $2\,\mathrm{rank}_s(E)$, and conversely. In this way one obtains the familiar interpretation of an abeloid space over S in terms of a "local system" R over S and a complex analytic vector-bundle structure on the real analytic bundle $R \otimes_{\mathbb{Z}} \mathbb{R}$ defined by R.

13.2. Formal completion along a fibre. We now fix a point $s \in S$, and shall work locally in a neighbourhood of s. For every integer $n \geqslant 0$, let S_n denote the n-th infinitesimal neighbourhood of s in S, and, for every analytic space X over S, put likewise $X_n = X \times_S S_n$. As n varies, the system of the X_n then forms an inductive system of analytic spaces over S, which we denote by $\widehat{X}$ and call the formal completion of X along X_s. It clearly depends functorially on X, and its formation commutes with finite projective limits; hence $X \longmapsto \widehat{X}$ takes groups to groups. Notice that $\widehat{X}$ in fact depends only on the restriction of X to an arbitrary open neighbourhood U of s in S.

Lemma 13.2.1. The data are those of 13.1, and suppose that the following conditions hold: a) $A = A^o$; b) R is constant in a neighbourhood of s; and c) R_s spans the complex vector space E_s. Let A' be a second smooth analytic group over S, and consider the map I 487

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{S\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad ,$$

where the notation is that of 13.2, and the analogous map obtained by restricting over a variable open neighbourhood U of s and passing to the inductive limit:

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{(S,s)\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad . \tag{13.2.1.1}$$

(Here (S, s) denotes the germ of the analytic space S at s.) Then the latter map is bijective.

Indeed, consider the exact sequence

$$0 \longrightarrow R' \longrightarrow E' \longrightarrow A'^o \longrightarrow 0$$

analogous to (13.1.3) and defined by A'. Since the homomorphisms $A \longrightarrow A'$ necessarily factor through A'^o, and the same holds after the base changes $U \longrightarrow S$ and $S_n \longrightarrow S$, we may assume that $A' = A'^o$. By the dictionary of 13.1.4, it remains to prove that giving a germ of a vector bundle homomorphism $E \longrightarrow E'$ which takes R into R' is equivalent to giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ which take R_n into R'_n.

If $\underline{\mathcal{O}}_s$ denotes the local ring of S at s, giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ is I ≈488 equivalent to giving a homomorphism of $\underline{\widehat{\mathcal{O}}}_s$-modules

$$\widehat{F} : \underline{\widehat{E}}_s \longrightarrow \underline{\widehat{E}}'_s,$$

where $\underline{E}_s$ and $\underline{E}'_s$ are the free $\underline{\mathcal{O}}_s$-modules obtained at s from the $\underline{\mathcal{O}}_S$-modules $\underline{E}$ and $\underline{E}'$, corresponding to E and E'. On the other hand, a germ of a vector bundle homomorphism $E \longrightarrow E'$ is equivalent to giving a homomorphism of $\underline{\mathcal{O}}_s$-modules

$$F : \underline{E}_s \longrightarrow \underline{E}'_s \quad .$$

This already shows that (13.2.1.1) is injective. For surjectivity, it is enough to prove that if $\widehat{F}$ is a homomorphism as above such that, for every n, the induced homomorphism $F_n : E_n \longrightarrow E'_n$ takes

R_n into R'_n, then $\widehat{F}$ comes from an F—that is, it takes $\underline{E}_s$ into $\underline{E'}_s$—and the corresponding germ of homomorphisms $E \longrightarrow E'$ takes R into R'.

By hypothesis c), $R_s = R_o$ spans E_s; it follows from Nakayama's lemma that the group R_o (identified either with the group of sections of $\widehat{R}$ over $\widehat{S}$, or with the group of germs of sections of R over S) generates $\underline{\widehat{E}}_s$. Thus, to prove that $\widehat{F}(\underline{E}_s) \subset \underline{E'}_s$, it is enough to prove that $\widehat{F}(R_o) \subset \underline{E'}_s$. But $\widehat{F}(R_o) \subset R'_o$, where R'_o is itself interpreted as the group of germs of sections of $R' \subset E'$ in a neighbourhood of s. This gives the asserted inclusion, and hence the existence of F. Moreover, we see that $F(R_o) \subset R'_o$, where R_o and R'_o are regarded as groups of germs of sections of E and E', respectively. By b), R_o generates R in a neighbourhood of s; it follows that $F(R) \subset R'$ in a neighbourhood of s, which completes the proof.

Corollary 13.2.2. If A satisfies conditions a), b), and c) of 13.2.1, it is determined up to unique isomorphism in a neighbourhood of s by the knowledge of the $\widehat{S}$-group $\widehat{A}$.

13.3. The analytic Raynaud extension. I≈**489**

We are now in a position to paraphrase, in analytic geometry, the construction of the Raynaud extension given in 7. Suppose that A is as in 13.1, with $A = A^o$,

and consider $R_o = R_s$, which is a free $\mathbb{Z}$-module of finite type. After restricting S to a sufficiently small open neighbourhood of s, and interpreting R_o as the group of germs of sections of R near s, one obtains a homomorphism

$$R_{oS} \longrightarrow R \subset E, \tag{13.3.1}$$

where R_{oS} denotes the constant analytic group over S with value R_o. On choosing the neighbourhood of s small enough that every section of $\underline{\mathcal{O}}_S$ whose germ at s is zero is itself zero, one sees that the preceding homomorphism is injective. Its image is therefore a constant analytic subgroup of R; it is plainly the largest constant subgroup of R, and is called the fixed part of R and denoted by R^f:

$$R_{oS} \simeq R^f \hookrightarrow R, \qquad \text{inducing an isomorphism } R^f_s \simeq R_s. \tag{13.3.2}$$

We may then consider the quotient group

$$A^\natural = E/R^f, \tag{13.3.3}$$

and, in view of (13.1.3), obtain a canonical exact sequence I≈**490**

$$0 \longrightarrow R/R^f \xrightarrow{\ i\ } A^\natural \longrightarrow A \longrightarrow 0. \tag{13.3.4}$$

Notice that, after restricting S to a neighbourhood of s, R^f is always closed in E, and hence $A^\natural$ is separated over S. Notice also that the homomorphism $A^\natural \longrightarrow A$ is étale, and therefore induces an isomorphism

$$\underline{\mathrm{Lie}}(A^\natural) \xrightarrow{\sim} \underline{\mathrm{Lie}}(A) = \underline{E}. \tag{13.3.5}$$

Moreover, for every $n \geqslant 0$, the canonical homomorphism $A^\natural \longrightarrow A$ induces an isomorphism $A^\natural_n \simeq A_n$ (since $R^f_n \xrightarrow{\sim} R_n$), and consequently an isomorphism

$$\widehat{A^\natural} \xrightarrow{\sim} \widehat{A}. \tag{13.3.6}$$

This last fact shows that $A^\natural$ does indeed play the role of the group, likewise denoted by $A^\natural$, that we considered in 7. Here it has been constructed without any hypothesis on the fibre $A_s = A_o$

of the form "an extension of an abeloid group by a torus". Such a hypothesis will nevertheless be necessary if we wish to recover on $A^\natural$ the important extension structure (7.2.3). By contrast, the homomorphism $A^\natural \longrightarrow A$ in (13.3.4) is essentially transcendental in nature and has no counterpart in the purely algebraic theory of § 7. In § 14 we shall return to the question of a purely algebraic interpretation of the homomorphism $i : R/R^f \longrightarrow A^\natural$ in (13.3.4).

Suppose first that we are given a complex subtorus

(13.3.7) $$T_o \subset A_o,$$

where by a "complex torus" we mean an analytic group isomorphic to a Cartesian power of $\mathbb{C}^*$. Let

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(13.3.8) $$T = T_o \times S$$

be the constant torus over S with value T_o. Proceeding as in 5.1, one obtains a unique homomorphism (indeed, an immersion) of inductive systems of analytic groups

$$\widehat{T} \longrightarrow \widehat{A^\natural},$$

which over S_o reduces to (13.3.7). By 13.2.2, this homomorphism comes from a unique germ of homomorphisms

(13.3.9) $$T \longrightarrow A^\natural,$$

and, after taking S small enough around s, we may suppose that it is defined over all of S. The corresponding homomorphism

$$\underline{\mathrm{Lie}}(T) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is injective over each S_n, hence over $\widehat{S}$, and therefore over $\underline{\mathcal{O}}_s$ by flat descent. Thus there is a neighbourhood of s in S, which we may take to be all of S, on which $\underline{\mathrm{Lie}}(T)$ identifies with a locally direct summand $\underline{E}^t$ of $\underline{E}$; denote the corresponding vector subbundle of E by E^t:

(13.3.10) $$\underline{\mathrm{Lie}}(T) \xrightarrow{\sim} \underline{E}^t \subset \underline{E} \simeq \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{\mathrm{Lie}}(A), \qquad E^t \subset E.$$

Likewise, let

(13.3.11) $$R^t_o \subset R_o, \qquad R^t = R^t_{oS} \subset R_{oS} \simeq R^f$$

denote respectively the sublattice of R_o defined by the subtorus T_o of A_o, and the corresponding constant analytic group over S. Then the homomorphism (13.3.9) corresponds simply to the homomorphism

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$$E^t/R^t_{oS} = E^t/R^t \longrightarrow E/R_{oS} = E/R^f$$

induced by the inclusions

$$E^t \hookrightarrow E, \qquad R^t \hookrightarrow R^f.$$

Using the fact that (13.3.9) induces an immersion on the fibre over s, we conclude that the homomorphism

$$R_o/R^t_o \longrightarrow E_o/E^t_o = (E/E^t)_o$$

is injective. It follows at once that

$$R_o/R_o^t \longrightarrow (E/E^t)_{s'}$$

is still injective for s' near s. Thus, after shrinking S around s, we may suppose that (13.3.9) is a closed immersion.

We are therefore led to introduce

(13.3.12) $$B = A^\natural/T \simeq (E/E^t)/(R^f/R^t),$$

so that $A^\natural$ acquires an extension structure

(13.3.13) $$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0,$$

where B is a relative analytic group over S, smooth and separated over S, with connected fibres. By construction we therefore have

(13.3.14) $$\underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t,$$

and the exponential exact sequence for B is

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(13.3.15) $$0 \longrightarrow R^f/R^t \longrightarrow E/E^t \longrightarrow B \longrightarrow 0.$$

Notice that

(13.3.16) $$R^f/R^t = (R_o/R_o^t)_S$$

is a constant analytic group over S. It follows from 13.1.5 that B is an abeloid group over S in a neighbourhood of s if and only if its fibre at s,

$$B_o = A_o/T_o,$$

is an abeloid analytic group; in that case A_o appears as an extension of such a group B_o by a torus T_o. Every way of writing A_o as an extension of an abeloid group by a complex torus $(*)$ therefore gives rise on A, in a neighbourhood of s, to a unique extension structure (13.3.13) of an abeloid group B by a torus T, inducing the given extension structure at s.

13.4. Interpretation in terms of Deligne's "mixed Hodge structures".

Let us first return to the dictionary of 13.1.4, which we shall reformulate in a slightly different way, better suited to introducing the "Hodge structures" relevant to the present situation. For this purpose, let $\underline{R}$ be the sheaf of sections of R over S, and introduce the $\underline{\mathcal{O}}_S$-module

(13.4.1) $$\mathcal{E}(A) = \mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \quad .$$

It is locally free of finite type when R is locally constant. (Note that $\mathcal{E}$ need not be coherent when R is not locally constant.) The homomorphism (13.1.3) $\underline{R} \longrightarrow \underline{E}$ then defines a canonical homomorphism

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(13.4.2) $$\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \longrightarrow \underline{E} \quad .$$

We denote its kernel by $\underline{F}(A)$, or simply by $\underline{F}$; thus there is an exact sequence

(13.4.3) $$0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \quad .$$

The most interesting case is that in which (13.4.2) is an epimorphism, that is, in which R_s spans the complex vector space E_s for every $s \in S$. We then have an exact sequence

(13.4.4) $$0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \longrightarrow 0 \quad .$$

If R is locally constant, this is a short exact sequence of locally free modules over S. Knowledge of the pair (E, R) is then equivalent to knowledge of the pair $(R, \underline{F})$, where R is a locally constant analytic group over S whose fibres are free $\mathbb{Z}$-modules of finite type, and $\underline{F}$ is a locally direct-summand submodule of $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$, subject in addition to

(13.4.5) $$\underline{F} \cap \underline{R} = 0 \quad .$$

From the standpoint of Hodge structures, $\mathcal{E}$ should be regarded as filtered by the "Hodge filtration"

(13.4.6) $$\mathrm{Fil}^{-1}(\mathcal{E}) = \mathcal{E}, \qquad \mathrm{Fil}^{0}(\mathcal{E}) = \underline{F}, \qquad \mathrm{Fil}^{1}(\mathcal{E}) = 0 \quad .$$

Notice that the exact sequence (13.4.3) still depends functorially on the analytic group A over S, and commutes with arbitrary base change.

Returning now to the conditions of 13.3, put I≈495

(13.4.7) $$\mathcal{E}^f = \underline{R}^f \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S, \qquad \mathcal{E}^t = \underline{R}^t \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \quad .$$

These modules are defined on a suitable neighbourhood of s, which we may suppose to be all of S, and define a filtration of $\mathcal{E}$, called the weight filtration,

(13.4.8) $$\mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0 \quad .$$

Here $\mathcal{E}^t$, $\mathcal{E}^f$, and $\mathcal{E}$ are to be regarded as having weights -2, -1, and 0, respectively; thus this is an increasing filtration. Like R^t, the module $\mathcal{E}^t$ depends on the choice of a subtorus T_o of $A_o = A_s$. Since R^f and R^t are locally constant, $\mathcal{E}^f$ and $\mathcal{E}^t$ are locally free; they may also be interpreted as

(13.4.9) $$\mathcal{E}^f = \mathcal{E}(A^\natural), \qquad \mathcal{E}^t = \mathcal{E}(T) \quad .$$

Applying the functoriality of (13.4.3), as A varies, to the homomorphisms

$$T \longrightarrow A^\natural \longrightarrow A$$

shows at once that $\underline{F}(A^\natural)$ and $\underline{F}(T)$ identify respectively with

(13.4.10) $$\underline{F}(A^\natural) \simeq \underline{F}^f \overset{\text{def}}{=} \underline{F} \cap \mathcal{E}^f, \qquad \underline{F}(T) = \underline{F}^t = \underline{F} \cap \mathcal{E}^t = 0 \quad .$$

The last equality follows immediately from the fact that $\underline{F}(T) = 0$ for a torus T: in this case the exact sequence of type (13.4.3) is $0 \longrightarrow 0 \longrightarrow \mathcal{E}(T) \longrightarrow \underline{\mathrm{Lie}}(T) \longrightarrow 0$. At the same time we see that the composite homomorphism

$$\mathcal{E}^t \longrightarrow \mathcal{E} \longrightarrow \underline{E}$$

induces an isomorphism

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$$\mathcal{E}^t \xrightarrow{\sim} \underline{E}^t \simeq \underline{\mathrm{Lie}}(T) \qquad . \tag{13.4.11}$$

Suppose now that R_s spans the complex vector space E_s, as happens in particular in the favourable case 13.3.16, where A_s is an extension of an abeloid group by a complex torus T_o. After restricting S to a suitable neighbourhood of s, the canonical homomorphism of type (13.4.2), with A replaced by $A^\natural$,

$$\mathcal{E}^f = \mathcal{E}(A^\natural) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is an epimorphism; a fortiori, so is (13.4.2). We therefore have the exact sequences (13.4.4) and

$$0 \longrightarrow \underline{F}^f \longrightarrow \mathcal{E}^f \longrightarrow \underline{E} \longrightarrow 0 \qquad . \tag{13.4.12}$$

In particular,

$$\underline{F} + \mathcal{E}^f = \mathcal{E} \qquad , \tag{13.4.13}$$

since $\mathcal{E}/\underline{F} \simeq \underline{E}$. Moreover, since $B = A^\natural/T$ is a quotient of $A^\natural$,

it satisfies the analogous surjectivity condition, and hence gives an exact sequence of type (13.4.4):

$$0 \longrightarrow \underline{F}(B) \longrightarrow \mathcal{E}(B) \longrightarrow \underline{\mathrm{Lie}}(B) \longrightarrow 0 \qquad . \tag{13.4.14}$$

Its interpretation in terms of the modules already introduced is immediate. Indeed, by (13.3.12),

$$\begin{cases} \mathcal{E}(B) \simeq (\underline{R}^f/\underline{R}^t) \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \simeq \mathcal{E}^f/\mathcal{E}^t, \\ \underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t. \end{cases} \tag{13.4.15}$$

It follows from (13.4.11) and the relation $\underline{F} \cap \mathcal{E}^t = 0$ in (13.4.10) that $\underline{F}(B) = \ker(\mathcal{E}^f/\mathcal{E}^t \longrightarrow \underline{E}/\underline{E}^t)$ I≈**497**
is isomorphic to $\underline{F}^f$ through the homomorphism

$$\underline{F}^f \simeq \underline{F}(A^\natural) \xrightarrow{\sim} \underline{F}(B) \qquad . \tag{13.4.16}$$

Let us summarize the properties just obtained for the weight filtration (13.4.8), and its relation with the Hodge filtration (13.4.6) of $\mathcal{E}$ defined by the submodule $\underline{F}$, in the favourable case 13.3.16 where A_s is an extension of an abeloid analytic group B_o by a torus T_o. We use this torus to define R^t, so that B is an abeloid analytic group over S.

Proposition 13.4.17. Under the preceding conditions, the Hodge and weight filtrations on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$,

$$\mathcal{E} \supset \underline{F}, \qquad \mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0, \tag{13.4.17.1}$$

satisfy the following conditions:

a) The weight filtration is obtained, by tensoring with $\underline{\mathcal{O}}_S$ over $\mathbb{Z}$, from the analogous filtration on $\underline{R}$

$$\underline{R} \supset \underline{R}^f \supset \underline{R}^t \supset 0,$$

where $\underline{R}^f$ and $\underline{R}^t$ are locally constant analytic groups over S, with values free $\mathbb{Z}$-modules of finite type, namely $R_o^f = R_s = \pi_1(A_s)$ and $R_o^t = \pi_1(T_o)$.

b) One has the relations

$$(*) \qquad \underline{F} \cap \mathcal{E}^t = 0, \qquad \underline{F} + \mathcal{E}^f = \mathcal{E} \qquad .$$

In other words, the filtration on $\mathcal{E}^t$ induced by the Hodge filtration, with the degree conventions of (13.4.6), is concentrated in degree -1, whereas the filtration on $\mathcal{E}/\mathcal{E}^f$ induced by the Hodge filtration is concentrated in degree 0. I≈498

c) Consider on

$$\mathcal{E}^{\mathrm{ab}} \overset{\mathrm{def}}{=} \mathcal{E}^f/\mathcal{E}^t \simeq \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S, \qquad \underline{R}^{\mathrm{ab}} = \underline{R}^f/\underline{R}^t \quad \text{(cf. a))},$$

the filtration induced by the Hodge filtration on $\mathcal{E}$. It is therefore defined by the submodule $\underline{F}^{\mathrm{ab}}$ of $\mathcal{E}^{\mathrm{ab}}$ which is the image of

$$\underline{F}^f \overset{\mathrm{def}}{=} \underline{F} \cap \mathcal{E}^f \longrightarrow \mathcal{E}^{\mathrm{ab}}.$$

The arrow is a monomorphism by the first relation $(*)$. Then the pair

$$(\underline{R}^{\mathrm{ab}}, \underline{F}^{\mathrm{ab}} \subset \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S = \mathcal{E}^{\mathrm{ab}})$$

is isomorphic to the pair $(\underline{R}(B), \underline{F}(B))$ associated with an abeloid group over S, as explained after (13.4.4). Thus

$$B \simeq \mathcal{E}^{\mathrm{ab}}/(F^{\mathrm{ab}} + R^{\mathrm{ab}}),$$

where the symbols without underlining denote the corresponding analytic groups over S.

Remark 13.4.18. In Deligne's terminology [4], the conclusion of the preceding proposition, after restriction to an open subset U of S on which R is locally constant, says that R, endowed with the two filtrations (13.4.17.1) on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$, is an "analytic family of mixed Hodge structures over U". The situation just described served as the model for Deligne's conjectures on Néron theory in higher dimensions, stated in [9, 9.8–9.13].

There is, however, an important feature of the algebraic theory studied in § 7 which is missing from the description in 13.4.17. Suppose that A is an abeloid group over an open subset U complementary to a nowhere-dense analytic subset Z of S—say $Z = \{s\}$ when $\dim S = 1$—so that, by (13.1.5), $R|U$ is a local system over U. Then the local system R/R^t over U ought to be constant; more precisely, it ought to be dual to the local system R'^f, where R' is the analytic group, ètale over S, which would correspond (as R corresponds to A) to a suitable analytic group A' over S such that $A'|U$ is the abeloid group dual to $A|U$. In the notation of § 5, the algebraic analogue is the duality between $T_\ell(A_K)/T_\ell(A_K)^t$ and $T_\ell(A'_K)^f$ in (5.5.9), which follows from the duality theorem 5.2.

I do not know whether such a statement holds under only the hypotheses of 13.4.17, or even whether the action of $\pi_1(U,t)$ on R_t is unipotent for $t \in U$. Nor do the naive methods used in the present paragraph seem capable of proving such a result, even under the additional hypothesis that $A|U$ is endowed with a "polarization", that is, with an invertible sheaf ample on every fibre. This is, however, something that the "semi-simple group people" apparently know how to prove, at least when S is nonsingular of dimension 1; see the report [9]. In the next subsection we shall show that everything works under relative algebraicity hypotheses on A, and at the same time relate the result to the construction of § 7. I≈499

13.5. The case of a relatively algebraic situation. We now suppose that A comes from a relative group scheme over S which is smooth and separated, and whose fibres are extensions of abelian schemes by tori. We denote this relative scheme by the same letter A. (We allow ourselves this abuse of notation because the functor $G \longmapsto G^{\mathrm{an}}$, from the category of relative group schemes of the type I≈500

under consideration to the category of analytic groups over S, is fully faithful, as would be easy to prove using results of relative GAGA type [10].) For everything concerning the notion of a relative scheme and the relations between relative schemes over S and analytic spaces over S, we refer to [10] (see also [2, Exp. 15]).

If $\underline{O}$ denotes the local ring of S at s, then the relative scheme A is determined near s by the ordinary scheme $A_{\underline{O}}$ over $\mathrm{Spec}(\underline{O})$ obtained from A by localization at s. It is then clear that the formal completion of $A_{\underline{O}}$ along its special fibre (isomorphic to $A_s = A_o$) is isomorphic to the formal completion of the analytic group A along A_s (13.2), where, by the abuse of notation introduced above, the $A_n = A \times_S S_n$ are identified with algebraic schemes over S. In the favourable case in which $\hat{A}$, viewed as a formal group scheme over $\underline{\hat{O}}$, is algebraizable (7.2.1), as is the case in particular when S is normal at s and A is quasi-projective over S (7.1.5), we therefore obtain a group scheme $A^{\natural}_{\underline{\hat{O}}}$ over $\mathrm{Spec}(\underline{\hat{O}})$, which is an extension of an abelian scheme $B_{\underline{\hat{O}}}$ by a torus. For simplicity, we suppose that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$ (as is indeed the case, we know, if $\underline{O}$, and hence $\underline{\hat{O}}$, is normal). We shall then prove:

I 501 Proposition 13.5.1. Under the preceding conditions, the analytic group $A^{\natural}$ of (13.3.3) comes, in a neighbourhood of s, from a relative group scheme determined up to a unique isomorphism, which is an extension of a projective relative abelian scheme over S by a torus. If $A^{\natural}_{\underline{O}}$ denotes the group scheme it defines over $\underline{O} = \underline{O}_{S,s}$, then the group scheme

$$A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \underline{\hat{O}}$$

is canonically isomorphic, with its extension structure, to the Raynaud extension $A^{\natural}_{\underline{\hat{O}}}$ introduced above.

The latter assertion follows immediately from the first, in view of 7.2.1, since, by (13.3.6), the formal completion of $A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \underline{\hat{O}}$ along its special fibre is $\hat{A}_{\underline{O}}$. To prove the algebraicity of $A^{\natural}$ near s, proceeding as in 7.2.1, we are immediately reduced to proving that of the abelian quotient B of $A^{\natural}$, which follows from the next lemma.

Lemma 13.5.2. Let B be an abeloid analytic group over the analytic space S, and let s be a point of S such that the completion $\hat{B}$ along the fibre B_s is algebraizable; that is, the $B_n = B \times_S S_n$ are algebraizable and $\hat{B}$, now viewed as a formal scheme over $\underline{\hat{O}}_{S,s}$, is algebraizable, or equivalently comes from an abelian scheme $B_{\underline{\hat{O}}}$ over $\underline{\hat{O}}$. Suppose moreover that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$. Then B is projective over S in a neighbourhood of s.

The hypothesis means that there is a polarization of the formal analytic space $\hat{B}$, that is, a sequence of polarizations p_n of the B_n which "glue together". The conclusion means that there is a polarization p of B over a neighbourhood of s. Let P be the analytic space over S of relative polarizations of B over S, which represents the functor

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$$S' \longmapsto \{\text{polarizations of } B_{S'} \text{ over } S'\}$$

on the category of analytic spaces S' over S. This is an open subfunctor of the Néron–Severi functor $\underline{\mathrm{NS}}_{B/S}$ over S. The simplest standard arguments, which we omit here, show that $\underline{\mathrm{NS}}_{B/S}$, and hence also P, is represented by an unramified analytic space over S. The hypothesis then says that there is a formal section p of P over S at s. Since P is unramified and therefore locally quasi-finite over S, it follows from [2, Exp. 18, no. 1] that every formal section of P over S at s comes from an analytic section near s, that is, from a polarization of B over a neighbourhood of s. This proves the assertion.

Remarks 13.5.3.

a) If in 13.5.2 we omit the hypothesis that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$,

we can still conclude that B is algebraizable near s. For this one uses an unpublished result (due independently to M. Artin and P. Deligne, using the theory of the sheaves $\underline{T}^i$ of VI 2) which asserts that, if X is proper over S and X_{red} is algebraizable, then X is algebraizable. This reduces us to the case where S is reduced. On the other hand, when S is normal, so that $\underline{\hat{O}}$ is normal, every

abelian scheme over $\underline{\hat{O}}$ is known to be projective over $\underline{\hat{O}}$ (see, for example, [23]), and 13.5.2 applies. To obtain the case of an arbitrary reduced S, one uses non-flat descent from the normalization S' of S, by means of the following (delicate) result, which we shall not prove here: a proper morphism of noetherian schemes $f : S' \longrightarrow S$ which is an epimorphism (that is, such that $\underline{O}_S \longrightarrow f_*(\underline{O}_{S'})$ is injective) is an effective descent morphism for the fibred category of relative abelian schemes.

b) It appears that the proof of 13.5.2 should also prove the analogous result for every analytic space B proper over S. Probably more delicate is the question whether an analytic space X proper over S whose formal completion $\hat{X}$ along X_s is algebraizable is itself algebraizable over S near s. We merely point out that here the notion of relative algebraizability of X over S should be taken in a broader sense than in [10]: near s, X should be required to come from an algebraic space over $\underline{O}_{S,s}$ in the sense of M. Artin [1]. I≈503

13.5.4. Let us place ourselves under the conditions of 13.5.1. For every integer $n > 0$, choose also the standard isomorphism of groups over $\mathbb{C}$

$$\mu_{n\mathbb{C}} \simeq (\mathbb{Z}/n\mathbb{Z})_{\mathbb{C}}, \qquad m \longmapsto \exp(2\pi i m/n) \qquad .$$

By passing to the inverse limit over the powers of a fixed prime ℓ, this gives a canonical isomorphism of ℓ-adic sheaves over $\mathbb{C}$,

$$\mathbb{Z}_\ell(1)_{\mathbb{C}} \simeq \mathbb{Z}_{\ell\mathbb{C}} \qquad ,$$

which we shall use to identify the two sides.

Applying the exact sequence of $\underline{\mathrm{Ext}}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to the exact sequence (13.1.3), we obtain a canonical isomorphism

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$$(13.5.5) \qquad {}_nA^{\mathrm{o}} \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \qquad ,$$

which, as n runs through the powers of a given prime ℓ, gives an isomorphism of inverse systems of groups over S,

$$(13.5.6) \qquad T_\ell(A^{\mathrm{o}}) \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad ,$$

where $\mathbb{Z}_\ell$ is regarded as the inverse system of the $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$. In the situation considered here, in which A comes from a separated group scheme over S, it is then immediately clear that the $\underline{R} \otimes \mathbb{Z}/n\mathbb{Z}$ may be regarded as relative schemes. This gives a meaning to the symbol $(\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}$ (whereas $\underline{R}_{\underline{O}}$ would in general have none!), and yields a canonical isomorphism

$$(13.5.7) \qquad T_\ell(A_{\underline{O}}) \simeq (\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

It follows immediately from the constructions of paragraph 2 and 13.3 that the preceding isomorphism carries the fixed part to the fixed part and the toric part to the toric part:

$$(13.5.8) \qquad T_\ell(A_{\underline{O}})^f \simeq (\underline{R}^f \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}, \qquad T_\ell(A_{\underline{O}})^t \simeq (\underline{R}^t \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

If A' is a relative analytic group over S satisfying the same hypotheses as A, and if $W_{\underline{O}}$ is a biextension of $(A_{\underline{O}}, A'_{\underline{O}})$ by $\mathbb{G}_{m\underline{O}}$, hence defines over a neighbourhood of s a biextension of (A, A') by $\mathbb{G}_{mS}$ (now using the notion in the complex-analytic sense, obtained by working in the site of analytic spaces over S for a suitable topology), then, by VIII 2.2, W defines pairings

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$$(13.5.9) \qquad (\underline{R} \otimes \mathbb{Z}_\ell) \times (\underline{R}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell) \simeq T_\ell(A) \times T_\ell(A') \longrightarrow \mathbb{Z}_\ell(1)_S \simeq \mathbb{Z}_{\ell S} \qquad ,$$

and it is again clear that the corresponding pairing

(13.5.10) $$T_\ell(A_{\underline{O}}) \times T_\ell(A'_{\underline{O}}) \longrightarrow \mathbb{Z}_\ell(1)_S$$

is precisely the one associated, by loc. cit., with the biextension $W_{\underline{O}}$ of $(A_{\underline{O}}, A'_{\underline{O}})$. It is also not difficult to see that, after restricting to a neighbourhood of s, the pairings (13.5.9) arise from a unique pairing (no longer depending on ℓ)

(13.5.11) $$\underline{R} \otimes \underline{R}' \longrightarrow \mathbb{Z} \quad .$$

With these preliminaries, the algebraic results concerning the $T_\ell(A_{\underline{O}})$ can be used to draw conclusions about the sheaf $\underline{R}$. Thus the pairing (13.5.11) is zero on $\underline{R}^t \times \underline{R}'^f$ and on $\underline{R}^f \times \underline{R}'^t$ (7.4.8). Moreover, if S is nonsingular of dimension 1 and, over $U = S - \{s\}$, A and A' are relative abelian schemes dual to one another through W, then $\underline{R}^t_U$ and $\underline{R}'^f_U$, as well as $\underline{R}^f_U$ and $\underline{R}'^t_U$, are exact annihilators of one another over a sufficiently small neighbourhood of s (7.4.8). Since (13.5.11) is a perfect duality over U, it follows that there is a duality over U between $\underline{R}/\underline{R}^t$ and $\underline{R}'^f$. In particular, $\underline{R}/\underline{R}^t$ is constant, and a fortiori the action of $\pi_1(U,t)$ on $\underline{R}_t$ is unipotent of level 2. Similarly, over U there is a perfect duality between $\underline{R}/\underline{R}^f$ and $\underline{R}'^t$. Since the latter is also isomorphic to the dual torus of the lattice M_S over S, where M is the group defined in (5.5.4), we obtain a canonical isomorphism

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(13.5.12) $$\underline{R}_U/\underline{R}^f_U \simeq M_U \quad ,$$

and the homomorphism i of (13.3.4) may be written as a homomorphism

(13.5.13) $$i : M_U \longrightarrow A^\natural \quad ,$$

which we shall encounter again in a slightly different context in the next paragraph.

14. Links with transcendental theory: the rigid-analytic case. The canonical homomorphism $\underline{M}_K \longrightarrow A^{\natural o}_K$.

14.1. We assume here the foundations of the theory of rigid-analytic spaces, for which one may consult [12], [13], and [34].

Let S be a complete trait, and let A_K be an abelian scheme over its field of fractions. In 8.1, by descent from the Raynaud extension in the semistable case, we constructed an extension $G_K = A^{\natural o}_K$ (8.1.1)

(14.1.1) $$0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0$$

of an abelian scheme B_K by a torus T_K. Considering the dual abelian scheme A'_K, we likewise have an extension $G'_K = A'^{\natural o}_K$

(14.1.2) $$0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0 \quad ,$$

where B'_K is moreover identified with the abelian scheme dual to B_K (8.2). As in (8.1.8), let $\underline{M}_K$ and $\underline{M}'_K$ denote the lattices over K dual to T'_K and T_K, respectively:

(14.1.3) $$\underline{M}_K = D(T'_K), \qquad \underline{M}'_K = D(T_K) \quad .$$

Thus the extension structures (14.1.2) and (14.1.1) may respectively be interpreted (VIII 3.7) as equivalent to the data of homomorphisms

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$$\underline{M}_K \xrightarrow{p_K} B_K, \qquad \underline{M}'_K \xrightarrow{p'_K} B'_K \qquad . \tag{14.1.4}$$

Let $W_K^{\natural}$ be the biextension of (B_K, B'_K) by $\mathbb{G}_{mK}$ which defines the canonical duality between these abelian schemes (8.2), and let E_K be the biextension of $(\underline{M}_K, \underline{M}'_K)$ by $\mathbb{G}_{mK}$ obtained by pulling $W_K^{\natural}$ back along the homomorphisms (14.1.4). One checks immediately that trivializations of E_K correspond canonically and bijectively to homomorphisms

$$i : \underline{M}_K \longrightarrow G_K \tag{14.1.5}$$

lifting $p_K : \underline{M}_K \longrightarrow B_K$, or, symmetrically, to homomorphisms

$$i' : \underline{M}'_K \longrightarrow G'_K \tag{14.1.6}$$

lifting $p'_K : \underline{M}'_K \longrightarrow B'_K$.

With this understood, write a superscript ^{an} for the rigid-analytic groups over K associated with the algebraic groups under consideration [13]. In [24] one defines a canonical exact sequence

$$0 \longrightarrow \underline{M}_K^{\mathrm{an}} \xrightarrow{i^{\mathrm{an}}} G_K^{\mathrm{an}} \xrightarrow{e} A_K^{\mathrm{an}} \longrightarrow 0 \qquad . \tag{14.1.7}$$

Here i^{an}, which may also be interpreted as a homomorphism $i : \underline{M}_K \longrightarrow G_K$ of ordinary group schemes, lifts the homomorphism p in (14.1.4), that is, it is of the form considered in (14.1.5). This exact sequence is the rigid-analytic analogue of the exact sequence (13.3.4). It depends functorially on A_K, and its formation is compatible with every change of traits $S' \longrightarrow S$.

The latter fact also shows that, in order to define and study (14.1.7), finite descent reduces us to I≈**508**
the case where A_K has semistable reduction. In this case the group schemes $T_K, G_K, B_K, \underline{M}_K, \ldots$ are the generic fibres of group schemes $T, G, B, M, \ldots$ over S, with G an extension of the abelian scheme B by the torus T (paragraph 7). We may then consider the formal completion $\hat{G}$ of G along its special fibre, and by construction there is a canonical isomorphism (7.2.6)

$$\hat{G} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}} \qquad . \tag{14.1.8}$$

Passing to the "generic fibres", which are rigid-analytic groups over K, gives a canonical isomorphism

$$\hat{G}^{\mathrm{an}} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}}^{\mathrm{an}} \qquad . \tag{14.1.9}$$

On the other hand, there are canonical homomorphisms (rigid-analytic open immersions, but not in general isomorphisms)

$$\hat{G}^{\mathrm{an}} \hookrightarrow G_K^{\mathrm{an}}, \qquad \widehat{A^{\mathrm{o}}}^{\mathrm{an}} \hookrightarrow A_K^{\mathrm{an}} \qquad . \tag{14.1.10}$$

The homomorphism e in (14.1.7) is uniquely characterized by the commutativity of the diagram

$$\begin{array}{ccc} \hat{G}^{\mathrm{an}} & \longrightarrow & \widehat{A^{\mathrm{o}}}^{\mathrm{an}} \\ \downarrow & & \downarrow \\ G_K^{\mathrm{an}} & \longrightarrow & A_K^{\mathrm{an}} \end{array} \quad , \tag{14.1.11}$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

Remarks 14.1.12.

a) Notice that the uniqueness of e making the preceding diagram commute is immediate, since the rigid-analytic space G_K^{an} is connected. The existence of e, on the other hand, is by no means trivial, and on this point we merely refer to the cited note of M. Raynaud. Moreover, the definition of i^{an} given in loc. cit. is likewise transcendental in nature, although it may be interpreted as an object of an essentially algebraic nature (14.1.5), once the construction of G itself, and hence of G_K, has been obtained through the constructions of formal geometry in paragraph 7. I know of no purely algebro-geometric description of it, but below (14.4) I shall propose an "arithmetic" characterization, obtained by reducing to a situation of finite type over the ring $\mathbb{Z}$. Thus, from the standpoint of algebraic geometry, the homomorphism (14.1.5) seems to be a more hidden piece of structure than those encountered in the preceding paragraphs.

b) As explained above, the homomorphism i is canonically associated with a well-determined trivialization of the biextension E_K of $(\underline{M}_K, \underline{M}'_K)$, or equivalently with a well-determined homomorphism (14.1.6) lifting p'_K. One then proves that the latter is precisely the homomorphism occurring in the exact sequence I≈509

$$0 \longrightarrow \underline{M}'^{\mathrm{an}}_K \xrightarrow{i'^{\mathrm{an}}} G'^{\mathrm{an}}_K \longrightarrow A'^{\mathrm{an}}_K \longrightarrow 0 \tag{14.1.13}$$

analogous to (14.1.7), associated with the abelian scheme A'_K dual to A_K (unless it is the opposite???).

14.2. Relations of the homomorphism $i : \underline{M}_K \longrightarrow G_K$ with monodromy. I510

Let n be an integer > 0. From the exact sequence (14.1.7), by means of the exact sequence of the $\underline{\mathrm{Ext}}^i(\mathbb{Z}/n\mathbb{Z}, -)$ on the site of rigid-analytic spaces over K, one then obtains a canonical exact sequence (taking account of the fact that $\underline{M}_K^{\mathrm{an}}$ is torsion-free and G_K^{an} is n-divisible):

$$0 \longrightarrow {}_nG_K^{\mathrm{an}} \longrightarrow {}_nA_K^{\mathrm{an}} \longrightarrow \underline{M}_{K,n}^{\mathrm{an}} \longrightarrow 0 \quad ,$$

where $\underline{M}_{K,n} \overset{\mathrm{dfn}}{=} \underline{M}_K/n\underline{M}_K$. Since all the terms occurring in this exact sequence come from finite group schemes over K, it may be rewritten as arising from an exact sequence

$$0 \longrightarrow {}_nG_K \longrightarrow {}_nA_K \longrightarrow \underline{M}_{K,n} \longrightarrow 0 \quad . \tag{14.2.1}$$

As n varies, these exact sequences behave in the expected way, and, on passing to the inverse limit over the powers of a prime number ℓ, one obtains the analogous exact sequence

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow 0 \quad . \tag{14.2.2}$$

But one already has such an exact sequence by purely algebraic means, thanks to the first formula (8.1.2) and to (8.1.7), which give

$$T_\ell(G_K) \xrightarrow{\sim} T_\ell(A_K)^{\mathrm{ef}} \subset T_\ell(A_K), \qquad T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}} \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad . \tag{14.2.3}$$

With this understood, I claim that the homomorphisms (14.2.3) arising from the algebraic theory are precisely those deduced from the exact sequence (14.2.2) arising from the transcendental theory. For the first homomorphism, which corresponds to the first homomorphism of (14.2.2), our assertion follows at once, after reduction to the semistable situation, from the commutativity of (14.1.11), since the vertical arrows there induce isomorphisms on the T_ℓ, while applying T_ℓ to the horizontal arrows gives, respectively, the arrow under consideration in (14.2.3) and that in (14.2.2). We shall admit the asserted compatibility for the two isomorphisms obtained between $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ and $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. Notice, moreover, that this compatibility (for a given ℓ) already suffices to characterize

uniquely, by transcendental means, the homomorphism (14.1.5) occurring in (14.1.7), interpreted as an isomorphism from $\underline{M}_K^{\mathrm{an}}$ onto $\mathrm{Ker}\, e$.

One may also say that the homomorphism e of (14.1.7) identifies the $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ with the ℓ-adic completions $\underline{R}_K \otimes \mathbb{Z}_\ell$ of a single lattice $\underline{R}_K = \mathrm{Ker}\, e$ over K, and that this "integral structure" on the preceding system of pro-ℓ groups is compatible, in the evident sense, with the integral structure arising from formulas (8.1.7) (orthogonality theorem 5.2). This then defines canonically an isomorphism I≈511

$$i : \underline{M}_K \xrightarrow{\sim} \mathrm{Ker}\, e \qquad .$$

Since the homomorphism i (14.1.5) enables one to reconstruct the usual extension structure on $T_\ell(A_K)$, namely (14.2.2), and hence also its structure as a blended extension, described in 9.6, by means of the exact sequence deduced from (14.1.1),

$$0 \longrightarrow \qquad T_\ell(T_K) \qquad \longrightarrow \quad T_\ell(G_K) \quad \longrightarrow \quad T_\ell(B_K) \quad \longrightarrow \quad 0$$

$$\underline{M}'_K \otimes \mathbb{Z}_\ell(1)$$

it also enables one, in the semistable case, to reconstruct the monodromy pairings (9.1.2) I≈512

(14.2.4) $$u_\ell : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_\ell \qquad ,$$

these having been defined precisely in terms of blended extensions in 9.6. Let us merely indicate the result of translating the construction of 9.6. For this purpose, observe that in the semistable case p_K and p'_K arise from homomorphisms I436

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$$p : \underline{M} \longrightarrow B, \qquad p' : \underline{M}' \longrightarrow B' \qquad ,$$

where B, B' are abelian schemes over S, dual to one another. Pulling back by (p, p') therefore gives a biextension E of $(\underline{M}, \underline{M}')$ by $\mathbb{G}_{mS}$ whose generic fibre is the biextension E_K already considered in 14.1.

With this said, suppose for simplicity that $\underline{M}$ and $\underline{M}'$ are constant, with values M and M'. For every pair $(m, m') \in M \times M'$, $E_{m,m'}$ is a torsor under $\mathbb{G}_{mS}$, and hence corresponds to an invertible module $L_{m,m'}$ over the ring V of S. It therefore admits a basis u_0, determined up to multiplication by a unit of V. Consequently one obtains a canonical "valuation" map

$$\underline{v} : E_{m,m'}(K) \longrightarrow \mathbb{Z}$$

which associates with every basis u of $L_{m,m'} \otimes_V K$, written $u = \lambda u_0$ ($\lambda \in K^*$), the element

$$\underline{v}(u) = \underline{v}(\lambda) \in \mathbb{Z} \qquad ,$$

where $\underline{v}$ on the right denotes the valuation associated with V (normalized in the usual way, so that its value group is $\mathbb{Z}$); the resulting integer is plainly independent of the choice of u_0. If φ is now a trivialization of E, one may consider the corresponding map

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(14.2.5) $$(m, m') \longmapsto \underline{v}(\varphi(m, m')) : M \times M' \longrightarrow \mathbb{Z} \qquad .$$

With this understood, one finds that the monodromy homomorphism (14.2.4) takes its values in $\mathbb{Z}$ and is precisely the homomorphism (14.2.5), in which φ is the splitting of E_K associated with the homomorphism i (14.1.5).

Remarks 14.2.6. In particular, one recovers (by means of the transcendental rigid-analytic theory) the fact that the monodromy pairing is integral and independent of ℓ (10.4 a)), proved by a different

(and likewise transcendental) method in 12.5. According to M. Raynaud, rigid-analytic theory also recovers the positivity assertion 10.4 b) (which is likewise contained in 12.5).

Remarks 14.2.7. Regard $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}$ as an extension of $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$ by $\underline{M}_K$. Using the homomorphism (14.1.5) $i : \underline{M}_K \longrightarrow G_K$, one deduces an extension $A_K^\flat$

$$0 \longrightarrow G_K \longrightarrow A_K^\flat \longrightarrow \underline{M}_K \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \qquad . \tag{14.2.8}$$

Observe moreover that, since G_K is a divisible group, the datum of such an extension is equivalent to that of a family of extensions of the form (14.2.2), namely those deduced from (14.2.8) by applying the T_ℓ. Since i is a monomorphism, the extensions in question which define (14.2.8) are precisely the extensions (14.2.2) arising from the exact sequence (14.2.1). As these latter extensions are defined by purely algebraic means, the same is true of the extension (14.2.8). In fact, the characterization just given shows at once that this extension is canonically isomorphic to the group, likewise denoted $A_K^\flat$, introduced in (8.2.3), in view of the isomorphism 11.11 from the group of connected components of the latter to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$.

14.3. Application to an arithmetic characterization of $i : \underline{M}_K \longrightarrow G_K$. I 514

The characterization we have in mind rests on the observation that, for every prime number ℓ, the extension (14.2.2) of $\underline{M}_K \otimes \mathbb{Z}_\ell$ by $T_\ell(G_K)$ deduced from i is known by purely algebraic means. In particular, the image of i under the canonical homomorphism

$$\mathrm{Hom}(\underline{M}_K, G_K) \longrightarrow \prod_\ell \mathrm{Ext}^1(\underline{M}_K \otimes \mathbb{Z}_\ell, T_\ell(G_K)) \qquad . \tag{14.3.1}$$

is known. In the case (to which one may always reduce by descent) in which $\underline{M}_K$ is constant, with value M, the preceding homomorphism may be interpreted as

$$\mathrm{Hom}(M, G_K(K)) \longrightarrow \prod_\ell \mathrm{Hom}(M, H^1(K, T_\ell(G_K))), \tag{14.3.2}$$

deduced from the canonical homomorphism

$$G_K(K) \longrightarrow \prod_\ell H^1(K, T_\ell(G_K)) \qquad , \tag{14.3.3}$$

which comes from the connecting homomorphisms in the Kummer exact sequences

$$0 \longrightarrow {}_nG_K \longrightarrow G_K \xrightarrow{n} G_K \longrightarrow 0 \qquad .$$

(The homomorphism (14.3.1) could in fact always be interpreted as the connecting homomorphism I ≈515 analogous to (14.3.3) for the twisted group $\check{M}_K \otimes G_K = \underline{\mathrm{Hom}}(\underline{M}_K, G_K)$.) The kernel of (14.3.3) plainly consists of the elements of $G_K(K)$ which are divisible by n for every integer $n \geq 1$, or, as we shall say, which are infinitely divisible. Thus this already determines i modulo an element of $\mathrm{Hom}(M, g)$, where g is the subgroup of $G_K(K)$ formed by its infinitely divisible elements:

$$g = \bigcap_{n \geq 1} nG_K(K) \qquad .$$

We now have the following result.

Lemma 14.3.4. Let V be a discrete valuation ring whose residue field k has characteristic $p > 0$ and is of finite type over its prime field, and let G_K be a group scheme of finite type over K, of unipotent rank zero if $\mathrm{char}\, K = 0$. Then the group g of infinitely divisible elements of $G_K(K)$ is zero; that is, (if G_K is divisible) the homomorphism (14.3.3) is injective.

After making a finite extension K'/K and replacing V by its normalization in K', we may suppose that G_K admits a composition series whose successive quotients are of one of the following types: a constant finite group, an abelian variety, $\mathbb{G}_{mK}$, or $\mathbb{G}_{aK}$. We are therefore immediately reduced to the case where G_K itself is of one of these types. The case of a constant finite group, or of $\mathbb{G}_{aK}$, is trivial. In the abelian case, G_K may be regarded as the generic fibre of its Néron model G, so we are reduced to showing that, if G is a scheme of finite type over S, the subgroup of $G(S)$ formed by the infinitely divisible points is zero.

This is true of the group $G(k)$, as follows at once from the usual dévissage of G_o and from the Mordell–Weil theorem [16] (which says that $G_o(k)$ is a finitely generated $\mathbb{Z}$-module if G_o is an abelian scheme over the field of finite type k). Consequently g is contained in the kernel $G(S)^{(1)}$ of $G(S) \longrightarrow G(k)$. The kernel of $G(S)^{(1)} \longrightarrow G(V/\underline{m}^{n+1})$ has a filtration whose quotients are isomorphic to vector groups over k (after supposing, as we may, that S is complete), and these are killed by p. It follows that g is contained in the kernel of the preceding homomorphism for every n, and hence that it is zero. The case $G_K = \mathbb{G}_{mK}$ reduces to the preceding one, since $G_K(K) = K^*$ is an extension of $\mathbb{Z}$ (which has no nonzero infinitely divisible elements) by $V^* = \mathbb{G}_m(S)$, and the latter satisfies the same condition by what we have just proved.

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Remarks 14.3.4.1. Notice that in 14.3.4 it is essential to require divisibility by all integers ≥ 1, and not only by the powers of a fixed prime ℓ; equivalently, one must consider the kernel of (14.3.3), rather than restricting the right-hand side to a single factor $H^1(K, T_\ell(G_K))$. Indeed, if G_K comes from a smooth group scheme G over S and S is henselian, every element of the kernel of $G(S) \longrightarrow G(k)$ is infinitely ℓ-divisible for every $\ell \neq p$! The argument proving 14.3.4 shows that the subgroup of $G_K(K)$ formed by the infinitely p-divisible elements (that is, the kernel of $G_K(K) \longrightarrow H^1(K, T_p(G_K))$ when G_K is p-divisible) is a finite group.

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14.3.5. The preceding considerations show that, when the residue field k has characteristic $p > 0$ and is of finite type in the absolute sense, the homomorphism i (14.1.5) is completely determined by its image under (14.3.1), that is, by the extensions (14.2.2). To deduce from this a principle characterizing i in the general case, with no restriction on k, observe that in paragraph 7 we were able to construct the scheme G over S under conditions considerably more general than those in which S is a complete trait and one has an abelian variety with semistable reduction over its field of fractions.

More generally, let S denote the spectrum of a normal noetherian ring V, separated and complete for an $\underline{m}$-adic topology, and consider a smooth quasi-projective group scheme A over S, with connected fibres. Suppose that $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$, is an extension of an abelian scheme B_o by an isotrivial torus T_o, and that A_U is an abelian scheme, where $U \subset S - S_o$ is a nonempty open subset. In 7.2.2 we then constructed a scheme G over S, an extension of an abelian scheme B by a torus T, characterized up to unique isomorphism by the isomorphism (14.1.8)

$$\hat{G} \xrightarrow{\sim} \hat{A} \tag{14.3.5.1}$$

on the formal completions along G_o and A_o. For every prime number ℓ one deduces an isomorphism (7.3.5)

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$$T_\ell(G) \xrightarrow{\sim} T_\ell(A)^f \quad , \tag{14.3.5.2}$$

and hence, on restricting to U, the structure of an extension of pro-ℓ Barsotti–Tate groups

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_\ell(G_U) & \longrightarrow & T_\ell(A_U) & \longrightarrow & T_\ell(A_U)/T_\ell(A_U)^f & \longrightarrow & 0 \\ & & \wr\downarrow & & & & & & \\ & & T_\ell(A_U)^f & & & & & & \end{array} \quad . \tag{14.3.5.3}$$

Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, together with a biextension W of (A, A') by $\mathbb{G}_m$ which over U induces a duality between the abelian schemes A_U and A'_U. (In fact, by VIII 7.1, giving W is equivalent to giving such a duality between A_U and A'_U.) Then W defines pairings

$$T_\ell(A_U) \times T_\ell(A'_U) \longrightarrow \mathbb{Z}_\ell(1) \qquad , \tag{14.3.5.4}$$

which give rise to the orthogonality theorem (7.4.8): $T_\ell(A_U)^f$ is the orthogonal of $T_\ell(A'_U)^t = T_\ell(T'_U)$, where T' is the toric part of the Raynaud extension $G' = A'^\natural$ associated with A'. Thus one again obtains an isomorphism

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq \underline{M}_U \otimes \mathbb{Z}_\ell \qquad , \tag{14.3.5.5}$$

where $\underline{M} = D(T')$ is the lattice over S dual to the torus T'. Consequently (14.3.5.3) and (14.3.5.5) give an extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow \underline{M}_U \otimes \mathbb{Z}_\ell \longrightarrow 0 \qquad . \tag{14.3.5.6}$$

With this understood, we may state the following conjecture.

Conjecture 14.4. The hypotheses being those just set out in 14.3.5, there exists a homomorphism I≈519

$$i : \underline{M}_U \longrightarrow G_U \tag{14.4.1}$$

such that, for every prime number ℓ, the extension (14.3.5.6) is isomorphic to the extension deduced from i as explained for (14.2.2). More precisely, there is one and only one way of associating with every situation of the type considered in 14.3.5 a homomorphism (14.4.1) satisfying the preceding condition, in such a way that the formation of i is compatible with every base change $S' \longrightarrow S$ associated with a continuous homomorphism $V \longrightarrow V'$ of noetherian adic rings.

14.4.2. It seems that the existence of i, compatibly with base change, follows from M. Raynaud's rigid-analytic theory, in view of the fact that he is developing it in a framework more general than that stated in 14.1, working, as here, over general noetherian adic rings. Moreover, to prove the existence and uniqueness asserted in 14.4, one is immediately reduced to considering only those S for which S_o is a scheme of finite type over $\mathbb{Z}$. Indeed, V is the filtered inductive limit of subrings V_i of finite type over $\mathbb{Z}$; these may be supposed normal by Nagata's theorem (EGA IV 7.8), and, for i sufficiently large, the situation (A, A', W) comes from an analogous situation over V_i, or alternatively over the separated completion of V_i for the topology defined by $m_i = \underline{m}V_i$. This gives the asserted reduction.

14.4.3. Observe now that $S - S_o$ is a Jacobson scheme (EGA IV 10.5.8). Thus, to see that two solutions i, i' of the problem, for a given situation (A, A', W), are equal, it is enough to prove that these homomorphisms are equal at every closed point η of U. Let S' be the normalization of the closure of η in S. Then S' is a complete trait, and the uniqueness result of 14.3.4 may be applied to the pullback to S' of the situation over S, since the residue field of S' is finite. Once uniqueness is thus established, it implies compatibility of the formation of i with base change, subject to existence. I 520

Remarks 14.5. It would remain to formulate with the desired generality, and to establish, a compatibility between the homomorphism i introduced in 14.4 by a rigid-analytic method and the analogous homomorphism (13.5.13) of complex-analytic theory—or, better still, to find a common generalization of them!

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Part II

SGA 7 II

II 1

INTERSECTIONS ON REGULAR SURFACES

by P. Deligne

In the first part of this exposé, after some preliminaries from intersection theory (contained in [5] ∪ [7]), we give Artin's proof of a special case of a theorem of Raynaud [9] (1.13), and an a priori proof of a theorem which Néron [8] had verified case by case (1.15). The theorems of [9] used in [2], and hence in IX, follow fairly easily from (1.13).

In the second part, continuing in the same vein, I explain how the "Italians" [1], [3] calculated intersection numbers on surfaces. The main results, 2.9 and 2.11, may also be found in Northcott [12] and [13]. To conclude, in 2.27, 2.30, and 2.34 I give a conjectural relation between the dimension of the versal formal moduli scheme of a singularity of a reduced curve, the invariant δ of the singularity, and an invariant measuring its symmetry.

This exposé will not be used in the remainder of the seminar.

1. Intersection numbers on arithmetic surfaces

1.0. Throughout this section, let X be a regular scheme of dimension 2, let Y be a reduced divisor on X, and let k be a subfield of $H^0(Y, \underline{O}_Y)$. We assume that Y is a curve (EGA II 7.4.1) proper over k. Here are three examples of this situation.

Example 1.1. Let (S, η, s) be a trait and let $f : X \longrightarrow S$ be a regular, proper, and flat S-scheme whose fibres are purely one-dimensional. We take $Y = (f^{-1}(s))_{\mathrm{red}}$ and $k = k(s)$. From 1.10 onward we shall restrict ourselves to this case.

Example 1.2. Let X_o be the spectrum of a normal local ring of dimension 2, let s be the closed II 2
point of X_o, and let $f : X \longrightarrow X_o$ be a regular connected X_o-scheme, proper over X_o, distinct from X_o but agreeing with X_o over $X_o \setminus \{s\}$. We take $Y = (f^{-1}(s))_{\mathrm{red}}$ and $k = k(s)$.

Example 1.3. We take X to be a complete regular surface over a field k, and Y to be a reduced curve on X.

Recall the following lemma (SGA 6 IV, Ex. 1.9).

Lemma 1.4. Let Z_o be a closed subscheme of a Noetherian scheme Z. Then the inclusion of Z_o in Z induces an isomorphism from the Grothendieck group of the category of coherent $\underline{O}_{Z_o}$-modules to the Grothendieck group of the category of coherent sheaves on Z with support in Z_o.

This lemma allows us to define the Euler–Poincaré characteristic $\chi(\mathcal{F})$ of a coherent sheaf $\mathcal{F}$ on X supported on Y so that:

(i) If $\mathcal{F}$ is a coherent sheaf of $\underline{O}_Y$-modules, then

$$\chi(\mathcal{F}) = \dim_k H^0(\mathcal{F}) - \dim_k H^1(\mathcal{F}). \tag{1.4.1}$$

(ii) χ is additive with respect to short exact sequences.

If the cohomology sheaves of a complex K of $\underline{O}_X$-modules are coherent sheaves supported on Y, we set

$$\chi(K) = \sum_i (-1)^i \chi(\underline{H}^i(K)). \tag{1.4.2}$$

If $\mathcal{F}$ is supported at a point y of Y, then

$$\chi(\mathcal{F}) = \mathrm{length}_{\underline{O}_y}(\mathcal{F}_y)[k(y) : k]. \tag{1.4.3}$$

Under the hypotheses of (1.1), (1.2), or (1.3), one has II ≈3

$$\chi(\mathcal{F}) = \mathrm{length}\, f_*\mathcal{F} - \mathrm{length}\, R^1 f_*\mathcal{F}. \tag{1.4.4}$$

Let D be a divisor ≥ 0 supported on Y, and let $\mathcal{L}$ be an invertible sheaf on D. There is then a unique way to define the degree $\deg_D(\mathcal{L})$ such that

$$\begin{cases} \deg_D(\mathcal{L}' \otimes \mathcal{L}'') = \deg_D(\mathcal{L}') + \deg_D(\mathcal{L}''), \\ \\ \text{for } \mathcal{L} \subset \underline{O}, \text{ one has } \deg_D(\mathcal{L}) = -\chi(\underline{O}/\mathcal{L}). \end{cases} \tag{1.4.5}$$

The Riemann–Roch formula on D is then

$$\chi(\mathcal{L}) = \chi(\underline{O}_D) + \deg_D(\mathcal{L}). \tag{1.4.6}$$

If $D = \sum n_i D_i$ is the decomposition of D into reduced irreducible divisors, then

$$\deg_D(\mathcal{L}) = \sum n_i \deg_{D_i}(\mathcal{L}). \tag{1.4.7}$$

Definition 1.5. Let D and E be two divisors ≥ 0 on X, and suppose that D is supported on Y. The intersection number of D and E (relative to the field k (1.0)) is defined by

$$(D, E) = \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) = \chi(\underline{O}_D \otimes \underline{O}_E) - \chi(\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E)).$$

Indeed, under the hypotheses made here, $\mathrm{Tor}^2(\underline{O}_D, \underline{O}_E) = 0$.

Proposition 1.6. Under the hypotheses of 1.5, one has:

(i) The intersection number (D, E) is additive in the divisors D and E; this allows us, by linearity, to define the intersection number of two divisors which are not necessarily positive.

(ii) $(D, E) = \deg_D(\underline{O}(E))$. II≈4

(iii) If D and E meet at only finitely many points, then $(D, E) = \chi(\underline{O}_D \otimes \underline{O}_E) \geq 0$, and equality can occur only if D and E are disjoint.

(iv) If D and E are supported on Y, then $(D, E) = (E, D)$.

(v) If N is the conormal sheaf of D, then

$$D^2 = (D, D) = -\deg_D(N).$$

(vi) If E, not necessarily positive, is the divisor of a meromorphic function on X, then $(D, E) = 0$.

Proof. The exact sequence

$$0 \longrightarrow \underline{O}(-E) \longrightarrow \underline{O}_X \longrightarrow \underline{O}_E \longrightarrow 0$$

gives, by (1.4.6),

$$\begin{aligned}(D, E) &\overset{\text{def}}{=} \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) \\ &= \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_X) - \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}(-E)) \\ &= \chi(\underline{O}_D) - \chi(\underline{O}(-E)|D) = -\deg_D(\underline{O}(-E)) = \deg_D(\underline{O}(E)).\end{aligned}$$

This proves (ii), (vi), and biadditivity, by (1.4.5) and (1.4.7). Under the hypotheses of (iii), one has $\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E) = 0$, which gives the conclusion. Under the hypotheses of (v), one has

$$\underline{O}_D \otimes \underline{O}_D = \underline{O}_D \quad \text{and} \quad \mathrm{Tor}^1(\underline{O}_D, \underline{O}_D) \simeq N \quad \text{(SGA 6 VII 2.5)},$$

hence

$$(D, D) = \chi(\underline{O}_D) - \chi(N) = -\deg_D(N).$$

Finally, (iv) follows from the symmetry of the tensor product.

Notice that (ii), (iv), and (vi) remain true without assuming the divisors positive.

Proposition 1.7. Let $(Y_i)_{1 \leq i \leq k}$ be the irreducible components of Y, f a meromorphic function on X, and II5

$$\mathrm{div}(f) = \sum_i n_i Y_i + Z$$

its divisor, with $|Z| \cap Y$ finite. Then, putting $D_i = n_i Y_i$, one has, for $(a_i) \in \mathbb{Q}^k$:

$$\left(\sum_i a_i D_i\right)^2 = -\sum_{i<j} (a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).$$

Proof (after [7]). By 1.6 (v), (vi),

$$\begin{aligned}\left(\sum_i a_i D_i\right)^2 &= \sum_i a_i \left(D_i, \sum_j a_j D_j\right) = \sum_i a_i \left(D_i, \sum_j a_j D_j - a_i \operatorname{div} f\right)\\ &= \sum_i a_i \left(D_i, \sum_{j\neq i}(a_j - a_i)D_j - a_i Z\right)\\ &= \sum_{i\neq j} a_i(a_j - a_i)(D_i, D_j) - \sum_i a_i^2 (D_i, Z)\\ &= -\sum_{i<j}(a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).\end{aligned}$$

Corollary 1.8. Under the hypotheses of Example 1.1, take for f a uniformizer π of S. Here $Z = 0$ and

$$\left(\sum_i k_i Y_i\right)^2 = -\sum_{i<j} \frac{1}{n_i n_j}(k_i n_j - k_j n_i)^2 (Y_i, Y_j);$$

thus the intersection form is negative, and if Y is connected, its kernel consists precisely of the multiples of $\operatorname{div}(\pi)$.

Corollary 1.9 (Mumford [7]). Under the hypotheses of Example 1.2, take for f a nonzero element of the maximal ideal of the local ring $\underline{O}_s$. Here $Z > 0$, so the intersection form is negative definite.

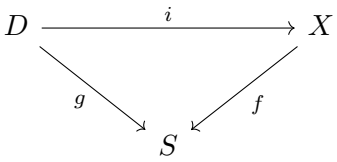
1.10. For the remainder of this section we restrict ourselves to the case considered in Example 1.1. II 6

Lichtenbaum [5] proved that, in this case, X is automatically projective over S, so that there is no difficulty in defining the complex $Rf^!\underline{O}_S$ ([4] III 8.7, p. 190). Since f has relative dimension 1 and X and S are regular (Gorenstein would suffice), there exists ([4] V 8.3 and V 9.1) an invertible sheaf $\omega_{X/S}$ on X, unique up to unique isomorphism, such that

$$Rf^!\underline{O}_S \simeq \omega_{X/S}[1] \quad . \tag{1.10.1}$$

We denote by K any divisor (effective or not) such that $\omega_{X/S} \simeq \underline{O}(K)$.

Let D be a divisor ≥ 0 supported on Y, and let N be its conormal sheaf:



One has

$$Rg^!\underline{O}_S = Ri^! Rf^!\underline{O}_S = Ri^!\omega_{X/S}[1] = \omega_{X/S}\big|_D \otimes N^\vee[0]. \tag{1.10.2}$$

By the duality theorem ([4] V 11.1, p. 210),

$$Rg_* Rg^!\underline{O}_S \simeq \mathrm{RHom}(Rg_*\underline{O}_D, \underline{O}_S) \quad . \tag{1.10.3}$$

Moreover, for every $\underline{O}_S$-module M supported at the closed point,

$$\chi(\mathrm{RHom}(M, \underline{O}_S)) = -\chi(M), \tag{1.10.4}$$

so that here, using (1.4.4) and (1.10.3), we obtain

$$\chi(Rg^!\underline{O}_S) = -\chi(\underline{O}_D) \quad .$$

Since, by (1.4.6),

$$\chi(Rg^!\underline{O}_S) = \chi(\underline{O}_D) + \deg_D(Rg^!\underline{O}_S),$$

we conclude from (1.10.2) that II≈7

$$-2\chi(\underline{O}_D) = \deg_D(Rg^!\underline{O}_S) = \deg_D(\omega_{X/S}|_D \otimes N^\vee) = \deg_D(\omega_{X/S}) - \deg_D(N),$$

which, by 1.6(ii) and (v), may be rewritten as follows.

Proposition 1.11. Under the hypotheses of (1.1), for every divisor $D \geq 0$ supported on Y, one has

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D+K) \quad .$$

1.12. We denote by d the greatest common divisor of the geometric multiplicities of the irreducible components of $X_s = f^{-1}(s)$; that is, the greatest common divisor of the lengths of the local rings at the generic points of the geometric special fibre of f. If X_η has a rational point, or equivalently if $f : X \longrightarrow S$ has a section x, then $d = 1$; more precisely, f is smooth at $x(s)$. Indeed, x is a closed immersion of regular schemes, hence a regular immersion, and EGA IV 17.12.1 applies.

Here is Artin's proof of a special case of a theorem of Raynaud [9], which only assumes that d is prime to the characteristic p of k.

Theorem 1.13. With the preceding notation, if Y is geometrically connected and $d = 1$, then X is cohomologically flat over S in dimension 0 (EGA III 7.8.1); more precisely (EGA III 7.6.9(ii)), one has

$$k \xrightarrow{\sim} H^0(X_s, \underline{O}_{X_s}).$$

It is easy to reduce to the case in which S is strictly local. Since Y is geometrically connected, $H^0(Y, \underline{O}_Y)$ is a purely inseparable extension of k; and since $(d, p) = 1$, at least one irreducible component of Y is

geometrically reduced. It follows readily that $H^0(Y, \underline{O}_Y) = k$. II≈8

Let D be a divisor on X such that

$$Y \leq D < X_s \quad .$$

Since $d = 1$, D is not a rational multiple of X_s; hence, by 1.8,

$$(D, X_s - D) = -(D, D) > 0 \quad .$$

The divisor Y therefore has an irreducible component E such that

$$0 < E \leq X_s - D \quad \text{and} \quad (D, E) > 0 \quad .$$

Consider the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \underline{O}(-D-E) & \longrightarrow & \underline{O} & \longrightarrow & \underline{O}_{D+E} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \varphi} & & \| & & \downarrow{\scriptstyle \psi} & & \\
0 & \longrightarrow & \underline{O}(-D) & \longrightarrow & \underline{O} & \longrightarrow & \underline{O}_D & \longrightarrow & 0
\end{array}$$

The snake lemma gives an isomorphism between $\mathrm{Ker}(\psi)$ and $\mathrm{Coker}(\varphi)$. This cokernel is precisely the restriction of $\underline{O}(-D)$ to E, whence an exact sequence

$$0 \longrightarrow \underline{O}(-D)|_E \longrightarrow \underline{O}_{D+E} \longrightarrow \underline{O}_D \longrightarrow 0 \quad .$$

By construction, $\deg_E(\underline{O}(-D)|_E) = -(D,E) < 0$, so that $H^0(\underline{O}(-D)|_E) = 0$ and the restriction map

$$0 \longrightarrow H^0(\underline{O}_{D+E}) \longrightarrow H^0(\underline{O}_D)$$

is injective. This allows us, by induction, to construct a sequence of divisors

$$Y = D_0 \leq D_1 \leq \cdots \leq D_n = X_s$$

such that $H^0(\underline{O}_{D_i})$ injects into $H^0(\underline{O}_{D_{i-1}})$. Thus $H^0(\underline{O}_{X_s}) = H^0(\underline{O}_Y) = k$, which completes the proof of 1.13.

The following fact is proved in [5] or [10]. II≈**9**

Theorem 1.14 (Castelnuovo, Enriques, ...). Under the hypotheses of (1.1), let D be a reduced irreducible divisor contained in Y. Suppose that $H^1(D, \underline{O}_D) = 0$ and that, on putting $k' = H^0(D, \underline{O}_D)$, one has

$$(D,D) = -[k' : k] \quad .$$

Then D is a projective line over k', and there exist a regular S-scheme X_1, a closed point x_1 of X_1 with residue field k', and an isomorphism from X to the S-scheme obtained from X_1 by blowing up x_1, under which D is identified with the fibre over x_1.

A divisor D having these properties is called an exceptional curve of the first kind.

I cannot resist the pleasure of proving the following proposition of Néron [8], Proposition 1′, p. 90.

Proposition 1.15. Suppose that X_η is an elliptic curve (in particular, that it has a section), and that X contains no exceptional curve of the first kind. Then, with the notation of (1.10.1), the canonical map

$$f^* f_* \omega_{X/S} \xrightarrow{\sim} \omega_{X/S}$$

is universally an isomorphism.

Since $d = 1$ (1.12), by (1.13) it is enough to prove that $\omega_{X/S}$ is isomorphic to $\underline{O}_X$. Since $\omega_{X/S}$ is trivial on the elliptic curve X_η, we know a priori that

$$\omega_{X/S} \simeq \underline{O}\Big(\sum_i n_i Y_i\Big),$$

where the Y_i are the irreducible components of Y; and since $\mathrm{div}(\pi) \sim 0$, we may assume the n_i II≈**10**
positive. If $\sum_i n_i Y_i$ is proportional to $\mathrm{div}(\pi)$, it is an integral multiple of $\mathrm{div}(\pi)$ and we are done. Otherwise,

$$\Big(\sum_i n_i Y_i\Big)^2 < 0,$$

there is an i such that

$$\Big(Y_i, \sum_j n_j Y_j\Big) < 0,$$

and the following lemma applies.

Lemma 1.16. Under the hypotheses of 1.1, suppose that Y is connected and that either Y is reducible or $\chi(\underline{O}_Y) \leq 0$. Then an irreducible component Y_i of Y is an exceptional curve of the first kind if and only if $(Y_i, K) < 0$.

Let $k' = H^0(Y_i, \underline{O}_{Y_i})$. The Riemann–Roch formula 1.11 gives

$$\chi(\underline{O}_{Y_i}) = [k' : k](1 - \dim_{k'} H^1(Y_i, \underline{O}_{Y_i})) = -\frac{1}{2}((Y_i, Y_i) + (Y_i, K)) \quad .$$

If Y_i is exceptional of the first kind, it follows that $(Y_i, K) = -[k' : k] < 0$. Conversely, if $(Y_i, K) < 0$, then $\chi(\underline{O}_{Y_i}) > 0$, which implies $\chi(\underline{O}_{Y_i}) = [k' : k]$ and that Y is reducible. Hence, by 1.8, $(Y_i, Y_i) < 0$. The negative integers (Y_i, Y_i) and (Y_i, K) are divisible by $[k' : k]$, and therefore necessarily

$$(Y_i, Y_i) = (Y_i, K) = -[k' : k] \quad .$$

Exercise. (a) Under the hypotheses of (1.1), suppose that X_η is a principal homogeneous space under an elliptic curve and that X contains no exceptional curve of the first kind. Show that on X the Riemann–Roch formula (1.11) takes the form

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D).$$

(b) Suppose in addition that the residue field $k(s)$ is algebraically closed. Show that the diagram of II≈11
irreducible components of X_s is a multiple of one of the diagrams considered by Néron [8], pp. 124–125, fourth column (cases a through c8). The diagram determines the genera of the reduced irreducible components, their multiplicities in X_s, and the nature and disposition of the singularities of $(X_s)_{\mathrm{red}}$.

First treat the case in which X_s is irreducible. Once this case has been excluded, prove that every reduced irreducible component is a projective line of self-intersection -2. If $X_s = \sum_i n_i D_i$ is the decomposition of X_s into reduced irreducible divisors, rewrite this condition as

$$(*) \qquad n_i = \frac{1}{2}\sum_{j \neq i} n_j (D_i\, D_j) \quad .$$

On the other hand, X_s is connected, so that:

(**) For any i and j, there exists a chain $i = i_1, i_2, \ldots, i_n = j$ with $(D_{i_k}\, D_{i_{k+1}}) \neq 0$.

Now find all solutions in integers of $(*) + (**)$. First show that, except in Néron's cases b2 and c2, the intersection numbers $(D_i\, D_j)$ are either zero or one.

2. Italian intersections

Let S' and S be two smooth projective surfaces over a field k, and let $f : S' \longrightarrow S$ be a birational morphism. If D and E are two divisors on S, then

$$(D, E) = (f^*D, f^*E),$$

as one sees, for example, by moving D and E.

It is therefore natural to seek a "birational" expression for the intersection number. I propose to II≈12
explain how the Italians proceeded. My principal source of information is [1], which contains a long bibliography. See also Manin [6] and Northcott [12], [13].

(2.0) Let S be the spectrum of a regular local ring of dimension 2, let s be the closed point of S, and let k be a subfield of $k(s)$ over which $k(s)$ is finite. Intersection numbers and Euler–Poincaré characteristics will be computed over k.

In this section, an S-scheme $p : S' \longrightarrow S$ will be called a birational transform of S if it is regular and connected, and if p is proper and induces an isomorphism $S' \setminus p^{-1}(s) \longrightarrow S \setminus \{s\}$. Everything said about S will also apply to the localizations of its birational transforms at a closed point.

(2.1) Let $f : (S'', p'') \longrightarrow (S', p')$ be a morphism of S-schemes between birational transforms of S. It is known [5] that f has a unique factorization

$$f = f_n \cdots f_1, \qquad f_i : S_{i+1} \longrightarrow S_i,$$

such that S_{i+1} is obtained from S_i by blowing up a finite set X_i of closed points, and $f_i(X_{i+1}) \subset X_i$. The set X_i is the set of points of S_i at which the projection of S'' onto S_i is not an isomorphism.

In particular, f induces an isomorphism over the complement of a finite subset of S' containing the generic points of the irreducible components of $p'^{-1}(s)$. Hence f^{-1} defines an injection from the set $\mathrm{Irr}(p'^{-1}(s))$ of irreducible components of $p'^{-1}(s)$ into $\mathrm{Irr}(p''^{-1}(s))$. II≈13

The birational transforms of S have no S-automorphisms. We write I for the set of S-isomorphism classes of birational transforms of S, ordered by domination. The set I is filtered to the right and has least element S.

Definition 2.2. The set S^* of points of S infinitely near to s is the direct-limit set

$$S^* = \varinjlim_I \mathrm{Irr}(p^{-1}(s)) \quad .$$

If an element $t \in S^*$ is represented on a birational transform S' of S by a complete reduced irreducible curve T, the field $H^0(T, \underline{O}_T)$ depends only on t, not on the choice of S'. It is called the residue field $k(t)$ of t.

Let S_1 be the blow-up of S along s. The projective line s^*, the inverse image of $\{s\}$ in S_1, is called the curvetta of s and defines an element s^* of S^* with residue field $k(s)$.

Let S' be a birational transform of S, let t be a closed point of S', and let S'_t be the spectrum of the local ring of S' at t. There is an evident injection $(S'_t)^* \hookrightarrow S^*$; in particular, t defines an element t^* of S^* with the same residue field. We write $(S')^*$ for the union in S^* of the $(S'_t)^*$ as t ranges over the closed points of S'.

The following proposition follows from the description 2.1 of the birational transforms of S.

Proposition 2.3. For every $t \in S^*$, there is a unique sequence of birational transforms $(S_i)_{0\le i\le n+1}$ of S and closed points $t_i \in S_i$ $(0 \le i \le n)$ such that II 14

(i) $S_0 = S$, and S_{i+1} is the blow-up of S_i along $\{t_i\}$;

(ii) $t_{i+1} \in t_i^*$ $(0 \le i < n)$, and $t = t_n^*$.

Moreover, for every birational transform (S', p') of S, the element t belongs to the image of $\mathrm{Irr}(p'^{-1}(s))$ in S^* if and only if S' dominates S_{n+1}.

The integer n is called the order of t. We write $S(t)$ (respectively $S_+(t)$) for the birational transform S_n (respectively S_{n+1}) of S. By abuse of notation, we also write t for the closed point t_n of S_n, and t^* for its inverse image t_n^* in S_{n+1}.

We equip S^* with the following order:

$$t_1 \prec t_2 \quad \Longleftrightarrow \quad S_+(t_2) \text{ dominates } S_+(t_1) \quad .$$

If $t_1 \prec t_2$, then $k(t_2)$ is an extension of $k(t_1)$. The ordered set S^* has least element s^*, and every bounded subset of S^* is finite and totally ordered (S^* is a "tree rooted at s^*"). Moreover, the order n of $t \in S^*$ is the distance in the tree S^* from s^* to t ($n = \#]s^*, t]$).

The following proposition makes the description 2.1 of the birational transforms of S explicit.

Proposition 2.4. The function assigning to a birational transform $p : S' \to S$ the subset $\mathrm{Irr}(p^{-1}(s))$ of S^* induces a bijection between I (2.1) and the finite subsets of S^* which, together with each element,

contain all its predecessors. II≈15

Let D be a divisor ≥ 0 on S, and let $t \in S^*$. We say that t belongs to D if the closed point t of $S(t)$ (2.3) belongs to the strict transform (the schematic closure of $D - \{s\}$) of D on $S(t)$.

Definition 2.5. Let $t, t' \in S^*$. We say that t' is proximate to t if $t' \in S_+(t)^*$ (2.2) and belongs to the curvetta t^* of t (2.3).

If t' is an immediate successor of t, then t' is proximate to t. If in a sequence $(t_i)_{0\le i\le n}$ every t_{i+1} is an immediate successor of t_i, and if t_n is proximate to t_0, then every t_i with $0 < i \le n$ is proximate to t_0, while t_n is not proximate to t_i for $0 < i < n-1$.

(2.6) Let A be a Noetherian local ring of dimension n, let T be its spectrum, let m be its maximal ideal, let $k = A/m$ be its residue field, and let $P = \sum_{0\le i<n} a_i X^i$ be its Hilbert–Samuel polynomial:

$$P(r) = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Recall that the multiplicity μ_s of the closed point s in T is defined by

$$a_{n-1} = \frac{\mu_s}{(n-1)!} \quad .$$

When $n = 1$, this simply says

$$\mu_s = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Geometrically, μ_s may be defined as the degree, in the projective space $\mathbf{P}_k(m/m^2)$, of the special fibre of the blow-up of T along the closed subscheme $\{s\}$.

Recall that if Y and Z are two closed subschemes of a scheme X, then the strict transform of Y II≈16
in the blow-up $\widetilde{X}$ of X along Z—the schematic closure of $Y \setminus Z$ in $\widetilde{X}$—is precisely the blow-up of Y along $Y \cap Z$.

Suppose that T is a closed subscheme of a regular scheme U. Let $p : U_1 \longrightarrow U$ be the blow-up of U along $\{s\}$, let s^* be the inverse image of s in U_1, and let T' be the strict transform of T in U_1. The preceding discussion shows that the multiplicity of T at s is the degree, in the projective space s^*, of the subscheme $s^* \cap T'$.

Suppose further that T is a divisor on U. Let T^* be its total transform in U_1 (the inverse-image divisor), and define ν by $T^* = \nu s^* + T'$. If X is a line in s^*, and Euler–Poincaré characteristics are calculated as alternating sums of dimensions over k, then

$$(X, s^*) = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{s^*}) = \deg_X \underline{O}(s^*) = -1, \tag{2.6.1}$$

and we have seen that

$$(X, T') = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T'}) = \deg_{s^*}(s^* \cap T') = \mu_s \quad . \tag{2.6.2}$$

The projection formula

$$Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T^*}) = Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} Lp^*\underline{O}_T) = Rp_*\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_T$$

and its corollary

$$(X, T^*) = \nu(X, s^*) + (X, T') = 0$$

give, by (2.6.1) and (2.6.2), the identity

$$\mu_s = \nu \quad . \tag{2.6.3}$$

Proposition 2.7. Let T be a divisor ≥ 0 on a regular scheme U, s a closed point of U, U_1 the II 17
blow-up of U along $\{s\}$, s^* the inverse image of s in U_1, T' the strict transform of T on U_1, and T^* its total transform.

The following three quantities are equal:

(i) the multiplicity, in Samuel's sense, of the point s on T;

(ii) the degree, in the projective space s^*, of $s^* \cap T'$;

(iii) the multiplicity of s^* in the divisor T^*.

2.8. Return to the hypotheses of 2.0. Let D be a divisor ≥ 0 on S and let $t \in S^*$. The multiplicity $\mu_t(D)$ of D at t is, by definition, the multiplicity at t of the strict transform of D on $S(t)$ (2.3). The support of the function $t \mapsto \mu_t(D)$ is the set of points infinitely near to s that belong to D (2.4). When D is regular, the numbers $\mu_t(D)[k(t) : k(s)]$ are zero or one: indeed, the strict transforms D' of D are isomorphic to D, and one applies the definition (2.6) of multiplicity.

Intersection numbers can be calculated in terms of multiplicities.

Theorem 2.9. If the divisors D and E meet only at s, then

$$(D,E)=\sum_{t\in S^*}\mu_t(D)\mu_t(E)[k(t):k],$$

where the sum is over the points infinitely near to s which belong to both D and E.

Proof. Let $p:S'\longrightarrow S$ be a birational transform of S. We denote the total transforms of D and E by D^* and E^*, and their strict transforms by D' and E'.

Lemma 2.9.1. (i) The total transform D^* is characterized by: II **18**

(a) D^*-D' is supported on $p^{-1}(s)$;

(b) if X is supported on $p^{-1}(s)$, then $(D^*,X)=0$.

(ii) $(D,E)=(D^*,E^*)=(D^*,E')=(D',E^*)$.

Assertion (i)(a) is immediate, and (i)(b) follows from 1.6(vi). That (a) and (b) determine D^* follows from 1.9. By construction, the cokernel of the map $\underline{O}_E\longrightarrow p_*\underline{O}_{E'}$ is finite; hence

$$\begin{aligned}(D^*,E')&=\chi(Lp^*\underline{O}_D\otimes\underline{O}_{E'})\\&=\chi(\underline{O}_D\otimes Rp_*\underline{O}_{E'})=\chi(\underline{O}_D\otimes\underline{O}_E)=(D,E),\end{aligned}$$

which, by (i), implies (ii).

Take for S' the blow-up of S at s. By (2.9.1), (2.6.3), and 2.7,

$$\begin{aligned}(D,E)&=(D^*,E')=(D'+\mu_s(D)s^*,E')\\&=(D',E')+\mu_s(D)(s^*,E')\\&=(D',E')+\mu_s(D)\mu_s(E)[k(s):k].\end{aligned}\tag{2.9.2}$$

Theorem 2.9 now follows by induction on (D,E), through repeated application of (2.9.2).

(2.10) Let A be a reduced Noetherian local ring of dimension one, let s be the closed point of $\mathrm{Spec}(A)$, let $\widetilde{A}$ be the normalization of A, and let k be a field over which $k(s)$ is finite. Suppose that $\widetilde{A}$ is finite over A (equivalently, that the completion $\widehat{A}$ is reduced). The δ-invariant of the singularity of $\mathrm{Spec}(A)$ at s is the length over k of the A-module $\widetilde{A}/A$.

Theorem 2.11. Let D be a reduced curve on S. Then: II **19**

(i) for every $t\in S^*$, with the notation of 2.8, one has

$$\mu_t(D)=\sum_{\substack{t'\ \text{proximate}\\ \text{to}\ t}}\mu_{t'}(D)[k(t'):k(t)]\quad .$$

(ii) Suppose that the normalization $\widetilde{D}$ of D is finite over D, and let $\delta=\chi(\underline{O}_{\widetilde{D}}/\underline{O}_D)$ be the invariant of the singularity of D at s. For all but finitely many $t\in S^*$, $\mu_t(D)$ is zero or one, and

$$\delta=\sum_{t\in S^*}\frac{1}{2}\mu_t(D)\big(\mu_t(D)-1\big)[k(t):k]\quad .$$

Let $p:S_1\longrightarrow S$ be the blow-up of S along $\{s\}$, let D^* be the total transform of D, and let D' be its strict transform. By (2.7),

$$D^*=D'+\mu_s(D)s^*\tag{2.11.1}$$

and

$$\mu_s(D)=(s^*,D')\,[k(s):k]^{-1}\quad .\tag{2.11.2}$$

The curvetta s^* is nonsingular. Thus, by 2.8 and Definition 2.5, formula 2.9, applied to the divisors s^* and D' on S_1, becomes

$$(s^*, D') = \sum_{\substack{t' \text{ proximate} \\ \text{to } s}} \mu_{t'}(D)[k(t') : k] \quad . \tag{2.11.3}$$

Assertion (i) for $s = t$ follows from (2.11.2) and (2.11.3); the general case is obtained by applying this result to the localization of $S(t)$ at t (2.3).

Let m be the maximal ideal defining $\{s\}$, and let $n \geq 0$. It is known (SGA VII 3.5) that

$$p_*\underline{O}_{S_1}(-ns^*) = m^n \quad \text{and} \quad R^i p_*\underline{O}_{S_1}(-ns^*) = 0 \quad (i > 0). \tag{2.11.4}$$

It follows from this and from the exact sequence II≈**20**

$$0 \longrightarrow \underline{O}_{S_1}(-ns^*) \longrightarrow \underline{O}_{S_1} \longrightarrow \underline{O}_{ns^*} \longrightarrow 0$$

that

$$p_*\underline{O}_{ns^*} = \underline{O}_S/m^n \quad \text{and} \quad R^i p_*\underline{O}_{ns^*} = 0 \quad (i > 0), \tag{2.11.5}$$

and that

$$\chi(\underline{O}_{ns^*}) = \chi(\underline{O}_S/m^n) = \frac{n(n+1)}{2}[k(s) : k] \quad . \tag{2.11.6}$$

Put $n = 0$ in (2.11.4). For every complex K of coherent sheaves on S, the projection formula gives

$$K \xrightarrow{\sim} Rp_*Lp^*K \qquad (\simeq K \overset{\mathbb{L}}{\otimes} Rp_*\underline{O}_{S_1}).$$

In particular, since $Lp^*\underline{O}_D = p^*\underline{O}_D = \underline{O}_{D^*}$, one has

$$p_*\underline{O}_{D^*} = \underline{O}_D \quad \text{and} \quad R^i p_*\underline{O}_{D^*} = 0 \quad (i > 0). \tag{2.11.7}$$

Put $E = \mu_s(D)s^*$. By (2.11.1), (2.9.1)(i)(b), and (2.6.1),

$$\begin{aligned} \chi(\underline{O}_{D' \cap E}) &= (D', E) = (D^* - E, E) = -(E, E) \\ &= -\mu_s(D)^2(s^*, s^*) = \mu_s(D)^2[k(s) : k]. \end{aligned} \tag{2.11.8}$$

The exact sequence

$$0 \longrightarrow \underline{O}_{D^*} \longrightarrow \underline{O}_{D'} \oplus \underline{O}_E \longrightarrow \underline{O}_{D' \cap E} \longrightarrow 0$$

gives, by direct image and (2.11.7), the exact sequence

$$0 \longrightarrow \underline{O}_D \longrightarrow p_*\underline{O}_{D'} \oplus p_*\underline{O}_E \longrightarrow p_*\underline{O}_{D' \cap E} \longrightarrow 0,$$

and hence

$$0 \longrightarrow p_*\underline{O}_E \longrightarrow p_*\underline{O}_{D' \cap E} \longrightarrow p_*\underline{O}_{D'}/\underline{O}_D \longrightarrow 0 \quad .$$

Consequently, by (2.11.6) and (2.11.8), II≈**21**

$$\begin{aligned} \chi(p_*\underline{O}_{D'}/\underline{O}_D) &= \mu_s(D)^2[k(s) : k] - \frac{1}{2}\mu_s(D)(\mu_s(D) + 1)[k(s) : k] \\ &= \frac{1}{2}\mu_s(D)(\mu_s(D) - 1)[k(s) : k] \quad . \end{aligned} \tag{2.11.9}$$

The schemes D and D' are reduced, differ only over s, and D' is finite over D. They therefore have the same normalization, and the δ-invariant of the singularity of D at s is the sum of $\chi(p_*\underline{O}_{D'}/\underline{O}_D)$ and the invariants δ of the singularities of D' at the points of $D' \cap s^*$. We prove 2.11(ii) by induction on δ. If $\mu_s(D) > 1$, the induction hypothesis applies to D', and 2.11(ii) follows from (2.11.9). If $\mu_s(D) = 1$, then $(D', s^*) = 1$ and $D' \cap s^*$ is a regular divisor on D'; by EGA IV 19.1.1, D' is regular. Formula (2.11.9) then shows that D is itself regular, and 2.11(ii) follows from 2.8.

2.12. Let $t \in S^*$, and let S' be a birational transform of S which dominates $S_+(t)$ (2.3). The total transform of t on S', again denoted by t^*, is the total transform on S' of the divisor t^* on $S_+(t)$.

Let $\nu : S^* \longrightarrow \mathbf{N}$ be a function of finite support, and suppose that, for every $t \in S^*$,

$$\nu(t) \geq \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \nu(t')[k(t') : k(t)] \tag{2.12.1}$$

(the proximate-point inequality). Let $f : S' \longrightarrow S$ be a birational transform of S which dominates $S_+(t)$ for each t in the support of ν, and let

$$\nu^* = \sum_{t \in S^*} \nu(t) t^*$$

be the corresponding divisor on S'.

We write $I(\nu)$ for the ideal of $\underline{O}_S$ obtained by direct image under f of the ideal of $\underline{O}_{S'}$ defining ν^*: II≈22

$$I(\nu) = f_*\underline{O}_{S'}(-\nu^*) \quad .$$

By (2.11.4), this ideal does not depend on the choice of S'. The ideals of S obtained in this way are those which Zariski [11] calls "complete."

Theorem 2.13. Under the preceding hypotheses, one has

$$\chi(\underline{O}_S/I(\nu)) = \sum_{t \in S^*} \frac{\nu(t)(\nu(t)+1)}{2}[k(t) : k],$$

and

$$R^i f_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0.$$

In old French, this is expressed briefly by saying: "requiring a plane curve to pass n times through a point t, whether proper or infinitely near, imposes $n(n+1)/2$ conditions, generally independent, on the coefficients of its equation."

We first prove the following lemma.

Lemma 2.14. Let $p : S_1 \longrightarrow S$ be the blow-up of S along $\{s\}$; let T be a closed subscheme of S_1 with ideal q; let n be an integer; let $J = \underline{O}_{S_1}(-s^*)$ be the ideal defining the divisor s^*; put $Y = \mathrm{Spec}(\underline{O}_{S_1}/J^n q)$; and let D be a regular divisor on S, with strict transform D' on S_1. Suppose that $T \cap s^*$ and $T \cap D'$ are finite.

(i) If $\mathrm{length}_{k(s)}(T \cap s^*) \leq n+1$, then

$$R^i p_*(qJ^n) = 0 \quad \text{for } i > 0,$$

and, if T is Artinian,

$$\mathrm{length}_{k(s)}(\underline{O}_S/p_*(qJ^n)) = \frac{n(n+1)}{2} + \mathrm{length}_{k(s)}(T).$$

(ii) If $\mathrm{length}_{k(s)}(T \cap s^*) \leq n$, then

II≈23

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = n + \mathrm{length}_{k(s)}(D' \cap T) \quad .$$

Consider the exact sequence on S_1

$$0 \longrightarrow J^n q \longrightarrow J^n \xrightarrow{u} J^n \otimes \underline{O}_T \longrightarrow 0 \quad . \tag{2.14.1}$$

To prove that $R^i p_*(J^n q) = 0$ for $i > 0$, it is enough, by (2.11.4), to show that $p_*(u)$ is surjective. By the theorem on formal functions, it suffices to prove that the maps

$$u_k : p_*(J^n \otimes \underline{O}/J^k) \longrightarrow p_*(J^n \otimes \underline{O}_T \otimes \underline{O}/J^k)$$

are surjective.

Write Gr_J for the associated graded object of the J-adic filtration. The module J^m/J^{m+1} is isomorphic to $\underline{O}_{s^*}(m)$; hence $R^1 p_*(J^m/J^{m+1}) = 0$, and the map

$$p_*(J^n/J^{n+k+1}) \longrightarrow p_*(J^n/J^{n+k})$$

is surjective. Therefore, to prove by induction on k that the maps u_k are surjective, it suffices to prove the surjectivity of

$$v_k : p_*(J^n \otimes \mathrm{Gr}^k_J(\underline{O})) \longrightarrow p_*(J^n \otimes \mathrm{Gr}^k_J(\underline{O}_T)) \qquad (k \geq 0).$$

The module $\mathrm{Gr}^k_J(\underline{O}_T)$ is a quotient of $J^k/J^{k+1} \otimes \underline{O}_T$, and thus has length at most $n+1$ over $k(s)$. The kernel N of the surjective map

$$v'_k : J^n \otimes \mathrm{Gr}^k_J(\underline{O}) \simeq \underline{O}_{s^*}(n+k) \longrightarrow J^n \otimes \mathrm{Gr}^k_J(\underline{O}_T)$$

is therefore an invertible sheaf on the projective line s^* of degree ≥ -1. Hence $R^1 p_* N = 0$, and $v_k = p_*(v'_k)$ is surjective.

If T is Artinian, (2.14.1) now gives

$$\mathrm{length}_{k(s)}\big(p_* J^n/p_*(J^n q)\big) = \mathrm{length}_{k(s)}(\underline{O}_T),$$

and 2.14(i) follows from (2.11.4) and (2.11.6). II≈**24**

Let $I_{D' \cap T}$ be the (principal) ideal of $\underline{O}_{D'}$ defining $D' \cap T$. Under the hypotheses (ii), the exact sequence

$$0 \longrightarrow I_{D' \cap T} \longrightarrow \underline{O}_{D' \cup T} \longrightarrow O_T \longrightarrow 0$$

shows that $D' \cup T$ satisfies hypothesis (i).

The projection p induces an isomorphism between D and D'. Under this isomorphism, the ideal of $\underline{O}_D$ defining the subscheme

$$D \cap \mathrm{Spec}(\underline{O}_S/p_*(qJ^n))$$

corresponds to the ideal of D' generated by the global sections of qJ^n on S_1. This ideal is contained in the ideal of D' defining $D' \cap Y$. It is in fact equal to it, because every section of $\underline{O}_{D' \cup Y}$ extends to a section of $\underline{O}_{S_1}$: if $I_{D' \cup Y}$ is the ideal defining $D' \cup Y$, the obstruction to extension lies in $R^1 p_*(I_{D' \cup Y})$. One has

$$\begin{aligned} I_{D' \cup Y} &= \underline{O}(-D') \cap qJ^n \\ &= (\underline{O}(-D') \cap J^n) \cap qJ^n \\ &= \underline{O}(-D')J^n \cap qJ^n \\ &= (\underline{O}(-D') \cap q)J^n, \end{aligned}$$

and by (i), applied to $D' \cup T$, there is no obstruction.

Thus

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = \mathrm{length}_{k(s)}(D' \cap Y) \quad . \tag{2.14.2}$$

The isomorphism

$$J^n \otimes (\underline{O}_{D'}/q\underline{O}_{D'}) \xrightarrow{\sim} \underline{O}_{D'}J^n/\underline{O}_{D'}qJ^n$$

gives the exact sequence

$$0 \longrightarrow J^n \otimes \underline{O}_{D'\cap T} \longrightarrow \underline{O}_{D'\cap Y} \longrightarrow \underline{O}_{D'\cap ns^*} \longrightarrow 0,$$

and hence

II ≈**25**

$$\text{(2.14.3)} \qquad \text{length}_{k(s)}(D' \cap Y) = n(D', s^*) + \text{length}_{k(s)}(D' \cap T).$$

Since $(D', s^*) = 1$, assertion 2.14(ii) follows from (2.14.2) and (2.14.3).

We now prove, by induction on the number of elements in the support of ν, the assertions of 2.13 and the following assertion:

$$\text{(2.14.4)} \qquad \text{if } D \text{ is a regular divisor on } S, \text{ then} \qquad \chi\Big(D \cap \text{Spec}(\underline{O}_S/I(\nu))\Big) = \sum_{\substack{t\in S^* \\ t \text{ on } D}} \nu(t)[k(t) : k].$$

If $\nu = 0$, then $\nu^* = 0$, $I(\nu) = \underline{O}_S$, and 2.13 and (2.14.4) are immediate (the second assertion of 2.13 follows by repeated application of (2.11.4) with $n = 0$). Otherwise $\nu(s) > 0$, and there is an S-morphism $g : S' \longrightarrow S_1$.

For each closed point a of S_1, let S_{1a} be the localization of S_1 at a, let ν_a be the restriction of ν to S^*_{1a}, let $S'_a = S' \times_{S_1} S_{1a}$, let $I(\nu_a) = g_*\underline{O}_{S'_a}(-\nu^*_a)$, and put

$$T_a = \text{Spec}(\underline{O}_{S_1}/I(\nu_a)), \qquad T = \coprod_a T_a,$$

where the union is finite and disjoint. Let q be the ideal of T. With the notation of 2.14, one has

$$\text{(2.14.5)} \qquad g_*\underline{O}_{S'}(-\nu^*) = qJ^{\nu(s)}.$$

The induction hypothesis applies to (S_{1a}, S'_a, ν_a). Hence

$$\text{(2.14.6)} \qquad R^ig_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0,$$

$$\text{(2.14.7)} \qquad \chi(T) = \sum_{t\in S^*\setminus\{s\}} \frac{\nu(t)(\nu(t)+1)}{2}[k(t) : k],$$

and, by (2.14.4) applied to the divisor s^* on S_1 and by (2.12.1),

$$\text{length}_{k(s)}(T \cap s^*) = \sum_{\substack{t \text{ proximate} \\ \text{to } s}} \nu(t)[k(t) : k(s)] \leq \nu(s).$$

This inequality allows us to apply 2.14. From (2.14.5) we obtain

II ≈**26**

$$R^ip_*g_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0$$

and

$$\text{length}_{k(s)}\big(\underline{O}_S/p_*g_*\underline{O}_{S'}(-\nu^*)\big) = \frac{\nu(s)(\nu(s)+1)}{2} + \text{length}_{k(s)}(T).$$

The assertions of 2.13 therefore follow from (2.14.6) and (2.14.7). Finally, 2.14(ii) gives

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}(\underline{O}_S/I(\nu))\Big) = \nu(s) + \mathrm{length}_{k(s)}(D' \cap T).$$

Applying (2.14.4) to D' gives

$$\chi\Big(D \cap \mathrm{Spec}(\underline{O}_S/I(\nu))\Big) = \nu(s)[k(s):k] + \sum_{\substack{t \text{ on } D \\ t \neq s}} \nu(t)[k(t):k]$$
$$= \sum_{\substack{t \in S^* \\ t \text{ on } D}} \nu(t)[k(t):k].$$

This completes the proof of 2.13.

(2.15) Fragments of the preceding discussion generalize to higher dimension. Let S be the spectrum of a regular local ring of dimension n, with closed point s, where $k(s)$ is a finite extension of k. We may again consider the set I of schemes $f : S' \longrightarrow S$ obtained from S by successive blow-ups of closed points, and put, as in 2.1,

$$S^* = \varinjlim_{S' \in I} \mathrm{Irr}(f^{-1}(s)).$$

The set S^* will again be called the set of points of S infinitely near to s.

If X is a closed subscheme of S and $t \in S^*$, the multiplicity of X at t is defined as in 2.8.

Let $f : S' \longrightarrow S$ be a composite of blow-up morphisms at closed points, let D be a divisor on S, and let $D^* = f^*D$ be its total transform on S'.

Let E be a curve on S, let E' be its strict transform on S', and suppose that $D \cap E$ is supported at s. The projection formula again gives II≈27

$$(D, E) = (D^*, E'). \tag{2.16}$$

Proceeding as in 2.9, one proves with the aid of (2.16) that

$$(D, E) = \sum_{t \in S^*} \mu_t(D)\mu_t(E)[k(t):k] \quad . \tag{2.17}$$

Define proximate points as in 2.5. Proceeding as in 2.11, one verifies that

$$\mu_t(E) = \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \mu_{t'}(E)[k(t'):k(t)] \quad . \tag{2.18}$$

On the other hand, formula 2.11(ii) is false in general (see the examples in 2.25).

2.19. Let S be a regular projective surface over a field k. Write S^* for the disjoint sum, taken over the closed points of S,

$$S^* = \coprod_{s \in S} \mathrm{Spec}(\underline{O}_{s,S})^*,$$

and call S^* the set of points infinitely near to the points of S.

Let I be the set of surfaces obtained from S by successive blow-ups of closed points. The set I is cofinal in the set of regular models of the function field of S, ordered by domination. For $i \in I$, let S_i denote the corresponding surface, equipped with $p_i : S_i \longrightarrow S$. We write $\mathrm{Div}^{\pm}(S_i)$ for the group of divisors, effective or not, on S_i.

Definition 2.20. (i) The group of "curves" on S is the inductive limit

$$\varinjlim_{i \in I} \mathrm{Div}^{\pm}(S_i).$$

(ii) The category of "invertible sheaves" on S is the inductive limit
(SGA 4 VI 6) of the categories of invertible sheaves on the $(S_i)_{i\in I}$. II≈28
These objects are birational invariants of S.

Example 2.21. (i) Every "curve" C on S defines an "invertible sheaf" $\underline{O}(C)$ on S.

(ii) Every divisor C on S defines a "curve" on S, denoted by C^*.

(iii) If $t \in S^*$, define the surface $S_+(t)$ and the divisor t^* on $S_+(t)$ as in 2.3; the "curve" defined by this divisor will again be denoted by t^*.

(2.22) One verifies easily that a "curve" C (respectively an "invertible sheaf" $\mathcal{L}$) can be written in one and only one way in the form

$$C = C_0^* - \sum_{t\in S^*} i_t t^* \qquad \text{respectively} \qquad \mathcal{L} = \mathcal{L}_0 \otimes \underline{O}\left(-\sum_{t\in S^*} i_t t^*\right),$$

where C_0 (respectively $\mathcal{L}_0$) is a divisor (respectively an invertible sheaf) on S, and where the integers i_t are almost all zero. We say that C (respectively $\mathcal{L}$) is the curve C_0 (respectively the invertible sheaf $\mathcal{L}_0$) endowed with the virtual multiplicities i_t.

The completion $\mathrm{Div}^{\pm}(S)\times\mathbb{Z}^{S^*}$ of the group $\mathrm{Div}^{\pm}(S)\times\mathbb{Z}^{(S^*)}$ of "curves" on S is again a birational invariant of S; it is called the group of "generalized curves" on S. On S, a "generalized curve" C is represented by a divisor C_0 endowed with a possibly infinite collection of virtual multiplicities.

"Generalized invertible sheaves" on S are defined in the same way.

Example 2.23. Suppose that k is perfect. Then the invertible sheaf $\Omega^2_{S/k}$, endowed with virtual II 29
multiplicity -1 at every point $t \in S^*$, is a birational invariant of S; denote it by ${\Omega^2_{S/k}}^*$.

Example 2.24. Let $f : S_1 \longrightarrow S_2$ be a dominant morphism of regular projective surfaces over k. The inverse image of a "curve" or an "invertible sheaf" on S_2 is defined without difficulty by passing to a model of S_2 on which this "curve" or "invertible sheaf" is an ordinary divisor or invertible sheaf. By passage to the limit (I have not checked whether this passage to the limit is possible), one similarly defines the inverse image of a "generalized curve" or a "generalized invertible sheaf." Suppose that k is perfect and that f is generically étale. The different $\mathfrak{d}(f)$ of f is then the effective divisor on S_1 characterized by

$$f^*\Omega^2_{S_2/k}(\mathfrak{d}(f)) \xrightarrow{\sim} \Omega^2_{S_1/k} \quad .$$

This different is not a birational invariant of f; to obtain a birational invariant one must consider the generalized coincidence curve

$$\mathfrak{d}^*(f) = \mathfrak{d}(f) + \sum_{t\in S_1^*} t^* - f^*\left(\sum_{t\in S_2^*} t^*\right),$$

or, briefly,

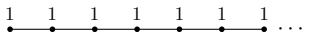
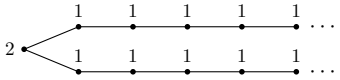
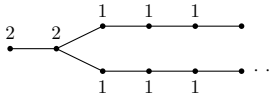
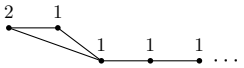
$$\mathfrak{d}^*(f) = {\Omega^2_{S_1/k}}^* - f^*{\Omega^2_{S_2/k}}^* \quad .$$

This "generalized curve" is supported on the closed subset of S_1 where f is not étale.

(2.25) Example. II≈30

Let k be an algebraically closed field and let E_n be affine n-space over k. Let C be a reduced curve in E_n having a singularity at the origin. With this singularity we associate its Enriques diagram (cf. [3]): its vertices are the points of C infinitely near the origin (2.4 and 2.15), and each vertex is joined by a line to the vertices proximate to it (2.5 and 2.15). This diagram determines the multiplicities of these points on C (2.11 and 2.18) and, when $n = 2$, the invariant δ (2.10 and 2.11) of the singularity of C at the origin.

For the simplest plane singularities one obtains the following table. A number next to a vertex is its multiplicity.

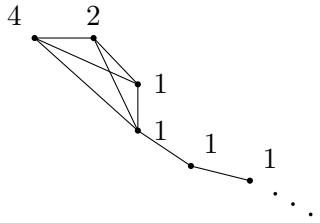
Sketch	Equation	δ	Enriques diagram
	$y = 0$	0	
	$xy = 0$	1	
	$y(y - x^2) = 0$	2	
	$y^2 = x^3$	1	

If $n > 2$, the invariant δ is no longer determined by this diagram. For example, the union of three concurrent coplanar lines (respectively, noncoplanar lines) has invariant $\delta = 3$ (respectively, $\delta = 2$), although the Enriques diagram is the same. I do not know whether this phenomenon also occurs for unibranch curves.

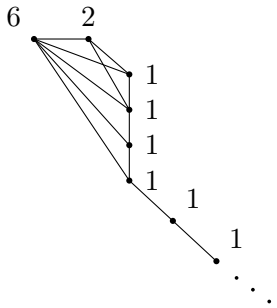
Here are two examples of diagrams in the case $n = 3$ which cannot occur for plane curves (cf. 2.5). We give, in order, the equations of the curve, the equations of the pure transform after blowing up the origin, parametric equations of the curve, the invariant δ, and the Enriques diagram.

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(2.25.1)
$$\begin{cases} \begin{cases} y^2 = xz^2, \\ z^3 = x^2, \end{cases} & \begin{cases} (y/z)^2 = (x/z)^3, \\ z = (x/z)^2, \end{cases} \\ \begin{cases} x = u^6, \\ y = u^7, \\ z = u^4, \end{cases} & \delta = 5 \end{cases}$$



(2.25.2)
$$\begin{cases} \begin{cases} y^2 z = x^3, \\ z^3 = y^2, \end{cases} & \begin{cases} (y/z)^2 = (x/z)^3, \\ z = (y/z)^2, \end{cases} \\ \begin{cases} x = u^8, \\ y = u^9, \\ z = u^6, \end{cases} & \delta = 10 \end{cases}$$



2.26. Theorem 2.27 below was stated as a false conjecture in the first version of this exposé (circulated by the IHES). It owes much to conversations with D. S. Rim and includes some of his results (D. S. Rim, "Torsion Differentials and Deformations," Theorem 2.7(i)).

Let C be a reduced projective curve over an algebraically closed field k, and let $s \in C(k)$. We shall write

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A_s for the local ring of C at s (or its completion, which would make no difference below);

$\widetilde{A}_s$ for the normalization of A_s;

$\delta(s)$ for $\dim_k(\widetilde{A}_s/A_s)$;

$m_0(s)$ for the dimension of the formal versal scheme T of deformations of $\mathrm{Spec}(A_s)$.

For an irreducible component E of T, put likewise $m_0^E(s) = \dim(E)$.

The module $\widetilde{\tau}$ of derivations of $\widetilde{A}_s$ and the module τ of derivations of A_s are torsion-free. If A_s is integral, they embed in the module of derivations of the fraction field of A_s; in general, they embed in the module of derivations of the total quotient ring of A_s (a product of fields). Moreover, through these embeddings, τ and $\widetilde{\tau}$ are commensurable: both $\tau/(\tau \cap \widetilde{\tau})$ and $\widetilde{\tau}/(\tau \cap \widetilde{\tau})$ are finite-dimensional over k. Set

$$m_1(s) = [\widetilde{\tau} : \tau] \underset{\text{def}}{=} \dim_k \frac{\widetilde{\tau}}{\tau \cap \widetilde{\tau}} - \dim_k \frac{\tau}{\tau \cap \widetilde{\tau}}.$$

Theorem 2.27. Let E be an irreducible component of the formal versal scheme T of deformations of A_s. If the deformation of A_s over the generic point e of E is smooth—that is, if the image of the formal non-smoothness subscheme (defined by a Jacobian ideal: VI 5), which is finite over T, does not contain e—then

$$3\delta(s) = m_0^E(s) + m_1(s).$$

The validity of 2.27 depends only on the completion of A_s. Replacing (C, s) by a pair (C', s) formally isomorphic to it at s, we may suppose that the following conditions hold: II ≈33

(a) s is the only singular point of C;

(b) C has no infinitesimal automorphisms.

Condition (b) ensures that the deformation functor of C is pro-representable. It is even effectively pro-representable, since there is no obstruction to lifting a prescribed ample invertible sheaf on C. Let $f : X_0 \longrightarrow T_0$ be the universal deformation of C, and let t_0 be the closed point of T_0, so that $C \xrightarrow{\sim} X_{0t_0}$.

Lemma 2.28. There exist a scheme T_1 of finite type over k, a point $t_1 \in T_1(k)$, and a proper flat morphism $f : X_1 \longrightarrow T_1$, whose fiber at t_1 is C, such that for every $u \in T_1(k)$ the completion of T_1 at u is a formal moduli scheme for the fiber X_{1u}.

Here is a direct proof of this corollary of general results of Artin. Let $\mathcal{L}$ be a very ample invertible sheaf on C such that $H^1(C, \mathcal{L}) = 0$, and put $\mathbb{P} = \mathbb{P}(H^0(C, \mathcal{L}))$. We have an embedding $v : C \hookrightarrow \mathbb{P}$. Let T_2 be the Hilbert scheme parametrizing the subvarieties of $\mathbb{P}$ having the same Hilbert polynomial as $v(C) \subset \mathbb{P}$, and let $t_2 \in T_2$ correspond to $v(C)$. The completion $\widehat{T}_2$ of T_2 at t_2 is smooth over T_0. The vertical tangent bundle of $\widehat{T}_2/T_0$ is induced by the subbundle L of the tangent bundle of T_2 which is the image of $\mathrm{Lie}(\underline{\mathrm{Aut}}(\mathbb{P}))$ under the action of $\underline{\mathrm{Aut}}(\mathbb{P})$ on T_2.

In a neighborhood of t_2, let $f_1, \ldots, f_n$, with $n = \dim \underline{\mathrm{Aut}}(\mathbb{P})$, be equations defining a subscheme T_1' of T_2 which passes through t_2 and whose completion at t_2 is étale over T_0. Take for T_1 a neighborhood of t_2 in T_1' on which the tangent bundle is transverse to L.

2.29. The morphism of functors II 34

$$(\text{deformations of } C) \longrightarrow (\text{deformations of } A_s)$$

corresponds to a morphism $p : T_0 \longrightarrow T$. By VI 2.10, p is smooth. Let t be the closed point of T. The Zariski tangent space of $p^{-1}(t)$ at t_0 classifies the locally trivial deformations of C over $k[\varepsilon]$, where $\varepsilon^2 = 0$. Putting $E_0 = p^{-1}(E)$, we obtain

$$\dim(E_0) = \dim(E) + \dim H^1(C, T^1_{C/k}). \tag{2.29.1}$$

Since X_0/T_0 is flat, all its fibers have the same arithmetic genus p, where $1 - p = \chi(\underline{O})$. By hypothesis, the fiber at the generic point of E_0 is smooth. The formal moduli scheme of a smooth proper curve being smooth of dimension $3p - 3$, Lemma 2.28 gives

$$\dim(E_0) = 3p - 3. \tag{2.29.2}$$

Let $\widetilde{C}$ be the normalization of C, of arithmetic genus $\widetilde{p} = p - \delta(s)$. We have

$$\begin{aligned} \dim H^1(C, T^1_{C/k}) &= -\chi(C, T^1_{C/k}) \\ &= -\chi(\widetilde{C}, T^1_{\widetilde{C}/k}) + m_1(s) \\ &= [3(p - \delta(s)) - 3] + m_1(s). \end{aligned} \tag{2.29.3}$$

Formulas (2.29.1), (2.29.2), and (2.29.3) prove 2.27.

Conjecture 2.30. Every reduced proper curve has a smooth deformation.

This conjecture implies, by 2.27 and 2.28, that the formal versal scheme of deformations of A_s is always equidimensional, of dimension $3\delta(s) - m_1(s)$.

2.31. Let C be a reduced proper curve over k, of arithmetic genus p. We shall write II **35**

M_0 for the dimension of the formal versal scheme T of deformations of C;

M_1 for the dimension of the Lie algebra of $\underline{\mathrm{Aut}}(C)$.

For an irreducible component E of T, write M_0^E for the dimension of E. The global analogue of 2.27 is the following.

Proposition 2.32. If the deformation of C over the generic point of E is smooth, then

$$3p - 3 = M_0^E - M_1.$$

Let $X \subset C(k)$ be a finite set of smooth points of C such that every vector field on C which vanishes at every point $x \in X$ is zero. The deformation functor of (C, X) is pro-representable and satisfies a variant of 2.28. The formal moduli scheme of (C, X) is smooth over T, of relative dimension $\#X - M_1$, and the analogue of 2.28 gives

$$M_0^E + (\#X - M_1) = 3p - 3 + \#X,$$

which proves 2.32.

Suppose now that k has characteristic zero. Then $M_1 = \dim \underline{\mathrm{Aut}}(C)$. Here is an analogous interpretation of $m_1(s)$.

Lemma 2.33. If k has characteristic zero, every derivation of A_s extends to a derivation of $\widetilde{A}_s$. II **36**

If D is a derivation of A_s, then

$$\exp(Dt) = \sum_{n \geq 0} \frac{D^n t^n}{n!}$$

is an automorphism of the completion at $(s,0)$ of $A_s \otimes_k k[t]$. By transport of structure, this automorphism acts on the normalization $(\widetilde{A}_s \otimes_k k[t])^\wedge$ of that ring. Reducing modulo t^2, we obtain an automorphism $1 + Dt$ of $\widetilde{A}_s^\wedge \otimes_k k[t]/(t^2)$, and D thereby extends to a derivation of $\widetilde{A}_s^\wedge$. Since it also extends to the fraction field of A_s (to the total quotient ring when A_s is not integral), it extends to $\widetilde{A}_s$.

Suppose that s is singular. A similar argument then shows that the vector field on $\mathrm{Spec}(\widetilde{A}_s)$ defined by D vanishes at every point above s.

Let c be an ideal of $\widetilde{A}_s$ contained in the conductor of $\widetilde{A}_s/A_s$. Put

$$\widetilde{G}_c = \underline{\mathrm{Aut}}(\widetilde{A}_s/c),$$

and let G_c be the subgroup of $\widetilde{G}_c$ which stabilizes the subalgebra A_s/c. If $d \subset c$, then $\widetilde{G}_d$ stabilizes c/d and maps onto $\widetilde{G}_c$. Notice that $g \in \widetilde{G}_c$ is determined by $g(t)$ for a uniformizer t, and that $g(t)$ may be any uniformizer. Consequently the pointed homogeneous space $\widetilde{G}_c/G_c$ is independent of c; denote it by $\widetilde{G}/G$.

Let $\widetilde{\tau}_0$ be the $\widetilde{A}_s$-module of vector fields on $\mathrm{Spec}(\widetilde{A}_s)$ which vanish at the points above s. When s is singular, one has

$$\dim_k(\widetilde{\tau}/\widetilde{\tau}_0) = \text{the number of branches},$$

and

$$\widetilde{\tau}_0/\tau \simeq \text{the tangent space at the origin of } \widetilde{G}/G.$$

Proposition 2.34. In characteristic zero, when s is singular, the integer $m_1(s)$ is the sum of the number of branches and the dimension of $\widetilde{G}/G$. II 37

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SGA 7

EXPOSÉ XI

COHOMOLOGY OF COMPLETE INTERSECTIONS

by P. DELIGNE

1. Cohomology of projective space and complete intersections

Let X be a nonsingular complex projective variety, purely of dimension n, and let $\underline{O}(1)$ be an ample invertible sheaf on X. By Akizuki and Nakano [1], Kodaira's vanishing theorem [4] implies that

$$(*) \qquad H^i(X, \Omega^j_X(-1)) = 0 \qquad \text{for } i + j < n.$$

Again by [1], this result implies the "weak Lefschetz theorem" in complex cohomology. We propose to prove, by brute force, that if X is a projective space over a field of arbitrary characteristic, then (*) remains true. This will allow us, following [1], to compare the Hodge and de Rham cohomology of complete intersections over a field of arbitrary characteristic with that of projective space.

We shall then spell out some consequences of the results obtained. In §2 we give qualitative results deduced from Hirzebruch's numerical formula (2.2). I wish to thank N. Katz for his help in the preparation of that section.

1.0. Let $\underline{E}$ be a locally free module of rank $r+1$ on a scheme S. We assume that $r > 0$. Recall that $\mathbb{P}(\underline{E})$ denotes the relative projective space over S of hyperplanes of $\underline{E}$. Let $p : \mathbb{P}(\underline{E}) \to S$ be the projection, and let

$$\eta \in H^0(S, R^1 p_* \Omega^1_{\mathbb{P}(\underline{E})/S})$$

be the first Chern class, in the Hodge sense, of the invertible sheaf $\underline{O}(1)$.

Recall that η is the image of the class of $\underline{O}(1)$ in $H^0(S, R^1 p_* \underline{O}^*)$ under the logarithmic differential II ≈**40**

$$df/f : \underline{O}^* \longrightarrow \Omega^1_{\mathbb{P}(\underline{E})/S}.$$

For every sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, put

$$\underline{F}(n) = \underline{F} \otimes \underline{O}(1)^{\otimes n} \qquad (n \in \mathbb{Z}).$$

Theorem 1.1. (i) The sheaves $R^i p_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ are locally free, and their formation is compatible with arbitrary base change.

(ii) For $0 \le i \le r$, the section η^i of $R^i p_*(\Omega^i_{\mathbb{P}(\underline{E})/S})$ defines an isomorphism from $\underline{O}_S$ onto this sheaf. Moreover,

$$R^i p_* \Omega^j_{\mathbb{P}(\underline{E})/S} = 0 \qquad \text{if } i \ne j \text{ or } i > r.$$

(iii) If $n \ne 0$, then, outside the following two cases, $R^i p_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ vanishes:

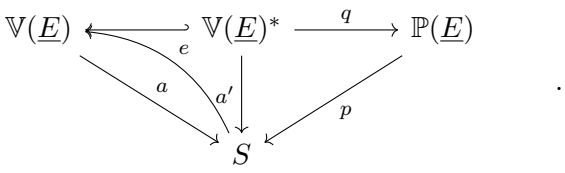
$$(\text{iii}_a) \qquad i = 0 \quad \text{and} \quad n > j,$$

$$(\text{iii}_b) \qquad i = r \quad \text{and} \quad n < j - r.$$

Let

$$\mathbb{V}(\underline{E}) = \mathrm{Spec}(\mathrm{Sym}^*\,\underline{E})$$

be the affine space defined by $\underline{E}$, let e be its zero section, and put $\mathbb{V}(\underline{E})^* = \mathbb{V}(\underline{E}) \setminus e(S)$. Let a and a' be the projections of $\mathbb{V}(\underline{E})$ and $\mathbb{V}(\underline{E})^*$ onto S, and let q be the projection of $\mathbb{V}(\underline{E})^*$ onto $\mathbb{P}(\underline{E})$:



.

The punctured affine space $\mathbb{V}(\underline{O}(1))^*$ over $\mathbb{P}(\underline{E})$ is canonically isomorphic to $\mathbb{V}(\underline{E})^*$. Consequently, for every quasi-coherent sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, there is a canonical isomorphism

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$$q_*q^*\underline{F} \simeq \bigoplus_{n\in\mathbb{Z}} \underline{F}(n). \tag{1.1.1}$$

This grading of $q_*q^*\underline{F}$ can also be obtained from the action on this sheaf of the group $\mathbb{G}_m$ of homotheties of $\mathbb{V}(\underline{E})$.

Relative vector fields on $\mathbb{V}(\underline{E})$ are identified with S-morphisms from $\mathbb{V}(\underline{E})$ to itself, the target being identified with the tangent space at the origin. The vector field X defined by the identity map of $\mathbb{V}(\underline{E})$ is invariant under homotheties. It is tangent to the fibers of q and defines a trivialization

$$X : \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow \underline{O}_{\mathbb{V}(\underline{E})^*}. \tag{1.1.2}$$

If the x_i form a basis of $\underline{E}$, then

$$X = \sum_i x_i\, dx^i.$$

Since $\Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})}$ has rank one, the exact sequence

$$0 \longrightarrow q^*\Omega^1_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow 0$$

induces exact sequences

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \otimes q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0.$$

Using (1.1.2), these become

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \xrightarrow{\iota_X} q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0. \tag{1.1.3}$$

Canonically,

$$\Omega^i_{\mathbb{V}(\underline{E})/S} = a^* \bigwedge^i \underline{E}. \tag{1.1.4}$$

The short exact sequences (1.1.3) are therefore obtained by splitting the following long exact sequence on $\mathbb{V}(\underline{E})^*$:

$$\cdots \longrightarrow a'^* \bigwedge^i \underline{E} \xrightarrow{\iota_X} a'^* \bigwedge^{i-1} \underline{E} \xrightarrow{\iota_X} a'^* \bigwedge^{i-2} \underline{E} \longrightarrow \cdots. \tag{1.1.5}$$

This long exact sequence is simply the restriction to $\mathbb{V}(\underline{E})^*$ of the Koszul resolution of $\underline{O}_{e(S)}$ on $\mathbb{V}(\underline{E})$. II≈42

Push the long exact sequence (1.1.5) forward to $\mathbb{P}(\underline{E})$ by the affine morphism q. By (1.1.1) and the formula $q_*a'^* = q_*q^*p^*$, one obtains a long exact sequence

$$\cdots \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^i \underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^{i-1} \underline{E}(n) \longrightarrow \cdots . \tag{1.1.6}$$

It splits into short exact sequences, the images of (1.1.3),

$$0 \longrightarrow \bigoplus_{n\in\mathbb{Z}} \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^i \underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0. \tag{1.1.7}$$

The exact sequences (1.1.7) are direct sums of the exact sequences

$$0 \longrightarrow \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow p^* \bigwedge^i \underline{E}(n-i) \xrightarrow{\iota_X} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0. \tag{1.1.8}$$

Recall (EGA III, 2.1.15, or FAC) that the sheaves $R^ip_*\underline{O}(k)$ are locally free and their formation is compatible with arbitrary base change, and that

$$R^ip_*\underline{O}(k) = 0 \quad \text{if } i \neq 0, r, \quad \text{or if } i = 0, k < 0, \quad \text{or if } i = r, k > -r-1, \tag{1.1.9}$$

and hence that $R^ip_*p^* \bigwedge^j \underline{E}(k)$ vanishes under the same conditions.

The exact sequences (1.1.8) give a resolution

$$0 \longrightarrow \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \xrightarrow{E} p^* \bigwedge^i \underline{E}(n-i) \xrightarrow{\iota_X} \cdots \longrightarrow p^* \bigwedge^0 \underline{E}(n) \longrightarrow 0 \tag{1.1.10}$$

of $\Omega^i_{\mathbb{P}(\underline{E})/S}(n)$. The spectral sequence defined by this resolution degenerates ($E_1 = E_\infty$) for degree reasons by (1.1.9); for $0 < j < r$, it gives

$$R^jp_*\Omega^i_{\mathbb{P}(\underline{E})/S}(n) = \mathcal{H}\left(\cdots \xrightarrow{\iota_X} p_* \bigwedge^{i-j} \underline{E}(n-i+j) \xrightarrow{\iota_X} \cdots \right). \tag{1.1.11}$$

Since $r \geq 1$, a depth argument shows that

$$\bigoplus_{n\in\mathbb{Z}} p_*p^* \bigwedge^i \underline{E}(n) = p_*q_*q^*p^* \bigwedge^i \underline{E} = a'_*a'^* \bigwedge^i \underline{E} = a_*a^* \bigwedge^i \underline{E}. \tag{1.1.12}$$

Thus $p_*p^* \bigwedge^i \underline{E}(n)$ is the homogeneous part of weight $n+i$ in $a_*a^* \bigwedge^i \underline{E}$, namely $\mathrm{Sym}^n(\underline{E}) \otimes \bigwedge^i \underline{E}$. II≈43 Remembering that (1.1.5) comes from the Koszul resolution, (1.1.11) shows that, for $0 < j < r$, the sheaf $R^jp_*\Omega^i_{\mathbb{P}(\underline{E})/S}(n)$ can be nonzero only when $i-j=0$ and $n-i+j=0$, that is, only when $n=0$ and $i=j$. In that case, moreover, $R^ip_*\Omega^i_{\mathbb{P}(\underline{E})/S}$ is isomorphic to $\underline{O}_S$.

For $j=0$, (1.1.10) also gives

$$p_*\Omega^i_{\mathbb{P}(\underline{E})/S}(n) = \ker\left(\iota_X : p_* \bigwedge^i \underline{E}(n-i) \longrightarrow p_* \bigwedge^{i-1} \underline{E}(n-i+1) \right), \tag{1.1.13}$$

and the same argument shows that ι_X is injective when $n - i \leq 0$.

By (1.1.8), with $n = 0$ and $i = r + 1$, one has

$$\Omega^r_{\mathbb{P}(\underline{E})/S} \simeq p^* \bigwedge^{r+1} \underline{E}(-r-1).$$

When S is the spectrum of a field, the vanishing assertions in 1.1(ii) and (iii) now follow from the preceding argument for $j \neq r$, and follow from it by duality ([2], V, 11.2, p. 211) for $j = r$. In particular, for any j and n there is at most one i for which $R^i p_* \Omega^j_{\mathbb{P}(\underline{E})/S}(n) \neq 0$. This proves (i), since $\Omega^j_{\mathbb{P}(\underline{E})/S}$ is flat over S (the hardest part is locating the reference in EGA III; we suggest 6.10.5, 7.4.1, and 7.5.5, followed by a passage to the limit).

Knowing that $R^i p_* \Omega^i_{\mathbb{P}(\underline{E})/S}$ is an invertible sheaf for $0 \leq i \leq r$, it remains only to prove that, when S is the spectrum of a field, $\eta^i \neq 0$. Indeed, η^r is the cohomology class of a rational point (the intersection of r hyperplanes), hence a basis of $H^r(\mathbb{P}^r, \Omega^r)$ (FGA, Exposé 149, no. 4).

1.2. The Euler–Poincaré characteristics of the $\Omega^i(n)$ may be calculated directly from (1.1.10) or (1.1.13). The rather unappetizing result is

$$\chi(\mathbb{P}^r_k, \Omega^j_{\mathbb{P}^r_k}(n)) = \sum_{0 \leq i \leq j} (-1)^{i-j} \binom{r+1}{i} \binom{r+n-i}{r}.$$

Proposition 1.3. Let $f : X \to S$ be a proper smooth morphism, let $\underline{W}$ be a locally free sheaf of rank c on X, let $\underline{F}$ be a coherent sheaf on X, let $d > 0$ be an integer, and let $s : \underline{W} \to \underline{O}_X$ be a section of $\underline{W}^\vee$. Assume that S is noetherian and that: II 44

(a) The closed subscheme H of X defined by the vanishing of s is smooth over S.

(b) Locally on X, if the s_i are the coordinates of s in a basis of $\underline{W}^\vee$, the s_i form both an $\underline{O}_X$-regular sequence and an $\underline{F}$-regular sequence.

(c) For every sequence of integers k_ℓ, not all zero, one has

$$R^i f_* \left(\bigotimes_\ell \bigwedge^{k_\ell} \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F} \right) = 0 \qquad \text{for } i + j < d.$$

Then:

(i) $$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \xrightarrow{\sim} R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \text{for } i + j < d - c,$$

(ii) $$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \hookrightarrow R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \text{for } i + j = d - c.$$

The conormal sheaf $N_{H/X}$ of H in X is isomorphic to the restriction of $\underline{W}$ to H, so the exact sequence

$$0 \longrightarrow N_{H/X} \longrightarrow \Omega^1_{X/S} \otimes \underline{O}_H \longrightarrow \Omega^1_{H/S} \longrightarrow 0 \tag{1.3.1}$$

defines a decreasing filtration on $\Omega^j_{X/S} \otimes \underline{O}_H$ whose successive quotients are

$$\mathrm{Gr}^k(\Omega^j_{X/S} \otimes \underline{O}_H) \simeq \bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S}. \tag{1.3.2}$$

It follows from (b) that one also has

$$\mathrm{Gr}^k(\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}) = \bigwedge^k \underline{W}\otimes \Omega^{j-k}_{H/S}\otimes \underline{F}. \tag{1.3.3}$$

We prove (i) and (ii) by induction on j, simultaneously for all $\underline{F}$ satisfying (b) and (c). The assertions are empty for $j<0$.

Suppose, then, that (i) and (ii) have been proved for $j'<j$. The induction hypothesis applies to II≈45
the sheaves $\bigwedge^k \underline{W}\otimes \underline{F}$ and, by (c), gives

$$R^i f_*\left(\bigwedge^k \underline{W}\otimes \Omega^{j-k}_{H/S}\otimes \underline{F}\right) = 0 \quad \text{for } k>0 \text{ and } i+j<d-c+k. \tag{1.3.4}$$

The filtration of $\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}$ considered above defines a spectral sequence

$$E_1^{p,i-p} = R^i f_*\left(\bigwedge^p \underline{W}\otimes \Omega^{j-p}_{H/S}\otimes \underline{F}\right) \Longrightarrow R^i f_*\left(\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}\right).$$

By (1.3.4), $E_1^{p,i-p}=0$ when $p>0$ and $i+j\le d-c$; hence

$$\begin{cases} R^i f_*\left(\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}\right) \xrightarrow{\sim} R^i f_*\left(\Omega^j_{H/S}\otimes \underline{F}\right) & \text{if } i+j<d-c,\\ R^i f_*\left(\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}\right) \hookrightarrow R^i f_*\left(\Omega^j_{H/S}\otimes \underline{F}\right) & \text{if } i+j=d-c. \end{cases} \tag{1.3.5}$$

By hypothesis (b), with a suitable differential, the sheaves $\bigwedge^k \underline{W}\otimes \Omega^j_{X/S}\otimes \underline{F}$ form a left Koszul resolution of $\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}$. This filtration defines a spectral sequence

$$E_1^{-p,q} = R^q f_*\left(\bigwedge^p \underline{W}\otimes \Omega^j_{X/S}\otimes \underline{F}\right) \Longrightarrow R^{q-p} f_*\left(\Omega^j_{X/S}\otimes \underline{O}_H\otimes \underline{F}\right). \tag{1.3.6}$$

By hypothesis (c), $E_1^{-p,q}=0$ for $p>0$ and $q+j<d$, and trivially for $p\notin[0,c]$. Consequently,

$$\begin{cases} R^i f_*\left(\Omega^j_{X/S}\otimes \underline{F}\right) \xrightarrow{\sim} R^i f_*\left(\Omega^j_{H/S}\otimes \underline{F}\right) & \text{if } i+j<d-c,\\ R^i f_*\left(\Omega^j_{X/S}\otimes \underline{F}\right) \hookrightarrow R^i f_*\left(\Omega^j_{H/S}\otimes \underline{F}\right) & \text{if } i+j=d-c. \end{cases} \tag{1.3.7}$$

Assertions (i) and (ii) now follow from (1.3.5) and (1.3.7).

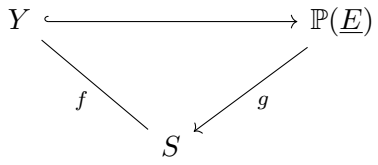
1.4. Let $\underline{E}$ be a locally free sheaf of rank $r+1$ on a scheme S, let $\mathbb{P}(\underline{E})$ be the corresponding II46
projective bundle, let $d\le r$ be an integer, and let $\underline{a}=(a_1,\dots,a_d)$ be a sequence of integers. A relative complete intersection of multidegree $\underline{a}$ in $\mathbb{P}(\underline{E})$ is a closed subscheme which has codimension d in every fiber and which, locally on S for the étale topology, is the intersection of d hypersurfaces of respective degrees $a_1,\dots,a_d$.

By 1.1, hypotheses (b) and (c) of 1.3 hold for $X/S=\mathbb{P}(\underline{E})/S$, $\underline{F}=\underline{O}_X$,

$$\underline{W} = \bigoplus_i \underline{O}(-a_i),$$

and a section s of $\underline{W}^\vee$ defining a relative complete intersection $Y\subset\mathbb{P}(\underline{E})$ of multidegree $\underline{a}$.

Theorem 1.5. Under the hypotheses of 1.4, let $n=r-d$ be the relative dimension of Y, and let $\eta\in H^0(S,R^1f_*\Omega^1_{Y/S})$ be the first Chern class of $\underline{O}(1)$ in the diagram



.

If Y is smooth over S, then:

(i) The sheaves $R^i f_* \Omega^j_{Y/S}$ are locally free, and their formation is compatible with arbitrary base change.

(ii) The spectral sequence

$$E_1^{p,q} = R^q f_* \Omega^p_{Y/S} \Longrightarrow R^{p+q} f_* (\Omega^*_{Y/S}),$$

relating Hodge and de Rham cohomology degenerates ($E_1 = E_\infty$); its abutment is locally free, and its formation is compatible with arbitrary base change.

(iii)$_a$ One has

$$R^i f_* \Omega^j_{Y/S} = 0 \qquad \text{if } i \neq j \text{ and } i + j \neq n.$$

(iii)$_b$ η^i is a basis of $R^i f_* \Omega^i_{Y/S}$ if $2i < n$. II≈**47**

(iii)$_c$ If $n < 2i \leq 2n$, and if α_i is the basis of $R^i f_* \Omega^i_{Y/S}$ dual to the basis η^{n-i} of $R^{n-i} f_* \Omega^{n-i}_{Y/S}$, then

$$\eta^i = \left(\prod_{h=1}^{d} a_h \right) \alpha_i.$$

(iii)$_d$ If $n = 2m$, then $\eta^m \neq 0$ (provided $S \neq \varnothing$), and the quotient $R^m f_* \Omega^m_{Y/S} / \underline{O}_S \eta^m$ is locally free.

It is enough to prove 1.5 in a "universal" case, which permits us to assume that S is smooth over $\mathrm{Spec}(\mathbb{Z})$. In this case (ii) follows from (i), since it holds over $\mathbb{C}$, and therefore over every field of characteristic zero, by Hodge theory. Moreover, since S is reduced, (i) follows from (iii), assumed proved when S is the spectrum of a field. Indeed, by (iii), for each j the functions

$$s \longmapsto \dim_{k(s)} H^i(Y_s, \Omega^j_{Y_s})$$

are constant on S, except perhaps when $i + j = n$; hence they are all constant, because their alternating sum is constant (EGA III, 7.9.2). The conclusion then follows from EGA III, 6.10.5, 7.6.9, and 7.7.5.

By 1.4 and 1.3, one has

$$(1.5.1) \qquad \begin{cases} R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \xrightarrow{\sim} R^i f_* \Omega^j_{Y/S} & \text{for } i + j < n, \\ R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \hookrightarrow R^i f_* \Omega^j_{Y/S} & \text{for } i + j = n. \end{cases}$$

When S is the spectrum of a field, assertions (iii)$_a$, (iii)$_b$, and (iii)$_d$ follow from (1.5.1) and Serre duality ([2], V, 11.2, p. 211). The cohomology class

η^n is the class of a zero-cycle of degree $\prod_{h=1}^d a_h$ (the degree of the complete intersection), so II≈**48**

$$\eta^n = \left(\prod_{h=1}^{d} a_h \right) \alpha_n.$$

By definition, $\eta^{n-i} \alpha_i = \alpha_n$, and (iii)$_c$ follows.

Thus (i) and (ii) hold; in the universal case, the assertions in (iii) follow from (i) and from (iii) in the case already proved, where S is the spectrum of a field.

Theorem 1.6. Keep the hypotheses and notation of 1.4 and 1.5, assume that Y is smooth, let ℓ be a prime invertible in all residue fields of S, and let $\eta_\ell \in H^0(S, R^2 f_* \mathbb{Z}_\ell(1))$ denote the ℓ-adic first Chern class of $\underline{O}(1)$ (SGA 5, VII). Then:

(i) $R^i f_* \mathbb{Z}_\ell = 0$ if i is odd and $i \neq n$.

(ii) If $2i < n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has η_ℓ^i as a basis.

(iii) If $n < 2i \leq 2n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has

$$\eta_\ell^i \Big/ \prod_{h=1}^{d} a_h$$

as a basis.

(iv) The $\mathbb{Z}_\ell$-sheaf $R^n f_* \mathbb{Z}_\ell$ is constructible, twisted constant, and torsion-free. If n is even, then $\eta_\ell^{n/2} \neq 0$ (provided $S \neq \varnothing$), and $R^n f_* \mathbb{Z}_\ell(n/2)/\mathbb{Z}_\ell \eta_\ell^{n/2}$ is torsion-free.

The $\mathbb{Z}_\ell$-sheaves $R^i f_* \mathbb{Z}_\ell$ are constructible and twisted constant, and their formation is compatible with arbitrary base change, by SGA 4, XVI, 2.2. It therefore suffices to prove 1.6 when S is the spectrum of an algebraically closed field. Repeated application of SGA 4, XIV, 3.3 then shows, for every m, that the canonical map

$$H_Y^q(\mathbb{P}(\underline{E}), \mathbb{Z}/\ell^m) \longrightarrow H^q(\mathbb{P}(\underline{E}), \mathbb{Z}/\ell^m)$$

is bijective for $q > r + d$ and surjective for $q = r + d$. From this and Poincaré duality (SGA 4, XVIII), one deduces (SGA 5, VII) that the restriction map II≈49

$$H^q(\mathbb{P}(\underline{E}), \mathbb{Z}_\ell) \longrightarrow H^q(Y, \mathbb{Z}_\ell)$$

is bijective for $q < n$, and injective with torsion-free cokernel for $q = n$. This proves (i) for $i < n$, as well as (ii) and (iv). As in 1.5, one checks that η_ℓ^n is $\prod_{h=1}^d a_h$ times the canonical class; (i) and (iii) then follow by Poincaré duality (SGA 4, XVIII).

1.7. If Y is a complete intersection of dimension at least three in a projective space $\mathbb{P}^r(k)$ over a field k, then $\mathrm{Pic}(Y)$ is known to be isomorphic to $\mathbb{Z}$ and to be generated by $\underline{O}(1)$ (SGA 2, XII, 3.7). In dimension two, when Y is smooth, one can still prove the following.

Theorem 1.8. Let Y be a smooth complete-intersection surface in the projective space $\mathbb{P}^r(k)$ over an algebraically closed field k. Then $\underline{\mathrm{Pic}}^\tau(Y) = 0$, and the class of $\underline{O}(1)$ is not divisible in the Néron–Severi group $\mathrm{NS}(Y)$.

Since $H^1(Y, \underline{O}) = 0$ by $1.5(\mathrm{iii})_a$, one knows that $\underline{\mathrm{Pic}}^0(Y) = 0$. The vanishing of $\underline{\mathrm{Pic}}^\tau(Y)$ then follows from the familiar lemma below, the fact that $H^2(Y, \mathbb{Z}_\ell(1))$ is torsion-free by 1.6(iv), and the fact that $H^0(Y, \Omega_Y^1) = 0$ by $1.5(\mathrm{iii})_a$.

Lemma 1.9. Let Y be a complete algebraic variety over an algebraically closed field k.

(i) If ℓ is prime to the characteristic of k, then the ℓ-torsion of $\mathrm{NS}(Y)$ identifies with the ℓ-torsion of $H^2(Y, \mathbb{Z}_\ell(1))$.

(ii) If k has nonzero characteristic p and Y is smooth, then the kernel of multiplication by p in $\mathrm{Pic}(Y)$ identifies with the subgroup of $H^0(Y, \Omega_Y^1)$ consisting of locally logarithmic differentials.

The Kummer exact sequences II≈50

$$0 \longrightarrow \mu_{\ell^n} \longrightarrow \mathbb{G}_m \xrightarrow{x \mapsto x^{\ell^n}} \mathbb{G}_m \longrightarrow 0$$

give exact sequences

$$0 \longrightarrow \mathrm{Pic}(Y)/\ell^n \mathrm{Pic}(Y) \longrightarrow H^2(Y, \mu_{\ell^n}) \longrightarrow H^2(Y, \mathbb{G}_m)_{\ell^n} \longrightarrow 0,$$

where

$$\mathrm{Pic}(Y)/\ell^n \mathrm{Pic}(Y) = \mathrm{NS}(Y) \otimes \mathbb{Z}/\ell^n.$$

Passing to the inverse limit gives

$$0 \longrightarrow \mathrm{NS}(Y) \otimes \mathbb{Z}_\ell \longrightarrow H^2(Y, \mathbb{Z}_\ell(1)) \longrightarrow \mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, H^2(Y, \mathbb{G}_m)) \longrightarrow 0.$$

The cokernel is torsion-free because $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible. This proves (i).

To prove (ii), it is enough to observe that the exact sequence

$$0 \longrightarrow \underline{O}_Y^* \xrightarrow{x \mapsto x^p} \underline{O}_Y^* \xrightarrow{df/f} \Omega_Y^1$$

defines a short exact sequence

$$0 \longrightarrow \underline{O}_Y^* \longrightarrow \underline{O}_Y^* \longrightarrow \Omega^1_{Y,\mathrm{loc\ log}} \longrightarrow 0,$$

and then to take the corresponding long exact cohomology sequence. This argument is due to Cartier, who proves moreover that, if C denotes the Cartier operator on $H^0(Y, \Omega_Y^1)$, then the locally logarithmic differentials form the kernel of $C - \mathrm{Id}$.

Suppose, for a contradiction, that the class of $\underline{O}(1)$ in $\mathrm{NS}(Y)$ has the form nu, with $n > 1$. If ℓ divides n and is prime to the characteristic, then in ℓ-adic cohomology one again has

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$$\eta_\ell = n\, c_1(u),$$

contradicting 1.6(iv).

Finally, if the characteristic p of k is nonzero and p divides n, then in Hodge cohomology one again has

$$\eta = n\, c_1(u) = 0,$$

contradicting $1.5(\mathrm{iii})_d$ and completing the proof.

2. Numerical results

2.1. Let $\underline{a} = (a_i)_{1 \le i \le d}$ be a sequence of integers ≥ 1, and let $V_n(\underline{a})$ denote a smooth complete intersection of dimension n and multidegree $\underline{a}$ in a projective space $\mathbb{P}^{n+d}$ over a field k. Put

$$h^{p,q}(\underline{a}) = \dim_k H^q(V_{p+q}(\underline{a}), \Omega^p)$$

and

$$h_0^{p,q}(\underline{a}) = h^{p,q}(\underline{a}) - \delta_{p,q}.$$

Thus the $h_0^{p,q}(\underline{a})$ are the Hodge numbers of the middle-dimensional primitive cohomology of a complete intersection of multidegree $\underline{a}$. They depend only on $\underline{a}$, p, and q, by 1.5. Set

$$H(\underline{a}) = \sum_{p,q \ge 0} h_0^{p,q}(\underline{a}) y^p z^q \in \mathbb{Z}[[y, z]]. \tag{2.1.1}$$

For $d = 1$ and $\underline{a} = (a)$, put $h^{p,q}(a) = h^{p,q}(\underline{a})$ and $H(a) = H(\underline{a})$. For $d = 0$, put $\sup(a_i) = 1$.

The following formula is the special case $k = 0$ of formula (2), p. 160, in Hirzebruch [3]:

II ≈**52**

$$\sum_{n=0}^{\infty} \sum_p \chi(V_n(\underline{a}), \Omega^p) y^p z^{n+d} = \frac{1}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + y(1-z)^{a_i}}. \tag{2.2}$$

We shall transform this into the following formula.

Theorem 2.3 (Hirzebruch). With the notation of 2.1, one has

$$H(\underline{a}) = \frac{1}{(1+y)(1+z)} \left[\prod_{i=1}^{d} \frac{(1+y)^{a_i} - (1+z)^{a_i}}{-(1+y)^{a_i} z + (1+z)^{a_i} y} - 1 \right].$$

By 1.5,

$$\chi(V_n(\underline{a}), \Omega^p) = (-1)^p + (-1)^{n-p} h_0^{p,n-p}(\underline{a}).$$

Consequently, the left-hand side I of (2.2) can be rewritten as

$$\begin{aligned} I &= \sum_{n=0}^{\infty} \sum_{p+q=n} \left((-1)^p + (-1)^q h_0^{p,q}(\underline{a})\right) y^p z^{p+q+d} \\ &= \sum_{p,q\geq 0} (-1)^p (yz)^p z^q z^d + \sum_{p,q\geq 0} (-1)^q h_0^{p,q}(\underline{a}) (yz)^p z^q z^d \\ &= z^d \left[\frac{1}{(1+yz)(1-z)} + H(\underline{a})(yz, -z) \right]. \end{aligned}$$

The right-hand side II of (2.2) can be rewritten as

$$II = \frac{z^d}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + yz(1-z)^{a_i}}.$$

After cancelling z^d in $I = II$, one obtains

$$H(\underline{a})(yz, -z) = \frac{1}{(1+yz)(1-z)} \left[\prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + (1-z)^{a_i} yz} - 1 \right],$$

and 2.3 follows by a change of variables.

Corollary 2.4. With the notation of 2.1: II **53**

(i) For $d = 1$, one has

$$\begin{aligned} H(a) &= -\frac{(1+y)^{a-1} - (1+z)^{a-1}}{(1+y)^a z - (1+z)^a y} \\ &= \frac{\displaystyle\sum_{i,j\geq 0} \binom{a-1}{i+j+1} y^i z^j}{\displaystyle 1 - \sum_{i,j\geq 1} \binom{a}{i+j} y^i z^j}. \end{aligned}$$

(ii) Writing $|P|$ for the number of elements of a subset P of $[1, d]$, one has

$$H(\underline{a}) = \sum_{\substack{P\subset[1,d] \\ P\neq\varnothing}} (1+y)^{|P|-1} (1+z)^{|P|-1} \prod_{i\in P} H(a_i).$$

By 2.3,

$$\begin{aligned} H(a) &= -\frac{1}{(1+y)(1+z)} \left[\frac{(1+y)^a - (1+z)^a}{(1+y)^a z - (1+z)^a y} + 1 \right] \\ &= -\frac{1}{(1+y)(1+z)} \left[\frac{(1+y)^a (1+z) - (1+z)^a (1+y)}{(1+y)^a z - (1+z)^a y} \right] \\ &= -\frac{(1+y)^{a-1} - (1+z)^{a-1}}{(1+y)^a z - (1+z)^a y}. \end{aligned}$$

Let N and D be the numerator and denominator in this last expression, so that $H(a) = -N/D$. One has II ≈ **54**

$$N = \sum_k \binom{a-1}{k}(y^k - z^k) = (y-z)\sum_{i,j\geq 0}\binom{a-1}{i+j+1}y^i z^j$$

and

$$\begin{aligned} D &= \sum_k \binom{a}{k}(y^k z - yz^k) = (z-y) + yz\sum_{k\geq 1}\binom{a}{k}(y^{k-1} - z^{k-1}) \\ &= (y-z)\left(-1 + yz\sum_{i,j\geq 0}\binom{a}{i+j+2}y^i z^j\right) \\ &= (y-z)\left(-1 + \sum_{i,j\geq 1}\binom{a}{i+j}y^i z^j\right). \end{aligned}$$

This proves (i).
Now invert the formula in 2.3 when $d = 1$. One finds

$$-\frac{(1+y)^a - (1+z)^a}{(1+y)^a z - (1+z)^a y} = 1 + (1+y)(1+z)H(a).$$

Substituting this identity into 2.3 gives

$$\begin{aligned} H(\underline{a}) &= \frac{1}{(1+y)(1+z)}\left[\prod_{i=1}^{d}(1 + (1+y)(1+z)H(a_i)) - 1\right] \\ &= \frac{1}{(1+y)(1+z)}\left[\sum_{\substack{P\subset[1,d]\\ P\neq\varnothing}}(1+y)^{|P|}(1+z)^{|P|}\prod_{i\in P}H(a_i)\right], \end{aligned}$$

which proves (ii).
Theorem 2.5. With the notation of 2.1:

(i) If $p+q = p'+q'$ and $p \leq p' \leq q' \leq q$, then

$$h_0^{p,q}(\underline{a}) \leq h_0^{p',q'}(\underline{a}).$$

(ii) If $p \leq q$, then $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$p > \left\lfloor \frac{p+q+d-\sum_i a_i}{\sup_i(a_i)} \right\rfloor.$$

Here $\lfloor\ \rfloor$ denotes the integer part of the quotient.

Let (*) be the following property of a formal power series $P = \sum a_{p,q}y^p z^q$: II ≈55

(*) the $a_{p,q}$ are nonnegative, $a_{p,q} = a_{q,p}$, and whenever $p+q = p'+q'$ and $p \leq p' \leq q' \leq q$, one has $a_{p,q} \leq a_{p',q'}$.

Assertion (i) follows at once from the formulas in 2.4, the identity $(1-x)^{-1} = \sum_{n\geq 0} x^n$, and the following lemma, whose proof is left to the reader.
Lemma 2.5.1. The set of formal power series satisfying (*) is stable under addition and multiplication.

We first prove (ii) in the special case of hypersurfaces ($d = 1$). For $p \leq q$, the desired formula can then be rewritten as

$$h_0^{p,q}(a) \neq 0 \quad \Longleftrightarrow \quad p \geq \left\lfloor \frac{p+q+1}{a} \right\rfloor \tag{2.5.2}$$

or, equivalently, $(p+1)a \geq p+q+2$. From the second formula in 2.4(i), one sees that $h_0^{p,q}(a)$ is nonzero if and only if there exist pairs (e,f) and (c_i,d_i) such that

$$\begin{cases} (e,f) \geq (0,0), & e+f \leq a-2, \\ (c_i,d_i) \geq (1,1), & c_i+d_i \leq a, \\ (p,q) = (e,f) + \sum_i (c_i,d_i). \end{cases} \tag{2.5.3}$$

For fixed $p+q$, the smallest possible p is obtained by taking $e = 0$ and every $c_i = 1$; it is the number of indices i. Thus it is the lower bound of those p for which II≈56

$$p \geq \frac{(p+q)-(a-2)}{a}, \qquad \text{i.e.} \qquad (p+1)a \geq p+q+2.$$

We now pass to the general case. If $\underline{a}'$ is obtained from $\underline{a}$ by deleting the a_i equal to one, then $H(\underline{a}) = H(\underline{a}')$. In view of the form of 2.5(ii), we may therefore assume that every $a_i > 1$. If $d = 0$, then $h_0^{p,q} = 0$ and the assertion is correct; we lose no generality by assuming $d > 0$ and $a_d = \sup_i(a_i)$.

By 2.4(ii) and (2.5.2), the condition $h_0^{p,q}(\underline{a}) \neq 0$ is equivalent to the existence of a nonempty subset P of $[1,d]$, a pair (e,f), and pairs $(c_i,d_i)_{i\in P}$ such that

$$\begin{cases} (|P|-1,|P|-1) \geq (e,f) \geq (0,0), \\ c_i,d_i \geq \left\lfloor \dfrac{c_i+d_i+1}{a_i} \right\rfloor, \qquad (c_i,d_i) \geq (0,0), \\ (p,q) = (e,f) + \sum_{i\in P} (c_i,d_i). \end{cases} \tag{2.5.4}$$

For fixed $n = p+q$, seek the smallest possible value of p. Put $n_i = c_i + d_i$. For such a p, conditions (2.5.4) become

$$\begin{cases} a_i = 2 \Longrightarrow n_i \text{ is even} \qquad \text{(otherwise (2.5.4) is impossible)}, \\ c_i \geq \left\lfloor \dfrac{n_i+1}{a_i} \right\rfloor, \qquad c_i \leq n_i, \\ (|P|-1,|P|-1) \geq (e,f) \geq (0,0), \\ n = e+f+\sum_{i\in P} n_i, \\ p = e + \sum_{i\in P} c_i. \end{cases} \tag{2.5.5}$$

This description first shows that the smallest p is obtained for $P = [1,d]$, to which case we henceforth restrict. If II≈57

$$\sum_i (a_i - 2) + (d-1) \geq n, \tag{2.5.6}$$

then a solution exists with $p = c_i = e = 0$. Otherwise, if $a_d \neq 2$, the best choice is $n_i = a_i - 2$ for $i \neq d$, so that $c_i = 0$ for $i \neq d$, $e = 0$, $f = d-1$, and

$$n_d = n - (d-1) - \sum_{i \neq d}(a_i - 2).$$

Then

$$p = \left\lfloor \frac{n - (d-1) - \sum_{i\neq d}(a_i - 2) + 1}{a_d} \right\rfloor, \tag{2.5.7}$$

and hence, for $p \leq q$, one has $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$p \geq \left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor. \tag{2.5.8}$$

Formula (2.5.8) reduces to (2.5.2) for $d = 1$ and is trivially true when (2.5.6) holds. If $d \geq 2$, (2.5.6) fails, and all the a_i equal two, the best choice is instead

$n_i = 0$ for $i \neq d$, $e = 0$, and $f = d-1$ or $d-2$, chosen congruent to n modulo two, with $n_d = n - f$. II≈**58**
If $n \equiv d - 1 \pmod 2$, this proves (2.5.7) as above. Otherwise, observe that

$$\left\lfloor \frac{n - (d-2) + 1}{2} \right\rfloor = \left\lfloor \frac{n - (d-1) + 1}{2} \right\rfloor,$$

so (2.5.7), and hence (2.5.8), remains valid when $a_d = 2$.

It remains to verify that (2.5.8) is equivalent to 2.5(ii). Indeed,

$$\left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor = \left\lfloor \frac{n + d - \sum_{i\neq d} a_i}{a_d} \right\rfloor = \left\lfloor \frac{p + q + d - \sum_i a_i}{a_d} \right\rfloor + 1.$$

2.6. If X is a complete intersection of dimension n in $\mathbb{P}^r$, it follows at once from (2.5.8) that the primitive part of $H^n(X)$ vanishes only when X is an odd-dimensional quadric.

Definition 2.7. If X is a proper smooth algebraic variety over $\mathbb{C}$, the Hodge level of $H^n(X)$ is the supremum of the numbers $q - p$ for which $p + q = n$ and $h^{p,q}(X) \neq 0$.

The Hodge level ℓ is therefore congruent to n modulo two, and is at most

$$\min(n, 2\dim(X) - n).$$

It is nonnegative unless $H^n(X) = 0$, in which case it is $-\infty$.

Corollary 2.8. Retain the notation of 2.1, put $k = \mathbb{C}$, and suppose that $a_d = \sup_i(a_i)$. Let ℓ be an integer with $0 \leq \ell < n$, congruent to n modulo two. The Hodge level of $H^n(V_n(\underline{a}))$ is at most ℓ if and only if

$$\sum_{i\neq d}(a_i - 1) \leq (\ell + 1) - (a_d - 2)\frac{n - \ell}{2}.$$

By (2.5.8), if the Hodge level is nonnegative, it is the maximum of zero and II≈**59**

$$n - 2\left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor.$$

The inequality

$$n - 2\left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor \leq \ell$$

may be rewritten as

$$\frac{n-\ell}{2} \leq \left\lfloor \frac{n - \sum_{i\neq d}(a_i - 1) + 1}{a_d} \right\rfloor,$$

or equivalently as

$$\sum_{i\neq d}(a_i - 1) \leq 1 + n - a_d\frac{n-\ell}{2} = \ell + 1 - (a_d - 2)\frac{n-\ell}{2}.$$

2.9. Using formula 2.8, it is easy, for small ℓ, to enumerate the multidegrees of complete intersections in $\mathbb{P}^r$ of dimension $n > 1$ whose H^n has level ℓ. The table given in the first version of this exposé was, however, incorrect; the table below is extracted from the more complete one given by M. Rapoport in his article "Complément à l'article de P. Deligne *La conjecture de Weil pour les surfaces K3*", *Inventiones Math.* 15, 227–236 (1972). For $\ell \leq 1$, the complete intersections $V_n(\underline{a})$ of dimension n and multidegree $\underline{a} = (a_1, \ldots, a_d)$, with $n > \ell$, $d \geq 1$, and $a_i \geq 2$, having Hodge level ℓ, are the following:

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$\ell = -\infty$ $\quad V_{2k+1}(2) \qquad (k \geq 0),$

$\ell = 0$ $\quad V_{2k}(2)\ (k \geq 1), \quad V_{2k}(2,2)\ (k \geq 1), \quad V_2(3),$

$\ell = 1$ $\quad V_{2k+1}(2,2)\ (k \geq 1), \quad V_{2k+1}(2,2,2)\ (k \geq 1),$
$V_3(3), \quad V_5(3), \quad V_3(3,2), \quad V_3(4).$

2.10. For small $\underline{a}$, the generating functions of the $h_0^{p,q}(\underline{a})$ are, by 2.4,

$$\begin{aligned} H(2) &= \frac{1}{1-yz}, \\ H(3) &= \frac{2+y+z}{1-3yz-y^2z-yz^2}, \\ H(2,2) &= \frac{y+z}{(1-yz)^2} + \frac{2}{(1-yz)^2} + \frac{1}{1-yz}, \end{aligned}$$

so that

$$\begin{aligned} h_0^{p,p}(2) &= 1 \quad , \\ h_0^{1,1}(3) &= 6 \quad , \\ h_0^{p,p}(2,2) &= 2p+3 \quad . \end{aligned}$$

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QUADRICS

by P. DELIGNE

0.0. The present exposé collects several well-known results about quadrics which we shall need in order to prove the Picard–Lefschetz formula.

0.1. Let m be a section of an $\mathcal{O}_S$-module M on a scheme S. We shall say that m is nowhere zero on S if, for every $s \in S$, the image of m in the $k(s)$-vector space M_s obtained from M by the base change $\{s\} \longrightarrow S$ is nonzero. We use the same terminology when $S = \mathrm{Spec}(A)$ and m is an element of an A-module M.

1. Quadratic forms

1.1. Let $S = \mathrm{Spec}(A)$ be an affine scheme, let V be a locally free A-module of rank n, and let Q be a quadratic form on V (Bourbaki, *Algèbre*, Ch. 9, §3, no. 4). Recall that Q is called nondegenerate if the associated bilinear form

$$\Phi(x, y) = Q(x + y) - Q(x) - Q(y)$$

is nondegenerate, that is, if it induces an isomorphism from V to its dual V^*. If 2 is not invertible in A, this can occur only when n is even (loc. cit.).

The quadratic forms on V form a projective A-module; a quadratic form Q is "nowhere zero" on S (0.1) if and only if the elements $Q(v)$ generate the unit ideal of A. II ≈**63**

The notation $\mathbb{P}(V)$ is used in the sense of EGA; recall that if A is a field, then $\mathbb{P}(V)$ is the projective space whose set of rational points is the quotient $(V^* \setminus \{0\})/A^*$.

The quadratic form Q identifies with a section of the invertible sheaf $\mathcal{O}(2)$ on the projective space $\mathbb{P}(V^*)$ over S. When Q is nowhere zero, the ("homogeneous") equation $Q = 0$ defines a subscheme of $\mathbb{P}(V^*)$, flat and purely of relative dimension $n - 2$ over S. This scheme is called the quadric defined by Q. The form Q is called ordinary if it is nowhere zero on S and the quadric that it defines is smooth over S.

The form Q is ordinary if and only if every form obtained from it by extending the scalars from A to a field is ordinary. If A is a field, then:

a) if n is even or $\mathrm{char}(A) \neq 2$, then Q is ordinary if and only if Q is nondegenerate;

b) if n is odd and $\mathrm{char}(A) = 2$, then Q is ordinary if and only if the kernel N of the associated (alternating) bilinear form Φ has dimension one and Q does not vanish on N.

Proposition 1.2. If Q is ordinary and $n = 2m$ (respectively, $n = 2m + 1$), then, locally for the étale topology on S, V has a basis $e_1, \ldots, e_n$ such that

$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m}$$

(respectively,

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$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m} + \lambda x_{2m+1}^2, \qquad (\lambda \text{ invertible})).$$

We prove 1.2 by induction on m. If $m = 0$, the assertion is clear. If $m > 0$, the quadric with equation $Q = 0$ is smooth over S and has nonempty geometric fibers; it therefore has sections locally

for the étale topology. Locally, such a section is defined by an element $e \in V$, nowhere zero on $\mathrm{Spec}(A)$, with $Q(e) = 0$. At no point of S does e belong to the kernel of Φ; hence, locally, one can find f' such that $\Phi(e, f') = 1$. If

$$f = -Q(f')e + f',$$

then

$$Q(e) = Q(f) = 0, \qquad \Phi(e, f) = 1.$$

If V_2 is the orthogonal complement of the submodule V_1 generated by e and f, then $V = V_1 \oplus V_2$, and the conclusion follows by applying the induction hypothesis to V_2.

1.3. For the definition and principal properties of the Clifford algebra of Q, we refer to Bourbaki, *Algèbre*, Ch. 9, §9. We denote by ρ the canonical map from V to $C(Q)$.

Lemma 1.3.1. Under the hypotheses of 1.1, for every $\lambda \in A$ there exists a unique homomorphism

$$[\lambda] \colon C^+(\lambda Q) \longrightarrow C^+(Q)$$

such that, for $x, y \in V$,

$$[\lambda](\rho(x)\rho(y)) = \lambda\rho(x)\rho(y).$$

This homomorphism is induced on passing to the quotient by the endomorphism $[\lambda]$ of the even part $T^+(V)$ of the tensor algebra of V, which acts as λ^r on $T^{2r}(V)$. Indeed, II ≈65

$$[\lambda](x \otimes x - \lambda Q(x)) = \lambda(x \otimes x - Q(x)).$$

Suppose that Q is nowhere zero on S. Lemma 1.3.1 then defines a transitive system of isomorphisms between the algebras $C^+(\lambda Q)$ for $\lambda \in A^*$.

Consequently, the algebra $C^+(Q)$ depends (up to unique isomorphism) only on V and on Q up to multiplication by a factor; in other words, $C^+(Q)$ depends only on V and on the subscheme of $\mathbb{P}(V^*)$ with equation $Q = 0$. Every homothety with invertible ratio is an automorphism of the pair $(V, X \subset \mathbb{P}(V^*))$; one checks that these automorphisms act trivially on $C^+(V, X)$. It follows that the algebra $C^+(V, X)$ depends only on the pair of schemes $X \subset \mathbb{P}(V^*)$ over A. We shall denote it by $C^+(X, \mathbb{P}(V^*))$.

1.4. Let V be a locally free A-module of rank $2m > 0$, and let Q be a nondegenerate quadratic form on V. Suppose given a decomposition $V = W_1 \oplus W_2$ into two totally singular subspaces. The form $Q(w_1 + w_2)$ then places W_1 and W_2 in perfect duality. Recall that, for $f \in W_2$ (identified with W_1^*), contraction by f is the endomorphism of degree -1 of $\bigwedge W_1$ defined by

$$f \lrcorner (e_1 \wedge \cdots \wedge e_p) = \sum_{i=1}^{p} (-1)^{i-1} \langle e_i, f \rangle e_1 \wedge \cdots \wedge \widehat{e_i} \wedge \cdots \wedge e_p.$$

Endow the A-module $\bigwedge W_1$ with the $\mathbb{Z}/2$-grading for which

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$$(\bigwedge W_1)^+ = \bigoplus_i \bigwedge^{2i} W_1, \qquad (\bigwedge W_1)^- = \bigoplus_i \bigwedge^{2i+1} W_1.$$

The algebra $\mathrm{End}(\bigwedge W_1)$ is then $\mathbb{Z}/2$-graded: its even (respectively, odd) endomorphisms are those that preserve the grading (respectively, have degree one modulo two).

We refer to Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, for the proof of the existence of a unique isomorphism s of $\mathbb{Z}/2$-graded algebras from $C(Q)$ to $\mathrm{End}(\bigwedge W_1)$ such that

$$s(\rho((e, 0))) = e \wedge (-), \qquad s(\rho((0, f))) = f \lrcorner (-).$$

Temporarily denote by $e(W_1, W_2)$ the element of $C^+(Q)$ for which $s(e(W_1, W_2))$ is the identity on $(\bigwedge W_1)^+$ and zero on $(\bigwedge W_1)^-$. The center $Z(Q)$ of $C^+(Q)$ is isomorphic to $A \oplus A$, this decomposition

being defined by the idempotents $e(W_1, W_2)$ and $1-e(W_1, W_2)$. Moreover, $C^+(Q)$ is a matrix algebra over $Z(Q)$.

From this and 1.2 one deduces, as in loc. cit.:

Proposition 1.5. Under the hypotheses of 1.1, if $n = 2m > 0$ and Q is nondegenerate, then the center $Z(Q)$ of $C^+(Q)$ is an étale algebra, locally free of rank two over A, and $C^+(Q)$ is an Azumaya algebra over $Z(Q)$.

1.6. Let V and Q be as in 1.4, and let W be a totally singular direct summand of V, locally free of rank m. Let W_1' be a complement of W, so that $V = W \oplus W_1'$. Every complement W_2' of W identifies with the graph of a map $u : W_1' \to W$. The form Φ places W and W_1' in duality, so u is uniquely determined by the bilinear form on W_1'

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$$B(x, y) = \Phi(x, u(y)),$$

and W_2' is totally singular if and only if

$$B(x, x) = -Q(x)$$

on W_1'. Since such forms B always exist, W has totally singular complements. Having chosen one of them, say W', the set of totally singular complements W_0' of W identifies with the set of alternating forms on W', or equivalently with the set of sections of the affine space $\mathbb{E}(\bigwedge^2 W')$ over S.

For each section x of $\mathbb{E}(\bigwedge^2 W')$, let $W_0'(x)$ be the corresponding complement of W and let $e(x) = e(W, W_0'(x))$. The function $x \mapsto e(x)$ is compatible with every extension of scalars and hence defines a morphism of S-schemes

$$e : \mathbb{E}(\bigwedge^2 W') \longrightarrow \mathrm{Spec}(Z(Q)).$$

The S-scheme $\mathbb{E}(\bigwedge^2 W')$ is flat with geometrically connected fibers, whereas $\mathrm{Spec}(Z(Q))$ is étale over S. The morphism e therefore factors through S, hence through a section of $\mathrm{Spec}(Z(Q))$ over S; equivalently, $e(x)$ is independent of x.

This justifies the following notation.

Notation 1.7. For W as above and any totally singular complement W' of W, denote by $e(W)$ the idempotent $e(W, W') \in Z(Q)$.

1.8. We saw in 1.3 that, for invertible λ, $C^+(Q)$ identifies with $C^+(\lambda Q)$. A continuity argument analogous to the preceding one shows that this identification carries

$$e(W) \in Z(Q) \qquad \text{to} \qquad e(W) \in Z(\lambda Q).$$

1.9. Recall that if B and C are two $\mathbb{Z}/2$-graded algebras, then multiplication in their graded tensor product $B \otimes' C$ is defined by the Koszul rule II 68

$$(b_1 \otimes c_1)(b_2 \otimes c_2) = (-1)^{\deg(c_1)\deg(b_2)} b_1 b_2 \otimes c_1 c_2$$

when the b_i and c_i are homogeneous.

If (V_1, Q_1) and (V_2, Q_2) are two quadratic modules, then

$$C(Q_1 \oplus Q_2) = C(Q_1) \otimes' C(Q_2)$$

(Bourbaki, loc. cit.).

If P and Q are two finite-type, locally free, $\mathbb{Z}/2$-graded A-modules, we identify the $\mathbb{Z}/2$-graded algebras $\mathrm{End}(P) \otimes' \mathrm{End}(Q)$ and $\mathrm{End}(P \otimes Q)$ by setting, for homogeneous u, v, p, q,

$$(u \otimes v)(p \otimes q) = (-1)^{\deg(v)\deg(p)} u(p) \otimes v(q).$$

Lemma 1.10. Let (V, Q) be the quadratic module which is the sum of two quadratic modules (V_1, Q_1) and (V_2, Q_2), where V_i is locally free of rank $2m_i$ and Q_i is nondegenerate. Let $V_i = W_i \oplus W_i'$ be a decomposition of V_i into two totally singular subspaces $(i = 1, 2)$. Then the diagram

$$\begin{array}{ccccc} C(Q) & & \xrightarrow{\quad\sim\quad} & & C(Q_1)\otimes' C(Q_2) \\ \Big\downarrow\wr & & & & \Big\| \\ \mathrm{End}(\bigwedge(W_1\oplus W_2)) & \simeq & \mathrm{End}(\bigwedge W_1\otimes\bigwedge W_2) & \simeq & \mathrm{End}(\bigwedge W_1)\otimes'\mathrm{End}(\bigwedge W_2) \end{array}$$

is commutative.

The verification is left to the reader. One deduces at once from this lemma that

$$e(W_1\oplus W_2) = e(W_1)e(W_2) + (1-e(W_1))(1-e(W_2)). \tag{1.10.1}$$

1.11. The following construction is borrowed from Micali and Villamayor. If $p : Z \longrightarrow S$ is a II**69** double étale covering of S, then Z may be regarded as a $\mathbb{Z}/2$-torsor over S. Denote the addition of torsors by $*$; for a section s of Z, let $e(s)$ be the idempotent of $p_*\mathcal{O}_Z$ that equals one on s and zero elsewhere. With the notation of 1.10, there is a unique isomorphism

$$\mathrm{Spec}(Z(Q_1)) * \mathrm{Spec}(Z(Q_2)) \longrightarrow \mathrm{Spec}(Z(Q)), \tag{1.11.1}$$

whose underlying addition morphism is

$$+ : \mathrm{Spec}(Z(Q_1)) \times_S \mathrm{Spec}(Z(Q_2)) \longrightarrow \mathrm{Spec}(Z(Q)),$$

and which, after every base change, satisfies

$$e(s+t) = e(s)e(t) + (1-e(s))(1-e(t)).$$

If $s(W)$ is the section of $\mathrm{Spec}(Z(Q))$ such that $e(s(W)) = e(W)$, then formula (1.10.1) becomes

$$s(W_1\oplus W_2) = s(W_1) + s(W_2).$$

Proposition 1.12. Let A be a field, let V be a vector space of dimension $2m > 0$ over A, let Q be a nondegenerate quadratic form on V, and let W_1, W_2 be two totally isotropic subspaces of V of dimension m. Then $e(W_1) = e(W_2)$ if and only if $\dim(W_1/(W_1\cap W_2))$ is even.

(Cf. Bourbaki, *Algèbre*, Ch. 9, §6, Exercise 28.) Let $e_1,\dots,e_m,f_1,\dots,f_m$ be a basis of V such II≈**70** that

a) $Q(\sum_i x_ie_i + \sum_i y_if_i) = \sum_i x_iy_i$;

b) W_1 is generated by the e_i;

c) W_2 is generated by $(e_i)_{1\le i\le s}$ and $(f_i)_{s<i\le m}$.

Then V is the sum of the $V_i = Ae_i + Af_i$, and

$$W_\alpha = \bigoplus_i (W_\alpha\cap V_i), \qquad \alpha = 1,2.$$

The addition formula 1.11 therefore reduces the proof to the case $\dim V = 2$. If $W_1 = W_2$, the assertion is clear. Otherwise, if $W_1 = Ae$ and $W_2 = Af$ with $\Phi(e,f) = 1$, then $e(W_1) = fe$ and $e(W_2) = ef = 1 - fe$.

2. Quadrics

2.1. Let k be an algebraically closed field and V a vector space of dimension $n+2$ over k. A smooth quadric of dimension n over k is a k-scheme X isomorphic to the subscheme of $\mathbb{P}(V^*)$ defined by the equation $Q = 0$, for some ordinary quadratic form Q over k.

Remark 2.2. For $n = 0, 1, 2$, X is a smooth quadric of dimension n over k if and only if

$$\begin{aligned} n = 0 &: X \simeq \mathrm{Spec}(k) \amalg \mathrm{Spec}(k), \\ n = 1 &: X \simeq \mathbb{P}^1_k, \\ n = 2 &: X \simeq \mathbb{P}^1_k \times \mathbb{P}^1_k. \end{aligned}$$

Lemma 2.3. With the notation of 2.1, suppose that X is defined by $Q = 0$. II **71**

(i) $\Omega^n_X \simeq \mathcal{O}_X(-n)$.

(ii)
$$\mathrm{Pic}(X) \simeq \begin{cases} 0, & n = 0, \\ \mathbb{Z}, & n = 1;\ \mathcal{O}_X(1) \text{ is the square of a generator}, \\ \mathbb{Z} \oplus \mathbb{Z}, & n = 2, \\ \mathbb{Z}, & n \geq 3;\ \mathcal{O}_X(1) \text{ is a generator}. \end{cases}$$

(iii) $H^i(X, \mathcal{O}_X) = 0$ for $i > 0$, and $H^1(X, \mathcal{O}_X(1)) = 0$.

(iv) The natural map
$$H^0(\mathbb{P}(V^*), \mathcal{O}_{\mathbb{P}(V^*)}(1)) = V^* \xrightarrow{\sim} H^0(X, \mathcal{O}_X(1))$$
is an isomorphism; and, for $n > 0$, $H^0(X, \mathcal{O}_X) = k$.

Assertion (i) follows from the fact that the conormal sheaf of X in $\mathbb{P}(V^*)$ is $\mathcal{O}_X(-2)$ and that the canonical sheaf of $\mathbb{P}^r$ is $\mathcal{O}(-r-1)$. Assertion (ii) follows from 2.2 when $n \leq 2$, and from SGA 2, XII, 3.7 when $n \geq 3$. Finally, (iii) and (iv) follow from FAC, no. 78.

Definition 2.4. A smooth quadric of dimension n over a scheme S is an S-scheme $f : X \longrightarrow S$, proper and smooth over S, whose geometric fibres are smooth quadrics.

For $n = 0$, a smooth quadric of dimension zero over S is simply a double étale covering of S. For $n = 1$, it is simply a Severi–Brauer variety of dimension one over S.

2.5. Let $\rho : X \longrightarrow S$ be a smooth quadric of dimension n over S. By 2.3(i), the inverse of the invertible sheaf $\Omega^n_{X/S}$ is ample.

By 2.3(iii), the Picard scheme $\mathrm{Pic}_{X/S}$ is smooth over S (because $H^2(X, \mathcal{O}_X) = 0$), unramified II≈**72** over S (because $H^1(X, \mathcal{O}_X) = 0$), and essentially proper over S (because X/S is smooth). Thus, locally on S for the étale topology, it is a constant discrete group having the structure described in 2.3(ii).

In particular, if X has a section, there is an invertible sheaf L on X such that $L^{\otimes(-n)}$ and $\Omega^n_{X/S}$ have the same image in $\mathrm{Pic}_{X/S}$; any two such sheaves are locally isomorphic on S. By 2.3(iii) and [1], 0.5, p. 19, $\rho_* L$ is locally free and its formation commutes with every base change. Hence $\rho_* L$ is locally free of rank $n+2$, the map $\rho^* \rho_* L \longrightarrow L$ is surjective, and the canonical map

$$X \longrightarrow \mathbb{P}(\rho_* L)$$

is a closed immersion. It identifies X with a relative divisor of degree two in $\mathbb{P}(\rho_* L)$; locally on S, X is the quadric in $\mathbb{P}(\rho_* L)$ defined by an ordinary quadratic form on $(\rho_* L)^*$, unique up to an invertible factor.

2.6. Suppose that $n > 0$. If L_1 and L_2 are two invertible sheaves of the type considered above, then locally on S there is an isomorphism $u : L_1 \xrightarrow{\sim} L_2$. It is unique up to an invertible factor, since $\rho_* \mathcal{O}_X = \mathcal{O}_S$ by 2.3(iv). Consequently all the isomorphisms $u : L_1 \xrightarrow{\sim} L_2$ induce the same isomorphism

$$\mathbb{P}(u) : \mathbb{P}(\rho_* L_1) \xrightarrow{\sim} \mathbb{P}(\rho_* L_2).$$

Thus $\mathbb{P}(\rho_* L_1)$ is canonically isomorphic to $\mathbb{P}(\rho_* L_2)$ and depends only on X/S, not on the choice of L. We denote it by $P(X)$.

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The projective space $P(X)$ is defined locally on S for the étale topology. By descent it defines a Severi–Brauer variety over S, again denoted $P(X)$, in which X is a relative divisor of degree two.

Locally on S, the pair $X \hookrightarrow P(X)$ defines a Clifford algebra $C^+(X, P(X))$ (1.3). Descent yields an algebra over S, which we denote simply by $C^+(X)$.

2.7. Suppose that $n = 2m > 0$. The centre Z of $C^+(X)$ is then an $\mathcal{O}_S$-algebra defining a double étale covering $Z(X)$ of S.

A generator of X is a subscheme D of X which is a linear subspace of dimension m in $P(X)$.

Locally on S, when $P(X) \simeq \mathbb{P}(V^*)$ and X is defined by $Q = 0$, a generator D of X is, by definition, identified with a locally direct-summand submodule $W(D)$ of V, totally singular and of rank $m+1$. By 1.8, the idempotent $e(W(D))$ in the centre of $C^+(X)$ depends only on $D \subset X$. Let $e(D)$ denote the corresponding section of $Z(X)$, namely the one on which $e(W(D))$ has value one.

The generators of X/S are the S-points of a scheme $\mathrm{Gen}(X)$ projective over S, and e defines a morphism of S-schemes

$$e : \mathrm{Gen}(X) \longrightarrow Z(X). \tag{2.7.1}$$

When $n = 0$, we set $Z(X) = X$, call a section of X a generator of X, and define e to be the identity.

Proposition 2.8. Under the hypotheses of 2.7, $p : \mathrm{Gen}(X) \longrightarrow S$ is projective and smooth, and (2.7.1) is its Stein factorization. II 74

The question is local for the étale topology on S. Locally, if D is a generator, then by 1.6 the generators D' disjoint from D are identified with the S-points of a suitable affine space over S. In this way $\mathrm{Gen}(X)$ is covered by open subschemes smooth over S.

To prove that the geometric fibres of e are geometrically connected, it is enough to consider the case in which S is the spectrum of an algebraically closed field k and X is defined by $Q = 0$ in $\mathbb{P}(V^*)$.

Let W_1 and W_2 be two totally isotropic subspaces of rank $m+1$ in V such that $e(W_1) = e(W_2)$, that is, such that $W_1/(W_1 \cap W_2)$ has even dimension (1.12). Proceeding as in 1.12, one obtains an orthogonal decomposition

$$V = V_0 \oplus \bigoplus_i V_i$$

such that

a) $W_\alpha = (W_\alpha \cap V_0) \oplus \bigoplus_i (W_\alpha \cap V_i)$ for $\alpha = 1, 2$;

b) $W_1 \cap V_0 = W_2 \cap V_0$;

c) $\dim V_i = 4$ and $V_i = (W_1 \cap V_i) \oplus (W_2 \cap V_i)$.

It remains to prove that $W_1 \cap V_i$ and $W_2 \cap V_i$ belong to the same connected family of maximal totally singular subspaces of V_i. In other words, the assertion reduces to the case of quadrics of dimension two. Such a quadric is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, and its generators are the subschemes $\{t\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{t\}$; the proposition is then evident.

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3. Cohomology of quadrics

3.1. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S. We write

$$\eta \in H^0\big(S, R^2 p_* \mathbb{Z}_\ell(1)\big)$$

for the "cohomology class of a hyperplane section." Locally on S for the étale topology, $P(X) \simeq \mathbb{P}(V^*)$; by definition, η is then the image in $H^0(S, R^2 p_* \mathbb{Z}_\ell(1))$ of the first Chern class

$$c_1(\mathcal{O}(1)) \in H^2(X, \mathbb{Z}_\ell(1)).$$

3.2. Suppose that $n = 2m$. A generator D of X then defines a cohomology class $c\ell'(D) \in H^n(X, \mathbb{Z}_\ell(m))$, whose image in $H^0(S, R^n p_* \mathbb{Z}_\ell(m))$ is denoted $c\ell(D)$. By 2.8, the latter depends only on the section $e(D)$ of $Z(X)$. We therefore obtain a morphism of étale sheaves on S

$$c\ell : Z(X) \longrightarrow R^n p_* \mathbb{Z}_\ell(m). \tag{3.2.1}$$

Theorem 3.3. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S.

(i) $R^{2i+1} p_* \mathbb{Z}_\ell = 0$.

(ii) If $0 \leq 2i < n$ (respectively, if $n < 2i \leq 2n$), then $R^{2i} p_* \mathbb{Z}_\ell(i)$ is canonically isomorphic to the constant sheaf $\mathbb{Z}_\ell$, and is generated by η^i (respectively, by $\eta^i/2$).

(iii) If $n = 2m$, the map deduced from (3.2.1),

$$\mathbb{Z}_\ell^{Z(X)} \longrightarrow R^n p_* \mathbb{Z}_\ell(m), \tag{3.3.1}$$

is an isomorphism. If α and β are two disjoint sections of $Z(X)$ (assumed isomorphic to $S \amalg S$, as is locally the case), then

(a) $\eta^m = c\ell(\alpha) + c\ell(\beta)$,

(b) $\begin{cases} \mathrm{Tr}(c\ell(\alpha)^2) = \mathrm{Tr}(c\ell(\beta)^2) = 1, & c\ell(\alpha)c\ell(\beta) = 0, \quad m \text{ even}, \\ c\ell(\alpha)^2 = c\ell(\beta)^2 = 0, & \mathrm{Tr}(c\ell(\alpha)c\ell(\beta)) = 1, \quad m \text{ odd}, \end{cases}$

(c) $\eta(c\ell(\alpha) - c\ell(\beta)) = 0$.

Assertions (i) and (ii) are recalled only for reference (XI, 1.6, supplemented by XI, 2.6 when $2i + 1 = n$ in (i)). II≈76

It is enough to verify (iii) when S is the spectrum of an algebraically closed field k (one could even take $S = \mathrm{Spec}(\mathbb{C})$).

Formula (a). We may suppose that X is the quadric in $\mathbb{P}^{2m+1}(k)$ with equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} = 0.$$

The class η^m is the cohomology class of the plane section defined by

$$x_i = 0 \qquad (0 \leq i < m),$$

and this cycle is the sum of the generators D_1, D_2 defined by II≈77

$$\begin{aligned} D_1 : &\quad x_i = 0 && (0 \leq i \leq m), \\ D_2 : &\quad x_i = 0 && (0 \leq i < m), \qquad x_{2m+1} = 0. \end{aligned}$$

By 1.12, $e(D_1) \neq e(D_2)$, and hence

$$\eta^m = c\ell(D_1) + c\ell(D_2) = c\ell(\alpha) + c\ell(\beta).$$

Formula (b). If generators D_1, D_2 are disjoint (respectively, meet transversally in one point), then $c\ell(D_1)c\ell(D_2) = 0$ (respectively, $\mathrm{Tr}(c\ell(D_1)c\ell(D_2)) = 1$). By 1.12, when m is even we have $e(D_1) \neq e(D_2)$ (respectively, $e(D_1) = e(D_2)$); when m is odd we have $e(D_1) = e(D_2)$ (respectively, $e(D_1) \neq e(D_2)$). Formula (b) follows.

Formula (c). This is clear when $n = 0$. Otherwise, by (ii), it is enough to prove

$$\eta^m(c\ell(\alpha) - c\ell(\beta)) = 0.$$

Indeed,

$$\mathrm{Tr}(\eta^m(c\ell(\alpha) - c\ell(\beta))) = \mathrm{Tr}\Big(c\ell(\alpha)^2 - c\ell(\beta)^2\Big) = 0.$$

Verification. When m is even,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\Big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\Big) = 1 + 1;$$

when m is odd,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\Big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\Big) = 2.$$

That (3.3.1) is an isomorphism now follows from the fact that the two bilinear forms in two variables

$$B(x, y) = x_1y_2 + x_2y_1, \qquad B(x, y) = x_1y_1 + x_2y_2$$

have discriminant ± 1.

Verification 3.4. Let X be a smooth quadric of dimension n over a finite field $\mathbb{F}_q$. In view of 3.3, II≈**78** the Lefschetz trace formula gives the classical formula

$$\#X(\mathbb{F}_q) = \begin{cases} \sum_{i=0}^{n} q^i, & n \text{ odd}, \\ \sum_{i=0}^{n} q^i + \varepsilon q^m, & n = 2m, \quad \varepsilon = \pm 1. \end{cases}$$

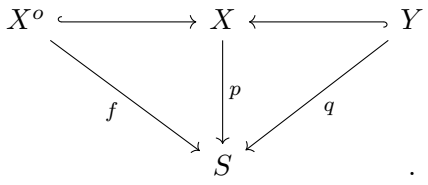
When n is even, $\varepsilon = 1$ if X is split, that is, if it has a rational generator, and $\varepsilon = -1$ otherwise.

3.5. Let $f : X \longrightarrow S$ be a smooth quadric of dimension $n = 2m$. The primitive part of $R^n f_*\mathbb{Z}_\ell(m)$ is the orthogonal complement of η^m; with the notation of 3.3, it is generated by $c\ell(\alpha) - c\ell(\beta)$. The primitive quotient of $R^n f_*\mathbb{Z}_\ell(m)$ is the quotient by $\mathbb{Z}_\ell\eta^m$. The primitive part and primitive quotient of $R^n f_*\mathbb{Z}_\ell$ are obtained by tensoring with $\mathbb{Z}_\ell(-m)$.

3.6. Let V be a locally free vector bundle of rank $n+2$ on S, and let X be the smooth quadric over S defined by an ordinary quadratic form Q on V. Let H moreover be a hyperplane in $\mathbb{P}(V^*)$ meeting X transversally, and put $X^o = X - H$. The schemes X and $Y = X \cap H$ are smooth quadrics over S, of dimensions n and $n - 1$. We may therefore use 3.3 to compute the cohomology of $X^o = X - Y$.

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(3.6.1)



.

If $n = 0$, then $Y = \varnothing$ and $X = X^o$. Suppose that $n > 0$. We have two long exact sequences dual to one another:

$$\cdots \xrightarrow{\partial} R^i f_!\mathbb{Z}_\ell \longrightarrow R^i p_*\mathbb{Z}_\ell \xrightarrow{r_i} R^i q_*\mathbb{Z}_\ell \longrightarrow \cdots \tag{3.6.2}$$

and

$$\cdots \xrightarrow{\partial} R^{i-2} q_*\mathbb{Z}_\ell(-1) \longrightarrow R^i p_*\mathbb{Z}_\ell \longrightarrow R^i f_*\mathbb{Z}_\ell \longrightarrow \cdots . \tag{3.6.3}$$

For n even, the restriction homomorphism r_i is an isomorphism for $i \neq n, 2n$; for n odd, it is an isomorphism for $i \neq n - 1, 2n$. If $n = 2m$ and α is the cohomology class of a generator, then

$$r_n(\alpha) = \tfrac{1}{2}\eta^m,$$

so r_n is surjective and its kernel is the primitive part of $R^n p_* \mathbb{Z}_\ell$. If $n = 2m+1$, then $r_{2m}(\eta^m) = \eta^m$, so r_{2m} is injective and its cokernel is the primitive quotient of $R^{2m} q_* \mathbb{Z}_\ell$. It follows that $R^i f_! \mathbb{Z}_\ell = 0$ for $i \neq n, 2n$, and that $R^i f_! \mathbb{Z}_\ell$ is a twisted constant free $\mathbb{Z}_\ell$-sheaf of rank one for $i = n, 2n$. By duality, or from (3.6.3), one likewise shows that $R^i f_* \mathbb{Z}_\ell = 0$ for $i \neq 0, n$, that $f_* \mathbb{Z}_\ell = \mathbb{Z}_\ell$, and that $R^n f_* \mathbb{Z}_\ell$ has rank one.

Suppose that $n = 2m$ and retain the notation of 3.3.

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$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^n f_! \mathbb{Z}_\ell(m) & \longrightarrow & R^n p_* \mathbb{Z}_\ell(m) & \longrightarrow & R^n q_* \mathbb{Z}_\ell(m) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varphi} & & \| & & & & \\ 0 & \longleftarrow & R^n f_* \mathbb{Z}_\ell(m) & \longleftarrow & R^n p_* \mathbb{Z}_\ell(m) & \longleftarrow & R^{n-2} q_* \mathbb{Z}_\ell(m-1) & \longleftarrow & 0. \end{array}$$

The sheaf $R^n f_! \mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, whose images in $R^n p_* \mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$. By 3.3(iii)(b),

$$\mathrm{Tr}(\delta^2) = (c\ell(\alpha) - c\ell(\beta))^2 = (-1)^m \cdot 2,$$

and $\pm\varphi(\delta)$ is twice the natural dual generators $\pm\delta'$ of $R^n f_* \mathbb{Z}_\ell(m)$, namely the opposite images of $c\ell(\alpha)$ and $c\ell(\beta)$.

Suppose now that $n = 2m+1$ and apply the notation of 3.3 to Y.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^{2m} p_* \mathbb{Z}_\ell(m) & \longrightarrow & R^{2m} q_* \mathbb{Z}_\ell(m) & \xrightarrow{\partial} & R^n f_! \mathbb{Z}_\ell(m) & \longrightarrow & 0 \\ & & & & & & \downarrow{\scriptstyle \varphi} & & \\ & & R^{2m+2} p_* \mathbb{Z}_\ell(m+1) & \longleftarrow & R^{2m} q_* \mathbb{Z}_\ell(m) & \longleftarrow & R^n f_* \mathbb{Z}_\ell(m+1) & \longleftarrow & 0. \end{array}$$

The sheaf $R^n f_! \mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, namely $\partial c\ell(\alpha)$ and $\partial c\ell(\beta)$. We have

$$\delta\,\varphi(\delta) = \delta^2 = (\partial c\ell(\alpha))^2 = \partial(c\ell(\alpha)\,\partial c\ell(\alpha)) = 0,$$

and therefore $\varphi(\delta) = 0$. On the other hand, $R^n f_* \mathbb{Z}_\ell(m+1)$ has two natural opposite generators $\pm\delta'$, whose images in $R^{2m} q_* \mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$, and $\mathrm{Tr}(\delta\delta') = \pm 1$.

These results are assembled in the following table.

Table 3.7. Cohomology of affine quadrics of dimension n, $X^o = X - Y$. II ≈**81**

A. The cohomology is torsion-free. The nonzero Betti numbers are

$$\begin{aligned} \text{ordinary cohomology:} \quad & b_0 = b_n = 1, \\ \text{cohomology with compact supports:} \quad & b_{2n} = b_n = 1, \end{aligned}$$

except that $b_0 = 2$ when $n = 0$.

B. If $n = 2m > 0$, then

$$\begin{aligned} R^n f_! \mathbb{Z}_\ell(m) &= \text{the primitive part of } R^{2m} p_* \mathbb{Z}_\ell(m), \\ R^n f_* \mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} p_* \mathbb{Z}_\ell(m). \end{aligned}$$

If $n = 2m+1$, then

$$\begin{aligned} R^n f_! \mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} q_* \mathbb{Z}_\ell(m), \\ R^n f_* \mathbb{Z}_\ell(m+1) &= \text{the primitive part of } R^{2m} q_* \mathbb{Z}_\ell(m). \end{aligned}$$

Locally, these groups have natural generators defined up to sign; δ denotes those for cohomology with compact supports, and δ' those for ordinary cohomology.

C. The map

$$\varphi : R^n f_! \mathbb{Z}_\ell \longrightarrow R^n f_* \mathbb{Z}_\ell$$

is zero when n is odd, and sends $\pm\delta$ to $\pm 2\delta'$ when $n > 0$ is even. Moreover,

$$\mathrm{Tr}(\delta\delta') = \pm 1, \qquad \mathrm{Tr}(\delta^2) = \begin{cases} 0, & n = 2m+1, \\ (-1)^m \cdot 2, & n = 2m. \end{cases}$$

Verification 3.8. The complex affine quadric S in $\mathbb{C}^n$ with equation $\sum_i z_i^2 = 1$ is diffeomorphic to the tangent bundle of the real sphere $S \cap \mathbb{R}^n$. This agrees with A and C.

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SGA 7 — EXPOSÉ XIII

The Formalism of Vanishing Cycles

by P. Deligne

II ≈**83**

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Introduction

In this exposé we take up again the formalism introduced in no. 2 of Grothendieck's Exposé I. The only new point is the introduction of the algebraic analogue of variation in the classical theory.

The topological picture I have in mind is as follows. Let X be an analytic space, let $D \subset \mathbb{C}$ be a disc centred at 0, put $D^* = D - \{0\}$, and let $f : X \longrightarrow D$ be a proper morphism (the nonproper case will be considered later). Replace D by a sufficiently small disc centred at 0, replace X by the inverse image of this small disc, and continue to denote the result by $f : X \longrightarrow D$. According to the complex-analytic variant of Thom's isotopy theorem, $f^{-1}(D^*)$ is a topological fibre bundle over D^*. I do not know whether one can always construct a pair (Γ, r) of the kind described below, but it seems plausible that one can.

(a) Γ is a locally coherent system of local trivializations of the bundle $f^{-1}(D^*)$ over D^*. In other words, Γ defines a $\pi_1(D^*)$-equivariant trivialization of the inverse image of X on a universal covering $\widetilde{D}^*$ of D^*.

(b) r is a retraction of X onto the special fibre $X_0 = f^{-1}(0)$, compatible with Γ: if, on a connected open subset U of D^*, Γ identifies $X_U = f^{-1}(U)$ with $X_1 \times U$, then $r = r(x_1, u)$ is independent of u.

Let (X, f, Γ, r) be given. Let $t \in D$ and $X_t = f^{-1}(t)$ be the "general fibre." The bundle $\varphi^* X$, obtained from X/D by pullback along

$$\varphi : [0,1] \longrightarrow D^*, \qquad x \longmapsto \exp(2\pi i x)\, t,$$

is trivialized by Γ. The monodromy $T : X_t \longrightarrow X_t$ is the homeomorphism

$$X_t = (\varphi^* X)_0 \xrightarrow[\Gamma]{\sim} (\varphi^* X)_1 = X_t.$$

Let $r_t : X_t \longrightarrow X_0$ be the restriction of r to X_t. Then $r_t T = r_t$. One reconstructs (X, f) from (X_t, X_0, T, r_t) as follows:

(a) identify the ends $X_t \times \{0\}$ and $X_t \times \{1\}$ of $X_t \times [0,1]$ using T, obtaining $f' : X' \longrightarrow S^1$;

(b) r_t defines $r' : X' \longrightarrow X_0$, and (X, f) is identified with the map of mapping cones

$$\mathrm{cone}(f') : \mathrm{cone}(r' : X' \longrightarrow X_0) \longrightarrow D = \mathrm{cone}(S^1 \longrightarrow \mathrm{pt}).$$

Put $\Psi = (r, f) : X \longrightarrow X_0 \times D$. The higher Leray direct images $R^i\Psi_*\mathbb{Z}$ satisfy the following condition: II≈84

(*) The space D^* is covered by open subsets U such that the restriction of the sheaf F to $X_0 \times U$ is pulled back from a sheaf on X_0.

Construction 0.1. The sheaves F on $X_0 \times D$ satisfying (*) are identified with systems (F_0, F_t, T, α) in which

a) F_0 is a sheaf on X_0;

b) F_t is a sheaf on X_0, endowed with an action of the generator T of $\pi_1(D^*, t)$;

c) α is a morphism from F_0 to the sheaf of T-invariants of F_t.

Let G be a sheaf on $X_0 \times D^*$ satisfying (*), and let $\widetilde{D}^*$ be the universal covering of D^*, with base point t. The pullback of G to $X_0 \times \widetilde{D}^*$ is, in one and only one way, pulled back from a sheaf G_t on X_0. The sheaf G_t is endowed with an action of $\pi_1(D^*, t)$, and G is obtained from $\mathrm{pr}_1^* G_t$ on $X_0 \times \widetilde{D}^*$ by taking the quotient by $\pi_1(D^*, t)$.

A sheaf F on $X_0 \times D$ is determined by

1) its restriction F_0 to X_0;

2) its restriction G to $X_0 \times D^*$;

3) a morphism $\alpha : F_0 \longrightarrow i^* j_* G$, where $i : X_0 \hookrightarrow X_0 \times D$ and $j : X_0 \times D^* \hookrightarrow X_0 \times D$.

If F satisfies (*), then G is described by (G_t, T) on X_0, and the asserted description of F follows.

For the sheaves $R^i\Psi_*\mathbb{Z}$ we have

$$(R^i\Psi_*\mathbb{Z})_0 = \begin{cases} \mathbb{Z}, & i = 0, \\ 0, & i > 0, \end{cases}$$

whereas, for $i > 0$, the $(R^i\Psi_*\mathbb{Z})_t$ deserve the name sheaves of vanishing cycles (cf. the discussion at the beginning of paragraph 2 of Exposé I).

This discussion is useful only after it is refined by passing to the derived category: the $R^i\Psi_*\mathbb{Z}$ are obtained from an object of $D^+(X_0 \times D)$, and indeed from an object $R\Psi_*\mathbb{Z}$ of the derived category of the category of abelian sheaves on $X_0 \times D$ satisfying (*). This object determines not only the sheaves of vanishing cycles and their monodromy, but also the classical "variation" morphisms (cf. 1.4 and XIV, 3.1). II≈85

We shall define its analogue in algebraic geometry. If S is a strictly henselian trait and $f : X \longrightarrow S$ is a morphism of schemes, we shall define an object $R\Psi(\mathbb{Z}/\ell^n)$ of the derived category of the category of abelian sheaves on the product topos $(X_0)_{\mathrm{et}} \times S_{\mathrm{et}}$. A sheaf on this topos may be interpreted as a system (F_0, F_t, T, α) in which

α) F_0 is a sheaf on X_0;

β) F_t is a sheaf on X_0, endowed with a continuous action T of the inertia group I;

γ) α is a morphism from F_0 to the sheaf of I-invariants of F_t.

We shall have to work in a slightly more general framework than that described above:

a) The construction of $R\Psi(\mathbb{Z}/\ell^n)$ does not require f to be proper. For a topological interpretation, one would repeat the preceding construction after replacing X by a neighbourhood of the special fibre X_0.

b) To construct $R\Psi(\mathbb{Z}/\ell^n)$ and prove its properties, one must consider not only the constant sheaf $\mathbb{Z}/\ell^n$, but also general sheaves and even complexes of sheaves.

c) When the residue field k of the henselian trait S is not assumed separably closed, the product topos $(X_0)_{\text{et}} \times S_{\text{et}}$ must be replaced by the fibred product topos

$$(X_0)_{\text{et}} \times_{\text{Spec}(k)_{\text{et}}} S_{\text{et}}.$$

The reader who dislikes 2-fibred products of topoi may take the Galois description of this topos in 1.2.4 as its definition.

0.2. Notation and terminology. II **86**

(0.2.1) trait: the spectrum of a discrete valuation ring.

(0.2.2) henselian: for a scheme, the spectrum of a henselian local ring.

(0.2.3) strictly henselian: for a ring, henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.2.4) geometric point of S: a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) =_{\text{def}} \bar{k}$ is a separable closure of $\kappa(x)$. We say that $\bar{x}$ is localized at x. By abuse of language, we shall sometimes also call a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ a geometric point when $\bar{k}$ is a separably closed extension of $\kappa(x)$; such a morphism factors uniquely through a geometric point of S in the proper sense.

(0.2.5) $s, \eta, \bar{s}, \bar{\eta}, \bar{S}$: If $S = \text{Spec}(V)$ is a henselian trait, we shall normally denote by s the closed point of S, by η its generic point, by $\bar{\eta}$ a generic geometric point of S (localized at η), and by $\bar{s}$ the corresponding geometric point localized at s (I, 0.0.3). Let $\bar{S}$ denote the spectrum of the normalization of V in $k(\bar{\eta})$; its residue field is an inseparable extension of $k(\bar{s})$. Write $\text{Gal}(\bar{\eta}/\eta)$ for the Galois group of $k(\bar{\eta})$ over $k(\eta)$, and similarly for s. There is an exact sequence

$$0 \longrightarrow I \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Gal}(\bar{s}/s) \longrightarrow 0,$$

where I is the inertia group.

(0.2.6) sheaf: on a scheme X, always means a sheaf on the étale site X_{et} of X.

(0.2.7) prime to: a torsion sheaf F on a scheme S (or a torsion abelian group F) is prime to the residue characteristics of S if, for every $s \in S$, multiplication by the characteristic exponent of $k(s)$ is an automorphism of F.

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1. Sheaves of sets

The sheaves of vanishing cycles will be defined as values of derived functors of the left exact functor Ψ defined in no. 3 for every sheaf of sets.

1.1. Galois sheaves.

1.1.1. Let Y be a scheme over a field k with separable closure $\bar{k}$, and put $\bar{Y} = Y \otimes_k \bar{k}$. Let G be a profinite group and let

$$u : G \longrightarrow \text{Gal}(\bar{k}/k)$$

be a continuous homomorphism from G to the Galois group of $\bar{k}$ over k. The Galois group $\text{Gal}(\bar{k}/k)$ acts on the left on $\bar{Y}$ by transport of structure. The group G acts on $\bar{Y}$ through u.

Let $\mathfrak{F}$ be a sheaf of sets on (the étale site of) $\bar{Y}$. An action of G on $\mathfrak{F}$ compatible with the action of G on $\bar{Y}$ is an action of G by automorphisms of the pair $(\bar{Y}, \mathfrak{F})$ which induces the given action on $\bar{Y}$. In other words, it is a system of isomorphisms

$$\sigma(g) : u(g)_*\mathfrak{F} \longrightarrow \mathfrak{F}$$

satisfying $\sigma(gh) = \sigma(g)\,\sigma(h)$. If G acts on $\mathfrak{F}$, then for every U étale over Y, with $\bar{U} = U \otimes_k \bar{k}$ étale over $\bar{Y}$, the group G acts on $\mathfrak{F}(\bar{U})$.

Definition 1.1.2. We say that G acts continuously on $\mathfrak{F}$ if, for every quasi-compact U étale over Y, the action of G on the discrete set $\mathfrak{F}(\bar{U})$ is continuous.

It would amount to the same thing to consider only affine U.

If $\mathfrak{F}$ is a sheaf of sets on Y, with inverse image $\bar{\mathfrak{F}}$ on $\bar{Y}$, then $\mathrm{Gal}(\bar{k}/k)$ acts on $\bar{\mathfrak{F}}$ by transport of structure, compatibly with its action on $\bar{Y}$.

Reminder 1.1.3.

(i) The action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{\mathfrak{F}}$ is continuous.

(ii) The functor

$$\mathfrak{F} \longmapsto \bar{\mathfrak{F}}$$

where $\bar{\mathfrak{F}}$ is endowed with the action of $\mathrm{Gal}(\bar{k}/k)$, is an equivalence from the category of sheaves of sets on Y to the category of sheaves of sets on $\bar{Y}$ endowed with a continuous action of $\mathrm{Gal}(\bar{k}/k)$ compatible with the action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{Y}$.

Write $\bar{k}$ as the direct limit of finite Galois extensions k_i. Let II ≈ **88**

$$\varphi_i : Y_i = Y \otimes_k k_i \longrightarrow Y, \qquad \bar{\varphi} : \bar{Y} \longrightarrow Y$$

be the projections. For U étale over Y, assumed affine, put $U_i = U \otimes_k k_i$. Then

$$\bar{\mathfrak{F}}(\bar{U}) = \varinjlim_i (\varphi_i^*\mathfrak{F})(U_i).$$

On $(\varphi_i^*\mathfrak{F})(U_i)$, the group $\mathrm{Gal}(\bar{k}/k)$ acts through $\mathrm{Gal}(k_i/k)$, which proves (i).

The functor in (ii) has as right adjoint the functor

$$\mathfrak{G} \longmapsto (\bar{\varphi}_*\mathfrak{G})^{\mathrm{Gal}(\bar{k}/k)}.$$

For every $\mathfrak{F}$ on Y, one has

$$\bar{\varphi}_*\bar{\varphi}^*\mathfrak{F} = \varinjlim_i \varphi_{i*}\varphi_i^*\mathfrak{F}.$$

Thus, to prove that

$$\mathfrak{F} \xrightarrow{\sim} (\bar{\varphi}_*\bar{\varphi}^*\mathfrak{F})^{\mathrm{Gal}(\bar{k}/k)},$$

it is enough to prove, for every i, that

$$\mathfrak{F} \xrightarrow{\sim} (\varphi_{i*}\varphi_i^*\mathfrak{F})^{\mathrm{Gal}(k_i/k)}.$$

It is even enough to prove that this arrow becomes an isomorphism after extension of scalars from k to k_i; then $\mathrm{Spec}(k_i)$ becomes a sum of copies of $\mathrm{Spec}(k)$ permuted simply transitively by $\mathrm{Gal}(k_i/k)$, and the assertion becomes trivial.

The exact functor in (ii) is therefore fully faithful. To conclude, it remains to verify that the arrow

$$\bar{\varphi}^*(\bar{\varphi}_*\mathfrak{G})^{\mathrm{Gal}(\bar{k}/k)} \longrightarrow \mathfrak{G}$$

is an epimorphism. Let $\bar{\varphi}_i : \bar{Y} \longrightarrow Y_i$. The hypothesis implies that

$$\varinjlim_i \bar{\varphi}_i^*(\bar{\varphi}_{i*}\mathfrak{G})^{\mathrm{Gal}(\bar{k}/k_i)} \longrightarrow \mathfrak{G}$$

is an epimorphism. II≈89

We have

$$\bar{\varphi}^*(\bar{\varphi}_*\mathfrak{G})^{\mathrm{Gal}(\bar{k}/k)} = \bar{\varphi}_i^*\left(\varphi_i^*\varphi_{i*}\left((\bar{\varphi}_{i*}\mathfrak{G})^{\mathrm{Gal}(\bar{k}/k_i)}\right)^{\mathrm{Gal}(k_i/k)}\right),$$

and the conclusion follows by observing, by the same Galois-descent argument as above, that for $\mathfrak{G}'$ on Y_i, endowed with an action of $\mathrm{Gal}(k_i/k)$ compatible with its action on Y_i, one has

$$\varphi_i^*(\varphi_{i*}\mathfrak{G}')^{\mathrm{Gal}(k_i/k)} \xrightarrow{\sim} \mathfrak{G}'.$$

1.2. The topos $Y \times_s S$

1.2.1.

Let S be a henselian local scheme with closed point $i : s \hookrightarrow S$. The functor

$$S' \longmapsto S' \times_S s = i^*S'$$

is an equivalence from the category of finite étale S-schemes S' to the category of finite étale s-schemes s' (spectra of étale $k(s)$-algebras). The inverse functor sp^*, from the étale site of s to the étale site of S, is a morphism of sites

$$\mathrm{sp} : S \longrightarrow s,$$

the specialization morphism (cf. SGA 4, VIII, 7). If s' is étale over s, one says that $\mathrm{sp}^*(s')$ is obtained from S by extension of the residue field. For a sheaf F on S, there is a functorial isomorphism

$$\mathrm{sp}_*F = i^*F. \tag{1.2.1.1}$$

For every open subscheme of S, $j : U \hookrightarrow S$, or for every point of S, $j : \eta \hookrightarrow S$, we shall again write sp for the composite morphism $\mathrm{sp} \circ j$. For a sheaf F on U (respectively on η), one has

$$\mathrm{sp}_*F = i^*j_*F. \tag{1.2.1.2}$$

1.2.2.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}, I$ be as in 0.2.5.

$$s \xhookrightarrow{i} S \xhookleftarrow{j} \eta \quad .$$

Here is the Galois-theoretic interpretation of sheaves on s, η, and S, and of the specialization morphism.

(a) Sheaves F on s (respectively on η) are identified, by $F \mapsto F_{\bar{s}}$ (respectively $F \mapsto F_{\bar{\eta}}$), with sets endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$) (SGA 4, VIII, 2.1, or 1.1.3).

(b) A sheaf F on S defines sheaves $F_s = i^*F$ and $F_\eta = j^*F$ on s and η, and an adjunction arrow $F \longrightarrow j_*j^*F$ inducing

$$\varphi : F_s = i^*F \longrightarrow i^*j_*j^*F = \mathrm{sp}_*F_\eta.$$

It is known (SGA 4, IV, 9.5.4) that the functor

$$F \longmapsto (F_s, F_\eta, \varphi)$$

is an equivalence from the category of sheaves on S to the category of triples (F_s, F_η, φ), where F_s is a sheaf on s, F_η is a sheaf on η, and $\varphi : F_s \longrightarrow i^*j_*F_\eta$.

Under the equivalence in (a), the functor i^*j_* is identified with the functor of "invariants under I". Thus sheaves on S are identified with triples consisting of a set $F_{\bar{s}}$ endowed with an action of $\mathrm{Gal}(\bar{s}/s)$, a set $F_{\bar{\eta}}$ endowed with an action of $\mathrm{Gal}(\bar{\eta}/\eta)$, and an equivariant morphism $\varphi : F_{\bar{s}} \longrightarrow F_{\bar{\eta}}$.

(c) Under (a) and (b), the morphisms sp, j, i, and $\mathrm{sp} = \mathrm{sp} \circ j$ are expressed as follows:

$$\begin{array}{lll}
\mathrm{sp} : S \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = (F_{\bar{s}}, F_{\bar{s}}, \mathrm{Id}), \\
& \mathrm{sp}_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
j : \eta \longrightarrow S : & j^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{\eta}}, \\
& j_*(F_{\bar{\eta}}) & = (F_{\bar{\eta}}^I, F_{\bar{\eta}}, \mathrm{inclusion}), \\
i : s \longrightarrow S : & i^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
& i_*(F_{\bar{s}}) & = (F_{\bar{s}}, \{0\}, \mathrm{projection}), \\
\mathrm{sp} = \mathrm{sp} \circ j : \eta \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = F_{\bar{s}}, \\
& \mathrm{sp}_*(F_{\bar{\eta}}) & = F_{\bar{\eta}}^I.
\end{array}$$

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1.2.3.

Let S be a henselian trait as in 1.2.2, and let Y be a scheme over s. We shall have to work with the 2-fiber product $Y \times_s S$ of the étale topoi of Y and S over the étale topos of s. By Giraud [1], this 2-fiber product exists, and it is readily seen to admit the Galois-theoretic description given below. Those who dislike 2-fiber products may take this description as the definition.

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Construction 1.2.4.

Sheaves F on $Y \times_s S$ are identified with triples (F_s, F_η, φ) in which:

(a) F_s is a sheaf on Y, or equivalently a sheaf $F_{\bar{s}}$ on $\bar{Y} = Y \otimes_s \bar{s}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ compatible with the action of $\mathrm{Gal}(\bar{s}/s)$ on $\bar{Y}$;

(b) F_η is a sheaf $F_{\bar{\eta}}$ on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$ compatible with the action (via 1.1.1) of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $\bar{Y}$;

(c) φ is an equivariant morphism from $F_{\bar{s}}$ to $F_{\bar{\eta}}$.

Similarly, a sheaf F_η on $Y \times_s \eta$ is an object as in 1.2.4(b). There is a diagram of topoi

(1.2.4.1)
$$\begin{array}{ccc} Y & & \\ \downarrow & \xleftarrow{\mathrm{sp}} & \\ s & & \end{array} \qquad\qquad \begin{array}{ccccc} Y & \xhookrightarrow{i} & Y \times_s S & \xhookleftarrow{j} & Y \times_s \eta \\ \downarrow & & \downarrow & & \downarrow \\ s & \xhookrightarrow{i} & S & \xhookleftarrow{j} & \eta \end{array} .$$

The topos $Y \times_s S$ is the union of the open subtopos $Y \times_s \eta$ and its closed complement Y. The formulas in 1.2.2(c) remain valid for $\mathrm{sp} : Y \times_s S \longrightarrow Y$ (also denoted pr_1), $j : Y \times_s \eta \hookrightarrow Y \times_s S$, $i : Y \hookrightarrow Y \times_s S$, and $\mathrm{sp} : Y \times_s \eta \longrightarrow Y$ (also denoted pr_1).

1.2.4.2. If 1.2.4 is taken as the definition of $Y \times_s S$, one must show by a direct construction that, up to unique isomorphism, $Y \times_s S$ does not depend on the choice of a separable closure $k(\bar{\eta})$ of $k(\eta)$. This is done by interpreting the F_η of 1.2.4(b) as a functor which, to each separable closure $k(\bar{\eta})$ of $k(\eta)$ defining a separable closure $k(\bar{s})$ of $k(s)$, associates a sheaf on $Y \otimes_{k(s)} k(\bar{s})$; the F_η of 1.2.4(b) is the restriction of this functor to $\{\bar{\eta}\}$.

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1.2.5.

Every geometric point $\bar{x}$ of $Y_{\bar{s}}$ defines a geometric point $(\bar{x}, \bar{\eta})$ of $Y \times_s \eta$, whose corresponding fiber functor is

$$F_\eta \longmapsto (F_{\bar{\eta}})_{\bar{x}}.$$

It defines geometric points $(\bar{x}, \bar{\eta})$ and $(\bar{x}, \bar{s})$ of $Y \times_s S$ by

$$F_{(\bar{x},\bar{\eta})} = (F_{\bar{\eta}})_{\bar{x}} \qquad \text{and} \qquad F_{(\bar{x},\bar{s})} = (F_{\bar{s}})_{\bar{x}}.$$

One verifies that all the points of the topoi $Y \times_s \eta$ and $Y \times_s S$ are obtained in this way. We shall not use this. One also verifies at once that these sets of points are conservative.

***Remark 1.2.6.**

The construction in 1.2.5 defines an equivalence from the category $\mathrm{Point}(Y \times_s \eta)$ (respectively $\mathrm{Point}(Y \times_s S)$) of points of $Y \times_s \eta$ (respectively of $Y \times_s S$) to the 2-fiber product

$$\mathrm{Point}(Y) \times_{\mathrm{Point}(s)} \mathrm{Point}(\eta).$$

This equivalence also follows from the fact that $Y \times_s \eta$ (respectively $Y \times_s S$) is a 2-fiber product of topoi.*

1.2.7.

By the functoriality of 2-fiber products, diagram (1.2.4.1) is functorial in Y and S. In particular:

a) Every morphism $f : Y \longrightarrow Y'$ of schemes over s induces a morphism of topoi from $Y \times_s S$ to $Y' \times_s S$. One has

$$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)), \tag{1.2.7.1}$$

and, when f is quasi-compact,

$$f_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_*F_{\bar{s}}, f_*F_{\bar{\eta}}, f_*(\varphi)). \tag{1.2.7.2}$$

Analogous formulas hold for $f : Y \times_s \eta \longrightarrow Y' \times_s \eta$.

b) Every surjective morphism of henselian traits $f : S' \longrightarrow S$ induces a morphism of topoi from $Y_{s'} \times_{s'} S'$ to $Y \times_s S$. Again writing f for the natural morphism $Y_{s'} = Y \otimes_{k(s)} k(s') \longrightarrow Y$, one has

$$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)). \tag{1.2.7.3}$$

When f is finite, f_* is expressed as an induced representation.

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1.2.8.

Let $f : Y \longrightarrow Y'$ be a separated morphism, locally of finite type, between s-schemes. Define a "direct image with proper support" functor

$$f_! : (\text{sheaves of pointed sets on } Y \times_s S) \longrightarrow (\text{sheaves of pointed sets on } Y' \times_s S)$$

by the formula

$$f_!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_!F_{\bar{s}}, f_!F_{\bar{\eta}}, f_!(\varphi)). \tag{1.2.8.1}$$

When f is a locally closed immersion, $f_!$ is called the extension-by-zero functor.

An analogous functor is defined for $Y \times_s \eta$.

1.2.9.

Let $f : Y \hookrightarrow Y'$ be a locally closed immersion of s-schemes. The functor $f_!$ has a right adjoint $f^!$, "sections with support in Y", satisfying

$$f^!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^! F_{\bar{s}}, f^! F_{\bar{\eta}}, f^!(\varphi)). \tag{1.2.9.1}$$

Suppose only that f is quasi-finite. The functor $f_!$ from abelian sheaves on $Y \times_s S$ to abelian sheaves on $Y' \times_s S$ still has a right adjoint $f^!$, given by (1.2.9.1) (cf. SGA 4, XVIII, 3.1.8). If f is étale, then $f^! = f^*$.

Analogous functors are defined for $Y \times_s \eta$.

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1.3. The functor Ψ

1.3.1.

Let S be a henselian trait as in 0.2.5, and let $p : X \longrightarrow S$. We propose to define a left exact functor Ψ from sheaves on X to sheaves on the topos $X_s \times_s S$ constructed in 1.2. In general, this functor is not the direct-image functor of a morphism of topoi.

We shall also define

$$\Psi_\eta : (\text{sheaves on } X_\eta) \longrightarrow (\text{sheaves on } X_s \times_s \eta).$$

1.3.2.

Put $\bar{X} = X \times_S \bar{S}$. There is a Cartesian diagram

$$\begin{array}{ccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow & & \downarrow & & \downarrow \\ X_s & \longrightarrow & X & \longleftarrow & X_\eta \end{array} \tag{1.3.2.1}$$

over the diagram

$$\begin{array}{ccccc} \bar{s} & \longrightarrow & \bar{S} & \longleftarrow & \bar{\eta} \\ \downarrow & & \downarrow & & \downarrow \\ s & \longrightarrow & S & \longleftarrow & \eta \end{array} .$$

Let F be a sheaf on X_η, with inverse image $F_{\bar{\eta}}$ on $X_{\bar{\eta}}$. Put

$$\Psi_\eta(F) = \bar{i}^* \bar{j}_* F_{\bar{\eta}}. \tag{1.3.2.2}$$

By transport of structure, this sheaf is endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$, compatible with the action of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $X_{\bar{s}}$. The functor

$$\Psi_\eta : (X_{\eta,\mathrm{et}})^\sim \longrightarrow (X_s \times_s \eta)^\sim \tag{1.3.2.3}$$

is left exact.

1.3.3.

Let F be a sheaf on X, and let F_η and F_s be its restrictions to X_η and X_s. Put

$$(\Psi(F))_\eta = \Psi_\eta(F_\eta) \tag{1.3.3.1}$$

and

$$(\Psi(F))_s = F_s. \tag{1.3.3.2}$$

Let $\bar{F}$ be the inverse image of F on $\bar{X}$, and let φ' be the adjunction morphism

$$\varphi' : \bar{F} \longrightarrow \bar{j}_*\bar{j}^*\bar{F}.$$

This morphism induces

$$\varphi : F_{\bar{s}} = \bar{i}^*\bar{F} \longrightarrow \bar{i}^*\bar{j}_*\bar{j}^*\bar{F} = (\Psi_\eta(F_\eta))_{\bar{\eta}}.$$

The triple

$$\Psi(F) = ((\Psi(F))_s, (\Psi(F))_\eta, \varphi)$$

is a sheaf on $X_s \times_s S$. The functor

$$\Psi : (X_{\text{et}})^\sim \longrightarrow (X_s \times_s S)^\sim \tag{1.3.3.3}$$

is left exact.

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1.3.4.

When there is no risk of confusion, we shall write

$$\Psi_\eta(F) = (\Psi(F))_\eta, \qquad \Psi_{\bar{\eta}}(F) = (\Psi(F))_{\bar{\eta}}, \qquad \text{and} \qquad \Psi_{\bar{\eta}}(F) = (\Psi_\eta(F))_{\bar{\eta}}.$$

1.3.5.

For the remainder of this section, we list functoriality properties of Ψ (cf. 1.2.7–1.2.9). They will be useful chiefly in their derived form (2.1).

Let $f : X \longrightarrow X'$ be a morphism of S-schemes, giving a diagram

$$\begin{array}{ccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} \\ X'_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X}' & \xleftarrow{\bar{j}} & X'_{\bar{\eta}} \end{array} \tag{1.3.5.1}$$

The ambiguity in the notation for the arrows simplifies the formulas and causes no confusion: only one interpretation makes sense of the composites below. Likewise, we write Ψ for the functor (1.3.3.3) for both X and X'.

The morphisms of functors described below for Ψ have variants for Ψ_η, which we generally omit to state explicitly.

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1.3.6.

The base-change morphism $\bar{i}^* f_* \longrightarrow f_* \bar{i}^*$ induces a morphism of functors

$$\Psi f_* \longrightarrow f_* \Psi, \tag{1.3.6.1}$$

which is an isomorphism when f is proper (SGA 4, XII, 5.1(i)).

The essential case is $X' = S$. One obtains a morphism

$$f_* \longrightarrow f_* \Psi \tag{1.3.6.2}$$

whose essential component is

$$\Gamma(X_{\bar{\eta}}, F) \longrightarrow \Gamma(X_{\bar{s}}, \Psi_{\bar{\eta}}(F)); \tag{1.3.6.3}$$

this morphism is an isomorphism when f is proper.

1.3.7.

The base-change morphism $f^*\bar{j}_* \longrightarrow \bar{j}_* f^*$ induces a morphism of functors

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.7.1}$$

1.3.8.

Suppose that f is quasi-finite. The functors $f_!$ commute with base change; the morphism $f_!\bar{j}_* \longrightarrow \bar{j}_* f_!$ induces a morphism of functors

$$f_!\Psi \longrightarrow \Psi f_!. \tag{1.3.8.1}$$

When f is finite, this is the inverse of (1.3.6.1).

1.3.9.

The situation is dual for $f^!$ (we suppose that f is quasi-finite when working with abelian sheaves, or that it is an immersion when working with sheaves of pointed sets). This functor commutes with direct images, and the morphism $\bar{i}^* f^! \longrightarrow f^! \bar{i}^*$ induces a morphism of functors

$$\Psi f^! \longrightarrow f^! \Psi. \tag{1.3.9.1}$$

When f is étale, this is the inverse of (1.3.7.1).

1.3.10.

Finally, a base change $S' \xrightarrow{f} S$ gives a base-change morphism

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.10.1}$$

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1.4. The variation

1.4.1.

Let S be a henselian trait as in 0.2.5, and let Y be a scheme over s. Let Λ be a ring. More generally, one could take for Λ a sheaf of rings on Y. For every object K of the derived category $D(Y \times_s S, \Lambda)$ of the category of sheaves of Λ-modules on $Y \times_s S$, we propose to define a variation morphism.

1.4.2.

A complex of sheaves of Λ-modules K on $Y \times_s S$ is identified with a triple (K_s, K_η, φ) in which

a) K_s (respectively K_η) is a complex of sheaves $K_{\bar{s}}$ (respectively $K_{\bar{\eta}}$) on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$);

b) φ is an equivariant morphism $\varphi : K_{\bar{s}} \longrightarrow K_{\bar{\eta}}$.

Such a complex K is always homotopic to a complex $K' = (K'_s, K'_\eta, \varphi')$ such that φ' is injective and the exact sequence

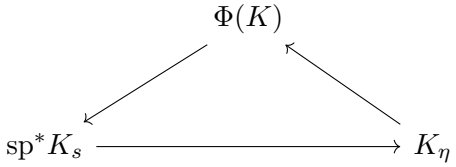
$$0 \longrightarrow K'_{\bar{s}} \longrightarrow K'_{\bar{\eta}} \longrightarrow \mathrm{coker}(\varphi') \longrightarrow 0 \tag{1.4.2.1}$$

is split degree by degree. It is enough to take $K'_s = K_s$ and to take for K'_η the direct sum of K_η and the cone of $\mathrm{sp}^* K_s$. Put

$$\Phi(K) = \mathrm{coker}(\varphi').$$

The exact sequence (1.4.2.1) defines a distinguished triangle

(1.4.2.2)



in the category $K(Y \times_s \eta, \Lambda)$ of complexes of sheaves of Λ-modules on $Y \times_s \eta$, up to homotopy. This triangle depends only on $K \in \mathrm{Ob}\, K(Y \times_s S, \Lambda)$.

The construction passes to the derived category and associates with every object K of $D(Y \times_s S, \Lambda)$ II≈98 a distinguished triangle of $D(Y \times_s \eta, \Lambda)$.

Triangle (1.4.2.2) is useful chiefly through the long exact hypercohomology sequence that it induces:

$$\cdots \longrightarrow H^i(\bar{Y}, K_{\bar{s}}) \longrightarrow H^i(\bar{Y}, K_{\bar{\eta}}) \longrightarrow H^i(\bar{Y}, \Phi(K)_{\bar{\eta}}) \longrightarrow \cdots .$$

1.4.3.

The inertia group I acts trivially on $K'_{\bar{s}}$. Hence, for every $\sigma \in I$, the endomorphism $\sigma - 1$ of $K'_{\bar{\eta}}$ (notation (1.4.2.1)) factors through a morphism

$$\mathrm{Var}(\sigma) : \mathrm{coker}(\varphi') \longrightarrow K'_{\bar{\eta}}.$$

This construction passes to the derived category: for $K \in \mathrm{Ob}\, D(Y \times_s S, \Lambda)$ and $\sigma \in I$, it defines the variation

(1.4.3.1) $$\mathrm{Var}(\sigma) : \Phi(K)_{\bar{\eta}} \longrightarrow K_{\bar{\eta}}.$$

Writing q for the arrow from K_η to $\Phi(K)$, one has

(1.4.3.2) $$\sigma = 1 + \mathrm{Var}(\sigma) q \quad \text{on } K_{\bar{\eta}},$$

(1.4.3.3) $$\sigma = 1 + q\,\mathrm{Var}(\sigma) \quad \text{on } \Phi(K)_{\bar{\eta}},$$

and

(1.4.3.4) $$\mathrm{Var}(\sigma\tau) = \mathrm{Var}(\sigma) q\, \mathrm{Var}(\tau) + \mathrm{Var}(\sigma) + \mathrm{Var}(\tau).$$

2. Vanishing cycles

2.1. The functor $R\Psi$

2.1.1.

Let S be a henselian trait as in 0.2.5, and let Λ be a torsion ring whose torsion is prime to the residual characteristic p of S (0.2.7). More generally, one could take for Λ a torsion sheaf whose torsion is prime to the residual characteristics; the most important case is $\Lambda = \mathbb{Z}/\ell^n$ with ℓ an invertible prime in $\mathcal{O}_S$.

Let X be a scheme over S. The functor Ψ (respectively Ψ_η) carries sheaves of Λ-modules on X II≈99 (respectively on X_η) to sheaves of Λ-modules on $X_s \times_s S$ (respectively on $X_s \times_s \eta$). Let $R\Psi$ and $R\Psi_\eta$ be the derived functors

$$R\Psi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s S, \Lambda),$$

$$R\Psi_\eta : D^+(X_\eta, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda),$$

and let $R\Phi$ be the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda).$$

2.1.2.

Consider the commutative diagram (1.3.2.1)

$$\begin{array}{ccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\ \downarrow & & \downarrow & & \downarrow \\ X_s & \xrightarrow{i} & X & \xleftarrow{j} & X_\eta \end{array}$$

If F is an injective sheaf of modules on X, its restriction to the open subscheme X_η is still injective. If F_η is injective on X_η, its inverse image $F_{\bar{\eta}}$ is acyclic. Thus the derived forms of (1.3.3.1) and (1.3.3.2) are

$$R\Psi(K)_s = i^*K \qquad (K \in \mathrm{Ob}\, D^+(X,\Lambda)), \tag{2.1.2.1}$$

$$R\Psi(K)_\eta = R\Psi_\eta(K_\eta) \qquad (K \in \mathrm{Ob}\, D^+(X,\Lambda)), \tag{2.1.2.2}$$

and

$$R\Psi_\eta(K)_{\bar{\eta}} = \bar{i}^* R\bar{j}_* K_{\bar{\eta}} \qquad (K \in \mathrm{Ob}\, D^+(X_\eta,\Lambda)). \tag{2.1.2.3}$$

This allows us to write, without conflict,

$$R\Psi_\eta(K) = (R\Psi(K))_\eta, \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi(K))_{\bar{\eta}}, \qquad \text{and} \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi_\eta(K))_{\bar{\eta}}.$$

With this notation, the distinguished triangle (1.4.2.2) becomes

(2.1.2.4)
$$\begin{array}{ccc} & R\Phi(K) & \\ \swarrow & & \nwarrow \\ \mathrm{sp}^* i^* K & \longrightarrow & R\Psi_\eta(K_\eta) \end{array} .$$

For $\sigma \in I$, there is a variation morphism II≈100

$$\mathrm{Var}(\sigma) : R\Phi(K)_{\bar{\eta}} \longrightarrow R\Psi_{\bar{\eta}}(K). \tag{2.1.2.5}$$

The sheaves $R^i\Phi(\Lambda)$, or more generally $R^i\Phi(K)$, are the sheaves of vanishing cycles.

2.1.3.

Let $(\bar{x},\bar{\eta})$ be the geometric point of $X_s \times_s \eta$ defined by a geometric point $\bar{x}$ of $X_{\bar{s}}$ (1.2.5). We compute the fiber of $R\Psi_\eta(K_\eta)$ at $(\bar{x},\bar{\eta})$ using (2.1.2.3) (cf. I, 2.3). The strict henselization $X_{(\bar{x})}$ of X at $\bar{x}$ is a scheme over the strict henselization S_{nr} of S, with generic point η_{nr}.

Proposition 2.1.4.

One has

$$(R\Psi(K_\eta))_{(\bar{x},\bar{\eta})} = R\Gamma\Big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\Big);$$

in particular,

$$R^i\Psi(K_\eta)_{(\bar{x},\bar{\eta})} = \mathbb{H}^i\Big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\Big).$$

Apply SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms). This gives the following restatement.

Restatement 2.1.5.

If f is smooth and F is a locally constant sheaf, then

$$R\Phi(F) = 0.$$

More generally, if $K \in \mathrm{Ob}\, D^+(X, \Lambda)$, then $R\Phi(K) = 0$ wherever f is smooth and the sheaves $\mathcal{H}^i(K)$ are locally constant. See also 2.1.11.

2.1.6.

Let $f : Y \longrightarrow Y'$ be a morphism of schemes over s. The functors f^*, f_*, $f^!$, and $f_!$ (1.2.7–1.2.9) have natural analogues in the derived category.

a) The functor f^* is exact and therefore derives trivially.

b) The functor f_* has a right derived functor Rf_*. When f is quasi-compact, one has

(2.1.6.1) $$(Rf_*K)_{\bar{s}} = Rf_{\bar{s}*}(K_{\bar{s}})$$

and

(2.1.6.2) $$(Rf_*K)_{\bar{\eta}} = Rf_{\bar{s}*}(K_{\bar{\eta}}).$$

c) When f is quasi-finite, the functor $f_!$ is exact and therefore derives trivially. In general, the derived functor of $f_!$ is pathological, and one proceeds as in SGA 4, XVII. Suppose that f is separated and of finite type and that Y' is noetherian. There is then [2] a factorization $f = \widetilde{f}\, j$, where j is an open immersion and $\widetilde{f}$ is proper, and one defines

$$Rf_! = R\widetilde{f}_* \circ j_!.$$

The arguments of SGA 4, XVII show that $Rf_!$ is independent of the choice of the factorization $\widetilde{f}\, j$. The analogues of (2.1.6.1) and (2.1.6.2) hold.

d) We shall use $Rf^!$ only for a quasi-finite morphism, in which case it is the derived functor of $f^!$. Again, the analogues of (2.1.6.1) and (2.1.6.2) hold.

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2.1.7.

We now review the functoriality properties of $R\Psi$, that is, we derive 1.3.5–1.3.10. Let $f : X \longrightarrow X'$ be a morphism of S-schemes, and retain the notation of 1.3.5.

The morphism (1.3.6.1) defines

(2.1.7.1) $$R\Psi\, Rf_* \longrightarrow Rf_*\, R\Psi.$$

By the proper base-change theorem (SGA 4, XII, 5.1), this morphism is an isomorphism if f is proper.

The morphism (1.3.7.1) defines

(2.1.7.2) $$f^* R\Psi \longrightarrow R\Psi f^*.$$

By the smooth base-change theorem (SGA 4, XVI, 1.2), this morphism is an isomorphism if f is smooth.

The isomorphism (2.1.7.1) for f proper and the morphism (1.3.8.1) define a morphism

(2.1.7.3) $$Rf_! R\Psi \longrightarrow R\Psi Rf_!,$$

which, when f is proper, is the inverse of (2.1.7.1).

When f is quasi-finite, (1.3.9.1) derives to

$$R\Psi Rf^! \longrightarrow Rf^! R\Psi, \tag{2.1.7.4}$$

which, when f is étale, is the inverse of (2.1.7.2).

Let $f : S' \longrightarrow S$ be a change of trait, and put $X' = X \times_S S'$: II≈**102**

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ S' & \xrightarrow{f} & S \end{array}$$

(for the ambiguity in the notation, see 1.3.5). Suppose that f is surjective. The base-change arrow (1.3.10.1) derives to

$$f^* R\Psi \longrightarrow R\Psi f^*. \tag{2.1.7.5}$$

The same statements hold for $R\Psi_\eta$.

2.1.8.

The essential case of (2.1.7.1) and (2.1.7.3) is $X' = S$. For $K \in \mathrm{Ob}\, D^+(X_{\bar\eta}, \Lambda)$ one then obtains morphisms

$$Rf_* \longrightarrow Rf_* R\Psi \tag{2.1.8.1}$$

and

$$Rf_! R\Psi \longrightarrow Rf_!, \tag{2.1.8.2}$$

hence

$$\mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \tag{2.1.8.3}$$

and

$$\mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \longrightarrow \mathbb{H}^i_c(X_{\bar\eta}, K), \tag{2.1.8.4}$$

and the commutativity of the diagrams

$$\begin{array}{ccc} (R^i f_* K)_{\bar s} & \xrightarrow{\mathrm{sp}} & \mathbb{H}^i(X_{\bar\eta}, K) \\ \downarrow & & \downarrow \\ \mathbb{H}^i(X_{\bar s}, K) & \longrightarrow & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.5}$$

and

$$\begin{array}{ccc} \mathbb{H}^i_c(X_{\bar s}, K) & \xrightarrow{\qquad\mathrm{sp}\qquad} & \mathbb{H}^i_c(X_{\bar\eta}, K) \\ & \searrow \quad \nearrow & \\ & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \end{array} \tag{2.1.8.6}$$

.

For $K \in \mathrm{Ob}\, D^+(X_\eta, \Lambda)$ and $\sigma \in I$, the following diagrams are commutative: II≈**103**

$$\begin{array}{ccc} \mathbb{H}^i(X_{\bar\eta}, K) & \xrightarrow{\qquad\sigma-1\qquad} & \mathbb{H}^i(X_{\bar\eta}, K) \\ \downarrow & & \downarrow \\ \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \xrightarrow{\mathrm{Var}(\sigma)} & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.7}$$

and

$$\begin{array}{ccc} \mathbb{H}^i_c(X_{\bar\eta}, K) & \xrightarrow{\quad\sigma-1\quad} & \mathbb{H}^i_c(X_{\bar\eta}, K) \\ \uparrow & & \uparrow \\ \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) & \longrightarrow \mathbb{H}^i_c(X_{\bar s}, R\Phi_{\bar\eta}(K)) \xrightarrow{\mathrm{Var}(\sigma)} & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{array} \tag{2.1.8.8}$$

When f is proper, (2.1.8.1) and (2.1.8.2), as well as (2.1.8.3) and (2.1.8.4), are mutually inverse isomorphisms. The distinguished triangle (2.1.2.4) gives a long exact sequence

$$\cdots \longrightarrow \mathbb{H}^i(X_{\bar s}, K) \longrightarrow \mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \longrightarrow \cdots . \tag{2.1.8.9}$$

Thus the vanishing complex $R\Phi(K)$ expresses the difference between the cohomologies of the geometric fibers $X_{\bar s}$ and $X_{\bar\eta}$, with coefficients in the inverse image of K. By (2.1.8.7), the variation $\mathrm{Var}(\sigma)$ determines the action of the inertia group I on $\mathbb{H}^i(X_{\bar\eta}, K)$. These facts underlie all applications of the method of vanishing cycles.

The properness hypothesis imposed above on f may be replaced by regularity-at-infinity conditions on f and K. For example, one has the following result (for $K \in \mathrm{Ob}\, D(X_{\bar\eta}, \Lambda)$). II≈104

Proposition 2.1.9.

Suppose that f admits a factorization $f = f_1 k$

$$X \xhookrightarrow{\quad k \quad} X_1 \xrightarrow{\quad f_1 \quad} S,$$

where f_1 is proper and k identifies X with the complement in X_1 of a relative normal-crossings divisor D (thus, by hypothesis, f_1 is smooth on a neighborhood of D). If, on a neighborhood U of D, the sheaves $\mathcal{H}^i(K)$ are locally constant on $U - (D \cup X_0)$ and tamely ramified along the normal-crossings divisor $D \cup X_0$, then (2.1.8.1) and (2.1.8.2) are isomorphisms.

The hypercohomology spectral sequence for K shows that it is enough to treat the case in which K consists of a sheaf F_η in degree zero. After replacing S by its normalization in a finite extension (which is permissible, as one sees directly from the definitions) and applying Abhyankar's lemma, we may suppose that F_η is the restriction to X_η of a sheaf F on X, locally constant on $U \cap X$ and tamely ramified along D. With the notation of 2.1.2, let $\overline{F}$ be the inverse image of F on $\overline{X}$, and write $\bar f : \overline{X} \longrightarrow \overline{S}$.

To prove that (2.1.8.1) is an isomorphism, it is enough to prove that

$$R\bar f_*(R\bar j_* \bar j^* \overline{F})$$

commutes with base change by $\bar s \longrightarrow \overline{S}$. In the distinguished triangle

$$\longrightarrow \overline{F} \longrightarrow R\bar j_* \bar j^* \overline{F} \longrightarrow \Delta \longrightarrow,$$

the sheaves $\mathcal{H}^i(\Delta)$ are supported on $X_{\bar s}$; for Δ, the base-change statement in question is tautological. It therefore remains to prove that $R\bar f_*(\overline{F})$ commutes with base change by $\bar s \longrightarrow \overline{S}$, or, in view of the base-change theorem for the proper morphism f_1, that $Rk_*(\overline{F})$ commutes with base change $X_{1\bar s} \longrightarrow X_1$. This is Lemma 2.1.10 below.

To prove that (2.1.8.2) is an isomorphism, it is enough to prove, for the sheaf F, that extension by zero $k_!$ commutes with the functor $R\bar j_*$; that is, on a neighborhood of D, II≈105

$$R\Phi(k_! F) = 0 \qquad \text{on } X_{1\bar s}.$$

This is Lemma 2.1.11 below.

Lemma 2.1.10.

Let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Formation of Rj_*F commutes with arbitrary base change $T \longrightarrow S$.

The question is local on X for the étale topology. We may therefore suppose that there is a morphism $g : X' \longrightarrow X$, where X' is the normalization of X in a finite étale surjective covering $g' : U' \longrightarrow U$, tamely ramified along D, such that g'^*F is constant on U'. By Abhyankar's lemma, we may further suppose that X' is smooth over S and that U' is the complement in X' of a relative normal-crossings divisor D'.

The sheaf F injects into $g'_*g'^*F$, and the cokernel again satisfies the hypotheses of the lemma. Iterating this construction (the same g continues to work) gives a resolution of F by sheaves of the form g'_*G, where G is constant and g' is as above. It therefore suffices to prove the assertion for such a sheaf, or, equivalently, for $(X'/S, D', G)$ with G constant on U'. This is a special case of SGA 5, II, 4.4.

Lemma 2.1.11.

Let S be a henselian trait, let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Then

$$R\Phi(j_!F) = 0.$$

The dévissage just used reduces us to the case in which F is constant, with value G. We may suppose that D is the union of smooth divisors $(D_i)_{i\in I}$. For $J \subset I$, let D_J be the intersection of the D_i for $i \in J$. The sheaf F has a resolution $F^\bullet$ whose terms are sums of sheaves G_J, constant on D_J and extended by zero to X. It is therefore enough to prove the lemma for one such sheaf, and this follows from (2.1.5), applied to D_J/S.

I conjecture that (2.1.7.5) is an isomorphism when S and S' are excellent. Here is a partial result (see also 2.4.2). II≈**106**

Proposition 2.1.12.

The morphism (2.1.7.5) is an isomorphism in the following cases:

(a) S is excellent and S' is its completion;
(b) S and S' have equal characteristic zero.

In case (b), we reduce to the situation in which S and S' are strictly local and a uniformizer of S is still a uniformizer of S' (extend the scalars without changing $\bar{S}' \longrightarrow \bar{S}$). In both cases (a) and (b), writing an overbar for normalization in a separable closure of the fraction field, one then has

$$\bar{S}' \simeq S' \times_S \bar{S}.$$

Returning to the definitions, it is enough to prove that $S' \longrightarrow S$ is universally locally acyclic (apply SGA 4, XVI, 1.1). This follows from I, 0.5.1 (based on Artin's version of Néron desingularization) and SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms).

2.1.13.

Suppose that $f : X \longrightarrow S$ is of finite type. The cohomological dimension (SGA 4, XVII, 1.2.9) of $R\Psi$ is then at most $\dim(X_\eta)$, as follows from 2.1.4 and SGA 4, XIV, 3.2 (cf. I, 4.2(i)). Thus $R\Psi$ extends to the entire derived category,

$$R\Psi : D(X, \Lambda) \longrightarrow D(X_s \times_s S, \Lambda),$$

and it carries D^- into D^-. Let $L \in \mathrm{Ob}\, D^-(\Lambda)$ be a bounded-above complex of right Λ-modules, and let $K \in \mathrm{Ob}\, D^-(X, \Lambda)$. There is then an isomorphism

$$L \overset{\mathbf{L}}{\otimes}_\Lambda R\Psi(K) \xrightarrow{\sim} R\Psi(L \overset{\mathbf{L}}{\otimes}_\Lambda K). \tag{2.1.13.1}$$

To verify (2.1.13.1), first replace L by a complex of free Λ-modules. Since $R\Psi$ is triangulated and has finite cohomological dimension, it is enough to verify the assertion after replacing L by one of its terms. Passage to the filtered colimit reduces us to the case in which L is a finite free module, and then to the tautological case $L = \Lambda$.

It follows from (2.1.13.1), by the usual argument (cf. SGA 4, XVII, 5.2.11), that if $K \in$ II≈**107**
$\mathrm{Ob}\, D(X_\eta, \Lambda)$ has finite Tor-dimension at most d, where $d \in \mathbb{Z}$ (SGA 4, XVII, 4.1.9), then $R\Psi_\eta(K)$ likewise has Tor-dimension at most d.

2.2. A compatibility (trace morphisms)

2.2.1.

Let S be a henselian trait, and let $u : X_s \longrightarrow Y_s$ be a separated, quasi-finite, flat morphism of finite presentation between s-schemes. For every abelian sheaf F on Y_s, there is a trace morphism

$$\mathrm{Tr}_u : u_! u^* F \longrightarrow F$$

(SGA 4, XVII, 6.2.3). If F is an abelian sheaf on $Y_s \times_s S$ or on $Y_s \times_s \eta$, this gives an analogous morphism $\mathrm{Tr}_u : u_! u^* F \longrightarrow F$.

Now let $u : X \longrightarrow Y$ be a separated, quasi-finite, flat morphism of finite presentation between S-schemes. We again write u for the induced morphism

$$X_s \times_s S \longrightarrow Y_s \times_s S,$$

and we use Ψ for the functors of 1.3.3, both for X and for Y.

Proposition 2.2.2.

For every abelian sheaf F on Y, the following diagram is commutative:

$$\begin{array}{ccccc} u_! u^* \Psi F & \xrightarrow{(1.3.7.1)} & u_! \Psi u^* F & \xrightarrow{(1.3.8.1)} & \Psi u_! u^* F \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \downarrow{\scriptstyle \mathrm{Tr}_u} \\ \Psi F & = & = & = & \Psi F. \end{array}$$

The verification is left to the reader. There is a variant for Ψ_η and for F on Y_η.

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2.2.3.

Let Λ be as in 2.1.1, and let $u : X \longrightarrow Y$ be a quasi-finite flat morphism between separated, finite-type, flat S-schemes:

$$\begin{array}{ccc} X & \xrightarrow{\quad u \quad} & Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$$

We again write f and g for the morphisms

$$X_s \times_s S \longrightarrow S, \qquad Y_s \times_s S \longrightarrow S,$$

and suppose that $\dim Y_\eta \leq n$. For every $K \in \mathrm{Ob}\, D(S, \Lambda)$, there is a trace morphism (SGA 4, XVIII, 2.9)

$$\mathrm{Tr}_f : Rf_! f^* K(n)[2n] \longrightarrow K.$$

Corollary 2.2.4.

The following diagram is commutative:

$$\begin{array}{ccccc} Rg_!u_!u^*R\Psi\, g^*K(n)[2n] & \xrightarrow{(1.3.7.1)} & Rf_!R\Psi\, f^*K(n)[2n] & \xrightarrow{(2.1.10.1)} & Rf_!f^*K(n)[2n] \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ Rg_!R\Psi\, g^*K(n)[2n] & \xrightarrow{(2.1.10.1)} & Rg_!g^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_g} & K \end{array}$$

If u is étale, (1.3.7.1) is an isomorphism.

With the arguments suppressed, the diagram (2.2.4) expands as follows:

$$\begin{array}{ccccccc} Rg_!u_!u^*R\Psi\, g^* & \longrightarrow & Rg_!R\Psi\, u_!u^*g^* & \longrightarrow & Rg_!u_!u^*g^* & \xrightarrow{\sim} & Rf_!f^* \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & (1) & \downarrow{\scriptstyle \mathrm{Tr}_u} & (2) & \downarrow{\scriptstyle \mathrm{Tr}_u} & (3) & \downarrow{\scriptstyle \mathrm{Tr}_f} \\ Rg_!R\Psi\, g^* & = & Rg_!R\Psi\, g^* & \longrightarrow & Rg_!g^* & \xrightarrow{\mathrm{Tr}_g} & K. \end{array}$$

The commutativity of (1) follows from 2.2.2; that of (2) is the functoriality of (2.1.10.2); and that of (3) is the transitivity of the trace (SGA 4, XVIII, 2.9, Variant 3).

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2.2.4.1.

There is an analogous result for Ψ_η and $K \in \mathrm{Ob}\, D(\eta, \Lambda)$.

2.2.5.

Let Φ be a subset of Y_s which is proper over s, and write Rg_Φ for the "direct image with support in Φ" functor. Let Tr_g denote the composite

$$\begin{aligned} \mathrm{Tr}_g : Rg_\Phi R\Psi g^*K(n)[2n] &\longrightarrow Rg_!R\Psi g^*K(n)[2n] \\ &\longrightarrow Rg_!g^*K(n)[2n] \longrightarrow K, \end{aligned} \tag{2.2.5.1}$$

and similarly for Ψ_η and $K \in D(\eta, \Lambda)$.

If Φ is a finite set of closed points, then

$$Rg_\Phi \circ u_! = Rf_{u^{-1}\Phi},$$

and hence there is a trace morphism

$$\mathrm{Tr}_u : Rf_{u^{-1}\Phi}u^* = Rg_\Phi u_!u^* \longrightarrow Rg_\Phi.$$

By 2.2.4, the following diagram is commutative:

$$\begin{array}{ccccc} Rf_{u^{-1}\Phi}u^*R\Psi\, g^*K(n)[2n] & \longrightarrow & Rf_{u^{-1}\Phi}R\Psi\, f^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_f} & K \\ \downarrow{\scriptstyle \mathrm{Tr}_u} & & & & \| \\ Rg_\Phi R\Psi\, g^*K(n)[2n] & & \xrightarrow[\mathrm{Tr}_g]{} & & K. \end{array}$$

Suppose that u is étale, let $x \in X_s$, and suppose that x and $y = u(x)$ have the same residue field. Then

$$Rf_{\{x\}} \xrightarrow{\sim} Rg_{\{y\}}.$$

This isomorphism, given by Tr_u, expresses the étale locality of cohomology with support at x. The diagram

$$\begin{array}{ccc} Rf_{\{x\}}R\Psi\, f^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_f} & K \\ \| & & \| \\ Rg_{\{y\}}R\Psi\, g^*K(n)[2n] & \xrightarrow{\mathrm{Tr}_g} & K \end{array} \tag{2.2.5.2}$$

is commutative.

Scholium 2.2.6.

The compatibility (2.2.5.2) says that the trace morphism (2.2.5.1), for $\Phi = \{y\}$, is a local invariant: it depends only on the henselization of the S-scheme X at x. The same holds for $R\Psi_\eta$.

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2.3. Finiteness theorem (in equal characteristic zero)

Theorem 2.3.1.

Let S be a henselian trait as in 0.2.5, let Λ be a noetherian torsion ring, and let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that S has equal characteristic zero. Then, for every constructible sheaf of Λ-modules F on X_η, the sheaves $R^i\Psi_{\bar\eta}(F)$ on $X_{\bar s}$ are constructible.

We begin by repeating the opening of the proof of SGA 4, XIV, 1.10. Apply SGA 4, IX, 2.14(ii). There is a resolution $F^\bullet$ of F

$$\begin{array}{ccccccccc} & & & & F & & & & \\ & & & & \downarrow & & & & \\ 0 & \longrightarrow & F^0 & \longrightarrow & F^1 & \longrightarrow & \cdots & & \end{array}$$

by sheaves of the form $\bigoplus_i p'_{i*}G_i$, where $p'_i : Y_{i\eta} \longrightarrow X_\eta$ is finite and G_i is a constructible constant sheaf. We may suppose that $Y_{i\eta}$ is reduced; let $p_i : Y_i \longrightarrow X$ be the normalization of X in $Y_{i\eta}$. Using the spectral sequence of the resolution $F^\bullet$ and applying (2.1.7.1) in the elementary case of the finite morphism p_i, we see that it suffices to prove the theorem for each pair (Y_i, G_i).

We next prove the theorem when F is the constant sheaf defined by a Λ-module M. Recall that Λ is a $\mathbb{Z}/n$-algebra for a suitable n. The module M may therefore be dévissé into modules M_i over $\Lambda/\ell_i\Lambda$, where each $\ell_i \mid n$ is a suitable prime. This reduces us to the case in which Λ is a $\mathbb{Z}/\ell$-algebra, with ℓ prime. Formula (2.1.13.1) then gives

$$M \otimes_{\mathbb{Z}/\ell} R^i\Psi_{\bar\eta}(\mathbb{Z}/\ell) \xrightarrow{\sim} R^i\Psi_{\bar\eta}(M).$$

It remains to treat the constant sheaf $F = \mathbb{Z}/\ell$.

Choose a descending sequence of closed subsets II≈**111**

$$X_\eta = X'_0 \supset \cdots \supset X'_n$$

such that $(X'_i - X'_{i+1})_{\mathrm{red}}$ is smooth over η, and let $X_i = \overline{X'_i}$ be the closure in X. The constant sheaf $\mathbb{Z}/\ell$ is dévissé into the constant sheaves $\mathbb{Z}/\ell$ on the strata $X'_i - X'_{i+1}$, extended by zero. Thus:

Reduction 1. It is enough to prove Theorem 2.3.1 for a sheaf $j_!\mathbb{Z}/\ell$, where $j : U \hookrightarrow X_\eta$ is the inclusion of a smooth open subset.

By resolution of singularities there is a morphism $g : X_1 \longrightarrow X$ such that

(a) g is proper and induces an isomorphism $g^{-1}(U) \xrightarrow{\sim} U$;
(b) X_1 is regular and $X_1 - g^{-1}(U)$ is a normal-crossings divisor in X_1, which is a union of regular divisors D_α.

Let $j_1 : g^{-1}(U) \hookrightarrow X_1$. Then

$$Rg_* j_{1!}\mathbb{Z}/\ell = j_!\mathbb{Z}/\ell.$$

Applying (2.1.7.1) and the finiteness theorem SGA 4, XIV, 1.1, we see that it suffices to prove Theorem 2.3.1 for $(X_1, j_{1!}\mathbb{Z}/\ell)$.

There is a resolution

$$0 \longrightarrow j_{1!}\mathbb{Z}/\ell \longrightarrow \mathbb{Z}/\ell \longrightarrow \bigoplus_{\alpha}(\mathbb{Z}/\ell)_{D_\alpha} \longrightarrow \bigoplus_{\alpha,\beta}(\mathbb{Z}/\ell)_{D_\alpha \cap D_\beta} \longrightarrow \cdots,$$

and hence:

Reduction 2. It is enough to prove Theorem 2.3.1 for the constant sheaf $\mathbb{Z}/\ell$ on a regular S-scheme X such that X_s is a normal-crossings divisor in X.

In this final case one uses the explicit computation I, 3.3, which in turn uses the purity theorem SGA 4, XIX, 3.2 and the characteristic-zero hypothesis.

2.4. Isolated singularities

2.4.1.

Throughout this section, for simplicity, we suppose that S is a strictly henselian trait. Let Λ be a noetherian torsion ring whose torsion is prime to the residual characteristic of S, let $f : X \longrightarrow S$ be a morphism of finite type, and let F be a constructible sheaf of Λ-modules on X. A point $x \in X$ is called a smooth point of (X, F) if f is smooth at x and F is locally constant in a neighborhood of x. Let Σ denote the set of points where (X, F) is not smooth, and let Σ_s be its special fiber. We know that $R\Phi(F) = 0$ outside Σ_s.

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Theorem 2.4.2.

At every isolated point x of Σ_s, the modules $R^i\Phi_{\bar\eta}(F)_x$ are Λ-modules of finite type and are invariant under change of trait (via 2.1.7.1).

In equal characteristic zero, these assertions were proved in 2.1.12 and 2.3.1 without assuming that x is isolated.

Let x be an isolated point of Σ_s. The theorem is local in a neighborhood of x, so we may suppose that f is proper. We then have the long exact sequence (2.1.8.9)

$$\cdots \longrightarrow H^i(X_s, F) \longrightarrow H^i(X_{\bar\eta}, F) \longrightarrow \mathbb{H}^i\big(X_s, R\Phi_{\bar\eta}(F)\big) \longrightarrow \cdots.$$

In this sequence, $H^i(X_s, F)$ and $H^i(X_{\bar\eta}, F)$ are invariant under change of trait and are Λ-modules of finite type (SGA 4, XIV, 1.1). The modules $\mathbb{H}^i(X_s, R\Phi_{\bar\eta}(F))$ therefore have the same properties. Recall the following lemma.

Lemma 2.4.3.

Let (T, Λ) be a ringed topos, let Z be a closed subtopos of T, and let $i : Z \hookrightarrow T$ be the inclusion. Then the functor

$$i_* : D(Z, \Lambda) \longrightarrow D(T, \Lambda)$$

is an equivalence from $D(Z, \Lambda)$ onto the full subcategory of $D(T, \Lambda)$ consisting of the complexes K for which $\mathcal{H}^i(K)$ is supported on Z.

Indeed, for such a complex K, the morphism $K \longrightarrow i_* i^* K$ is a quasi-isomorphism.

We may consequently regard $R\Phi_{\bar\eta}(F)$ as an object of

$$D^b(\Sigma_s,\Lambda) = D^b(\Sigma_s - \{x\},\Lambda) \oplus D^b(\{x\},\Lambda).$$

Passing to a direct summand, we find that

$$\mathbb{H}^i\big(\{x\}, R\Phi_{\bar\eta}(F)\big) = R^i\Phi_{\bar\eta}(F)_x$$

has the properties stated in 2.4.2. This completes the proof.

Remark 2.4.4.

The same proof shows that, if K is constructible in $D^+(X,\Lambda)$ and x is isolated in the support of $R\Phi_{\bar\eta}(K)$, then the cohomology modules of the stalk $R\Phi_{\bar\eta}(K)_x$ are of finite type over Λ. If, after a change of trait, x remains an isolated point of the support, these modules are invariant under that change of trait.

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2.4.5.

Let $x \in X_s$ be an isolated point where (X,F) is not smooth. More generally, let $K \in \mathrm{Ob}\, D^b(X,\Lambda)$, and let x be an isolated point of the support of $R\Phi(K)$. Let $i : \{x\} \hookrightarrow X_s$ be the inclusion. The complex $i_*i^*R\Phi(K)$, which is isomorphic to $R\Phi(K)$ in a neighborhood of x, is a direct summand of $R\Phi(K)$ by 2.4.3. Thus, for $\sigma \in I$, the variation morphism induces

$$\mathrm{Var}(\sigma) : i_*i^*R\Phi_{\bar\eta}(K) \longrightarrow R\Psi_{\bar\eta}(K),$$

that is,

$$\mathrm{Var}(\sigma) : \big(R\Phi_{\bar\eta}(K)\big)_x \longrightarrow R\Gamma_{\{x\}}\big(R\Psi_{\bar\eta}(K)\big). \tag{2.4.5.1}$$

2.4.6.

Suppose that the support Σ of $R\Phi(K)$ consists of finitely many isolated points. For every $\sigma \in I$, the endomorphism $\sigma - 1$ of $R\Psi_{\bar\eta}(K)$ is then the composite displayed in the following diagram:
(2.4.6.1)

$$\begin{array}{ccc}
R\Psi_{\bar\eta}(K) & \longrightarrow & R\Phi_{\bar\eta}(K) = \bigoplus_{x\in\Sigma} i_{x*}\big(R\Phi_{\bar\eta}(K)\big)_x \\
\Big\downarrow{\scriptstyle \sigma-1} & & \Big\downarrow{\scriptstyle \bigoplus i_{x*}\,\mathrm{Var}(\sigma)} \\
R\Psi_{\bar\eta}(K) & \longleftarrow & \bigoplus_{x\in\Sigma} i_{x*}R\Gamma_{\{x\}}R\Psi_{\bar\eta}(K)
\end{array}$$

If f is proper, applying H^i to (2.4.6.1) shows that the endomorphism $\sigma - 1$ of $\mathbb{H}^i(X_{\bar\eta}, K)$ is the

composite
(2.4.6.2)

$$\begin{array}{ccccc}
\mathbb{H}^i(X_{\bar\eta},K) & \xrightarrow{\sim} & \mathbb{H}^i\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longrightarrow & \mathbb{H}^i\big(X_s,R\Phi_{\bar\eta}(K)\big)=\bigoplus_{x\in\Sigma}R^i\Phi_{\bar\eta}(K)_x \\
\Big\downarrow{\scriptstyle \sigma-1} & & & & \Big\downarrow{\scriptstyle \bigoplus \mathrm{Var}(\sigma)} \\
\mathbb{H}^i(X_{\bar\eta},K) & \xleftarrow{\sim} & \mathbb{H}^i\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longleftarrow & \bigoplus_{x\in\Sigma}\mathbb{H}^i_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big)
\end{array}$$

This describes the action of I on $\mathbb{H}^i(X_{\bar\eta},K)$; its essential ingredients are the variation morphisms (2.4.5.1), which are local on X. We also have the long exact sequence

$$\longrightarrow \mathbb{H}^i(X_s,K)\longrightarrow \mathbb{H}^i(X_{\bar\eta},K)\longrightarrow \bigoplus_{x\in\Sigma}R^i\Phi_{\bar\eta}(K)_x\longrightarrow\cdots. \tag{2.4.6.3}$$

The same argument applies when f is not necessarily proper, provided that (X,K) satisfies 2.1.9. The diagram (2.4.6.5) below defines a morphism

$$\mathrm{Var}(\sigma): R\Gamma(X_{\bar\eta},K)\longrightarrow R\Gamma_c(X_{\bar\eta},K) \tag{2.4.6.4}$$

such that the endomorphism $\sigma-1$ of $R\Gamma(X_{\bar\eta},K)$ (respectively, $R\Gamma_c(X_{\bar\eta},K)$) is the composite $\alpha\,\mathrm{Var}(\sigma)$ (respectively, $\mathrm{Var}(\sigma)\alpha$), where

$$\alpha: R\Gamma_c(X_{\bar\eta},K)\longrightarrow R\Gamma(X_{\bar\eta},K).$$

(2.4.6.5)

$$\begin{array}{ccccc}
R\Gamma(X_{\bar\eta},K) & \xrightarrow{\sim} & R\Gamma\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longrightarrow & \bigoplus_{x\in\Sigma}\big(R\Phi_{\bar\eta}(K)\big)_x \\
\Big\downarrow{\scriptstyle \mathrm{Var}(\sigma)} & & & & \Big\downarrow{\scriptstyle \bigoplus} \\
R\Gamma_c(X_{\bar\eta},K) & \xrightarrow[\sim]{} & R\Gamma_c\big(X_s,R\Psi_{\bar\eta}(K)\big) & \longleftarrow & \bigoplus_{x\in\Sigma}R\Gamma_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big)
\end{array}\quad.$$

In this case we also have the long exact sequences II≈**114**
(2.4.6.6)

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & \mathbb{H}^i_c(X_s,K) & \longrightarrow & \mathbb{H}^i_c(X_{\bar\eta},K) & \longrightarrow & \bigoplus_{x\in\Sigma}\big(R^i\Phi_{\bar\eta}(K)\big)_x & \longrightarrow & \cdots \\
 & & \Big\downarrow & & \Big\downarrow & & \Big\| & & \\
\cdots & \longrightarrow & \mathbb{H}^i(X_s,K) & \longrightarrow & \mathbb{H}^i(X_{\bar\eta},K) & \longrightarrow & \bigoplus_{x\in\Sigma}R^i\Phi_{\bar\eta}(K)_x & \longrightarrow & \cdots
\end{array}$$

2.4.7.

Let $X_{s(x)}$ be the henselization of X_s at x, and put

$$X^*_{s(x)}=X_{s(x)}-\{x\}.$$

The long hypercohomology exact sequences of $(X_{s(x)}, \{x\})$, with coefficients in K_s, $R\Psi_{\bar{\eta}}(K)$, and $R\Phi_{\bar{\eta}}(K)$, give the following commutative diagram of long exact sequences:
(2.4.7.1)

$$\begin{array}{ccccccccc}
 & & 0 & \longrightarrow & R^{n-1}\Phi_{\bar{\eta}}(K)_x & \longrightarrow & R^{n-1}\Phi_{\bar{\eta}}(K)_x & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
\longrightarrow & \mathbb{H}^{n-1}(X^*_{s(x)}, K) & \xrightarrow{j} & \mathbb{H}^n_{\{x\}}(X_s, K) & \longrightarrow & \underline{H}^n(K)_x & \longrightarrow & \mathbb{H}^n(X^*_{s(x)}, K) & \longrightarrow \\
 & \| & & \downarrow & & \downarrow & & \| & \\
\longrightarrow & \mathbb{H}^{n-1}(X^*_{s(x)}, K) & \xrightarrow[①]{j} & \mathbb{H}^n_{\{x\}}(X_s, R\Psi_{\bar{\eta}}(K)) & \longrightarrow & R^n\Psi_{\bar{\eta}}(K)_x & \longrightarrow & \mathbb{H}^n(X^*_{s(x)}, K) & \longrightarrow \\
 & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\
 & 0 & \longrightarrow & R^n\Phi_{\bar{\eta}}(K)_x & = & R^n\Phi_{\bar{\eta}}(K)_x & \longrightarrow & 0 &
\end{array}$$

The image of the arrow ① is contained in the inertia invariants.

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SGA 7 — EXPOSÉ XIV

Comparison with Transcendental Theory

by P. Deligne

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Introduction

To compare the theory of vanishing cycles in Exposé XIII with its transcendental analogue over $\mathbb{C}$, we proceed in two stages.

(a) We give a transcendental formalism modeled on that of Exposé XIII and prove a comparison theorem by the usual methods (already used to prove the finiteness theorem in XIII, 2.3).

(b) In the case of isolated singularities, we explain in more concrete terms the meaning of the transcendental formalism introduced here.

Besides the comparison theorem, this exposé contains a proof of the Picard–Lefschetz formula in the transcendental case.

In §4 we give a result preliminary to the Hodge theory of vanishing cycles. That section will not be used later in the seminar. II≈**117**

1. Transcendental formalism of vanishing cycles

1.1. The fundamental group of the punctured disc: sign conventions

We shall have to consider punctured discs D^* and their fundamental group $\mathbb{Z}$. Since this group is abelian, sign ambiguities may arise. We eliminate them by forgetting that it is abelian.

1.1.1.

Let X be a nice connected topological space and let $x \in X$. The elements of the fundamental group are homotopy classes of loops based at x. If $\alpha, \beta \in \pi_1(X, x)$ are represented by loops a, b, then $\alpha\beta$ is represented by the loop obtained by traversing b and then a.

1.1.2.

Let F be a locally constant sheaf on X. For every path $a : [0,1] \longrightarrow X$, the inverse image a^*F on $[0,1]$ is locally constant, hence constant, and therefore gives a bijection

$$a : F_{a(0)} \xleftarrow{\sim} H^0([0,1], a^*F) \xrightarrow{\sim} F_{a(1)}.$$

This bijection depends only on the homotopy class of a. Through this construction, $\pi_1(X,x)$ acts on the left on F_x. The functor $F \longmapsto F_x$ is an equivalence from the category of locally constant sheaves on X to the category of sets equipped with an action of $\pi_1(X,x)$.

1.1.3.

As x varies, the groups $\pi_1(X,x)$ are the fibers of a locally constant sheaf of groups $\Pi_1(X)$, which acts on every locally constant sheaf F. If $\pi_1(X,x)$ is abelian, then $\Pi_1(X)$ is constant; its value is denoted by $\pi_1(X)$.

1.1.4.

The group $\pi_1(X,x)$ acts on the universal covering

$$p : (\widetilde{X}, \widetilde{x}) \longrightarrow (X,x).$$

Regarded as an étalé space over X, and hence as a sheaf (the sheaf of its local sections), this covering corresponds under 1.1.2 to the set $\pi_1(X,x)$ with distinguished element e, on which $\pi_1(X,x)$ acts by left translation. The group also acts on this π_1-set by automorphisms of π_1-sets: an element $\sigma \in \pi_1(X,x)$ acts by

$$g \longmapsto g\sigma^{-1}.$$

This construction identifies

$$\pi_1(X,x) = \mathrm{Aut}_X(\widetilde{X}).$$

If F is locally constant on X, the bijection II≈118

$$H^0(\widetilde{X}, p^*F) \longrightarrow (p^*F)_{\widetilde{x}} = F_x \tag{1.1.4.1}$$

commutes with the action of $\pi_1(X,x)$: on the left the action is transport of structure through its action on $\widetilde{X}$, while on the right it is the action of 1.1.2. The equivalence of 1.1.2 may therefore be reformulated by saying that

$$F \longmapsto (\widetilde{X} \longmapsto H^0(\widetilde{X}, p^*F))$$

is an equivalence from the category of locally constant sheaves on X to the category of functors

$$(\text{universal coverings of } X) \longrightarrow (\text{Sets}).$$

1.1.5.

If $\pi_1(X,x)$ is abelian and hence independent of x, the action of 1.1.4 of $\pi_1(X)$ on $\widetilde{X}$ is independent of the base point: $\pi_1(X)$ acts on every universal covering $\widetilde{X}$ of X.

One must bear in mind that this action, which is the only one we shall use, corresponds to the inverse of the action in 1.1.3.

1.1.6.

Let D be a disc, that is, a topological space homeomorphic to the standard disc

$$\{z \in \mathbb{C} : |z| < 1\}.$$

Let $0 \in D$ and put $D^* = D - \{0\}$. Suppose that D is equipped with the structure of an analytic manifold, and hence is oriented. The fundamental group $\pi_1(D^*)$ is then canonically isomorphic to $\mathbb{Z}$. For the standard disc, with center 0 and $t \in D^*$, the positive generator of $\pi_1(D^*,t)$ is represented by the loop

$$x \longmapsto \exp(2\pi i x)t.$$

1.1.7.

Let $z : D \longrightarrow \mathbb{C}$ be a uniformizing parameter, so that $(dz)_0 \neq 0$, and let $\widetilde{D}^*$ be a universal covering of D^*. Write $\log z$ for any function on $\widetilde{D}^*$ such that

$$\exp(\log z) = z.$$

The pair $(\widetilde{D}^*, \log z)$ is unique up to unique isomorphism.

For $\alpha \in \mathbb{C}$, put II≈**119**

$$z^\alpha = \exp(\alpha \log z).$$

Let T be the positive generator of $\pi_1(D^*)$, acting on $\widetilde{D}^*$ as in 1.1.5 and on functions on $\widetilde{D}^*$ by transport of structure. Then

$$T \log z = \log z + 2\pi i, \tag{1.1.7.1}$$

and

$$Tz^\alpha = \exp(2\pi i\alpha) z^\alpha. \tag{1.1.7.2}$$

1.1.8.

Let D be a disc, let $0 \in D$, and put $D^* = D - \{0\}$. If $D' \subset D$ is a smaller disc containing 0, restriction to D'^* is an equivalence from the category of locally constant sheaves on D^* to the category of locally constant sheaves on D'^*. Indeed, for $t \in D'^*$,

$$\pi_1(D'^*, t) \xrightarrow{\sim} \pi_1(D^*, t).$$

Suppose now that D is equipped with a differentiable-manifold structure. Let T be the tangent space to D at 0, and put $T^* = T - \{0\}$. For $D' \subset D$ as above and an embedding $f : D' \hookrightarrow T$ satisfying $f(0) = 0$ and $(df)_0 = \mathrm{id}$, the composite

$$(\text{restriction to } D')^{-1} \circ f^*$$

is an equivalence $[f]$ from the category of locally constant sheaves on T^* to the category of locally constant sheaves on D^*. Up to canonical isomorphism it depends only on the germ of f at 0, subject to $(df)_0 = \mathrm{id}$. A path from one such germ to another defines an isomorphism of equivalences, and homotopic paths define the same isomorphism. Since the space of such germs is contractible—by the retraction to a fixed germ f_0 given by $(1 - \lambda)f + \lambda f_0$—we obtain an equivalence, well defined up to unique isomorphism, from locally constant sheaves on T^* to locally constant sheaves on D^*.

1.1.9.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering. We write

$$p : \widetilde{D} \longrightarrow D$$

for the topological space over D obtained from $\widetilde{D}^*$ by adjoining a point 0, in such a way that:

(a) $\widetilde{D}^*$ is open in $\widetilde{D}$;

(b) $p : \widetilde{D} \longrightarrow D$ extends $p : \widetilde{D}^* \longrightarrow D^*$ and $p(0) = 0$;

(c) the sets $p^{-1}(U)$, as U ranges over the neighborhoods of 0 in D, form a neighborhood basis of 0 in $\widetilde{D}$.

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1.1.10.

Let X be a nonsingular complex algebraic curve, let $0 \in X$, and let D be a disc neighborhood of 0 in X. The ring of germs at 0 of holomorphic functions on D which are algebraic over the field $K(X)$ of rational functions on X is the henselization $\mathcal{O}_0^h$ of the local ring $\mathcal{O}_0$ of X at 0. Let $\widetilde{D}^*$ be a universal covering of $D^* = D - \{0\}$. The field of germs at $0 \in \widetilde{D}$ of meromorphic functions on $\widetilde{D}^*$ which are algebraic over $K(X)$ is an algebraic closure $\overline{K}$ of $K(X)$, and hence also of the fraction field K_0^h of $\mathcal{O}_0^h$.

The group $\pi_1(D^*)$ of D-automorphisms of $\widetilde{D}^*$ (1.1.5) acts by transport of structure on $\overline{K}$, giving

$$\begin{array}{ccc} \pi_1(D^*) & \longrightarrow & \mathrm{Gal}(\overline{K}/K_0^h) \\ \| & & \| \\ \mathbb{Z} & & \widehat{\mathbb{Z}}(1) \end{array} \qquad (1.1.10.1)$$

It follows from (1.1.7.2) that this diagram is compatible with the natural inclusion, over $\mathbb{C}$, of $\mathbb{Z}$ into $\widehat{\mathbb{Z}}(1)$. This inclusion is the composite of $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ and the isomorphism

$$\widehat{\mathbb{Z}} \xrightarrow{\sim} \widehat{\mathbb{Z}}(1),$$

which is the inverse limit of the isomorphisms

$$\mathbb{Z}/n \longrightarrow \mu_n, \qquad k \longmapsto \exp\left(\frac{2\pi i}{n}k\right).$$

1.1.11.

Every sheaf F on $\mathrm{Spec}(K_0^h)$ defines a locally constant sheaf on D^*. We use the equivalence of 1.1.4 and, to each universal covering $\widetilde{D}^*$ of D^*, associate $F_{\overline{K}}$ for the field $\overline{K}$ constructed in 1.1.10.

1.2. Sheaves of sets

1.2.1.

Let D be a disc (1.1.6), let $0 \in D$, and put $D^* = D - \{0\}$. We denote by $\mathcal{D}^*$ the topos of locally constant sheaves of sets on D^*. By 1.1.8 this topos does not "change" if D is replaced by a smaller disc D' containing 0, nor, when D is endowed with the structure of a differentiable manifold, if D is replaced by its tangent space at 0.

For $t \in D^*$, the functor $F \longmapsto F_t$ identifies this topos with the topos of sets equipped with an action of $\pi_1(D^*)$ (1.1.2). By 1.1.4, the functor II≈121

$$F \longmapsto (\widetilde{D}^* \longmapsto H^0(\widetilde{D}^*, p^*F))$$

identifies it with the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\text{Sets}),$$

and these identifications are related by (1.1.4.1).

1.2.2.

Let F be a sheaf on D whose restriction to D^* is locally constant. There is a unique morphism of sheaves

$$\underline{F_0} \xrightarrow{\alpha} F$$

whose map on the stalk at 0 is the identity. As explained in XIII, 0.1 (with $X_0 = (P^t)$), the functor

$$F \longmapsto (F_0, F|_{D^*}, \alpha)$$

identifies these sheaves with triples consisting of a set F_0, a locally constant sheaf F_η on D^*, and a morphism

$$\alpha : (\text{constant sheaf with value } F_0 \text{ on } D^*) \longrightarrow F_\eta.$$

1.2.3.

We denote by $\mathcal{D}$ the topos of sheaves of sets on D whose restriction to D^* is locally constant. The inclusion of $\mathcal{D}$ into the topos of sheaves on D is the inverse-image functor of a morphism of topoi $D \longrightarrow \mathcal{D}$, which fits into the commutative diagram

$$\begin{array}{ccccc} \{0\} & \hookrightarrow & D & \hookleftarrow & D^* \\ \| & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\beta} \\ \{0\} & \hookrightarrow & \mathcal{D} & \hookleftarrow & \mathcal{D}^* \end{array} \quad .$$

As in 1.2.1, one checks that this topos does not change when D is replaced by a smaller disc or by the tangent space to D at 0.

1.2.4.

Let Y be a topological space. The product topos $Y \times \mathcal{D}$ has the following three descriptions (for their equivalence, see XIII, 0.1).

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(a) $Y \times \mathcal{D}$ is the topos of sheaves F on $Y \times D$ satisfying

$(*)$ D^* is covered by open subsets U such that $F|_{Y\times U}$ is the inverse image of a sheaf on Y.

(b) Let $t \in D^*$. The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_t, α) in which

$\alpha)$ F_0 is a sheaf on Y;
$\beta)$ F_t is a sheaf on Y equipped with an action of $\pi_1(D^*)$;
$\gamma)$ $\alpha : F_0 \longrightarrow F_t$ is a morphism whose image is contained in the invariants.

(c) The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_η, α) in which

$\alpha')$ F_0 is a sheaf on Y;
$\beta')$ F_η is a functor, with values denoted by $F_{\bar\eta}$,

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y);$$

$\gamma')$ α is a morphism from F_0 (regarded as the constant functor with value F_0 from the universal coverings of D^* to the sheaves on Y) to F_η.

It is immaterial here whether this topos is literally the product topos of Y and $\mathcal{D}$. Likewise, we define $Y \times \mathcal{D}^*$, at our choice, as

(a)* the topos of sheaves on $Y \times D^*$ satisfying $(*)$;
(b)* for $t \in D^*$, the topos of sheaves F_t on Y equipped with an action of $\pi_1(D^*)$;
(c)* the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y).$$

1.2.5.

The topos $Y \times \mathcal{D}$ has the same functoriality properties in Y as the topoi $Y \times_s S$ of Exposé XIII, and we shall not list them. To fix notation, we merely record the diagram

$$\begin{array}{ccccccc} Y & & Y & \xrightarrow{\ i\ } & Y \times \mathcal{D} & \xleftarrow{\ j\ } & Y \times \mathcal{D}^* \\ \downarrow & \xleftarrow{\ \text{sp}\ } & \downarrow & & \downarrow & & \downarrow \\ (P^t) & & \{0\} & \xrightarrow{\ i\ } & \mathcal{D} & \xleftarrow{\ j\ } & \mathcal{D}^* \end{array}$$

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1.2.6.

Let K be a complex of sheaves of abelian groups on $Y \times \mathcal{D}$. As in XIII, 1.4.2, one obtains from K a distinguished triangle in $K(Y \times \mathcal{D}^*)$:

$$\begin{array}{ccc} & \Phi(K) & \\ \swarrow & & \nwarrow \\ \mathrm{sp}^* K_0 & \longrightarrow & K_\eta \end{array}$$

This construction passes to the derived category and defines a functor

$$D(Y \times \mathcal{D}) \longrightarrow (\text{distinguished triangles of } D(Y \times \mathcal{D}^*)).$$

Suppose that D is oriented (for example, complex analytic), and let T be the positive generator of $\pi_1(D^*)$ (1.1.6). For $K \in \mathrm{Ob}\, D(Y \times \mathcal{D})$, we define, as in XIII, 1.4.3, the variation $V = \mathrm{Var}(T)$,

$$V : \Phi(K) \longrightarrow K_\eta.$$

The endomorphism $T - \mathrm{Id}$ of K_η is the composite

$$T - \mathrm{Id} : K_\eta \longrightarrow \Phi(K) \longrightarrow K_\eta.$$

1.2.7.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. We shall define a functor

$$\Psi : (\text{sheaves on } X) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}).$$

Let $\widetilde{D}^*$ be a universal covering of D^*, let $\widetilde{D}$ be as in 1.1.9, and put

$$\overline{X} = X \times_D \widetilde{D}, \qquad X^* = X - X_0 = X \times_D D^*, \qquad \overline{X}^* = \overline{X} - X_0 = X \times_D \widetilde{D}^*.$$

There is a commutative diagram

$$\begin{array}{ccccc} X_0 & \xhookrightarrow{\bar{i}} & \overline{X} & \xhookleftarrow{\bar{j}} & \overline{X}^* \\ \| & & \downarrow p & & \downarrow p' \\ X_0 & \xhookrightarrow{i} & X & \xhookleftarrow{j} & X^* \end{array} \tag{1.2.7.1}$$

cartesian over the diagram

$$\begin{array}{ccccc} \{0\} & \xhookrightarrow{\bar{i}} & \widetilde{D} & \xhookleftarrow{\bar{j}} & \widetilde{D}^* \\ \| & & \downarrow p & & \downarrow p' \\ \{0\} & \xhookrightarrow{i} & D & \xhookleftarrow{j} & D^* \end{array} \tag{1.2.7.2}$$

For a sheaf F on X, we set (with the notation of 1.2.4(c)) II≈**124**

$$\Psi(F) = \big(i^*F,\ \bar{i}^*\bar{j}_*((jp')^*F),\ \alpha\big),$$

where

$$\begin{aligned} \alpha : i^*F = \bar{i}^* p^* F &\longrightarrow \bar{i}^*\bar{j}_*((jp')^*F) \\ &= \bar{i}^*\bar{j}_*\bar{j}^* p^* F \end{aligned}$$

is obtained, after applying $\bar{i}^*$, from the adjunction map $\mathrm{Id} \longrightarrow \bar{j}_*\bar{j}^*$.

We also define a functor

$$\Psi_\eta : (\text{sheaves on } X^*) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}^*)$$

by

$$\Psi_\eta(F) = \bar{i}^*\bar{j}_*(p'^* F).$$

1.2.8.

The functoriality properties of Ψ are the same as those of its analogue studied in XIII, 1.3.5–1.3.9.

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1.3. The functor $R\Psi$

1.3.1.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. The functors $R\Psi$ and $R\Psi_\eta$ are the derived functors of Ψ and Ψ_η:

$$R\Psi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}),$$

$$R\Psi_\eta : D^+(X^*) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

We denote by $R\Phi$ the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

The properties of these functors are exactly analogous to those of the functors with the same names in XIII, 2.1. Their proofs are much more elementary, since in the topological setting the base-change theorems for a proper morphism or for a smooth morphism (that is, one locally isomorphic to $\mathbb{R}^n \times S \longrightarrow S$) are easy to prove. The minor changes required are as follows.

1.3.2.

The functor $Rf_!$ is defined only for a separated morphism between locally compact spaces, and is the derived functor of the direct-image functor with proper supports $f_!$.

1.3.3.

For a proper (and separated) morphism $f : X \longrightarrow D$, the fundamental isomorphism (XIII, (2.1.8.3)) must be rewritten as follows. Let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering of D^*. For every neighborhood U of 0 in D, put

$$\widetilde{U}^* = p^{-1}(U), \qquad X_{\widetilde{U}^*} = X \times_D \widetilde{U}^*.$$

For $K \in \mathrm{Ob}\, D^+(f^{-1}(D^*))$, one has

$$\varinjlim_U \mathbb{H}^i(X_{\widetilde{U}^*}, K) \xrightarrow{\sim} \mathbb{H}^i(X_0, R\Psi_{\bar{\eta}}(K)). \tag{1.3.3.1}$$

This isomorphism is especially useful when $X^* = f^{-1}(D^*)$ is a topological fiber bundle over D^*. As a disc $U \subset D$ shrinks to 0, the Leray spectral sequence for $X_{\widetilde{U}^*} \longrightarrow \widetilde{U}^*$ then shows that, for every locally constant sheaf Λ on X^* and every $t \in \widetilde{U}^*$,

$$H^i(X_{\widetilde{U}^*}, \Lambda) \xrightarrow{\sim} H^i(X_t, \Lambda).$$

Thus, for $t \in D^*$,

$$H^i(X_t, \Lambda) \xrightarrow{\sim} H^i(X_0, R\Psi_t(\Lambda)), \tag{1.3.3.2}$$

provided that X/D is proper, X^* is a fiber bundle, and Λ is locally constant.

In practice, to verify that X^* is a fiber bundle over D^* one uses Thom's isotopy theorem, one form of which is stated below. For the proof of this theorem, none of whose consequences will be used in the subsequent exposés, we refer to Whitney and Mather. II ≈**126**

1.3.4. Preliminary.

Let $f : X \longrightarrow Y$ be a smooth morphism (by definition, one with no critical point) of differentiable manifolds. A connection ∇ on X/Y is a vector subbundle of the tangent bundle of X complementary to $\ker(df)$. For a vector field u on Y, there is a unique vector field $\nabla(u)$ on X which lifts u and lies in this subbundle. The connection is called integrable if

$$\nabla([u,v]) = [\nabla(u), \nabla(v)].$$

When f is proper, an integrable connection gives local trivializations over Y of X/Y,

$$\alpha : f^{-1}(U) \xrightarrow{\sim} X_0 \times U,$$

compatible with projection to U and satisfying $d\alpha(\nabla(u)) = (0, u)$. When f is not proper, the same result holds under suitable conditions at infinity; for example, it is enough that $\nabla(u)g$ vanish for a suitable exhaustion function g.

1.3.5. Thom's isotopy theorem.

Let

$$D = \{z \in \mathbb{C} : |z| < 1\}, \qquad D^* = D - \{0\},$$

and let $f : X \longrightarrow D$ be a proper morphism of analytic spaces. After replacing D by a smaller disc and X by its inverse image, there exists a stratification Σ of $X^* = f^{-1}(D^*)$ having the following properties:

(a) *Σ is a filtration*

$$X^* = F_0 \supset F_1 \supset \cdots \supset F_n \supset F_{n+1} = \varnothing$$

of X^* by closed subsets such that the closure $\overline{F}_i$ of F_i in X is an analytic subspace of X.

(b) *The open strata $F_i - F_{i+1}$ are smooth over D^*.*

(c) *Σ satisfies Whitney's conditions (a) and (b).*

For such a stratification Σ, there are integrable connections ∇_i on the $F_i - F_{i+1}$ such that II≈**127**

(i) *locally on D^*, ∇_i defines a trivialization of $F_i - F_{i+1}$;*

(ii) *these trivializations glue to give local trivializations of X^*/D^*.*

Finally, if (Y_α) is a finite family of closed analytic subspaces of X, then, after shrinking D further, one can choose Σ so that every Y_α is a union of connected components of open strata $(F_i - F_{i+1})$. A connection ∇ as above then locally trivializes $(X, (Y_\alpha))$.

1.3.6.

The isomorphism (1.3.3.2) remains valid for hypercohomology with values in a complex of sheaves K when X is proper over D and the following condition holds:

$(*)$ For every $i \in \mathbb{Z}$, the space D^* is covered by open subsets U for which there exist an isomorphism compatible with projection to U,

$$\varphi : f^{-1}(U) \xrightarrow{\sim} Y \times U,$$

and a sheaf G on Y such that, on $f^{-1}(U)$, $\underline{H}^i(K)$ is isomorphic to $\varphi^* \operatorname{pr}_1^* G$.

Under this hypothesis, the triangle of 1.2.6 gives a long exact sequence

$$\cdots \longrightarrow \mathbb{H}^i(X_0, K) \longrightarrow \mathbb{H}^i(X_t, K) \longrightarrow \mathbb{H}^i(X_0, R\Phi_t(K)) \longrightarrow \cdots . \tag{1.3.6.1}$$

By Thom's theorem, condition $(*)$ holds (after shrinking D) whenever X admits a complex analytic stratification such that all the $\underline{H}^i(K)$ are locally constant on every open stratum.

1.4. A special case

This section will not be used later in the seminar. It formalizes a construction sketched in the Introduction to Exposé XIII.

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1.4.1.

Let D be an oriented disc, let $0 \in D$, and put $D^* = D - \{0\}$. A topological space $\mathfrak{X}^*$ over $\mathfrak{D}^*$ is a functor

$$(\text{universal coverings of } D^*) \longrightarrow (\text{topological spaces}).$$

After a universal covering $\widetilde{D}^*$ has been chosen, this is a topological space X_η equipped with an action of $\pi_1(D^*)$; when $\widetilde{D}^*$ is the universal covering of (D^*, t), we denote it by $\mathfrak{X}^*_t$.

A sheaf on $\mathfrak{X}^*$ is a functor

$$(\text{universal coverings of } D^*) \longrightarrow (\text{topological spaces equipped with a sheaf}),$$

or, equivalently, a $\pi_1(D^*)$-equivariant sheaf on X_η.

To $\mathfrak{X}^*$ we associate the topological space

$$\beta^*\mathfrak{X}^* = X_\eta \times \widetilde{D}^*/\pi_1(D^*)$$

over D^*. Every sheaf F on $\mathfrak{X}^*$ descends to a sheaf β^*F on $\beta^*\mathfrak{X}^*$, giving a commutative diagram of topoi

$$\begin{array}{ccc} \beta^*\mathfrak{X}^* & \longrightarrow & \mathfrak{X}^* \\ \downarrow & & \downarrow \\ D^* & \xrightarrow{\beta} & \mathfrak{D}^*. \end{array}$$

1.4.2.

A topological space $\mathfrak{X}$ over $\mathfrak{D}$ consists of

(a) a topological space $\mathfrak{X}^*$ over $\mathfrak{D}^*$;

(b) a topological space X_0;

(c) a $\pi_1(D^*)$-invariant morphism

$$\mathrm{sp} : X_\eta \longrightarrow X_0,$$

where $\pi_1(D^*)$ acts trivially on X_0.

A sheaf on $\mathfrak{X}$ consists of a sheaf F_0 on X_0, a sheaf on $\mathfrak{X}^*$ given by F_η, and an equivariant morphism

$$\mathrm{sp} : \mathrm{sp}^*F_0 \longrightarrow F_\eta \qquad (\text{or } F_0 \longrightarrow \mathrm{sp}_*F_\eta).$$

There is a commutative diagram

$$\begin{array}{ccccc} X_0 & \hookrightarrow & \mathfrak{X} & \hookleftarrow & \mathfrak{X}^* \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \hookrightarrow & \mathfrak{D} & \hookleftarrow & \mathfrak{D}^* \end{array}$$

and, in addition, a morphism of topoi

$$\Psi : \mathfrak{X} \longrightarrow X_0 \times \mathfrak{D}.$$

It is obtained from the functoriality of the preceding construction, applied to

$$(X_\eta, X_0, \mathrm{sp}) \longrightarrow (X_0, X_0, \mathrm{Id}),$$

and satisfies

$$\Psi(F_0, F_\eta, \mathrm{sp}) = (F_0, \mathrm{sp}_* F_\eta, \mathrm{sp}).$$

This morphism induces

$$\Psi_\eta : \mathfrak{X}^* \longrightarrow X_0 \times \mathfrak{D}^*.$$

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1.4.3.

To a topological space $\mathfrak{X}$ over $\mathfrak{D}$ we associate a topological space $X = \beta^*\mathfrak{X}$ over D as follows. To $X^* = \beta^*\mathfrak{X}^*$ we adjoin the fiber X_0 over 0. A fundamental system of neighborhoods of $x_0 \in X_0$ is formed by the sets

$$V \cup (\mathrm{sp}^{-1}(V) \times \widetilde{U}^*/\pi_1(D^*)),$$

where V is a neighborhood of x_0 in X_0, U is a neighborhood of 0 in D, and $\widetilde{U}^* = U \times_D \widetilde{D}^*$. Let j (respectively i) denote the inclusion of X^* (respectively X_0) into X. For a sheaf F on $\mathfrak{X}^*$, one has

$$i^* j_* \beta^* F = (\mathrm{sp}_* F_\eta)^{\pi_1(D^*)}.$$

It follows that there is a commutative diagram of topoi

$$\begin{array}{ccc} X & \xrightarrow{\beta} & \mathfrak{X} \\ \downarrow & & \downarrow \\ D & \xrightarrow{\beta} & \mathfrak{D} \end{array}$$

such that, for every sheaf F on $\mathfrak{X}$, there is an isomorphism

$$\Psi(F) \xrightarrow{\sim} \Psi(\beta^* F). \tag{1.4.3.1}$$

Proposition 1.4.4.

The map (1.4.3.1) derives to an isomorphism

$$R\Psi \xrightarrow{\sim} R\Psi \circ \beta^*$$

between functors from $D^+(\mathfrak{X})$ to $D^+(X_0 \times \mathfrak{D})$.

Indeed, let $\overline{j}$ be the inclusion of II≈**130**

$$\overline{X}^* = X \times_D \widetilde{D}^* \simeq X_\eta \times \widetilde{D}^*$$

into $\overline{X} = X \times_D \widetilde{D}$. The proposition reduces to checking that, for every flabby sheaf of abelian groups F on X_η, the sheaf $\mathrm{pr}_1^* F$ is acyclic for $\overline{j}_*$. This in turn reduces (with $B = \widetilde{D}^*$) to the following familiar assertion, proved by a homotopy argument. If F is a sheaf of abelian groups on a space Y and B is a disc, then

$$H^i(Y, F) \xrightarrow{\sim} H^i(Y \times B, \mathrm{pr}_1^* F).$$

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1.4.5.

Proposition 1.4.4 gives a method for computing $R\Psi$ for sheaves on X of the form $\beta^* F$, reducing the computation of $R\Psi$ essentially to that of $R\,\mathrm{sp}_*$.

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1.4.6.

Later we shall consider the case in which X_η is locally compact and X_0 is obtained from X_η by contracting a compact subset $V \subset X_\eta$ to a point 0, with sp the contraction map and the action of $\pi_1(D^*)$ trivial outside V. Proper base change for sp then gives, for $K \in \mathrm{Ob}\, D^+(\mathfrak{X}^*)$,

$$R\Psi_\eta(\beta^* K)_0 = R\Gamma(V, K) \tag{1.4.6.1}$$

and

$$R\Gamma_{\{0\}} R\Psi_\eta(\beta^* K) = R\Gamma_V(X_\eta, K), \tag{1.4.6.2}$$

the latter being cohomology with supports.

For K on $\mathfrak{X}$, suppose that

$$\mathrm{sp} : \mathrm{sp}^* K_0 \longrightarrow K_\eta$$

is a quasi-isomorphism outside V. Then $R\Phi(K) = 0$ outside 0. In particular, for a constant sheaf Λ on $\mathcal{X}$,

$$R^i\Phi(\Lambda) = \widetilde{H}^i(V, \Lambda), \qquad \text{supported at } 0,$$

where the right-hand side is reduced cohomology.

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1.4.7.

The case of interest to us is that in which X_η is a differentiable manifold and $V \subset X_\eta$ is a submanifold with boundary and interior V°. Then $X - \{0\}$ is a differentiable manifold obtained by gluing $\beta^* X_\eta$ to the constant bundle over D with fiber $X_\eta - V$. The map

$$H^i_c(V^\circ, \Lambda) \xrightarrow{\sim} H^i_V(X_\eta, \Lambda) \tag{1.4.7.1}$$

is an isomorphism. These groups are also denoted by $H^i(V \bmod \partial V, \Lambda)$. The isomorphism (1.4.7.1) defines an isomorphism

$$R\Gamma_c(V^\circ, \Lambda) \xrightarrow{\sim} R\Gamma_{\{0\}} R\Psi_\eta(\Lambda), \tag{1.4.7.2}$$

and the variation is defined, by the procedure of XIII, 2.4.5 and through (1.4.7.2), as a morphism whose source is the reduced cohomology of V:

$$\mathrm{Var} : R\widetilde{\Gamma}(V, \Lambda) \longrightarrow R\Gamma_c(V^\circ, \Lambda). \tag{1.4.7.3}$$

The automorphism induced by T on the pair $(V, \partial V)$ is again denoted by T, and its restriction ∂T to ∂V is the identity. The variation (1.4.7.3) should coincide with the classical variation morphism (3.1.9) attached to such an automorphism T. A statement of this kind will be proved in 3.1.10.

2. The comparison theorem

2.1.

For every scheme X of finite type over $\mathbb{C}$, we denote by X_{cl} the space $X(\mathbb{C})$ endowed with its usual topology.

Let S be a smooth curve over $\mathbb{C}$, that is, a smooth separated scheme of finite type over $\mathbb{C}$, pure of dimension 1. Let $0 \in S(\mathbb{C})$, let $f : X \longrightarrow S$ be a separated S-scheme of finite type, and let Λ be a noetherian torsion ring (for example $\mathbb{Z}/n\mathbb{Z}$). Put $X_0 = f^{-1}(0)$.

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2.2.

We denote by S^h the henselized henselian trait of S at 0, and by η the generic point of S^h. There is a functor

$$\Psi : (\text{sheaves on } X) \xrightarrow{\text{restriction}} (\text{sheaves on } X \times_S S^h) \xrightarrow{\Psi} (\text{sheaves on } X_0 \times S^h).$$

Its derived functor is the composite

$$R\Psi : D(X, \Lambda) \longrightarrow D(X \times_S S^h, \Lambda) \xrightarrow{R\Psi} D(X_0 \times S^h, \Lambda).$$

2.3.

Let $D \subset S_{\text{cl}}$ be a disc containing 0. The composite functor Ψ_{cl} (or simply Ψ)

$$\Psi_{\text{cl}} : (\text{sheaves on } X_{\text{cl}}) \xrightarrow{\text{restriction}} (\text{sheaves on } X_{\text{cl}} \times_{S_{\text{cl}}} D) \xrightarrow{\Psi} (\text{sheaves on } (X_0)_{\text{cl}} \times \mathcal{D})$$

is independent, up to unique isomorphism, of the choice of D; recall that $\mathcal{D}$ does not change when D is replaced by a smaller disc. Its derived functor is the composite

$$R\Psi_{\text{cl}} : D(X_{\text{cl}}, \Lambda) \longrightarrow D(f_{\text{cl}}^{-1}(D), \Lambda) \xrightarrow{R\Psi} D((X_0)_{\text{cl}} \times \mathcal{D}, \Lambda).$$

2.4.

Likewise, we define

$$R\Psi_\eta : D(X - X_0, \Lambda) \longrightarrow D(X_0 \times \eta, \Lambda),$$
$$R\Psi_{\eta\,\text{cl}} : D((X - X_0)_{\text{cl}}, \Lambda) \longrightarrow D((X_0)_{\text{cl}} \times \mathcal{D}^*, \Lambda).$$

2.5.

The construction of 1.1.11 defines a morphism of topoi from $\mathcal{D}^*$ to η, which extends to

$$\varepsilon : \mathcal{D} \longrightarrow S^h.$$

Together with $\varepsilon : (X_0)_{\text{cl}} \longrightarrow X_0$ (SGA 4, XI, 4), this morphism defines

$$\varepsilon : (X_0)_{\text{cl}} \times \mathcal{D} \longrightarrow X_0 \times S^h.$$

If $\widetilde{D}^*$ is a universal covering of D^* and $k(\bar{\eta})$ is the corresponding algebraic closure (1.1.10) of $k(\eta)$, then

$$\varepsilon^*(F_0, F_\eta, \alpha) = (\varepsilon^* F_0, \varepsilon^* F_{\bar{\eta}}, \varepsilon^* \alpha).$$

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2.6.

Let $\widetilde{D}^*$ be a universal covering of D^*, let $k(\bar{\eta})$ be the corresponding algebraic closure of $k(\eta)$, and let $\overline{S}$ be the normalization of S^h in $k(\bar{\eta})$. Write $\overline{S}$ as an inverse limit of affine S-schemes S_i. For every i, after replacing D by a smaller disc D_i, there is a natural morphism of S_{cl}-spaces

$$a_i : \widetilde{D}_i \longrightarrow (S_i)_{\text{cl}}.$$

Let $j_i : U_i \hookrightarrow S_i$ be the open complement in S_i of the image 0 of the closed point of $\overline{S}$, and let $i : 0 \hookrightarrow S_i$.

Let F be a sheaf on X, and put $F_{\mathrm{cl}} = \varepsilon^* F$. We again denote by j_i the inclusion of $X \times_S U_i$ into $X \times_S S_i$, by i the inclusion of X_0 into $X \times_S S_i$, and by F the inverse image of F on $X \times_S S_i$. Then

$$\Psi(F) = \left(i^* F,\ \varinjlim_i i^* j_{i*} j_i^* F,\ \varinjlim_i \varphi_i \right).$$

The functors ε^* and inverse image commute. The morphism

$$\varepsilon^* j_{i*} \longrightarrow (j_{i,\mathrm{cl}})_* \varepsilon^*$$

therefore gives

$$c_i : (i^* j_{i*} j_i^* F)_{\mathrm{cl}} \longrightarrow i^* j_{i*} j_i^* (F_{\mathrm{cl}}).$$

Let $\overline{j}$ (respectively $\overline{j}_i$) be the inclusion of $\widetilde{D}^*$ (respectively $\widetilde{D}_i^*$) into $\widetilde{D}$ (respectively $\widetilde{D}_i$), as well as the induced inclusions

$$X \times_S \widetilde{D}^* \hookrightarrow X \times_S \widetilde{D}, \qquad X \times_S \widetilde{D}_i^* \hookrightarrow X \times_S \widetilde{D}_i.$$

The adjunction map $\mathrm{Id} \longrightarrow a_{i*} a_i^*$ defines

$$i^* j_{i*} j_i^* F_{\mathrm{cl}} \longrightarrow i^* \overline{j}_{i*} \overline{j}_i^* \overline{F}_{\mathrm{cl}} = i^* \overline{j}_* \overline{j}^* \overline{F}_{\mathrm{cl}}.$$

Passing to the limit gives

$$c_\eta : \Psi_{\bar{\eta}}(F)_{\mathrm{cl}} \longrightarrow \Psi_{\mathrm{cl}}(F_{\mathrm{cl}}).$$

This morphism defines the comparison morphism

$$c : \varepsilon^* \Psi(F) \longrightarrow \Psi_{\mathrm{cl}}(\varepsilon^* F). \tag{2.6.1}$$

Similarly, one defines

$$c_\eta : \varepsilon^* \Psi_\eta(F) \longrightarrow \Psi_\eta(\varepsilon^* F). \tag{2.6.2}$$

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2.7.

The morphisms (2.6.1) and (2.6.2) derive to

$$c : \varepsilon^* R\Psi \longrightarrow R\Psi_{\mathrm{cl}} \circ \varepsilon^*$$

and

$$c_\eta : \varepsilon^* R\Psi_\eta \longrightarrow R\Psi_{\eta\,\mathrm{cl}} \circ \varepsilon^*.$$

Theorem 2.8.

Let $K \in \mathrm{Ob}\, D(X, \Lambda)$ (respectively, $K \in \mathrm{Ob}\, D(X - X_0, \Lambda)$) be a complex of sheaves of Λ-modules whose cohomology sheaves $\underline{H}^i(K)$ are constructible. Then the comparison morphism

$$c : \varepsilon^* R\Psi(K) \longrightarrow R\Psi_{\mathrm{cl}}(\varepsilon^* K)$$

(respectively,

$$c : \varepsilon^* R\Psi_\eta(K) \longrightarrow R\Psi_{\eta\,\mathrm{cl}}(\varepsilon^* K))$$

is an isomorphism.

It suffices to prove the assertion for $R\Psi_\eta$. Moreover, both sides of 2.7 are triangulated functors of finite cohomological dimension, so it is enough to treat the case in which K is a single sheaf F placed in degree 0. The proof is exactly analogous to that of the finiteness theorem XIII, 2.3.1. Since $R\Psi$ and $R\Psi_{\mathrm{cl}}$ have the same variance properties, the dévissages used there (with permission to shrink S) reduce the problem to the case in which X is regular, $f^{-1}(0)$ is a normal crossings divisor in X, and $F = \mathbb{Z}/\ell\mathbb{Z}$. One then compares Exposé I, 3.3 with its transcendental analogue.

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3. Isolated singularities

3.1. The manifold of vanishing cycles

3.1.1.

Let

$$D = \{z \in \mathbb{C} \mid |z| < 1\},$$

let X be an analytic space with $0 \in X$, and let $f : X \to D$ be a morphism such that $f(0) = 0$. We suppose that f is smooth away from 0 (in particular, X is smooth away from 0). We intend to study the vanishing cycles near 0; hence there is no loss of generality in supposing that X is embedded in $\mathbb{C}^n \times D \subset \mathbb{C}^{n+1}$ compatibly with the projection to D, and in such a way that 0 maps to 0.

3.1.2.

Let $\delta(x)$ be a real-analytic function on X (the restriction to X of a real-analytic function on $\mathbb{C}^{n+1}$) such that, for $x \in X_0 = f^{-1}(0)$,

(a) $\delta(x) \geq 0$;

(b) $\delta(x) = 0$ if and only if $x = 0$.

For example, we may take δ to be the square of the Euclidean distance from x to 0 in $\mathbb{C}^{n+1}$. The critical locus of the restriction of δ to X_0 is real analytic and therefore, in a suitable neighborhood of 0, has only finitely many connected components. Replacing X by an open neighborhood of 0 and applying Sard's theorem, we may consequently obtain the following situation:

(c) for a suitable $r > 0$, the space $X_0 \cap \delta^{-1}([0, r])$ is compact;

(d) on X_0, δ has no critical value in $(0, r]$.

After shrinking X further, we then have:

(e) there exists a C^∞ function $\tau(s)$, $0 \leq s \leq r$, with $\tau(0) = 0$ and $\tau'(s) > 0$ for $s > 0$, such that, for $0 < s \leq r$ and $0 \leq |t|^2 \leq \tau(s)$, the fiber $X_t = f^{-1}(t)$ is transverse to the real hypersurface $\delta(x) = s$ in X, and the box

$$B_{t,d} = \{x \in X \mid \delta(x) \leq d \text{ and } |f(x)|^2 < t\}$$

is proper over the open disc D_t of radius $t^{1/2}$.

Let $L = X_0 \cap \delta^{-1}(r)$. By (c) and (d), there is a homeomorphism from $X_0 \cap \delta^{-1}([0, r])$ to the cone II≈136

$$L \times [0, r]/(L \times \{0\} \text{ contracted to a point})$$

which transforms δ into the second coordinate and is C^∞ away from the origin.

3.1.3.

Let $d \leq r$ and $t \leq \tau(d)$. Away from 0, $B_{t,d}$ is a manifold with corners, and the projection $f : B_{t,d} \to D_t$ is a fibration. In particular:

(a) over $D_t^* = D_t - \{0\}$, $B_{t,d}$ induces a C^∞ bundle of compact manifolds with boundary;

(b) the boundary $\delta(x) = d$ of this bundle extends to a C^∞ bundle $\partial B_{t,d}$ of compact manifolds over D_t.

Let $y \in D_t^*$. We call the manifold with boundary $V = f^{-1}(y)$ the manifold of vanishing cycles. Since the disc is contractible, the bundle $\partial B_{t,d}$ over D_t is isomorphic to the constant bundle with fiber ∂V, and the space of bundle isomorphisms

$$\alpha : \partial B_{t,d} \xrightarrow{\sim} \partial V \times D_t$$

that restrict to the identity on the fiber over y is contractible. Choose one, denoted by α. Lift the vector field $\partial/\partial\theta$ on $S_{|y|}$, the circle of radius $|y|$, to a vector field u on $f^{-1}(S_{|y|})$ which is compatible with α on the boundary, so that

$$d\alpha(u) = (0, \partial/\partial\theta).$$

The space of such lifts is contractible. Integrating u once around the circle in the positive direction defines a diffeomorphism T of V onto itself which is the identity on ∂V; this is the monodromy transformation. It is well defined up to isotopy relative to ∂V.

In a sense whose precise formulation is left to the reader, (V, T) is independent of the choices of t and d. I do not know whether it is independent of the choice of δ. This is nevertheless the case if we restrict δ to be the restriction to X of a real-analytic function g on $\mathbb{C}^{n+1}$ such that, on the Zariski tangent space of X_0 at 0, one has $dg = 0$ and $d^2g > 0$ (positive definite). Indeed, for δ and δ' of this kind, r can be chosen as in 3.1.2(c),(d), uniformly for every $\lambda\delta + (1-\lambda)\delta'$, $0 \leq \lambda \leq 1$.

3.1.4.

Let (V_1, T_1) be obtained from (V, T) by adjoining a collar:

$$V_1 - V^\circ \simeq \partial V \times [0,1], \qquad T_1|_{V_1 - V^\circ} = \mathrm{id}\,.$$

Define the following space over D_t (cf. 1.4.3):

$$f' : W \longrightarrow D_t.$$

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(a) Let $\widetilde{D}_t^*$ be a universal covering of D_t^* with base point y, and let W^* be the bundle over D_t^* obtained as the quotient of $V_1 \times \widetilde{D}_t^*$ by the equivalence relation generated by

$$(v, y) \sim (T_1 v, e^{2\pi i} y).$$

(b) Since $T_1|_{V_1 - V^\circ}$ is trivial, W^* can be glued to the trivial bundle over D_t with fiber $V_1 - V$, giving a space W_1 over D_t.

(c) Put $W = W_1 \cup \{0\}$ and $f'(0) = 0$. Give W the unique topology inducing that of W_1, for which f' is continuous and, for $t' < t$, $f'^{-1}(\overline{D}_{t'})$ is the one-point compactification of $W_1 \cap f'^{-1}(\overline{D}_{t'})$.

Equip $W - \{0\}$ with the integrable connection of 1.3.4 for which the trivialization, supplied by (a), of the inverse image of W on $\widetilde{D}^*$ is horizontal.

Proposition 3.1.5.

There exists a homeomorphism

$$\Psi : B_{t,d} \xrightarrow{\sim} W$$

compatible with the projections to D_t and carrying 0 to 0; this homeomorphism may be chosen to be a diffeomorphism away from 0.

This result will not be used in the remainder of the seminar.

Let $d_0 < d_1 \leq r$, let $t_0 \leq \tau(d_0)$, and let $y \in D_{t_0}^*$. We have

$$B_{t_0,d_0} \subset B_{t_0,d_1},$$

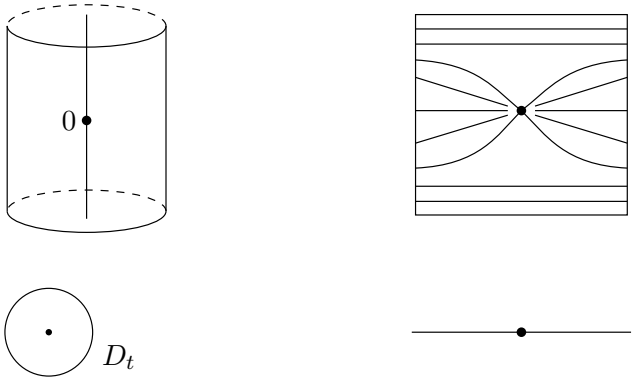
and $B_{t_0,d_1} - B^{\circ}_{t_0,d_0}$ is a bundle over D_{t_0}. Trivialize it and, over the circle $S_{|y|}$, lift $\partial/\partial\theta$ compatibly with this trivialization. Let T_1 be the corresponding endomorphism of

$$V_1 = f^{-1}(y) \cap B_{t_0,d_1}, \qquad V_0 = f^{-1}(y) \cap B_{t_0,d_0}.$$

Then $(V_0 \subset V_1, T_1)$ is isomorphic to $(V \subset V_1, T_1)$ of 3.1.4. Let W be constructed from $(V_0 \subset V_1, T_1)$. We have $f'^{-1}(y) = V_1$ canonically. The proof will show that Ψ can be chosen to induce the identity

$$f^{-1}(y) \xrightarrow{\sim} f'^{-1}(y).$$

The following drawings of f and of its restriction to $f^{-1}(\mathbb{R})$ summarize the proof.



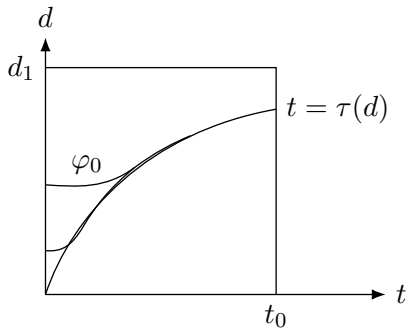
Let $B^+ \subset B_{t_0,d_1}$ be the set of points x such that II≈**138**

$$|f(x)|^2 \leq \tau(\delta(x)).$$

Let φ_n be a sequence of C^∞ functions from $[0, t_0]$ to $(0, d_1]$ such that, for a sequence $t_0 > r_1 > r_2 > \cdots$ tending to 0,

(a) for $|t|^2 \geq r_i$, φ_i is the inverse function of τ;

(b) for $|t|^2 < r_i$ and $j > i$, $\varphi_j(t) < \varphi_i(t)$.



The space $B^+ - \{0\}$ is the union of the bundles B_i^+ over D_t defined by

$$\delta(x) \geq \varphi_i(|f(x)|^2).$$

A standard argument, using the fact that δ has no critical points on the fibers of $B^+ - \{0\} \to D_t$, shows that there is a diffeomorphism $\alpha_i : B_i^+ \to B_{i+1}^+$, compatible with the projection to D_t and equal to the identity outside a small neighborhood of the subspace

$$\delta(x) = \varphi_i(|f(x)|^2), \qquad |f(x)|^2 \leq r_i,$$

of B_i^+. Composing the α_i and passing to the limit gives a diffeomorphism

$$\alpha : B_0^+ - \{x \mid f(x) = 0,\ \delta(x) = \varphi_0(0)\} \xrightarrow{\sim} B^+ - \{0\}.$$

Transport a trivialization of the bundle B_0^+ over D_t by α to obtain an integrable connection ∇' on $B^+ - \{0\}$. We may then:

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(a) extend the restriction of this connection to $f^{-1}(S_{|y|}) \cap B^+$ to a connection on $f^{-1}(S_{|y|})/S_{|y|}$; this extension gives T_1;

(b) extend to B_{t_0,d_1} the lift, supplied by ∇' on $B^+ - \{0\}$, of the radial vector field on D_t.

There is then a unique integrable connection ∇ on B_{t_0,d_1} extending ∇', (a), and (b), and a unique horizontal diffeomorphism from $B_{t_0,d_1} - \{0\}$ to $W - \{0\}$ that is the identity on $f^{-1}(y)$. This is the required diffeomorphism.

3.1.6. A generalization.

The preceding discussion generalizes to the case in which $f : X \to D$ is a morphism such that $0 \in X$ is an isolated non-smooth point in X_0. After shrinking X and D, the non-smooth locus of f is then finite over D. The manifold of vanishing cycles V has a finite set Σ of critical points; $V - \Sigma$ is a differentiable manifold with boundary (its boundary is compact, although the manifold is not). The monodromy transformation T is the identity on the boundary and stabilizes Σ. Finally, Proposition 3.1.5 remains valid: the fibered homeomorphism Ψ is a diffeomorphism away from the critical locus, which it carries to the part of W arising from Σ.

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3.1.7.

Proposition 3.1.5 allows us to use the calculations of 1.4. We obtain:

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(A) The vanishing-cycle groups $(R^i\Phi(\Lambda))_0$ are the reduced cohomology groups, with coefficients in Λ, of the manifold of vanishing cycles V.

(B) The groups $H^i_{\{0\}}(X_0, R\Psi(\Lambda))$ are the cohomology groups with proper support of V°, with coefficients in Λ.

(C) The variation

$$\mathrm{Var} : \widetilde{H}^i(V, \Lambda) \longrightarrow H^i_c(V^\circ, \Lambda) = H^i(V \bmod \partial V, \Lambda)$$

is the classical variation defined by the automorphism T of V, which is the identity on ∂V.

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3.1.8.

Assertions (A), (B), and (C) are in fact more elementary than Proposition 3.1.5. Here is a direct proof and construction which spares us from checking that the isomorphisms in (A) and (B) are independent of the choices made in the construction of 3.1.5.

Let $d \leq r$ and $t \leq \tau(d)$, and let B be the open box II ≈**140**

$$B = \{x \mid \delta(x) < d \text{ and } |f(x)|^2 < t\}.$$

Let $\widetilde{D}^*$ be a universal covering of D^*, and consider the diagram

$$\begin{array}{ccccc} \{0\} & \xhookrightarrow{\bar{\imath}} & \widetilde{D}_t & \xhookleftarrow{\bar{\jmath}} & \widetilde{D}_t^* \\ \| & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ \{0\} & \xhookrightarrow{i} & D_t & \xhookleftarrow{j} & D_t^* \end{array} \tag{3.1.8.1}$$

obtained from (1.2.7.2) by restriction to D_t. Let

$$\begin{array}{ccccc} B_0 & \xhookrightarrow{\bar{\imath}} & \overline{B} & \xhookleftarrow{\bar{\jmath}} & \overline{B}^* \\ \| & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ B_0 & \xhookrightarrow{i} & B & \xhookleftarrow{j} & B^* \end{array} \tag{3.1.8.2}$$

be the inverse image of $f : B \to D$ over (3.1.8.1). We again write f for the projection from (3.1.8.2) to (3.1.8.1).

(A) By definition, $R^i\Psi_\eta(\Lambda)_0$ is the direct limit, as U runs through the neighborhoods of 0 in X, of the cohomology groups

$$H^i(U \times_D \widetilde{D}^*, \Lambda).$$

Let U run through the fundamental system of neighborhoods of 0 formed by boxes B as above. The fiber product $U \times_D \widetilde{D}^* = \overline{B}^*$ is a bundle with fiber V° over the contractible space $\widetilde{D}_t^*$. Hence the direct system of the groups $H^i(\overline{B}^*, \Lambda)$ is constant, with value

$$H^i(V^\circ, \Lambda) = H^i(V, \Lambda).$$

The resulting isomorphisms

$$R^i\Psi_\eta(\Lambda)_0 \simeq H^i(V, \Lambda) \tag{3.1.8.3}$$

may be refined to the following composite isomorphism (cf. (B) below):

$$\begin{array}{ccccccc} R\Psi_\eta(\Lambda)_0 & \xleftarrow{\sim} & R\Gamma(B_0, R\Psi_\eta(\Lambda)) & \xleftarrow{\sim} & (Rf_*R\bar{\jmath}_*\Lambda)_0 & \xleftarrow{\sim} & R\Gamma(\widetilde{D}_t, Rf_*R\bar{\jmath}_*\Lambda) \\ & & & & & & \| \\ R\Gamma(V, \Lambda) & \xrightarrow{\sim} & R\Gamma(V^\circ, \Lambda) & \xleftarrow{\sim} & R\Gamma(\overline{B}^*, \Lambda) & & \end{array} \tag{3.1.8.4}$$

The complex $R\Phi(\Lambda)_0$ is defined by the distinguished triangle

$$\Lambda \longrightarrow (R\Psi_\eta\Lambda)_0 \longrightarrow R\Phi(\Lambda)_0 \longrightarrow .$$

Consequently, the groups $R^i\Phi(\Lambda)_0$ are the reduced cohomology groups of V.

(B) Let $\mathfrak{F}$ be the family of closed subsets of $\overline{B}^*$ whose closure in $\overline{B}$ is proper over $\widetilde{D}$. The arrows in the following diagram are isomorphisms: II ≈141

$$\begin{array}{ccccc} R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) & \underset{①}{\xrightarrow{\sim}} & R\Gamma_c(B_0, R\Psi_\eta(\Lambda)) & \underset{②}{\xleftarrow{\sim}} & (Rf_!R\bar{\jmath}_*\Lambda)_0 \\ & & & & \uparrow{\scriptstyle ③}\wr \\ R\Gamma_c(V^\circ, \Lambda) & \underset{⑤}{\xleftarrow{\sim}} & R\Gamma_{\mathfrak{F}}(\overline{B}^*, \Lambda) & \underset{④}{=\!=} & R\Gamma(\widetilde{D}_t, Rf_!R\bar{\jmath}_*\Lambda) \end{array} \tag{3.1.8.5}$$

Indeed, ① is an isomorphism because B_0 is a cone (3.1.2) and $R^i\Psi_\eta(\Lambda)$ is constant away from 0; ② by the proper base-change theorem; ③ because $Rf_!R\bar{\jmath}_*\Lambda$ is constant away from 0; ④ is the derived-category version of the spectral sequence of a composite functor; and ⑤ because $\widetilde{D}^*$ is contractible.

The isomorphism (3.1.8.5) supplies the desired isomorphisms

$$R^i\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) = H^i_c(V^\circ, \Lambda). \tag{3.1.8.6}$$

We shall verify (C) in 3.1.10.

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3.1.9.

Cf. [3]. Let X be a topological space, let $i : Y \hookrightarrow X$ be a closed subset, let T be an automorphism of X inducing the identity on Y, and let Λ be an abelian group. To define the variation Var, choose a T-equivariant resolution Λ^*_X of Λ on X, a resolution Λ^*_Y of the constant sheaf Λ on Y on which, by convention, T acts trivially, and an equivariant morphism

$$\varphi : \Lambda^*_X \longrightarrow i_*\Lambda^*_Y$$

resolving the evident morphism $\varphi_0 : \Lambda_X \to i_*\Lambda_Y$. Suppose that the sheaves Λ^i_X and Λ^i_Y are acyclic for the functors to be considered. Passing to global sections gives

$$\Gamma(\varphi) : \Gamma(\Lambda^*_X) \longrightarrow \Gamma(\Lambda^*_Y).$$

For simplicity, suppose that every $\Gamma(\varphi)^i$ is surjective; otherwise, make it surjective by adding the cone of $\Gamma(\Lambda^*_Y)$ to $\Gamma(\Lambda^*_X)$. Put

$$R\Gamma(X \bmod Y, \Lambda) = \mathrm{Ker}(\Gamma(\varphi)).$$

Under reasonable hypotheses, if $\mathfrak{F}$ is the family of closed subsets of X disjoint from Y, there is a quasi-isomorphism

$$\Gamma_{\mathfrak{F}}(X - Y, \Lambda^*_X) \xrightarrow{\approx} R\Gamma(X \bmod Y, \Lambda).$$

The endomorphism $T-1$ of $\Gamma(\Lambda^*_X)$, where T acts by transport of structure, is zero on Λ (the constants) and factors through $\mathrm{Ker}(\Gamma(\varphi))$. Hence, in the derived category and under the same hypotheses, it gives a variation morphism

$$\mathrm{Var} : R\widetilde{\Gamma}(X, \Lambda) \longrightarrow R\Gamma_{\mathfrak{F}}(X - Y, \Lambda).$$

Without these hypotheses, one obtains instead

$$\mathrm{Var} : R\widetilde{\Gamma}(X, \Lambda) \longrightarrow R\Gamma(X \bmod Y, \Lambda).$$

This description simplifies when T is the identity on a neighborhood of Y and acts as the identity on Λ^*_X in that neighborhood. The endomorphism $T - 1$ of $\Gamma(\Lambda^*_X)$ then factors through $\Gamma_{\mathfrak{F}}(\Lambda^*_X)$ and supplies the variation. II≈143

Proposition 3.1.10.

Under the isomorphisms of 3.1.8(A),(B), the variation

$$(R\Phi_\eta(\Lambda))_0 \longrightarrow R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda))$$

is equal to the variation of 3.1.9 defined by the automorphism T of V, which is the identity on ∂V.

The reader is invited to draw a small picture rather than read the proof. Let ${}_fB$ be the closed box of 3.1.2 and let ${}_fC \subset {}_fB$ be a collar along the boundary; thus ${}_fC \simeq \partial_f B \times [0,1]$ as a bundle over D_t, through an isomorphism carrying $\partial_f B \times \{1\}$ identically onto $\partial_f B$. Let

$${}_f\alpha : {}_fC \xrightarrow{\sim} (\partial V \times [0,1]) \times D_t$$

be a trivialization of the bundle ${}_fC$ over D_t whose fiber at $y \in D_t$ induces the identity

$$\partial V = \partial(f^{-1}(y)) \xrightarrow{\sim} \partial V \times \{1\}.$$

Put $C = {}_f C \cap B$ and let

$$\alpha : C \xrightarrow{\sim} (\partial V \times [0,1)) \times D_t$$

be induced by ${}_f\alpha$. We retain the notation of (3.1.8.2).

Let R be the equivalence relation on $\overline{B}$ defined by xRy if either $x = y$, or $p(x), p(y) \in C$ and $\alpha(p(x))$, $\alpha(p(y))$ have the same image in $\partial V \times [0,1)$. Put

$${}_1\overline{B} = \overline{B}/R, \qquad {}_1\overline{B}^* = \overline{B}^*/R, \qquad {}_1\overline{C}^* = p^{-1}(C)/R.$$

There is a commutative diagram

$$\begin{array}{ccccc} B_0 & \xhookrightarrow{\bar{\imath}} & \overline{B} & \xhookleftarrow{\bar{\jmath}} & \overline{B}^* \\ \| & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle q'} \\ B_0 & \xhookrightarrow{\bar{\imath}} & {}_1\overline{B} & \xhookleftarrow{{}_1\bar{\jmath}} & {}_1\overline{B}^* \end{array}$$

Let Λ^* be a $\pi_1(D^*)$-equivariant resolution of Λ by acyclic sheaves on $\overline{B}^*$. By definition, the variation is computed from the morphism of complexes

$$\Lambda[0] \longrightarrow \bar{\imath}^* \bar{\jmath}_*(\Lambda^*).$$

Since $\Lambda \xrightarrow{\approx} Rq_*\Lambda$ and $\Lambda \xrightarrow{\approx} Rq'_*\Lambda$, it may equally be computed using an equivariant resolution of Λ on ${}_1\overline{B}^*$ and the morphism

$$\Lambda[0] \longrightarrow \bar{\imath}^* {}_1\bar{\jmath}_*(\Lambda^*).$$

Under (A) and (B), the variation is thereby identified with the composite morphism

$$\begin{array}{ccccc} R\Gamma(V,\Lambda) & \xrightarrow{\sim} & R\Gamma(V^\circ,\Lambda) & \xleftarrow[①]{\sim} & R\Gamma({}_1\overline{B}^*,\Lambda) \\ & & & & \downarrow{\scriptstyle \mathrm{Var}} \\ & & R\Gamma_c(V^\circ,\Lambda) & \xleftarrow[②]{\sim} & R\Gamma({}_1\overline{B}^* \bmod {}_1\overline{C}^*,\Lambda) \end{array} \tag{3.1.10.1}$$

where ① and ② are induced by the inclusion of V° in ${}_1\overline{B}^*$, and where Var is the variation of 3.1.9 for ${}_1\overline{B}^*$ and the action of the positive generator of $\pi_1(D^*)$. This group acts on ${}_1\overline{B}^*$ through the action described in 1.1.5 and acts trivially on ${}_1\overline{C}^*$.

Choose a connection ∇ on B^*/D^* compatible with the trivialization of C. This connection II≈**144**
trivializes $\overline{B}^*$ and, on passing to the quotient, defines a map $u : {}_1\overline{B}^* \to V$. For T defined as in 3.1.3, the diagram

$$\begin{array}{ccc} {}_1\overline{B}^* & \xrightarrow{u} & V \\ {\scriptstyle T}\downarrow & & \downarrow{\scriptstyle T} \\ {}_1\overline{B}^* & \xrightarrow{u} & V \end{array}$$

commutes. The map u induces isomorphisms in cohomology, so Proposition 3.1.10 follows from (3.1.10.1) and the functoriality of variation.

Proposition 3.1.10 is meant to justify an explicit calculation to be made later in which, as here, II **143**
T is the identity on a neighborhood of ∂V. We shall take a cohomology class c represented by a current C and use the fact that $\mathrm{Var}(c)$ is represented by the current $TC - C$, compactly supported in V°.

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3.1.11.

In practice, T is often calculated in two steps.

(a) Fix t and d with $t \leq \tau(d)$, choose $y \in D_t^*$, and choose a connection ∇ on $f^{-1}(S_{|y|}) \cap B_{t,d}$. This connection defines a diffeomorphism T_0 of

$$V_0 = f^{-1}(y) \cap B_{t,d}$$

onto itself, which need not be the identity on the boundary.

(b) Choose a trivialization of $\partial B_{t,d}$ over D_t. Integrating ∇ gives a diffeomorphism $\varphi(\theta)$ from ∂V to

$$f^{-1}(e^{2\pi i\theta} y) \cap \partial B_{t,d},$$

which is identified with ∂V by the trivialization. For $0 \leq \theta \leq 1$, $\varphi(\theta)$ is a path in $\mathrm{Diff}(\partial V)$ joining the identity to ∂T_0.

The resulting pair (T_0, φ) depends on the arbitrary choices made. Since the space of these choices is contractible, it is independent of them up to homotopy. For a transformation T of the kind considered in 3.1.3, $(T, \text{constant path})$ is a pair of the kind considered here.

3.1.12.

The map

$$\mathrm{Diff}(V) \longrightarrow \mathrm{Diff}(\partial V)$$

is a fibration. Consequently, the assignment $T \mapsto (T, \text{constant path})$ induces a bijection between:

a) diffeomorphisms $T : V \to V$ satisfying $\partial T = \mathrm{id}$, up to isotopy; and

b) pairs (T_0, φ), where $T_0 : V \xrightarrow{\sim} V$ and φ is a homotopy class of paths from id to ∂T_0, up to isotopy.

A pair (T_0, φ) calculated by 3.1.11(a),(b) therefore determines T. Concretely, calculate T from (T_0, φ) as follows. II≈146

(α) Choose a collar embedding

$$[0,1] \times \partial V \xhookrightarrow{\alpha} V, \qquad \alpha(1, x) = x,$$

and a representative $\varphi : [0,1] \to \mathrm{Diff}(\partial V)$ such that its extension by id for negative parameters is C^∞. On the collar put

$$\beta(x) = \mathrm{pr}_1\, \alpha^{-1}(x).$$

Thus β has no critical point, takes values in $[0,1]$, and equals 1 on ∂V.

(β) Take $T = T_0$ away from the collar and, on the collar, set

$$T(x) = T_0\big(\varphi(\beta(x))^{-1}(x)\big).$$

3.1.13.

Here is a variant of 3.1.11(b) for calculating the homotopy class of the path φ. On every ray $\arg(z) = \theta$, choose a trivialization of the bundle $\partial B_{t,d}$ which varies continuously with θ. Use the trivialization over the ray through y to identify

$$(\partial V)_0 = f^{-1}(0) \cap \partial B_{t,d}$$

with ∂V. Let

$$\Psi_\theta : V \longrightarrow f^{-1}(e^{i\theta} y) \cap B_{t,d}$$

be obtained by integrating ∇. Transporting $\partial \Psi_\theta$ along the rays gives a diffeomorphism of $(\partial V)_0$, hence of ∂V, onto itself. This is $\varphi(\theta)$.

3.2. The Picard–Lefschetz formula (transcendental case)

3.2.1.

We shall make the results of 3.1 explicit for the map

$$f = \sum_{i=1}^{n+1} z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

For δ take the function $\sum |z_i|^2$, and take the box $B = B_{a,b}$ defined by

$$\left|\sum z_i^2\right| \leq a^2, \qquad \sum |z_i|^2 \leq b^2,$$

where $0 < a < b$. It is only to simplify the notation that we use a closed box rather than a box as in 3.1.2.

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3.2.2.

The manifold of vanishing cycles V has equations

$$\begin{array}{rcl} \sum z_i^2 & = & a^2, \\ \sum |z_i|^2 & \leq & b^2. \end{array} \tag{3.2.2.1}$$

Write $z = (z_i)$ in terms of its real and imaginary parts, $z = x + iy$. The preceding equations become

$$\begin{array}{rcll} x^2 - y^2 & = & a^2, & \\ x \cdot y & = & 0 & (\text{i.e. } x \perp y \text{ in } \mathbb{R}^{n+1}), \\ x^2 + y^2 & \leq & b^2. & \end{array} \tag{3.2.2.2}$$

Introduce on V the new coordinate system

$$X = \frac{x}{|x|}, \qquad Y = y\left(\frac{b^2 - a^2}{2}\right)^{-1/2}.$$

Then $x = X(a^2 + y^2)^{1/2}$, and (3.2.2.2) becomes

$$\begin{array}{rcl} X^2 & = & 1, \\ X \cdot Y & = & 0, \\ Y^2 & \leq & 1. \end{array} \tag{3.2.2.3}$$

This identifies V with the tangent ball bundle of the sphere S^n.

3.2.3.

The map

$$\theta \longmapsto \exp(2\pi i\theta)a$$

identifies $\mathbb{R}$ with the universal covering of the circle S_a in $\mathbb{C}$ having radius a and center 0. The pullback to $\mathbb{R}$ of the bundle B is isomorphic to $\mathbb{R} \times V$. We trivialize it by means of the isomorphism

$$v \longmapsto \exp(\pi i\theta)v$$

from V to the fiber of B over $\exp(2\pi i\theta)a$. Let T_0 be the map furnished by this trivialization as in 3.1.11. In the coordinates (3.2.2.3),

$$T_0(X, Y) = (-X, -Y).$$

3.2.4.

The bundle over the disc D_a of radius a which forms the boundary of the "bundle" B has equations

$$\sum |z_i|^2 = b^2, \qquad f(z) \in D_a,$$

the projection to the disc being f. We shall describe a trivialization of the inverse image of this bundle over each ray

$$e^{2\pi i\theta}\mathbb{R}^+ \cap D_a.$$

Over the ray $\mathbb{R}^+ \cap D_a$, take the isomorphism with $(\mathbb{R}^+ \cap D_a) \times \partial V$ which, in the coordinates (3.2.2.3) on ∂V, is given by II≈148

$$z = x + iy \longmapsto (f(z), X, Y) = \left(f(z), \frac{x}{|x|}, \frac{y}{|y|}\right).$$

Over $(e^{2\pi i\theta}\mathbb{R}^+) \cap D_a$, use the image of this trivialization under multiplication by $e^{\pi i\theta}$. Apply 3.1.13 and use on ∂V the coordinates (X, Y) satisfying

$$X^2 = Y^2 = 1, \qquad X \cdot Y = 0. \tag{3.2.4.1}$$

If $\varphi(\theta)$ takes (X, Y) to (X_θ, Y_θ), one finds

$$X_\theta + iY_\theta = e^{\pi i\theta}(X + iY) \qquad (0 \le \theta \le 1). \tag{3.2.4.2}$$

3.2.5.

In summary, the manifold of vanishing cycles V is the tangent ball bundle of S^n. The monodromy transformation is represented by (T_0, φ), where T_0 is the differential of the antipodal map and φ is the following path from Id to ∂T_0: under $\varphi(\theta)$, each unit tangent vector is parallel-transported along the geodesic which it determines through a distance $+\pi\theta$ (the plus sign means "in the direction in which the vector points"). At $\theta = 1$ one reaches the antipodal vector.

By 3.1.12 this gives the following description of T. Let $\varphi : [0,1] \to [0,1]$ be a C^∞ map which is 0 near 0 and 1 near 1. The map T moves u, along the geodesic which u determines, through the distance

$$-\pi(1 - \varphi(|u|)).$$

Near $|u| = 1$ this is the identity; near $|u| = 0$ it is the antipodal map.

3.2.7.

Let W be a differentiable manifold of dimension d, and let

$$p : T(W) \longrightarrow W$$

be its tangent bundle, regarded as a manifold. If $(x_1, \ldots, x_d)$ is a local coordinate system on W, then

$$(x_1, \ldots, x_d, dx_1, \ldots, dx_d)$$

is a local coordinate system on $T(W)$. Let ε' be the orientation of $T(W)$ defined by this coordinate system. It is independent of the choice of the x_i, and is therefore globally defined.

Suppose that W is orientable. The submanifold $W \subset T(W)$ (the zero section), endowed with an orientation (respectively, with an orientation of its normal bundle), defines an element w of $H_d(T(W))$ (respectively, of $H_c^d(T(W))$). The self-intersection of $W \subset T(W)$ is the Euler–Poincare characteristic: II≈149

$$\int w \wedge w = \chi(W)$$

for the orientation ε'.

For the manifold of vanishing cycles V, its complex structure defines an orientation ε. It is related to the orientation ε' of the tangent bundle of S^n by

$$\varepsilon = (-1)^{\frac{n(n-1)}{2}}\varepsilon'.$$

3.2.8.

The vanishing cycle δ is well defined up to sign. For $n \neq 0$, it is the element of $H_c^n(V^\circ)$ defined by the submanifold $S^n \subset V^\circ$. For $n = 0$, the space V consists of two points, and δ is their difference. With the complex orientation of V, 3.2.7 gives

$$(\delta, \delta) = (-1)^{\frac{n(n-1)}{2}}\big(1 + (-1)^n\big). \tag{3.2.8.1}$$

3.2.9.

Put

$$\widetilde{H}_c^i(V^\circ) = \mathrm{Ker}\Big(H_c^i(V^\circ) \xrightarrow{\mathrm{Tr}} \mathbb{Z}\Big);$$

this is $H_c^i(V^\circ)$ for $i \neq 2n$. Then

$$\begin{aligned} \widetilde{H}_c^i(V^\circ) &= 0 \quad \text{for } i \neq n, \\ \widetilde{H}_c^n(V^\circ) &= \mathbb{Z}, \quad \text{generated by } \delta. \end{aligned}$$

Dually,

$$\begin{aligned} \widetilde{H}^i(V) &= 0 \qquad\qquad \text{for } i \neq n, \\ \widetilde{H}^n(V) &\xrightarrow[\sim]{(\,\cdot\,,\delta)} \mathbb{Z}. \end{aligned}$$

For $n \neq 0$, the image of $\delta \in H_c^n(V^\circ)$ in $H^n(V)$ is zero when n is odd, and is divisible by two (and no further) when n is even.

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3.2.10.

Let us calculate the variation

$$\mathrm{Var} : \widetilde{H}^n(V) \longrightarrow H_c^n(V^\circ).$$

We first use the orientation ε' and suppose that $n \neq 0$. Let α be an orientation of the tangent bundle of S^n, and let $\delta \in H_c^n(V^\circ)$ be defined by $S^n \subset V^\circ$ together with the orientation α of its normal bundle. The cocycle $\delta' \in H^n(V)$ such that $(\delta', \delta) = 1$ is represented by a fiber of the fibration of V° over S^n. The normal bundle of this fiber is the constant bundle with value the tangent space at one point of S^n, and it is endowed with the orientation α.

This assertion is independent of the sign convention chosen to define the map

$$(\text{submanifold} + \text{orientation of its normal bundle}) \longrightarrow \text{cohomology}.$$

Indeed, changing the convention changes δ to $-\delta$ and δ' to $-\delta'$. The assertion is valid when

$$\mathrm{Tr} : H_c^{2n}(V, \mathbb{R}) \longrightarrow \mathbb{R}$$

is defined as follows:

(α) since Ω_V^* is a soft resolution of $\underline{\mathbb{R}}$, $\Gamma_c(\Omega_V^*)$ represents $R\Gamma_c(V, \mathbb{R})$, whence an isomorphism

$$H^i\big(\Gamma_c(V, \Omega_V^*)\big) \simeq H_c^i(V, \mathbb{R})$$

(the iterated coboundary is not used);

(β) Tr is integration.

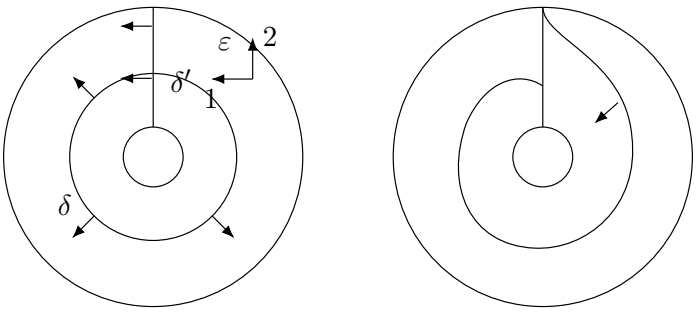
The class $T(\delta')$ coincides with δ' in a neighborhood of ∂V, so that $\mathrm{Var}(\delta') = T(\delta') - \delta'$ has compact support in V°. We have

$$T(\delta') - \delta' \sim -\delta, \qquad \text{and hence}$$

$$\mathrm{Var}(\delta') = -(\delta', \delta)\,\delta.$$

Here is the picture for $n = 1$. We draw the tangent bundle of S^1 so that vectors pointing in the chosen counterclockwise orientation of S^1 point outward.

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δ, δ' and ε $\qquad$ $\mathrm{Var}(\delta')$

For the orientation ε, we therefore have

$$\mathrm{Var}(x) = -\frac{\varepsilon}{\varepsilon'}(x, \delta)\,\delta.$$

Theorem 3.2.11.

The group $\widetilde{H}^n_c(V^\circ)$ is isomorphic to $\mathbb{Z}$, generated by the vanishing cycle δ. The group $\widetilde{H}^n(V)$ is also isomorphic to $\mathbb{Z}$ and is dual to the preceding group. All the other groups $\widetilde{H}^i_c(V^\circ)$ and $\widetilde{H}^i(V)$ vanish.

The variation is given by

$$\mathrm{Var}(x) = -(-1)^{\frac{n(n-1)}{2}}(x, \delta)\,\delta, \qquad \text{and}$$

$$(\delta, \delta) = (-1)^{\frac{n(n-1)}{2}}(1 + (-1)^n).$$

According to the congruence class of n modulo 4, we have the following table:

n mod 4	0	1	2	3
(δ, δ)	2	0	-2	0
$\mathrm{Var}(x) = \pm(x, \delta)\delta$	$-$	$-$	$+$	$+$
$T(\delta) = (-1)^{n+1}\delta$	$-\delta$	δ	$-\delta$	δ

As a check, we indeed have

$$T(\delta) = \delta + \mathrm{Var}(\delta).$$

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4. De Rham cohomology

4.1.

Let D be a complex analytic disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $\widetilde{D}^*$ be a universal covering of D^*. Let X be an analytic space, let $f : X \longrightarrow D$ be a morphism, and consider the diagram of

(1.2.7.1):

(4.1.1)
$$\begin{array}{ccccc} X_0 & \xrightarrow{\bar{i}} & \bar{X} & \xleftarrow{\bar{j}} & \bar{X}^* \\ \| & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ X_0 & \xrightarrow{i} & X & \xleftarrow{j} & X^* \end{array}$$

Suppose that $X^* = X - X_0$ is nonsingular (that is, smooth). Its covering $\bar{X}^*$ is therefore smooth. Denote by $\Omega^*_{\bar{X}^*}$ the De Rham complex of sheaves of holomorphic differential forms on $\bar{X}^*$. It is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincare lemma), and hence gives a morphism

(4.1.2)
$$\bar{j}_*\Omega^*_{\bar{X}^*} \longrightarrow R\bar{j}_*\mathbb{C}$$

in $D^+(\bar{X}, \mathbb{C})$.

Proposition 4.2.

The morphism (4.1.2) is an isomorphism.

It is clear that (4.1.2) is a quasi-isomorphism at every point of $\bar{X}^* \subset \bar{X}$. It is therefore enough to show that every point $x \in X_0$ has a fundamental system of neighborhoods U in $\bar{X}$ such that

$$H^q(\bar{j}^{-1}(U), \Omega^p) = 0 \qquad \text{for } q > 0.$$

As V runs through a fundamental system of neighborhoods of x in X, the sets $p^{-1}(V)$ form a fundamental system of neighborhoods of x in $\bar{X}$. In particular, one may take the $p^{-1}(V)$ for V a Stein neighborhood of x in X. If V is Stein, then $V - X_0$ is still Stein (since $X_0 \cap V$ is defined by one equation in V), as is its covering

$$p'^{-1}(V - X_0) = \bar{j}^{-1}p^{-1}(V).$$

Consequently, for every coherent sheaf $\mathfrak{F}$ on $\bar{X}^*$,

$$H^q(\bar{j}^{-1}p^{-1}(V), \mathfrak{F}) = 0 \qquad \text{for } q > 0,$$

which proves the proposition.

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Corollary 4.3.

The isomorphism (4.1.2) induces an isomorphism

(4.3.1)
$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*} \xrightarrow{\sim} R\Psi_\eta(\mathbb{C}).$$

Variant 4.4.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a complex local system). The quasi-isomorphism

$$V \longrightarrow \Omega^*_{X^*}(V)$$

of the Poincare lemma induces a quasi-isomorphism

(4.4.1)
$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}(p'^*V) \xrightarrow{\sim} R\Psi_\eta(V).$$

We shall show that (4.3.1) remains true when $\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}$ is replaced by a subcomplex defined by growth and monodromy conditions.

4.5.

Let $f : X^* \longrightarrow D^*$ be a topological space over D^*, let F be a sheaf of $f^\circ\mathcal{O}_{D^*}$-modules on X^*, and let $\bar{F}$ be its inverse image on

$$\bar{X}^* = X^* \times_{D^*} \widetilde{D}^*.$$

Let $z : D \hookrightarrow \mathbb{C}$ be a holomorphic embedding with $z(0) = 0$. We shall always be free to shrink D below; for a sufficiently small disc the existence of z is trivial. Choose on $\widetilde{D}^*$ a function $\log z$ such that $\exp(\log z) = z$, and put

$$z^\alpha = \exp(\alpha \log z) \qquad (\alpha \in \mathbb{C}).$$

The group $\pi_1(D^*)$ acts, by transport of structure, on $H^0(\bar{X}^*, \bar{F})$. An element $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if its transforms under $\pi_1(D^*)$ span a finite-dimensional complex vector space $V(f)$. Let T be the positive generator of $\pi_1(D^*)$. The section f is of finite determination if and only if there is a nonzero polynomial $P \in \mathbb{C}[X]$ such that $P(T)(f) = 0$. The polynomial of least degree with this property is then both the minimal polynomial and the characteristic polynomial of T acting on $V(f)$.

We say that f is of unipotent finite determination if there is an A such that II≈154

$$(T-1)^A f = 0.$$

On the space of these sections define

$$N = \frac{\log T}{2\pi i}$$

by the finite sum

$$N(f) = -\frac{1}{2\pi i} \sum_{n>0} \frac{(1-T)^n}{n} f.$$

If f (respectively, g) is a section of unipotent finite determination in $H^0(D^*, \mathcal{O})$ (respectively, in $H^0(\bar{X}^*, \bar{F})$), then

$$N(fg) = N(f)g + fN(g),$$

since the logarithm of an automorphism is a derivation.

We say that f is of quasi-unipotent finite determination if there are A and $B > 0$ such that

$$(T^B - 1)^A f = 0.$$

Examples 4.6.

For $X = D^*$ and $F = \mathcal{O}$, one has

$$\big(T - \exp(2\pi i\alpha)\big)^{k+1} (z^\alpha (\log z)^k) = 0,$$

$$N(\log z) = 1,$$

and

$$N\left(\frac{(\log z)^k}{k!}\right) = \frac{(\log z)^{k-1}}{(k-1)!}, \qquad (0 \text{ for } k = 0).$$

Proposition 4.7.

A section $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if and only if it can be written

$$f = \sum_{\alpha,k} f_{\alpha,k}\, z^\alpha \frac{(\log z)^k}{k!}, \tag{4.7.1}$$

where $0 \le \mathrm{Re}(\alpha) < 1$, $k \in \mathbb{N}$, and the $f_{\alpha,k} \in H^0(X^*, F)$ are almost all zero. For f of finite determination, the decomposition (4.7.1) is unique.

By 4.6, every section of the form (4.7.1) is of finite determination. Conversely, let f be of finite determination, with $P(T)(f) = 0$. Write II ≈155

$$P(X) = \prod_{\beta\in\mathbb{C}} (X-\beta)^{n'(\beta)} = \prod_{0\le \mathrm{Re}(\alpha)<1} \left(X - \exp(2\pi i\alpha)\right)^{n(\alpha)},$$

where $n(\alpha) = n'(\exp(2\pi i\alpha))$ is almost always zero. Decomposing the $\mathbb{C}[X]$-module $V(f)$ into its isotypic components, one finds a unique decomposition

$$f = \sum_\alpha f'_\alpha$$

such that

$$\left(T - \exp(2\pi i\alpha)\right)^{n(\alpha)} f'_\alpha = 0.$$

Choose a polynomial $Q_\alpha \in \mathbb{C}[X]$ congruent to 1 modulo $\left(X - \exp(2\pi i\alpha)\right)^{n(\alpha)}$ and to 0 modulo $\left(X - \exp(2\pi i\alpha')\right)^{n(\alpha')}$ for every $\alpha' \neq \alpha$. Then $f'_\alpha = Q_\alpha(T)(f)$. Put

$$f_\alpha = z^{-\alpha} f'_\alpha = z^{-\alpha} Q_\alpha(T)(f). \tag{4.7.2}$$

Since

$$T(z^{-\alpha} f'_\alpha) = \exp(-2\pi i\alpha) z^{-\alpha} T(f'_\alpha),$$

we have

$$(T-1)^{n(\alpha)} f_\alpha = 0.$$

If

$$f_\alpha = \sum_k f_{\alpha,k} \frac{(\log z)^k}{k!}, \qquad f_{\alpha,k} \in H^0(X^*, F),$$

then 4.6 and the identity $\sum_j (-1)^j \binom{k}{j} = \delta_{0,k}$ give

$$f_{\alpha,k} = \sum_{j\ge 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+k} f_\alpha, \tag{4.7.3}$$

which proves uniqueness.

Conversely, if f_α is a section of unipotent finite determination, the $f_{\alpha,k}$ defined by (4.7.3) vanish for large k (because $N^k f_\alpha = 0$ for large k) and are invariant under T; hence they identify with sections of F on X^*. Equivalently, we must show that $N f_{\alpha,k} = 0$. Apply to $N^k f_\alpha$ the identity

$$\begin{aligned} N \sum_{j\ge 0} (-1)^j \frac{(\log z)^j}{j!} N^j &= \sum_{j>0} (-1)^j \frac{(\log z)^{j-1}}{(j-1)!} N^j \\ &\quad + \sum_{j\ge 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+1} = 0. \end{aligned}$$

Finally,

$$\begin{aligned} \sum_{k\ge 0} f_{\alpha,k} \frac{(\log z)^k}{k!} &= \sum_{j,k\ge 0} (-1)^j \frac{(\log z)^{j+k}}{j!k!} N^{j+k} f_\alpha \\ &= \sum_{n\ge 0} \left(\sum_j (-1)^j \binom{n}{j} \right) \frac{(\log z)^n}{n!} N^n f_\alpha = f_\alpha, \end{aligned}$$

which proves 4.7.

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Remark 4.8.1.

The condition $\mathrm{Re}(\alpha) \in [0,1)$ in 4.7 may be replaced by $\alpha \in C$, where C is any fundamental domain for $\mathbb{C} \longrightarrow \mathbb{C}/\mathbb{Z}$.

Remark 4.8.2.

Let f be of finite determination, with decomposition (4.7.1). For $0 \leq \mathrm{Re}(\alpha) < 1$ and for α such that some $f_{\alpha,k} \neq 0$, put

$$m(\alpha) = 1 + \sup\{k \mid f_{\alpha,k} \neq 0\}.$$

If all the $f_{\alpha,k}$ vanish, put $m(\alpha) = 0$. The characteristic polynomial of T acting on $V(f)$ is

$$\mathrm{Pc}(T, V(f)) = \prod_{\alpha} (X - \exp(2\pi i \alpha))^{m(\alpha)}.$$

In particular, f is of quasi-unipotent finite determination if and only if the α in (4.7.1) may all be taken in $\mathbb{Q}$.

4.9.

Return to the more restrictive hypotheses of 4.1, and let F be a coherent sheaf on X whose restriction to X^* is locally free.

Proposition 4.10.

Let f be a section of finite determination of the inverse image $\bar{F}$ of F on $\bar{X}^*$. Then f has moderate growth along X_0 if and only if the $f_{\alpha,k}$ in the decomposition (4.7.1) are meromorphic along X_0.

For the definition of "moderate growth," we refer to [1], 2.10. It is clear from (4.7.1), (4.7.2), and (4.7.3) that f has moderate growth if and only if the $f_{\alpha,k}$ do. For sections of F on X^*, moderate growth along X_0 is known to be equivalent to being meromorphic along X_0.

For $X = D$ and $F = \mathcal{O}$, 4.10 has the following elementary corollary. II≈157

Corollary 4.11.

Let K be the field of germs at 0 of meromorphic functions on D, and let K^m (respectively, K^{mu} and K^{mqu}) be the K-algebra of germs at 0 of holomorphic functions on $\widetilde{D}^*$ which have moderate growth at 0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Then the following are bases over K:

$$\begin{aligned} K^m: &\quad z^\alpha (\log z)^k, \quad 0 \leq \mathrm{Re}(\alpha) < 1,\ k \in \mathbb{N}, \\ K^{mu}: &\quad (\log z)^k, \quad k \in \mathbb{N}, \\ K^{mqu}: &\quad z^\alpha (\log z)^k, \quad \alpha \in \mathbb{Q},\ 0 \leq \alpha < 1,\ k \in \mathbb{N}. \end{aligned}$$

Under the hypotheses of 4.9, denote by $\Psi_\eta^m(F)$ (respectively, $\Psi_\eta^{mu}(F)$ and $\Psi_\eta^{mqu}(F)$) the subsheaf of $\bar{i}^* \bar{j}_* \bar{F}$ consisting of sections of $\bar{F}$, near points of X_0, which have moderate growth along X_0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Let

$$F^m \subset j_* j^* F$$

be the sheaf on X of sections of $j_* j^* F$ which are meromorphic along X_0. Proposition 4.10 has the following corollary, independent of the arbitrary choices of z and $\log z$.

Corollary 4.12.

The natural morphisms

$$\text{(i)} \qquad K^m \otimes_K i^*F^m \longrightarrow \Psi_\eta^m(F),$$
$$\text{(ii)} \qquad K^{mu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mu}(F),$$
$$\text{(iii)} \qquad K^{mqu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mqu}(F)$$

are isomorphisms.

The main result of this section is the following.

Theorem 4.13.

The natural morphism induced by (4.3.1),

$$\Psi_\eta^{mqu}(\Omega_X^*) \longrightarrow R\Psi_\eta(\mathbb{C}), \tag{4.13.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

We shall in fact prove the more general results 4.15 and 4.17. Their statements require a few preliminaries which a reader interested only in 4.13 may omit. II≈158

4.14.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a local system). Let

$$\mathcal{V} = V \otimes_{\mathbb{C}} \mathcal{O}$$

be the corresponding locally free $\mathcal{O}$-module, and let ∇ be the integrable connection on $\mathcal{V}$ for which the local sections of V are horizontal. It is known that $\mathcal{V}$ admits an extension $\widetilde{\mathcal{V}}$ to a coherent sheaf on X such that ∇ is regular along X_0 relative to this extension. It is also known that any two such extensions are meromorphically equivalent ([1], II, 5.7). Thus the sheaf $\Psi_\eta^m(\widetilde{\mathcal{V}})$, and its variants Ψ^{mu} and Ψ^{mqu}, does not depend on the chosen extension; we denote it by $\Psi_\eta^m(\mathcal{V})$. Likewise define

$$\Psi_\eta^m(\Omega_{X^*}^p(V)) = \Psi_\eta^m(\Omega_X^p \otimes \mathcal{V}),$$

and its variants Ψ^{mu} and Ψ^{mqu}.

Proposition 4.15.

The natural morphism induced by (4.4.1),

$$\Psi_\eta^m(\Omega_{X^*}^*(V)) \longrightarrow R\Psi_\eta(V), \tag{4.15.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

4.16.

A local system V as in 4.14 is said to be unipotent (respectively, quasi-unipotent) along X_0 if, for every morphism u from the standard disc D_0 to X such that $u^{-1}(X_0) = \{0\}$, the monodromy transformation of the local system u^*V on D_0^* is unipotent (respectively, quasi-unipotent).

Proposition 4.17.

If V is quasi-unipotent, the natural morphism induced by (4.4.1),

$$\Psi_\eta^{mqu}(\Omega_{X^*}^*(V)) \longrightarrow R\Psi_\eta(V), \tag{4.17.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

Theorem 4.13 is the special case of 4.17 in which $V = \underline{\mathbb{C}}$. II 157 II≈159

4.18.

We shall first treat the case in which X is smooth and $(X_0)_{\mathrm{red}}$ is a normal-crossings divisor in X. Since the theorems are local, we may suppose that f is the map

$$f(\underline{z}) = \prod_{i=1}^{d} z_i^{n_i}, \qquad 0 < d \leq e, \quad n_i > 0,$$

from $X = D^e$ to D, where this time D denotes the standard disc $D = \{z \in \mathbb{C} \mid |z| < 1\}$. It is enough to prove that the fiber at $0 \in D^e$ of (4.15.1) (respectively, (4.17.1)) is a quasi-isomorphism.

Let U be a neighborhood of 0 in D^e of the form $|z_i| < a$, where $0 < a \leq 1$. Define U_0, $\bar{U}$, U^*, and $\bar{U}^*$ by taking the inverse image of U in (4.1.1). The cohomology of U^* or $\bar{U}^*$ is independent of a, and hence

$$H^*(\bar{U}^*, V) \xrightarrow{\sim} \big(R^*\Psi_\eta(V)\big)_0.$$

The fundamental group $\mathbb{Z}^d$ of $X^* = (D^*)^d \times D^{e-d}$ (which is also that of U^*) is abelian. Therefore V has a finite filtration whose successive quotients are rank-one local systems. By devissage, we may thus suppose that V has rank one.

For $\widetilde{\mathcal{V}}$ as in 4.14, write $j_*^m\Omega_{X^*}^*(V)$ for the subcomplex of $j_*\Omega_{X^*}^*(V)$ whose degree-p term is $(\Omega_X^p \otimes \widetilde{\mathcal{V}})^m$. By [1], II 6.4.1 and 6.8, the natural map induced by the quasi-isomorphism $V \xrightarrow{\approx} \Omega_{X^*}^*(V)$ (the holomorphic Poincare lemma),

$$j_*^m\Omega_{X^*}^*(V) \longrightarrow Rj_*V, \tag{4.18.1}$$

is a quasi-isomorphism. Thus

$$H^*\big((j_*^m\Omega_{X^*}^*(V))_0\big) \xrightarrow{\sim} (R^*j_*V)_0 \xleftarrow{\sim} H^*(U^*, V). \tag{4.18.2}$$

This will be the only analytic ingredient in this part 4.18 of the proof.

Lemma 4.18.3.

For a rank-one V, one has

(i) if V is nontrivial: $H^*(U^*, V) = 0$,

(ii) if $V = \underline{\mathbb{C}}$: $H^*(U^*, V) = \overset{*}{\Lambda}\mathbb{C}^d$. II≈160

Indeed, U^* has the homotopy type of a torus.

Let η be the generator of $H^1(D^*, \mathbb{Z})$, and also its inverse image in $H^1(U^*, \mathbb{Z})$.

Lemma 4.18.4.

For a rank-one V, the natural map

$$H^*(U^*, V) \xrightarrow[(4.18.1)]{\sim} H^*\big((j_*^m\Omega_{X^*}^*(V))_0\big) \longrightarrow H^*\big(\Psi_\eta^{mu}(\Omega_{X^*}^*(V))_0\big)$$

identifies $H^*(\Psi_\eta^{mu}(\Omega_{X^*}^*(V))_0)$ with $H^*(U^*, V)/\eta H^*(U^*, V)$.

Use 4.12(ii) and the identity $d\log f = \sum_i n_i\, dz_i/z_i$. The complex $\Psi_\eta^{mu}(\Omega_{X^*}^*(V))_0$ identifies with the simple complex $\underline{\underline{s}}K$ associated (using direct sums) with the double complex K whose terms are

$$K^{-p,n+p} = \frac{(\log f)^p}{p!} \cdot (j_*^m\Omega_{X^*}^n(V))_0.$$

This complex lies in a bad quadrant ($p \leq 0, q \geq 0, p+q \geq 0$). Nevertheless, $\underline{\underline{s}}K$ is the inductive limit of the subcomplexes $\underline{\underline{s}}\,\sigma_{p \geq p_0}(K)$ whose terms are

$$(\underline{\underline{s}}\,\sigma_{p\geq p_0}K)^n = \bigoplus_{\substack{p+q=n \\ p\geq p_0}} K^{pq},$$

and this suffices to justify the argument below.

The spectral sequence for the filtration by the first degree is

$$E_1^{-p,n+p} = H^n(U^*, V) \qquad (p \geq 0),$$

with

$$d_1 = 2\pi i\, \eta\wedge : H^n(U^*, V) \longrightarrow H^{n+1}(U^*, V).$$

If V is nontrivial, then $E_1 = 0$ by 4.18.3, and 4.18.4 reduces to $0 = 0$. Suppose that $V = \underline{\mathbb{C}}$. The complexes (E_1, d_1) are then Koszul complexes, possibly truncated, so that, by acyclicity of the Koszul complex,

$$E_2^{pq} = 0 \qquad \text{for } p \neq 0,$$

and

$$E_2^{0n} = H^n(U^*, V)/\eta H^{n-1}(U^*, V),$$

which proves 4.18.4.

For $\beta \in \mathbb{C}^*$, let V_β denote the rank-one local system on D^* whose inverse image on $\widetilde{D}^*$ is trivialized and whose monodromy is β. We use the same notation for its inverse image on U^*. On $\bar{U}^*$, the inverse image of $V \otimes V_\beta$ is canonically isomorphic to that of V. II≈161

Lemma 4.18.5.

For a rank-one V, the map

$$\bigoplus_{\beta\in\mathbb{C}^*} H^*(U^*, V \otimes V_\beta)/\eta H^*(U^*, V \otimes V_\beta) \longrightarrow H^*(\bar{U}^*, V)$$

is bijective.

Let k be the greatest common divisor of the n_i. The space $\bar{U}^*$ has k connected components, indexed by the kth roots of unity. On the component $\bar{U}^*_\xi$ indexed by ξ, the inverse image of the function $z^{1/k}$ on $\widetilde{D}^*$ coincides with the inverse image of the function

$$\xi \prod_{i=1}^{d} z_i^{n_i/k}$$

on U^*. The component $\bar{U}^*_\xi$ has the homotopy type of a torus, and the sequence

$$0 \longrightarrow \pi_1(\bar{U}^*_\xi) \longrightarrow \pi_1(U^*) \longrightarrow \pi_1(D^*)$$

is exact. Write $\varphi : \mathbb{Z}^d \to \mathbb{Z}$ for the map $\pi_1(U^*) \to \pi_1(D^*)$. If $\ker(\varphi)$ acts nontrivially on V, both sides of 4.18.5 vanish. Otherwise, after replacing V by a suitable $V \otimes V_\beta$, we may suppose that V is trivial, that is, $V = \mathbb{C}$. In complex cohomology we then have

$$H^0(\bar{U}^*) \otimes H^*(U^*)/\eta H^*(U^*) \xrightarrow{\sim} H^*(\bar{U}^*).$$

The result follows on observing that V_β is trivial on U^* when $\beta^k = 1$, that

$$\bigoplus_{\beta^k=1} H^0(U^*, V_\beta) = \bigoplus_{\beta^k=1} \mathbb{C} \xrightarrow{\sim} H^0(\bar{U}^*, \mathbb{C}),$$

and that $H^*(U^*, V_\beta) = 0$ when $\beta^k \neq 1$.

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Remark 4.18.6.

If V is quasi-unipotent, the sum over $\mathbb{C}^*$ in 4.18.5 may be replaced by a sum over the roots of unity; the proof is the same.

Applying 4.12 gives a commutative diagram

$$\begin{array}{ccccc} H^*\Psi_\eta^m(\Omega^*_{X^*}(V))_0 & \xleftarrow[4.12]{\sim} & \bigoplus_\beta H^*\Psi_\eta^{mu}(\Omega^*_{X^*}(V\otimes V_\beta)) & \xleftarrow[4.18.4]{\sim} & \bigoplus_\beta H^*(U^*, V\otimes V_\beta)/\eta H^*(U^*, V\otimes V_\beta) \\ \downarrow & & & & \wr \downarrow 4.18.5 \\ H^*R\Psi(V)_0 & \xleftarrow{\qquad\qquad\sim\qquad\qquad} & & & H^*(\bar{U}^*, V) \end{array}$$

from which 4.15 follows. Proposition 4.17 is obtained by replacing Ψ^m with Ψ^{mqu}, the sum over β with a sum over the roots of unity, and 4.18.5 with 4.18.6.

4.19.

Let us deduce 4.15 and 4.17 from the special case 4.18. Let $u : Y \longrightarrow X$ be a resolution of (X, X_0), that is, a proper morphism inducing an isomorphism $Y^* \xrightarrow{\sim} X^*$, such that Y is smooth and $(Y_0)_{\mathrm{red}}$ is a normal-crossings divisor in Y. If V is quasi-unipotent along X_0, its inverse image u^*V is quasi-unipotent along Y_0.

For every coherent sheaf F on X, denote by F^m, or by $j_*^m j^*F$, the subsheaf of j_*j^*F formed by the meromorphic sections. If f is an equation for X_0, then $j_*^m j^*F$ is the inductive limit of the system

$$F \xrightarrow{f} F \xrightarrow{f} F \longrightarrow \cdots . \tag{4.19.1}$$

The following lemma is proved in [2].

Lemma 4.19.2.

For every coherent sheaf F on X, one has

$$F^m \xrightarrow{\sim} u_*(u^*F)^m$$

and $R^iu_*(u^*F)^m = 0$ for $i > 0$.

By Grauert's theorem, the $R^iu_*u^*F$ are coherent sheaves. The kernels and cokernels of $F \longrightarrow u_*u^*F$, as well as the $R^iu_*u^*F$ for $i > 0$, vanish outside X_0 and are therefore locally annihilated by a power of f. Since u is proper, R^iu_* commutes with filtered inductive limits, in particular with the inductive limit (4.19.1), and (4.19.2) follows. II≈163

From 4.19.2 and 4.12 one deduces that, under the hypotheses just stated and still denoting by u the map $Y_0 \to X_0$ induced by u, one has the following.

Lemma 4.19.3.

The morphisms

$$\Psi_\eta^m(F) \longrightarrow u_*\Psi_\eta^m(u^*F) \longrightarrow Ru_*\Psi_\eta^m(u^*F)$$

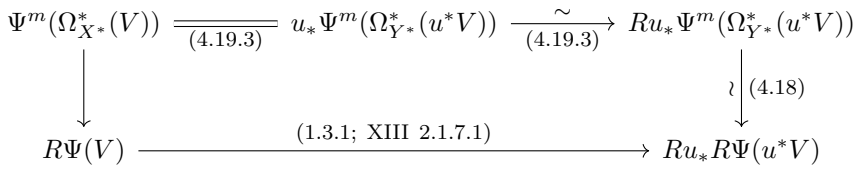
are isomorphisms.

That the second arrow is an isomorphism means that

$$R^iu_*\Psi_\eta^m(u^*F) = 0 \qquad (i > 0).$$

This follows from 4.12 and from $R^iu_*(i^*u^*F^m) = 0$ for $i > 0$, which in turn follows from $R^iu_*(u^*F^m) = 0$ for $i > 0$ and the proper base-change theorem for u.

To prove 4.15, it remains to consider the commutative diagram



The proof of 4.17 is the same, with Ψ^m replaced everywhere by Ψ^{mqu}.

Remark 4.20.

Theorem 4.13 gives an incarnation of the monodromy theorem at the level of complexes.

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SGA 7 — EXPOSÉ XV

The Picard–Lefschetz Formula

by P. Deligne

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1. Ordinary quadratic singularities (normal forms)

In this section we take up the results of VI 6 again, adopting a "henselian" rather than a "formal" point of view. We give normal forms for ordinary quadratic singularities and their deformations, into which they may be put locally for the etale topology.

These results follow easily from VI 6, the implicit-function theorem (in Tougeron's form), and R. Elkik's theorem on versal henselian moduli of isolated singularities. At the time of the oral seminar, the latter was not yet available; a weaker theorem, covering only the case of complete intersections (which was enough for us), had been proved and will not be published.

1.1. Reminders

1.1.1.

Let $f : X \longrightarrow S$ be a morphism, with X locally describable by p equations in relative affine space $\mathbb{A}_S^{n+p}$ of dimension $n+p$ over S. The Jacobian ideal $J^n(X/S)$ was defined in VI 5. If X is the closed subscheme of $\mathbb{A}_S^{n+p}$ defined by the equations $f_i = 0$ $(1 \leq i \leq p)$, it is the ideal sheaf generated by the $p \times p$ minors of the Jacobian matrix $(\partial f_i/\partial x_j)$. The following theorem is well known. A special case of its formal analogue is proved in Bourbaki, *Algèbre commutative*, III, §4, no. 6, Corollary 3 to Theorem 2; the C^∞ analogue is proved in Tougeron, *Idéaux de fonctions différentiables* (thesis, Rennes, May 1967); the case considered here is treated in Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHES 36 (Zariski's volume), Lemma 5.10.

Theorem 1.1.2 (Tougeron–Artin).

Let S be the spectrum of a henselian local ring A, let $f : X \to S$ be a morphism as in 1.1.1, let a and I be two proper ideals of A, and let δ' be an ideal sheaf of X contained in $J^n(X/S)$. Suppose that I is of finite type. Let s be a section of f over S/I, that is, $s \in X(A/I)$. Denote by $s^*\delta'^2$ the ideal of A inverse to the ideal $s^*\delta'^2$ of A/I.

If $(s^*\delta'^2)\,a \supset I$, then there is a section $s_1 \in X(A)$ of f congruent to s modulo $(s^*\delta')\,a$.

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1.1.3.

Let S be the spectrum of a henselian local ring, with closed point s. Let X_0' be a scheme of finite type over $k(s)$, smooth away from a closed point x_0, and let X_0 be the henselization of X_0' at x_0. We are interested in systems

$$\begin{array}{ccccc} X_0 & \hookrightarrow & X & & \\ \downarrow & & \downarrow & & \\ s & \hookrightarrow & T & \longrightarrow & S \end{array}$$

of the following kind:

(a) T is henselian local and essentially of finite presentation over S (I 0.0.5); the morphism $T \to S$ is local and induces an isomorphism of residue fields;

(b) X is henselian local, flat, and essentially of finite presentation over T;

(c) the special fiber of X/T is equipped with a $k(s)$-isomorphism to X_0.

These are the henselian S-deformations of X_0. A morphism from (X,T) to (X',T') is a commutative diagram of S-schemes

$$\begin{array}{ccc} X & \longrightarrow & X' \\ \downarrow & & \downarrow \\ T & \longrightarrow & T' \end{array}$$

which induces the identity of X_0 after passage to the special fibers. A deformation X/T is called versal (minimally versal) if:

(a) for every morphism $X''/T'' \longrightarrow X'/T'$ for which $T'' \to T'$ is a closed immersion, the map

$$\mathrm{Hom}(X'/T', X/T) \longrightarrow \mathrm{Hom}(X''/T'', X/T)$$

is surjective. In particular, on taking $X''/T'' = X_0/s$, one has $\mathrm{Hom}(X'/T', X/T) \neq \varnothing$;

(b) for $T' = \mathrm{Spec}(k(s)[\varepsilon]/(\varepsilon^2))$, the "inverse-image" map

$$\mathrm{Hom}(T', T) \longrightarrow \{\text{isomorphism classes of deformations of } X_0 \text{ over } T'\}$$

is bijective.

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Theorem 1.1.4 (R. Elkik).

There exists a versal henselian S-deformation of X_0 (and it is unique up to a nonunique isomorphism). If S is noetherian, a deformation X/T is such a deformation if and only if X induces over the completion $\widehat{T}$ of T a versal formal S-deformation of X_0 (a minimally formal versal deformation; VI 1.9 and VI 4).

For the proof of this delicate theorem, we refer to R. Elkik (ENS seminar, 1971/72).

1.2. Ordinary quadratic singularities

Let y be a closed point of a scheme Y of finite type over a field k of characteristic p, and let n be the dimension of Y at y.

Definition 1.2.1 (= VI 6.6).

(i) If k is algebraically closed, y is called an ordinary quadratic point of Y if the completion $\widehat{\mathcal{O}}_{Y,y}$ of the local ring of Y at y is isomorphic to the quotient of $k[[x_1, \ldots, x_{n+1}]]$ by the ideal generated by a single formal power series f which can be written

$$f(\underline{X}) = Q(\underline{X}) + \text{terms of order } > 2, \tag{1.2.1.1}$$

where $Q(\underline{X})$ is an ordinary quadratic form.

(ii) If $\bar{k}$ is an algebraic closure of k, y is called an ordinary quadratic point of Y if the points of $\bar{Y} = Y \otimes_k \bar{k}$ lying over y are ordinary quadratic points of $\bar{Y}$.

1.2.2.

Replacing "ordinary quadratic form" by "nondegenerate quadratic form" in this definition gives the notion of a nondegenerate quadratic point (VI 6.0). A point y is a nondegenerate quadratic point of Y if and only if it is an ordinary quadratic point and either $p \neq 2$ or n is odd.

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Example 1.2.3.

Let $Q(\underline{X})$ be a nondegenerate quadratic form in $n+1$ variables over k. Let Y_0 be the quadratic cone in $\mathbb{A}_k^{n+1}$ with equation $Q = 0$. Then the origin, denoted by y_0, is a nondegenerate quadratic point of Y_0.

Example 1.2.4.

Suppose that $p = 2$ and $n = 2m$. Let $a \in k$, and let Q be a polynomial

$$Q(\underline{X}) = (X_0^2 - a) + \sum_{0<i\leq j\leq 2m} a_{ij} x_i x_j.$$

Suppose that the quadratic form in $2m$ variables $\sum_{0<i\leq j} a_{ij} x_i x_j$ is nondegenerate, and denote by y_0 the closed point of $\mathbb{A}_k^{n+1}$ with coordinates $(\sqrt{a}, 0, \ldots, 0)$. Then y_0 is an ordinary quadratic point of the closed subscheme $Y_0 \subset \mathbb{A}_k^{n+1}$ with equation $Q = 0$.

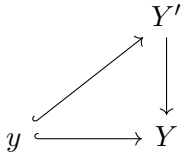
Example 1.2.5.

Let k' be a finite separable extension of k, and let Y_0 be a k'-scheme of the kind considered in 1.2.3 (respectively, 1.2.4). If Y_0 is regarded as a k-scheme, then y_0 is a nondegenerate (respectively, ordinary) quadratic point of Y_0. The extension k'/k identifies with the largest separable subextension of $k(y_0)/k$.

Theorem 1.2.6.

Suppose that y is an ordinary quadratic point of the k-scheme Y, and let k' be the largest separable subextension of $k(y)/k$. There then exist a k'-scheme $Y_0 \subset \mathbb{A}_{k'}^{n+1}$ of the kind in 1.2.3 or 1.2.4 and a k-isomorphism φ between the henselization $Y_{(y)}$ of Y at y and the henselization $Y_{0(y_0)}$ of Y_0 at y_0.

First observe that there is an etale neighborhood Y' of y in Y



which is a k'-scheme. Indeed, $Y_{(y)}$ is canonically a k'-scheme. Replacing (k, Y, y) by (k', Y', y) reduces us to the case $k = k'$, that is, to the case in which $k(y)/k$ is radicial.

The $k(y)$-vector space $(\Omega^1_{Y/k})_y$ has dimension $n+1$. It is enough to check this after extending II≈170 scalars from $k(y)$ to its algebraic closure and completing at y, where it is immediate. A neighborhood of y in Y is therefore a closed subscheme of a smooth k-scheme Z of dimension $n+1$. Moreover, after replacing Y and Z by suitable neighborhoods of y, the ideal in $\mathcal{O}_Z$ defining Y is principal (the same argument as above), say with equation f.

Lemma 1.2.7.

Suppose that y is a nondegenerate quadratic point of Y and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $J^n_{Y/k}$ (1.1.1) is generated by one element, supported at y, and of rank one over k. In particular, $k(y) = k$.

It is enough to prove the first assertion after passing to an algebraic closure of k and completing at y. It is then immediate by direct calculation: with the notation of 1.2.1, Nakayama's lemma shows that the $\partial f/\partial x_i$ (and hence, a fortiori, f and the $\partial f/\partial x_i$) generate the maximal ideal.

Let us prove 1.2.6 when y is nondegenerate. We may and shall suppose that $k(y) = k$, that Y is defined by an equation f in a smooth scheme Z, and that there is an etale morphism

$$x : Z \longrightarrow \mathbb{A}^{n+1}_k$$

(a "system of local coordinates x_i") taking y to 0. There is then a nondegenerate quadratic form Q in $n+1$ variables such that, at y,

$$f = Q(x_i) + \text{terms of order } > 2.$$

Let Y_0 be defined by Q as in 1.2.3.

On Y, the system (x_i) is an "approximate solution" of the equation $Q(x_i) = 0$: if $\mathfrak{m}$ is the ideal defining y, then

$$Q(x_i) \in \mathfrak{m}^3 \qquad \text{on } Y.$$

Since Q is nondegenerate, the ideal of $\mathbb{A}^{n+1}_k$ generated by the $\partial Q/\partial x_i$ is the maximal ideal at 0, and its inverse image under $x = (x_i) : Y \to \mathbb{A}^{n+1}_k$ is $\mathfrak{m}$. By the implicit-function theorem 1.1.2 (applied with $a = \mathfrak{m}$ and $\delta' = J$), there are functions x'_i on the henselization $Y_{(y)}$, congruent to the x_i modulo $\mathfrak{m}^2$, which satisfy $Q(x'_i) = 0$. They give the required isomorphism between $Y_{(y)}$ and $Y_{0(y_0)}$.

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Lemma 1.2.8.

Suppose that y is a degenerate ordinary quadratic point ($p = 2$ and n even), and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $T^1_{Y/k}$ is generated by one element, supported at y, and of rank two over k. In particular, either $k(y) = k$, or $k(y)$ is isomorphic to $k(\sqrt{a})$ for some $a \in k$, $a \notin k^2$.

Proceed as in 1.2.7. With the notation of 1.2.1, it is enough to check that

$$\dim_k \left(k[[x_1, \ldots, x_{n+1}]]/(f, \partial f/\partial x_i)\right) = 2.$$

In suitable coordinates $x_0, x_1, \ldots, x_{2m}$, the formal power series f can be written

$$f = x_0^2 + \sum_{i=1}^{m} x_i x_{i+m} + R, \qquad \operatorname{ord}(R) \geq 3.$$

Let $\mathfrak{n}'$ be the ideal generated by f and the $\partial f/\partial x_i$, let $\mathfrak{n}$ be the ideal generated by x_0^2 and the x_i for $i \neq 0$, and let $\mathfrak{q}$ be the maximal ideal. We prove that $\mathfrak{n} = \mathfrak{n}'$. Plainly $\mathfrak{n}' \subset \mathfrak{n}$, and by Nakayama it is enough to check that

$$\mathfrak{n}' + \mathfrak{n}\mathfrak{q} = \mathfrak{n}.$$

Modulo $\mathfrak{nq}$, one has $f \equiv x_0^2$; modulo $\mathfrak{nq} + (x_0^2)$, one has

$$\frac{\partial f}{\partial x_i} \equiv x_{i+m}, \qquad \frac{\partial f}{\partial x_{i+m}} \equiv x_i \qquad (1 \leq i \leq m),$$

which proves the assertion.

Under the hypotheses of 1.2.8, if $k = k(y)$, we shall let a denote any square in k.

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Lemma 1.2.9.

Let V be a vector bundle on a scheme S, and let T be a closed subscheme of $\mathbb{A}_S(V)$ locally free of rank two over S. Then T is contained in one and only one line in $\mathbb{A}_S(V)$.

The question is local for the fppf topology on S, so we may suppose that T/S has a section. By translation, we may suppose that T contains the zero section. Let $\mathfrak{n}$ be the ideal of T defining 0; it is locally free of rank one over S. Every homogeneous linear form $v \in V$ on $\mathbb{A}_S(V)$ defines an element of $\mathfrak{n}$, and hence an epimorphism $V \twoheadrightarrow \mathfrak{n}$. Its kernel defines the required line.

1.2.10.

In particular, if A is a radicial rank-two closed subscheme of affine space $\mathbb{A}_k^{n+1}$ over a field k of characteristic 2, there is an $a \in k$ such that, in suitable affine coordinates, A has equations

$$x_0^2 - a = 0, \qquad x_i = 0 \quad (1 \leq i \leq n).$$

It is enough to take the x_i which vanish on the line defined by A.

1.2.11.

Let us prove 1.2.6 for a degenerate y ($p = 2$ and $n = 2m$). We may and shall suppose that $k(y) = k(\sqrt{a})$, that Y is defined by an equation f in a smooth scheme Z, and that there is a system of local coordinates at y

$$x = (x_i)_{0 \leq i \leq 2m} : Z \longrightarrow \mathbb{A}_k^{2m+1}$$

taking y to the closed point $(\sqrt{a}, 0, \ldots, 0)$, such that, in a neighborhood of y,

$$J^n_{Y/k} \simeq \mathcal{O}_Y/(x_0^2 - a, x_1, \ldots, x_{2m}).$$

After replacing f by a suitable multiple, one has

$$f = (x_0^2 - a) + \sum_{1 \leq i \leq j \leq 2m} a_{ij} x_i x_j + R,$$

where

(a) the quadratic form in $2m$ variables $\sum a_{ij} x_i x_j$ is nondegenerate;

(b) after extending scalars to $\bar{k}$, R has order at least 3 in $x_0 - \sqrt{a}$ and the x_i.

Let Y_0 be defined, as in 1.2.4, by

$$Q = x_0^2 - a + \sum a_{ij} x_i x_j.$$

On Y, the system (x_i) may again be regarded as an approximate solution of $Q(X_i) = 0$. This time, II≈**173**
the ideal δ, the inverse image under x of the ideal generated by the $\partial Q/\partial X_i$, satisfies $\mathcal{O}_Y/\delta \simeq J^n_{Y/k}$; at y it is

$$\delta = (x_0^2 - a, (x_i)_{1 \leq i}).$$

Let $\mathfrak{q}$ be the maximal ideal defining y. To conclude as in the nondegenerate case by applying the implicit-function theorem, it remains to verify that

$$Q(x_i) \equiv 0 \pmod{\delta^2\mathfrak{q}} \qquad \underline{\text{on } Y}. \tag{1.2.11.1}$$

Let $\bar{\mathfrak{q}}$ be the ideal defining y after scalar extension to an algebraic closure of k. One checks that

$$\delta^2\bar{\mathfrak{q}} \cap \mathcal{O}_{Y,y} = \delta^2\mathfrak{q}.$$

It is therefore enough to verify (1.2.11.1) after extending scalars to $\bar{k}$; we may suppose that $a = 0$. At y one then has

$$\delta^2\mathfrak{q} = (x_0^5, (x_0^3x_i)_{1\le i}, (x_0x_ix_j)_{1\le i,j}, (x_ix_jx_k)_{1\le i,j,k}).$$

It follows that $\delta^2\mathfrak{q} \supset \mathfrak{q}^5$ and that

$$x_0^2 f \equiv x_0^4 \pmod{\delta^2\mathfrak{q}}, \qquad \delta^2\mathfrak{q} + (x_0^2 f) \supset \mathfrak{q}^4,$$

$$x_0 f \equiv x_0^3 \pmod{\delta^2\mathfrak{q} + \mathfrak{q}^4}, \qquad \delta^2\mathfrak{q} + (x_0 f) \supset \mathfrak{q}^3.$$

Finally,

$$Q(\chi_i) \equiv 0 \pmod{\delta^2\mathfrak{q} + (f)} \qquad \underline{\text{on } Z},$$

which is (1.2.11.1).

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Remark 1.2.12.

Let Q be a not necessarily homogeneous, nonzero quadratic form in $n+1$ variables, and suppose that the closed subscheme $Y \subset \mathbb{A}_k^{n+1}$ with equation Q has an ordinary quadratic singularity at a point y. A nonhomogeneous linear change of variables then puts Y into the form 1.2.3 or 1.2.4. In the nondegenerate case $k(y) = k$, and it is enough to put y at the origin. In the degenerate case it is enough that the support of $T^1_{Y/k}$ lie on the line $x_i = 0$ $(1 \le i \le 2m)$ (cf. 1.2.9).

1.3. Moduli of quadratic singularities

The following proposition follows from R. Elkik's theorem 1.1.4 and from explicit calculations left to the reader (and essentially contained in VI 6).

Proposition 1.3.1.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $Y_0 \subset \mathbb{A}^{n+1}_{k(s)}$ be a scheme of the kind described in 1.2.3 or 1.2.4.

(i) Suppose that Y_0 is of the kind described in 1.2.3, defined by the equation

$$Q(\underline{X}) = \sum_{i\le j} \bar{a}_{ij}x_ix_j = 0 \qquad (\bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ to elements a_{ij} of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}^{n+1}_T$ with equation

$$\sum a_{ij}x_ix_j - b = 0.$$

(ii) Suppose that Y_0 is of the kind described in 1.2.4, defined by the equation

$$Q(\underline{x}) = (x_0^2 - \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0 \qquad (\bar{a}, \bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ and $\bar{a}$ to elements a_{ij} and a of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b, c\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}_T^{n+1}$ with equation

$$(x_0^2 - a) + \sum a_{ij} x_i x_j + b x_0 + c = 0.$$

The following corollary follows at once from 1.2.6 and 1.3.1. II≈**175**

Corollary 1.3.2.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $f : X \to S$ be a flat morphism of finite presentation. Suppose that, at the closed point x of X_s, the scheme X_s has an ordinary quadratic singularity, that $k(x)$ is radicial over $k(s)$, and that X_s has dimension n.

(i) If x is nondegenerate, there exist a nondegenerate quadratic form Q in $n+1$ variables with coefficients in A, and an element b in the maximal ideal of A, such that the henselized S-scheme of X at x is isomorphic to the henselization at the origin of the closed subscheme of $\mathbb{A}_S^{n+1}$ with equation $Q - b = 0$, as in 1.3.1(i).

(ii) If x is degenerate ($n = 2m$, $\mathrm{char}(k(s)) = 2$), there exists a nonhomogeneous quadratic form in $n+1$ variables with coefficients in A

(1.3.2.1) $$Q(\underline{X}) = x_0^2 + b x_0 + c + \sum_{1 \le i \le j \le 2m} a_{ij} x_i x_j,$$

such that b lies in the maximal ideal, the quadratic form $\sum a_{ij} x_i x_j$ in $2m$ variables is nondegenerate, and the henselized S-scheme of X at x is isomorphic to the henselization at $(\sqrt{c}, 0, \ldots, 0)$ of the closed subscheme of $\mathbb{A}_S^{n+1}$ with equation $Q = 0$.

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Remark 1.3.3.

In case (i), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at O of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$\sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = \sum a_{ij} x_i x_j.$$

In case (ii), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at $(\sqrt{a}, 0, \ldots, 0)$ of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$(x_0^2 + \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = x_0^2 + b x_0 + c + \sum a_{ij} x_i x_j, \qquad c \equiv \bar{a} \pmod{\mathfrak{m}_A}.$$

In both cases, the isomorphism promised in 1.3.2 may be chosen to extend φ.

Corollary 1.3.4.

Let $f : X \to S$ be as in 1.3.2. There is a neighborhood U of x in X such that the nonsmooth points of f contained in U are ordinary quadratic points of their respective fibers.

2. Computation of vanishing cycles in a standard ordinary quadratic case

2.1. Preliminary: the cohomology of a cone

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2.1.1.

Let $Y \subset \mathbb{P}^r$ be a projective variety over an algebraically closed field k. We use the following notation.

The affine cone over Y is denoted by $X \subset \mathbb{A}^{r+1}$, and its vertex by O;

$X_{(O)}$ is the henselization of X at O;

$X^* = X - \{O\}$ and $X^*_{(O)} = X_{(O)} - \{O\}$;

$X_1 \subset \mathbb{P}^{r+1}$ is the projective cone over Y, so $X = X_1 - Y$;

$\widetilde{X}$, $\widetilde{X}_{(O)}$, and $\widetilde{X}_1$ are the blowups of X, $X_{(O)}$, and X_1 at O;

Y_0 is the inverse image of O in $\widetilde{X}$, $\widetilde{X}_{(O)}$, or $\widetilde{X}_1$;

$h : \widetilde{X}_1 \to Y$ is the natural projection;

$y_0, y_\infty : Y \to \widetilde{X}_1$ are the sections of h whose images are Y_0 and Y;

F is an abelian torsion group of order prime to the characteristic of k. All cohomology groups considered below have coefficients in F, which will not be displayed.

Our purpose in this subsection is to relate the cohomology of $X^*_{(O)}$ to that of Y.

Proposition 2.1.2.

There are isomorphisms

$$\text{(i)} \qquad H^i(X) \xrightarrow{\sim} H^i(\{O\}) = \begin{cases} F, & i = 0, \\ 0, & i > 0; \end{cases}$$

$$\text{(ii)} \qquad H^i_{\{O\}}(X) \xrightarrow{\sim} H^i_c(X).$$

This proposition is proved by a homotopy argument, which we now formalize.

Let k be an algebraically closed field, let U and V be two k-schemes, let Λ be a torsion ring of order prime to the characteristic of k, and let

$$K \in \operatorname{Ob} D^+(U, \Lambda), \qquad L \in \operatorname{Ob} D^+(V, \Lambda).$$

By a "morphism" f from (U, K) to (V, L) we mean a pair consisting of a morphism $f : U \to V$ and a morphism $\varphi : f^*L \to K$. Such a morphism induces

$$f^* : H^*(V, L) \longrightarrow H^*(U, K).$$

We say that two "morphisms" f_0 and f_1 are homotopic if there exist a connected k-scheme T of finite type, two points 0 and 1 of T, and a "morphism" H from $(U \times T, \operatorname{pr}_1^* K)$ to (V, L) whose fibers at 0 and 1 induce f_0 and f_1. II≈**178**

Lemma 2.1.3.

If f_0 is homotopic to f_1, then $f_0^* = f_1^*$.

There is a sequence of points $0 = x_0, x_1, \ldots, x_n = 1$ of T, smooth connected curves Γ_i, and morphisms $a_i : \Gamma_i \to T$ such that x_i and x_{i+1} belong to the image of a_i. To construct the Γ_i, normalize a connected one-dimensional closed subscheme of T containing 0 and 1. This reduces us to the case in which T is a smooth curve. Consider the commutative square

$$\begin{array}{ccc} U \times T & \xrightarrow{\mathrm{pr}_1} & U \\ {\scriptstyle \mathrm{pr}_2}\downarrow & & \downarrow{\scriptstyle f} \\ T & \xrightarrow[t]{} & k. \end{array}$$

By smooth base change, applied to t, one has

$$t^* R f_* K \xrightarrow{\sim} R\,\mathrm{pr}_{2*}(\mathrm{pr}_1^* K).$$

The map f_i^* $(i = 0, 1)$ therefore factors through

$$\begin{array}{ccccc} H^n(V, L) & \xrightarrow{H^*} & H^n(U \times T, \mathrm{pr}_1^* K) & \longrightarrow & H^0\big(T, R^n\,\mathrm{pr}_{2*}(\mathrm{pr}_1^* K)\big) \\ & & & \nearrow\!\!\sim & \\ & & H^0\big(T, t^* H^n(U, K)\big) & & \downarrow{\scriptstyle \text{fiber at } i} \\ & & & \nwarrow\!\!\sim & \\ & & & & H^n(U, K) \end{array}$$

and is consequently independent of i.

2.1.3.

Let us prove 2.1.2. Scalar multiplication gives a homotopy between the identity map of X and the constant map from X to X with value O; this proves (i).

Using scalar multiplications whose factors tend to infinity, one sees that Y is a deformation retract of X_1^*. Assertion (ii) then follows from the five lemma and the diagram II≈**179**

$$\begin{array}{ccccccccc} \xrightarrow{\partial} & H^i_{\{O\}}(X) & \longrightarrow & H^i(X_1) & \longrightarrow & H^i(X_1^*) & \xrightarrow{\partial} & \\ & \downarrow & & \| & & \downarrow\wr & & \\ \xrightarrow{\partial} & H^i_c(X) & \longrightarrow & H^i(X_1) & \longrightarrow & H^i(Y) & \xrightarrow{\partial} & . \end{array}$$

Corollary 2.1.4.

$$H^i(X^*) \xrightarrow{\sim} H^i(X^*_{(O)}).$$

Indeed, apply the five lemma to

$$\begin{array}{ccccccc} \longrightarrow & H^i_{\{O\}}(X) & \longrightarrow & H^i(X) & \longrightarrow & H^i(X^*) & \longrightarrow \\ & \downarrow\wr & & \wr\downarrow{\scriptstyle (2.1.2(i))} & & \downarrow & \\ \longrightarrow & H^i_{\{O\}}(X_{(O)}) & \longrightarrow & H^i(X_{(O)}) & \longrightarrow & H^i(X^*_{(O)}) & \longrightarrow . \end{array}$$

2.1.5.

The Leray spectral sequences for the restrictions of h to $\widetilde{X}_1 - Y_0$ and $\widetilde{X}_1 - Y$ show that the restriction maps

$$H^i(\widetilde{X}_1 - Y_0) \xrightarrow{\sim} H^i(Y), \qquad H^i(\widetilde{X}) \xrightarrow{\sim} H^i(Y_0) = H^i(Y)$$

are isomorphisms. Under these isomorphisms, the long exact cohomology sequences of the pairs $(\widetilde{X}_1 - Y_0, Y)$ and $(\widetilde{X}_1 - Y, Y_0)$ form the rows of the diagram

$$\begin{array}{ccccccccc} H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow{\eta} & H^i(Y) & \longrightarrow & H^i(X^*) & \longrightarrow \\ \| & & \| & & \big\vert{-1} & & \| & \\ H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow[-\eta]{} & H^i(Y) & \longrightarrow & H^i(X^*) & \longrightarrow . \end{array} \tag{2.1.5.1}$$

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Lemma 2.1.6.

The arrows denoted by η and $-\eta$ in (2.1.5.1) are cup products with the cohomology class of a hyperplane section. Diagram (2.1.5.1) is commutative.

The first assertion follows because η (respectively $-\eta$) is the restriction to Y (respectively to $Y_0 \simeq Y$) of the cohomology class of the invertible sheaf $\mathcal{O}(Y)$ (respectively $\mathcal{O}(Y_0)$) on $\widetilde{X}_1 - Y_0$ (respectively $\widetilde{X}_1 - Y$).

Use the Leray spectral sequence for h. The two rows of (2.1.5.1) are obtained by applying the functor $H(Y, -)$ to the distinguished triangles

$$y_{\infty *} R y_\infty^! F \longrightarrow R(h|\widetilde{X}_1 - Y_0)_* F \xrightarrow{①} R(h|X^*)_* F \xrightarrow{②}$$

and

$$y_{0*} R y_0^! F \longrightarrow R(h|\widetilde{X}_1 - Y)_* F \longrightarrow R(h|X^*)_* F \longrightarrow .$$

The nonzero cohomology sheaves of the complexes at the vertices of these triangles occur in degrees (2), (0), and (0 and 1). The arrows ① and ② are therefore determined by their effect on cohomology sheaves. This reduces us to the case in which Y is a point, which is left to the reader.

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2.1.7.

The long exact sequence of the pair $(\widetilde{X}_{(O)}, Y_0)$ is a local analogue of the second row of (2.1.5.1). One has

$$H^i(\widetilde{X}_{(O)}) \xrightarrow{\sim} H^i(Y_0)$$

by proper base change for the morphism $\widetilde{X}_{(O)} \to X_{(O)}$. The long exact sequence in question may therefore be written

$$\begin{array}{ccccccccc} H^{i-1}(X^*_{(O)}) & \longrightarrow & H^{i-2}(Y_0)(-1) & \xrightarrow{-\eta} & H^i(Y_0) & \longrightarrow & H^i(X^*_{(O)}) \\ \wr\big\uparrow & & \| & & \| & & \wr\big\vert & \\ H^{i-1}(X^*) & \longrightarrow & H^{i-2}(Y)(-1) & \xrightarrow{-\eta} & H^i(Y) & \longrightarrow & H^i(X^*) \end{array} , \tag{2.1.7.1}$$

the second row of this commutative diagram being the second row of (2.1.5.1).

The following technical lemma will be used in 2.2.7.

Lemma 2.1.8.

$$\begin{array}{ccccccccc} H^{n-1}(Y_0) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}) & \longrightarrow & H^{n-1}(X^*) & \longrightarrow & H^n_{\{x\}}(X) & \longrightarrow & H^n_c(X_s) \\ \| & & & & & & & & \| \\ H^{n-1}(Y) & & & & \xrightarrow{\quad\partial\quad} & & & & H^n_c(X_s) \end{array}$$

is anticommutative.

By 2.1.7, this is equivalent to proving the commutativity of

$$\begin{array}{ccc} H^{n-1}(X_1 - \{O\}) & \xrightarrow{\partial} & H^n_{\{O\}}(X_1) = H^n_{\{O\}}(X) \\ \downarrow \wr & & \downarrow \\ H^{n-1}(Y) & \xrightarrow{\partial} & H^n_c(X). \end{array}$$

This is general nonsense.

2.2. Vanishing cycles

2.2.1.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}$ have the meaning of XIII, 0.2.5. Put $\Lambda = \mathbb{Z}/k$, where k is prime to the residual characteristic of S. Let $\mathbb{A}^{n+1}_S$ be standard affine $(n+1)$-space over S, and let X be the closed subscheme of $\mathbb{A}^{n+1}_S$ defined by a quadratic equation $Q = 0$ which is nonzero modulo a uniformizer:

$$Q(\underline{x}) = \sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i + c.$$

Regard $\mathbb{A}^{n+1}_S$ as the complement in $\mathbb{P}^{n+1}_S$ of the hyperplane at infinity H. Then X is obtained from the quadric $X_1 \subset \mathbb{P}^{n+1}_S$ with homogeneous equation

$$\sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i z + c z^2 = 0 \tag{2.2.1.1}$$

by removing $Y = X_1 \cap H$.

We make the following assumptions: II≈182

(a) Y is a smooth quadric over S, that is, the quadratic form $\sum_{i \le j} a_{ij} x_i x_j$ is ordinary;

(b) the quadric X_s is singular, that is, $X_{\bar{s}}$ is a quadratic cone.

Let x_0 denote the singular point of X_s, and let $f : X \to S$ be the projection.

Remark 2.2.2.

If x_0 is a rational point, a translation puts it at the origin of the coordinates. If $x_0 = 0$, the b_i and c are divisible by a uniformizer.

The subscheme A of X_s defined by the equations $\partial Q / \partial x_i$ is concentrated at x_0. Except in the exceptional case where $k(s)$ has characteristic 2 and $n+1$ is even, this scheme has degree one over $k(s)$, and hence x_0 is rational. In the exceptional case it is ramified over $k(s)$ and has rank 2, so that $k(x_0)$ is either $k(s)$ or a purely inseparable quadratic extension of $k(s)$. The assertions about A are checked most easily after passing to the algebraic closure $k(\bar{s})$ of $k(s)$.

Proposition 2.2.3.

Under the assumptions of 2.2.1, the morphisms

$$H^i(X_{\bar{\eta}}, \Lambda) \xrightarrow{\sim} \mathbb{H}^i\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xrightarrow{\sim} R^i\psi_{\bar{\eta}}(\Lambda)_{x_0}$$

$$H^i_c(X_{\bar{\eta}}, \Lambda) \xleftarrow{\sim} \mathbb{H}^i_c\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xleftarrow{\sim} \mathbb{H}^i_{\{x_0\}}\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big)$$

are isomorphisms.

That the left-hand arrows are isomorphisms follows from 2.1.8.6 and 2.1.10.5; the assertion for the right-hand arrows is a corollary of 2.1.2. Indeed, there is a distinguished triangle II≈183

$$(\Lambda \text{ on } X_{\bar{s}})[0] \longrightarrow R\psi_{\bar{\eta}}(\Lambda) \longrightarrow R\Phi_{\bar{\eta}}(\Lambda),$$

and 2.1.2 applies to $(\Lambda \text{ on } X_{\bar{s}})$, whereas $R\Phi_{\bar{\eta}}(\Lambda)$ is supported at x_0.

Corollary 2.2.4.

If the geometric generic fiber $X_{\bar\eta}$ is singular, that is, if it is still a quadratic cone, then all the sheaves $R^i\Phi(\Lambda)$ vanish.

By 2.1.2,

$$\Lambda = H^0(X_{\bar s},\Lambda) \xrightarrow{\sim} H^0(X_{\bar\eta},\Lambda), \qquad 0 = H^i(X_{\bar s},\Lambda) = H^i(X_{\bar\eta},\Lambda) \quad (i>0).$$

We then apply the long exact sequence XIII, 2.1.8.9 (valid by XIII, 2.1.9).

2.2.5.

Suppose now that

$(*)$ the generic fiber X_η is smooth.

For simplicity we shall also assume that S is strictly henselian. The sheaves of vanishing cycles are easily computed with the aid of 2.2.3 and XII, 3.7. The cohomology with coefficients in Λ considered here follows from the ℓ-adic cohomology of the cited passage by the universal-coefficient formula (cf. 2.1.13). The results are as follows.

A. Cohomology without supports.

One has $R^i\psi_{\bar\eta}(\Lambda) = 0$ for $i \neq 0, n$.

1) If $n \neq 0$: $\psi_{\bar\eta}(\Lambda) = \Lambda$. Moreover, $R^n\psi_{\bar\eta}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

2) For arbitrary n: even when $n = 0$, one checks that $R^i\Phi_{\bar\eta}(\Lambda) = 0$ for $i \neq n$, and that $R^n\Phi_{\bar\eta}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

B. Cohomology with support at x_0. II ≈184

One has

$$H^i_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda)) = 0 \qquad (i \neq n, 2n).$$

There is, moreover, a trace morphism

$$\mathrm{tr} : H^{2n}_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(n))) \longrightarrow \Lambda. \tag{2.2.5.1}$$

If $n \neq 0$: the morphism (2.2.5.1) is an isomorphism. Moreover,

$$H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(n)))$$

is (noncanonically) isomorphic to Λ.

C. Duality. For

$$a \in R^n\psi_{\bar\eta}(\Lambda)_{x_0}, \qquad b \in H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(n))),$$

put $(a,b) = \mathrm{Tr}(a \wedge b)$, where Tr is the morphism XIII, 2.2.6

$$\mathrm{Tr} : H^{2n}_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(n))) \longrightarrow \Lambda.$$

It follows from Poincaré duality on $X_{\bar\eta}$ and the isomorphisms 2.2.3 that (a,b) gives a perfect duality between the free Λ-modules

$$R^n\psi_{\bar\eta}(\Lambda)_{x_0} \quad \text{and} \quad H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(n))).$$

D. $n = 2m > 0$.

By XII, 3.7, the groups

$$H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(m))) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. Let φ be the natural map from the first of these groups to the second. The signs of δ and δ' may be normalized so that

$$(\delta', \delta) = 1; \tag{2.2.5.2}$$

then

$$\varphi(\delta) = (-1)^m\, 2\, \delta'. \tag{2.2.5.3}$$

Let Z be the separable quadratic extension of $k(\eta)$ which is the center of the even part of the Clifford algebra of the quadratic form (2.2.1.1), and let ϵ be the corresponding character, of order 1 or 2, of the inertia group I. For $\sigma \in I$, one has II≈185

$$\sigma\delta = \epsilon(\sigma)\delta \tag{2.2.5.4}$$

and

$$\sigma\delta' = \epsilon(\sigma)\delta'. \tag{2.2.5.5}$$

Formulas (2.2.5.2), (2.2.5.3), and (2.2.5.4), in cohomology with coefficients in $\mathbb{Z}/2k$, imply that the variation is

$$\mathrm{Var}(\sigma)(a) = \frac{\epsilon(\sigma) - 1}{2}\,(-1)^m\,(a, \delta)\,\delta. \tag{2.2.5.6}$$

E. $n = 2m + 1$—Vanishing cycles.

By XII, 3.7, the groups

$$H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(m))) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m+1))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. With the notation of D, their signs may be normalized so that

$$(\delta', \delta) = 1; \tag{2.2.5.7}$$

then

$$\varphi(\delta) = 0. \tag{2.2.5.8}$$

F. $n = 2m + 1$—The variation.

The inertia group I acts trivially on the cohomology of X_η. This follows from XII, 3.7 and from the fact that I acts trivially on the cohomology of Y_η, since Y is proper and smooth over S.

For σ in the inertia group one necessarily has

$$D(\sigma)(\delta') = \lambda(\sigma)\delta, \qquad \lambda(\sigma) \in \Lambda(1). \tag{2.2.5.9}$$

The additivity formula (XIII, 1.4.3.4) becomes

$$\lambda(\sigma\tau) = \lambda(\sigma) + \lambda(\tau).$$

Let

$$\epsilon : I \longrightarrow \Lambda(1) = \mu_k$$

be the homomorphism such that

$$\sigma\big(\sqrt[k]{t}\big) = \epsilon(\sigma)\sqrt[k]{t}$$

for a uniformizer t. In view of the structure of the tame inertia group, λ is necessarily a multiple of ϵ. Formula (2.2.5.9) may therefore be rewritten

$$\mathrm{Var}(\sigma)(a) = \lambda_X\epsilon(\sigma)(a, \delta)\delta \tag{2.2.5.10}$$

for a suitable $\lambda_X \in \Lambda$ (depending on X/S). We shall determine λ_X later.

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2.2.6.

We shall give the cycles δ and δ' of D and E a description which is local for the ètale topology.

The case of even n is easy. If $2 \nmid k$ ($\Lambda = \mathbb{Z}/k$), then $\pm\delta$ is characterized by (2.2.5.3) and (2.2.5.4), that is, by

$$(\delta, \delta) = (-1)^m\, 2. \tag{2.2.6.1}$$

In general, $\pm\delta$ is the image of any cycle $\widetilde{\delta}$, in cohomology with coefficients in $\mathbb{Z}/2^a k$ for a sufficiently large, which satisfies (2.2.6.1).

Suppose that n is odd. Then x_0 is a rational point (2.2.2). Let $X_{s(0)}$ be the henselization of X_s at x_0, let $\widetilde{X}_{s(0)}$ be its blowup at x_0, let Y_0 be the inverse image of x_0 in $\widetilde{X}_{s(0)}$ (a quadric), and put

$$X^*_{s(0)} = X_{s(0)} - \{x_0\} = \widetilde{X}_{s(0)} - Y_0.$$

Consider the composite map (cf. XIII, 2.4.6.1)

$$\begin{array}{ccccc} H^{n-1}(Y_0, \Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(0)}, \Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(0)}, \Lambda(m)) \\ & & & & \downarrow \\ & & H^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(m))) & \longleftarrow & H^n_{\{x_0\}}(X_s, \Lambda(m)) \end{array} \tag{2.2.6.2}$$

Lemma 2.2.7.

The map (2.2.6.2) identifies $\mathbb{H}^n_{\{x_0\}}(X_s, R\psi_{\bar\eta}(\Lambda(m)))$ with the primitive quotient of the middle-dimensional cohomology of the $2m$-dimensional quadric Y_0; the two classes $\pm\delta$ are the images of the natural generators of this primitive quotient.

The cycles $\pm\delta$ are characterized by their images in $H^n_c(X_{\bar\eta}, \Lambda(m))$. Using (2.1.7.1), it is therefore enough to prove that the composite map II≈187

$$\begin{array}{ccccc} H^{n-1}(Y_0, \Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_s, \Lambda(m)) & \longrightarrow & H^{n-1}(X^*_s, \Lambda(m)) \\ & & & & \downarrow \\ H^n_c(X_{\bar\eta}, \Lambda(m)) & \xleftarrow{\text{sp}} & H^n_c(X_s, \Lambda(m)) & \longleftarrow & H^n_{\{x_0\}}(X_s, \Lambda(m)) \end{array}$$

sends the distinguished generators of the primitive quotient of the first group to the distinguished generators of the last group.

Apply 2.1.8. Up to sign, the map may be rewritten

$$\begin{array}{ccc} H^{n-1}(Y_s, \Lambda(m)) & \xrightarrow{\partial} & H^n_c(X_s, \Lambda(m)) \\ \| & & \downarrow \\ H^{n-1}(Y_{\bar\eta}, \Lambda(m)) & \longrightarrow & H^n_c(X_{\bar\eta}, \Lambda(m)), \end{array}$$

and 2.2.7 follows.

3. The Picard–Lefschetz formula

3.1. Preliminary results

3.1.1.

Let S be a henselian trait and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n. Suppose that the special fiber X_s is smooth except at a finite set of points Σ', at which X_s has an ordinary quadratic singularity.

Let Σ be the set of points $x \in \Sigma'$ such that the generic fiber X_η is smooth in a neighborhood of x. Finally, let Λ be a torsion ring whose order is prime to the residual characteristic of S. II≈188

Proposition 3.1.2.

(i) The sheaves $R^i\Phi(\Lambda)$ vanish for $i \neq n$.

(ii) The sheaf $R^n\Phi(\Lambda)$ is zero outside Σ. Its restriction to Σ is a sheaf of Λ-modules of rank one.

(iii) For every geometric point $\bar{x}$ of Σ, the pairing (a, b) of 2.2.5(C) puts the Λ-modules

$$\left(R^n\psi(\Lambda)\right)_{\bar{x}} \quad \text{and} \quad H^n_{\{\bar{x}\}}(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda))$$

in duality. They are free of rank one if $n \neq 0$, and of rank two if $n = 0$.

The assertions of 3.1.2 are local for the ètale topology. We may therefore treat only the case where S is strictly henselian and X is the hypersurface in relative affine space $\mathbb{A}^{n+1}_S$ defined by a quadratic equation satisfying 2.2.1(a) and (b) (apply 1.3.2).

The morphism f is smooth at every point of $X_s - \Sigma'$, hence $R\Phi = 0$ outside Σ'. By 2.2.4, $R\Phi$ even vanishes outside Σ. We may now assume 2.2.5(*); assertions (ii) and (iii) then follow from 2.2.5(A), (B), and (C).

3.2. Even relative dimension

We retain the notation of 3.1 and take $n = 2m$ to be even. For simplicity we suppose that S is strictly henselian (the general case would follow by descent).

Proposition 3.2.1.

(i) For $x \in \Sigma$, the group $H^n_{\{x\}}(X_s, R\psi_{\bar{\eta}}(\Lambda(m)))$ has a natural generator δ, well defined up to sign, characterized by being natural in Λ and by satisfying

$$(\delta, \delta) = (-1)^m\, 2.$$

For $n = 0$, add the condition $\mathrm{Tr}(\delta) = 0$.

(ii) For a suitable character, independent of Λ,

$$\varepsilon_x : I \longrightarrow \{\pm 1\}$$

of the inertia group I, the variation is

$$\mathrm{Var}(\sigma)(a) = (-1)^m \left(\frac{\varepsilon_x(\sigma) - 1}{2}\right)(a, \delta)\delta$$

for $a \in R^n\Phi_{\bar{\eta}}(\Lambda(m))$. Consequently,

$$\sigma(\delta) = \varepsilon_x(\sigma)\delta.$$

Passing to the inverse limit, it is enough to treat the universal case $\Lambda = \mathbb{Z}_\ell$. Up to sign, only one δ can then satisfy the conditions in (i). As in 3.1.2, we reduce to the case 2.2.5(*) and apply 2.2.5(D). II≈189

Complement 3.2.2.

Suppose that, at $x \in \Sigma$, the henselization $X_{(x)}$ of X at x is isomorphic to the henselization at x_0 of the projective scheme Q with equation

$$\sum_{i \le j} a_{ij} x_i x_j = 0.$$

Then the character ε_x is defined by the separable quadratic extension $Z(C^+(Q))$ of the fraction field. This follows from 2.2.5(D).

3.2.3.

In residual characteristic different from 2, Complement 3.2.2 can be made more explicit. In that case I has a unique nontrivial character ε of order 2. For a uniformizer t, one has

$$\sigma(\sqrt{t}) = \varepsilon(\sigma)\sqrt{t}.$$

On the other hand, $X_{(x)}$ is then isomorphic to the henselization at 0 of an affine scheme

$$\sum a_{ij} x_i x_j = b,$$

where b lies in the maximal ideal and $Q(x) = \sum a_{ij} x_i x_j$ is nondegenerate. The ideal (b) defines the image of the subscheme of $X_{(x)}$ which is the support of $J^n_{X_{(x)}/S}$. The center of the Clifford algebra is isomorphic to

$$k(\eta)\left(\left((-1)^{m+1} \cdot 2b \det(a_{ij})\right)^{1/2}\right)$$

(Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, final remark), whence

$$\varepsilon_x = \varepsilon^{v(b)}.$$

The variation is zero if b has even valuation; otherwise $\varepsilon_x = \varepsilon$.

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3.3. Odd relative dimension

To prove the Picard–Lefschetz formula in odd relative dimension, we shall have to reduce to the transcendental case treated in XIV, 3.2.1.1. The specialization argument required for this forces us to work over a base which is no longer one-dimensional. We shall use ad hoc arguments adapted to the precise case we need: I do not have a usable theory over a base of dimension greater than one, and I have preferred not to cover my ignorance with generalities.

3.3.1.

Let $S = \mathrm{Spec}(A)$ be a strictly local noetherian scheme, and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n, whose special fiber X_s has no nonsmooth points other than nondegenerate quadratic points. Let s be the closed point of S, let X_s be the special fiber, and let $x_0 \in X_s$ be a singular point.

By 1.3.2(i), in a neighborhood of x_0 the scheme X/S is locally isomorphic to an S-scheme defined in A^{n+1}_S by an equation $Q - b = 0$, where b belongs to the maximal ideal and Q is a nondegenerate quadratic form. The ideal (b) of A depends only on $(X/S, x_0)$: in a neighborhood of x there is a section

$$s \in X(A/(b))$$

of X over $\mathrm{Spec}(A/(b))$ such that

$$J^n_{X/S} \simeq \mathcal{O}_{s(\mathrm{Spec}(A/b))}.$$

We denote this section by s_x, and denote by $b(x)$ any generator of (b).

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3.3.2.

Suppose that S is normal, and let $b \in A$. If n is invertible in A and $b \neq 0$, write $\varepsilon_b^{(n)}$ for the following homomorphism from the Galois group of the fraction field of A to $\mathbb{Z}/(n)(1)$:

$$\varepsilon_b^{(n)}(\sigma) = (\sigma \sqrt[n]{b})/\sqrt[n]{b}.$$

One has

$$\big(\varepsilon_b^{(nm)}\big)^m = \varepsilon_b^{(n)}.$$

If Λ is a $\mathbb{Z}/n$-algebra, we again denote by ε_b the composite

$$\mathrm{Gal} \longrightarrow \mathbb{Z}/(n)(1) \longrightarrow \Lambda(1).$$

It is independent of n.

The character ε_b factors through

$$\pi_1\big(S - (\text{the subscheme } b = 0)\big).$$

3.3.3.

Suppose that S is a strictly henselian trait and that n is odd, $n = 2m+1$. Let x be a singular point of X_s in a neighborhood of which X_η is smooth, and let X_1/S be a subscheme of $\mathbb{A}_S^{n+1}$ with equation $Q - b = 0$, as in 1.3.2(i). The assumption on x means that b, although zero modulo the maximal ideal, is not zero.

Let $X_{s(x)}$ be the henselization of X_s at x, let $\widetilde{X}_{s(x)}$ be the blowup of X_s at x, let Y be the quadric which is the inverse image of x, and put

$$X^*_{s(x)} = X_{s(x)} - \{x\} = \widetilde{X}_{s(x)} - Y.$$

The composite morphism

$$\begin{array}{ccccc} H^{n-1}(Y,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(x)},\Lambda(m)) \\ & & & & \downarrow \\ & & H^n_{\{x\}}(X_s, R\psi_{\bar\eta}(\Lambda(m))) & \longleftarrow & H^n_{\{x\}}(X_s,\Lambda(m)) \end{array}$$

identifies the latter group with the primitive quotient (XII, 3.5) of the cohomology of the quadric Y. Indeed, this follows from 2.2.7 in a locally isomorphic situation. The images of the two natural generators of this quotient (the cohomology classes of the rulings of Y) are the two basis vectors

$$\pm\delta \in H^n_{\{x\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(m))\big).$$

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3.3.4.

Return to the more general assumptions of 3.3.1, with $n = 2m+1$. For every singular point x of X_s, we again denote by $\pm\delta_x$ the sections of $R^n f_!\Lambda(m)$ over S which are the images of the generators of the primitive quotient of $H^{n-1}(Y,\Lambda(m))$ (with the same notation as above) under the composite morphism

$$\begin{array}{ccccc} H^{n-1}(Y,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(x)},\Lambda(m)) \\ & & & & \downarrow \\ H^0(S, R^n f_!\Lambda(m)) & \xleftarrow{\sim} & H^n_c(X_s,\Lambda(m)) & \longleftarrow & H^n_{\{x\}}(X_s,\Lambda(m)) \end{array}$$

The class $\pm\delta_x$ is the vanishing cycle at x.

When S is a trait, the class

$$\pm\delta \in H^n_c(X_{\bar\eta}, \Lambda(m))$$

is the image under XIII, 2.1.8.4 of the class of 3.3.3

$$\pm\delta \in H^n_{\{x\}}(X_s, R\psi_{\bar\eta}(\Lambda(m))).$$

We shall prove the following two results simultaneously.

Proposition 3.3.5.

Let S be a henselian trait and let X/S be as in 3.3.3. The variation is given by

$$\mathrm{Var}(\sigma)(a) = (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta)\delta.$$

Proposition 3.3.6.

Let S be strictly local and regular, and let $f : X \to S$ be a proper flat morphism of relative dimension $n = 2m+1$ such that X_s has exactly one nonsmooth point x, assumed to be a nondegenerate quadratic point. Suppose also that $b(x)$ is a parameter. Then Galois acts on the cohomology of the geometric generic fiber by

$$\sigma(a) = a + (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta_x)\delta_x.$$

(A) Let k be an algebraically closed field of characteristic 0, and let S be the henselization of $\mathrm{Spec}(k[T])$ at 0. We prove 3.3.5 when $b(x)$ is a uniformizer.

By the Lefschetz principle, we reduce to the case $k = \mathbb{C}$. Since the question is local, we may II≈193
suppose that $f : X \to S$ is obtained by base change from the map

$$\sum_i z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

The construction of $\pm\delta$ given above has a transcendental analogue in integral cohomology. Via XIV, 2.1, the class $\pm\delta$ therefore corresponds to a transcendental vanishing cycle, the image of

$$\pm\delta_{\mathrm{cl}} \in H^n_{\{0\}}(X_0, R\psi_{\mathrm{cl}\eta}(\mathbb{Z})).$$

By XIV, 2.1 and the corresponding property of the algebraic cycles $\pm\delta$, the image of $\pm\delta_{\mathrm{cl}}$ in

$$H^n_{\{0\}}(X_0, R\psi_{\mathrm{cl}\eta}(\mathbb{Z})) \otimes \mathbb{Z}_\ell$$

is a generator of this group. Thus $\pm\delta_{\mathrm{cl}}$ are the two generators of $H^n_{\{0\}}(X_0, R\psi_{\mathrm{cl}\eta}(\mathbb{Z}))$, and coincide with the vanishing cycles of XIV, 3.2. Formula 3.3.5 consequently follows from XIV, 3.2.11. Strictly speaking, one should have checked compatibility between the isomorphism of XIV, 2.1 and cup products and traces.

(B) We prove 3.3.6 for X/S as in (A), with X/S proper in addition. This follows from XIII, 2.4.6.2.

(C) Let p be a prime number, and let S_0 be the strict henselization of $\mathrm{Spec}(\mathbb{Z}[T])$ at (p,T). We prove 3.3.6 for X/S_0 such that $b(x) = T$.

The restriction of the sheaf $R^i f_*\Lambda$ to the strict henselization of S_0 at (T) is governed by (B). This sheaf is locally constant on the complement of the smooth divisor $T = 0$, which has mixed characteristic. Assertion (C) therefore follows from (B) and Abhyankar's lemma.

(D) In at least one case to which (C) applies, δ_x is not a torsion element in ℓ-adic cohomology.

Let S_1 be the strict henselization of $\mathrm{Spec}(\mathbb{Z})$ at (p), and let $V \subset \mathbb{P}^r/S_1$ be a smooth projective II≈194
variety over S_1, purely of relative dimension $n+1$. Suppose that there is a Lefschetz pencil of

hyperplane sections for the special fiber V_p of V/S_1 (XVII, 2.5). This pencil is defined by a line d' in the special fiber of the projective space $\check{\mathbb{P}}^{,r}/S_1$ dual to $\mathbb{P}^r/S_1$. Lift d' to a line d in $\check{\mathbb{P}}^{,r}/S_1$, and let $X \to d$ be the pencil of hyperplane sections of V parametrized by d (XVIII, 3.1). The exceptional locus Δ of d is ètale over S_1 (XVII, 6). After ètale localization at a point of $\Delta_p \subset d$, we therefore obtain a situation of type (C).

To check that δ_x is not torsion, it is enough to do so after extending scalars from S_1 to $\mathbb{C}$. Take $V = \mathbb{P}^{n+1}$ with the projective embedding defined by $\mathcal{O}(N)$. For $N \geq 2$, XVII, 2.5 applies (it is valid for every N in this particular case). We then use the following lemma.

Lemma 3.3.7.

For a Lefschetz pencil of hypersurfaces of sufficiently large degree N in $\mathbb{P}^{n+1}(\mathbb{C})$ ($N \geq 3$ suffices), a vanishing cycle in $\mathbb{Q}_\ell$-cohomology is nonzero.

This lemma follows from the theory of Lefschetz pencils, for example from XVIII, 6.6.1. There is no circularity in using here this result of XVIII, based on the "Picard–Lefschetz formula," since (A) is already available. One may instead use the more direct, purely transcendental variant XIX, no. 4, which depends only on XIV, 3.2.

(E) We prove 3.3.5 in residual characteristic p for $(X/S, x \in X_s)$.

Let X_0/S_0 be as in (D). By 1.3.2(i), there is a morphism $g : S \to S_0$ such that the henselization at x_0 of

$$g^*X_0 = X_0 \times_{S_0} S$$

is S-isomorphic to the henselization of X at x. It is therefore enough to prove 3.3.5 for g^*X_0. By base change from (C) for X_0/S_0, one obtains

$$\sigma(a) = a - (-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$, for σ in the inertia group. Let π be a uniformizer and compare this formula with (2.2.5.9), where $\varepsilon = \varepsilon_\pi$. We find

$$\lambda_X \varepsilon_\pi(\sigma)(a, \delta_x)\delta_x = -(-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$. Since δ_x is not torsion, we may cancel $(a, \delta_x)\delta_x$ from this identity. Formula (2.2.5.9), with the resulting value of λ_X, is precisely 3.3.5.

(F) Proof of 3.3.6 in the general case. II≈**195**

The general case of 3.3.6 reduces to the general case of 3.3.5 just as (B) and (C) reduce to (A).

3.4. Summary

Propositions 3.1.2, 3.2.1, 3.2.2, 3.2.3, and 3.3.5 constitute the "Picard–Lefschetz formula." This formula is the algebraic analogue of XIV, 3.2.11, but takes very different forms according to the parity of the relative dimension.

In the most interesting case, where X/S is proper, the results may be summarized as follows.

Theorem 3.4 (Picard–Lefschetz formula).

Let S be a strictly henselian trait and let $f : X \to S$ be a proper flat morphism, purely of relative dimension $n = 2m$ or $2m + 1$. Suppose that X_η is smooth and that X_s has only ordinary quadratic singularities. Let Σ be the singular locus of X_s.

(i) To every $x \in \Sigma$ is attached a vanishing cycle

$$\pm\delta_x \in H^n(X_{\bar{\eta}})(m).$$

The δ_x are pairwise orthogonal. When n is even,

$$(\delta_x, \delta_x) = (-1)^m, 2.$$

(ii) There are isomorphisms

$$H^i(X_s) \xrightarrow{\sim} H^i(X_{\bar{\eta}})$$

for $i \neq n, n+1$, and an exact sequence

$$0 \longrightarrow H^n(X_s) \longrightarrow H^n(X_{\bar{\eta}}) \xrightarrow{((\,\cdot\,,\delta_x))_{x\in\Sigma}} \bigoplus_{x\in\Sigma} \Lambda(n-m) \longrightarrow H^{n+1}(X_s) \longrightarrow H^{n+1}(X_{\bar{\eta}}) \longrightarrow 0.$$

(iii) There are defined local characters of the inertia group I:

$$\begin{aligned} n \text{ even}: &\quad \varepsilon_x : I \longrightarrow \{\pm 1\}, \\ n \text{ odd}: &\quad \varepsilon_x : I \longrightarrow \Lambda(1). \end{aligned}$$

When X is regular at x, these characters are surjective; when x is nondegenerate, they are respectively the unique character of order 2 and the canonical character.

The action of the inertia group I is given by

$$n = 2m \qquad \sigma(a) = a + (-1)^m \sum_{x\in\Sigma} \frac{\varepsilon_x(\sigma) - 1}{2} (a, \delta_x)\delta_x,$$

$$n = 2m+1 \qquad \sigma(a) = a - (-1)^m \sum_{x\in\Sigma} \varepsilon_x(\sigma)(a, \delta_x)\delta_x.$$

In particular,

$$\begin{aligned} n \text{ even}: &\quad \sigma(\delta_x) = \varepsilon_x(\sigma)\delta_x, \\ n \text{ odd}: &\quad \sigma(\delta_x) = \delta_x. \end{aligned}$$

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SGA 7 — EXPOSÉ XVI

The Milnor Formula

by P. Deligne

In this exposé, Milnor's formula [3] (see also [2]) is transposed to characteristic p. In certain cases this formula makes it possible to compute the dimensions of spaces of vanishing cycles. The proof is global in nature and treats only the "geometric" case in which the base is a henselian trait of equal characteristic with perfect residue field.

1. Statement of the problem

1.1.

Let S be a regular scheme, purely of dimension one, let s be a closed point of S with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, let x_0 be a closed point of the fiber X_s, and let $\bar{x}_0$ be a geometric point of X localized at x_0. Suppose that

(a) x_0 is an isolated point at which f is not smooth: there are an integer n and a neighborhood U of x_0, purely of relative dimension n, such that $U - \{x_0\}$ is smooth over S;

(b) X is regular at x_0.

Under these assumptions, X is a relative complete intersection over S at x_0. We refer to VI, 3.10 for the definition of $D^1_{X/S}$. On the neighborhood U one has

$$D^1_{X/S} = \mathcal{H}^1(R\mathcal{H}om(L\Omega_{X/S}, \mathcal{O}_X)) = \mathcal{E}xt^1(\Omega^1_{X/S}, \mathcal{O}_X);$$

on this neighborhood, $D^1_{X/S}$ has finite length and support contained in $\{x_0\}$.

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Definition 1.2.

The Milnor number $\mu = \mu(X/S, \bar{x}_0)$ of X/S at $\bar{x}_0$ is the length of the $\mathcal{O}_{X,x_0}$-module defined by $D^1_{X/S}$.

Example 1.3.

Take S to be a smooth curve over an algebraically closed field k. After replacing X by a neighborhood of x_0, we may suppose that X is smooth over k. The cotangent complex then reduces to the following resolution of $\Omega^1_{X/S}$:

$$L\Omega_{X/S} = [f^*\Omega^1_{S/k} \longrightarrow \Omega^1_{X/k}].$$

Choose a local coordinate $t : S \to \mathbb{E}^1_k$ on S at s (with t étale at s), again denote by f the composite $t \circ f$, and let $(x_i)_{1 \le i \le n+1}$ be a system of local coordinates on X at x_0. At x_0, the module $f^*\Omega^1_{S/k}$ then has basis df, while $\Omega^1_{X/k}$ has basis the dx_i. Thus $D^1_{X/S}$ is identified with the quotient of $\mathcal{O}_X$ by the ideal generated by the $\partial f/\partial x_i$, and

$$\mu = \dim_k\left(\mathcal{O}_{X,x_0}/\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_{n+1}}\right)\right). \tag{1.3.1}$$

This is the definition given by Milnor.

Example 1.4.

Suppose that X is the closed subscheme of $\mathbb{E}_S^{n+1}$ defined by an equation g, and let $i : X \hookrightarrow \mathbb{E}_S^{n+1}$ be the inclusion. The cotangent complex then reduces to

$$[(g)/(g)^2 \longrightarrow i^*\Omega^1_{\mathbb{E}^{n+1}/S}],$$

and $D^1_{X/S}$ is isomorphic to the quotient of $\mathcal{O}_X$ by the ideal generated by $(\partial g/\partial x_i)_{1\leq i\leq n+1}$. Equivalently, it is the quotient of $\mathcal{O}_{\mathbb{E}^{n+1}}$ by the ideal generated by g and the $\partial g/\partial x_i$.

Example 1.5.

Let S be the spectrum of a discrete valuation ring V with perfect residue field and fraction field K, and let X be the normalization of S in a separable extension K' of K. By 1.4 and [5], III, 7.14, p. 68, $D^1_{X/S}$ is then isomorphic to the quotient of $\mathcal{O}_X$ by the different $\mathfrak{d}_{X/S}$. If $\bar{x}_0$ is a geometric point of X_s, localized at $x_0 \in X_s$, its Milnor number is therefore

$$\mu = \text{length}(\mathcal{O}_{X,x_0}/\mathfrak{d}_{X/S}). \tag{1.5.1}$$

Let $\bar{s}$ be a geometric point of S localized at the closed point s, and let $d_{K'/K} \in V$ generate the discriminant ideal, with valuation $v(d_{K'/K})$. It follows from (1.5.1) that

$$v(d_{K'/K}) = \sum_{x\in X(\bar{s})} \mu(X/S, x). \tag{1.5.2}$$

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1.6.

Retain the assumptions of 1.1. Let $\bar{s}$ be the geometric point of S which is the image of $\bar{x}_0$, let $S_{(\bar{s})}$ be the spectrum of the strict henselization of $\mathcal{O}_{S,s}$ at $\bar{s}$, and let $X_{(\bar{s})}$ be the inverse image of X over $S_{(\bar{s})}$. Let $\bar{\eta}$ be the spectrum of an algebraic closure of the fraction field $k(\eta)$ of $S_{(\bar{s})}$, and put

$$I = \text{Gal}(k(\bar{\eta})/k(\eta)).$$

Let ℓ be a prime number different from the residual characteristic of S. In a neighborhood of $\bar{x}_0$ in $X_{\bar{s}}$, the sheaves of vanishing cycles

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell) = R^i\Phi_{\bar{\eta}}(\mathbb{Z}/\ell)$$

relative to the constant sheaf $\mathbb{Z}/\ell$ on $X_{(\bar{s})}$ are concentrated at $\bar{x}_0$. Moreover, the inertia group I acts on the $\mathbb{Z}/\ell$-vector spaces

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0}.$$

These spaces are finite-dimensional (XIII, 2.4.2).

1.7.

Let I_1 be a finite quotient of the inertia group. Recall that the Swan character Sw of I_1 is the difference $a - u$ between the Artin character a and the character u of the augmentation representation. Recall also that, up to isomorphism, there is a unique projective $\mathbb{Z}_\ell[I_1]$-module $S(I_1)$ with character Sw (see [6]).

Let V be a finite-dimensional $\mathbb{Z}/\ell$-vector space with an action $\rho : I \to \text{GL}(V)$, and suppose that ρ factors through I_1. The wild dimension of V is then II≈**200**

$$\dim \text{Sw}(V) = \dim_{\mathbb{Z}/\ell} \text{Hom}_{I_1}(S(I_1) \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, , V)\,. \tag{1.7.1}$$

This integer is independent of the choice of I_1, since for a normal subgroup H of I_1 one has $S(I_1/H) \sim S(I_1)_H$.

We further put

$$\dim \text{tot}(V) = \dim \text{Sw}(V) + \dim_{\mathbb{Z}/\ell}(V)$$

and call this number the total dimension of V.

Definition 1.8.

With the terminology of 1.6 and 1.7, the number of vanishing cycles of X/S at $\bar{x}_0$ (respectively the wild number of vanishing cycles, respectively the total number of vanishing cycles) is the integer

$$(-1)^n \dim \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$(-1)^n \dim \mathrm{Sw}\, \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \mathrm{Sw}\, \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$\begin{aligned}(-1)^n \dim \mathrm{tot}\, \Phi_{\bar{x}_0} &\underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \\ &\quad \cdot \dim \mathrm{tot}\, \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0} \\ &= (-1)^n (\dim \Phi_{\bar{x}_0} + \dim \mathrm{Sw}\, \Phi_{\bar{x}_0}).\end{aligned}$$

Conjecture 1.9.

Under the assumptions of 1.1, the Milnor number μ of X/S at $\bar{x}_0$ is equal to the total number of vanishing cycles of X/S at $\bar{x}_0$.

Remark 1.10.

Let S be a henselian trait with imperfect residue field and let $f : X \to S$ be a morphism which is smooth except at $x_0 \in X_s$. Suppose that X is regular at x_0. Definition 1.2 of the Milnor number μ still makes sense under these assumptions, but I no longer have a geometric interpretation of μ to propose.

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1.11.

For the remainder of this section, suppose that $S = \mathrm{Spec}(V)$ is a henselian trait with algebraically closed residue field and closed point s, that $f : X \to S$ is a flat morphism of finite type, purely of relative dimension n, that X is regular, that x_0 is a closed point of X_s, and that $X - \{x_0\}$ is smooth over S.

This entails no loss of generality.

Proposition 1.12.

If $n = 0$ (that is, if X/S is finite), Conjecture 1.9 holds for $(X/S, x_0)$.

We may suppose that X is the normalization of S in a finite extension K' of the fraction field K. By 1.5, $\mu = v(d_{K'/K})$.

The extension K' is defined by an open subgroup H of the inertia group I. Let $H' \subset H$ be an open normal subgroup of I, and put $I' = I/H'$. Let $s_{I/H}$ be the character of the permutation representation of I' on I/H, let $a_{I'}$ be the Artin character, let $\mathrm{Sw}_{I'}$ be the Swan character, and let $u_{I'}$ be the character of the augmentation representation. By Artin's "Führerdiskriminantenproduktformel," the discriminant ideal is the Artin conductor of $s_{I/H}$ ([5], VI, 3, Corollary 1, p. 111):

$$\mu = (a_{I'}, s_{I/H}) = |I'|^{-1} \sum_{i \in I'} a_{I'}(i) s_{I/H}(i).$$

One has

$$a_{I'} = u_{I'} + \mathrm{Sw}_{I'}, \qquad (u_{I'}, s_{I/H}) = \#(I/H) - 1,$$

and

$$\begin{aligned}(\mathrm{Sw}_{I'}, s_{I/H}) &= \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, (\mathbb{Z}/\ell)^{I/H}\Big) \\ &= \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, \mathrm{Coker}(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H})\Big)\,.\end{aligned}$$

Consequently,

$$\mu = \#(I/H) - 1 + \dim \mathrm{Sw}\,\mathrm{Coker}(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}).$$

One checks immediately that $\Phi^i_{x_0} = 0$ for $i \neq 0$ and that

$$\Phi^0_{x_0} = \mathrm{Coker}(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H});$$

the formula follows.

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Proposition 1.13.

If X_s has an ordinary quadratic singularity at x_0, Conjecture 1.9 holds for $(X/S, x_0)$.

Case 1: $n = \dim(X_s) = 2m - 1$.

The problem is local for the étale topology on X. We may therefore suppose that X is the closed subscheme of $\mathbb{E}^{n+1}_S$ defined by an equation

$$f = \sum_{i=1}^{m} x_i x_{i+m} + a = 0,$$

where a lies in the maximal ideal of V and x_0 is the origin of the coordinates. Moreover, X is regular at x_0 if and only if a is a uniformizer. By 1.4,

$$\mu = \mathrm{length}\left(\mathcal{O}_{\mathbb{E}^{2m}_S, x_0} / \left(f, \frac{\partial f}{\partial x_i}\right)\right) = 1.$$

By the ordinary-quadratic vanishing-cycle calculation of Exposé XV, $\Phi^i_{x_0} = 0$ for $i \neq n$, while $\Phi^n_{x_0}$ has dimension one and inertia acts trivially on it. The wild number of vanishing cycles is therefore zero and the total number is 1, which verifies 1.9.

Case 2: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) \neq 2$. II ≈**203**

Under these assumptions we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + x^2_{2m+1} + a = 0,$$

where a is a uniformizer. Again $\mu = 1$, $\Phi^i_{x_0} = 0$ for $i \neq n$, and $\Phi^n_{x_0}$ has dimension one. This time the inertia group acts by ± 1 on $\Phi^n_{x_0}$, hence tamely. The wild number of vanishing cycles is again zero, and 1.9 is verified.

Case 3: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) = 2$.

This time we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + (x^2_{2m+1} + a x_{2m+1} + b) = 0,$$

where a and b lie in the maximal ideal. The scheme X is regular if and only if b is a uniformizer, and its generic fiber is smooth if and only if $a^2 - 4b \neq 0$.

The Milnor number of X/S is the same as that of $X_0 \subset \mathbb{E}^1_S$ defined by

$$x^2 + ax + b = 0. \tag{1}$$

Likewise, $\Phi^i_{x_0} = 0$ for $i \neq n$, $\Phi^n_{x_0}$ has dimension one, and I acts on it by ± 1; the index-two subgroup of I acting trivially corresponds to the extension of K defined by (1). The total number of vanishing cycles of X/S at x_0 therefore coincides with the one for X_0/S, and 1.9 in this case follows from 1.12. The Milnor number is given by the discriminant formula of 1.5:

$$\mu = v(a^2 - 4b). \tag{1.13.1}$$

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2. The "geometric" case

We shall prove that Conjecture 1.9 holds in equal characteristic. The key geometric result is the following. Its possible use was suggested to me by conversations with B. Saint-Donat.

Proposition 2.1.

Let S be a smooth connected projective curve over an algebraically closed field k, let X be a smooth proper variety over k, purely of dimension $n+1$, and let $f : X \to S$ be a morphism which is smooth away from a finite set F of points of X. Then the Euler–Poincaré characteristics in étale cohomology of S, X, and the geometric generic fiber $X_{\bar\eta}$ of f satisfy

$$\chi(X) = \chi(S)\chi(X_{\bar\eta}) - (-1)^n \sum_{x\in F} \mu(X/S, x). \tag{2.1.1}$$

The Euler–Poincaré characteristic of X is the trace (or integral over X) of the top Chern class of the tangent bundle:

$$\chi(X) = \int c_{n+1}(\Theta_X) = (-1)^{n+1} \int c_{n+1}(\Omega^1_X).$$

Write

$$\Omega^1_X = \left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \otimes f^*\Omega^1_S.$$

The total Chern class of the invertible sheaf $f^*\Omega^1_S$ is the inverse image of the total Chern class of Ω^1_S on S:

$$c^*(f^*\Omega^1_S) = 1 + f^*c^1(S, \Omega^1_S),$$

and the cup powers of c^1 vanish:

$$\left(f^*c^1(S,\Omega^1_S)\right)^n = f^*\left(c^1(S,\Omega^1_S)^n\right) = 0 \qquad (n > 1).$$

It follows that

$$\begin{aligned} c^{n+1}(\Omega^1_X) = c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S,\Omega^1_S) \\ + c^{n+1}\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right). \end{aligned}$$

If S has genus g, then $c_1(S,\Omega^1_S)$ is $(2g-2)$ times the cohomology class of any closed point s of S. Consequently, when X_s is nonsingular, II≈**205**

$$\begin{aligned} \mathrm{I} &= \int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S,\Omega^1_S) \\ &= (2g-2)\int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*\,\mathrm{cl}(s) \\ &= (2g-2)\int_{X_s} c^n\left(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\right). \end{aligned}$$

Moreover, on X_s the sheaf $f^*\Omega^1_S$ is isomorphic to $\mathcal{O}_{X_s}$, and $\Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{O}_{X_s}$ occurs in the exact sequence

$$0 \longrightarrow f^*\Omega^1_S \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_X \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_{X_s} \longrightarrow 0.$$

Hence

$$c^n\left(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\right) = c^n(X_s, \Omega^1_{X_s}),$$

and

$$\mathrm{I} = (2g-2)\int_{X_s} c^n(X_s, \Omega^1_{X_s}) = (-1)^{n+1}\chi(S)\chi(X_s).$$

The canonical map II≈**206**

$$f^* : f^*\Omega^1_S \longrightarrow \Omega^1_X$$

is identified with a section a of $\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}$. After choosing systems of local coordinates on S and X, thereby identifying this bundle with $\mathcal{O}_X^{n+1}$, the coordinates of a are the partial derivatives $\partial f/\partial x_i$. Thus a vanishes only on F, and by 1.3 the multiplicity of its zero at $x \in F$ is the Milnor number $\mu(X/S, x)$.

Recall that if a section a of a vector bundle G of rank d on a smooth proper k-scheme X, purely of dimension d, vanishes only at isolated points, then $\int_X c^d(G)$ is the number of zeros of a, counted with multiplicity. Here

$$\mathrm{II} = \int_X c^{n+1}\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) = \sum_{x \in F} \mu(X/S, x).$$

Collecting terms gives

$$\chi(X) = (-1)^{n+1}(\mathrm{I} + \mathrm{II}) = \chi(S)\chi(X_s) - (-1)^n \sum_{x \in F} \mu(X/S, x),$$

which proves 2.1, since $\chi(X_s) = \chi(X_{\bar\eta})$.

Corollary 2.2.

Under the preceding assumptions, if $n(X/S, x)$ is the total number of vanishing cycles of X/S at x (1.8), then

$$\sum_{x \in F} \mu(X/S, x) = \sum_{x \in F} n(X/S, x).$$

The Leray spectral sequence for f gives

$$\begin{aligned}\chi(X) &= \sum_{p,q} (-1)^{p+q} \dim_{\mathbb{Z}/\ell} H^p(S, R^q f_* \mathbb{Z}/\ell) \\ &= \sum_q (-1)^q \chi(S, R^q f_* \mathbb{Z}/\ell).\end{aligned}$$

For every constructible sheaf G of $\mathbb{Z}/\ell$-modules on S, one has ([4] or SGA 5, X, 7.2)

$$\chi(S, G) = \chi(S) \dim_{\mathbb{Z}/\ell}(G_{\bar\eta}) - \sum_{s \in S} \varepsilon_s(G),$$

where, for each $s \in S$, after identifying $\bar\eta$ with the algebraic closure of the fraction field of the henselization of S at s,

$$\varepsilon_s(G) = \dim(G_{\bar\eta}) - \dim(G_s) + \dim \mathrm{Sw}(G_{\bar\eta}).$$

For $s \in S$, the exact sequences XIII, 2.4.6.3, II≈**207**

$$\cdots \longrightarrow (R^q f_* \mathbb{Z}/\ell)_s \longrightarrow (R^q f_* \mathbb{Z}/\ell)_{\bar\eta} \longrightarrow \bigoplus_{x \in F(s)} \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x \longrightarrow \cdots$$

give

(a)
$$\begin{aligned}&\sum_q (-1)^q \left(\dim(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \dim(R^q f_* \mathbb{Z}/\ell)_s\right) \\ &\qquad = \sum_{x \in F(s)} \sum_q (-1)^q \dim \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x,\end{aligned}$$

and

(b)
$$\sum_q (-1)^q \dim \mathrm{Sw}(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} = \sum_{x \in F(s)} \sum_q (-1)^q \dim \mathrm{Sw}\, \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x.$$

Consequently,

$$\text{(c)} \qquad \sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell) = \sum_{x \in F(s)} \dim \operatorname{tot} \Phi_x.$$

The Euler–Poincaré formula therefore becomes

$$\begin{aligned}
\chi(X) &= \sum_q (-1)^q \chi(S, R^q f_* \mathbb{Z}/\ell) \\
&= \sum_q (-1)^q \left(\chi(S) \dim (R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \sum_s \varepsilon_s(R^q f_* \mathbb{Z}/\ell) \right) \\
&= \chi(S) \sum_q (-1)^q \dim (R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \sum_s \sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell) \\
&= \chi(S)\chi(X_{\bar\eta}) - \sum_s \sum_{x \in F(s)} \dim \operatorname{tot} \Phi_x \\
&= \chi(S)\chi(X_{\bar\eta}) - (-1)^n \sum_{x \in F} n(X/S, x).
\end{aligned}$$

Comparison with (2.1.1) proves 2.2.

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Corollary 2.3.

Under the preceding assumptions and with the preceding notation, suppose that $x_0 \in F$ and that, for every $x \in F$ distinct from x_0, the fiber of f through x has an ordinary quadratic singularity at x. Then

$$\mu(X/S, x_0) = n(X/S, x_0).$$

For $x \neq x_0$, Proposition 1.13 gives

$$\mu(X/S, x) = n(X/S, x),$$

and it remains to apply 2.2.

Theorem 2.4.

Let S be a henselian trait of equal characteristic with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, and let $\bar x$ be a geometric point of X whose image in X is a closed point x of X_s at which X/S is not smooth, and which is isolated as such. Suppose that X is regular at x. Then the total number (1.8) of vanishing cycles of X/S at $\bar x$ coincides with the Milnor number $\mu(X/S, x)$.

Let S' be the henselian trait obtained from S by extending the residue field $k(s)$ to its separable closure, and put $X' = X \times_S S'$. The assertion of 2.4 for X/S is equivalent to the assertion for X'/S'. Since $k(s)$ is already assumed perfect, we may therefore suppose that the residue field of S is algebraically closed.

For such an S, let S' be its completion and put $X' = X \times_S S'$. The Milnor number at x is the same for X/S and X'/S', and by XIII, 2.4.2 the same is true of the total number of vanishing cycles. We may thus suppose in addition that S is complete, hence isomorphic to $k[[t]]$, with k algebraically closed. If X_s has dimension n at x, we then have:

Proposition 2.5.

There exist a smooth projective variety Y over k, purely of dimension $n+1$, a morphism $f : Y \to \mathbb{P}^1_k$ which is smooth outside a finite set F of points of Y, and a point $y_0 \in F$ such that:

(a) if $y \in F$ and $y \neq y_0$, then y is an ordinary quadratic singular point of the fiber of f through y;

(b) $f(y_0) = 0$, and if $a : S \to \mathbb{P}^1_k$ is the natural identification of S with the completion of $\mathbb{P}^1_k$ at 0, there is an S-isomorphism between the henselization of X at x and the henselization of a^*Y at y_0.

Let us show that 2.5 implies 2.4. By 2.3 and 2.5(a), the Milnor number of $Y/\mathbb{P}^1_k$ at y_0 coincides with the total number of vanishing cycles. As above, XIII, 2.4.2 gives the same result for a^*Y/S, hence also for $(X/S, x)$, which by 2.5(b) is locally isomorphic to $(a^*Y/S, y_0)$ for the étale topology. II≈209

We prove 2.5. As a $k[[t]]$-algebra, the completion of the local ring $\mathcal{O}_{X,x}$ is isomorphic to a quotient

$$\widehat{\mathcal{O}}_{X,x} = k[[t, x_1, \ldots, x_{n+1}]]/(f).$$

The implicit-function theorem XV, 1.1.2 (that is, [1], Lemma 5.10) then shows that there is an integer N such that condition 2.5(b) holds whenever

$$\widehat{\mathcal{O}}_{Y,y_0} \simeq k[[t, x_1, \ldots, x_{n+1}]]/(f_1)$$

as $k[[t]]$-algebras, with $f_1 - f$ in the Nth power $\mathfrak{m}^N$ of the maximal ideal of $k[[t, x_1, \ldots, x_{n+1}]]$.

Let (t_0, t_1) and $(x_0, \ldots, x_{n+1})$ be the usual homogeneous coordinate systems on $\mathbb{P}^1$ and $\mathbb{P}^{n+1}$. We take $Y/\mathbb{P}^1$ to be the subvariety of $\mathbb{P}^1 \times \mathbb{P}^{n+1}$ defined by an equation

$$g(t_0, t_1; x_0, \ldots, x_{n+1}) = 0, \tag{2.5.1}$$

homogeneous of degree d' in (t_0, t_1) and of degree d'' in the x_i. More intrinsically, g is a section of

$$\mathcal{O}(d', d'') = \mathrm{pr}_1^*\, \mathcal{O}(d') \otimes \mathrm{pr}_2^*\, \mathcal{O}(d'').$$

Condition (b) holds if

$$g(1, t; 1, x_1, \ldots, x_{n+1}) \equiv f \pmod{\mathfrak{m}^N}. \tag{2.5.2}$$

Fix d' and d''. The sections g of type (2.5.1) are the k-rational points of an affine space A. For $u \in \mathbb{P}^1 \times \mathbb{P}^{n+1}$, let $B(u, r)$ denote the affine space whose points are the r-jets at u of sections of $\mathcal{O}(d', d'')$. Put $u_0 = (1, 0; 1, 0, \ldots, 0)$. If $u \neq u_0$, then for sufficiently large d' and d'' the always-linear restriction map II≈210

$$A \longrightarrow B(u_0, N) \times B(u, 2) \tag{2.5.3}$$

is surjective.

Let $C(u) \subset B(u, 2)$ be the set of 2-jets which are the 2-jets either of the zero germ or of a germ $\widetilde{h}$ such that

(a) u lies on the variety V defined by $\widetilde{h} = 0$;

(b) u is neither a smooth point nor an ordinary quadratic point of the fiber through u of $V/\mathbb{P}^1$.

It is readily checked that $C(u)$ is closed of codimension $n + 3$ in $B(u, 2)$.

Let D be the affine subspace of A consisting of the g which satisfy (2.5.2). For $u \neq u_0$, let $D(u)$ be the subspace of D consisting of the zero g and of those g such that $g(u) = 0$ and u is neither a smooth point nor an ordinary quadratic point of the fiber through u of the variety V defined by $g = 0$. If d' and d'' are sufficiently large for (2.5.3) to be surjective, then $D(u)$ has codimension $n + 3$ in D. The union of the $D(u)$ for $u \neq u_0$ therefore has codimension at least

$$1 = (n + 3) - (n + 2),$$

and a general element g of D lies in none of the $D(u)$. It solves the problem.

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SGA 7 — EXPOSÉ XVII

Lefschetz Pencils: Existence Theorem

by N. M. Katz

0. Introduction

In the present exposé (which, in the oral seminar, came before Exposés XIV and XV devoted to the Picard–Lefschetz formula), we introduce the "Lefschetz pencils" of hyperplane sections of a smooth projective scheme and give conditions for their existence. In the following exposé we make an elementary cohomological study of them, based on the known structure of the cohomology of blown-up varieties. The fact that certain hypotheses on Lefschetz pencils which we shall introduce are often satisfied will follow from the Picard–Lefschetz formula of Exposé XV, which is less elementary in nature than the results of the present exposé and the next. II ≈**213**

Some developments of the present exposé and the next will be given elsewhere in a more general setting (general base schemes, quasi-projectivity hypotheses in place of projectivity, and so forth): in particular, the precohomological study of pencils of hyperplane sections, and especially of Lefschetz pencils, in EGA V, and the cohomological study of blown-up schemes in Jouanolou's Exposé VII of SGA 5. The reader will also find in EGA V proofs in which intrinsic diagram chasing replaces the use of coordinates favored by the writer of the present exposé.

I wish to thank P. Deligne and P. Griffiths for the help they gave me in conceiving this exposé. Throughout it, k denotes an algebraically closed field unless expressly stated otherwise.

1. A reminder on quadratic singularities (cf. VI, 6)

1.1.

Let Y be a k-scheme pure of dimension $n-1$.

A closed point $y \in Y$ is called an ordinary quadratic singularity of Y if:

(a) the completion $\widehat{O}_{Y,y}$ of the local ring of Y at y is isomorphic to

$$k[[T_1, \ldots, T_n]]/(Q(T)),$$

the quotient of the formal power-series ring in $n = 1 + \dim Y$ variables by the ideal generated by an element $Q(T)$;

(b) Q begins in degree two, and the subvariety of $\mathbb{P}_k^{n-1}$ defined by the vanishing of the homogeneous degree-two part of $Q(T)$ is smooth over k when $n \geq 2$, while the degree-two part of $Q(T)$ is nonzero when $n = 1$.

If the characteristic of k is different from two, or if n is even, condition (b) is equivalent to:

(b′) Q begins in degree two and its Hessian matrix

$$\left(\frac{\partial^2 Q}{\partial T_i \partial T_j}(0)\right)_{1 \leq i,j \leq n}$$

is invertible.

This follows from the Jacobian criterion for smoothness.

The standard form for the homogeneous degree-two part Q_2 of a series Q satisfying (b), obtainable after a linear change of variables, is II≈214

$$(1.1.1)\qquad \begin{cases} Q_2(T) = \sum_{i=1}^{n/2} T_{2i-1}T_{2i} & \text{if } n \text{ is even,} \\ Q_2(T) = T_n^2 + \sum_{i=1}^{(n-1)/2} T_{2i-1}T_{2i} & \text{if } n \text{ is odd.} \end{cases}$$

This follows from the theory of quadratic forms (Bourbaki, *Algèbre*, Ch. IX).

1.2.

A closed point $y \in Y$ satisfying (a) and (b′) is called a nondegenerate quadratic singularity. Since the determinant of the Hessian matrix is identically zero in characteristic two when n is odd, nondegenerate quadratic singularities presented in n variables exist only when n is even or the characteristic of k is different from two; in those cases they are precisely the ordinary quadratic singularities.

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2. Lefschetz pencils: statement of the results

2.1.

Let P^r denote projective r-space over k, $\check{\mathbb{P}}^r$ the dual projective space of hyperplanes in $\mathbb{P}^r$, and $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ the variety of lines in $\check{\mathbb{P}}^r$. A point $D \in \mathrm{Gr}(1, \check{\mathbb{P}}^r)$ is called a pencil of hyperplanes in $\mathbb{P}^r$. Regarding D as a line in $\check{\mathbb{P}}^r$, we write, by abuse of notation, $D = \{H_t\}_{t\in D}$, where the H_t are the hyperplanes of $\mathbb{P}^r$ which are the points of the line D.

The axis of D is the codimension-two linear subvariety of $\mathbb{P}^r$ which is the intersection $H_0 \cap H_\infty$ of any two distinct members of the pencil. Regarding this axis as a point of $\mathrm{Gr}(r-2, \mathbb{P}^r)$, the variety of $(r-2)$-dimensional linear subvarieties of $\mathbb{P}^r$, the map sending D to its axis is the familiar orthogonality isomorphism

$$\mathrm{Gr}(1, \check{\mathbb{P}}^r) \xrightarrow{\sim} \mathrm{Gr}(r-2, \mathbb{P}^r).$$

2.2.

Let X be a proper, smooth, irreducible k-scheme of dimension $n \geq 1$, equipped with an embedding $X \hookrightarrow \mathbb{P}^r$. A pencil $D = \{H_t\}_{t\in D}$ of hyperplanes in $\mathbb{P}^r$ is called a Lefschetz pencil (relative to this embedding) if:

(a) the axis of D meets X transversely (EGA IV, 17.13.7);

(b) there is a dense open subset $U \subset D$ such that, for every $t \in U$, the hyperplane H_t meets X transversely;

(c) for $t_0 \in D - U$, the hyperplane H_{t_0} meets X transversely except at one point, which is an ordinary quadratic singular point of $X \cap H_{t_0}$.

We shall see below in 3.2.1 that the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$.

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2.3.

The embedding $X \hookrightarrow \mathbb{P}^r$ is called a Lefschetz embedding if the Lefschetz pencils form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. [22]

[22] More precisely, they form a dense open subset in the set of closed points of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. We shall permit ourselves such abuses of language below.

2.4.

Recall the notion of a multiple of a given embedding $X \hookrightarrow \mathbb{P}^r$. It is the composite of that embedding with the d-th Veronese embedding

$$S_d : \mathbb{P}^r \hookrightarrow \mathbb{P}^{\binom{r+d}{d}-1}, \qquad (X_0, \ldots, X_r) \longmapsto (\ldots, X^W, \ldots),$$

where

$$X^W = X_0^{W_0} \cdots X_r^{W_r} \quad \text{and} \quad W_0 + \cdots + W_r = d.$$

Thus a "section of X by a hypersurface of degree d" for the given embedding—in English, a hypersurface section of degree d—is simply a hyperplane section for the d-th multiple of that embedding.

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Theorem 2.5.

Let X be irreducible, proper, and smooth over k, of dimension $n \geq 1$, and equipped with an embedding $i : X \hookrightarrow \mathbb{P}^r$. Then:

2.5.1. every d-th multiple of i, with $d \geq 2$, is a Lefschetz embedding;

2.5.2. if k has characteristic zero, the given embedding i is a Lefschetz embedding.

The proof is divided into several pieces, which follow.

3. The dual variety

3.1.

Let I be the ideal in $\mathbb{P}^r$ defining X. Write $N = I/I^2$ for the conormal sheaf, locally free of rank $r-n$ on X, and let $\mathbb{P}(N)$ be the projective bundle over X formed by the lines in $\check{N}$. (When $X = \mathbb{P}^r$, the bundle $\mathbb{P}(N)$ is empty.) The bundle $\mathbb{P}(N)$ identifies with the subvariety of $X \times \check{\mathbb{P}}^r$ formed by the pairs (x, H) for which H is tangent to X at x, by the map

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r.$$

More explicitly,

$$\text{(3.1.1)} \qquad \mathbb{P}(I/I^2)_x \ni F \longmapsto \left(x, \text{ the hyperplane with equation } \sum_i \frac{\partial F}{\partial X_i}(x) T_i = 0 \right),$$

where F denotes a local section of I/I^2. Define

$$\text{(3.1.2)} \qquad \varphi : \mathbb{P}(N) \longrightarrow \check{\mathbb{P}}^r$$

to be the composite

$$\mathbb{P}(N) \xrightarrow{\nu} X \times \check{\mathbb{P}}^r \xrightarrow{\mathrm{pr}_2} \check{\mathbb{P}}^r.$$

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3.1.3.

Let $\check{X}$ denote the image of φ. It is the set of hyperplanes which do not meet X transversely, and is called the dual variety of X for the embedding $i : X \hookrightarrow \mathbb{P}^r$.

Proposition 3.1.4.

The dual variety $\check{X}$ is irreducible and proper, of dimension at most $r-1$.

Proof. It is the image under φ of $\mathbb{P}(N)$, which is itself irreducible of dimension $r-1$ if $X \neq \mathbb{P}^r$, and empty if $X = \mathbb{P}^r$.

Remark 3.1.5.

The interpretation of $\mathbb{P}(N)$ can be made more precise as follows. Let $\mathcal{Y} \subset X \times \check{\mathbb{P}}^r$ be the incidence subscheme whose fiber over every point $H \in \check{\mathbb{P}}^r(k)$ is $X \cap H$. The scheme $\mathcal{Y}$ is plainly smooth of dimension $n + r - 1$, being a projective bundle of relative dimension $r - 1$ over X. Thus $\mathcal{Y} \to \check{\mathbb{P}}^r$ makes $\mathcal{Y}$ a relative complete intersection of virtual relative dimension $(n + r - 1) - r = n - 1$.

The Jacobian subscheme

$$J_{n-1}(\mathcal{Y}/\check{P}^r)$$

(VI, 5.2) is then precisely $\mathbb{P}(N)$. Formation of the Jacobian subscheme of a relative complete intersection is compatible with arbitrary base change. Consequently, for every line $D \subset \check{\mathbb{P}}^r$, the Jacobian subscheme $J_{n-1}(\widetilde{X}/D)$, where

$$\widetilde{X} = \mathcal{Y} \times_{\check{\mathbb{P}}^r} D,$$

is precisely $\mathbb{P}(N) \times_{\check{\mathbb{P}}^r} D$.

Proposition 3.2.

The subset $B \subset \check{X}$ formed by the hyperplanes H such that $X \cap H$ has exactly one singular point, and that point is an ordinary quadratic singularity, is open.

This is a special case of XV, 1.3.2, based on R. Elkik's theory of "henselian modules" (XV, 1.1.4). II≈**219**
The case in which one restricts to nondegenerate quadratic points—that is, where $\mathrm{char}(k) \neq 2$ or $n = \dim X$ is even—is also immediate from 3.3 below, whose proof does not use 3.2.

Corollary 3.2.1.

Under the hypotheses of 3.2, let $F = \check{X} - B$, a closed subset of $\check{X}$. Then the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$, and the following conditions are equivalent:

3.2.2. the given embedding $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding;

3.2.3. there exists a Lefschetz pencil of hyperplanes for the given embedding;

3.2.4. the codimension of F in $\check{\mathbb{P}}^r$ is at least two.

Proof. A line $D \subset \check{\mathbb{P}}^r$ is a Lefschetz pencil if and only if:

(3.2.5) the axis of D meets X transversely;

(3.2.6) D is not contained in $\check{X}$;

(3.2.7) D does not meet F.

Conditions (3.2.5) and (3.2.6) define open subsets of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$; since F is closed, so does (3.2.7). Hence the Lefschetz pencils form an open subset.

Because $\dim \check{X} \leq r - 1$, the open subset in (3.2.6) is nonempty and therefore dense. The open subset in (3.2.5) is likewise nonempty and dense. Indeed, choose a hyperplane H meeting X transversely; such an H exists because $\dim \check{X} \leq r - 1$. Since $\dim (X \check{\cdot} H) \leq r - 1$, there is a hyperplane G meeting $X \cap H$ transversely, and the line D through H and G then has an axis meeting X transversely. Thus the embedding is Lefschetz if and only if the open subset in (3.2.7) is nonempty, hence dense. Since F is closed, this occurs if and only if F has codimension at least two in $\check{\mathbb{P}}^r$ (cf. EGA V, or Mumford, *Introduction to Algebraic Geometry*, p. 88, Cor. 3).

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Corollary 3.2.8.

Under the hypotheses of 3.2, suppose that $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding. Let H_0 be a hyperplane meeting X transversely, and let $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ be the subvariety of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ formed by the lines through H_0. Then the Lefschetz pencils through H_0 form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$.

Proof. By 3.2.1 these pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$. The argument just given to prove that the open subsets (3.2.5) and (3.2.6) are nonempty also shows that their intersections with this subvariety are nonempty. It remains to prove that some lines through H_0 do not meet F.

Choose a hyperplane $T \subset \check{\mathbb{P}}^r$ not containing H_0. There is an isomorphism II≈**221**

$$\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0} \xleftarrow{\ \sim\ } T$$

$$\text{the line through } H_0 \text{ and } \tau \longleftarrow\!\shortmid\ \tau$$

so $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ "is" a projective space of dimension $r-1$. The lines in it which pass through a point of F form a closed subset of dimension at most $r-2$, being the image of the morphism

$$F \longrightarrow \mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$$

$$f \longmapsto \text{the line through } H_0 \text{ and through } f.$$

Proposition 3.3.

The morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is unramified at a closed point (x_0, H) if and only if x_0 is a nondegenerate quadratic singularity (1.2) of $X \cap H$. In particular, the subset of $\mathbb{P}(N)$ formed by the points (x, H) for which x is a nondegenerate quadratic singularity of $X \cap H$ is open.

Proof of Proposition 3.3. We may restrict to the case $X \neq \mathbb{P}^r$. The two properties asserted to be equivalent are independent of the particular choice of coordinates, so we may describe them using coordinates chosen in a very special way. Since X is a smooth subvariety of $\mathbb{P}^r$ of dimension $n \geq 1$, choose homogeneous coordinates $X_0, \ldots, X_r$ so that:

3.3.1. the point $x_0 \in \mathbb{P}^r$ is $(1, 0, \ldots, 0)$;

3.3.2. putting $x_i = X_i/X_0$ for $i = 1, \ldots, r$, so that $\widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ identifies with $k[[x_1, \ldots, x_r]]$, the ideal defining the trace of X in $\operatorname{Spec} \widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ is generated by elements g_i $(i = 1, \ldots, r-n)$ which, in the completion, have the form

$$g_i = x_{n+i} - h_i(x_1, \ldots, x_n), \qquad i = 1, \ldots, r-n,$$

where the h_i are power series in $x_1, \ldots, x_n$ alone and lie in the square of the maximal ideal of $k[[x_1, \ldots, x_n]]$;

3.3.3. under (3.1.1), the hyperplane H comes from g_{r-n}.

Put $V = \operatorname{Spec}(\widehat{\mathcal{O}}_{X, x_0})$. It follows from 3.3.2 that there is a trivialization

$$(I/I^2)|_V \simeq \mathcal{O}_V^{r-n}$$

given by

$$(\Lambda_1, \ldots, \Lambda_{r-n}) \longmapsto \sum_{i=1}^{r-n} \Lambda_i g_i. \tag{3.3.4}$$

Together with 3.3.3 this gives a neighborhood W of (x_0, H) in $\mathbb{P}(N)|_V$ and an isomorphism $V \times \mathbb{A}^{r-n-1} \simeq W$, given by

$$\begin{aligned} V \times \mathbb{A}^{r-n-1} &\longrightarrow \mathbb{P}(I/I^2)|_V, \\ (v, (\lambda_1, \ldots, \lambda_{r-n-1})) &\longmapsto \left(v, \left[g_{r-n} + \sum_{i=1}^{r-n-1} \lambda_i g_i\right]\right), \end{aligned} \tag{3.3.5}$$

where $\lambda_i = \Lambda_i/\Lambda_{r-n}$.

In particular, the completion of the local ring of $\mathbb{P}(N)$ at (x_0, H) identifies with

$$k[[x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_{r-n-1}]].$$

We now express φ in these coordinates at the level of completions:

$$\varphi(x, \lambda) = \text{the hyperplane with equation } \sum_{i=0}^{r} \varphi_i(x, \lambda) T_i = 0. \tag{3.3.6}$$

Since this image hyperplane contains the point x, whose homogeneous coordinates are

$$x = (1, x_1, \ldots, x_n, h_1(x), \ldots, h_{r-n}(x)), \tag{3.3.7}$$

we have

$$\varphi_0(x, \lambda) = -\sum_{i=1}^{n} x_i \varphi_i(x, \lambda) - \sum_{j=1}^{r-n} h_j(x) \varphi_{n+j}(x, \lambda) \tag{3.3.8}$$

in $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Moreover, by (3.1.1), if

$$D_i = \frac{\partial}{\partial x_i} \in \mathrm{Der}_k(\widehat{\mathcal{O}}_{\mathbb{P}^r,x_0}, \widehat{\mathcal{O}}_{\mathbb{P}^r,x_0}),$$

then for $1 \leq i \leq r$,

$$\varphi_i(x, \lambda) = D_i \left(g_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j g_j\right)(x) \quad \text{in } \widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}. \tag{3.3.9}$$

Recalling that $g_i = x_{n+i} - h_i(x_1, \ldots, x_n)$, one obtains

$$\varphi_i(x, \lambda) = -\frac{\partial}{\partial x_i} \left(h_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j h_j\right), \qquad 1 \leq i \leq n, \tag{3.3.10}$$

$$\varphi_{n+j}(x, \lambda) = \lambda_j, \qquad 1 \leq j \leq r-n-1, \tag{3.3.11}$$

and

$$\varphi_r(x, \lambda) = 1. \tag{3.3.12}$$

Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Since $h_i \equiv 0 \pmod{\mathfrak{m}^2}$ for $i = 1, \ldots, r-n$, we find II≈222

$$\varphi_0(x, \lambda) \equiv 0 \pmod{\mathfrak{m}^2} \tag{3.3.13}$$

and

$$\varphi_i(x, \lambda) \equiv -\frac{\partial h_{r-n}}{\partial x_i} \pmod{\mathfrak{m}^2}, \qquad i = 1, \ldots, n. \tag{3.3.14}$$

Putting (3.3.11)–(3.3.14) together, the Jacobian matrix of φ at (x_0, H) has the block form

$$J_{(x_0,H)}(\varphi) = \left(\begin{array}{c|c} 0_{1\times n} & 0_{1\times(r-n-1)} \\ \hline -\operatorname{Hess}(h_{r-n})(0) & 0_{n\times(r-n-1)} \\ \hline 0_{(r-n-1)\times n} & I_{r-n-1} \\ \hline 0_{1\times n} & 0_{1\times(r-n-1)} \end{array}\right). \tag{3.3.15}$$

Thus φ is unramified at (x_0, H) if and only if the Hessian matrix

$$\left(\frac{\partial^2 h_{r-n}}{\partial x_i \partial x_j}(0)\right)_{1\le i,j\le n}$$

is invertible. On the other hand, 3.3.2 and 3.3.3 show that the completion of the local ring of $X \cap H$ at x_0 is

$$k[[x_1, \ldots, x_n]]/(h_{r-n}(x_1, \ldots, x_n)), \tag{3.3.16}$$

and the result follows from Definition 1.2.

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Example 3.4.

Suppose that X is a hypersurface in $\mathbb{P}^r$, defined by a homogeneous equation $F(X_0, \ldots, X_r) = 0$. Then $\mathbb{P}(N) = X$, and in homogeneous coordinates $\varphi : X \to \check{\mathbb{P}}^r$ is given by

$$\varphi(X_0, \ldots, X_r) = \left(\frac{\partial F}{\partial X_0}, \ldots, \frac{\partial F}{\partial X_r}\right). \tag{3.4.1}$$

This is the Gauss morphism.

Here is an example in which φ is everywhere ramified. Let p be the characteristic of k, let $r \ge 1$, and take the hypersurface X with equation

$$\sum_{i=0}^{r} X_i^{p^r+1} = 0. \tag{3.4.2}$$

Then

$$\varphi(X_0, \ldots, X_r) = (X_0^{p^r}, \ldots, X_r^{p^r}). \tag{3.4.3}$$

Thus $\check{X}$ is reduced to a point if $p = 0$, while if $p > 0$ one has $\check{X} \simeq X$ and φ is the Frobenius morphism. It follows from 3.3 (or from 3.5.0 below) that if $p > 2$, this embedding is not a Lefschetz embedding.

We now return to the general case.

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Proposition 3.5.

Suppose that the morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is generically unramified (that is, is not everywhere ramified). Then φ induces a birational morphism from $\mathbb{P}(N)$ onto $\check{X}$. More precisely, the following three open subsets of $\check{X}$ are identical:

(a) the largest open subset on which $\check{X}$ is smooth;

(b) the largest open subset $V \subset \check{X}$ such that φ induces an isomorphism $\varphi^{-1}(V) \xrightarrow{\sim} V$;

(c) the open subset (cf. 3.2) formed by those hyperplanes H for which $X \cap H$ has exactly one singular point, that point being a nondegenerate quadratic singularity.

Corollary 3.5.0.

Suppose that $\mathrm{char}(k) \neq 2$ or that $\dim X$ is even. The given embedding is a Lefschetz embedding if and only if one of the following two conditions holds:

(a) φ is generically unramified;

(b) $\dim \check{X} \leq r-2$.

Indeed, sufficiency follows from criterion (3.2.4). In case (a), Proposition 3.5 identifies F with the singular locus of $\check{X}$, so

$$\dim F < \dim \check{X} = r-1;$$

in case (b), the criterion is immediate. Conversely, suppose that D is a Lefschetz pencil and $\dim \check{X} = r-1$. There is a point $H \in D \cap \check{X}$, and 3.3 together with Definition 2.2(c) shows that φ is unramified at the points of $\varphi^{-1}(H)$. Hence it is generically unramified.

Proof of Proposition 3.5. The hypothesis implies

$$\dim \check{X} = \dim \mathbb{P}(N) = r-1,$$

and therefore 3.2 shows that $\check{X}$ is an irreducible hypersurface in $\check{\mathbb{P}}^r$.

Let R be the ramification locus of φ. By hypothesis it is a closed subset of $\mathbb{P}(N)$ of dimension at most $r-2$; hence $\varphi(R)$ is a closed subset of $\check{X}$ of dimension at most $r-2$. Let V' be the dense open subset II≈**228**

$$V' = (\check{X} - \varphi(R)) \cap (\check{X} - \mathrm{Sing}(\check{X})) \tag{3.5.1}$$

of $\check{X}$, and denote its complement by F:

$$F = \varphi(R) \cup \mathrm{Sing}(\check{X}). \tag{3.5.1.1}$$

We first prove that φ induces an isomorphism $\varphi^{-1}(V') \simeq V'$ by constructing an inverse $\psi : V' \to \varphi^{-1}(V')$. Regard $\varphi^{-1}(V')$ as embedded in $X \times V'$ by (3.1.1), so that φ is the second projection. Constructing ψ is then equivalent to constructing a morphism $\psi_1 : V' \to \mathbb{P}^r$ which makes the diagram

(3.5.2)

$$\begin{array}{ccccc} \mathbb{P}(N) & & \supset \varphi^{-1}(V') & \xrightarrow{\varphi} & V' \\ \downarrow & & & & \downarrow \\ & & & & \psi_1 \\ \downarrow & & & & \downarrow \\ X & \hookrightarrow & & & \mathbb{P}^r \end{array}$$

commute; one then puts $\psi(v) = (\psi_1(v), v) \in \mathbb{P}^r \times V'$.

For any hypersurface in projective space, the Gauss morphism (3.4.1) is defined at every smooth point. Apply this to the hypersurface $\check{X} \subset \check{\mathbb{P}}^r$. Since V' lies in the smooth locus V of $\check{X}$, the Gauss morphism gives

$$\psi_1 : V' \longrightarrow \check{\check{\mathbb{P}}}^r \simeq \mathbb{P}^r. \tag{3.5.3}$$

It remains to check that (3.5.2) commutes. Let (x_0, H_0) be a closed point of $\varphi^{-1}(V')$. We must show that x_0, viewed as a hyperplane in $\check{\mathbb{P}}^r$, is tangent to $\check{X}$ at H_0: in other words, tangency of H_0 to X at x_0 implies tangency of x_0 to $\check{X}$ at H_0. Since φ is unramified at (x_0, H_0) and $\check{X}$ is smooth at $H_0 = \varphi(x_0, H_0)$, its differential is an isomorphism

$$d\varphi : T_{\mathbb{P}(N),(x_0,H_0)} \xrightarrow{\sim} T_{\check{X},H_0}. \tag{3.5.4}$$

The first part of the proof is therefore completed by the following lemma.

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Lemma 3.6 (Euler).

Let (x_0, H_0) be a closed point of $\mathbb{P}(N)$, and let T_L denote the tangent bundle of the hyperplane $L \subset \check{\mathbb{P}}^r$ defined by x_0. Then, inside $T_{\check{\mathbb{P}}^r, H_0}$,

$$d\varphi\big(T_{\mathbb{P}(N),(x_0,H_0)}\big) \subset T_{L,H_0}.$$

Proof. Choose homogeneous coordinates $X_0, \dots, X_r$ on $\mathbb{P}^r$ in which x_0 is the point $\underline{A} = (A_0, \dots, A_r)$. A point of T_{X,x_0} is a point of $\mathbb{P}^r$ with values in $k[\epsilon]/(\epsilon^2)$ of the form

$$\underline{A} + \epsilon \underline{B}, \tag{3.6.1}$$

such that, for every local section H of I,

$$\sum_{i=0}^{r} B_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0. \tag{3.6.2}$$

Choose also a local section H of I which gives rise to H_0 by (3.1.1). Since $\mathbb{P}(N)$ is locally trivial over X, a point of $T_{\mathbb{P}(N),(x_0,H_0)}$ can be represented as

$$(\underline{A} + \epsilon \underline{B},\ H + \epsilon F), \tag{3.6.3}$$

where F is a local section of I and $\underline{A}, \underline{B}$ are as above. Its image under φ is the hyperplane over $k[\epsilon]/(\epsilon^2)$ with equation

$$\sum_{i=0}^{r} \frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) T_i = 0. \tag{3.6.4}$$

Thus the assertion is equivalent to

$$\sum_{i=0}^{r} A_i \frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) = 0. \tag{3.6.5}$$

Because $\epsilon^2 = 0$, Taylor's formula gives

$$\frac{\partial (H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon \underline{B}) = \frac{\partial H}{\partial X_i}(\underline{A}) + \epsilon \frac{\partial F}{\partial X_i}(\underline{A}) + \epsilon \sum_{j=0}^{r} B_j \frac{\partial^2 H}{\partial X_j \partial X_i}(\underline{A}). \tag{3.6.6}$$

It is therefore enough to verify

$$\sum_{i=0}^{r} A_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0, \tag{3.6.7}$$

$$\sum_{i=0}^{r} A_i \frac{\partial F}{\partial X_i}(\underline{A}) = 0, \tag{3.6.8}$$

and

$$\sum_{i,j=0}^{r} A_i B_j \frac{\partial^2 H}{\partial X_i \partial X_j}(\underline{A}) = 0. \tag{3.6.9}$$

Here H and F are represented by homogeneous functions of degree zero, so Euler's formula gives

$$\sum_i X_i \frac{\partial H}{\partial X_i} = 0, \qquad \sum_i X_i \frac{\partial F}{\partial X_i} = 0. \tag{3.6.10}$$

Since $\partial H/\partial X_j$ is homogeneous of degree -1, the same formula gives

$$\sum_i X_i \frac{\partial^2 H}{\partial X_i \partial X_j} = -\frac{\partial H}{\partial X_j}.$$

Consequently, (3.6.9) becomes

$$-\sum_j B_j \frac{\partial H}{\partial X_j}(\underline{A}) = 0, \tag{3.6.11}$$

which is precisely (3.6.2).

This proves that φ induces an isomorphism II≈231

$$\varphi^{-1}(V') \xrightarrow{\sim} V', \tag{3.6.12}$$

whose inverse is ψ.

Now observe that the morphism II≈232

$$\begin{cases} \psi : V' \longrightarrow \mathbb{P}^r \times V' \subset \mathbb{P}^r \times \check{X}, \\ \psi(v) = (\psi_1(v), v) \end{cases} \tag{3.6.13}$$

extends to a morphism

$$\begin{cases} \widetilde{\psi} : V \longrightarrow \mathbb{P}^r \times V, \\ \widetilde{\psi}(v) = (\widetilde{\psi}_1(v), v), \end{cases} \tag{3.6.14}$$

where

$$\widetilde{\psi}_1 : V \longrightarrow \mathbb{P}^r \tag{3.6.15}$$

is the Gauss morphism, now regarded as defined on all of V.

We have just proved

$$\psi(V') \subset \mathbb{P}(N). \tag{3.6.16}$$

Since V' is dense in V and $\mathbb{P}(N)$ is closed in $\mathbb{P}^r \times \check{X}$, this implies

$$\widetilde{\psi}(V) \subset \mathbb{P}(N), \tag{3.6.17}$$

and in fact plainly

$$\widetilde{\psi}(V) \subset \varphi^{-1}(V). \tag{3.6.18}$$

Again by continuity, the identities

$$\psi \circ \varphi = \mathrm{id}_{\varphi^{-1}(V')}, \qquad \varphi \circ \psi = \mathrm{id}_{V'}$$

extend to

$$\widetilde{\psi} \circ \varphi = \mathrm{id}_{\varphi^{-1}(V)}, \qquad \varphi \circ \widetilde{\psi} = \mathrm{id}_V .$$

Thus the open subsets in 3.5(a) and 3.5(b) are identical.

We next show that the open subsets in 3.5(a) and 3.5(c) are identical. By 3.3, the latter is the II≈233 subset $U \subset \check{X}$ defined by

$$U = \left\{ H \in \check{X} \;\middle|\; \begin{array}{l} \varphi^{-1}(H) \text{ consists of exactly one point, and} \\ \varphi \text{ is unramified at that point} \end{array} \right\}. \tag{3.6.19}$$

The scheme $\varphi^{-1}(U)$ is normal, since it is smooth, and the induced morphism

$$\varphi : \varphi^{-1}(U) \longrightarrow U$$

is birational, as already proved, radicial by the definition of U, and proper. It is therefore the normalization of U. In particular, U is geometrically unibranch. Since this morphism is unramified, EGA IV, 18.10.1 shows that it is étale; EGA IV, 18.10.18 then shows that it is an open immersion. It is consequently an isomorphism, and U is smooth. This proves $3.5(c) \subset 3.5(a)$. Formula (3.6.19) makes the inclusion $3.5(b) \subset 3.5(c)$ clear, and the equality $3.5(a) = 3.5(b)$ completes the proof.

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3.7.

Theorem 2.5 is true when $\mathrm{char}(k) \neq 2$, and also when X has even dimension.
Proof. In view of 3.5.0 and 3.3, it is enough to verify the following assertion: II ≈234

(3.7.1) For every closed point $x_0 \in X$ and every integer $d \geq 2$, there is a hypersurface H of degree d, tangent to X at x_0, such that x_0 is an ordinary quadratic singularity of $X \cap H$.

Choose homogeneous coordinates $X_0, \ldots, X_r$ so that $x_0 = (1, 0, \ldots, 0)$ and the functions X_i/X_0, $i = 1, \ldots, n$, are local coordinates on X near x_0. One may take H to be the hypersurface with equation

$$H = X_0^{d-2} \sum_{i=1}^{n} X_i^2 \qquad \text{if } \mathrm{char}(k) \neq 2, \tag{3.7.1.1}$$

or

$$H = X_0^{d-2} \sum_{i=1}^{n/2} X_i X_{n/2+i} \qquad \text{if } n \text{ is even.} \tag{3.7.1.2}$$

When n is odd (under the first hypothesis), the equivalent standard choice recorded in the source is

$$H = X_0^{d-2} \left(X_n^2 + \sum_{i=1}^{\lfloor n/2 \rfloor} X_i X_{\lfloor n/2 \rfloor + i} \right). \tag{3.7.1.3}$$

For assertion 2.5.2, note that either φ is generically unramified, in which case 3.5.0 applies, or it is everywhere ramified. In characteristic zero the latter case implies $\dim \check{X} \leq r - 2$, so criterion (3.2.4) is then automatically satisfied.

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4. The general case

We shall sketch another method of proving 2.5, valid in every characteristic, assuming the following statement, which is close to 3.1 and is a special case of XV, 1.1.4.

(4.1) The subset of $\mathbb{P}(N)$ formed by the points (x, H) such that x is an ordinary quadratic singularity of $X \cap H$ is open.

Observe that 4.1 follows from 3.3, except when $\mathrm{char}(k) = 2$ and $\dim X$ is odd.

4.1.1.

Let R_1 be the closed complement in $\mathbb{P}(N)$ of the open set 4.1, and let

$$F_1 = \varphi(R_1) \subset \check{X} \subset \check{\mathbb{P}}^r$$

be its image.

Lemma 4.1.2.

If the open set 4.1 is nonempty, then $\mathrm{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$.
Proof. By hypothesis,

$$\dim R_1 < \dim \mathbb{P}(N) = r - 1,$$

and $F_1 = \varphi(R_1)$.

4.1.3.

Let F_2 be the subset of $\check{X}$ formed by the hyperplanes which are tangent to X at at least two points. Notice that F_2 is not closed, but that

$$\overline{F_2} \subset F_1 \cup F_2. \tag{4.1.4}$$

This follows from the fact that $F_1 \cup F_2 = F$ (cf. 3.2) is closed. (Inclusion (4.1.4) "says" that a confluence of two singularities is not an ordinary quadratic singularity.) In any case, the construction in 4.2.4 will show that F_2 is constructible.

To prove 2.5.1 we shall apply criterion 3.2.4. For this we need to know that

$$\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2 \quad \text{and} \quad \operatorname{codim}_{\check{\mathbb{P}}^r}(F_2) \geq 2.$$

On the one hand, 3.7.1 and 4.1.2 show that, after replacing the given embedding by its d-th multiple, $d \geq 2$, one has $\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$.

Thus 2.5.1 will follow once we know the following proposition. II≈**236**

Proposition 4.2.

Let X be a smooth irreducible subvariety of $\mathbb{P}^r$, of dimension $n \geq 1$. For every integer $d \geq 2$, the subset F_2 of

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

formed by the hypersurfaces of degree d in $\mathbb{P}^r$ which are tangent to X at at least two points has codimension at least 2 in $\check{\mathbb{P}}^N$.

4.2.1.

Begin with the special case $d = 2$ and X a linear subvariety of $\mathbb{P}^r$. It is enough to find a line in $\check{\mathbb{P}}^N$ which does not meet F (cf. 3.2.4). Choose homogeneous coordinates $X_0, \ldots, X_r$ on $\mathbb{P}^r$ so that X is defined by

$$X_{n+1} = \cdots = X_r = 0,$$

and homogeneous coordinates (λ, μ) on $\mathbb{P}^1$.

For n even, $n = 2k$, take the pencil of quadrics II≈**237**

$$H_{(\lambda,\mu)} = \lambda X_0^2 + (\lambda X_k + \mu X_0)X_{2k} + \sum_{i=1}^{k-1} (\lambda X_i + \mu X_{i+1})X_{k+i}. \tag{4.2.2}$$

For n odd, $n = 2k + 1$, take the pencil

$$H_{(\lambda,\mu)} = (\lambda X_1 + \mu X_0)X_0 + \sum_{i=1}^{k} (\lambda X_{2i} + \mu X_i)X_{2i+1}. \tag{4.2.3}$$

In both cases one checks that $X \cap H_{(\lambda,\mu)}$ is smooth for $(\lambda, \mu) \neq (0, 1)$, while $X \cap H_{(0,1)}$ has just one singular point, which is moreover an ordinary quadratic singularity. Thus the pencil does not meet F.

4.2.4.

Let $\operatorname{Btg}^d(X)$ be the subvariety of

$$\check{\mathbb{P}}^N \times ((X \times X) - \Delta(X \times X))$$

formed by triples (H, x_1, x_2) such that H is a hypersurface of degree d in $\mathbb{P}^r$, tangent to X at x_1 and at x_2, with $x_1 \neq x_2$. The image of $\mathrm{Btg}^d(X)$ under pr_1 is the set F_2 of Proposition 4.2, so it is enough to prove

(4.2.5) $$\dim \mathrm{Btg}^d(X) \leq N - 2 \quad \text{if } d \geq 3, \text{ or if } d = 2 \text{ and } X \text{ is not linear.}$$

For this purpose, let $\mathrm{Lin}(X \times X)$ be the subvariety of $(X \times X) - \Delta(X \times X)$ given by

(4.2.6) $$\begin{aligned} \mathrm{Lin}(X \times X) &= \{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad \text{the line spanned by } x_1, x_2 \text{ is tangent to } X \text{ at } x_1\} \\ &= \{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad x_2 \in X \cap \mathrm{Tg}_X(x_1)\}. \end{aligned}$$

Clearly:

(4.2.7) If X is not a linear subvariety of $\mathbb{P}^r$, then II≈**238**

$$\dim \mathrm{Lin}(X \times X) < 2 \dim X.$$

Consequently, to prove (4.2.5) it is enough to show that the fibers of the morphism

(4.2.8) $$\mathrm{Btg}^d(X) \xrightarrow{\mathrm{pr}_{2,3}} (X \times X) - \Delta(X \times X),$$

regarded as subvarieties of $\check{\mathbb{P}}^N$, have

(4.2.9) $$\begin{cases} \text{codimension at least } 2 \dim X + 1 & \text{over } \mathrm{Lin}(X \times X), \\ \text{codimension at least } 2 \dim X + 2 & \text{over } (X \times X) - \Delta(X \times X) - \mathrm{Lin}(X \times X). \end{cases}$$

Statement (4.2.9) follows from the next proposition by taking L_i to be the tangent space to X at x_i, for $i = 1, 2$.

Proposition 4.3.

Let x_1 and x_2 be two distinct closed points of $\mathbb{P}^r$, and let L_1 and L_2 be two linear subspaces of $\mathbb{P}^r$ such that $x_1 \in L_1$ and $x_2 \in L_2$. In the space

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

of hypersurfaces of degree d, the linear subspace formed by the hypersurfaces tangent to L_1 at x_1 and to L_2 at x_2 has codimension

$$\begin{cases} \dim L_1 + \dim L_2 + 2, & \text{if } d \geq 3, \text{ or if } d = 2 \text{ and } L_1 \cap L_2 \text{ does not} \\ & \text{contain the line } (x_1, x_2) \text{ through } x_1, x_2, \\ \dim L_1 + \dim L_2 + 1, & \text{if } d = 2 \text{ and } L_1 \cap L_2 \text{ contains the line } (x_1, x_2). \end{cases}$$

Proof. For $d \geq 3$, one must prove that $\dim L_1 + \dim L_2 + 2$ linear conditions are independent. It is therefore enough to treat the case $L_1 = L_2 = \mathbb{P}^r$, that is, to determine how many independent conditions make a hypersurface singular at both x_1 and x_2. Choose homogeneous coordinates so that

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0).$$

The hypersurface with equation $\sum_W A_W X^W = 0$ is singular at $(1, 0, \ldots, 0)$ if and only if all the coefficients of the monomials $X_0^{d-1} X_i$, $i = 0, \ldots, r$, vanish. It is singular at $(0, 1, 0, \ldots, 0)$ if and only if all the coefficients of the monomials $X_1^{d-1} X_i$, $i = 0, \ldots, r$, vanish. For $d \geq 3$ all these monomials are distinct, which proves the assertion in this case.

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For $d = 2$, the same calculation gives

$$\text{codim} \geq \dim L_1 + \dim L_2 + 1,$$

because the monomial X_0X_1 is counted twice.

Suppose first that $L_1 \cap L_2$ does not contain the line (x_1, x_2). We reduce to the case $L_1 = \mathbb{P}^r$ and L_2 is a hyperplane in $\mathbb{P}^r$ which does not contain (x_1, x_2). Choose homogeneous coordinates $X_0, \ldots, X_r$ so that

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0), \qquad L_2 = (X_0 = 0).$$

A quadric $\sum_{i \leq j} A_{ij}X_iX_j$ is singular at $(1, 0, \ldots, 0)$ if and only if $A_{0j} = 0$ for $j = 0, \ldots, r$, and is tangent to $X_0 = 0$ at $(0, 1, \ldots, 0)$ if and only if $A_{1j} = 0$ for $j = 1, \ldots, r$. Together these are

$$2r + 1 = \dim L_1 + \dim L_2 + 2$$

independent conditions.

Finally, suppose that $L_1 \cap L_2$ contains the line (x_1, x_2). We reduce to the case $L_1 = L_2 = (x_1, x_2)$. II ≈240
In homogeneous coordinates with

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0),$$

the quadric $\sum_{i \leq j} A_{ij}X_iX_j$ passes through $(1, 0, \ldots, 0)$ and is tangent there to L_1 if and only if $A_{00} = A_{01} = 0$. It passes through $(0, 1, 0, \ldots, 0)$ and is tangent there to L_2 if and only if $A_{11} = A_{01} = 0$. Thus there are altogether only

$$3 = \dim L_1 + \dim L_2 + 1$$

conditions.

5. The degree of the dual variety by an "elementary" method

(Cf. XVIII, 3.2 for another method when $\dim X$ is even or $\text{char}(k) \neq 2$.)

We continue to use the notation of 2.2: thus $X \hookrightarrow \mathbb{P}^r$ is a smooth irreducible subvariety of dimension n.

5.1.

We first consider the morphism (3.1.1)

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r. \tag{5.1.0}$$

By definition, the conormal sheaf $\check{N}$ occurs in the exact sequence of coherent sheaves on X

$$0 \longrightarrow \check{N} \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \Omega^1_X \longrightarrow 0. \tag{5.1.1}$$

On the other hand, writing differential forms on $\mathbb{P}^r$ in homogeneous coordinates gives an exact sequence of sheaves on $\mathbb{P}^r$

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k} \longrightarrow \mathcal{O}_{\mathbb{P}^r}(-1)^{r+1} \longrightarrow \mathcal{O}_{\mathbb{P}^r} \longrightarrow 0. \tag{5.1.2}$$

Its restriction to X is therefore the exact sequence

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \mathcal{O}_X(-1)^{r+1} \longrightarrow \mathcal{O}_X \longrightarrow 0. \tag{5.1.3}$$

Combining (5.1.1) and (5.1.3) gives an inclusion

$$\check{N} \hookrightarrow \mathcal{O}_X(-1)^{r+1}, \tag{5.1.4}$$

which, by transposition, gives a surjection

$$\mathcal{O}_X(1)^{r+1} \longrightarrow N. \tag{5.1.5}$$

Hence there is a closed immersion

$$\alpha : \mathbb{P}(N) \longrightarrow \mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r, \tag{5.1.6}$$

which is precisely (5.1.0).

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5.2. The idea of the calculation.

From now on suppose that $\check{X} = \varphi(\mathbb{P}(N))$ has dimension $r-1$, so that it is a hypersurface in $\check{\mathbb{P}}^r$. Its degree is naturally calculated by means of Chern classes [1].

For every smooth projective variety T, let $\mathrm{CH}(T)$ denote its Chow ring, graded by codimension. For $z \in \mathrm{CH}(T)$, write

$$\int_T z$$

for the degree of its zero-dimensional component. The class in $\mathrm{CH}^1(T)$ of an invertible sheaf $\mathcal{L}$ on T will also be denoted by $\mathcal{L}$ whenever no confusion can arise (for example, under the integral sign).

Let L be the class of a line in $\mathrm{CH}(\check{\mathbb{P}}^r)$. Then

$$\deg(\check{X}) = \int_{\check{\mathbb{P}}^r} L \cdot \check{X}. \tag{5.2.1}$$

Set-theoretically, $\check{X}$ is the image of $\mathbb{P}(N)$ under φ (3.1.2). From the viewpoint of the proper push-forward

$$\varphi_* : \mathrm{CH}(\mathbb{P}(N)) \longrightarrow \mathrm{CH}(\check{\mathbb{P}}^r),$$

however, one has

$$\varphi_*[\mathbb{P}(N)] = \deg(\varphi)[\check{X}].$$

Consequently, (5.2.1) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\check{\mathbb{P}}^r} L \cdot \varphi_*[\mathbb{P}(N)]. \tag{5.2.2}$$

Here, by abuse of notation, φ in $\deg(\varphi)$ denotes the morphism $\mathbb{P}(N) \to \check{X}$ induced by (3.1.2); its degree is a nonnegative integer, taken to be zero precisely when $\dim \check{X} < r-1 = \dim \mathbb{P}(N)$.

The projection formula rewrites (5.2.2) as

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \varphi^*(L). \tag{5.2.3}$$

Let $\eta \in \mathrm{CH}(\check{\mathbb{P}}^r)$ be the class of a hyperplane. Since $L = \eta^{r-1}$, equation (5.2.3) becomes II ≈**243**

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\varphi^*(\eta)\big)^{r-1}. \tag{5.2.4}$$

We now calculate $\varphi^*(\eta)$ concretely.

Let ξ (respectively τ) denote the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(N)}(1)$ on $\mathbb{P}(N)$ (respectively $\mathcal{O}_{\mathbb{P}(\mathcal{O}_X(1)^{r+1})}(1)$ on $\mathbb{P}(\mathcal{O}_X(1)^{r+1})$). Functoriality gives, by (5.1.6),

$$\alpha^*(\tau) = \xi. \tag{5.2.5}$$

Under the identification

$$\mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r,$$

and writing $H = \mathcal{O}_X(1)$, one plainly has

$$\tau = \mathrm{pr}_1^*(H) \otimes \mathrm{pr}_2^*(\eta). \tag{5.2.6}$$

Let $\pi : \mathbb{P}(N) \to X$ be the structural morphism. Then

$$\pi = \mathrm{pr}_1 \circ\alpha, \qquad \varphi = \mathrm{pr}_2 \circ\alpha.$$

Applying α^* to (5.2.6) and using (5.2.5), we obtain

$$\xi = \alpha^*(\tau) = \pi^*(H) \otimes \varphi^*(\eta). \tag{5.2.6$'$}$$

Thus, in $\mathrm{CH}(\mathbb{P}(N))$,

$$\varphi^*(\eta) = \xi - \pi^*(H), \tag{5.2.7}$$

and (5.2.3) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\xi - \pi^*(H)\big)^{r-1}. \tag{5.2.8}$$

The right-hand side of (5.2.8) is calculated by means of the following lemma (cf. 5.3.3).

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Lemma 5.3.

Let X be a smooth connected quasi-projective variety of dimension n, and let M be a locally free sheaf of finite rank a on X. Let ξ denote the invertible sheaf $\mathcal{O}_{\mathbb{P}(M)}(1)$ on $\mathbb{P}(M)$, and let $\pi : \mathbb{P}(M) \to X$ be the structural morphism. Recall that $\mathrm{CH}(\mathbb{P}(M))$ is a free $\mathrm{CH}(X)$-module (via π^*), with basis $1, \xi, \dots, \xi^{a-1}$, where ξ satisfies

$$\xi^a - C_1\xi^{a-1} + C_2\xi^{a-2} - C_3\xi^{a-3} + \cdots = 0, \tag{5.3.1}$$

the C_i being the Chern classes of M in $\mathrm{CH}(X)$. Put $C(M) = 1 + C_1 + C_2 + \cdots$, the total Chern class of M; it is an invertible element of $\mathrm{CH}(X)$. Then:

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(i) one has

$$\pi_*\left(\frac{\xi^{a-1}}{1+\xi}\right) = \frac{1}{C(M)}; \tag{5.3.2}$$

(ii) let $F(T) \in \mathrm{CH}(X)[T]$ be a polynomial such that the coefficient of T^i is homogeneous of degree $n + a - 1 - i$. Thus $F(\xi)$ has maximal degree $n + a - 1 = \dim \mathbb{P}(M)$ in $\mathrm{CH}(\mathbb{P}(M))$. Then

$$\int_{\mathbb{P}(M)} F(\xi) = (-1)^{a-1} \int_X \frac{F(-1)}{C(M)}. \tag{5.3.3}$$

Proof. We first deduce (5.3.3) from (5.3.2). Write

$$\frac{1}{C(M)} = \sum_{j\geq 0} b_j, \qquad b_0 = 1, \qquad b_j \in \mathrm{CH}^j(X), \qquad b_j = 0 \text{ for } j < 0. \tag{5.3.4}$$

Then (5.3.2) is equivalent to

$$\pi_*(\xi^{a-1+j}) = (-1)^j b_j \qquad \text{whenever } a - 1 + j \geq 0, \tag{5.3.5}$$

since $\pi_*(\xi^q) = 0$ for $q < a - 1$ for dimensional reasons.

By linearity it suffices to prove (5.3.3) when $F(T) = uT^{a-1+j}$, with $u \in \mathrm{CH}^{n-j}(X)$ homogeneous. II≈246
By (5.3.5) and the projection formula,

$$\begin{aligned}\int_{\mathbb{P}(M)} F(\xi) &= \int_{\mathbb{P}(M)} \pi^*(u)\xi^{a-1+j} \\ &= \int_X u\,\pi_*(\xi^{a-1+j}) = (-1)^j \int_X u\,b_j.\end{aligned} \tag{5.3.6}$$

On the other hand,

$$\begin{aligned}(-1)^{a-1}\int_X \frac{F(-1)}{C(M)} &= (-1)^{a-1}\int_X \frac{(-1)^{a-1+j}u}{C(M)} \\ &= (-1)^j \int_X \sum_i u\,b_i.\end{aligned} \tag{5.3.7}$$

Because $\deg u = n - j$ and $\deg b_i = i$, one has

$$\int_X u\,b_i = 0 \qquad \text{for } i \neq j.$$

Thus (5.3.7) reduces to

$$(-1)^{a-1}\int_X \frac{F(-1)}{C(M)} = (-1)^j \int_X u\,b_j, \tag{5.3.8}$$

which proves the asserted implication.

It remains to prove (5.3.2), or equivalently (5.3.5). II≈247

For dimensional reasons,

$$\pi_*(\xi^j) = 0 \qquad \text{if } j < a-1,$$

whereas

$$\pi_*(\xi^{a-1}) = 1_X$$

for the geometric reason that ξ^{a-1} meets each fiber in one point. The projection formula therefore gives

$$\pi_*\left(\sum_{i=0}^{a-1} \pi^*(U_i)\xi^i\right) = U_{a-1}. \tag{5.3.9}$$

By (5.3.9), assertion (5.3.5) follows from the next lemma, in which ξ is replaced by T and C_i by S_i, in view of (5.3.1).

Lemma 5.4.

Let $a > 0$ be an integer and let $A = \mathbb{Z}[[X_1, \ldots, X_a]]$ be the ring of formal power series in a indeterminates. Let B be the A-algebra

$$B = A[[T]] \Big/ \left(\prod_{i=1}^{a}(T - X_i)\right).$$

Thus B is a free A-module with basis $1, T, \ldots, T^{a-1}$:

$$B = AT^{a-1} \oplus AT^{a-2} \oplus \cdots \oplus A. \tag{5.4.1}$$

Let $S_i \in A$ be the i-th elementary symmetric function

$$S_i = \sum_{1 \leq \alpha_1 < \cdots < \alpha_i \leq a} X_{\alpha_1} \cdots X_{\alpha_i},$$

so that

$$\sum_{i=0}^{a}(-1)^i S_i T^{a-i} = 0 \qquad \text{in } B. \tag{5.4.2}$$

The element $1 + S_1 + \cdots + S_a$ is invertible in A; write its inverse as

$$\frac{1}{\sum_{i=0}^{a} S_i} = \sum_{j\geq 0} b_j, \qquad b_j \text{ homogeneous of degree } j \text{ in } A, \qquad b_0 = 1. \tag{5.4.3}$$

For every integer $j \geq 0$, define a homogeneous element $a_j \in A$ of degree j by writing, using (5.4.1),

$$T^{a-1+j} = a_j T^{a-1} + ?\, T^{a-2} + ?\, T^{a-3} + \cdots + ?, \qquad ? \in A. \tag{5.4.4}$$

Then, for every $j \geq 0$,

$$a_j = (-1)^j b_j. \tag{5.4.5}$$

Proof. In view of the definition (5.4.3) of the b_j, it is enough to prove

$$\begin{cases} a_0 = 1, \\ \sum_{i=0}^{a}(-1)^i S_i\, a_{j-i} = 0 \quad \text{if } j > 0. \end{cases} \tag{5.4.6}$$

Clearly $a_0 = 1$. For a fixed $j > 0$, the second relation is obtained by taking the coefficient of T^{a-1}, using (5.4.1), in the relation

$$\sum_{i=0}^{a}(-1)^i S_i T^{a-1+j-i} = 0 \qquad \text{in } B. \tag{5.4.7}$$

This relation is obtained by multiplying (5.4.2) by T^{j-1}, which proves the lemma. II ≈**248** II **249**

Corollary 5.5.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{(1+H)^{r-1}}{C(N)}. \tag{5.5.1}$$

Proof. Apply (5.3.3), with $M = N$ the normal sheaf of $X \hookrightarrow \mathbb{P}^r$, to (5.2.8).

For a smooth quasi-projective variety S, write

$$C(S) \overset{\text{def}}{=} C(T_S) \tag{5.5.2}$$

for its total Chern class, where T_S denotes the tangent bundle of S.

Dualizing the exact sequence (5.1.2) gives

$$C(\mathbb{P}^r) = (1+H)^{r+1} \qquad \text{in } \mathrm{CH}(\mathbb{P}^r). \tag{5.5.3}$$

Dualizing the exact sequence (5.1.1) gives

$$C(N)\, C(X) = C(\mathbb{P}^r)|_X \qquad \text{in } \mathrm{CH}(X). \tag{5.5.4}$$

Combining (5.5.3) and (5.5.4), formula (5.5.1) becomes:

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Corollary 5.6.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{C(X)}{(1+H)^2}. \tag{5.6.1}$$

5.7.

For a smooth projective variety S, define its Euler–Poincaré characteristic by

$$\chi(S) = \int_S C(S). \tag{5.7.1}$$

Proposition 5.7.2.

Let Y be a smooth hyperplane section of X, and let Δ be a smooth section of X by a codimension-two linear subspace of $\mathbb{P}^r$. Then

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} [\chi(X) + \chi(\Delta) - 2\chi(Y)]. \tag{5.7.3}$$

Proof. Let $i : Y \hookrightarrow X$ be the inclusion. One has II ≈**251**

$$N_{Y/X} = H|_Y = i^*H.$$

The exact sequence

$$0 \longrightarrow T_Y \longrightarrow i^*T_X \longrightarrow i^*H \longrightarrow 0$$

therefore gives

$$C(Y) = i^*\left(\frac{C(X)}{1+H}\right). \tag{5.7.4}$$

Hence

$$\chi(Y) = \int_Y i^*\left(\frac{C(X)}{1+H}\right),$$

and the projection formula gives

$$\chi(Y) = \int_X i_*i^*\left(\frac{C(X)}{1+H}\right) = \int_X \frac{H\,C(X)}{1+H}. \tag{5.7.5}$$

Iterating this calculation for Δ, one obtains

$$\begin{aligned} \chi(\Delta) &= \int_Y \frac{H\,C(Y)}{1+H} \\ &= \int_Y i^*\left(\frac{H}{1+H} \cdot \frac{C(X)}{1+H}\right) = \int_X \frac{H^2 C(X)}{(1+H)^2}. \end{aligned} \tag{5.7.6}$$

It follows that

$$\begin{aligned} \chi(X) - 2\chi(Y) + \chi(\Delta) &= \int_X C(X)\left[1 - \frac{2H}{1+H} + \frac{H^2}{(1+H)^2}\right] \\ &= \int_X \frac{C(X)}{(1+H)^2}. \end{aligned}$$

Together with (5.6.1), this proves the proposition.

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6. The case of a general base

6.1.

Let S be a scheme and let $\bar{s}$ be a geometric point of S, defined by an algebraic closure of the residue field $k(s)$ for a point $s \in S$. Let $\mathbb{P}^r_S$ be projective r-space over S, let $\check{\mathbb{P}}^r_S$ be its dual projective space, and let

$$I \subset \mathbb{P}^r_S \times_S \check{\mathbb{P}}^r_S$$

be the incidence variety. Let $D \subset \check{\mathbb{P}}^r_S$ be a line, defined by an S-valued point of the appropriate Grassmannian, and let Δ be the axis of the corresponding pencil of hyperplanes (cf. 2.1).

Let X be a proper smooth S-scheme, pure of dimension $n \geq 1$, equipped with a projective embedding $X \hookrightarrow \mathbb{P}^r_S$. Put

$$\widetilde{X} = (X \times_S D) \cap I.$$

Thus $\widetilde{X}$ is the scheme over D whose fibers are the hyperplane sections of X parametrized by D; let $u : \widetilde{X} \to D$ be the projection.

We use a subscript $\bar{s}$ to denote base change by $\bar{s} \to S$. Suppose that $\Delta_{\bar{s}}$ is transverse to $X_{\bar{s}}$. Then, over a neighborhood U of s in S, Δ is transverse to X. In what follows we replace S by this neighborhood and write $U = S$. The scheme $\widetilde{X}$ is then smooth over S, and a neighborhood in $\widetilde{X}$ of the inverse image of Δ is smooth over D.

Suppose moreover that one of the hyperplane sections of $X_{\bar{s}}$ belonging to the pencil $D_{\bar{s}}$ is smooth. After again replacing S by a suitable neighborhood of s, the closed subscheme $F \subset \widetilde{X}$ defined by the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ is finite and flat over S.

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6.2.

The exceptional locus of the pencil D is the relative Cartier divisor E of D/S defined by the finite flat $\mathcal{O}_D$-module $u_*\mathcal{O}_F$. Locally on D, if

$$\mathcal{O}_D^m \xrightarrow{v} \mathcal{O}_D^m \longrightarrow u_*\mathcal{O}_F \longrightarrow 0$$

is a presentation of $u_*\mathcal{O}_F$, then E has equation $\det(v) = 0$. Formation of E commutes with arbitrary base change.

The divisor $E_{\bar{s}}$ on $D_{\bar{s}}$ can be written

$$E_{\bar{s}} = \sum_P n(P)[P],$$

where $n(P)$ is the sum of the colengths of the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ at the critical points of u mapping to P.

Proposition 6.3.

Suppose that $D_{\bar{s}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{s}}$ and that $\operatorname{char} k(s) \neq 2$ or n is even. There is a neighborhood U of s in S such that:

(a) for every geometric point $\bar{t}$ of U, $D_{\bar{t}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{t}}$;

(b) over U, the exceptional locus E is étale over S.

To prove (a) and (b), it is enough to observe that, under these hypotheses, $D_{\bar{s}}$ satisfies condition 2.2(c) if and only if the divisor $E_{\bar{s}}$ is reduced. Invoking 3.3, one could use only the easier implication from left to right.

6.4 (exceptional case).

The proposition is false in the exceptional case just mentioned. Nevertheless, if S has characteristic two and n is odd, there is a neighborhood U of s over which (a) holds, and one can define a divisor E' étale over U such that $E = 2E'$. We shall not use this result.

References

[1] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. Math. France **86** (1958), 137–154.

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SGA 7 — EXPOSÉ XVIII

Cohomological Study of Lefschetz Pencils

by N. Katz[23]

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Introduction

In the present exposé we give the elementary cohomological structure of Lefschetz pencils. Sections 1–4 give the "well-known" facts concerning the cohomological structure of projective bundles and blow-ups. In Section 5 we carry out the study named in the title, assuming that X satisfies the "Lefschetz vache theorem" (5.2) and that the pencil D in question satisfies a certain condition (A), local in nature on D (5.3). The principal results are 5.6, 5.7, and 5.8.7; the last plays an essential role in the proof of Griffiths's theorem in Exposé XX. In the final section, which is distinctly less elementary, we use Mme Raynaud's Exposé XIII of SGA 1 on the tame fundamental group, and the Picard–Lefschetz formula (XVI), to prove:

(1) condition (A) is often satisfied (6.3, 6.4);

(2) except when $\dim(X)$ is odd and the characteristic is 2, the monodromy group of a Lefschetz pencil acts irreducibly on the "vanishing cohomology," and transitively (up to sign) on the set of "local vanishing cycles" associated, via the Picard–Lefschetz formula, with the various "critical points" of the pencil (6.6 and 6.7).

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1. The cohomology of a projective bundle

Theorem 1.1 (Deligne).

Let $f : X \longrightarrow S$ be a proper morphism, let ℓ be a prime number invertible on S, let $u \in H^2(X, \mathbb{Z}_\ell(1))$ (respectively, $u \in H^2(X, \mathbb{Q}_\ell(1))$ be a cohomology class on X, and let $\mathfrak{F}$ be a $\mathbb{Z}_\ell$-sheaf (respectively, a $\mathbb{Q}_\ell$-sheaf) on X. Suppose that there is an integer n such that, for every integer j and every integer $i > 0$, the map

$$R^{n-i}f_*(\mathfrak{F}(j)) \xrightarrow{\wedge u^i} R^{n+i}f_*(\mathfrak{F}(j+i)) \tag{1.1.1}$$

[23] Based on succinct notes of Grothendieck.

is an isomorphism of $\mathbb{Z}_\ell$-sheaves (respectively, $\mathbb{Q}_\ell$-sheaves) on S. Then, for every morphism $g : S \longrightarrow T$, the Leray spectral sequence

$$E_2^{p,q} = R^p g_* R^q f_*(\mathfrak{F}) \Longrightarrow R^{p+q}(gf)_*(\mathfrak{F})$$

degenerates.

The proof is found in [1, Prop. 2.4]. Notice that, by the proper base-change theorem (SGA 4 XII, 5.2), the hypothesis can be checked fibre by fibre.

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Theorem 1.2 (cf. SGA 5 VII, 2.2.1).

Let $\mathcal{E} \longrightarrow X$ be a vector bundle, locally free of rank $1 + r$. Let $f : \mathbb{P}(\mathcal{E}) \longrightarrow X$ denote the projective bundle defined by $\mathcal{E}$, and let $u \in H^2(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell(1))$ be the first Chern class of the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. For ℓ a prime number invertible on X, one has:

a) the spectral sequence

(1.2.1) $$E_2^{p,q} = H^p\big(X, R^q f_* \mathbb{Z}_\ell\big) \Longrightarrow H^{p+q}\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big)$$

degenerates;

b) the map

(1.2.2) $$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big) \longrightarrow H^q\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big), \qquad (\tau_{q-2j})_j \longmapsto \sum_j f^*(\tau_{q-2j}) \wedge u^j$$

is an isomorphism.

Proof. The fibres of f are projective spaces of dimension r, and one knows that

(1.2.3) $$R^i f_* \mathbb{Z}_\ell \xleftarrow[\sim]{\wedge^{i/2} u} \begin{cases} \mathbb{Z}_\ell\left(-\dfrac{i}{2}\right), & \text{if } i \text{ is even and } 0 \le i \le 2r, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 1.1 applies to $(f, \ell, u, \mathbb{Z}_\ell, r)$, which proves (a). By (1.2.3),

(1.2.4) $$E_2^{a,b} = \begin{cases} H^a\left(X, \mathbb{Z}_\ell\left(-\dfrac{b}{2}\right)\right), & \text{if } b \text{ is even and } 0 \le b \le 2r, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the graded object associated with $H^q(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell)$ for the filtration coming from the spectral sequence is

$$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big).$$

If this direct sum is equipped with the filtration

$$F^i = \bigoplus_{\substack{0 \le j \le r \\ q-2j \ge i}} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big),$$

the map (1.2.2) is compatible with the filtrations and induces the identity on the associated graded objects. This proves (b).

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2. The cohomology of blown-up varieties (cf. SGA 5 VII, 8)

2.1.

Let X be a smooth scheme over a field k, and let $\Delta \subset X$ be an everywhere smooth subscheme of codimension $1+r$. It is known (EGA IV, 19.4.9) that there is a smooth scheme $\widetilde{X}$, equipped with a projective morphism

$$f : \widetilde{X} \longrightarrow X, \tag{2.1.1}$$

such that, on setting

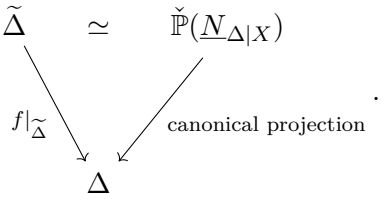
$$\widetilde{\Delta} = f^{-1}(\Delta) = \Delta \times_X \widetilde{X}, \tag{2.1.2}$$

f induces an isomorphism

$$f|_{\widetilde{X}-\widetilde{\Delta}} : \widetilde{X} - \widetilde{\Delta} \xrightarrow{\sim} X - \Delta, \tag{2.1.3}$$

and there is a Δ-isomorphism between $\widetilde{\Delta}$ and $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$, the projective bundle of lines in the normal bundle $\underline{N}_{\Delta/X}$ of Δ in X. Thus one has a commutative diagram

(2.1.4)



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2.1.5.

In the special case

$$X = \mathrm{Spec}(A), \qquad \Delta = \mathrm{Spec}(A/I), \qquad I = (g_0, \ldots, g_r),$$

$\widetilde{X}$ is obtained by the Hopf construction (cf. EGA IV, 19.4.11). One defines a morphism from $X - \Delta$ to $\mathbb{P}^r$ by

$$x \in X - \Delta \longmapsto \big(g_0(x), \ldots, g_r(x)\big) \in \mathbb{P}^r$$

and regards its graph Γ as embedded in $X \times \mathbb{P}^r$. Then $\widetilde{X}$ is defined to be the schematic closure of Γ in $X \times \mathbb{P}^r$, and the morphism $f : \widetilde{X} \longrightarrow X$ is the composite

$$\widetilde{X} \hookrightarrow X \times \mathbb{P}^r \xrightarrow{\mathrm{pr}_1} X.$$

2.1.6.

Similarly (cf. again EGA IV, 19.4.11), if X is projective and Δ is the intersection of X with $1+r$ hypersurfaces, say with equations $G_i = 0$, $i = 0, \ldots, r$, one again considers the morphism

$$X - \Delta \longrightarrow \mathbb{P}^r, \qquad x \longmapsto \big(G_0(x), \ldots, G_r(x)\big),$$

and obtains $\widetilde{X}$ as the closure of its graph in $X \times \mathbb{P}^r$. Concretely, in terms of homogeneous coordinates $\lambda_0, \ldots, \lambda_r$ on $\mathbb{P}^r$, $\widetilde{X}$ is the subscheme of $X \times \mathbb{P}^r$ defined by the equations

$$\lambda_i G_j(X) - \lambda_j G_i(X), \qquad 0 \leq i < j \leq r. \tag{2.1.6.1}$$

These equations show that the point (x, λ) lies in $\widetilde{\Delta}$ if and only if $x \in \Delta$; in other words, $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$ is the trivial projective bundle

(2.1.6.2)
$$\begin{array}{ccc} \check{\mathbb{P}}(\underline{N}_{\Delta/X}) & \simeq & \Delta \times \mathbb{P}^r \\ & \searrow^{\text{canonical projection}} \quad \swarrow_{\mathrm{pr}_1} & \\ & \Delta & \end{array} .$$

One has the following commutative diagram:

(2.1.6.3)
$$\begin{array}{ccccccc} \Delta \times \mathbb{P}^r & \xrightarrow{\sim} & \widetilde{\Delta} & \hookrightarrow & \widetilde{X} & \hookrightarrow & X \times \mathbb{P}^r \\ & \searrow^{\mathrm{pr}_1} & \downarrow{\scriptstyle f|_{\widetilde{\Delta}}} & & \downarrow{\scriptstyle f} & \swarrow_{\mathrm{pr}_1} & \\ & & \Delta & \hookrightarrow & X & & \end{array} ,$$

and the composite inclusion $\Delta \times \mathbb{P}^r \hookrightarrow X \times \mathbb{P}^r$ is obtained from $\Delta \hookrightarrow X$ by base change.

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Theorem 2.2.

Let X be smooth over a field k, let Δ be an everywhere smooth subscheme of codimension $1 + r$, and let $f : \widetilde{X} \longrightarrow X$ be the blow-up of X along Δ. Then:

(2.2.1) The Leray spectral sequence

$$E_2^{p,q} = H^p(X, R^q f_* \mathbb{Z}_\ell) \Longrightarrow H^{p+q}(\widetilde{X}, \mathbb{Z}_\ell)$$

degenerates.

(2.2.2) There is an additive isomorphism

$$\begin{aligned} H^q(\widetilde{X}, \mathbb{Z}_\ell) &\xleftarrow{\sim} H^q(X, \mathbb{Z}_\ell) \oplus \frac{H^q(\widetilde{\Delta}, \mathbb{Z}_\ell)}{\operatorname{Im} H^q(\Delta, \mathbb{Z}_\ell)} \\ &\simeq \left(\bigoplus_{j=1}^{r} H^{q-2j}(\Delta, \mathbb{Z}_\ell(-j)) \right) \oplus H^q(X, \mathbb{Z}_\ell). \end{aligned}$$

Proof. Let II ≈**261**

(2.2.3)
$$g : \widetilde{\Delta} \longrightarrow \Delta$$

be the restriction of f to $\widetilde{\Delta}$, and let

(2.2.4)
$$\begin{cases} j : \Delta \hookrightarrow X, \\ i : \widetilde{\Delta} \hookrightarrow \widetilde{X} \end{cases}$$

be the inclusions, so that one has the commutative diagram

(2.2.5)
$$\begin{array}{ccc} \widetilde{X} & \xleftarrow{\ i\ } & \widetilde{\Delta} \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xleftarrow{\ j\ } & \Delta \end{array} .$$

It induces a homomorphism between the Leray spectral sequences for the morphisms f and g, with the sheaf $\mathbb{Z}_\ell$ on $\widetilde{X}$ and $\widetilde{\Delta}$:

(2.2.6)
$$E_s^{p,q}(f) \xrightarrow{(i^*, j^*)} E_s^{p,q}(g).$$

On the other hand, since f induces an isomorphism from $\widetilde{X}-\widetilde{\Delta}$ to $X-\Delta$, the proper base-change theorem gives

$$R^q f_*\mathbb{Z}_\ell \simeq \begin{cases} \mathbb{Z}_\ell, & q=0, \\ j_*R^q g_*\mathbb{Z}_\ell, & q>0. \end{cases} \tag{2.2.7}$$

Hence

$$E_2^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_2^{p,q}(g) \qquad (q\neq 0). \tag{2.2.8}$$

We shall prove by induction on s that

$$E_s^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_s^{p,q}(g) \qquad (q\neq 0,\ s\geq 2) \tag{2.2.9}$$

and

$$d_s^{p,q}(f)=0 \qquad (s\geq 2). \tag{2.2.10}$$

For the moment, admit the formula

$$d_s^{p,s-1}(f)=0 \qquad (s\geq 2,\ \text{for every } p). \tag{2.2.11}$$

Begin with $s=2$. We have a commutative diagram II≈**262**

$$\begin{array}{ccc} E_2^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_2^{p,q}(g) \\ \downarrow{\scriptstyle d_2^{p,q}(f)} & & \downarrow{\scriptstyle 0=d_2^{p,q}(g)} \\ E_2^{p+2,q-1}(f) & \xrightarrow{(i^*,j^*)} & E_2^{p+2,q-1}(g) \end{array} . \tag{2.2.12}$$

For $q\leq 0$, one has $E_2^{p+2,q-1}(f)=0$, hence $d_2^{p,q}(f)=0$. For $q=1$, formula (2.2.11) gives $d_2^{p,1}(f)=0$. For $q>1$, the horizontal arrows in (2.2.12) are isomorphisms, while $d_2^{p,q}(g)=0$ by 1.2 and 2.1.4. This proves (2.2.10) for $s=2$, and (2.2.9) for $s=2$ is exactly (2.2.8).

Suppose (2.2.9) and (2.2.10) are known for some $s\geq 2$. Hypothesis (2.2.10) implies $E_s^{p,q}(f)\simeq$ II≈**263** $E_{s+1}^{p,q}(f)$, and by 1.2 one has $E_s^{p,q}(g)\simeq E_{s+1}^{p,q}(g)$, so there is a commutative diagram

$$\begin{array}{ccc} E_s^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_s^{p,q}(g) \\ \wr\downarrow & & \downarrow\wr \\ E_{s+1}^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p,q}(g) \end{array} , \tag{2.2.13}$$

which proves (2.2.9) for $s+1$, in view of (2.2.9) for s. To deduce (2.2.10) for $s+1$, consider the commutative diagram

$$\begin{array}{ccc} E_{s+1}^{p,q}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p,q}(g) \\ \downarrow{\scriptstyle d_{s+1}^{p,q}(f)} & & \downarrow{\scriptstyle 0=d_{s+1}^{p,q}(g)} \\ E_{s+1}^{p+s+1,q-s}(f) & \xrightarrow{(i^*,j^*)} & E_{s+1}^{p+s+1,q-s}(g) \end{array} . \tag{2.2.14}$$

For $q\leq s-1$, one has $E_{s+1}^{p+s+1,q-s}(f)=0$, hence $d_{s+1}^{p,q}(f)=0$. For $q=s$, one has (2.2.11), and for $q>s$ the horizontal arrows in (2.2.14) are isomorphisms and $d_{s+1}^{p,q}(g)=0$ by 1.2. This gives the desired conclusion.

It remains to verify (2.2.11), which says that the composite edge homomorphism

$$E_2^{p,0}(f) \longrightarrow E_3^{p,0}(f) \longrightarrow E_4^{p,0}(f) \longrightarrow \cdots \longrightarrow E_\infty^{p,0}(f) \tag{2.2.15}$$

is an isomorphism; that is, that

$$f^* : H^p(X, \mathbb{Z}_\ell) \longrightarrow H^p(\widetilde{X}, \mathbb{Z}_\ell) \tag{2.2.16}$$

is injective. But it has a left inverse, namely the Gysin homomorphism (SGA 5 IV)

$$f_* : H^p(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^p(X, \mathbb{Z}_\ell), \tag{2.2.17}$$

which indeed satisfies

$$f_* f^* = \mathrm{id}. \tag{2.2.18}$$

This follows from the projection formula and the fact that $f_*(1) = 1$, since f is birational and therefore has degree one. This proves (2.2.1).

As for (2.2.2), passing to cohomology in diagram (2.2.5) gives a commutative diagram II ≈**264**

$$\begin{array}{ccc} H^p(\widetilde{X}, \mathbb{Z}_\ell) & \xrightarrow{\quad i^* \quad} & H^p(\widetilde{\Delta}, \mathbb{Z}_\ell) \\ \uparrow{\scriptstyle f^*} & & \uparrow{\scriptstyle g^*} \\ H^p(X, \mathbb{Z}_\ell) & \xrightarrow{\quad j^* \quad} & H^p(\Delta, \mathbb{Z}_\ell) \end{array} , \tag{2.2.19}$$

hence a morphism

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{f^* H^p(X, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^p(\Delta, \mathbb{Z}_\ell)}. \tag{2.2.20}$$

Let F^i denote the filtrations on $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$ and $H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)$ coming from the Leray spectral sequences. Then (2.2.20) can be rewritten as

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{X}, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}. \tag{2.2.21}$$

By (2.2.9), the map induced by (2.2.21) on associated graded objects is an isomorphism, so (2.2.20) is an isomorphism. On the other hand, (2.2.18) gives a direct-sum decomposition

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f_* \oplus (1 - f^* f_*)}{\overset{f^* + \text{inclusion}}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus \operatorname{Ker} f_* \ , \tag{2.2.22}$$

and (1.2.2) gives the direct-sum decomposition

$$\frac{H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^\bullet(\Delta, \mathbb{Z}_\ell)} \xleftarrow{\sim} \bigoplus_{1 \le j \le r} H^{\bullet - 2j}(\Delta, \mathbb{Z}_\ell(-j)). \tag{2.2.23}$$

Combining the isomorphism (2.2.20) and decompositions (2.2.22) and (2.2.23) gives (2.2.2), as required.

Remark 2.2.24.

Let us sketch a second, distinctly simpler proof of (2.2.2), valid when X is proper over k. More generally, without any hypothesis on X, one obtains the corresponding result (2.2.26) in cohomology with proper supports. The exact sequences of cohomology with proper supports, denoted $H_c^\bullet$, for the pairs $(\widetilde{X}, \widetilde{\Delta})$ and (X, Δ) form the commutative diagram

$$
\begin{array}{lccccccc}
(2.2.25) & \longrightarrow H_c^i(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{X},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) \\
& \wr\uparrow & & \uparrow f^* & & \uparrow g^* & & \wr\uparrow \\
(2.2.26) & \longrightarrow H_c^i(X-\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(X,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(X-\Delta,\mathbb{Z}_\ell)
\end{array}
$$

The quotient of the acyclic complex (2.2.25) by the acyclic subcomplex (2.2.26) gives the acyclic complex

$$
(2.2.27) \qquad \longrightarrow 0 \longrightarrow \frac{H_c^i(\widetilde{X},\mathbb{Z}_\ell)}{f^*H_c^i(X,\mathbb{Z}_\ell)} \longrightarrow \frac{H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell)}{g^*H_c^i(\Delta,\mathbb{Z}_\ell)} \longrightarrow 0 \longrightarrow .
$$

Thus, without considering spectral sequences, one finds that (2.2.20) is an isomorphism. The arguments just given then apply and yield (2.2.2).

Corollary 2.3.

Set

$$
b_q(X) = \dim_{\mathbb{Q}_\ell} H^q(X,\mathbb{Q}_\ell), \qquad \chi(X) = \sum_q (-1)^q b_q(X).
$$

Then

$$
(2.3.1) \qquad b_q(\widetilde{X}) = b_q(X) + \sum_{j=1}^{r} b_{q-2j}(\Delta),
$$

$$
(2.3.2) \qquad \chi(\widetilde{X}) = \chi(X) + r\,\chi(\Delta).
$$

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3. The blow-up associated with a pencil

3.1.

Let X be projective, smooth, and irreducible over an algebraically closed field k, of dimension n, and let $D = \{H_\tau\}_{\tau\in\mathbb{P}^1}$ be a pencil of hypersurfaces whose axis meets X transversely, in Δ. We consider the blow-up of X along Δ, which we shall now describe explicitly.

In terms of equations, let (λ, μ) be homogeneous coordinates on $\mathbb{P}^1$, and let F and G be two linear forms such that $F = G = 0$ defines the axis of the pencil. The pencil is written

$$
(3.1.1) \qquad \lambda F - \mu G.
$$

By definition, the subscheme Δ is the intersection of X with the linear subvariety $G = F = 0$. By 2.1.6,

$$
(3.1.2) \qquad \widetilde{X} = \{(x,(\lambda,\mu)) \in X \times \mathbb{P}^1 \mid \lambda F - \mu G = 0\};
$$

it is the scheme-theoretic closure, in $X \times \mathbb{P}^1$, of the graph of the map

$$X - \Delta \longrightarrow \mathbb{P}^1$$

(3.1.3)

$$x \longrightarrow (G(x), F(x)).$$

The great advantage of $\widetilde{X}$ is that it is projective (being embedded in $X \times \mathbb{P}^1$) and smooth (because Δ is smooth) over k, and admits a morphism ρ to $\mathbb{P}^1$, namely

$$\rho : \widetilde{X} \longrightarrow X \times \mathbb{P}^1 \xrightarrow{\mathrm{pr}_2} \mathbb{P}^1, \tag{3.1.4}$$

whose fibers are the hyperplane sections of X by the members of our pencil:

$$\rho^{-1}(t) = X_t \overset{\text{def}}{=} X \cdot H_t.$$

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3.2. The degree of the dual variety: topological method

Suppose in addition that the pencil of 3.1 is such that

$$X_t \text{ is smooth for every } t \text{ in a nonempty open subset of } \mathbb{P}^1, \tag{3.2.1}$$

and

(3.2.2)

if X_t is singular, its only singularities are nondegenerate quadratic singularities (XVII 1.1).

These conditions hold in particular for a Lefschetz pencil (XVII 2.21) if $\operatorname{char} k \neq 2$ or if X has even dimension.

Let η be the generic point of $\mathbb{P}^1$, let $\bar{\eta}$ be a geometric point above it, let $n = \dim X$, and let χ be the Euler–Poincaré characteristic with coefficients in $\mathbb{Q}_\ell$. The Euler–Poincaré formula XVI 2.1 applied to ρ gives

$$\chi(\widetilde{X}) = \chi(\mathbb{P}^1)\,\chi(X_{\bar{\eta}}) + (-1)^n \sum_{\substack{\text{singular points} \\ \text{of the fibers}}} 1, \tag{3.2.3}$$

which, in view of (2.2.2), can be rewritten

$$(-1)^n \left[\begin{array}{c} \text{total number of} \\ \text{singular points} \end{array} \right] = \chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}}). \tag{3.2.4}$$

Notice that the quantity $\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})$ is independent of the particular choice of pencil. It depends only on the embedding of X in some $\mathbb{P}^r$, and indeed only on the Chern class of $\underline{\mathcal{O}}_X(1)$; see XVII 5.7.5 and 5.7.6.

3.2.5.

Let us make this explicit in a concrete case. Take $X = \mathbb{P}^2$ and let $\{H_\tau\}$ be a pencil of plane curves of degree d satisfying 3.1, (3.2.1), and (3.2.2). Then

$$\begin{aligned} &\chi(X) = 3 && \text{(by 1.2)}, \\ &\chi(X_{\bar{\eta}}) = -d(d-3) && \text{(see SGA 5 VII 7.3)}, \\ &\chi(\Delta) = d^2, \end{aligned} \tag{3.2.6}$$

whence

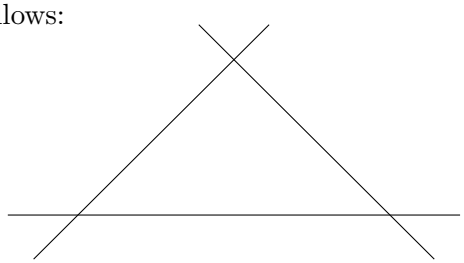
(3.2.7) the total number of singular points is $3(d-1)^2$.

Take the special case $d = 3$ and consider the family of elliptic curves (in characteristic different from 3)

$$X^3 + Y^3 + Z^3 + 3\mu XYZ. \tag{3.2.8}$$

The four singular fibers lie over $\mu^3 = -1$ and $\mu = \infty$, respectively, and each singular fiber is the union of three lines as follows:

(3.2.9)



,

It has exactly three singular points, all nondegenerate quadratic singularities. Altogether this gives twelve singular points and verifies (3.2.7) in this example.

Proposition 3.2.10.

Let $X \hookrightarrow \mathbb{P}^r$ be a Lefschetz embedding (XVII 2.3), and suppose that $n = \dim X$ is even, or that $\operatorname{char} k \neq 2$. Then every Lefschetz pencil, regarded as a line in $\check{\mathbb{P}}^r$, meets $\check{X}$ in

$$(-1)^n[\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})] \tag{3.2.11}$$

points, and meets it transversely (EGA IV 17.13.7).

Proof. By hypothesis, every singularity of a fiber of the Lefschetz pencil is a nondegenerate quadratic singularity. Formula (3.2.4) therefore shows that the number of singular fibers in the pencil D—each containing exactly one singular point—is the number (3.2.11). Now the fiber X_τ is singular if and only if $\tau \in \check{X} \cap D$. II ≈270

To see that the intersection is transverse, observe that (3.2.11) is independent of the particular Lefschetz pencil. Since the generic pencil is a Lefschetz pencil, (3.2.11) is the degree of $\check{X}$; that is, its degree as a hypersurface in $\check{\mathbb{P}}^r$, taken by convention to be 0 if $\dim \check{X} \leq r-2$. If a line D meets a hypersurface of degree $d > 0$ in d distinct points x_i $(1 \leq i \leq d)$, the intersection must be transverse. Indeed, the local intersection multiplicity

$$\mu_i = \operatorname{length}_{x_i}(D \cap \check{X})$$

is at least 1, while $\sum_{i=1}^{d} \mu_i = d$. Hence $\mu_i = 1$ for every i; equivalently, $D \cap \check{X}$ is smooth, which is precisely transversality.

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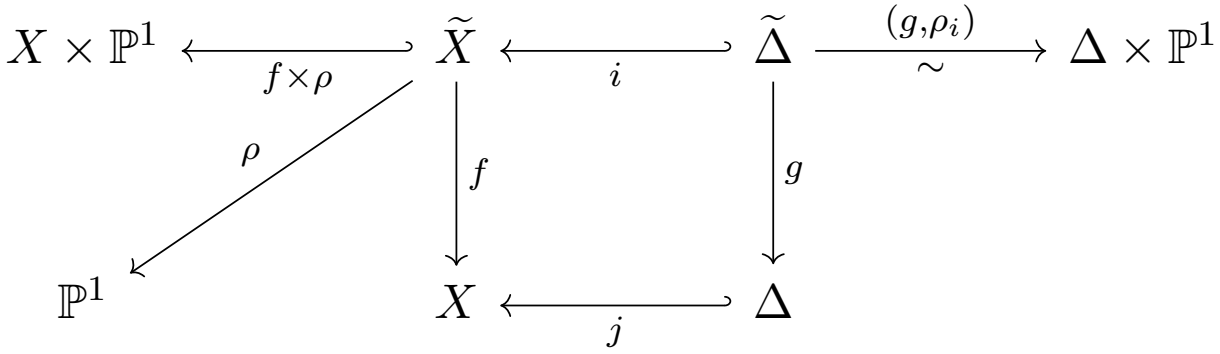
Remark 3.2.12.

For a Lefschetz pencil D already assumed to meet $\check{X}$ transversely (a condition satisfied in any case by sufficiently general D), the result 3.2.10 may also be regarded as the conjunction of XVII 5.7.2 and the fact that $\deg \varphi = 1$ contained in XVII 3.5. Of course, the fact proved here that every Lefschetz pencil D meets $\check{X}$ transversely—by means of the Euler–Poincaré formula XVI 2.1—may also be verified directly and elementarily, using the fact that the axis of D meets X transversely.

4. Cohomology of the blow-up associated with a pencil

4.1.

We return to the situation of 3.1, summarized by the commutative diagram (see 2.1.6.3):
(4.1.1)



We begin by making (2.2.2) more explicit in this case.

Proposition 4.2.

Under the hypotheses of 3.1:
(4.2.1) The homomorphisms of $\mathbb{Z}_\ell$-modules

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f^* + \mathrm{incl}}{\overset{f_* \oplus (1 - f^* f_*)}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus \mathrm{Ker}\Big(f_* : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \to H^\bullet(X, \mathbb{Z}_\ell)\Big)$$

are mutually inverse isomorphisms.
(4.2.2) The homomorphisms of $\mathbb{Z}_\ell$-modules II ≈**272**

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f^* + i_* g^*}{\overset{f_* \oplus (-g_* i^*)}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1))$$

are mutually inverse isomorphisms.
(4.2.3) The algebra structure on

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1)),$$

transported by (4.2.2) from the cup product on $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$, is expressed, for

$$a, b \in H^\bullet(X, \mathbb{Z}_\ell), \qquad x, y \in H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1)),$$

by the formulas

(4.2.3.1) $$(0 \oplus x) \wedge (0 \oplus y) = -j_*(xy) \oplus 2xy\, c_1(\underline{\mathcal{O}}_\Delta(1)),$$

(4.2.3.2) $$(a \oplus 0) \wedge (b \oplus 0) = ab \oplus 0,$$

(4.2.3.3) $$(a \oplus 0) \wedge (0 \oplus y) = 0 \oplus j^*(a) y,$$

(4.2.3.4) $$(0 \oplus x) \wedge (b \oplus 0) = 0 \oplus x j^*(b).$$

In particular, if

$$a \oplus x \in H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet - 2}(\Delta, \mathbb{Z}_\ell(-1))$$

and

$$b \oplus y \in H^{2n-\bullet}(X, \mathbb{Z}_\ell) \oplus H^{2n-2-\bullet}(\Delta, \mathbb{Z}_\ell(-1)),$$

then

$$(a \oplus x) \wedge (b \oplus y) = ab - j_*(xy). \tag{4.2.3.5}$$

In other words, the total intersection matrix of $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$ is

$$\left(\begin{array}{c|c} \begin{array}{c}\text{total intersection}\\ \text{matrix on } X\end{array} & 0 \\ \hline 0 & -\begin{array}{c}\text{total intersection}\\ \text{matrix on } \Delta\end{array} \end{array}\right).$$

Proof. We have already proved (4.2.1); see (2.2.22). For (4.2.2), first observe that, since $g : \widetilde{\Delta} \to \Delta$ is a projective-line bundle, the sequence

$$0 \longrightarrow H^\bullet(\Delta, \mathbb{Z}_\ell) \xrightarrow{g^*} H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell) \xrightarrow{g_*} H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \longrightarrow 0 \tag{4.2.4}$$

is exact, by (1.2.2). Together with (2.2.26), this gives the commutative diagram of exact sequences (4.2.5)

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^\bullet(X, \mathbb{Z}_\ell) & \xrightarrow{f^*} & H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) & \xrightarrow{g_* i^*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle j^*} & & \downarrow{\scriptstyle i^*} & & \| & & \\ 0 & \longrightarrow & H^\bullet(\Delta, \mathbb{Z}_\ell) & \xrightarrow{g^*} & H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell) & \xrightarrow{g_*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longrightarrow & 0. \end{array}$$

In view of the splitting (4.2.1), this yields the isomorphism

$$f_* \oplus (-g_* i^*) : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{\sim} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)). \tag{4.2.6}$$

The transpose of (4.2.5), after the degree reversal $\bullet \mapsto 2n - \bullet$, gives the commutative diagram of exact sequences (4.2.7)

$$\begin{array}{ccccccccc} 0 & \longleftarrow & H^\bullet(X, \mathbb{Z}_\ell) & \xleftarrow{f_*} & H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) & \xleftarrow{i_* g^*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longleftarrow & 0 \\ & & \uparrow{\scriptstyle j_*} & & \uparrow{\scriptstyle i_*} & & \| & & \\ 0 & \longleftarrow & H^{\bullet-4}(\Delta, \mathbb{Z}_\ell(-2)) & \xleftarrow{g_*} & H^{\bullet-2}(\widetilde{\Delta}, \mathbb{Z}_\ell(-1)) & \xleftarrow{g^*} & H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) & \longleftarrow & 0 \end{array}$$

and hence the isomorphism

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow[\sim]{f^* + i_* g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell). \tag{4.2.8}$$

It remains to prove that the composite

$$\begin{aligned} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) &\xrightarrow{f^* + i_* g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \\ &\xrightarrow{f_* \oplus (-g_* i^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \end{aligned} \tag{4.2.9}$$

is the identity. Now

$$f_* f^* = \mathrm{id} \quad \text{on } H^\bullet(X, \mathbb{Z}_\ell) \quad \text{by (2.2.18)},$$

$$g_* i^* f^* = 0 \quad \text{by (4.2.5)}, \qquad f_* i_* g^* = 0 \quad \text{by (4.2.7)}.$$

It therefore suffices to prove

$$-g_* i^* i_* g^*(x) = x \quad \text{for } x \in H^\bullet(\Delta, \mathbb{Z}_\ell(-1)). \tag{4.2.10}$$

To verify (4.2.10), first observe that $i : \widetilde{\Delta} \hookrightarrow \widetilde{X}$ is the inclusion of a smooth subvariety of codimension one. Thus (SGA 5 VII)

$$i^* i_*(x) = c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\, x, \tag{4.2.11}$$

and

$$i_*(x) i_*(y) = i_*\Big(xy\, c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big). \tag{4.2.12}$$

Write $\xi_X = \underline{\mathcal{O}}_X(1)$, $\xi_\Delta = \underline{\mathcal{O}}_\Delta(1)$, and $\xi_{\mathbb{P}^1} = \underline{\mathcal{O}}_{\mathbb{P}^1}(1)$. By EGA II 8.1.7, if $\widetilde{\mathfrak{J}}$ is the ideal of $\widetilde{\Delta}$ in $\widetilde{X}$, then

$$\begin{aligned} \check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} &\simeq \widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}^{,2} \simeq \widetilde{\mathfrak{J}} \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} \\ &\simeq \underline{\mathcal{O}}_{\widetilde{X}}(1) \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} = \underline{\mathcal{O}}_{\widetilde{\Delta}}(1). \end{aligned} \tag{4.2.13}$$

Here $\widetilde{\Delta}$ is viewed as $\mathbb{P}(\mathfrak{J}/\mathfrak{J}^2)$, where $\mathfrak{J}$ is the ideal of Δ in X. Since Δ is the intersection of X with two given hyperplanes, there is a canonical isomorphism

$$\mathfrak{J}/\mathfrak{J}^2 \simeq \underline{\mathcal{O}}_\Delta^{,2}(-1),$$

which induces

$$\widetilde{\Delta} = \mathbb{P}(\mathfrak{J}/\mathfrak{J}^2) \simeq \mathbb{P}(\underline{\mathcal{O}}_\Delta^{,2}(-1)) \xrightarrow[\sim]{\alpha} \mathbb{P}(\underline{\mathcal{O}}_\Delta^{,2}) \simeq \Delta \times \mathbb{P}^1.$$

The canonical isomorphism α of EGA II 4.1.4 carries the tautological line bundle to

$$(\rho i)^* \underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^* \underline{\mathcal{O}}_\Delta(-1).$$

Consequently, (4.2.13) becomes

$$\check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} \simeq (\rho i)^* \underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^* \underline{\mathcal{O}}_\Delta(-1). \tag{4.2.14}$$

Lemma 4.3.

One has

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}}) = g^* c_1(\xi_\Delta) - (\rho i)^* c_1(\xi_{\mathbb{P}^1}).$$

Return now to the proof of (4.2.10). By (4.2.11) and Lemma 4.3, it is enough to prove

$$g_*(g^*(x)\,[(\rho i)^* c_1(\xi_{\mathbb{P}^1}) - g^* c_1(\xi_\Delta)]) = x.$$

By (4.2.4), $g_* g^* = 0$, while

$$\widetilde{\Delta} \xrightarrow[\sim]{g \times \rho_i} \Delta \times \mathbb{P}^1,$$

so

$$g_*(g^*(x)(\rho i)^* c_1(\xi_{\mathbb{P}^1})) = x, \tag{4.3.1}$$

and hence

$$g_*\Big(g^*(x) c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) = -x. \tag{4.3.2}$$

We now compute the algebra structure on

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$$

transported from the cup product by (4.2.2). Let a, b belong to $H^\bullet(X, \mathbb{Z}_\ell)$ and x, y to $H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$. Begin with (4.2.3.1), and write

$$(0 \oplus x) \wedge (0 \oplus y) = A \oplus B.$$

By (4.2.12) and (4.3.2),

$$\begin{aligned} A &= f_*((i_*g^*x)(i_*g^*y)) \\ &= f_*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) \\ &= j_*g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) = -j_*(xy), \end{aligned}$$

and

$$\begin{aligned} B &= -g_*i^*((i_*g^*x)(i_*g^*y)) \\ &= -g_*i^*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) \\ &= -g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2\Big) \qquad \text{by (4.2.11).} \end{aligned}$$

Lemma 4.3 gives

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2 = g^*\Big(c_1(\xi_\Delta)^2\Big) - 2g^*c_1(\xi_\Delta)(\rho i)^*c_1(\xi_{\mathbb{P}^1}),$$

because $c_1(\xi_{\mathbb{P}^1})^2 = 0$. Therefore, by (4.2.4) and (4.3.1),

$$B = 0 + 2xy\, c_1(\xi_\Delta),$$

which proves (4.2.3.1). Formula (4.2.3.2) expresses the fact that f^* is an algebra homomorphism.

To prove (4.2.3.3), the projection formula shows that $\operatorname{Ker} f_*$ is stable under multiplication by $\operatorname{im} f^*$. Hence

$$(a \oplus 0) \wedge (0 \oplus y) = 0 \oplus C,$$

where

$$C = -g_*i^*(f^*(a)i_*g^*(y))\,. \tag{4.3.3}$$

Using (4.2.11) and (4.3.2),

$$\begin{aligned} C &= -g_*\Big(i^*f^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) \\ &= -g_*\Big(g^*j^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) \\ &= j^*(a)y. \end{aligned}$$

This proves (4.2.3.3). Finally, (4.2.3.4) follows from (4.2.3.3) by graded commutativity of the cup product.

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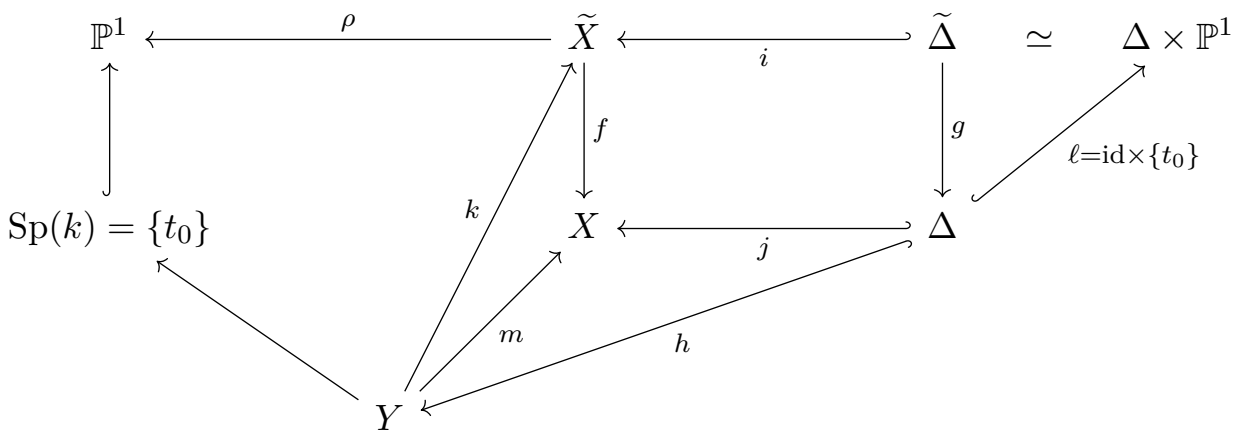
5. Cohomological study of the morphism $\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$ associated with a pencil

5.1. The restriction homomorphism $H^*(\widetilde{X}) \longrightarrow H^*(Y)$

Let Y denote one of the smooth fibers of $\rho : \widetilde{X} \to \mathbb{P}^1$, say

$$Y = \rho^{-1}(t_0) = X \cdot H_{t_0}.$$

We have the commutative diagram



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Proposition 5.1.1.

Under the hypotheses of (3.1), with Y denoting a smooth fiber:

(5.1.2) The restriction homomorphism II ≈**278**

$$k^* : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^\bullet(Y, \mathbb{Z}_\ell)$$

is expressed (via (4.2.2)) by

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{m^*+h_*} H^\bullet(Y, \mathbb{Z}_\ell).$$

(5.1.3) The Gysin map II ≈**279**

$$k_* : H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \longrightarrow H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$$

is expressed by

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{m_*\oplus(-h^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)).$$

Proof. The smooth subschemes $\widetilde{\Delta}$ and Y of $\widetilde{X}$ intersect transversely in Δ; in other words, the square of smooth schemes

(5.1.4)
$$\begin{array}{ccc} \Delta & \xrightarrow{\ell} & \widetilde{\Delta} \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle i} \\ Y & \xrightarrow[k]{} & \widetilde{X} \end{array}$$

is Cartesian. It therefore implies the formula [SGA 5, IV]

(5.1.5) $$h_*\ell^* = k^*i_*, \qquad \ell_*h^* = i^*k_*.$$

The composite

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{f^*+i_*g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{k^*} H^\bullet(Y, \mathbb{Z}_\ell)$$

is the sum of $k^*f^* = m^*$ and $k^*i_*g^* = h_*\ell^*g^*$ by (5.1.5). Since $g\ell = \mathrm{id}_\Delta$, this proves (5.1.2).

The composite II ≈**280**

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{k_*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{f_*\oplus(-g_*i^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$$

is the sum of

$$f_*k_* = m_* \qquad \text{and} \qquad -g_*i^*k_* = -g_*\ell_*h^* = -h^*,$$

which proves (5.1.3).

Corollary 5.1.6.

Under the hypotheses of 5.1.1, for $i \leq n-1$, the images of

$$m^* : H^i(X, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

and of

$$k^* : H^i(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

are equal.

Proof. By the weak Lefschetz theorem for hyperplane sections (SGA 5, VII 7.1), for $i \leq n-1$ the map

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \tag{5.1.7}$$

is an isomorphism. Thus, by (5.1.2), the image of k^* is the sum of the image of m^* and the image of the composite

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{h_*} H^i(Y, \mathbb{Z}_\ell).$$

But the projection formula gives

$$\begin{aligned} h_* j^*(x) &= h_*\big(h^* m^*(x)\big) = m^*(x) h_*(1) = \xi_Y \, m^*(x) \\ &= m^*(x \xi_X), \qquad \text{since } m^*(\xi_X) = \xi_Y. \end{aligned}$$

Consequently $\operatorname{Im} k^* \subset \operatorname{Im} m^*$, and the reverse inclusion follows directly from (5.1.2). This proves the corollary.

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5.2. The hard Lefschetz theorem (LV)

5.2.1.

Let S be projective of dimension n over the algebraically closed field k, and let $S \hookrightarrow \mathbb{P}$ be a projective embedding. Denote by

$$u \in H^2(S, \mathbb{Z}_\ell(1))$$

the cohomology class of a hyperplane section. Thus u is the pullback along $S \hookrightarrow \mathbb{P}$ of the first Chern class of $\mathcal{O}_{\mathbb{P}}(1)$ in $H^2(\mathbb{P}, \mathbb{Z}_\ell(1))$. Write

$$L_S : H^\bullet(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{\bullet+2}(S, \mathbb{Q}_\ell(j+1))$$

for multiplication by u.

5.2.2.

We say that S (equipped with u) satisfies (LV), the hard Lefschetz theorem, if, for every (or for one) integer j and every integer $i > 0$, the maps

$$L_S^i : H^{n-i}(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+i}(S, \mathbb{Q}_\ell(i+j))$$

are isomorphisms.

5.2.2.1.

It is conjectured that this condition holds when S is smooth. It is true when k has characteristic zero [6]. The specialization theorem [SGA 4, XVI 2.2] therefore gives the same conclusion when S, together with its projective embedding, lifts to characteristic zero. In particular, it holds for a smooth complete intersection. It is also known for a smooth S of dimension at most two [2], and for an abelian variety, either directly (Lieberman [4] and [2]) or by Mumford's theorem [5] that a polarized abelian variety lifts, together with its polarization, to characteristic zero.

Recall that, for a projective variety S and $i \leq n$, the primitive part of cohomology is defined by II≈282

$$\text{(5.2.2.2)} \qquad \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) = \mathrm{Ker}\Big(L^{n-i+1} : H^i(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{2n-i+2}(S, \mathbb{Q}_\ell(j+n-i+1))\Big)\,.$$

If S satisfies (LV), then for $i \leq n$ one has the decomposition

$$\text{(5.2.2.3)} \qquad H^i(S, \mathbb{Q}_\ell(j)) \xleftarrow[\sim]{1+L} \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) \oplus H^{i-2}(S, \mathbb{Q}_\ell(j-1)).$$

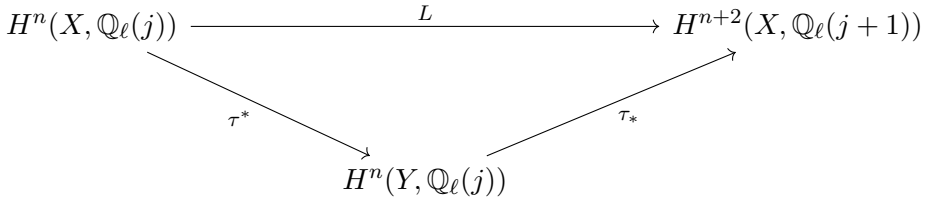
Proposition 5.2.3.

Let X be smooth and projective and suppose that it satisfies (LV). Let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then

$$\text{(5.2.3.1)} \qquad \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) = \mathrm{Ker}(\tau^* : H^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^n(Y, \mathbb{Q}_\ell(j)))\,.$$

Proof. The projection formula for $\tau : Y \to X$ gives the commutative diagram

(5.2.3.2)



By duality and weak Lefschetz,

$$\tau_* : H^n(Y, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+2}(X, \mathbb{Q}_\ell(j+1))$$

is the transpose of an isomorphism and hence is itself an isomorphism. The assertion follows.

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Lemma 5.2.4.

Let X be smooth and projective, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then X satisfies (LV) if and only if the following two conditions hold:

5.2.4.1. Y satisfies (LV).

5.2.4.2. The restriction of the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell)$ to the subspace $\mathrm{Im}\, H^{n-1}(X, \mathbb{Q}_\ell)$ is nondegenerate.

Proof. We have

$$\text{(5.2.4.3)} \qquad \begin{cases} L_X = \tau_* \tau^*, \\ L_Y = \tau^* \tau_* \qquad \text{(by 4.2.11)}, \end{cases}$$

and consequently

$$\text{(5.2.4.4)} \qquad L_X^{i+1} = \tau_* L_Y^i \tau^*.$$

In the commutative diagram

$$\begin{array}{ccc} H^{n-i-1}(X,\mathbb{Q}_\ell(j)) & \xrightarrow{L_X^{i+1}} & H^{n+i+1}(X,\mathbb{Q}_\ell(j+i+1)) \\ \downarrow{\scriptstyle \tau^*} & & \uparrow{\scriptstyle \tau_*} \\ H^{n-1-i}(Y,\mathbb{Q}_\ell(j)) & \xrightarrow{L_Y^{i}} & H^{n-1+i}(Y,\mathbb{Q}_\ell(j+i)) \end{array} \tag{5.2.4.5}$$

τ^* is an isomorphism for $i \geq 1$ by weak Lefschetz, and τ_* is an isomorphism by duality. Hence X satisfies (LV) if and only if Y satisfies (LV) and

$$L_X : H^{n-1}(X,\mathbb{Q}_\ell(j)) \longrightarrow H^{n+1}(X,\mathbb{Q}_\ell(j+1)) \tag{5.2.4.6}$$

is an isomorphism.

It remains to identify (5.2.4.6) with 5.2.4.2. Let $a, b \in H^{n-1}(X,\mathbb{Q}_\ell(j))$. By Poincaré duality on II≈284
X, (5.2.4.6) is equivalent to nondegeneracy of the bilinear form

$$\begin{aligned} H^{n-1}(X,\mathbb{Q}_\ell(j)) \times H^{n-1}(X,\mathbb{Q}_\ell(j)) &\longrightarrow H^{2n}(X,\mathbb{Q}_\ell(2j+1)), \\ (a,b) &\longmapsto a\, L_X(b). \end{aligned} \tag{5.2.4.7}$$

But by (5.2.4.3) and the projection formula,

$$aL_X(b) = a\tau_*\tau^*(b) = \tau_*(\tau^*(a)\tau^*(b)).$$

This proves the equivalence of (5.2.4.7) and 5.2.4.2.

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Corollary 5.2.5.

Let X be smooth and projective of dimension n, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Suppose that X satisfies (LV). Then, for $q \geq n$, the map

$$\tau^* : H^q(X,\mathbb{Q}_\ell(j)) \longrightarrow H^q(Y,\mathbb{Q}_\ell(j))$$

is surjective.

Proof. By (5.2.4.3), for $i \geq 0$ one has

$$L_Y^{i+1} = \tau^* L_X^i \tau_*,$$

and hence the commutative diagram

$$\begin{array}{ccc} H^{n-i}(X,\mathbb{Q}_\ell(j+1)) & \xrightarrow{L_X^{i}} & H^{n+i}(X,\mathbb{Q}_\ell(j+i+1)) \\ \uparrow{\scriptstyle \tau_*} & & \downarrow{\scriptstyle \tau^*} \\ H^{n-i-2}(Y,\mathbb{Q}_\ell(j)) & \xrightarrow[\sim]{L_Y^{i+1}} & H^{n+i}(Y,\mathbb{Q}_\ell(j+i+1)). \end{array}$$

This gives the required result.

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Lemma 5.2.6.

Let $f : F \to B$ be a smooth proper morphism with B connected. Let $\mathcal{O}_F(1)$ be a relatively ample invertible sheaf on F, and equip each geometric fiber $F_{\bar{b}}$ with the ample invertible sheaf induced by $\mathcal{O}_F(1)$. If one geometric fiber satisfies (LV), then so do all the others.

Proof. Let $\xi_F \in H^2(F, \mathbb{Z}_\ell(1))$ be the first Chern class of $\mathcal{O}_F(1)$. This class and its powers $\xi_F^i \in H^{2i}(F, \mathbb{Z}_\ell(i))$ induce, by cup product, the homomorphisms

$$\xi_F^i : R^{n-i} f_* \mathbb{Q}_\ell(j) \longrightarrow R^{n+i} f_* \mathbb{Q}_\ell(j+i), \tag{5.2.6.1}$$

where n is the relative dimension of the fibers. By proper base change [SGA 4, XII 5.2], all geometric fibers satisfy (LV) if and only if all the maps (5.2.6.1) are isomorphisms. By the specialization theorem [SGA 4, XVI 2.2], the sheaves $R^i f_* \mathbb{Q}_\ell(j)$ are twisted constant sheaves. Thus the maps (5.2.6.1) are isomorphisms if and only if they are so over one given geometric point.

Corollary 5.2.7.

Let X be smooth and projective. If one smooth hyperplane section satisfies (LV), then every smooth hyperplane section does.

Proof. The universal family of smooth hyperplane sections of X satisfies the hypotheses of Lemma 5.2.6.

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5.3. Condition (A)

5.3.1.

Let

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

be the "fibration" associated with a pencil whose axis meets X transversely and whose generic fiber is assumed to be smooth.

Let U be a nonempty open subset of $\mathbb{P}^1$ over which ρ is smooth, and let

$$\nu : U \hookrightarrow \mathbb{P}^1 \tag{5.3.1.1}$$

be the inclusion. Since the sheaves $R^i \rho_* \mathbb{Q}_\ell(j)$ are twisted constant over U, the sheaves

$$\nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j)$$

are seen to be essentially independent of the choice of U; the canonical "adjunction" homomorphisms

$$R^i \rho_* \mathbb{Q}_\ell(j) \longrightarrow \nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j) \tag{5.3.2}$$

are likewise independent of the choice of U.

5.3.3.

Observe that, for an arbitrary pencil of hyperplane sections, the weak Lefschetz theorem implies that, for $i \leq n-2$, the immersion

$$\widetilde{X} \overset{(f \times \rho)}{\hookrightarrow} X \times \mathbb{P}^1$$

gives an isomorphism of sheaves on $\mathbb{P}^1$

$$R^i \rho_* \mathbb{Q}_\ell(j) \overset{\sim}{\longleftarrow} H^i(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1}, \qquad i \leq n-2.$$

In other words, $R^i \rho_* \mathbb{Q}_\ell(j)$ is constant for $i \leq n-2$ and, a fortiori, the maps (5.3.2) are isomorphisms for $i \leq n-2$.

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5.3.4.

The local theory of vanishing cycles, and in particular the Picard–Lefschetz formula [SGA 7, XV 3.4], shows that, for a Lefschetz pencil, the homomorphisms (5.3.2) are isomorphisms for $i \neq n$, and are surjective for $i = n$.

5.3.5.

We say that ρ, or the pencil from which it arises, satisfies condition (A) if all the homomorphisms (5.3.2) are isomorphisms (equivalently, if this holds for every i with $j = 0$).

We shall see below (6.3, 6.4) that condition (A) has a strong tendency to hold.

Lemma 5.4.

Suppose that one smooth hyperplane section of X (and hence every such section, by Corollary 5.2.7) satisfies (LV)—for example, suppose that X satisfies (LV), cf. Lemma 5.2.4—and suppose that ρ satisfies condition (A). Then every fiber (including every singular fiber) satisfies (LV), and the sheaves $R^i\rho_*\mathbb{Q}_\ell(j)$ are constant for $i \neq n-1$.

Proof. Let

$$L : R^i\rho_*\mathbb{Q}_\ell(j) \longrightarrow R^{i+2}\rho_*\mathbb{Q}_\ell(j+1)$$

denote multiplication by $f^*c_1(\mathcal{O}_X(1)) \in H^2(\widetilde{X}, \mathbb{Z}_\ell(1))$; fiber by fiber, this is the operation considered in (5.2.1). By hypothesis, over U we have an isomorphism

$$\nu^*(L^i) : \nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i) \tag{5.4.1}$$

for every $i > 0$ and every j. Applying the functor ν_* gives the isomorphisms

$$(\nu_*\nu^*)L^i : \nu_*\nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu_*\nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.2}$$

In view of (A), these identify with the morphisms

$$L^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.3}$$

The conclusion follows from 5.3.3.

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5.4.4.

Let Y be a smooth fiber of $\rho : \widetilde{X} \to \mathbb{P}^1$. By weak Lefschetz, the restriction map

$$H^{n-1}(X, \mathbb{Q}_\ell(j)) \overset{m^*}{\hookrightarrow} H^{n-1}(Y, \mathbb{Q}_\ell(j))$$

is injective. We saw in 5.2.4 that, if X satisfies (LV), then the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ remains nondegenerate when restricted to $m^*H^{n-1}(X, \mathbb{Q}_\ell(j))$. The orthogonal complement of $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ is therefore a complementary subspace to $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$; following the classical terminology, it is called the vanishing-cohomology space of Y.

Let $\nu : U \hookrightarrow \mathbb{P}^1$ be any nonempty open subset over which ρ is smooth. The corresponding subsheaf of $\nu_*\nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)$, the "vanishing-cohomology sheaf" $E^{n-1}\rho_*\mathbb{Q}_\ell(j)$ on $\mathbb{P}^1$, is defined by

$$E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \nu_*\left\{\begin{array}{l}\text{the orthogonal complement in } \nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j) \text{ of the} \\ \text{constant subsheaf } H^{n-1}(X, \mathbb{Q}_\ell(j))_U\end{array}\right\}. \tag{5.4.4.1}$$

It is clear that this sheaf is independent of the choice of U.

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Lemma 5.5.

Suppose that X satisfies (LV) and that the maps (5.3.2) are isomorphisms for $i = n-1$—for example, suppose that ρ arises from a Lefschetz pencil, cf. 5.3.4. Then there is a direct-sum decomposition

$$R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_*\mathbb{Q}_\ell(j). \tag{5.5.1}$$

Proof. By (LV), over U there is a decomposition

$$\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j), \tag{5.5.2}$$

to which we apply ν_*. This proves the assertion.

Theorem 5.6.

Let X be a smooth, irreducible, projective variety satisfying (LV) (5.2.2). Let $\{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely, in X_η and Δ respectively. Suppose that the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ satisfies condition (A). Then:

(5.6.1) the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \Longrightarrow H^{p+q}(\widetilde{X}, \mathbb{Q}_\ell)$$

degenerates.

(Note.) This remains valid if, instead of assuming that X satisfies (LV), one assumes only that a smooth hyperplane section Y does.

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5.6.2.

Let η denote the generic point of $\mathbb{P}^1$, $\bar{\eta}$ the geometric generic point, $X_{\bar{\eta}}$ the geometric generic fiber of ρ, and π the Galois group of $k(\bar{\eta})/k(\eta)$.

For $q \neq n-1$ there are canonical isomorphisms

$$E_2^{0,q} = H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell), \tag{5.6.2.1}$$

$$E_2^{2,q} = H^2(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)), \tag{5.6.2.2}$$

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) = 0, \qquad p \neq 0, 2. \tag{5.6.2.3}$$

For $q = n-1$ one has

$$E_2^{0,n-1} = H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell), \tag{5.6.2.4}$$

$$E_2^{2,n-1} = H^2(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1))^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell(-1)). \tag{5.6.2.5}$$

(The missing value $E_2^{1,n-1}$ will be found in 5.7 below).

Proof. By (5.4.3), Theorem 1.1 applies and gives (5.6.1). The first three formulas (5.6.2.1)–(5.6.2.3) follow from the constancy of $R^q\rho_*\mathbb{Q}_\ell$ for $q \neq n-1$ (Lemma 5.4); for (5.6.2.3) one also uses, of course, the simple connectedness of $\mathbb{P}^1$.

The first isomorphism in (5.6.2.4) follows from

$$R^{n-1}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell.$$

Thus

$$H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^0(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell),$$

and $\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell$ is a "twisted constant" sheaf on U, hence corresponds to the π-module $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$. The second isomorphism in (5.6.2.4) follows from degeneration together with Corollary 5.1.6.

We turn to (5.6.2.5). Let $F^\bullet$ denote the filtration on $H^\bullet(\widetilde{X}, \mathbb{Q}_\ell)$ defined by the spectral sequence. By degeneration, $F^2 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) = E_2^{2,n-1}$ and $\mathrm{gr}^1 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) = E_2^{1,n} = 0$ by (5.6.2.3). Consequently (cf. 5.1.0),

$$\begin{aligned} E_2^{2,n-1} &= F^1 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \\ &= \mathrm{Ker}\Big(H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^{n+1}\rho_* \mathbb{Q}_\ell)\Big) \\ &\overset{(A)}{=} \mathrm{Ker}\Big(k^* : H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)\Big)\,. \end{aligned} \tag{5.6.3}$$

By (5.1.2), k^* may be written as

$$H^{n+1}(X, \mathbb{Q}_\ell) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \xrightarrow{m^* + h_*} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell). \tag{5.6.4}$$

Using (5.2.2.3), this becomes

$$\begin{aligned} \mathrm{Prim}^{n-1}(X, \mathbb{Q}_\ell(-1)) \oplus H^{n-3}(X, \mathbb{Q}_\ell(-2)) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \\ \xrightarrow{m^* L + m^* L^2 + h_*} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell). \end{aligned} \tag{5.6.5}$$

Now the map

$$H^{n-3}(X, \mathbb{Q}_\ell(-2)) \xrightarrow{m^* L^2} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$

is an isomorphism by (LV), since $m^* L_X^2 = L_{X_{\bar{\eta}}}^2 m^*$ (cf. 5.2.4 and 5.2.5). Hence the kernel of (5.6.5) identifies, by $\mathrm{pr}_1 \oplus \mathrm{pr}_3$, with

$$\mathrm{Prim}^{n-1}(X, \mathbb{Q}_\ell(-1)) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)).$$

By (LV) and weak Lefschetz,

$$H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \underset{\sim}{\xleftarrow{L}} H^{n-3}(\Delta, \mathbb{Q}_\ell(-2)) \underset{\sim}{\xleftarrow{}} H^{n-3}(X, \mathbb{Q}_\ell(-2)).$$

Together with (5.2.2.3), this gives

$$E_2^{2,n-1} \simeq H^{n-1}(X, \mathbb{Q}_\ell(-1)). \tag{5.6.6}$$

The other isomorphism in (5.6.2.5) follows by twisting (5.6.2.4).

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Remark 5.6.7.

The assertion (5.6.1) can be proved under substantially weaker hypotheses, as Deligne explained to me:

Theorem 5.6.8.

Let X be smooth of dimension n over an algebraically closed field k, and let

$$\rho : X \longrightarrow \mathbb{P}^1_k$$

be a projective morphism whose generic fiber is smooth. Suppose that:

(5.6.8.1) the geometric generic fiber, equipped with the restriction of the relatively ample invertible sheaf $\mathcal{O}_X(1)$, satisfies (LV);

(5.6.8.2) the maps (5.3.2) are isomorphisms for $i \neq n$ and are surjective for $i = n$ (a condition satisfied in particular when ρ arises from a Lefschetz pencil, cf. 5.3.4). Then the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q \rho_* \mathbb{Q}_\ell) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell)$$

degenerates.

Remark. The same theorem, with the same proof, holds with $\mathbb{P}^1_k$ replaced by any smooth, irreducible, projective curve.

Proof. For the proof we identify the sheaf $\mathbb{Q}_\ell(1)$ with $\mathbb{Q}_\ell$; this is possible because k is algebraically closed. We may plainly suppose that X is connected and that ρ is surjective, hence flat with fibers of dimension $n-1$. Let II≈294

$$\xi_X \in H^2(X, \mathbb{Q}_\ell)$$

be the first Chern class of $\mathcal{O}_X(1)$, and let

$$\nu : U \hookrightarrow \mathbb{P}^1$$

be the inclusion of an open subset over which ρ is smooth. By (5.6.8.1) and Lemma 5.2.6, the cup-product maps

$$\xi_X^i : \nu^* R^{n-1-i}\rho_*\mathbb{Q}_\ell \longrightarrow \nu^* R^{n-1+i}\rho_*\mathbb{Q}_\ell$$

are isomorphisms for $i \geq 1$. Applying ν_* and using (5.6.8.2) gives, for $i \geq 2$, the isomorphisms

$$\xi_X^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell, \tag{5.6.8.3}$$

and, for $i = 1$, the isomorphism

$$\xi_X : R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.4}$$

The latter factors as

$$R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\xi_X} R^n\rho_*\mathbb{Q}_\ell \xrightarrow{\text{adjunction}} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.5}$$

Consequently there is a canonical decomposition

$$R^n\rho_*\mathbb{Q}_\ell \xleftarrow[\sim]{\xi_X+\text{incl}} R^{n-2}\rho_*\mathbb{Q}_\ell \oplus \text{Local}^n\,\mathbb{Q}_\ell, \tag{5.6.8.6}$$

where

$$\text{Local}^n\,\mathbb{Q}_\ell = \text{Ker}(R^n\rho_*\mathbb{Q}_\ell \longrightarrow \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell)\,. \tag{5.6.8.7}$$

The sheaf $\text{Local}^n\,\mathbb{Q}_\ell$ is concentrated at the points of $\mathbb{P}^1$ where the fiber of ρ is not smooth. Combining this fact with (5.6.8.3), we obtain

$$\xi_X^{i+1} : H^j(\mathbb{P}^1, R^{n-2-i}\rho_*\mathbb{Q}_\ell) \xrightarrow{\sim} H^j(\mathbb{P}^1, R^{n+i}\rho_*\mathbb{Q}_\ell), \qquad i \geq 0, \quad j \geq 1. \tag{5.6.8.8}$$

We now prove degeneration. Since $E_2^{p,q} = 0$ unless $p = 0, 1, 2$, it is enough to prove that all the differentials

$$d_2 : H^0(\mathbb{P}^1, R^i\rho_*\mathbb{Q}_\ell) \longrightarrow H^2(\mathbb{P}^1, R^{i-1}\rho_*\mathbb{Q}_\ell)$$

vanish. For $q \leq n-1$, set

$$\text{Prim}(E_2^{p,q}) = \text{Ker}\Big(\xi_X^{n-q} : E_2^{p,q} \longrightarrow E_2^{p,2n-q}\Big)\,. \tag{5.6.8.9}$$

Equations (5.6.8.3) and (5.6.8.6), in particular the injectivity of

$$\xi_X : H^p(\mathbb{P}^1, R^{n-2}\rho_*\mathbb{Q}_\ell) \longrightarrow H^p(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell),$$

give, for $q \leq n-1$, the decomposition

$$E_2^{p,q} \xleftarrow[\sim]{1+\xi_X+\xi_X^2+\cdots} \text{Prim}(E_2^{p,q}) \oplus \text{Prim}(E_2^{p,q-2}) \oplus \text{Prim}(E_2^{p,q-4}) \oplus \cdots\,. \tag{5.6.8.10}$$

To prove that

$$d_2^{p,q} = 0 \qquad (q \leq n-1),$$

consider the commutative diagram

$$\begin{array}{ccc} \mathrm{Prim}(E_2^{p,q}) & \xrightarrow{\xi_X^{n-q}=0} & E_2^{p,2n-q} \\ \downarrow{\scriptstyle d_2^{p,q}} & & \downarrow{\scriptstyle d_2^{p,2n-q}} \\ E_2^{p+2,q-1} & \xrightarrow[\sim]{\xi_X^{n-1-(q-1)}} & E_2^{p+2,2n-q-1}. \end{array}$$

The lower arrow is an isomorphism by (5.6.8.8). The diagram therefore shows that $d_2^{p,q}$ kills $\mathrm{Prim}(E_2^{p,q})$ for $q \leq n-1$, and (5.6.8.10) then shows that d_2 kills $E_2^{p,q}$ for $q \leq n-1$. By (5.6.8.3), it also kills $E_2^{p,q}$ for $q \geq n+1$. It remains only to prove that

$$d_2^{0,n} : E_2^{0,n} \longrightarrow E_2^{2,n-1}$$

is zero, or equivalently that

$$\dim E_3^{2,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.12}$$

We have already proved

$$\dim E_2^{p,q} = \dim E_3^{p,q} \qquad \text{for } (p,q) \neq (0,n), (2,n-1), \tag{5.6.8.13}$$

and for degree reasons

$$E_3^{p,q} = E_\infty^{p,q} \qquad \text{for every pair } (p,q). \tag{5.6.8.14}$$

It follows that

$$b_i(X) = \dim H^i(X, \mathbb{Q}_\ell) = \sum_{p+q=i} \dim E_3^{p,q}.$$

In view of (5.6.8.13), we therefore have

$$b_{n-1}(X) = \dim E_2^{0,n-1} + \dim E_2^{1,n-2} + \dim E_2^{2,n-3} \tag{5.6.8.15}$$

and

$$b_{n+1}(X) = \dim E_2^{0,n+1} + \dim E_2^{1,n} + \dim E_3^{2,n-1}. \tag{5.6.8.16}$$

Poincaré duality gives

$$b_{n-1}(X) = b_{n+1}(X). \tag{5.6.8.17}$$

First, by (5.6.8.8),

$$\dim E_2^{1,n} = \dim E_2^{1,n-2}.$$

Thus (5.6.8.17) shows that the proof of (5.6.8.12) is complete once we know

$$\dim E_2^{2,n-3} = \dim E_2^{0,n+1}, \tag{5.6.8.18}$$

$$\dim E_2^{0,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.19}$$

These two formulas follow from (5.6.8.2), Poincaré duality on the smooth fibers of ρ, and the following lemma applied to $\mathbb{P}^1$ and the sheaves $R^{n-3}\rho_*\mathbb{Q}_\ell$ and $R^{n-1}\rho_*\mathbb{Q}_\ell$.

Lemma 5.6.9.

Let D be a smooth, connected, proper curve over an algebraically closed field k. Let $i : \eta \hookrightarrow D$ be the inclusion of the generic point, and let F be a constructible $\mathbb{Q}_\ell$-sheaf on D such that $F \xrightarrow{\sim} i_* i^* F$. Then

$$H^2(D, F) \simeq H^0(D, \check{F}(-1))^\vee,$$

where $\check{F} \simeq i_*((i^*F)^\vee)$.

Corollary 5.6.10.

(Uses Picard–Lefschetz.) Let $\rho : \widetilde{X} \to \mathbb{P}^1$ be the "fibration" arising from a Lefschetz pencil, and suppose that the smooth fibers satisfy (LV). Then

$$H^i(X_{\bar\eta}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\Big(H^i(X, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar\eta}, \mathbb{Q}_\ell)\Big). \tag{5.6.10.1}$$

Proof. By 5.3.4, which uses the Picard–Lefschetz formula, Theorem 5.6.8 applies and gives

$$H^i(X_{\bar\eta}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\Big(H^i(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar\eta}, \mathbb{Q}_\ell)\Big).$$

It remains to check that $H^i(X, \mathbb{Q}_\ell)$ and $H^i(\widetilde{X}, \mathbb{Q}_\ell)$ have the same image in $H^i(X_{\bar\eta}, \mathbb{Q}_\ell)$. This holds for $i \le n-1$ by Corollary 5.1.6, and for $i \ge n$ by Corollary 5.2.5.

We now turn to the most important computation:

Theorem 5.7.

Under the hypotheses of 5.6, set

$$E^{n-2}(\Delta, \mathbb{Q}_\ell(j)) = \text{the orthogonal complement in } H^{n-2}(\Delta, \mathbb{Q}_\ell(j)) \text{ of the image of } H^{n-2}(X, \mathbb{Q}_\ell(j)).$$

(This is the "vanishing" part of the cohomology of Δ; cf. 5.4.4.) Then there is a direct-sum decomposition (cf. 4.2)

$$E_2^{1,n-1} = H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \xleftarrow[f^*+i_*g^*]{\sim} \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus E^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \tag{5.7.1}$$

Proof. By degeneration, II≈**299**

$$E_2^{1,n-1} \simeq F^1H^n(\widetilde{X}, \mathbb{Q}_\ell)/F^2H^n(\widetilde{X}, \mathbb{Q}_\ell).$$

By (A),

$$F^1H^n(\widetilde{X}, \mathbb{Q}_\ell) = \ker\Big(k^* : H^n(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^n(X_{\bar\eta}, \mathbb{Q}_\ell)\Big), \tag{5.7.2}$$

which, by (5.1.2), identifies with

$$\ker\Big(H^n(X, \mathbb{Q}_\ell) \oplus H^{n-2}(\Delta, \mathbb{Q}_\ell(-1)) \xrightarrow{m^*+h_*} H^n(X_{\bar\eta}, \mathbb{Q}_\ell)\Big).$$

Under (LV), k^* can also be written as

$$\mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus H^{n-2}(X, \mathbb{Q}_\ell(-1)) \oplus H^{n-2}(\Delta, \mathbb{Q}_\ell(-1)) \xrightarrow{m^*+m^*L+h_*} H^n(X_{\bar\eta}, \mathbb{Q}_\ell), \tag{5.7.3}$$

and the middle arrow

$$H^{n-2}(X, \mathbb{Q}_\ell(-1)) \xrightarrow{m^*L} H^n(X_{\bar\eta}, \mathbb{Q}_\ell) \tag{5.7.4}$$

is an isomorphism (cf. 5.2.3 and (5.2.2.3)). Thus, denoting by

$$\wp : H^n(X, \mathbb{Q}_\ell) \longrightarrow \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \tag{5.7.5}$$

the projection onto the primitive part, we obtain

$$F^1H^n(\widetilde{X}, \mathbb{Q}_\ell) \xrightarrow[\wp f_* \oplus (-g_* i^*)]{\sim} \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus H^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \tag{5.7.6}$$

By (5.6.2.2) and degeneration,

$$F^2H^n(\widetilde{X}, \mathbb{Q}_\ell) \simeq E_2^{2,n-2} \simeq H^{n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)). \tag{5.7.7}$$

On the other hand, the inclusion of $F^2H^n(\widetilde{X}, \mathbb{Q}_\ell) = H^{n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1))$ in $H^n(\widetilde{X}, \mathbb{Q}_\ell)$ is the Gysin map k_*.

Strictly speaking, the Gysin map is defined only as a map

$$H^{i-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)) \longrightarrow H^i(X \times_k k(\bar{\eta}), \mathbb{Q}_\ell),$$

but, in view of the canonical isomorphism (SGA 4 XII 5.4)

$$H^i(X, \mathbb{Q}_\ell) \xrightarrow{\sim} H^i(X \otimes_k K, \mathbb{Q}_\ell)$$

for every algebraically closed field K/k, we hope the reader will allow us such abuses. By (5.1.3), the Gysin map is expressed as

$$H^{n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)) \xrightarrow{m_* \oplus (-h^*)} H^n(X, \mathbb{Q}_\ell) \oplus H^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \tag{5.7.8}$$

By weak Lefschetz,

$$m_*H^{n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)) = m_*m^*H^{n-2}(X, \mathbb{Q}_\ell(-1)) = L_X H^{n-2}(X, \mathbb{Q}_\ell(-1))$$

is the orthogonal complement of the primitive part. Hence we have the commutative diagram

$$\begin{array}{ccc} F^2H^n(\widetilde{X}, \mathbb{Q}_\ell) & \hookrightarrow & F^1H^n(\widetilde{X}, \mathbb{Q}_\ell) \\ \wr \uparrow k_* & & \wr \downarrow \wp f_* \oplus (-g_* i^*) \\ H^{n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(-1)) & \xrightarrow{0 \oplus h^*} & \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus H^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \end{array} \tag{5.7.9}$$

This proves (5.7.1), as required.

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Remark 5.7.10.

In 5.7, hypothesis (A) may be replaced by the weaker assumption that, for $q = n$, the maps

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \tag{5.7.11}$$

and

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) \tag{5.7.12}$$

have the same kernel, as our proof shows. Without this condition, the conclusion can fail; cf. 5.9.

5.8. The Griffiths homomorphism

5.8.1.

Let $f : T \to S$ be a morphism of schemes. The Leray spectral sequence defines a filtration (F^i) on the groups $H^\bullet(T, \mathbb{Q}_\ell(j))$. Define

$$\mathrm{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = F^1 H^\bullet(T, \mathbb{Q}_\ell(j)). \tag{5.8.1.1}$$

Because the Leray spectral sequence lies in the first quadrant, we have

$$E_\infty^{0,\bullet} \hookrightarrow E_2^{0,\bullet}, \tag{5.8.1.2}$$

and hence

$$\mathrm{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = \ker\Big(H^\bullet(T, \mathbb{Q}_\ell(j)) \longrightarrow H^0(S, R^\bullet f_* \mathbb{Q}_\ell(j))\Big). \tag{5.8.1.3}$$

For the same reason,

$$E_\infty^{1,\bullet} \hookrightarrow E_2^{1,\bullet}. \tag{5.8.1.4}$$

The composite

$$\begin{array}{ccc} \mathrm{Prim}^i(T/S, \mathbb{Q}_\ell(j)) = F^1 H^i(T, \mathbb{Q}_\ell(j)) & \longrightarrow & \mathrm{gr}^1 H^i(T, \mathbb{Q}_\ell(j)) = E_\infty^{1,i-1} \\ & & \big\downarrow \\ & & H^1(S, R^{i-1} f_* \mathbb{Q}_\ell(j)) = E_2^{1,i-1} \end{array} \tag{5.8.1.5}$$

will be denoted

$$U^i : \mathrm{Prim}^i(T/S, \mathbb{Q}_\ell(j)) \longrightarrow H^1(S, R^{i-1} f_* \mathbb{Q}_\ell(j)). \tag{5.8.1.6}$$

5.8.2.

Now let X be projective, smooth, and irreducible, and let $D = \{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely. Let $U \xrightarrow{\nu} \mathbb{P}^1$ be a nonempty open subset over which the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ is smooth, and put $\widetilde{X}|_U = \widetilde{X} \times_{\mathbb{P}^1} U$. We have the commutative diagram

$$\begin{array}{ccccc} & & \widetilde{X} & \xleftarrow{\alpha} & \widetilde{X}|_U \\ & \swarrow^{f} & \big\downarrow{\scriptstyle \rho} & & \big\downarrow{\scriptstyle \rho|_U} \\ X & & \mathbb{P}^1 & \xleftarrow[\nu]{} & U. \end{array} \tag{5.8.2.1}$$

Since $\rho|_U : \widetilde{X}|_U \to U$ is proper and smooth, the specialization theorem for cohomology (SGA 4 XVI 2.2) gives

$$\mathrm{Prim}^\bullet(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) = \ker\Big(H^\bullet(\widetilde{X}|_U, \mathbb{Q}_\ell(j)) \longrightarrow H^\bullet(X_{\bar\eta}, \mathbb{Q}_\ell(j))\Big). \tag{5.8.2.2}$$

Consequently, by (5.2.3.1), we have:

Proposition 5.8.2.3.

Under the hypotheses of 5.8.2, for $x \in H^n(X, \mathbb{Q}_\ell(j))$, the following conditions are equivalent:

$$x \in \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff \alpha^* f^*(x) \in \mathrm{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)). \tag{5.8.2.4}$$

If, in addition, ρ satisfies hypothesis (A) of 5.3, then

$$x \in \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff f^*(x) \in \mathrm{Prim}^n(\widetilde{X}/\mathbb{P}^1, \mathbb{Q}_\ell(j)). \tag{5.8.2.5}$$

Proof. Formula (5.8.2.4) is proved. For (5.8.2.5), hypothesis (A) implies that

$$H^0(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell(j)) \xrightarrow[\nu^*]{\sim} H^0(U, \nu^* R^n \rho_* \mathbb{Q}_\ell(j)).$$

Thus $f^*(x)$ is primitive relative to $\mathbb{P}^1$ if and only if $\alpha^* f^*(x)$ is primitive relative to U.

Definition 5.8.3.

In the situation of 5.8.2, the composite homomorphism

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(5.8.3.1)
$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\alpha^* f^*} & \mathrm{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) \\ & \searrow^{\mathrm{Grif}} & \downarrow{\scriptstyle U^n} \\ & & H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \end{array}$$

is called the Griffiths homomorphism (for X, relative to the given pencil D).

Adjoin to diagram (5.8.2.1) the constant family over U with fiber X:

(5.8.3.2)
$$\begin{array}{ccccc} & & \widetilde{X} & \xleftarrow{\alpha} & \widetilde{X}|_U & \xrightarrow{f\times\rho|_U} & X \times U \\ & \swarrow^{f} & \downarrow & & \downarrow{\scriptstyle \rho|_U} & \swarrow_{\mathrm{pr}_2} & \\ X & & \mathbb{P}^1 & \xleftarrow{\nu} & U & & \end{array} .$$

Theorem 5.8.4.

Suppose that X satisfies (LV) (5.2.2), and that ρ satisfies hypothesis (A) (5.3). Then, for every nonempty smoothness open $U \xrightarrow{\nu} \mathbb{P}^1$ of ρ, the Griffiths homomorphism "modulo the fixed part", which is by definition the composite

(5.8.4.1)
$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad\mathrm{Grif}\qquad} & H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(U, \nu^* E^{n-1}\rho_* \mathbb{Q}_\ell(j)) \simeq \dfrac{H^1(U, \nu^* R^{n-1}\rho_* \mathbb{Q}_\ell(j))}{(f\times\rho|_U)^* H^1(U, H^{n-1}(X, \mathbb{Q}_\ell(j))_U)} \end{array}$$

is injective.

Proof. By (5.5.1), over $\mathbb{P}^1$ we have

$$R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_*\mathbb{Q}_\ell(j), \tag{5.8.4.2}$$

and over U we have

$$\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j). \tag{5.8.4.3}$$

This permits us to rewrite (5.8.4.1) as indicated:

$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\text{Grif}} & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(U, \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array} \tag{5.8.4.4}$$

By (5.7.1), the map (defined only by virtue of hypothesis (A))

$$u : \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \tag{5.8.4.5}$$

is injective. Since $\mathbb{P}^1$ is simply connected, decomposition (5.8.4.2) gives

$$H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq H^1(\mathbb{P}^1, E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \tag{5.8.4.6}$$

We can therefore complete (5.8.4.4) to a commutative square:

$$\begin{array}{ccc} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\text{Grif}} & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ \downarrow{\scriptstyle u} & & \downarrow \\ H^1(\mathbb{P}^1, E^{n-1}\rho_*\mathbb{Q}_\ell(j)) & \xrightarrow{\nu^*} & H^1(U, \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array} \tag{5.8.4.7}$$

It remains only to see that the lower arrow is injective. This follows from

$$E^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq \nu_*\nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j) \tag{5.8.4.8}$$

and the following well-known lemma.

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Lemma 5.8.5.

Let $\nu : A \to B$ be a morphism of topoi, and let F be a sheaf on A. Then the canonical map

$$H^1(B, \nu_* F) \longrightarrow H^1(A, F) \tag{5.8.5.1}$$

is injective.

Proof. In a first-quadrant spectral sequence,

$$E_2^{1,0} \simeq E_\infty^{1,0}. \tag{5.8.5.2}$$

Thus, for the Leray spectral sequence

$$E_2^{p,q} = H^p(B, R^q\nu_* F) \Longrightarrow H^{p+q}(A, F),$$

we have

$$E_2^{1,0} = H^1(B, \nu_* F) \simeq E_\infty^{1,0} = F^1 H^1(A, F). \tag{5.8.5.3}$$

Taking $F = \nu^* E$, where E is a sheaf on B, gives

$$H^1(B, \nu_*\nu^* E) \hookrightarrow H^1(A, \nu^* E). \tag{5.8.5.4}$$

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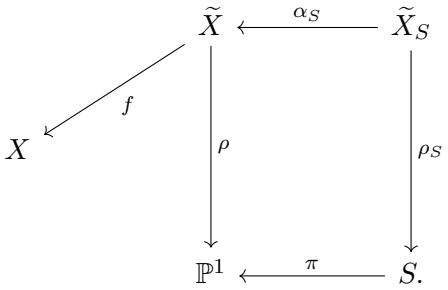
5.8.6.

Let $L/k(\eta)$ be a finite extension of the rational function field of $\mathbb{P}^1$, and put

(5.8.6.1) $\qquad S = \text{the normalization of } \mathbb{P}^1 \text{ in } L.$

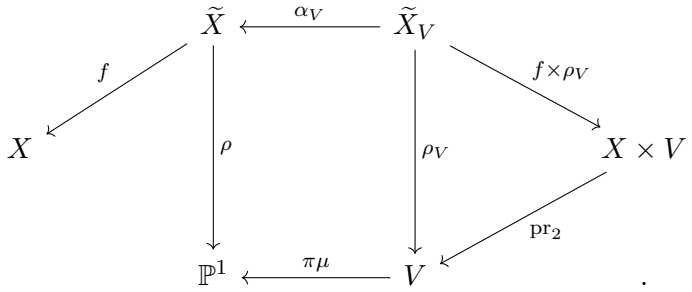
Make the base change $S \to \mathbb{P}^1$ in situation (5.8.3.2):

(5.8.6.1 bis)



For every open immersion $V \overset{\mu}{\hookrightarrow} S$ over which ρ_S is smooth, base change by $V \xrightarrow{\pi\mu} \mathbb{P}^1$ in (5.8.3.2) gives

(5.8.6.2)



.

We still have (cf. (5.8.2.3))

(5.8.6.3) $\qquad \alpha_V^* f^* \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \subset \operatorname{Prim}^n(\widetilde{X}_V/V, \mathbb{Q}_\ell(j)),$

so that we may again speak of the Griffiths homomorphism

(5.8.6.4) $\qquad \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \xrightarrow{\text{Grif}} H^1(V, R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j)).$

Corollary 5.8.7.

With the preceding notation and under the hypotheses of 5.8.4, the Griffiths homomorphism "modulo the fixed part," that is, the composite

(5.8.7.1)

$$\begin{array}{ccc} \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) & \xrightarrow{\qquad\qquad\text{Grif}\qquad\qquad} & H^1(V, R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(V, (\pi\mu)^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq \dfrac{H^1(V, R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j))}{(f\times\rho_V)^* H^1(V, H^{n-1}(X, \mathbb{Q}_\ell(j))_V)} \end{array}$$

is injective.

Proof. By the specialization theorem for cohomology,

$$R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j) = (\pi\mu)^* R^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Thus decomposition (5.8.4.2) rewrites (5.8.7.1) as indicated:

$$\begin{array}{ccc} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) & \xrightarrow{\text{Grif}} & H^1(V,(\pi\mu)^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ & & \downarrow \\ & & H^1(V,(\pi\mu)^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array}$$

It is therefore enough to prove the injectivity of

$$\mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \tag{5.8.7.3}$$

for arbitrarily small open immersions $V \overset{\mu}{\hookrightarrow} S$. We are reduced to the case

$$\begin{cases} U \overset{\nu}{\hookrightarrow} \mathbb{P}^1 & \text{is an open subset over which } \rho \text{ is smooth,} \\ V \xrightarrow{\pi_U} U & \text{is the normalization of } U \text{ in } L. \end{cases}$$

From the cartesian square

$$\begin{array}{ccc} V & \xhookrightarrow{\mu} & S \\ \downarrow{\scriptstyle \pi_U} & & \downarrow{\scriptstyle \pi} \\ U & \xhookrightarrow{\nu} & \mathbb{P}^1 \end{array}$$

we obtain

$$\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Consequently, (5.8.7.3) factors as

$$\begin{array}{ccc} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) & \longrightarrow & H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \\ & & \downarrow{\scriptstyle \pi_U^*} \\ & & H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{array}$$

By (5.8.4), it is enough to verify that

$$\pi_U^* : H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j))$$

is injective. Since $\pi_U : V \to U$ is finite and flat, trace theory [SGA 4 XVII 6.2] supplies, for every $\mathbb{Q}_\ell$-sheaf F on U (for example $F = \nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)$), a map

$$\mathrm{tr}_{\pi_U} : H^1(V,\pi_U^*F) \longrightarrow H^1(U,F)$$

such that

$$\mathrm{tr}_{\pi_U} \circ \pi_U^* = \deg(\pi_U)\ \mathrm{id}\,.$$

It follows that $H^*(U,F) \to H^*(V,\pi_U^*F)$ is injective, completing the proof.

5.9. An example in which hypothesis (A) is not satisfied.

Let X be a smooth quadric surface in $\mathbb{P}^3$. Then X satisfies (LV) (5.2.4). Take a pencil (a Lefschetz pencil, if desired) of hyperplanes whose generic member and axis meet X transversely. Then (as Deligne pointed out to me) the projection

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

does not satisfy (A). Indeed, $R^2\rho_*\mathbb{Q}_\ell$ is not constant: the singular fibers are singular conics, hence reducible, and in a Lefschetz pencil each is the union of two lines meeting at one point (this is the definition of a Lefschetz pencil in this simple case). Thus

$$\dim_{\mathbb{Q}_\ell} H^2(\text{singular fiber}, \mathbb{Q}_\ell) = 2,$$

whereas

$$\dim_{\mathbb{Q}_\ell} H^2(\text{smooth fiber}, \mathbb{Q}_\ell) = 1.$$

Moreover, the conclusion of Theorem 5.8.4 is false here. Let $U \overset{\nu}{\hookrightarrow} \mathbb{P}^1$ be a nonempty open subset over which ρ is smooth. Since the fibers of ρ over U are smooth curves of genus zero, we have

$$\nu^* R^1\rho_*\mathbb{Q}_\ell = 0$$

and hence

$$H^1(U, \nu^* R^1\rho_*\mathbb{Q}_\ell) = 0.$$

Consequently, the Griffiths homomorphism

$$\mathrm{Prim}^2(X, \mathbb{Q}_\ell) \xrightarrow{\mathrm{Grif}} H^1(U, \nu^* R^1\rho_*\mathbb{Q}_\ell)$$

is zero. But on a quadric surface,

$$\dim_{\mathbb{Q}_\ell} \mathrm{Prim}^2(X, \mathbb{Q}_\ell) = 1.$$

Likewise, the conclusion of 5.7 fails here.

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6. Application of the local theory of vanishing cycles (Picard–Lefschetz)

Proposition 6.1.

Let Z be a hypersurface in projective space $\mathbb{P}^r_k$, $r \geq 2$, where k is an algebraically closed field. Let

$$i_o : D_o \hookrightarrow \mathbb{P}^r$$

be the inclusion of a line which meets Z only in its smooth locus and meets it transversely. Choose a geometric point $\bar{a}$ of $D_o - D_o \cap Z$, and denote by

$$\pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

the quotient of $\pi_1(D_o - D_o \cap Z, \bar{a})$ classifying the étale coverings of $D_o - D_o \cap Z$ which are tamely ramified at every point of $D_o \cap Z$. Then:

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6.1.1.

The canonical homomorphism

(6.1.1.1) $$i_{o*} : \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a})$$

is surjective.

6.1.2.

Every choice of a geometric point $\bar{x}_o \in D_o \cap Z$ and of a "path" in D_o from $\bar{x}_o$ to $\bar{a}$ determines a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.2.1}$$

whose image is "the" inertia group at $\bar{x}_o$. Composing this morphism with i_{o*} gives a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}). \tag{6.1.2.2}$$

If Z is irreducible, all these homomorphisms are conjugate to one another.

Proof. Let T be the subscheme of the Grassmannian formed by the lines in $\mathbb{P}^r$ which

(i) pass through $\bar{a}$;

(ii) meet Z only in its smooth locus, and meet it there transversely.

Let η be its generic point, and let

$$i_\eta : D_\eta \longrightarrow \mathbb{P}^r \otimes_k k(\eta)$$

be the corresponding line (thus a line "joining $\bar{a}$ to the generic point of $\mathbb{P}^r_k$"). Choose also a geometric point $\bar{\eta}$ over η, and write $D_{\bar{\eta}}$ for the corresponding "geometric" line. Let $o \in T$ be the point corresponding to the given line D_o.

By [SGA 1, XIII 2.8], there is a specialization homomorphism (depending on a path in T from $\bar{\eta}$ to o, and hence defined up to conjugacy)

$$\pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.3}$$

such that the following assertions hold.

6.1.4.

The local-monodromy homomorphisms (6.1.2.1)

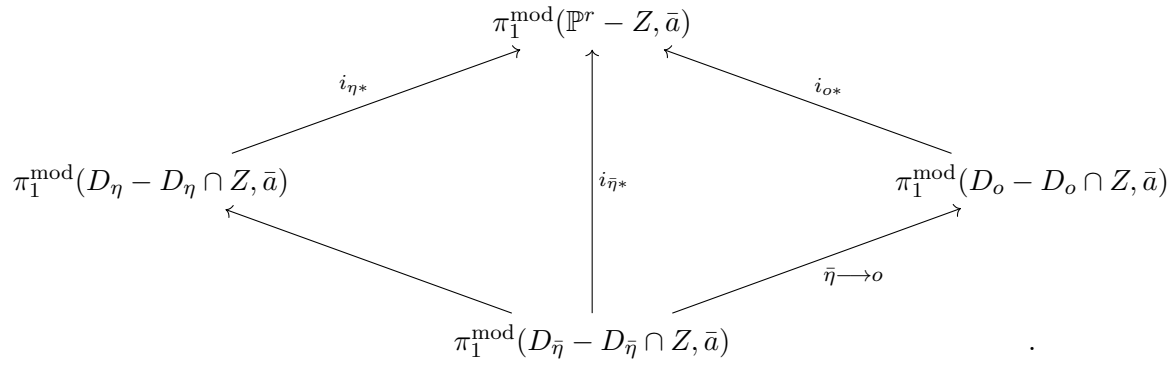
$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

are obtained by composing the specialization arrow (6.1.3) with the local monodromy homomorphisms for $D_{\bar{\eta}}$:

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}). \tag{6.1.4.1}$$

6.1.5.

The following diagram is commutative:



Assume this for the moment, and first prove 6.1.1. By the commutativity of 6.1.5, it is enough to prove that

$$i_{\bar\eta *} : \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar a) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar a)$$

is surjective. For this, it is enough to prove that the analogous homomorphism of the ordinary fundamental groups

$$i_{\bar\eta *} : \pi_1(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar a) \longrightarrow \pi_1(\mathbb{P}^r - Z, \bar a) \tag{6.1.6}$$

is surjective. In other words, for every étale covering $E \to \mathbb{P}^r - Z$ with E connected (and hence irreducible, since $\mathbb{P}^r - Z$ is smooth and therefore unibranch [EGA IV, 18.10.3]), its inverse image

$$E_{\bar\eta} = E \times_{\mathbb{P}^r - Z} (D_{\bar\eta} - D_{\bar\eta} \cap Z)$$

over $D_{\bar\eta} - D_{\bar\eta} \cap Z$ must remain geometrically irreducible. It is enough to use the following variant of Bertini's theorem ([EGA V, citation left blank in the source]) for the composite $E \to \mathbb{P}^r - Z \hookrightarrow \mathbb{P}^r$ and $s = 1$.

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6.1.6.1. Bertini's theorem.

Let k be a field, let $\mathbb{P}^r = \mathbb{P}^r_k$, and let $f : X \to \mathbb{P}^r$ be a nonconstant morphism, where X is a geometrically irreducible algebraic scheme. For every integer $s < \dim \overline{f(X)}$, let L_t denote the generic linear subspace of codimension s in $\mathbb{P}^r$. Then $X_t = f^{-1}(L_t)$ is geometrically irreducible.

We now prove 6.1.2. By 6.1.4 and the commutativity of (6.1.5), it is enough to prove that all the local-monodromy homomorphisms (6.1.3) for $D_{\bar\eta}$,

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar a), \tag{6.1.7.1}$$

become conjugate after composition with the homomorphism

$$\pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar a) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar a). \tag{6.1.7.2}$$

To prove this, observe that for every finite Galois extension K of $k(\eta)$, with

$$k(\eta) \subset K \subset k(\bar\eta),$$

the scheme

$$D_K \overset{\mathrm{def}}{=} D_\eta \times_{k(\eta)} K$$

is an étale covering of D_η. Thus $D_K - D_K \cap Z$ is a tame covering of $D_\eta - D_\eta \cap Z$, and hence $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar a)$ acts on the scheme $D_K \cap Z$. We take K sufficiently large that every point of $D_K \cap Z$ is K-rational; this is possible because, by hypothesis, $D_\eta \cap Z$ is étale over $k(\eta)$.

It remains to prove that $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar a)$ acts transitively on the points of $D_K \cap Z$. This II≈**316**
will prove that the composite local-inertia homomorphisms

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar a)$$

attached to the various points of $D_K \cap Z$ are conjugate. Now the action on $D_K \cap Z$ factors through the canonical quotient

$$\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar a) \longrightarrow \pi_1(\eta, \bar\eta) = \mathrm{Gal}\big(k(\bar\eta)/k(\eta)\big).$$

The action of $\mathrm{Gal}(k(\bar\eta)/k(\eta))$ on $D_K \cap Z$ is the one compatible with the inclusion

$$D_K \cap Z \hookrightarrow \mathbb{P}^r_k \times_k K,$$

where the Galois group acts on $\mathbb{P}^r_k \times_k K$ only through the second factor. It follows that the desired assertion is equivalent to saying that $D_\eta \cap Z$ has only one point, that is, that it is the spectrum of a field. In elementary terms, this is precisely the following Bertini-type fact (take an affine equation $F = 0$ for Z and suppose that $\bar a$ is the origin).

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Fact 6.1.8.

Let k be an algebraically closed field, let $r \geq 2$, and let $F(X_1, \ldots, X_r) \in k[X_1, \ldots, X_r]$ be irreducible. Then

$$G(T) = F(X_1T, \ldots, X_rT) \in k(X_1, \ldots, X_r)[T]$$

is irreducible as a polynomial in T with coefficients in $k(X_1, \ldots, X_r)$.

This completes the proof of 6.1.2.

Remark 6.1.9.

Another method of proving 6.1.2, valid under considerably more general conditions, would be simply to establish a general compatibility for "monodromy homomorphisms": the composite homomorphisms in 6.1.2 are conjugate to the monodromy homomorphism at the generic point of Z, and therefore are conjugate to one another.

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Corollary 6.1.10.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$. Let D be a Lefschetz pencil for X, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the natural representation of $\pi_1(D - D \cap \check{X}, \bar{\eta})$ on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ factors through $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$. If this representation is nontrivial, then $D \cap \check{X} \neq \varnothing$, and the inertia group at every point of $D \cap \check{X}$ acts nontrivially.

Proof.

If $\dim(\check{X}) \leq r - 2$, then $D \cap \check{X}$ is empty and there is nothing to prove, since $\mathbb{P}^1$ is simply connected. Otherwise $\check{X}$ is a hypersurface in $\mathbb{P}^r$ [XVII, 3.1.4], and 6.1.1 applies by 3.2.10 and XVII 3.5, with $Z = \check{X}$ and $D_o = D$. The fact that the action of π_1 on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is tame was established in the local theory by the Picard–Lefschetz formula [XV, 3.4], recalled below. Finally, the nontrivial fact that $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$ is generated by the inertia groups at the various points of $D \cap \check{X}$ [SGA 1, XIII 2.10] gives the required conclusion.

We shall now use the preceding result to study condition (A) of (5.3.5). First recall the following consequence of the Picard–Lefschetz formula [XV, 3.4].

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Proposition 6.2.

Let X be a smooth connected projective variety of dimension n over the algebraically closed field k. Let D be a Lefschetz pencil, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the pencil D satisfies condition (A) if and only if the inertia group at every point of $D \cap \check{X}$ acts nontrivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

Theorem 6.3.

Under the hypotheses of 6.2, the following assertions hold.

(6.3.1) If $n = \dim X$ is odd, condition (A) is satisfied by every Lefschetz pencil for X.

(6.3.2) Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$. Then the following conditions are equivalent:

(i) D does not satisfy condition (A) of (5.3.5);

(ii) $\pi_1(D - D \cap \check{X})$ acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$, and $D \cap \check{X} \neq \varnothing$;

(iii) there is a point of $D \cap \check{X}$ whose inertia group acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$;

(iv) the sheaf $R^{n-1}\rho_*\mathbb{Q}_\ell$ is constant on D, and $D \cap \check{X}$ is nonempty;

(v) the restriction $R^{n-1}\rho_*\mathbb{Q}_\ell|_U$ is constant on U, where U is a nonempty open subset of D over which ρ is smooth, and ρ is not smooth.

(6.3.3) If, in addition, X satisfies condition (LV) of (5.2.2), conditions (i)–(v) are also equivalent to: II≈**320**

(vi) the restriction map

$$H^{n-1}(X, \mathbb{Q}_\ell) \longrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$

is surjective (and hence is an isomorphism by the weak Lefschetz theorem).

Proof.

Assertion 6.3.1 follows from XV 3.4. For the equivalences in 6.3.2, (i) is equivalent to (iii) by 6.2, (ii) to (iii) by 6.1.10, (ii) to (v) tautologically, and (iv) to (v) by 5.3.4. Finally, if X satisfies condition (LV), 5.6.10 gives the equivalence of (ii) and (vi).

Remark 6.3.4.

Condition (A) does not depend on the choice of the Lefschetz pencil D, but only on the chosen (Lefschetz) immersion $i : X \longrightarrow \mathbb{P}^r$. This is clear from (6.3.1) when $\dim X$ is odd. When $\dim X$ is even, condition (A) also says that the local system on $\mathbb{P}^r - \check{X}$ formed by the groups $H^{n-1}(X \cap H_s, \mathbb{Q}_\ell)$ is constant. This follows from criterion (ii), from the fact that the local system in question is tamely ramified at the generic point of $\check{X}$, and from the surjectivity in (6.1.1).

Corollary 6.4.

Under the hypotheses of 6.2, suppose that $n = \dim X$ is even or $\mathrm{char}(k) \neq 2$, and that X satisfies condition (LV). Then there is an integer d_o such that, for $d \geq d_o$, every Lefschetz pencil of sections of X by hypersurfaces of degree d satisfies condition (A).

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Proof.

By the equivalence (i)$\Longleftrightarrow$(vi) of (6.3.2), it is enough to prove that, if $Y(d)$ denotes a smooth hypersurface section of X of degree d, then

$$\dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \longrightarrow \infty \qquad \text{as } d \longrightarrow \infty. \tag{6.4.1}$$

Indeed, we have the following precise result.

Lemma 6.4.2.

Let X be a smooth connected projective variety of dimension $n \geq 1$ over an algebraically closed field k, and let $Y(d)$ denote a hypersurface section of degree d. Then there is a polynomial $f(T) \in \mathbb{Z}[T]$ such that

$$f(T) = \deg(X)T^n + \text{a polynomial of degree at most } n-1, \tag{6.4.2.1}$$

$$f(d) = \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \qquad (d \geq 1). \tag{6.4.2.2}$$

Proof.

By XVII 5.7.5, if $C(X)$ denotes the total Chern class of X and H the first Chern class of a hyperplane, the Euler–Poincaré characteristic of $Y(d)$ is

$$\chi(Y(d)) = \int_X \frac{dH\, C(X)}{1+dH} = \sum_{i=1}^{n} (-1)^{i-1} d^i \int_X H^i C_{n-i}. \tag{6.4.2.3}$$

On the other hand, Poincaré duality and weak Lefschetz give

$$\chi(Y(d)) = (-1)^{n-1} \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) + 2 \sum_{i\geq 1} (-1)^{n-1-i} \dim H^{n-1-i}(X, \mathbb{Q}_\ell). \tag{6.4.2.4}$$

Combining (6.4.2.3) and (6.4.2.4) gives the result.

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6.5. A reminder on the Picard–Lefschetz formula (XV 3.4)

Let S be the spectrum of a strictly henselian valuation ring, with generic point η, geometric generic point $\bar{\eta}$, and closed point s. Let

$$\rho : X \longrightarrow S$$

be a proper morphism, where X is a connected regular scheme of dimension n. Suppose that ρ is smooth away from a single point x in the special fiber and that x is an ordinary quadratic singularity of that fiber.

Let I denote the inertia group of S at s. By XV 3.4, I acts trivially on $H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ for $i \neq n-1$, and its action on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is described as follows.

6.5.1.

If $n = 2m+1$, then I acts through a quotient isomorphic to $\mathbb{Z}/2\mathbb{Z}$ (if $\mathrm{char}(k) \neq 2$, there is only one such quotient). There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

determined uniquely up to sign and called the vanishing cycle at s, such that

$$\delta \cdot \delta = (-1)^m 2 \quad \text{in} \quad H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(n-1)) \simeq \mathbb{Q}_\ell. \tag{6.5.1.1}$$

The nontrivial element σ of $\mathbb{Z}/2\mathbb{Z}$ acts on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the reflection

$$\sigma x = x - (-1)^m (x \cdot \delta)\delta. \tag{6.5.1.2}$$

Here $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m)$, so $(x \cdot \delta)\delta$ defines an element of $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

6.5.2.

If $n = 2m+2$, then I acts through its canonical quotient $\mathbb{Z}_\ell(1)$. There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

canonically determined up to sign and called the vanishing cycle at s, such that $\sigma \in \mathbb{Z}_\ell(1)$ acts on $x \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the transvection

$$\sigma(x) = x - (-1)^m \sigma\, (x \cdot \delta)\delta. \tag{6.5.2.1}$$

Indeed, $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m-1)$. Multiplication by $\sigma \in \mathbb{Z}_\ell(1)$ therefore gives an element $\sigma(x \cdot \delta) \in \mathbb{Q}_\ell(-m)$, whose product with $\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$ lies in $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

We return to the global situation. II ≈**323**

Theorem 6.6.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that n is even or that $\mathrm{char}(k) \neq 2$. Let D be a Lefschetz pencil for X, let η be its generic point, let $\bar{\eta}$ be a geometric point over η, and let $s_1, \ldots, s_N$ be the points of $D \cap \check{X}$.

For each i, choose an isomorphism ℓ_i between $\bar{\eta}$ and the geometric generic point of S_i, the strict henselization of D at s_i. Via ℓ_i, let I_i be the inertia group at s_i acting on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(j))$, and let δ_i be the vanishing cycle at s_i defined in 6.5. Thus II ≈**324**

$$\delta_i \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)), \qquad m = \begin{cases} (n-1)/2, & n \text{ odd}, \\ n/2 - 1, & n \text{ even}. \end{cases}$$

Choose the ℓ_i so that the subgroups

$$I_i \subset \pi = \pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$$

topologically generate π, as is possible by SGA 1, XIII 2.10. Then:

6.6.1. If X satisfies condition (LV), the vanishing cohomology space

$$E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \overset{\mathrm{def}}{=} H^{n-1}(X, \mathbb{Q}_\ell(m))^\perp \subset H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$$

(cf. 5.4.4) is generated by the vanishing cycles δ_i, $i = 1, \ldots, N$.

6.6.2. The vanishing cycles δ_i are conjugate, up to sign, under the global monodromy group $\pi_1(D - D \cap \check{X})$.

Proof.

For 6.6.1, the Picard–Lefschetz formula 6.5 and the fact that the I_i generate $\pi = \pi_1(D - D \cap \check{X})$ give

$$H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))^\pi = \langle \delta_1, \ldots, \delta_N \rangle^\perp \quad \text{in } H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)). \tag{6.6.3}$$

By 5.6.10,

$$H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))^\pi = \mathrm{Im}\big(H^{n-1}(X, \mathbb{Q}_\ell(m)) \longrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))\big). \tag{6.6.4}$$

Taking orthogonals in (6.6.3) proves 6.6.1.

For 6.6.2, note simply that the vanishing cycle δ_i is determined up to sign by the representation of the inertia group at s_i. Apply 6.1.2, which gives conjugacy in $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta})$ of the composite inertia homomorphisms

$$\begin{array}{ccc} \mathbb{Z}_\ell(1) & \longrightarrow & \pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta}) \\ & & \downarrow \\ & & \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta}) \end{array} .$$

Consequently, the vanishing cycles determined by these homomorphisms are conjugate up to sign by elements of $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta})$. The proof is completed by 6.1.1, which gives the surjection

$$\pi_1^{\mathrm{mod}}(D - D \cap \check{X}) \longrightarrow \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}).$$

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Corollary 6.7.

Under the hypotheses of 6.6, suppose that X satisfies condition (LV). Then:

6.7.1. The representation of $\pi_1(D - D \cap \check{X})$ on $E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is absolutely irreducible.

6.7.2. If n is even, every finite-index subgroup of $\pi_1(D - D \cap \check{X})$ acts absolutely irreducibly on $E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

Proof.

We may replace the module in question by its twist

$$E = E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)).$$

Let $\bar{E} = E \otimes_{\mathbb{Q}_\ell} \bar{\mathbb{Q}}_\ell$, and choose $0 \neq \chi \in \bar{E}$. To prove 6.7.1, we must show that the $\bar{\mathbb{Q}}_\ell$-linear subspace of $\bar{E}$ generated by χ and its conjugates under π is all of $\bar{E}$. By 6.6.1 and 6.6.2, it is enough to prove that this subspace contains one of the vanishing cycles δ_i.

Since X satisfies (LV), the restriction of the cup-product form to

$$H^{n-1}(X, \mathbb{Q}_\ell(m)) \longhookrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$$

is nondegenerate; hence so is its restriction to the orthogonal $E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$. By 6.6.1, there is therefore at least one i such that

$$\chi \cdot \delta_i \neq 0.$$

For this i, 6.5 supplies an element σ_i of the inertia group at s_i such that

$$\sigma_i(\chi) - \chi = \text{a nonzero multiple of } \delta_i. \tag{6.7.3}$$

This proves 6.7.1.

For 6.7.2, $n-1$ is odd, so $\delta_i \cdot \delta_i = 0$. Iterating 6.5.2 gives, for every k,

$$\sigma_i^k(\chi) = \chi + k\,\sigma_i(\chi \cdot \delta_i)\delta_i. \tag{6.7.4}$$

Thus a $\mathbb{Q}_\ell$-subspace M of E is stable under the action of π if and only if it is stable under a finite-index subgroup: both conditions are equivalent to

$$\chi \in M \quad \Longrightarrow \quad \delta_i \in M \quad \text{for every } i \text{ with } \chi \cdot \delta_i \neq 0.$$

The assertion therefore reduces to 6.7.1.

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SGA 7 — EXPOSÉ XIX

The Noether Theorem

by P. Deligne

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1. Statement of the theorems

The present exposé updates Lefschetz's proof [3, p. 108] of the Noether theorem (1.2) stated below.

1.1.

Let k be an algebraically closed field of characteristic p. Let $d \geq 1$ and let

$$\underline{a} = (a_i)_{1 \leq i \leq d}$$

be a sequence of integers at least 1. A complete intersection V of dimension n and multidegree $\underline{a}$ in the projective space $\mathbb{P}^{n+d}$ over k will be called generic if, after an extension of k, it is projectively isomorphic to a complete intersection defined by equations whose coefficients are algebraically independent over the prime field. In what follows, unless expressly stated otherwise, we suppose that $d \geq 1$ and $a_i \geq 2$.

Theorem 1.2.

Let $V \subset \mathbb{P}^{2+d}$ be a generic complete-intersection surface of multidegree $\underline{a}$. Suppose that

$$\underline{a} \neq (2), (3), (2,2).$$

Then every (reduced) curve on V is the scheme-theoretic intersection of V with a hypersurface of $\mathbb{P}^{2+d}$.

Let us show that the theorem follows from its corollary

$$\mathrm{Pic}(\mathbb{P}^{2+d}) = \mathbb{Z} \xrightarrow{\sim} \mathrm{Pic}(V). \tag{1.2.1}$$

If C is a curve on V, (1.2.1) implies that $\mathcal{O}(C) \simeq \mathcal{O}(n)$ for a suitable n. If H is a hyperplane section, then

$$n = \left(\textstyle\prod_i a_i\right)^{-1}(C, H) > 0.$$

It remains to prove that the linear system induced on V by $|nH|$ is complete. Write V as the intersection of hypersurfaces H_i of degree a_i, and put $V_i = H_1 \cap \cdots \cap H_i$. Induction on i shows that the linear system induced by $|nH|$ on V_i is complete: in the exact sequence

$$H^0(V_i, \mathcal{O}(n)) \longrightarrow H^0(V_{i+1}, \mathcal{O}(n)) \longrightarrow H^1(V_i, \mathcal{O}(n - a_{i+1})),$$

the last group vanishes (FAC).

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Theorem 1.3.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of nonzero even dimension $2n$ and multidegree $\underline{a}$. Suppose that one of the following conditions holds:

(a) k is not of characteristic 2, and $(2n;\underline{a})$ is distinct from $(2n;2)$, $(2n;2,2)$, and $(2;3)$;

(b) the Hodge number $h^{2n,0}$ is nonzero, i.e. $\sum_i (a_i - 1) > 2n$.

Then every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of the class η^n of a linear section.

Let us deduce (1.2.1) from 1.3. The exceptional cases in 1.2 are exactly those for which $\sum_i (a_i - 1) \leq 2$; hence 1.3(b) applies to the surface V of 1.2. We know that $\mathrm{Pic}^\tau(V) = 0$ (XI 1.8), so $\mathrm{Pic}(V)$ injects into $H^2(V, \mathbb{Q}_\ell(1))$. By 1.3 every element of $\mathrm{Pic}(V)$ is a rational multiple of the class η of $\mathcal{O}(1)$, and (XI 1.8) gives the conclusion.

It is clear that the validity of 1.2 and 1.3 does not depend on the chosen generic complete intersection, but only on p, n, and $\underline{a}$.

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Theorem 1.4.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of dimension $2n+1$ over k. Let $\ell \subset \check{\mathbb{P}}$ be a Lefschetz pencil of hyperplane sections of X. If k has characteristic 2, suppose in addition that ℓ is a generic Lefschetz pencil.

Suppose that X satisfies condition (LV) (XVIII 5.2.2), and let V be the generic hyperplane section in the pencil ℓ. One of the following two conditions holds:

(a) no nonzero cohomology class in the vanishing part $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ of $H^{2n}(V, \mathbb{Q}_\ell(n))$ is algebraic;

(b) the algebraic cohomology classes generate $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$.

Let us make precise the meaning of "generic" in 1.4.

(a) Let G be the Grassmannian of lines in the projective space $\check{\mathbb{P}}$ dual to $\mathbb{P}$. By a "geometric point" of G we mean a morphism from the spectrum of an algebraically closed field to G. Every geometric point $\bar{s} \to G$ defines a pencil ℓ of hyperplane sections of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call ℓ generic if $\bar{s}$ lies over the generic point of G.

(b) Let ℓ be a pencil of hyperplane sections (a line in $\check{\mathbb{P}}$). Every geometric point $\bar{s} \to \ell$ defines a hyperplane section of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call this section generic if $\bar{s}$ lies over the generic point of ℓ.

Let $\bar{s}$ be as above, with image the generic point s of ℓ, and let $S \subset \ell$ be the finite set of points giving singular hyperplane sections. The monodromy group $\pi_1(\ell - S, \bar{s})$, a quotient of $\mathrm{Gal}(k(\bar{s})/k(s))$, acts on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$. If k is not of characteristic 2, this action is irreducible (XVIII 6.7); in Section 2 below we shall show that the assertion remains true without a characteristic hypothesis for a generic pencil. The subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by algebraic cohomology classes is stable under monodromy (that is, under Galois). It is therefore either zero or the whole space, which proves 1.4.

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Corollary 1.5.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of even dimension $2n$ and multidegree $\underline{a}$. Then either

(a) every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of η^n; or

(b) $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic cohomology classes.

If (b) holds, then for every complete intersection of dimension $2n$ and multidegree $\underline{a}$ over an algebraically closed field of characteristic p, the group $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic classes.

Apply 1.4 with X a generic complete intersection of dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$, and with ℓ a generic pencil of sections of X by hypersurfaces of degree a_d (cf. XVII 2.4). One could equally take X of multidegree $(a_1, \ldots, a_d)$ and ℓ a generic pencil of hyperplane sections. The final assertion follows because an algebraic cohomology class remains algebraic under specialization.

2. A complement to XVIII

Proposition 2.1.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of odd dimension $2n+1$ over an algebraically closed field k, and let $\ell \subset \check{\mathbb{P}}$ be a generic Lefschetz pencil of hyperplane sections of X. Let s be a geometric point of ℓ, let H_s be the corresponding hyperplane section, let $S \subset \ell$ be the set of exceptional values of the pencil, and let $\pi_1(\ell - S, s)$ be the monodromy group. Suppose that X satisfies condition (LV). Then the representation of $\pi_1(\ell - S, s)$ on the vanishing cohomology

$$\mathrm{Ev}(H_s, \mathbb{Q}_\ell) \subset H^{2n}(H_s, \mathbb{Q}_\ell)$$

is absolutely irreducible.

Recall that $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$, and that this image equals the subgroup of monodromy invariants (XVIII 5.6.10).

Proof.

The projective line $\mathbb{P}^1$ is simply connected. Hence $\pi_1(\ell - S, s)$ is topologically generated by the inertia subgroups at the points of S and their conjugates. Its image in $\mathrm{GL}(H^{2n}(H_s, \mathbb{Q}_\ell))$ is therefore topologically generated by the transformations

$$x \longmapsto x - (-1)^n (x \cdot \delta)\delta,$$

where δ is conjugate to a vanishing cycle. Consequently, the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the subspace generated by the vanishing cycles and their conjugates; thus $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is generated by those cycles and their conjugates.

For a generic Lefschetz pencil, the arguments of XVIII §6 show that the vanishing cycles attached to the different points of S are conjugate to one another. II≈**332**

Let $W \subset \mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ be a nonzero invariant subspace. There exist a conjugate δ of a vanishing cycle and an element $v \in W$ such that $(v, \delta) \neq 0$; this is where condition (LV) is used. Since

$$v - (-1)^n (v \cdot \delta)\delta \in W,$$

we have $\delta \in W$. The subspace W, containing one vanishing cycle and all its conjugates, is therefore $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$. The argument remains valid after extending the coefficient field $\mathbb{Q}_\ell$, proving absolute irreducibility.

Remark 2.2.

If condition (LV) is omitted, the preceding arguments still show that every invariant subspace of $H^{2n}(H_s, \mathbb{Q}_\ell)$ either contains $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ or is contained in the image of $H^{2n}(X, \mathbb{Q}_\ell)$. The same assertion holds for X of even dimension (and for every Lefschetz pencil).

3. Proof of 1.3

3.1.

Under the hypotheses of the theorem, it remains to prove that 1.5(b) does not hold. The case $h^{0,2n} \neq 0$ will be treated by Katz in XXI 4.1 (the condition $h^{0,2n} \neq 0$ is equivalent to

$$\sum_i a_i - (2n + r + 1) \geq 0,$$

this number being the integer m such that $\Omega_V^{2n} \simeq \mathcal{O}(m)$).

The results of the present exposé will not be used again in the remainder of the seminar, so there is no vicious circle.

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3.2.

When $k = \mathbb{C}$, one proves 1.3 by observing that if 1.5(b) holds, the cohomology of V is purely of type (n, n), since algebraic cohomology is of type (n, n). This occurs only in the exceptional cases excluded in 1.3 (XI 2.9). By the Lefschetz principle, 1.3 is therefore true in characteristic zero.

Proposition 3.3.

Assume the hypotheses of 1.4, and suppose that (b) holds. Let $\mathrm{Ev}(V, \mathbb{Q})$ be the $\mathbb{Q}$-vector subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by the algebraic cohomology classes.

(a) There is an isomorphism

$$\mathrm{Ev}(V, \mathbb{Q}) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \mathrm{Ev}(V, \mathbb{Q}_\ell(n)).$$

On $\mathrm{Ev}(V, \mathbb{Q})$ the intersection form is rational and definite, of sign $(-1)^n$.

(b) The vanishing cycles and their conjugates belong to $\mathrm{Ev}(V, \mathbb{Q})$; there they form an irreducible root system of type A, D, or E.

(c) The monodromy group (the image of $\pi_1(\ell - S, \bar{s})$ in $\mathrm{GL}(\mathrm{Ev}(V, \mathbb{Q}))$) is finite; it is the Weyl group of the root system in (b).

Apart from definiteness of the intersection form, (a) follows because the intersection form is nondegenerate on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ (by (LV)) and rational on $\mathrm{Ev}(V, \mathbb{Q})$.

Condition 1.4(b) says equivalently that there are algebraic cycles on V, modulo numerical equivalence, which are not invariant under monodromy; this condition is therefore independent of ℓ. Let x be such an algebraic cycle, and let δ be the vanishing cycle associated with $s_0 \in S$ and with a path c from $\bar{s}$ (which defines V) to s_0. For a suitable c, $(x, \delta) \neq 0$. The Galois transform

$$\sigma(x) = x - (-1)^n (x, \delta)\delta$$

is again algebraic; hence $(x, \delta)\delta$ is algebraic, the image of a cycle y independent of ℓ. The rational number

$$\frac{(-1)^n}{2}(y, y) = (x, \delta)^2$$

is consequently independent of ℓ. It is a square in every $\mathbb{Q}_\ell$, hence a square in $\mathbb{Q}$; thus, up to sign, (x, δ) belongs to $\mathbb{Q}$ and is independent of ℓ. This shows not only that $\delta \in \mathrm{Ev}(V, \mathbb{Q})$, but also that the algebraic cycle modulo numerical equivalence

$$\pm\delta = \frac{\sigma(x) - x}{(x, \delta)}$$

is independent of ℓ.

An algebraic cycle on a variety V is always defined over a finite extension of any field of definition of V. It follows that $\pi_1(\ell - S, s)$ acts on $\mathrm{Ev}(V, \mathbb{Q})$ through a finite group. This group is generated by the reflections associated with the vanishing cycles δ and their conjugates (XVIII 6.6.1 and Section 2). The cycles δ and their conjugates therefore form a root system whose Weyl group is the monodromy group. Moreover, all roots of this root system are conjugate under the Weyl group; it is of type A, D, or E. II ≈334

Up to a scalar, the intersection form is the unique invariant form under the Weyl group. It is therefore definite. Its sign is determined by computing

$$(\delta, \delta) = 2(-1)^n.$$

Proposition 3.4.

If $k = \mathbb{C}$, then under the hypotheses of 1.4 the following conditions are equivalent:

(i) the monodromy group is finite;

(ii) the vanishing cohomology $\mathrm{Ev}(V)$ is purely of type (n, n).

Let $\widetilde{X} \to \ell$ be the family whose fibers are the hyperplane sections in the pencil ℓ, and let V_s be the fiber at $s \in \ell(\mathbb{C})$. The $\mathrm{Ev}(V_s)$ form an algebraic variation of Hodge structures on ℓ. By the fixed-part theorem [2, 4.1.2], applied to a finite covering of ℓ (cf. [2, 4.1.3.3]), (i) implies that the Hodge structure on $\mathrm{Ev}(V_s)$ is locally constant. The monodromy group therefore acts by automorphisms of the Hodge structure. For a Picard–Lefschetz transformation T, $\mathrm{Im}(T - 1)$ is a rank-one Hodge substructure, hence is of type (n, n) and is generated by a vanishing cycle δ. Since the vanishing cycles are of type (n, n), all of $\mathrm{Ev}(V_s)$ is of that type.

Conversely, if $\mathrm{Ev}(V)$ is purely of type (n, n), the intersection form on $\mathrm{Ev}(V)$ is definite and the monodromy group is finite (cf. [2, 4.1.3.3]).

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Remark 3.5.

If the equivalent conditions of 3.4 hold, the argument of 3.3 shows that the vanishing cycles and their conjugates form a root system of type A, D, or E, whose Weyl group is the monodromy group.

3.6.

Let us prove 1.3 when k has finite characteristic different from 2. Let $S = \mathrm{Spec}(\mathcal{V})$ be the spectrum of a discrete valuation ring of mixed characteristic with residue field k, and let $X \subset \mathbb{P}_S^{2n+d}$ be a complete intersection of relative dimension $2n + 1$ and multidegree $(a_1, \ldots, a_{d-1})$ over S. Let ℓ be a Lefschetz pencil of sections of X by hypersurfaces of degree a_d (in the special fiber, ℓ_s is a Lefschetz pencil for X_s). Let

$$f : \widetilde{X} \longrightarrow \ell \simeq \mathbb{P}_S^1$$

be the family whose fibers are the sections in the pencil. The exceptional locus $T \subset \ell$ is ètale over S (XVII 6; here one uses $p \neq 2$). The sheaf $R^{2n} f_* \mathbb{Q}_\ell$ is locally constant on $\ell - T$; it is therefore tamely ramified, and its monodromy is the same in characteristic p as in characteristic zero. If 1.5(b) holds, it follows from 3.3(c) and 3.4 that the cohomology group $H^{2n}(V)$ of complex complete intersections of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n, n). This occurs only in the cases excluded by 1.3 (XI 2.9).

4. A second complement to XVIII

In this section we work over $\mathbb{C}$ and with integral, classical cohomology. Using methods closer to Lefschetz's, we revisit some of the results of XVIII. There they were proved only after tensoring with $\mathbb{Q}$ (or rather with $\mathbb{Q}_\ell$, which amounts to the same thing).

4.1.

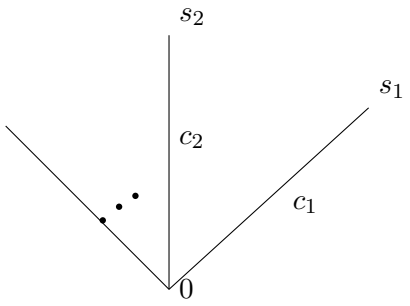
Let $X \subset \mathbb{P}$ be a smooth complex projective variety, pure of dimension $n+1$, and let ℓ be a Lefschetz pencil of hyperplane sections of X, with axis A. We also write ℓ for the parameter space of the pencil, which is isomorphic to $\mathbb{P}^1$. Let $S \subset \ell$ be the set of exceptional values. For $s \in \ell$, write H_s for the corresponding hyperplane section of X (of dimension n).

Let $T(A)$ be a closed tubular neighborhood of $A \cap X$ in X, whose boundary is transverse to every H_s. II ≈**336**

Choose two points $0, \infty \in \ell - S$, and let $\alpha : \ell \to \mathbb{P}^1(\mathbb{C})$ be an orientation-preserving diffeomorphism that sends 0 to 0, ∞ to ∞, and S to the circle of radius one centered at 0.

The composite αf of α with the projection to ℓ makes $X - T(A)^\circ$ a singular fibration in manifolds with boundary over $\mathbb{P}^1(\mathbb{C})$ (one singularity above each point of $\alpha(S)$). Write c_s for the straight segment joining 0 to $\alpha(s)$.

Suppressing α from the notation, the picture is



Near $s \in S$ and the singular point x_s of $H_s \subset X$, suitable local coordinates put us in the situation studied in XIV 3.2. If α is chosen holomorphic near s, one can even choose local coordinates on X and ℓ in which f is written $\sum z_i^2$ and $\alpha^{-1}(c_s)$ is given by "the z_i are real and positive." Above the intersection of a neighborhood of s with $\alpha^{-1}(c_s) - \{s\}$, consider the bundle of real spheres $\Delta'_s \subset X$ defined by "the z_i are real." As $t \to s$, the fiber Δ'_{st} of Δ'_s at t tends to x_s.

Let Δ_s denote the union of $\{x_s\}$ and a sphere bundle over $\alpha^{-1}(c_s) - \{s\}$, contained in $X - T(A)$ and extending Δ'_s. Then Δ_s is an embedded disk in X whose boundary $H_0 \cap \Delta_s$ is a sphere representing the vanishing cycle δ_s at s, transported to H_0 along the path $\alpha^{-1}(c_s)$.

Apply Morse theory to the function $|\alpha f|^2$ to study how the homotopy type of

$$\{x \in X \mid x \in T(A) \text{ or } |\alpha f(x)|^2 \leq r\}$$

changes as r increases. Put

$$X^- = \{x \in X \mid x \in T(A) \text{ or } \alpha f(x) \neq \infty\}.$$

One finds that the inclusions

$$H_0 \cup \bigcup_{s \in S} \Delta_s \hookrightarrow T(A) \cup H_0 \cup \bigcup_{s \in S} \Delta_s \hookrightarrow X^-$$

are homotopy equivalences. Morse theory also shows that $X - X^-$ can be deformation-retracted onto a subcomplex of real dimension at most n. Altogether this gives the following result.

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Theorem 4.2.

There exists a compact subcomplex Σ of real dimension at most n in $H_\infty^{\text{aff}} = H_\infty - A$ such that the inclusion

$$H_0 \cup \bigcup_{s \in S} \Delta_s \longrightarrow X - \Sigma$$

is a homotopy equivalence.

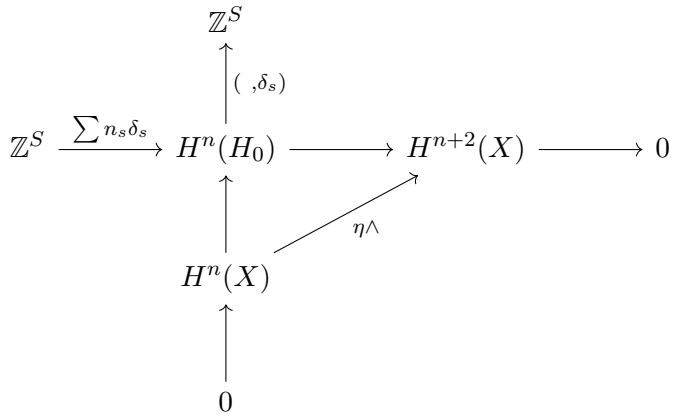
4.3.

Removing Σ has no effect on the cohomology, homology, or homotopy groups of X in degrees $i \leq n$. Since $H_0 \cup \bigcup_{s\in S} \Delta_s$ is obtained from H_0 by attaching $(n+1)$-disks along spheres representing the vanishing cycles, homology gives

$$\mathbb{Z}^S \longrightarrow H_n(H_0) \longrightarrow H_n(X) \longrightarrow 0.$$

The analogous exact sequence in cohomology, together with the sequence deduced from it by Poincaré duality, forms the commutative diagram

(4.3.1)



5. The exceptional cases of the Noether theorem

In this section we continue to work over $\mathbb{C}$, with integral cohomology.

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5.1.

Let $2n$, d, and $\underline{a} = (a_1, \ldots, a_d)$, with $d \geq 1$ and $a_i \geq 2$, be such that the middle cohomology of smooth complete intersections V of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n, n). In other words, either $n = 0$, or $(2n; \underline{a})$ is one of $(2n; 2)$, $(2n; 2, 2)$, and $(2; 3)$ (XI 2.9).

Suppose that V is obtained from a complete intersection $X \subset \mathbb{P}^{2n+d}$ (respectively, $X \subset \mathbb{P}^{2n+d+1}$) of dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$ (respectively, $(a_1, \ldots, a_d)$) by taking a section by a hypersurface of degree a_d (respectively, a hyperplane section) belonging to a Lefschetz pencil ℓ.

By (4.3.1), the vanishing part $\mathrm{Ev}(V)$ of $H^{2n}(V)$, which equals its primitive part, is generated over $\mathbb{Z}$ by the vanishing cycles. It is the orthogonal complement of the power η^n of the cohomology class η of a hyperplane section.

Since the intersection form on $H^{2n}(V)$ is unimodular and η^n is not by, with $b > 1$, for another cohomology class y (XI 1.6(iv)), the discriminant of the intersection form on $\mathrm{Ev}(V)$ is

$$(\eta^n, \eta^n) = \prod_i a_i.$$

Let R be the root system in $H^{2n}(V)$ formed by the vanishing cycles and their conjugates. We have the following information about R:

(a) R is empty or is of type A, D, or E;

(b) the rank of R is given in XI 1.10;

(c) its discriminant is $\prod_i a_i$. In the tables, this discriminant is the order of the center of the corresponding simply connected complex simple group. The argument by which we computed the discriminant shows that this center, dual to the quotient of the weight lattice by the root lattice, is cyclic.

In every case this information determines R completely if $n \neq 0$. One finds: II≈**339**

Proposition 5.2.

(i) If V is an even-dimensional quadric, R is of type A_1.

(ii) If V is a cubic surface, R is of type E_6.

(iii) If V has dimension $2n$ and is the intersection of two quadrics, R is of type D_{2n+3}.

5.3.

Suppose that $n \neq 0$. The set R^1 of vectors of length ± 2 in $\mathrm{Ev}(V)$ is also a root system and has the same properties (a), (b), and (c) as R. Hence $R = R^1$. Let π_1 be the fundamental group of the parameter space of all complete intersections $V \subset \mathbb{P}^{2n+d}$ of the type under consideration. The image of π_1 in $\mathrm{GL}(\mathrm{Ev}(V))$ is the Weyl group of R. In every case it is the automorphism group of the triple

$$\big(H^{2n}(V), \eta^n, \text{intersection form}\big).$$

5.4.

With R replaced by R^1 from 5.3, assertion 5.2(ii) is well known [1], and for small values of n, 5.2(iii) was known, at least to Manin ($n = 0$ is trivial; a surface that is the intersection of two quadrics is a del Pezzo surface). A detailed study of the intersection of two quadrics was made by M. Reid [4]. He deduces 5.2(iii) from a study of the geometry of the n-dimensional linear subspaces on an intersection of two quadrics $V \subset \mathbb{P}^{2n+2}$.

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SGA 7 — EXPOSÉ XX

The Griffiths Theorem

by N. Katz

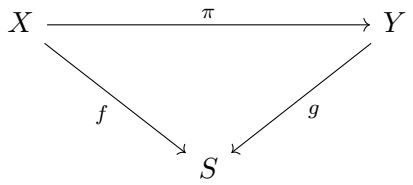
0. Introduction

We prove Griffiths's theorem [1], which gives a systematic (though nonconstructive) way of "finding" algebraic cycles that are $\mathbb{Q}_\ell$-cohomologous to zero, but no multiple of which is algebraically equivalent to zero. The proof given here (due to Grothendieck) is the translation into purely algebraic terms of Griffiths's original, more or less transcendental, proof. In both proofs, the irreducibility theorem XVIII 6.7 is a key point (cf. 3.1.6 and 3.2).

1. The formalism of primitive classes

Throughout what follows, a prime number ℓ is fixed. All schemes under consideration are assumed to have characteristic different from ℓ. Consider the situation

(1.0.1)



in which we assume

(1.0.3) $\qquad f,\ g \quad$ smooth of relative dimensions $\quad d_1$ and d_2,

and

(1.0.4) $\qquad \pi \quad$ proper.

Since X and Y are smooth over S, the properness of π gives a Gysin morphism (SGA 5, Exposé IV)

$$\pi_* : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q-2(d_1-d_2)}(Y, \mathbb{Q}_\ell(i - d_1 + d_2)). \tag{1.0.5}$$

Localizing on S in the ètale topology, we likewise obtain a morphism of sheaves on S

$$\pi_* : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2), \tag{1.0.6}$$

and, on applying the functor $H^p(S, -)$, a morphism

$$\pi_* : H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2)). \tag{1.0.7}$$

Definition 1.0.8.

Let $(E_r^{p,q}, d_r^{p,q})$ and $('E_r^{p,q}, 'd_r^{p,q})$ be two spectral sequences in a given abelian category, and let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$. The shift of $('E_r^{p,q}, 'd_r^{p,q})$ by (a, b) is the spectral sequence $(''E_r^{p,q}, ''d_r^{p,q})$ given by

$$\begin{cases} ''E_r^{p,q} = 'E_r^{p+a,q+b}, \\ ''d_r^{p,q} = 'd_r^{p+a,q+b}. \end{cases} \tag{1.0.9}$$

1.0.10.

A morphism of spectral sequences from $(E_r^{p,q}, d_r^{p,q})$ to $({}''E_r^{p,q}, {}''d_r^{p,q})$ is called a morphism of bidegree (a,b) from $(E_r^{p,q}, d_r^{p,q})$ to $({}'E_r^{p,q}, {}'d_r^{p,q})$.

The following lemma is an immediate consequence of the theory of the Gysin morphism (SGA 5, Exposé IV).

Lemma 1.1.

There is a canonical morphism π_* of bidegree $(0, -2(d_1 - d_2))$ between the Leray spectral sequences

$$R^p\alpha_* R^q f_* \mathbb{Q}_\ell(i) \Longrightarrow R^{p+q}(\alpha f)_* \mathbb{Q}_\ell(i) \tag{1.1.0}$$

and

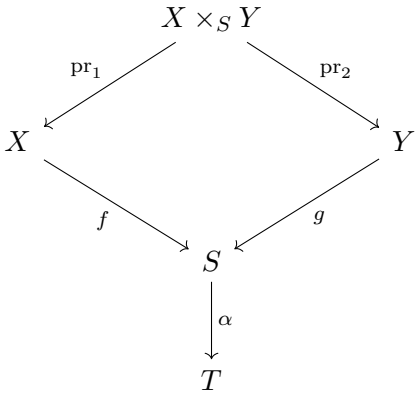
$$R^p\alpha_* R^q g_* \mathbb{Q}_\ell(i - d_1 + d_2) \Longrightarrow R^{p+q}(\alpha g)_* \mathbb{Q}_\ell(i - d_1 + d_2), \tag{1.1.1}$$

which is given on the E_2 term by (1.0.7), and on the abutment by (1.0.5).

1.2.

Now consider the situation

(1.2.1)



in which S, X, f, Y, g are as in (1.0.1)–(1.0.4), but now f is also assumed proper. Let

$$z \in H^a(X \times_S Y, \mathbb{Q}_\ell(b)). \tag{1.2.2}$$

Then z induces a map, again denoted by z,

$$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i + b - d_1)) \tag{1.2.3}$$

by the formula

$$z(x) = \mathrm{pr}_{2*}(z \wedge \mathrm{pr}_1^*(x)), \tag{1.2.4}$$

where pr_{2*} is the Gysin homomorphism. By ètale localization on S, the class z likewise defines

$$z : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1), \tag{1.2.5}$$

and, after applying $H^p(S, -)$, maps

$$H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1)). \tag{1.2.6}$$

Lemma 1.3.

The class z defines a morphism of bidegree $(0, a-2d_1)$ between the Leray spectral sequences

$$H^p(S, R^q f_*\mathbb{Q}_\ell(i)) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell(i)) \tag{1.2.7}$$

and

$$H^p(S, R^q g_*\mathbb{Q}_\ell(i+b-d_1)) \Longrightarrow H^{p+q}(Y, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.2.8}$$

which is given on the E_2 terms by (1.2.6), and on the abutments by (1.2.3).

Proof. In view of (1.2.4), and since "multiplication by z" defines a morphism of bidegree $(0, a)$ between the Leray spectral sequences

$$H^p(S, R^q(f\circ \mathrm{pr}_1)_*\mathbb{Q}_\ell(i)) \Longrightarrow H^{p+q}(X \times_S Y, \mathbb{Q}_\ell(i)) \tag{1.2.9}$$

$$\Big\downarrow z$$

$$H^p\big(S, R^{q+a}(f\circ \mathrm{pr}_1)_*\mathbb{Q}_\ell(i+b)\big) \Longrightarrow H^{p+q+a}(X \times_S Y, \mathbb{Q}_\ell(i+b)), \tag{1.2.10}$$

it suffices to apply Lemma 1.1 to the situation

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{\quad\mathrm{pr}_2\quad} & Y \\ & {\scriptstyle f\circ \mathrm{pr}_1}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array} \tag{1.2.11}$$

.

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Corollary 1.4.

In the situation (1.2.1)–(1.2.2), the morphism (1.2.3)

$$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i+b-d_1)) \tag{1.4.1}$$

takes primitive classes (XVIII 5.8.1) to primitive classes, and the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i)) & \xrightarrow{\ z\ } & \mathrm{Prim}^{q+a-2d_1}(Y/S, \mathbb{Q}_\ell(i+b-d_1)) \\ \Big\downarrow{\scriptstyle u_f} & & \Big\downarrow{\scriptstyle u_g} \\ H^1(S, R^{q-1}f_*\mathbb{Q}_\ell(i)) & \xrightarrow{\ z\ } & H^1(S, R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1)). \end{array} \tag{1.4.2}$$

Here u_f and u_g are the canonical homomorphisms of XVIII 5.8.1.

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1.5.

Let $\bar\eta$ be a geometric point of S (lying over a point η of S), and let

$$z_{\bar\eta} \in H^a(X_{\bar\eta} \times_{k(\bar\eta)} Y_{\bar\eta}, \mathbb{Q}_\ell(b)) \tag{1.5.1}$$

be the class induced by z on the corresponding geometric fiber of

$$X \times_S Y \longrightarrow S. \tag{1.5.2}$$

Proposition 1.5.3.

Suppose that f and g are proper, that S is connected, and let $q \in \mathbb{Z}$. If the class $z_{\bar\eta}$ induces the zero homomorphism

$$0 = z_{\bar\eta} : H^{q-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1-1}(Y_{\bar\eta}, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.5.3}$$

then, for every $x \in \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i))$, one has

$$u_g(z(x)) = 0 \quad \text{in} \quad H^1(S, R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1)). \tag{1.5.4}$$

Proof. By the commutativity of (1.4.2), it is enough to see that z induces the zero homomorphism II≈347

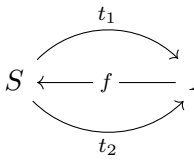
$$z : R^{q-1}f_*\mathbb{Q}_\ell(i) \xrightarrow{0} R^{q+a-2d_1-1}g_*\mathbb{Q}_\ell(i+b-d_1). \tag{1.5.5}$$

Indeed, since f and g are proper and smooth, the sheaves $R^\bullet f_*\mathbb{Q}_\ell(\cdot)$ and $R^\bullet g_*\mathbb{Q}_\ell(\cdot)$ are twisted constant sheaves (SGA 4, XVI 2.2), hence are equivalent to their values at $\bar\eta$, regarded as modules under $\pi_1(S, \bar\eta)$.

1.6. A construction of Manin

Suppose that two sections of $f : X \longrightarrow S$ are given:

(1.6.0)



,

and associate with each its cohomology class

$$\mathrm{cl}(t_i) = t_{i*}(1) \in H^{2d_1}(X, \mathbb{Q}_\ell(d_1)). \tag{1.6.1}$$

If the fibers of $f : X \longrightarrow S$ are geometrically connected (more generally, if, for $s \in S$, $t_1(s)$ and $t_2(s)$ belong to the same connected component of X_s), it is clear that

$$\mathrm{cl}(t_1) - \mathrm{cl}(t_2) \in \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)). \tag{1.6.2}$$

We denote by

$$u_f(t_1 - t_2) \in H^1(S, R^{2d_1-1}f_*\mathbb{Q}_\ell(d_1)) \tag{1.6.3}$$

its image under

$$u_f : \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)) \longrightarrow H^1(S, R^{2d_1-1}f_*\mathbb{Q}_\ell(d_1)). \tag{1.6.4}$$

(It is an analogue $\mu(t_1 - t_2)$ of this element that Manin studies in [5].)

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1.6.5.

Now let z be an algebraic cycle on $X \times_S Y$, of codimension at least b on each fiber of $X \times_S Y$,[24] which determines a cohomology class (SGA 5, Exposé IV)

$$\mathrm{cl}(z) \in H^{2b}(X \times_S Y, \mathbb{Q}_\ell(b)). \tag{1.6.6}$$

[24] If S is smooth over a scheme T (for example $T = \mathrm{Spec}\, k$, with k a field), it would be enough to suppose that z has codimension at least b on each fiber over T.

This class defines a map (cf. (1.2.3))

$$\mathrm{cl}(z) : H^{2d_1}(X, \mathbb{Q}_\ell(d_1)) \longrightarrow H^{2b}(Y, \mathbb{Q}_\ell(b)), \tag{1.6.7}$$

and, if the cycles $z(t_1)$ and $z(t_2)$ are defined, the compatibility of the cup product with intersections of cycles (SGA 5, Exposé IV) gives

$$\mathrm{cl}(z)\big(\mathrm{cl}(t_1) - \mathrm{cl}(t_2)\big) = \mathrm{cl}\big(z(t_1) - z(t_2)\big). \tag{1.6.8}$$

The commutativity of (1.4.2) gives us

$$\mathrm{cl}(z)\big(u_f(t_1 - t_2)\big) = u_g\big(\mathrm{cl}(z(t_1) - z(t_2))\big). \tag{1.6.9}$$

As a corollary of Proposition 1.5.3, we therefore have:

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Proposition 1.7.

Let t_1 and t_2 be two sections of $f : X \longrightarrow S$, with f assumed to have connected fibers, and let z be an algebraic cycle on $X \times_S Y$ of codimension b, such that the algebraic cycle

$$z(t_1) - z(t_2) \tag{1.7.1}$$

is defined. Then its cohomology class is primitive (XVIII 5.8.1). If the algebraic cycle $z_{\bar\eta}$ on $X_{\bar\eta} \times Y_{\bar\eta}$ induces the zero homomorphism

$$z_{\bar\eta} : H^{2d_1-1}(X_{\bar\eta}, \mathbb{Q}_\ell(d_1)) \xrightarrow{0} H^{2b-1}(Y_{\bar\eta}, \mathbb{Q}_\ell(b)), \tag{1.7.2}$$

then

$$u_g\big(z(t_1) - z(t_2)\big) = 0 \quad \text{in} \quad H^1(S, R^{2b-1}g_*\mathbb{Q}_\ell(b)), \tag{1.7.3}$$

where u_g is the homomorphism of XVIII 5.8.1 occurring in 1.4.

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2. A reminder on level

(cf. [2], 9.7 and 10.1).

2.0.1.

Let X be proper and smooth over an algebraically closed field k. We saw in (1.2.3) that, for every

$$T \quad \text{proper, smooth, and connected over } k, \text{ of dimension } d, \tag{2.0.2}$$

and every

$$\text{algebraic cycle } z \text{ of codimension } b \text{ on } T \times X, \tag{2.0.3}$$

the cohomology class

$$\mathrm{cl}(z) \in H^{2b}(T \times X, \mathbb{Q}_\ell(b)) \tag{2.0.4}$$

gives rise to maps

$$\mathrm{cl}(z) : H^q(T, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+2b-2d}(X, \mathbb{Q}_\ell(i + b - d)). \tag{2.0.5}$$

The coniveau filtration (N^j) on $H^q(X, \mathbb{Q}_\ell(i))$ is defined by

$$N^j H^q(X, \mathbb{Q}_\ell(i)) = \sum_{(T,z)} \mathrm{Im}\Big(\mathrm{cl}(z) : H^{q-2j}(T, \mathbb{Q}_\ell(i-j)) \longrightarrow H^q(X, \mathbb{Q}_\ell(i))\Big), \tag{2.0.6}$$

where the sum extends over the pairs

$$(T, z), \qquad \begin{array}{l} T/k \text{ proper, smooth, and connected, of dimension } d, \\ z \text{ an algebraic cycle on } T \times X \text{ of codimension } j+d. \end{array} \tag{2.0.7}$$

It should be noted that one sometimes speaks of the increasing filtration F^j "by level", defined by

$$F^j H^q(X, \mathbb{Q}_\ell(i)) = N^{\left[\frac{q-j}{2}\right]} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.8}$$

Replacing (T, z) by $(\mathbb{P}^1 \times T, \{\mathrm{pt}\} \times z)$, one sees that

$$N^j H^q(X, \mathbb{Q}_\ell(i)) \subset N^{j-1} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.9}$$

Obviously, one has:

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Proposition 2.1.

Let X be proper, smooth, and geometrically connected over a field k, let $\bar{k}$ be an algebraic closure of k, and put $X_{\bar{k}} = X \otimes_k \bar{k}$. Then $\mathrm{Gal}(\bar{k}/k)$, acting on $H^q(X_{\bar{k}}, \mathbb{Q}_\ell(i))$, preserves the coniveau filtration.

Standard arguments (cf. [2]) prove the following statements.

Proposition 2.2.

The level filtration is invariant under an extension k'/k of algebraically closed fields.

Proposition 2.3.

Let S be a reduced and irreducible scheme, with generic point η, and let

$$\pi : X \longrightarrow S$$

be a proper and smooth morphism with geometrically connected fibers. If, for given integers j and q, one has on the geometric generic fiber $X_{\bar{\eta}}$

$$H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell), \tag{2.3.1}$$

then there exists a nonempty open subset $U \subset S$ such that, at every point $u \in U$, one has on the geometric fiber $X_{\bar{u}}$

$$H^q(X_{\bar{u}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{u}}, \mathbb{Q}_\ell). \tag{2.3.2}$$

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3. Application to Lefschetz pencils

3.1.

From now on, we fix

$$k, \quad \text{an algebraically closed field,} \tag{3.1.0}$$

and

$$X \subset \mathbb{P}^r, \quad \text{a projective, smooth, connected } k\text{-scheme of even dimension } n, \text{ satisfying (LV) (XVIII 5.2.2).} \tag{3.1.1}$$

Take

$$D = \{H_t\}_{t\in\mathbb{P}^1}, \quad \text{a Lefschetz pencil of hypersurfaces of degree } d, \tag{3.1.2}$$

with generic fiber

$$X_\eta, \qquad \eta = \text{the generic point of } \mathbb{P}^1. \tag{3.1.3}$$

Put (cf. XVIII 5.4.4)

$$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta), \quad \text{the inclusion,} \tag{3.1.4}$$

and

$$E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = \big(m^* H^{n-1}(X, \mathbb{Q}_\ell(i))\big)^\perp \subset H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)). \tag{3.1.5}$$

We saw in XVIII 6.7.2 that the Picard–Lefschetz formula implies

$$\begin{gathered}\text{If } n \text{ is even, every finite-index subgroup of } \mathrm{Gal}(k(\bar\eta)/k(\eta)) \\ \text{acts irreducibly on } E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).\end{gathered} \tag{3.1.6}$$

Proposition 3.2.

Let $D = \{H_t\}$ be a Lefschetz pencil, and suppose that $n = 2m$ is even. Then either

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \subset m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1} H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)), \tag{3.2.1}$$

or else:

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(3.2.2)

For every curve C proper and smooth over $k(\bar\eta)$ and every algebraic cycle z on $C \times X_{\bar\eta}$ of codimension m, the induced homomorphism (1.2.3)

$$\mathrm{cl}(z) : H^1(C, \mathbb{Q}_\ell(1 + i - m)) \longrightarrow H^{2m-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \tag{3.2.3}$$

has its image in $m^* H^{2m-1}(X, \mathbb{Q}_\ell(i))$.

Proof. Suppose that the image of such a homomorphism (3.2.3) is not contained in $m^* H^{n-1}(X, \mathbb{Q}_\ell(i))$, or, equivalently, that its image in $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))/H^{n-1}(X, \mathbb{Q}_\ell(i))$ is nonzero. This image is evidently stable under the action of a finite-index subgroup of $\mathrm{Gal}(k(\bar\eta)/k(\eta))$. The irreducibility (3.1.6) of $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$ therefore says that

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + \mathrm{Im}(3.2.3).$$

By definition,

$$\mathrm{Im}(3.2.3) = \mathrm{cl}(z)\big(H^1(C, \mathbb{Q}_\ell(1 + i - m))\big) \subset N^{m-1} H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).$$

Consequently,

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1} H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)),$$

which is precisely the first alternative.

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3.3. Condition (B)

3.3.1. We shall say that the inclusion

$$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$$

satisfies condition (B) if there exists

(3.3.2) $\qquad V_{\bar\eta}$, an algebraic cycle of codimension $n-1$ on $X_{\bar\eta} \times (X \times_k k(\bar\eta))$,

such that the map it defines

$$V_{\bar\eta} : H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X, \mathbb{Q}_\ell(i)) \tag{3.3.3}$$

is a left inverse of

$$m^* : H^{n-1}(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)), \tag{3.3.4}$$

that is, such that

$$V_{\bar\eta}\, m^* = \mathrm{id}. \tag{3.3.5}$$

3.4.

Condition (B) is satisfied if the operation Λ of [4, 1.4.2.1] is algebraic (cf. [4, 2.12(ii)]). It is conjectured that Λ is always algebraic, and hence that condition (B) always holds; this is known in many cases [4]. It holds trivially when X is a complete intersection of even dimension, since in that case we know (IX 1.6) that

$$H^{n-1}(X, \mathbb{Q}_\ell(i)) = 0. \tag{3.4.1}$$

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4. Griffiths's theorem

Theorem 4.1.

Let X be projective, smooth, and connected over k, of even dimension $n = 2m \geq 4$. Let D be a Lefschetz pencil of sections by hypersurfaces of degree d. Assume that:

(4.1.0) X satisfies (LV) (XVIII 5.2.2);
(4.1.1) the pencil D satisfies hypothesis (A) (XVIII 5.3);
(4.1.2) the pencil D satisfies alternative (3.2.2), that is, in the setting of Proposition 3.2, the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ is not of level ≤ 1;
(4.1.3) the inclusion $m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$ satisfies hypothesis (B) (3.3).

Then, if an algebraic cycle z of codimension m on X has a restriction to $X_{\bar\eta}$ which, as an algebraic cycle, satisfies

$$z_{\bar\eta} = z|_{X_{\bar\eta}} \quad \text{is algebraically equivalent to zero,} \tag{4.1.4}$$

then

$$\mathrm{cl}(z) = 0 \quad \text{in } H^{2m}(X, \mathbb{Q}_\ell(m)); \tag{4.1.5}$$

in other words, z is $\mathbb{Q}_\ell$-homologous to zero.

4.2. A remark on the case where X is a complete intersection

As already noted (cf. XVIII 5.2.5 and (3.4)), hypotheses (4.1.0) and (4.1.3) then hold automatically. We saw in XVIII 6.3.4 and 6.4 that hypothesis (4.1.1) holds for a pencil of hypersurfaces of degree $d > 2n$. Moreover, in the next exposé we shall prove condition (4.1.2) for a "general" pencil of hypersurfaces of degree $d > 2n$ (XXI 5.2). Consequently, a "general" pencil of hypersurfaces of degree $d > 2n$ satisfies all the hypotheses of Theorem 4.1.

4.3.

Returning to the case of arbitrary X, it seems very plausible that, for d sufficiently large, all the conditions of 4.1 hold for D, at least when D is sufficiently general. We have already seen this for (4.1.1) (XVIII 6.4), and we noted the conjecture that (4.1.0) and (4.1.3) always hold (with (4.1.0) known in every case when $\mathrm{char}(k) = 0$). As for condition (4.1.2), Hodge theory proves it for large d when $\mathrm{char}(k) = 0$, while heuristic arguments, given in the appendix to the next exposé, make the same conclusion plausible when $\mathrm{char}(k) > 0$.

More generally, these arguments—Hodge theory and, respectively, crystalline cohomology—prove, II ≈356
and respectively make plausible, that for a smooth projective variety X of dimension n, and for every Lefschetz pencil D of hypersurfaces of sufficiently large degree d (respectively, every sufficiently general pencil D of hypersurfaces of degree d), the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ has level exactly $n-1$; that is, it is not of level $\leq n-3$, or equivalently is not of coniveau 1.

4.4. Proof of Griffiths's theorem

Let z be an algebraic cycle of codimension m on X such that

$$z_{\bar\eta} = z|_{X_{\bar\eta}} \quad \text{is algebraically equivalent to zero.} \tag{4.4.1}$$

By definition, there exist

$$\text{a proper, smooth, connected curve } C_{\bar\eta} \text{ over } k(\bar\eta), \tag{4.4.2}$$

$$t_i(\bar\eta) : \operatorname{Spec} k(\bar\eta) \longrightarrow C_{\bar\eta}, \qquad i = 1, 2, \tag{4.4.3}$$

and

$$\text{an algebraic cycle } P_{\bar\eta} \text{ of codimension } m \text{ on } C_{\bar\eta} \times X_{\bar\eta}, \tag{4.4.4}$$

such that

$$z_{\bar\eta} = P_{\bar\eta}(t_1(\bar\eta)) - P_{\bar\eta}(t_2(\bar\eta)) \quad \text{on } X_{\bar\eta}. \tag{4.4.5}$$

Regard $P_{\bar\eta}$ as an algebraic correspondence from $C_{\bar\eta}$ to $X_{\bar\eta}$. It defines a cohomological correspondence, again denoted by $P_{\bar\eta}$, or equivalently a homomorphism

$$P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \tag{4.4.6}$$

Likewise, consider the algebraic correspondence $V_{\bar\eta}$ of (3.3.2), from $X_{\bar\eta}$ to $X \otimes_k k(\bar\eta)$, supplied by hypothesis (B) in (4.1.3). It defines

$$V_{\bar\eta} : H^*(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)) \longrightarrow H^*(X \otimes_k k(\bar\eta), \mathbb{Q}_\ell(i+m-1)). \tag{4.4.7}$$

The composite algebraic correspondence $Q_{\bar\eta} = V_{\bar\eta} \circ P_{\bar\eta}$ from $C_{\bar\eta}$ to $X \otimes_k k(\bar\eta)$ therefore defines

$$Q_{\bar\eta} = V_{\bar\eta} \circ P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X \otimes_k k(\bar\eta), \mathbb{Q}_\ell(i+m-1)). \tag{4.4.8}$$

Consider the inclusion

$$m : X_{\bar\eta} \longrightarrow X \otimes_k k(\bar\eta).$$

It gives the algebraic correspondence m^* from $X \otimes_k k(\bar\eta)$ to $X_{\bar\eta}$, hence the composite correspondence $m^* Q_{\bar\eta}$ from $C_{\bar\eta}$ to $X_{\bar\eta}$, defining

$$m^* Q_{\bar\eta} = m^* V_{\bar\eta} P_{\bar\eta} : H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \tag{4.4.9}$$

Now consider the algebraic correspondence

$$\begin{aligned} R_{\bar\eta} = P_{\bar\eta} - m^* Q_{\bar\eta} = (\mathrm{id} - m^* V_{\bar\eta}) P_{\bar\eta} : \\ H^*(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{*+2(m-1)}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)). \end{aligned} \tag{4.4.10}$$

Since $V_{\bar\eta}$ is a left inverse of m^*, $\mathrm{id} - m^* V_{\bar\eta}$ is a projector on $H^*(X_{\bar\eta}, -)$ whose image is a complement of

$$\mathrm{Im}\big(m^* : H^*(X \otimes_k k(\bar\eta), -) \longrightarrow H^*(X_{\bar\eta}, -)\big).$$

By hypothesis (4.1.2), the image of

$$R_{\bar\eta} : H^1(C_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{2m-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i+m-1)) \tag{4.4.11}$$

lies in $\mathrm{Im}\, m^*$. Consequently,

(4.4.12) the homomorphism (4.4.11) is zero.

Choose a finite extension II≈**358**

$$L/k(\eta)$$

over which the objects

$$C_{\bar\eta}, \quad t_1(\bar\eta), \quad t_2(\bar\eta), \quad Q_{\bar\eta}, \quad R_{\bar\eta}$$

are already defined, and let U be a nonempty open subset of the normalization of $\mathbb{P}^1$ in L. After shrinking U, there is a proper smooth relative curve

$$C \longrightarrow U$$

with geometrically connected fibers and geometric generic fiber $C_{\bar\eta}$, together with two sections

$$t_i : U \longrightarrow C, \qquad i = 1, 2,$$

whose values at $\bar\eta$ are $t_i(\bar\eta)$.

Let II≈**359**

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

be the projection of XVIII 3.1.4 whose fibers are the members of the pencil, put

$$\widetilde{X}_U = \widetilde{X} \times_{\mathbb{P}^1} U,$$

and let

$$M : \widetilde{X}_U \hookrightarrow X \times U$$

be the canonical immersion into the constant family over U with fiber X. Set

$$z_U = M^* \,\mathrm{pr}_1^*(z),$$

the algebraic cycle on $\widetilde{X}_U$ induced by z.

After shrinking U once more, there are an algebraic cycle R on $C \times_U \widetilde{X}_U$ whose value on $C_{\bar\eta} \times X_{\bar\eta}$ is $R_{\bar\eta}$, and an algebraic cycle Q on $C \times X \times U$ whose value on $C_{\bar\eta} \times (X \otimes_k k(\bar\eta))$ is $Q_{\bar\eta}$, such that

$$z_U = R(t_1) - R(t_2) + (\mathrm{id} \times M)^*\big[Q(t_1) - Q(t_2)\big]. \tag{4.4.13}$$

To prove the theorem it suffices, by the injectivity of the Griffiths homomorphism "modulo the fixed part" (XVIII 5.8.7),

$$\mathrm{Prim}^{2m}(X,\mathbb{Q}_\ell(m)) \xrightarrow{\text{grif}} \frac{H^1(U,R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m))}{M^*H^1(U,H^{2m-1}(X,\mathbb{Q}_\ell(m)))},$$

to show that the image of z_U under

$$u_0 : \mathrm{Prim}^{2m}(\widetilde{X}_U/U,\mathbb{Q}_\ell(m)) \longrightarrow H^1(U,R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m))$$

lies in $M^*H^1(U,H^{2m-1}(X,\mathbb{Q}_\ell(m)))$.

In view of (4.4.13) and the commutativity of (1.4.2), we have the commutative diagram

$$\begin{array}{ccccc}
\mathrm{Prim}^2(C/U,\mathbb{Q}_\ell(1)) & \xrightarrow{Q} & \mathrm{Prim}^{2m}(X\times U/U,\mathbb{Q}_\ell(m)) & \xrightarrow{M^*} & \mathrm{Prim}^{2m}(\widetilde{X}_U/U,\mathbb{Q}_\ell(m)) \\
 & & \downarrow{\scriptstyle u_{\mathrm{pr}_2}} & & \downarrow{\scriptstyle u_0} \\
 & & H^1(U,H^{2m-1}(X,\mathbb{Q}_\ell(m))) & \xrightarrow{M^*} & H^1(U,R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m)),
\end{array}$$

so it is enough to prove

$$u_0(R(t_1)-R(t_2)) = 0 \quad \text{in} \quad H^1(U,R^{2m-1}\rho_{U*}\mathbb{Q}_\ell(m)).$$

This follows from (4.4.12), by 1.7.

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5. The Griffiths group

5.0.

Let S be proper and smooth over an algebraically closed field k. Denote by

(5.0.1) $Z^i(S)$ the group of algebraic cycles of codimension i on S; II ≈**361**

(5.0.2) $Z^i_{\mathrm{alg}}(S)$ the subgroup of $Z^i(S)$ consisting of cycles algebraically equivalent to zero; and, for every prime number $\ell \neq \mathrm{char}(k)$,

(5.0.3) $Z^i_{\ell\text{-coh}}(S)$ the subgroup of $Z^i(S)$ consisting of cycles whose cohomology class in $H^{2i}(S,\mathbb{Q}_\ell(i))$ is zero.

It is conjectured in [4] that this group is independent of ℓ; this is known at least when $\mathrm{char}(k)=0$. Set

$$\mathrm{Grif}^i_\ell(S) = Z^i_{\ell\text{-coh}}(S)/Z^i_{\mathrm{alg}}(S). \tag{5.0.4}$$

If, moreover, S is projective and is equipped with a projective embedding, set

(5.0.5) $Z^{i,\mathrm{prim}}(S)$ equal to the subgroup of $Z^i(S)$ consisting of primitive cycles (that is, cycles whose cohomology class is primitive), and

(5.0.6) $\mathrm{Prim}^{2i}_{\mathrm{alg}}(S,\mathbb{Q}_\ell(i))$ equal to the $\mathbb{Q}$-vector subspace of $H^{2i}(S,\mathbb{Q}_\ell(i))$ spanned by the cohomology classes of primitive algebraic cycles.

Corollary 5.1.

Under the hypotheses of Theorem 4.1, one has

$$\dim_{\mathbb{Q}}(\mathrm{Grif}^m_\ell(X_{\bar\eta})\otimes_{\mathbb{Z}}\mathbb{Q}) \geq \dim_{\mathbb{Q}}\Big(\mathrm{Prim}^{2m}_{\mathrm{alg}}(X,\mathbb{Q}_\ell(m))\Big). \tag{5.1.1}$$

Indeed, by (4.1.5), the kernel of the composite map

$$\mathbb{Q}\otimes_{\mathbb{Z}} Z^{m,\mathrm{prim}}(X) \xrightarrow{\text{res. to } X_{\bar\eta}} \mathbb{Q}\otimes_{\mathbb{Z}} Z^m_{\ell\text{-coh}}(X_{\bar\eta}) \longrightarrow \mathrm{Grif}^m_\ell(X_{\bar\eta})$$

is contained in $Z^m_{\ell\text{-coh}}(X)\otimes_{\mathbb{Z}}\mathbb{Q}$. Consequently, $\mathrm{Prim}^{2m}_{\mathrm{alg}}(X,\mathbb{Q}_\ell(m))$ is a quotient of a subspace of $\mathrm{Grif}^m_\ell(X_{\bar\eta})\otimes_{\mathbb{Z}}\mathbb{Q}$, which proves the assertion.

Remark 5.2.

For X a quadric hypersurface in $\mathbb{P}^{2m+1}$, one has

$$\dim_{\mathbb{Q}_\ell}\Big(\mathrm{Prim}^{2m}_{\mathrm{alg}}(X,\mathbb{Q}_\ell(m))\Big)=1.$$

In particular, when $m=2$, for X a quadric in $\mathbb{P}^5$, the section of X by a general hypersurface of degree >5 contains a curve which is $\mathbb{Q}_\ell$-homologous to zero but of which no nonzero multiple is algebraically equivalent to zero.

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EXPOSÉ XXI

THE LEVEL OF THE COHOMOLOGY OF COMPLETE INTERSECTIONS

by N. Katz

with an Appendix by P. Deligne (§5)

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0. Introduction

Let X be a smooth projective variety of dimension $n \geq 3$, let $Y(d)$ be a section by a hypersurface of degree d, and let $E^{n-1}(Y(d), \mathbb{Q}_\ell)$ be the "vanishing" cohomology of $Y(d)$ (XVIII 5.4.4). The aim of this exposé is to prove that, for d sufficiently large and $Y(d)$ sufficiently general, it is not true that the whole of $E^{n-1}(Y(d), \mathbb{Q}_\ell)$ "comes" from the groups H^i, $i < n-1$, of smooth proper algebraic varieties by means of algebraic correspondences. In characteristic zero this follows trivially from the Hodge bigrading on complex cohomology and from the fact that, in coherent cohomology,

$$\dim H^{n-1}(Y(d), \mathcal{O}_{Y(d)}) \longrightarrow \infty \qquad \text{as } d \longrightarrow \infty.$$

In characteristic p, the method again relates the coherent group $H^{n-1}(Y(d), \mathcal{O}_{Y(d)})$ to the étale group $H^{n-1}_{\text{et}}(Y(d), \mathbb{Q}_\ell)$. To do so, one works over a finite base field. The two are then linked by the zeta function of $Y(d)$: this function is expressed in terms of ℓ-adic cohomology, while its "reduction modulo p" is expressed in terms of coherent cohomology (cf. XXII 3.9). For complete intersections this method works without difficulty, owing to the detailed knowledge of both kinds of cohomology (étale and coherent) in that case. It yields the main result, Theorem 4.2, as a consequence of a result on Hasse–Witt operations, Theorem 1.4 (see 1.0).

There are heuristic arguments due to A. Grothendieck, based on the yoga of crystalline cohomol- II ≈**365**
ogy, which make the general statement plausible for every smooth projective X, by essentially the same method.

Even in the special case of complete intersections, the proof makes essential use in 2.2 of Theorem 2.1 on the integrality of the eigenvalues of Frobenius, recently proved by P. Deligne and proved here in the Appendix (§5).

1. The Hasse–Witt matrix of a complete intersection

Definition 1.0.

A proper scheme X over a field k of characteristic exponent p is called *special* if the additive p-linear operation

$$\mathfrak{F}_p : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad g \longmapsto g^p, \tag{1.0.1}$$

induces a nilpotent operation

$$\mathfrak{F}_p : H^{\dim X}(X, \mathcal{O}_X) \longrightarrow H^{\dim X}(X, \mathcal{O}_X). \tag{1.0.2}$$

Equivalently, if $M = \dim_k H^{\dim X}(X, \mathcal{O}_X)$, the p^M-linear operation

$$\mathfrak{F}_p^M = \mathfrak{F}_{p^M} : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad g \longmapsto g^{p^M}, \tag{1.0.3}$$

annihilates $H^{\dim X}(X, \mathcal{O}_X)$.

Recall from [2] that, for every $i \in \mathbb{Z}$, the homomorphism II ≈**366**

$$H^i(X, \mathcal{O}_X) \xrightarrow{\mathfrak{F}_p} H^i(X, \mathcal{O}_X)$$

induced by (1.0.1) is classically called the *Hasse–Witt homomorphism* on $H^i(X, \mathcal{O}_X)$. Notice that, when $p = 1$, saying that X is non-special means simply that $H^{\dim X}(X, \mathcal{O}_X) \neq 0$.

Theorem 1.1.

Fix a characteristic exponent p, a dimension $n \geq 1$, an integer $r \geq 1$, and a multidegree $(d_1, \ldots, d_r)$. If

$$\sum_{i=1}^{r} d_i \geq (n + r) + 1, \tag{1.1.0}$$

then, among all complete intersections of dimension n in $\mathbb{P}^{n+r}$ having multidegree $(d_1, \dots, d_r)$ in characteristic p, a "sufficiently general" member is non-special in the sense of 1.0.

Proof. Let

$$f : X_U \longrightarrow U \tag{1.1.1}$$

denote the "universal family" of complete intersections in $\mathbb{P}^{n+r}$ of multidegree $(d_1, \dots, d_r)$ in characteristic p, together with their defining equations. Thus U is an open subset of a product of r projective spaces over $\mathbb{F}_p$, of impressive dimension.

The $\mathcal{O}_U$-module II ≈367

$$R^n f_* \mathcal{O}_{X_U} \tag{1.1.2}$$

is locally free, and its formation commutes with every base change (XI 1.5(i)). By (1.1.0), its rank M is nonzero (XI 2.5). When $p = 1$, this proves the theorem for every complete intersection satisfying (1.1.0).

Suppose now that $p \geq 2$. The iterated Hasse–Witt operation

$$\mathfrak{F}_p^M = \mathfrak{F}_{p^M} : R^n f_* \mathcal{O}_{X_U} \longrightarrow R^n f_* \mathcal{O}_{X_U} \tag{1.1.3}$$

is p^M-linear over $\mathcal{O}_U$. Since $R^n f_* \mathcal{O}_{X_U}$ is locally free, (1.1.3) is represented locally on U by an (M, M)-matrix H with entries in $\mathcal{O}_U$. Consequently,

$$\{u \in U \mid X_u \text{ is non-special}\} \quad \text{is open in } U. \tag{1.1.4}$$

Indeed, this is the set of u for which $H(u) \neq 0$. Since U is irreducible, it is therefore enough to find one value of u for which X_u is non-special; we shall do so over a finite field.

Proposition 1.1.5.

Let X be a complete intersection over $\mathbb{F}_q$. If X has no $\mathbb{F}_q$-rational points, then X is non-special.

Proof. It is plainly enough to prove, for a complete intersection X, the implication

$$X \text{ special} \quad \Longrightarrow \quad \operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv 1 \pmod{p}. \tag{1.1.5.1}$$

By the trace formula, proved in the following Exposé XXII, 3.2,

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv \sum_{i \geq 0} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big) \pmod{p}.$$

Now $H^i(X, \mathcal{O}_X) = 0$ unless $i = 0$ or $i = n = \dim X$, while $H^0(X, \mathcal{O}_X) \simeq \mathbb{F}_q$ (XI 1.5(iii)); hence this becomes

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) \equiv 1 + (-1)^n \operatorname{tr}\big(\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)\big) \pmod{p}.$$

If X is special, the operation of $\mathfrak{F}_q$ on $H^n(X, \mathcal{O}_X)$ is nilpotent by definition and therefore has trace zero. This proves (1.1.5.1), and hence the proposition.

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Proposition 1.2.

For every power q of p, if

$$\sum_{i=1}^{r} d_i \geq n + r + 1,$$

there exists a complete intersection in $\mathbb{P}^{n+r}_{\mathbb{F}_q}$ of multidegree $(d_1, \dots, d_r)$ having no $\mathbb{F}_q$-rational point.

Proof. Begin with the case $r = 1$. For every $d \geq n+2$, we must find a homogeneous form of degree d,

$$H(X_1, \ldots, X_{n+2}) \in \mathbb{F}_q[X_1, \ldots, X_{n+2}], \tag{1.2.1}$$

such that

$$(\alpha_1, \ldots, \alpha_{n+2}) \in (\mathbb{F}_q)^{n+2}, \qquad H(\alpha_1, \ldots, \alpha_{n+2}) = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_{n+2} = 0. \tag{1.2.2}$$

Set $\mathbb{E} = \mathbb{F}_{q^d}$, and choose elements $\beta_1, \ldots, \beta_{n+2}$ of $\mathbb{E}$ that are linearly independent over $\mathbb{F}_q$. Take

$$H(X_1, \ldots, X_{n+2}) = N_{\mathbb{E}/\mathbb{F}_q}\left(\sum_{i=1}^{n+2} \beta_i X_i\right). \tag{1.2.3}$$

In the general case, since

$$\sum_{i=1}^{r} d_i \geq n + r + 1, \tag{1.2.4}$$

we can set $a_0 = 0$ and choose r integers

$$1 \leq a_1 < a_2 < \cdots < a_r = n + r + 1 \tag{1.2.5}$$

such that

$$d_i \geq a_i - a_{i-1} \qquad (1 \leq i \leq r). \tag{1.2.6}$$

For each $i = 1, \ldots, r$, choose a homogeneous form of degree d_i,

$$H_i(X_{a_{i-1}+1}, \ldots, X_{a_i}) \in \mathbb{F}_q[X_{a_{i-1}+1}, \ldots, X_{a_i}],$$

in the $a_i - a_{i-1}$ displayed variables and having property (1.2.2). The required complete intersection is defined by the homogeneous forms

$$H_1(X_1, \ldots, X_{a_1}), \quad \ldots, \quad H_r(X_{a_{r-1}+1}, \ldots, X_{a_r}).$$

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Theorem 1.3.

Let X be an irreducible complete intersection of dimension $n \geq 1$ over a field k of characteristic exponent p. For every integer $d \geq n+1$, a sufficiently general section of X by a hypersurface of degree d is non-special in the sense of 1.0.

Proof. The sections $X.H_u$ of X by hypersurfaces of degree d are parametrized by the lines in $H^0(X, \mathcal{O}_X(d))$, that is, by

$$U = \mathbb{P}(H^0(X, \mathcal{O}_X(d))). \tag{1.3.1}$$

Over U we have the "universal section of X by a hypersurface of degree d,"

$$f : X.H_u \longrightarrow U. \tag{1.3.2}$$

Let U' be the open subset of those $u \in U$ for which $\dim(X_u) = n - 1$. By XI 1.5(i) and 2.5, the restriction

$$R^{n-1} f_* \mathcal{O}_{X.H_u}\Big|_{U'} \tag{1.3.3}$$

is locally free, its formation commutes with every base change, and its rank M is at least one. If $p = 1$, this proves the theorem for every section by a hypersurface of degree at least $n + 1$.

Suppose therefore that $p > 1$. The iterated Hasse–Witt operation $\mathfrak{F}_{p^M}$ is again represented locally by an $M \times M$ matrix with entries in $\mathcal{O}_U$. Consequently,

$$\{u \in U' \mid X.H_u \text{ is non-special}\} \quad \text{is open in } U'. \tag{1.3.4}$$

It remains only to prove that the set in (1.3.4) is nonempty.

First consider the case

$$k = \mathbb{F}_q. \tag{1.3.5}$$

After replacing $\mathbb{F}_q$ by a finite extension, we may suppose that

$$\operatorname{card}\big(X(\mathbb{F}_q)\big) > 0 \tag{1.3.6}$$

and that

$$\begin{aligned} &\text{there are homogeneous coordinates} \\ &\quad X_1, \dots, X_{n+r+1} \text{ in the ambient space } \mathbb{P}^{n+r}_k \\ &\text{such that } X \text{ does not meet the linear subvariety} \\ &\quad X_1 = \cdots = X_{n+1} = 0. \end{aligned} \tag{1.3.7}$$

Choose a homogeneous form of degree d,

$$H(X_1, \dots, X_{n+1}) \in \mathbb{F}_q[X_1, \dots, X_{n+1}], \tag{1.3.8}$$

having property (1.2.2), which is possible because $d \geq n + 1$. Let H also denote the hypersurface in $\mathbb{P}^{n+r}$ defined by $H(X_1, \dots, X_{n+1}) = 0$. By construction, $X.H$ has no $\mathbb{F}_q$-rational point. In view of (1.3.6), this also shows that $X \not\subset H$; since X is irreducible, $X.H$ has dimension $n - 1$, so $H \in U'$. Proposition 1.1.5 now shows that $X.H$ is non-special.

We now pass to the general case. Looking at the equations defining X, we may suppose that k is finitely generated over $\mathbb{F}_p$. There is then a subring $B \subset k$, finitely generated as an algebra over $\mathbb{F}_p$ and having fraction field k, such that II ≈372

$$X \subset \mathbb{P}^{n+r}_k \tag{1.3.9}$$

is the generic fiber of a relative complete intersection $\widetilde{X} \subset \mathbb{P}^{n+r}_B$ which, fiber by fiber, has the same multidegree as X and is geometrically irreducible.

Write

$$\rho : \widetilde{X} \longrightarrow S = \operatorname{Spec}(B) \tag{1.3.10}$$

for the projection,

$$E = \rho_* \mathcal{O}_{\widetilde{X}}(d) \tag{1.3.11}$$

for the locally free $\mathcal{O}_S$-module, and

$$\pi : T = \check{\mathbb{P}}(E) \longrightarrow S \tag{1.3.12}$$

for the associated bundle of lines in $\mathbb{E}$. The generic fiber of (1.3.12) is precisely

$$U = \mathbb{P}\big(H^0(X, \mathcal{O}_X(d))\big).$$

Over T we have the universal section of $\widetilde{X} \subset \mathbb{P}^{n+r}_B$ by hypersurfaces of degree d, denoted

$$\alpha : \widetilde{X}.H_T \longrightarrow T. \tag{1.3.13}$$

These constructions fit into the following diagram of cartesian squares:

$$\begin{array}{ccc} \widetilde{X}.H_T & \longleftarrow & X.H_u = (\widetilde{X}.H_T) \times_T U \\ \downarrow{\scriptstyle \alpha} & & \downarrow \\ T & \longleftarrow & U = T \times_S \mathrm{Sp}(k) \\ \downarrow{\scriptstyle \pi} & & \downarrow \\ S & \longleftarrow & \mathrm{Sp}(k) \end{array} \tag{1.3.14}$$

The $\mathcal{O}_T$-module

$$R^{n-1}\alpha_*\mathcal{O}_{\widetilde{X}.H_T} \tag{1.3.15}$$

is locally free over the open subset $T' \subset T$ consisting of the t for which $\dim \alpha^{-1}(t) = n-1$, and its formation there commutes with every base change. Since T and U have the same generic point, it is enough, in order to show that (1.3.4) is nonempty, to prove that the open subset of T'

$$\{t \in T' \mid \widetilde{X}_{\pi(t)}.H_t \text{ is non-special}\} \tag{1.3.16}$$

is nonempty. But the already established case (1.3.5) shows that, for every closed point $s \in S$, the fiber $\pi^{-1}(s)$ has nonempty intersection with the open set (1.3.16). This proves the theorem.

Theorem 1.4.

Let $X \subset \mathbb{P}^N$ be a projective Cohen–Macaulay scheme, equidimensional of dimension $n \geq 1$, over a field k. There is an integer d_0 such that, for every $d \geq d_0$, if $i : X.H \longrightarrow X$ denotes the inclusion of a section by a general hypersurface of degree d, then the action of

$$\mathfrak{F}_p \quad \text{on} \quad \frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{i^*H^{n-1}(X, \mathcal{O}_X)} \tag{1.4.0}$$

is non-nilpotent.

Proof. As above, we reduce to the case $k = \mathbb{F}_q$ and to the problem of finding just one H—for any fixed sufficiently large degree for which $\dim H^{n-1}(X.H, \mathcal{O}_{X.H})$ is independent of H; see (1.4.4)—for which (1.4.0) holds. After enlarging q, we may suppose that the irreducible components X_i of X are geometrically irreducible, that

$$\mathrm{card}\big(X_i(\mathbb{F}_q)\big) \geq 2, \tag{1.4.1}$$

and that

$$\begin{gathered} \text{there are projective coordinates } X_0, \ldots, X_N \text{ such that} \\ X \text{ does not meet the linear subvariety } X_0 = \cdots = X_n = 0. \end{gathered} \tag{1.4.2}$$

By a vanishing theorem of Serre ([1], or SGA 2 XII 1.4), there is an integer d_1 such that, for every $d \geq d_1$,

$$H^i(X, \mathcal{O}_X(-d)) = 0 \qquad \text{if } i \neq n = \dim X. \tag{1.4.3}$$

Thus, if H is a hypersurface of degree $d \geq d_1$, then (1.4.3) gives

$$\begin{aligned} H^i(X, \mathcal{O}_X) &\xrightarrow{\sim} H^i(X.H, \mathcal{O}_{X.H}) \qquad \text{if } i \leq n-2, \\ H^{n-1}(X, \mathcal{O}_X) &\hookrightarrow H^{n-1}(X.H, \mathcal{O}_{X.H}) \quad \text{injectively.} \end{aligned} \tag{1.4.4}$$

We prove the theorem with $d_0 = \max(N+1, d_1)$. For every $d \geq d_0$, we must find a hypersurface H of degree d, defined over $\mathbb{F}_q$, which contains no X_i and for which

$$\operatorname{tr}\left(\mathfrak{F}_q \;\middle|\; \frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{H^{n-1}(X, \mathcal{O}_X)}\right) \neq 0. \tag{1.4.5}$$

By the trace formula (XXII 3.2.1), modulo p we have

$$\begin{aligned} \operatorname{card}\big((X.H)(\mathbb{F}_q)\big) &\equiv \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X.H, \mathcal{O}_{X.H})\big) \\ &\overset{(1.4.4)}{\equiv} \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big) \\ &\quad + (-1)^{n-1} \operatorname{tr}\left(\mathfrak{F}_q \;\middle|\; \left(\frac{H^{n-1}(X.H, \mathcal{O}_{X.H})}{H^{n-1}(X, \mathcal{O}_X)}\right)\right). \end{aligned} \tag{1.4.6}$$

Let $a \in \mathbb{Z}/p\mathbb{Z}$ denote the quantity

$$a = \sum_{i=0}^{n-1} (-1)^i \operatorname{tr}\big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\big).$$

It is therefore enough, for every $d \geq d_0$, to find a hypersurface H of degree d, defined over $\mathbb{F}_q$, which does not contain X and such that

$$\operatorname{card}\big((X.H)(\mathbb{F}_q)\big) \not\equiv a \pmod{p}. \tag{1.4.7}$$

There are two cases. If $a \neq 0$, it is enough to take an H over $\mathbb{F}_q$ for which $(X.H)(\mathbb{F}_q)$ is empty, just as in (1.3.8). Such an H satisfies (1.4.7) and contains no X_i, because $\operatorname{card}(X_i(\mathbb{F}_q)) \geq 2$ by hypothesis. II≈376

If $a = 0$, it is enough to find an H such that $\operatorname{card}((X.H)(\mathbb{F}_q)) = 1$. Then (1.4.7) holds and $X_i \not\subset H$, since $\operatorname{card}(X_i(\mathbb{F}_q)) \geq 2$. Choose projective coordinates $X_0, \ldots, X_N$ such that $(1, 0, \ldots, 0)$ is a point of X. Take H to have equation $H(X_1, \ldots, X_N) = 0$, where H is a homogeneous form satisfying (cf. (1.2.2))

$$\alpha_1, \ldots, \alpha_N \in \mathbb{F}_q, \quad H(\alpha_1, \ldots, \alpha_N) = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_N = 0. \tag{1.4.8}$$

The hypersurface defined by H in $\mathbb{P}^N$ then has only the point $(1, 0, \ldots, 0)$ over $\mathbb{F}_q$; a fortiori, $X.H$ has only this point over $\mathbb{F}_q$. This proves the theorem.

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2. Coniveau over a finite field of definition

2.0.

Let X be a separated scheme of finite type over $\mathbb{F}_q$. For every prime number $\ell \neq \operatorname{char}(\mathbb{F}_q)$, write

$$H_c^i(\overline{X}, \mathbb{Q}_\ell) \tag{2.0.1}$$

for the compactly supported cohomology groups of $\overline{X} = X \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. These are finite-dimensional $\mathbb{Q}_\ell$-vector spaces on which the Galois group $\operatorname{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts. Let

$$\varphi_q \in \operatorname{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \tag{2.0.2}$$

denote the Frobenius automorphism. It is known (SGA 5 XIII 3.2.1) that the zeta function of X (XXII 3.0.1) is given by

$$Z(t, X/\mathbb{F}_q) = \prod_{i \geq 0} \det\Big(1 - t\varphi_q^{-1} \mid H_c^i(\overline{X}, \mathbb{Q}_\ell)\Big)^{(-1)^{i+1}}. \tag{2.0.3}$$

Deligne has recently proved, generalizing a result of Washnitzer for smooth projective X, the following theorem, which is proved in the appendix below (5.5.3).

Theorem 2.1.

The eigenvalues of φ_q^{-1} on $H_c^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers; if $i > \dim X$, they are divisible, as algebraic integers, by $q^{i-\dim X}$.

In particular, tautologically, we obtain:

(2.1 bis). The eigenvalues of φ_q^{-1} on $H_c^i(\overline{X}, \mathbb{Q}_\ell(-j))$ are algebraic integers divisible by q^j. II≈**378**

Corollary 2.2.

Let X be proper, smooth, and geometrically connected over $\mathbb{F}_q$. Then the eigenvalues of φ_q^{-1} on $N^j H^q(\overline{X}, \mathbb{Q}_\ell)$ (notation of XX 2.0.6) are algebraic integers divisible by q^j.

Proof. The assertion is invariant under a finite extension $\mathbb{F}_{q^M}/\mathbb{F}_q$, which amounts to replacing the eigenvalues under consideration by their M-th powers. After making such a base change, the definition gives finitely many projective, smooth, geometrically connected schemes T_i over $\mathbb{F}_q$, of dimensions d_i, and algebraic cycles Z_i on $T_i \times_{\mathbb{F}_q} X$ of codimension $j + d_i$, such that the homomorphism of $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$-modules

$$\bigoplus_i H^{q-2j}(\overline{T}_i, \mathbb{Q}_\ell(-j)) \xrightarrow{\oplus_i \mathrm{Cl}(Z_i)} N^j H^q(\overline{X}, \mathbb{Q}_\ell)$$

is surjective. The conclusion follows from (2.1 bis). II≈**379**

3. Application to complete intersections

3.0.

Let X be a smooth complete intersection of dimension $n \geq 1$ in $\mathbb{P}^N$ over a field k. For $\ell \neq \mathrm{char}(k)$, weak Lefschetz and duality give (XI 1.6)

$$H^i(\overline{X}, \mathbb{Q}_\ell) = \begin{cases} 0 & \text{if } i \text{ is odd and } i \neq n, \\ \mathbb{Q}_\ell\Big(-\dfrac{i}{2}\Big) & \text{if } i \text{ is even, } i \leq 2n, \text{ and } i \neq n, \\ H^n(\overline{X}, \mathbb{Q}_\ell) & \text{if } i = n. \end{cases} \tag{3.0.1}$$

For $k = \mathbb{F}_q$ this gives the following formula for the zeta function:

$$Z(t, X/\mathbb{F}_q) \prod_{i=0}^{n} (1 - q^i t) = \det\Big(1 - t\varphi_q^{-1} \mid \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell)\Big)^{(-1)^{n+1}}. \tag{3.0.2}$$

Here we have used the primitive decomposition and (3.0.1).

Corollary 3.0.3.

Under the hypotheses of 3.0, with $k = \mathbb{F}_q$, there is a congruence modulo p

$$\det\Big(1 - t\varphi_q^{-1} \mid H^n(\overline{X}, \mathbb{Q}_\ell)\Big) \equiv \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)).$$

Proof. By the congruence formula (XXII 3.2.1), and since

$$H^i(X, \mathcal{O}_X) \simeq \begin{cases} \mathbb{F}_q & i = 0, \\ 0 & 1 \leq i \leq n-1, \\ 0 & i \geq n+1, \end{cases} \tag{3.0.4}$$

we have

$$Z(t, X/\mathbb{F}_q) \equiv (1-t)^{-1} \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X))^{(-1)^{n+1}} \pmod{p}. \tag{3.0.5}$$

Consequently, (3.0.5) and (3.0.2) give

$$\det\Big(1 - t\varphi_q^{-1} \mid \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell)\Big) \equiv \det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)) \pmod{p}. \tag{3.0.6}$$

The desired conclusion follows because

$$H^n(\overline{X}, \mathbb{Q}_\ell) \simeq \begin{cases} \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell) & \text{if } n \text{ is odd,} \\ \mathrm{Prim}^n(\overline{X}, \mathbb{Q}_\ell) \oplus \mathbb{Q}_\ell\Big(-\dfrac{n}{2}\Big) & \text{if } n \text{ is even.} \end{cases} \tag{3.0.7}$$

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Corollary 3.1.

Let X be a smooth complete intersection of dimension $n \geq 1$ over $\mathbb{F}_q$. If

$$H^n(\overline{X}, \mathbb{Q}_\ell) = N^1 H^n(\overline{X}, \mathbb{Q}_\ell), \tag{3.1.0}$$

then

$$\mathfrak{F}_p \quad \text{is nilpotent on} \quad H^n(X, \mathcal{O}_X). \tag{3.1.1}$$

Proof. By Corollary 2.2, (3.1.0) implies

$$\det\Big(1 - t\varphi_q^{-1} \mid H^n(\overline{X}, \mathbb{Q}_\ell)\Big) \equiv 1 \pmod{p} \qquad \text{(indeed, even modulo } q\text{)}. \tag{3.1.2}$$

Together with Corollary 3.0.3, this gives

$$\det(1 - t\mathfrak{F}_q \mid H^n(X, \mathcal{O}_X)) = 1. \tag{3.1.3}$$

Thus $\mathfrak{F}_q$ is nilpotent on $H^n(X, \mathcal{O}_X)$, and hence so is $\mathfrak{F}_p$. This proves the corollary.

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Proposition 3.2.

Let X be a smooth complete intersection of dimension $n \geq 1$ over a field k. If

$$H^n(\overline{X}, \mathbb{Q}_\ell) = N^1 H^n(\overline{X}, \mathbb{Q}_\ell), \tag{3.2.1}$$

then X is special; that is,

$$\mathfrak{F}_p \quad \text{is nilpotent on} \quad H^n(X, \mathcal{O}_X). \tag{3.2.2}$$

Proof. We may suppose that k is finitely generated over $\mathbb{F}_p$ (the case $p = 0$ being trivial), and that there is a finitely generated $\mathbb{F}_p$-subalgebra B of k such that X/k is the generic fiber of a smooth relative complete intersection $\widetilde{X}$:

$$\begin{array}{ccc} \widetilde{X} & \longleftarrow & X = \widetilde{X}_\eta \\ \Big\downarrow{\scriptstyle \pi} & & \Big\downarrow \\ \mathrm{Spec}(B) = S & \longleftarrow & \eta = \mathrm{Sp}(k)\ . \end{array} \tag{3.2.3}$$

The B-module $R^n\pi_*(\mathcal{O}_{\widetilde{X}})$ is locally free, and its formation commutes with every base change. Consequently the set of points

$$\{\, s \in S \mid X_s \text{ is special} \,\} \tag{3.2.4}$$

is closed in S.

By (XX 2.3), after shrinking S we may suppose that II≈382

$$\forall s \in S, \qquad H^n(X_{\bar{s}}, \mathbb{Q}_\ell) = N^1 H^n(X_{\bar{s}}, \mathbb{Q}_\ell). \tag{3.2.5}$$

By Corollary 3.1, X_s is special for every closed point s of S. It follows that the closed subset (3.2.4) is all of S and hence, in particular, that $X = \widetilde{X}_\eta$ is special.

4. Conclusions

Combining Theorem 1.1 with Proposition 3.2 gives the following result.

Theorem 4.1.

Fix a characteristic p, a dimension n, and a multidegree $(d_1, \ldots, d_r)$. Let

$$X_V \longrightarrow V \tag{4.1.0}$$

denote the universal smooth complete intersection of dimension n and multidegree $(d_1, \ldots, d_r)$ in characteristic p. If

$$\sum_{i=1}^{r} d_i \geq n + r + 1, \tag{4.1.1}$$

then there is a dense open subset $U \subset V$ such that

$$\text{for every } u \in U, \qquad H^n(X_{\bar{u}}, \mathbb{Q}_\ell) \neq N^1 H^n(X_{\bar{u}}, \mathbb{Q}_\ell). \tag{4.1.2}$$

Combining Theorem 1.3 with Proposition 3.2 likewise gives the following. II≈383

Theorem 4.2.

Let X be a smooth complete intersection of dimension $n \geq 1$ over a field k. Fix an integer

$$d \geq n + 1. \tag{4.2.1}$$

Let

$$X \cdot H_u \longrightarrow U \tag{4.2.2}$$

denote the universal family of sections of X by hypersurfaces of degree d that are smooth of dimension $n-1$. There is a dense open subset $T \subset U$ such that

$$\text{for every } t \in T, \qquad H^{n-1}(X \cdot H_{\bar{t}}, \mathbb{Q}_\ell) \neq N^1 H^{n-1}(X \cdot H_{\bar{t}}, \mathbb{Q}_\ell). \tag{4.2.3}$$

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5. Appendix: Integrality Theorems

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By P. Deligne

5.0.

Let p be a prime number, q a power of p, $\mathbb{F}_q$ a field with q elements, E the completion of a number field at a place dividing a prime number $\ell \neq p$, and T a set of prime numbers (for example, $T = \varnothing$).

An element α of an algebraic closure of E is called *T-integral* if it is algebraic over $\mathbb{Q}$ and integral over $\mathbb{Z}[(1/r)_{r\in T}]$. An endomorphism F of a finite-dimensional E-vector space is called T-integral if its eigenvalues are T-integral.

Let k be a finite field with q' elements, let $\bar{k}$ be an algebraic closure of k, and let

$$\mathfrak{G} = \mathrm{Gal}(\bar{k}/k)$$

be the Galois group of k. We denote by $f \in \mathfrak{G}$ the Frobenius substitution

$$f(x) = x^{q'},$$

and by F the "geometric" Frobenius, the inverse of f in $\mathfrak{G}$. A representation ρ of $\mathfrak{G}$ on a finite-dimensional E-vector space is called T-integral if $\rho(F)$ is T-integral.

If X is a scheme of finite type over $\mathbb{F}_q$ and x is a closed point of X, we denote by F_x the geometric Frobenius element of the Galois group of $k(x)$. If $\bar{x}$ is a geometric point of X lying over x and $\underline{G}$ is an E-sheaf on X, then F_x acts, by transport of structure, on the fiber $G_{\bar{x}}$.

Definition 5.1.

A constructible E-sheaf $\underline{G}$ on a scheme X of finite type over $\mathbb{F}_q$ is called T-integral if, for every geometric point $\bar{x}$ of X lying over a closed point x of X, F_x is a T-integral automorphism of $\underline{G}_{\bar{x}}$.

If $a : X \longrightarrow \mathrm{Spec}(\mathbb{F}_q)$ is a scheme of finite type over $\mathbb{F}_q$ and $\underline{G}$ is a constructible E-sheaf on II ≈385
X, we write $\underline{H}^*(X, \underline{G})$ for the E-sheaves $R^*a_*\underline{G}$ on $\mathrm{Spec}(\mathbb{F}_q)_{\acute{e}t}$. If $\overline{\mathbb{F}}_q$ is an algebraic closure of $\mathbb{F}_q$ and $\overline{X} = X \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$, these sheaves identify with the cohomology groups $H^*(\overline{X}, \underline{G})$ regarded as Galois modules, that is, endowed with an action of the geometric Frobenius F. We use the same convention for cohomology with proper support.

Lemma 5.2.1.

Let $\underline{G}$ be a constructible E-sheaf on a scheme X of finite type over $\mathbb{F}_q$, of dimension at most n.

(i) There is a closed subset Y of X, of dimension 0, such that the evident map

$$\underline{H}^0(X, \underline{G}) \longrightarrow \underline{H}^0(Y, \underline{G})$$

is injective.

(ii) If X is separated and U is an open subset of X whose complement Z has dimension less than n, there is a closed subset Y of U, of dimension 0, and a surjective map

$$\underline{H}^0(Y, \underline{G})(-n) \longrightarrow \underline{H}^{2n}_c(X, \underline{G}).$$

Proof. Assertion (i) is evident. We prove (ii). After replacing X by X_{red} and shrinking U, we may suppose that U is smooth and that $\underline{G}|_U$ is twisted constant. Since $\dim Z < n$, we have

$$0 = \underline{H}^{2n-1}_c(Z, \underline{G}) \longrightarrow \underline{H}^{2n}_c(U, \underline{G}) \xrightarrow{\sim} \underline{H}^{2n}_c(X, \underline{G}) \longrightarrow \underline{H}^{2n}_c(Z, \underline{G}) = 0,$$

and Poincaré duality (SGA 4, Exposé XVIII, and [2]) gives

$$\underline{H}^{2n}_c(U, \underline{G}) \simeq \underline{H}^0(U, \underline{G}^\vee(n))^\vee.$$

The conclusion follows from (i).

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Theorem 5.2.2.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension at most n, and let $\underline{G}$ be a constructible T-integral E-sheaf. Then the endomorphisms F and $q^{n-i}F$ of $\underline{H}^i_c(X, \underline{G})$ are T-integral.

Proof. The case $\dim X \leq 0$ is trivial, and the cases $i = 0$ and $i = 2n$ follow from Lemma 5.2.1. Let $Z(X, \underline{G}, t)$ be the formal power series

$$Z(X, \underline{G}, t) = \prod_{x \in X} \det\Big(1 - F_x t^{\deg(x)} \,;\, \underline{G}_x\Big)^{-1} \in E[[t]].$$

By SGA 5, Exposé XV, 3.2, one has

$$Z(X, \underline{G}, t) = \prod_i \det\Big(1 - Ft \,;\, \underline{H}^i_c(X, \underline{G})\Big)^{(-1)^{i+1}}. \tag{5.2.3}$$

We first prove the theorem for $n = 1$. Then $\underline{H}^i_c(X, \underline{G}) = 0$ for $i > 2$ (SGA 4, Exposé X, 4.3, or Exposé XVII, 5.2.8.1). By hypothesis the coefficients of $Z(X, \underline{G}, t)$ are T-integral. By the cases $i = 0$ and $i = 2n = 2$ already established, the product of $Z(X, \underline{G}, t)$ with

$$\det\Big(1 - Ft \,;\, \underline{H}^0_c(X, \underline{G})\Big) \det\Big(1 - Ft \,;\, \underline{H}^2_c(X, \underline{G})\Big)$$

still has T-integral coefficients. This product is precisely $\det(1 - Ft \,;\, \underline{H}^1_c(X, \underline{G}))$, proving the theorem for $n = 1$.

We now proceed by induction on $n = \dim X$. Let U be a Zariski-open subset of X whose complement Z has dimension less than n, and suppose there is a morphism $f : U \longrightarrow Y$ to a curve, with fibers of dimension at most $n - 1$:

$$\begin{array}{ccccc} U & \xhookrightarrow{j} & X & \xhookleftarrow{i} & Z \\ {\scriptstyle f}\downarrow & & & & \\ Y & & & & \end{array} \quad .$$

The exact sequence

$$0 \longrightarrow j_! j^* \underline{G} \longrightarrow \underline{G} \longrightarrow i_! i^* \underline{G} \longrightarrow 0$$

induces exact sequences

$$\underline{H}^i_c(U, \underline{G}) \longrightarrow \underline{H}^i_c(X, \underline{G}) \longrightarrow \underline{H}^i_c(Z, \underline{G}).$$

By induction it is therefore enough to verify the theorem for $\underline{G}$ on U. Applying the induction hypothesis to the fibers of f, together with proper base change, shows that the sheaves $R^i f_! \underline{G}$ and $R^i f_! \underline{G}(i - (n-1))$ on Y are T-integral. The Leray spectral sequence for f is

$$E_2^{pq} = \underline{H}^p_c(Y, R^q f_! \underline{G}) \Longrightarrow \underline{H}^{p+q}_c(U, \underline{G}).$$

By what precedes and by the theorem already proved for Y, both E_2^{pq} and $E_2^{pq}(p + q - n)$ are T-integral. The theorem follows.

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Corollary 5.3.

Let X be a scheme of finite type over $\mathbb{F}_q$, of dimension at most 1, let U be an open subset of X, let $j : U \hookrightarrow X$ be the inclusion, and let $\underline{G}$ be a constructible E-sheaf on U.

(a) If $\underline{G}$ is T-integral, then $j_* \underline{G}$ is T-integral.

(b) If $G^\vee_x$ is T-integral for every $x \in U$, then $(j_* \underline{G})^\vee_x$ is T-integral for every $x \in X$.

Proof. We reduce first to the case where X is affine, and then to the case where X is projective. II≈388
Define $\underline{H}$ by the exact sequence

$$0 \longrightarrow j_!\underline{G} \longrightarrow j_*\underline{G} \longrightarrow \underline{H} \longrightarrow 0.$$

If Y is the finite support of $\underline{H}$, the corresponding cohomology sequence gives an exact sequence

$$\underline{H}^0(U,\underline{G}) \longrightarrow \underline{H}^0(Y,\underline{H}) \longrightarrow \underline{H}^1_c(U,\underline{G}).$$

Lemma 5.2.1 and Theorem 5.2.2 show that the two outer terms are T-integral. Hence $\underline{H}^0(Y,\underline{H})$ is T-integral, which proves (a).

Let X' be the normalization of the pure one-dimensional part of X_{red}, and choose an open subset U' of U, dense in the smooth one-dimensional locus of U_{red}, such that $\underline{G}|_{U'}$ is twisted constant:

$$\begin{array}{ccccc} U' & \overset{j'}{\hookrightarrow} & X' & & \\ \downarrow & & \downarrow p & & \\ U & \overset{j}{\hookrightarrow} & X & \hookleftarrow & Y \end{array} \quad ,$$

Then

$$(p_*j'_*\underline{G})|_Y \simeq (j_*\underline{G})|_Y.$$

It is therefore enough to treat the case in which X is smooth of dimension one and $\underline{G}|_U$ is twisted constant. The sheaf $\underline{G}^\vee$ is then T-integral, so we already know that $(j_*\underline{G}^\vee)_x$ is T-integral for every $x \in X$.

Let $\mathfrak{G}_x$ denote the Galois group of the fraction field of $\widehat{\mathcal{O}}_{X,x}$, and let $\mathfrak{I}_x$ be the inertia group. The II≈389
sheaf $\underline{G}$ defines a representation of $\mathfrak{G}_x$, again denoted by $\underline{G}$. The $\mathfrak{G}_x/\mathfrak{I}_x = \mathrm{Gal}(\overline{k(x)}/k(x))$-module $(j_*\underline{G}^\vee)_x$ (respectively, $(j_*\underline{G})^\vee_x$) identifies with the module of invariants (respectively, coinvariants) in $\underline{G}^\vee$. It remains to prove:

Lemma 5.3.1.

Let K be a local field with residue field $\mathbb{F}_q$, let G be its Galois group, let I be the inertia group, and let ρ be a continuous representation of G on an n-dimensional E-vector space V (cf. 5.0). Let V^I and V_I denote respectively the space of invariants and the space of coinvariants under I, regarded as G/I-modules. Put $r = \dim_E V^I$, and let $\alpha_1, \ldots, \alpha_r$ be the eigenvalues of F on V^I. Then there are integers $m_i \geq 0$ such that the eigenvalues of F on V_I are

$$\alpha_i q^{m_i} \qquad (1 \leq i \leq r).$$

Proof. Let I' be the largest subgroup of I of order prime to ℓ. The group $\rho(I')$ is finite, since it is a compact subgroup of $\mathrm{GL}(V)$ of order prime to ℓ. Hence

$$V_I \simeq (V^{\rho(I')})_I.$$

After replacing V by $V^{I'}$, we may therefore suppose that I' acts trivially on V. As G/I-modules one has

$$I/I' \simeq \mathbb{Z}_\ell(1).$$

Let T be a topological generator of I/I'. For each α, let $V^{(\alpha)}$ be the generalized eigenspace of T on V with eigenvalue α. Then

$$V = \bigoplus_\alpha V^{(\alpha)},$$

and the decomposition

$$V = V^{(1)} \oplus \bigoplus_{\alpha \neq 1} V^{(\alpha)}$$

is stable under G and compatible with duality. We may thus replace V by $V^{(1)}$ and suppose that I acts unipotently.

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Lemma 5.3.2.

Let V be a finite-dimensional vector space over a field k of characteristic 0, let I be a one-parameter additive group, let $\rho : I \longrightarrow \mathrm{GL}(V)$ be a representation, and let N be an infinitesimal generator of I. Set

$$V^{I,k} = V^I \cap \mathrm{Im}(N^k)$$

and

$$V_{I,k} = \mathrm{Im}\Big(\mathrm{Ker}(N^{k+1}) \longrightarrow V_I\Big).$$

Then N^k induces an isomorphism

$$N^k : V_{I,k}/V_{I,k-1} \xrightarrow{\sim} V^{I,k}/V^{I,k+1},$$

and defines canonical isomorphisms

$$V_{I,k}/V_{I,k-1} \otimes I^{\otimes k} \simeq V^{I,k}/V^{I,k+1}.$$

Proof. Via N, representations V of I identify with $k[T]$-modules killed by a power of T. It is enough to verify Lemma 5.3.2 when V corresponds to an indecomposable $k[T]$-module $k[T]/T^k$, where the assertion is immediate.

We now finish the proof of Lemma 5.3.1. Lemma 5.3.2 supplies increasing and decreasing filtrations $V_{I,k}$ and $V^{I,k}$ of V_I and V^I, together with isomorphisms of G/I-modules

$$V^{I,k}/V^{I,k+1} \simeq (V_{I,k}/V_{I,k-1})(k).$$

The assertion follows.

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Theorem 5.4.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension n, and let $\underline{G}$ be a constructible E-sheaf on X. Suppose that, for every $x \in X$, the endomorphism F_x of $G_x^\vee$ is T-integral. Then the endomorphisms $q^i F^\vee$ and $q^n F^\vee$ of $\underline{H}^i_c(X, \underline{G})^\vee$ are T-integral.

Proof. As in Theorem 5.2.2, we reduce to the case in which X is a smooth curve and $\underline{G}$ is twisted constant on X. Let j be the inclusion of X in a smooth projective curve X' compactifying it.

By Poincaré duality, $\underline{H}^i(X', j_*\underline{G})$ is dual to $\underline{H}^{2-i}(X', j_*\underline{G}^\vee(1))$. By Corollary 5.3(a), Theorem 5.2.2 applies to $j_*\underline{G}^\vee$:

$$\begin{aligned}
\underline{H}^0(X', j_*\underline{G}^\vee) &\simeq \underline{H}^2(X', j_*\underline{G})^\vee(-1) && \text{is } T\text{-integral},\\
\underline{H}^1(X', j_*\underline{G}^\vee) &\simeq \underline{H}^1(X', j_*\underline{G})^\vee(-1) && \text{is } T\text{-integral},\\
\underline{H}^2(X', j_*\underline{G}^\vee)(1) &\simeq \underline{H}^0(X', j_*\underline{G})^\vee && \text{is } T\text{-integral}.
\end{aligned}$$

This proves the theorem for $j_*\underline{G}$.

Define $\underline{H}$ by the exact sequence

$$0 \longrightarrow j_!\underline{G} \longrightarrow j_*\underline{G} \longrightarrow \underline{H} \longrightarrow 0.$$

If Y is the support of $\underline{H}$, the long exact cohomology sequence provides

$$\underline{H}^0_c(X, \underline{G}) \hookrightarrow \underline{H}^0(X', j_*\underline{G}) \quad ,$$

$$\underline{H}^0(Y, \underline{H}) \longrightarrow \underline{H}^1_c(X, \underline{G}) \longrightarrow \underline{H}^1(X', j_*\underline{G}) \quad ,$$

$$\underline{H}^2_c(X, \underline{G}) \xrightarrow{\sim} \underline{H}^2(X', j_*\underline{G}) \quad ,$$

Thus the theorem for $(X, \underline{G})$ follows from the theorem for $(X', j_*\underline{G})$.

5.5.

The preceding method of fibration into curves also gives Lang–Weil estimates [3] for the complex absolute values of the eigenvalues of Frobenius.

Let X be a scheme of finite type over $\mathbb{F}_q$, let $\underline{G}$ be a constructible E-sheaf on X, and let $\lambda \in \mathbb{R}$. II≈**392**
Suppose that, for every point $x \in X$, with $\#k(x) = q_x$, the eigenvalues of F_x acting on G_x are algebraic numbers α all of whose complex absolute values satisfy

$$|\alpha| \leq q_x^{\lambda}.$$

We shall then write

$$\log |\underline{G}| \leq \lambda.$$

Lemma 5.5.1.

Let X be a curve over $\mathbb{F}_q$, by which we mean here a separated scheme X of finite type over $\mathbb{F}_q$ and of dimension at most 1. Let $\underline{G}$ be a constructible E-sheaf on X such that $\log |\underline{G}| \leq \lambda$, with $\lambda \in \mathbb{R}$. Then the complex absolute values of the eigenvalues α of F acting on $\underline{H}^1_c(X, G)$ satisfy

$$|\alpha| \leq q^{\lambda+1}.$$

Proof. Let σ be a complex embedding of E, or simply a complex embedding of the algebraic closure of $\mathbb{Q}$ in E. It is enough to verify that, for every σ, the zeros β of the rational function $\sigma Z(X, \underline{G}, t)$ satisfy

$$|\beta^{-1}| \leq q^{\lambda+1}.$$

Indeed, Lemma 5.2.1 already shows that, if α is an eigenvalue of Frobenius acting on $\underline{H}^0_c(X, \underline{G})$ or on $\underline{H}^2_c(X, \underline{G})$, then $|\sigma\alpha| \leq q^{\lambda+1}$.

We verify that the series II≈**393**

$$\log \sigma Z(X, \underline{G}, t) = \sum_k \frac{t^k}{k} \sum_{x \in X(\mathbb{F}_{q^k})} \sigma \operatorname{Tr}(F, x^* \underline{G})$$

converges for $|t| < q^{-\lambda-1}$. Representing X as a ramified covering of $\mathbb{P}^1$, one obtains the elementary estimate

$$\#X(\mathbb{F}_{q^k}) \leq Aq^k.$$

On the other hand, the hypothesis gives

$$|\sigma \operatorname{Tr}(F, x^* \underline{G})| \leq B(q^k)^{\lambda}.$$

Thus the coefficient c_k of t^k in $\log \sigma Z(X, \underline{G}, t)$ satisfies

$$|c_k| \leq \frac{1}{k} C(q^{1+\lambda})^k,$$

and the assertion follows.

Theorem 5.5.2.

Let X be a separated scheme of finite type over $\mathbb{F}_q$, of dimension at most n, and let G be a constructible E-sheaf on X such that $\log |G| \leq \lambda$, with $\lambda \in \mathbb{R}$. Then the complex absolute values of the eigenvalues α of Frobenius acting on $\underline{H}^i_c(X, \underline{G})$ satisfy

$$|\alpha| \leq q^{\lambda+i} \qquad \text{and} \qquad |\alpha| \leq q^{\lambda+n}.$$

Proof. If $n \leq 0$, the theorem is trivial. If $n = 1$, its cases $i = 0$ and $i = 2$ follow from Lemma 5.2.1, while the case $i = 1$ follows from Lemma 5.5.1. We then proceed by induction on n, as in the proof of Theorem 5.2.2.

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Corollary 5.5.3.

Let X be a separated scheme of finite type and dimension n over $\mathbb{F}_q$. Then the eigenvalues α of the geometric Frobenius endomorphism F of $\underline{H}^i_c(X, \mathbb{Q}_\ell)$ are algebraic integers. Moreover:

(i) for $\ell' \neq p$, α is an ℓ'-adic unit;

(ii) for $i > n$, α is divisible by q^{i-n};

(iii) for $i \leq n$ (respectively, $i \geq n$), $q^i\alpha^{-1}$ (respectively, $q^n\alpha^{-1}$) is an algebraic integer;

(iv) for $0 < i$, the complex absolute values of α satisfy

$$|\alpha| \leq q^{i-\frac{1}{2}} \qquad \text{and} \qquad |\alpha| \leq q^n.$$

Proof. By Theorem 5.2.2, α and $q^{n-i}\alpha$ are algebraic integers. Assertions (i)–(iii) then follow from Theorem 5.4. For $i > n$, assertion (iv) follows from Theorem 5.5.2. To obtain (iv) when $i \leq n$, we must combine Theorem 5.5.2 with the Riemann hypothesis for curves.

We argue by induction on n, the case $n = 0$ being trivial. If U is an open subset of X whose complement Z has dimension less than n, the exact sequence

$$\underline{H}^i_c(U, \mathbb{Q}_\ell) \longrightarrow \underline{H}^i_c(X, \mathbb{Q}_\ell) \longrightarrow \underline{H}^i_c(Z, \mathbb{Q}_\ell)$$

shows that it is enough to prove the desired assertion for U. After first replacing X by X_{red}, this reduces us to the case in which X is smooth and affine and there is a smooth morphism $f : X \longrightarrow S$ satisfying:

(a) $\dim S \leq n - 1$;

(b) the fibers of f are purely one-dimensional.

Since the fibers of f are affine curves, one has

$$R^0 f_! \mathbb{Q}_\ell = 0.$$

By Weil [4],

$$\log|R^1 f_! \mathbb{Q}_\ell| \leq \frac{1}{2}, \tag{5.5.3.1}$$

and Lemma 5.2.1 applied to the fibers of f gives

$$\log|R^2 f_! \mathbb{Q}_\ell| \leq 1. \tag{5.5.3.2}$$

The Leray spectral sequence for f reduces here to a long exact sequence; the relevant part is

$$\underline{H}^{i-2}_c(S, R^2 f_! \mathbb{Q}_\ell) \longrightarrow \underline{H}^i_c(X, \mathbb{Q}_\ell) \longrightarrow \underline{H}^{i-1}_c(S, R^1 f_! \mathbb{Q}_\ell).$$

By (5.5.3.1) (respectively, (5.5.3.2)) and Theorem 5.5.2, the complex absolute values of the eigenvalues α of Frobenius on $\underline{H}^{i-1}_c(S, R^1 f_! \mathbb{Q}_\ell)$ (respectively, on $\underline{H}^{i-2}_c(S, R^2 f_! \mathbb{Q}_\ell)$) satisfy

$$|\alpha| \leq q^{i-1+\frac{1}{2}} = q^{i-\frac{1}{2}}$$

(respectively,

$$|\alpha| \leq q^{i-2+1} = q^{i-1} \leq q^{i-\frac{1}{2}}).$$

The corollary follows.

5.6.

To obtain results analogous to the preceding ones for cohomology without supports, it seems necessary to have resolution of singularities available. In dimension at most 1, the problem is solved by Corollary 5.3.

Theorem 5.6.

Let N be an integer, and suppose that strong resolution of singularities in the sense of Hironaka is available for schemes of finite type over $\mathbb{F}_q$ of dimension at most N (at present one may therefore take $N = 2$ [1]). Let $f : X \longrightarrow Y$ be a morphism of schemes of finite type over $\mathbb{F}_q$, and let $\underline{G}$ be a constructible T-integral E-sheaf on X. If $\dim X \leq N$, then the E-sheaves $R^i f_* \underline{G}$ on Y are T-integral.

Proof. We argue by induction on the dimension $n \leq N$ of X. Let $j : U \hookrightarrow X$ be an open subset of X, smooth and affine, such that $\underline{G}|_U$ is twisted constant and the dimension of its complement Z is less than n. By resolution of singularities, the sheaves $R^i j_*(j^* \underline{G})$ are constructible (SGA 5, Exposé I; [2]); for $i > 0$ they are supported on Z, and for $i = 0$, $j_*(j^* \underline{G})$ differs from $\underline{G}$ only on Z. The induction hypothesis therefore shows that it is enough to prove the theorem for the sheaf $j^* \underline{G}$ and the morphisms j and $f \circ j$. Thus we may further suppose that X is smooth and affine and that $\underline{G}$ is twisted constant, that is, comes from a continuous representation of the fundamental group of X on a finite-dimensional E-vector space.

Let R be the ring of integers of E, let λ be a uniformizer, let e be the absolute ramification index, II ≈**397**
and choose an integer $b > e/(\ell - 1)$. Since X is smooth, hence normal, and its fundamental group is profinite, there is a twisted-constant R-sheaf $\underline{G}^\circ$ such that

$$\underline{G} \simeq \underline{G}^\circ \otimes_R E.$$

Let $p : X' \longrightarrow X$ be the étale covering that trivializes the locally constant sheaf $\underline{G}^\circ \otimes_R R/\lambda^b$. The sheaf $\underline{G}$ is a direct summand of $p_* p^* \underline{G}$, so it is enough to prove the theorem for $p^* \underline{G}$ and $f \circ p$. It remains to treat the case in which the following conditions hold:

(∗) X is smooth and affine, and there is a twisted-constant R-sheaf $\underline{G}^\circ$ such that $\underline{G} \simeq \underline{G}^\circ \otimes_R E$ and $\underline{G}^\circ \otimes_R R/\lambda^b$ is constant.

Factor f as a proper morphism followed by an open immersion, $f = \bar{f} \circ j$:

$$\begin{array}{ccc} X & \xrightarrow{\ j\ } & X' \\ {\scriptstyle f}\big\downarrow & \swarrow_{\bar{f}} & \\ Y & & \end{array} \quad .$$

The theorem has already been proved for $\bar{f}$ (5.2), so it is enough to prove it for j. Applying resolution of singularities to X', we may suppose that X' is smooth and that X is the complement in X' of a normal-crossings divisor which is a union of smooth divisors. Hypothesis (∗) implies that $\underline{G}$ is tamely ramified along this divisor. We then have:

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Lemma 5.6.1.

Let X be a smooth scheme of finite type over $\mathbb{F}_q$, let D be a normal-crossings divisor in X which is a union of smooth divisors D_i, let $j : U \hookrightarrow X$ be the complement of D in X, and let $\underline{G}$ be a twisted-constant E-sheaf on U, tamely ramified along D. If $\underline{G}$ is T-integral, then the sheaves $R^i j_* \underline{G}$ are T-integral.

We shall admit the following result (cf. SGA 1, Exposé XIII).

Lemma 5.6.2.

(i) Under the hypotheses of Lemma 5.6.1, let D' be a component of D, let D'' be the union of the other components, and put $U' = X - D''$ and $D'^{\circ} = D' \cap U'$:

$$\begin{array}{ccccc} & & D'^{\circ} & \xhookrightarrow[j'_D]{} & D' \\ & & \downarrow{\scriptstyle i^{\circ}} & & \downarrow{\scriptstyle i} \\ U & \xhookrightarrow[j_1]{} & U' & \xhookrightarrow[j']{} & X \end{array} .$$

Then the base-change map

$$i^* R j'_* (R j_{1*} \underline{G}) \longrightarrow R j'_{D*}\, i^{\circ *} (R j_{1*} \underline{G})$$

is an isomorphism, and the sheaves $i^{\circ *} R j_{1*} \underline{G}$ are twisted constant and tamely ramified along $D' - D'^{\circ}$.

(ii) If C is a smooth curve on X, transverse to D at a smooth point x of D,

$$\begin{array}{ccc} C \cap U & \xrightarrow[j_c]{} & C \\ \downarrow{\scriptstyle k'} & & \downarrow{\scriptstyle k} \\ U & \xrightarrow[j]{} & X \end{array} ,$$

then the base-change map

$$k^* R j_* \underline{G} \longrightarrow R j_{c*}\, k'^* \underline{G}$$

is an isomorphism at x.

Let us deduce Lemma 5.6.1 from Lemma 5.6.2 by induction on the dimension, temporarily admitting Lemma 5.6.1 for curves. Lemma 5.6.2(ii) and this hypothesis show that the sheaves $R^i j_* \underline{G}$ are T-integral at every smooth point of D. In particular, in the notation of Lemma 5.6.2(i), the sheaves $R^i j_{1*} \underline{G}$ are T-integral. Lemma 5.6.2(i), together with the induction hypothesis applied to $i^{\circ *} R j_{1*} \underline{G}$ and to j'_D, therefore shows that the sheaves $i^* R^i j_* \underline{G}$ are T-integral, whichever component D' of D is chosen. This proves Lemma 5.6.1. II ≈**399**

It remains to handle curves. Under the hypotheses of Lemma 5.6.1, let x be a point of $X - U$, let $\mathfrak{G}$ be the Galois group of the fraction field of $\widehat{\mathcal{O}}_{x,X}$, and let $\mathfrak{I}$ be the inertia group. Denote again by $\underline{G}$ the representation of $\mathfrak{G}$ defined by the sheaf, and let $\bar{x}$ be a geometric point of X localized at x. There are isomorphisms of Galois $\mathfrak{G}/\mathfrak{I}$-modules

$$(j_* \underline{G})_{\bar{x}} \simeq \underline{G}^{\mathfrak{I}},$$

$$(R^1 j_* \underline{G})_{\bar{x}} \simeq \underline{G}_{\mathfrak{I}}(-1),$$

and

$$(R^i j_* \underline{G})_{\bar{x}} = 0 \qquad \text{for } i \geq 2.$$

The assertion follows from Corollary 5.3(a) and Lemma 5.3.1.

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EXPOSÉ XXII

A CONGRUENCE FORMULA FOR THE ZETA FUNCTION

by N. Katz

0. Introduction

Let X be a proper scheme over $\mathbb{F}_q$. Its zeta function, regarded as an element of $\mathbb{Z}[[T]]$, may be reduced modulo p; we wish to compute it modulo p. We obtain an expression of the kind one might have expected from fixed-point formulas of "Woods Hole" type: an alternating product of characteristic polynomials of a suitable "Frobenius" acting on any one of the following three cohomology spaces:

1) coherent cohomology $H^i(X, \mathcal{O}_X)$ (3.1.1);

2) de Rham cohomology $H^i_{\mathrm{DR}}(X) = \mathbb{H}^i(X, \Omega^\bullet_X)$ (3.1.1);

3) étale cohomology $H^i_{\mathrm{et}}(X, \mathbb{Z}/p\mathbb{Z})$ (3.1.2).

This expression should be viewed as an improvement of a special case of the Woods Hole trace formula, namely that of the structure sheaf and the Frobenius endomorphism. That case is indeed a corollary of the congruence formula (cf. 3.2.1).

The method of this exposé is completely naïve. It consists in reducing to the case of a hypersurface and then "reducing modulo p" Dwork's computation of the actual zeta function of a hypersurface. By "reducing" carefully, one obtains the improvement 5.0.6, due to Ax [1] (and also to Dwork in the smooth case [4, §7]), of the well-known Warning–Chevalley theorem [8].

Another method, entirely cohomological and in the spirit of SGA 5, which also applies to "twisted coefficients," has just been found by Deligne and ought to appear in the new edition of SGA 5. It rests on the trace formula of Woods Hole type and on Deligne's "symmetric Künneth formula" (SGA 4, XVII 5.5.21). II≈**402**

1. A reminder on "p-linear" operations

1.0.

Let V be a finite-dimensional vector space over a field k of characteristic $p > 0$, and let q be a power of p. A map

$$\varphi : V \longrightarrow V \tag{1.0.1}$$

is called *q-linear* if it is additive and satisfies

$$\varphi(\lambda v) = \lambda^q \varphi(v) \qquad \text{for every } \lambda \in k,\ v \in V. \tag{1.0.2}$$

1.0.3.

Notice that the n-th iterate φ^n of a q-linear endomorphism is q^n-linear and that, if k is not perfect, the image $\varphi(V)$ of a q-linear endomorphism need not be a k-vector subspace.

1.0.4.

For every extension K/k, a q-linear endomorphism φ of V gives rise to a unique q-linear endomorphism φ_K of $V_K = V \otimes_k K$ satisfying

$$\varphi_K(\mu \otimes v) = \mu^q \varphi(v) \qquad \text{for } \mu \in K,\ v \in V. \tag{1.0.5}$$

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1.0.6.

A q-linear endomorphism φ of V is called *semisimple* if its image spans V over k, and *nilpotent* if some iterate is zero. Define the following φ-stable k-subspaces:

$$V_{\mathrm{ss}} = \bigcap_{n \geq 1} \langle \varphi^n(V) \rangle_k, \tag{1.0.7}$$

$$V_{\mathrm{nilp}} = \bigcup_{n \geq 1} \mathrm{Ker}(\varphi^n : V \longrightarrow V). \tag{1.0.8}$$

Clearly, the restriction of φ to V_{ss} is semisimple, while its restriction to V_{nilp} is nilpotent. If k is perfect, then

$$V = V_{\mathrm{ss}} \oplus V_{\mathrm{nilp}}. \tag{1.0.9}$$

If k is not perfect, however, (1.0.9) is false, since there exist injective q-linear endomorphisms which are not semisimple. For an arbitrary extension K/k, one has

$$(V_K)_{\mathrm{ss}} = V_{\mathrm{ss}} \otimes_k K. \tag{1.0.10}$$

Proposition 1.1.

Let φ be a q-linear endomorphism of a finite-dimensional vector space V over a separably closed field k. Suppose that φ is semisimple, and put

$$V^{1-\varphi} = \text{the } \mathbb{F}_q\text{-vector space of fixed points of } \varphi = \mathrm{Ker}(1 - \varphi).$$

Then

$$V \xleftarrow{\sim} (V^{1-\varphi}) \otimes_{\mathbb{F}_q} k. \tag{1.1.1}$$

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Corollary 1.1.2.

There is an isomorphism

$$V^{1-\varphi^n} \xleftarrow{\sim} (V^{1-\varphi}) \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}.$$

Proof of Proposition 1.1. (Lang, cf. [6], p. 119.) Let $\underline{e}$ be a k-basis of V, and write

$$\varphi(\underline{e}) = A\underline{e}, \qquad A \in \mathrm{GL}(n,k). \tag{1.1.3}$$

Let

$$F : \mathrm{GL}(n,k) \longrightarrow \mathrm{GL}(n,k) \tag{1.1.4}$$

denote the map which raises every coordinate to the q-th power. For $g \in \mathrm{GL}(n,k)$, the basis $g\underline{e}$ of V consists of fixed points of φ if and only if

$$F(g)Ag^{-1} = 1. \tag{1.1.5}$$

Let $\bar{k}$ be an algebraic closure of k. The group $\mathrm{GL}(n,\bar{k})$ acts on the variety $\mathrm{GL}(n,\bar{k})$ by

$$\begin{cases} \mathrm{GL}(n,\bar{k}) \times \mathrm{GL}(n,\bar{k}) \longrightarrow \mathrm{GL}(n,\bar{k}), \\ (g,A) \longmapsto F(g)Ag^{-1}. \end{cases} \tag{1.1.6}$$

For fixed A, the map

$$g \longmapsto F(g)Ag^{-1} \tag{1.1.7}$$

is étale, as one sees by computing its tangent map. Its orbits are therefore open. Since $\mathrm{GL}(n,\bar{k})$ is irreducible, there is only one orbit; in other words, the map

$$\begin{aligned} \mathrm{GL}(n,\bar{k}) &\longrightarrow \mathrm{GL}(n,\bar{k}), \\ g &\longrightarrow F(g^{-1})\, g \end{aligned} \tag{1.1.8}$$

is surjective. Since it is also étale and k is separably closed, the map II ≈**405**

$$\begin{aligned} \mathrm{GL}(n,k) &\longrightarrow \mathrm{GL}(n,k), \\ g &\longrightarrow F(g^{-1})\, g \end{aligned} \tag{1.1.9}$$

is surjective. Thus V has a k-basis consisting of fixed points of φ. Since φ is q-linear, every fixed point of φ lies in the $\mathbb{F}_q$-vector space spanned by such a k-basis. This proves Proposition 1.1.

Corollary 1.1.10.

Let φ be a q-linear endomorphism of a finite-dimensional vector space V over a separably closed field k. There is a canonical isomorphism

$$V_{\mathrm{ss}} \xleftarrow{\sim} V^{1-\varphi} \otimes_{\mathbb{F}_q} k. \tag{1.1.11}$$

Indeed, it is enough to apply Proposition 1.1 to V_{ss}.

Proposition 1.2.

Let φ be a q-linear endomorphism of a finite-dimensional vector space V over a separably closed field k. Then the additive map

$$1 - \varphi : V \longrightarrow V$$

is surjective.

Proof. There is an exact sequence of abelian groups

$$0 \longrightarrow V_{\mathrm{ss}} \longrightarrow V \longrightarrow V/V_{\mathrm{ss}} \longrightarrow 0.$$

on which $1 - \varphi$ acts. By the snake lemma, it is enough to treat V_{ss}, which is immediate from Proposition 1.1 by using a basis consisting of fixed points, and V/V_{ss}, on which φ is nilpotent and hence $1 - \varphi$ is invertible. II≈**406**

2. Coherent cohomology and étale cohomology

Proposition 2.0.

Let $\bar{X}$ be a proper scheme over a separably closed field k, and let

$$\overline{\mathfrak{F}}_p : H^{\cdot}(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow H^{\cdot}(\bar{X}, \mathcal{O}_{\bar{X}}) \tag{2.0.1}$$

denote the p-linear map induced by

$$\overline{\mathfrak{F}}_p : \mathcal{O}_{\bar{X}} \longrightarrow \mathcal{O}_{\bar{X}}, \qquad \overline{\mathfrak{F}}_p(f) = f^p. \tag{2.0.2}$$

Denote by

$$H^{\cdot}_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$$

the étale cohomology groups with coefficients in the constant sheaf $\mathbb{Z}/p\mathbb{Z}$. Then there are canonical isomorphisms

$$H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \simeq (H^i(\bar{X}, \mathcal{O}_{\bar{X}}))^{1-\overline{\mathfrak{F}}_p}. \tag{2.0.3}$$

Proof. There is an exact sequence of sheaves on $\bar{X}_{\mathrm{et}}$:

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathcal{O}_{\bar{X}} \xrightarrow{1-\mathfrak{F}_p} \mathcal{O}_{\bar{X}} \longrightarrow 0, \tag{2.0.4}$$

which gives an exact cohomology sequence

$$\cdots \longrightarrow H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i_{\mathrm{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \xrightarrow{1-\mathfrak{F}_p} H^i_{\mathrm{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow \cdots . \tag{2.0.5}$$

One has $H^i_{\mathrm{et}}(\bar{X}, \mathcal{O}_{\bar{X}}) \simeq H^i(\bar{X}, \mathcal{O}_{\bar{X}})$; this is a finite-dimensional k-vector space because $\bar{X}$ is proper. Applying Proposition 1.2 gives exact sequences

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$$0 \longrightarrow H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \xrightarrow{1-\mathfrak{F}_p} H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \longrightarrow 0, \tag{2.0.6}$$

which proves the assertion. Proposition 1.1.10 now gives the following corollary of Proposition 2.0.

Corollary 2.1.

For $\bar{X}$ proper over a separably closed field k, there is a canonical isomorphism

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}})_{\mathrm{ss}} \xleftarrow{\sim} H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k. \tag{2.1.1}$$

2.2.

Let now $f : X \longrightarrow \mathrm{Spec}(\mathbb{F}_q)$ be a proper scheme over $\mathbb{F}_q$. Let k be an algebraic closure of $\mathbb{F}_q$, and put

$$\bar{X} = X \times_{\mathbb{F}_q} k. \tag{2.2.1}$$

The étale sheaf $R^i f_*(\mathbb{Z}/p\mathbb{Z})$ on $\mathrm{Spec}(\mathbb{F}_q)$ identifies with $H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$, viewed as a $\mathrm{Gal}(k/\mathbb{F}_q)$-module. Let

$$\mathfrak{F}_q : H^i(X, \mathcal{O}_X) \longrightarrow H^i(X, \mathcal{O}_X) \tag{2.2.2}$$

denote the linear operation (it is q-linear, but the ground field is $\mathbb{F}_q$) induced by

$$\mathfrak{F}_q : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad \mathfrak{F}_q(f) = f^q. \tag{2.2.3}$$

Proposition 2.2.4.

Let $\varphi_q \in \mathrm{Gal}(k/\mathbb{F}_q)$ be the "Frobenius" element $\varphi_q(\lambda) = \lambda^q$, and let $i \in \mathbb{Z}$. There is an identity of characteristic polynomials

$$\det_{\mathbb{F}_p}\Big(1 - t\varphi_q^{-1} \mid H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\Big) = \det_{\mathbb{F}_q}\Big(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\Big). \tag{2.2.4.1}$$

Here the left-hand side is the characteristic polynomial of the element φ_q^{-1} of $\pi_1(\mathrm{Spec}(\mathbb{F}_q))$ acting on the Galois module $H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})$.

Proposition 2.2.4 is a corollary of the following result.

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Proposition 2.2.5.

Let X be proper over $\mathrm{Spec}(\mathbb{F}_q)$. There is a canonical isomorphism

$$H^i(X, \mathcal{O}_X)_{\mathrm{ss}} \otimes_{\mathbb{F}_q} k \simeq H^i_{\mathrm{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k \tag{2.2.6}$$

which carries the automorphism induced by $\mathfrak{F}_q$ on the left-hand side to the automorphism induced by φ_q^{-1} on the right-hand side. The latter may also be interpreted as $\mathrm{fr}_q^{X*} \otimes_{\mathbb{F}_p} k$, where $\mathrm{fr}_q^X : \bar{X} \longrightarrow \bar{X}$ is the Frobenius k-endomorphism (cf. SGA 5, XVIII 2.3.4).

Proof. Since

$$H^i(\bar{X}, \mathcal{O}_{\bar{X}}) \simeq H^i(X, \mathcal{O}_X) \otimes_{\mathbb{F}_q} k, \tag{2.2.7}$$

Proposition 1.0.10 gives

(2.2.8) $$H^i(\bar{X},\mathcal{O}_{\bar{X}})_{\mathrm{ss}} \simeq H^i(X,\mathcal{O}_X)_{\mathrm{ss}} \otimes_{\mathbb{F}_q} k.$$

Thus the isomorphism (2.2.6) is obtained by composing (2.2.8) with (2.1.1).
In view of (2.2.7), the q-linear operation

(2.2.9) $$\mathfrak{F}_q : H^i(X,\mathcal{O}_X) \longrightarrow H^i(X,\mathcal{O}_X)$$

has a q-linear extension (cf. 1.0.5)

(2.2.10) $$\overline{\mathfrak{F}}_q : H^i(\bar{X},\mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X},\mathcal{O}_{\bar{X}}),$$

and also a k-linear extension, again denoted by $\mathfrak{F}_q$,

(2.2.11) $$\mathfrak{F}_q : H^i(\bar{X},\mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X},\mathcal{O}_{\bar{X}}).$$

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(2.2.12) $$\varphi_q : H^i(\bar{X},\mathcal{O}_{\bar{X}}) \longrightarrow H^i(\bar{X},\mathcal{O}_{\bar{X}})$$

denote the q-linear operation whose fixed points are $H^i(X,\mathcal{O}_X)$. Under (2.2.7), the operators $\mathfrak{F}_q$, φ_q, and $\overline{\mathfrak{F}}_q$ may also be written respectively as $\mathfrak{F}_q \otimes \mathrm{id}_k$, $\mathrm{id} \otimes \varphi_q$, and $\mathfrak{F}_q \otimes \varphi_q$. They therefore commute with one another, and

(2.2.13) $$\overline{\mathfrak{F}}_q = \varphi_q \mathfrak{F}_q = \mathfrak{F}_q \circ \varphi_q.$$

They all commute with $\overline{\mathfrak{F}}_p$, so $\mathfrak{F}_q$ and φ_q act on both $H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q}$ and $H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_p}$. By (2.2.13), $\mathfrak{F}_q$ and φ_q are inverse to one another on $H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q}$. Hence the action of $\mathfrak{F}_q$ on

$$H^i(\bar{X},\mathcal{O}_{\bar{X}})_{\mathrm{ss}} \simeq H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \otimes_{\mathbb{F}_q} k \qquad \text{(cf. 1.1.11)}$$

is inverse to the action of $\varphi_q \otimes \mathrm{id}_k$ on

$$H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \otimes_{\mathbb{F}_q} k.$$

Writing

(2.2.14) $$H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_q} \simeq H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_p} \otimes_{\mathbb{F}_p} \mathbb{F}_q,$$

the automorphism φ_q of the left-hand side of (2.2.14) becomes, via (2.2.14), the automorphism induced by φ_q on $H^i(\bar{X},\mathcal{O}_{\bar{X}})^{1-\overline{\mathfrak{F}}_p} \otimes_{\mathbb{F}_p} \mathbb{F}_q$. Consequently the automorphism induced by φ_q on $H^i(\bar{X},\mathcal{O}_{\bar{X}})_{\mathrm{ss}}$ becomes, via (2.1.1), the automorphism induced by φ_q^{-1} on $H^i_{\mathrm{et}}(\bar{X},\mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{F}_p} k$. This proves Proposition 2.2.5.

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3. The congruence formula: statements and elementary equivalences

I wish to thank Grothendieck and Deligne for explaining to me the variants (3.1.2), (3.2.1)–(3.2.2), and (3.3.1) of (3.1.1).

3.0.

Let X be a scheme of finite type over $\mathbb{F}_q$. The degree of a closed point $y \in X$ is defined to be the degree of its residue field $\mathbb{F}_q(y)/\mathbb{F}_q$. The zeta function of $X/\mathbb{F}_q$, as an element of $\mathbb{Z}[[t]]$, is defined by the formula (cf. also SGA 5, XIII 3.1)

$$Z(t, X/\mathbb{F}_q) = \prod_y (1 - t^{\deg(y)})^{-1}. \tag{3.0.1}$$

By the "zeta function modulo p" we shall always mean the image of $Z(t, X/\mathbb{F}_q)$ in $\mathbb{F}_q[[t]]$ under the "reduction modulo $p\mathbb{Z}[[t]]$" map

$$\mathbb{Z}[[t]] \longrightarrow \mathbb{F}_q[[t]].$$

Theorem 3.1.

Let X be a proper scheme over $\mathbb{F}_q$. Then

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\Big(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\Big)^{(-1)^{i+1}} \pmod{p}, \tag{3.1.1}$$

or, equivalently by Proposition 2.2.4,

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\Big(1 - t\varphi_q^{-1} \mid H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\Big)^{(-1)^{i+1}} \pmod{p}. \tag{3.1.2}$$

Corollary 3.2.

Let X be a proper scheme over $\mathbb{F}_q$. Then

$$\sum_{i \geq 0} (-1)^i \operatorname{Tr}\Big(\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\Big) \equiv \#X(\mathbb{F}_q) \pmod{p}, \tag{3.2.1}$$

or, equivalently by Proposition 2.2.4,

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$$\sum_{i \geq 0} (-1)^i \operatorname{Tr}\Big(\varphi_q^{-1} \mid H^i_{\text{et}}(\bar{X}, \mathbb{Z}/p\mathbb{Z})\Big) \equiv \#X(\mathbb{F}_q) \pmod{p}. \tag{3.2.2}$$

Indeed, take the coefficients of t in the congruences (3.1.1) and (3.1.2).

Theorem 3.3.

Let X be a proper scheme over $\mathbb{F}_q$, and denote by $H^i_{\text{DR}}(X/\mathbb{F}_q)$ the $\mathbb{F}_q$-vector spaces $\mathbb{H}^i(X, \Omega^\bullet_{X/\mathbb{F}_q})$, on which $\mathfrak{F}_q$ acts. Then

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\Big(1 - t\mathfrak{F}_q \mid H^i_{\text{DR}}(X/\mathbb{F}_q)\Big)^{(-1)^{i+1}} \pmod{p}. \tag{3.3.1}$$

Indeed, consider the spectral sequence

$$E_1^{p,q} = H^q(X, \Omega^p_{X/\mathbb{F}_q}) \Longrightarrow H^{p+q}_{\text{DR}}(X/\mathbb{F}_q). \tag{3.3.2}$$

The operator $\mathfrak{F}_q$ acts on it through the endomorphism $\mathfrak{F}_q = (\mathrm{Fr}_q^X)^*$ induced by $\mathrm{Fr}_q^X : X \longrightarrow X$, and kills $E_1^{p,q}$ for $p \neq 0$. It follows that (3.3.1) is equivalent to (3.1.1).

Let us add a small complement. Since formation of the "semisimple part" (cf. 1.0.7) is an exact functor, (3.3.2) gives a spectral sequence

$$(E_1^{p,q})_{\mathrm{ss}} \Longrightarrow H_{\mathrm{DR}}^{p+q}(X/\mathbb{F}_q)_{\mathrm{ss}}. \tag{3.3.3}$$

Since $(E_1^{p,q})_{\mathrm{ss}} = 0$ for $p \neq 0$, we obtain for every i an isomorphism

$$H_{\mathrm{DR}}^i(X/\mathbb{F}_q)_{\mathrm{ss}} \xrightarrow{\sim} H^i(X, \mathcal{O}_X)_{\mathrm{ss}} \tag{3.3.4}$$

that commutes with $\mathfrak{F}_q$. Hence

$$\det\Big(1 - t\mathfrak{F}_q \mid H_{\mathrm{DR}}^i(X/\mathbb{F}_q)\Big) = \det\Big(1 - t\mathfrak{F}_q \mid H^i(X, \mathcal{O}_X)\Big). \tag{3.3.5}$$

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Theorem 3.4.

Let X be a separated scheme of finite type over $\mathbb{F}_q$. Denoting by H_c^i étale cohomology with proper supports (SGA 4, XVII 5.1.9(iii)), one has

$$Z(t, X/\mathbb{F}_q) \equiv \prod_{i \geq 0} \det\Big(1 - t\varphi_q^{-1} \mid H_c^i(\bar{X}, \mathbb{Z}/p\mathbb{Z})\Big)^{(-1)^{i+1}} \pmod{p}. \tag{3.4.1}$$

We begin by proving the equivalence of (3.4.1) and (3.1.1). For the moment, write

$$Z(X) \quad \text{in place of } Z(t, X/\mathbb{F}_q) \tag{3.4.2}$$

$$Z'(X) \quad \text{in place of} \quad \prod \det\Big(1 - t\varphi_q^{-1} \mid H_c^i(\bar{X}, \mathbb{Z}/p\mathbb{Z})\Big)^{(-1)^{i+1}}. \tag{3.4.3}$$

Lemma 3.5.

If $\{U_i\}$ is a finite covering of X by open subschemes, then

$$Z'(X) = \prod_{j \geq 0} \prod_{1 \leq i_0 < \cdots < i_j \leq n} Z'(U_{i_0} \cap \cdots \cap U_{i_j})^{(-1)^j}. \tag{3.5.1}$$

Proof. This follows immediately from the localization spectral sequence (SGA 4, XVII 6.2.10)

$$E_1^{-a,b} = \bigoplus_{i_0 < \cdots < i_a} H_c^b(\bar{U}_{i_0} \cap \cdots \cap \bar{U}_{i_a}, \mathbb{Z}/p\mathbb{Z}) \Longrightarrow H_c^{-a+b}(\bar{X}, \mathbb{Z}/p\mathbb{Z}).$$

It may also be obtained by induction on n, using Lemma 3.6 below.

Lemma 3.6.

If U is an open subscheme of X and Y is the closed subscheme $X - U$, then

$$Z'(X) = Z'(U)Z'(Y). \tag{3.6.1}$$

Proof. This follows from the exact cohomology sequence

$$\longrightarrow H^i_c(\bar{U}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i_c(\bar{X}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow H^i_c(\bar{Y}, \mathbb{Z}/p\mathbb{Z}) \longrightarrow . \tag{3.6.2}$$

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Lemma 3.7 (Möbius).

Let B be a finite set and let $\{A_i\}$ be a finite covering by subsets. Then

$$\operatorname{card}(B) = \sum_{j\geq 0}(-1)^j \sum_{i_0<\cdots<i_j} \operatorname{card}(A_{i_0} \cap \cdots \cap A_{i_j}). \tag{3.7.1}$$

Proof. There is a Leray spectral sequence for the rational cohomology of the discrete space B and the open covering $\{A_i\}$,

$$E_1^{p,q} = \bigoplus_{i_0<\cdots<i_q} H^p(A_{i_0} \cap \cdots \cap A_{i_q}, \mathbb{Q}) \Longrightarrow H^{p+q}(B, \mathbb{Q}),$$

and one computes the Euler–Poincaré characteristic of B.

Lemma 3.8.

If $\{U_i\}$ is a finite covering of X by subschemes, then

$$Z(X) = \prod_{j\geq 0} \prod_{i_0<\cdots<i_j} Z(U_{i_0} \cap \cdots \cap U_{i_j})^{(-1)^j}. \tag{3.8.1}$$

Proof. One has

$$\operatorname{card}(X(\mathbb{F}_{q^n})) = \sum_{r|n} r.\operatorname{card}\{y \in X \mid \deg(y) = r\}, \tag{3.8.2}$$

which gives

$$Z(X) = \exp\left(\sum_{n\geq 1} \operatorname{card}(X(\mathbb{F}_{q^n}))\frac{t^n}{n}\right). \tag{3.8.3}$$

The conclusion follows from (3.7.1).

Lemma 3.9.

Suppose that formula (3.4.1), or equivalently (3.1.2) and hence (3.1.1), holds when X is a projective hypersurface. Then it holds for every separated scheme X of finite type.

Proof. By Lemmas 3.8 and 3.6, (3.4.1) holds for the complement $\mathbb{P}^n - H$ of a hypersurface. Since II≈**414**

$$\bigcap(\mathbb{P}^n - H_i) = \mathbb{P}^n - \bigcup_i H_i$$

is again the complement of a hypersurface, the open covering $\{\mathbb{P}^n - H_i\}$ of $\mathbb{P}^n - \bigcap H_i$ allows us to apply Lemmas 3.8 and 3.5 and deduce (3.4.1) for $\mathbb{P}^n - \bigcap H_i$. By Lemmas 3.8 and 3.6, (3.4.1) then holds for $\bigcap H_i$, and hence for every projective X. Another application of Lemmas 3.8 and 3.5 gives (3.4.1) for every quasi-projective X, in particular for every affine X. Finally, if X is separated and of finite type, choose a finite covering by affine open subschemes and apply Lemmas 3.8 and 3.5.

4. The calculation of the zeta function of a hypersurface by Dwork's method

4.0.

Let

$$f(X) = f(X_1, \dots, X_{n+2}) \in \mathbb{F}_q[X] \tag{4.0.1}$$

be homogeneous of degree d, and let its zero locus define the hypersurface $V \subset \mathbb{P}^{n+1}$. Denote by V^* the open subset of V on which $\prod_{i=1}^{n+2} X_i$ is invertible, and by V^*_{aff} the affine cone over V^*.

Choose a nontrivial additive character

$$\chi : \mathbb{F}_{q^\nu} \longrightarrow \mu_p(C), \tag{4.0.2}$$

where C is a fixed algebraically closed field of characteristic zero (for example $C = \mathbb{C}$). Thus

$$\text{for } x \in \mathbb{F}_{q^\nu}, \qquad \sum_{y \in \mathbb{F}_{q^\nu}} \chi(xy) = \begin{cases} q^\nu, & x = 0, \\ 0, & x \neq 0. \end{cases} \tag{4.0.3}$$

For $x = (x_1, \dots, x_{n+2}) \in (\mathbb{F}^*_{q^\nu})^{n+2}$ we therefore have

$$1 + \sum_{x_0 \in \mathbb{F}^*_{q^\nu}} \chi(x_0 f(x)) = \sum_{x_0 \in \mathbb{F}_{q^\nu}} \chi(x_0 f(x)) = \begin{cases} q^\nu, & f(x) = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{4.0.4}$$

Summing (4.0.4) over $x \in (\mathbb{F}^*_{q^\nu})^{n+2}$ gives II≈**415**

$$\operatorname{card} V^*_{\text{aff}}(\mathbb{F}_{q^\nu}) = \frac{1}{q^\nu} \sum_{x \in (\mathbb{F}^*_{q^\nu})^{n+2}} \left\{ 1 + \sum_{x_0 \in \mathbb{F}^*_{q^\nu}} \chi(x_0 f(x)) \right\}. \tag{4.0.5}$$

Since

$$\operatorname{card} V^*_{\text{aff}}(\mathbb{F}_{q^\nu}) = (q^\nu - 1) \operatorname{card} V^*(\mathbb{F}_{q^\nu}), \tag{4.0.6}$$

formula (4.0.5) becomes

$$(q^\nu - 1)\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = \frac{(q^\nu-1)^{n+2}}{q^\nu} + \frac{1}{q^\nu}\sum_{(x_0,x)\in(\mathbb{F}_{q^\nu}^*)^{n+3}} \chi(x_0 f(x)). \tag{4.0.7}$$

Write $f(X)$ explicitly as a sum of monomials,

$$f(X) = \sum a_w X^w, \tag{4.0.8}$$

where $w = (w_1, \dots, w_{n+2})$, $\sum w_i = d$, and $X^w := X_1^{w_1} \cdots X_{n+2}^{w_{n+2}}$. Then (4.0.7) may be rewritten as

$$(q^\nu - 1)\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = \frac{(q^\nu-1)^{n+2}}{q^\nu} + \frac{1}{q^\nu}\sum_{(x_0,x)\in(\mathbb{F}_{q^\nu}^*)^{n+3}} \prod_w \chi(a_w x_0 x^w). \tag{4.0.9}$$

4.1.

We must now make explicit our "additive character with values in characteristic zero." One would like simply to take

$$\chi : x \longmapsto \exp\Big(2\pi i \ \operatorname{tr}_{\mathbb{F}_{q^\nu}/\mathbb{F}_p}(x)/p\Big) = \xi_p^{\operatorname{tr}_{\mathbb{F}_{q^\nu}/\mathbb{F}_p}(x)}. \tag{4.1.0}$$

Let II≈**416**

$$\Omega = \text{an algebraic closure of } \mathbb{Q}_p, \tag{4.1.1}$$

$$\operatorname{ord} : \Omega^* \longrightarrow \mathbb{Q} \qquad \text{be the valuation normalized by } \operatorname{ord}(p) = 1, \tag{4.1.2}$$

$$\xi_p \in \Omega \qquad \text{be a primitive } p\text{-th root of unity.} \tag{4.1.3}$$

One checks (cf. [2]) that there exists

$$\pi \in \mathbb{Q}_p(\xi_p), \qquad \operatorname{ord}(\pi) = \frac{1}{p-1}, \tag{4.1.4}$$

such that

$$\sum_{n\geq 0} \frac{\pi^{p^n}}{p^n} = 0. \tag{4.1.5}$$

(Notice also that the left-hand side converges in $\mathbb{Q}_p(\xi_p)$.)
Define the Artin–Hasse exponential

$$E(X) = \exp\left(\sum_{n\geq 0} \frac{X^{p^n}}{p^n}\right) \in \mathbb{Q}_p[[X]], \tag{4.1.6}$$

which, miraculously, lies in $\mathbb{Z}_p[[X]]$. Define

$$\Theta(X) = E(\pi X). \tag{4.1.7}$$

Its fundamental properties are summarized in the following lemma.

Lemma 4.2.

$$\Theta(X) = \sum B_n X^n, \qquad B_0 = 1, \quad B_1 = \pi, \quad \operatorname{ord}(B_n) \geq \frac{n}{p-1}. \tag{4.2.1}$$

$$\Theta(1) = \xi_p, \qquad \text{a primitive } p\text{-th root of unity.} \tag{4.2.2}$$

Let $t \in \Omega$, and suppose that, for a given integer $a > 0$,

$$t = t^{p^a}; \tag{4.2.3}$$

that is, t is a Teichmüller representative in $\mathbb{W}(\mathbb{F}_{p^a}) \subset \Omega$. Then $\sum_{j=0}^{a-1} t^{p^j} \in \mathbb{Z}_p$, so the formal series

$$(\Theta(X))^{\sum_{j=0}^{a-1} t^{p^j}}$$

is defined, and one has the identity of formal power series II≈**417**

$$\prod_{j=0}^{a-1} \Theta(t^{p^j} X) = (\Theta(X))^{\sum_{j=0}^{a-1} t^{p^j}}. \tag{4.2.3.1}$$

Proof. Since $E(X) \in \mathbb{Z}_p[[X]]$, (4.2.1) follows from the definition. For (4.2.2), we have

$$\log(\Theta(1)) = 0 \qquad \text{by the choice of } \pi \text{ in (4.1.5)}, \tag{4.2.4}$$

and

$$\operatorname{ord}(\Theta(1) - 1) = \frac{1}{p-1} \qquad \text{by (4.2.1)}. \tag{4.2.5}$$

By (4.2.4), there is an integer $n \geq 0$ such that $\Theta(1)^{p^n} = 1$ (this is the characterization of the kernel of the logarithm), and (4.2.5) implies that $n = 1$.

As for (4.2.3.1), after taking logarithms it follows by an immediate calculation, once one knows that $\sum_{j=0}^{a-1} t^{p^j} \in \mathbb{Z}_p$. But the t^{p^j}, $0 \leq j \leq a-1$, are precisely the conjugates over $\mathbb{Z}_p$ of $t \in \mathbb{W}(\mathbb{F}_{p^a})$; hence

$$\sum_{j=0}^{a-1} t^{p^j} = \operatorname{Tr}_{\mathbb{W}(\mathbb{F}_{p^a})/\mathbb{Z}_p}(t) \in \mathbb{Z}_p.$$

Evaluating the two sides of (4.2.3.1) at $X = 1$ gives the following result.

Corollary 4.3.

Let $t \in \Omega$ satisfy $t^{p^a} = t$, and denote its residue class by $\bar{t} \in \mathbb{F}_{p^a}$. Then

$$\prod_{j=0}^{a-1} \Theta(t^{p^j}) = \xi_p^{\operatorname{tr}_{\mathbb{F}_{p^a}/\mathbb{F}_p}(\bar{t})}. \tag{4.3.1}$$

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4.4.

Recall that the residue field of the ring of integers of $\mathbb{Q}_p(\xi_p)$ is $\mathbb{F}_p$. Let

(4.4.0) $$K = \text{the unramified extension of } \mathbb{Q}_p(\xi_p) \text{ with residue field } \mathbb{F}_q,$$

and let

(4.4.1) $$\tau \in \mathrm{Gal}(K/\mathbb{Q}_p(\xi_p))$$

be the "Frobenius", that is, the automorphism reducing to the Frobenius automorphism of $\mathbb{F}_q/\mathbb{F}_p$, $x \longmapsto x^p$.

Let

(4.4.2) $$F(X) = \sum A_W X^W \in K[X_1, \ldots, X_{n+2}]$$

be the lifting of $f(X)$ whose coefficients satisfy $A_W^q = A_W$. Putting (4.3.1) and (4.0.9) together, and taking into account the definition (4.1.0) of χ, one obtains

(4.4.3) $$\begin{cases} (q^\nu - 1)\,\mathrm{card}\, V^*(\mathbb{F}_{q^\nu}) - \dfrac{(q^\nu-1)^{n+2}}{q^\nu} \\ \quad = \dfrac{1}{q^\nu} \displaystyle\sum_{(X_0,X)\in(\mu_{q^\nu-1}(\Omega))^{n+3}} \prod_W \prod_{j=0}^{\mathrm{ord}(q^\nu)-1} \Theta\Big(A_W^{p^j} X_0^{p^j} X^{p^j W}\Big). \end{cases}$$

Define

$$H(X) = H(X_0, \ldots, X_{n+2}) \in K[[X_0, \ldots, X_{n+2}]]$$

by

(4.4.4) $$H(X) = \prod_W \Theta\Big(A_W X_0 X_1^{W_1} \cdots X_{n+2}^{W_{n+2}}\Big).$$

We let $\mathrm{Gal}(K/\mathbb{Q}_p(\xi_p))$ act on $K[[X_0, \ldots, X_{n+2}]]$ through its action on the coefficients; the componentwise action of $\sigma \in \mathrm{Gal}(K/\mathbb{Q}_p(\xi_p))$ will be denoted

(4.4.5) $$G(X) \longmapsto G^\sigma(X).$$

In particular, II≈**419**

$$H^{\tau^j}(X) = \prod_W \Theta\Big(A_W^{\tau^j} X_0 X_1^{W_1} \cdots X_{n+2}^{W_{n+2}}\Big).$$

Since $A_W^{\tau^j} = A_W^{p^j}$, formula (4.4.3) may therefore be rewritten as

(4.4.6) $$\begin{aligned} (q^\nu - 1)\,\mathrm{Card}\, V^*(\mathbb{F}_{q^\nu}) &= \frac{(q^\nu-1)^{n+2}}{q^\nu} \\ &\quad + \frac{1}{q^\nu} \sum_{(X_0,X)\in(\mu_{q^\nu-1}(\Omega))^{n+3}} \prod_{j=0}^{\mathrm{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j}). \end{aligned}$$

4.5.

We shall now express the sum in (4.4.6) as a trace.

For each $b \in \mathbb{Q}$, define a p-adic Banach K-algebra

(4.5.1) $L(b) =$ the subring of $K[[X_0, \dots, X_{n+2}]]$ consisting of the series

which can be written

$$\sum_{dU_0 = \sum_{i \geq 1} U_i} B_U X_0^{U_0} X_1^{U_1} \cdots X_{n+2}^{U_{n+2}} \qquad (U = (U_0, \dots, U_{n+2})) \tag{4.5.2}$$

and which satisfy

$$\inf_U \big(\mathrm{ord}(B_U) - bU_0\big) > -\infty. \tag{4.5.3}$$

It is a p-adic Banach algebra for the order function (the negative of the logarithm of the norm)

$$\inf_U \big(\mathrm{ord}(B_U) - bU_0\big) = \mathrm{ord}\Big(\sum B_U X^U\Big). \tag{4.5.4}$$

It follows from (4.4.4) and (4.2.1) that

$$H(X) \in L\Big(\frac{1}{p-1}\Big).$$

Let us introduce some operators. For every integer $k > 0$, define a continuous operator between Banach spaces over K by II≈**420**

$$\begin{cases} \psi_k : L(b) \longrightarrow L(kb), \\ \psi_k\Big(\sum B_U X^U\Big) = \sum B_{kU} X^U. \end{cases} \tag{4.5.5}$$

For every pair (b, b') with $b' < b$, there is a "restriction" operator, which is completely continuous (that is, a uniform limit of finite-rank operators):

$$\begin{cases} L(b) \xrightarrow{\text{restriction}} L(b'), \\ \sum B_U X^U \longmapsto \sum B_U X^U. \end{cases} \tag{4.5.6}$$

Each element G of the algebra $L(b)$ gives the continuous endomorphism of multiplication by G,

$$\begin{aligned} G : L(b) &\longrightarrow L(b) \\ \eta &\longmapsto \eta G. \end{aligned} \tag{4.5.7}$$

These operators are related by the following lemma, whose proof is found in [2] (see also [4]).

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Lemma 4.5.8 (Dwork trace lemma).

Let $b \in \mathbb{Q}$, $b > 0$, let $k \in \mathbb{Z}$, $k \geq 2$, and let $G \in L(b/k)$. Then the composite

$$L(b) \xrightarrow{\text{restriction}} L(b/k) \xrightarrow{\times G} L(b/k) \xrightarrow{\psi_k} L(b),$$

denoted by $\psi_k \circ G$, is completely continuous, and its trace (cf. [4]) is given by

$$\operatorname{tr}(\psi_k \circ G) = \frac{1}{(k-1)^{n+2}} \sum_{X \in (\mu_{k-1}(\Omega))^{n+3}} G(X). \tag{4.5.9}$$

Apply this lemma with

$$\begin{cases} k = q^\nu, \\ G(X) = \displaystyle\prod_{j=0}^{\operatorname{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j}) \in L\Big(\frac{p}{(p-1)q^\nu}\Big), \\ b = \dfrac{1}{p-1}. \end{cases}$$

Formula (4.4.6) becomes

$$\begin{aligned} (q^\nu - 1) \operatorname{card} V^*(\mathbb{F}_{q^\nu}) &= \frac{(q^\nu-1)^{n+2}}{q^\nu} \\ &\quad + \frac{(q^\nu-1)^{n+3}}{q^\nu} \operatorname{tr}\left(\psi_{q^\nu} \prod_{j=0}^{\operatorname{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j}) \,\middle|\, L\Big(\frac{1}{p-1}\Big)\right). \end{aligned} \tag{4.5.10}$$

Let $L^0(b)$ be the ideal of the elements of $L(b)$ which "vanish at zero", that is, which have no constant term. It is stable under $\psi_k \circ G$, and evidently

$$\operatorname{tr}(\psi_k \circ G) = G(0) + \operatorname{tr}\Big(\psi_k \circ G \,\Big|\, L^0\Big). \tag{4.5.11}$$

Since $H(0) = 1$, formula (4.5.10) may be rewritten as

$$\begin{aligned} \operatorname{card} V^*(\mathbb{F}_{q^\nu}) &= (q^\nu - 1)^{n+1} \\ &\quad + \frac{(q^\nu-1)^{n+2}}{q^\nu} \operatorname{tr}\left(\psi_{q^\nu} \circ \prod_{j=0}^{\operatorname{ord}(q^\nu)-1} H^{\tau^j}(X^{p^j}) \,\middle|\, L^0\Big(\frac{1}{p-1}\Big)\right). \end{aligned} \tag{4.5.12}$$

Consider now the τ^{-1}-linear operator II≈**422**

$$\tau^{-1} \circ \psi_p \circ H(X) : L(b) \longrightarrow L(b). \tag{4.5.13}$$

Its iterates are

$$(\tau^{-1} \circ \psi_p \circ H(X))^a = \tau^{-a} \circ \psi_{p^a} \circ H(X) H^\tau(X^p) \cdots H^{\tau^{a-1}}(X^{p^{a-1}}), \tag{4.5.14}$$

and, if $\operatorname{ord}(q) \mid a$, then $\tau^{-a} = \mathrm{id}$. Define the τ^{-1}-linear operation

$$\begin{cases} \alpha : L^0\Big(\dfrac{1}{p-1}\Big) \longrightarrow L^0\Big(\dfrac{1}{p-1}\Big), \\ \qquad\qquad \alpha = \dfrac{1}{p}\, \tau^{-1} \circ \psi_p \circ H(X). \end{cases} \tag{4.5.15}$$

Then (4.5.12) takes the form

$$\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = (q^\nu - 1)^{n+1} + (q^\nu - 1)^{n+2} \operatorname{tr}\Big((\alpha)^{\operatorname{ord}(q^\nu)} \Big| L^0\Big(\frac{1}{p-1}\Big)\Big). \tag{4.5.16}$$

Put

$$S = \{1, \dots, n+2\}. \tag{4.5.17}$$

For every nonempty subset $A \subset S$, define the subspace

(4.5.18)
$L^A(b) =$ the set of series $\sum B_U X^U \in L(b)$ such that, for every monomial X^U with $B_U \neq 0$, the indices i for which $U_i > 0$ are exactly 0 and the $j \in A$.

Then

$$L^0(b) = \bigoplus_{\substack{A \subset S \\ A \neq \varnothing}} L^A(b). \tag{4.5.19}$$

And, for every nonempty subset $B \subset S$, one has II≈**423**

$$\bigoplus_{A \supset B} L^A(b) = L(b) \cap \left(\prod_{i \in B} X_i \right) K[[X]]. \tag{4.5.20}$$

Clearly every subspace $\bigoplus_{A \supset B} L^A(b)$ is stable under α. It follows that the same is true of $\bigoplus_{A \supsetneq B} L^A(b)$, which is a sum of subspaces of the preceding type. Define

(4.5.21) $\alpha_B =$ the τ^{-1}-linear endomorphism induced by α on $\dfrac{\displaystyle\bigoplus_{A \supset B} L^A\Big(\frac{1}{p-1}\Big)}{\displaystyle\bigoplus_{A \supsetneq B} L^A\Big(\frac{1}{p-1}\Big)} \simeq L^B\Big(\frac{1}{p-1}\Big).$

Explicitly, α_B is given by

$$\alpha_B = \frac{\tau^{-1}}{p}\, \psi_p\, H_B(X), \tag{4.5.22}$$

an operation under which L^B is stable, where $H_B(X)$ is obtained from $H(X)$ by substituting $X_i = 0$ for $i \notin \{0\} \cup B$.

Rewriting (4.5.16) once more gives

$$\operatorname{card} V^*(\mathbb{F}_{q^\nu}) = (q^\nu - 1)^{n+1} + (q^\nu - 1)^{n+2} \sum_{\substack{A \subset S \\ A \neq \varnothing}} \operatorname{tr}\Big((\alpha_A)^{\operatorname{ord}(q^\nu)} \Big). \tag{4.5.23}$$

For $A \subset S$, $A \neq \varnothing$, define the hypersurface

$$V_A \subset \mathbb{P}^{\operatorname{card}(A)-1} \simeq \mathbb{P}^{n+1} \cap \{X_i = 0 \text{ for every } i \notin A\} \tag{4.5.24}$$

by $f_A(X)$, the polynomial obtained from $f(X)$ by substituting $X_i = 0$ for $i \notin A$.

By compatibility of the formation of the series H from $f(X)$ with substitution of 0 for some of the variables, (4.5.23) gives, for every $A \subseteq S$, $A \neq \varnothing$, II≈**424**

$$\operatorname{card}(V_A^*(\mathbb{F}_{q^\nu})) = (q^\nu - 1)^{\operatorname{card}(A)-1} + (q^\nu - 1)^{\operatorname{card}(A)} \sum_{B \subset A} \operatorname{tr}\Big((\alpha_B)^{\operatorname{ord}(q^\nu)}\Big). \tag{4.5.25}$$

On the other hand,

$$\operatorname{card} V(\mathbb{F}_{q^\nu}) = \sum_{\substack{A \subseteq S \\ A \neq \varnothing}} \operatorname{card} V_A^*(\mathbb{F}_{q^\nu}). \tag{4.5.26}$$

After an easy summation, (4.5.25) therefore gives

$$\begin{aligned} \operatorname{card} V(\mathbb{F}_{q^\nu}) &= 1 + q^\nu + q^{2\nu} + \cdots + q^{\nu(\operatorname{card}(S)-1)} \\ &\quad + \sum_{\substack{B \subseteq S \\ B \neq \varnothing}} (q^\nu - 1)^{\operatorname{card}(B)} q^{\nu \operatorname{card}(S-B)} \operatorname{tr}\Big((\alpha_B)^{\operatorname{ord}(q^\nu)}\Big). \end{aligned} \tag{4.5.27}$$

Since α, and hence the α_B, are τ^{-1}-linear, the operators $\alpha_B^{\operatorname{ord}(q)}$ are K-linear. Define the characteristic series

$$\Delta_B(t) \in 1 + tK[[t]]$$

by

$$\Delta_B(t) = \det\Big(1 - t\alpha_B^{\operatorname{ord}(q)} \;\Big|\; L^B\Big(\frac{1}{p-1}\Big)\Big). \tag{4.5.28}$$

Introduce the following endomorphisms of $1 + tK[[t]]$:

$$\varphi : g(t) \longmapsto g^\varphi(t) = g(qt) \tag{4.5.29}$$

and

$$\delta = 1 - \varphi : g(t) \longmapsto g^\delta(t) = \frac{g(t)}{g(qt)}. \tag{4.5.30}$$

Taking account of the identities II≈**425**

$$Z(t, V/\mathbb{F}_q) = \exp\left(\sum_{\nu \geq 1} \operatorname{card}(V(\mathbb{F}_{q^\nu}))\frac{t^\nu}{\nu}\right) \tag{4.5.31}$$

and

$$\det\Big(1 - t\alpha_B^{\operatorname{ord}(q)}\Big) = \exp\left(-\sum_{\nu \geq 1} \operatorname{tr}\Big((\alpha_B)^{\nu \operatorname{ord} q}\Big)\frac{t^\nu}{\nu}\right), \tag{4.5.32}$$

formula (4.5.27) can be rewritten as

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1}(1 - q^i t) = \prod_{j=1}^{n+2} \left(\prod_{\substack{B \subset S \\ \operatorname{card}(B)=j}} \Delta_B(t)\right)^{-(-\delta)^j \varphi^{n+2-j}}. \tag{4.5.33}$$

4.6.

We now study the integrality of the series Δ_B. Put

(4.6.1) $\mathcal{O}$ = the ring of integers of K,

and define a filtration on $1 + t\mathcal{O}[[t]]$ by

(4.6.2) $$F^i(1 + t\mathcal{O}[[t]]) = 1 + q^i t\mathcal{O}[[q^i t]] = \varphi^i(1 + t\mathcal{O}[[t]]).$$

One has

(4.6.3) $$\varphi(F^i) \subset F^{i+1},$$

(4.6.4) $$\delta(F^i) \subset F^i,$$

(4.6.5) δ induces the identity on F^i/F^{i+1}.

Lemma 4.6.6.

One has $\Delta_B(t) \in 1 + t\mathcal{O}[[t]]$; moreover, if an integer j satisfies

(4.6.7) $$jd < \mathrm{card}(B),$$

then

(4.6.8) $$\Delta_B(t) \in F^j(1 + t\mathcal{O}[[t]]).$$

Proof. The Banach spaces $L^B\left(\frac{1}{p-1}\right)$ are all of type $B(I)$ in the sense of Serre [7]; that is, they admit orthonormal bases, namely the monomials II≈426

(4.6.9) $$\pi^{U_0} X_0^{U_0} X_1^{U_1} \cdots X_{n+2}^{U_{n+2}}$$

which belong to them. To say that this is an orthonormal basis means, by definition, that

(4.6.10) $$\mathrm{ord}\Big(\sum C_U \pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}\Big) = \min_U \mathrm{ord}(C_U).$$

Let

(4.6.11) $\mathbf{R}_B$ = the $\mathcal{O}$-submodule of $L^B\left(\dfrac{1}{p-1}\right)$ formed by the elements of order ≥ 0.

To prove the lemma, it is enough to show

(4.6.12) $$jd < \mathrm{card}(B) \quad \Longrightarrow \quad (\alpha_B)^{\mathrm{ord}(q)} R_B \subset q^j R_B,$$

or, a fortiori, to show

(4.6.13) $$jd < \mathrm{card}(B) \quad \Longrightarrow \quad \alpha_B(R_B) \subset p^j R_B.$$

By (4.5.22),

$$\alpha_B = \frac{\tau^{-1}}{p}\,\psi_p \circ H_B(X). \tag{4.6.14}$$

Since

$$H(X) \in L\Big(\frac{1}{p-1}\Big), \qquad \operatorname{ord}(H(X)) \geq 0, \tag{4.6.15}$$

we have

$$H_B(X) \cdot R_B \subset R_B \tag{4.6.16}$$

and, plainly,

$$\tau^{-1}(R_B) = R_B. \tag{4.6.17}$$

Consequently, (4.6.13) follows from II ≈**427**

$$jd < \operatorname{card}(B) \quad \Longrightarrow \quad \psi_p(R_B) \subset p^{j+1} R_B. \tag{4.6.18}$$

We verify (4.6.18) monomial by monomial. We have

$$\begin{cases} \psi_p\Big(\pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}\Big) = 0, & \text{unless } p \mid U_i \text{ for every } i, \\ \text{and} \\ \psi_p\Big(\pi^{pU_0} X_0^{pU_0} \cdots X_{n+2}^{pU_{n+2}}\Big) = \pi^{(p-1)U_0} \cdot \underbrace{\pi^{U_0} X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}}_{\in R_B}. \end{cases} \tag{4.6.19}$$

The conditions for $X_0^{U_0} \cdots X_{n+2}^{U_{n+2}}$ to belong to $L^B\big(\frac{1}{p-1}\big)$ are

$$\begin{cases} dU_0 = \sum_{i \geq 1} U_i, \\ U_i > 0 \Longleftrightarrow i \in B, \qquad i \geq 1. \end{cases} \tag{4.6.20}$$

In particular,

$$dU_0 \geq \operatorname{card}(B). \tag{4.6.21}$$

Hence

$$\operatorname{ord}\Big(\pi^{(p-1)U_0}\Big) = U_0 \geq \frac{\operatorname{card}(B)}{d} > j, \tag{4.6.22}$$

and, since $U_0 \in \mathbb{Z}$,

$$\operatorname{ord}\Big(\pi^{(p-1)U_0}\Big) = U_0 \geq j + 1. \tag{4.6.23}$$

Together with (4.6.19), this proves (4.6.18).

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5. A theorem of Ax [1] (see also [4], Section 7)

Theorem 5.0.

Let V be a hypersurface in $\mathbb{P}^{n+1}$ of degree $d > 0$, defined over $\mathbb{F}_q$, and let j be an integer. If

$$jd < n + 2, \tag{5.0.1}$$

then

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) \in 1 + q^j t\mathbb{Z}[[q^j t]], \tag{5.0.2}$$

and the reciprocal of every zero or pole of

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) \tag{5.0.3}$$

is of the form q^j times an algebraic integer.

Theorem 5.0.bis.

Let V_{aff} be a hypersurface in $\mathbb{A}^{n+2}$ defined by a homogeneous form of degree d, over $\mathbb{F}_q$. If

$$jd < n + 2, \tag{5.0.4}$$

then

$$Z(t, V_{\text{aff}}/\mathbb{F}_q) \in 1 + q^j t\mathbb{Z}[[q^j t]]. \tag{5.0.5}$$

In particular,

$$\operatorname{card}(V_{\text{aff}}(\mathbb{F}_q)) \equiv 0 \mod q^j, \tag{5.0.6}$$

and consequently

$$\operatorname{card}(V(\mathbb{F}_q)) \geq \frac{q^j - 1}{q - 1} > 0, \tag{5.0.6 bis}$$

which improves the Chevalley–Warning theorem [8].

Proof. The implication (5.0.5) ⇒ (5.0.6) is clear on taking the coefficient of t in the zeta function. II≈**429**
Assertions (5.0.5) and (5.0.2) are equivalent, since

$$\operatorname{card} V_{\text{aff}}(\mathbb{F}_{q^\nu}) = 1 + (q^\nu - 1) \operatorname{card} V(\mathbb{F}_{q^\nu}), \tag{5.0.7}$$

whence

$$Z(t, V_{\text{aff}}/\mathbb{F}_q) = \frac{Z(qt, V/\mathbb{F}_q)}{Z(t, V/\mathbb{F}_q)(1 - t)} \tag{5.0.8}$$

and

$$Z(t, V/\mathbb{F}_q) = \prod_{i\geq 0} \left[(1 - q^i t) Z(q^i t, V_{\mathrm{aff}}/\mathbb{F}_q)\right]^{-1}. \tag{5.0.9}$$

Assertions (5.0.2) and (5.0.3) are likewise equivalent, by the theorem of Fatou [3] and the rationality of the zeta function. Since

$$Z(t, V/\mathbb{F}_q) \in \mathbb{Z}[[t]], \tag{5.0.10}$$

it is enough to prove

$$Z(t, V/\mathbb{F}_q) \prod_{i=0}^{n+1} (1 - q^i t) \in F^j(1 + t\mathcal{O}[[t]]). \tag{5.0.11}$$

In view of (4.5.33) and (4.6.4)–(4.6.5), it is enough to show, for $\varnothing \neq B \subset \{1, \ldots, n+2\}$, that

$$\Delta_B(t) \in F^{j+\mathrm{card}(B)-(n+2)}. \tag{5.0.12}$$

We may finish the proof by applying Lemma 4.6.6 once we have proved

$$jd < n+2 \quad \Longrightarrow \quad (j + \mathrm{card}(B) - (n+2))d < \mathrm{card}(B). \tag{5.0.13}$$

This may be rewritten as

$$jd < n+2 \quad \Longrightarrow \quad (j - (n+2))d < \mathrm{card}(B)(1-d). \tag{5.0.14}$$

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$$\inf_{\varnothing \neq B \subset \{1,\ldots,n+2\}} \mathrm{card}(B)(1-d) = (n+2)(1-d),$$

so it is enough to prove

$$jd < n+2 \quad \Longrightarrow \quad (j - (n+2))d < (n+2)(1-d), \tag{5.0.15}$$

which is true.

Remark 5.1.

Condition

$$jd < n+2 \tag{5.1.1}$$

admits a cohomological interpretation. Let V be a <u>smooth</u> hypersurface of degree d and dimension n. As in XI, 2.1, put, for $a + b = n$,

$$h_0^{a,b} = \dim H^b(V, \underline{\Omega}_V^a) - \delta_{a,b} = \dim E^{a,b}(V), \tag{5.1.2}$$

where $E^{a,b}(V)$ is the bidegree-(a,b) component of the vanishing Hodge cohomology (compare XVII, 5.4.4)

$$E^{a,b}(V) = \mathrm{Coker}\Big(H^b(\mathbb{P}^{n+1}, \underline{\Omega}^a_{\mathbb{P}^{n+1}}) \hookrightarrow H^b(V, \underline{\Omega}^a_V)\Big)\,. \tag{5.1.3}$$

Then, by XI, 2.8,

$$jd < n+2 \quad \Longleftrightarrow \quad \Big[h_0^{a,n-a} = 0 \quad \text{for } a < j \text{ and } a > n-j\Big]\,. \tag{5.1.4}$$

In other words, the condition says that the Hodge level (XI, 2.7) of the vanishing de Rham cohomology of V,

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$$E^n_{\mathrm{DR}}(V) = \mathrm{Coker}\Big(H^n_{\mathrm{Hdg}}(\mathbb{P}^{n+1}) \hookrightarrow H^n_{\mathrm{Hdg}}(V)\Big)\,,$$

is at most $n-2j$, or equivalently that its coniveau is at least j. On the other hand, the conclusion also says that the eigenvalues of the Frobenius $\mathfrak{F}_q$ acting on the vanishing ℓ-adic cohomology ($\ell \neq p$) are algebraic integers divisible by q^j (XXI, A.2). Thus Ax's result 5.0 supports the following conjecture, suggested by the philosophy of the level of motives and relating the classical Hodge and Tate conjectures [5].

Conjecture 5.2.

Let $f : X \to S$ be a proper smooth morphism, with S integral, of finite type over $\mathrm{Spec}(\mathbb{Z})$, and dominant over $\mathrm{Spec}(\mathbb{Z})$; let i, j be integers. Suppose that the Hodge coniveau of H^i of the generic fiber of f is at least j. Then, for every point $s \in S$, the coniveau (XX, 2.0.6) of $H^i(X_s, \mathbb{Q}_\ell)$ is at least j. Hence (XX, 2.2), if s is a closed point of S, the eigenvalues of Frobenius acting on $H^i(X_s, \mathbb{Q}_\ell)$ are algebraic integers divisible by q^j, where $q = \mathrm{card}(k(s))$.

6. End of the proof of the congruence formula

Taking account of (4.6.6) and (4.6.3)–(4.6.4), formula (4.5.33) gives

$$Z(t, V/\mathbb{F}_q)(1-t) \equiv \Delta_S(t)^{(-1)^{n+1}} \mod 1 + qt\mathcal{O}[[t]]. \tag{6.0.1}$$

We now calculate $\Delta_S(t)$ modulo $\pi\mathcal{O}[[t]]$. Define, as in (4.6.11),

$$W_S = R_S \otimes_{\mathcal{O}} \mathbb{F}_q. \tag{6.0.2}$$

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$$\bar{\alpha}_S : W_S \longrightarrow W_S \tag{6.0.3}$$

denotes the τ^{-1}-linear operation induced by α_S;

(6.0.4) $R_S(i)$ the submodule of R_S formed by the elements of degree i in X_0,

and

$$W_S(i) = R_S(i) \otimes_{\mathcal{O}} \mathbb{F}_q. \tag{6.0.5}$$

Proposition 6.0.6.

One has

$$\alpha_S(R_S) \subset R_S(1) + pR_S.$$

Proof. We have the factorization (4.6.14)

$$\alpha_S = \tau^{-1} \circ \frac{1}{p}\,\psi_p \circ H(X). \tag{6.0.7}$$

We know (see (4.6.15)) that

$$H(X) \cdot R_S \subset R_S, \tag{6.0.8}$$

and plainly

$$\tau^{-1}(R_S(1) + pR_S) = R_S(1) + pR_S. \tag{6.0.9}$$

It is therefore enough to verify

$$\frac{1}{p}\,\psi_p(R_S) \subset R_S(1) + pR_S, \tag{6.0.10}$$

which is clear from (4.6.19).

Corollary 6.0.11.

One has

$$\bar{\alpha}_S W_S \subset W_S(1).$$

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Corollary 6.0.12.

One has the congruence

$$\Delta_S(t) \equiv \det\Big(1 - t(\bar{\alpha}_S)^{\mathrm{ord}(q)} \;\Big|\; W_S(1)\Big) \pmod{\pi\mathcal{O}[[t]]}.$$

Proposition 6.1.

The τ^{-1}-linear operation $\bar{\alpha}_S : W_S(1) \to W_S(1)$ is equal to

$$\tau^{-1}\psi_p \circ (X_0 f(X))^{p-1}. \tag{6.1.1}$$

Proof. Write

$$H(X) = 1 + \sum_{i \geq 1} H_i(X), \tag{6.1.2}$$

where

$$H_i(X) \text{ has degree } i \text{ in } X_0. \tag{6.1.3}$$

Again by (4.6.19),

$$\frac{1}{p}\,\psi_p R_S(i) \subset pR_S \qquad \text{if } i \neq p. \tag{6.1.4}$$

Since

$$H_i(X)\cdot R_S(j) \subset R_S(i+j), \tag{6.1.5}$$

we have

$$\left(\alpha_S - \frac{\tau^{-1}}{p}\,\psi_p H_{p-1}(X)\right)(R_S(1)) \subset pR_S. \tag{6.1.6}$$

Consequently,

$$\left[(\alpha_S)^{\mathrm{ord}(q)} - \left(\frac{\tau^{-1}}{p}\,\psi_p \cdot H_{p-1}(X)\right)^{\mathrm{ord}(q)}\right] R_S(1) \subset pR_S. \tag{6.1.7}$$

It remains to calculate $H_{p-1}(X)$. From (4.4.4),

$$\begin{aligned} H(X) &= \prod_W \Theta(A_W X_0 X^W) \\ &= \prod_W \exp\left(\sum_{n\geq 0} \frac{(\pi A_W X_0 X^W)^{p^n}}{p^n}\right) \\ &= \prod_W \prod_{n\geq 0} \exp\left(\frac{(\pi A_W X_0 X^W)^{p^n}}{p^n}\right) \\ &\equiv \prod_W \exp(\pi A_W X_0 X^W) \pmod{\text{terms of degree } \geq p} \\ &= \exp\Big(\pi \sum A_W X_0 X^W\Big) \pmod{\text{terms of degree } \geq p} \\ &= \exp(\pi X_0 F(X)) \pmod{\text{terms of degree } \geq p}. \end{aligned} \tag{6.1.8}$$

It follows that II≈**434**

$$H_{p-1}(X) \equiv \frac{\pi^{p-1} X_0^{p-1}}{(p-1)!}\,F(X)^{p-1}. \tag{6.1.9}$$

One easily checks that

$$\frac{\pi^{p-1}}{p} \equiv -1 \pmod{p}, \tag{6.1.10}$$

and hence

$$\frac{1}{p}\,H_{p-1}(X) \equiv (X_0 f(X))^{p-1} \pmod{p}. \tag{6.1.11}$$

This completes the proof of Proposition 6.1.

Corollary 6.1.12.

The linear operator

$$(\bar{\alpha}_S)^{\mathrm{ord}(q)} : W_S(1) \longrightarrow W_S(1)$$

is precisely

$$\psi_q \circ (X_0 f(X))^{q-1}.$$

Proof. By (6.1.1), II≈**435**

$$\begin{aligned}(\bar{\alpha}_S)^{\mathrm{ord}(q)} &= \left(\tau^{-1}\psi_p \circ (X_0 f(X))^{p-1}\right)^{\mathrm{ord}(q)} \\ &= \psi_q \circ \prod_{j=0}^{\mathrm{ord}(q)-1} \left(X_0^{p^j} f^{\tau^j}(X^{p^j})\right)^{p-1} \\ &= \psi_q \circ \prod_{j=0}^{\mathrm{ord}(q)-1} (X_0 f(X))^{p^j(p-1)} \\ &= \psi_q \circ (X_0 f(X))^{q-1}.\end{aligned}$$

Corollary 6.1.13.

Modulo p, one has the congruence

$$Z(t, V/\mathbb{F}_q) \equiv \frac{\det\left(1 - t\,\psi_q(X_0 f(X))^{q-1} \;\middle|\; W_S(1)\right)^{(-1)^{n+1}}}{1-t}.$$

6.2.

It remains to calculate the cohomology groups of $\mathcal{O}_V$. Put

(6.2.1) $\qquad \mathfrak{F}_q = \text{the operation of raising to the } q\text{-th power in } \mathcal{O}_V.$

There is an exact sequence of sheaves on $\mathbb{P}^{n+1} = \mathbb{P}$,

(6.2.2) $$0 \longrightarrow \mathcal{O}_{\mathbb{P}}(-d) \xrightarrow{f} \mathcal{O}_{\mathbb{P}} \longrightarrow \mathcal{O}_V \longrightarrow 0,$$

and the following diagram is commutative:

(6.2.3) $$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{O}_{\mathbb{P}}(-d) & \xrightarrow{f} & \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_V & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f^{q-1}\mathfrak{F}_q} & & \downarrow{\scriptstyle \mathfrak{F}_q} & & \downarrow{\scriptstyle \mathfrak{F}_q} & & \\ 0 & \longrightarrow & \mathcal{O}_{\mathbb{P}}(-d) & \xrightarrow{f} & \mathcal{O}_{\mathbb{P}} & \longrightarrow & \mathcal{O}_V & \longrightarrow & 0. \end{array}$$

Taking account of II≈**436**

(6.2.4) $$\begin{cases} H^0(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}) \simeq \mathbb{F}_q, \\ H^i(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}) = 0, & i \neq 0, \\ H^i(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) = 0, & i \neq 0, n+1, \end{cases}$$

the long exact cohomology sequence associated with (6.2.2) gives

$$\begin{cases} H^0(V,\mathcal{O}_V) \simeq \mathbb{F}_q, \\ H^i(V,\mathcal{O}_V) = 0, & i \neq 0, n, \\ H^n(V,\mathcal{O}_V) \simeq H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)). \end{cases} \tag{6.2.5}$$

The commutative diagram (6.2.3) therefore gives

$$\det\Big(1 - t\mathfrak{F}_q \;\Big|\; H^0(V,\mathcal{O}_V)\Big) = 1 - t \tag{6.2.6}$$

and

$$\det\Big(1 - t\mathfrak{F}_q \;\Big|\; H^n(V,\mathcal{O}_V)\Big) = \det\Big(1 - t(f^{q-1}\mathfrak{F}_q) \;\Big|\; H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))\Big)\,. \tag{6.2.7}$$

It remains only to prove

$$\begin{aligned} &\det\Big(1 - t(f^{q-1}\mathfrak{F}_q) \;\Big|\; H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))\Big) \\ &\qquad = \det\Big(1 - t\,\psi_q \cdot (X_0 f(X))^{q-1} \;\Big|\; W_S(1)\Big) \end{aligned} \tag{6.2.8}$$

which is a consequence of the following proposition.

Proposition 6.3.

There is an isomorphism

$$W_S(1) \simeq \mathrm{Hom}_{\mathbb{F}_q}\Big(H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)), \mathbb{F}_q\Big) \tag{6.3.1}$$

with respect to which the operator

$$\psi_q \circ (X_0 f(X))^{q-1} : W_S(1) \longrightarrow W_S(1) \tag{6.3.2}$$

is the transpose of II≈**437**

$$f(X)^{q-1}\mathfrak{F}_q : H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) \longrightarrow H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)). \tag{6.3.3}$$

Proof. Computing $H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d))$ by means of the standard covering of $\mathbb{P}^{n+1}$ gives

$$\begin{aligned} H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) = {} & \text{the quotient of the } \mathbb{F}_q\text{-vector space with basis} \\ & X_1^{W_1} \cdots X_{n+2}^{W_{n+2}} \qquad \text{with } W_i \in \mathbb{Z} \text{ satisfying } \sum W_i = -d, \\ & \text{by the } \mathbb{F}_q\text{-subspace with basis those monomials} \\ & X_1^{W_1} \cdots X_{n+2}^{W_{n+2}} \text{ for which } \exists i \text{ with } W_i \geq 0. \end{aligned} \tag{6.3.4}$$

On the other hand, $W_S(1) = R_S(1) \otimes_{\mathcal{O}} \mathbb{F}_q$, and

$$\begin{aligned} R_S(1) = {} & \text{the free } \mathcal{O}\text{-module with basis the monomials} \\ & (\pi X_0) X_1^{U_1} \cdots X_{n+2}^{U_{n+2}} \text{ satisfying } \sum U_i = d, \quad U_i > 0 \ \forall_i. \end{aligned} \tag{6.3.5}$$

The pairing

$$(\ ,\ ) : W_S(1) \times H^{n+1}(\mathbb{P}^{n+1}, \mathcal{O}_{\mathbb{P}}(-d)) \longrightarrow \mathbb{F}_q \tag{6.3.6}$$

defined by

$$\Big(\text{class of } \pi X_0 X_1^{U_1} \cdots X_{n+2}^{U_{n+2}}, X_1^{W_1} \cdots X_{n+2}^{W_{n+2}}\Big) = \begin{cases} 1, & U_i = -W_i \text{ for every } i, \\ 0, & \text{otherwise,} \end{cases} \tag{6.3.7}$$

is plainly a perfect duality of $\mathbb{F}_q$-vector spaces. One checks immediately that $\psi_q \cdot (X_0 f(X))^{q-1}$ is the transpose of $f(X)^{q-1} \cdot \mathfrak{F}_q$, which proves Proposition 6.3.

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